Weil Conjectures

The complete blueprint

Appendices included

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Let X be a smooth projective variety of dimension d over $\mathbf{F}_q$. The Weil conjectures assert that its zeta function, which counts the points of X over all the fields $\mathbf{F}_{q^m}$, is a rational function, that it satisfies a functional equation, and that its zeros lie on the circles $|T| = q^{-i/2}$ with i odd and its poles on those with i even, an analogue of the Riemann hypothesis. The points over $\mathbf{F}_{q^m}$ are the fixed points of the mth power of Frobenius. Weil therefore expected the first two statements to follow from a Lefschetz fixed point formula and Poincaré duality in a cohomology theory with coefficients of characteristic zero. These notes construct such a theory and prove all three statements.

The theory is ℓ-adic cohomology. Étale cohomology works best with the torsion coefficients $\mathbf{Z}/\ell^n$, but traces must be taken in characteristic zero. On the pro-étale site of Bhatt and Scholze, $\mathbf{Z}_\ell$ and $\mathbf{Q}_\ell$ define sheaves of rings just as $\mathbf{Z}/\ell^n$ does, so one set of definitions and one formalism of six operations serve every coefficient ring. The trace formula and Poincaré duality then give rationality and the functional equation.

The Riemann hypothesis follows from the Deligne–Laumon theorem, which bounds the absolute values of Frobenius eigenvalues, the weights, on the cohomology of sheaves. We follow Laumon's proof: fibring by curves reduces the theorem to curves, where the ℓ-adic Fourier transform and a positivity argument that Deligne took from Rankin bound the weights.

The documents speak some of the most important definitions and theorems in our lean formalization. They quote some foundational results from algebraic geometry and from étale cohomology with torsion coefficients, each stated precisely. We assume a first course in algebraic geometry and some familiarity with derived categories and singular cohomology, but no étale cohomology.

1 Introduction

These notes prove the Weil conjectures, which concern the number of solutions of polynomial equations over finite fields. Weil saw that rationality and the functional equation would follow from a cohomology theory for varieties over finite fields with a Lefschetz fixed point formula and Poincaré duality, while the deepest conjecture, an analogue of the Riemann hypothesis, needs in addition a bound on the eigenvalues of Frobenius. Following this plan, we construct such a theory, ℓ-adic cohomology, and prove the trace formula and Poincaré duality for it. Then we prove the bound, which is the Deligne–Laumon theorem.

1.1 The Weil conjectures

Let X_0 be a smooth projective geometrically connected curve of genus g over the finite field $\mathbf{F}_q$, and let N_m be the number of its points with coordinates in $\mathbf{F}_{q^m}$. The projective line has exactly $1 + q^m$ such points, and the *Hasse–Weil bound* says that every curve of genus g comes close to this count:

$$|N_m - (1 + q^m)| \leq 2g\, q^{m/2}.$$

Hasse proved it for elliptic curves [Has36] and Weil for all curves [Wei48a].

To state Weil's conjectures in higher dimension we put all the numbers N_m into one generating function. For a separated scheme X_0 of finite type over $\mathbf{F}_q$, its *zeta function* is

$$Z(X_0, T) = \exp\Big(\sum_{m \geq 1} \#X_0(\mathbf{F}_{q^m}) \frac{T^m}{m}\Big) \in 1 + T\mathbf{Q}[[T]].$$

A closed point x of X_0 with residue field $\mathbf{F}_{q^{\deg x}}$ accounts for exactly $\deg x$ points of $X_0(\mathbf{F}_{q^m})$ when $\deg x$ divides m, and for none otherwise. Hence $\#X_0(\mathbf{F}_{q^m}) = \sum_{\deg x | m} \deg x$, so

$$\sum_{m \geq 1} \#X_0(\mathbf{F}_{q^m}) \frac{T^m}{m} = \sum_{x \in |X_0|} \sum_{k \geq 1} \frac{T^{k \deg x}}{k} = - \sum_{x \in |X_0|} \log(1 - T^{\deg x}).$$

Only finitely many closed points have a given degree, so exponentiating gives the Euler product

$$Z(X_0, T) = \prod_{x \in |X_0|} (1 - T^{\deg x})^{-1},$$

whose factors are power series with integer coefficients. So the coefficients of $Z(X_0, T)$ are integers. For example, $\#\mathbf{P}^1(\mathbf{F}_{q^m}) = 1 + q^m$ gives $Z(\mathbf{P}^1, T) = 1/((1-T)(1-qT))$. Set $T = q^{-s}$. Since the residue field of x has $N(x) = q^{\deg x}$ elements, the Euler product becomes

$$\zeta_{X_0}(s) = Z(X_0, q^{-s}) = \prod_{x \in |X_0|} (1 - N(x)^{-s})^{-1},$$

the analogue of Riemann's

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1},$$

in which the closed points of $\operatorname{Spec} \mathbf{Z}$ have residue fields with p elements. Riemann conjectured that the nontrivial zeros of $\zeta(s)$ lie on the line $\operatorname{Re} s = \frac{1}{2}$. Weil conjectured the following.

1. *Rationality.* $Z(X_0, T)$ is a rational function of T.
2. *Functional equation.* If X_0 is smooth and proper of pure dimension d, then $Z(X_0, 1/(q^d T)) = \pm q^{d\chi/2} T^\chi Z(X_0, T)$, where $\chi = \sum_i (-1)^i b_i$ is the Euler characteristic, with b_i as in (4). For a curve of genus g it equals $2 - 2g$.
3. *Riemann hypothesis.* If X_0 is smooth and proper of pure dimension d, then

$$Z(X_0, T) = \frac{P_1(T) P_3(T) \cdots P_{2d-1}(T)}{P_0(T) P_2(T) \cdots P_{2d}(T)},$$

   where $P_i \in \mathbf{Z}[T]$ has constant term 1, and its reciprocal roots (the numbers α with $P_i(1/\alpha) = 0$) are algebraic integers all of whose complex conjugates have absolute value $q^{i/2}$.
4. *Betti numbers.* The degree b_i of P_i behaves like a Betti number. For instance, $b_i = b_{2d-i}$, as for Poincaré duality on a compact manifold of real dimension $2d$. More precisely, suppose that X_0 comes from characteristic zero: there is a smooth projective variety Y over a number field K and a prime $\mathfrak{p}$ of K with residue field $\mathbf{F}_q$ such that Y extends to a smooth projective scheme over the local ring $\mathcal{O}_{K,\mathfrak{p}}$ whose special fibre is X_0. Then b_i is the ith Betti number of the compact complex manifold $Y(\mathbf{C})$, for any embedding $K \subset \mathbf{C}$. For a curve of genus g this gives $b_0 = b_2 = 1$ and $b_1 = 2g$.

For a curve the conjectures recover the Hasse–Weil bound. Here $Z(X_0,T) = P_1(T)/((1-T)(1-qT))$ with P_1 of degree $2g$, and statement (3) says that $P_1(T) = \prod_{j=1}^{2g}(1-\alpha_j T)$ with $|\alpha_j| = q^{1/2}$. Comparing the logarithmic derivatives of both sides with the definition of Z gives

$$N_m = 1 + q^m - \sum_{j=1}^{2g} \alpha_j^m,$$

and since each $|\alpha_j^m| = q^{m/2}$, the Hasse–Weil bound follows. This is also why (3) is called the Riemann hypothesis. In the variable s with $T = q^{-s}$, the zeros of $P_1(q^{-s})$ satisfy $|q^{-s}| = q^{-1/2}$, that is, they lie on the line $\operatorname{Re} s = \frac{1}{2}$, exactly where Riemann conjectured the zeros of $\zeta(s)$ to lie.

We follow one example through the notes, the elliptic curve

$$C_0 \colon\ y^2 = x^3 - x \qquad \text{over } \mathbf{F}_3.$$

Counting points, including the point at infinity, gives $N_1 = 4$, $N_2 = 16$ and $N_3 = 28$. Suppose the Weil conjectures hold for C_0. Then $Z(C_0,T) = P_1(T)/((1-T)(1-3T))$, where P_1 has degree $b_1 = 2g = 2$. Its leading coefficient is $q = 3$, because by duality it is the determinant of Frobenius on H^1 (Example 10.23). So $P_1 = 1 - aT + qT^2$, and comparing the coefficients of T in Z gives $a = 1 + q - N_1 = 0$. Hence $P_1(T) = 1 + 3T^2$, whose reciprocal roots $\pm i\sqrt{3}$ have absolute value $3^{1/2}$. Statements (1)–(4) alone would not fix the leading coefficient: $1 - 3T^2$ also satisfies them, with the sign $-$ in (2), and it is the count $N_2 = 16$ that excludes it. As a check, the formula $N_m = 1 + 3^m - \big((i\sqrt{3})^m + (-i\sqrt{3})^m\big)$ gives 16 and 28 for $m = 2, 3$. We return to this curve after duality, after the trace formula and at the end (Examples 10.23, Example 12.32 and Example 19.9).

1.2 Weil cohomology

Write $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let $F : X \to X$ be the *Frobenius*: if X_0 is cut out of an affine space by equations with coefficients in $\mathbf{F}_q$, then F raises each coordinate to its q-th power. A point of X is fixed by F^m exactly when its coordinates lie in $\mathbf{F}_{q^m}$, so $\#X_0(\mathbf{F}_{q^m})$ is the number of fixed points of F^m. Weil's idea was to count these fixed points as in topology.

Suppose that we had a cohomology theory with compact supports, assigning to X finite-dimensional vector spaces $H_c^i(X)$ over a field of characteristic zero on which F acts. Compact supports are needed because the trace formula below has to hold for every X_0, proper or not: point counts are additive when X_0 is cut into an open subscheme and its complement, and compactly supported cohomology has the matching long exact sequence. For proper X_0 we write $H^i(X) = H_c^i(X)$; for the conjectures, which concern smooth proper varieties, this is the only cohomology we need. Suppose that the theory satisfies the following.

- (a) *Finiteness.* $H_c^i(X) = 0$ unless $0 \le i \le 2\dim X$.
- (b) *Trace formula.* The fixed points of F^m are counted by a Lefschetz formula,

$$\#X_0(\mathbf{F}_{q^m}) = \sum_i (-1)^i \operatorname{Tr}(F^m \mid H_c^i(X)).$$

- (c) *Poincaré duality.* For smooth proper X_0 of pure dimension d, cup product gives a perfect pairing between H^i and H^{2d-i}, and $\langle Fx, Fy\rangle = q^d\langle x, y\rangle$. The factor q^d is the degree of F. It is the scalar by which F acts on the one-dimensional space H^{2d} when X is connected.
- (d) *Purity.* For smooth proper X_0, the eigenvalues of F on $H^i(X)$ are algebraic integers all of whose complex conjugates have absolute value $q^{i/2}$.

Then the conjectures follow. For rationality we compare logarithms. For an endomorphism F of a finite-dimensional vector space V we have

$$\log\det(1-TF\mid V)^{-1}=\sum_{m\geq 1}\operatorname{Tr}(F^m\mid V)\frac{T^m}{m},$$

as one sees by triangulating F: if F has eigenvalues $\alpha_1,\dots,\alpha_b$, both sides equal $\sum_j\sum_m \alpha_j^m T^m/m$. Substituting the trace formula (b) into the definition of Z therefore gives

$$\log Z(X_0,T)=\sum_{m\geq 1}\sum_i(-1)^i\operatorname{Tr}\big(F^m\mid H_c^i(X)\big)\frac{T^m}{m}=\sum_i(-1)^{i+1}\log\det\big(1-TF\mid H_c^i(X)\big),$$

and so

$$Z(X_0,T)=\prod_i\det\big(1-TF\mid H_c^i(X)\big)^{(-1)^{i+1}},$$

a finite product by (a). In particular $Z(X_0,T)$ is rational, and for smooth proper X_0 the polynomial $P_i(T)$ of the conjectures is $\det(1-TF\mid H^i(X))$. By duality, the eigenvalues in degree $2d-i$ are the numbers q^d/α for the eigenvalues α in degree i, and this gives the functional equation. Purity gives the Riemann hypothesis.

Purity also shows that the factors do not depend on the cohomology theory. Indeed, the factors in different degrees have reciprocal roots of different absolute values, so no cancellation can occur in the alternating product. Each factor is therefore determined by $Z(X_0,T)$ alone, and so is its degree b_i.

Which cohomology could this be? On an irreducible scheme a constant sheaf is flasque, so its Zariski cohomology vanishes in positive degrees. It cannot see the $2g$ independent loops of a curve of genus g. Coherent cohomology sees $H^1(C,\mathcal{O}_C)$, of dimension g. But this is a vector space over $\overline{\mathbf{F}}_q$, and traces on it determine numbers of points only modulo p.

An argument of Serre (see [Den22]) shows that the coefficient field cannot be $\mathbf{Q}$, $\mathbf{R}$ or $\mathbf{Q}_p$ either. Let C be a supersingular elliptic curve over $\overline{\mathbf{F}}_p$. Then $D=\operatorname{End}(C)\otimes\mathbf{Q}$ is a quaternion algebra over $\mathbf{Q}$, and $D\otimes\mathbf{R}$ and $D\otimes\mathbf{Q}_p$ are still division algebras [Deu41]. Endomorphisms of C would act on $H^1(C)$, a vector space of dimension $2g=2$ over the coefficient field K. This gives a map of four-dimensional K-algebras $D\otimes K\to M_2(K)$. The map is injective because $D\otimes K$ is simple, hence it is an isomorphism. That is impossible if $D\otimes K$ is a division algebra. For $\ell\neq p$, on the other hand, $D\otimes\mathbf{Q}_\ell\simeq M_2(\mathbf{Q}_\ell)$, and the argument does not exclude $\mathbf{Q}_\ell$.

The theory that works is étale cohomology. Its idea is to replace the open subsets of the Zariski topology by étale maps $U\to X$, which play the role of small open neighbourhoods in topology: like a covering space, an étale map is a local isomorphism in the sense of the implicit function theorem. A sheaf in this topology assigns a group to every étale map $U\to X$, compatibly with restriction, and étale cohomology is the derived functor of global sections. Because there are many more étale maps than open subsets, a sheaf can now have nonzero cohomology in positive degrees.

For a curve we can already see the missing loops in degree one. Let C be a smooth projective connected curve of genus g over $\overline{\mathbf{F}}_q$, and let n be prime to q. Three étale sheaves are involved: $\mathbf{G}_m$, which sends $U\to C$ to the unit group $\mathcal{O}(U)^\times$; the same sheaf with the nth power map $t\mapsto t^n$; and its kernel μ_n, the sheaf of nth roots of unity. The sequence

$$0 \to \mu_n \to \mathbf{G}_m \xrightarrow{t \mapsto t^n} \mathbf{G}_m \to 0$$

is exact in the étale topology. Indeed, surjectivity of a map of sheaves only has to hold locally, and an nth root of a unit a exists on the étale cover defined by $t^n = a$, which is étale because n is invertible. (In the Zariski topology the last map is not surjective.) As in the Zariski topology, $H^1(C_{ét}, \mathbf{G}_m)$ is the Picard group of C (Hilbert's Theorem 90). The global units of C are $\overline{\mathbf{F}}_q^\times$, in which nth roots exist, so $H^0(\mathbf{G}_m) \xrightarrow{n} H^0(\mathbf{G}_m)$ is surjective. The long exact sequence therefore identifies $H^1(C_{ét}, \mu_n)$ with the kernel of multiplication by n on $H^1(C_{ét}, \mathbf{G}_m) = \mathrm{Pic}(C)$.

To compute this kernel we use the degree. It gives an exact sequence $0 \to \mathrm{Pic}^0(C) \to \mathrm{Pic}(C) \xrightarrow{\deg} \mathbf{Z} \to 0$, and since $\mathbf{Z}$ has no torsion, every line bundle killed by n has degree zero. So $\mathrm{Pic}(C)[n] = \mathrm{Pic}^0(C)[n]$. The group $\mathrm{Pic}^0(C)$ is the group of points of the *Jacobian* J of C, an abelian variety of dimension g (Section C.4), and on an abelian variety of dimension g the n-torsion points form a group $(\mathbf{Z}/n)^{2g}$ when n is prime to the characteristic (Proposition B.14). Altogether,

$$H^1(C_{ét}, \mu_n) \simeq \mathrm{Pic}(C)[n] = J[n] \simeq (\mathbf{Z}/n)^{2g}$$

(Proposition D.3), which is the size of H^1 of a compact Riemann surface of genus g with coefficients $\mathbf{Z}/n$. Now let $n = \ell^k$ and let $k \to \infty$. The groups $J[\ell^k]$, with the maps $J[\ell^{k+1}] \to J[\ell^k]$ given by multiplication by ℓ, form the *Tate module* $T_\ell J = \varprojlim_k J[\ell^k]$, free of rank $2g$ over $\mathbf{Z}_\ell$ (Proposition B.14). On the cohomology side these maps correspond to $\mu_{\ell^{k+1}} \to \mu_{\ell^k}$, $z \mapsto z^\ell$, so the identification above passes to the limit:

$$\varprojlim_k H^1(C_{ét}, \mu_{\ell^k}) \simeq T_\ell J$$

(Proposition C.17). In other words, the Tate module is the first cohomology of C with coefficients in the inverse system $(\mu_{\ell^k})_k$, which is written $\mathbf{Z}_\ell(1)$ (we come back to what this limit means below). Weil used $T_\ell J$ in his proof for curves, in place of an H^1 that was not yet available.

Frobenius acts on both sides. Let π be the endomorphism of J induced by Frobenius; it raises the coordinates of torsion points to the q-th power, and under the isomorphism above it is the action of the arithmetic Frobenius $\sigma : z \mapsto z^q$ in $\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ on the torsion points. The map F on cohomology is pullback along the Frobenius of C, which is the inverse of this Galois action on $\overline{\mathbf{F}}_q$-points (we compare the two Frobenius maps in Section 12.1). Hence F acts on $\varprojlim_k H^1(C_{ét}, \mu_{\ell^k})$ by π^{-1}. Weil showed [Wei48b] that the characteristic polynomial of π on $T_\ell J$ has integer coefficients, and that the numerator of the zeta function is

$$P_1(T) = \det(1 - T\pi \mid T_\ell J).$$

Because of the twist, F acts on the untwisted group $H^1(C, \mathbf{Q}_\ell)$ by $q\pi^{-1}$ (Section 10.5). Its eigenvalues q/α run through the eigenvalues α of π again, by the functional equation, so also $P_1(T) = \det(1 - TF \mid H^1(C, \mathbf{Q}_\ell))$, as predicted in Section 1.1. For the curve C_0 above, π acts on $T_\ell J$ with characteristic polynomial $T^2 + 3$, and indeed $P_1(T) = 1 + 3T^2$.

The example suggests how to obtain a theory with coefficients of characteristic zero: compute étale cohomology with the finite coefficients $\mathbf{Z}/\ell^k$, where the theory works, take the inverse limit over k to obtain $\mathbf{Z}_\ell$-modules, and finally invert ℓ. The limit has to be taken with care, as $\mathrm{Spec}\,\mathbf{F}_q$ already shows. On it, étale sheaves are sets with a continuous action of the Galois group $\widehat{\mathbf{Z}}$, and H^1 with coefficients in an abelian group A is the group of continuous homomorphisms $\widehat{\mathbf{Z}} \to A$. For $A = \mathbf{Z}/\ell^k$ this is $\mathbf{Z}/\ell^k$, and the limit of these groups is $\mathbf{Z}_\ell$. But the étale topology sees $\mathbf{Z}_\ell$

only as a discrete group, and a continuous homomorphism from the compact group $\widehat{\mathbf{Z}}$ to the discrete torsion-free group $\mathbf{Z}_\ell$ is zero, so the constant sheaf $\mathbf{Z}_\ell$ has $H^1 = 0$. What one wants, therefore, is the limit of the cohomology groups, not the cohomology of the limit sheaf. After inverting ℓ this gives a theory with coefficients in $\mathbf{Q}_\ell$. However, the eigenvalues of Frobenius lie in an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$ rather than in $\mathbf{Q}_\ell$ itself, and so we also allow coefficients in $\overline{\mathbf{Q}}_\ell$, where characteristic polynomials split.

The resulting theory is ℓ-adic cohomology, for a prime ℓ not dividing q. It was constructed by Grothendieck and his collaborators, who proved (a), (b) and (c) for it in SGA 4 and SGA 5 [SGA4], [SGA5]. Purity (d) is the hard part, and Deligne proved it [Del74], [Del80]. The general form of his theorem, on weights of $Rf_!$, was later reproved by Laumon with the ℓ-adic Fourier transform [Lau87] (see also [Kat88], [KW01]); we call it the Deligne–Laumon theorem. Our proof uses the Fourier transform as Laumon does, but it is organized differently in places.

Of statement (4) we prove that $b_i = b_{2d-i}$. We also prove that $b_i = \dim H^i(X, \mathbf{Q}_\ell)$ for every ℓ not dividing q; in particular this dimension does not depend on ℓ (Theorem 19.7). The comparison with the Betti numbers of a lift to characteristic zero needs the comparison theorem between étale and singular cohomology, which we do not prove.

1.3 The function–sheaf dictionary

The proof of purity does not treat one variety at a time. It cuts a variety into pieces and fibres it over other varieties, and point counts behave simply under both operations. If $Z_0 \subset X_0$ is closed with complement U_0, then $\#X_0(\mathbf{F}_q) = \#U_0(\mathbf{F}_q) + \#Z_0(\mathbf{F}_q)$, and for a morphism $f : X_0 \to Y_0$,

$$\#X_0(\mathbf{F}_q) = \sum_{y \in Y_0(\mathbf{F}_q)} \#f^{-1}(y)(\mathbf{F}_q).$$

On the right we sum a function on Y_0 that is not constant. To follow such sums with cohomology, we replace functions by sheaves.

The sheaves in question are *constructible*: there is a decomposition of X into finitely many locally closed pieces on each of which the sheaf is locally constant with finite stalks. After passing to the limit, the stalks are finite-dimensional $\mathbf{Q}_\ell$-vector spaces instead. A *Frobenius structure* on a constructible sheaf K on X is an isomorphism $F^*K \simeq K$, and a sheaf with a Frobenius structure is a *Weil sheaf*. The same definitions apply to complexes of sheaves (Section 10.5). Let x be a point of X_0 over $\mathbf{F}_{q^m}$. The corresponding point $\bar{x}$ of X is fixed by F^m, so the stalk of $(F^m)^*K$ at $\bar{x}$ is $K_{\bar{x}}$. Iterating the Frobenius structure gives an isomorphism $(F^m)^*K \simeq K$, and on the stalk at $\bar{x}$ it becomes an endomorphism Frob_x of $K_{\bar{x}}$. Its trace defines the *trace function*

$$t_K : X_0(\mathbf{F}_{q^m}) \longrightarrow \mathbf{Q}_\ell, \qquad x \longmapsto \mathrm{Tr}(\mathrm{Frob}_x \mid K_{\bar{x}}),$$

where for a complex the trace is the alternating sum of the traces on the cohomology groups. The constant sheaf has trace function 1.

Operations on functions correspond to operations on sheaves. For example, pulling back a function along f corresponds to the inverse image f^*K, and the product of two functions to the tensor product. Similarly, summing a function over the fibres of f corresponds to the direct image with compact supports $Rf_!K$, whose stalk at a point $\bar{y}$ is the compactly supported cohomology of K on the fibre over $\bar{y}$. The resulting identity $t_{Rf_!K}(y) = \sum_{f(x)=y} t_K(x)$ is a form of the trace formula (Corollary 12.26). When f is the map to a point, it is (b) with coefficients in K. The one further operation we need is a Fourier transform, explained in Section 1.6. This correspondence between functions and sheaves is Grothendieck's *function–sheaf dictionary* [SGA4½, Sommes trig.]:

functions on $X_0(\mathbf{F}_{q^m})$	Weil complexes on X
$t \mapsto t \circ g$	g^*K
$t \cdot t'$	$K \otimes^{\mathbf{L}} L$
$t \mapsto \big(y \mapsto \sum_{f(x)=y} t(x)\big)$	$Rf_!K$
$t = t\vert_U + t\vert_Z$	$j_!j^*K \to K \to i_*i^*K \to$
$\sum_x t(x)$	$R\Gamma_c(X, K)$ with its Frobenius
$t \mapsto \big(y \mapsto -\sum_x t(x)\psi(\mathrm{Tr}_{\mathbf{F}_{q^m}/\mathbf{F}_q}(xy))\big)$	$Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[1]$

Proposition 12.28 states and proves the first five lines. The last line is built from the first three, as we explain in Section 1.6. Since $Rf_!$ of a sheaf is a complex, we work in derived categories throughout.

1.4 The six operations

Duality has no simple counterpart for functions, and it needs two more operations: the internal Hom of sheaves and the *exceptional inverse image* $f^!$, a right adjoint of $Rf_!$. With f^*, Rf_*, $Rf_!$ and $\otimes$ these are Grothendieck's *six operations*. Part I of these notes, roughly their first third, constructs them for all the coefficient rings we need. To see why $f^!$ is the right tool, consider Poincaré duality for a smooth variety X of pure dimension d over an algebraically closed field k, with coefficients $\Lambda = \mathbf{Z}/\ell^n$. It should say that cup product

$$H^i_c(X, \Lambda) \times H^{2d-i}(X, \Lambda(d)) \longrightarrow H^{2d}_c(X, \Lambda(d)) \longrightarrow \Lambda$$

is a perfect pairing. Here $\Lambda(d) = \mu_{\ell^n}^{\otimes d}$ is the *Tate twist* (Definition 2.53). The twist is needed because Frobenius does not act trivially on top cohomology. For instance, the map $F : \mathbf{P}^1 \to \mathbf{P}^1$ has degree q. It pulls the class of a point back to q times itself, so F acts on $H^2(\mathbf{P}^1, \Lambda)$ by q. With our conventions F acts on $\Lambda(1)$ by q^{-1} (Section 2.12), and therefore it acts trivially on $H^2(\mathbf{P}^1, \Lambda(1))$. In general the twist (d) removes the factor q^d from the pairing.

For an oriented manifold V of dimension r there is a clean proof, due to Verdier. Let $a : V \to \mathrm{pt}$ be the map to a point. The functor $Ra_!$, cohomology with compact supports, has a right adjoint $a^!$, and the adjunction gives, for every sheaf G on V,

$$R\mathrm{Hom}\big(R\Gamma_c(V, G), \Lambda\big) \simeq R\mathrm{Hom}(G, a^!\Lambda).$$

This isomorphism comes from the adjunction alone and holds for every V; what makes it a duality theorem is a computation of $a^!\Lambda$. It is a complex of sheaves on V, and its restriction to an open subset U is the corresponding complex for U. So it can be computed in a chart $\mathbf{R}^r$. There its only cohomology sheaf sits in degree $-r$, with stalks $\mathrm{Hom}(H^r_c(\mathbf{R}^r, \Lambda), \Lambda) \simeq \Lambda$. An orientation identifies these stalks consistently, and this gives $a^!\Lambda \simeq \Lambda[r]$. For $G = \Lambda$ the displayed isomorphism then reads $R\mathrm{Hom}(R\Gamma_c(V, \Lambda), \Lambda) \simeq R\mathrm{Hom}(\Lambda, \Lambda[r]) = R\Gamma(V, \Lambda)[r]$. Taking cohomology in degree $-i$ and using that Λ is a field or, for $\Lambda = \mathbf{Z}/\ell^n$, a self-injective ring, this says $\mathrm{Hom}(H^i_c(V, \Lambda), \Lambda) \simeq H^{r-i}(V, \Lambda)$, which is Poincaré duality for V.

We follow the same route. For the structure morphism a of a smooth variety X of dimension d we show $a^!\Lambda \simeq \Lambda_X(d)[2d]$, again by a local computation (*smooth purity*). Poincaré duality follows.

The main theorem of Part I is that $f^!$ exists and carries constructible complexes to constructible complexes (Theorem 5.2). Existence is a general fact. Indeed, to give a right adjoint of $Rf_!$ is to find, for each N, an object $f^!N$ with $\mathrm{Hom}(M, f^!N) = \mathrm{Hom}(Rf_!M, N)$, naturally in M, and Brown's representability theorem provides such objects for an exact functor between triangulated

categories that commutes with arbitrary direct sums. It needs one hypothesis on the source: it must be *compactly generated*, that is, built by shifts, cones and direct sums from a set of objects M for which $\mathrm{Hom}(M, -)$ commutes with direct sums. Such objects are called compact; in the derived category of a ring, for example, the perfect complexes are the compact objects. Constructible complexes are such objects, but they are not closed under direct sums. We therefore apply the theorem to the larger category built from them in this way.

It remains to show that $f^!$ carries constructible complexes to constructible complexes, and this is geometric. Locally f is a closed immersion followed by a projection from affine space, and affine space is open in a product of projective lines, so two cases remain. For the projection $\mathbf{P}^1_Y \to Y$, we compute $f^!$ explicitly, whereas for a closed immersion i with open complement j, a distinguished triangle computes $i^!K$ from K and Rj_*j^*K. This case therefore comes down to the theorem of Deligne that the ordinary direct image Rj_* preserves constructible complexes.

The six operations must also be available for $\mathbf{Z}_\ell$- and $\mathbf{Q}_\ell$-coefficients, compatibly with reduction modulo ℓ^n. This is a problem, because inverse limits behave badly on the étale site: on $\operatorname{Spec}\mathbf{F}_q$, as we saw in Section 1.2, the constant sheaf $\mathbf{Z}_\ell$ has $H^1 = 0$, although the limit of the groups $H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}/\ell^k)$ is $\mathbf{Z}_\ell$. For this reason the classical theory works with compatible systems of sheaves modulo ℓ^n and builds the derived categories by hand [Eke90]. We use instead the pro-étale site of Bhatt and Scholze [BS15]. Its objects over X include, besides the étale maps $U \to X$, inverse limits $\varprojlim_i U_i \to X$ of affine étale maps, for example the spectrum of an algebraic closure of $\mathbf{F}_q$ over $\operatorname{Spec}\mathbf{F}_q$, and its coverings are the families that are jointly surjective in a strong sense. Every étale sheaf defines a sheaf on it, and every bounded below complex of étale sheaves has the same cohomology on the two sites (Chapter 2). The gain is that $\mathbf{Z}_\ell$, with its topology, becomes a sheaf whose cohomology is the limit: on the pro-étale site $H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}_\ell) = \mathbf{Z}_\ell$ (Example 2.57). On it, $\mathbf{Z}_\ell$ is a sheaf of rings whose sections are continuous functions to $\mathbf{Z}_\ell$, so that an ℓ-adic sheaf is simply a sheaf of modules over it, and we use the usual derived category of these modules, with no inverse systems. In this setting all the geometric arguments are made with coefficients $\mathbf{Z}/\ell^n$, where the classical theory applies, and the pro-étale site then carries them over to $\mathbf{Z}_\ell$, $\mathbf{Q}_\ell$ and $\overline{\mathbf{Q}}_\ell$.

1.5 The trace formula

Part II proves Poincaré duality and the trace formula. Why should the Lefschetz formula hold for Frobenius? For a self-map φ of a compact manifold V whose fixed points are transverse, the Lefschetz fixed point formula reads

$$\sum_i (-1)^i \operatorname{Tr}(\varphi^* \mid H^i(V, \mathbf{Q})) = \sum_{\varphi(x)=x} \operatorname{sign}\det(1 - d\varphi_x),$$

so each fixed point x is counted with the sign of $\det(1 - d\varphi_x)$. In our case the differential of Frobenius is zero, because $d(u^q) = qu^{q-1}du = 0$ in characteristic p. So every fixed point should count with $+1$. On a curve one can see this directly. Let u be a local coordinate at a fixed point, and v the same coordinate on the second factor of $X \times X$. In these coordinates the graph of F^m is $v = u^{q^m}$, and it meets the diagonal $v = u$ with multiplicity one.

The Grothendieck–Lefschetz trace formula states that for every Weil complex K,

$$\operatorname{Tr}(F^m \mid R\Gamma_c(X, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} t_K(y)$$

(Theorem 12.24). For the constant sheaf it is axiom (b). In Example 12.32 we apply it to the elliptic curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$ of Section 1.1, where it reads $N_m = 1 - \operatorname{Tr}(F^m \mid H^1) + 3^m$.

Since $N_1 = 4$, the trace of F on H^1 is 0. Duality gives $\det(F \mid H^1) = 3$, and we recover $Z(C_0, T) = (1 + 3T^2)/((1 - T)(1 - 3T))$.

We deduce the trace formula, as in SGA 5 [SGA5, III, III B, XV], from a general Lefschetz formula. A *correspondence* on X is a pair of maps $c_1, c_2 : C \to X$. For instance, the graph of a map φ is a correspondence with $c_1 = \mathrm{id}$ and $c_2 = \varphi$. To act on the cohomology of K, a correspondence needs a map $u : c_2^* K \to c_1^! K$, and for the graph of F this is the Frobenius structure $F^* K \to K$. For such data the Lefschetz–Verdier formula (Theorem 11.13) holds when X and C are proper. It writes the trace on cohomology as a sum, over the fixed points, of *local terms*, numbers computed from K and u near each fixed point. In general, however, a local term is not the trace on the stalk.

To apply this to Frobenius on a curve X, which need not be proper, we compactify X by adding finitely many points and extend K by zero. The added points are again fixed by Frobenius, but there K is no longer locally constant: its stalk is zero, while nearby stalks need not be. The main computation of Part II shows that even so, at every fixed point y of Frobenius on a curve, the local term is the trace of Frob_y on $K_{\bar{y}}$ (Proposition 12.16). To prove this, we cut K into its restrictions to y and to the complement of y. Local terms are additive for this decomposition, and for the piece supported at y the local term is the trace. It therefore suffices to show that the local term vanishes when $K_{\bar{y}} = 0$.

Here $dF = 0$ is used once more. Because of it, at (y, y) the graph $v = u^{q^m}$ is tangent to the axis $v = 0$, that is, to $X \times \{y\}$, while the diagonal $v = u$ is not. Blowing up the point (y, y) of $X \times X$ therefore separates the two curves. After the blow-up the local term is a map between cohomology groups supported near two disjoint curves meeting the exceptional divisor, and such a map factors through the stalk of K at y; so it vanishes when $K_{\bar{y}} = 0$ (Lemma 12.15). (In Chapter 12 we use instead the transposed graph $\{(F^m x, x)\}$, which is tangent to $\{y\} \times X$.)

Curves suffice for the general case. Indeed, the trace formula is compatible with $Rf_!$, and a variety can be fibred in curves over a variety of smaller dimension.

1.6 The Deligne–Laumon theorem

Part III proves purity. Fix an embedding $\iota : \overline{\mathbf{Q}}_\ell \to \mathbf{C}$, and write q_x for the number of elements of the residue field of a closed point x of X_0. A Weil sheaf with coefficients in $\overline{\mathbf{Q}}_\ell$ is *pure of weight* w if every eigenvalue α of Frobenius on every stalk satisfies $|\iota(\alpha)| = q_x^{w/2}$, and *mixed* if it has a finite filtration whose quotients are pure. Similarly, a complex has *weights at most* w if each cohomology sheaf $\mathcal{H}^i$ is mixed with weights at most $w + i$ (Definition 13.1). The shift by i is chosen so that both the constant sheaf, pure of weight 0, and its cohomology complex $R\Gamma(X, \overline{\mathbf{Q}}_\ell)$, whose H^i should have weight i, have weights at most 0.

The *Deligne–Laumon theorem* (Theorem 13.11), proved by Deligne in Weil II [Del80] and reproved by Laumon with the Fourier transform [Lau87], says that if K has weights at most w, so does $Rf_! K$ for every morphism f of separated $\mathbf{F}_q$-schemes of finite type. It gives only an upper bound. For the constant sheaf on a smooth proper X_0, duality supplies the lower one. With f the map to a point, the theorem says that the eigenvalues on H^i satisfy $|\iota(\alpha)| \le q^{i/2}$. Since Poincaré duality pairs α on H^i with q^d/α on H^{2d-i}, the same bound in degree $2d - i$ gives $|\iota(\alpha)| \ge q^{i/2}$. The two bounds together are purity. Theorem 19.7 deduces the Weil conjectures from it.

The proof has four steps, each of which has a model in classical mathematics.

Step 1: from varieties to curves. The theorem is stated for all sheaves, not only the constant one, because the proof is an induction on dimension. Away from a closed subset, a variety is fibred in curves over a variety of smaller dimension, much as one computes the cohomology of a fibre bundle by the Leray spectral sequence from the cohomology of the fibres and of the base.

Along such a fibration, however, the direct image of the constant sheaf is no longer constant, so the induction needs the theorem for all sheaves. It thereby reduces to curves, provided the direct images that occur along the way are mixed (Theorem 18.8). On a curve the statement says: if $\mathcal{F}$ is a lisse sheaf, pure of weight w, on a smooth curve U_0, then $H^1_c(U, \mathcal{F})$ has weights at most $w+1$ (Theorem 17.15). Since every curve, after removing finitely many points, maps to the affine line by a finite map, it is enough to treat open subsets of $\mathbf{A}^1$. The proof on curves combines two ideas: Rankin's positivity argument, which Deligne used in Weil I and Weil II [Del74], [Del80], and the Fourier transform, which Laumon [Lau87] used in place of Deligne's monodromy arguments.

Step 2: positivity bounds the eigenvalues at points. The model is Rankin's method for the coefficients of modular forms [Ran39]: the Dirichlet series $\sum |\tau(n)|^2 n^{-s}$ has nonnegative coefficients, so the position of its first pole bounds each individual coefficient. Here the role of the Dirichlet series is played by an L-function. Let $\mathcal{G}$ be a *lisse* sheaf on an affine curve U_0, that is, a locally constant one. Such a sheaf is the same as a representation of the fundamental group, and its *monodromy* is the image of that representation. Call $\mathcal{G}$ *real* if the characteristic polynomials of Frobenius on its stalks have real coefficients (Definition 14.21). Its L-function is the product, over closed points x, of the *Euler factors* $\det(1 - \mathrm{Frob}_x\, T^{\deg x} \mid \mathcal{G}_{\bar{x}})^{-1}$. On an affine curve $H^0_c = 0$ and $H^i_c = 0$ for $i > 2$, so the trace formula shows that the L-function equals $\det(1 - TF \mid H^1_c)/\det(1 - TF \mid H^2_c)$. Two facts make this useful.

- Consider an even tensor power $\mathcal{G}^{\otimes 2m}$. The logarithm of an Euler factor has coefficients $\mathrm{Tr}(\mathrm{Frob}^n_x \mid \mathcal{G}_{\bar{x}})^{2m}/n$, and these are nonnegative because the traces are real. Each Euler factor therefore has nonnegative coefficients bounded by those of the whole product. In particular it converges wherever the product does (Lemma 14.24).
- The product has poles only where $\det(1 - TF \mid H^2_c)$ vanishes, and H^2_c is easy to compute: it is the largest quotient of a stalk on which the monodromy acts trivially, twisted by (-1).

The second fact bounds the eigenvalues on H^2_c of $\mathcal{G}^{\otimes 2m}$, and the first turns this into a bound at every point. To state the result, let β be the largest *determinant weight* of the irreducible constituents of $\mathcal{G}$; the determinant weight of a constituent of rank r is the weight of its determinant divided by r (Definition 14.7). The eigenvalues on H^2_c of $\mathcal{G}^{\otimes 2m}$ have absolute value at most $q^{m\beta+1}$. Indeed, by the second fact H^2_c is a quotient of a stalk of $\mathcal{G}^{\otimes 2m}$ on which the geometric monodromy acts trivially, twisted by (-1). On such a quotient Frobenius acts through one-dimensional pieces, each a constituent of $\mathcal{G}^{\otimes 2m}$, and the determinant weight of every constituent of $\mathcal{G}^{\otimes 2m}$ is at most $2m\beta$ (Proposition 14.19). So these eigenvalues have absolute value at most $q^{m\beta}$, and the twist (-1) multiplies them by q. The product therefore converges for $|T| < q^{-m\beta-1}$, and by the first fact the Euler factor at x converges there too. This factor has a pole where $T^{\deg x} = \alpha^{-2m}$, for each eigenvalue α of Frob_x on $\mathcal{G}_{\bar{x}}$. Hence $|\iota(\alpha)|^{2m} \le q_x^{m\beta+1}$ for every m, and taking $2m$-th roots and letting $m \to \infty$ gives the intermediate result:

every Frobenius eigenvalue α at a point of a real lisse sheaf satisfies $|\iota(\alpha)|^2 \le q_x^{\beta}$

(Proposition 14.25).

Step 3: rank one, and purity of real sheaves. The bound of Step 2 is only useful if the determinants have weights, that is, if lisse sheaves of rank one are pure. A rank-one sheaf is a character of the fundamental group, the analogue of a Hecke character, and it is pure as soon as its restriction to the geometric fundamental group has finite image. We show that this image is finite. If it were not, the logarithm of a power of the character would give a nonzero class in H^1 of the curve fixed by Frobenius. This class cannot come from the removed points, since Frobenius acts on those classes by powers of q. So it comes from the compactification, and it gives a nonzero vector in the Tate module of its Jacobian fixed by Frobenius. Such a vector is a compatible

system of rational ℓ-power torsion points of unbounded order, which is impossible because the Jacobian has only finitely many rational points (Section 14.1). With the determinants under control, Step 2 bounds the eigenvalues of $\mathcal{G}$ by its largest determinant weight. To bound those of a single constituent we apply Step 2 to exterior powers of $\mathcal{G}$, in which that constituent is multiplied by the determinants of the others, which are now known; this isolates its weight (as in the example $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ of Chapter 14). Applied also to the dual, which gives the lower bound, it yields the second intermediate result:

each irreducible constituent of a real lisse sheaf on a curve is pure

(Theorem 14.26).

Step 4: the Fourier transform turns cohomology into a stalk. Steps 2 and 3 bound eigenvalues at the points of a curve, but we must bound eigenvalues on its cohomology. The Fourier transform turns one into the other. Its model is the discrete Fourier transform $\hat{t}(y) = \sum_{x\in\mathbf{F}_q} t(x)\psi(xy)$ of a function t on $\mathbf{F}_q$, where ψ is a nontrivial additive character. Two properties matter. The value $\hat{t}(0)$ is the sum of all values of t, so a sum over the line becomes a single value. And Plancherel's formula $\sum_y |\hat{t}(y)|^2 = q\sum_x |t(x)|^2$ holds.

The classical example is the Gauss sum. Let χ be a nontrivial multiplicative character, extended by $\chi(0) = 0$, and let $g = \sum_x \chi(x)\psi(x)$ be its Gauss sum. Substituting $x \mapsto x/y$ shows that $\hat{\chi}(y) = \overline{\chi}(y)g$ for $y \neq 0$, so the transform has the same absolute value $|g|$ at all $q-1$ nonzero points, and $\hat{\chi}(0) = 0$. Plancherel now reads $(q-1)|g|^2 = q(q-1)$, and we find $|g| = \sqrt{q}$. In other words, a function of absolute value 1 is carried to one of absolute value $\sqrt{q}$, that is, weight 0 to weight 1. This is the shape of the whole argument: first the transform is shown to have a single absolute value (to be pure), and then Plancherel finds which one.

Grothendieck's dictionary turns the transform into an operation on sheaves on the affine line. Read the formula for $\hat{t}$ from the inside out: pull t back to the plane along $p_1(x,y) = x$, multiply by the function $\psi(xy)$, and sum over the fibres of $p_2(x,y) = y$. Each step is a line of the table in Section 1.3. So if $\mathcal{K}_\psi$ is a sheaf on the plane with trace function $\psi(xy)$ and we put

$$\mathrm{FT}_\psi\, K = Rp_{2!}(p_1^* K \otimes \mathcal{K}_\psi)[1],$$

then for $y \in \mathbf{F}_q$ the table gives

$$t_{\mathrm{FT}_\psi K}(y) = -\,t_{Rp_{2!}(p_1^*K\otimes\mathcal{K}_\psi)}(y) = -\sum_{x\in\mathbf{F}_q} t_{p_1^*K}(x,y)\, t_{\mathcal{K}_\psi}(x,y) = -\sum_{x\in\mathbf{F}_q} t_K(x)\,\psi(xy) = -\,\hat{t}_K(y).$$

The sign comes from the shift [1], which changes the sign of every trace.

Now let M be an irreducible lisse sheaf, pure of weight w, on a dense open $j : U_0 \to \mathbf{A}^1$, and let j_*M be its direct image to $\mathbf{A}^1$. Its compactly supported cohomology lives in degrees 1 and 2; as in Step 2, H^2_c is easy, and the group we must bound is $H^1_c(\mathbf{A}^1, j_*M)$. Just as $\hat{t}(0)$ is the sum of t, the stalk at 0 of the Fourier transform of j_*M is the compactly supported cohomology of j_*M. The sheaves $\mathcal{L}_{\psi(ax)}$ with trace function $x \mapsto \psi(ax)$ are exceptions, as the characters $\psi(ax)$ are for the discrete transform: their transform is concentrated at one point. Suppose that no constituent of M over $\overline{\mathbf{F}}_q$ is of this form. Then the transform is again of the form j'_*M' for a lisse sheaf M' on a dense open of the dual line (Proposition 16.17). The hypothesis excludes in particular constant constituents (the case $a = 0$), so the monodromy of M has no coinvariants and $H^2_c(\mathbf{A}^1, j_*M) = 0$. This gives the third intermediate result:

the stalk of j'_*M' at 0 is exactly $H^1_c(\mathbf{A}^1, j_*M)$.

It remains to find the weight of M', and here the two halves of the Gauss sum argument return. To show that M' is pure we use Step 3: we twist M to weight 0 and add its dual, whose eigenvalues are the complex conjugates. The transform of the result is again a real sheaf, and it maps onto a twist of M', which is irreducible. So a twist of M' is an irreducible constituent of a real sheaf, hence pure by Step 3, and therefore M' is pure (Proposition 17.4). The weight of M' is then found from the sheaf version of Plancherel's formula, which compares the sums of $|t_M(x)|^2$ and $|t_{M'}(y)|^2$ over $\mathbf{F}_{q^m}$ as m grows: it is $w+1$ (Theorem 17.5). By Proposition 15.5, the stalk of j'_*M' at 0 has weights at most $w+1$, and this is the bound we want. For genus-one curves over $\mathbf{F}_3$, the resulting purity gives $1 \le N_1 \le 7$ (Example 19.9).

1.7 Outline

The main theorem is the Weil conjectures for smooth proper varieties over $\mathbf{F}_q$ (Theorem 19.7), with the statement on Betti numbers as above. The proof follows Weil's plan: it constructs a cohomology theory satisfying the properties (a)–(d) of a Weil cohomology listed in Section 1.2. Each part uses the one before it.

- Part I (Chapters 2–Chapter 8) constructs the six operations for ℓ-adic coefficients (Theorem 5.2). It ends with smooth base change and the Künneth formula.
- Part II (Chapters 9–Chapter 12) proves smooth purity (Theorem 9.3), Verdier and Poincaré duality (Theorem 10.10), which is (c), and the trace formula, which is (b). It ends with rationality and the functional equation. As for (a), the groups H^i_c are finite because $Rf_!$ preserves constructibility. They vanish outside $[0, 2\dim X]$, and Theorem 10.15 treats the smooth proper case used for the Weil conjectures.
- Part III (Chapters 13–Chapter 19) defines weights, proves purity of real sheaves on curves, introduces the Fourier transform and proves the Deligne–Laumon theorem, first on curves and then in general. It ends with the Weil conjectures.

Most appendices prove results from algebraic geometry that the main text uses: Nagata compactification and Chow's lemma, abelian varieties, the Picard scheme of a curve, the cohomology of curves with torsion coefficients, and facts on algebraic groups. Appendix E is of a different kind. It is not a preliminary but a second, independent theory of local terms, by specialization to the normal cone, which computes the local terms of contracting correspondences; the main line of the proof does not use it.

1.8 Prerequisites and reading guide

We assume a first course in algebraic geometry: schemes, quasi-coherent sheaves and their cohomology, curves and divisors, and étale morphisms; some homological algebra (derived functors, derived and triangulated categories); and singular cohomology with Poincaré duality and the Lefschetz fixed point formula, which serve as models. We do not assume étale cohomology. The results from the étale theory with torsion coefficients that we need [SGA4], [SGA4½] are stated where they are used, with references.

On a first reading one can take the foundations on trust and follow the main line:

- the statements of Chapter 3 and of the main theorem of Part I, Theorem 5.2;
- the base change map and the statement of proper base change (Theorem 4.33);
- Chapter 6 on why $f^!$ preserves constructibility with torsion coefficients;
- smooth purity with finite coefficients (Section 9.2), which is where the twist (d), the shift $[2d]$ and the trace map come from;

- Chapter 10 on duality;
- Frobenius and Weil complexes (Section 10.5); in Chapter 12, the local calculation at a fixed point of Frobenius on a curve;
- Chapters 13–Chapter 19, beginning with the plan in the opening of Part III.

The passages between coefficient rings (most of which are applications of Proposition 7.12), the coherence of the various comparison maps, smooth base change (Chapter 8), and the pro-étale foundations of Chapter 2 can be read when needed.

1.9 Conventions

We fix a prime ℓ and a field k that is finite or algebraically closed, of characteristic different from ℓ. Unless said otherwise, schemes are separated and of finite type over k, and morphisms are k-morphisms. The hypothesis on k is needed to bound the ℓ-cohomological dimension of k (Lemma 2.46), while the constructions of Sections 2.12 and Section 3.1 make sense for arbitrary schemes. Finally, a citation such as [Stacks, 0F0C] refers to the tag of a result in the Stacks Project [Stacks].

On the pro-étale site we regard every coefficient ring Λ as a condensed ring, that is, as a sheaf of rings on profinite sets, which remembers the topology of Λ. (On the étale site, which we use only with finite coefficients, a finite ring is simply a ring.) The coefficient sheaf Λ_X on X is then the pullback of Λ along the map from X to a point (Section 2.12). For a finite ring A this pullback is the classical constant sheaf.

Derived categories are the classical ones. Namely, $D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is the derived category of the abelian category of sheaves of Λ_X-modules. On these categories all functors are derived, and we write them f^*, Rf_*, $Rf_!$ and $f^!$. Unless another category is named, we have $f^* : D(Y_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ and $Rf_* : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$, whereas the categories on which $Rf_!$ and $f^!$ act are fixed where these functors are defined. To avoid set-theoretic problems, we take sizes with respect to a fixed universe that is large enough to contain the w-contractible covers of Theorem 2.13.

The table below lists the main notation, together with the technical terms (lisse, Weil structure, pure, mixed, real) whose meaning in these notes is fixed by a definition, and the place where each is defined. The Frobenius F of the Introduction, which raises coordinates to their q-th powers, is written Φ_X in the body of the notes. A Frobenius structure $F^*K \simeq K$ becomes there a Weil structure $\mathrm{Fr}_X^* K \simeq K$ for the geometric Frobenius Fr_X, and Lemma 12.3 shows that the two give the same operators.

Notation

symbol	meaning	defined in
$X_{\text{pro\'et}}$	the pro-étale site	Definition 2.4
Λ_X, $\Lambda_X(d)$	coefficient sheaf, Tate twists	Definition 2.53
$D^b_c(X,\Lambda)$, $D^b_{\text{ctf}}(X,\Lambda)$	constructible complexes	Definition 3.1
lisse	locally of constant finite presentation	Definition 3.2
K_n	reduction modulo π_E^n	Definition 2.61
cd_ℓ	cohomological dimension	Definition 2.45
$R\underline{\Gamma}_Z$	local cohomology	Definition 2.59
$\text{Loc}(D^b_{\text{ctf}}(X,\Lambda))$	localizing subcategory	Definition 5.1
$Rf_!$, $f^!$	exceptional operations	Theorem 5.2
ω_f	relative dualizing complex	Definition 9.4
Tr_f	trace map	Definition 9.24
$\mathbb{D}_X$	Verdier dual	Definition 10.2
Fr_X, Weil structure	geometric Frobenius, $\text{Fr}_X^* K \simeq K$	Definition 10.18
Φ_X	q-power Frobenius	Definition 12.2
c, $\text{Fix}(c)$	correspondences	Definition 11.5
$T_c(u)$	action of a correspondence	Definition 11.11
$\text{Tr}_c(u)$, LT	trace class, local terms	Definition 11.9
pure, mixed	ι-weights	Definition 13.1
dw_ι	determinant weight	Definition 14.7
real	real Weil sheaves	Definition 14.21
$j_*\mathcal{F}$	middle extension	Definition 15.1
FT_ψ	Fourier transform	Chapter 16
$T_\ell A$, $V_\ell A$	Tate module	Definition B.13

Part I The six operations

On the étale site the constant sheaf $\mathbf{Z}_\ell$ on $\operatorname{Spec}\mathbf{F}_q$ has $H^1 = 0$, although $\varprojlim_n H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}/\ell^n) = \mathbf{Z}_\ell$ (Section 1.2). Moreover, Frobenius eigenvalues live in $\overline{\mathbf{Q}}_\ell$, but SGA 4 constructs $Rf_!$ and $f^!$ and proves the finiteness theorems only for $\mathbf{Z}/\ell^n$ [SGA4], [SGA4½]. So the theory we need is not the one SGA 4 provides. This part has to overcome two difficulties. The first concerns the coefficients: we must pass to the limit over n without losing cohomology. Classically one does this with systems (F_n) and makes every derived construction by hand [Eke90]. We work instead on the pro-étale site of Bhatt and Scholze (Chapter 2). There $\mathbf{Z}_\ell$ is a sheaf of rings with the expected cohomology, and one definition serves all the coefficient rings

$$A,\ \mathbf{Z}_\ell,\ \mathcal{O}_E,\ \mathbf{Q}_\ell,\ E,\ \overline{\mathbf{Q}}_\ell,$$

compatibly with every change of coefficients between them (Section 7.2).

What the rest of the notes needs from this part is the table of Section 1.3: for every morphism f of schemes of finite type over k, the operations f^*, Rf_*, $Rf_!$, $f^!$, $\otimes^{\mathbf{L}}$ and $R\mathcal{H}om$, with their adjunctions, base change and projection formulas, and the finiteness statement that each carries constructible complexes to constructible complexes (Theorem 5.2 and Proposition 7.1). The later parts use these operations as their basic tools. For example, the trace formula of Part II sums trace functions over the fibres of f by applying $Rf_!$, and Poincaré duality comes from the computation of $f^!\Lambda$ for the structure morphism f of a smooth variety. Similarly, the Fourier transform of Part III is built from f^*, $\otimes^{\mathbf{L}}$ and $Rf_!$.

The second difficulty is to show that $f^!$ preserves constructible complexes, and it would be there even with finite coefficients. It arises because $f^!$ is given only by its universal property, and

already for a closed immersion i with open complement j its value is not obvious: we know only that $i^!K$ sits in a triangle $i^!K \to i^*K \to i^*Rj_*j^*K$. To see that it is constructible we need to know that Rj_* preserves constructibility, and this is Deligne's generic base change theorem. Accordingly, we construct $f^!$ in Chapter 5 and prove that it preserves constructible complexes in Chapter 6, as sketched in Section 1.4.

The two difficulties are handled separately, so that each geometric argument is made only once. The geometric inputs, namely proper base change (Chapter 4) and the computation of $f^!$ on charts (Chapter 6), are proved with torsion coefficients on the étale site, where SGA 4 applies. They are then transferred to ℓ-adic coefficients without being proved again. Two facts make this transfer possible (Chapters 3 and Chapter 7). First, a constructible complex over $\mathcal{O}_E$ is determined by its reductions modulo π_E^n (Proposition 3.11), so its properties can be read off from torsion complexes. Second, whether a natural map between two such constructions is an isomorphism can be checked with finite coefficients (Proposition 7.12). For example, smooth base change for $\mathbf{Z}_\ell$, E and $\overline{\mathbf{Q}}_\ell$ follows from smooth base change for finite rings in this way (Section 8.8).

2 The pro-étale site

On the étale site the constant sheaf $\mathbf{Z}_\ell$ forgets that $\mathbf{Z}_\ell$ is profinite. The pro-étale site $X_{\text{pro}\acute{e}t}$ of Bhatt and Scholze [BS15] repairs this. Its objects are the weakly étale X-schemes, and its coverings are fpqc coverings (Definition 2.4).

We need three properties of this site.

- (a) *Limits.* For a tower $\cdots \to F_2 \to F_1$ of surjections of sheaves, cohomology commutes with the limit: $R\Gamma(X, \varprojlim_n F_n) \simeq R\varprojlim_n R\Gamma(X, F_n)$.
- (b) *Topology.* The sheaf $\mathbf{Z}_\ell$ remembers the topology of $\mathbf{Z}_\ell$.
- (c) *Comparison.* Sheaves coming from the étale site keep their étale cohomology, so that the results of SGA 4 still apply to them.

Property (b) can be seen on a point. Over a separably closed field the topos of $X_{\text{pro}\acute{e}t}$ is, up to a set-theoretic cutoff, that of condensed sets (Proposition 2.50, proved in Subsection 2.12.1 below), and there the sections of $\mathbf{Z}_\ell$ over a profinite set S are all continuous maps $S \to \mathbf{Z}_\ell$, not only the locally constant ones.

	on $X_{\acute{e}t}$	on $X_{\text{pro}\acute{e}t}$
objects	étale X-schemes	weakly étale X-schemes
local objects	strictly henselian local rings	w-contractible rings
topos of a geometric point	sets	condensed sets
the sheaf $\mathbf{Z}_\ell$ on a point	$\mathbf{Z}_\ell$ with discrete topology	continuous maps to $\mathbf{Z}_\ell$

Property (a) fails on $X_{\acute{e}t}$, for the following reason. A surjection of sheaves is surjective on sections only after refining the cover. A countable product of surjections needs one refinement for each factor, and these infinitely many refinements need not have a common refinement among étale covers. So a countable product of surjections need not be surjective, $R^1\varprojlim_n F_n$ need not vanish, and $\varprojlim_n F_n$ differs from $R\varprojlim_n F_n$. For $F_n = \mathbf{Z}/\ell^n$ on $\operatorname{Spec}\mathbf{F}_q$ this is the lost H^1. On $X_{\text{pro}\acute{e}t}$ there are enough objects on which no refinement is needed: the spectra of w-contractible rings, roughly families of strictly henselian local rings parametrized by an extremally disconnected profinite set. Every cover of such a spectrum splits, so every surjection of sheaves is surjective on its sections there, and limits of sheaves can be computed there as limits of sets and modules.

However, building these objects takes two theorems from outside algebraic geometry: Gleason's theorem on extremally disconnected spaces and Olivier's theorem on weakly étale maps. Once we have them, property (a) follows in a few lines (Proposition 2.55). In particular, over $\mathbf{F}_q$ it gives back $H^1 = \mathbf{Z}_\ell$ (Example 2.57).

For property (c) we compare the two sites through the pullback functor ν_X^* from sheaves on $X_{ét}$ to sheaves on $X_{\text{pro}ét}$. Its image consists of the sheaves that we call *classical*: a sheaf on $X_{\text{pro}ét}$ is classical if its sections over a limit $U = \varprojlim_i U_i$ of affine étale X-schemes are the colimit of its sections over the U_i (Definition 2.30). In other words, a classical sheaf is determined by its values on étale X-schemes, as an étale sheaf is. The comparison has four parts.

- ν_X^* is an equivalence from étale sheaves onto classical sheaves (Theorem 2.32).
- It preserves the cohomology of bounded below complexes (Theorem 2.35).
- It commutes with direct images along quasi-compact quasi-separated morphisms (Theorem 2.38).
- For a finite coefficient ring, a constructible sheaf on $X_{\text{pro}ét}$ is classical, and ν_X^* is an equivalence from $D_c^b(X_{ét}, \Lambda)$ onto $D_c^b(X, \Lambda)$ (Theorem 2.43). So with finite coefficients the two sites have the same constructible complexes.

Later, $f^!$ is built by Brown representability on the unbounded derived category, so we also need the comparison for unbounded complexes. With finite coefficients it holds as well, because affine schemes étale over a scheme of finite type over k have uniformly bounded cohomological dimension. Each cohomology group of an unbounded complex therefore depends only on a bounded below truncation, where the comparison is already known (Proposition 2.49). The sheaf $\mathbf{Z}_\ell$, however, is not classical (Example 2.33). This is what the computation of $H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}_\ell)$ in Section 1.2 needs: H^1 comes out as $\mathbf{Z}_\ell$ only when sections see the topology of $\mathbf{Z}_\ell$. So the comparison applies to the $\mathbf{Z}/\ell^n$ but not to $\mathbf{Z}_\ell$ itself, and we reach $\mathbf{Z}_\ell$ from them by derived completion (Section 2.14).

Throughout this chapter, schemes are arbitrary unless a result states a finiteness hypothesis, because objects of $X_{\text{pro}ét}$ are in general neither Noetherian nor of finite type, even when X is a variety. The standing assumption that schemes are of finite type over k resumes in Chapter 3.

2.1 Weakly étale morphisms

The objects we want are inverse limits of étale X-schemes. Being such a limit is a condition on a presentation, which is awkward to work with, so we take as objects the weakly étale X-schemes of the following definition. This class is defined by flatness conditions. For this reason it is stable under base change and composition, and membership in it is local for the Zariski topology (Lemma 2.3). We lose little by the change: after a faithfully flat refinement every weakly étale scheme becomes a limit of étale schemes (Theorem 2.14).

Typical examples are the schemes that the étale site can only reach through colimits. The strict henselization $\operatorname{Spec}\mathcal{O}^{\text{sh}}_{X,\bar{x}}$ is the limit of the affine étale neighbourhoods of $\bar{x}$, and on $X_{ét}$ it appears only through the stalk at $\bar{x}$. On $X_{\text{pro}ét}$ it is an object. Similarly $\operatorname{Spec}\overline{\mathbf{F}}_q = \varprojlim_m \operatorname{Spec}\mathbf{F}_{q^m}$ is an object over $\operatorname{Spec}\mathbf{F}_q$, and for a profinite set $S = \varprojlim S_i$ the scheme $X \times S = \varprojlim_i \coprod_{S_i} X$ is an object over X. Through the last family the site sees profinite sets, and hence the topology of $\mathbf{Z}_\ell$.

Definition 2.1 (Ind-étale, pro-étale and weakly étale morphisms)**.** A ring map $R \to B$ is *ind-étale* if $B = \varinjlim_\alpha B_\alpha$ for a filtered system of étale R-algebras B_α. A *pro-étale affine over X* is a scheme $U = \varprojlim_i U_i$ that is the limit of a cofiltered system of affine schemes U_i étale over X. If $X = \operatorname{Spec}(R)$ is affine, the pro-étale affines over X are the schemes

$$\operatorname{Spec}(B) = \varprojlim_{\alpha} \operatorname{Spec}(B_\alpha) \longrightarrow X$$

with $R \to B$ ind-étale.

A scheme morphism $u : U \to X$ is *weakly étale* if u and its diagonal $\Delta_u : U \to U \times_X U$ are flat.

The three classes of ring maps are related as follows.

Proposition 2.2 (Étale, ind-étale and weakly étale ring maps)**.** *Let $R \to B$ be a ring map.*

1. *If $R \to B$ is étale, it is ind-étale; if it is ind-étale, it is weakly étale.*
2. *If $R \to B$ is weakly étale and of finite presentation, it is étale.*

Proof. The first implication in (1) is immediate from the definition, since an étale algebra is the colimit of the constant system on itself. For the second, a filtered colimit of étale algebras is flat, and its multiplication map is the filtered colimit of the multiplication maps $B_\alpha \otimes_R B_\alpha \to B_\alpha$, which are flat; see [Stacks, 092A].

For (2) we show that $R \to B$ is unramified, which together with flatness and finite presentation means that it is étale. The multiplication map $B \otimes_R B \to B$ is surjective, and it is flat by assumption. It is of finite presentation because B is of finite presentation over R. Hence the diagonal of $\operatorname{Spec}(B) \to \operatorname{Spec}(R)$ is a closed immersion that is flat and of finite presentation. Its ideal I is then finitely generated with $I = I^2$, hence generated by an idempotent, so the diagonal is also an open immersion. A morphism locally of finite type whose diagonal is an open immersion is unramified, and this proves (2). ∎

For instance, $\mathbf{F}_q \to \overline{\mathbf{F}}_q$ is ind-étale, as the colimit of the finite fields $\mathbf{F}_{q^m}$, but it is not étale, because $\overline{\mathbf{F}}_q$ is not of finite type over $\mathbf{F}_q$.

In (2) the flatness of the diagonal takes the place of unramifiedness; flatness of $R \to B$ alone would not suffice. The converse of the second implication in (1) fails in general: a weakly étale map need not be a filtered colimit of étale maps. It does become ind-étale after a faithfully flat refinement (Theorem 2.14 below, and [Stacks, 097Z]). Geometrically, a weakly étale scheme over X is covered, in the pro-étale topology, by a limit of étale X-schemes.

Lemma 2.3 (Stability)**.** *Weakly étale morphisms are stable under base change and composition, and the property is local on source and target for the Zariski topology. Every X-morphism between two weakly étale X-schemes is weakly étale. Ind-étale ring maps are stable under base change and composition. On affine charts, a morphism* $\operatorname{Spec}(B) \to \operatorname{Spec}(R)$ *is weakly étale if and only if both*

$$R \longrightarrow B, \qquad B \otimes_R B \longrightarrow B$$

are flat.

Proof. Since weakly étale means that a morphism and its diagonal are flat, each assertion about weakly étale morphisms comes down to writing the relevant diagonal in terms of diagonals and base changes that we already know to be flat.

Base change preserves flatness of a morphism and its diagonal. For a composite $V \to U \to X$, the morphism $V \to X$ is flat as a composite of flat morphisms. Its diagonal $\Delta_{V/X}$ is $\Delta_{V/U}$

followed by $V \times_U V \to V \times_X V$, and the second map is a base change of $\Delta_{U/X}$. Both maps are flat, and hence so is $\Delta_{V/X}$.

Let $h : U \to V$ be an X-morphism between weakly étale X-schemes. We factor h as the graph $U \to U \times_X V$ followed by the projection $U \times_X V \to V$. The graph is a base change of the diagonal of V/X, which is flat, and the projection is a base change of $U \to X$, which is also flat. Their composite h is therefore flat. Similarly, the diagonal $\Delta_h : U \to U \times_V U$ is the base change of $\Delta_{U/X}$ along the monomorphism $U \times_V U \to U \times_X U$, so it is flat as well. Finally, Zariski locality follows from the fact that flatness is Zariski local.

On affine charts, the diagonal of $\operatorname{Spec}(B) \to \operatorname{Spec}(R)$ is the closed immersion $\operatorname{Spec}(B) \to \operatorname{Spec}(B \otimes_R B)$ given by the multiplication map, so the morphism is weakly étale exactly when $R \to B$ and $B \otimes_R B \to B$ are flat. By Zariski locality on source and target, this describes weakly étale morphisms on any affine charts.

For ind-étale ring maps, base change is clear because tensor products commute with filtered colimits. For composition, let $R \to B$ and $B \to C$ be ind-étale, with $B = \varinjlim_\alpha B_\alpha$. Each étale B-algebra C' occurring in a presentation of C is of finite presentation over B, hence is the base change of an étale algebra C'_α over some B_α. Since B_α is étale over R, so is C'_α, and C' is the filtered colimit of the étale R-algebras $C'_\alpha \otimes_{B_\alpha} B_\beta$ for $\beta \geq \alpha$. Thus C is a filtered colimit of ind-étale R-algebras, and such a colimit is again ind-étale [Stacks, 097L]. ∎

Definition 2.4 (The small pro-étale site)**.** The objects of $X_{\text{pro}\acute{e}t}$ are weakly étale X-schemes, with all X-morphisms as arrows. A family $\{U_i \to U\}_{i \in I}$ is a covering if it is an fpqc covering. Explicitly, for every affine open $W \subset U$ there must be finitely many indices $i_1, \ldots, i_m$ and affine opens $V_j \subset U_{i_j} \times_U W$ with

$$W = \bigcup_{j=1}^{m} \operatorname{image}(V_j \to W).$$

The indices may depend on W, and I may be infinite. Equivalently one may use finitely many quasi-compact opens above each W. See [Stacks, 0989, 098A].

The finite refinement condition is part of the topology, and a jointly surjective family that does not satisfy it is not a covering. We record two consequences of the definition.

Proposition 2.5 (Coverings in the pro-étale site)**.** *Let U be an object of $X_{\text{pro}\acute{e}t}$.*

1. *If U is affine, every covering of U can be refined by a finite affine covering, whose disjoint union is affine and faithfully flat over U.*
2. *Every étale covering and every Zariski covering of U is a pro-étale covering.*

Proof. Part (1) is the covering condition of Definition 2.4 with $W = U$. For (2), recall that flat morphisms locally of finite presentation are open. By quasi-compactness we then find the required finite refinements; see [Stacks, 098B]. ∎

By (2) the pro-étale topology is finer than the étale topology. By (1), over an affine object a covering may always be replaced by a single faithfully flat map. For example, the single map $\operatorname{Spec} \overline{\mathbf{F}}_q \to \operatorname{Spec} \mathbf{F}_q$ is a pro-étale covering, although it is not an étale one. These coverings define a topos with the expected functoriality in X.

Proposition 2.6 (The topos and its functoriality)**.** *The above coverings define a subcanonical site. Its category of sheaves* $\mathrm{Sh}(X_{\text{pro\'et}})$ *is a Grothendieck topos. Every scheme morphism* $f : X \to Y$ *induces a geometric morphism*

$$f_{\text{pro\'et}} : \mathrm{Sh}(X_{\text{pro\'et}}) \longrightarrow \mathrm{Sh}(Y_{\text{pro\'et}}), \qquad f^{-1} \dashv f_*.$$

Here $f^{-1} : \mathrm{Sh}(Y_{\text{pro\'et}}) \to \mathrm{Sh}(X_{\text{pro\'et}})$ *preserves finite limits and all colimits, and is exact on abelian sheaves. For a second morphism* $g : Y \to Z$ *there is a canonical isomorphism* $(g \circ f)^{-1} \simeq f^{-1} \circ g^{-1}$.

Proof. There are two things to check: that the coverings form a subcanonical topology, and that base change of objects is a morphism of sites.

Identity covers, composites of covers, and base changes of covers are again covers. For the underlying morphisms this is Lemma 2.3, and for the finite refinement condition it is part of the fpqc covering axioms. Representable presheaves satisfy descent for these covers by fpqc descent for morphisms of schemes, so they are sheaves, and the topology is subcanonical.

For $f : X \to Y$, the functor $V \mapsto V \times_Y X$ sends weakly étale Y-schemes to weakly étale X-schemes by Lemma 2.3. It preserves fibre products and the final object, hence finite limits, and it carries covering families to covering families because fpqc coverings are stable under base change. Such a functor defines a geometric morphism, whose inverse image f^{-1} is left exact and, as a left adjoint, preserves all colimits. Exactness on abelian sheaves follows, and compatibility with composition holds because $(V \times_Z Y) \times_Y X \simeq V \times_Z X$ canonically. This is the small-site version of the constructions in [Stacks, 0988]. ∎

2.2 w-local and w-contractible rings

To compute limits of sheaves by taking sections, we want objects over which every cover splits, so that sheafification changes nothing there. On the étale site the spectrum of a strictly henselian local ring, such as the étale local ring of a variety at a geometric point, has this property, but such spectra do not cover X. Instead, the rings we need are families of strictly henselian local rings over a profinite set of components. Moreover, we require this set to be extremally disconnected, which means that the closure of every open subset is open, because under this condition a cover splits. Namely, over each component of $\mathrm{Spec}(R)$ we choose a component of the cover lying above it, and extremal disconnectedness lets us make this choice continuously. The model case is the ring of locally constant functions from a profinite set to a separably closed field (Example 2.9), which is a family of copies of one field indexed by the profinite set. Below we define these rings, the w-contractible rings, by a hierarchy of local conditions, and we show that every ring has a faithfully flat ind-étale cover by one of them (Theorem 2.13).

Recall that spectral spaces are quasi-compact and sober, with a basis of quasi-compact opens closed under finite intersections, and that the space of connected components of a spectral space is profinite.

Definition 2.7 (w-local, w-strictly local and w-contractible rings)**.** A topological space is *extremally disconnected* if the closure of every open subset is open. A spectral space T is *w-local* if its set T^c of closed points is closed and every point specializes to exactly one closed point. A ring R is w-local if $\mathrm{Spec}(R)$ is w-local. It is *w-strictly local* if it is w-local and $R_{\mathfrak{m}}$ is strictly henselian for every maximal ideal $\mathfrak{m}$. It is *w-contractible* if it is w-strictly local and the profinite space $\pi_0(\mathrm{Spec}(R))$ is extremally disconnected.

Closedness of T^c does not follow from uniqueness of closed specializations, which is why the definition requires it separately. The basic properties of these notions are as follows.

Proposition 2.8 (Connected components of w-local spaces and rings)**.**

1. *If T is a w-local spectral space, the natural map $T^c \to \pi_0(T)$ is a homeomorphism.*
2. *If R is w-local, the component of $\operatorname{Spec}(R)$ containing a closed point $\mathfrak{m}$ is the image of $\operatorname{Spec}(R_{\mathfrak{m}})$. In particular, if R is w-strictly local, every component is the spectrum of a strictly henselian local ring.*
3. *A strictly henselian local ring is w-contractible.*

Proof. For (1), each component consists of the generalizations of its unique closed point, and T^c is compact. Part (2) follows, and (3) holds because the spectrum of a strictly henselian local ring has one closed point and one component. See [Stacks, 096A, 0968, 0969]. ■

So a w-strictly local ring is a family of strictly henselian local rings indexed by the profinite set of its components, and a w-contractible ring is such a family indexed by an extremally disconnected profinite set.

Example 2.9 (Locally constant functions)**.** Let k' be a separably closed field, let $S = \varprojlim_i S_i$ be a profinite set written as a limit of finite sets, and let $R = LC(S, k')$ be the ring of locally constant functions $S \to k'$. Then $R = \varinjlim_i k'^{S_i}$, and $\operatorname{Spec}(k'^{S_i})$ is the finite discrete set S_i, so $\operatorname{Spec}(R) = \varprojlim_i S_i = S$: the point $s \in S$ is the maximal ideal of functions vanishing at s. Every point is closed and is its own component, so R is w-local with $\pi_0(\operatorname{Spec}(R)) = S$. The local ring at s is the colimit of the local rings k' of the k'^{S_i} at the image of s, that is, k', which is strictly henselian. Thus R is w-strictly local: it is the family of copies of $\operatorname{Spec} k'$ indexed by S. It is w-contractible exactly when S is extremally disconnected. In this example the definitions are statements about S alone.

The constructions below change the set of components while keeping the local rings, and for this we use the following class of maps. A ring map $R \to A$ is a *local isomorphism* if $\operatorname{Spec}(A)$ is covered by principal opens $D(g)$, each mapping isomorphically onto an open subscheme of $\operatorname{Spec}(R)$ [Stacks, 096E], and it is *ind-Zariski* if it is a filtered colimit of local isomorphisms [Stacks, 096M]. For instance, $R \to R \times R$ is a local isomorphism. Taking the colimit over the finite quotients S_i of a profinite set S, we see that $R \to \varinjlim_i R^{S_i} = LC(S, R)$ is ind-Zariski. Its spectrum is $\operatorname{Spec}(R) \times S$. A localization $R \to R_{\mathfrak{p}}$ at a prime is ind-Zariski as well, as the colimit of the R_f with $f \notin \mathfrak{p}$. The property we use is that an ind-Zariski map does not change local rings: each local ring of A is a local ring of R.

We use extremally disconnected spaces through their projectivity. In Proposition 2.12 we use it to split a surjection of component spaces, and in Theorem 2.13 we cover a profinite component space by an extremally disconnected one (Lemma 2.11).

Proposition 2.10 (Gleason's theorem)**.** *A compact Hausdorff space S is extremally disconnected if and only if it is projective with respect to continuous surjections of compact Hausdorff spaces: for every continuous surjection $T_1 \to T_2$ of compact Hausdorff spaces, every continuous map $S \to T_2$ lifts to a continuous map $S \to T_1$.*

Proof. This is the projectivity theorem for compact Hausdorff spaces; see [Stacks, 08YN]. Here we sketch the implication from extremal disconnectedness to projectivity, which is the direction used later. To lift a map $S \to T_2$ along a surjection $T_1 \to T_2$, one pulls the surjection back to S and chooses a minimal closed subspace of the fibre product that still surjects onto S. Then extremal disconnectedness forces this irreducible surjection to be a homeomorphism, and its inverse gives the lift. ■

To use the proposition we need enough extremally disconnected spaces. They come from Stone–Čech compactifications of discrete sets.

Lemma 2.11 (Extremally disconnected covers)**.** *Every compact Hausdorff space T, in particular every profinite space, admits a continuous surjection $S \to T$ with S compact Hausdorff and extremally disconnected. Every extremally disconnected compact Hausdorff space is profinite.*

Proof. Write T^δ for the underlying set of T with the discrete topology and take $S = \beta(T^\delta)$, its Stone–Čech compactification. The Stone–Čech compactification of a discrete space is extremally disconnected, and the identity $T^\delta \to T$ extends uniquely to a continuous surjection $\beta(T^\delta) \to T$; see [Stacks, 090D]. An extremally disconnected compact Hausdorff space is totally disconnected, since the closure of an open set is clopen, and hence it is profinite. ∎

A finite discrete space is extremally disconnected, while the Cantor set $\{0,1\}^{\mathbf{N}}$ is not. To see this, let V_n be the clopen set of sequences beginning with 0^n1, and let U be the union of the V_n for odd n. The closure of U is $U \cup \{000\cdots\}$, and it is not open, because every neighbourhood of $000\cdots$ contains some V_n with n even, which is disjoint from the closure. More generally, infinite extremally disconnected compact spaces have no nontrivial convergent sequences, and therefore they are never metrizable. In practice they arise as Stone–Čech compactifications.

Over a w-contractible ring an ind-étale cover splits at each closed point by strict henselianity, and Gleason's theorem lets us choose these local splittings continuously in the point.

Proposition 2.12 (Splitting ind-étale covers)**.** *If R is w-contractible, every faithfully flat ind-étale ring map $R \to B$ has an R-algebra retraction $B \to R$.*

Proof (Proof outline). We first replace B by an ind-Zariski B-algebra, so that its spectrum is w-local and its closed points lie exactly over the closed points of $\operatorname{Spec}(R)$; such a refinement does not change the local rings. Let $\mathfrak{q} \subset B$ lie over a closed point $\mathfrak{m}$ of $\operatorname{Spec}(R)$. Since $R_\mathfrak{m}$ is strictly henselian, the ind-étale local homomorphism $R_\mathfrak{m} \to B_\mathfrak{q}$ is an isomorphism. Hence every component of $\operatorname{Spec}(B)$ maps isomorphically onto the component of $\operatorname{Spec}(R)$ below it, and a retraction amounts to a continuous choice of one component above each component of $\operatorname{Spec}(R)$. Such a choice exists because, by Proposition 2.10, the surjection $\pi_0(\operatorname{Spec}(B)) \to \pi_0(\operatorname{Spec}(R))$ of profinite component spaces has a section. Given such a section, we restrict B to the corresponding closed set of components by a filtered colimit of localizations at idempotents. The result is an R-algebra isomorphic to R, and this gives the retraction. For details see [Stacks, 097D, 09AZ, 0982]. ∎

The proposition would be of little use if w-contractible rings were rare. The next theorem shows that every ring has a faithfully flat ind-étale cover by one, so that, after a single refinement, every affine object of $X_{\text{pro\'et}}$ is replaced by one over which covers split.

Theorem 2.13 (Existence of contractible covers)**.** *Every ring R admits a faithfully flat ind-étale algebra D that is w-contractible.*

Proof. The construction is that of [BS15, Theorem 2.4.9] and [Stacks, 0983]. We first make the local rings strictly henselian, and then replace the profinite space of components by an extremally disconnected one surjecting onto it, which does not change the local rings.

In the first stage we construct a faithfully flat ind-étale R-algebra C that is w-strictly local. To do so we first replace R by an ind-Zariski algebra whose spectrum is w-local, and then take a filtered colimit of étale neighbourhoods of the closed points, which makes the local rings at

the closed points strictly henselian. This is the construction of simultaneous strict localizations in [Stacks, 097X]. Each step is a filtered colimit of étale algebras, so C is ind-étale over R by Lemma 2.3.

For the second stage, choose a surjection $S \to \pi_0(\operatorname{Spec}(C))$ with S extremally disconnected using Lemma 2.11, and realize this map by an ind-Zariski C-algebra D such that

$$|\operatorname{Spec}(D)| \simeq |\operatorname{Spec}(C)| \times_{\pi_0(\operatorname{Spec}(C))} S$$

as topological spaces. Concretely, write S as a limit of finite quotients, use finite products of C, restrict to the relevant closed sets of components by idempotent localizations, and take a filtered colimit; see [Stacks, 097D]. This construction identifies local rings, is surjective when S is, and preserves w-locality. Hence $C \to D$ is faithfully flat and the local rings at closed points of D remain strictly henselian. The space of components of $\operatorname{Spec}(D)$ is S, so D is w-contractible, and the composite $R \to C \to D$ is ind-étale and faithfully flat. ■

To compute on $X_{\text{pro\'et}}$ we want the limits of étale affines to form a basis (Proposition 2.15), so that étale sheaves can be evaluated on every object by a colimit. Objects of $X_{\text{pro\'et}}$ are only weakly étale, while the covers above are ind-étale, and the next theorem bridges the two.

Theorem 2.14 (Pro-étale refinements of weakly étale maps)**.** *If $R \to B$ is weakly étale, there is a faithfully flat ind-étale map $B \to B'$ such that $R \to B'$ is ind-étale.*

Proof. Weakly étale and ind-étale maps differ only in how local rings are glued. Namely, after strict localization of the base a weakly étale map identifies local rings, that is, for every prime $\mathfrak{q}$ of B lying over $\mathfrak{p} \subset R$ the induced map of local rings $R_{\mathfrak{p}} \to B_{\mathfrak{q}}$ is an isomorphism. Such a map becomes ind-Zariski after refinement.

Applying the construction of simultaneous strict localizations to the pair $R \to B$ we obtain compatible ind-étale refinements $R \to C$ and $B \to D$, with both spectra w-local and with closed points of D mapping to closed points of C. The map $C \to D$ is still weakly étale by Lemma 2.3, and the local rings of C at closed points are strictly henselian. Olivier's rigidity theorem says that a weakly étale local map out of a strictly henselian local ring is an isomorphism; see [Stacks, 092Z]. Applying it at the closed points of D, which lie over closed points of C, we get isomorphisms of local rings there. Every point of the w-local $\operatorname{Spec}(D)$ is a generalization of a closed point, and its local ring is a localization of the local ring at that closed point. Hence $C \to D$ identifies all local rings. For a ring map that identifies local rings there is a faithfully flat ind-Zariski refinement of the target over which the map becomes ind-Zariski [Stacks, 097G]. Applied to $C \to D$, this gives a faithfully flat ind-Zariski map $D \to D'$ such that $C \to D'$ is ind-Zariski, and we take $B' = D'$.

The algebra B' has the two required properties. The map $B \to B'$ is the composite $B \to D \to D'$ of two faithfully flat ind-étale maps, and $R \to B'$ is the composite $R \to C \to D'$ of an ind-étale and an ind-Zariski map. Both composites are ind-étale by Lemma 2.3. For details see [Stacks, 097Z]. ■

2.3 Affine bases

To compute with sheaves on $X_{\text{pro\'et}}$ it suffices to evaluate on two kinds of affines. Every object of $X_{\text{pro\'et}}$ is covered by pro-étale affines (Theorem 2.14), and these are in turn covered by w-contractible affines (Theorem 2.13), over which taking sections is exact.

The two kinds of affines are used for different purposes. On a pro-étale affine $U = \varprojlim_i U_i$, a sheaf pulled back from $X_{ét}$ can be evaluated by a colimit, and for this reason the comparison theorems of Section 2.7 are proved on such affines. On a w-contractible affine, by contrast, every epimorphism of sheaves is surjective on sections, so the homological results of the next three sections are proved there.

In Lemma 2.16 we turn the splitting of ring maps in Proposition 2.12 into a statement about sheaves. The idea is this. A finite cover of a w-contractible affine has a section. Such a section amounts to a decomposition of the affine into finitely many clopen pieces, each mapping into one member of the cover. Local lifts along an epimorphism can then be glued over these pieces without any further refinement.

Proposition 2.15 (Pro-étale affines form a basis)**.** *Every object of $X_{\text{proét}}$ has a pro-étale covering by pro-étale affines over X. These form a full subsite whose category of sheaves, of sets or of abelian groups, is equivalent to $\mathrm{Sh}(X_{\text{proét}})$ by restriction.*

Proof. The first assertion is the refinement theorem applied chart by chart, and the second is then formal.

Given an object U of $X_{\text{proét}}$, cover X by affine opens and cover their inverse images in U by affine opens. These Zariski coverings are pro-étale coverings, so it suffices to cover each chart. On such a chart the map is a weakly étale ring map $R \to B$. By Theorem 2.14 there is a single faithfully flat cover $\operatorname{Spec}(B') \to \operatorname{Spec}(B)$ such that $\operatorname{Spec}(B')$ is a limit of affine schemes étale over $\operatorname{Spec}(R)$, hence over X. The equivalence of categories of sheaves follows from the comparison lemma for sites; see [BS15, Lemma 4.2.4]. ∎

Over $X = \operatorname{Spec} \mathbf{F}_q$, for example, the weakly étale X-scheme $\operatorname{Spec} \overline{\mathbf{F}}_q$ is already a pro-étale affine, and X itself is covered by it. Applying Theorem 2.13 to the charts of such a basis, we pass from pro-étale affines to w-contractible ones. The next two results show that the spectra of w-contractible rings have the splitting property for sheaves and that they cover every object.

Lemma 2.16 (Sections of epimorphisms onto w-contractible affines)**.** *Let $U = \operatorname{Spec}(D)$ be an object of $X_{\text{proét}}$ with D w-contractible. Then every epimorphism of sheaves $F \to h_U$ has a section. Equivalently, h_U is projective in $\mathrm{Sh}(X_{\text{proét}})$. More generally, for an object W of $X_{\text{proét}}$ the following are equivalent: every epimorphism of sheaves of sets $F \to h_W$ admits a section; h_W is projective in $\mathrm{Sh}(X_{\text{proét}})$; every epimorphism $F \to G$ induces a surjection $F(W) \to G(W)$.*

Proof. For the equivalence of the three conditions on an object W: by the Yoneda lemma, h_W is projective exactly when every epimorphism $F \to G$ induces a surjection $F(W) = \operatorname{Hom}(h_W, F) \to \operatorname{Hom}(h_W, G) = G(W)$. If every epimorphism onto h_W splits, pull an epimorphism $F \to G$ back along a section $h_W \to G$; the resulting epimorphism onto h_W splits, so the section lifts to F. Conversely, if every epimorphism is surjective on sections over W, then id_{h_W} lifts along any epimorphism onto h_W.

To see that covers of U split, note that by Theorem 2.14 a faithfully flat weakly étale D-algebra can be refined to a faithfully flat ind-étale D-algebra. This refinement splits by Proposition 2.12, and hence so does the original map. Now, given a cover of the affine U, we refine it by a finite affine cover $\{V_j \to U\}$. Its disjoint union $\coprod_j V_j$ is a single faithfully flat weakly étale affine cover, which therefore has a section $s : U \to \coprod_j V_j$. The sets $s^{-1}(V_j)$ then form a finite clopen partition of U, and on $s^{-1}(V_j)$ the inclusion into U factors through V_j.

If $F \to h_U$ is an epimorphism, then by the definition of an epimorphism of sheaves id_U lifts after passing to a covering of U. We refine this covering by the clopen partition just constructed, so

that on each piece $s^{-1}(V_j)$ we have a lift. Since the pieces are disjoint and cover U, the sheaf condition glues these lifts with no compatibility to check, and the result is a global lift of id_U, that is, a section. The same argument, after pulling back an arbitrary epimorphism along a map from h_U, proves projectivity. See [Stacks, 098F, 0990]. ∎

Theorem 2.17 (Enough affine weakly contractible objects)**.** *Every object of $X_{\text{pro\'et}}$ has a covering by affines $U = \mathrm{Spec}(D)$ with D w-contractible.*

Proof. Cover the given object by pro-étale affines $\mathrm{Spec}(B)$ (Proposition 2.15), and for each of them choose, by Theorem 2.13, a faithfully flat ind-étale B-algebra D that is w-contractible. Then $\mathrm{Spec}(D) \to \mathrm{Spec}(B)$ is weakly étale by Proposition 2.2, so $\mathrm{Spec}(D)$ is an object of $X_{\text{pro\'et}}$ by Lemma 2.3, and as a faithfully flat map of affines it is a covering. Composing these coverings gives the result. ∎

2.4 Weakly contractible objects

On a topological space we check exactness at points, and on the étale site at geometric points. On $X_{\text{pro\'et}}$, instead, we check exactness, and compute limits and colimits, by evaluating on weakly contractible objects. Over these objects we can compute in the ordinary derived categories of sheaves of modules by taking sections. Until Section 2.7, $\mathcal{R}$ denotes a sheaf of rings on $X_{\text{pro\'et}}$, for instance the pullback of a condensed ring (Section 2.12).

The main result is Corollary 2.20. On a weakly contractible affine W, taking sections is exact and commutes with all limits and colimits of module sheaves. Moreover, these functors, for all such W, are jointly conservative.

Stalk functors behave worse. Indeed, a stalk functor is exact and commutes with colimits, but not with infinite products. Moreover, products of sheaves on $X_{\acute{e}t}$ need not be exact either, because a section of a product of surjections lifts in each factor only after its own refinement of the cover, and infinitely many refinements cannot in general be made at once. Evaluation at W avoids both problems: it commutes with products because limits of sheaves are computed on sections, and it is exact because covers of W split.

In the simplest case, X a geometric point, the site is that of profinite sets (Proposition 2.50, proved in Subsection 2.12.1 below). The weakly contractible affines are then the extremally disconnected ones, among them $\beta(T^\delta)$ for a set T. Indeed, a surjection of profinite sets onto such an S has a section by Gleason's theorem.

Definition 2.18 (Weak contractibility)**.** An object W of $X_{\text{pro\'et}}$ is *weakly contractible* if every epimorphism of sheaves of sets $F \to h_W$ admits a section, where h_W is the representable sheaf associated to W. By Lemma 2.16, equivalently h_W is projective in $\mathrm{Sh}(X_{\text{pro\'et}})$, or every epimorphism $F \to G$ induces a surjection $F(W) \to G(W)$.

This is the categorical definition of [BS15, §3.2].

Example 2.19 (Weak contractibility over a point)**.** Let X be a geometric point, so that the affine objects of $X_{\text{pro\'et}}$ are the profinite sets (Proposition 2.50 below). A continuous surjection $S \to T$ of profinite sets is a covering, so $h_S \to h_T$ is an epimorphism, and by the Yoneda lemma a section of it is a continuous map $T \to S$ that is a section of the surjection. If T is finite, such a section exists: choose a preimage of each point; it is continuous because T is discrete. If $T = \{1, 2, 3, \ldots\} \cup \{\infty\}$ is a convergent sequence, it need not exist. Let $S = T_0 \amalg T_1$, where T_0 is the convergent sequence $\{2, 4, 6, \ldots\} \cup \{\infty_0\}$ and T_1 is $\{1, 3, 5, \ldots\} \cup \{\infty_1\}$, and map $S \to T$

by the identity on integers and $\infty_0, \infty_1 \mapsto \infty$. A section must send each integer n to its only preimage n, and then the even integers converge to ∞_0 while the odd ones converge to ∞_1, so no value at ∞ makes the section continuous. Hence T is not weakly contractible. In agreement with Gleason's theorem, T is not extremally disconnected: the set of odd integers is open, and its closure, which contains ∞, is not open, because every neighbourhood of ∞ contains even integers.

Note that the splitting condition concerns a covering family as a whole. If W is disconnected, a lift of id_{h_W} along a covering family may use different members of the family on different clopen pieces of W.

By Lemma 2.16, the spectrum of a w-contractible ring is weakly contractible. By Theorem 2.17, such affines cover every object of $X_{\mathrm{pro\acute{e}t}}$. For abelian sheaves we use these two facts in the following form.

Corollary 2.20 (Evaluation on weakly contractible objects)**.** *Let W be a weakly contractible object of $X_{\mathrm{pro\acute{e}t}}$, for instance $W = \mathrm{Spec}(D)$ with D w-contractible.*

1. *The functor $F \mapsto F(W)$ from abelian sheaves on $X_{\mathrm{pro\acute{e}t}}$ to abelian groups is exact. Hence $H^q(W, F) = 0$ for $q > 0$, and $R\Gamma(W, K) \simeq K(W)$ for every complex K, with the right side computed term by term.*
2. *If W is affine, evaluation at W preserves all limits and all colimits of $\mathcal{R}$-modules.*
3. *Exactness of complexes and isomorphisms of $\mathcal{R}$-module sheaves can be tested by evaluation on all affine weakly contractible objects of $X_{\mathrm{pro\acute{e}t}}$.*

Proof. All three parts follow from the fact that every epimorphism onto a sheaf is surjective on sections over W, together with the fact that such W cover every object. Exactness and conservativity follow directly, and for colimits we need in addition that W is quasi-compact.

(1) Taking sections is left exact for any object of a site, since it preserves kernels. By weak contractibility every epimorphism of sheaves is surjective on sections over W, so the sections functor is also right exact. An exact functor carries acyclic complexes to acyclic complexes, and so it agrees with its derived functor, also for unbounded complexes. In particular $H^q(W, F) = 0$ for $q > 0$.

(2) Limits of sheaves are computed on sections, so only colimits need an argument. We first consider a direct sum $\bigoplus_i M_i$. The direct sum sheaf is the sheafification of the presheaf direct sum, so a section over W is, locally on W, a finite sum of sections of the M_i. Since W is affine, hence quasi-compact as an object of the site, a finite covering suffices, and a single finite set of indices i serves for all members of this covering. By the sheaf condition, the section is then a section of the corresponding finite subsum, and finite sums commute with evaluation. It follows that evaluation preserves arbitrary direct sums [Stacks, 0935]. An arbitrary colimit of modules is the cokernel of a map between direct sums, and evaluation preserves cokernels by (1). Hence it preserves all colimits.

(3) Suppose that a sequence $F' \to F \to F''$ with composite zero becomes exact on all affine weakly contractible W. A local section of the kernel of $F \to F''$ is, after restriction to a covering by such W, the image of a section of F', because these objects cover every object. Hence it lies in the image sheaf, and the sequence is exact. Applied to kernels and cokernels of a map, this criterion shows that a map which is an isomorphism on all such W is an isomorphism, and applied in each degree it gives the statement for complexes; see [Stacks, 0945]. ∎

The quasi-compactness used here is quasi-compactness as an object of the site: coverings admit finite refinements. Affine objects of $X_{\text{proét}}$ have this property. The affine weakly contractible objects are quasi-compact, quasi-separated and projective, and they cover every object.

Cohomology on other objects need not vanish. For instance, a fibre product of two weakly contractible affines need not be weakly contractible, and for this reason the Čech complex of a cover by them does not in general compute cohomology.

2.5 Repleteness and limits

We need inverse limits to pass from torsion to adic coefficients, for example to form $\varprojlim_n \mathbf{Z}/\ell^n$, and the first use of weakly contractible objects is to show that such limits are well behaved. In a general topos a limit of epimorphisms need not be an epimorphism, because the local lifts at each stage may require ever finer covers. On $X_{\text{proét}}$, however, this does not happen, since surjectivity and exactness can be checked on sections over weakly contractible objects, and there no refinement is needed. As a result, the limit of a tower of epimorphisms maps epimorphically onto each term (Proposition 2.22). For the same reason products are exact (Proposition 2.23).

Exactness of products means that derived limits can be computed as for abelian groups. Namely, the derived limit of a tower is the fibre of $1 - T$ on the product. Its cohomology is therefore given by the Milnor sequence of Proposition 2.24, and $R^p \varprojlim$ vanishes for $p \geq 2$. For a tower of epimorphisms, such as $(\mathbf{Z}/\ell^n)_X$, moreover $R^1 \varprojlim$ vanishes, and the derived limit is the ordinary limit. In the next section we also deduce left completeness from the Milnor sequence. With these facts we compute the cohomology of $\mathbf{Z}_\ell$ as continuous cohomology (Proposition 2.55).

Definition 2.21 (Repleteness)**.** A topos $\mathcal{T}$ is *replete* if, for every inverse sequence $\cdots \to F_2 \to F_1 \to F_0$ of sheaves of sets with epimorphic transition maps, the projection $\varprojlim_n F_n \to F_m$ is an epimorphism for every m.

Proposition 2.22 (Repleteness of the pro-étale topos)**.** *For every scheme X, the topos $\mathrm{Sh}(X_{\text{proét}})$ is replete.*

Proof. Evaluate the sequence on a weakly contractible W. Since every epimorphism of sheaves is surjective on sections over W, the transition maps $F_{n+1}(W) \to F_n(W)$ are surjective. Given a section at stage m, we choose lifts successively at stages $m+1, m+2, \ldots$ (by countable choice), while the earlier stages are determined by the transition maps. This gives a compatible section of $\varprojlim_n F_n(W)$. Since limits are computed on sections, the projection is surjective on W.

An arbitrary U is covered by affine weakly contractible objects, so every section of F_m over U lifts to $\varprojlim_n F_n$ on each member of such a cover. Hence the projection is an epimorphism of sheaves; see [BS15, §§3.1–3.2]. ∎

For sheaves of modules the same lifting argument, applied coordinatewise rather than along a tower, shows that products are exact. This lets us compute derived limits by termwise products of complexes.

Proposition 2.23 (Exact products)**.** *Arbitrary set-indexed products are exact in $\mathrm{Mod}(X_{\text{proét}}, \mathcal{R})$. Products in $D(X_{\text{proét}}, \mathcal{R})$ are computed by taking termwise products of representative complexes, and their cohomology sheaves are the products of the cohomology sheaves.*

Proof. By Corollary corollary-evaluation-weakly-contractible-objects, exactness of sheaves can be checked on affine weakly contractible W, and by part (1) evaluation at such W is exact. Since

limits are computed on sections, a product of short exact sequences therefore evaluates on W to a product of short exact sequences of $\mathcal{R}(W)$-modules, which is exact because by the axiom of choice we may lift a tuple coordinate by coordinate. This proves the first assertion.

The derived product is the termwise product of K-injective representatives (complexes I such that maps from acyclic complexes to I are null-homotopic, the unbounded analogue of bounded-below injective complexes). Since products are exact, a product of quasi-isomorphisms is a quasi-isomorphism: its cone is the product of the cones, which is acyclic. We may therefore replace the K-injective representatives by any quasi-isomorphic ones. Hence termwise products compute the product in $D(X_{\text{pro\'et}}, \mathcal{R})$, and they commute with cohomology because products are exact; see [Stacks, 0947, 08TB]. ∎

Derived limits of countable towers have an explicit model. For a countable tower of modules, write $t_n : M_{n+1} \to M_n$ and let $T : \prod_n M_n \to \prod_n M_n$ be $T((m_n)_n) = (t_n(m_{n+1}))_n$. For a tower (K_n) in $D(X_{\text{pro\'et}}, \mathcal{R})$, represented by a tower of complexes $K_{n+1} \to K_n$, the same formula defines T termwise, and the derived inverse limit is the homotopy fibre of $1 - T$

$$R\varprojlim_n K_n = \operatorname{fib}\Big(\prod_n K_n \xrightarrow{1-T} \prod_n K_n\Big) = \operatorname{Cone}(1-T)[-1]. \tag{2.1}$$

The cone is formed on the level of complexes, so that the construction is functorial in the tower. By Proposition 2.23 the termwise products are the derived products, and so, by [Stacks, 0940], this cone agrees with the derived inverse limit.

Proposition 2.24 (The Milnor sequence)**.** *For a countable tower of $\mathcal{R}$-modules, $R^p\varprojlim = 0$ for $p > 1$, and*

$$R^1\varprojlim_n M_n = \operatorname{coker}\left(\prod_n M_n \xrightarrow{1-T} \prod_n M_n\right).$$

For a tower (K_n) of complexes there is a short exact sequence of sheaves

$$0 \longrightarrow R^1\varprojlim_n H^{q-1}(K_n) \longrightarrow H^q(R\varprojlim_n K_n) \longrightarrow \varprojlim_n H^q(K_n) \longrightarrow 0. \tag{2.2}$$

If the transitions of a tower of modules are epimorphisms, its derived limit is its ordinary limit in degree zero.

Proof. Everything follows from the triangle (2.1). Indeed, for a tower of modules, $R\varprojlim_n M_n$ is the fibre of a map between two sheaves in degree 0, so it has cohomology only in degrees 0 and 1, namely the kernel and the cokernel of $1 - T$. Here the kernel is $\varprojlim_n M_n$, and the cokernel is $R^1\varprojlim_n M_n$, which proves the first two assertions. Similarly, for a tower of complexes, the long exact cohomology sequence of (2.1) reads

$$\prod_n H^{q-1}(K_n) \xrightarrow{1-T} \prod_n H^{q-1}(K_n) \to H^q(R\varprojlim_n K_n) \to \prod_n H^q(K_n) \xrightarrow{1-T} \prod_n H^q(K_n),$$

where we used that by Proposition 2.23 the cohomology of each product is the product of the cohomology sheaves. The cokernel on the left is $R^1\varprojlim_n H^{q-1}(K_n)$ and the kernel on the right is $\varprojlim_n H^q(K_n)$, which gives (2.2).

Suppose now that the transition maps of a tower of modules are epimorphisms. We show that $1 - T$ is an epimorphism; then $R^1\varprojlim_n M_n = 0$ and the derived limit is $\varprojlim_n M_n$. By Corollary

2.20 it suffices to check surjectivity on sections over every weakly contractible W, where the maps t_n are surjective. Given (a_n), choose m_0 arbitrarily and then, recursively, m_{n+1} with $t_n(m_{n+1}) = m_n - a_n$. Then $(1-T)(m_n)_n = (a_n)_n$. See [Stacks, 0947]. ∎

Example 2.25 (A nonzero $R^1\varprojlim$)**.** Without surjective transition maps the derived limit can be nonzero in degree one. To see this, work over a geometric point and evaluate on the final object, which is weakly contractible, so that we may compute with abelian groups. Let $M_n = \mathbf{Z}$ for all n, with transition maps multiplication by ℓ. Then the maps $M_n \to \mathbf{Z}$, $m \mapsto \ell^n m$, form a map to the constant tower $\mathbf{Z}$ whose cokernel is the tower $(\mathbf{Z}/\ell^n)$. The constant tower has limit $\mathbf{Z}$ and no $R^1\varprojlim$, and $\varprojlim_n M_n = 0$, because an element of M_0 in the image of every M_n is divisible by every power of ℓ. The long exact sequence of $R\varprojlim$ therefore reads

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathbf{Z}_\ell \longrightarrow R^1 \varprojlim_n M_n \longrightarrow 0,$$

so the sections of $R^1\varprojlim_n M_n$ over the point form $\mathbf{Z}_\ell/\mathbf{Z}$, an uncountable group. If the M_n are the groups H^{q-1} of a tower of complexes, this group appears in H^q of the derived limit by (2.2).

More generally, the $R^1\varprojlim$ term vanishes if, on every affine weakly contractible W, the tower of section modules satisfies the Mittag–Leffler condition. To see this, note that evaluation on W preserves cokernels (Corollary 2.20). So the cokernel of $1-T$ evaluated on W is the cokernel of $1-T$ on the tower $(M_n(W))$, which vanishes by the Mittag–Leffler lemma [Stacks, 07KV]. As evaluation on these W detects vanishing, the $R^1\varprojlim$ term vanishes.

2.6 Left completeness and hypercovers

An unbounded complex on $X_{\text{pro\'et}}$ is determined by its bounded-below truncations, and its cohomology is computed from theirs by a derived limit. On the étale site of a scheme of infinite cohomological dimension this can fail, whereas on $X_{\text{pro\'et}}$ it holds because products are exact, so that the Milnor sequence applies.

Left completeness (Proposition 2.26) lets us treat an unbounded complex as the derived limit of bounded below ones, to which injective resolutions and the comparison theorems apply. In this way we prove the unbounded comparison of Proposition 2.49.

For bounded below complexes we also give an explicit model of cohomology. On a quasi-compact quasi-separated object U, Proposition 2.28 provides a hypercover all of whose terms W_r are affine weakly contractible. Since these terms have no higher cohomology, the hypercover spectral sequence collapses, and $R\Gamma(U, K)$ is the total complex of the sections $K^b(W_r)$. This model has a consequence we use later. Namely, in each total degree only finitely many terms occur, and evaluation on each W_r commutes with filtered colimits. Hence cohomology commutes with filtered colimits of uniformly bounded below complexes (Corollary 2.29). We use this to show that cohomology and direct images commute with inverting ℓ and with extension of scalars to $\overline{\mathbf{Q}}_\ell$, the colimit over the finite extensions E' of E.

Proposition 2.26 (Left completeness)**.** *The standard t-structure on $D(X_{\text{pro\'et}}, \mathcal{R})$ is left complete: for every complex K the natural map*

$$K \longrightarrow R \varprojlim_{n \geq 0} \tau_{\geq -n} K$$

is an isomorphism.

Proof. We compute the cohomology sheaves of the derived limit with the Milnor sequence and find that they are those of K.

Represent K by a complex and form its canonical truncation tower, with maps $K \to \tau_{\geq -n}K$ and $\tau_{\geq -(n+1)}K \to \tau_{\geq -n}K$. For fixed q, the sheaves $H^q(\tau_{\geq -n}K)$ and $H^{q-1}(\tau_{\geq -n}K)$ are eventually constant, with identity transition maps, equal to the corresponding cohomology sheaves of K, so the R^1 term in (2.2) vanishes and the limit term is $H^q(K)$. As the natural map induces this identification in every degree, it is an isomorphism; see Stacks, [0947](https://stacks.math.columbia.edu/tag/0947). ∎

To compute the cohomology of an unbounded complex from its truncations, we need derived functors on the whole unbounded derived category that commute with derived inverse limits.

Proposition 2.27 (Unbounded derived functors preserve products and limits)**.** *Let U be an object of $X_{\text{pro\'et}}$ and $f : X \to Y$ a morphism of schemes. The functor $R\Gamma(U, -)$, and the functor Rf_* from $D(X_{\text{pro\'et}}, \mathcal{R})$ to the derived category of sheaves of $f_*\mathcal{R}$-modules on $Y_{\text{pro\'et}}$, are defined on the unbounded derived categories. They preserve products and derived inverse limits.*

Proof. K-injective resolutions of arbitrary complexes exist in the Grothendieck category of $\mathcal{R}$-modules [Stacks, 079P], which defines both functors. Both are right adjoints on derived categories: $R\Gamma(U, -)$ is right adjoint to the functor sending a complex of abelian groups M to $\mathcal{R}[h_U] \otimes^{\mathbf{L}}_{\mathbf{Z}} M$, where $\mathcal{R}[h_U]$ is the free $\mathcal{R}$-module on h_U, and Rf_* is right adjoint to the derived pullback of modules. Right adjoints preserve products. Since (2.1) is built from products and a cone, an exact functor preserving products also preserves derived inverse limits [Stacks, 0940]. ∎

For bounded-below complexes, weakly contractible objects provide explicit cochain complexes computing cohomology.

Proposition 2.28 (Weakly contractible hypercovers)**.** *Let U be a quasi-compact, quasi-separated object of $X_{\text{pro\'et}}$. There is a hypercover $W_\bullet \to U$ whose term in every simplicial degree is one affine weakly contractible object. If $K \in D^+(X_{\text{pro\'et}}, \mathcal{R})$ is represented by a complex $K^\bullet$ vanishing below degree a, then*

$$R\Gamma(U, K) \simeq \operatorname{Tot}\big(K^b(W_r)\big)_{r\geq 0,\ b\geq a}. \tag{2.3}$$

The cosimplicial differential is the alternating sum of pullbacks along face maps, combined with $(-1)^r$ times the differential of $K^\bullet$. In total degree q this complex has the finite sum $\bigoplus_{0\leq r\leq q-a} K^{q-r}(W_r)$.

Proof. We build a hypercover whose terms have no higher cohomology, so that the hypercover spectral sequence collapses to the complex of sections. By quasi-compactness each term can be taken to be a single affine.

We start with a weakly contractible cover of U, and at each stage we cover the matching object by weakly contractible affines, as for fibre products above. The matching objects are finite limits of quasi-compact, quasi-separated schemes, so they are again quasi-compact and quasi-separated and are covered by finitely many affine weakly contractible objects. Taking finite disjoint unions, we get a single affine term in each degree; see [Stacks, 09A1].

For a sheaf M, the hypercover spectral sequence has $E_1^{r,s} = H^s(W_r, M)$, and the rows with $s > 0$ vanish by Corollary corollary-evaluation-weakly-contractible-objects. The remaining row is the cosimplicial complex of sections. For bounded-below $K^\bullet$, choose a bounded-below injective

resolution $K^\bullet \to I^\bullet$. Since evaluation on W_r is exact, $K^\bullet(W_r) \to I^\bullet(W_r)$ is a quasi-isomorphism. After a shift, both double complexes lie in the first quadrant, so their total complexes are quasi-isomorphic, and hypercover descent for $I^\bullet$ identifies the second with $R\Gamma(U, K)$. See [Stacks, 09A2]. ∎

For a single sheaf this is the cochain complex $M(W_0) \to M(W_1) \to M(W_2) \to \cdots$. It may be nonzero in every degree, so the hypercover gives no bound on cohomological dimension. For unbounded K, by left completeness (Proposition 2.26) and since $R\Gamma(U, -)$ commutes with derived inverse limits, we get $R\Gamma(U, K) \simeq R\varprojlim_n R\Gamma(U, \tau_{\geq -n}K)$, and the hypercover complex computes each term.

Corollary 2.29 (Continuity of cohomology)**.** *Suppose U is quasi-compact and quasi-separated and $(K_i^\bullet)$ is a filtered diagram of complexes of $\mathcal{R}$-modules vanishing below one fixed degree a. Then for every q,*

$$\varinjlim_i H^q(U, K_i^\bullet) \simeq H^q(U, \varinjlim_i K_i^\bullet).$$

If $f : X \to Y$ is quasi-compact and quasi-separated, the analogous map $\varinjlim_i R^q f_* K_i^\bullet \to R^q f_*(\varinjlim_i K_i^\bullet)$ is an isomorphism.

Proof. We compute both sides with one hypercover by weakly contractible affines, chosen independently of i. Because of the common lower bound a, each total degree q of (2.3) is a finite sum, of the terms $K_i^{q-r}(W_r)$ with $0 \leq r \leq q - a$, and the range of r is the same for all i.

By Corollary 2.20, sections on each W_r commute with filtered colimits, hence so does the finite sum in each total degree of (2.3), and the hypercover complex of $\varinjlim_i K_i^\bullet$ is the filtered colimit of the hypercover complexes of the $K_i^\bullet$. The first assertion follows because filtered colimits of modules are exact and so commute with cohomology.

The second assertion reduces to the first. Recall that $R^q f_* K$ is the sheaf associated to the presheaf $V \mapsto H^q(X \times_Y V, K)$, and that the affine objects V of $Y_{\text{pro\'et}}$ form a basis. For such V the object $X \times_Y V$ is quasi-compact and quasi-separated, so the first assertion applies to it, and sheafification commutes with colimits. ∎

In particular, $H^q(U, -)$ commutes with direct sums of sheaves of modules. Quasi-compactness of U is needed already for H^0. Let X be a geometric point and $U = \coprod_{n \in \mathbf{N}} X$, and consider the direct sum $\bigoplus_{i \in \mathbf{N}} \mathbf{Z}$ of copies of the constant sheaf. Its sections over U are $\prod_n \bigoplus_i \mathbf{Z}$, which contains the element whose nth component is the nth basis vector, whereas $\bigoplus_i H^0(U, \mathbf{Z}) = \bigoplus_i \prod_n \mathbf{Z}$ consists of the elements with only finitely many nonzero indices i.

For direct sums of unbounded complexes the common lower bound is replaced by uniform bounds on cohomological dimension (Proposition 2.49).

2.7 Classical sheaves

From here until Section 2.13, Λ denotes a discrete commutative ring and Λ_X its constant sheaf on $X_{\text{pro\'et}}$. To use the torsion arguments of étale cohomology, we must know which pro-étale sheaves come from étale sheaves. The main result answers this: étale sheaves embed fully faithfully, and the essential image consists of the sheaves whose sections on a limit $\varprojlim U_i$ are the colimit of the sections on the U_i. In particular, the reductions $\mathbf{Z}/\ell^n$ lie in this image, while the ℓ-adic coefficient sheaves themselves, such as $\mathbf{Z}_{\ell,X}$, do not (Example 2.33).

The colimit condition is forced on us by the way étale sheaves extend to limits. If F is an étale sheaf and $\bar{x}$ a geometric point, the only reasonable value of the pullback of F on the strict henselization $\operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{X,\bar{x}} = \varprojlim (U, \bar{u})$ is $\varinjlim F(U) = F_{\bar{x}}$. In this way stalks of étale sheaves become sections of pro-étale sheaves. Proposition 2.31 defines the pullback ν_X^* by this rule, and Theorem 2.32 characterizes its image by it. The part that is not formal is cohomology, because pro-étale covers are not étale covers. In the next section we show that they create no new cohomology classes for classical sheaves.

Definition 2.30 (Classical pro-étale sheaves)**.** A sheaf F on $X_{\mathrm{pro\acute{e}t}}$ is *classical* if, for every pro-étale affine presentation $U = \varprojlim_i U_i$ with U_i affine and étale over X, its restriction maps induce a bijection

$$\varinjlim_i F(U_i) \xrightarrow{\sim} F(U). \tag{2.4}$$

A complex is classical if all its cohomology sheaves are classical.

Proposition 2.31 (The comparison morphism and its inverse image)**.** *Inclusion of étale objects defines a morphism of topoi*

$$\nu_X : X_{\mathrm{pro\acute{e}t}} \longrightarrow X_{\acute{e}t}.$$

Its direct image $\nu_{X,*} : \mathrm{Sh}(X_{\mathrm{pro\acute{e}t}}) \to \mathrm{Sh}(X_{\acute{e}t})$ *is restriction to the étale subsite. Its inverse image* $\nu_X^* : \mathrm{Sh}(X_{\acute{e}t}) \to \mathrm{Sh}(X_{\mathrm{pro\acute{e}t}})$ *is exact and has the description*

$$(\nu_X^* F)(h : V \to X) = \Gamma(V_{\acute{e}t}, h_{\acute{e}t}^* F), \tag{2.5}$$

where $h_{\acute{e}t}^* : \mathrm{Sh}(X_{\acute{e}t}) \to \mathrm{Sh}(V_{\acute{e}t})$ *is the étale inverse image along the weakly étale map* h*. Both functors carry sheaves of* Λ*-modules to sheaves of* Λ*-modules, so they restrict to functors between* $\mathrm{Mod}(X_{\acute{e}t}, \Lambda)$ *and* $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$*. In particular, for* $U = \varprojlim_i U_i$ *as above, we have*

$$(\nu_X^* F)(U) = \varinjlim_i F(U_i). \tag{2.6}$$

Proof. The morphism ν_X is formal. For the formula, we write down the right-hand side of (2.5), check that it is a sheaf, and check that it has the universal property of the left adjoint of $\nu_{X,*}$.

Since an étale morphism is weakly étale and an étale covering is a pro-étale covering, the inclusion is a continuous functor preserving finite limits, and it defines ν_X. For (2.5), we define the right-hand side for every weakly étale V. Since pro-étale coverings are fpqc coverings, fpqc descent for étale sheaves shows that this rule is already a sheaf, so no sheafification is needed. If we pull back the restriction of a pro-étale sheaf G to $X_{\acute{e}t}$, we get the counit $\nu_X^* \nu_{X,*} G \to G$. On an étale object V the rule returns $F(V)$, and this is the unit $F \to \nu_{X,*} \nu_X^* F$. The two adjunction identities follow by restriction and pullback, so the rule is ν_X^* [Stacks, 0GLZ].

For an affine limit U, formula (2.6) follows from the fact that sections of an étale sheaf commute with affine inverse limits [Stacks, 099S]. ∎

For $X = \operatorname{Spec} \mathbf{F}_q$ and $U = \operatorname{Spec} \overline{\mathbf{F}}_q$, formula (2.6) says that $(\nu_X^* F)(U)$ is the colimit of the $F(\operatorname{Spec} \mathbf{F}_{q^m})$, that is, the underlying set of the Galois module corresponding to F. Formula (2.6) also shows at once that every sheaf $\nu_X^* F$ is classical. The next theorem says that there are no other classical sheaves.

Theorem 2.32 (Classical sheaves as the essential image)**.** *The functor ν_X^* is fully faithful. Its essential image is exactly the classical sheaves of Definition 2.30. For modules it is an exact equivalence between* $\mathrm{Mod}(X_{\acute{e}t}, \Lambda)$ *and the full subcategory of classical Λ_X-module sheaves on $X_{\mathrm{pro\acute{e}t}}$.*

Proof. A left adjoint is fully faithful exactly when its unit is an isomorphism, and the essential image is then the set of objects on which the counit is an isomorphism. We check both on pro-étale affines, where (2.6) computes ν_X^*.

Evaluating (2.5) on an étale object V gives $F(V)$, so $F \xrightarrow{\sim} \nu_{X,*}\nu_X^* F$, which proves full faithfulness. Moreover, every $\nu_X^* F$ is classical by (2.6). Conversely, let G be classical. On a pro-étale affine the counit $\nu_X^*\nu_{X,*}G \to G$ is

$$(\nu_X^*\nu_{X,*}G)(U) = \varinjlim_i G(U_i) \longrightarrow G(U),$$

which is an isomorphism by (2.4). Since pro-étale affines form a covering basis (Proposition 2.15), the counit is an isomorphism of sheaves. ∎

Example 2.33 (A coefficient sheaf outside the classical part)**.** Let X be a geometric point and let $S = \mathbf{Z}_\ell$ be regarded as a profinite object of $X_{\mathrm{pro\acute{e}t}}$. The completed coefficient sheaf $\mathbf{Z}_{\ell,X}$ has sections $\mathrm{Cont}(S, \mathbf{Z}_\ell)$ on S, among them the identity function. The pullback ν_X^* from $X_{\acute{e}t}$ of the constant sheaf with value the *discrete* abelian group $\mathbf{Z}_\ell$ has only locally constant functions as sections, and these have finite image on the compact space S. Hence the identity function does not come from any finite stage of the profinite presentation of S, and $\mathbf{Z}_{\ell,X}$ is not classical. Its reductions $\mathbf{Z}/\ell^n$, on the other hand, are classical. So étale arguments apply at each finite level, while the coefficient sheaf itself is defined once on the pro-étale site.

2.8 Comparison with the étale site

For a bounded-below complex K on $X_{\acute{e}t}$, the pro-étale cohomology of $\nu_X^* K$ is the étale cohomology of K (Theorem 2.35). Bounds on cohomological dimension are needed only for unbounded complexes (Section 2.11). With this theorem, statements of SGA 4 about the cohomology of torsion sheaves, such as $\mathbf{Z}/\ell^n$ on a variety, can be quoted on the pro-étale site.

The theorem rests on two facts. A pro-étale cover of an affine can be refined to a limit of étale covers, and sections of a classical sheaf on a limit are the colimit of sections. It follows that the Čech complex of a pro-étale cover is a filtered colimit of étale Čech complexes. Because of this, an injective étale sheaf stays acyclic after pullback (Lemma 2.34), and resolving by injectives gives the theorem. For non-classical coefficients the comparison fails, as the computation of $H^1(\mathrm{Spec}\,\mathbf{F}_q, \mathbf{Z}_\ell)$ in Section 1.2 shows.

It follows that the cohomology of a pro-étale affine is the colimit of the cohomology of its étale approximations (Corollary 2.36). Moreover, classical complexes in D^+ are exactly the pullbacks of étale ones (Corollary 2.37), and this is what we need for proper base change on the pro-étale site.

Lemma 2.34 (Acyclicity of pulled-back injectives)**.** *If I is an injective sheaf of Λ-modules on $X_{\acute{e}t}$, then $R^q\nu_{X,*}\nu_X^* I = 0$ for $q > 0$.*

Proof. The sheaf $R^q\nu_{X,*}\nu_X^* I$ is associated to $V \mapsto H^q(V_{\mathrm{pro\acute{e}t}}, \nu_X^* I)$, so it suffices to show that positive cohomology vanishes on affine étale objects V. The restriction of I to $V_{\acute{e}t}$ is injective,

since restriction has an exact left adjoint, namely extension by zero. Hence we may assume X affine. By Theorem 2.14, every pro-étale covering of an affine object is refined by a finite covering by affines that are ind-étale over it. By the Čech acyclicity criterion for a basis, it therefore suffices to show that the Čech complexes of $\nu_X^* I$ for such coverings have no positive cohomology.

As in [Stacks, 099U], we claim that the Čech complex of such a covering is a filtered colimit of Čech complexes of étale coverings. Write each covering map as an inverse limit of affine étale maps over the object being covered. By (2.6), the Čech complex of the covering is the filtered colimit of the Čech complexes of these étale approximations. Each finitely presented approximation descends to a common étale stage over the base, becomes a covering after increasing that stage, and its nerve descends by taking fibre products. Each augmented Čech complex at a finite stage is obtained by applying $\mathrm{Hom}(-, I)$ to the exact augmented complex of free sheaves of Λ-modules on the nerve, so it has no positive cohomology because I is injective. Filtered colimits are exact, so the same holds for the pro-étale covering, and the criterion implies the required vanishing. ■

Since injective sheaves compute derived functors, the lemma suffices to compare derived direct images, and we obtain the comparison of cohomology by resolving a bounded-below complex by injectives.

Theorem 2.35 (Bounded-below cohomology comparison)**.** *For $K \in D^+(X_{ét}, \Lambda)$ the unit of the adjunction between $\nu_X^* : D^+(X_{ét}, \Lambda) \to D^+(X_{\mathrm{pro}ét}, \Lambda)$ and $R\nu_{X,*}$ is an isomorphism*

$$K \xrightarrow{\sim} R\nu_{X,*}\nu_X^* K. \tag{2.7}$$

Consequently $\nu_X^* : D^+(X_{ét}, \Lambda) \to D^+(X_{\mathrm{pro}ét}, \Lambda)$ is fully faithful, and there is a canonical isomorphism

$$R\Gamma(X_{\mathrm{pro}ét}, \nu_X^* K) \simeq R\Gamma(X_{ét}, K).$$

Proof. We compute $R\nu_{X,*}\nu_X^* K$ with an injective resolution of K on the étale site. By the lemma its pullback is still acyclic for $\nu_{X,*}$, and by full faithfulness on sheaves the result is the resolution itself.

Choose a bounded-below injective resolution $K \to I^\bullet$. Since ν_X^* is exact, $\nu_X^* I^\bullet$ represents $\nu_X^* K$, and by Lemma 2.34 its terms are $\nu_{X,*}$-acyclic, so $R\nu_{X,*}\nu_X^* K$ is computed by the complex $\nu_{X,*}\nu_X^* I^\bullet$. By Theorem 2.32 the unit $I^q \to \nu_{X,*}\nu_X^* I^q$ is an isomorphism for every q, so this complex is $I^\bullet$, which proves (2.7). Since $\Gamma(X_{\mathrm{pro}ét}, -) = \Gamma(X_{ét}, -) \circ \nu_{X,*}$, the statement on global sections follows by composing the derived functors; see [Stacks, 099V, 099W]. ■

Applied to a pro-étale affine, the theorem computes its cohomology from the étale schemes of which it is the limit.

Corollary 2.36 (Cohomology of a pro-étale affine)**.** *Let $K \in D^+(X_{ét}, \Lambda)$, and let $U = \varprojlim_i U_i$ be a pro-étale affine with U_i affine and étale over X. Then there is an isomorphism*

$$R\Gamma(U_{\mathrm{pro}ét}, (\nu_X^* K)|_U) \simeq \varinjlim_i R\Gamma((U_i)_{ét}, K|_{U_i}). \tag{2.8}$$

Proof. By the theorem applied to U, the left side is $R\Gamma(U_{ét}, K|_U)$, so the assertion is about étale cohomology alone. For a sheaf, continuity of étale cohomology under affine limits [Stacks, 03Q6] identifies this with the colimit in each degree. For a bounded-below complex, only finitely many

terms of the hypercohomology spectral sequence contribute to each total degree, and filtered colimits are exact. ■

To apply statements proved on the étale site, such as proper base change, on $X_{\text{pro}\acute{e}t}$, we also need the essential image of ν_X^* on D^+. It consists of the classical complexes. Write $D^+_{\text{class}}(X_{\text{pro}\acute{e}t}, \Lambda)$ for the full subcategory of $D^+(X_{\text{pro}\acute{e}t}, \Lambda)$ of classical complexes.

Corollary 2.37 (Bounded-below classical complexes)**.** *Classical module sheaves form a weak Serre subcategory, closed under kernels, cokernels and extensions. There is an equivalence*

$$D^+_{\text{class}}(X_{\text{pro}\acute{e}t}, \Lambda) \simeq D^+(X_{\acute{e}t}, \Lambda),$$

with inverse functors $R\nu_{X,*}$ and ν_X^*.

Proof. We need the first assertion so that "classical complex" defines a triangulated subcategory, and for extensions we use the vanishing of $R^1\nu_{X,*}$ from the theorem. The equivalence then follows by comparing cohomology sheaves.

For kernels and cokernels the first assertion follows from exactness and full faithfulness of ν_X^*. Indeed, a map between classical sheaves is ν_X^* of a map of étale sheaves, and ν_X^* carries its kernel and cokernel to the kernel and cokernel. Given an extension $0 \to \nu_X^* F \to G \to \nu_X^* F' \to 0$, we apply $\nu_{X,*}$. Since $R^1\nu_{X,*}\nu_X^* F = 0$ by Theorem 2.35, the result $0 \to F \to \nu_{X,*}G \to F' \to 0$ is exact, and the counit maps its pullback to the original extension. This map is the identity on the outer terms, so by the five lemma $G \simeq \nu_X^*\nu_{X,*}G$ is classical.

The functor ν_X^* lands in classical complexes and is fully faithful on D^+ by Theorem 2.35, so the equivalence amounts to the counit $\nu_X^* R\nu_{X,*}K \to K$ being an isomorphism for K in D^+_{class}. By (2.7), every classical sheaf $\nu_X^* F$ is $\nu_{X,*}$-acyclic, so the hypercohomology spectral sequence for $R\nu_{X,*}K$ degenerates and $H^q(R\nu_{X,*}K) = \nu_{X,*}H^q(K)$. On H^q the counit is therefore the counit $\nu_X^*\nu_{X,*}H^q(K) \to H^q(K)$ for a classical sheaf, which is an isomorphism by Theorem 2.32; see [Stacks, 09B1]. ■

2.9 Direct images

Proper base change is proved on the étale site (Chapter 4), while $Rf_!$ is defined by direct images on the pro-étale site. To transfer proper base change, and to compute these direct images, we compare direct images on the two sites.

Let $g : X \to Y$ be a morphism of schemes. Base change of objects of the sites defines a commutative square of topoi

$$\begin{array}{ccc} X_{\text{pro}\acute{e}t} & \xrightarrow{\nu_X} & X_{\acute{e}t} \\ {\scriptstyle g_{\text{pro}\acute{e}t}}\downarrow & & \downarrow{\scriptstyle g_{\acute{e}t}} \\ Y_{\text{pro}\acute{e}t} & \xrightarrow{\nu_Y} & Y_{\acute{e}t}. \end{array}$$

It induces an isomorphism $g^*_{\text{pro}\acute{e}t}\nu_Y^* \simeq \nu_X^* g^*_{\acute{e}t}$ of functors $\text{Mod}(Y_{\acute{e}t}, \Lambda) \to \text{Mod}(X_{\text{pro}\acute{e}t}, \Lambda)$, compatible with composition. Since all four functors are exact, it is also an isomorphism of functors $D(Y_{\acute{e}t}, \Lambda) \to D(X_{\text{pro}\acute{e}t}, \Lambda)$. For direct images the comparison needs a hypothesis on g. The comparison map $\nu_Y^* Rg_{\acute{e}t,*} \to Rg_{\text{pro}\acute{e}t,*}\nu_X^*$ of functors $D^+(X_{\acute{e}t}, \Lambda) \to D^+(Y_{\text{pro}\acute{e}t}, \Lambda)$ is defined as the adjoint of the composite of $g^*_{\text{pro}\acute{e}t}\nu_Y^* Rg_{\acute{e}t,*} \simeq \nu_X^* g^*_{\acute{e}t} Rg_{\acute{e}t,*}$ with ν_X^* applied to the counit of the adjunction $g^*_{\acute{e}t} \dashv Rg_{\acute{e}t,*}$.

Evaluated on a pro-étale affine $V = \varprojlim_i V_i$ over Y, the comparison map is the map from $\varinjlim_i H^q(X \times_Y V_i, F)$ to $H^q(X \times_Y V, F)$. So the theorem below is the continuity of étale cohomology under affine limits, and like it, it needs g to be quasi-compact. Without this the map fails to be surjective already for $q = 0$. For example, let Y be affine and connected, $X = \coprod_{n \in \mathbf{N}} Y$ and F the constant sheaf $\mathbf{Z}/2$. Then g_*F is the product of countably many copies of $\mathbf{Z}/2$ on Y. On $V = Y \times S$ with S an infinite profinite set, the pro-étale side has as sections all sequences of locally constant functions $S \to \mathbf{Z}/2$, while the classical side only has the sequences that factor through a common finite quotient of S.

Theorem 2.38 (Comparison of direct images)**.** *If g is quasi-compact and quasi-separated, the canonical natural transformation*

$$\nu_Y^* R g_{\text{ét},*} K \longrightarrow R g_{\text{proét},*} \nu_X^* K \tag{2.9}$$

is an isomorphism for $K \in D^+(X_{\text{ét}}, \Lambda)$.

Proof. We work over an affine open of Y and evaluate on the covering basis $V = \varprojlim_i V_i$, with V_i affine and étale over Y. By (2.6) and Theorem 2.35, for a sheaf F the map in degree q is the continuity map

$$\varinjlim_i H^q_{\text{ét}}(X \times_Y V_i, F) \longrightarrow H^q_{\text{ét}}(X \times_Y V, F).$$

Since g is quasi-compact and quasi-separated, so are its base changes, and the map is an isomorphism by continuity of étale cohomology under affine limits [Stacks, 03Q6]. Using bounded-below resolutions, or the convergent hypercohomology spectral sequences, we pass from sheaves to K; see [Stacks, 09A6]. ∎

Since every bounded-below classical complex is of the form $\nu_X^* K$ (Corollary 2.37), the theorem shows that, for g quasi-compact and quasi-separated, $Rg_{\text{proét},*}$ carries bounded-below classical complexes to classical complexes.

In later arguments we transport maps built from units, counits and composition from the étale to the pro-étale site, and for this we record how the comparison map behaves under these operations. Write $\beta_g : g^*_{\text{proét}} \nu_Y^* \simeq \nu_X^* g^*_{\text{ét}}$ for the isomorphism above and c_g for the map (2.9).

Lemma 2.39 (Compatibilities of the comparison of direct images)**.** *Let $g : X \to Y$ be a morphism of schemes.*

1. *For $L \in D^+(Y_{\text{ét}}, \Lambda)$, the composite $\nu_Y^* L \to \nu_Y^* R g_{\text{ét},*} g^*_{\text{ét}} L \xrightarrow{c_g} R g_{\text{proét},*} \nu_X^* g^*_{\text{ét}} L$ of ν_Y^* of the unit with c_g equals the unit $\nu_Y^* L \to R g_{\text{proét},*} g^*_{\text{proét}} \nu_Y^* L$ followed by $R g_{\text{proét},*}(\beta_g)$.*
2. *For $K \in D^+(X_{\text{ét}}, \Lambda)$, the composite $g^*_{\text{proét}} \nu_Y^* R g_{\text{ét},*} K \xrightarrow{g^* c_g} g^*_{\text{proét}} R g_{\text{proét},*} \nu_X^* K \to \nu_X^* K$ with the counit equals ν_X^* of the counit $g^*_{\text{ét}} R g_{\text{ét},*} K \to K$ precomposed with β_g.*
3. *For a second morphism $h : Y \to Z$, under the identifications $R(h \circ g)_* \simeq R h_* R g_*$ on both sites, $c_{h \circ g}$ is the composite*

$$\nu_Z^* R h_{\text{ét},*} R g_{\text{ét},*} K \xrightarrow{c_h} R h_{\text{proét},*} \nu_Y^* R g_{\text{ét},*} K \xrightarrow{R h_{\text{proét},*}(c_g)} R h_{\text{proét},*} R g_{\text{proét},*} \nu_X^* K.$$

Proof. Part (2) is the definition of c_g as an adjoint. For (1), by (2) and the naturality of β_g, the adjoint of the composite in (1) is β_g followed by ν_X^* of the composite $g^*_{\text{ét}} L \to g^*_{\text{ét}} R g_{\text{ét},*} g^*_{\text{ét}} L \to g^*_{\text{ét}} L$, which is the identity by the triangle identity. Hence the adjoint is β_g, whose adjoint is the map

on the right of (1). For (3), the isomorphisms β are compatible with composition, $\beta_{h\circ g}$ being β_g composed with $g^*_{\mathrm{pro\acute{e}t}}(\beta_h)$, and the adjoint of a composite of maps defined as in (2) is the composite of their adjoints. ∎

In the comparison for constructible sheaves below we argue by induction on strata, using the functors of localization triangles. The next proposition identifies these functors on the two sites.

Proposition 2.40 (Comparison for immersions)**.**

1. *For an open immersion $j : U \hookrightarrow X$, there is a canonical isomorphism*
$$\nu_X^* j_{\acute{e}t,!} \simeq j_{\mathrm{pro\acute{e}t},!}\nu_U^* \quad \textit{of functors } \mathrm{Mod}(U_{\acute{e}t}, \Lambda) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda), \tag{2.10}$$
and hence of functors $D(U_{\acute{e}t}, \Lambda) \to D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$.
2. *For a closed immersion $i : Z \hookrightarrow X$, the map (2.9) is an isomorphism $\nu_X^* Ri_{\acute{e}t,*} \simeq Ri_{\mathrm{pro\acute{e}t},*}\nu_Z^*$ of functors $D^+(Z_{\acute{e}t}, \Lambda) \to D^+(X_{\mathrm{pro\acute{e}t}}, \Lambda)$, and $i^*_{\mathrm{pro\acute{e}t}}\nu_X^* \simeq \nu_Z^* i^*_{\acute{e}t}$. For $K \in D^+(X_{\acute{e}t}, \Lambda)$, these isomorphisms and (2.10) carry ν_X^* of the localization triangle $j_{\acute{e}t,!}j^*_{\acute{e}t}K \to K \to Ri_{\acute{e}t,*}i^*_{\acute{e}t}K \to$ to a distinguished triangle $j_{\mathrm{pro\acute{e}t},!}j^*_{\mathrm{pro\acute{e}t}}\nu_X^*K \to \nu_X^*K \to Ri_{\mathrm{pro\acute{e}t},*}i^*_{\mathrm{pro\acute{e}t}}\nu_X^*K \to$ whose first two maps are the counit and the unit of the adjunctions $j_! \dashv j^*$ and $i^* \dashv Ri_*$ on $X_{\mathrm{pro\acute{e}t}}$.*
3. *Let $f : X \to Y$ be a separated morphism of finite type with a factorization $f = \bar{f} \circ j$, where j is an open immersion and $\bar{f}$ is proper. Then there is a canonical isomorphism $\nu_Y^* R\bar{f}_{\acute{e}t,*}j_{\acute{e}t,!} \simeq R\bar{f}_{\mathrm{pro\acute{e}t},*}j_{\mathrm{pro\acute{e}t},!}\nu_X^*$ of functors $D^+(X_{\acute{e}t}, \Lambda) \to D^+(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$.*

Proof. For (1), both functors are left adjoint to restriction followed by $\nu_{U,*}$, and this also identifies the adjunction maps [Stacks, 09AD]. As the functors are exact, the identity passes to derived categories. In (2), the first identity is Theorem 2.38, and the second is the comparison of inverse images. Since ν_X^* is exact, it carries the localization triangle to a distinguished triangle. Its first map is carried to the counit on $X_{\mathrm{pro\acute{e}t}}$ because the isomorphism of (1) identifies the adjunction maps, and its second map to the unit by Lemma lemma-comparison-direct-images-compatibilities, applied to $g = i$ and $L = K$. Part (3) follows from (1) and Theorem 2.38 for the proper, hence quasi-compact and quasi-separated, map $\bar{f}$. ∎

2.10 Constructible sheaves

The six operations act on constructible complexes, and their finiteness theorems are proved on the étale site. To use these theorems we show that constructible pro-étale sheaves are classical, so that they correspond to constructible étale sheaves, and that the comparison preserves finite Tor amplitude.

A sheaf of modules on $X_{\mathrm{pro\acute{e}t}}$ is locally constant if it is trivialized by a pro-étale covering, and it is of finite presentation if its constant models are finitely presented Λ-modules. Building on this, for Noetherian Λ, a sheaf is constructible if, locally on X, there is a finite partition into constructible locally closed strata such that it is locally constant of finite type on each stratum. We write $\mathrm{Cons}(X, \Lambda)$ for the category of constructible sheaves and $D^b_c(X, \Lambda)$ for the bounded complexes with constructible cohomology sheaves. A complex K has *Tor amplitude in* $[a, b]$ if $K \otimes^{\mathbf{L}}_{\Lambda_X} N$ has cohomology only in $[a, b]$ for every sheaf N of Λ_X-modules, and $D^b_{\mathrm{ctf}}(X, \Lambda) \subset D^b_c(X, \Lambda)$ consists of the complexes of finite Tor amplitude. For a discrete ring these notions agree with Definition 3.1, which is stated for all coefficient rings in Section 3.2 below. These conditions refer only to pro-étale covers, so the main step is to descend torsors from pro-étale to étale covers.

A priori there could be more pro-étale local systems than étale ones, since the covers are larger. For finite groups, however, there are not. Indeed, a torsor under a finite group is described by a cocycle on one pro-étale affine cover. That cover is a limit of étale covers and the group is

classical, so the cocycle is defined at a finite stage. For $\mathbf{Z}_\ell$, by contrast, there are more local systems, and this is the reason for using the pro-étale site. For example, over $\operatorname{Spec}\mathbf{F}_q$, a rank-one $\mathbf{Z}_\ell$-local system on which Frobenius acts by a unit $u \in \mathbf{Z}_\ell^\times$ of infinite order is trivialized by a pro-étale cover but by no étale one. Our comparison therefore concerns only finite coefficients, and we study ℓ-adic local systems through their reductions.

What we use later is the case of a finite ring A. For such A, Theorem 2.43 and Corollary 2.44 identify the category $D^b_{\mathrm{ctf}}(X, A)$ on the pro-étale site with the classical one.

Lemma 2.41 (Torsors under classical groups)**.** *Every pro-étale torsor under a classical sheaf of groups is classical, and hence has a trivializing étale covering.*

Proof. Write the group as $\nu_X^* G_0$. We may work on an affine open of X. Using Theorem 2.14 and finite affine refinements, we refine a trivializing covering to a single faithfully flat ind-étale affine $U \to X$, and we write $U = \varprojlim_i U_i$ with U_i affine étale over X. A trivialization over U determines a transition element in $\nu_X^* G_0(U \times_X U)$. By continuity for affine limits it descends to $G_0(U_i \times_X U_i)$ for some i, and the cocycle identity on the triple product holds after enlarging i. Since $U \to X$ is surjective, each $U_i \to X$ is an étale covering. Gluing the trivial torsor along this cocycle, we get an étale torsor whose pullback is isomorphic to the given torsor. Hence the given torsor is classical by Theorem 2.32, and it is trivialized by the étale covering $U_i \to X$. See [Stacks, 099X]. ∎

The hypothesis that the group is classical is essential. For instance, the group $\mathbf{Z}_\ell^\times$ is not classical, and the rank-one $\mathbf{Z}_\ell$-local system on $\operatorname{Spec}\mathbf{F}_q$ described above is a $\mathbf{Z}_\ell^\times$-torsor with no étale trivialization. On the other hand, for locally constant sheaves with finitely presented constant models over a discrete ring, the relevant group is the automorphism group of the model, which is discrete, and the lemma applies.

Proposition 2.42 (Locally constant sheaves of finite presentation)**.** *Every locally constant pro-étale Λ_X-module sheaf of finite presentation is classical. In particular, it has a trivializing étale covering, and ν_X^* identifies the étale local systems of finite presentation with these pro-étale ones.*

Proof. A locally constant sheaf with model M is the same as a torsor under the automorphism sheaf of the constant sheaf $\underline{M}$, so the proposition follows from the lemma once we know that this automorphism sheaf is classical.

Decompose X into the open pieces on which the constant model has a fixed isomorphism type, and fix a finitely presented module M. A presentation $\Lambda^m \to \Lambda^n \to M \to 0$ identifies the internal endomorphism sheaf of $\underline{M}$ with the kernel of $\underline{M}^n \to \underline{M}^m$. Since the constant sheaf functor is exact, this kernel is the constant sheaf on $\operatorname{End}_\Lambda(M)$, and so the automorphism sheaf of $\underline{M}$ is the constant discrete group $\operatorname{Aut}_\Lambda(M)$. Hence $\mathit{Isom}(\underline{M}, F)$ is a torsor under this classical group, and by Lemma 2.41 it has sections étale locally. Since F is obtained from this torsor by twisting $\underline{M}$, it is the pullback of the corresponding twist of the constant étale sheaf, and so it is classical. This is the descent argument of [Stacks, 099Y]. ∎

To pass from a single stratum to arbitrary constructible sheaves, note that for a finite ring A the proposition applies to every finite local system of A-modules. A constructible sheaf is built from such local systems on the strata of a stratification, by extension by zero and extensions. Both operations preserve classical sheaves. For extension by zero this follows from the comparisons of Section 2.9. For extensions it is Corollary 2.37, which uses that $R^1\nu_{X,*}$ vanishes on classical sheaves.

Theorem 2.43 (Comparison for constructible sheaves)**.** *Let X be quasi-compact and quasi-separated. For a discrete Noetherian Λ, every constructible pro-étale Λ_X-module sheaf is classical. The functor ν_X^* gives an equivalence from constructible étale sheaves of Λ-modules to* $\mathrm{Cons}(X,\Lambda)$*, and an equivalence $D^b_c(X_{ét},\Lambda)\to D^b_c(X,\Lambda)$ of bounded constructible derived categories.*

Proof. The preceding proposition treats the case of a single stratum. For a finite stratification we argue locally, by induction on the number of strata. Once the sheaves are known to be classical, the equivalences follow from full faithfulness of ν_X^* and its version for D^+.

Given a point, we shrink to an affine open on which the stratum containing it is closed, with inclusion i and open complement j. The exact sequence

$$0\longrightarrow j_!j^*F\longrightarrow F\longrightarrow i_*i^*F\longrightarrow 0$$

reduces the assertion to fewer strata, because on pro-étale sheaves of modules the functors $j_!$ and i_* preserve classical sheaves, by (2.10) and Theorem 2.38, and classical sheaves are closed under extensions (Corollary 2.37). Since ν_X^* is fully faithful, sheaves that are classical on the members of a Zariski covering are classical, so the local statement suffices. The derived statement follows from Corollary 2.37 and bounded truncation; see [Stacks, 09AK, 09AL]. ∎

The six operations act on complexes of finite Tor amplitude, so we also need to know that the equivalence just proved respects this condition. A priori the pro-étale condition is stronger, since it requires a Tor bound against all pro-étale modules, most of which are not classical.

Corollary 2.44 (Finite Tor amplitude under comparison)**.** *Let X be quasi-compact and quasi-separated and let Λ be a discrete Noetherian ring. The equivalence $\nu_X^*: D^b_c(X_{ét},\Lambda)\to D^b_c(X,\Lambda)$ of Theorem 2.43 restricts to an equivalence*

$$D^b_{\mathrm{ctf}}(X_{ét},\Lambda)\simeq D^b_{\mathrm{ctf}}(X,\Lambda).$$

Proof. For one direction we represent an étale complex by a bounded flat complex and pull it back, and for the other we use the fact that ν_X^* detects vanishing.

The functor ν_X^* is exact (Proposition 2.31), and since $\nu_X^*\Lambda=\Lambda_X$ it is symmetric monoidal on sheaves of modules. The pullback of a flat sheaf of modules is flat [Stacks, 05VD]. An étale complex of Tor amplitude in $[a,b]$ is represented by a complex of flat sheaves in degrees $[a,b]$, so its pullback is a complex of flat pro-étale sheaves in the same degrees and has Tor amplitude in $[a,b]$ against every pro-étale module, not only against pulled-back ones. Conversely, let K be an étale complex. Its Tor amplitude can be read off from $K\otimes^{\mathbf{L}}_{\Lambda}M$ for arbitrary étale sheaves of modules M. The inverse image of this complex is $\nu_X^*K\otimes^{\mathbf{L}}_{\Lambda_X}\nu_X^*M$, and since ν_X^* is exact and fully faithful, it detects vanishing of cohomology. Hence a Tor bound on ν_X^*K implies the same bound on K. ∎

In particular, the corollary applies to every finite commutative ℓ-primary ring A.

2.11 Unbounded complexes

Proper base change needs the comparison with the étale site only for bounded-below complexes. Brown representability, however, takes place in the unbounded derived category, so we extend the comparison to unbounded complexes. The difficulty is that a morphism from a sheaf into an unbounded complex K involves in principle all the cohomology sheaves of K at once, and so

does a cohomology sheaf of Rg_*K. Both depend on only finitely many of them if Ext groups and cohomology of sheaves vanish above a fixed degree, uniformly in the sheaf. The smallest such degree is the cohomological dimension defined below.

Without such a bound, Hom out of the constant sheaf need not commute with direct sums. For example, take $\ell = 2$ and $S = \operatorname{Spec}\mathbf{R}$. Here étale cohomology is the cohomology of $\mathrm{Gal}(\mathbf{C}/\mathbf{R}) = \mathbf{Z}/2$, and $H^q(\mathbf{Z}/2, \mathbf{F}_2) = \mathbf{F}_2$ for every $q \geq 0$. Thus $\mathrm{cd}_2(S) = \infty$, and $\mathrm{Hom}(\mathbf{F}_2, -)$ does not commute with the direct sum $\bigoplus_{n\geq 0} \mathbf{F}_2[n]$ (see the opening of Chapter 5).

Over a field of finite cohomological dimension, the étale cohomology of schemes of finite type is uniformly bounded. In that case every computation in a fixed degree can be carried out on a bounded-below truncation. We use the bounds in this way in Proposition 2.49 below and, later, in the proofs of compactness and of the preservation of direct sums by Rp_* (Lemmas 5.6 and Lemma 5.17).

Definition 2.45 (Cohomological dimension)**.** For a qcqs scheme S and a qcqs morphism $g : X \to Y$, the *ℓ-cohomological dimensions* are

$$\mathrm{cd}_\ell(S) = \sup\{q : H^q(S_{\text{proét}}, F) \neq 0 \text{ for some classical } \ell\text{-power torsion sheaf } F\},$$
$$\mathrm{cd}_\ell(g) = \sup\{q : R^q g_* F \neq 0 \text{ for some classical } \ell\text{-power torsion sheaf } F \text{ on } X_{\text{proét}}\}.$$

For a field k' we write $\mathrm{cd}_\ell(k') = \mathrm{cd}_\ell(\operatorname{Spec} k')$.

By Theorems 2.35 and Theorem 2.38, these are the usual ℓ-cohomological dimensions computed on the étale sites. So the classical bounds apply. If k' is a field with $\mathrm{cd}_\ell(k') < \infty$, then $\mathrm{cd}_\ell(S)$ and $\mathrm{cd}_\ell(g)$ are finite for all schemes S and morphisms g of finite type over k' [Stacks, 0F10]. For our field k the hypothesis holds by the following lemma.

Lemma 2.46 (Cohomological dimension of the base field)**.** *If k' is an algebraically closed field, then $\mathrm{cd}_\ell(k') = 0$. If k' is a finite field, then $\mathrm{cd}_\ell(k') \leq 1$.*

Proof. Étale cohomology of $\operatorname{Spec} k'$ is Galois cohomology [Stacks, 03QQ]. For algebraically closed k' the Galois group is trivial. For finite k' it is $\widehat{\mathbf{Z}}$, which has cohomological dimension one (see Serre, *Galois cohomology*, II.§3). ∎

Over $\overline{\mathbf{F}}_q$, for example, a variety of dimension n has ℓ-cohomological dimension at most $2n$, and the curve $\mathbf{P}^1$ attains this bound, since $H^2(\mathbf{P}^1, \mathbf{Z}/\ell) \neq 0$. Over $\mathbf{F}_q$, however, one more degree is possible. Indeed, for $\ell \neq p$ the Hochschild–Serre spectral sequence gives $H^3(\mathbf{P}^1_{\mathbf{F}_q}, \mu_\ell) \simeq H^1(\operatorname{Spec}\mathbf{F}_q, H^2(\mathbf{P}^1_{\overline{\mathbf{F}}_q}, \mu_\ell)) = H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}/\ell) = \mathbf{Z}/\ell$.

To extend the comparison theorems to unbounded complexes with these bounds, we need two facts. First, the bounds hold uniformly over the affine objects on which we take sections (Lemma 2.47). Second, on the pro-étale side a cohomology group of an unbounded complex in a fixed degree only sees a bounded-below truncation (Lemma 2.48).

Lemma 2.47 (Uniform bounds)**.** *Let X and Y be of finite type over k, let A be a finite commutative ℓ-primary ring, let $g : X \to Y$ be of finite type, and put $d = \dim(X) + \mathrm{cd}_\ell(k)$. There is an integer c, depending only on X, such that for every classical sheaf F of A-modules on $X_{\text{proét}}$, every pro-étale affine U over X and every pro-étale affine V over Y,*

$$H^q(U, F) = 0 \quad \text{for } q > d, \qquad H^q(X \times_Y V, F) = 0 \quad \text{for } q > c.$$

Moreover $\mathrm{cd}_\ell(g) < \infty$.

Proof. We write both cohomologies as colimits of étale cohomology of schemes of finite type over k and bound these uniformly. Write $U = \varprojlim_i U_i$ and $V = \varprojlim_i V_i$ with U_i affine étale over X and V_i affine étale over Y, and write $F = \nu_X^* F_{ét}$ (Theorem 2.32). By Theorem 2.35, applied to the schemes U and $X \times_Y V$, the cohomology of F on them is the étale cohomology of $F_{ét}$. As $U = \varprojlim_i U_i$ and $X \times_Y V = \varprojlim_i X \times_Y V_i$ are limits of qcqs schemes with affine transition maps, [Stacks, 03Q6] identifies this with the colimit of the étale cohomology on the U_i, respectively on the $X \times_Y V_i$. Each U_i is affine of finite type over k with $\dim(U_i) \leq \dim(X)$, so by [Stacks, 0F0V] we have $\mathrm{cd}_\ell(U_i) \leq d$ for sheaves of A-modules. To bound cohomology on $X \times_Y V_i$ independently of i, we choose a finite affine open cover of X, with r members. Since X and Y are separated over k, its intersections and their pullbacks to the affine V_i are affine; each pullback is étale over the corresponding intersection and has dimension at most $\dim(X)$. The alternating Čech complex has length $r-1$, so the Čech spectral sequence gives the bound $c = d + r - 1$. The last assertion is [Stacks, 0F10], which applies because g is of finite type over a field with $\mathrm{cd}_\ell(k) < \infty$ (Lemma 2.46). ∎

Lemma 2.48 (Truncation on the pro-étale site)**.** *Let A be a finite commutative ℓ-primary ring, let W be an object of $X_{\mathrm{pro\acute{e}t}}$, and let e be an integer such that $H^q(W, F) = 0$ for $q > e$ and every classical sheaf F of A-modules on $X_{\mathrm{pro\acute{e}t}}$. Let $L \in D(X_{\mathrm{pro\acute{e}t}}, A)$ have classical cohomology sheaves, with $H^q(L) = 0$ for $q > b$. Then $H^j R\Gamma(W, L) = 0$ for $j > b + e + 1$.*

Proof. By left completeness (Proposition 2.26), $L \simeq R\varprojlim_n \tau_{\geq -n} L$, and derived sections preserve derived inverse limits, so $R\Gamma(W, L) \simeq R\varprojlim_n R\Gamma(W, \tau_{\geq -n} L)$. Each $\tau_{\geq -n} L$ is bounded with classical cohomology sheaves in degrees at most b, so by the hypothesis and the hypercohomology spectral sequence, $R\Gamma(W, \tau_{\geq -n} L)$ has no cohomology above $b + e$. The Milnor sequence for this tower of complexes of abelian groups, proved as in Proposition 2.24, expresses H^j of the derived limit through $\varprojlim^1$ of H^{j-1} and $\varprojlim$ of H^j of the terms. Both vanish for $j > b + e + 1$. ∎

Proposition 2.49 (Unbounded comparison for finite coefficients)**.** *Let X and Y be of finite type over the base field k, let A be a finite commutative ℓ-primary ring, and let $g : X \to Y$ be of finite type. Then $\nu_X^* : D(X_{ét}, A) \to D(X_{\mathrm{pro\acute{e}t}}, A)$ is fully faithful and preserves arbitrary direct sums. The map (2.9), now a natural transformation of functors $D(X_{ét}, A) \to D(Y_{\mathrm{pro\acute{e}t}}, A)$, is an isomorphism for every $K \in D(X_{ét}, A)$.*

Proof. By the two lemmas, both sides in a fixed degree depend only on a bounded-below truncation of K, where Theorems 2.35 and Theorem 2.38 apply. Let d and c be as in Lemma 2.47, fix $K \in D(X_{ét}, A)$ and a degree m, and consider the triangle $\tau_{<N} K \to K \to \tau_{\geq N} K$ for an integer N chosen below.

Full faithfulness of ν_X^* means that the unit $K \to R\nu_{X,*}\nu_X^* K$ is an isomorphism, and it suffices to check this on the maps $H^m R\Gamma(U, K) \to H^m R\Gamma(U, \nu_X^* K)$ for affine étale U over X. We first reduce to bounded-below complexes. On the étale side, $\Gamma(U, -)$ has cohomological dimension at most d on sheaves of A-modules, and by [Stacks, 07K7] the group $H^m R\Gamma(U, K)$ depends only on $\tau_{\geq N} K$ as soon as $N \leq m - d$. On the pro-étale side, since ν_X^* is exact it commutes with truncations, and Lemma 2.48, applied with $e = d$ to $L = \nu_X^* \tau_{<N} K$ and $b = N - 1$, shows that H^m and H^{m+1} of $R\Gamma(U, L)$ vanish as soon as $N < m - d$. Hence for such N both sides are unchanged when K is replaced by $\tau_{\geq N} K$, and for this bounded-below complex the map is an isomorphism by Theorem 2.35.

For direct images, we evaluate (2.9) on a pro-étale affine V over Y; these objects form a basis of $Y_{\mathrm{pro\acute{e}t}}$. As before, we bound the contribution of $\tau_{<N} K$ on both sides. On the étale side, the complex $Rg_{ét,*}\tau_{<N} K$ has cohomology sheaves in degrees at most $N - 1 + \mathrm{cd}_\ell(g)$, so $\nu_Y^* Rg_{ét,*}\tau_{<N} K$ has

classical cohomology in the same range, and Lemma 2.48 on Y, with $e = \dim(Y)+\mathrm{cd}_\ell(k)$ (Lemma 2.47 for Y), shows that its derived sections over V vanish above degree $N+\mathrm{cd}_\ell(g)+\dim(Y)+\mathrm{cd}_\ell(k)$. On the other side, $R\Gamma(V, Rg_{\mathrm{pro\acute{e}t},*}\nu_X^*\tau_{<N}K) = R\Gamma(X\times_Y V, \nu_X^*\tau_{<N}K)$, and the same lemma with $W = X\times_Y V$ and $e = c$ gives vanishing above degree $N + c$. Therefore, choosing N so small that H^m and H^{m+1} vanish on both sides, both sides of (2.9), evaluated on V in degree m, are unchanged when K is replaced by $\tau_{\geq N}K$, and for this bounded-below complex the map is an isomorphism by Theorem 2.38.

The functor ν_X^* preserves direct sums because it is an exact left adjoint. ■

Since by Theorem 2.32 the classical sheaves of A-modules are the essential image of ν_X^* on sheaves, and ν_X^* commutes with direct sums of sheaves, classical sheaves of A-modules are closed under arbitrary direct sums in $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, A)$.

2.12 Coefficient rings

We want pullback along morphisms and change of coefficients to be compatible without any bookkeeping. For this we define each coefficient ring once, on a point, and pull it back to every scheme along a single morphism of topoi. This gives A_X, $\mathcal{O}_{E,X}$, E_X and $\overline{\mathbf{Q}}_{\ell,X}$. Here A is a finite commutative ring killed by a power of ℓ, with the discrete topology, and A_X is the usual constant sheaf. For the point we take the site of profinite sets described below. By Proposition 2.50 it is the pro-étale site of a separably closed field. Unlike the étale site of the field, it keeps the topology of the coefficient rings.

2.12.1 Condensed rings

The constant sheaf with value the abstract ring $\mathbf{Z}_\ell$ forgets the topology of $\mathbf{Z}_\ell$, and on $\operatorname{Spec}\mathbf{F}_q$ it has $H^1 = 0$ (Section 1.2). We therefore need a notion of coefficient ring on a geometric point that retains the topology. The pro-étale site of a separably closed field is the site of profinite sets (Proposition 2.50 below). So a coefficient ring must at least be a sheaf of rings on profinite sets, that is, a condensed ring. The continuous maps from profinite sets into $\mathbf{Z}_\ell$ form such a sheaf. With it, H^1 of $\mathbf{F}_q$ becomes $\mathbf{Z}_\ell$ (Example 2.57).

Let $*_{\mathrm{pro\acute{e}t}}$ be the site of profinite sets, with finite jointly surjective families as covers. A *condensed ring* is a sheaf of rings on $*_{\mathrm{pro\acute{e}t}}$. A topological ring V defines a condensed ring $\underline{V}$, whose sections over S are the continuous maps $S \to V$. This is a sheaf because a surjection of profinite sets is a quotient map.

Proposition 2.50 (The pro-étale site of a separably closed field)**.** *Let Ω be a separably closed field. The category of pro-étale affines over $\operatorname{Spec}\Omega$ is equivalent to the category of profinite sets, with finite jointly surjective families as covers. Consequently its topos is, up to a set-theoretic cutoff, the topos of condensed sets. A discrete ring A defines the classical constant sheaf, while a profinite or locally profinite coefficient ring defines its sheaf of continuous functions on profinite sets.*

Proof. Since a finitely presented étale Ω-algebra is a finite product of copies of Ω, filtered colimits of such algebras correspond, under Spec, to inverse limits of finite sets. Under this correspondence, faithfully flat covering maps become surjections of profinite sets, and by quasi-compactness every cover can be refined by a finite one, so the sites are equivalent. The topoi then agree because the pro-étale affines form a basis and sheaves are determined by their values on them. Finally, the last assertion is the definition of the condensed ring $\underline{R}$ of a topological ring R given above,

whose sections over a profinite set S are the continuous maps $S \to R$; for discrete R these are the locally constant functions, which are the sections of the classical constant sheaf. ■

The étale topos of Ω is the topos of sets, whereas by the proposition its pro-étale topos is the topos of condensed sets. The final object of $*_{\text{pro\'et}}$ is weakly contractible, so taking global sections is exact. However, a condensed sheaf is not determined by its global sections. For classical sheaves geometric stalks still suffice, whereas a general sheaf of modules on the pro-étale site has to be evaluated on all weakly contractible affines (Corollary 2.20). For example, the completed coefficient sheaf $\mathbf{Z}_{\ell,X}$ of Example 2.33 is of this general kind: its sections on a profinite set S are all continuous functions $S \to \mathbf{Z}_\ell$.

Fix an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$. Throughout, E denotes a finite extension of $\mathbf{Q}_\ell$ inside $\overline{\mathbf{Q}}_\ell$, with ring of integers $\mathcal{O}_E$ and uniformizer π_E. For $\mathcal{O}_E$ and E the ℓ-adic topologies define condensed rings $\underline{\mathcal{O}_E}$ and $\underline{E}$. For $\overline{\mathbf{Q}}_\ell$, however, the topology causes a problem. To see why, recall that classically a $\overline{\mathbf{Q}}_\ell$-sheaf is an E-sheaf for some finite E with scalars extended, and all arguments with lattices and reduction modulo π_E take place over that E. They cannot be made over $\overline{\mathbf{Q}}_\ell$ itself, because $\overline{\mathbf{Q}}_\ell$ is not complete and its valuation ring is not a discrete valuation ring. But the sheaf of all continuous maps to $\overline{\mathbf{Q}}_\ell$ has sections that are defined over no finite E (see below), so it would lose this reduction. To keep it available, we take instead the filtered colimit over the finite coefficient fields. Using the valuation topology on $\overline{\mathbf{Q}}_\ell$, we put

$$\underline{\overline{\mathbf{Q}}_\ell}(S) = \{a : S \to \overline{\mathbf{Q}}_\ell \text{ continuous} : a(S) \subset E \text{ for some finite } E/\mathbf{Q}_\ell\} = \varinjlim_E \underline{E}(S).$$

This is already a sheaf, because a cover of a profinite set can be refined by a finite one, and finitely many finite extensions of $\mathbf{Q}_\ell$ lie in a common one. Since filtered colimits of sheaves on $*_{\text{pro\'et}}$ are computed sectionwise, $\underline{\overline{\mathbf{Q}}_\ell} = \varinjlim_E \underline{E}$ as condensed rings.

This condensed ring is smaller than $\underline{V}$ for $V = \overline{\mathbf{Q}}_\ell$ with the valuation topology. Indeed, algebraic numbers of unbounded degree can tend to 0. Such a sequence is a continuous map from a convergent sequence to $\overline{\mathbf{Q}}_\ell$ with image in no finite extension. With our choice, Lemma 3.23 shows that every constructible $\overline{\mathbf{Q}}_\ell$-complex and every morphism between such complexes descends to a finite E.

2.12.2 Coefficient sheaves

A profinite set S defines a pro-étale scheme $\operatorname{Spec} LC(S, \mathbf{Z})$ over $\mathbf{Z}$, where $LC(S, \mathbf{Z})$ is the ring of locally constant functions. The functor $S \mapsto \operatorname{Spec} LC(S, \mathbf{Z})$ is continuous and preserves finite limits, as one checks on finite quotients of S, so it induces a morphism of topoi $\operatorname{Sh}((\operatorname{Spec} \mathbf{Z})_{\text{pro\'et}}) \to \operatorname{Sh}(*_{\text{pro\'et}})$ [Stacks, 00X6]. Composing with the structure morphism of X, we obtain

$$p_X : \operatorname{Sh}(X_{\text{pro\'et}}) \longrightarrow \operatorname{Sh}(*_{\text{pro\'et}}),$$

and for a condensed ring Λ we write $\Lambda_X = p_X^*\Lambda$. For a morphism $f : X \to Y$ we have $p_Y \circ f = p_X$, and hence a canonical isomorphism $f^{-1}\Lambda_Y \simeq \Lambda_X$, compatible with identities and composition. Likewise, a morphism $\Lambda \to \Lambda'$ of condensed rings pulls back to $\Lambda_X \to \Lambda'_X$, which defines extension of scalars $\Lambda'_X \otimes_{\Lambda_X} -$ from Λ_X-modules to Λ'_X-modules on $X_{\text{pro\'et}}$. Thus both pullback along morphisms and change of coefficients are obtained from p_X^*.

The next lemma identifies these rings with the completed coefficient rings of classical ℓ-adic cohomology.

Lemma 2.51 (Comparison with completed coefficients)**.** *Write $\mathcal{O}_{E,X}$, E_X and $\overline{\mathbf{Q}}_{\ell,X}$ for the pullbacks to $X_{\text{pro\'et}}$ of $\underline{\mathcal{O}_E}$, $\underline{E}$ and $\underline{\overline{\mathbf{Q}}_\ell}$. There are canonical isomorphisms of sheaves of rings*

$$\mathcal{O}_{E,X} \simeq \varprojlim_n (\mathcal{O}_E/\pi_E^n)_X, \qquad E_X \simeq \mathcal{O}_{E,X}[1/\ell], \qquad \overline{\mathbf{Q}}_{\ell,X} \simeq \varinjlim_E E_X,$$

where $(\mathcal{O}_E/\pi_E^n)_X$ is the constant sheaf. They are compatible with inverse image along every morphism of schemes.

Proof. The functor p_X^* commutes with colimits, so the second and third isomorphisms reduce to identities on the point. The first is an inverse limit, and since p_X^* need not commute with inverse limits, we use instead that $\underline{\mathcal{O}_E}$ is representable.

On the point, $\underline{\mathcal{O}_E}$ is represented by the profinite set $\mathcal{O}_E = \varprojlim_n \mathcal{O}_E/\pi_E^n$, so its pullback to X is represented by the pro-étale X-scheme $\varprojlim_n \coprod_{\mathcal{O}_E/\pi_E^n} X$. By the Yoneda lemma the sheaf this scheme represents is $\varprojlim_n (\mathcal{O}_E/\pi_E^n)_X$, and the ring structures match.

A continuous map from a profinite set to E has compact image, and hence lands in $\ell^{-m}\mathcal{O}_E$ for some m. Thus $\underline{E} = \underline{\mathcal{O}_E}[1/\ell]$ on the point. Since inverting ℓ is a filtered colimit and p_X^* commutes with colimits, we obtain the second isomorphism. In the same way the third follows from $\underline{\overline{\mathbf{Q}}_\ell} = \varinjlim_E \underline{E}$. All three isomorphisms are obtained by applying p_X^*, and $p_Y \circ f = p_X$ for every morphism $f : X \to Y$, so they are compatible with pullback. ∎

The first isomorphism describes $\mathcal{O}_{E,X}$ as a limit of its finite quotients. To compute derived reductions modulo π_E^n we also need these quotients as cokernels of multiplication by π_E^n, which the next corollary provides.

Corollary 2.52 (Finite quotients and their transition maps)**.** *For every $n \geq 1$ there is a canonical exact sequence on $X_{\text{pro\'et}}$*

$$0 \longrightarrow \mathcal{O}_{E,X} \xrightarrow{\pi_E^n} \mathcal{O}_{E,X} \longrightarrow (\mathcal{O}_E/\pi_E^n)_X \longrightarrow 0.$$

Thus the quotient is represented by the two-term complex $[\mathcal{O}_{E,X} \xrightarrow{\pi_E^n} \mathcal{O}_{E,X}]$ in degrees $-1, 0$. Under this representation, reduction from level $n+1$ to level n is the chain map (π_E, id). These resolutions and maps commute with inverse image along every morphism of schemes.

Proof. It suffices to prove exactness on the point, since p_X^* is exact and inverse image along $f : X \to Y$ satisfies $p_Y \circ f = p_X$. Now, on a profinite set S, multiplication by π_E^n is injective on continuous functions $S \to \mathcal{O}_E$, and its image is the kernel of reduction. Moreover, reduction is surjective because a continuous function $S \to \mathcal{O}_E/\pi_E^n$ takes finitely many values, and choosing a representative of each gives a continuous lift. Finally, the chain map (π_E, id) commutes with the differentials because $\pi_E^n \pi_E = \pi_E^{n+1}$, and its degree-zero component is the identity, so it induces the quotient projection. ∎

Using the two-term resolution we see that, for every complex K of $\mathcal{O}_{E,X}$-modules, the derived reduction $K \otimes^{\mathbf{L}} (\mathcal{O}_E/\pi_E^n)_X$ is the cone of $\pi_E^n : K \to K$ (Lemma 2.62).

We call the rings $A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$ the *admissible* coefficient rings. Here $E = \mathbf{Q}_\ell$ is allowed, so that $\mathbf{Z}_\ell$ and $\mathbf{Q}_\ell$ are the case $E = \mathbf{Q}_\ell$ of $\mathcal{O}_E$ and E. For $E = \mathbf{Q}_\ell$ we write $\mathbf{Z}_{\ell,X} = \mathcal{O}_{\mathbf{Q}_\ell,X}$ and $\mathbf{Q}_{\ell,X} = (\mathbf{Q}_\ell)_X$. By the comparison lemma above,

$$\mathbf{Z}_{\ell,X} \simeq \varprojlim_n (\mathbf{Z}/\ell^n)_X, \qquad \mathbf{Q}_{\ell,X} \simeq \mathbf{Z}_{\ell,X}[1/\ell].$$

For a condensed ring Λ we write $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the category of Λ_X-modules. Its derived category is $D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$.

Duality and the trace map need, besides the constant sheaf, its twists by roots of unity. The reason is already visible on $\mathbf{P}^1$ over an algebraically closed field k, for n invertible in k. The Kummer sequence $0 \to \mu_n \to \mathbf{G}_m \xrightarrow{n} \mathbf{G}_m \to 0$ gives an isomorphism $H^2(\mathbf{P}^1, \mu_n) \simeq \mathrm{Pic}(\mathbf{P}^1)/n = \mathbf{Z}/n$, given by the degree of a line bundle. This isomorphism involves no choice, and it sends the class of a point to 1. An identification of $H^2(\mathbf{P}^1, \mathbf{Z}/n)$ with $\mathbf{Z}/n$, on the other hand, requires an isomorphism $\mu_n \simeq \mathbf{Z}/n$, that is, a choice of a primitive nth root of unity. Frobenius does not respect such a choice.

To see this, let k be an algebraic closure of $\mathbf{F}_q$. The arithmetic Frobenius acts on $\mu_{\ell^n}(k)$ by $\zeta \mapsto \zeta^q$. The geometric Frobenius is the inverse of the arithmetic Frobenius in $\mathrm{Gal}(k/\mathbf{F}_q)$. Hence it acts on $\Lambda(1)$ by q^{-1}, and on $H^2(\mathbf{P}^1, \Lambda) \simeq \Lambda(-1)$ by q. This agrees with $\#\mathbf{P}^1(\mathbf{F}_q) = 1 + q$.

Over $\mathbf{C}$ the twist is $2\pi i\mathbf{Z}$. The exponential sequence identifies $H^2(\mathbf{P}^1(\mathbf{C}), 2\pi i\mathbf{Z})$ with $\mathbf{Z}$. An identification of $H^2(\mathbf{P}^1(\mathbf{C}), \mathbf{Z})$ with $\mathbf{Z}$, by contrast, is a choice of orientation, and the complex orientation depends on the choice of i.

With twists, the trace map of $\mathbf{P}^1$ (Proposition 6.7) and smooth purity $f^! \simeq f^*(d)[2d]$ (Theorem 9.3) are canonical and commute with Frobenius (Proposition 10.20). This is where the factor q^d in Poincaré duality (c) comes from.

Definition 2.53 (Tate twists)**.** Let X be a scheme on which ℓ is invertible, and let $\mu_{\ell^n,X}$ be the sheaf of ℓ^nth roots of unity on $X_{\mathrm{pro\acute{e}t}}$. Then it is locally isomorphic to $(\mathbf{Z}/\ell^n\mathbf{Z})_X$. For a finite ring A with $\ell^r A = 0$ put $A_X(1) = \mu_{\ell^r,X} \otimes_{\mathbf{Z}/\ell^r\mathbf{Z}} A_X$, which by the Kummer maps is independent of r. Put $\mathcal{O}_{E,X}(1) = \varprojlim_n (\mathcal{O}_E/\ell^n)_X(1)$, and obtain $E_X(1)$ and $\overline{\mathbf{Q}}_{\ell,X}(1)$ by inverting ℓ and passing to the colimit over E; equivalently, $\overline{\mathbf{Q}}_{\ell,X}(1) = \overline{\mathbf{Q}}_{\ell,X} \otimes_{\mathbf{Z}_{\ell,X}} \mathbf{Z}_{\ell,X}(1)$ with $\mathbf{Z}_{\ell,X}(1) = \varprojlim_n \mu_{\ell^n,X}$. Each $\Lambda_X(1)$ is locally free of rank one. For $d \geq 0$, $\Lambda_X(d)$ is the d-fold tensor power, $\Lambda_X(-d)$ its dual, and $K(d) = K \otimes^{\mathbf{L}}_{\Lambda_X} \Lambda_X(d)$.

2.12.3 Inverse image

Every base change argument below compares a module over Λ_Y with its pullback to X, and for this we want the pullback of a Λ_Y-module to be its inverse image as a sheaf, with no tensor product. For coherent sheaves this fails, since $f^*M = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}M$ and $f^{-1}\mathcal{O}_Y \neq \mathcal{O}_X$ in general. For our coefficient sheaves the two notions agree, because all coefficient sheaves are pulled back from the point. The same holds for their Tate twists, where ℓ is invertible.

The second part of the next lemma says that extension of scalars to a finite extension E'/E is a finite direct sum. We use this, for instance, in the proof of Proposition 4.40.

Lemma 2.54 (Inverse image of the coefficient sheaves)**.** *Let $f : X \to Y$ be a morphism of schemes.*

1. *For every admissible Λ, the canonical map*

$$f^{-1}\Lambda_Y \longrightarrow \Lambda_X$$

is an isomorphism of sheaves of rings. If ℓ is invertible on Y, the same holds for its Tate twists, including the twists of a finite ring A.

2. *If $E \subset E'$ is a finite extension, then $\mathcal{O}_{E',X}$ is a finite free $\mathcal{O}_{E,X}$-module and E'_X is a finite free E_X-module.*

Proof. (1) For every condensed ring Λ, the identity $p_Y \circ f = p_X$ shows that

$$f^{-1}\Lambda_Y = f^{-1}p_Y^*\Lambda \simeq p_X^*\Lambda = \Lambda_X,$$

and this proves the assertion for the coefficient rings themselves.

For the twists, ℓ is invertible on Y, hence on X, as Definition 2.53 requires. The torsion twists μ_{ℓ^n} are then represented by finite étale group schemes, and these commute with base change. Their inverse limit is represented by the inverse limit of these schemes, so that its inverse image is represented by the base change [Stacks, 09A5]. This proves the assertion for $\mathcal{O}_E(1)$. For a finite A killed by ℓ^n, its twist is obtained from the twist at finite level by extending scalars from $\mathbf{Z}/\ell^n\mathbf{Z}$, so inverse image commutes with it as well. The other twists are obtained by inverting ℓ, extending scalars, and taking tensor powers and duals.

(2) An $\mathcal{O}_E$-basis of $\mathcal{O}_{E'}$ is an isomorphism of topological $\mathcal{O}_E$-modules $\mathcal{O}_{E'} \simeq \mathcal{O}_E^{[E':E]}$, hence of condensed modules $\underline{\mathcal{O}_{E'}} \simeq \underline{\mathcal{O}_E}^{[E':E]}$. Pull it back along p_X, and invert ℓ for the fields. ∎

By (1), the pullback of modules over these sheaves, defined in Section 3.1 below, is the exact functor f^{-1}.

2.13 Continuous cohomology

The comparison theorems above concern discrete coefficients, whereas the integral and rational ℓ-adic coefficient rings of Section 2.12 are defined directly on the pro-étale site, because the discrete constant sheaves with the same values have the wrong cohomology (Example 2.57). The main result of this section, Proposition 2.55, says that pro-étale cohomology with coefficients $\mathbf{Z}_\ell$ is the classical continuous cohomology, that is, the derived limit of cohomology with coefficients $\mathbf{Z}/\ell^n$.

The sheaf $\mathbf{Z}_{\ell,X}$ is the inverse limit of the classical sheaves $(\mathbf{Z}/\ell^n)_X$, with surjective transition maps (Lemma 2.51). For such a limit, the cohomology is the derived limit of the cohomology of the terms. When the terms are classical, this is continuous étale cohomology.

Proposition 2.55 (Continuous cohomology as pro-étale cohomology)**.** *Let X be a scheme and (F_n) an inverse system of abelian sheaves on $X_{\text{proét}}$ with epimorphic transition maps. Put $F = \varprojlim_n F_n$ in the pro-étale topos. Then, in the derived category of abelian groups,*

$$R\Gamma(X_{\text{proét}}, F) \simeq R\varprojlim_n R\Gamma(X_{\text{proét}}, F_n).$$

Proof. Since the transition maps are epimorphisms, $F \simeq R\varprojlim_n F_n$ by the last assertion of Proposition 2.24. Moreover, $R\Gamma(X_{\text{proét}}, -)$ commutes with derived inverse limits since it is a right adjoint. ∎

When the F_n come from the étale site, as the sheaves $(\mathbf{Z}/\ell^n)_X$ do, each term on the right is étale cohomology by the comparison theorem, and the proposition identifies pro-étale cohomology of the limit with continuous étale cohomology $R\varprojlim_n R\Gamma(X_{ét}, F_n)$.

Corollary 2.56 (Continuous étale cohomology)**.** *In Proposition 2.55, suppose that $F_n = \nu_X^* F_{n,ét}$ for an inverse system $(F_{n,ét})$ of abelian sheaves on $X_{ét}$ with epimorphic transition maps. Then $R\Gamma(X_{\text{pro}ét}, F)$ is the continuous étale cohomology of $(F_{n,ét})$: the right derived functor of $(G_n) \mapsto \Gamma(X_{ét}, \varprojlim_n G_n)$ on the abelian category of inverse systems, evaluated at $(F_{n,ét})$.*

Proof. By Theorem 2.35, each $R\Gamma(X_{\text{pro}ét}, F_n)$ is $R\Gamma(X_{ét}, F_{n,ét})$. The derived functor of $(G_n) \mapsto \Gamma(X_{ét}, \varprojlim_n G_n)$ is the composite of the derived functors of sections and of inverse limit, so its value at $(F_{n,ét})$ is $R\varprojlim_n R\Gamma(X_{ét}, F_{n,ét})$ [BS15, Proposition 5.6.2]. ∎

Applying the Milnor sequence (2.2) to the tower of complexes $R\Gamma(X_{\text{pro}ét}, F_n)$ and using the proposition, we obtain for every q a natural exact sequence

$$0 \longrightarrow R^1 \varprojlim_n H^{q-1}(X_{\text{pro}ét}, F_n) \longrightarrow H^q(X_{\text{pro}ét}, F) \longrightarrow \varprojlim_n H^q(X_{\text{pro}ét}, F_n) \longrightarrow 0. \tag{2.11}$$

The first term can be nonzero, because the transition maps between the cohomology groups need not be surjective even if those between the sheaves F_n are. It vanishes if the groups $H^{q-1}(X_{\text{pro}ét}, F_n)$ satisfy the Mittag–Leffler condition. This holds for instance if they are finite, since then descending chains of images stabilize at each level.

Example 2.57 (The first cohomology of a finite field)**.** For $X = \operatorname{Spec} \mathbf{F}_q$ and $\ell \nmid q$,

$$H^1(X_{\text{pro}ét}, \mathbf{Z}_{\ell,X}) \simeq \mathbf{Z}_\ell, \qquad H^1(X_{\text{pro}ét}, \mathbf{Q}_{\ell,X}) \simeq \mathbf{Q}_\ell.$$

At each finite level the first cohomology is $\operatorname{Hom}_{\text{cont}}(\widehat{\mathbf{Z}}, \mathbf{Z}/\ell^n) = \mathbf{Z}/\ell^n$, and the groups in degree zero are finite, so that their $\varprojlim^1$ vanishes. Hence the sequence (2.11) for $F_n = (\mathbf{Z}/\ell^n)_X$ proves the integral statement. Since X is qcqs, its pro-étale cohomology commutes with filtered colimits [Stacks, 09A3], in particular with inverting ℓ, and the rational statement follows. For the constant étale sheaf with value the discrete group $\mathbf{Q}_\ell$, on the other hand, the first cohomology is $\operatorname{Hom}_{\text{cont}}(\widehat{\mathbf{Z}}, \mathbf{Q}_\ell) = 0$, because a continuous homomorphism from a profinite group to a discrete torsion-free group has finite, hence zero, image.

2.14 Extension by zero and completion

The integral coefficient sheaves are inverse limits. To restrict integral complexes to the strata of a stratification, and to extend them by zero, we therefore need the functors $j_!$, j^*, i^* and i_* to commute with limits. On the pro-étale site this holds, even for inverse image along a closed immersion, and from this we deduce the localization triangles and construct the local cohomology functor. With these tools we can argue with closed supports before the exceptional inverse image and proper base change are available.

That i^* commutes with limits is special to the pro-étale site. On $X_{ét}$ the stalk of i^*F at a point of Z is the stalk of F. This is a colimit, and colimits do not commute with infinite products. On $X_{\text{pro}ét}$, by contrast, the value of i^*M on a weakly contractible affine over Z is the value of M on one affine object of $X_{\text{pro}ét}$, henselian along Z (Proposition 2.58), and evaluation on one object preserves limits. For $\mathcal{O}_{E,X} = \varprojlim (\mathcal{O}_E/\pi_E^n)_X$ this means that i^* applied to the limit is the limit of the restrictions $(\mathcal{O}_E/\pi_E^n)_Z$. The same holds for any integral complex given as a derived limit of its reductions.

A complex over $\mathcal{O}_E$ is determined by its reductions modulo π_E^n when it is derived complete (Definition 2.61 below). Combined with the computation of the cohomology of these reductions

on the étale site, this is how we reduce integral statements to finite coefficients throughout the notes.

Proposition 2.58 (Extension by zero and restriction to a closed subscheme)**.** *Let X be quasi-compact and quasi-separated, $j : U \hookrightarrow X$ a quasi-compact open immersion, and $i : Z \hookrightarrow X$ its closed complement. For a sheaf of rings $\mathcal{R}$ on $X_{\text{proét}}$, we use its inverse images as the coefficient rings on U and Z. Thus i^* goes from $\mathcal{R}$-module sheaves on $X_{\text{proét}}$ to $i^*\mathcal{R}$-module sheaves on $Z_{\text{proét}}$, i_* goes back, and j^*, $j_!$ relate $\mathcal{R}$-module sheaves on $X_{\text{proét}}$ and $j^*\mathcal{R}$-module sheaves on $U_{\text{proét}}$. Then i^* is exact and preserves all small limits and colimits; i_* is exact and fully faithful. The functors j^* and $j_!$ are exact, and every $\mathcal{R}$-module sheaf M has the exact sequence*

$$0 \longrightarrow j_!j^*M \longrightarrow M \longrightarrow i_*i^*M \longrightarrow 0.$$

Moreover $j_!$ preserves products and derived inverse limits of module complexes. Finally, for a morphism $g : X' \to X$ with base changes $j' : U' \hookrightarrow X'$ and $g' : U' \to U$, the canonical map $j'_!g'^* \to g^*j_!$ is an isomorphism.

Proof. The special feature of the pro-étale site is that i^* preserves limits; we prove this first, deduce exactness of $j_!$ from the localization sequence, obtain the statement about products and derived limits from a triangle expressing $j_!$ through Rj_*, i^* and i_*, and check base change by adjunction.

Step 1: the functors i^* and i_*. The functor i^* preserves limits, because evaluating i^*M on a weakly contractible affine over Z is the same as evaluating M on an affine object of $X_{\text{proét}}$, and evaluation commutes with limits. The assertion is local on X. Let V be a weakly contractible object of the affine ind-étale basis of Z. It determines an affine object $v(V)$ in the corresponding basis of X, henselian along its inverse image of Z: the functor v is the fully faithful left adjoint to base change of these affine bases, and $v(V) \times_X Z \simeq V$ [Stacks, 09A9]. The pullback of a sheaf, evaluated on V, is its evaluation on $v(V)$, because every covering of a weakly contractible object splits, so that sheafification does not change values there. Since evaluation commutes with limits and these objects detect isomorphisms, i^* preserves limits. Being a left adjoint, it also preserves colimits. Since it preserves both limits and colimits, it is exact. Lifting covers along v shows that i_* is exact and that $i^*i_* = \mathrm{id}$ and $j^*i_* = 0$. Finally, the localization sequence is exact on the pro-étale site [Stacks, 09AH], that is, the adjunction map $M \to i_*i^*M$ is surjective with kernel $j_!j^*M$.

Step 2: exactness of j^* and $j_!$. The functor j^* is restriction to $U_{\text{proét}}$, hence exact. The functor $j_!$ is exact because its composites $j^*j_!$ and $i^*j_!$ are exact and the pair (j^*, i^*) detects exactness. Here the first composite is $j^*j_! = \mathrm{id}$, and the second is $i^*j_! = 0$. Indeed, in the localization sequence of $j_!N$ the first map $j_!j^*j_!N \to j_!N$ is an isomorphism since $j^*j_!N = N$. Hence $i_*i^*j_!N = 0$, and $i^*j_!N = 0$ because i_* is fully faithful. The pair (j^*, i^*) detects exactness, because both functors are exact, hence commute with taking homology of a complex of sheaves, while a sheaf M with $j^*M = 0$ and $i^*M = 0$ is zero by the localization sequence.

Step 3: products and derived inverse limits. For $K \in D(U_{\text{proét}}, j^*\mathcal{R})$, applying the localization sequence to a complex representing Rj_*K and using $j^*Rj_*K \simeq K$, we obtain the triangle

$$j_!K \longrightarrow Rj_*K \longrightarrow i_*i^*Rj_*K \longrightarrow j_!K[1].$$

The last two terms preserve products and derived inverse limits, because Rj_* and i_* are right adjoints and i^* is exact and preserves limits by Step 1; hence so does their fibre $j_!$. As $j_!$ is exact and products are exact, the corresponding statements about limits of sheaves of modules follow.

Step 4: base change. The map $j'_! g'^* \to g^* j_!$ is an isomorphism because its right adjoint is. Indeed, $j'_! g'^*$ is left adjoint to $g'_* j'^*$ and $g^* j_!$ is left adjoint to $j^* g_*$, and the right adjoint of the map is the base change map $j^* g_* \to g'_* j'^*$, which is an isomorphism since both sides send a sheaf on $X'_{\text{pro\'et}}$ to its restriction to the objects lying over U'. The same holds on derived categories, since the coefficient rings are inverse images, so that g^*, g'^* and $j'^!$ are exact. The argument with the functor v is [BS15, §6.1]; see also [Stacks, 09AA, 09AH, 09BM] and [Stacks, 09BP]. ∎

For arguments with closed supports we need a right adjoint of i_* before $i^!$ is constructed by Brown representability. We build it from i^*, j^* and Rj_* alone. We keep the notation of Proposition 2.58.

Definition 2.59 (Local cohomology)**.** The *local cohomology functor* $R\underline{\Gamma}_Z : D(X_{\text{pro\'et}}, \mathcal{R}) \to D(Z_{\text{pro\'et}}, i^*\mathcal{R})$ is the right derived functor of the functor of sections supported on Z,

$$F \longmapsto i^* \ker(F \longrightarrow j_* j^* F),$$

where $F \to j_* j^* F$ is the unit of $j^* \dashv j_*$.

Lemma 2.60 (Local cohomology is right adjoint to i_*)**.** *The functor $R\underline{\Gamma}_Z$ is right adjoint to the exact fully faithful functor $i_* : D(Z_{\text{pro\'et}}, i^*\mathcal{R}) \to D(X_{\text{pro\'et}}, \mathcal{R})$. It is computed by*

$$R\underline{\Gamma}_Z K \simeq i^* \operatorname{fib}(K \longrightarrow Rj_* j^* K),$$

where $K \to Rj_* j^* K$ is the unit of $j^* \dashv Rj_*$.

Proof. We first compute $R\underline{\Gamma}_Z$. Represent K by a K-injective complex J with injective terms, so that $Rj_* j^* K$ is represented by $j_* j^* J$. The unit $J \to j_* j^* J$ is surjective in each degree: for an injective $\mathcal{R}$-module I and an object V, a section of $j_* j^* I$ over V is a map $j_! j^* \mathcal{R}_V \to I$, and it extends along the monomorphism $j_! j^* \mathcal{R}_V \to \mathcal{R}_V$. Hence the kernel of $J \to j_* j^* J$, taken termwise, represents the fibre of the unit, and applying the exact functor i^* gives $R\underline{\Gamma}_Z K \simeq i^* \operatorname{fib}(K \to Rj_* j^* K)$.

Let $F = \operatorname{fib}(K \to Rj_* j^* K)$. Since $j^* i_* M = 0$, we have $\operatorname{Hom}(i_* M, Rj_* j^* K) = \operatorname{Hom}(j^* i_* M, j^* K) = 0$ in all degrees, so maps $i_* M \to K$ are the same as maps $i_* M \to F$. The restriction $j^* F$ vanishes, because j^* of the unit is an isomorphism, so the localization sequence gives $F \simeq i_* i^* F$, and $\operatorname{Hom}(i_* M, i_* i^* F) = \operatorname{Hom}(M, i^* F)$ since i_* is fully faithful. ∎

Once $i^!$ exists, it agrees with $R\underline{\Gamma}_Z$ on the localizing category on which $i^!$ is defined, since both are right adjoints of i_* there.

We treat integral coefficients through their reductions modulo powers of π_E. This is possible for complexes that are recovered from their reductions, the derived complete ones. Derived completeness is the analogue for complexes of the condition $M = \varprojlim_n M/\pi_E^n M$ on a module. For example, $\mathcal{O}_{E,X}$ itself is derived complete. The sheaf E_X is not, since all its reductions vanish.

Definition 2.61 (Derived adic completeness)**.** Let $E/\mathbf{Q}_\ell$ be finite, with uniformizer π_E, and let $\mathcal{O}_{E,X} = p_X^* \underline{\mathcal{O}_E}$ and $(\mathcal{O}_E/\pi_E^n)_X$ be the integral and finite coefficient sheaves of Section 2.12. A complex $K \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ is *derived complete* if its canonical map

$$K \longrightarrow R\varprojlim_n (K \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} (\mathcal{O}_E/\pi_E^n)_X)$$

is an isomorphism.

We write $K_n = K \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} (\mathcal{O}_E/\pi_E^n)_X$ for the reductions of K. The canonical transition maps $K_{n+1} \to K_n$ induce isomorphisms $K_{n+1} \otimes^{\mathbf{L}}_{(\mathcal{O}_E/\pi_E^{n+1})_X} (\mathcal{O}_E/\pi_E^n)_X \simeq K_n$, by associativity of the derived tensor product. We call (K_n), with these isomorphisms, the *tower of reductions* of K.

Lemma 2.62 (Reduction and derived Nakayama)**.** *Let $K \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$. Then K_n is the cone of $\pi_E^n : K \to K$. If K is derived complete and $K_1 = 0$, then $K = 0$. Consequently a morphism between derived complete complexes is an isomorphism whenever the induced map on K_1 is an isomorphism.*

Proof. The first assertion follows from the two-term resolution of $(\mathcal{O}_E/\pi_E^n)_X$ in Corollary 2.52. Tensoring K with the short exact sequence $0 \to \pi_E^n\mathcal{O}_E/\pi_E^{n+1} \to \mathcal{O}_E/\pi_E^{n+1} \to \mathcal{O}_E/\pi_E^n \to 0$, whose first term is isomorphic to $\mathcal{O}_E/\pi_E$, gives a triangle $K_1 \to K_{n+1} \to K_n$. By induction $K_1 = 0$ implies $K_n = 0$ for all n. Then the derived limit of the K_n vanishes, and $K = 0$ by completeness.

For the last assertion, let $u : K \to L$ be a morphism of derived complete complexes and C its cone. Derived completeness is preserved by cones, since both sides of the completion map are exact functors of K, so C is derived complete. Reduction modulo π_E is exact, so C_1 is the cone of $u_1 : K_1 \to L_1$, which vanishes when u_1 is an isomorphism. Hence $C = 0$, and u is an isomorphism. ∎

Over a point, for $K = \mathcal{O}_E$ placed in degree 0, the reductions are $K_n = \mathcal{O}_E/\pi_E^n$, and the lemma is the classical Nakayama lemma for the complete local ring $\mathcal{O}_E$. The hypothesis of completeness cannot be dropped, since $E \neq 0$ but $E/\pi_E = 0$.

Let K be derived complete. Since $R\Gamma(X_{\text{pro\'et}}, -)$ is a right adjoint, it commutes with the derived limit, and therefore

$$R\Gamma(X_{\text{pro\'et}}, K) \simeq R\varprojlim_n R\Gamma(X_{\text{pro\'et}}, K_n).$$

When the K_n are classical bounded-below complexes, each term on the right can be computed on the étale site by Theorem 2.35. In this way we compute the cohomology of K from its tower of reductions (K_n).

In Section 3.2 we define constructible complexes over $\mathcal{O}_E$, and we show that for Noetherian X they are exactly the derived complete complexes K whose reduction K_1 is constructible (Proposition 3.11). Derived complete complexes can equally be studied over the constant sheaf with value the discrete ring $\mathcal{O}_E$ (Lemma lemma-integral-models).

The rational coefficient sheaves are obtained by inverting ℓ and passing to the colimit over E: $E_X \simeq \mathcal{O}_{E,X}[1/\ell]$ and $\overline{\mathbf{Q}}_{\ell,X} \simeq \varinjlim_E E_X$ (Lemma 2.51). Chapter 3 also compares constructible complexes over E_X and $\overline{\mathbf{Q}}_{\ell,X}$ with the classical adic categories. It does so by lattices and by descent to a finite extension.

3 Constructible complexes

A lisse E-sheaf is the analogue of a local system: pro-étale locally it is isomorphic to E_X^r for some r (Definition 3.2 below). Inside such a sheaf L, a lattice is a locally free $\mathcal{O}_E$-subsheaf L_0 with $L_0[1/\ell] = L$, and the first point of this chapter is that a lattice need not exist. For an example, glue 0 and ∞ on $\mathbf{P}^1$ to obtain a nodal curve X. Then the infinite chain Y of projective lines,

with ∞ on each line glued to 0 on the next, maps to X by an étale covering. Moreover, the group $\mathbf{Z}$ acts on Y by translation, simply transitively on the fibres, so $Y \to X$ is an étale $\mathbf{Z}$-torsor. Using this torsor and letting the generator of $\mathbf{Z}$ act on E by multiplication by ℓ, we obtain a lisse E-sheaf of rank one. It has no lattice, because ℓ is not a unit of $\mathcal{O}_E$, so that multiplication by ℓ does not carry any lattice onto itself. The sheaf comes from the character $n \mapsto \ell^n$ of $\mathbf{Z}$, and since this character does not extend continuously to $\widehat{\mathbf{Z}}$, the sheaf has no classical counterpart as a system (F_n) (see the discussion before Lemma 3.24; the example is in [BS15]).

This matters because every geometric input to the next chapters (proper base change, the cohomology of curves, Deligne's generic base change) is a theorem about torsion sheaves, whereas a rational sheaf has no reduction modulo ℓ: since E_X is ℓ-divisible, we have $E_X \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E = 0$. To use the torsion theory we must therefore find integral objects inside rational ones, and then pass to their torsion reductions. The example above shows that this cannot be done globally.

Classically each coefficient ring comes with its own theory:

coefficients	classical constructible objects
finite ring A	constructible étale sheaves of A-modules
$\mathbf{Z}_\ell$, $\mathcal{O}_E$	systems (F_n) of constructible $\mathcal{O}_E/\pi_E^n$-sheaves with $F_{n+1} \otimes \mathcal{O}_E/\pi_E^n \simeq F_n$
$\mathbf{Q}_\ell$, E	the same systems up to isogeny, that is, with ℓ inverted
$\overline{\mathbf{Q}}_\ell$	a colimit of the categories for all finite $E/\mathbf{Q}_\ell$

On the pro-étale site, by contrast, one local condition covers all of them. Namely, a Λ_X-module is constructible if, on the strata of a finite stratification of X, it is pro-étale locally of the form $\Lambda_X \otimes_{\Lambda(*)} P$ for a finitely presented module P over $\Lambda(*)$, the ring of sections over the point (Definition 3.1). For finite A these are the classical constructible sheaves, while for E and $\overline{\mathbf{Q}}_\ell$ they are lisse on the strata. For $\mathcal{O}_E$, on the other hand, the modules P may have torsion, for example $P = \mathcal{O}_E/\pi_E$.

The six operations will act on $D^b_{\mathrm{ctf}}(X, \Lambda)$, bounded complexes with constructible cohomology and finite Tor amplitude. For Λ not a finite ring the Tor condition is automatic, that is, $D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda)$ (Proposition 3.12). For a finite ring, however, it matters: over $\mathbf{Z}/\ell^2$ the constructible sheaf $\mathbf{F}_\ell$ has infinite Tor dimension (Section 3.2 below).

Over $\mathcal{O}_E$ the passage to torsion is a criterion: for Noetherian X, a complex is constructible exactly when it is derived complete and its reduction modulo π_E is constructible (Proposition 3.11). The criterion is not formal, because the local models of the reductions modulo π_E^n must be made compatible in n (Proposition 3.10, Section 3.3). This description is due to Bhatt and Scholze [BS15].

Over E and $\overline{\mathbf{Q}}_\ell$, as the nodal curve shows, lattices exist only after stratifying, and this is enough. After descending from $\overline{\mathbf{Q}}_\ell$ to a finite extension E of $\mathbf{Q}_\ell$ and refining a stratification, a lisse sheaf has a lattice on each stratum (Lemma 3.24), and the reductions of the lattice modulo π_E^n are classical torsion sheaves (Lemma 3.25).

On a connected normal scheme no stratification is needed: there a lattice exists globally, and lisse E-sheaves are continuous representations of $\pi_1(X, \bar{x})$ (Proposition 13.10, proved in Part III).

Dévissage along strata also shows that Hom out of a constructible complex commutes with filtered colimits of complexes bounded below by a common degree (Lemma 3.22); this continuity is used, together with descent to a finite E, in Chapter 5.

Throughout this chapter schemes are again separated and of finite type over k, as in our conventions.

3.1 Derived functors

Four of the six operations, f^*, Rf_*, $\otimes^{\mathbf{L}}$ and internal Hom, exist for every condensed ring Λ on the whole unbounded derived categories by the usual resolutions. The remaining two need more work: $Rf_!$ is built from proper direct image and extension by zero (Section 5.4), and its right adjoint comes from Brown representability.

Sheaves of Λ_X-modules can be viewed as module objects over the ring object Λ_X in $\mathrm{Sh}(X_{\mathrm{pro\acute{e}t}}, \mathrm{Ab})$, with the tensor product over Λ_X given by the coequalizer of the two actions $M \otimes \Lambda_X \otimes N \rightrightarrows M \otimes N$. The category of Λ_X-modules is Grothendieck abelian. Hence complexes have K-injective resolutions, which define Rf_* and $R\mathcal{H}om$ [Stacks, 07A5, 08J7]. They also have K-flat resolutions, which define $\otimes^{\mathbf{L}}$, and $K \otimes^{\mathbf{L}} -$ is left adjoint to $R\mathcal{H}om(K, -)$ [Stacks, 06YL].

The pullback of a Λ_Y-module M along $f : X \to Y$ is $f^*M = \Lambda_X \otimes_{f^{-1}\Lambda_Y} f^{-1}M$. Since $f^{-1}\Lambda_Y \simeq \Lambda_X$ (Lemma 2.54), this is just $f^{-1}M$. Therefore $f^* : \mathrm{Mod}(Y_{\mathrm{pro\acute{e}t}}, \Lambda) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is exact, and on derived categories it is left adjoint to $Rf_* : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$. For an open immersion $j : U \to X$, extension by zero $j_! : \mathrm{Mod}(U_{\mathrm{pro\acute{e}t}}, \Lambda) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is exact and left adjoint to j^*.

Constructible sheaves will be built from lisse sheaves on the strata of a stratification, each extended by zero to all of X (Lemma 3.18 and Corollary 3.20). We fix this extension by zero now. Let $a : S \hookrightarrow X$ be a locally closed immersion, for instance the inclusion of a stratum. Then S is open in its closure, so a factors as $a = i \circ j$, with $j : S \hookrightarrow \overline{S}$ the open immersion into the closure and $i : \overline{S} \hookrightarrow X$ the closed immersion. We put

$$a_! = i_* j_! : \mathrm{Mod}(S_{\mathrm{pro\acute{e}t}}, \Lambda) \longrightarrow \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda).$$

For a Λ_S-module L, the sheaf $a_!L$ restricts to L on S, and its restrictions to $\overline{S} \setminus S$ and to $X \setminus \overline{S}$ are zero. Indeed $a^* a_! = j^* i^* i_* j_! = \mathrm{id}$, the sheaf $j_!L$ vanishes on $\overline{S} \setminus S$, and i_* of a sheaf vanishes on $X \setminus \overline{S}$ (Proposition 2.58, which applies because the open immersions involved are quasi-compact, X being Noetherian). The same proposition shows that $j_!$ and i_* are exact, so $a_!$ is exact. It therefore acts on complexes term by term and defines $a_! : D(S_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ with no derived functor needed. Once $Rf_!$ is constructed (Section 5.4), this functor is $Ra_!$.

3.2 Constructibility

With finite coefficients a constructible sheaf is, on the strata of a finite stratification, locally constant with finite stalks. The six operations, however, will act on constructible complexes of finite Tor amplitude for every coefficient ring, so we need a definition of constructibility that covers all coefficient rings at once. We give it by a local condition (Definition 3.1). This condition has two parts, a stratification and a local model that replaces local constancy, and the complexes we use must in addition have finite Tor amplitude. As we explain below, each of the three conditions answers a specific need. For $\mathcal{O}_E$ we then show that constructibility can be tested on the reduction modulo π_E (Proposition 3.11). For a morphism $g : Z \to X$ and a Λ_X-module M we write $M|_Z = g^*M$, a Λ_Z-module.

A stratification cannot be avoided, already for sheaves coming from geometry: for $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$, the sheaf $j_!\Lambda$ is zero at the origin and Λ elsewhere, so it is not locally constant on $\mathbf{A}^1$.

Local constancy itself has to be rephrased. For instance, for $\Lambda = \mathbf{Z}_\ell$, "locally constant" with value a $\mathbf{Z}_\ell$-module would mean locally a constant sheaf with value the discrete group, but on $\operatorname{Spec} \mathbf{F}_q$ such a sheaf has $H^1 = 0$ (Section 1.2), whereas the answer we want is $\mathbf{Z}_\ell$. We therefore ask instead that M be locally of the form $\Lambda_U \otimes_{\Lambda(*)} P$, with P a finitely presented module over the ring $\Lambda(*)$ of sections over the point, as in the opening of this chapter. This keeps the ℓ-adic

topology. For finite Λ it is local constancy, and for E it says that M is locally free of finite rank. In this formulation, finite presentation of P replaces finiteness of the stalks.

Finally, the six operations should commute with change of coefficients, and this needs finite Tor amplitude. Over $\mathbf{Z}/\ell^2$ the sheaf $\mathbf{F}_\ell$ is constructible, but $\mathbf{F}_\ell \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^2} \mathbf{F}_\ell$ has cohomology $\mathbf{F}_\ell$ in every degree ≤ 0. With the Tor condition imposed, by contrast, the category D^b_{ctf} is stable under change of coefficients and under $\otimes^{\mathbf{L}}$, and for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ the Tor condition is automatic (Proposition 3.12).

Definition 3.1 (Constructible sheaves)**.** Let Λ be a commutative condensed ring.

1. Consider the following condition on a Λ_X-module M: there is a pro-étale cover $\{U_i \to X\}$ and finitely presented modules P_i over the ring $\Lambda(*)$ of sections over the point such that
$$M|_{U_i} \simeq \Lambda_{U_i} \otimes_{\Lambda(*)} P_i.$$
Here $\Lambda(*)$ and P_i are constant sheaves on U_i, acting through $\Lambda(*) \to \Lambda_{U_i}$. Equivalently, on each U_i the module M has a presentation by a finite matrix with entries in $\Lambda(*)$.
2. For qcqs X, a Λ_X-module M is *constructible* if there is a finite stratification of X by constructible locally closed subschemes such that on each stratum S the restriction $M|_S$ satisfies the condition of (1). On an arbitrary scheme this condition is imposed locally on affine opens.

We write $\mathrm{Cons}(X, \Lambda) \subset \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the full subcategory of constructible sheaves, and $D^b_c(X, \Lambda) \subset D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the full subcategory of bounded complexes with constructible cohomology sheaves. A complex K has *Tor amplitude in* $[a, b]$ if $K \otimes^{\mathbf{L}}_{\Lambda_X} N$ has cohomology only in $[a, b]$ for every sheaf N of Λ_X-modules. We write $D^b_{\mathrm{ctf}}(X, \Lambda)$ for the full subcategory of $D^b_c(X, \Lambda)$ of complexes of finite Tor amplitude.

The modules that satisfy the condition of (1) on all of X are the analogue of local systems, and they are the building blocks on the strata.

Definition 3.2 (Lisse sheaves)**.** Let Λ be a commutative condensed ring. A Λ_X-module M is *lisse* if it satisfies the condition of Definition definition-constructible-sheaves on X itself, without a stratification. A complex $K \in D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ is *lisse* if its cohomology sheaves are lisse.

For the individual coefficient rings the definitions take a familiar form.

Remark 3.3 (Constructible sheaves for the individual coefficient rings)**.**

1. For Noetherian X, every locally closed subset is constructible, so in Definition definition-constructible-sheaves any finite stratification by locally closed subschemes may be used.
2. For finite A and qcqs X, $D^b_{\mathrm{ctf}}(X, A)$ is the pullback of the usual étale category $D^b_{\mathrm{ctf}}(X_{\acute{e}t}, A)$ by Corollary 2.44.
3. If $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, the ring $\Lambda(*)$ is a field, so each P_i is a finite-dimensional vector space, and a lisse module is locally free of finite rank. In other words, a constructible E- or $\overline{\mathbf{Q}}_\ell$-sheaf is one that is lisse on the strata of a finite stratification [BS15, Definition 6.8.6].
4. If $\Lambda = \mathcal{O}_E$, locally on a stratum a constructible $\mathcal{O}_{E,X}$-module is $P \otimes_{\mathcal{O}_E} \mathcal{O}_{E,X}$ for a finitely generated $\mathcal{O}_E$-module P. For example, $\mathcal{O}_{E,X}/\pi_E$ is constructible.
5. For the admissible coefficients, the constructible sheaves form an abelian category. For finite A this is the classical theory of sheaves with finite stalks, and the other cases are Lemma 3.18, Corollary 3.20 and Lemma 5.19, which we prove in Subsection 5.4.2.

6. For finite A, a lisse A_X-module is the same as a locally constant sheaf with finite stalks. For $\Lambda = \mathcal{O}_E$ a lisse module may have torsion, such as $\mathcal{O}_{E,X}/\pi_E$; the lisse $\mathcal{O}_{E,X}$-modules without torsion are the locally free ones of finite rank, since a finitely generated torsion free $\mathcal{O}_E$-module is free.

Part (5) is special to the admissible coefficients. For a general condensed ring, kernels of maps between constructible sheaves can fail to be constructible, and then $D^b_c(X, \Lambda)$ is not triangulated.

3.3 Integral constructible complexes

Constructibility over $\mathcal{O}_E$ is defined by a local condition, but in practice we know a complex $K \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ through its reductions $K_n = K \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} (\mathcal{O}_E/\pi_E^n)_X$ of Definition 2.61. The criterion below says that K is constructible exactly when it is derived complete and K_1 is constructible. Write $\mathbf{F}_E = \mathcal{O}_E/\pi_E$ for the residue field, so that $K_1 \in D(X_{\text{pro\'et}}, \mathbf{F}_E)$.

To prove the criterion we must produce, on each stratum, one complex of finite free $\mathcal{O}_E$-modules that is locally a model for all the K_n at once. Each K_n separately is easy to control: its bounds and its stratification are inherited from K_1 by dévissage along the powers of π_E, and on a stratum it is locally isomorphic to a constant perfect complex. The difficulty is that these local isomorphisms are chosen independently at each level, so they have to be made compatible before we pass to the limit. On a weakly contractible cover we do this by a Mittag–Leffler argument, which needs a uniform rate. Such a rate exists because a bounded complex of finite free modules over the discrete valuation ring $\mathcal{O}_E$ has finitely generated cohomology, and the torsion of this cohomology is killed by a fixed power of π_E. Throughout, we use the étale site only at the finite levels, through Theorem 2.43.

Example 3.4 (The criterion over a point)**.** On the spectrum of an algebraically closed field, take for K the sheaf attached to an $\mathcal{O}_E$-module. The computation is then one with modules, and it shows where the uniform rate comes from. Resolving $\mathcal{O}_E/\pi_E^e$ by $[\mathcal{O}_E \xrightarrow{\pi_E^e} \mathcal{O}_E]$ in degrees $-1, 0$ and reducing modulo π_E, we find

$$\mathcal{O}_E \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathbf{F}_E = \mathbf{F}_E, \qquad \mathcal{O}_E/\pi_E^e \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathbf{F}_E = [\mathbf{F}_E \xrightarrow{0} \mathbf{F}_E] = \mathbf{F}_E \oplus \mathbf{F}_E[1] \quad (e \geq 1),$$

since π_E^e becomes zero modulo π_E. For $P = \mathcal{O}_E^r \oplus \bigoplus_{i=1}^s \mathcal{O}_E/\pi_E^{e_i}$ the reduction K_1 has $H^0 = \mathbf{F}_E^{r+s}$ and $H^{-1} = \mathbf{F}_E^s$. Thus $H^{-1}(K_1)$ counts the cyclic torsion summands. For $K = \bigoplus_{n\geq 1} \mathcal{O}_E/\pi_E^n$ we get $K_1 = \bigoplus_{n\geq 1}(\mathbf{F}_E \oplus \mathbf{F}_E[1])$. This is not constructible, because its cohomology is an infinite-dimensional vector space in degrees 0 and -1. In other words, a constructible K_1 allows only finitely many torsion summands. For P the single power $\pi_E^{\max e_i}$ then kills all the torsion, and this is the uniform rate.

The completeness hypothesis cannot be dropped either. The module E is π_E-divisible, so $E \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E^n = 0$ for all n. Hence $K_1 = 0$ is constructible, yet E is not a finitely generated $\mathcal{O}_E$-module. The criterion does not apply because E is not derived complete: its derived completion $R\varprojlim_n (E \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E^n)$ is zero.

Lemma 3.5 (Bounds for the finite reductions)**.** *Let X be Noetherian and let $K \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ be derived complete with $K_1 \in D^b_c(X, \mathbf{F}_E)$. Choose $[a, b]$ containing the cohomology of K_1. Then every K_n lies in $D^b_c(X, \mathcal{O}_E/\pi_E^n)$ and has Tor amplitude in $[a, b]$, and K has cohomology only in $[a, b]$. If the cohomology sheaves of K_1 are locally constant on a stratum, then so are those of every K_n.*

Proof. Everything about K_n is obtained from K_1 by induction on n, using that K_n is an extension of K_{n-1} by K_1. The Tor bound needs an extra argument with minimal resolutions, and the bound on K itself then follows from the Milnor sequence.

The filtration of $\mathcal{O}_E/\pi_E^n$ by the powers of π_E has graded pieces $\mathbf{F}_E$. Tensoring its short exact sequences with K over $\mathcal{O}_{E,X}$, we get triangles $K_1 \to K_n \to K_{n-1}$, up to the identification $\pi_E^{n-1}\mathcal{O}_E/\pi_E^n \simeq \mathbf{F}_E$. Moreover, complexes with constructible cohomology in $[a,b]$ are closed under extensions in triangles, by the long exact cohomology sequence and the fact that constructible sheaves form a weak Serre subcategory. Induction on n now shows that K_n is bounded in $[a,b]$ with constructible cohomology. Since locally constant finite sheaves of modules also form a weak Serre subcategory, the same induction works on a fixed stratum, and a single stratification works for all n. By the comparison theorem for constructible sheaves (Theorem 2.43), these finite sheaves and complexes are classical, that is, pulled back from the étale site.

For the Tor bound we use $K_n \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^n} \mathbf{F}_E \simeq K_1$. At an étale geometric point $\bar{x}$, take a bounded above resolution of the stalk of K_n by finite free modules over the Artinian local ring $\mathcal{O}_E/\pi_E^n$. Cancelling a two-term identity summand whenever a differential matrix has a unit entry, we obtain a minimal resolution, whose differentials have entries in (π_E), so they vanish after tensoring with $\mathbf{F}_E$. Since stalks commute with derived tensor products, the minimal resolution tensored with $\mathbf{F}_E$ computes the stalk of K_1 at $\bar{x}$, with zero differential. Its rank in degree q is therefore $\dim_{\mathbf{F}_E} H^q(K_1)_{\bar{x}}$. This vanishes outside $[a,b]$, and we conclude that the stalk has a bounded finite free resolution in $[a,b]$. Such a bound on stalks gives the same Tor bound on the étale site, and Corollary 2.44 carries it to the pro-étale site, for Tor against arbitrary sheaves of modules there.

By completeness $K = R\varprojlim K_n$, so the Milnor sequence (2.2) gives the lower bound and the vanishing above $b+1$. For $H^{b+1}(K)$ we use the top degree, where $H^b(K_n) = H^b(K_{n+1}) \otimes_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n$, so the transition maps are epimorphisms. Their $R^1\varprojlim$ vanishes by the last assertion of Proposition 2.24, and this means that $H^{b+1}(K) = 0$. ∎

To make the local trivializations of the K_n compatible, we need two facts about complexes over $\mathcal{O}_E$: a compatible system of perfect complexes over the rings $\mathcal{O}_E/\pi_E^n$ comes from a single bounded finite free complex, and the images of the cohomology at finite levels stabilize at a uniform rate.

Lemma 3.6 (Perfect complexes over $\mathcal{O}_E$)**.** *For $n \geq 1$ let $M_n \in D(\mathcal{O}_E/\pi_E^n)$ be perfect, with maps $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1} \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n \simeq M_n$. Then there is a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules with isomorphisms $P^\bullet/\pi_E^n \simeq M_n$ in $D(\mathcal{O}_E/\pi_E^n)$, compatible with the transition maps.*

Proof. We use that a minimal complex over the local ring $\mathcal{O}_E/\pi_E^n$ is determined up to isomorphism of complexes, not only up to homotopy, so that the transition maps can be made into equalities of matrices and passed to the limit.

Represent each M_n by a minimal finite free complex, obtained by cancelling unit entries in the differential matrices. Its reduction modulo π_E has zero differential, so its rank in degree q is $\dim_{\mathbf{F}_E} H^q(M_1)$, which is independent of n and nonzero only in finitely many degrees. The given isomorphism $M_{n+1}/\pi_E^n \simeq M_n$ between minimal complexes is represented by a chain homotopy equivalence f, with homotopy inverse g. Since both complexes have zero differential modulo π_E, the relations $gf - 1 = dh + hd$ and $fg - 1 = dh' + h'd$ show that f is an isomorphism modulo π_E in each degree. By Nakayama's lemma its matrices are then invertible over $\mathcal{O}_E/\pi_E^n$. Lifting them to invertible matrices over $\mathcal{O}_E/\pi_E^{n+1}$ and changing bases, we may assume that the differential at level $n+1$ reduces exactly to the differential at level n, with the given transition in the derived category. Induction on n gives compatible matrices, and their limits define $P^\bullet$. Its differential

squares to zero because it does so modulo every π_E^n, and the identifications with the M_n are compatible by construction. ∎

The second fact is used in the proof of Lemma 3.9 for the internal Hom complex $(P^\bullet)^\vee \otimes P^\bullet$, whose H^0 at level n consists of the maps $P^\bullet/\pi_E^n \to P^\bullet/\pi_E^n$ in the derived category. There it says that a map at level n which lifts far enough lifts to all levels, and how far is enough depends only on the torsion of the cohomology.

Lemma 3.7 (Images of cohomology at finite levels)**.** *Let $Q^\bullet$ be a bounded complex of finite free $\mathcal{O}_E$-modules, and let π_E^e kill the torsion of $H^{q+1}(Q^\bullet)$. Then for $m \geq n + e$*

$$\operatorname{im}(H^q(Q^\bullet/\pi_E^m) \to H^q(Q^\bullet/\pi_E^n)) = \operatorname{im}(H^q(Q^\bullet) \to H^q(Q^\bullet/\pi_E^n)).$$

Such an e exists, since $\mathcal{O}_E$ is a Noetherian discrete valuation ring and $H^{q+1}(Q^\bullet)$ is finitely generated.

Proof of Lemma 3.7. The two-term resolution of $\mathcal{O}_E/\pi_E^n$ gives an exact sequence

$$0 \to H^q(Q^\bullet)/\pi_E^n \to H^q(Q^\bullet/\pi_E^n) \to H^{q+1}(Q^\bullet)[\pi_E^n] \to 0.$$

For the transition from level m to level n, the induced map on the right-hand terms is multiplication by π_E^{m-n}, because the corresponding map of resolutions is the identity in degree zero and multiplication by π_E^{m-n} in degree -1. If $m - n \geq e$, this map is zero, since $H^{q+1}(Q^\bullet)[\pi_E^m]$ consists of torsion elements and π_E^e kills them. The image of $H^q(Q^\bullet/\pi_E^m)$ therefore lies in the left-hand term $H^q(Q^\bullet)/\pi_E^n$ at level n, which is the image of $H^q(Q^\bullet)$. The reverse inclusion holds because the map from $H^q(Q^\bullet)$ factors through level m. ∎

The bound $m \geq n + e$ cannot be improved. To see this, take $Q^\bullet = [\mathcal{O}_E \xrightarrow{\pi_E^e} \mathcal{O}_E]$ in degrees $0, 1$, so that $H^0(Q^\bullet) = 0$ and $H^1(Q^\bullet) = \mathcal{O}_E/\pi_E^e$. For $m \geq e$ the group $H^0(Q^\bullet/\pi_E^m)$ is $\pi_E^{m-e}\mathcal{O}_E/\pi_E^m$, and its image at level n is generated by π_E^{m-e} modulo π_E^n. This image is zero exactly when $m \geq n + e$, in agreement with the image of $H^0(Q^\bullet) = 0$.

At a single finite level the local structure is a statement about the étale site: a perfect complex with locally constant cohomology is locally constant. On a connected scheme its local model is moreover the same at every point.

Lemma 3.8 (Locally constant perfect complexes)**.** *Let S be a connected Noetherian scheme, let A be a finite local ring, and let $L \in D^b(S_{ét}, A)$ have finite Tor amplitude and cohomology sheaves that are locally constant with finite stalks. Let $\bar{s}$ be a geometric point of S and let M be a bounded complex of finite free A-modules with $M \simeq L_{\bar{s}}$ in $D(A)$. Then L is étale locally on S isomorphic to the constant complex $\underline{M}$.*

Proof. We first show that a model of a stalk spreads out to a neighbourhood, and then use connectedness to see that the model is the same everywhere.

Every stalk $L_{\bar{t}}$ has finite Tor amplitude and finite cohomology, hence is perfect over the local ring A, that is, represented by a bounded complex $M_{\bar{t}}$ of finite free A-modules. An isomorphism $M_{\bar{t}} \simeq L_{\bar{t}}$ is then a section at $\bar{t}$ of the zeroth cohomology sheaf of $R\mathcal{H}om(\underline{M_{\bar{t}}}, L)$; here the stalks of this sheaf are computed stalkwise, since $\underline{M_{\bar{t}}}$ is a bounded complex of finite free modules. We extend the section to an étale neighbourhood of $\bar{t}$ and lift it there to a morphism $\underline{M_{\bar{t}}} \to L$. Its cone has bounded locally constant finite cohomology and zero stalk at $\bar{t}$. Hence its finitely many

cohomology sheaves vanish after shrinking the neighbourhood, and L is isomorphic to $\underline{M_{\bar{t}}}$ there. In particular, the stalks of L at all geometric points over the image of this neighbourhood are isomorphic to $M_{\bar{t}}$ in $D(A)$.

For each isomorphism class c in $D(A)$, let $S_c \subset S$ be the set of points at whose geometric points the stalk of L lies in c. By the previous paragraph and openness of étale morphisms, each S_c is open. Moreover, the sets S_c are disjoint and cover S. Since S is connected, $S = S_c$ for the class c of M. Thus every point has an étale neighbourhood on which $L \simeq \underline{M}$; compare [Stacks, 09BI]. ∎

On a weakly contractible object the local isomorphisms of the finite levels become global, and the only remaining problem is to choose them compatibly. For this we use the uniform rate of Lemma 3.7.

Lemma 3.9 (Integral models on weakly contractible schemes)**.** *Let W be an affine weakly contractible scheme, let $K \in D(W_{\text{proét}}, \mathcal{O}_E)$ be derived complete, and let $P^\bullet$ be a bounded complex of finite free $\mathcal{O}_E$-modules. Suppose that for every $n \geq 1$ there is an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n$ in $D(W_{\text{proét}}, \mathcal{O}_E/\pi_E^n)$. Then $K \simeq \mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet$.*

Proof. The hypothesis gives an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n$ at each level n, but these isomorphisms need not be compatible with the reduction maps, so they do not assemble to a map into $K = R\varprojlim_n K_n$. What we need is a compatible family of maps, that is, an element of the inverse limit of the Hom groups between the reductions. Such an element lifts to a map $\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet \to K$, because K is derived complete. To find one that is an isomorphism at level 1, we show that these Hom groups satisfy the Mittag–Leffler condition with a rate e independent of n. Derived Nakayama then shows that the lift is an isomorphism.

Consider the tower of complexes

$$R\mathcal{H}om_{(\mathcal{O}_E/\pi_E^n)_W}\big((P^\bullet/\pi_E^n)_W, K_n\big) \in D(W_{\text{proét}}, \mathcal{O}_E/\pi_E^n),$$

with transition maps induced by the reduction maps $K_{n+1} \to K_n$, and the tower of groups $\mathrm{Hom}_{D(W_{\text{proét}}, \mathcal{O}_E/\pi_E^n)}((P^\bullet/\pi_E^n)_W, K_n)$ obtained by applying $H^0 R\Gamma(W, -)$. Since $P^\bullet$ is finite free, the first tower is the tower of reductions of $(\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} (P^\bullet)^\vee) \otimes^{\mathbf{L}}_{\mathcal{O}_{E,W}} K$.

These Hom groups can be infinite, so the Mittag–Leffler condition has to come from a uniform bound. Let $Q^\bullet = (P^\bullet)^\vee \otimes_{\mathcal{O}_E} P^\bullet$ and choose e such that π_E^e kills the torsion of $H^1(Q^\bullet)$. Let $N \geq m \geq n + e$ and choose one isomorphism $(P^\bullet/\pi_E^N)_W \simeq K_N$. Its reductions to levels m and n identify the three internal Hom complexes at levels $N \to m \to n$ with the constant complexes $Q^\bullet/\pi_E^N \to Q^\bullet/\pi_E^m \to Q^\bullet/\pi_E^n$. Their zeroth cohomology sheaves are therefore the constant sheaves on the groups $H^0(Q^\bullet/\pi_E^N) \to H^0(Q^\bullet/\pi_E^m) \to H^0(Q^\bullet/\pi_E^n)$. By Lemma 3.7 the images from levels N and m at level n agree, as subsheaves. Since W is weakly contractible, taking sections over W is exact, and so it commutes with taking cohomology and images. Hence the images in the Hom group at level n from levels N and m coincide as well. Since e does not depend on n, m or N, we obtain the Mittag–Leffler condition: for every n the images at level n are constant from level $n + e$ on.

To lift, note that each Hom group contains an isomorphism, and its reduction to level 1 is again an isomorphism. The stable image at level 1 is the image from level $1 + e$, so it contains an invertible class. By the Mittag–Leffler property this class lifts to a compatible family $(\varphi_n)_n$ in the inverse limit of the Hom groups. The Milnor sequence gives a surjection

$$H^0 R\varprojlim_n R\Gamma(W, R\mathcal{H}om_{(\mathcal{O}_E/\pi_E^n)_W}((P^\bullet/\pi_E^n)_W, K_n))$$
$$\longrightarrow \varprojlim_n \mathrm{Hom}_{D(W_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n)}((P^\bullet/\pi_E^n)_W, K_n),$$

whose source is $\mathrm{Hom}_{D(W_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)}(\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet, K)$. This holds because derived sections commute with inverse limits, K is complete, and tensoring with the finite free dual of $P^\bullet$ commutes with inverse limits. The family (φ_n) therefore lifts to a morphism $\mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet \to K$ whose first reduction is an isomorphism. Since both objects are complete, derived Nakayama (Lemma 2.62) shows that this morphism is an isomorphism. Here completeness of the source follows termwise from $\mathcal{O}_{E,W} = \varprojlim (\mathcal{O}_E/\pi_E^n)_W$. Indeed, the transition maps are epimorphisms, so this limit agrees with the derived limit by Proposition 2.24. ∎

Proposition 3.10 (Local structure of integral constructible complexes)**.** *Let X be Noetherian and let $K \in D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ be derived complete with $K_1 \in D_c^b(X, \mathbf{F}_E)$. Then there is a finite stratification of X such that on each stratum S, pro-étale locally,*

$$K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$$

for a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules.

Proof. We first trivialize each K_n locally, separately for each n, and then make the trivializations compatible. To begin, choose a finite stratification on which all cohomology sheaves of K_1 are locally constant, and refine it by connected components; there are finitely many of these, since the strata are Noetherian.

For every stratum S the restriction $K|_S$ is again derived complete with constructible first reduction. Indeed, restriction to an open subscheme preserves limits, and so does restriction to a closed subscheme on the pro-étale site (Proposition 2.58). Moreover, both restrictions are exact and commute with the two-term cones defining the reductions. From now on we fix a connected stratum S.

Local trivializations at each finite level. By Lemma 3.5, each $K_n|_S$ is classical and of finite Tor amplitude, with locally constant finite cohomology, so we may work on $S_{\acute{e}t}$. Fix a geometric point $\bar{s}$. Then the stalk complex at $\bar{s}$ has a minimal finite free model M_n in the common interval $[a, b]$. By Lemma 3.8 the complex $K_n|_S$ is étale locally isomorphic to the constant complex $(M_n)_S$.

On stalks at $\bar{s}$, the reduction maps $K_{n+1} \to K_n$ induce maps $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1} \otimes^{\mathbf{L}}_{\mathcal{O}_E/\pi_E^{n+1}} \mathcal{O}_E/\pi_E^n \simeq M_n$, because the étale stalk functor is exact and symmetric monoidal. By Lemma 3.6 there is a bounded finite free complex $P^\bullet$ with $P^\bullet/\pi_E^n \simeq M_n$. Back on $S_{\mathrm{pro\acute{e}t}}$, we get for each n local isomorphisms $(P^\bullet/\pi_E^n)_S \simeq K_n|_S$ in $D(S_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n)$, chosen separately for each n.

The integral isomorphism. Choose an affine weakly contractible cover $W \to S$. For each n, the local isomorphism becomes an isomorphism $(P^\bullet/\pi_E^n)_W \simeq K_n|_W$ on all of W, since by weak contractibility a covering by trivializing charts splits. Moreover, the restriction $K|_W$ is derived complete, and its reductions are the $K_n|_W$. By Lemma 3.9 we conclude that $K|_W \simeq \mathcal{O}_{E,W} \otimes_{\mathcal{O}_E} P^\bullet$. ∎

The local structure is the hard half of the following criterion.

Proposition 3.11 (The integral constructibility criterion)**.** *Let X be Noetherian. The following conditions on $K \in D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ are equivalent:*

1. $K \in D^b_c(X, \mathcal{O}_E)$;
2. K *is derived complete and* $K_1 \in D^b_c(X, \mathbf{F}_E)$;
3. *there is a finite stratification of* X *such that on each stratum* S*, pro-étale locally,* $K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ *for a bounded complex* $P^\bullet$ *of finite free* $\mathcal{O}_E$*-modules;*
4. $K \in D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$.

In the classical adic setting, this description of the integral constructible category is [BS15, Proposition 6.8.11] and [Stacks, 09C7].

Proof of Proposition 3.11. The substantial implication is (2)⇒(3), which is Proposition 3.10. The others hold because the complexes $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ are complete, have finite Tor amplitude, and have constructible reductions. These properties survive extension by zero from strata. We prove (1)⇒(2)⇒(3)⇒(1) and (3)⇔(4).

(1) implies (2). We first treat a single constructible sheaf, locally on a stratum, then extend by zero from the strata, and finally pass to complexes.

A finitely generated $\mathcal{O}_E$-module is a finite sum of copies of $\mathcal{O}_E$ and of modules $\mathcal{O}_E/\pi_E^r$, so its sheaf has a two-term finite free resolution. The coefficient sheaf $\mathcal{O}_{E,U_i}$ on a member U_i of the cover in Definition definition-constructible-sheaves is derived complete. Indeed, it is the limit of its finite quotients by Lemma 2.51, and since the transition maps between these quotients are surjective, this limit agrees with the derived limit by Proposition 2.24. As completeness is preserved by finite sums and cones, the modules $\mathcal{O}_{E,U_i} \otimes_{\mathcal{O}_E} P_i$ of Definition definition-constructible-sheaves are complete, and their reductions are bounded locally constant finite complexes. We may check these assertions on a pro-étale cover, because restriction to an object is a localization of the topos, and it preserves limits and the cones defining the reductions.

For a constructible sheaf, choose a finite stratification as in the definition. Then for a stratum $a : S \hookrightarrow X$, factored as $a = i \circ j$ as in Section 3.1, the open immersion j is quasi-compact because X is Noetherian. Extension by zero $a_! = i_* j_!$ preserves completeness, because it is exact, commutes with the cones defining the reductions, and preserves limits. Here, for $j_!$ the last point is the assertion about limits in Proposition 2.58, see [Stacks, 09BP], while for i_* it holds because i_* is exact and a right adjoint. Using the finite filtration by open and closed pieces, we see that the whole sheaf is complete with constructible finite reductions. Applying this to the finitely many cohomology sheaves of K and using the bounded Postnikov triangles, we obtain (2).

(2) implies (3). This is Proposition 3.10.

(3) implies (1). Each of the finitely many strata is quasi-compact, so the local isomorphisms of (3) can be taken on a finite covering by quasi-compact pro-étale objects, and their bounds give a common cohomological interval. Since the map from the constant sheaf with value the discrete ring $\mathcal{O}_E$ to $\mathcal{O}_{E,S}$ is flat [Stacks, 099L], the cohomology of $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ is $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} H^q(P^\bullet)$. Here $H^q(P^\bullet)$ is finitely generated over $\mathcal{O}_E$, so these sheaves satisfy the condition of Definition definition-constructible-sheaves. This proves (1).

(3) is equivalent to (4). The complexes $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ have finite Tor amplitude, and since the stratification is finite there is a common interval. Since the restrictions to the strata are exact and jointly conservative, they detect vanishing after tensoring with any sheaf of modules, and (3) implies (4). Conversely, (4) implies (1) by definition of D^b_{ctf}, and hence (3) by the implications already proved. ∎

Lemma 3.25 compares $D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$ with the classical category of constructible $\mathcal{O}_E$-complexes [Stacks, 09C1], which is defined with modules over the constant sheaf with value the discrete ring $\mathcal{O}_E$.

3.4 Finite Tor amplitude

For the integral and rational rings the finite Tor amplitude condition is automatic.

Proposition 3.12 (Constructible complexes have finite Tor amplitude for ℓ-adic coefficients)**.** *For $\Lambda = \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, we have*

$$D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda).$$

Proof. For $\mathcal{O}_E$, and hence for $\mathbf{Z}_\ell$, this is the equivalence of (1) and (4) in the criterion above. For the fields we show that every constructible sheaf is flat, which bounds the Tor amplitude of a bounded complex by its cohomological amplitude.

For a field $\Lambda(*)$, a constructible sheaf M is, on each stratum and locally on a cover $\{U_i\}$, of the form $\Lambda_{U_i} \otimes_{\Lambda(*)} P_i$ with P_i a finite-dimensional vector space, so it is locally a finite free Λ_{U_i}-module. Restriction to the strata is exact and the restrictions are jointly conservative, as in the proof of the criterion. Hence every constructible sheaf has Tor amplitude in $[0, 0]$. Finally, a bounded complex whose finitely many cohomology sheaves have finite Tor amplitude has finite Tor amplitude, by the triangles of its canonical filtration. ∎

For a finite ring that is not regular the condition is needed, as the $\mathbf{Z}/\ell^2$-module $\mathbf{F}_\ell$ of Section 3.2 shows. Moreover, finite Tor amplitude is a condition on the whole complex, not on its cohomology sheaves. The category D^b_{ctf} is thick, but in general it is not stable under standard truncations. For example, for $A = \mathbf{Z}/\ell^2$ the complex $[A \xrightarrow{\ell} A]$ is perfect, whereas its nonzero cohomology modules are $\mathbf{F}_\ell$, which are not perfect. For this reason, the compactness proofs for finite A work with bounded flat constructible models, and standard truncations are taken in D^b_c. The models exist by the next lemma.

Lemma 3.13 (Flat models and generators)**.** *Every $C \in D^b_{\mathrm{ctf}}(X, A)$ on $X_{\mathrm{pro\acute{e}t}}$ is represented by a bounded complex of flat constructible A-sheaves. A flat constructible sheaf has a finite stratification on whose strata it is locally of the form $\underline{M}$ with M a finite projective A-module. Consequently, $D^b_{\mathrm{ctf}}(X, A)$ is generated, using shifts, finite extensions and direct summands, by sheaves $a_!L$, where $a : S \hookrightarrow X$ is locally closed and L is locally of the form $\underline{M}$ with M a finite projective A-module.*

Proof. The flat model exists on the étale site by a classical result, and we pull it back. The generators then come from two filtrations: the stupid filtration of the model, and the filtration of each term by strata.

By Corollary 2.44, $C = \nu_X^* C_{\acute{e}t}$ with $C_{\acute{e}t} \in D^b_{\mathrm{ctf}}(X_{\acute{e}t}, A)$, where $\nu_X^* : D(X_{\acute{e}t}, A) \to D(X_{\mathrm{pro\acute{e}t}}, A)$ is the comparison inverse image. By [Stacks, 0F4M] the complex $C_{\acute{e}t}$ has a bounded flat model, and since ν_X^* is exact on sheaves, its pullback is a bounded flat constructible model on $X_{\mathrm{pro\acute{e}t}}$. We choose a common stratification on which its finitely many terms are locally constant, and their stalks are then finite projective, because a finite flat module over the Noetherian ring A is finite projective. If F is a flat constructible sheaf, then the exact sequences for the successive open and closed strata express F as a finite extension of the sheaves $a_!L$ attached to its strata. Similarly, the stupid filtration of the bounded flat complex expresses C as a finite extension of shifts of its *terms*, and shifts and finite extensions preserve finite Tor amplitude. The same argument proves the corresponding statement on $X_{\acute{e}t}$, which we use in the proof of Proposition 5.14 before passing to the pro-étale site. ∎

3.5 Field coefficients

Until the end of this section, $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, so that $D^b_{\mathrm{ctf}} = D^b_c$ and we can argue with constructible cohomology sheaves and standard truncations. Recall (after Definition 3.2) that a Λ_X-module is constructible exactly when it is lisse, that is locally free of finite rank for the pro-étale topology, on each stratum of a finite stratification of X. By dévissage we reduce statements about $D^b_c(X, \Lambda)$ to the simplest constructible sheaves, namely lisse sheaves extended by zero from a stratum, the sheaves $a_!L$ of Section 3.1. This dévissage first splits a complex into its cohomology sheaves by truncation, and then filters each constructible sheaf by strata.

For this dévissage we need constructible sheaves to form an abelian category (Lemma 3.18). In the proof we reduce a morphism of lisse E-sheaves to a morphism of integral lattices, and for $\overline{\mathbf{Q}}_\ell$ we first descend the morphism to a finite E. To prepare for this, in the next two lemmas we construct the lattices locally and carry out the descent.

Definition 3.14 (Lattices)**.** Let $E/\mathbf{Q}_\ell$ be finite and let L be a lisse E_X-module. A *lattice* in L is a lisse $\mathcal{O}_{E,X}$-submodule $M \subset L$ with $M[1/\ell] = L$.

Lemma 3.15 (Lattices in lisse E-sheaves)**.** *Let $E/\mathbf{Q}_\ell$ be finite and let L be a lisse E_X-module. The sheaf on $X_{\mathrm{pro\acute{e}t}}$ of lattices in L is represented by a separated surjective étale X-scheme, with a universal lattice. In particular, L admits a lattice étale locally on X.*

Proof. Where L is free of rank r, a lattice is the same as a lattice in E^r, so the sheaf of lattices is locally the constant sheaf on the discrete set $\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)$. The representing scheme is obtained by descending the corresponding disjoint unions of copies of the trivializing objects.

The rank of L is a locally constant function on $|X|$. Indeed, on a trivializing covering the subsets of constant rank are open, closed and saturated, and they descend by fpqc descent of open subsets. So X is the disjoint union of the open and closed subsets on which the rank is constant. Since the sheaf of lattices restricts to the sheaf of lattices on each of them, we may assume that L has constant rank r.

We now construct the representing scheme. On a pro-étale affine V where L is free of rank r, the sheaf of lattices is the constant sheaf on the set

$$\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)$$

of lattices in E^r, which is discrete because $\mathrm{GL}_r(\mathcal{O}_E)$ is an open subgroup of $\mathrm{GL}_r(E)$. To glue these constant sheaves, recall from Lemma 2.51 that on an affine object, sections of E_X are continuous functions to E, with bounded denominators by quasi-compactness. A change of basis on $V \times_X V$ is therefore a continuous function to $\mathrm{GL}_r(E)$. Since the stabilizer of a lattice is open, the image of each fixed lattice under this function is locally constant on $V \times_X V$. Now consider the disjoint union of copies of $V \times_X V$ indexed by the lattices in E^r. On each component, the preimages of the target components are then open and closed and form a disjoint open cover, so the induced map from this disjoint union to itself is a morphism of schemes. In this way the cocycle of the transition matrices defines a descent datum of schemes for these disjoint unions of copies of V.

For affine $V = \operatorname{Spec} B$, the disjoint union of copies of V indexed by a set I is ind-quasi-affine, since its finite subunions are affine open subschemes and together they cover it. Since descent along fpqc coverings is effective for ind-quasi-affine morphisms [Stacks, 0APK], the descent datum above is effective. Taking a finite affine trivializing covering of each affine open of X and gluing, we therefore obtain a scheme representing the sheaf of lattices. This scheme is separated and

étale over X by [Stacks, 02KU, 02VN], and it is surjective because every free module has a lattice. By descent of sheaves of modules, this scheme carries a universal lattice. ∎

For example, in rank one the lattices in E are the $\pi_E^m \mathcal{O}_E$ for $m \in \mathbf{Z}$, so the discrete set of lattices is $E^\times/\mathcal{O}_E^\times \simeq \mathbf{Z}$. Hence for $L = E_X$ the representing scheme is $\coprod_{m\in\mathbf{Z}} X$.

Example 3.16 (The scheme of lattices on the nodal curve)**.** Let $E = \mathbf{Q}_\ell$ and let L be the sheaf on the nodal curve X from the start of this chapter, obtained from E_Y on the chain of lines $Y \to X$ by letting the generator of $\mathbf{Z}$ act by ℓ. Over Y the lattices in $L|_Y = E_Y$ are the constant lattices $\ell^m \mathbf{Z}_\ell$, so over Y the scheme of lattices is $\coprod_{m\in\mathbf{Z}} Y$. The generator of $\mathbf{Z}$ moves the copy indexed by m to the copy indexed by $m+1$ while translating Y. Hence the scheme of lattices over X, which is the quotient by $\mathbf{Z}$, is a single copy of Y. It is surjective and étale over X, as the lemma says, but it has no section. Indeed, a section would trivialize the $\mathbf{Z}$-torsor $Y \to X$, whereas Y is connected and $\coprod_{\mathbf{Z}} X$ is not. This is the absence of a global lattice seen geometrically, and the universal lattice lives on Y, where L becomes constant.

With lattices we pass from E to $\mathcal{O}_E$. For $\overline{\mathbf{Q}}_\ell$ we first need to come down to a finite extension E, and on a quasi-compact scheme this only involves finitely many matrix entries.

Lemma 3.17 (Descent of lisse $\overline{\mathbf{Q}}$-sheaves)**.** *A finite collection of lisse $\overline{\mathbf{Q}}_{\ell,X}$-modules and morphisms between them is obtained by extension of scalars from lisse E_X-modules and their morphisms, for one finite extension $E/\mathbf{Q}_\ell$.*

Proof. The data are described by finitely many matrices with entries in $\overline{\mathbf{Q}}_{\ell,X}$ over finitely many quasi-compact objects, and each entry already lies in some E_X. We make this precise.

Since X is quasi-compact, we may choose a finite covering by quasi-compact pro-étale affines on which all the modules are free; such refinements exist by [Stacks, 0988]. Their pairwise intersections $U_i \times_X U_j$ are quasi-compact since X is quasi-separated, and quasi-separated since they are subschemes of the affine schemes $U_i \times U_j$. On a quasi-compact quasi-separated object, taking sections commutes with filtered colimits, because coverings admit finite quasi-compact refinements and the sheaf condition involves only finite limits [Stacks, 0739]. By Lemma 2.51 we have $\overline{\mathbf{Q}}_{\ell,X} = \varinjlim_E E_X$. Hence the finitely many transition matrices, their inverses, and the matrices of the given morphisms have entries in E_X for a single finite E. Moreover, their cocycle and compatibility identities already hold over E_X, since $E_X \to \overline{\mathbf{Q}}_{\ell,X}$ is injective. Descent of sheaves of modules now gives the required lisse modules and morphisms. ∎

With these two lemmas we can show that constructible sheaves with field coefficients form an abelian category, and obtain the filtration by extensions by zero of lisse sheaves that the dévissage uses.

Lemma 3.18 (Abelian category of constructible sheaves)**.** *Let $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$.*

1. $\mathrm{Cons}(X,\Lambda)$ *is closed under kernels, cokernels and extensions in* $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}},\Lambda)$*. In particular it is an abelian category, and its inclusion into* $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}},\Lambda)$ *is exact.*
2. $\mathrm{Cons}(X,\Lambda)$ *is essentially small.*
3. *Every $F \in \mathrm{Cons}(X,\Lambda)$ has a finite filtration $0 = F_0 \subset F_1 \subset \cdots \subset F_m = F$ whose subquotients F_k/F_{k-1} are of the form $a_!L$, with $a : S \hookrightarrow X$ a locally closed immersion and L a lisse Λ_S-module.*

If X is a curve, part (3) amounts to an exact sequence $0 \to j_!L \to F \to i_*i^*F \to 0$, where j is the inclusion of a dense open subset on which F is lisse, $L = j^*F$, and i is the inclusion of the finitely many closed points of the complement.

Proof of Lemma 3.18. Part (1) is the substantial one. All three closure properties are local on a common stratification, so it suffices to treat lisse sheaves on a single stratum, and there the only difficulty is the kernel of a morphism. Parts (2) and (3), by contrast, are formal.

(1) Given finitely many constructible sheaves, we refine their stratifications to a common one. Then all three closure properties can be checked on a single stratum S.

Extensions. Let $0 \to F' \to F \to F'' \to 0$ be exact with F' and F'' lisse on S. After passing to a cover, F' and F'' are free. Since $F \to F''$ is surjective, after a further cover a basis of F'' lifts to sections of F. Hence the sequence splits locally, and F is locally free of finite rank.

Kernels and cokernels. Locally, a morphism of lisse sheaves is a matrix of continuous functions to E, and a priori its kernel need not be locally free. Once we choose lattices, however, the morphism becomes a constant matrix on a suitable pro-étale cover, because its reductions are maps of finite locally constant sheaves. The kernel and cokernel of a constant matrix over E are constant of finite rank, and since being lisse is a local property for the pro-étale topology, this suffices.

First let $\Lambda = E$ and let $f : L \to L'$ be a morphism of lisse sheaves on S. By Lemma 3.15 we may replace S by a connected quasi-compact affine étale neighbourhood V on which L and L' contain lattices M and N. On a finite quasi-compact pro-étale covering trivializing both lattices, the matrix entries of f have a common denominator, so $h = \pi_E^m f$ maps M into N for some $m \geq 0$.

The reductions $M_n = M/\pi_E^n M$ and $N_n = N/\pi_E^n N$ are finite locally constant étale sheaves, and $M = \varprojlim M_n$, $N = \varprojlim N_n$ by Lemma 2.51, checked where M and N are free. Fix a geometric point $\bar{v}$ of V. Then the completed fibres $M_{\bar{v}} = \varprojlim (M_n)_{\bar{v}}$ and $N_{\bar{v}} = \varprojlim (N_n)_{\bar{v}}$ are finite free $\mathcal{O}_E$-modules with continuous $\pi_1(V, \bar{v})$-actions, and the reductions of h give an equivariant map $h_{\bar{v}}$ between them. To make h constant, note that each M_n and N_n becomes constant on a connected finite étale cover of the connected scheme V, and we may choose these covers to form a tower. Their inverse limit $V' \to V$ is a connected pro-étale covering, so maps between constant sheaves on V' are constant. On V' the lattices M and N are identified with the constant sheaves attached to $M_{\bar{v}}$ and $N_{\bar{v}}$, and h with the constant matrix $h_{\bar{v}}$. Hence on V' the kernel and the cokernel of $f = \pi_E^{-m} h$ are the constant sheaves attached to the kernel and the cokernel of $h_{\bar{v}} \otimes_{\mathcal{O}_E} E$. In particular they are lisse.

For $\Lambda = \overline{\mathbf{Q}}_\ell$, we descend L, L' and f to a finite extension $E/\mathbf{Q}_\ell$ by Lemma 3.17. Since the functor $\overline{\mathbf{Q}}_{\ell,X} \otimes_{E_X} -$ is exact, it takes the kernel and cokernel over E to those of f. Finally, restriction to strata is exact, so the kernel and cokernel of a morphism of constructible sheaves on X are constructible.

(2) We choose the sites in a fixed universe. Up to isomorphism, a constructible sheaf is then determined by a finite stratification and, on each stratum, descent data of finite rank on a covering, and these data form a set.

(3) By Noetherian induction we refine the stratification of F so that there is an open stratum $j : U \hookrightarrow X$ with closed complement $i : Z \hookrightarrow X$ that is again a union of strata. The exact sequence $0 \to j_! j^* F \to F \to i_* i^* F \to 0$ separates the stratum U from the rest, and here $j^* F$ is lisse. Applying the same step to $i^* F$ on Z and iterating, we obtain a finite filtration of F whose subquotients are $a_!(F|_S)$ for the strata $a : S \hookrightarrow X$, and each $F|_S$ is lisse. ∎

Example 3.19 (A running example)**.** We use the same few schemes throughout:

$$\{0\} \xrightarrow{i} \mathbf{A}^1_k \xleftarrow{j} \mathbf{G}_{m,k}, \qquad \mathbf{A}^1_k \xrightarrow{j_\infty} \mathbf{P}^1_k.$$

The sheaf $j_!\Lambda_{\mathbf{G}_m}$ is constructible for the stratification $\mathbf{A}^1 = \mathbf{G}_m \sqcup \{0\}$. Its stalk is Λ at a geometric point of $\mathbf{G}_m$ and 0 at the origin, so it is not lisse on $\mathbf{A}^1$, and the sequence of part (3) is the localization sequence

$$0 \to j_!\Lambda_{\mathbf{G}_m} \to \Lambda_{\mathbf{A}^1} \to i_*\Lambda \to 0.$$

We return to this example for compact supports (after Lemma 6.4), for $i^!$ (Example 6.25) and for the trace (Section 9.6).

Exercises. (1) Compute the stalks of $a_!L$ for a locally closed immersion a and a lisse L. (2) For $K = \Lambda_X[1]$, in which degree is the nonzero cohomology sheaf? (3) Show that $j_!\Lambda_{\mathbf{G}_m}$ has no nonzero global sections on $\mathbf{A}^1_k$.

For complexes we get the following. Since constructible sheaves form an abelian subcategory, complexes with constructible cohomology are stable under cones, and the filtration of part (3) together with the truncation triangles generates them from the sheaves $a_!L$.

Corollary 3.20 (Generators of D^b_c with field coefficients)**.** *Let $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$. Then $D^b_c(X, \Lambda)$ is a triangulated subcategory of $D(X_{\text{pro\'et}}, \Lambda)$, and every object of $D^b_c(X, \Lambda)$ is obtained by finitely many cones from objects $a_!L[n]$, where $a : S \hookrightarrow X$ is a locally closed immersion, L is a lisse Λ_S-module and $n \in \mathbf{Z}$.*

Proof. The first assertion is closure under cones, which we check on cohomology sheaves. Let $K \to K' \to K'' \to K[1]$ be a distinguished triangle with K, K' in $D^b_c(X, \Lambda)$. By the long exact cohomology sequence, each $\mathcal{H}^i(K'')$ is an extension of $\ker(\mathcal{H}^{i+1}K \to \mathcal{H}^{i+1}K')$ by $\operatorname{coker}(\mathcal{H}^iK \to \mathcal{H}^iK')$. Both are constructible by Lemma lemma-constructible-sheaves, and hence so is $\mathcal{H}^i(K'')$. Closure under shifts is clear, since shifts only renumber the cohomology sheaves.

For the second assertion, we argue by induction on the number of nonzero cohomology sheaves of K. If i is the largest degree with $\mathcal{H}^i(K) \neq 0$, the truncation triangle $\tau_{<i}K \to K \to \mathcal{H}^i(K)[-i] \to$ exhibits K as the cone of a map $\mathcal{H}^i(K)[-i-1] \to \tau_{<i}K$, and $\tau_{<i}K$ has fewer nonzero cohomology sheaves. This reduces the claim to a single shifted constructible sheaf, and for it the filtration from Lemma lemma-constructible-sheaves yields finitely many short exact sequences, that is, distinguished triangles, with terms $a_!L$. ∎

3.6 Continuity of Ext

A morphism of $\overline{\mathbf{Q}}_\ell$-complexes should be defined over some finite extension E of $\mathbf{Q}_\ell$. Since $\overline{\mathbf{Q}}_{\ell,X} = \varinjlim_E E_X$, this amounts to asking that Hom out of a constructible complex commute with a filtered colimit, where a filtered colimit of complexes means the termwise colimit of a filtered diagram of complexes of sheaves. We prove this continuity for complexes with a common lower bound. In this form it is the pro-étale analogue of the classical fact that cohomology of a quasi-compact quasi-separated scheme commutes with filtered colimits. Besides the descent to a finite coefficient field below, continuity is used in two places. The first is the uniform bound for Ext groups, which is proved first for integral lattices, so we need continuity for $\mathcal{O}_E$ as well. The second is the compactness argument for E and $\overline{\mathbf{Q}}_\ell$ (Section 5.2).

For cohomology and higher direct images the statement is Corollary 2.29, which holds for every sheaf of rings. To reach Hom, recall that a constructible sheaf has a finite filtration by sheaves $a_!L$ with L lisse on a stratum S (Lemma lemma-constructible-sheaves). Using sections with support in a closed subscheme, we turn Hom out of $a_!L$ into cohomology on S of a complex built from the target by Rj_* and restrictions, and the comparison then goes through a hypercohomology

spectral sequence. Because of the common lower bound, only finitely many of its terms contribute to a given degree, and each of these terms commutes with the colimit.

Lemma 3.21 (Sections with closed support)**.** *Let Λ be admissible, let $i : Z \hookrightarrow X$ be a closed immersion with open complement $j : V \hookrightarrow X$, and let $R\underline{\Gamma}_Z : D(X_{\text{pro\'et}}, \Lambda) \to D(Z_{\text{pro\'et}}, \Lambda)$ be the local cohomology functor (Definition 2.59),*

$$R\underline{\Gamma}_Z K = i^* \operatorname{fib}(K \longrightarrow Rj_*j^*K).$$

For a sheaf M of Λ_X-modules on $X_{\text{pro\'et}}$, let $\underline{\Gamma}_Z(M) = i^* \ker(M \to j_*j^*M)$, the sheaf on Z of sections of M with support in Z.

1. *$R\underline{\Gamma}_Z$ is the right derived functor of $\underline{\Gamma}_Z$. Consequently the adjunction of Lemma 2.60 holds in the form $R\mathrm{Hom}_X(i_*N, K) = R\mathrm{Hom}_Z(N, R\underline{\Gamma}_Z K)$.*
2. *Let $a = i \circ u : S \hookrightarrow X$ with $u : S \hookrightarrow Z$ an open immersion, and let L be a finite locally projective Λ_S-module with dual $L^\vee$. Then naturally in $K \in D(X_{\text{pro\'et}}, \Lambda)$*

$$R\mathrm{Hom}_X(a_!L, K) = R\Gamma\big(S, L^\vee \otimes u^* R\underline{\Gamma}_Z K\big). \tag{3.1}$$

Proof. Lemma 2.60 gives the adjunction between i_* and $R\underline{\Gamma}_Z$ on derived categories. What is new here is that $R\underline{\Gamma}_Z$ is the right derived functor of $\underline{\Gamma}_Z$, and this upgrades the adjunction to $R\mathrm{Hom}$ complexes and yields formula (3.1). For this we show two things: $\underline{\Gamma}_Z$ is right adjoint to the exact functor i_* on sheaves, and $M \to j_*j^*M$ is surjective for injective M. The second means that on injective sheaves the kernel of this map is also its fibre.

We use that i^* and i_* are exact, that $i^*i_* = \mathrm{id}$, and that every sheaf M sits in the localization sequence $0 \to j_!j^*M \to M \to i_*i^*M \to 0$ [Stacks, 09AA, 09AH, 09BM]. The last fact shows that a sheaf whose restriction to V vanishes is of the form i_* of a sheaf on Z. In particular $\ker(M \to j_*j^*M) = i_*\underline{\Gamma}_Z(M)$.

Adjunction on sheaves. The functor $\underline{\Gamma}_Z$ is right adjoint to $i_* : \mathrm{Mod}(Z_{\text{pro\'et}}, \Lambda) \to \mathrm{Mod}(X_{\text{pro\'et}}, \Lambda)$. Indeed, a map $i_*N \to M$ composes to zero with $M \to j_*j^*M$, because $j^*i_*N = 0$, so it factors through $\ker(M \to j_*j^*M) = i_*\underline{\Gamma}_Z(M)$. Since i_* is fully faithful, such factorizations are the maps $N \to \underline{\Gamma}_Z(M)$. Moreover, as i_* is exact, its right adjoint $\underline{\Gamma}_Z$ preserves K-injective complexes.

Surjectivity on injectives. For an injective sheaf I the unit $I \to j_*j^*I$ of the adjunction $j^* \dashv j_*$ on $\mathrm{Mod}(X_{\text{pro\'et}}, \Lambda)$ is surjective, because it has a right inverse. To construct it, extend the inclusion $j_!j^*I \to I$ along $j_!j^*I \hookrightarrow j_*j^*I$ to a map $s : j_*j^*I \to I$, which is possible because I is injective. Then the composite $j_*j^*I \xrightarrow{s} I \to j_*j^*I$ restricts to the identity on V. Since j_* is fully faithful, an endomorphism of j_*j^*I is determined by its restriction to V, and so this composite is the identity.

Comparison with the fibre. The right derived functor of $\underline{\Gamma}_Z$ agrees with $i^* \operatorname{fib}(K \to Rj_*j^*K)$. To see this, represent K by a K-injective complex $I^\bullet$ with injective terms [Stacks, 079P]. Then $\underline{\Gamma}_Z(I^\bullet)$ computes the right derived functor of $\underline{\Gamma}_Z$ at K. On the other side, the complex $j^*I^\bullet$ is K-injective, because the left adjoint $j_!$ of j^* is exact. Hence $j_*j^*I^\bullet$ represents Rj_*j^*K. By the surjectivity just proved, the sequences

$$0 \to i_*\underline{\Gamma}_Z(I^n) \to I^n \to j_*j^*I^n \to 0$$

are exact termwise, and they define a distinguished triangle $i_*\underline{\Gamma}_Z(I^\bullet) \to K \to Rj_*j^*K \to i_*\underline{\Gamma}_Z(I^\bullet)[1]$. Applying i^* and using $i^*i_* = \mathrm{id}$ identifies $\underline{\Gamma}_Z(I^\bullet)$ with $i^* \operatorname{fib}(K \to Rj_*j^*K)$.

Part (1). With $I^\bullet$ as above, the complex $\underline{\Gamma}_Z(I^\bullet)$ is K-injective by the adjunction on sheaves, and it represents $R\underline{\Gamma}_Z K$ by the comparison with the fibre. The adjunction on complexes therefore gives

$$R\mathrm{Hom}_X(i_*N, K) = \mathrm{Hom}^\bullet(i_*N, I^\bullet) = \mathrm{Hom}^\bullet(N, \underline{\Gamma}_Z(I^\bullet)),$$

whose right-hand side is $R\mathrm{Hom}_Z(N, R\underline{\Gamma}_Z K)$.

Formula (3.1). Since $a_! = i_* u_!$, part (1), the adjunction $u_! \dashv u^*$ and the identity $R\mathcal{H}om(L, -) = L^\vee \otimes -$ for the finite locally projective sheaf L give $R\mathrm{Hom}_X(i_* u_! L, K) = R\mathrm{Hom}_Z(u_! L, R\underline{\Gamma}_Z K) = R\Gamma(S, L^\vee \otimes u^* R\underline{\Gamma}_Z K)$. ∎

Formula (3.1) expresses Hom out of $a_! L$ through cohomology on the stratum S and the functors i^*, u^* and Rj_*. Each of these commutes with filtered colimits of uniformly bounded below complexes, and from this we deduce the continuity of Ext.

Lemma 3.22 (Continuity of Ext)**.** *Let $(K_\alpha^\bullet)$ be a filtered diagram of complexes of Λ_X-modules that all vanish below a fixed degree, and let M be one of the following:*

1. *an object of $D_c^b(X, \Lambda)$, where $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$;*
2. *a finite extension of sheaves $a_! L_0$, with $a : S \hookrightarrow X$ locally closed and L_0 a finite locally free $\mathcal{O}_{E,S}$-module, where $\Lambda = \mathcal{O}_E$.*

Then for every integer n the natural map

$$\varinjlim_\alpha \mathrm{Hom}(M, K_\alpha^\bullet[n]) \longrightarrow \mathrm{Hom}(M, (\varinjlim_\alpha K_\alpha^\bullet)[n])$$

is an isomorphism.

Applied to sheaves, which are complexes in degree 0, the lemma says that for $M \in \mathrm{Cons}(X, \Lambda)$, $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, the functors $\mathrm{Ext}^n(M, -)$ commute with filtered colimits of Λ_X-modules.

Proof of Lemma 3.22. We reduce to a single generator $a_! L$ and then use (3.1) to turn Hom into cohomology on the stratum S, where Corollary 2.29 applies. The common lower bound is used at the end, to compare the abutments of two spectral sequences.

In both cases M is obtained by finitely many extensions and shifts from sheaves $a_! L$, with $a : S \hookrightarrow X$ a locally closed immersion and L a finite locally free Λ_S-module: in case (1) by Corollary 3.20, and in case (2) by assumption. Both sides are cohomological functors of M. By the five lemma, the class of objects M for which the claim holds is therefore closed under extensions and shifts, so it suffices to treat $M = a_! L$, with $a = i \circ u : S \hookrightarrow Z \hookrightarrow X$ as in Lemma 3.21. The only property of L used below is that $L^\vee \otimes -$ is exact.

By (3.1), we must show that the cohomology of $R\Gamma(S, L^\vee \otimes u^* R\underline{\Gamma}_Z K)$ commutes with the colimit, where $K = \varinjlim_\alpha K_\alpha^\bullet$. To this end we apply the exact functor $L^\vee \otimes u^* i^*(-) : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(S_{\mathrm{pro\acute{e}t}}, \Lambda)$ to the triangle $i_* R\underline{\Gamma}_Z K \to K \to Rj_* j^* K \to$ and to the corresponding triangles for the $K_\alpha^\bullet$. By the long exact sequence and the five lemma, it suffices to treat the two terms

$$R\Gamma(S, L^\vee \otimes u^* i^* K) \quad \text{and} \quad R\Gamma(S, L^\vee \otimes u^* i^* Rj_* j^* K).$$

For the first term, Corollary 2.29 on S applies, because the functor $L^\vee \otimes u^* i^*(-)$ is exact, commutes with colimits and preserves the lower bound. For the second term, consider the hypercohomology spectral sequence

$$E_2^{p,r} = H^p(S, L^\vee \otimes u^* i^* R^r j_*(j^* K)) \Longrightarrow H^{p+r}(S, L^\vee \otimes u^* i^* R j_* j^* K)$$

which is functorial in K. The sheaves $R^r j_*(j^* K_\alpha)$ commute with the colimit by the second part of Corollary 2.29, and $H^p(S, -)$ does by its first part. Hence, after passing to the colimit, the map of spectral sequences is an isomorphism on E_2. The abutments agree as well, since by the common lower bound only finitely many $E_2^{p,r}$ contribute to a given total degree. ∎

3.7 Descent to a finite extension

The lattice arguments below need the discrete valuation ring $\mathcal{O}_E$ with its uniformizer π_E, but the valuation ring of $\overline{\mathbf{Q}}_\ell$ is not of this kind. Since $\overline{\mathbf{Q}}_\ell$ is the filtered union of the finite extensions E of $\mathbf{Q}_\ell$, we show instead that every object and every morphism of $D_c^b(X, \overline{\mathbf{Q}}_\ell)$ is defined over some finite E.

For $C \in D(X_{\text{pro\'et}}, E)$ and a finite extension E'/E, the extensions of scalars are the complexes $C_{E'} = E'_X \otimes_{E_X} C$ and $C_{\overline{\mathbf{Q}}_\ell} = \overline{\mathbf{Q}}_{\ell,X} \otimes_{E_X} C$. These tensor products need not be derived, since E'_X and $\overline{\mathbf{Q}}_{\ell,X}$ are flat over E_X.

The classical counterpart is the fact that a continuous representation of a profinite group on $\overline{\mathbf{Q}}_\ell^r$, with the ℓ-adic topology, takes values in $\mathrm{GL}_r(E)$ for some finite $E/\mathbf{Q}_\ell$. Similarly, a constructible complex, or a morphism between two of them, is described by finitely many matrices with entries in $\overline{\mathbf{Q}}_{\ell,X}$, and on a quasi-compact scheme finitely many such entries already lie in a single E_X. We use the next lemma whenever a statement for $\overline{\mathbf{Q}}_\ell$ is reduced to one for a finite E, as in Lemma lemma-integral-models and in the uniform Ext bound (Lemma 5.6).

Lemma 3.23 (Descent to a finite coefficient field)**.**

1. *For $C, D \in D_c^b(X, E)$ and every integer r, extension of scalars induces an isomorphism*
$$\varinjlim_{E' \supset E} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E')}(C_{E'}, D_{E'}[r]) \xrightarrow{\sim} \mathrm{Hom}_{D(X_{\text{pro\'et}}, \overline{\mathbf{Q}}_\ell)}(C_{\overline{\mathbf{Q}}_\ell}, D_{\overline{\mathbf{Q}}_\ell}[r]),$$
the colimit running over finite extensions E' of E in $\overline{\mathbf{Q}}_\ell$.
2. *Every object of $D_c^b(X, \overline{\mathbf{Q}}_\ell)$ is isomorphic to $K_{\overline{\mathbf{Q}}_\ell}$ for some $K \in D_c^b(X, E)$ and some finite $E/\mathbf{Q}_\ell$.*

Proof. Morphisms descend because $\overline{\mathbf{Q}}_{\ell,X}$ is the filtered colimit of the E'_X and Hom out of a constructible complex commutes with this colimit (Lemma 3.22). Objects are built from lisse sheaves by finitely many extensions, each classified by a morphism, and so they descend as well.

(1) Extension of scalars $E'_X \otimes_{E_X} -$ is left adjoint to restriction of scalars $D(X_{\text{pro\'et}}, E') \to D(X_{\text{pro\'et}}, E)$. Hence $\mathrm{Hom}_{D(X_{\text{pro\'et}}, E')}(C_{E'}, D_{E'}[r]) = \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{E'}[r])$, and similarly for $\overline{\mathbf{Q}}_\ell$. Represent D by a bounded complex $D^\bullet$. Then the complexes $D^\bullet \otimes_{E_X} E'_X$ form a filtered system of complexes of E_X-modules, all vanishing below the lower bound of $D^\bullet$, and their colimit is $D^\bullet \otimes_{E_X} \overline{\mathbf{Q}}_{\ell,X}$, which represents $D_{\overline{\mathbf{Q}}_\ell}$. Applying Lemma 3.22 to $C \in D_c^b(X, E)$, we get

$$\varinjlim_{E'} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{E'}[r]) \xrightarrow{\sim} \mathrm{Hom}_{D(X_{\text{pro\'et}}, E)}(C, D_{\overline{\mathbf{Q}}_\ell}[r]),$$

which is the claim after the adjunction.

(2) First let K be a constructible $\overline{\mathbf{Q}}_{\ell,X}$-module, with filtration $N_1 \subset \cdots \subset N_m = K$ as in Lemma lemma-constructible-sheaves. We descend N_k by induction on k. The subquotient $N_k/N_{k-1} = a_! L$

descends, because L does by Lemma 3.17 and $a_!$ commutes with extension of scalars. Suppose N_{k-1} and N_k/N_{k-1} descend to E. Then the extension $0 \to N_{k-1} \to N_k \to N_k/N_{k-1} \to 0$ gives a distinguished triangle, classified by a morphism $N_k/N_{k-1} \to N_{k-1}[1]$. By (1) this morphism descends after a finite extension of E. The fibre of the descended morphism, that is, its cone shifted by $[-1]$, is a model of N_k.

For a bounded complex K, we argue in the same way by induction on the number of nonzero cohomology sheaves. Let i be the largest degree with $\mathcal{H}^i(K) \neq 0$, and consider the truncation triangle $\tau_{<i}K \to K \to \mathcal{H}^i(K)[-i] \to$. Once $\tau_{<i}K$ and $\mathcal{H}^i(K)$ have models over a finite E, the connecting morphism $\mathcal{H}^i(K)[-i-1] \to \tau_{<i}K$ descends after a finite extension by (1), and the cone of the descended morphism is a model of K. Each step replaces E by a finite extension, so after finitely many steps K is defined over a single finite E. ∎

3.8 Lattices after stratification

We pass from statements with coefficients $\mathcal{O}_E/\pi_E^n$ to statements with coefficients E through integral lattices. A lisse E_X-module may fail to have a lattice on all of X, but it has one on each stratum of a finite stratification (Lemma 3.24), and this is enough for dévissage arguments. In particular, from the stratified statement we obtain the integral generators of Lemma lemma-integral-models. Lemma 3.24 is also the first step in the proofs of the uniform Ext bound and of the degree bound for proper direct image in Chapter 5. On a connected normal scheme, by contrast, there is a global lattice, and lisse E-sheaves are representations of $\pi_1(X, \bar{x})$; we prove this where it is used, in Proposition 13.10.

On the nodal curve X of Example 3.16, the sheaf L has no global lattice, and one stratification step repairs this. On the open stratum $X \smallsetminus \{\text{node}\} \simeq \mathbf{G}_m$ the torsor is trivial, so L is constant there and has a lattice. The node, with its reduced structure, is the spectrum of a field, which is normal, and hence the restriction of L to the node has a lattice as well.

Lemma 3.24 (Lattices after a finite stratification)**.** *Let L be a lisse E_X-module. There is a finite stratification of X by locally closed subschemes S such that each $L|_S$ contains a lisse $\mathcal{O}_{E,S}$-submodule L_0 with $L_0[1/\ell] = L|_S$.*

In the proof we first find a lattice after a surjective étale base change and then push it down by the trace. For the first step we use that the lattices in E^r form a discrete set on which $\mathrm{GL}_r(E)$ acts continuously, which makes the sheaf of lattices in L an étale X-scheme. For the second step the base change must be finite étale, which we arrange by restricting to the strata of a finite stratification.

Proof of Lemma 3.24. Let $Y \to X$ be the étale X-scheme representing lattices in L (Lemma 3.15), with its universal lattice.

Pushing down by the trace. Choose finitely many quasi-compact affine opens of this scheme whose images cover X, and let $p : V \to X$ be their disjoint union. Then p is quasi-compact, separated, surjective and étale, and the universal lattice gives a lattice $M_0 \subset p^*L$. By [Stacks, 07RY, 07RZ], X has a finite stratification by locally closed subschemes over each of which p is finite étale and surjective. On such a stratum S, let $\mathrm{Tr} : p_*p^*L \to L$ be the trace of the finite étale map p over S, and define $L_0 \subset L|_S$ as the image of the composite

$$p_*M_0 \longrightarrow p_*p^*L \xrightarrow{\mathrm{Tr}} L.$$

To see that L_0 is a lattice, we split the finite étale cover locally on S into d copies of S. Then p_*M_0 is a direct sum of d lattices $M_1, \dots, M_d$ in L, and the trace is their sum, so $L_0 = M_1 + \cdots + M_d$.

If we further trivialize L pro-étale locally, each M_k becomes a locally constant function to the discrete set of lattices in E^r, and after a common refinement each M_k is the constant sheaf attached to a lattice $N_k \subset E^r$. Then L_0 is the constant sheaf attached to $N_1 + \cdots + N_d$, which is a finite free $\mathcal{O}_E$-lattice spanning E^r. Hence L_0 is lisse and $L_0[1/\ell] = L|_S$. ∎

3.9 Integral models

To deduce statements over E and $\overline{\mathbf{Q}}_\ell$ from statements over $\mathcal{O}_E$, we need integral objects that generate $D_c^b(X, E)$ and $D_c^b(X, \overline{\mathbf{Q}}_\ell)$. We also want to know that our integral constructible complexes are those of the Stacks Project, so that results proved there apply, and we prove both in the next lemma. Recall from Proposition 3.11 in Section 3.3 that an object lies in $D_c^b(X, \mathcal{O}_E) = D_{\mathrm{ctf}}^b(X, \mathcal{O}_E)$ precisely when it is derived complete and its reduction modulo π_E is a bounded constructible complex pulled back from $X_{ét}$. The Stacks Project works instead with the category $D_{\mathrm{cons}}(X, \mathcal{O}_E)$ of [Stacks, 09C1], built from modules over the constant sheaf $\mathcal{O}_E$ rather than over $\mathcal{O}_{E,X}$.

Lemma 3.25 (Integral models)**.**

1. *Restriction of scalars along the map from the constant sheaf $\mathcal{O}_E$ to $\mathcal{O}_{E,X}$ is an equivalence between the derived complete objects of $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ and the derived complete complexes of modules over the constant sheaf $\mathcal{O}_E$.*
2. *An object K of $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ lies in $D_c^b(X, \mathcal{O}_E)$ if and only if its restriction of scalars lies in $D_{\mathrm{cons}}(X, \mathcal{O}_E)$.*
3. *$D_c^b(-, \mathcal{O}_E)$ is preserved by $f^* : D(Y_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E) \to D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ for every morphism $f : X \to Y$, by $\otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}}$ on $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$, and by $a_! : D(S_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E) \to D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ for every locally closed immersion $a : S \hookrightarrow X$.*
4. *Every object of $D_c^b(X, E)$ is obtained by finitely many cones and shifts from objects $(a_! L_0)[1/\ell]$, where $a : S \hookrightarrow X$ is a locally closed immersion and L_0 is a lisse $\mathcal{O}_{E,S}$-module that is locally free of finite rank. Every object of $D_c^b(X, \overline{\mathbf{Q}}_\ell)$ is obtained in the same way from the objects $\overline{\mathbf{Q}}_{\ell,X} \otimes_{\mathcal{O}_{E,X}} a_! L_0$, for some finite $E/\mathbf{Q}_\ell$.*

In (4) each L_0 satisfies the condition of Definition definition-constructible-sheaves, so it lies in $\mathrm{Cons}(S, \mathcal{O}_E)$. By (3) the objects $a_! L_0$ then lie in $D_c^b(X, \mathcal{O}_E)$. Their reductions modulo π_E^n are bounded constructible complexes with coefficients in $\mathcal{O}_E/\pi_E^n$, which are pulled back from $X_{ét}$ (Corollary 2.44).

Proof of Lemma 3.25. In (1) and (2) we compare two coefficient sheaves related by a flat map that is an isomorphism modulo every π_E^n, so they have the same derived complete objects. For (3) we use the two characterizations of integral constructibility, while for (4) we combine the dévissage of Corollary 3.20 with the lattices of Lemma 3.24.

(1) and (2). The map from the constant sheaf $\mathcal{O}_E$ to $\mathcal{O}_{E,X}$ is flat and induces isomorphisms modulo every π_E^n [Stacks, 099L]. Hence by [Stacks, 099I] restriction of scalars is an equivalence on derived complete objects, which is (1). By [Stacks, 09C1], an object of $D_{\mathrm{cons}}(X, \mathcal{O}_E)$ is a derived complete complex whose reduction modulo π_E is constructible of finite Tor dimension over $\mathbf{F}_E$. Since over the field $\mathbf{F}_E$ every bounded complex has finite Tor dimension, this description agrees with the description of $D_c^b(X, \mathcal{O}_E)$ in Proposition 3.11, and together with (1) this gives (2).

(3) By Proposition 3.11, an object of $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ lies in $D_c^b(X, \mathcal{O}_E)$ if and only if it satisfies condition (2) there: it is derived complete with constructible reduction modulo π_E. Equivalently, it satisfies condition (3): there is a finite stratification such that on each stratum S, pro-étale locally, $K|_S \simeq \mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P^\bullet$ for a bounded complex $P^\bullet$ of finite free $\mathcal{O}_E$-modules; here the

implication from (2) to (3) is Proposition 3.10. For each of the three operations we therefore check whichever of these two conditions is convenient.

Inverse image. By Lemma 2.54, f^* sends finite free $\mathcal{O}_{E,Y}$-modules to finite free $\mathcal{O}_{E,X}$-modules, so it preserves the local description of condition (3).

Extension by zero. The functor $a_! = i_* j_!$ preserves derived completeness. Indeed, $j_!$ for a quasi-compact open immersion j preserves limits by [Stacks, 09BP], and i_* is exact and preserves limits by [Stacks, 09AA, 09BM]. Moreover, both functors commute with reduction modulo π_E and preserve constructibility of the reductions, so $a_!$ preserves condition (2).

Tensor product. Let $K, M \in D_c^b(X, \mathcal{O}_E)$, and choose a stratification for K as above. Derived completeness and constructibility of the reduction modulo π_E are both preserved by cones, so $D_c^b(X, \mathcal{O}_E)$ is closed under cones. Moreover, the localization triangles along the strata exhibit K as a finite iterated extension of the objects $a_! a^* K$, for the strata $a : S \hookrightarrow X$, and the functor $- \otimes^{\mathbf{L}} M$ preserves these triangles. It therefore suffices to show that $(a_! a^* K) \otimes^{\mathbf{L}} M$ lies in $D_c^b(X, \mathcal{O}_E)$.

For this we use the projection formula for a locally closed immersion: for $L \in D(S_{\text{pro\'et}}, \mathcal{O}_E)$ and $M \in D(X_{\text{pro\'et}}, \mathcal{O}_E)$ the canonical map

$$(a_! L) \otimes^{\mathbf{L}} M \longrightarrow a_!(L \otimes^{\mathbf{L}} a^* M)$$

is an isomorphism. Indeed, write $a = i \circ j$ with j open and i closed, so that $a_! = i_* j_!$. On S the map restricts to the identity of $L \otimes^{\mathbf{L}} a^* M$, while on every complementary stratum both sides restrict to zero. Since restriction to the strata is exact and jointly conservative by [Stacks, 09AD], the map is an isomorphism.

Given the parts on inverse image and extension by zero, we are reduced to showing that $a^* K \otimes^{\mathbf{L}} a^* M$ lies in $D_c^b(S, \mathcal{O}_E)$. Tensoring with $a^* K$ preserves derived completeness, because $a^* K$ is pro-étale locally a bounded complex of finite free modules. Moreover, the reduction modulo π_E of the tensor product is the tensor product of the reductions, and this is constructible.

(4) For $\overline{\mathbf{Q}}_\ell$, we first descend to a finite E by Lemma 3.23. By Corollary 3.20, every object of $D_c^b(X, E)$ is obtained by finitely many cones and shifts from sheaves $a_! L$ with L lisse. We refine the stratum S by a finite stratification as in Lemma 3.24. This replaces $a_! L$ by finitely many extensions of such sheaves, and on each new stratum L contains a lattice L_0 with $L_0[1/\ell] = L$, so that $a_! L = (a_! L_0)[1/\ell]$. It remains to see that L_0 is integrally constructible. The reduction of L_0 modulo π_E^n is a locally constant sheaf of finitely presented $\mathcal{O}_E/\pi_E^n$-modules, so it comes from the étale site by [Stacks, 099Y]. Thus $L_0 \in D_c^b(S, \mathcal{O}_E)$, and $a_! L_0 \in D_c^b(X, \mathcal{O}_E)$ by (3). ∎

On $\mathbf{A}^1$, part (4) is visible in the running example. There the constant sheaf $E_{\mathbf{A}^1}$ is the cone of a map $i_* E[-1] \to j_! E_{\mathbf{G}_m}$, and up to shift both terms are of the form $(a_! L_0)[1/\ell]$ with L_0 the constant lattice $\mathcal{O}_E$ on the stratum.

3.10 Inverting ℓ

Passing from $\mathcal{O}_E$ to E means inverting ℓ, and on derived categories this is a Verdier localization.

Proposition 3.26 (Inverting ℓ on derived categories)**.** *Extension of scalars $E_X \otimes_{\mathcal{O}_{E,X}} - : D(X_{\text{pro\'et}}, \mathcal{O}_E) \to D(X_{\text{pro\'et}}, E)$ identifies $D(X_{\text{pro\'et}}, E)$ with the Verdier quotient of $D(X_{\text{pro\'et}}, \mathcal{O}_E)$ by the complexes K with $H^q(K)[1/\ell] = 0$ for all q.*

Proof. A functor with a fully faithful right adjoint is the Verdier localization at its kernel, so we show that restriction of scalars is fully faithful and compute the kernel.

Inverting ℓ is a filtered colimit along multiplication by ℓ, hence exact. Therefore $E_X = \mathcal{O}_{E,X}[1/\ell]$ is flat over $\mathcal{O}_{E,X}$, and $H^q(E_X \otimes K) = H^q(K)[1/\ell]$. Restriction of scalars is right adjoint to extension of scalars, and the counit is an isomorphism, because ℓ already acts invertibly on an E_X-module. Restriction is therefore fully faithful, and extension of scalars is the Verdier localization at its kernel. By the formula for $H^q(E_X \otimes K)$, this kernel consists of the complexes K with $H^q(K)[1/\ell] = 0$ for all q. ∎

The kernel contains more than the complexes killed by a power of ℓ, for example $\bigoplus_{n\geq 1} \mathcal{O}_{E,X}/\ell^n$. It does so even among constructible sheaves: on $\bigsqcup_{n\geq 1} \operatorname{Spec} k$, the sheaf equal to $\mathcal{O}_E/\ell^n$ on the n-th point is killed by inverting ℓ but by no power of ℓ. When the underlying space of X is Noetherian, however, this cannot happen in $D_c^b(X, \mathcal{O}_E)$, and there inverting ℓ amounts to tensoring the Hom groups with E.

Proposition 3.27 (Inverting ℓ)**.** *Assume that the underlying topological space of X is Noetherian (for instance, that X is of finite type over a field). A complex in $D_c^b(X, \mathcal{O}_E)$ that is killed by inverting ℓ is killed by a fixed power of ℓ, and extension of scalars induces an equivalence*

$$D_c^b(X, \mathcal{O}_E)[\ell^{-1}] \xrightarrow{\sim} D_c^b(X, E),$$

where the source category has the objects of $D_c^b(X, \mathcal{O}_E)$ and Hom groups tensored with E.

Proof. Both assertions are [BS15, Proposition 6.8.14(2)]. We prove the first, which is where the Noetherian hypothesis enters. Let $K \in D_c^b(X, \mathcal{O}_E)$. By Definition 3.1, there is a finite stratification such that the restriction of each of the finitely many cohomology sheaves of K to each stratum S is, on quasi-compact pro-étale charts U, of the form $\mathcal{O}_{E,U} \otimes_{\mathcal{O}_E} P$ with P finitely generated. Suppose that $K[1/\ell] = 0$. Extension of scalars to E_X is exact, so $E_U \otimes_{\mathcal{O}_E} P = 0$, and each such P is torsion. The underlying space of X is Noetherian, so each stratum is quasi-compact and finitely many charts suffice. With finitely many P, one power of ℓ kills the restrictions of all cohomology sheaves of K to all strata. By the localization sequences of Proposition 2.58, a larger power kills every cohomology sheaf of K. Since K is bounded, a further power kills K itself, by induction along its truncation triangles.

Full faithfulness says that for $K, L \in D_c^b(X, \mathcal{O}_E)$ every morphism $E_X \otimes K \to E_X \otimes L$ is ℓ^{-c} times the image of a morphism $K \to L$ for some $c \geq 0$, and that a morphism $K \to L$ becomes zero after extension of scalars only if some power of ℓ kills it. For essential surjectivity, recall from Lemma lemma-integral-models that $D_c^b(X, E)$ is generated by objects coming from $D_c^b(X, \mathcal{O}_E)$. By the description of the Hom groups, a connecting morphism between them lifts to an integral one after multiplication by a power of ℓ. Hence finite cones of the generators also have integral models. ∎

4 Proper base change

The function–sheaf dictionary (Section 1.3, Proposition 12.28) turns the identity $\#X_0(\mathbf{F}_q) = \sum_y \#f^{-1}(y)(\mathbf{F}_q)$ into a statement about sheaves, and for this the trace function of $Rf_!K$ at a point y must be the sum of t_K over the fibre $f^{-1}(y)$. This requires the stalks of $Rf_!K$ to be the cohomology of the fibres, but nothing in the definition of the derived direct image says so.

Question 4.1 (Stalks of higher direct images along a proper map)**.** For f proper and F a torsion sheaf on X, is the stalk $(R^q f_* F)_{\bar{y}}$ equal to $H^q(X_{\bar{y}}, F)$?

For a proper map $f : X \to Y$ of locally compact Hausdorff spaces the answer is yes. Indeed, the sets $f^{-1}(V)$, for V a neighbourhood of y, form a basis of neighbourhoods of the compact fibre, and the cohomology of a compact subset is the colimit of the cohomology of its neighbourhoods.

In the étale topology, however, the stalk of $R^q f_* F$ at a geometric point $\bar{y}$ is the cohomology of $X \times_Y Y_{(\bar{y})}$, where $Y_{(\bar{y})}$ is the spectrum of the strict henselization of Y at $\bar{y}$. This scheme is much larger than the fibre $X_{\bar{y}}$. For instance, if f is the identity of a curve and $\bar{y}$ lies over a closed point, then it is the spectrum of a strictly henselian local ring of dimension one, not a point.

Properness cannot be dropped. For example, let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ be the inclusion, $i : \{0\} \hookrightarrow \mathbf{A}^1$ the origin, and A the constant sheaf on $\mathbf{G}_m$ with value a nonzero finite abelian group. Then the sheaf $i^* j_* A$ is nonzero, while the base change of $\mathbf{G}_m$ to the origin is empty.

For proper f, nevertheless, the answer is yes. More generally, pulling a proper family back to a new base and taking its cohomology can be done in either order. Precisely, for a cartesian square with f proper, $g : Y' \to Y$ arbitrary and $X' = X \times_Y Y'$, the base change map

$$g^* R f_* K \longrightarrow R f'_* g'^* K$$

is an isomorphism for every bounded below classical complex K with torsion cohomology sheaves (Theorem 4.33). In particular, taking for g a geometric point gives the stalk formula (Corollary 4.34). The theorem is due to Artin [SGA4, XII–XIII], and unlike smooth base change (Chapter 8), it holds for torsion of every order, including torsion divisible by the residue characteristics.

The content of the theorem is already visible in degree zero, where it is a statement about connectedness. Indeed, sections of the constant sheaf $\{0, 1\}$ are open and closed subsets, and for a proper scheme over a henselian local ring, the open and closed subsets correspond to those of the closed fibre. When X is finite over a henselian local ring R, this says that idempotents of the finite R-algebra lift from its reduction modulo the maximal ideal, which is one form of Hensel's lemma. Theorem 4.8 extends the statement to sections of every étale sheaf of sets. Higher degrees then reduce to degree zero and the projective line, and from $\mathbf{P}^1$ the general case follows by formal steps, through $(\mathbf{P}^1)^n$, $\mathbf{P}^n$ and Chow's lemma (Proposition 4.32).

We make these calculations on the étale site, where the classical tools are available, and transport them to classical complexes on the pro-étale site by the comparison theorems of Chapter 2. For $\mathcal{O}_E$ we pass to the limit over the reductions modulo π_E^n (Proposition 4.38). Since pullback need not preserve derived completeness, at this stage the statement for $\mathcal{O}_E$ involves the completed inverse image. Once we know that Rp_* preserves constructibility (Proposition 4.40), however, the completion is unnecessary, and we obtain the ordinary base change isomorphism on D^b_{ctf} for every admissible coefficient ring.

We use proper base change three times later. First, it computes the stalks of $Rf_!$ in the trace formula (Corollary 12.26). Second, it makes $Rf_! = Rp_* j_!$ independent of the compactification (Proposition 5.26). Finally, it bounds the Tor amplitude of Rp_* with finite coefficients (Proposition 4.35).

Convention. For proper base change, schemes are arbitrary unless an additional hypothesis is stated. The theorem concerns the classical part of $D^+(T_{\mathrm{pro\acute{e}t}}, \Lambda)$, for a discrete commutative ring Λ, but the intermediate calculations (acyclicity, curves, Chow's lemma) are made on $T_{\acute{e}t}$, with abelian sheaves or modules over its discrete constant ring. There we write $D^+(T_{\acute{e}t}, \Lambda)$ for the bounded below derived category. Inverse image of étale sheaves along $u : S \to T$ is written $u^* : \mathrm{Sh}(T_{\acute{e}t}) \to \mathrm{Sh}(S_{\acute{e}t})$, as for module sheaves; it is exact, and so is its extension to

$D^+(T_{\acute{e}t}, \Lambda) \to D^+(S_{\acute{e}t}, \Lambda)$. The derived direct image is written Ru_*. Torsion always means torsion of the underlying abelian sheaves.

Proper base change holds for torsion coefficients of every characteristic. Accordingly, until Section 4.7, ℓ denotes an arbitrary prime, which may equal a residue characteristic. From that section on, we return to the standing finite-type hypotheses and the standing condition on ℓ.

4.1 The base change map

Proper base change asserts that a specific map, from the pullback of the direct image to the direct image of the pullback, is an isomorphism. We define this map and describe it on sections, so that we can compute it on stalks.

From Definition 4.14 on, we fix the coefficient ring. Integral and rational coefficients are treated in Section 4.8. Results for abelian sheaves apply to sheaves of modules, by the following lemma.

Lemma 4.2 (Modules and abelian sheaves)**.** *For a sheaf of Λ-modules on a site, its cohomology and its higher direct images along a morphism of sites agree with those of its underlying abelian sheaf. The same holds for Ru_* of bounded below complexes.*

Proof. An injective sheaf of modules is acyclic for the section functor on every object of the site, although it need not be injective as an abelian sheaf [Stacks, 03FD]. Hence an injective resolution of modules computes the cohomology of the underlying abelian sheaf, on every object. Applying this to the objects over the target gives the statement for $R^q u_*$, and bounded below resolutions give it for Ru_*. ∎

Consider a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y. \end{array} \tag{4.1}$$

Since $gf' = fg'$, the composition isomorphisms of inverse images yield an isomorphism $f'^* g^* \simeq g'^* f^*$ on sheaves. Inverse images are exact, so the same isomorphism exists on bounded below derived categories.

Definition 4.3 (Underived base change map)**.** Let F be a sheaf on $X_{\acute{e}t}$. The *underived base change map*

$$g^* f_* F \longrightarrow f'_* g'^* F \tag{4.2}$$

is the adjoint, under $f'^* \dashv f'_*$, of

$$f'^* g^* f_* F \simeq g'^* f^* f_* F \xrightarrow{g'^* \varepsilon_f} g'^* F,$$

where $\varepsilon_f : f^* f_* \to \mathrm{id}$ is the counit.

Equivalently, first apply the unit $g^* f_* F \to f'_* f'^* g^* f_* F$, and then apply f'_* to the displayed composite. This construction is natural in F, and it applies equally to sheaves of Λ-modules.

On sections the map has the following simple description. A section of g^*f_*F over an étale $V \to Y'$ is represented by an étale $W \to Y$, a map $V \to Y' \times_Y W$, and a section over $X \times_Y W$. Given such data, the base change map restricts this section along

$$X' \times_{Y'} V \longrightarrow X \times_Y W,$$

and it is the sheafification of this map of presheaves.

Definition 4.4 (Derived base change map)**.** For $K \in D^+(X_{ét}, \Lambda)$, the *base change map*

$$g^*Rf_*K \longrightarrow Rf'_*g'^*K \tag{4.3}$$

is the adjoint, under $f'^* \dashv Rf'_*$, of

$$f'^*g^*Rf_*K \simeq g'^*f^*Rf_*K \xrightarrow{g'^*\varepsilon_f} g'^*K,$$

where $\varepsilon_f : f^*Rf_* \to \mathrm{id}$ now denotes the derived counit. The morphism is natural in K.

The derived adjunction exists because inverse image is exact and its right adjoint preserves injectives. Note that the construction of the base change map uses only that the square commutes; the cartesian hypothesis is needed only when we ask whether the map is an isomorphism.

To compute (4.3), choose a bounded below injective resolution $K \to I^\bullet$ and an injective replacement $g'^*I^\bullet \to J^\bullet$. Then the base change map is represented by the morphism of complexes

$$g^*f_*I^\bullet \longrightarrow f'_*g'^*I^\bullet \longrightarrow f'_*J^\bullet. \tag{4.4}$$

Here the first arrow is the underived base change map in each degree. By exactness of g^* the source is g^*Rf_*K, and the last complex computes $Rf'_*g'^*K$. The formula does not depend on the choices, since comparison maps between injective resolutions are unique up to homotopy. Moreover, it is the adjoint map of Definition 4.4, because under adjunction the first arrow corresponds to the square isomorphism followed by the counit, and the replacement map is the canonical passage to the derived direct image.

Base change maps for composite squares are composites of base change maps.

Lemma 4.5 (Identity and pasting)**.** *Let $K \in D^+(X_{ét}, \Lambda)$. The base change map is the identity for an identity square, and is compatible with horizontal and vertical pasting of commutative squares, in particular of cartesian squares. More explicitly, if $h : Y'' \to Y'$ is a second base change, completed by maps $h' : X'' \to X'$ and $f'' : X'' \to Y''$ to a commutative square with $f'h' = hf''$, the base change map for the pasted square, with bottom map gh, is the composite*

$$h^*g^*Rf_*K \longrightarrow h^*Rf'_*g'^*K \longrightarrow Rf''_*h'^*g'^*K$$

of h^* applied to the base change map of (4.1) and the base change map of the second square, under the canonical composition isomorphisms.

For composable maps $X \xrightarrow{f} Y \xrightarrow{g} Z$ and a map $h : Z' \to Z$, let $g' : Y' \to Z'$, $h_Y : Y' \to Y$ and $f' : X' \to Y'$, $h_X : X' \to X$ complete g, h and f, h_Y to commutative squares (for instance $Y' = Y \times_Z Z'$ and $X' = X \times_Z Z'$). Then the base change map for the pasted square with left map $g'f'$ and right map gf is the composite

$$h^*Rg_*Rf_*K \longrightarrow Rg'_*h_Y^*Rf_*K \longrightarrow Rg'_*Rf'_*h_X^*K$$

of the base change map for g at Rf_*K and Rg'_* applied to the base change map for f at K.

Proof. Each base change map is a unit followed by a square isomorphism and a counit. When two squares are pasted, the square isomorphisms compose, and the triangle identities cancel the inner unit and counit. In the vertical case we also use that direct image preserves injectives, so that $R(gf)_* \simeq Rg_*Rf_*$ is computed on the same resolution. ∎

On the pro-étale site we use the same construction, but here Λ may be any coefficient ring pulled back from the point, not only a torsion ring. For such Λ inverse images are exact (Lemma 2.54), and the adjunction $f^* \dashv Rf_*$ exists on the unbounded derived categories.

Definition 4.6 (Pro-étale base change map)**.** Let $K \in D(X_{\text{pro}\acute{e}t}, \Lambda)$. The *base change map*

$$g^*Rf_*K \longrightarrow Rf'_*g'^*K \tag{4.5}$$

in $D(Y'_{\text{pro}\acute{e}t}, \Lambda)$ is the adjoint transpose, for $f'^* \dashv Rf'_*$, of the composite

$$f'^*g^*Rf_*K \simeq g'^*f^*Rf_*K \xrightarrow{g'^*\varepsilon_f} g'^*K,$$

where $\varepsilon_f : f^*Rf_* \to \mathrm{id}$ is the counit of $f^* \dashv Rf_*$.

Equivalently, with $\eta_{f'}$ the unit of $f'^* \dashv Rf'_*$, it is the composite

$$g^*Rf_*K \xrightarrow{\eta_{f'}} Rf'_*f'^*g^*Rf_*K \simeq Rf'_*g'^*f^*Rf_*K \xrightarrow{Rf'_*g'^*\varepsilon_f} Rf'_*g'^*K.$$

This is the pro-étale form of the base change map of Definition 4.4. It is defined for every cartesian square (4.1), and the same construction applies to any commutative square. Since proper and smooth base change are proved on the étale site and then transported to the pro-étale site, we need to know that the two maps correspond.

Lemma 4.7 (Naturality, pasting and comparison)**.** *The maps (4.5) form a natural transformation $g^*Rf_* \to Rf'_*g'^*$. It is the identity for an identity square, and for a horizontal or vertical pasting of two commutative squares the base change map of the pasted square is the composite of the two base change maps, in the form displayed in Lemma 4.5. For a discrete ring Λ and $K \in D^+(X_{\acute{e}t}, \Lambda)$, with f and f' qcqs, the comparison isomorphisms of Theorem 2.38 identify the base change map of ν_X^*K with the inverse image on $Y'_{\text{pro}\acute{e}t}$ of the étale base change map of K.*

Proof. Every assertion holds because the base change map is assembled from units, counits and composition isomorphisms for inverse images, and each of the operations in the lemma respects these ingredients. Naturality in K is immediate, since the counit ε_f is a natural transformation.

By definition, the base change map is the adjoint transpose of a counit preceded by the composition isomorphism for inverse images. Since the counit for a composite adjunction is the composite of the two counits, taking adjoint transposes proves pasting, in either order of composing the two directions of a rectangle. Similarly, the assertion about identity squares follows from the triangle identities, as in Lemma 4.5. Finally, the comparisons of sites of Theorem 2.38 identify the composition maps for inverse images and the units and counits (Lemma 2.39), and hence also the base change maps constructed from them. ∎

4.2 Base change in degree zero

For a proper morphism, the underived base change map is an isomorphism for every étale sheaf of sets, with no torsion hypothesis. The reason is that, for a proper scheme over a henselian pair, restriction of sections to the closed fibre is bijective (Theorem 4.8).

Recall that a pair (R, I), with I an ideal of the ring R, is *henselian* if I is contained in the Jacobson radical of R and Hensel's lemma holds for it: every factorization modulo I of a monic polynomial into coprime monic factors lifts to a factorization in R [Stacks, 09XE]. A local ring with its maximal ideal is a henselian pair exactly when it is henselian. For example, a complete local ring is henselian.

Theorem 4.8 (Henselian comparison in degree zero)**.** *Let (R, J) be a henselian pair, let $Z \to \operatorname{Spec} R$ be proper, and let $i : Z_0 = Z \times_R \operatorname{Spec}(R/J) \hookrightarrow Z$. For every sheaf of sets F on $Z_{ét}$, restriction is a bijection*

$$\Gamma(Z, F) \xrightarrow{\sim} \Gamma(Z_0, i^*F).$$

The comparison is natural in F.

The theorem is [Stacks, 0A0C]. At heart it is a statement about connectedness: every irreducible closed subset of a scheme proper over a henselian pair meets the closed fibre in a nonempty connected subset. This follows from Stein factorization and the connectedness properties of henselian pairs [Stacks, 0A0B]; over a general affine base, Stein factorization only produces an algebra integral over R, but that suffices. A topological argument on spectral spaces then turns connectedness into bijectivity of restriction for Zariski sheaves.

Passing to étale sheaves uses finite covers, which are harmless here: a finite cover of Z remains proper over R, so the connectedness statement also holds after every finite cover. Given an étale covering of Z, we choose a finite surjection $Z' \to Z$ that factors through the covering Zariski locally on Z', and sections then descend along $Z' \times_Z Z' \rightrightarrows Z'$ [Stacks, 09ZF].

Example 4.9 (Why the pair must be henselian)**.** Let k be a field of characteristic $\neq 2$ and let $R = k[t]_{(t)}$, the local ring of $\mathbf{A}^1_k$ at the origin, with maximal ideal $J = (t)$. The finite R-algebra $B = R[x]/(x^2 - 1 - t)$ is a domain, because $1 + t$ is not a square in $k(t)$. Hence $Z = \operatorname{Spec} B$ is connected and $\Gamma(Z, \mathbf{Z}/2) = \mathbf{Z}/2$. The closed fibre, on the other hand, is $\operatorname{Spec} k[x]/(x^2 - 1)$, which consists of the two points $x = \pm 1$, and so $\Gamma(Z_0, \mathbf{Z}/2) = (\mathbf{Z}/2)^2$. Thus the conclusion of the theorem fails for (R, J). The reason is that (R, J) is not henselian: the factorization $x^2 - 1 = (x - 1)(x + 1)$ modulo t does not lift. In the henselization, by contrast, and in $k[[t]]$, the element $1 + t$ has the square root $1 + t/2 - t^2/8 + \cdots$. There the factorization lifts, and Z splits into two copies of the base, so both sides become $(\mathbf{Z}/2)^2$.

The stalks of f_*F are sections over geometric fibres. Indeed, a stalk is the set of sections over the strict localization, and the henselian comparison identifies these with the sections over the closed fibre.

Lemma 4.10 (Sections over a geometric fibre)**.** *Let $f : X \to Y$ be proper, let F be an étale sheaf of sets on X, and let $\bar{y} : \operatorname{Spec} \Omega \to Y$ be a geometric point. Then restriction is a bijection*

$$(f_*F)_{\bar{y}} \xrightarrow{\sim} \Gamma(X_{\bar{y}}, F|_{X_{\bar{y}}}). \tag{4.6}$$

Proof. Let $y \in Y$ be the image of $\bar{y}$, let $Y_{(\bar{y})} = \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{Y,\bar{y}}$ and $X_{(\bar{y})} = X \times_Y Y_{(\bar{y})}$. In degree zero the stalk is computed on the strict localization, that is,

$$(f_*F)_{\bar{y}} \simeq \Gamma(X_{(\bar{y})}, F|_{X_{(\bar{y})}}).$$

For quasi-compact quasi-separated f this holds by continuity of étale sections ([Stacks, 03Q9], whose proof in degree zero applies to sheaves of sets). To recall the argument: by definition, the stalk $(f_*F)_{\bar{y}}$ is the filtered colimit of $\Gamma(X \times_Y V, F|)$ over pointed affine étale neighbourhoods V of $\bar{y}$, where $F|$ denotes the inverse image of F. Since the limit of these neighbourhoods is $Y_{(\bar{y})}$, there is a natural map from the colimit to the sections over $X_{(\bar{y})}$, and we must see that it is bijective. A section over the limit is represented by finitely many covering objects and finitely many agreement conditions, because $X_{(\bar{y})}$ is quasi-compact and quasi-separated. Since all these data are of finite presentation, they descend to one neighbourhood. This proves surjectivity, and the same descent for two representatives proves injectivity.

The local ring of $Y_{(\bar{y})}$ is strictly henselian, with separably closed residue field $\kappa(y)^{\mathrm{sep}} \subseteq \Omega$. Thus $\mathcal{O}^{\mathrm{sh}}_{Y,\bar{y}}$ with its maximal ideal is a henselian pair, and $X_{(\bar{y})}$ is proper over it with closed fibre $X_{\kappa(y)^{\mathrm{sep}}}$. Theorem 4.8 therefore identifies the sections on the right with those over $X_{\kappa(y)^{\mathrm{sep}}}$. These in turn equal $\Gamma(X_{\bar{y}}, F|_{X_{\bar{y}}})$, because extending a separably closed ground field from $\kappa(y)^{\mathrm{sep}}$ to Ω does not change the sections of the pullback of an étale sheaf [Stacks, 0A3I]. ∎

Proposition 4.11 (Proper base change in degree zero)**.** *For (4.1) with f proper, and for every étale sheaf of sets F on X, the underived base change map (4.2)*

$$g^* f_* F \longrightarrow f'_* g'^* F$$

is an isomorphism.

Proof. We show that both stalks are the sections over the same geometric fibre. To this end, choose $\bar{y}' : \operatorname{Spec} \Omega \to Y'$ and use the composite $\bar{y} = g\bar{y}'$ as the geometric point of Y. Since properness is preserved by base change, Lemma 4.10 applies to f' as well as to f. Moreover, the cartesian square identifies

$$X'_{\bar{y}'} = X \times_Y \operatorname{Spec} \Omega = X_{\bar{y}}$$

and identifies the two inverse image sheaves on it. Thus the stalks of both sides of the desired map are the same set of sections. It remains to see that the map is the identity. The underived base change map sends a section represented over $X \times_Y V$ to its restriction to the base change of the neighbourhood, and hence to its restriction to this same fibre. In other words, the map on stalks is the identity under the displayed identifications. Since geometric stalks detect isomorphisms, this proves the assertion. ∎

4.3 Torsion sheaves and acyclicity

In degree zero we needed no hypothesis on the sheaf. In higher degrees we do: proper base change fails for the constant sheaf $\mathbf{Z}$.

Example 4.12 (Proper base change fails for $\mathbf{Z}$)**.** Let R be a strictly henselian discrete valuation ring with uniformizer t and algebraically closed residue field of characteristic $\neq 2$, and let $X \subset \mathbf{P}^2_R$ be the closure of the affine curve $y^2 = x^3 - x^2 + t$. Then X is regular (at the origin of the

closed fibre the equation has the linear term t), while its closed fibre X_s is the nodal cubic $y^2 = x^2(x-1)$. Let $f : X \to \operatorname{Spec} R$ be the structure map. Since R is strictly henselian, the stalk of $R^1 f_* \mathbf{Z}$ at the closed point is $H^1(X, \mathbf{Z})$.

This group is zero. Indeed, X is normal and connected, so $\mathbf{Z}$-torsors on X correspond to continuous homomorphisms $\pi_1(X) \to \mathbf{Z}$, and the image of such a homomorphism is a compact subgroup of $\mathbf{Z}$, hence zero. On the closed fibre, however, $H^1(X_s, \mathbf{Z}) = \mathbf{Z}$, generated by the loop through the node. This means that the base change map from the stalk of $R^1 f_* \mathbf{Z}$ at the closed point to $H^1(X_s, \mathbf{Z})$ is not surjective.

With $\mathbf{Z}/n$ in place of $\mathbf{Z}$, the class of the loop is realized by the cyclic étale cover of X_s of degree n obtained by unwinding the node, and because R is henselian, this cover lifts to X. This is the phenomenon the torsion hypothesis captures.

We therefore restrict to torsion sheaves.

Definition 4.13 (Torsion sheaves)**.** A sheaf F of abelian groups, or of Λ-modules, is *torsion* if every geometric stalk is a torsion abelian group. Equivalently,

$$F = \varinjlim_{n \geq 1} F[n], \qquad F[n] = \ker(n : F \to F),$$

where the positive integers are ordered by divisibility.

If Λ is torsion as an abelian group, it is killed by the additive order of 1, so every sheaf of Λ-modules is torsion. The two descriptions of torsion in Definition 4.13 agree. Indeed, the colimit is taken in sheaves, and geometric stalks commute with colimits and finite limits and detect isomorphisms, so the equivalence reduces to the case of abelian groups.

Torsion is a local condition on sections: every section is killed by an integer after passing to an étale covering. However, the integer may vary between members of the covering. For example, the constant sheaf $\mathbf{Q}/\mathbf{Z}$ on a nonempty scheme is torsion, but on the disjoint union of countably many copies of the scheme its sections include $(1/2^n)_{n \geq 1}$, which has infinite order.

Definition 4.14 (The PBC property)**.** For a morphism $f : X \to Y$, write $\mathrm{PBC}_\Lambda(f)$ if the base change map (4.3) is an isomorphism for every base change $g : Y' \to Y$ and every $K \in D^+(X_{\acute{e}t}, \Lambda)$ with torsion cohomology sheaves.

From now until the end of Section 4.5, Λ denotes a commutative ring that is torsion as an abelian group, unless another ring is specified. For such Λ and proper f we prove $\mathrm{PBC}_\Lambda(f)$ in Propositions 4.26 and Proposition 4.32, and Theorem 4.33 then removes the condition on Λ.

To pass from degree zero to higher degrees we apply the degree-zero result to an injective resolution $F \to I^\bullet$ of a sheaf F on X. By Proposition 4.11 the base change map is an isomorphism $g^* f_* I^m \simeq f'_* g'^* I^m$ in each degree. Here the complex $g^* f_* I^\bullet$ computes $g^* R f_* F$. The complex $f'_* g'^* I^\bullet$, on the other hand, is f'_* applied to a resolution of $g'^* F$ that need not be injective, since g'^* does not preserve injectives. It still computes $R f'_* g'^* F$ if every $g'^* I^m$ is f'_*-acyclic. We call this condition injective acyclicity, and Theorem 4.17 shows that it is equivalent to proper base change. Its advantage is that, unlike the base change statement, injective acyclicity is a vanishing condition on single sheaves.

Definition 4.15 (Injective acyclicity)**.** For a ring Λ, write $\mathrm{IA}_\Lambda(f)$ if for every cartesian square (4.1) and every injective sheaf I of Λ-modules on $X_{\acute{e}t}$ which is torsion, one has

$$R^q f'_* g'^* I = 0 \qquad (q > 0).$$

In general $g'^* I$ is not injective, and injective acyclicity only asks that it be f'_*-acyclic. By the next lemma this suffices to compute Rf'_* of the pullback of an injective resolution term by term.

Lemma 4.16 (Acyclic resolutions)**.** *Let $v : \mathcal{A} \to \mathcal{B}$ be a left exact additive functor between abelian categories with enough injectives. If $F \to M^\bullet$ is a resolution and each M^m is v-acyclic, then*

$$\mathcal{H}^q(vM^\bullet) \simeq R^q v F.$$

More generally, for a bounded below complex $M^\bullet$ of v-acyclic objects, the canonical comparison $vM^\bullet \to Rv(M^\bullet)$ is an isomorphism in $D^+(\mathcal{B})$.

Proof. This is Leray's acyclicity lemma [Stacks, 015E], and the first assertion is the case of a resolution of F. We recall the argument, which compares $M^\bullet$ with an injective replacement through the cone.

In a bounded below exact complex of v-acyclic objects the cycles are again v-acyclic, by induction along the short exact sequences of cycles, so the complex stays exact after applying v. We apply this to the cone of a quasi-isomorphism $M^\bullet \to J^\bullet$ into a bounded below complex of injectives. This cone is bounded below and exact, and its terms are v-acyclic, being finite direct sums of terms of $M^\bullet$ and of $J^\bullet$. It follows that $vM^\bullet \to vJ^\bullet$ is a quasi-isomorphism, and the target computes $Rv(M^\bullet)$. ∎

To pass from sheaves killed by an integer to all torsion sheaves, we use continuity of étale cohomology: if $u : T \to S$ is quasi-compact and quasi-separated, then

$$\varinjlim_i R^q u_* F_i \xrightarrow{\sim} R^q u_* (\varinjlim_i F_i) \qquad (q \geq 0) \tag{4.7}$$

for every filtered system of abelian étale sheaves on T. The same holds after any base change of u [Stacks, 09Z1]. Indeed, $R^q u_* F$ is the sheafification of $V \mapsto H^q(T \times_S V, F|_{T \times_S V})$ on affine étale $V \to S$, and the étale cohomology of the quasi-compact and quasi-separated scheme $T \times_S V$ commutes with filtered colimits of abelian sheaves [Stacks, 03Q5].

The following criterion reduces proper base change in higher degrees to injective acyclicity, a vanishing statement that can be checked on curves.

Theorem 4.17 (PBC and injective acyclicity)**.** *Let $f : X \to Y$ be proper. The following are equivalent:*

- *(i)* $\mathrm{PBC}_\Lambda(f)$ *holds.*
- *(ii) For every sheaf F of Λ-modules and every base change (4.1), the maps induced by (4.3) on cohomology sheaves*
$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F$$
are isomorphisms for all $q \geq 0$.
- *(iii)* $\mathrm{IA}_\Lambda(f)$ *holds.*

Proof. We prove (i)⇒(ii)⇒(iii)⇒(ii)⇒(i). The step (iii)⇒(ii) is the main one: it applies the degree-zero isomorphism termwise to an injective resolution.

(i)⇒(ii)⇒(iii). Condition (ii) is the case of (i) where F sits in degree zero. Suppose (ii) holds, and let I be an injective sheaf of Λ-modules. Then $R^q f'_* g'^* I \simeq g^* R^q f_* I$ by (ii). The right hand side vanishes for $q > 0$ because I is injective, and this is (iii).

(iii)⇒(ii). Let F be a sheaf of Λ-modules. We compare the two sides of the base change map through an injective resolution $F \to I^\bullet$ by sheaves of Λ-modules. The complex $f'_* g'^* I^\bullet$ computes $Rf'_* g'^* F$. Indeed, since g'^* is exact, $g'^* I^\bullet$ is a resolution of $g'^* F$, and its terms are f'_*-acyclic by (iii), so Lemma 4.16 applies. Similarly, the complex $g^* f_* I^\bullet$ computes $g^* R f_* F$, because inverse image is exact. Between the two, Proposition 4.11 gives an isomorphism

$$g^* f_* I^m \xrightarrow{\sim} f'_* g'^* I^m$$

in each degree, and these form an isomorphism of complexes by naturality. By (4.4) this isomorphism is the derived base change map for F, which proves (ii).

(ii)⇒(i). Let $K \in D^+(X_{\acute{e}t}, \Lambda)$. For bounded K we argue by induction on the cohomological length. A complex of length one is a shifted sheaf, which (ii) covers, while a longer one is, by a canonical truncation triangle, an extension of two complexes of smaller length. Since the base change map is a morphism of triangles, the five lemma on cohomology sheaves shows that it is an isomorphism for K. An unbounded K reduces to the bounded case, because a given cohomology sheaf only depends on a bounded truncation. More precisely, fix a degree m and choose $N \geq m$, so that $\tau_{\leq N} K$ is bounded and has the desired comparison. Both functors $g^* R f_*$ and $R f'_* g'^*$ send $D^{\geq N+1}(X_{\acute{e}t}, \Lambda)$ into $D^{\geq N+1}(Y'_{\acute{e}t}, \Lambda)$, since inverse images are exact and derived direct images are left t-exact. Applying them to

$$\tau_{\leq N} K \longrightarrow K \longrightarrow \tau_{\geq N+1} K \xrightarrow{+1}$$

we see that the comparison for K in degree m is the comparison for $\tau_{\leq N} K$, hence an isomorphism. As m was arbitrary, (i) follows. ∎

4.4 The projective line

Let $\pi : \mathbf{P}^1_S \to S$ be the projection. A geometric stalk of $R^q \pi_* F$ is the cohomology of $\mathbf{P}^1$ over a strictly henselian local ring, and proper base change for π says that it is the cohomology of the geometric fibre, and that this group does not change under base change.

4.4.1 Henselian pairs and curves

The model case is a complete discrete valuation ring R. Think of $\operatorname{Spec} R$ as a small disc around its closed point, and of $\mathbf{P}^1_R$ as a family of projective lines over the disc. In topology such a family retracts onto the central fibre, and we expect that, for torsion sheaves, the cohomology of $\mathbf{P}^1_R$ is that of its closed fibre. Besides the comparison in degree zero of Theorem 4.8, the proof uses two theorems about henselian pairs from the Stacks Project. The first is the affine analogue of what we want to prove.

Theorem 4.18 (Gabber's affine analogue of proper base change)**.** *Let (B, J) be a henselian pair and let G be a torsion abelian sheaf on $\operatorname{Spec}(B)_{\acute{e}t}$. Then restriction induces isomorphisms*

$$H^q(\operatorname{Spec} B, G) \xrightarrow{\sim} H^q(\operatorname{Spec}(B/J), G|_{\operatorname{Spec}(B/J)}) \qquad (q \geq 0).$$

This is [Stacks, 09ZI]; Gabber proves it by an effacement argument on affines. The second fact concerns finite étale covers of a proper scheme. In degree one it says that $\mathbf{Z}/n$-torsors, like sections, are determined by the closed fibre.

Theorem 4.19 (Finite étale covers over a proper henselian pair)**.** *Let (B, J) be a henselian pair, let Y be proper over B, and put $Y_0 = Y \times_{\operatorname{Spec} B} \operatorname{Spec}(B/J)$. Then restriction is an equivalence between the categories of finite étale schemes over Y and over Y_0,*

$$F\acute{E}t_Y \xrightarrow{\sim} F\acute{E}t_{Y_0}.$$

Consequently, for a finite abelian group M,

$$H^1(Y, \underline{M}) \xrightarrow{\sim} H^1(Y_0, \underline{M}). \tag{4.8}$$

Proof. The equivalence is [Stacks, 0GS2], proved by algebraization, approximation, and descent of finite presentation. For the consequence, note that the groups in (4.8) classify M-torsors, which are finite étale schemes with an action of M satisfying the equations of a simply transitive action. The equivalence transports these schemes, their actions and the equations, so it induces a bijection on isomorphism classes of M-torsors. ∎

The last step of the proof, the change of separably closed ground field, uses the following facts about curves over an algebraically closed field, which we also need later. For torsion divisible by the characteristic, proper and affine curves behave differently, which is why some of the assertions require C to be proper.

Theorem 4.20 (Cohomology of curves)**.** *Let k be algebraically closed, let C be a separated scheme of finite type over k with $\dim C \leq 1$, and let F be a classical torsion abelian sheaf on $C_{\text{proét}}$. With cohomology taken on $C_{\text{proét}}$, the following assertions hold:*

- (C_1) *$H^q(C, F) = 0$ for $q > 2$.*
- (C_2) *If C is affine, $H^q(C, F) = 0$ for $q > 1$.*
- (C_3) *If $\operatorname{char} k = p > 0$ and F is p-primary, $H^q(C, F) = 0$ for $q > 1$.*
- (C_4) *If F is constructible with finite stalks and prime to $\operatorname{char} k$, all $H^q(C, F)$ are finite.*
- (C_5) *If C is proper and F is constructible with finite stalks, all $H^q(C, F)$ are finite.*
- (C_6) *If k'/k is an extension of algebraically closed fields and F is prime to $\operatorname{char} k$, the pullback maps $H^q(C, F) \to H^q(C_{k'}, F|)$ are isomorphisms.*
- (C_7) *If C is proper, the maps in (C_6) are isomorphisms for every torsion F.*
- (C_8) *For any open $U \subseteq C$, restriction $H^2(C, F) \to H^2(U, F|_U)$ is surjective.*

In characteristic zero, the condition that F be prime to the characteristic in (C_4) and (C_6) is automatic.

Proof of Theorem 4.20. The proof is in Appendix D. For a smooth proper connected curve and n invertible in k, one computes $H^q(C, \mu_n)$ with the Kummer sequence and the Jacobian (Appendix C), while for p-primary coefficients one uses the Artin–Schreier sequence instead. The case of general torsion sheaves is then reduced to these cases by normalization, removal of finitely many points and a finite étale cover of degree prime to the torsion order. ∎

The final step for the projective line compares cohomology over the residue fields of two strict localizations, which are separably closed but in general not algebraically closed. For this we need the following form of (C_7).

Lemma 4.21 (Separably closed field extension)**.** *Let $k \subseteq k'$ be separably closed fields, let C be proper over k with $\dim C \leq 1$, and let F be a classical torsion abelian sheaf on $C_{\text{proét}}$. Then the natural map*

$$H^q(C, F) \xrightarrow{\sim} H^q(C_{k'}, F|) \qquad (q \geq 0)$$

is an isomorphism.

Proof. A purely inseparable extension does not change the étale topos, which allows us to replace k and k' by their algebraic closures and then apply (C_7).

By the classical comparison we may compute on the étale site. Choose compatible algebraic closures $\bar{k} \subseteq \bar{k}'$. Since $\bar{k}/k$ and $\bar{k}'/k'$ are purely inseparable, $C_{\bar{k}} \to C$ and $C_{\bar{k}'} \to C_{k'}$ are universal homeomorphisms, and hence they induce equivalences of small étale topoi [Stacks, 03SI]. This reduces the claim to the pullback from $C_{\bar{k}}$ to $C_{\bar{k}'}$, which is an isomorphism by (C_7) of Theorem 4.20. Since all these maps are pullback maps, the comparison square commutes [Stacks, 0A5E]. ∎

Remark 4.22 (Why properness is needed in (C_7))**.** Properness in (C_7) is needed when the torsion order is divisible by the characteristic, and this case arises in Proposition 4.26, where the field is extended for arbitrary torsion sheaves. For example, in characteristic p,

$$H^1(\mathbf{A}^1_k, \mathbf{F}_p) = k[t]/\{h^p - h : h \in k[t]\},$$

and extending an algebraically closed ground field adds new classes. Indeed, we have $c^p t^{pm} \equiv ct^m$, and every element of k is of the form $h^p - h$. It follows that every class has a unique representative $\sum_{p \nmid m} a_m t^m$, and the same holds over an algebraically closed $k' \supset k$. Hence for $a \in k' \smallsetminus k$ the class of at does not come from $k[t]$.

4.4.2 Injectives on the closed fibre

Let R be a henselian local ring with maximal ideal $\mathfrak{m}$ and residue field κ, and let $i : \mathbf{P}^1_\kappa \hookrightarrow \mathbf{P}^1_R$ be the inclusion of the closed fibre. For a sheaf G on $\mathbf{P}^1_{R,ét}$ write $G_0 = i^*G$. To compare the cohomology of $\mathbf{P}^1_R$ with that of its closed fibre, we apply the comparison in degree zero of Theorem 4.8 to an injective resolution. This works once we know that the restriction I_0 of an injective sheaf I is acyclic on $\mathbf{P}^1_\kappa$ (Lemma 4.24). However, an injective sheaf is too large to work with directly, so we approximate it by constructible sheaves and embed these into direct images along finite morphisms, using three standard finiteness facts about small étale topoi of quasi-compact quasi-separated schemes.

Lemma 4.23 (Finiteness facts on $\mathbf{P}^1_R$)**.**

1. *A sheaf of $\mathbf{F}_\ell$-modules on $\mathbf{P}^1_{R,ét}$ is a filtered colimit of constructible ones, and cohomology on $\mathbf{P}^1_R$ and on $\mathbf{P}^1_\kappa$ commutes with filtered colimits.*

2. *A constructible sheaf G of $\mathbf{F}_\ell$-modules on $\mathbf{P}^1_{R,ét}$ admits a monomorphism*

$$G \hookrightarrow \bigoplus_{a=1}^{N} (h_a)_* \underline{M_a}, \qquad h_a : Y_a \to \mathbf{P}^1_R \text{ finite and finitely presented}, \tag{4.9}$$

with M_a finite $\mathbf{F}_\ell$-modules.

3. *For a finite morphism $h : Y \to \mathbf{P}^1_R$ and an abelian sheaf G on $Y_{ét}$, we have*

$$R^q h_* G = 0 \quad (q > 0), \qquad i^* h_* G \xrightarrow{\sim} (h_0)_* i_Y^* G. \tag{4.10}$$

Here $h_0 : Y_0 = Y \times_{\mathbf{P}^1_R} \mathbf{P}^1_\kappa \to \mathbf{P}^1_\kappa$ and $i_Y : Y_0 \to Y$ are the base changes of h and i, and the second map is the underived base change map (4.2).

Proof. Part (1) is [Stacks, 03SA] and [Stacks, 03Q5], and part (2) is [Stacks, 09Z7]. Part (3) is the case $u = h$, $W = \mathbf{P}^1_R$ of Lemma 4.27 below, which is proved directly from the stalk description of finite direct images [Stacks, 03QP]. ∎

The sheaves on the right of (4.9) are direct images along finite morphisms, and by Lemma lemma-finiteness-small-etale-topoi such direct images are exact functors and commute with restriction to the closed fibre.

Lemma 4.24 (Acyclicity on the closed fibre)**.** *If I is an injective sheaf of $\mathbf{F}_\ell$-modules on $\mathbf{P}^1_{R,ét}$, then $H^q(\mathbf{P}^1_\kappa, I_0) = 0$ for every $q > 0$.*

Proof. We prove vanishing on affine charts by Gabber's theorem, deduce the case $q \geq 2$ by Mayer–Vietoris, and treat $q = 1$ by lifting classes to finite torsors over $\mathbf{P}^1_R$, where injectivity of I kills them.

Step 1: affine opens. Let $W = \operatorname{Spec} B$ be an affine open of $\mathbf{P}^1_R$ and set $J = \mathfrak{m}B$. We show that $I_0|_{W_0}$ has no cohomology in positive degrees. Since the pair (B, J) need not be henselian, we pass to its henselization (B^h, JB^h), for which $B^h/JB^h = B/J$. Moreover, B^h is the filtered colimit of the étale B-algebras B' equipped with $B'/JB' \simeq B/J$. Gabber's theorem (Theorem 4.18) and continuity of étale cohomology give

$$H^q(W_0, I_0|_{W_0}) \simeq H^q(\operatorname{Spec} B^h, I|) \simeq \varinjlim_{B'} H^q(\operatorname{Spec} B', I|). \tag{4.11}$$

Every group on the right vanishes for $q > 0$, because the restriction of an injective sheaf to an étale object is injective: the restriction functor has an exact left adjoint, namely extension by zero in the localized site.

Step 2: degrees $q \geq 2$. The two standard affine charts $U, V \subset \mathbf{P}^1_R$ have affine intersection, so (4.11) shows that the restriction of I to each of U_0, V_0 and $(U \cap V)_0$ has no cohomology in positive degrees. The Mayer–Vietoris sequence on $\mathbf{P}^1_\kappa$ therefore gives $H^q(\mathbf{P}^1_\kappa, I_0) = 0$ for $q \geq 2$ (this is the case of two charts in [Stacks, 0A51]). In degree one the same sequence only identifies $H^1(\mathbf{P}^1_\kappa, I_0)$ with the cokernel of $\Gamma(U_0, I_0) \oplus \Gamma(V_0, I_0) \to \Gamma((U \cap V)_0, I_0)$, and nothing said so far shows that this cokernel vanishes.

Step 3: degree one. Let $\xi \in H^1(\mathbf{P}^1_\kappa, I_0)$. Following [Stacks, 0A5H], we show that $\xi = 0$. First write I as a filtered colimit of constructible $\mathbf{F}_\ell$-sheaves. Since pullback and H^1 on $\mathbf{P}^1_\kappa$ commute with this colimit, ξ is the image of a class $\zeta \in H^1(\mathbf{P}^1_\kappa, G_0)$ for some constructible G with a map $G \to I$. Next choose (4.9) and put $P = \bigoplus_a (h_a)_* \underline{M_a}$. Since I is injective, the map $G \to I$ extends to $P \to I$. Let ζ' be the image of ζ in $H^1(\mathbf{P}^1_\kappa, P_0)$.

The class ζ' lifts to $H^1(\mathbf{P}^1_R, P)$. Indeed, (4.10) and the Leray spectral sequence give

$$H^1(\mathbf{P}^1_\kappa, P_0) \simeq \bigoplus_a H^1((Y_a)_0, \underline{M_a}), \qquad H^1(\mathbf{P}^1_R, P) \simeq \bigoplus_a H^1(Y_a, \underline{M_a}).$$

Each Y_a is finite over $\mathbf{P}^1_R$, hence proper over R, so the comparison (4.8) for finite torsors (Theorem 4.19) lifts ζ'. This lift maps to zero in $H^1(\mathbf{P}^1_R, I)$ because I is injective. Now the commutative square of restriction maps

$$\begin{array}{ccc} H^1(\mathbf{P}^1_R, P) & \longrightarrow & H^1(\mathbf{P}^1_R, I) \\ \downarrow & & \downarrow \\ H^1(\mathbf{P}^1_\kappa, P_0) & \longrightarrow & H^1(\mathbf{P}^1_\kappa, I_0) \end{array}$$

shows that the image of ζ' in $H^1(\mathbf{P}^1_\kappa, I_0)$ is zero, and this image is ξ, because $G \to P \to I$ is the given map $G \to I$. Hence $\xi = 0$. ∎

With this acyclicity, an injective resolution on $\mathbf{P}^1_R$ restricts to an acyclic resolution on the closed fibre. Applying the degree-zero comparison of Theorem 4.8 term by term, we obtain the comparison in all degrees. Since the lemma only concerns sheaves killed by a prime ℓ, we then extend the result to all torsion sheaves by dévissage.

Proposition 4.25 (Cohomology of $\mathbf{P}^1$ over henselian rings)**.** *Let R be a henselian local ring with residue field κ. For every torsion abelian sheaf F on $\mathbf{P}^1_{R,ét}$, restriction induces isomorphisms, natural in F, with $F_0 = i^*F$,*

$$H^q(\mathbf{P}^1_R, F) \xrightarrow{\sim} H^q(\mathbf{P}^1_\kappa, F_0) \qquad (q \geq 0).$$

Proof. We treat sheaves killed by a prime first, where Lemma 4.24 applies, and then reach all torsion sheaves by extensions, direct sums and filtered colimits.

Suppose first that $\ell F = 0$ for a prime ℓ. Take an injective resolution $F \to I^\bullet$ in sheaves of $\mathbf{F}_\ell$-modules on $\mathbf{P}^1_{R,ét}$. Since i^* is exact, $F_0 \to I_0^\bullet$ is again a resolution, and by Lemma 4.24 its terms are $\Gamma(\mathbf{P}^1_\kappa, -)$-acyclic. Moreover, Theorem 4.8 shows that the two complexes of sections agree term by term:

$$\Gamma(\mathbf{P}^1_R, I^\bullet) \xrightarrow{\sim} \Gamma(\mathbf{P}^1_\kappa, I_0^\bullet).$$

The left complex computes $R\Gamma(\mathbf{P}^1_R, F)$, and the right computes $R\Gamma(\mathbf{P}^1_\kappa, F_0)$. Hence the natural restriction map between them is a quasi-isomorphism.

By the long exact cohomology sequences and the five lemma, the sheaves for which the statement holds form a class closed under extensions. Since a sheaf killed by ℓ^a has a finite filtration with quotients killed by ℓ, and a sheaf killed by an integer n is the finite direct sum of its primary parts, the class contains all sheaves of bounded torsion. An arbitrary torsion F is the colimit $\varinjlim_n F[n]$, with positive integers ordered by divisibility, and the comparison passes to this colimit, because the cohomology of the quasi-compact quasi-separated schemes $\mathbf{P}^1_R$ and $\mathbf{P}^1_\kappa$ commutes with filtered colimits [Stacks, 03Q5]. Finally, all maps used are restriction maps, which gives naturality in F. ∎

For example, let R be strictly henselian and n invertible in R. Then the residue field κ is separably closed, and passing to its algebraic closure does not change the étale topos [Stacks, 03SI]. Hence $H^2(\mathbf{P}^1_\kappa, \mu_n) \simeq \mathbf{Z}/n$, generated by the Kummer class of $\mathcal{O}(1)$ (Proposition D.3, in Appendix D), and the proposition says that $H^2(\mathbf{P}^1_R, \mu_n)$ is also $\mathbf{Z}/n$, generated by the class of $\mathcal{O}(1)$.

4.4.3 Arbitrary base change

Let $\pi : \mathbf{P}^1_S \to S$ be the projection and $\bar{s}$ a geometric point of S. The stalk of $R^q\pi_*F$ at $\bar{s}$ is the cohomology of $\mathbf{P}^1$ over the strict localization at $\bar{s}$, by (4.12) below. By Proposition 4.25 it is therefore the cohomology of $\mathbf{P}^1_{k_s}$, where k_s is the separably closed residue field. Along a base change $g : T \to S$ the residue field grows from k_s to k_t, so on stalks the base change map becomes the pullback from $\mathbf{P}^1_{k_s}$ to $\mathbf{P}^1_{k_t}$, which is an isomorphism by Lemma 4.21. This is the second place where properness enters: for $\mathbf{A}^1$ and p-torsion coefficients the corresponding map is not surjective.

The stalk formula holds for every quasi-compact quasi-separated morphism $f : Z \to S$. For a geometric point $\bar{s}$ of S and an abelian sheaf G on $Z_{\acute{e}t}$ we have

$$(R^q f_* G)_{\bar{s}} \simeq H^q(Z \times_S S_{(\bar{s})}, G|), \qquad S_{(\bar{s})} = \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{S,\bar{s}}. \tag{4.12}$$

Indeed, the stalk is the filtered colimit of $H^q(Z \times_S V, G|)$ over pointed étale neighbourhoods $(V, \bar{v})$ of $\bar{s}$, and by continuity of étale cohomology this colimit is the cohomology of the indicated limit scheme [Stacks, 03Q9].

Proposition 4.26 (Proper base change for $\mathbf{P}^1$)**.** *Let S be a scheme and Λ a commutative ring. In the cartesian square*

$$\begin{array}{ccc} \mathbf{P}^1_T & \xrightarrow{g'} & \mathbf{P}^1_S \\ {\scriptstyle \pi'}\downarrow & & \downarrow{\scriptstyle \pi} \\ T & \xrightarrow{g} & S, \end{array}$$

the base change map

$$g^* R\pi_* K \longrightarrow R\pi'_* g'^* K$$

is an isomorphism for every $K \in D^+(\mathbf{P}^1_{S,\acute{e}t}, \Lambda)$ with torsion cohomology sheaves. In particular, $\mathrm{IA}_\Lambda(\pi)$ *holds.*

Proof. For a torsion sheaf we compare the two sides on geometric stalks. Both stalks are the cohomology of a projective line over a separably closed field, and the base change map becomes the pullback along a field extension. Bounded below complexes then follow by truncation.

Let F be a torsion sheaf. Choose a geometric point $\bar{t} : \operatorname{Spec} \Omega \to T$ and put $\bar{s} = g\bar{t}$. Then the strict localizations fit into a natural map $T_{(\bar{t})} \to S_{(\bar{s})}$ of strictly henselian local schemes. Write k_s, k_t for their residue fields; these are separably closed, and the map induces an inclusion $k_s \hookrightarrow k_t$. For each $q \geq 0$, using (4.12) and Proposition 4.25, we obtain the commutative diagram

$$\begin{array}{ccc} (g^*R^q\pi_*F)_{\bar t} & \longrightarrow & (R^q\pi'_*g'^*F)_{\bar t} \\ \sim\downarrow & & \downarrow\sim \\ H^q(\mathbf{P}^1_{S_{(\bar s)}}, F|) & \longrightarrow & H^q(\mathbf{P}^1_{T_{(\bar t)}}, F|) \\ \sim\downarrow & & \downarrow\sim \\ H^q(\mathbf{P}^1_{k_s}, F|) & \longrightarrow & H^q(\mathbf{P}^1_{k_t}, F|). \end{array}$$

Here the top arrow is the stalk at $\bar t$ of the map induced by (4.3) on qth cohomology sheaves, while the middle and bottom arrows are the natural pullback maps. The top square commutes by the definition of the base change map: a section over $\mathbf{P}^1_V$, for a pointed étale neighbourhood $V \to S$, is pulled back to $\mathbf{P}^1_{V\times_S T}$, and passing to the limits over the two strict localizations gives exactly the middle arrow. The bottom square commutes by naturality of restriction to the closed fibres.

The vertical arrows are isomorphisms: the upper ones by (4.12), applied to π and to π'; here $\mathbf{P}^1_{S_{(\bar s)}}$ and $\mathbf{P}^1_{T_{(\bar t)}}$ are the base changes of $\mathbf{P}^1_S$ and $\mathbf{P}^1_T$ to the strict localizations. The lower ones are isomorphisms by Proposition 4.25, applied to the strictly henselian local rings of $S_{(\bar s)}$ and $T_{(\bar t)}$. Hence the top arrow is an isomorphism as soon as the bottom arrow is, and the bottom arrow is an isomorphism by Lemma 4.21, applied to $C = \mathbf{P}^1_{k_s}$ and the extension $k_s \subseteq k_t$. Thus every geometric stalk of $g^*R^q\pi_*F \to R^q\pi'_*g'^*F$ is an isomorphism. As geometric stalks are conservative, the base change map for F is an isomorphism in $D^+(T_{\acute et}, \mathbf{Z})$. Moreover, cohomology and higher direct images of sheaves of Λ-modules agree with those of their underlying abelian sheaves (Lemma 4.2). Hence the base change map for F is also an isomorphism in $D^+(T_{\acute et}, \Lambda)$.

Now let $K \in D^+(\mathbf{P}^1_{S,\acute et}, \Lambda)$ have torsion cohomology and fix $m \in \mathbf{Z}$. The two functors $g^*R\pi_*$, $R\pi'_*g'^* : D^+(\mathbf{P}^1_{S,\acute et}, \Lambda) \to D^+(T_{\acute et}, \Lambda)$ send $D^{\geq m+1}$ into $D^{\geq m+1}$, since inverse image is exact and derived direct image is left t-exact. Using the truncation triangle for $\tau_{\leq m}K$, we can therefore reduce the assertion in degree m to the bounded complex $\tau_{\leq m}K$. By induction on the length, using the truncation triangles and naturality of the base change map, we reduce further to the shifted torsion sheaves $H^a(K)[-a]$. For these the assertion has already been proved.

For $\mathrm{IA}_\Lambda(\pi)$, let I be a torsion injective sheaf of Λ-modules. Then $R^q\pi_*I = 0$ for $q > 0$, and the case of sheaves gives $R^q\pi'_*g'^*I \simeq g^*R^q\pi_*I = 0$. ∎

4.5 Proper morphisms

Proper base change for an arbitrary proper morphism follows from the case of the projective line (Proposition 4.26) by formal steps. Finite morphisms satisfy proper base change directly, and Lemma 4.29 shows that the property is stable under composition and descends along proper surjections. Descent along a proper surjection $u : X \to Y$ works because every injective sheaf on Y is a direct summand of u_*J for an injective sheaf J on X, and the hypotheses on u and on the composite keep u_*J acyclic after base change.

The steps are then as follows. First, composition gives the projection of $(\mathbf{P}^1)^n$. Then descent along the finite surjection $(\mathbf{P}^1)^n \to \mathbf{P}^n$ gives the projection of $\mathbf{P}^n$, and with it every projective morphism. Finally, by Chow's lemma every proper morphism is dominated by a projective one through a proper surjection, so a last descent covers all proper morphisms.

Lemma 4.27 (Finite morphisms and base change)**.** *Let $u : V \to W$ be finite. The formation of u_* on abelian sheaves commutes with arbitrary base change: for every base change $W' \to W$, the underived base change map (4.2) is an isomorphism. Consequently, for every constant coefficient ring Λ, the derived base change map (4.3) is an isomorphism for every $K \in D^+(V_{\acute et}, \Lambda)$.*

Proof. By Lemma D.1, u_* is exact, and its stalks are given by the stalk formula there. In a cartesian square over $W' \to W$, a geometric point of W' and its image in W have the same geometric fibre. Therefore the underived base change map is the identity on the corresponding factors of the stalk formula, hence an isomorphism. Since both direct images in the square are exact, their derived functors and the base change map are computed term by term. Finally, the same argument applies to sheaves of modules. ∎

For instance, if $u : \operatorname{Spec} k' \to \operatorname{Spec} k$ is a separable extension of degree d and $\bar{w}$ is a geometric point, the formula of Lemma D.1 says that $(u_*F)_{\bar{w}}$ is the product of the d stalks of F at the d embeddings of k' into Ω over k.

Example 4.28 (A ramified double cover)**.** Let k be algebraically closed of characteristic $\neq 2$, let $u : \mathbf{A}^1_k \to \mathbf{A}^1_k$ be $x \mapsto x^2$, and let A be a finite ring. Over a point $t \neq 0$ the fibre consists of the two points $\pm\sqrt{t}$, whereas over $t = 0$ it is the single point $x = 0$, with a nonreduced structure that the étale topology does not see. By Lemma D.1, therefore,

$$(u_*A)_{\bar{t}} = A \oplus A \quad (t \neq 0), \qquad (u_*A)_{\bar{0}} = A,$$

and the rank of the stalks jumps at the origin, so u_*A is not lisse. It is, however, constructible for the stratification $\mathbf{G}_m \sqcup \{0\}$. Base change to the origin gives the stalk A, the cohomology of the fibre, as Lemma 4.27 predicts. For the open immersion $j : \mathbf{G}_m \to \mathbf{A}^1$, by contrast, the stalk of j_*A at 0 is A while the fibre is empty.

Stability under composition and descent along proper surjections are best proved on the level of injective acyclicity, since by Theorem 4.17 it is equivalent to proper base change and it lends itself to a spectral sequence argument.

Lemma 4.29 (Composition and cancellation)**.** *Let $X \xrightarrow{u} Y \xrightarrow{v} Z$ be proper morphisms. Then proper base change is stable under composition, and it descends along a proper surjection:*

- *(i) If* $\mathrm{PBC}_\Lambda(u)$ *and* $\mathrm{PBC}_\Lambda(v)$ *hold, then* $\mathrm{PBC}_\Lambda(vu)$ *holds.*
- *(ii) If u is surjective and* $\mathrm{PBC}_\Lambda(u)$ *and* $\mathrm{PBC}_\Lambda(vu)$ *hold, then* $\mathrm{PBC}_\Lambda(v)$ *holds.*

Proof. Theorem 4.17 reduces both parts to the condition IA_Λ of injective acyclicity, and we read off this vanishing from the Leray spectral sequence of the base-changed composite. For (ii) the hypotheses do not apply directly to an injective sheaf on Y, so we first realize it as a direct summand of u_*J with J injective on X.

To set up the spectral sequence, fix an arbitrary base change $Z' \to Z$. Write $g_X : X' \to X$ and $g_Y : Y' \to Y$ for the projections, and $u' : X' \to Y'$, $v' : Y' \to Z'$ for the induced maps. The direct image u'_* preserves injectives, because its left adjoint u'^* on sheaves of Λ-modules is exact. Hence for every sheaf J of Λ-modules on X we have the Grothendieck spectral sequence for the composite direct image [Stacks, 0734]:

$$R^p v'_* R^q u'_* g_X^* J \implies R^{p+q}(v'u')_* g_X^* J. \tag{4.13}$$

Part (i). Let J be an injective sheaf of Λ-modules on X. We show $R^m(v'u')_* g_X^* J = 0$ for $m > 0$, which is $\mathrm{IA}_\Lambda(vu)$. For this it suffices that the E_2-terms of (4.13) vanish except for $p = q = 0$, and they do. Indeed, $R^q u'_* g_X^* J = 0$ for $q > 0$ by $\mathrm{IA}_\Lambda(u)$. For $q = 0$, proper base change in degree zero (Proposition 4.11) gives

$$u'_* g_X^* J \simeq g_Y^* u_* J,$$

and the sheaf u_*J is injective, so $R^p v'_* g_Y^* u_* J = 0$ for $p > 0$ by $\mathrm{IA}_\Lambda(v)$. The abutment therefore vanishes in positive degrees.

Part (ii), the splitting. Let I be an injective sheaf of Λ-modules on Y. Choose a monomorphism of sheaves

$$u^* I \hookrightarrow J$$

with J injective on X, and let $\iota : I \to u_*J$ be its adjoint. The map ι is a monomorphism. To see this, given a geometric point $\bar{y}$, choose a geometric point $\bar{x}$ above it, which exists because u is surjective. Then the composite

$$I_{\bar{y}} \xrightarrow{\iota_{\bar{y}}} (u_*J)_{\bar{y}} \longrightarrow J_{\bar{x}}$$

is the stalk of the chosen monomorphism, under $(u^*I)_{\bar{x}} = I_{\bar{y}}$, so it is injective. As I is an injective sheaf, there is then a map of sheaves $s : u_*J \to I$ with $s\iota = \mathrm{id}_I$. In other words, I is a direct summand of u_*J.

Part (ii), the vanishing. We first show $R^p v'_* g_Y^* u_* J = 0$ for $p > 0$. The hypotheses $\mathrm{IA}_\Lambda(u)$ and $\mathrm{IA}_\Lambda(vu)$ give

$$R^q u'_* g_X^* J = 0 \quad (q > 0), \qquad R^m (v'u')_* g_X^* J = 0 \quad (m > 0),$$

so the spectral sequence (4.13) collapses to its row $q = 0$. This means that $R^p v'_* u'_* g_X^* J \simeq R^p (v'u')_* g_X^* J = 0$ for $p > 0$. Since $u'_* g_X^* J \simeq g_Y^* u_* J$ by proper base change in degree zero, we obtain

$$R^p v'_* g_Y^* u_* J \simeq R^p v'_* u'_* g_X^* J = 0 \qquad (p > 0).$$

Now $g_Y^* I$ is a direct summand of $g_Y^* u_* J$, by applying g_Y^* to the splitting $s\iota = \mathrm{id}_I$, and as $R^p v'_*$ is additive, $R^p v'_* g_Y^* I$ is a direct summand of $R^p v'_* g_Y^* u_* J$. Hence $R^p v'_* g_Y^* I = 0$ for $p > 0$, which is $\mathrm{IA}_\Lambda(v)$. ∎

Part (i) already gives proper base change for $(\mathbf{P}^1_S)^n \to S$, a composite of projections of projective lines. To reach $\mathbf{P}^n_S$ we use part (ii), and for this we need a finite surjection from $(\mathbf{P}^1_S)^n$ onto $\mathbf{P}^n_S$ over S.

Lemma 4.30 (A finite cover of projective space)**.** *For every scheme S and every $n \geq 1$ there is a finite surjective S-morphism*

$$\sigma_n : (\mathbf{P}^1_S)^n \longrightarrow \mathbf{P}^n_S.$$

Proof. Recall that $\mathbf{P}^n$ parametrizes binary forms of degree n up to scalars, and the map σ_n multiplies n linear forms; over a field it is the quotient of $(\mathbf{P}^1)^n$ by the symmetric group. Explicitly, we use the coefficients of a product of binary linear forms:

$$\prod_{i=1}^{n} (x_i T + y_i U) = \sum_{j=0}^{n} F_j T^{n-j} U^j, \qquad \sigma_n\big(([x_i : y_i])_i\big) = [F_0 : \cdots : F_n].$$

Every F_j is homogeneous of degree one in each pair (x_i, y_i), and the corresponding sections of $\mathcal{O}(1, \dots, 1)$ have no common zero. Indeed, at a geometric point every linear factor is nonzero, and hence so is their product. Therefore the sections define the stated morphism over any S, and the construction commutes with base change.

For example, on the chart $F_0 \neq 0$ of the target, whose preimage is the chart where every x_i is invertible, the coordinates $c_j = F_j/F_0$ pull back to the elementary symmetric polynomials in the $w_i = y_i/x_i$, as in the classical description of $\mathbf{A}^n$ as the quotient of $\mathbf{A}^n$ by the symmetric group.

Finiteness of σ_n follows from properness and finiteness of the fibres. Since the source is proper over S and the target is separated over S, the morphism σ_n is proper. For the fibres, recall that over an algebraically closed field, a nonzero binary form of degree n is a product of n linear forms $x_i T + y_i U$, and these are unique up to order and up to multiplying each by a nonzero constant. Here a factor with $x_i = 0$ is a multiple of U and corresponds to the point $[0 : 1]$. In other words, the form determines the multiset of points $[x_i : y_i] \in \mathbf{P}^1$. Hence every geometric fibre of σ_n is nonempty, and σ_n is surjective. Moreover, each geometric fibre has at most $n!$ geometric points, namely the orderings of this multiset. Since a scheme of finite type over a field with finitely many geometric points is zero dimensional, σ_n is quasi-finite, and a proper quasi-finite morphism is finite [Stacks, 02LS]. ∎

For $n = 1$ the map σ_1 is the identity of $\mathbf{P}^1_S$. For $n = 2$ it sends $([x_1 : y_1], [x_2 : y_2])$ to $[x_1x_2 : x_1y_2 + y_1x_2 : y_1y_2]$, and over a field it is the quotient by swapping the two factors, of degree 2.

Lemma 4.31 (Projective space)**.** *If* $\mathrm{PBC}_\Lambda(\mathbf{P}^1_S \to S)$ *holds for every scheme* S*, then* $\mathrm{PBC}_\Lambda(\mathbf{P}^n_S \to S)$ *holds for every scheme* S *and every* $n \geq 0$*.*

Proof. The case $n = 0$ is the identity. For $n \geq 1$, the projection $(\mathbf{P}^1_S)^n \to S$ is a composite of n projections of projective lines over successive bases. Hence it satisfies PBC_Λ by the hypothesis and Lemma lemma-composition-cancellation. The finite surjection σ_n of Lemma 4.30 is a morphism over S, and it satisfies PBC_Λ by Lemma 4.27. Cancelling σ_n by Lemma lemma-composition-cancellation, we obtain PBC_Λ for $\mathbf{P}^n_S \to S$. ∎

We can now treat projective morphisms, since they factor as a closed immersion followed by the projection of a projective space. By Chow's lemma an arbitrary proper morphism is dominated by a projective one through a proper surjection, and the cancellation of Lemma lemma-composition-cancellation then passes proper base change to the proper morphism.

Proposition 4.32 (Étale proper base change)**.** *If* $\mathrm{PBC}_\Lambda(\mathbf{P}^1_S \to S)$ *holds for every scheme* S*, then* $\mathrm{PBC}_\Lambda(f)$ *holds for every proper morphism of schemes* f*.*

Proof. Let $f : X \to Y$ be proper. We first reduce to the case that Y is affine, as Chow's lemma requires. Let $Y = \bigcup U$ be an affine open cover, and assume PBC_Λ for each restriction $f^{-1}(U) \to U$. Given a base change $Y' \to Y$, the inverse images U' of the sets U then cover Y'. Since higher direct images commute with restriction to open subschemes, the restriction of the base change map to U' is the base change map of $f^{-1}(U) \to U$ along $U' \to U$, hence an isomorphism. As isomorphisms can be checked on geometric stalks, the base change map over Y' is an isomorphism.

Now Y is affine, hence quasi-compact and quasi-separated, so Theorem A.25 with $S = Y$ gives a proper surjection $\pi : \widetilde{X} \to X$ and closed immersions $i_Y : \widetilde{X} \hookrightarrow \mathbf{P}^n_Y$ and $i_X = (\pi, i_Y) : \widetilde{X} \hookrightarrow \mathbf{P}^n_X$ with $f\pi$ equal to i_Y followed by the projection. Here we need Chow's lemma over a base that is affine but need not be Noetherian; we prove this form in Appendix A. The two maps π and

$f\pi$ factor as closed immersions i_X and i_Y followed by projections of projective spaces. Closed immersions are finite, so they satisfy PBC_Λ by Lemma 4.27, and projections of projective spaces satisfy PBC_Λ by the hypothesis and Lemma 4.31. By stability under composition (Lemma lemma-composition-cancellation), both π and $f\pi$ therefore satisfy PBC_Λ. As π is surjective, we may cancel it by Lemma lemma-composition-cancellation, and this gives PBC_Λ for f. ∎

4.6 Pro-étale complexes

The six operations live on the pro-étale site, where we also meet torsion sheaves of modules over rings that are not torsion, such as $\mathcal{O}_E$. So far $\mathrm{PBC}_\Lambda(f)$ is known on the étale site only when Λ is torsion (Propositions 4.26 and Proposition 4.32). We therefore drop the condition on Λ, keeping torsion cohomology sheaves, and pass to classical pro-étale complexes by the comparison theorems of Chapter 2.

The reduction to torsion rings is possible because a torsion sheaf of Λ-modules is a filtered colimit of sheaves killed by an integer n. Such a sheaf has a finite filtration with quotients killed by primes p, and these quotients are modules over $\Lambda/p\Lambda$. Moreover, the cohomology of a sheaf of modules is that of its underlying abelian sheaf, so the base change map does not depend on which ring we regard as the coefficients.

For the passage to the pro-étale site we use two facts: a classical complex in D^+ is ν_X^* of an étale one (Corollary 2.37), and ν^* commutes with g^* and with direct images along proper maps (Theorem 2.38). Taking for g a geometric point then shows that the stalks of Rf_*K are the cohomology of the geometric fibres (Corollary 4.34).

Theorem 4.33 (Pro-étale proper base change)**.** *Let Λ be a commutative ring, let $f : X \to Y$ be a proper morphism of schemes, and let $g : Y' \to Y$ be arbitrary. Set $X' = X \times_Y Y'$, with projections $f' : X' \to Y'$ and $g' : X' \to X$. Let $K \in D^+_{\mathrm{class}}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ have torsion cohomology sheaves (as abelian sheaves). Then the base change map (4.5) of Definition 4.6*

$$g^* Rf_* K \longrightarrow Rf'_* g'^* K$$

is an isomorphism in $D^+_{\mathrm{class}}(Y'_{\mathrm{pro\acute{e}t}}, \Lambda)$.

Suppose Λ is killed by a positive integer, for instance that Λ is a finite ring. Then every complex of Λ_X-modules has torsion cohomology sheaves, and the theorem applies to every $K \in D^+_{\mathrm{class}}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$.

Proof of Theorem 4.33. We work on the étale site, where Propositions 4.26 and Proposition 4.32 apply to the torsion rings $\Lambda/p\Lambda$. By dévissage we reach all torsion sheaves of Λ-modules, and the comparison theorems then bring us back to the pro-étale site.

By Corollary 2.37, we may write $K = \nu_X^* K_{\acute{e}t}$ with $K_{\acute{e}t} \in D^+(X_{\acute{e}t}, \Lambda)$, and the cohomology sheaves of $K_{\acute{e}t}$ are torsion. Indeed, ν_X^* is exact, commutes with colimits and is fully faithful on sheaves, and therefore detects the condition $F = \varinjlim_n F[n]$ of Definition 4.13. We must show that the base change map for $K_{\acute{e}t}$ is an isomorphism in $D^+(Y'_{\acute{e}t}, \Lambda)$, and then pass to the pro-étale site.

We reduce to the rings $\Lambda/p\Lambda$, p prime. These are killed by p, so $\mathrm{PBC}_{\Lambda/p\Lambda}(f)$ holds by Propositions 4.26 and Proposition 4.32. Now a sheaf F of Λ-modules with $pF = 0$ is a sheaf of $\Lambda/p\Lambda$-modules, and restriction of scalars along $\Lambda \to \Lambda/p\Lambda$ is exact and conservative. Moreover, it commutes with inverse images, derived direct images and base change maps, since both derived direct images are computed on underlying abelian sheaves (Lemma 4.2). Hence the base change map for F is an isomorphism.

If $nF = 0$, we argue by induction on the number of prime factors of n. For a prime $p \mid n$, the outer terms of $0 \to F[p] \to F \xrightarrow{p} pF \to 0$ are killed by p and by n/p, so the five lemma applies to the long exact cohomology sequences. Finally, a torsion sheaf is the filtered colimit $\varinjlim_n F[n]$. Since (4.7) applies to the qcqs morphisms f and f', the base change map is an isomorphism for every torsion sheaf of Λ-modules. The truncation argument at the end of the proof of Theorem 4.17 then passes from sheaves to bounded below complexes with torsion cohomology, in particular to $K_{ét}$.

To return to the pro-étale site we apply $\nu_{Y'}^*$. Since f and f' are proper, hence qcqs, the comparison of direct images in Theorem 2.38 applies. Using it together with the comparison of inverse images $g_{\mathrm{pro}ét}^* \nu_Y^* \simeq \nu_{Y'}^* g_{ét}^*$, we identify $\nu_{Y'}^* g^* Rf_* K_{ét}$ and $\nu_{Y'}^* Rf'_* g'^* K_{ét}$ with $g^* Rf_* K$ and $Rf'_* g'^* K$ in $D^+(Y'_{\mathrm{pro}ét}, \Lambda)$. By the last assertion of Lemma 4.7, under these identifications the base change map of K is the inverse image on $Y'_{\mathrm{pro}ét}$ of the étale base change map of $K_{ét}$. We have shown that the latter is an isomorphism. ∎

Taking for g a geometric point, we answer the question at the start of the chapter. The only additional point is to make sense of a stalk on the pro-étale site, where the stalk is the value of $\bar{y}^* Rf_* K$ on the point $\operatorname{Spec}\Omega$, whose topos is that of condensed sets. We check that for a classical complex this value is the étale stalk.

Corollary 4.34 (Cohomology of geometric fibres)**.** *Under the hypotheses of Theorem 4.33, for every geometric point $\bar{y} : \operatorname{Spec}(\Omega) \to Y$ there is a canonical isomorphism*

$$(Rf_* K)_{\bar{y}} \simeq R\Gamma\big(X_{\bar{y},\mathrm{pro}ét}, K|_{X_{\bar{y}}}\big).$$

For a classical torsion sheaf F of Λ-modules this says

$$(R^q f_* F)_{\bar{y}} \simeq H^q_{\mathrm{pro}ét}(X_{\bar{y}}, F|_{X_{\bar{y}}}) \qquad (q \geq 0).$$

Proof. We apply the theorem to $g = \bar{y}$ and take global sections over the point. The right side becomes the cohomology of the fibre, and we check that the left side is the étale stalk.

Take $Y' = \operatorname{Spec}(\Omega)$ in the theorem and apply the functor of global sections on $Y'_{\mathrm{pro}ét}$. By Proposition 2.50 the topos of $Y'_{\mathrm{pro}ét}$ is that of condensed sets, where global sections is evaluation at the point. This functor is exact, because every surjection of profinite sets onto a point has a section. On the right we therefore get the cohomology of the fibre. On the left, $(Rf_* K)_{\bar{y}}$ means $\Gamma(Y'_{\mathrm{pro}ét}, \bar{y}^* Rf_* K)$, and by the proof of the theorem we have $Rf_* K \simeq \nu_Y^* Rf_* K_{ét}$. Since $\bar{y}^* \nu_Y^* \simeq \nu_{Y'}^* \bar{y}^*$, the complex $\bar{y}^* Rf_* K$ is the constant condensed complex on the étale stalk $(Rf_* K_{ét})_{\bar{y}}$, so its global sections are that stalk. Finally, by the theorem on the étale site, this stalk is the étale cohomology of the fibre. ∎

4.7 Finite coefficients

Let A be a finite admissible coefficient ring and $p : X \to Y$ a proper morphism of separated schemes of finite type over the standing field k. For $C \in D^b_{\mathrm{ctf}}(X, A)$, proper base change identifies the geometric stalks of $Rp_* C$ with the cohomology of the fibres. From this identification and the classical finiteness theorem we obtain that $Rp_* C$ is constructible, with a bound on its Tor amplitude. Both facts are needed for the passage to integral and rational coefficients below, whereas direct sums in the unbounded localizing categories are treated in Section 5.3.

Proposition 4.35 (Proper direct image for finite coefficients)**.** *Let A be a finite admissible ring, and let $p : X \to Y$ be a proper morphism of separated schemes of finite type over k. Then Rp_**

maps $D^b_{\mathrm{ctf}}(X,A)$ into $D^b_{\mathrm{ctf}}(Y,A)$. More precisely, if $C \in D^b_{\mathrm{ctf}}(X,A)$ has Tor amplitude in $[a,b]$ and $\mathrm{cd}_\ell(X_{\bar y}) \le d$ for all geometric points $\bar y$ of Y, then Rp_*C has Tor amplitude in $[a, b+d]$.

Proof. By Corollary 2.44 the étale and pro-étale categories D^b_{ctf} agree, and by Theorem 2.38 so do the proper direct images of their bounded below complexes. We can therefore transport bounded constructibility and the Tor bound from the étale site.

On the étale site, let $C \in D^b_{\mathrm{ctf}}(X_{ét}, A)$. The proper direct image of a constructible torsion sheaf is constructible [Stacks, 0GL0], and $\mathrm{cd}_\ell(p) < \infty$ [Stacks, 0F10]. Hence $Rp_*C \in D^b_c(Y_{ét}, A)$.

To check finite Tor amplitude, suppose that C has Tor amplitude $[a,b]$, and let d be an integer with $\mathrm{cd}_\ell(X_{\bar y}) \le d$ for all geometric points $\bar y$ of Y. Since derived tensor products are computed stalkwise, it suffices to show that each stalk $(Rp_*C)_{\bar y}$ has Tor amplitude in $[a, b+d]$ as a complex of A-modules. In other words, $(Rp_*C)_{\bar y} \otimes^{\mathbf{L}}_A N = ((Rp_*C) \otimes^{\mathbf{L}}_A N_Y)_{\bar y}$ should have cohomology in $[a, b+d]$ for every A-module N. By proper base change and the projection formula for constant modules [Stacks, 0F12], this stalk is

$$R\Gamma\big(X_{\bar y}, C|_{X_{\bar y}} \otimes^{\mathbf{L}}_A N_{X_{\bar y}}\big).$$

The complex $C|_{X_{\bar y}} \otimes^{\mathbf{L}}_A N_{X_{\bar y}}$ has cohomology in $[a,b]$, because restriction to the fibre preserves Tor amplitude. Since $R\Gamma(X_{\bar y}, -)$ preserves $D^{\ge a}$ and has cohomological dimension at most d, it keeps the lower bound and raises the upper bound by at most d. Thus $Rp_*C \in D^b_{\mathrm{ctf}}(Y_{ét}, A)$, with Tor amplitude in $[a, b+d]$. ∎

4.8 ℓ-adic coefficients

Pro-étale proper base change (Theorem 4.33) concerns classical complexes with torsion cohomology sheaves. For a finite ring A it therefore applies, with $\Lambda = A$, to every $K \in D^b_{\mathrm{ctf}}(X,A)$ for schemes of finite type over k, because bounded constructible complexes on quasi-compact quasi-separated schemes are classical (Theorem 2.43). For $\mathcal{O}_E$, however, the theorem gives proper base change only at each finite level $\mathcal{O}_E/\pi_E^n$, and we must pass to the limit. Later statements of this kind make this passage once and for all, by the reduction principle of Section 7.3; here we make it directly.

The difficulty is completeness: the inverse image $g^* : D(Y_{\mathrm{pro}ét}, \mathcal{O}_E) \to D(Y'_{\mathrm{pro}ét}, \mathcal{O}_E)$, which is exact on sheaves of modules, need not preserve derived completeness. We therefore complete after pulling back.

Definition 4.36 (Completed inverse image)**.** Let $g : Y' \to Y$ be a morphism. For $M \in D(Y_{\mathrm{pro}ét}, \mathcal{O}_E)$, the *completed inverse image* is

$$\hat g^* M = R\varprojlim_n g^*(M \otimes^{\mathbf{L}}_{\mathcal{O}_{E,Y}} (\mathcal{O}_E/\pi_E^n)_Y) \in D(Y'_{\mathrm{pro}ét}, \mathcal{O}_E).$$

The natural map $g^*M \to \hat g^* M$ is the completion map of g^*M, since g^* commutes with the reductions. For constructible M the completion does nothing. Indeed, let Y and Y' be of finite type over k and $M \in D^b_{\mathrm{ctf}}(Y, \mathcal{O}_E)$. Then g^*M lies in $D^b_c(Y', \mathcal{O}_E)$ by Lemma lemma-integral-models, hence it is derived complete by Proposition 3.11 ((1)⇒(2)), and $\hat g^* M = g^* M$. However, until we know that proper direct image preserves constructibility, we can only state base change with the completed inverse image.

In the proof we use the following map, which also appears in the projection formula.

Definition 4.37 (Projection formula map)**.** Let $p : X \to Y$ be a morphism, $K \in D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ and $M \in D(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$. The *projection formula map*

$$Rp_*K \otimes^{\mathbf{L}} M \longrightarrow Rp_*(K \otimes^{\mathbf{L}} p^*M) \tag{4.14}$$

is the adjoint, under $p^* \dashv Rp_*$, of the composite $p^*(Rp_*K \otimes^{\mathbf{L}} M) \simeq p^*Rp_*K \otimes^{\mathbf{L}} p^*M \to K \otimes^{\mathbf{L}} p^*M$ of the monoidal isomorphism of p^* and the counit $p^*Rp_*K \to K$ tensored with p^*M.

Proposition 4.38 (Proper base change for complete complexes)**.** *Let $p : X \to Y$ be proper, let $g : Y' \to Y$ be any morphism, and let $p' : X' \to Y'$ and $g' : X' \to X$ be the projections from $X' = X \times_Y Y'$. Fix a finite extension $E/\mathbf{Q}_\ell$. Suppose that $K \in D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$ is derived π_E-complete and that each reduction*

$$K_n = K \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} (\mathcal{O}_E/\pi_E^n)_X$$

belongs to $D^+_{\mathrm{class}}(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n)$. Then Rp_*K is derived π_E-complete, and there are canonical isomorphisms

$$\hat{g}^*Rp_*K \simeq R\varprojlim_n g^*Rp_*K_n, \qquad Rp'_*\hat{g}'^*K \simeq R\varprojlim_n Rp'_*g'^*K_n,$$

and under them the derived limit of the base change maps $g^*Rp_*K_n \to Rp'_*g'^*K_n$ is an isomorphism $\hat{g}^*Rp_*K \xrightarrow{\sim} Rp'_*\hat{g}'^*K$.

Proof. Theorem 4.33 applies at each finite level, so the point is that Rp_* and the completed inverse images are compatible with the passage to the derived limit of the reductions.

The reductions of Rp_*K are controlled by the projection formula. The projection formula map (4.14) for $M = (\mathcal{O}_E/\pi_E^n)_Y$ is a map

$$(Rp_*K) \otimes^{\mathbf{L}} (\mathcal{O}_E/\pi_E^n)_Y \longrightarrow Rp_*K_n,$$

which we regard in $D(Y_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n)$. We claim that it is an isomorphism. Since restriction of scalars to $\mathcal{O}_E$ is conservative, we may check this after forgetting the $\mathcal{O}_E/\pi_E^n$-structure. Now the ring $\mathcal{O}_E/\pi_E^n$ has the free resolution $[\mathcal{O}_E \xrightarrow{\pi_E^n} \mathcal{O}_E]$, so by Corollary 2.52 the source is the cone of π_E^n on Rp_*K, and the target is Rp_* of the cone of π_E^n on K. These agree because Rp_* is triangulated and $\mathcal{O}_E$-linear. Moreover, the resolutions for $n+1$ and n are related by the chain map (π_E, id), and applying Rp_* to it shows that these comparisons form an isomorphism of towers. Since all these constructions can be made on complexes with functorial resolutions, the comparisons define a map in the derived category of towers of Definition 2.61.

Since K is derived complete, it is the derived limit of its reductions K_n, and since the functor Rp_* is a right adjoint, it commutes with derived limits, and we obtain

$$Rp_*K \simeq Rp_*R\varprojlim_n K_n \simeq R\varprojlim_n Rp_*K_n.$$

By the reduction comparisons, the right side is the derived limit of the reductions $(Rp_*K) \otimes^{\mathbf{L}} (\mathcal{O}_E/\pi_E^n)_Y$, that is, the derived completion of Rp_*K. Hence the completion map of Rp_*K is an isomorphism, and Rp_*K is derived complete. The definition of completed inverse image and the same right-adjoint property for Rp'_* now give

$$\hat{g}^* Rp_* K \simeq R\varprojlim_n g^* Rp_* K_n,$$
$$Rp'_* \hat{g}'^* K \simeq R\varprojlim_n Rp'_* g'^* K_n.$$

The base change maps $g^* Rp_* K_n \to Rp'_* g'^* K_n$ are natural in K_n, so they form a map of towers, and each is an isomorphism by Theorem 4.33. Its cone therefore has zero cohomology at every level, and applying $R\varprojlim$ gives the asserted isomorphism. ∎

When the pullbacks are already complete, the completed base change isomorphism is the ordinary one.

Lemma 4.39 (Completed and ordinary base change)**.** *In the setting of Proposition 4.38, the square formed by the base change map $g^* Rp_* K \to Rp'_* g'^* K$ of (4.5), the completed base change isomorphism $\hat{g}^* Rp_* K \to Rp'_* \hat{g}'^* K$, and the completion maps $g^* Rp_* K \to \hat{g}^* Rp_* K$ and $Rp'_* g'^* K \to Rp'_* \hat{g}'^* K$ commutes. In particular, if $g^* Rp_* K$ and $g'^* K$ are derived complete, then the base change map $g^* Rp_* K \to Rp'_* g'^* K$ is an isomorphism.*

Proof. Both maps are built from the counit of $p^* \dashv Rp_*$, as in Definition 4.6, and this counit is natural with respect to the quotient maps and the chain maps (π_E, id) relating the reductions. Hence the square commutes.

Now suppose that $g^* Rp_* K$ and $g'^* K$ are derived complete. Then the first completion map is an isomorphism, and the second is Rp'_* applied to the completion map of $g'^* K$, so it is an isomorphism as well. By the commutative square, the base change map is then a composite of isomorphisms. ∎

We will show that $Rp_* K$ is again constructible when K is constructible. Once this is known, the completed inverse images in Proposition 4.38 agree with g^* and g'^*, and the proposition becomes the ordinary base change isomorphism

$$g^* Rp_* K \xrightarrow{\sim} Rp'_* g'^* K \qquad (K \in D_c^b(X, \mathcal{O}_E)).$$

4.9 Proper direct image

Let $p : X \to Y$ be proper. We need to know, for every admissible coefficient ring, that Rp_* preserves D^b_{ctf} and satisfies proper base change and the projection formula there, because the construction of $Rf_! = Rp_* \circ j_!$ in Section 5.4 starts from these facts. For $Y = \operatorname{Spec} k$ with k algebraically closed, they say that the cohomology groups $H^i(X, \overline{\mathbf{Q}}_\ell)$ of a proper X are finite-dimensional and vanish outside a bounded range of degrees.

For finite coefficients, constructibility and the Tor bound are Proposition 4.35, and the other admissible rings reduce to this case.

For $\mathcal{O}_E$, the complex $Rp_* K$ is derived complete, and its reduction modulo π_E is Rp_* of the reduction of K, so the integral constructibility criterion (Proposition 3.11) and derived Nakayama (Lemma 2.62) reduce everything to torsion coefficients. For E and $\overline{\mathbf{Q}}_\ell$ it suffices to treat the generators $(a_! L_0)[1/\ell]$. Indeed, on bounded below complexes Rp_* commutes with inverting ℓ and with extension of scalars to finite extensions and to $\overline{\mathbf{Q}}_\ell$, since both are filtered colimits (Corollary 2.29).

Proposition 4.40 (Proper direct image of constructible complexes)**.** *Let $p : X \to Y$ be proper and let Λ be admissible.*

1. *$Rp_* : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$ maps $D^b_{\mathrm{ctf}}(X, \Lambda)$ into $D^b_{\mathrm{ctf}}(Y, \Lambda)$.*
2. *For a cartesian square with $g : Y' \to Y$, $g' : X' \to X$ and $p' : X' \to Y'$, the base change map (4.5) $g^* Rp_* \to Rp'_* g'^*$ is an isomorphism of functors $D^b_{\mathrm{ctf}}(X, \Lambda) \to D^b_{\mathrm{ctf}}(Y', \Lambda)$.*
3. *The projection formula map (4.14) is an isomorphism for $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ and $M \in D^b_{\mathrm{ctf}}(Y, \Lambda)$.*

Proof. Finite coefficients. Part (1), including the Tor bound, is Proposition 4.35, and proper base change holds by Theorem 4.33 with $\Lambda = A$. The projection formula, finally, is the classical torsion formula [Stacks, 0F0G] in $D(Y_{\acute{e}t}, A)$, transported to $D(Y_{\mathrm{pro\acute{e}t}}, A)$ by pullback from the étale site.

Integral coefficients. Let $K \in D^b_c(X, \mathcal{O}_E)$. We first show that $Rp_* K \in D^b_c(Y, \mathcal{O}_E)$, by computing its reduction as in (4.15) below. By Proposition 3.11 and Lemma 3.5, K is derived complete with bounded classical finite reductions. Hence Proposition 4.38 applies. It shows that $Rp_* K$ is derived complete and gives a canonical comparison of reductions

$$\rho_K : (Rp_* K) \otimes^{\mathbf{L}} \mathcal{O}_{E,Y}/\pi_E \xrightarrow{\sim} Rp_*(K \otimes^{\mathbf{L}} \mathcal{O}_{E,X}/\pi_E). \tag{4.15}$$

The target of ρ_K is constructible and bounded. Indeed, it is the direct image of a torsion complex pulled back from $X_{\acute{e}t}$ [Stacks, 09B1], so it is the pullback to $Y_{\mathrm{pro\acute{e}t}}$ of the direct image in $D(Y_{\acute{e}t}, \mathbf{F}_E)$ [Stacks, 09A6]. The latter is constructible by the finiteness theorem for proper direct image with torsion coefficients [Stacks, 0GL0], and bounded because p has finite cohomological dimension [Stacks, 0F10]. Now Proposition 3.11 gives $Rp_* K \in D^b_c(Y, \mathcal{O}_E)$.

Proper base change holds by Lemma 4.39, which applies because $g^* Rp_* K$ and $g'^* K$ are derived complete, as inverse image preserves integral constructibility (Lemma lemma-integral-models).

The projection formula is tested modulo π_E in the same way. Let C be the cone of the projection formula map for $K \in D^b_c(X, \mathcal{O}_E)$ and $M \in D^b_c(Y, \mathcal{O}_E)$. Its source $Rp_* K \otimes^{\mathbf{L}} M$ and target $Rp_*(K \otimes^{\mathbf{L}} p^* M)$ are integrally constructible, by Lemma lemma-integral-models and the fact just proved that Rp_* maps $D^b_c(X, \mathcal{O}_E)$ into $D^b_c(Y, \mathcal{O}_E)$. As objects of D^b_c they are derived complete, and hence so is C. Derived Nakayama (Lemma 2.62) therefore reduces us to showing $C/\pi_E = 0$. In other words, we must show that the reduction of the projection formula map is an isomorphism.

This reduction is the torsion projection formula map, which is an isomorphism by [Stacks, 0F0G]. To identify the two maps, note that the sources and targets correspond by ρ_K and associativity of extension of scalars, and the maps agree because both are adjoint to the counit $p^* Rp_* K \to K$, as is ρ_K.

Rational coefficients. Here Rp_* commutes with the operations that produce the generators of Lemma lemma-integral-models from integral complexes. For $K \in D^b_c(X, \mathcal{O}_E)$, the complex $K[1/\ell]$ is the colimit of $K \xrightarrow{\ell} K \xrightarrow{\ell} \cdots$, and since this system has a common lower bound, Corollary 2.29 shows that

$$Rp_*(K[1/\ell]) \simeq (Rp_* K)[1/\ell].$$

Likewise Rp_* commutes with extension of scalars to a finite E'/E, which is a finite direct sum by Lemma lemma-inverse-image-coefficient-sheaves. By Corollary 2.29 again, it also commutes with the filtered colimit $\overline{\mathbf{Q}}_{\ell,X} \otimes_{E_X} - = \varinjlim_{E'} E'_X \otimes_{E_X} -$. By the integral case, it follows that Rp_* maps the generators into $D^b_c(Y, E)$ and $D^b_c(Y, \overline{\mathbf{Q}}_\ell)$. Since Rp_* is exact and $D^b_c(Y, E)$ and $D^b_c(Y, \overline{\mathbf{Q}}_\ell)$ are triangulated (Corollary 3.20), Rp_* preserves $D^b_c(-, E)$ and $D^b_c(-, \overline{\mathbf{Q}}_\ell)$.

For proper base change and the projection formula it is enough to treat the generators. Indeed, both sides of each map are exact functors in each variable, so the objects for which the map is

an isomorphism form a triangulated subcategory. On the generators, the maps are obtained from the integral isomorphisms by inverting ℓ and extending scalars. These operations commute with Rp_* by the preceding paragraph, and with g^* and $\otimes^{\mathbf{L}}$ because they are given by tensoring with the coefficient sheaves. Moreover, the maps so obtained are the base change and projection formula maps, since these scalar comparisons commute with the counits of $p^* \dashv Rp_*$. ■

5 Construction of $f^!$

Verdier duality, Poincaré duality and the trace formula all rest on a right adjoint $f^!$ of $Rf_!$. Indeed, Verdier duality (Chapter 10) is duality with respect to the dualizing complex $a_X^!\Lambda$, where $a_X : X \to \operatorname{Spec} k$ is the structure map, and Poincaré duality follows from it once smooth purity (Theorem 9.3) computes $f^!$ for smooth morphisms. Similarly, the local terms of the trace formula (Chapter 12) are built from $f^!$ for the two projections of a correspondence.

The functor $Rf_!$ itself is easy to write down. A separated morphism of finite type $f : X \to Y$ has a compactification $f = p \circ j$, with $j : X \hookrightarrow \overline{X}$ an open immersion and $p : \overline{X} \to Y$ proper (Theorem A.2), and we put $Rf_! = Rp_* \circ j_!$. This definition stays within constructible complexes because, for every admissible Λ, the functor Rp_* maps $D^b_{\mathrm{ctf}}(\overline{X}, \Lambda)$ into $D^b_{\mathrm{ctf}}(Y, \Lambda)$. The resulting functor satisfies proper base change and the projection formula (Proposition 4.40).

The right adjoint of $Rf_!$ is another matter. The factor $j_!$ has the right adjoint j^*, but for Rp_* with p proper no formula for a right adjoint is available. Even where a formula exists, it is not computed pointwise: for a closed immersion i the functor $i^!$ takes sections with support in the closed subscheme, so the value of $i^!K$ at a point depends on K near the point, not only on its stalk there.

Grothendieck and Hartshorne built the coherent $f^!$ from local formulas and then had to glue them. Neeman noticed, instead, that $D_{\mathrm{qc}}(X)$ is generated by compact perfect complexes and that Rf_* preserves direct sums, so that Brown representability produces $f^!$ with no formula. We follow his approach, with D^b_{ctf} in place of the perfect complexes.

The theorem we use is the following. Let $\mathcal{T}$ be a triangulated category with arbitrary direct sums, and suppose that $\mathcal{T}$ is compactly generated: there is a set $\mathcal{G}$ of compact objects G (that is, $\operatorname{Hom}(G, -)$ commutes with direct sums) such that $K = 0$ whenever $\operatorname{Hom}(G[n], K) = 0$ for all $G \in \mathcal{G}$ and $n \in \mathbf{Z}$. Then every exact functor out of $\mathcal{T}$ that preserves direct sums has an exact right adjoint [Stacks, 0A8F, 0A8G]. The category $D^b_{\mathrm{ctf}}(X, \Lambda)$ is not closed under infinite direct sums. We therefore apply the theorem to the localizing category $\operatorname{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ that it generates (Section 5.1). The result is Theorem 5.31: for every admissible Λ and every separated morphism of finite type $f : X \to Y$ there is an exact functor

$$f^! : \operatorname{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda)) \to \operatorname{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda)), \qquad \operatorname{Hom}(Rf_!M, N) \simeq \operatorname{Hom}(M, f^!N).$$

For finite coefficients nothing is lost by passing to the localizing category, since $\operatorname{Loc}(D^b_{\mathrm{ctf}}(X, A))$ is equivalent to the classical derived category $D(X_{\acute{e}t}, A)$ (Proposition 5.15).

Everything therefore rests on the two hypotheses of the theorem: that the objects of $D^b_{\mathrm{ctf}}(X, \Lambda)$ are compact in $\operatorname{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$, and that Rp_* preserves direct sums for proper p. Neither is formal, because the summands of a direct sum may live in degrees going off to $-\infty$. If the Ext groups out of a sheaf do not vanish in high degrees, compactness fails even for the constant sheaf. Here is an example, outside our conventions because $\mathbf{R}$ is neither finite nor algebraically closed. Take $\ell = 2$ and $X = \operatorname{Spec} \mathbf{R}$. Then sheaves on $X_{\acute{e}t}$ are modules over $\operatorname{Gal}(\mathbf{C}/\mathbf{R}) = \mathbf{Z}/2$, and $\operatorname{Ext}^n(\mathbf{F}_2, \mathbf{F}_2) = H^n(\mathbf{Z}/2, \mathbf{F}_2) = \mathbf{F}_2$ for all $n \geq 0$. Since products of modules are exact and the

complexes $\mathbf{F}_2[n]$ live in distinct degrees, the sum $\bigoplus_{n\geq 0}\mathbf{F}_2[n]$ in $D(X_{ét},\mathbf{F}_2)$ is also the product. Hence

$$\mathrm{Hom}\Big(\mathbf{F}_2,\bigoplus_{n\geq 0}\mathbf{F}_2[n]\Big)=\prod_{n\geq 0}\mathbf{F}_2\neq\bigoplus_{n\geq 0}\mathbf{F}_2=\bigoplus_{n\geq 0}\mathrm{Hom}(\mathbf{F}_2,\mathbf{F}_2[n]).$$

By our conventions (Section 1.9) the base field is finite or algebraically closed, so its ℓ-cohomological dimension is finite (Lemma 2.46). For finite coefficients this rules out such examples. In that case both hypotheses reduce to the fact that classical étale cohomology commutes with direct sums of unbounded complexes (Section 5.3). For $\Lambda=E$ or $\overline{\mathbf{Q}}_\ell$ a bound on the base is not enough, and we prove instead a bound on Ext that depends only on the source (Lemma 5.6).

5.1 Existence and finiteness of $f^!$

We construct $f^!$ by Brown representability, as outlined at the beginning of this chapter, without writing it down. Brown representability needs a category with arbitrary direct sums, and $D^b_{\mathrm{ctf}}(X,\Lambda)$ is not closed under them. We therefore enlarge it to the smallest category that has them and show that $D^b_{\mathrm{ctf}}(X,\Lambda)$ compactly generates it.

Definition 5.1 (The localizing subcategory generated by $D^b_{ctf}(X,\Lambda)$)**.** For admissible Λ, let $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ be the smallest full triangulated subcategory of $D(X_{\mathrm{pro}ét},\Lambda)$ containing $D^b_{\mathrm{ctf}}(X,\Lambda)$ and closed under arbitrary direct sums.

For integral and rational coefficients, this is also $\mathrm{Loc}(D^b_c(X,\Lambda))$, by Proposition 3.12.

For $\mathcal{O}_E$ the objects of $D^b_{\mathrm{ctf}}(X,\mathcal{O}_E)$ are derived complete, but the localizing subcategory they generate also contains complexes that are not. For this reason the proof of Theorem theorem-exceptional-inverse-image must show, for integral constructible K, that $f^!K$ is derived complete.

Theorem 5.2 (The exceptional inverse image)**.** *Let $\Lambda\in\{A,\mathbf{Z}_\ell,\mathcal{O}_E,\mathbf{Q}_\ell,E,\overline{\mathbf{Q}}_\ell\}$ be an admissible coefficient ring and $f:X\to Y$ a morphism of separated schemes of finite type over k.*

1. *The objects of $D^b_{\mathrm{ctf}}(X,\Lambda)$ are compact in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ and a small skeleton compactly generates it.*
2. *Compactifications define an exact functor $Rf_!:\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))\to\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$, preserving direct sums and sending $D^b_{\mathrm{ctf}}(X,\Lambda)$ into $D^b_{\mathrm{ctf}}(Y,\Lambda)$.*
3. *The functor $Rf_!$ has an exact right adjoint $f^!:\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))\to\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$, obtained from Brown representability.*
4. *This right adjoint sends $D^b_{\mathrm{ctf}}(Y,\Lambda)$ into $D^b_{\mathrm{ctf}}(X,\Lambda)$, so its adjunction restricts to D^b_{ctf}.*

By Proposition 3.12, the theorem holds with D^b_c in place of D^b_{ctf} for $\Lambda=\mathbf{Z}_\ell,\mathcal{O}_E,\mathbf{Q}_\ell,E,\overline{\mathbf{Q}}_\ell$.

For proper p we may use the trivial compactification, so that $Rp_!=Rp_*$. By part (4), the functor $Rp_*:D^b_{\mathrm{ctf}}(X,\Lambda)\to D^b_{\mathrm{ctf}}(Y,\Lambda)$ then has the right adjoint $p^!$.

Parts (1)–(3) are proved in this chapter and part (4) in Chapter 6. For smooth f, smooth purity (Chapter 9) gives a global isomorphism $f^!\simeq f^*(d)[2d]$, compatible with trace and composition.

5.2 Compactness of constructible complexes

Brown representability produces $f^!$ once the bounded constructible complexes are known to be compact, that is, once $\mathrm{Hom}(C,-)$ commutes with direct sums for constructible C. The difficulty is that the summands may have cohomology in degrees going off to $-\infty$, so that $\mathrm{Hom}(C,-)$ sees all the groups $\mathrm{Ext}^n(C,-)$ at once. For instance, over $\operatorname{Spec}\mathbf{R}$ these groups never stop, and we saw at the beginning of this chapter that compactness fails there. By contrast, over a finite field they stop, as the following example shows.

Example 5.3 (Compactness over a finite field)**.** Let $X = \operatorname{Spec}\mathbf{F}_q$ and $\Lambda = E$. Then $\mathrm{Ext}^n(E,E) = H^n(X_{\mathrm{pro\acute{e}t}},E)$ is E for $n = 0,1$ and vanishes for $n \geq 2$. To see the vanishing, consider first $\mathbf{Z}_\ell$ and the Milnor sequence (2.11) with $F_m = (\mathbf{Z}/\ell^m)_X$. Since the field $\mathbf{F}_q$ has ℓ-cohomological dimension one, we have $H^n(X,\mathbf{Z}/\ell^m) = 0$ for $n \geq 2$ and all m, while the finite groups $H^1(X,\mathbf{Z}/\ell^m)$ have vanishing $\varprojlim^1$. The Milnor sequence therefore gives $H^n(X_{\mathrm{pro\acute{e}t}},\mathbf{Z}_{\ell,X}) = 0$ for $n \geq 2$. From here, inverting ℓ and using Example 2.57 for $n = 1$ gives the claim for $\mathbf{Q}_\ell$; for $E \neq \mathbf{Q}_\ell$ we tensor with E over $\mathbf{Q}_\ell$.

As in the example over $\mathbf{R}$, products are exact (Proposition 2.23) and the complexes $E[n]$ live in distinct degrees. Hence $\bigoplus_{n\geq 0} E[n] = \prod_{n\geq 0} E[n]$, and

$$\mathrm{Hom}\Big(E, \bigoplus_{n\geq 0} E[n]\Big) = \prod_{n\geq 0} \mathrm{Ext}^n(E,E) = E \oplus E = \bigoplus_{n\geq 0} \mathrm{Hom}(E,E[n]).$$

The product equals the sum because only two factors are nonzero. Over $\mathbf{R}$ every factor is nonzero, and the product is larger.

The general argument rests on the same mechanism: a bound d_C, valid for all constructible targets at once, beyond which $\mathrm{Ext}^q(C,-)$ vanishes. Until the end of Section 5.2, Λ is E or $\overline{\mathbf{Q}}_\ell$, and we show that bounded constructible complexes are compact in $\mathrm{Loc}(D^b_c(X,\Lambda))$. Since $D^b_{\mathrm{ctf}}(X,\Lambda) = D^b_c(X,\Lambda)$, these are the proposed compact generators. The advantage of working with D^b_c is that we can use truncations and the constructible heart, which is abelian by Lemma 3.18. As for notation, Hom and Ext without decoration are taken in $D(X_{\mathrm{pro\acute{e}t}},\Lambda)$, also when both arguments are constructible sheaves, and a subscript indicates the coefficient ring where it changes, as in the lattice arguments over $\mathcal{O}_E$.

By dévissage it suffices to show, for a constructible sheaf C and a family (K_λ) of objects of $\mathrm{Loc}(D^b_c(X,\Lambda))$, that $\bigoplus_\lambda \mathrm{Hom}(C,K_\lambda) \to \mathrm{Hom}(C,\bigoplus_\lambda K_\lambda)$ is bijective. Besides the uniform bound $\mathrm{Ext}^q(C,N) = 0$ for $q > d_C$ and all constructible N (Lemma 5.6 below), we use that $\mathrm{Ext}^q(C,-)$ commutes with filtered colimits of complexes that vanish below a fixed degree, in particular of sheaves (Lemma 3.22, proved in Section 3.6).

5.2.1 A uniform bound for Ext

The idea of the bound is to count cohomological dimensions modulo π_E^n, add one degree for the derived limit over n, and invert π_E by passing to a colimit. For torsion coefficients the count is direct. By (3.1), $R\mathrm{Hom}(a_!L,N)$ is the cohomology on the stratum S of a complex obtained from N by restriction and Rj_*. Each of these steps raises degrees by a bounded amount that does not depend on N. With coefficients E we cannot count in this way, because E-sheaves are not torsion and cd_ℓ only controls torsion sheaves. This is why we choose lattices on the strata. Inverting π_E at the end then needs the continuity of Ext from Section 3.6.

The bound d_C on $\mathrm{Ext}^q(C,-)$ is a sum of the cohomological dimensions cd_ℓ of Definition 2.45. In the proof we compute $R\mathrm{Hom}(a_!L_0,K)$, for a derived complete $K \in D(X_{\mathrm{pro\acute{e}t}},\mathcal{O}_E)$, as a derived

inverse limit over the reductions modulo π_E^n. For this, the functors in (3.1) have to commute with derived inverse limits, and we use the next lemma with $\Lambda = \mathcal{O}_E$.

Lemma 5.4 (Limits and sections with support)**.** *Let Λ be admissible, let $i : Z \hookrightarrow X$ be a closed immersion with open complement $j : V \hookrightarrow X$, and let $u : S \hookrightarrow Z$ be an open immersion. The functors $R\underline{\Gamma}_Z : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(Z_{\mathrm{pro\acute{e}t}}, \Lambda)$ and $u^* R\underline{\Gamma}_Z : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(S_{\mathrm{pro\acute{e}t}}, \Lambda)$ are left t-exact and commute with derived inverse limits of sequences, and so does tensoring with a finite locally free Λ_S-module.*

Proof. The functor $R\underline{\Gamma}_Z$ is left t-exact because it is right adjoint to the exact functor i_*. Restricting along the open immersion u is exact, so $u^* R\underline{\Gamma}_Z$ is left t-exact as well.

For the limits, recall that the derived inverse limit of a sequence (K_n) is the fibre of the map $\prod_n K_n \to \prod_n K_n$ given by the identity minus the transition maps. It therefore suffices to show that the functors preserve products and fibres. Restriction to an open is exact and preserves limits, so this holds for j^* and u^*. It holds for i^* by [Stacks, 09AA], and for Rj_* because Rj_* is a right adjoint. It follows for $R\underline{\Gamma}_Z = i^* \operatorname{fib}(\mathrm{id} \to Rj_* j^*)$. Finally, tensoring with a finite locally free sheaf L is exact and is right adjoint to tensoring with $L^\vee$, so it preserves products as well. ∎

The main case is that of lattices. There we count cohomological dimensions for the reductions modulo π_E^n, and we pass to the limit by Lemma 5.4.

Lemma 5.5 (Ext bound for lattices)**.** *Let $i : Z \hookrightarrow X$ be a closed immersion with open complement $j : X \setminus Z \hookrightarrow X$, let $u : S \hookrightarrow Z$ be an open immersion, put $a = i \circ u$, and let L_0 be a lisse $\mathcal{O}_{E,S}$-module that is locally free of finite rank. Then for every locally closed immersion $b : T \hookrightarrow X$ and every lisse $\mathcal{O}_{E,T}$-module M_0 that is locally free of finite rank,*

$$\operatorname{Ext}^q_{\mathcal{O}_E}(a_! L_0, b_! M_0) = 0 \quad (q > \mathrm{cd}_\ell(j) + \mathrm{cd}_\ell(S) + 2). \tag{5.1}$$

Proof. We first bound the complex in (3.1) for the torsion reductions of $b_! M_0$ by cohomological dimensions, and then pass to the derived limit, which costs one degree. Both cohomological dimensions are finite by [Stacks, 0F10] and Lemma 2.46.

By Lemma lemma-integral-models and (4), $b_! M_0$ lies in $D^b_c(X, \mathcal{O}_E)$, so it is derived complete. Since $b_!$ is exact and M_0 is π_E-torsion free, its reductions are

$$b_! M_0 \otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}} \mathcal{O}_{E,X}/\pi_E^n = b_!(M_0/\pi_E^n).$$

These are constructible torsion sheaves. They are pulled back from $X_{\acute{e}t}$, because extension by zero and closed direct image commute with this pullback [Stacks, 09AF, 09A6].

Torsion targets. We bound the complex that appears in (3.1) when the target is one of the torsion sheaves $b_!(M_0/\pi_E^n)$. By [Stacks, 09A6], Rj_* in $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n)$ of a sheaf pulled back from the étale site is the pullback of its direct image in $D(X_{\acute{e}t}, \mathcal{O}_E/\pi_E^n)$. On the étale site the degrees are controlled by the cohomological dimension, and hence $Rj_* j^* b_!(M_0/\pi_E^n)$ has cohomology in degrees $[0, \mathrm{cd}_\ell(j)]$. The fibre of the map from a sheaf to this complex lives in degrees $[0, \mathrm{cd}_\ell(j)+1]$, so the fibre formula of Lemma lemma-sections-support-closed-subscheme shows that

$$R\underline{\Gamma}_Z\, b_!(M_0/\pi_E^n) \in D^{[0,\mathrm{cd}_\ell(j)+1]}(Z_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E/\pi_E^n),$$

and its cohomology sheaves are torsion sheaves pulled back from the étale site, though not necessarily constructible. Restricting to S and tensoring with the locally free lisse sheaf $L_0^\vee/\pi_E^n$

does not change this range of degrees. Since the cohomology of S with coefficients in a torsion sheaf vanishes above $\mathrm{cd}_\ell(S)$, the hypercohomology spectral sequence on S gives

$$H^q R\Gamma\big(S, (L_0^\vee/\pi_E^n) \otimes u^* R\underline{\Gamma}_Z\, b_!(M_0/\pi_E^n)\big) = 0 \quad (q > \mathrm{cd}_\ell(j) + \mathrm{cd}_\ell(S) + 1), \tag{5.2}$$

for every n, b and M_0.

Integral targets. For the lattice $b_!M_0$ itself we write the relevant complex as a derived limit of its reductions modulo π_E^n. Consider the complex $L_0^\vee \otimes u^* R\underline{\Gamma}_Z\, b_!M_0$. By Lemma 5.4, the functors that form it commute with derived inverse limits of sequences. They also commute with reduction modulo π_E^n, which is the cone of multiplication by π_E^n. Since $b_!M_0$ is derived complete, so is this complex, and its reductions are the complexes in (5.2). In other words, the complex is the derived limit of its reductions, and we may compute its cohomology on S from theirs. Derived global sections preserve products and cones, and hence derived inverse limits. Using (3.1) with $\mathcal{O}_E$ coefficients, that is, for $R\mathrm{Hom}$ in $D(X_{\mathrm{pro\acute{e}t}}, \mathcal{O}_E)$, we get

$$R\mathrm{Hom}_X(a_!L_0, b_!M_0) = R\Gamma\big(S, L_0^\vee \otimes u^* R\underline{\Gamma}_Z\, b_!M_0\big) = R\varprojlim_n R\Gamma\big(S, (L_0^\vee \otimes u^* R\underline{\Gamma}_Z\, b_!M_0) \otimes^{\mathbf{L}} \mathcal{O}_E/\pi_E^n\big).$$

A derived inverse limit of a sequence raises an upper bound on cohomology by at most one. Indeed, it is the fibre of the difference of the identity and the transition maps on the product, and products of abelian groups are exact. This is the Milnor sequence [Stacks, 08TB]. Applied to the bound (5.2) for the reductions, it gives (5.1). ∎

Lemma 5.6 (Uniform Ext bound)**.** *For every $C \in \mathrm{Cons}(X, \Lambda)$ there is an integer $d_C \geq 0$ such that*

$$\mathrm{Ext}^q(C, N) = 0 \qquad \textit{for all } q > d_C \textit{ and all } N \in \mathrm{Cons}(X, \Lambda).$$

Suppose $C = a_!L$, with a factored as $S \hookrightarrow Z \hookrightarrow X$ and j the open complement of Z, and suppose that L contains a lisse $\mathcal{O}_E$-lattice. By Lemma 3.24, this holds after refining S. In this case the proof gives $d_C = \mathrm{cd}_\ell(j) + \mathrm{cd}_\ell(S) + 2$. One of the two extra degrees comes from the cone defining $R\underline{\Gamma}_Z$, and the other from a derived inverse limit over the reductions modulo π_E^n.

Proof of Lemma 5.6. We reduce to extensions by zero of lisse sheaves on strata, prove the bound for $\Lambda = E$ by choosing lattices, applying Lemma 5.5 and inverting π_E, and reduce $\Lambda = \overline{\mathbf{Q}}_\ell$ to finite extensions of E.

Reduction to strata. Bounds pass to extensions in either variable. Indeed, if $0 \to C' \to C \to C'' \to 0$ is exact, then in the long exact sequence $\mathrm{Ext}^q(C'', N) \to \mathrm{Ext}^q(C, N) \to \mathrm{Ext}^q(C', N)$ the outer groups vanish above the larger of the two bounds for C' and C'', and so the middle one does too. Hence a bound that holds for every subquotient of a finite filtration of the source holds for the source, if we take the maximum. The same sequence in the second variable shows that a bound that holds for every subquotient of a finite filtration of the target holds for the target, with the same value. In view of Lemma lemma-constructible-sheaves, we may therefore assume $C = a_!L$, with $a = i \circ u : S \hookrightarrow Z \hookrightarrow X$ as in Lemma 3.21 and L lisse. It suffices to find an integer d, depending only on a and L, with $\mathrm{Ext}^q(a_!L, b_!M) = 0$ for $q > d$, for all locally closed $b : T \hookrightarrow X$ and lisse M on T. Let $j : X \setminus Z \hookrightarrow X$ be the open complement of Z. When L contains a lisse $\mathcal{O}_E$-lattice, we show that $d = \mathrm{cd}_\ell(j) + \mathrm{cd}_\ell(S) + 2$ works.

Lattices. First take $\Lambda = E$. Since torsion arguments need integral structures, we choose lattices. After refining the stratifications (Lemma 3.24), L and M contain lisse $\mathcal{O}_E$-lattices L_0 and M_0. Refining is harmless, because it replaces $a_!L$ and $b_!M$ by finite extensions of sheaves of the same form, and the first step handles finite extensions. Moreover, the refinement of the source stratum depends only on L, and that of the target only on M. So the bound obtained for the refined source strata is still independent of N.

The lattices are locally free of finite rank over $\mathcal{O}_E$. Indeed, being lisse, they are locally of the form $\mathcal{O}_{E,U} \otimes_{\mathcal{O}_E} P$ with P finitely generated (Definition 3.2). As subsheaves of E-modules they are torsion free, so P is free. Lemma 5.5 now gives $\operatorname{Ext}^q_{\mathcal{O}_E}(a_!L_0, b_!M_0) = 0$ for $q > \operatorname{cd}_\ell(j) + \operatorname{cd}_\ell(S) + 2$.

Inverting π_E. Inverting π_E is a filtered colimit. By continuity of Ext out of $a_!L_0$, the Ext groups over E are therefore colimits of groups that all vanish in the same range. In detail, we have $a_!L = a_!L_0[1/\pi_E]$ and $b_!M = (b_!M_0)[1/\pi_E] = \varinjlim(b_!M_0 \xrightarrow{\pi_E} b_!M_0 \xrightarrow{\pi_E} \cdots)$. The adjunction between extension and restriction of scalars gives

$$R\operatorname{Hom}_E(a_!L, b_!M) = R\operatorname{Hom}_{\mathcal{O}_E}(a_!L_0, (b_!M_0)[1/\pi_E]).$$

By Lemma lemma-continuity-ext-out-constructible-complex the cohomology of the right side is the colimit of the groups $\operatorname{Ext}^q_{\mathcal{O}_E}(a_!L_0, b_!M_0)$. Hence (5.1) implies the bound for E.

The case $\Lambda = \overline{\mathbf{Q}}_\ell$. The bound for E depends only on the strata and on cd_ℓ, not on E, so it passes to the colimit over finite extensions. We first fix everything that depends on L alone, before the target is given. By Lemma 3.17, applied to L alone, there are a finite extension $E_0/\mathbf{Q}_\ell$ and a lisse E_0-sheaf L' with $L = L'_{\overline{\mathbf{Q}}_\ell}$. After refining S (Lemma 3.24), L' contains a lisse $\mathcal{O}_{E_0}$-lattice L'_0. As in the case $\Lambda = E$, the refinement replaces $a_!L$ by a finite extension of sheaves of the same form, and it depends only on L. For every finite extension $E' \supset E_0$, the sheaf $L'_0 \otimes_{\mathcal{O}_{E_0}} \mathcal{O}_{E'}$ is a lisse $\mathcal{O}_{E'}$-lattice in $(L')_{E'}$ on the same strata. The case $\Lambda = E'$ therefore gives $\operatorname{Ext}^q_{E'}((a_!L')_{E'}, N') = 0$ for $q > \operatorname{cd}_\ell(j) + \operatorname{cd}_\ell(S) + 2$, for each refined stratum S and every $N' \in \operatorname{Cons}(X, E')$. This bound does not depend on E' or on N'.

Now let b and M be given. By Lemma 3.17 again, applied to M, there are a finite extension $E \supset E_0$ (enlarge the field given by the lemma to contain E_0) and a lisse E-sheaf M' with $M = M'_{\overline{\mathbf{Q}}_\ell}$. The field E depends on M, but neither the strata of the source nor the lattice L'_0 do. Then $a_!L = (a_!L')_{\overline{\mathbf{Q}}_\ell}$ and $b_!M = (b_!M')_{\overline{\mathbf{Q}}_\ell}$, and by Lemma lemma-descent-finite-coefficient-field,

$$\operatorname{Ext}^q_{\overline{\mathbf{Q}}_\ell}(a_!L, b_!M) = \varinjlim_{E' \supset E} \operatorname{Ext}^q_{E'}((a_!L')_{E'}, (b_!M')_{E'}).$$

Every group on the right vanishes for $q > \operatorname{cd}_\ell(j) + \operatorname{cd}_\ell(S) + 2$ by the previous paragraph, and hence so does the colimit. ∎

Example 5.7 (The bound over a finite field)**.** The proof gives an explicit bound, which is far from sharp. For instance, for $X = \operatorname{Spec} \mathbf{F}_q$ and $C = N = E$ we have $Z = S = X$. The complement of Z is then empty, so $R\underline{\Gamma}_Z$ is the identity. Moreover $\operatorname{cd}_\ell(\mathbf{F}_q) = 1$, and so the formula, read with $\operatorname{cd}_\ell(j) = 0$ for the empty open, gives $d_C = 0 + 1 + 2 = 3$, while $\operatorname{Ext}^q(E, E)$ vanishes already for $q > 1$ (Example 5.3). The two wasted degrees do no harm, because the arguments below only need some bound independent of N.

5.2.2 Ind-constructible sheaves

The targets in the compactness statement are direct sums of objects of $\operatorname{Loc}(D^b_c(X, \Lambda))$, and their cohomology sheaves need not be constructible. Already the sheaf $\bigoplus_{n \in \mathbf{N}} \Lambda_X$ is not constructible,

since its stalks are not finitely generated. Such sheaves are, however, filtered colimits of constructible sheaves, and both continuity and the uniform bound for Ext groups extend to these colimits.

Recall that the ind-category $\mathrm{Ind}(\mathcal{C})$ of a small category $\mathcal{C}$ has as objects the filtered diagrams in $\mathcal{C}$ ("formal filtered colimits"), with $\mathrm{Hom}(\varinjlim_\alpha C_\alpha, \varinjlim_\beta D_\beta) = \varprojlim_\alpha \varinjlim_\beta \mathrm{Hom}(C_\alpha, D_\beta)$. Proposition 5.9 below says that for constructible sheaves the formal and the actual colimits agree.

The dévissage arguments need a class of sheaves that is abelian and closed under extensions and direct sums, and on which $\mathrm{Ext}^q(C, -)$ behaves as on constructible sheaves. By Proposition 5.9, the category $\mathrm{IndCons}(X, \Lambda)$ has the first three properties, and Corollary 5.10 gives the last.

Definition 5.8 (Ind-constructible sheaves)**.** An *ind-constructible sheaf* on X is a sheaf of Λ_X-modules on $X_{\mathrm{pro\acute{e}t}}$ that is a filtered colimit, computed in $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$, of constructible sheaves. We write $\mathrm{IndCons}(X, \Lambda) \subset \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ for the full subcategory of ind-constructible sheaves.

Proposition 5.9 (Ind-constructible sheaves)**.**

1. *The functor* $\mathrm{Ind}(\mathrm{Cons}(X, \Lambda)) \to \mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ *sending a formal filtered colimit to the actual colimit is fully faithful, with essential image* $\mathrm{IndCons}(X, \Lambda)$.
2. $\mathrm{IndCons}(X, \Lambda)$ *is closed in* $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ *under filtered colimits, direct sums, kernels, cokernels and extensions. In particular it is a Grothendieck abelian category and its inclusion is exact.*

Proof. The proof uses one property of constructible sheaves: maps out of them commute with filtered colimits. Because of this, the formal colimits of the ind-category behave like actual colimits. Each closure property in (2) then reduces to the same property of $\mathrm{Cons}(X, \Lambda)$, once the objects and maps involved are written as filtered colimits of constructible sheaves and maps between them.

Finite presentation. Constructible sheaves are finitely presented in $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$. Indeed, Lemma 3.22 in degree zero shows that $\mathrm{Hom}(C, -)$ commutes with filtered colimits for C constructible. By Lemma lemma-constructible-sheaves,(2), the category $\mathrm{Cons}(X, \Lambda)$ is essentially small, abelian, and closed under extensions. We fix a small skeleton, so that the ind-category in (1) is defined.

Full faithfulness. For filtered diagrams (C_α) and (D_β) of constructible sheaves, finite presentation implies

$$\mathrm{Hom}(\varinjlim_\alpha C_\alpha, \varinjlim_\beta D_\beta) = \varprojlim_\alpha \varinjlim_\beta \mathrm{Hom}_{\mathrm{Cons}(X,\Lambda)}(C_\alpha, D_\beta),$$

which is the Hom set in the ind-category. This proves (1), since the essential image is $\mathrm{IndCons}(X, \Lambda)$ by definition.

Canonical presentations. The closure properties below involve several ind-constructible sheaves at once, so to compare their presentations we use a presentation that involves no choices. Namely, for $B \in \mathrm{IndCons}(X, \Lambda)$, let $J(B)$ be the category of constructible sheaves C with a map $C \to B$; we show that $B = \varinjlim_{J(B)} C$.

The category $J(B)$ is filtered. Indeed, two objects $C_1 \to B$ and $C_2 \to B$ both map to $C_1 \oplus C_2 \to B$, and two parallel arrows $C \rightrightarrows C'$ over B are equalized by the cokernel of their difference, which is constructible and still maps to B. Now write $B = \varinjlim_\alpha C_\alpha$. By finite presentation, every $C \to B$

factors through some C_α, and two factorizations agree at a later stage. This means that the given diagram is cofinal in $J(B)$, and hence $B = \varinjlim_{J(B)} C$.

Filtered colimits and direct sums. Let $B = \varinjlim_\lambda B_\lambda$ be a filtered colimit with $B_\lambda \in \mathrm{IndCons}(X, \Lambda)$. To present B, consider the category of pairs $(\lambda, C \to B_\lambda)$ with C constructible. It is filtered: we move finitely many indices to a common upper bound, and then use direct sums and cokernels in $J(B_\lambda)$. Moreover, the colimit of the C over this category is $\varinjlim_\lambda \varinjlim_{J(B_\lambda)} C = B$. A direct sum is the filtered colimit of its finite subsums, so it lies in $\mathrm{IndCons}(X, \Lambda)$ as well.

Kernels and cokernels. Let $f : B \to B'$ be a map in $\mathrm{IndCons}(X, \Lambda)$. It suffices to write f as a filtered colimit of maps v of constructible sheaves. Since filtered colimits are exact, $\ker f$ and $\operatorname{coker} f$ are then the filtered colimits of the constructible sheaves $\ker v$ and $\operatorname{coker} v$. Let P_f be the category of commutative squares

$$\begin{array}{ccc} C & \xrightarrow{v} & D \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B', \end{array} \qquad C, D \in \mathrm{Cons}(X, \Lambda).$$

This category is filtered, because finite colimits of such squares are formed by finite colimits of the arrows v in the abelian category $\mathrm{Cons}(X, \Lambda)$. We claim that the functor $P_f \to J(B)$, $(v) \mapsto (C \to B)$, is cofinal. Indeed, every $C \to B$ is the source of a square, namely $v = \mathrm{id}_C$ with $C \to B'$ the composite. If $C \to C_1 \xrightarrow{v_1} D_1$ and $C \to C_2 \xrightarrow{v_2} D_2$ are two squares under the same $C \to B$, then both map to the square with source $C_1 \amalg_C C_2$ and target $D_1 \amalg_C D_2$. Here the pushouts are taken in $\mathrm{Cons}(X, \Lambda)$, and they map to B and B' because the two maps from C to B, and the two maps from C to B', agree.

Similarly, the functor $P_f \to J(B')$, $(v) \mapsto (D \to B')$, is cofinal. Every $D \to B'$ is the target of a square, namely $v : 0 \to D$. If $C_1 \xrightarrow{v_1} D_1$ and $C_2 \xrightarrow{v_2} D_2$ are two squares under the same $D \to B'$, then both map to the square with source $C_1 \oplus C_2$ and target $D_1 \amalg_D D_2$. The two cofinality statements give $f = \varinjlim_{P_f} v$, and exactness of filtered colimits gives

$$\ker f = \varinjlim_{P_f} \ker v, \qquad \operatorname{coker} f = \varinjlim_{P_f} \operatorname{coker} v,$$

both of which lie in $\mathrm{IndCons}(X, \Lambda)$.

Extensions. Let $0 \to N \to M \to B \to 0$ be exact in $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$, with $N = \varinjlim_\beta D_\beta$ and $B = \varinjlim_\alpha C_\alpha$ ind-constructible. We show that M is a filtered colimit of extensions of constructible sheaves. First, the pullback of the extension along $C_\alpha \to B$ is ind-constructible. Its class lies in $\mathrm{Ext}^1(C_\alpha, N)$, and Lemma 3.22 gives

$$\mathrm{Ext}^1(C_\alpha, N) = \varinjlim_\beta \mathrm{Ext}^1(C_\alpha, D_\beta).$$

So the pulled-back extension is pushed out from an extension $M_{\alpha\beta}$ of C_α by some D_β. Since $\mathrm{Cons}(X, \Lambda)$ is closed under extensions, $M_{\alpha\beta}$ and each of its pushouts along $D_\beta \to D_{\beta'}$ are constructible. Their colimit over β' is the pullback of M to C_α. Second, since filtered colimits are exact, M is the colimit over α of these pullbacks. By closure under filtered colimits, $M \in \mathrm{IndCons}(X, \Lambda)$.

Grothendieck. The category $\mathrm{IndCons}(X, \Lambda)$ is Grothendieck abelian in the sense of [Stacks, 079A]. Indeed, by the above its colimits exist and are computed in $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$, so filtered colimits are exact. The direct sum of the objects of the skeleton of $\mathrm{Cons}(X, \Lambda)$ is a generator. ∎

Since every ind-constructible sheaf is a filtered colimit of constructible ones, the continuity of $\mathrm{Ext}^q(C,-)$ lets us extend the uniform bound of Lemma 5.6 to the whole category $\mathrm{IndCons}(X,\Lambda)$. Writing a direct sum as the filtered colimit of its finite subsums, we also see that $\mathrm{Ext}^q(C,-)$ commutes with direct sums of ind-constructible sheaves.

Corollary 5.10 (Ext from a constructible sheaf into ind-constructible sheaves)**.** *Let $C \in \mathrm{Cons}(X,\Lambda)$, with d_C as in Lemma 5.6. Then $\mathrm{Ext}^q(C,B) = 0$ for all $q > d_C$ and all $B \in \mathrm{IndCons}(X,\Lambda)$, and each functor $\mathrm{Ext}^q(C,-)$ commutes with direct sums of objects of $\mathrm{IndCons}(X,\Lambda)$.*

Proof. Write $B = \varinjlim_\alpha B_\alpha$ as a filtered colimit of constructible sheaves, which is possible by Proposition 5.9. Sheaves are complexes concentrated in degree 0, so they vanish below a common degree, and Lemma 3.22 applies. It shows that $\mathrm{Ext}^q(C,B) = \varinjlim_\alpha \mathrm{Ext}^q(C,B_\alpha)$. Each group on the right vanishes for $q > d_C$ by Lemma 5.6, and so does the colimit.

For the second claim, let (B_λ) be a family in $\mathrm{IndCons}(X,\Lambda)$. Its direct sum is the filtered colimit of the finite subsums, and $\mathrm{Ext}^q(C,-)$ commutes with finite direct sums because it is additive. Applying Lemma 3.22 once more, to the diagram of finite subsums, we obtain $\mathrm{Ext}^q(C,\bigoplus_\lambda B_\lambda) = \bigoplus_\lambda \mathrm{Ext}^q(C,B_\lambda)$. ∎

Objects of $\mathrm{Loc}(D^b_c(X,\Lambda))$ can be unbounded, and we do not know a priori that their truncations stay in $\mathrm{Loc}(D^b_c(X,\Lambda))$. For this reason, here and in Section 5.4, we carry out truncation arguments among complexes with ind-constructible cohomology sheaves.

Corollary 5.11 (The derived category of ind-constructible sheaves)**.** *The full subcategory*

$$D_{\mathrm{IndCons}}(X,\Lambda) = \{K \in D(X_{\mathrm{pro\acute{e}t}},\Lambda) : H^n(K) \in \mathrm{IndCons}(X,\Lambda) \text{ for every } n\}$$

of $D(X_{\mathrm{pro\acute{e}t}},\Lambda)$ is triangulated, closed under direct sums, and stable under the truncations $\tau_{\le n}$, $\tau_{\ge n}$. In particular $\mathrm{Loc}(D^b_c(X,\Lambda)) \subset D_{\mathrm{IndCons}}(X,\Lambda)$.

Proof. In the long exact cohomology sequence of a triangle, each cohomology sheaf of the third term is an extension of a kernel by a cokernel of maps between cohomology sheaves of the other two. Since $\mathrm{IndCons}(X,\Lambda)$ is closed under kernels, cokernels and extensions (Proposition 5.9), the third term lies in $D_{\mathrm{IndCons}}(X,\Lambda)$ whenever the other two do. Closure under shifts is clear. As direct sums of sheaves are exact, H^n commutes with direct sums. Since $\mathrm{IndCons}(X,\Lambda)$ is closed under direct sums, it follows that $D_{\mathrm{IndCons}}(X,\Lambda)$ is closed under direct sums. Truncation only removes cohomology sheaves and keeps the others. Hence $D_{\mathrm{IndCons}}(X,\Lambda)$ is a localizing subcategory of $D(X_{\mathrm{pro\acute{e}t}},\Lambda)$ containing $D^b_c(X,\Lambda)$. It therefore contains the smallest one, $\mathrm{Loc}(D^b_c(X,\Lambda))$. ∎

5.2.3 Rational coefficients

On ind-constructible sheaves, $\mathrm{Ext}^q(C,-)$ vanishes above d_C and commutes with direct sums (Corollary 5.10). Because the bound d_C is uniform, maps from a constructible sheaf C into an unbounded complex depend only on its cohomology sheaves in degrees $[-d_C,0]$. This is the step that fails over $\mathbf{R}$ in the example at the beginning of this chapter, where $\mathrm{Ext}^n(\mathbf{F}_2,\mathbf{F}_2) \neq 0$ for all n. Since the window has the same length for a whole family of targets, compactness follows from the case of sheaves by induction on the length of the window.

Lemma 5.12 (Reduction to finitely many degrees)**.** *Let $C \in \mathrm{Cons}(X,\Lambda)$, with d_C as in Lemma 5.6, and let $K \in D_{\mathrm{IndCons}}(X,\Lambda)$ and $m \in \mathbf{Z}$. The truncation maps induce a natural isomorphism*

$$\mathrm{Hom}(C, K[m]) \simeq \mathrm{Hom}(C, (\tau_{\geq m-d_C}\tau_{\leq m}K)[m]). \tag{5.3}$$

Proof. Cutting off the cohomology above degree m is harmless for degree reasons alone. To remove the cohomology below degree $m - d_C$ we use the uniform bound: for bounded K it kills the tail layer by layer, and for K unbounded below we write K as the derived limit of its truncations and control the resulting $\varprojlim^1$ term.

Replacing K by $K[m]$, we may assume $m = 0$. The truncation $\tau_{>0}K$ lies in $D^{\geq 1}$, and C is a sheaf in degree 0, so there are no nonzero maps from C to an object of $D^{\geq 1}$. Hence $\mathrm{Hom}(C, \tau_{>0}K[s]) = 0$ for $s = 0, -1$. In the long exact sequence of $\mathrm{Hom}(C, -)$ for the truncation triangle $\tau_{\leq 0}K \to K \to \tau_{>0}K \to$, these are the two groups adjacent to $\mathrm{Hom}(C, K)$, and so $\mathrm{Hom}(C, K) = \mathrm{Hom}(C, \tau_{\leq 0}K)$. To remove the cohomology in degrees $\leq -d_C - 1$ from $K' = \tau_{\leq 0}K$, we use the triangle $\tau_{\leq -d_C-1}K' \to K' \to \tau_{\geq -d_C}K' \to$. It suffices to show that

$$\mathrm{Hom}(C, L) = 0 \qquad \text{for every } L \in D_{\mathrm{IndCons}}(X, \Lambda) \text{ with } H^q(L) = 0 \text{ for } q > -d_C - 1, \tag{5.4}$$

applied to $L = \tau_{\leq -d_C-1}K'$ and to $L = \tau_{\leq -d_C-1}K'[1]$.

We first prove (5.4) for bounded L. Such an L is a finite extension of its shifted cohomology sheaves $H^q(L)[-q]$, and by the long exact Hom sequences it suffices to show that each of them receives no nonzero map from C. A layer $H^q(L)[-q]$ with $q \leq -d_C - 1$ contributes $\mathrm{Hom}(C, H^q(L)[-q]) = \mathrm{Ext}^{-q}(C, H^qL)$. Since $-q > d_C$ and $H^q(L)$ is ind-constructible, this group vanishes by Corollary 5.10. For L unbounded below we cannot argue in finitely many steps, and we use instead that the derived category is left complete.

The pro-étale site has a basis of weakly contractible objects (Theorem 2.17). For this reason Proposition 2.23 applies: products of Λ_X-modules are exact, products in $D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ are computed termwise, and

$$L = R\varprojlim_n \tau_{\geq -n}L$$

for every L. Concretely, if $L^\bullet$ is a complex representing L, the limit is the derived inverse limit of the tower of good truncations $\tau_{\geq -n}L^\bullet$ with their canonical transition maps. Since $\mathrm{Hom}(C, -)$ takes products to products, the triangle defining $R\varprojlim$ yields the Milnor exact sequence

$$0 \longrightarrow \varprojlim_n{}^1 \mathrm{Hom}(C, \tau_{\geq -n}L[-1]) \longrightarrow \mathrm{Hom}(C, L) \longrightarrow \varprojlim_n \mathrm{Hom}(C, \tau_{\geq -n}L) \longrightarrow 0.$$

Each $\tau_{\geq -n}L$ is bounded with cohomology in degrees $\leq -d_C - 1$, and hence the right-hand term vanishes by the bounded case. For the left-hand term, the triangle

$$H^{-n-1}(L)[n+1] \longrightarrow \tau_{\geq -n-1}L \longrightarrow \tau_{\geq -n}L \longrightarrow H^{-n-1}(L)[n+2]$$

gives an exact sequence

$$\mathrm{Ext}^n(C, H^{-n-1}L) \to \mathrm{Hom}(C, \tau_{\geq -n-1}L[-1]) \to \mathrm{Hom}(C, \tau_{\geq -n}L[-1]) \to \mathrm{Ext}^{n+1}(C, H^{-n-1}L).$$

The outer terms vanish for $n > d_C$ by Corollary 5.10. The transition maps are therefore isomorphisms for $n > d_C$, so the tower is eventually constant and its $\varprojlim^1$ vanishes. This proves (5.4). ∎

The window $[m - d_C, m]$ in (5.3) depends on C and m only, not on K, and so we can treat a whole family (K_λ) at once in the compactness theorem.

Theorem 5.13 (Compactness with rational coefficients)**.** *For $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, every object of $D^b_c(X, \Lambda)$ is compact in $\mathrm{Loc}(D^b_c(X, \Lambda))$. More precisely, for $C \in D^b_c(X, \Lambda)$ and every family (K_λ) of objects of $D_{\mathrm{IndCons}}(X, \Lambda)$, the natural map*

$$\bigoplus_\lambda \mathrm{Hom}(C, K_\lambda) \longrightarrow \mathrm{Hom}\Big(C, \bigoplus_\lambda K_\lambda\Big)$$

is an isomorphism.

Proof. The idea is that, by Lemma 5.12, maps out of a constructible sheaf depend only on the cohomology of the target in $d_C + 1$ consecutive degrees, uniformly in the family. For targets with cohomology in a fixed bounded interval the claim follows from the case of ind-constructible sheaves by induction on the length.

The first assertion follows from the second, since $\mathrm{Loc}(D^b_c(X, \Lambda)) \subset D_{\mathrm{IndCons}}(X, \Lambda)$ (Corollary 5.11) and direct sums in $\mathrm{Loc}(D^b_c(X, \Lambda))$ are those of $D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$.

First let $C \in \mathrm{Cons}(X, \Lambda)$, and suppose that all K_λ have cohomology in a fixed interval $[a, b]$. We prove the claim, for all shifts of the K_λ at once, by induction on $b - a$. If $a = b$, then $K_\lambda = B_\lambda[-a]$ with $B_\lambda \in \mathrm{IndCons}(X, \Lambda)$, and the claim is that $\mathrm{Ext}^q(C, -)$ commutes with direct sums of objects of $\mathrm{IndCons}(X, \Lambda)$, which is Corollary 5.10. In general, the triangles $\tau_{\le b-1} K_\lambda \to K_\lambda \to H^b(K_\lambda)[-b] \to$ are compatible with direct sums, because H^n commutes with direct sums. We apply $\mathrm{Hom}(C, -)$ to these triangles and take the direct sum over λ; since direct sums of abelian groups are exact, the result is again a long exact sequence. It maps to the long exact sequence of the summed triangle. The outer terms are of the form covered by the induction hypothesis (cohomology in $[a, b-1]$ and in $[b, b]$), and so the five lemma gives the claim for cohomology in $[a, b]$.

For arbitrary K_λ, we apply Lemma 5.12 with $m = 0$, which allows us to replace each K_λ, and $\bigoplus_\lambda K_\lambda$, by its truncation to degrees $[-d_C, 0]$. Since the interval is the same for the whole family and truncation commutes with direct sums, the bounded case applies.

From constructible sheaves C we pass to complexes by dévissage. For fixed (K_λ), the objects C for which the map is an isomorphism are closed under shifts. They are closed under cones by the five lemma, applied to the long exact sequences of $\mathrm{Hom}(-, K_\lambda)$ summed over λ and of $\mathrm{Hom}(-, \bigoplus_\lambda K_\lambda)$. A bounded constructible complex is a finite extension of shifts of its constructible cohomology sheaves, and hence it belongs to this class. ∎

Lemma 5.30 combines this theorem with the corresponding statements for the other coefficient rings and shows that a small skeleton of $D^b_{\mathrm{ctf}}(X, \Lambda)$ is a set of compact generators of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$.

5.3 Compactness for finite coefficients

In this section the coefficient ring is a finite admissible ring A. Proper direct image preserves D^b_{ctf} by Proposition 4.35, and we now show that the constructible generators are compact and that proper direct image preserves direct sums on the localizing categories. In both cases we reduce to the classical étale theory, using bounded flat constructible models and the fact that étale cohomology commutes with direct sums of unbounded complexes.

Finite Tor amplitude is a property of the complex, not of its cohomology sheaves: the cohomology sheaves of an object of D^b_{ctf} can have infinite Tor amplitude (Section 3.4). The property can, however, be read off from a flat model. This is why we work with the bounded flat models of Lemma 3.13.

Proposition 5.14 (Compactness for finite coefficients)**.** *Every object of $D^b_{\mathrm{ctf}}(X, A)$ is compact in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))$, and a small skeleton of $D^b_{\mathrm{ctf}}(X, A)$ is a set of compact generators. Moreover, $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))^c = D^b_{\mathrm{ctf}}(X, A)$.*

Proof. The plan is to prove compactness on the étale site against arbitrary objects of $D(X_{ét}, A)$, where the commutation of cohomology with direct sums that we need is classical, and then transfer the result to $X_{\mathrm{pro}ét}$ by ν_X^*.

Recall that compact objects are closed under shifts, finite extensions and direct summands. Hence, by Lemma 3.13, it suffices to treat $a_!L$. To do so, factor $a = i \circ u$ with $i : Z \hookrightarrow X$ closed and $u : S \hookrightarrow Z$ open, and let $j : X \setminus Z \hookrightarrow X$ be the complementary open immersion. The idea is then to write $R\mathrm{Hom}(a_!L, -)$ as a composite of functors each of which commutes with direct sums. Since L has finite projective stalks, it is dualizable with dual $L^\vee = \underline{\mathrm{Hom}}_A(L, A_S)$, and the étale form of Lemma lemma-sections-support-closed-subscheme shows that for $K \in D(X_{ét}, A)$ we have

$$R\mathrm{Hom}_X(a_!L, K) = R\Gamma\big(S, L^\vee \otimes_A u^* R\underline{\Gamma}_Z K\big), \qquad R\underline{\Gamma}_Z K = i^* \operatorname{fib}\big(K \to Rj_*j^*K\big),$$

where $R\underline{\Gamma}_Z$ denotes sections with support in Z. We claim that every functor on the right, viewed as a functor between étale derived categories, preserves arbitrary direct sums. For Rj_* and $R\Gamma(S, -)$, this is [Stacks, 0F11], which says that a derived direct image of finite cohomological dimension commutes with direct sums of unbounded complexes. It applies to A-modules because derived direct image and global sections commute with forgetting the coefficient ring [Stacks, 0D6J]. The remaining functors are inverse images, tensor products with finite projective sheaves, and a fibre, and these also preserve direct sums: inverse images are left adjoints, tensor products commute with direct sums, and the fibre of a map between functors that preserve direct sums again preserves them. Hence $R\mathrm{Hom}_X(a_!L, -)$ commutes with direct sums. Taking H^0, we see that $\mathrm{Hom}(a_!L, -)$ commutes with direct sums, that is, $a_!L$ is compact. By Lemma 3.13 it follows that every object of $D^b_{\mathrm{ctf}}(X_{ét}, A)$ is compact.

On the pro-étale site, Corollary 2.44 and Proposition 2.49 show that the inverse image $\nu_X^* : D(X_{ét}, A) \to D(X_{\mathrm{pro}ét}, A)$ is exact, fully faithful and preserves direct sums. Such a functor identifies the localizing subcategories generated by D^b_{ctf} on the two sites. Compactness against all étale complexes therefore implies compactness in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))$.

For the last assertion we have to show that there are no other compact objects. Since constructibility and finite Tor amplitude are preserved by direct summands, the category $D^b_{\mathrm{ctf}}(X, A)$ is thick. It is essentially small, because it is a full subcategory of the essentially small bounded constructible category, and by definition a skeleton generates $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))$. In a triangulated category with direct sums generated by a set of compact objects, the compact objects form the thick closure of that set [Stacks, 09SM, 09SN]. In our case this closure is $D^b_{\mathrm{ctf}}(X, A)$. ∎

For finite coefficients the localizing category is a familiar one: the next proposition shows that it is the classical unbounded derived category of the étale site. In particular, $f^!$ with finite coefficients is defined on all of $D(X_{ét}, A)$. The proposition also shows that $Rf_!$, once it is defined as $Rp_*j_!$ by a compactification in Section 5.4, is computed by the classical functor $Rp_{ét,*}j_{ét,!}$ on the étale site.

Proposition 5.15 (Comparison with the étale derived category)**.** *For a finite admissible ring A, the comparison functor $\nu_X^* : D(X_{ét}, A) \to D(X_{\mathrm{proét}}, A)$ identifies*

$$D(X_{ét}, A) = \mathrm{Loc}(D^b_{\mathrm{ctf}}(X_{ét}, A)) \simeq \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A)).$$

It preserves arbitrary direct sums. For every morphism $g : X \to Y$ of schemes of finite type over k and every $K \in D(X_{ét}, A)$, the comparison map (2.9) $\nu_Y^* Rg_{ét,*} K \to Rg_{\mathrm{proét},*} \nu_X^* K$ is an isomorphism, and ν^* commutes with inverse images and, by (2.10), with extension by zero along open immersions.

Proof. There are two statements: ν_X^* identifies the two localizing categories, and on the étale side the localizing category is everything. The first follows formally from the comparison results for the pro-étale site. For the second we exhibit compact generators of $D(X_{ét}, A)$ that lie in D^b_{ctf}.

An exact fully faithful functor that preserves direct sums identifies the localizing subcategories generated by corresponding generators. In our case, Proposition 2.49 shows that ν_X^* is such a functor on unbounded complexes and that it is compatible with direct images, and on D^b_{ctf} it is the equivalence of Corollary 2.44. Finally, inverse images commute with ν^* because both composites are inverse images along the same morphism of topoi.

For the equality with the whole étale derived category, let $u : U \to X$ range over a basis of quasi-compact separated étale objects. The free representable module sheaves $u_! A_U$ are flat constructible, hence belong to $D^b_{\mathrm{ctf}}(X_{ét}, A)$. We have $R\mathrm{Hom}_X(u_! A_U, K) = R\Gamma(U, K)$. It follows that their shifts detect zero objects of $D(X_{ét}, A)$. Indeed, if $H^n(U, K) = 0$ for all such U and all n, then every cohomology sheaf of K vanishes, being the sheafification of $U \mapsto H^n(U, K)$. These objects are also compact in all of $D(X_{ét}, A)$. The reason is that $R\Gamma(U, -)$ commutes with direct sums, since $\mathrm{cd}_\ell(U) < \infty$, as in the étale part of the proof of Proposition 5.14. A set of compact objects whose shifts detect zero objects generates a triangulated category with direct sums as a localizing subcategory. Hence these objects generate the entire étale derived category. ∎

Proper direct image already preserves D^b_{ctf} by Proposition 4.35. Its extension to the localizing categories therefore only needs preservation of arbitrary direct sums. By the comparison just proved, this reduces to the unbounded étale statement.

Corollary 5.16 (Proper direct image on the localizing categories)**.** *Let A be a finite admissible ring and let $p : X \to Y$ be proper. The functor Rp_* restricts to an exact functor*

$$Rp_* : \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A)) \longrightarrow \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, A))$$

that preserves arbitrary direct sums.

Proof. On unbounded étale derived categories, proper direct image preserves arbitrary direct sums by [Stacks, 0F0D]. Forgetting coefficients gives the A-linear statement, and Proposition 5.15 transports it to the pro-étale localizing categories.

Since Rp_* maps $D^b_{\mathrm{ctf}}(X, A)$ into $D^b_{\mathrm{ctf}}(Y, A)$ by Proposition 4.35, the objects $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))$ whose direct image belongs to $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, A))$ form a triangulated subcategory closed under direct sums and containing $D^b_{\mathrm{ctf}}(X, A)$, that is, all of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, A))$. ∎

5.4 Direct image with compact support

The trace formula needs a relative version of cohomology with compact supports, whose stalks are the groups H^i_c of the fibres. Consider first a locally compact Hausdorff space X with an open embedding $j : X \hookrightarrow \overline{X}$ into a compact space. A section of $j_!F$ is a section of F that vanishes near $\overline{X} \setminus X$, and $H^i_c(X, F) = H^i(\overline{X}, j_!F)$. For example, for $X = \mathbf{R}$ and $\overline{X}$ the circle, we find $H^1_c(\mathbf{R}, \mathbf{Z}) = \mathbf{Z}$.

The definition by sections with compact support has no direct analogue for schemes, because every scheme of finite type over k is quasi-compact, and the algebraic substitute for compactness is properness, which is a property of a morphism and not of a subset. The two-step form, however, does carry over. In it, the functor $j_!$ imposes compact support along the open immersion, while for a proper map p, as for a map of spaces with compact fibres, the direct image with compact supports is the ordinary direct image Rp_*. By proper base change, the result does not depend on the compactification (Proposition 5.26). Proper base change also identifies the stalks of $Rf_!$ with the compactly supported cohomology of the fibres (Corollary 12.26).

Let $f : X \to Y$ be a separated morphism of finite type. For a compactification $f = p \circ j$, with j an open immersion and p proper, we define $Rf_! = Rp_* \circ j_!$. We must show that this functor preserves direct sums. In the following subsection we take $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, so that $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda)) = \mathrm{Loc}(D^b_c(X, \Lambda))$.

5.4.1 Direct sums

To apply Brown representability, we need $Rf_!$ to preserve arbitrary direct sums. Since $j_!$ is a left adjoint, it does. The question is therefore whether Rp_* preserves direct sums for proper p. For complexes with a common lower and upper bound, this would follow from finiteness of $\mathrm{cd}_\ell(p)$. However, the summands of a direct sum in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ need not have any common bound. As in the proof of compactness, we use a uniform bound instead. Let d bound the fibre dimensions of p. We show that $R^q p_*$ vanishes on $\mathrm{IndCons}(X, \Lambda)$ for $q > 2d + 1$. This means that $\mathcal{H}^q(Rp_*K)$ depends only on the cohomology of K in degrees $[q - 2d - 1, q]$.

We prove the bound on sections over affine weakly contractible objects and then pass to colimits. For this we use the continuity of pro-étale cohomology, Corollary 2.29, applied to sheaves, which are complexes vanishing below degree 0. It says that for a quasi-compact and quasi-separated morphism p the functors $R^q p_*$ on sheaves of modules commute with filtered colimits. Since a direct sum is the filtered colimit of its finite subsums, they also commute with direct sums.

We also use the following description of $R^q p_*$. If $V \to Y$ is an affine weakly contractible object and $X_V = X \times_Y V$, then $\Gamma(V, -)$ is exact (Corollary corollary-evaluation-weakly-contractible-objects), and hence

$$(R^q p_* M)(V) = H^q((X_V)_{\mathrm{pro\acute{e}t}}, M|_{X_V}). \tag{5.5}$$

Since such V form a basis of $Y_{\mathrm{pro\acute{e}t}}$, a sheaf on $Y_{\mathrm{pro\acute{e}t}}$ vanishes if its sections over all of them vanish.

Lemma 5.17 (A uniform bound for proper direct image)**.** *Let $p : X \to Y$ be proper, and let d be an integer bounding the dimensions of all geometric fibres of p (for instance $d = \dim X$). For $M \in \mathrm{IndCons}(X, \Lambda)$,*

$$R^q p_* M = 0 \quad (q > 2d + 1), \qquad R^q p_* M \in \mathrm{IndCons}(Y, \Lambda) \quad (q \geq 0),$$

and $R^q p_* : \mathrm{IndCons}(X, \Lambda) \to \mathrm{IndCons}(Y, \Lambda)$ *preserves arbitrary direct sums.*

By a stratification we reduce the lemma to the extension by zero of a lisse $\mathcal{O}_E$-lattice on a stratum. Its reductions modulo π_E^n are torsion sheaves, for which the classical bound $2d$ holds. In the passage to the limit over n, the $\varprojlim^1$ term raises the bound by one, to $2d+1$. The remaining steps, inverting ℓ, extending scalars to $\overline{\mathbf{Q}}_\ell$ and passing to ind-objects, are filtered colimits. By Corollary 2.29, the bound survives each of them.

Proof of Lemma 5.17. Reduction to a lattice. The vanishing statement is stable under extensions: if $R^q p_*$ vanishes for $q > 2d+1$ on two sheaves, it vanishes on every extension of one by the other. By Lemma lemma-constructible-sheaves, every $M \in \mathrm{Cons}(X, \Lambda)$ is a finite extension of sheaves $a_! L$, with $a : S \hookrightarrow X$ a locally closed immersion and L lisse. Moreover, for $\Lambda = \overline{\mathbf{Q}}_\ell$, L is the scalar extension $\overline{\mathbf{Q}}_{\ell,X} \otimes_E L_E$ of a lisse E_S-module L_E for some finite $E/\mathbf{Q}_\ell$ (Lemma 3.17). For $\Lambda = E$, we simply put $L_E = L$. In both cases we may refine S by a finite stratification, since this replaces $a_! L$ by a finite extension of sheaves of the same form. By Lemma 3.24, after such a refinement L_E contains a lattice L_0, locally free of finite rank over $\mathcal{O}_{E,S}$, with $L_0[1/\ell] = L_E$. Hence it is enough to prove the bound for $a_! L_0$, and then to pass to $a_! L_E = (a_! L_0)[1/\ell]$ and, if $\Lambda = \overline{\mathbf{Q}}_\ell$, to its scalar extension $a_! L$.

In the passage to the limit below we use that the sheaf $a_! L_0$ is derived π_E-complete. To see this, note that a is a closed immersion composed with a quasi-compact open immersion. Closed pushforward is exact and preserves products [Stacks, 09BM], and extension by zero is exact and preserves limits [Stacks, 09BP]. Therefore both functors preserve the derived inverse limit that expresses the completeness of L_0.

The torsion bound. Over the objects V that compute $R^q p_*$, the reductions $a_!(L_0/\pi_E^n)$ have cohomology only up to degree $2d$. Fix an affine weakly contractible $V \to Y$ and let $p_V : X_V \to V$ be the base change of p, which is proper with the same bound d. Since L_0 is flat, $a_! L_0 \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n$ is the sheaf $a_!(L_0/\pi_E^n)$, which is the pullback to the pro-étale site of a classical constructible étale sheaf of $\mathcal{O}_E/\pi_E^n$-modules. Since the scheme V is not Noetherian, we use the proper vanishing theorem in the generality of [Stacks, 095U]: if a proper morphism has geometric fibres of dimension at most d, then its higher direct images of torsion sheaves vanish in degrees $q > 2d$. With the comparison [Stacks, 09A6] of étale and pro-étale direct images, this theorem for p_V says that $R^q p_{V,*}$ of this sheaf vanishes for $q > 2d$. As V is weakly contractible, $\Gamma(V, -)$ is exact on all pro-étale sheaves [Stacks, 0991]. Hence $H^q(X_V, -) = \Gamma(V, R^q p_{V,*}(-))$, and

$$H^q(X_V, a_!(L_0/\pi_E^n)) = 0 \quad (q > 2d).$$

The limit. Derived completeness lets us pass from the reductions to $a_! L_0$. Since restriction to X_V is exact and commutes with products, hence with derived inverse limits, completeness gives $(a_! L_0)|_{X_V} \simeq R\varprojlim_n a_!(L_0/\pi_E^n)|_{X_V}$. Derived global sections, too, commute with derived inverse limits, since a product of K-injective complexes is K-injective and sections preserve products. The Milnor exact sequence of Proposition 2.24

$$0 \to \varprojlim{}^1_n H^{q-1}(X_V, a_!(L_0/\pi_E^n)) \to H^q(X_V, a_! L_0) \to \varprojlim_n H^q(X_V, a_!(L_0/\pi_E^n)) \to 0$$

then shows that $H^q(X_V, a_! L_0) = 0$ for $q > 2d+1$, since for such q both outer terms vanish by the torsion bound.

Back to Λ. The remaining passages are filtered colimits. By Corollary 2.29, cohomology on the quasi-compact and quasi-separated scheme X_V commutes with them. For instance, the sheaf $(a_! L_0)[1/\ell]$ is the filtered colimit of $a_! L_0$ along multiplication by ℓ, and so it has the same

vanishing. For $\Lambda = \overline{\mathbf{Q}}_\ell$, scalar extension from E to $\overline{\mathbf{Q}}_\ell$ is the filtered colimit of the extensions to finite $E' \supset E$. Since E' is finite free over E, each of these is, as a sheaf of abelian groups, a finite direct sum of copies of the original sheaf. Hence the continuity lemma again gives the bound. As V was an arbitrary affine weakly contractible object, formula (5.5) shows that $R^q p_*(a_! L) = 0$ for $q > 2d+1$. By the first step, $R^q p_* M = 0$ for $q > 2d+1$ and every $M \in \mathrm{Cons}(X, \Lambda)$.

Passage to $\mathrm{IndCons}(X, \Lambda)$. For constructible M we have $R^q p_* M \in \mathrm{Cons}(Y, \Lambda)$, because Rp_* maps $D^b_c(X, \Lambda)$ into $D^b_c(Y, \Lambda)$ by Proposition 4.40. Moreover, every object of $\mathrm{IndCons}(X, \Lambda)$ is a filtered colimit of constructible sheaves, and $R^q p_*$ commutes with filtered colimits and direct sums by Corollary 2.29. All three assertions for $\mathrm{IndCons}(X, \Lambda)$ therefore follow. ∎

The bound $2d+1$ is not sharp. For example, for the structure map of a smooth projective curve over $\bar{k}$, where $d = 1$, the lemma only says that cohomology vanishes above degree 3, while in fact it vanishes above degree 2. The extra degree does no harm, because the arguments below only need a bound that does not depend on the sheaf. As in the compactness argument, such a bound passes from sheaves to unbounded complexes.

Lemma 5.18 (Proper direct image of unbounded complexes)**.** *Let $p : X \to Y$ be proper, and let d bound the dimensions of the geometric fibres of p. For integers a and b, write $D^{\leq b}_{\mathrm{IndCons}}(X, \Lambda)$ and $D^{\geq a}_{\mathrm{IndCons}}(X, \Lambda)$ for the objects of $D_{\mathrm{IndCons}}(X, \Lambda)$ (Corollary 5.11) with cohomology in degrees $\leq b$, respectively $\geq a$.*

1. *Rp_* maps $D_{\mathrm{IndCons}}(X, \Lambda)$ into $D_{\mathrm{IndCons}}(Y, \Lambda)$, with $Rp_* D^{\leq b}_{\mathrm{IndCons}}(X, \Lambda) \subset D^{\leq b+2d+1}_{\mathrm{IndCons}}(Y, \Lambda)$ and $Rp_* D^{\geq a}_{\mathrm{IndCons}}(X, \Lambda) \subset D^{\geq a}_{\mathrm{IndCons}}(Y, \Lambda)$.*
2. *For $K \in D_{\mathrm{IndCons}}(X, \Lambda)$ and every integer q, the truncation maps $K \leftarrow \tau_{\leq q} K \to \tau_{\geq q-2d-1} \tau_{\leq q} K$ induce an isomorphism in* $\mathrm{IndCons}(Y, \Lambda)$

$$\mathcal{H}^q(Rp_* K) \simeq \mathcal{H}^q(Rp_* \tau_{\geq q-2d-1} \tau_{\leq q} K). \tag{5.6}$$

3. *$Rp_* : D_{\mathrm{IndCons}}(X, \Lambda) \to D_{\mathrm{IndCons}}(Y, \Lambda)$ preserves arbitrary direct sums.*

Proof. For bounded below K everything is read off from the hypercohomology spectral sequence, which has only $2d+2$ nonzero columns. A general K, in turn, is the derived limit of its truncations $\tau_{\geq -n} K$, and the column bound shows that in each fixed degree the tower of direct images is eventually constant. The window (5.6) and the preservation of direct sums follow from this.

Bounded below K. The hypercohomology spectral sequence

$$E_2^{u,v} = R^u p_* \mathcal{H}^v(K) \Longrightarrow \mathcal{H}^{u+v}(Rp_* K)$$

converges. By Lemma 5.17, its only nonzero columns are $0 \leq u \leq 2d+1$, and their terms lie in $\mathrm{IndCons}(Y, \Lambda)$. So each $\mathcal{H}^n(Rp_* K)$ has a finite filtration whose graded pieces are subquotients of objects of $\mathrm{IndCons}(Y, \Lambda)$. Since this category is closed under kernels, cokernels and extensions by Proposition 5.9, we get $\mathcal{H}^n(Rp_* K) \in \mathrm{IndCons}(Y, \Lambda)$. Both amplitude bounds can be read off from the columns.

Arbitrary K. By Proposition 2.26, $K \simeq R\varprojlim_n \tau_{\geq -n} K$. Since Rp_* is a right adjoint, it preserves products and hence derived inverse limits. Thus $Rp_* K \simeq R\varprojlim_n Rp_* \tau_{\geq -n} K$, and it suffices to see that in each fixed degree the tower $Rp_* \tau_{\geq -n} K$ stabilizes. The fibre of $\tau_{\geq -n-1} K \to \tau_{\geq -n} K$ is $\mathcal{H}^{-n-1}(K)[n+1]$, a single ind-constructible sheaf in degree $-n-1$. By the bounded below case its direct image has cohomology only in degrees $[-n-1, -n-1+2d+1] = [-n-1, 2d-n]$. In particular it has none in degrees $q-1$, q, $q+1$ once $n > 2d+1-q$. For fixed q, the long exact sequences then show that the transition maps of the towers $\mathcal{H}^q(Rp_* \tau_{\geq -n} K)$ and

$\mathcal{H}^{q-1}(Rp_*\tau_{\geq -n}K)$ are isomorphisms for n large. Since an eventually constant tower has vanishing $\varprojlim^1$, the Milnor sequence of Proposition 2.24 gives

$$\mathcal{H}^q(Rp_*K) \simeq \mathcal{H}^q(Rp_*\tau_{\geq -n}K) \quad \text{for all sufficiently large } n.$$

The bounded below case, applied to the right-hand side, shows that $\mathcal{H}^q(Rp_*K) \in \mathrm{IndCons}(Y,\Lambda)$, with the bound $b+2d+1$ if $K \in D^{\leq b}_{\mathrm{IndCons}}(X,\Lambda)$. The lower bound needs no new argument, because an object of $D^{\geq a}_{\mathrm{IndCons}}$ is bounded below.

The isomorphism (5.6). It comes from the bounds, applied to the truncation triangles

$$\tau_{\leq q}K \to K \to \tau_{\geq q+1}K, \qquad \tau_{\leq q-2d-2}K \to \tau_{\leq q}K \to \tau_{\geq q-2d-1}\tau_{\leq q}K.$$

The direct image of $\tau_{\geq q+1}K$ lives in degrees $\geq q+1$, and that of $\tau_{\leq q-2d-2}K$ in degrees $\leq q-1$. The long exact cohomology sequences of the two triangles therefore yield isomorphisms $\mathcal{H}^q(Rp_*\tau_{\leq q}K) \simeq \mathcal{H}^q(Rp_*K)$ and $\mathcal{H}^q(Rp_*\tau_{\leq q}K) \simeq \mathcal{H}^q(Rp_*\tau_{\geq q-2d-1}\tau_{\leq q}K)$.

Direct sums, assertion (3). The window (5.6) reduces us to families with cohomology in a common finite interval, and for such a family (M_α) we show by induction on the length of the interval that the map $\bigoplus_\alpha Rp_*M_\alpha \to Rp_*(\bigoplus_\alpha M_\alpha)$ is an isomorphism. When the length is zero, each M_α is a shifted object of $\mathrm{IndCons}(X,\Lambda)$, and the claim is the assertion about direct sums in Lemma 5.17, in each degree. Then, for the inductive step, we take the direct sum of the truncation triangles that split off the top cohomology sheaf and apply the five lemma to the long exact cohomology sequences.

Now let (K_α) be arbitrary and fix q. Since direct sums of sheaves are exact, cohomology and truncations commute with direct sums. The objects $\tau_{\geq q-2d-1}\tau_{\leq q}K_\alpha$ have cohomology in the common interval $[q-2d-1,q]$. Applying (5.6) to each K_α and to $\bigoplus_\alpha K_\alpha$, and the first paragraph to the truncated family, we see that $\bigoplus_\alpha Rp_*K_\alpha \to Rp_*(\bigoplus_\alpha K_\alpha)$ is an isomorphism on $\mathcal{H}^q$. Since q is arbitrary, the map is an isomorphism. ∎

5.4.2 Integral coefficients

For $\mathcal{O}_E$ the rational arguments for compactness (Theorem 5.13) and for direct sums (Lemma 5.18) go through with one change. Integral constructible sheaves are no longer locally free on strata; for example, $\mathcal{O}_{E,X}/\pi_E$ is constructible. But $\mathcal{O}_E$ is a discrete valuation ring, so they have free resolutions of length one. The estimates only use the resulting Tor amplitude $[-1,0]$.

Lemma 5.19 (Structure of integral constructible complexes)**.**

1. *$D^b_c(X,\mathcal{O}_E)$ is a triangulated subcategory of $D(X_{\mathrm{pro\acute{e}t}},\mathcal{O}_E)$ of bounded objects, stable under truncation; its objects in degree 0 form the category $\mathrm{Cons}(X,\mathcal{O}_E)$, an essentially small abelian subcategory of $\mathrm{Mod}(X_{\mathrm{pro\acute{e}t}},\mathcal{O}_E)$ closed under extensions.*
2. *For $C \in \mathrm{Cons}(X,\mathcal{O}_E)$ there is a finite stratification on whose strata C is pro-étale locally $\mathcal{O}_{E,S} \otimes_{\mathcal{O}_E} P$, with P finitely generated.*
3. *On each such stratum C is dualizable of Tor amplitude $[-1,0]$, and its dual has Tor amplitude $[0,1]$.*

Proof. The conditions of Proposition 3.11, derived completeness and constructible reduction, are stable under shifts and cones, so $D^b_c(X,\mathcal{O}_E)$ is triangulated. By Proposition 3.10, on the strata of a finite stratification an object is pro-étale locally of the form $\mathcal{O}_{E,S} \otimes P^\bullet$ with $P^\bullet$ a bounded complex of finite free $\mathcal{O}_E$-modules. Since $\mathcal{O}_{E,S}$ is flat over $\mathcal{O}_E$, we get $H^i = \mathcal{O}_{E,S} \otimes H^i(P^\bullet)$ on

such a stratum. This local form of the cohomology sheaves shows that the truncations stay in D^b_c. It also gives (2), and then the abelian part of (1). In (1), essential smallness holds because a constructible sheaf is determined up to isomorphism by a stratification, local presentations, descent data and finitely many extension classes. Finally, for (3) we use that a finitely generated module over the discrete valuation ring $\mathcal{O}_E$ has a free resolution $0 \to \mathcal{O}_E^r \to \mathcal{O}_E^s \to P \to 0$ of length one. ∎

For instance, $\mathcal{O}_{E,X}/\pi_E$ is the cokernel of $\pi_E : \mathcal{O}_{E,X} \to \mathcal{O}_{E,X}$, so it has Tor amplitude $[-1, 0]$. Its dual $R\mathcal{H}om(\mathcal{O}_{E,X}/\pi_E, \mathcal{O}_{E,X}) = (\mathcal{O}_{E,X}/\pi_E)[-1]$ has Tor amplitude $[0, 1]$, as in (3).

The dévissage in the proof of compactness also needs that the operations it uses preserve $D^b_c(-, \mathcal{O}_E)$.

Lemma 5.20 (Inverse image and tensor product)**.** *Inverse image, $\otimes^{\mathbf{L}}_{\mathcal{O}_{E,X}}$ without completion, extension by zero along a quasi-compact open immersion and direct image along a closed immersion preserve $D^b_c(-, \mathcal{O}_E)$; on D^b_c inverse image agrees with the completed inverse image of the Stacks Project.*

Proof. The preservation statements follow from Lemma lemma-integral-models, since closed direct image is exact and preserves limits [Stacks, 09AA, 09BM]. The two inverse images agree on objects that are pro-étale locally of the form $\mathcal{O}_{E,S} \otimes P$ with P finitely generated, and by the local structure of Lemma lemma-structure-integral-constructible-complexes and (2) they therefore agree on D^b_c. ∎

With these two lemmas, the integral case of the two hypotheses of Brown representability can be checked by following the rational proofs.

Proposition 5.21 (Compactness and proper direct image)**.** *Every object of $D^b_{\mathrm{ctf}}(X, \mathcal{O}_E) = D^b_c(X, \mathcal{O}_E)$ is compact in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \mathcal{O}_E))$, and for proper $p : X \to Y$ the functor Rp_* preserves direct sums of objects of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \mathcal{O}_E))$.*

Proof. The rational proofs (Theorem 5.13, Lemmas 5.12 and Lemma 5.18) go through once we supply for $\mathcal{O}_E$ the four inputs they use: the abelian structure of Lemma lemma-structure-integral-constructible-complexes, continuity of $\mathrm{Ext}^i(C, -)$ for bounded below systems, a bound d_C with $\mathrm{Ext}^i(C, N) = 0$ for $i > d_C$ and N constructible, and a bound on the amplitude of Rp_*.

For continuity, take $C = a_!L$ on a stratum. By Lemma lemma-sections-support-closed-subscheme, $R\mathrm{Hom}(a_!L, -)$ is $R\Gamma(S, -)$ of the dual of L tensored with $u^*R\underline{\Gamma}_Z$. The proof of that formula uses only that L is dualizable, which holds by Lemma lemma-structure-integral-constructible-complexes, and so continuity follows from Corollary 2.29.

The bound d_C comes from counting degrees as in Lemma 5.6. Namely, the reductions of N lie in $[-1, 0]$, and the dual of L adds one degree. Next, the functors $R\Gamma$ and $R\underline{\Gamma}_Z$ add the cohomological dimensions of Definition 2.45. Finally, the derived limit over the reductions adds one more [Stacks, 08TB].

The same count bounds Rp_*. For fibres of dimension $\leq d$, the torsion bound [Stacks, 095U] gives $2d$, and the derived limit adds one. Hence Rp_* of a constructible sheaf lies in $[0, 2d + 1]$. ∎

These are the facts about $\mathcal{O}_E$ that we use in the construction of $Rf_!$ below.

5.4.3 Construction of $Rf_!$

From now on, Λ is any admissible coefficient ring. Lemmas 5.17 and Lemma 5.18 only treat E and $\overline{\mathbf{Q}}_\ell$, but we use them only through the fact that Rp_* preserves direct sums for proper p. For finite and integral coefficients this fact was proved separately, and the next proof collects the references.

Proper base change shows, by one and the same argument, that $Rp_* \circ j_!$ does not depend on the compactification and is compatible with composition and base change. The comparison maps are maps out of an extension by zero $j_!K$. They are therefore determined by their restriction to X, where they are the identity. On the closed complement of X, proper base change shows that both sides vanish, and so the map is an isomorphism. For instance, if $h : Z' \to Z$ is proper and $h^{-1}(X) = X$, then $Rh_* j'_! K$ restricted to $Z \setminus X$ is the direct image of the restriction of $j'_! K$ to $Z' \setminus X$, which is zero. Since proper base change is only known on D^b_{ctf}, we first show that all the functors involved preserve direct sums, so that it suffices to check isomorphisms on D^b_{ctf} (Lemma 5.23).

That both factors of $Rp_* \circ j_!$ preserve the localizing categories comes from the definition of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ in the following form: if the objects of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ with some property form a triangulated subcategory that is closed under direct sums and contains $D^b_{\mathrm{ctf}}(X, \Lambda)$, then every object of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ has the property.

Proposition 5.22 (Proper direct image and extension by zero)**.** *For every admissible Λ, if $p : X \to Y$ is proper, $Rp_* : D(X_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(Y_{\mathrm{pro\acute{e}t}}, \Lambda)$ restricts to an exact functor $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$ that preserves direct sums. If $j : U \to X$ is an open immersion, $j_! : D(U_{\mathrm{pro\acute{e}t}}, \Lambda) \to D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$ restricts to an exact functor $\mathrm{Loc}(D^b_{\mathrm{ctf}}(U, \Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ that preserves direct sums and maps $D^b_{\mathrm{ctf}}(U, \Lambda)$ into $D^b_{\mathrm{ctf}}(X, \Lambda)$.*

Proof. For each functor we need two facts: it preserves direct sums, and it maps D^b_{ctf} into D^b_{ctf}. The rest then follows from the definition of the localizing categories, in the form recalled before the proposition.

For Rp_* with E and $\overline{\mathbf{Q}}_\ell$, Lemma lemma-proper-direct-image-unbounded-complexes shows that Rp_* preserves direct sums on $D_{\mathrm{IndCons}}(X, \Lambda) \supset \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$, and Proposition 4.40 shows that Rp_* preserves $D^b_{\mathrm{ctf}} = D^b_c$. For finite A, these two facts are Proposition 4.35 and Corollary 5.16; for $\mathcal{O}_E$, they are Proposition 5.21 and Proposition 4.40. In each case, the objects $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ with $Rp_*K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$ form a triangulated subcategory closed under direct sums and containing $D^b_{\mathrm{ctf}}(X, \Lambda)$, and hence they form all of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$.

Extension by zero is exact and left adjoint to j^*, so it preserves direct sums. It preserves constructibility, since we may add the closed complement as a stratum on which the sheaf is zero. For finite A it also preserves finite Tor amplitude, as one checks on stalks, while for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ we have $D^b_{\mathrm{ctf}} = D^b_c$, so constructibility suffices; for $\mathcal{O}_E$ this is Lemma 5.20. With this, the localizing-subcategory argument shows that $j_!$ restricts as stated. ∎

The proper base change map and the comparison maps between compactifications are natural transformations between exact functors on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-, \Lambda))$ that preserve direct sums. To show that such a transformation is an isomorphism, it suffices to check this on $D^b_{\mathrm{ctf}}(-, \Lambda)$.

Lemma 5.23 (Checking isomorphisms on constructible complexes)**.** *Let Λ be admissible, let $F_1, F_2 : \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$ be exact functors preserving direct sums, and let $\eta : F_1 \to F_2$ be a natural transformation of exact functors. If η_K is an isomorphism for every $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$, then it is an isomorphism for every $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$.*

Proof. The objects K for which η_K is an isomorphism are closed under shifts and cones by the five lemma, and under direct sums because F_1 and F_2 preserve them. Since they contain $D^b_{\mathrm{ctf}}(X,\Lambda)$, they form all of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$. ∎

For a cartesian square with p proper, the base change map $g^*Rp_* \to Rp'_*g'^*$ of (4.5) is a map of functors $D(X_{\mathrm{pro\acute{e}t}},\Lambda) \to D(Y'_{\mathrm{pro\acute{e}t}},\Lambda)$. The lemma lets us pass from $D^b_{\mathrm{ctf}}(X,\Lambda)$, where it is an isomorphism, to $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$.

Proposition 5.24 (Proper base change on the localizing categories)**.** *Let Λ be admissible. Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow p' & & \downarrow p \\ Y' & \xrightarrow{g} & Y \end{array}$$

with p proper. The base change map $g^*Rp_*K \to Rp'_*g'^*K$ is an isomorphism for every $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$.

Proof. On $D^b_{\mathrm{ctf}}(X,\Lambda)$ the base change map is an isomorphism by Proposition 4.40, so by Lemma 5.23 it suffices to show that g^*Rp_* and $Rp'_*g'^*$ are exact functors $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y',\Lambda))$ preserving direct sums. For Rp_* and Rp'_* this is Proposition 5.22, so it remains to treat inverse image. It is exact and preserves direct sums, and it sends $D^b_{\mathrm{ctf}}(-,\Lambda)$ into $D^b_{\mathrm{ctf}}(-,\Lambda)$. Indeed, for finite A this holds because of bounded flat constructible representatives, and for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ it holds because pullback preserves the local description of constructible sheaves on strata (Lemma 5.20 in the integral case). Given this, the localizing-subcategory argument before Proposition 5.22 shows that inverse image also preserves $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,\Lambda))$. ∎

Remark 5.25 (Compactifications)**.**

1. By Nagata's theorem (Theorem A.2), every separated morphism of finite type $f : X \to Y$ with Y qcqs has a compactification, that is, a factorization $f = p \circ j$ with $j : X \to \overline{X}$ a quasi-compact open immersion and $p : \overline{X} \to Y$ proper.
2. In comparisons we may replace $\overline{X}$ by the scheme-theoretic closure of X, so that X is dense, and by Lemma A.5 this does not change $Rp_*j_!$.

The comparisons between compactifications below are the proper direct images of the unique maps out of $j_!K$ that restrict to the identity on X. They are unique because, for an open immersion $j : X \to Z$, the functor $j_!$ is left adjoint to j^* on the pro-étale derived categories, so that a map out of $j_!K$ is determined by its restriction to X.

Proposition 5.26 (Construction and independence of $Rf_!$)**.** *Let Λ be admissible and let $f : X \to Y$ be separated of finite type, with compactification $f = p \circ j$ as above. The functor*

$$Rf_! = Rp_* \circ j_! : \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$$

is exact, preserves direct sums, and maps $D^b_{\mathrm{ctf}}(X,\Lambda)$ into $D^b_{\mathrm{ctf}}(Y,\Lambda)$. Two compactifications give canonically isomorphic functors, and these isomorphisms satisfy the cocycle condition.

Proof. For a single compactification there is nothing new to prove, and two compactifications are compared through a third one that refines both, by the argument described at the start of this subsection.

The properties of $Rp_* \circ j_!$ for a single compactification follow from Proposition 5.22 and the preservation of $D^b_{\mathrm{ctf}}(-,\Lambda)$ by proper direct image (Proposition 4.40).

For independence it suffices to compare a compactification with a refinement of it. Indeed, the schematic closure of X in a compactification is a refinement of it, since its inverse image over X is X. Moreover, any two dense compactifications have a common refinement (Proposition A.7).

Refinements. Let $j : X \to Z$ and $j' : X \to Z'$ be compactifications over Y, and let $h : Z' \to Z$ be a refinement (Definition A.1), that is, a proper Y-morphism with $h \circ j' = j$ and $h^{-1}(X) = X$. To construct the comparison map, note that restriction to the open subscheme X commutes with Rh_*. This holds because $h^{-1}(X) = X$ and restriction is exact and preserves K-injectives, its left adjoint $j_!$ being exact. As $j'^* j'_! K = K$, we get $j^* Rh_* j'_! K \simeq K$. By adjunction this isomorphism corresponds to a map

$$e_{h,K} : j_! K \longrightarrow Rh_* j'_! K \tag{5.7}$$

whose restriction to X is the identity. We claim that it is an isomorphism for $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$. By Proposition 5.22, both sides are exact functors of K preserving direct sums, so by Lemma 5.23 we may assume that $K \in D^b_{\mathrm{ctf}}(X,\Lambda)$, where proper base change is available.

By the exact sequence for an open and its closed complement, applied in each degree [Stacks, 09AH], a morphism in $D(Z_{\mathrm{pro\acute{e}t}},\Lambda)$ that is an isomorphism on X and on $Z \setminus X$ is an isomorphism. On X our map is the identity, so it remains to see that on $Z \setminus X$ both sides vanish. For $j_! K$ this holds by definition of extension by zero, while for $Rh_* j'_! K$ we use proper base change on D^b_{ctf} (Proposition 5.24) for the inclusion of $Z \setminus X$ into Z. It shows that the restriction of $Rh_* j'_! K$ is the direct image of the restriction of $j'_! K$ to $h^{-1}(Z \setminus X) = Z' \setminus X$, which is zero. This proves the claim. Applying Rp_* and using $R(p \circ h)_* = Rp_* Rh_*$, we obtain an isomorphism between the two candidate functors $Rf_!$, namely $Rp_* j_!$ and $R(ph)_* j'_!$.

If $Z'' \xrightarrow{k} Z' \xrightarrow{h} Z$ are two successive refinements, with $j'' : X \to Z''$ the open immersion, then $Rh_*(e_{k,K}) \circ e_{h,K}$ and $e_{hk,K}$ are maps from $j_! K$ to $R(hk)_* j''_! K$ that both restrict to the identity on X. By the observation before the proposition, they are equal.

Common refinements. Two arbitrary compactifications need not be related by a refinement, but they have a common one. Indeed, given dense compactifications Z_1 and Z_2, Proposition A.7 provides a dense compactification Z with proper maps $h_i : Z \to Z_i$ satisfying $h_i^{-1}(X) = X$, and similarly for finitely many compactifications. The refinement isomorphisms for h_1 and h_2 then compose to an isomorphism between the functors defined by Z_1 and Z_2, and transitivity of the refinement maps on a common refinement shows that this comparison does not depend on the chosen refinement and satisfies the cocycle condition. ∎

Example 5.27 (A singular compactification of the line)**.** Over k, let $C \subset \mathbf{P}^2$ be the cuspidal cubic $zy^2 = x^3$, with cusp $c = (0:0:1)$. Its normalization is the map $h : \mathbf{P}^1 \to C$, $(s:u) \mapsto (su^2 : u^3 : s^3)$. It sends $u = 0$ to the cusp and the affine line $u \neq 0$ isomorphically onto $C \setminus \{c\}$. So $X = \mathbf{A}^1$ has the two compactifications $\mathbf{P}^1$ and C. Since $h^{-1}(C \setminus \{c\}) = \mathbf{A}^1$, the map h is a refinement. The refinement map $j_!\Lambda \to Rh_* j'_!\Lambda$ is the identity on X, and at the cusp both stalks vanish: for the right side, proper base change shows that the stalk of $Rh_* j'_!\Lambda$ at c is the cohomology of $j'_!\Lambda$ on the single point $h^{-1}(c) = \{\infty\}$, where $j'_!\Lambda$ is zero. So the singularity of C does not affect $R\Gamma_c(\mathbf{A}^1,\Lambda)$.

To make $Rf_!$ functorial in f, and for the base change theorems later on, we need composition and base change isomorphisms. The base change map is defined as follows. Let $f : X \to Y$ be separated of finite type with compactification $f = p \circ j$, let $v : Y' \to Y$ be a morphism, and let $f' : X' \to Y'$, $u : X' \to X$, $p' : \overline{X}' \to Y'$, $j' : X' \to \overline{X}'$ and $v' : \overline{X}' \to \overline{X}$ be the base changes, so that $f' = p' \circ j'$ is a compactification. For $K \in D(X_{\text{pro\'et}}, \Lambda)$, the isomorphism $j'^* v'^* j_! K \simeq u^* j^* j_! K = u^* K$ corresponds under $j'_! \dashv j'^*$ to a map $j'_! u^* K \to v'^* j_! K$. This map is an isomorphism: it is the identity on the open X', and both sides vanish on the closed complement $v'^{-1}(\overline{X} \setminus X) = \overline{X}' \setminus X'$ [Stacks, 09AH]. The base change map for $Rf_!$ is the composite

$$v^* Rp_* j_! K \longrightarrow Rp'_* v'^* j_! K \simeq Rp'_* j'_! u^* K \tag{5.8}$$

of the base change map (4.5) for p and Rp'_* of the inverse of this isomorphism.

Proposition 5.28 (Composition and base change for $Rf_!$)**.** *For every admissible Λ, the following statements hold on* $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-, \Lambda))$ *and restrict to* $D^b_{\mathrm{ctf}}(-, \Lambda)$*.*

1. *For separated morphisms of finite type $X \xrightarrow{f} Y \xrightarrow{g} Z$ there is a canonical isomorphism $Rg_! \circ Rf_! \simeq R(gf)_!$. These isomorphisms are associative and compatible with the identity $R(\mathrm{id})_! = \mathrm{id}$.*
2. *Let $v : Y' \to Y$ be a morphism, $f' : X' \to Y'$ the base change of f along v, and $u : X' \to X$ the projection. The base change map (5.8) is an isomorphism $v^* Rf_! \simeq Rf'_! u^*$. It does not depend on the compactification, and it is compatible with composing cartesian squares and with the composition isomorphisms of (1).*

Proof. For composition, the only new ingredient is a map that moves an extension by zero past a proper direct image. It is an isomorphism by the same argument as for refinements: it restricts to the identity on an open, and both sides vanish on the closed complement by proper base change. For base change of $Rf_!$ we combine proper base change with the elementary fact that extension by zero commutes with pullback.

The composition isomorphism. Choose compactifications $f = p \circ j$ and $g = q \circ j_Y$, with $j : X \to \overline{X}$ and $j_Y : Y \to \overline{Y}$ open and p, q proper. In the composite $Rq_* j_{Y!} Rp_* j_!$, the middle factor $j_{Y!} Rp_*$ applies a proper direct image before an extension by zero, and we want to interchange the two. To do this, compactify the morphism $j_Y \circ p : \overline{X} \to \overline{Y}$ as $\overline{X} \xrightarrow{j'} W \xrightarrow{r} \overline{Y}$, with j' open, r proper and $\overline{X}$ dense in W. Then $r^{-1}(Y) = \overline{X}$, because the open immersion $\overline{X} \to r^{-1}(Y)$ is a morphism from a proper Y-scheme to a separated Y-scheme, hence proper, and its image is closed, open and dense.

The interchange map is defined as follows. For M in $D(\overline{X}_{\text{pro\'et}}, \Lambda)$, restriction to the open $Y \subset \overline{Y}$ identifies $j_Y^* Rr_* j'_! M$ with $Rp_* M$, because $r^{-1}(Y) = \overline{X}$, the restriction of r over Y is p, and $j'_! M$ restricts to M on $\overline{X}$. By the adjunction $j_{Y!} \dashv j_Y^*$, we obtain a map

$$\iota_M : j_{Y!} Rp_* M \longrightarrow Rr_* j'_! M, \tag{5.9}$$

which is the unique map restricting to the identity of $Rp_* M$ over Y.

For $M \in D^b_{\mathrm{ctf}}(\overline{X}, \Lambda)$, this map is an isomorphism. Indeed, over Y it is the identity, and over $\overline{Y} \setminus Y$ both sides vanish. For the right-hand side, the vanishing follows from proper base change on D^b_{ctf} (Proposition 5.24) for the inclusion of $\overline{Y} \setminus Y$ into $\overline{Y}$, since $j'_! M$ vanishes on $r^{-1}(\overline{Y} \setminus Y) = W \setminus \overline{X}$. Both sides preserve direct sums on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(\overline{X}, \Lambda))$ (Proposition 5.22). Hence Lemma 5.23 shows that ι_M is an isomorphism for all $M \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(\overline{X}, \Lambda))$. For $M = j_! K$ it gives

$$Rq_*j_{Y!}Rp_*j_!K \simeq Rq_*Rr_*j'_!j_!K = R(qr)_*(j'j)_!K,$$

and the right side is $R(gf)_!K$ computed with the compactification $gf = (qr) \circ (j'j)$.

Verification of canonicity and associativity. Every map used is characterized by its restriction to an open subscheme, since the refinement maps (5.7) and the maps ι_M restrict to identities. For example, replacing any of the chosen compactifications by a refinement commutes with ι_M, because both composites restrict to the same identity and are then equal by adjunction. For associativity, given three composable morphisms, we choose common refinements (by graph closure) of the finitely many compactifications that occur in the composites of two of them. On these, each of the two bracketings is built from composites of proper direct images, composites of extensions by zero, and maps of the form ι_M. The first two kinds of pieces are associative: the isomorphisms $Rq_*Rr_* \simeq R(qr)_*$ are associative because they are the isomorphisms of derived functors of composites of left exact functors (equivalently, they are adjoint to the composition isomorphisms of the inverse images), and composition of extensions by zero is associative by adjunction. Moreover, interchanging two adjacent open and proper steps in either order gives the same map by the uniqueness of ι_M. So the two bracketings agree. Similarly, the unit conditions are the special case where one compactification is the identity. Finally, transitivity of the refinement maps carries these equalities over to the originally chosen compactifications.

Base change. The base change map for $Rf_! = Rp_*j_!$ is built from one for each factor, so we treat $j_!$ and Rp_* separately. Since open immersions and proper morphisms are stable under base change, we may pull back a compactification $f = p \circ j$ along v to a compactification $f' = p' \circ j'$, where now j' and p' denote the base changes of j and p. For $j_!$, extension by zero commutes with inverse image, as shown before the proposition. Combining this with proper base change on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(\overline{X}, \Lambda))$ (Proposition 5.24), we obtain the isomorphism $v^*Rp_*j_! \simeq Rp'_*v'^*j_! \simeq Rp'_*j'_!u^*$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$. This isomorphism is compatible with change of compactification because the refinement maps are characterized by their restriction to the open subscheme. It is also compatible with composing two cartesian squares, because both factors are. For the first factor, a map between extensions by zero from an open is determined by its restriction to that open, since $\mathrm{Hom}(j_!B, j_!B') = \mathrm{Hom}(B, B')$. For the second, a base change map is the adjoint transpose of a counit, and the counit of a composite adjunction is the composite of the counits. Finally, compatibility with the composition isomorphisms of (1) follows in the same way as canonicity: both composites are built from refinement maps, maps ι_M and proper base change maps, and they agree because their restrictions to the open part are the same identity. ∎

On a space, a section with compact support is in particular a section, and so compactly supported cohomology maps to ordinary cohomology. The algebraic counterpart is a map $Rf_! \to Rf_*$, which is an isomorphism for proper f, and we need it to be compatible with the composition maps just constructed: in Section 6.1 this is one of the compatibilities that the comparison with the étale site has to respect.

Proposition 5.29 (Forgetting supports and composition)**.** *There is a natural map $\eta_f : Rf_! \to Rf_*$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$. For a compactification $f = pj$ it is*

$$Rp_*j_! \longrightarrow Rp_*Rj_* = Rf_*,$$

where $j_! \to Rj_*$ is adjoint to the inverse of the counit $j^*Rj_* \xrightarrow{\sim} \mathrm{id}$. It is independent of the compactification, and the square

$$\begin{array}{ccc} Rg_!Rf_! & \xrightarrow{c^!_{g,f}} & R(gf)_! \\ {\scriptstyle Rg_*(\eta_f)\circ(\eta_g Rf_!)}\downarrow & & \downarrow{\scriptstyle \eta_{gf}} \\ Rg_*Rf_* & \xrightarrow{c^*_{g,f}} & R(gf)_* \end{array}$$

commutes. Here $c^!_{g,f}$ is the composition map of Proposition 5.28, and $c^*_{g,f}$ is the composition map for ordinary direct images. If f is proper, η_f is an isomorphism; under these isomorphisms the two composition maps agree whenever f and g are proper.

Proof. All maps involved are maps out of an extension by zero, so each of them is determined by its restriction to the open subscheme, where it is an identity or an inverse counit. We use this repeatedly.

Independence of the compactification reduces to a refinement $h : Z' \to Z$. The two maps $j_!K \to Rh_*Rj'_*K = Rj_*K$ obtained from (5.7) and from $j_!K \to Rj_*K$ have the same adjoint under $j_! \dashv j^*$, namely the inverse of $j^*Rj_*K \to K$. Hence they are equal. Applying Rp_* gives independence.

The square is checked on the compactifications that define $c^!_{g,f}$ and the interchange map (5.9). Apart from composition and naturality squares for ordinary direct images, it comes down to comparing the two maps

$$j_{Y!}Rp_*M \longrightarrow Rr_*j'_!M \longrightarrow Rr_*Rj'_*M = Rj_{Y*}Rp_*M, \qquad j_{Y!}Rp_*M \longrightarrow Rj_{Y*}Rp_*M.$$

Their adjoints under $j_{Y!} \dashv j^*_Y$ are both the inverse counit for $j^*_Y \dashv Rj_{Y*}$, because the interchange map restricts to the identity on Y. So the two maps agree. Applying Rq_* with $M = j_!K$, and using naturality for $j_!K \to Rj_*K$, we get the displayed square for the chosen compactifications. The refinement calculation extends it to any choices. If f is proper, the compactification with $j = \mathrm{id}$ has η_f equal to the identity, and the last assertion follows from the square. ∎

5.5 Brown representability

Let Λ be an admissible coefficient ring and fix N in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$. Since the functor $Rf_!$ preserves direct sums (Proposition 5.26), the functor $M \mapsto \mathrm{Hom}(Rf_!M, N)$ turns direct sums into products. If the objects of $D^b_{\mathrm{ctf}}(X,\Lambda)$ are compact generators of the localizing category, Brown representability produces an object $f^!N$ representing it. The next lemma checks this hypothesis for every admissible Λ; for $\mathcal{O}_E$ it uses the compactness proved in Subsection 5.4.2 by reduction to torsion coefficients.

Compactness enters a second time. The functor $Rf_!$ also maps $D^b_{\mathrm{ctf}}(X,\Lambda)$ into $D^b_{\mathrm{ctf}}(Y,\Lambda)$, so it sends the compact generators to compact objects. For this reason the adjoint $f^!$ preserves direct sums (Theorem 5.31).

Lemma 5.30 (Hypotheses of Brown representability)**.** *For every admissible Λ, the category $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ is locally small and triangulated, has arbitrary direct sums, and has split idempotents. The thick category $D^b_{\mathrm{ctf}}(X,\Lambda)$ is essentially small, and a small skeleton is a set of compact generators of $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$.*

Proof. The set-theoretic conditions are general facts about derived categories of Grothendieck abelian categories. Compactness, on the other hand, was proved separately for each kind of coefficients, and generation follows from the definition of the localizing category.

As the derived category of a Grothendieck abelian category of module sheaves, the ambient category is locally small, and so is its full subcategory $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,\Lambda))$. Moreover, the latter has direct sums by definition, so its idempotents split [Stacks, 05QW]. Finally, the subcategory $D^b_{\mathrm{ctf}}(X,\Lambda)$ is thick, since constructibility and finite Tor amplitude are preserved by shifts, cones and direct summands.

To see that $D^b_{\mathrm{ctf}}(X,\Lambda)$ is essentially small, note first that the constructible sheaves in the heart have a set of representatives for each admissible Λ. A bounded complex with such cohomology is built from its cohomology sheaves by finitely many extensions, formed with truncations in the ambient category. Each extension step is classified by an element of a Hom set of the ambient category, and these Hom sets are sets because the ambient category is locally small. Hence these complexes form an essentially small category, of which $D^b_{\mathrm{ctf}}(X,\Lambda)$ is a full subcategory.

Compactness for finite A is Proposition 5.14. For E and $\overline{\mathbf{Q}}_\ell$ it is Theorem 5.13, and for $\mathcal{O}_E$ it is Proposition 5.21; in the integral and rational cases, $D^b_{\mathrm{ctf}} = D^b_c$.

For generation, suppose that $K \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ receives no nonzero maps from shifts of objects of a small skeleton. The objects M with $\mathrm{Hom}(M[n],K)=0$ for all n form a localizing subcategory containing D^b_{ctf}. Hence this subcategory contains K. Then $\mathrm{id}_K = 0$, and so $K = 0$. ∎

With the hypotheses checked, the theorem below follows from Brown representability in the form recalled at the start of the chapter.

Theorem 5.31 (Existence of the exceptional inverse image)**.** *Let $\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$ be any admissible coefficient ring. For every separated morphism of finite type $f : X \to Y$, the coproduct-preserving functor $Rf_! : \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$ has an exact right adjoint*

$$f^! : \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda)) \longrightarrow \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda)), \qquad \mathrm{Hom}(Rf_!M, N) \simeq \mathrm{Hom}(M, f^!N),$$

for $M \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ and $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$. It is unique up to unique isomorphism compatible with the adjunction, and it preserves arbitrary direct sums.

Proof. We construct $f^!N$ one object at a time as the object representing maps out of $Rf_!(-)$ into N, and then check direct sums on the compact generators.

For $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$, the functor

$$\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))^{\mathrm{op}} \longrightarrow \mathrm{Ab}, \qquad M \longmapsto \mathrm{Hom}(Rf_!M, N)$$

is cohomological and sends direct sums to products. By Lemma 5.30, $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ is compactly generated, and Brown representability [Stacks, 0A8F] says that on such a category every cohomological functor sending direct sums to products is representable, say by $f^!N$. By the Yoneda lemma, these objects form a right adjoint, and a right adjoint of an exact functor of triangulated categories is exact [Stacks, 0A8G].

To see that $f^!$ preserves direct sums, we test the canonical map $\bigoplus_\alpha f^!N_\alpha \to f^!(\bigoplus_\alpha N_\alpha)$ on a compact generator $G \in D^b_{\mathrm{ctf}}(X,\Lambda)$ and its shifts. On the source, compactness of G and adjunction give

$$\mathrm{Hom}\Big(G[n], \bigoplus_\alpha f^!N_\alpha\Big) = \bigoplus_\alpha \mathrm{Hom}(G[n], f^!N_\alpha) = \bigoplus_\alpha \mathrm{Hom}(Rf_!G[n], N_\alpha).$$

On the target, adjunction and compactness of $Rf_!G \in D^b_{\mathrm{ctf}}(Y,\Lambda)$ give the same group. The map therefore induces bijections on $\mathrm{Hom}(G[n],-)$ for all compact generators G and all n. Applying the detection property to its cone, we see that it is an isomorphism. ∎

5.6 Composition and restriction

Brown representability gives $f^!$ one morphism at a time. To compute it in Chapter 6, however, we factor f locally. The two are linked by uniqueness of adjoints: an isomorphism between composites of left adjoints, such as $Rg_! \circ Rf_! \simeq R(gf)_!$ or $Rf_! = Rp_* \circ j_!$, gives an isomorphism between the right adjoints, composed in the opposite order. Since $j^! = j^*$ for an open immersion j, it follows that $f^!$ is local on X and on Y. For example, a smooth f is locally the composite of an étale map and a projection $\mathbf{A}^n_Y \to Y$. By composition and locality, $f^!$ is then determined by its values on these two kinds of maps.

Base change for $f^!$ does not follow in this way. Passing to right adjoints in the base change isomorphism $v^* Rf_! \simeq Rf'_! u^*$ relates $f^! Rv_*$ and $Ru_* f'^!$. These involve direct images, not the inverse images $u^* f^!$ and $f'^! v^*$ that one wants to compare. We prove base change for smooth morphisms in Proposition 6.9. That $f^!$ preserves D^b_{ctf} is proved in Theorems 6.20 and Theorem 6.27.

Proposition 5.32 (Composition, open restriction and compactifications for $f^!$)**.** *For every admissible Λ, the adjoints $f^!$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,\Lambda))$ have the following canonical compatibilities.*

1. *For composable $X \xrightarrow{f} Y \xrightarrow{g} Z$ there are canonical isomorphisms $(gf)^! \simeq f^! g^!$, associative and compatible with the identity.*
2. *For an open immersion $j : U \to X$, $j^! \simeq j^*$.*
3. *Let $v : V \to Y$ be an open immersion, $X_V = f^{-1}(V)$ with inclusion $u : X_V \to X$, and $f_V : X_V \to V$ the restriction of f. Then $u^* f^! \simeq f_V^! v^*$.*
4. *If $f = p \circ j$ is a compactification, then $f^! \simeq j^* \circ p^!$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$.*

In particular, $f^!$ can be computed locally on X and on Y for open coverings.

Proof. Each statement is the right adjoint of a statement about lower shriek functors already proved, and we conclude by uniqueness of adjoints.

(1) For $M \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ and $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Z,\Lambda))$,

$$\mathrm{Hom}(M, f^! g^! N) \simeq \mathrm{Hom}(Rf_! M, g^! N) \simeq \mathrm{Hom}(Rg_! Rf_! M, N) \simeq \mathrm{Hom}(R(gf)_! M, N),$$

where the last step holds by Proposition 5.28. By the Yoneda lemma, $f^! g^! \simeq (gf)^!$. Passing to right adjoints transforms the associativity and identity constraints of Proposition proposition-composition-base-change-rf-shriek into those for $f^!$.

(2) For an open immersion, $Rj_! = j_!$ (take the trivial compactification $p = \mathrm{id}$), and $j_! \dashv j^*$ on the localizing categories; here we use that j^* preserves the localizing categories, as inverse images do (see the proof of Proposition 5.24). By uniqueness of right adjoints, $j^! \simeq j^*$.

(3) Since $f \circ u = v \circ f_V$, parts (1) and (2) show that $u^* f^! \simeq (fu)^! = (v f_V)^! \simeq f_V^! v^*$. For an open $j : U \to X$, parts (1) and (2) also give $j^* f^! \simeq (fj)^!$. These two isomorphisms show that $f^!$ is local on X and on Y.

(4) On $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,\Lambda))$, j^* is right adjoint to $j_!$ and $p^!$ to $Rp_* = Rp_!$, so $j^* \circ p^!$ and $f^!$ are both right adjoint to $Rp_* \circ j_! = Rf_!$, and they are isomorphic by uniqueness of right adjoints. ∎

Once we know that $f^!$ preserves $D^b_{\mathrm{ctf}}(-,\Lambda)$ (Chapter 6), these isomorphisms restrict to the full subcategories $D^b_{\mathrm{ctf}}(-,\Lambda)$, and the same statements hold for the adjunction between $D^b_{\mathrm{ctf}}(X,\Lambda)$ and $D^b_{\mathrm{ctf}}(Y,\Lambda)$.

6 Constructibility of $f^!$

Brown representability gave us $f^!$ (Theorem 5.31), but only as a functor

$$f^! : \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda)) \longrightarrow \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$$

for $f : X \to Y$ separated of finite type and Λ an admissible coefficient ring. It tells us nothing about the values of $f^!$. A priori, $f^!$ of a constructible complex is just some object of the large category on the right, possibly unbounded and with cohomology sheaves that are not constructible.

The rest of the notes needs more than this. For example, Verdier duality (Chapter 10) says that $\mathbb{D}_X = R\mathcal{H}om(-, a^!_X\Lambda)$ is an involution of $D^b_{\mathrm{ctf}}(X,\Lambda)$, and this makes sense only if the dualizing complex $a^!_X\Lambda$ lies in $D^b_{\mathrm{ctf}}(X,\Lambda)$. Similarly, the local terms of the trace formula are defined through the isomorphism $R\mathcal{H}om(c_2^*K, c_1^!K) \simeq c^!(K \boxtimes \mathbb{D}_X K)$ for a correspondence $c = (c_1, c_2)$ (Lemma 11.7), which also needs $f^!$ to preserve constructible complexes. In this chapter we prove what both need (part (4) of Theorem 5.2): $f^!$ maps $D^b_{\mathrm{ctf}}(Y,\Lambda)$ into $D^b_{\mathrm{ctf}}(X,\Lambda)$. A universal property cannot give this, so we have to compute $f^!$, and the computation is local.

By Proposition 5.32, the functor $f^!$ is compatible with composition, equals j^* for an open immersion j, and is local on X and on Y. We may therefore assume that Y is affine and that f is a closed immersion into some $\mathbf{A}^n_Y$ followed by the projection to Y. Since $\mathbf{A}^n_Y$ is open in $(\mathbf{P}^1_Y)^n$, and $(\mathbf{P}^1_Y)^n \to Y$ is a composite of n projections $\mathbf{P}^1_T \to T$, two cases remain.

For the projection $p : \mathbf{P}^1_T \to T$ we can write the adjunction down, and we find $p^! = p^*(1)[2]$ (Proposition 6.7). Here the counit is the trace $Rp_*\Lambda(1)[2] \to \Lambda$, and the unit is the cycle class of the diagonal in $\mathbf{P}^1_T \times_T \mathbf{P}^1_T$. The triangle identities then hold because the diagonal meets each fibre of either projection in a single point. For a closed immersion $i : Z \hookrightarrow X$ with open complement $j : U \hookrightarrow X$, on the other hand, the functor $i^!$ takes sections with support in Z, and it sits in the localization triangle

$$i_*i^!K \longrightarrow K \longrightarrow Rj_*j^*K \longrightarrow .$$

Let K be constructible. Then j^*K is constructible, and since $i^*i_* \simeq \mathrm{id}$ and i_* preserves constructibility, $i^!K$ is constructible exactly when the third term Rj_*j^*K is. So the constructibility of $f^!$ comes down to one question: does direct image along an open immersion preserve bounded constructible complexes and, with finite coefficients, finite Tor amplitude? Proper base change does not help here, since j is not proper.

The answer is not formal, because Rj_*, and with it $i^!$, sees more than the stalks of K on Z: it sees the cohomology of K on a punctured neighbourhood of Z. For the origin $i : 0 \hookrightarrow \mathbf{A}^1$ over an algebraically closed field, the stalk of $Rj_*\Lambda$ at 0 is the cohomology of a punctured disc, which the Kummer covers of $\mathbf{G}_m$ compute to be $\Lambda \oplus \Lambda(-1)[-1]$, and the triangle gives $i^!\Lambda \simeq \Lambda(-1)[-2]$, whereas $i^*\Lambda = \Lambda$ (Example 6.25 below).

Over a field the answer is Deligne's finiteness theorem in SGA $4\frac{1}{2}$ [SGA4½], and we follow his proof. Its induction on dimension leads to fibrations over bases of positive dimension, so Deligne proves a statement in families over a base S. In families, however, the formation of Rj_* need not commute with base change. Let $S = \mathbf{A}^1$ with coordinate t, let $Y = \mathbf{A}^2$ with coordinates (t, x), viewed over S, and let j be the inclusion of $X = Y \setminus \{(0,0)\}$. By purity (Theorem 9.3, proved in Chapter 9 below), the punctured strict henselization of Y at the origin has cohomology Λ in degree 0 and $\Lambda(-2)$ in degree 3 only. Hence $R^1j_*\Lambda = 0$. On the fibre over $t = 0$, however, j becomes $\mathbf{G}_m \hookrightarrow \mathbf{A}^1$, whose $R^1j_*\Lambda$ is $\Lambda(-1)$ at the origin. Since over $t \neq 0$ the map j is an isomorphism, base change fails only at $t = 0$.

Deligne's generic base change theorem says that this is the general picture. Let $j : X \hookrightarrow Y$ be an open immersion of schemes of finite type over an integral base S of finite type over k, and let F be a constructible $\mathbf{F}_\ell$-sheaf on X. Then there is a nonempty open $S_0 \subset S$ over which Rj_*F is bounded constructible and its formation commutes with base change along every $T \to S_0$ of finite type (Theorem 6.17; Deligne proves more). Over $S = \operatorname{Spec} k$ the only nonempty open is S itself, so we obtain the finiteness and base change of Rj_* over a field (Corollary 6.18).

The projective line returns in the proof of the generic theorem. There we must control Rf_*L for smooth f and locally constant L, and we do so through the formula for $f^!$ of a smooth morphism (Section 6.3).

For finite coefficients (Theorem 6.20) we also bound the Tor amplitude of Rj_*K: it grows by at most $\operatorname{cd}_\ell(j)$ (Proposition 6.19). For integral and rational coefficients we have $D^b_{\mathrm{ctf}} = D^b_c$, and we reduce modulo a uniformizer. This works because an integral complex is constructible if it is derived complete with constructible reduction (Proposition 3.11). Only the adjunction for the projective line resists this reduction. Its unit and counit must be constructed as single morphisms with $\mathcal{O}_E$-coefficients (Proposition 6.22). The result is Theorem 6.27.

6.1 Conventions

Until Section 6.6 the coefficient ring is a finite commutative ℓ-primary ring, which we denote by A. With such coefficients nothing is lost by computing on the étale site. For X of finite type over k, the comparison functor

$$\nu_X^* : D(X_{\acute{e}t}, A) \longrightarrow D(X_{\mathrm{pro}\acute{e}t}, A)$$

is fully faithful (Proposition 2.49), and it identifies $D(X_{\acute{e}t}, A)$ with the localizing category $\operatorname{Loc}(D^b_{\mathrm{ctf}}(X, A))$ on which $f^!$ was constructed (Proposition 5.15). It also identifies the constructible complexes on the two sites (Theorem 2.43). In the following sections we therefore work in $D(X_{\acute{e}t}, A)$, and every result is transported to the pro-étale site by ν_X^*.

This transport is harmless because ν^* identifies all the operations we use with their étale counterparts, compatibly with units and counits (Lemma 2.39). These operations are the cohomology (Theorem 2.35); the direct images and their base change maps, also for unbounded complexes (Theorem 2.38 and Proposition 2.49); and the inverse images, tensor products, extension by zero and the localization triangles (Section 2.9). Moreover, by Corollary 2.44, finite Tor amplitude can be checked on the étale site.

We write $D^b_c(X, A)$ also for the bounded constructible objects of $D(X_{\acute{e}t}, A)$, which correspond to those of the pro-étale category. For a separated morphism of finite type $f : X \to S$, the functor $Rf_!$ and its right adjoint $f^!$ of Theorem 5.31 correspond to the functors $Rf_!$ and $f^!$ between $D(X_{\acute{e}t}, A)$ and $D(S_{\acute{e}t}, A)$, and the two adjunctions correspond. Accordingly, from now on g^*, Rf_*, $Rf_!$, $f^!$ and $\otimes^{\mathbf{L}}$ denote the functors on the étale derived categories.

Below we apply Proposition 5.28, stated on $\operatorname{Loc}(D^b_{\mathrm{ctf}}(-, A))$, to the étale functors. This is legitimate because the comparison identifies the composition maps of Proposition 5.28 on the two sites: the refinement maps (5.7) and the open–proper interchange (5.9) are characterized by their restriction to the open. The comparison also preserves the counits defining base change and the inverse counits defining forgetting supports (Proposition 5.29). Taking right adjoints, we see that it identifies the composition maps for $f^!$ as well.

Except in Lemmas 6.3, Lemma 6.4 and Lemma 6.13, which allow more general bases as stated there, all schemes and base changes are of finite type over k.

We use three classical theorems about the étale site.

Theorem 6.1 (Results from the étale theory)**.** *Let $f : X \to S$ be a morphism of schemes of finite type over k.*

1. *(Proper base change) If f is proper, then for every morphism of schemes $g : S' \to S$, not necessarily of finite type over k, and every $K \in D(X_{\acute{e}t}, A)$, the base change map $g^* Rf_* K \to Rf'_* g'^* K$ is an isomorphism, where $f' : X' \to S'$ and $g' : X' \to X$ are the projections of $X' = X \times_S S'$. In particular, the geometric stalks of $Rf_* K$ are the cohomology of the geometric fibres.*
2. *(Finiteness) If f is separated, then $Rf_!$ maps $D^b_c(X, A)$ into $D^b_c(S, A)$.*
3. *(Cohomological dimension) The functors Rf_* and, if f is separated, $Rf_!$ have finite cohomological amplitude on torsion sheaves. In particular $\mathrm{cd}_\ell(f) < \infty$ in the sense of Definition 2.45.*

Proof. For bounded below complexes, (1) is Theorem 4.33 together with the comparison of direct images (Theorem 2.38); for unbounded complexes it is [Stacks, 0F0C], whose finiteness hypothesis on f is part (3). Part (2) is [Stacks, 0GL0]. As for part (3), for Rf_* it is [Stacks, 0F10], and for $Rf_!$ we write $f = \bar{f} \circ j$ with $\bar{f}$ proper and j an open immersion; then $Rf_! = R\bar{f}_* j_!$ with $j_!$ exact, and [Stacks, 0F10] applies to $\bar{f}$. Finally, the last assertion follows by the comparison of direct images, since classical sheaves are of the form $\nu_X^* F$. ∎

Definition 6.2 (The base change map for $f^!$, finite coefficients)**.** Let $f : X \to S$ be separated and of finite type, let $g : S' \to S$ be a morphism, let $X' = X \times_S S'$, and let $f' : X' \to S'$, $g' : X' \to X$ be the projections. For $N \in D(S_{\acute{e}t}, A)$, let $\varepsilon_N : Rf_! f^! N \to N$ be the counit of $Rf_! \dashv f^!$. The *base change map*

$$\beta_N : g'^* f^! N \longrightarrow f'^! g^* N \tag{6.1}$$

is the transpose, for f', of the morphism

$$Rf'_! g'^* f^! N \xrightarrow{\sim} g^* Rf_! f^! N \xrightarrow{g^* \varepsilon_N} g^* N, \tag{6.2}$$

where the first arrow is the inverse of the base change isomorphism for $Rf_!$ (Proposition proposition-composition-base-change-rf-shriek).

It is an isomorphism for smooth f (Proposition 6.9). The global isomorphism $f^! \simeq f^*(d)[2d]$ for smooth f of relative dimension d, and its compatibility with composition, are treated with smooth purity in Chapter 9. Here we need only its restriction to charts $U \to \mathbf{A}^d_S$.

6.2 Duality for the projective line

Let $p : \mathbf{P}^1_S \to S$ be the projection. We construct the adjunction $Rp_* \dashv p^*(1)[2]$ from two maps: the counit is a trace $Rp_* A(1)[2] \to A$, obtained from the cohomology of $\mathbf{P}^1$, and the unit is the cycle class of the diagonal in $\mathbf{P}^1_S \times_S \mathbf{P}^1_S$, a special case of the cycle class of an effective Cartier divisor. Both triangle identities say that pushing the class of the diagonal forward along either projection $\mathbf{P}^1 \times \mathbf{P}^1 \to \mathbf{P}^1$ and applying the trace gives 1. This holds because the diagonal meets each fibre of either projection in a single point.

In topology the same two maps appear. For instance, for a bundle $E \to B$ with fibre S^2, integration over the fibre $H^2(S^2) \to \mathbf{Z}$ is the counit of Poincaré duality on the fibres, and the unit is the Thom class of the diagonal of $E \times_B E$, which is Poincaré dual to the identity

correspondence. In étale cohomology, by contrast, the fundamental class of a point on $\mathbf{P}^1$ comes from a line bundle through the Kummer sequence. So both maps are built from line bundles: the trace comes from $c_1(\mathcal{O}(1))$, and the class of the diagonal comes from the divisor $x_0y_1 - x_1y_0 = 0$.

For every other morphism, we shall compute $f^!$ from this adjunction, from $j_! \dashv j^*$ for open immersions and from localization.

Lemma 6.3 (The cycle class of a Cartier divisor)**.** *Let Λ be a finite commutative ℓ-primary ring, and let $i : D \hookrightarrow Z$ be an effective Cartier divisor on a scheme Z on which ℓ is invertible. There is a morphism*

$$c_D : i_*\Lambda_D \longrightarrow \Lambda_Z(1)[2] \quad \textit{in } D(Z_{\acute{e}t}, \Lambda),$$

and hence, via ν_Z^*, in $D(Z_{\mathrm{pro\acute{e}t}}, \Lambda)$, whose composite with $\Lambda_Z \to i_*\Lambda_D$ is the Kummer first Chern class $c_1(\mathcal{O}_Z(D))$. It is compatible with change of coefficients and with every base change along which the pullback of the divisor is still an effective Cartier divisor.

Proof. The map c_D will be the boundary map of the extension of $i_*\mathbf{Z}_D$ by $\mathbf{G}_m$ formed by meromorphic functions with poles along D, reduced modulo ℓ^r; this extension is the étale version of the divisor sequence $0 \to \mathcal{O}_Z \to \mathcal{O}_Z(D) \to \mathcal{O}_Z(D)|_D \to 0$.

We construct c_D in $D(Z_{\acute{e}t}, \Lambda)$. The functor ν_Z^*, which commutes with inverse images and with change of coefficients, then transports it and its compatibilities to $Z_{\mathrm{pro\acute{e}t}}$.

Let $j : Z \setminus D \hookrightarrow Z$ be the open complement. Let $\mathcal{M}_D \subset j_*\mathbf{G}_m$ be the subsheaf which, on an open where D has a regular equation t, consists of the products ut^{-m} with u a unit and m a locally constant integer on D; equivalently, it is generated by $\mathbf{G}_m$ and t^{-1}. Since replacing t by a unit multiple gives the same subsheaf, $\mathcal{M}_D$ is globally defined. Sending t^{-1} to 1, we obtain a short exact sequence of abelian sheaves on $Z_{\acute{e}t}$

$$\Sigma_D : \ 0 \longrightarrow \mathbf{G}_m \longrightarrow \mathcal{M}_D \longrightarrow i_*\mathbf{Z}_D \longrightarrow 0, \tag{6.3}$$

whose exactness can be checked on geometric stalks. Away from D, the function t is a unit, so there $\mathcal{M}_D = \mathbf{G}_m$, and both the quotient and the stalk of $i_*\mathbf{Z}_D$ are zero. At a point of D, on the other hand, no nonzero power of a regular nonunit t is a unit, and therefore the exponent m in ut^{-m} is well defined. The kernel of the map $ut^{-m} \mapsto m$ is then $\mathbf{G}_m$, and the map is surjective because exponents lift locally to powers of t^{-1}. For example, for $D = \{0\} \subset Z = \mathbf{A}^1 = \operatorname{Spec} k[t]$, the sections of $\mathcal{M}_D$ near the origin are the functions ut^{-m}, that is, the rational functions whose only zero or pole is at 0, and the map records the order of the pole, $m = -\operatorname{ord}_0$.

The boundary of Σ_D is a morphism $i_*\mathbf{Z}_D \to \mathbf{G}_m[1]$ in $D(Z_{\acute{e}t}, \mathbf{Z})$, to which we apply $- \otimes^{\mathbf{L}}_{\mathbf{Z}} \mathbf{Z}/\ell^r\mathbf{Z}$. The source becomes $i_*\mathbf{Z}/\ell^r\mathbf{Z}$. For the target, the Kummer sequence [Stacks, 03PL] shows that the cone of multiplication by ℓ^r on $\mathbf{G}_m$ has the single cohomology sheaf μ_{ℓ^r} in degree -1, and hence the target becomes $\mu_{\ell^r}[2]$. This is c_D for $\mathbf{Z}/\ell^r\mathbf{Z}$. For a finite $\mathbf{Z}/\ell^r\mathbf{Z}$-algebra Λ, we apply $- \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^r} \Lambda$. Since μ_{ℓ^r} is locally free of rank one over $\mathbf{Z}/\ell^r\mathbf{Z}$, the target becomes $\Lambda_Z(1)[2]$.

To identify the composite with c_1, we cover Z by opens with local equations t_α. The local lifts of the section 1 of $i_*\mathbf{Z}_D$ through Σ_D are the t_α^{-1}. Their ratios are the transition functions of $\mathcal{O}_Z(D)$; this fixes the sign convention. Hence the boundary of Σ_D followed by the Kummer boundary is the class of $\mathcal{O}_Z(D)$. Changing a local equation multiplies a lift by a unit, and this changes the cocycle by a coboundary.

Naturality holds because the local equations, the units and the sequence Σ_D pull back, compatibly with the boundary maps. On the source we use the identity $g^*i_* = i'_*g_D^*$ for a base change $g : Z' \to Z$ with $i' : D' \hookrightarrow Z'$ and $g_D : D' \to D$; it can be checked on geometric stalks. ∎

The lemma will be applied twice: to the section at infinity of $\mathbf{P}^1_S$, which gives the trace, and to the diagonal of $\mathbf{P}^1_S \times_S \mathbf{P}^1_S$, which gives the unit. The first application enters the computation of Rp_*A.

Lemma 6.4 (Cohomology of the projective line and the trace)**.** *Let Λ be a finite commutative ℓ-primary ring, let S be a Noetherian scheme on which ℓ is invertible, and let $p : \mathbf{P}^1_S \to S$ be the projection. The unit $\Lambda_S \to Rp_*\Lambda_{\mathbf{P}^1_S}$ and the class $c_1(\mathcal{O}_{\mathbf{P}^1_S}(1))$ define an isomorphism*

$$\sigma_S : \Lambda_S \oplus \Lambda_S(-1)[-2] \xrightarrow{\sim} Rp_*\Lambda_{\mathbf{P}^1_S} \quad \text{in } D^+(S_{\acute{e}t}, \Lambda), \tag{6.4}$$

compatible with base change in the sense that for $g : S' \to S$ with S' Noetherian, the base change map $g^*Rp_*\Lambda \to Rp'_*\Lambda$ carries $g^*\sigma_S$ to $\sigma_{S'}$. Let

$$\mathrm{Tr}_p : Rp_*\Lambda_{\mathbf{P}^1_S}(1)[2] \longrightarrow \Lambda_S \tag{6.5}$$

be the morphism given by the projection onto the summand Λ_S of the twist of σ_S. Then Tr_p is compatible with every Noetherian base change.

Proof. By the comparison theorems for the cohomology and the proper direct image of bounded below complexes (Theorems 2.35 and Theorem 2.38), it suffices to prove the assertions on σ_S and Tr_p in $D^+(S_{\acute{e}t}, \Lambda)$ for Noetherian S.

Both components of σ_S are globally defined. The second comes from Lemma 6.3 applied to the section at infinity, whose line bundle is $\mathcal{O}(1)$. Geometric stalks detect isomorphisms, and by proper base change the stalk of $Rp_*\Lambda$ at a geometric point $\bar{s}$ is $R\Gamma(\mathbf{P}^1_{\bar{s}}, \Lambda)$. It therefore suffices to check that σ_S is an isomorphism on a geometric fibre, first for $\Lambda = \mathbf{Z}/\ell^r\mathbf{Z}$. By the genus zero case of Proposition D.3, the cohomology lives only in degrees 0 and 2, with $H^0 = \mathbf{Z}/\ell^r\mathbf{Z}$ and $H^2 = (\mathbf{Z}/\ell^r\mathbf{Z})(-1)$. The Kummer description sends a line bundle to its degree modulo ℓ^r. As $\mathcal{O}(1)$ has degree one, its class generates H^2, and σ_S is an isomorphism. The unit and the class of $\mathcal{O}(1)$ commute with base change. Therefore so do σ_S, its inverse, and Tr_p.

For a finite $\mathbf{Z}/\ell^r\mathbf{Z}$-algebra Λ, the projection formula [Stacks, 0GL5] gives $Rp_*\Lambda = Rp_*(\mathbf{Z}/\ell^r\mathbf{Z}) \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^r} \Lambda$. Under this identification, σ_S for Λ is σ_S for $\mathbf{Z}/\ell^r\mathbf{Z}$ tensored with Λ. It is therefore again an isomorphism compatible with base change. ∎

Over $S = \operatorname{Spec} \bar{k}$, (6.4) reads $R\Gamma(\mathbf{P}^1_{\bar{k}}, A) \simeq A \oplus A(-1)[-2]$. Here the shift $[-2]$ places the second summand in degree 2, so that it is H^2, and the twist (-1) records how Galois acts on H^2: the Kummer map identifies H^2 with $\mathrm{Pic}(\mathbf{P}^1) \otimes \mu_{\ell^r}^{\otimes -1}$, that is, with $A(-1)$. Twisting by (1) and shifting by [2] therefore moves this summand to degree 0 with trivial twist. The map Tr_p is the projection onto this summand, and it sends the class of a point to 1. It is the counit of $Rp_* \dashv p^*(1)[2]$ in Proposition 6.7. After its lift to $\mathcal{O}_E$ in Subsection 6.7.1, it is also the trace map Tr_p of $\mathbf{P}^1$ in Section 9.6.

As a check of the twist, let us work over $\mathbf{F}_q$. The geometric Frobenius acts by 1 on H^0 and by q on $H^2 = A(-1)$. The alternating sum of the traces is $1 + q = \#\mathbf{P}^1(\mathbf{F}_q)$, as in Corollary 12.31.

Example 6.5 (Compact supports on the running example)**.** We use the open immersion $j_\infty : \mathbf{A}^1 \to \mathbf{P}^1$ of Example 3.19, base changed to $\bar{k}$, and we let $i_\infty : \{\infty\} \to \mathbf{P}^1_{\bar{k}}$ be the inclusion of the complementary point.

By definition, $R\Gamma_c(\mathbf{A}^1, A) = R\Gamma(\mathbf{P}^1, j_{\infty!}A)$. The localization triangle $j_{\infty!}A \to A_{\mathbf{P}^1} \to i_{\infty *}A$ and (6.4) show that

$$R\Gamma_c(\mathbf{A}^1_{\bar{k}}, A) \simeq \mathrm{fib}(A \oplus A(-1)[-2] \to A) \simeq A(-1)[-2],$$

since restriction to ∞ is an isomorphism on H^0. In other words, $H^2_c(\mathbf{A}^1) = A(-1)$ is the only nonzero compactly supported cohomology group. The trace on it is the same Tr_p.

Exercise. Using the localization sequence of Example 3.19, show that $H^1_c(\mathbf{G}_{m,\bar{k}}, A) \simeq A$ and $H^2_c(\mathbf{G}_{m,\bar{k}}, A) \simeq A(-1)$, and that these are the only nonzero groups.

The adjunction is built from the trace Tr_p of (6.5) and the cycle class of the diagonal provided by Lemma 6.3. As in all of this part of the section, we compute on the étale sites (Section 6.1). Let $p : \mathbf{P}^1_S \to S$ be the projection, let $p_1, p_2 : \mathbf{P}^1_S \times_S \mathbf{P}^1_S \to \mathbf{P}^1_S$ be the two projections and let $\Delta : \mathbf{P}^1_S \hookrightarrow \mathbf{P}^1_S \times_S \mathbf{P}^1_S$ be the diagonal. For $i = 1, 2$, the projection p_i is a projective line over the other factor, with trace Tr_{p_i} given by (6.5). Both triangle identities rest on the following computation.

Lemma 6.6 (The class of the diagonal of $\mathbf{P}^1_S$)**.** *Let S be of finite type over k, and let p_i and Tr_{p_i} be as above. The diagonal Δ is a relative effective Cartier divisor in $\mathbf{P}^1_S \times_S \mathbf{P}^1_S$ over S, and it remains one after every base change $S' \to S$. Let*

$$\delta : \Delta_* A_{\mathbf{P}^1_S} \longrightarrow A_{\mathbf{P}^1_S \times_S \mathbf{P}^1_S}(1)[2]$$

be its cycle class (Lemma 6.3). Then for $i = 1, 2$ the composite

$$s_i : A_{\mathbf{P}^1_S} = Rp_{i*}\Delta_* A_{\mathbf{P}^1_S} \xrightarrow{Rp_{i*}\delta} Rp_{i*} A_{\mathbf{P}^1_S \times_S \mathbf{P}^1_S}(1)[2] \xrightarrow{\mathrm{Tr}_{p_i}} A_{\mathbf{P}^1_S} \tag{6.6}$$

is the identity.

Proof. Let x_0, x_1 and y_0, y_1 be the pullbacks along p_1 and p_2 of the homogeneous coordinates of $\mathbf{P}^1_S$. Then Δ is the zero locus of the section $x_0y_1 - x_1y_0$ of $p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$. On the open where x_0 and y_0 are invertible, in the affine coordinates $x = x_1/x_0$ and $y = y_1/y_0$, this equation becomes $y - x$. It is a nonzerodivisor, and the quotient $\mathcal{O}_S[x, y]/(y - x) \simeq \mathcal{O}_S[x]$ is flat over S. The open where x_1 and y_1 are invertible is treated in the same way, and these two opens cover Δ. Hence Δ is a relative effective Cartier divisor, and it remains one after every base change.

To show $s_i = 1$, note that both sides are endomorphisms of a sheaf placed in degree zero. We have

$$\mathrm{Hom}_{D((\mathbf{P}^1_S)_{\acute{e}t}, A)}(A_{\mathbf{P}^1_S}, A_{\mathbf{P}^1_S}) = \mathrm{Hom}_{\mathrm{Mod}((\mathbf{P}^1_S)_{\acute{e}t}, A)}(A_{\mathbf{P}^1_S}, A_{\mathbf{P}^1_S}),$$

so s_i is a morphism of sheaves and is determined by its geometric stalks. By proper base change and the compatibility of δ and Tr_{p_i} with base change, the stalk of (6.6) at a geometric point $\bar{x}$ of $\mathbf{P}^1_S$ is computed on the fibre $p_i^{-1}(\bar{x}) \simeq \mathbf{P}^1_{\bar{x}}$. On this fibre the equation $x_0y_1 - x_1y_0$ restricts to a nonzero linear form, so the diagonal cuts out a single reduced point, whose line bundle is $\mathcal{O}(1)$. By Lemma 6.3, the class of this point composed with the unit is $c_1(\mathcal{O}(1))$. In turn, the trace Tr_p sends $c_1(\mathcal{O}(1))$ to 1, since by definition (6.5) it is the projection onto the summand of (6.4) spanned by this class. Thus the stalk of s_i at every geometric point $\bar{x}$ sends 1 to 1, and so $s_i = 1$. ∎

We build the unit and counit of the adjunction from δ and from the trace. For $M \in D((\mathbf{P}^1_S)_{\acute{e}t}, A)$, let

$$\begin{aligned}\eta_M : M &= Rp_{2*}\Delta_* M \\ &\longrightarrow Rp_{2*}(p_1^* M(1)[2]) \xrightarrow{\sim} p^* Rp_* M(1)[2].\end{aligned} \tag{6.7}$$

Here the first arrow is Rp_{2*} applied to $\delta \otimes p_1^* M$, where we use the projection formula $\Delta_* M = \Delta_* A \otimes^{\mathbf{L}} p_1^* M$, and the second arrow is the proper base change isomorphism $p^* Rp_* \simeq Rp_{2*} p_1^*$. For $N \in D(S_{\acute{e}t}, A)$, let

$$\varepsilon_N : Rp_*(p^* N(1)[2]) \xrightarrow{\sim} N \otimes^{\mathbf{L}}_A Rp_* A_{\mathbf{P}^1_S}(1)[2] \xrightarrow{1 \otimes \mathrm{Tr}_p} N, \tag{6.8}$$

where the first arrow is the projection formula. These maps exist for unbounded complexes by [Stacks, 0F0C, 0GL5].

Proposition 6.7 (The adjunction for the projective line)**.** *Let S be of finite type over k, and let $p : \mathbf{P}^1_S \to S$ be the projection. The functors $Rp_* : D((\mathbf{P}^1_S)_{\acute{e}t}, A) \to D(S_{\acute{e}t}, A)$ and $p^*(1)[2] : D(S_{\acute{e}t}, A) \to D((\mathbf{P}^1_S)_{\acute{e}t}, A)$ form an adjunction*

$$Rp_* \ \dashv \ p^*(1)[2], \tag{6.9}$$

with unit (6.7) and counit (6.8). For every morphism $g : S' \to S$ with S' of finite type over k, the unit and counit are compatible with base change along g. They are also compatible with change of finite coefficients.

Proof. The projection formula and proper base change reduce each of the two triangle identities to the identity $s_i = 1$ of Lemma 6.6, tensored with the identity of a coefficient complex.

First triangle identity. This is the equation $\varepsilon_{Rp_* M} \circ Rp_*(\eta_M) = 1_{Rp_* M}$. Using composition of direct images, $pp_1 = pp_2$, proper base change and the projection formula, this composite equals

$$\begin{aligned} Rp_* M &= R(pp_1)_* \Delta_* M \\ &\longrightarrow R(pp_1)_* p_1^* M(1)[2] \\ &\xrightarrow{\sim} Rp_*(M \otimes^{\mathbf{L}}_A Rp_{1*} A_{\mathbf{P}^1_S \times_S \mathbf{P}^1_S}(1)[2]) \\ &\xrightarrow{Rp_*(1_M \otimes \mathrm{Tr}_{p_1})} Rp_* M. \end{aligned} \tag{6.10}$$

Here the first arrow is Rp_* of the first arrow of (6.7), with $R(pp_2)_*$ rewritten as $R(pp_1)_*$. The middle isomorphism is the projection formula for p_1. It replaces the proper base change isomorphism of (6.7), after Rp_*, followed by the projection formula isomorphism of (6.8). The last arrow replaces $1 \otimes \mathrm{Tr}_p$ in (6.8) by Tr_{p_1}, the base change of Tr_p along p (Lemma 6.4). In this form, the trace is applied to $Rp_{1*}\delta$, that is, to the factor $\Delta_* A$, and M only appears as a coefficient.

This rewriting is an equality of maps because the projection formula isomorphism is compatible with proper base change. To see this, recall that the proper base change map is the adjoint transpose of the counit of $g^* \dashv Rg_*$ for the relevant pullback g. The projection map is the adjoint transpose of the same counit tensored with the identity, and the counit of a composite adjunction is the composite of the counits. Therefore the two routes from $R(pp_2)_* p_1^* M = R(pp_1)_* p_1^* M$ to $Rp_* M \otimes Rp_* A(1)[2]$ have the same adjoint transpose, as in the pasting Lemma 4.5.

On the route through (6.10) we identify $Rp_{1*}A = p^*Rp_*A$ by proper base change and then use the projection formula for p. By naturality of the projection formula in its second variable, the first arrow of (6.10) is then $1_M \otimes Rp_{1*}\delta$ under the projection identification. The composite is therefore $Rp_*(1_M \otimes s_1)$, which is 1_{Rp_*M} by Lemma 6.6.

Second triangle identity. Here the roles of the two projections are exchanged: we must show that $p^*(\varepsilon_N)(1)[2] \circ \eta_{p^*N(1)[2]}$ is the identity. Take $M = p^*N(1)[2]$ in (6.7). Since N and the Tate twist are pulled back from S, both p_1^*M and p_2^*M are $(pp_1)^*N(1)[2] = (pp_2)^*N(1)[2]$. The projection formula moves this common factor outside Rp_{2*}, and $\eta_{p^*N(1)[2]}$ becomes $1 \otimes Rp_{2*}\delta$. By the base change compatibility of (6.5), $p^*(\varepsilon_N)(1)[2]$ becomes $1 \otimes \mathrm{Tr}_{p_2}$, applied to the target of $Rp_{2*}\delta$. We obtain

$$p^*(\varepsilon_N)(1)[2] \circ \eta_{p^*N(1)[2]} = 1_{p^*N(1)[2]} \otimes (\mathrm{Tr}_{p_2} \circ Rp_{2*}\delta) = 1_{p^*N(1)[2]} \otimes s_2 = 1_{p^*N(1)[2]}. \tag{6.11}$$

All shifts moved in (6.10) and (6.11) are even, so no Koszul signs appear.

The unit is constructed from δ and proper base change, and the counit from (6.5) and the projection formula. All of these ingredients are compatible with base change and change of coefficients, and hence so are η and ε. Transported by ν^*, the same then holds for the corresponding adjunction on the pro-étale categories. ∎

6.3 Smooth morphisms

Let $f : X \to S$ be smooth of relative dimension d. Extending the formula $p^! = p^*(1)[2]$ for $p : \mathbf{P}^1_S \to S$, we show in Proposition 6.9 that $f^!N \simeq f^*N \otimes^{\mathbf{L}} \omega_f$ with ω_f locally isomorphic to $A(d)[2d]$. We need this formula in the proof of generic constructibility below, because it must control Rf_*L for a locally constant sheaf L. Indeed, we know nothing about Rf_*L directly, whereas the finiteness theorem and proper base change control $Rf_!$. With the formula, however, the adjunction $Rf_! \dashv f^!$ becomes a duality between Rf_*L and $Rf_!$ of the twisted dual of L (Lemma 6.10). On a manifold, Poincaré duality exchanges ordinary and compactly supported cohomology in the same way.

For example, let $f : \mathbf{G}_m \to \operatorname{Spec} \bar{k}$ and $L = A = \mathbf{F}_\ell$. The localization triangle for the origin of $\mathbf{A}^1$ and $R\Gamma_c(\mathbf{A}^1_{\bar{k}}, A) \simeq A(-1)[-2]$ give $R\Gamma_c(\mathbf{G}_m, A) \simeq \mathrm{fib}(A(-1)[-2] \to A)$. Hence $H^1_c = A$ and $H^2_c = A(-1)$ are the only nonzero groups, and $R\Gamma_c(\mathbf{G}_m, A(1)[2])$ has cohomology $A(1)$ in degree -1 and A in degree 0. Its dual has A in degree 0 and $A(-1)$ in degree 1. This is indeed $R\Gamma(\mathbf{G}_m, A)$, with $H^1 = A(-1)$ given by the Kummer torsors.

The duality must also commute with base change. This requires the base change isomorphism $g'^*f^! \simeq f'^!g^*$ for smooth f, and for it the local form $\omega_f|_U \simeq A_U(d)[2d]$ on charts suffices. A global identification of ω_f with $A_X(d)[2d]$, by contrast, needs a trace map for f; it is part of smooth purity (Chapter 9). For the local form we use that, Zariski locally on X and S, a smooth morphism $f : X \to S$ factors as an étale morphism to some $\mathbf{A}^d_S$ followed by the projection to S, and the next lemma treats these two kinds of morphisms.

Lemma 6.8 (Affine space and étale localization)**.** *Let S be of finite type over the base field k.*

1. *For the projection $p_d : \mathbf{A}^d_S \to S$, the right adjoint $p_d^!$ of $Rp_{d!} : D((\mathbf{A}^d_S)_{ét}, A) \to D(S_{ét}, A)$ is $p_d^*(d)[2d]$. Its counit $Rp_{d!}p_d^*N(d)[2d] \to N$ is compatible with base change: for a morphism $g : S' \to S$ of finite type over k, with base change $p'_d : \mathbf{A}^d_{S'} \to S'$, the pullback g^* of the counit for N is, under the base change isomorphism for $Rp_{d!}$, the counit for g^*N. The same holds for the unit, which is determined by the counit through the triangle identities.*

2. *Let $f : X \to S$ be separated of finite type, let $N \in D(S_{ét}, A)$, and let $u : U \to X$ be separated, étale and of finite type. There is a canonical isomorphism*

$$u^* f^! N \simeq (fu)^! N \tag{6.12}$$

under which the counit of $R(fu)_! \dashv (fu)^!$ is the composite

$$R(fu)_! u^* f^! N \simeq Rf_! u_! u^* f^! N \longrightarrow Rf_! f^! N \longrightarrow N$$

of the composition isomorphism, the counit of $u_! \dashv u^*$ and the counit of $Rf_! \dashv f^!$. For a second separated étale morphism $u' : U' \to U$ of finite type, the isomorphism (6.12) for uu' is the composite of u'^* of (6.12) for u and (6.12) for u' and fu.

Part (2) generalizes $j^! = j^*$ from open immersions to étale maps. For $d = 1$, k algebraically closed and $S = \operatorname{Spec} k$, part (1) says that $\operatorname{Hom}(R\Gamma_c(\mathbf{A}^1, F), M) = \operatorname{Hom}(F, M_{\mathbf{A}^1}(1)[2])$. For $F = A$ both sides are $\operatorname{Hom}(A, M(1)[2])$, consistent with the example above: the left side because $R\Gamma_c(\mathbf{A}^1_{\bar{k}}, A) \simeq A(-1)[-2]$, and the right side because $R\Gamma(\mathbf{A}^1_{\bar{k}}, A) = A$.

Proof of Lemma 6.8. Both parts follow formally from uniqueness of right adjoints. Part (1) comes from (6.9) and $j_! \dashv j^*$ for $\mathbf{A}^1 \subset \mathbf{P}^1$, and part (2) from $u_! \dashv u^*$ for étale u.

(1) Let $j : \mathbf{A}^1_S \hookrightarrow \mathbf{P}^1_S$ be the open immersion, so that $Rp_{1!} = Rp_* j_!$ for $d = 1$. Since $j_! \dashv j^*$ and $Rp_* \dashv p^*(1)[2]$ by (6.9), composing right adjoints gives $p_1^! = j^* p^*(1)[2]$. This equals $p_1^*(1)[2]$ because $p \circ j = p_1$. For general d, write p_d as a composite of d projections $r_1, \dots, r_d$ of affine lines. By the composition isomorphism for $Rf_!$ (Proposition proposition-composition-base-change-rf-shriek) we have $Rp_{d!} \simeq Rr_{1!} \circ \cdots \circ Rr_{d!}$. Hence $p_d^!$ is the composite of the right adjoints of the $Rr_{i!}$, which is $p_d^*(d)[2d]$ by the case $d = 1$. Under this isomorphism, the counit of $Rp_{d!} \dashv p_d^*(d)[2d]$ is the composite of the d counits for the affine lines. Each of these is a trace (6.5) composed with the counit of $j_! \dashv j^*$. By the associativity in Proposition proposition-composition-base-change-rf-shriek, the composite does not depend on how the d factors are grouped. Each factor and the composition isomorphism commute with base change (Proposition proposition-composition-base-change-rf-shriek), and therefore so does the counit. The unit is determined by the counit through the triangle identities, and it commutes with base change as well.

(2) For an étale u, the functor $Ru_!$ is extension by zero. By its elementary construction [Stacks, 03S2], it is an exact functor $u_! : \operatorname{Mod}(U_{ét}, A) \to \operatorname{Mod}(X_{ét}, A)$ left adjoint to u^*. As both functors are exact, they need not be derived, and the adjunction extends to derived categories. For every $M \in D(U_{ét}, A)$,

$$\operatorname{Hom}(M, u^* f^! N) = \operatorname{Hom}(u_! M, f^! N) = \operatorname{Hom}(Rf_! u_! M, N) = \operatorname{Hom}(R(fu)_! M, N),$$

where the last step is the composition isomorphism of Proposition proposition-composition-base-change-rf-shriek. The Yoneda lemma therefore gives (6.12). Taking $M = u^* f^! N$ and the identity on the left, we see that the counit of $R(fu)_! \dashv (fu)^!$ is the composite of the counits as stated. For the cocycle condition, the isomorphism for uu' and the composite of those for u and u' both correspond under the Yoneda lemma to the bijections obtained from the composition isomorphism for $f \circ u \circ u'$. These bijections agree by associativity of the composition isomorphisms (Proposition proposition-composition-base-change-rf-shriek). ∎

Applied along local charts $U \xrightarrow{e} \mathbf{A}^d_S \to S$, the two parts of the lemma compute $f^!$ locally on X for every smooth f. The proposition below turns this into two global statements, a tensor formula and base change.

Proposition 6.9 (The exceptional inverse image of a smooth morphism)**.** *Let $f : X \to S$ be separated, smooth and of finite type, of pure relative dimension d, and let $\omega_f = f^!A_S \in D(X_{ét}, A)$ be its dualizing complex. Let $N \in D(S_{ét}, A)$.*

1. *On an open $U \subset X$ with $f|_U = p_d \circ e$, where $e : U \to \mathbf{A}^d_S$ is separated and étale, we have $(f^!N)|_U \simeq e^*p_d^*N(d)[2d]$, with counit the sum over the fibres of e followed by the counit of Lemma lemma-affine-space-etale-localization. In particular $\omega_f|_U \simeq A_U(d)[2d]$, and $\omega_f[-2d]$ is a locally constant sheaf of free A-modules of rank one.*
2. *The canonical map*
$$\mu_N : f^*N \otimes^{\mathbf{L}}_A \omega_f \longrightarrow f^!N \tag{6.13}$$
is an isomorphism.
3. *Let $g : S' \to S$ be a morphism, let $X' = X \times_S S'$, and let $f' : X' \to S'$, $g' : X' \to X$ be the projections. Then the base change map $\beta_N : g'^*f^!N \to f'^!g^*N$ of Definition 6.2 is an isomorphism. It identifies $g'^*\omega_f$ with $\omega_{f'}$ and is compatible with μ_N.*

Proof. The maps μ_N and β_N are defined globally as adjoint transposes. Since being an isomorphism is local on X, respectively on X', we show that on a chart both maps become identifications of inverse images under the explicit identification (6.14). For this we follow the counits through the factorization $U \to \mathbf{A}^d_V \to V$.

We write $\varepsilon_N : Rf_!f^!N \to N$ for the counit of $Rf_! \dashv f^!$ (for $f = p$ this is (6.8)). The map μ_N is the transpose of the morphism

$$Rf_!(f^*N \otimes^{\mathbf{L}} \omega_f) \xrightarrow{\sim} N \otimes^{\mathbf{L}} Rf_!\omega_f \xrightarrow{1\otimes\varepsilon_{A_S}} N,$$

where the first arrow is the projection formula. The map β_N is the transpose, for f', of the morphism (6.2) (Definition 6.2).

Local charts. The computation of $f^!$ on a chart also proves (1). Zariski locally on X and S, a smooth morphism is an étale map to affine space followed by the projection [Stacks, 054L]. Thus every point of X has an affine open neighbourhood $u : U \hookrightarrow X$ mapping into an affine open $v : V \hookrightarrow S$, together with a separated étale map $e : U \to \mathbf{A}^d_V$ such that $fu = vp_de$, where $p_d : \mathbf{A}^d_V \to V$ is the projection. By Lemma 6.8 and uniqueness of right adjoints, we have

$$u^*f^!N = e^*p_d^*v^*N(d)[2d], \qquad u^*\omega_f = A_U(d)[2d]. \tag{6.14}$$

In particular, ω_f has a single nonzero cohomology sheaf, in degree $-2d$, which is locally free of rank one. This proves (1), because the counit on U is the composite of the étale counit for e and the counit for p_d. The first of these is the sum over the fibres.

μ_N is an isomorphism. On each such U, under (6.14), the source and the target of μ_N are both $e^*p_d^*v^*N(d)[2d]$. We claim that the map is the identification obtained by reassociating tensor product and twist. It suffices to compare the transposes. Both are the projection formula followed by the counit for v, the d counits for the affine lines, and the étale counit for e. They agree by the formula for the counit of a composite in Lemma 6.8. Hence μ_N is an isomorphism on X.

β_N is an isomorphism. For each type of chart, under the explicit description of $f^!$ as a twisted inverse image, β_N is the evident identification of inverse images. For the projective line, the transpose of (6.2) is the usual identification $g'^*p^*N(1)[2] = p'^*g^*N(1)[2]$ of pullbacks. This holds because the unit (6.7) and counit (6.8) commute with base change. By restriction to the affine line and composition, the same holds for affine space. For a separated étale map u, the transpose

is again the usual identification of inverse images. Indeed, the stalk of $u_!M$ at a geometric point is the direct sum of the stalks of M at the geometric points over it. This description is compatible with base change, and under it the counit $u_!u^*N \to N$ is the sum of the identity maps of the stalks of N.

The cases combine, because base change for $Rf_!$ is compatible with composition (Proposition proposition-composition-base-change-rf-shriek) and the counit of a composite is the composite of the counits. The open immersions u and v are étale, so the étale case covers them as well. Substituting these facts into (6.2), we see that the restriction of β_N to $U \times_X X'$ is the composite of the étale and affine-space cases just treated. Hence β_N is an isomorphism on a cover of X', and therefore it is an isomorphism.

Compatibilities. Taking $N = A_S$, we see that β_N identifies $g'^*\omega_f$ with $\omega_{f'}$, because $g^*A_S = A_{S'}$. The compatibility of μ_N with β_N says that two maps $g'^*(f^*N \otimes^{\mathbf{L}} \omega_f) \to f'^!g^*N$ agree. Here the first is μ_N followed by β_N, and the second is the identification $g'^*(f^*N \otimes^{\mathbf{L}} \omega_f) = f'^*g^*N \otimes^{\mathbf{L}} \omega_{f'}$, coming from $g'^*f^* = f'^*g^*$ and the case $N = A_S$, followed by μ_N for f'. Their transposes are maps $Rf'_!g'^*(f^*N \otimes^{\mathbf{L}} \omega_f) \to g^*N$. Both are given by the projection formula followed by the counit, once before and once after the base change isomorphism for $Rf_!$. They agree for two reasons. First, the projection formula isomorphism is compatible with the base change isomorphism for $Rf_!$. Second, by the definition (6.2) of β_N, the counit for f' composed with $Rf'_!$ of β_N is g^* of the counit for f under that isomorphism. ■

In Step 2 of the proof of Theorem 6.17, the next lemma is applied to $f = bj$, with $b : Y \to S$ projective and $j : X \hookrightarrow Y$ open.

Lemma 6.10 (Smooth morphisms and locally constant sheaves)**.** *Let S be integral and let $f : X \to S$ be separated, smooth and of finite type. For a finite locally constant $\mathbf{F}_\ell$-sheaf L on $X_{\acute{e}t}$, there is a nonempty open $S_0 \subset S$ with the following property. For the restriction $f_{S_0} : X_{S_0} \to S_0$, the complex $Rf_{S_0,*}L$ is bounded with finite locally constant cohomology sheaves, and its formation commutes with base change. Namely, for every k-scheme $v : T \to S_0$ of finite type, with base change $f_T : X_T \to T$ and $v_X : X_T \to X_{S_0}$, the canonical base change map $v^*Rf_{S_0,*}L \to Rf_{T,*}v_X^*L$ is an isomorphism.*

Proof. We write Rf_*L as the dual of $Rf_!N$ for a suitable constructible N. The point is that constructibility and base change are known for $Rf_!N$, and over a dense open of S the cohomology sheaves of $Rf_!N$ are locally constant, and there dualizing preserves both properties.

By Proposition 2.42, the locally constant sheaf L comes from a locally constant sheaf on the étale site, and we work with the latter.

Since X is the disjoint union of finitely many open pieces of pure relative dimension, we may assume that f has pure relative dimension d. Let $\omega_f = f^!\mathbf{F}_{\ell,S}$ as in Proposition 6.9, and let

$$N = R\mathcal{H}om_X(L, \mathbf{F}_{\ell,X}) \otimes^{\mathbf{L}} \omega_f.$$

By Proposition 6.9, ω_f is locally $\mathbf{F}_{\ell,X}(d)[2d]$. Hence N is bounded constructible and $R\mathcal{H}om_X(N, \omega_f) = L$.

Duality. Consider the pairing

$$Rf_!N \otimes^{\mathbf{L}} Rf_*L \longrightarrow Rf_!(N \otimes^{\mathbf{L}} L) \longrightarrow Rf_!\omega_f \xrightarrow{\varepsilon_{\mathbf{F}_{\ell,S}}} \mathbf{F}_{\ell,S}, \tag{6.15}$$

where the first arrow is the projection formula followed by the counit $f^*Rf_*L \to L$, and the second is evaluation. Its adjoint is a map

$$\delta_L : Rf_*L \longrightarrow R\mathcal{H}om_S(Rf_!N, \mathbf{F}_{\ell,S}), \tag{6.16}$$

and we claim that it is an isomorphism. Since the map is an adjoint transpose, we test it on Hom groups out of a family of generators, where adjunctions turn one side into the other. To this end we write $L = R\mathcal{H}om_X(N, f^!\mathbf{F}_{\ell,S})$, and test δ_L on the objects $P[n]$, where $P = u_!\mathbf{F}_\ell$ and $u : V \to S$ is quasi-compact, separated and étale. The objects $u_!\mathbf{F}_\ell$ are constructible of finite Tor amplitude, and their shifts detect isomorphisms in $D(S_{ét}, \mathbf{F}_\ell)$. Since $f^*P \otimes^{\mathbf{L}} N$ lies in $D^b_{\mathrm{ctf}}(X, \mathbf{F}_\ell)$, we may use tensor-Hom adjunction, the adjunction $Rf_! \dashv f^!$ of Theorem 5.31 between $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \mathbf{F}_\ell))$ and $\mathrm{Loc}(D^b_{\mathrm{ctf}}(S, \mathbf{F}_\ell))$, and the projection formula. With Hom groups in $D(S_{ét}, \mathbf{F}_\ell)$, we then obtain

$$\mathrm{Hom}(P, Rf_*R\mathcal{H}om(N, f^!\mathbf{F}_{\ell,S})) = \mathrm{Hom}(Rf_!(f^*P \otimes^{\mathbf{L}} N), \mathbf{F}_{\ell,S}) = \mathrm{Hom}(P \otimes^{\mathbf{L}} Rf_!N, \mathbf{F}_{\ell,S}).$$

By tensor-Hom adjunction, the last group is $\mathrm{Hom}(P, R\mathcal{H}om_S(Rf_!N, \mathbf{F}_{\ell,S}))$. Unwinding the adjunctions, one sees that the composite bijection is the map induced by δ_L, for every shift of P. Hence δ_L is an isomorphism.

Shrinking the base. Since δ_L is an isomorphism, it now suffices to understand the dual of $Rf_!N$. By the finiteness theorem for $Rf_!$ [Stacks, 0GL0, 0G2A], the complex $Rf_!N$ is bounded constructible. A constructible sheaf on the integral scheme S is locally constant on a nonempty open subset. Thus each of the finitely many nonzero cohomology sheaves of $Rf_!N$ is locally constant on a nonempty open of S. Let S_0 be the intersection of these opens; it is nonempty because S is irreducible.

Over the field $\mathbf{F}_\ell$, a finite locally constant sheaf is locally free of finite rank. Its dual is exact, and evaluation identifies it with its double dual. We now argue by induction on the number of nonzero cohomology sheaves, using that the relevant canonical maps form morphisms of the truncation triangles. In this way we see that a bounded complex with finite locally constant cohomology is dualizable, that its dual is again of this kind, and that forming the dual commutes with every pullback. Through δ_L, this proves the first assertion over S_0.

Base change. The duality δ_L exchanges Rf_*L with the dual of $Rf_!N$, and $Rf_!N$ satisfies base change. We show that the duality is compatible with base change; then Rf_*L satisfies base change too. Let $v : T \to S_0$ be a k-scheme of finite type over S_0, and let $f_T : X_T \to T$ and $v_X : X_T \to X$ be the base changes. First, base change for $Rf_!$ and (6.1) identify v_X^*N, $v^*Rf_!N$ and $v_X^*\omega_f$ with the corresponding objects for f_T. Next, the base change map $v^*Rf_*L \to Rf_{T,*}v_X^*L$ is characterized by the fact that it carries the pullback of the counit $f^*Rf_*L \to L$ to the counit for f_T. Hence the pullback of (6.15) is the pairing for f_T, composed with this base change map. Finally, the complex $v^*Rf_!N$ is dualizable with dual $v^*R\mathcal{H}om(Rf_!N, \mathbf{F}_{\ell,S})$. Therefore the base change map for Rf_*L is identified with the dual of the base change isomorphism for $Rf_!N$, and it is an isomorphism. ∎

Example 6.11 (Why the base must be shrunk)**.** Let k be algebraically closed, $S = \mathbf{A}^1$ with coordinate t, and let $X \subset \mathbf{A}^1_S$ be the complement of the two sections $x = 0$ and $x = t$, with $f : X \to S$ the projection and $L = \mathbf{F}_\ell$. The morphism f is smooth. For $t \neq 0$ its fibre is $\mathbf{A}^1$ minus two points, that is, a projective line minus three points. Its H^1 is $\mathbf{F}_\ell(-1)^{\oplus 2}$. The fibre over $t = 0$, however, is $\mathbf{G}_m$, with $H^1 = \mathbf{F}_\ell(-1)$. On $S_0 = \mathbf{G}_m$ the lemma applies, but no open containing the origin works. Suppose R^1f_*L were locally constant near 0 and commuted with base change to the points of S. Then its stalks at 0 and at nearby t would be the H^1 of the fibres, which have different ranks. What happens is that the two missing points collide at $t = 0$, and Rf_* cannot see this from the generic fibre.

6.4 Coordinate projections

To control Rj_*F for an open immersion $j : X \hookrightarrow Y$ over S, with Y projective over S, we cover most of Y by opens where the induction hypothesis applies and treat the rest separately. On a chart Y_α, a coordinate function $t : Y_\alpha \to \mathbf{A}^1_S$ lets us view $X \cap Y_\alpha$ over the new base $\mathbf{A}^1_S$, over which its generic fibre has one dimension less. Hence the induction hypothesis in Theorem 6.17 gives constructibility and base change over a dense open of $\mathbf{A}^1_S$. The preimages of these opens, however, do not cover Y. Take for instance $Y = \mathbf{P}^1_S$ with the two standard charts: there a point is missed only if each of the coordinates x and $1/x$ defined there takes an excluded value. After shrinking S, the leftover set is therefore a closed subset finite over S. Lemma 6.13 shows that the same holds in general.

On the leftover finite part we cannot argue by dimension again. We do not have to, since a finite direct image is exact and conservative. This means that on a subset finite over S, the constructibility of a sheaf can be read off from its direct image to S. Lemma 6.12 turns this into a criterion. If a complex is already constructible away from such a subset, a proper direct image to S detects whether it is constructible everywhere. In the smooth case this direct image is $R(bj)_*F$, which Lemma 6.10 controls. For this reason the induction passes through the smooth case first.

Lemma 6.12 (Detection by a proper direct image)**.** *Let $b : Y \to S$ be proper, let $j : U \hookrightarrow Y$ be an open immersion with reduced closed complement $i : Z \hookrightarrow Y$, and suppose that $bi : Z \to S$ is finite.*

1. *If $K \in D^+(Y_{ét}, \mathbf{F}_\ell)$, $j^*K \in D^b_c(U, \mathbf{F}_\ell)$, and $Rb_*K \in D^b_c(S, \mathbf{F}_\ell)$, then $K \in D^b_c(Y, \mathbf{F}_\ell)$.*
2. *If $\varphi : K \to K'$ is a morphism in $D^+(Y_{ét}, \mathbf{F}_\ell)$ such that $j^*\varphi$ and $Rb_*\varphi$ are isomorphisms, then φ is an isomorphism.*

Proof. Using the localization triangle we split K into a part on U, which is known, and a part on Z. Since Z is finite over S, the part on Z is detected by its direct image to S. We compute this direct image from Rb_*K and the known part.

We work on the étale sites, in $D^+(-_{ét}, \mathbf{F}_\ell)$ and $\mathrm{Mod}(-_{ét}, \mathbf{F}_\ell)$.

Finite direct images. Let $h : Z \to S$ be finite. Then h_* is exact and conservative, and $Rh_*M \in D^b_c$ implies $M \in D^b_c$. The first two properties hold because, for a sheaf F of $\mathbf{F}_\ell$-modules on Z, h_*F has stalks

$$(h_*F)_{\bar{s}} = \prod_{\bar{z} \in |Z_{\bar{s}}|} F_{\bar{z}} \tag{6.17}$$

by [Stacks, 03QP]. Suppose h_*F is constructible. The counit $h^*h_*F \to F$ is surjective, since on stalks it is a projection from the finite product (6.17). Its source is constructible, and a quotient of a constructible sheaf is constructible [Stacks, 09BH]. Hence F is constructible. For a complex M we have $H^n(Rh_*M) = h_*H^n(M)$ because h_* is exact, so the statement for complexes follows from conservativity and the sheaf case in each degree.

The localization triangle. The exact sequence for an open and its closed complement [Stacks, 095L], applied termwise to K, gives a triangle

$$j_!j^*K \longrightarrow K \longrightarrow i_*i^*K \longrightarrow j_!j^*K[1]. \tag{6.18}$$

Part (1). The first term $j_!j^*K$ is bounded constructible, because j^*K is and extension by zero preserves constructibility. Its direct image $Rb_*j_!j^*K$ is bounded constructible by the finiteness

theorem [Stacks, 0GL0, 0G2A], which applies since $Rb_* = Rb_!$ for proper b. Applying Rb_* to (6.18), we obtain a triangle whose first two terms, $Rb_*j_!j^*K$ and Rb_*K, are bounded constructible. Hence the third term $Rb_*i_*i^*K = R(bi)_*i^*K$ is bounded constructible too. The first step, applied to $h = bi$, now gives $i^*K \in D^b_c(Z, \mathbf{F}_\ell)$. As i_* is exact and preserves constructibility, (6.18) shows that $K \in D^b_c(Y, \mathbf{F}_\ell)$.

Part (2). Let M be a cone of φ; we must show $M = 0$. We have $j^*M = 0$ because $j^*\varphi$ is an isomorphism. By (6.18), this means $M = i_*i^*M$. Because $Rb_*\varphi$ is an isomorphism, we also have $0 = Rb_*M = R(bi)_*i^*M$. Since $(bi)_*$ is exact, the cohomology sheaves of $R(bi)_*i^*M$ are the sheaves $(bi)_*H^n(i^*M)$. Conservativity of $(bi)_*$ then shows that every $H^n(i^*M)$ vanishes, and so $M = i_*i^*M = 0$. ∎

The next lemma produces such a finite complement. Projections to coordinate lines lower the dimension of the generic fibre, and the subset that they do not cover becomes finite over a smaller base.

Lemma 6.13 (Coordinate projections)**.** *Let S be an integral Noetherian affine scheme with function field $k(S)$, and let $Y \to S$ be projective. Choose a finite affine open cover $Y = \bigcup_\alpha Y_\alpha$ and closed immersions $Y_\alpha \hookrightarrow \mathbf{A}^{m_\alpha}_S$ with $m_\alpha \geq 1$, and write $t_{\alpha r} : Y_\alpha \to \mathbf{A}^1_S$ $(1 \leq r \leq m_\alpha)$ for the coordinate functions.*

1. *If $X \subset Y$ is open and $\dim X_{k(S)} \leq n$, then for every α, r the generic fibre of $t_{\alpha r} : X \cap Y_\alpha \to \mathbf{A}^1_S$ has dimension at most $n - 1$.*
2. *Suppose given a dense open $V_{\alpha r} \subset \mathbf{A}^1_S$ for each α, r, and let*

$$G = \bigcup_{\alpha, r} t_{\alpha r}^{-1}(V_{\alpha r}) \subset Y.$$

After shrinking S, the reduced closed complement $Z = Y \setminus G$ is finite over S.

Proof. Part (1) is a transcendence degree count. For part (2), note that a point of Z has all its coordinates in the finite sets of excluded values. After shrinking S, these sets are cut out by monic polynomials, and monic equations in every coordinate force finiteness.

(1) We use that the dimension of an integral scheme of finite type over a field is the transcendence degree of its function field. Consider an irreducible component of $X_{k(S)} \cap Y_\alpha$ with function field L. If $t_{\alpha r}$ is algebraic over $k(S)$ on it, the component does not meet the generic fibre of $t_{\alpha r}$, as it does not dominate $\mathbf{A}^1_{k(S)}$. Otherwise $t_{\alpha r}$ is transcendental, and

$$\operatorname{trdeg}_{k(S)(t_{\alpha r})} L = \operatorname{trdeg}_{k(S)} L - 1 \leq n - 1.$$

The left-hand side is the dimension of the corresponding component of the generic fibre of $t_{\alpha r}$, and every component of that generic fibre arises in this way.

(2) Let t be the coordinate of $\mathbf{A}^1_S$. The ideal of the complement of $V_{\alpha r}$ contains a nonzero polynomial $P_{\alpha r} \in \mathcal{O}(S)[t]$. Indeed, since $V_{\alpha r}$ is dense, its complement is a proper closed subset of the integral scheme $\mathbf{A}^1_S$, and its ideal is nonzero. After inverting the leading coefficients of these finitely many polynomials, we may assume that all of them are monic. The function $P_{\alpha r}(t_{\alpha r})$ is zero in the coordinate ring of the reduced affine scheme $Z \cap Y_\alpha$, because it vanishes at every point, $t_{\alpha r}$ mapping $Z \cap Y_\alpha$ into the complement of $V_{\alpha r}$. This ring is therefore a quotient of

$$\mathcal{O}(S)[t_{\alpha 1}, \ldots, t_{\alpha m_\alpha}] / (P_{\alpha 1}(t_{\alpha 1}), \ldots, P_{\alpha m_\alpha}(t_{\alpha m_\alpha})),$$

which is generated as an $\mathcal{O}(S)$-module by the monomials whose exponent in each variable is less than the degree of the corresponding polynomial. Each $Z \cap Y_\alpha$ is thus finite over S, and $Z \to S$ has finite fibres. As Z is closed in the projective scheme Y, the morphism $Z \to S$ is also proper, and a proper morphism with finite fibres is finite [Stacks, 02LS]. ■

6.5 Generic base change

Let $j : X \hookrightarrow Y$ be an open immersion of schemes over an integral base S, and let F be a constructible $\mathbf{F}_\ell$-sheaf on X. For such data, Deligne's generic base change theorem says that over a dense open of S the complex Rj_*F is bounded constructible and its formation commutes with base change. In particular, over a field it gives the finiteness of Rj_* (Corollary 6.18), and this is all that the computation of $f^!$ requires. On the other hand, the punctured plane over the t-line, discussed at the beginning of this chapter, shows that we cannot ask for more than a dense open.

In characteristic p the generic fibre need not contain a smooth dense open. For example, let $X = \mathbf{A}^1_{\mathbf{F}_p}$ map to the t-line by $x \mapsto x^p$. The generic fibre is $\operatorname{Spec} \mathbf{F}_p(t^{1/p})$, which is not smooth over $\mathbf{F}_p(t)$. For this reason Step 3 of the proof of Theorem 6.17 below allows a finite universal homeomorphism of the base. Such a map changes neither the étale topoi nor the conclusion.

We fix notation for base change. With j as above and a morphism $T \to S$, we write $j_T : X_T \hookrightarrow Y_T$ for the base change and $g_X : X_T \to X$, $g_Y : Y_T \to Y$ for the projections. The base change map $g_Y^* Rj_*K \to Rj_{T,*} g_X^* K$ is then the map (4.5).

The proof of Theorem 6.17 is an induction on the dimension of the generic fibre, in three steps. The next three lemmas are the parts of this proof that do not use the induction hypothesis, and each serves one step. First, Lemma 6.14 is Step 2: if X is smooth and F locally constant, it extends the conclusion from an open G with complement finite over S, provided by Step 1, across that complement. Next, Lemma 6.15 allows Step 3 to replace S by a finite universal homeomorphism, as the example above requires. Lemma 6.16 then provides, after this replacement, the dense open V of Step 3, smooth with F locally constant, whose complement has a smaller generic fibre.

Lemma 6.14 (Extension across a finite subset)**.** *Let S be integral and affine, let $b : Y \to S$ be projective, and let $j : X \hookrightarrow Y$ be an open immersion with X smooth over S. Let F be a finite locally constant $\mathbf{F}_\ell$-sheaf on $X_{ét}$. Suppose that there is an open $G \subset Y$ whose reduced complement $i : Z \hookrightarrow Y$ is finite over S, such that $(Rj_*F)|_G \in D^b_c(G, \mathbf{F}_\ell)$ and such that for every k-scheme $T \to S$ of finite type the base change map $g_Y^* Rj_*F \to Rj_{T,*} g_X^* F$ is an isomorphism on G_T. Then there is a nonempty open $S_0 \subset S$ such that $Rj_{S_0,*}(F|_{X_{S_0}}) \in D^b_c(Y_{S_0}, \mathbf{F}_\ell)$ and such that for every k-scheme $T \to S_0$ of finite type the base change map $g_Y^* Rj_{S_0,*}(F|_{X_{S_0}}) \to Rj_{T,*} g_X^* F$ is an isomorphism.*

Proof. Lemma 6.10 controls the direct image $R(bj)_*F$ of Rj_*F to S. Then Lemma 6.12 turns this, together with the hypothesis on G, into control of Rj_*F along Z. Throughout, we work on the étale sites.

For constructibility we check the hypotheses of part (1) of Lemma 6.12 for $K = Rj_*F$ and $U = G$. The complex Rj_*F lies in $D^+(Y_{ét}, \mathbf{F}_\ell)$ because F is a sheaf. Its restriction to G is in D^b_c by assumption, and $Z = Y \setminus G$ is finite over S. It remains to treat the direct image. Since $bj : X \to S$ is smooth and F is locally constant, we may apply Lemma 6.10. After shrinking S to a nonempty open S_0, the complex $Rb_*Rj_*F = R(bj)_*F$ lies in $D^b_c(S_0, \mathbf{F}_\ell)$, and its formation commutes with base change along every $T \to S_0$ of finite type over k. Part (1) of Lemma 6.12 then gives $Rj_*F \in D^b_c(Y_{S_0}, \mathbf{F}_\ell)$.

For base change we use part (2) of the lemma instead. Let $T \to S_0$ be a k-scheme of finite type, and let φ be the base change map $g_Y^* Rj_*F \to Rj_{T,*} g_X^* F$. By assumption it is an isomorphism

on G_T. Proper base change for b and the compatibility of base change maps with composition [Stacks, 0E46] identify $Rb_{T,*}\varphi$ with the base change map for $R(bj)_*F$, which is an isomorphism. Since b_T is proper and $b_T i_T$ is finite, part (2) of Lemma 6.12 applies over T. It shows that φ is an isomorphism. ∎

Lemma 6.15 (Universal homeomorphisms of the base)**.** *Let S be integral, let $j : X \hookrightarrow Y$ be an open immersion of S-schemes, and let F be a constructible $\mathbf{F}_\ell$-sheaf on $X_{ét}$. Let $S' \to S$ be a finite universal homeomorphism with S' integral, let $X' = (X_{S'})_{\mathrm{red}}$ and $Y' = (Y_{S'})_{\mathrm{red}}$, let $j' : X' \hookrightarrow Y'$ be the open immersion, and let F' be the pullback of F to X'. Suppose that there is a nonempty open $U' \subset S'$ such that $Rj'_{U',*}(F'|_{X'_{U'}}) \in D^b_c(Y'_{U'}, \mathbf{F}_\ell)$ and such that for every k-scheme $T' \to U'$ of finite type the base change map for j' along T' is an isomorphism. Then there is a nonempty open $U \subset S$ with the same two properties for j and F.*

Proof. A finite universal homeomorphism is a homeomorphism of underlying spaces, so we may remove from S the closed image of $S' \setminus U'$. The remaining open U is nonempty, and its inverse image S'_U lies in U'. Over U the morphisms $X'_U \to X_U$ and $Y'_U \to Y_U$ are finite universal homeomorphisms, and pullback along them is therefore an equivalence of étale topoi [Stacks, 03SI]. Such equivalences preserve constructibility in both directions and commute with Rj_*. Hence pullback along these morphisms detects boundedness and constructibility, and we get $Rj_{U,*}(F|_{X_U}) \in D^b_c(Y_U, \mathbf{F}_\ell)$.

For base change, let $T \to U$ be a k-scheme of finite type, with projections $g_X : X_T \to X$ and $g_Y : Y_T \to Y$, and let

$$\beta_T : g_Y^* Rj_* F \longrightarrow Rj_{T,*} g_X^* F$$

be the base change map for j along T. Put $T' = T \times_U S'_U$. Then $v : T' \to T$ is again a universal homeomorphism, and so are the induced maps $v_X : X'_{T'} \to X_T$ and $v_Y : Y'_{T'} \to Y_T$, each a thickening followed by a base change of v. Let $\beta'_{T'} : g_{Y'}^* Rj'_* F' \to Rj'_{T',*} g_{X'}^* F'$ be the base change map for j' along T'. Base change maps compose [Stacks, 0EZR, 0EZS], and $v_Y^* Rj_{T,*} \simeq Rj'_{T',*} v_X^*$ because v_X^* and v_Y^* are equivalences of étale topoi [Stacks, 03SI]. Hence $v_Y^* \beta_T$ is identified with $\beta'_{T'}$. The map $\beta'_{T'}$ is an isomorphism by assumption, so $v_Y^* \beta_T$ is one. Since v_Y^* is an equivalence, it is conservative, and therefore β_T is an isomorphism. ∎

Lemma 6.16 (A smooth dense open after a universal homeomorphism)**.** *Let S be integral, let X be an S-scheme all of whose irreducible components dominate S, let $n = \dim X_{k(S)}$, and let F be a constructible $\mathbf{F}_\ell$-sheaf on $X_{ét}$. After shrinking S to a nonempty open, there are a finite universal homeomorphism $S' \to S$ with S' integral and a dense open $V \subset (X_{S'})_{\mathrm{red}}$ such that V is smooth over S', the pullback of F to V is locally constant, and the generic fibre of $(X_{S'})_{\mathrm{red}} \setminus V$ over S' has dimension less than n.*

Proof. By [Stacks, 0550], after shrinking S, there is a finite universal homeomorphism $S' \to S$ with S' integral, such that $(X_{S'})_{\mathrm{red}} \to S'$ is flat of finite presentation with geometrically reduced generic fibre. This generic fibre has a dense smooth open [Stacks, 056V]. By the fibrewise criterion [Stacks, 01V9], its generic points therefore lie in the smooth locus of $(X_{S'})_{\mathrm{red}} \to S'$. After shrinking the base once more to remove non-dominant components, this smooth locus contains a dense open of $(X_{S'})_{\mathrm{red}}$. Since F is constructible, there is a smaller dense open V on which the pullback of F is locally constant. Its complement contains no generic point of a component of the generic fibre. Hence the generic fibre of the complement has dimension less than n. ∎

Theorem 6.17 (Generic constructibility and base change for Rj_*)**.** *Let S be integral, and let $j : X \hookrightarrow Y$ be an open immersion of S-schemes. For a constructible $\mathbf{F}_\ell$-sheaf F on $X_{ét}$, there is a nonempty open $S_0 \subset S$ such that*

$$Rj_{S_0,*}(F|_{X_{S_0}}) \in D^b_c(Y_{S_0}, \mathbf{F}_\ell), \tag{6.19}$$

and such that for every k-scheme $T \to S_0$ of finite type, with base change $j_T : X_T \hookrightarrow Y_T$ and projections $g_X : X_T \to X_{S_0}$ and $g_Y : Y_T \to Y_{S_0}$, the base change map

$$\beta_T : g_Y^* Rj_{S_0,*}(F|_{X_{S_0}}) \longrightarrow Rj_{T,*} g_X^*(F|_{X_{S_0}}) \tag{6.20}$$

is an isomorphism.

Proof. We work with the étale sheaf corresponding to F, in $D^+(-_{ét}, \mathbf{F}_\ell)$, and prove (6.19) and β_T together by induction on $n = \dim X_{k(S)}$, allowing at each stage all integral bases S of finite type over k. We first handle the complement of a finite set by passing to the base $\mathbf{A}^1_S$, then the smooth, locally constant case, and finally the general case by dévissage.

Preliminary reductions. Several modifications of the data are harmless, and we will make them freely in the induction. First, the assertions extend from open to locally closed immersions with the same source. Indeed, such an immersion factors as an open immersion into its scheme-theoretic closure followed by a closed immersion. Since closed direct image is exact, preserves constructibility, and commutes with arbitrary pullback [Stacks, 0959], the assertions pass along this factorization. We will use the induction hypothesis in this form. Next, replacing Y by the scheme-theoretic closure of X, we may assume that X is dense in Y. We may also shrink S to be affine, and we may remove the closures of the images of the irreducible components of Y that do not dominate S, since there are finitely many of them and none contains the generic point of S. Finally, the assertions are local on Y. Since we only use finite affine covers, the finitely many resulting opens of S can be intersected.

Base case. If the generic fibre is empty or zero-dimensional, then after shrinking S the immersion j becomes an isomorphism or Y becomes empty. Suppose first that $X_{k(S)}$ is empty. By density $Y_{k(S)}$ is empty. On a finite affine cover, a coordinate algebra that vanishes over $k(S)$ has its unit killed by a nonzero element of $\mathcal{O}(S)$. Hence Y becomes empty after shrinking S. If $n = 0$, the dense open $X_{k(S)}$ of the zero-dimensional Noetherian space $Y_{k(S)}$ is all of $Y_{k(S)}$. The same argument applied to $Y \setminus X$ shows that j becomes an isomorphism after shrinking S.

Now let $n \geq 1$, and assume the theorem, including β_T, for generic fibres of dimension less than n.

Step 1: for Y projective over S, both assertions hold on an open $G \subset Y$ whose complement is finite over S. We replace the base S by the larger base $\mathbf{A}^1_S$, over which the generic fibre is smaller, so that the induction hypothesis applies. To do this, choose charts and coordinates as in Lemma 6.13. Via $t_{\alpha r}$, we regard $X \cap Y_\alpha \hookrightarrow Y_\alpha$ as an open immersion of schemes over the integral base $\mathbf{A}^1_S$, which is of finite type over k. By part (1) of that lemma, its generic fibre has dimension less than n. Therefore the induction hypothesis, applied over the base $\mathbf{A}^1_S$ to the restriction of F, gives a dense open $V_{\alpha r} \subset \mathbf{A}^1_S$ over which both assertions hold.

Since Rj_* commutes with restriction to the open Y_α, the assertions hold for Rj_*F on $t_{\alpha r}^{-1}(V_{\alpha r})$. They continue to hold after a base change $T \to S$ with T of finite type over k, as we see by applying the hypothesis to the inverse image of $V_{\alpha r}$ in $\mathbf{A}^1_T$. Let G be the finite union of the opens $t_{\alpha r}^{-1}(V_{\alpha r})$. Then on G the map β_T is an isomorphism, and the complex Rj_*F is bounded constructible. For the boundedness, note that there are finitely many pieces, and therefore the cohomology lies in a common finite range of degrees. Finally, part (2) of the lemma shows that, after shrinking S, the reduced complement $Z = Y \setminus G$ is finite over S.

Step 2: the theorem holds in dimension n when X is smooth over S and F is locally constant. We may assume that Y is projective over S. Indeed, let Y_β be a member of a finite affine cover of Y, embed it in some $\mathbf{A}^m_S$ over the affine base, and let $\overline{Y}$ be the closure of $X \cap Y_\beta$ in $\mathbf{P}^m_S$. Then $X \cap Y_\beta$ is open and dense in $\overline{Y}$, which is projective over S. Moreover $X \cap Y_\beta$ is still smooth over S, with generic fibre of dimension at most n. The result for $X \cap Y_\beta \subset \overline{Y}$ gives it on Y_β, because Rj_* and its base change maps commute with restriction to opens of the target.

For Y projective, Step 1 gives an open $G \subset Y$ with complement finite over S on which both assertions hold. Since X is smooth over S and F is locally constant, Lemma 6.14 applies. It shows that the assertions hold on all of Y after shrinking S.

Step 3: the general case in dimension n. We find a dense open $V \subset X$ to which Step 2 applies, and treat the difference between F and Rv_*v^*F, which lives on the smaller set $X \setminus V$, by the induction hypothesis. Let $S' \to S$ and V be as in Lemma 6.16. Lemma 6.15 shows that it suffices to prove the theorem for $(X_{S'})_{\mathrm{red}} \hookrightarrow (Y_{S'})_{\mathrm{red}}$ and the pullback of F over S'. Its generic fibre still has dimension n because $S' \to S$ is a universal homeomorphism. We therefore replace S by S' and all schemes by these reductions. Then $V \subset X$ is a dense open, smooth over S, on which F is locally constant. The generic fibre of $X \setminus V$ has dimension less than n.

Let $v : V \hookrightarrow X$ be the open immersion and $i : X \setminus V \hookrightarrow X$ its reduced closed complement. By Step 2, after shrinking the base, both v and jv have bounded constructible direct images of v^*F. Their base change maps are isomorphisms after every base change $T \to S$ with T of finite type over k. Complete the unit of $v^* \dashv Rv_*$ to a triangle

$$\Delta_F : \; F \longrightarrow Rv_*v^*F \longrightarrow D \longrightarrow F[1]. \tag{6.21}$$

The first term is the constructible sheaf F, and the second is bounded constructible by Step 2. Hence D is bounded constructible too. Indeed, by the long exact cohomology sequence, its cohomology sheaves are extensions of kernels by cokernels of maps between constructible sheaves, and these are constructible [Stacks, 09BH]. Moreover $v^*D = 0$, as the unit is an isomorphism on V. The localization triangle (6.18) for v and i now shows that

$$D = i_*i^*D.$$

The locally closed immersion $ji : X \setminus V \to Y$ has generic fibre of dimension less than n. We apply the induction hypothesis to each of the finitely many constructible cohomology sheaves of i^*D, together with their truncation triangles. This gives

$$R(ji)_*i^*D = Rj_*D \in D^b_c(Y, \mathbf{F}_\ell),$$

with base change maps that are isomorphisms. Now apply Rj_* to the triangle Δ_F. In the resulting triangle, the middle term $R(jv)_*v^*F$ is bounded constructible by Step 2, and the third term Rj_*D was treated just now. Hence Rj_*F is bounded constructible.

To see that the base change map β_T is an isomorphism, we compare the triangle Δ_F with its analogue over T. Let $g_X : X_T \to X$, $g_Y : Y_T \to Y$ and $g_V : V_T \to V$ be the projections, and $v_T : V_T \to X_T$ the base change of v. The base change map

$$\beta_v : g_X^*Rv_*v^*F \longrightarrow Rv_{T,*}g_V^*v^*F$$

is an isomorphism by Step 2, and it carries g_X^* of the first arrow of the triangle Δ_F to the unit $g_X^*F \to Rv_{T,*}v_T^*g_X^*F$ of $v_T^* \dashv Rv_{T,*}$. Since cones are not functorial, we have to choose

the triangles compatibly, and we take as triangle for this unit the pullback of the triangle Δ_F, transported through β_v. Now the base change map

$$\beta_j : g_Y^* Rj_* \longrightarrow Rj_{T,*} g_X^*$$

is a natural transformation of exact functors. It therefore induces a morphism from g_Y^* of Rj_* of the triangle Δ_F to $Rj_{T,*}$ of this triangle. On the second term it is β_j at Rv_*v^*F followed by $Rj_{T,*}\beta_v$. Since base change maps compose [Stacks, 0EZR, 0EZS], this composite is the base change map for jv, an isomorphism by Step 2. On the third term it is β_j at D. It is an isomorphism because, composed with $Rj_{T,*}$ of the base change isomorphism for the closed immersion i [Stacks, 0959], it is the base change map for ji at i^*D treated above. Hence, by the five lemma for triangles, it is an isomorphism on the first term, which is β_T.

This completes the induction. ∎

The only nonempty open of $\operatorname{Spec} k$ is $\operatorname{Spec} k$ itself, and hence over a field the theorem holds without any shrinking.

Corollary 6.18 (Open direct image over a field)**.** *Let $j : X \hookrightarrow Y$ be an open immersion. Then*

$$Rj_* : D_c^b(X, A) \longrightarrow D_c^b(Y, A), \tag{6.22}$$

and for every k-scheme T of finite type, with projections $g_X : X_T \to X$ and $g_Y : Y_T \to Y$, and every $K \in D_c^b(X, A)$, the base change map $g_Y^* Rj_* K \to Rj_{T,*} g_X^* K$ is an isomorphism.

Proof. We argue on the étale site. The theorem gives the case of $\mathbf{F}_\ell$-sheaves, and we pass to $\mathbf{Z}/\ell^r$-sheaves by filtrations, to A-sheaves by forgetting the A-structure, and to complexes by truncation.

For a constructible $\mathbf{F}_\ell$-sheaf this is Theorem 6.17 with $S = \operatorname{Spec} k$. A constructible sheaf F killed by ℓ^r has the finite filtration $F \supset \ell F \supset \cdots \supset \ell^r F = 0$. Its successive quotients $\ell^i F/\ell^{i+1}F$ are constructible and killed by ℓ, that is, constructible $\mathbf{F}_\ell$-sheaves. Each short exact sequence in the filtration is a triangle in the derived category, and Rj_* carries it to a triangle. By induction on r, the long exact cohomology sequences of these triangles give (6.22). The base change maps form morphisms between the resulting triangles, and they are isomorphisms by the five lemma.

For a finite ℓ-primary ring A, we apply this to the underlying $\mathbf{Z}/\ell^r\mathbf{Z}$-sheaf and carry the A-action along all maps. This is possible for two reasons. First, $Rj_* : D^+(X_{ét}, A) \to D^+(Y_{ét}, A)$ commutes with the forgetful functors to $D^+(-_{ét}, \mathbf{Z}/\ell^r\mathbf{Z})$ [Stacks, 0D6J]. Second, an A-sheaf is constructible if and only if its underlying sheaf is. To see this, consider a stratum where the underlying sheaf is finite locally constant. There the action of each of the finitely many elements of A is a section of the finite locally constant sheaf of endomorphisms, hence constant étale locally. Therefore the A-module is locally constant on that stratum. Truncation triangles extend both conclusions from sheaves to bounded constructible complexes. ∎

6.6 Finite coefficients

Let A be a finite commutative ℓ-primary ring and f a separated morphism of finite type. We show that the right adjoint $f^!$ of Theorem 5.31 preserves $D^b_{\mathrm{ctf}}(-, A)$ (Theorem 6.20). By the reduction at the start of this chapter and by the previous section, we only have to show that direct image along an open immersion j preserves finite Tor amplitude. The upper end of the

Tor amplitude grows by at most the ℓ-cohomological dimension $\mathrm{cd}_\ell(j)$ of Definition 2.45. This dimension is finite for morphisms of finite type over k.

Finite Tor amplitude cannot be checked on cohomology sheaves, since these may have infinite Tor dimension over A: for $A = \mathbf{Z}/\ell^2$, already the constant sheaf $\mathbf{F}_\ell$ has infinite Tor dimension. We therefore obtain the Tor bounds below from flat models. By [Stacks, 0F4M], every object of $D^b_{\mathrm{ctf}}(X, A)$ is represented by a bounded complex of constructible flat A-sheaves. On the strata of a finite stratification, such a model is locally a bounded complex of finite projective A-modules.

Proposition 6.19 (Open direct image preserves finite Tor amplitude)**.** *Let $j : U \hookrightarrow X$ be an open immersion. Then $Rj_* : D(U_{\mathrm{pro\acute{e}t}}, A) \to D(X_{\mathrm{pro\acute{e}t}}, A)$ restricts to*

$$Rj_* : D^b_{\mathrm{ctf}}(U, A) \longrightarrow D^b_{\mathrm{ctf}}(X, A).$$

More precisely, if K has Tor amplitude in $[a, b]$ and $d \geq \mathrm{cd}_\ell(j)$, then Rj_*K has Tor amplitude in $[a, b + d]$.

Proof. We test Tor amplitude by tensoring with modules. By a projection formula for constant modules we may move the tensor product inside Rj_*. There the complex $K \otimes^{\mathbf{L}} \underline{M}$ has amplitude $[a, b]$, and Rj_* adds at most d degrees.

The complex Rj_*K is bounded constructible by Corollary 6.18, and $\mathrm{cd}_\ell(j)$ is finite by Theorem theorem-results-from-etale-theory. To bound the Tor amplitude, we work on the étale site, where geometric points are conservative. By Corollary 2.44, we have $K \simeq \nu_U^* K_{\acute{e}t}$ for some $K_{\acute{e}t} \in D^b_{\mathrm{ctf}}(U_{\acute{e}t}, A)$. By Proposition 2.49, the pro-étale pullback of $Rj_{\acute{e}t,*}K_{\acute{e}t}$ is Rj_*K.

We first prove the projection formula for constant coefficients. Namely, for every A-module M, viewed as a constant sheaf, the projection morphism

$$(Rj_*K) \otimes^{\mathbf{L}}_A \underline{M}_X \xrightarrow{\sim} Rj_*(K \otimes^{\mathbf{L}}_A \underline{M}_U) \tag{6.23}$$

is an isomorphism. Recall that for a scheme V with $\mathrm{cd}_\ell(V) < \infty$, the projection formula of [Stacks, 0F12] identifies $R\Gamma(V, K) \otimes^{\mathbf{L}}_A M$ with $R\Gamma(V, K \otimes^{\mathbf{L}}_A \underline{M})$. Moreover, a map of complexes of sheaves is an isomorphism if it induces isomorphisms on derived sections over a basis of the site. Since the sections of Rj_* over V are the sections over $V \times_X U$, we apply the projection formula to every quasi-compact étale $V \to X$ and to $V \times_X U$. This gives isomorphisms between the derived sections of the two sides of the étale version of (6.23), compatibly with the projection morphism. Hence the étale version holds. Finally, inverse image is exact and symmetric monoidal, and it commutes with unbounded direct image, so the pullback of the étale isomorphism is (6.23).

On the étale site it suffices to bound Tor against constant A-modules. Indeed, let G be an étale sheaf of A-modules and $\bar{x}$ a geometric point. The stalk of $Rj_{\acute{e}t,*}K_{\acute{e}t} \otimes^{\mathbf{L}}_A G$ at $\bar{x}$ is that of $Rj_{\acute{e}t,*}K_{\acute{e}t} \otimes^{\mathbf{L}}_A \underline{M}$ with $M = G_{\bar{x}}$, and geometric stalks are conservative. Now $K \otimes^{\mathbf{L}}_A \underline{M}_U$ has cohomology in $[a, b]$. Hence the étale version of the right-hand side of (6.23) lies in $[a, b + d]$. By the projection formula, so does $Rj_{\acute{e}t,*}K_{\acute{e}t} \otimes^{\mathbf{L}}_A \underline{M}$. In other words, $Rj_{\acute{e}t,*}K_{\acute{e}t}$ has Tor amplitude in $[a, b + d]$.

To pass to $X_{\mathrm{pro\acute{e}t}}$, we pull back a bounded flat constructible model of $Rj_{\acute{e}t,*}K_{\acute{e}t}$ in degrees $[a, b+d]$ [Stacks, 0F4M]. Since pullback preserves flatness, its pullback to $X_{\mathrm{pro\acute{e}t}}$ is again a complex of flat sheaves in degrees $[a, b + d]$. Hence its tensor product with any sheaf of A-modules on $X_{\mathrm{pro\acute{e}t}}$ is computed termwise and lies in degrees $[a, b + d]$. This argument also covers the sheaves on $X_{\mathrm{pro\acute{e}t}}$ that are not pulled back from $X_{\acute{e}t}$, for which geometric stalks would not detect the Tor amplitude. ∎

For $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ over $\bar{k}$ and $K = A$, the stalk of Rj_*A at the origin is the cohomology of the punctured strict henselization. It has cohomology A in degree 0 and $A(-1)$ in degree 1, both free. It is therefore isomorphic to $A \oplus A(-1)[-1]$. The Tor amplitude grows from $[0,0]$ to $[0,1]$, and the bound of the proposition, with $d = \mathrm{cd}_\ell(j) = 1$, is attained.

Combining the proposition with the computation of $p^!$ for affine space, we obtain the main theorem for finite coefficients.

Theorem 6.20 ($f^!$ with finite coefficients)**.** *For every separated morphism $f : X \to Y$ of finite type, the right adjoint $f^! : \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,A)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,A))$ of Theorem 5.31 maps $D^b_{\mathrm{ctf}}(Y,A)$ into $D^b_{\mathrm{ctf}}(X,A)$. Consequently, its adjunction restricts to*

$$Rf_! : D^b_{\mathrm{ctf}}(X,A) \;\rightleftarrows\; D^b_{\mathrm{ctf}}(Y,A) : f^!.$$

Proof. Locally on X and Y, the morphism f is a closed immersion followed by the projection from an affine space. We therefore treat these two cases and then glue.

Closed immersions. Let $i : Z \hookrightarrow X$ be a closed immersion with open complement $j : U \hookrightarrow X$, and let $R\underline{\Gamma}_Z$ be the local cohomology functor of Section 2.14. We show that $R\underline{\Gamma}_Z$ preserves D^b_{ctf} and that it is $i^!$. For $K \in D^b_{\mathrm{ctf}}(X,A)$, there is a triangle

$$i_*R\underline{\Gamma}_Z K \longrightarrow K \longrightarrow Rj_*j^*K \longrightarrow i_*R\underline{\Gamma}_Z K[1].$$

By Proposition 6.19, the two terms on the right lie in D^b_{ctf}, and hence so does $i_*R\underline{\Gamma}_Z K$. Since $i^*i_* = \mathrm{id}$ and i^* preserves D^b_{ctf}, we get $R\underline{\Gamma}_Z K \in D^b_{\mathrm{ctf}}(Z,A)$. Explicitly, if K has Tor amplitude in $[a,b]$, then $R\underline{\Gamma}_Z K$ has Tor amplitude in $[a, b + \mathrm{cd}_\ell(j) + 1]$. The functor $R\underline{\Gamma}_Z : D(X_{\mathrm{pro\acute{e}t}}, A) \to D(Z_{\mathrm{pro\acute{e}t}}, A)$ is right adjoint to i_* (Lemma lemma-sections-support-closed-subscheme), and $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,A))$ is a full subcategory of $D((-)_{\mathrm{pro\acute{e}t}}, A)$. By uniqueness of right adjoints, $R\underline{\Gamma}_Z K$ is therefore $i^!K$ in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Z,A))$.

Projections. Let $p : \mathbf{A}^d_Y \to Y$ be the projection. From the adjunction $Rq_* \dashv q^*(1)[2]$ for the projective line $q : \mathbf{P}^1_Y \to Y$ (Proposition 6.7) and from Lemma 6.8, we deduce that $p^! \simeq p^*(-)(d)[2d]$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,A))$. This functor preserves D^b_{ctf}, since pullback preserves a bounded constructible flat model and the Tate twist is locally free of rank one.

For example, for the origin $i : 0 \hookrightarrow \mathbf{A}^1$ over $\bar{k}$ and $K = A$, the triangle gives $i^!A \simeq A(-1)[-2]$ (Example 6.25 below). Its Tor amplitude $[2,2]$ lies in the predicted interval $[0, 0+1+1]$.

Gluing. After restricting to affine opens in X and Y, a morphism of finite type factors as a closed immersion into an affine space over Y followed by its projection. Since $f^!$ is compatible with composition and restriction to opens (Proposition 5.32), the restriction of $f^!K$ is obtained by applying the two functors just considered. Thus $f^!K$ is bounded constructible of finite Tor amplitude locally on X. A finite affine cover of the quasi-compact scheme X gives a single cohomological interval and a single interval of Tor amplitude valid everywhere. Finally, $Rf_!$ preserves D^b_{ctf} by Proposition 4.35. Both sides of the adjunction therefore lie in D^b_{ctf}, and its Hom bijections restrict. ∎

6.7 ℓ-adic coefficients

For $\Lambda = \mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ we have $D^b_{\mathrm{ctf}} = D^b_c$, and we show that $f^!$ preserves D^b_c. The argument of Section 6.6 uses two geometric facts, namely the finiteness of Rj_* for an open immersion and the adjunction for the projective line, and we deduce both from the case of torsion coefficients. This is possible because an integral complex lies in D^b_c if it is derived complete with constructible

reduction modulo π_E (Proposition 3.11), and rational complexes are built from integral ones (Lemma lemma-integral-models).

For Rj_* this reduction is immediate, since Rj_* commutes with $R\varprojlim$ and with cones (Proposition 6.21). For the projective line it is not, because the unit and counit are morphisms, and a compatible family of morphisms modulo ℓ^n need not come from a morphism of the limits. We therefore construct the class of the diagonal and the trace once, at the level of inverse systems (Subsection 6.7.1). Besides this, the integral case uses the structure of integral constructible complexes and the fact that proper direct image preserves direct sums (Subsection 5.4.2).

Proposition 6.21 (Direct image along an open immersion)**.** *Let $j : U \hookrightarrow X$ be a quasi-compact open immersion. Then $Rj_* : D(U_{\text{proét}}, \Lambda) \to D(X_{\text{proét}}, \Lambda)$ maps $D^b_c(U, \Lambda)$ into $D^b_c(X, \Lambda)$ for $\Lambda = \mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$.*

Proof. For $\mathcal{O}_E$ we use the criterion of Proposition 3.11, which asks for derived completeness and constructible reduction. Let $K \in D^b_c(U, \mathcal{O}_E)$. Since Rj_* commutes with $R\varprojlim$ [Stacks, 099L], Rj_*K is derived complete. Since Rj_* also commutes with cones, the reduction of Rj_*K modulo π_E is Rj_* of the reduction of K. That reduction comes from $U_{ét}$. Its direct image is therefore the pullback of the étale one [Stacks, 09A6], which is bounded constructible by Corollary 6.18. Both conditions of the criterion hold, and so $Rj_*K \in D^b_c(X, \mathcal{O}_E)$.

For E and $\overline{\mathbf{Q}}_\ell$ we reduce to the integral case. Since Rj_* is exact, the objects K with $Rj_*K \in D^b_c$ form a triangulated subcategory. It therefore suffices to treat the generators of Lemma lemma-integral-models. These come from integral objects by inverting ℓ and by a colimit over finite extensions of E. Both operations are filtered colimits of systems with a common lower bound, and Rj_* commutes with such colimits (Corollary 2.29). Hence Rj_* of a generator is obtained from Rj_* of an integral object by the same operations, and it lies in D^b_c. ∎

6.7.1 The projective line

With torsion coefficients, $Rq_* \dashv q^*(1)[2]$ is built from the cycle class $\delta : \Delta_*\Lambda \to \Lambda(1)[2]$ of the diagonal of $\mathbf{P}^1_Y \times_Y \mathbf{P}^1_Y$, the trace $\mathrm{Tr}_q : Rq_*\Lambda(1)[2] \to \Lambda$ and two identities between them (Proposition 6.7). The classes modulo ℓ^n, however, are morphisms in different derived categories, compatible under reduction. Since morphisms in a derived category are not computed levelwise, such a family need not give a morphism after $R\varprojlim$. We therefore construct the class once, in the derived category of inverse systems of sheaves, and then pass to the limit.

Proposition 6.22 (The ℓ-adic cycle class of the diagonal)**.** *Let $q : \mathbf{P}^1_Y \to Y$ be the projection, Δ the diagonal of $\mathbf{P}^1_Y \times_Y \mathbf{P}^1_Y$ and p_1, p_2 its projections. There are a morphism $\delta_{\mathcal{O}_E} : \Delta_*\mathcal{O}_E \to \mathcal{O}_E(1)[2]$ reducing to the cycle classes of Lemma 6.3 and a trace $\mathrm{Tr}_q : Rq_*\mathcal{O}_E(1)[2] \to \mathcal{O}_E$ lifting those of Lemma 6.4, with $\mathrm{Tr}_{p_i} \circ Rp_{i*}(\delta_{\mathcal{O}_E}) = 1$ for $i = 1, 2$. Extending scalars gives δ_Λ and Tr_q with the same identities for $\Lambda = E, \overline{\mathbf{Q}}_\ell$. All are compatible with base change along separated morphisms of finite type $Y' \to Y$.*

Proof. The cycle class comes from a single map defined with $\mathbf{Z}$-coefficients, the boundary of the divisor sequence. Hence all its reductions are reductions of one map of systems.

The diagonal is an effective Cartier divisor, which gives a boundary morphism $\Delta_*\mathbf{Z} \to \mathbf{G}_m[1]$ (Lemma 6.3). Tensoring with the system $(\mathbf{Z}/\ell^n)_n$ and using the Kummer sequence, we get one morphism of systems $\Delta_*\mathbf{Z}/\ell^\bullet \to \mathbf{Z}/\ell^\bullet(1)[2]$. Its level n is the class of Lemma 6.3 for $\mathbf{Z}/\ell^n$. The transition maps between levels are part of the data, and this is what a family of separately constructed classes would lack. Next we take $R\varprojlim$. The site is replete with exact products (Propositions 2.22 and Proposition 2.23). Hence $R\varprojlim \circ (- \otimes \mathbf{Z}/\ell^\bullet)$ is derived ℓ-completion [Stacks,

099M], and the limit $\Delta_*\mathbf{Z}_\ell \to \mathbf{Z}_\ell(1)[2]$ is the completion of the boundary map of Lemma 6.3. It is linear over the completed coefficient sheaf by [Stacks, 099I]. We obtain the trace in the same way from the decomposition of $Rq_*\mathbf{Z}/\ell^\bullet$.

Each $\mathrm{Tr}_{p_i} \circ Rp_{i*}(\delta)$ is an endomorphism of $\mathbf{Z}/\ell^\bullet$ in degree 0, that is, a family of endomorphisms of the levels. These are the identity by Proposition 6.7. We then extend scalars to $\mathcal{O}_E$, invert ℓ and take the colimit over finite extensions (Corollary 2.29). Base change holds because the Cartier and Kummer maps are natural and pullback commutes with $R\varprojlim$ here (Lemma 2.54). ■

To turn δ_Λ and Tr_q into an adjunction we repeat the torsion calculation. It used the projection formula and proper base change on the categories where Rq_* and $q^*(1)[2]$ act. Proper base change on $\mathrm{Loc}(D^b_{\mathrm{ctf}})$ is Proposition 5.24, and we extend the projection formula in the same way.

Proposition 6.23 (Projection formula with rational coefficients)**.** *For $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ is closed under $\otimes^{\mathbf{L}}$, and for proper $p : X \to Y$ the map $(Rp_*M) \otimes^{\mathbf{L}} N \to Rp_*(M \otimes^{\mathbf{L}} p^*N)$ is an isomorphism for $M \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$ and $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$.*

Proof. Both claims are first checked on D^b_c and then extended. Tensor products and both sides of the projection map commute with direct sums and cones in each variable. Hence, fixing all variables but one, the objects satisfying a claim form a localizing subcategory. It suffices that this subcategory contains D^b_c (Lemma 5.23). By the dévissage of Corollary 3.20, D^b_c is closed under $\otimes^{\mathbf{L}}$, since $a_!L \otimes^{\mathbf{L}} b_!M$ is the extension by zero of a lisse sheaf on the intersection of the two strata. The projection formula on D^b_c is Proposition 4.40. ■

Now the torsion proof of the adjunction for $\mathbf{P}^1$ can be run with δ_Λ and Tr_q, once $s_1 = s_2 = 1$ is known.

Proposition 6.24 (The projective line, rational coefficients)**.** *For $\Lambda = E$ or $\overline{\mathbf{Q}}_\ell$, the functor $q^*(1)[2] : \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda)) \to \mathrm{Loc}(D^b_{\mathrm{ctf}}(\mathbf{P}^1_Y,\Lambda))$ is right adjoint to Rq_*. Hence $q^! \simeq q^*(1)[2]$ for the right adjoint $q^!$ of Theorem 5.31.*

Proof. Define the unit and counit by the formulas (6.7) and (6.8) of Proposition 6.7, using $\Delta_*M \simeq p_1^*M \otimes^{\mathbf{L}} \Delta_*\Lambda$ [Stacks, 09AD] and the maps δ_Λ and Tr_q in place of their torsion versions. The proof of the identities $s_1 = s_2 = 1$ computed stalks of torsion sheaves, but apart from these identities the torsion calculation of the triangle identities used only the projection formula and proper base change. These hold here by Propositions 6.23 and Proposition 5.24. Thus the calculation reduces the triangle identities to the identities $\mathrm{Tr}_{p_i} \circ Rp_{i*}(\delta_\Lambda) = 1$ of Proposition 6.22. Finally, uniqueness of adjoints gives the last claim. ■

6.7.2 Finiteness of $f^!$

For admissible Λ and a closed immersion $i : Z \hookrightarrow X$ with complement j, the local cohomology functor $R\underline{\Gamma}_Z$ is right adjoint to $i_* = Ri_!$ (Lemma lemma-sections-support-closed-subscheme). Hence $i^! \simeq R\underline{\Gamma}_Z$, and $i^!$ is computed by the localization triangle

$$i_*i^!K \longrightarrow K \longrightarrow Rj_*j^*K \longrightarrow i_*i^!K[1].$$

Example 6.25 (Local cohomology at the origin of the affine line)**.** Let $X = \mathbf{A}^1_k$, k algebraically closed, Z the origin and $U = \mathbf{G}_{m,k}$. Then $Rj_*\Lambda_U$ has $\mathcal{H}^0 = \Lambda_X$ and $\mathcal{H}^1 = i_*\Lambda(-1)$, from the

Kummer covers $t \mapsto t^{\ell^n}$. The triangle gives $i^!\Lambda_X \simeq \Lambda(-1)[-2]$, as Theorem 9.3, applied to the structure maps of Z and X, predicts.

For $\mathcal{O}_E$ we also need the projection formula on the localizing categories, which follows from the case of D_c^b and the results of Subsection 5.4.2.

Lemma 6.26 (Projection formula with integral coefficients)**.** *For proper $p : X \to Y$ the map $(Rp_*M) \otimes^{\mathbf{L}} N \to Rp_*(M \otimes^{\mathbf{L}} p^*N)$ is an isomorphism for $M \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \mathcal{O}_E))$ and $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \mathcal{O}_E))$.*

Proof. On D_c^b this is Proposition 4.40. Both sides make sense on the localizing categories, because $M \otimes^{\mathbf{L}} p^*N$ lies in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \mathcal{O}_E))$. For M and N in D_c^b this is Lemma 5.20, and since the tensor product commutes with direct sums and cones in each variable, the argument of Proposition 6.23 extends it to the localizing categories. Indeed, fix one variable. Both sides are then exact and preserve direct sums in the other; for the variable M this holds because Rp_* preserves direct sums (Proposition 5.21). Therefore Lemma 5.23, applied first in M and then in N, extends the isomorphism from D_c^b to the localizing categories. ∎

We can now repeat the argument for finite coefficients with $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$.

Theorem 6.27 ($f^!$ with ℓ-adic coefficients)**.** *Let $\Lambda = \mathcal{O}_E$, E or $\overline{\mathbf{Q}}_\ell$, and let $f : X \to Y$ be separated of finite type. Then $f^!$ maps $D^b_{\mathrm{ctf}}(Y, \Lambda)$ into $D^b_{\mathrm{ctf}}(X, \Lambda)$, and this restriction is right adjoint to $Rf_! : D^b_{\mathrm{ctf}}(X, \Lambda) \to D^b_{\mathrm{ctf}}(Y, \Lambda)$.*

Proof. As for finite coefficients, we factor f locally into a closed immersion and a projection from affine space, and treat the two factors separately.

Choose finite affine covers of X and Y and closed immersions $U \hookrightarrow \mathbf{A}^d_V$. Here $u : U \to X$ and $v : V \to Y$ are the inclusions of the affine opens, and $i : U \hookrightarrow \mathbf{A}^d_V$ and $p : \mathbf{A}^d_V \to V$ are the two factors. Then $u^*f^! \simeq i^!p^!v^*$ (Proposition 5.32), and being in D_c^b is local on X, so it suffices to show that $p^!$ and $i^!$ preserve D_c^b.

For E and $\overline{\mathbf{Q}}_\ell$ we argue as in Section 6.6. For p we use Proposition 6.24, the adjunction $j_! \dashv j^*$ for $\mathbf{A}^1_V \subset \mathbf{P}^1_V$, and Proposition 5.32 applied d times. They give $p^! \simeq p^*(d)[2d]$, which preserves D_c^b. For i, let j' be the inclusion of the open complement of U in $\mathbf{A}^d_V$. Then the localization triangle before Example 6.25, with j' in place of j, shows that $i^!K \in D_c^b(U, \Lambda)$ for $K \in D_c^b(\mathbf{A}^d_V, \Lambda)$. Indeed, its third term $Rj'_*j'^*K$ is in D_c^b by Proposition 6.21, and $i^*i_* = \mathrm{id}$.

For $\mathcal{O}_E$ that argument applies once its ingredients are available with integral coefficients. They are as follows. First, the adjunction for the projective line comes from Proposition 6.22, with the triangle identities verified as in Proposition 6.24. Next, the projection formula is Lemma 6.26, and inverse image and tensor product preserve $D_c^b(-, \mathcal{O}_E)$ by Lemma 5.20. Moreover, the finiteness of Rj_* and proper base change are Propositions 6.21 and Proposition 5.24. We also use that derived completeness is local on X; it is part of the criterion for D_c^b with $\mathcal{O}_E$-coefficients. Finally, $Rf_!$ preserves D_c^b (Proposition 5.26), and so the adjunction restricts to D_c^b. ∎

6.8 Proof of the theorem

The parts of Theorem 5.2 were proved separately for each kind of coefficient ring, and we collect the references.

Proof of Theorem 5.2. Part (1) is Lemma 5.30. Its compactness statement was proved separately for each kind of coefficient ring: for finite A in Proposition 5.14, for E and $\overline{\mathbf{Q}}_\ell$ in Theorem 5.13, and for $\mathcal{O}_E$ in Proposition 5.21.

Part (2) is Proposition 5.26. It rests on the fact that proper direct image preserves direct sums, which was proved for finite A in Corollary 5.16, for E and $\overline{\mathbf{Q}}_\ell$ in Lemma 5.18, and for $\mathcal{O}_E$ in Proposition 5.21 (see Proposition 5.22).

Part (3) is Theorem 5.31.

Part (4) is Theorem 6.20 for finite A and Theorem 6.27 for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$. The second theorem includes $\mathbf{Z}_\ell$ and $\mathbf{Q}_\ell$ as the case $E = \mathbf{Q}_\ell$; recall that $D^b_{\mathrm{ctf}} = D^b_c$ in the integral and rational cases. For the last assertion, note that the categories $D^b_{\mathrm{ctf}}(X, \Lambda)$ and $D^b_{\mathrm{ctf}}(Y, \Lambda)$ are full subcategories of the localizing categories, and $Rf_!$ and $f^!$ preserve them. Hence the Hom bijections of the adjunction restrict to them. ∎

7 The six operations on D^b_{ctf}

Smooth base change and Verdier biduality will be proved with torsion coefficients, as were proper base change and Deligne's theorem on Rj_*. The Weil conjectures, however, need them over $\overline{\mathbf{Q}}_\ell$, where Frobenius eigenvalues live. This chapter provides the principle that carries every such statement across. A canonical map built from the six operations is an isomorphism for all admissible Λ as soon as it is one for every finite ring (Proposition 7.12).

For $\mathcal{O}_E$ both sides are derived complete, and by derived Nakayama it suffices to check the map modulo π_E. For instance, let K be in $D^b_{\mathrm{ctf}}(X, \mathcal{O}_E)$. Both sides of the biduality map $K \to \mathbb{D}_X\mathbb{D}_X K$ are derived complete. Modulo π_E, moreover, the map is biduality with $\mathbf{F}_E$-coefficients, which is an isomorphism. Hence the biduality map is an isomorphism. From there we pass to E and $\overline{\mathbf{Q}}_\ell$: every object over E is built by cones from integral models with ℓ inverted, and every object over $\overline{\mathbf{Q}}_\ell$ is defined over a finite extension E (Proposition 7.11). In this way smooth base change (Theorem 8.11) and Verdier biduality (Theorem 10.8) pass from torsion to ℓ-adic coefficients.

The principle needs two things, and neither is formal. First, both sides must lie in $D^b_{\mathrm{ctf}}(-, \Lambda)$. Thus all six operations must preserve these categories, and not only $Rf_!$ and $f^!$, which we now control (Theorem 5.2, Section 6.8). Second, each operation must commute with change of coefficients. For f^* and $\otimes^{\mathbf{L}}$ this is clear, since a change of coefficients is itself a tensor product. For a right adjoint, however, it is a real statement. For example, consider a map of finite rings $A \to B$; then the isomorphism $R\Gamma(X, K) \otimes^{\mathbf{L}}_A B \simeq R\Gamma(X, K \otimes^{\mathbf{L}}_A B)$ is a projection formula, and it uses finite cohomological dimension. Next consider $\mathcal{O}_E \to E$. Since E is the colimit of the modules $\pi_E^{-n}\mathcal{O}_E$, the comparison says that Rf_* commutes with a filtered colimit. This holds because the terms are uniformly bounded below.

Internal Hom causes a further difficulty, because it hides the exceptional inverse image. For a closed immersion i we have

$$R\mathcal{H}om(i_*\Lambda, K) \simeq i_* i^! K.$$

Before we can say anything about $R\mathcal{H}om(K, L) \otimes^{\mathbf{L}} \Lambda'$, we must therefore know how $i^!$ behaves under change of coefficients. The comparison map for $f^!$ is defined by adjunction, and to show that it is an isomorphism we compute $f^!$ once more, for a closed immersion and for $\mathbf{P}^1$. For $\mathbf{P}^1$, for instance, it comes down to the compatibility of the cycle class of the diagonal and of the trace with change of coefficients (Proposition 7.10).

The first requirement needs no construction: on the pro-étale site the six operations are the ordinary derived functors of sheaves of modules, and we only check that they preserve $D^b_{\mathrm{ctf}}(-,\Lambda)$ (Proposition 7.1). For torsion coefficients on the étale site, these finiteness theorems are in SGA 4 and SGA $4\frac{1}{2}$ [SGA4], [SGA4½]. In general, the work is in Rf_* and in $R\mathcal{H}om$ (Section 7.1). There, a compactification reduces Rf_* to proper direct image and Rj_*, while the formula above and its analogue for open immersions express $R\mathcal{H}om$ through Ra_*, $a^!$ and tensor products. Tensor products, finally, are handled by the local description of a complex of finite Tor amplitude: on the strata of a finite stratification it is locally $\Lambda_S \otimes P$ with P perfect (Section 3.4).

From now on we work in D^b_{ctf} without further comment and pass between $\mathcal{O}_E$, E, $\overline{\mathbf{Q}}_\ell$ and finite rings whenever a statement is easier for one of them, as in the trace formula and in the proof of the Weil conjectures.

7.1 Stability of D^b_{ctf}

Pullback and tensor product are local, and they preserve the description of a complex of D^b_{ctf} on the strata of a stratification as $\Lambda_S \otimes P$ with P perfect. The direct images and internal Hom are not local. For the direct images a compactification $f = pj$ helps. Since $Rf_* = Rp_*Rj_*$, it suffices to know proper direct image and direct image along an open immersion. Internal Hom hides both of these operations and also $f^!$. For an open immersion j we have $R\mathcal{H}om(j_!\Lambda, K) = Rj_*j^*K$, and for a closed immersion i we have the formula $R\mathcal{H}om(i_*\Lambda, K) \simeq i_*i^!K$ recalled at the beginning of this chapter. If a is the locally closed immersion of a stratum and P is a perfect complex on it, the formula

$$R\mathcal{H}om(a_!P, K) \simeq Ra_*(P^\vee \otimes^{\mathbf{L}} a^!K)$$

combines the two. Every constructible complex is built from objects $a_!P$ by finitely many cones. By this formula, internal Hom is therefore expressed through Ra_*, $a^!$ and tensor products. This is one reason we needed $f^!$ on D^b_{ctf} first. The result is Proposition 7.1.

Proposition 7.1 (The six operations on D^b_{ctf})**.** *For every admissible $\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, the operations f^*, Rf_*, $Rf_!$, $f^!$, $\otimes^{\mathbf{L}}$ and $R\mathcal{H}om$ preserve $D^b_{\mathrm{ctf}}(-,\Lambda)$ for morphisms of schemes of finite type over k. Their adjunctions are the restrictions of the adjunctions $f^* \dashv Rf_*$ and $- \otimes^{\mathbf{L}} K \dashv R\mathcal{H}om(K,-)$ on $D(-_{\mathrm{pro\acute{e}t}},\Lambda)$ or of the adjunction $Rf_! \dashv f^!$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-,\Lambda))$.*

Proof. We treat the operations in the order in which they depend on each other: first the local operations f^* and $\otimes^{\mathbf{L}}$, then the two direct images, and finally internal Hom, which we express through the others.

Pullback and tensor product preserve the property of being, on the strata of a finite stratification, locally $\Lambda_S \otimes P$ with P a perfect complex. For pullback this is clear, since the strata pull back to a stratification, while for tensor products we take a common refinement of the two stratifications, and on each of its strata we use that $P \otimes^{\mathbf{L}} Q$ is perfect when P and Q are. Similarly, extension by zero preserves finite Tor amplitude, since it is exact and satisfies the projection formula $a_!K \otimes^{\mathbf{L}} N \simeq a_!(K \otimes^{\mathbf{L}} a^*N)$. Concretely, for finite A these statements can be checked on the bounded flat constructible models of Lemma 3.13. For $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$, on the other hand, they follow from Lemma 5.20 and from Definition 3.1 of constructible sheaves: on each stratum, such a sheaf is locally $\Lambda_X \otimes_{\Lambda(*)} P$ with P finitely generated over the discrete valuation ring or field $\Lambda(*)$. Over such a ring a finitely generated module has a finite free resolution of length at most one. Hence P is perfect.

For the direct images we use a compactification to separate the proper part from the open part. Proper direct image preserves $D^b_{\mathrm{ctf}}(-,\Lambda)$ by Proposition 4.40. Direct image along an open immersion preserves it by Proposition 6.19 for finite A and Proposition 6.21 for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$. If $f = pj$ is a compactification, then $Rf_* = Rp_*Rj_*$ and $Rf_! = Rp_*j_!$. Each of these composites consists of functors already known to preserve D^b_{ctf}, which proves both assertions on direct images. That $f^!$ preserves $D^b_{\mathrm{ctf}}(-,\Lambda)$ is part of Theorem 5.2.

For internal Hom we reduce to a first argument supported on a single stratum, where $R\mathcal{H}om$ can be written in terms of Ra_*, $a^!$ and $\otimes^{\mathbf{L}}$. This is possible because the functor $R\mathcal{H}om(-,K)$ sends triangles in the first argument to triangles, and D^b_{ctf} is closed under cones and shifts. Hence, for a triangle $M' \to M \to M''$, the complex $R\mathcal{H}om(M,K)$ lies in D^b_{ctf} whenever $R\mathcal{H}om(M',K)$ and $R\mathcal{H}om(M'',K)$ do. Using a finite stratification and finite cones, we may therefore reduce to the case where the first argument is $a_!P$. Here $a : S \hookrightarrow X$ is locally closed and P is a locally constant perfect complex on S. For finite A this reduction follows from the bounded flat model and its stupid filtration, and for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ it follows from the same description of constructible sheaves on strata. By tensor-Hom adjunction and the projection formula for $a_!$ we have

$$R\mathcal{H}om_X(a_!P, K) \simeq Ra_*R\mathcal{H}om_S(P, a^!K) \simeq Ra_*(P^\vee \otimes^{\mathbf{L}} a^!K).$$

Here $P^\vee = R\mathcal{H}om_S(P,\Lambda_S)$, and the second identity holds because P is dualizable. The last term belongs to $D^b_{\mathrm{ctf}}(-,\Lambda)$ by the cases already established, namely those of $a^!$, of tensor product and of Ra_*.

Finally, since the inclusions are full, the Hom bijections of the ambient categories restrict to the stated adjunctions. ∎

Example 7.2 (Internal Hom on the affine line)**.** Over an algebraically closed k, let $j : U = \mathbf{G}_{m,k} \hookrightarrow X = \mathbf{A}^1_k$ and let i be the inclusion of the origin. The localization sequence $0 \to j_!\Lambda_U \to \Lambda_X \to i_*\Lambda \to 0$ shows that Λ_X is built from $j_!\Lambda_U$ and $i_*\Lambda$, and the formula above computes $R\mathcal{H}om$ out of each piece. By Example 6.25,

$$R\mathcal{H}om(j_!\Lambda_U, \Lambda_X) = Rj_*\Lambda_U, \qquad R\mathcal{H}om(i_*\Lambda, \Lambda_X) = i_*i^!\Lambda_X \simeq i_*\Lambda(-1)[-2].$$

The first has $\mathcal{H}^0 = \Lambda_X$ and $\mathcal{H}^1 = i_*\Lambda(-1)$. In the second, the local Ext sheaves from the skyscraper sheaf $i_*\Lambda$ to the constant sheaf vanish except in degree two. Both results are constructible but not lisse. The first is computed by a nonproper direct image, and the second by $i^!$.

7.2 Change of coefficients

Smooth base change for E and $\overline{\mathbf{Q}}_\ell$ is deduced from the integral case by extending scalars (Chapter 8), and later proofs often argue in the same way, through the reduction principle of Proposition 7.12. For this the six operations must commute with extension of scalars. For pullback, extension by zero and tensor product this is formal, since extension of scalars is itself a tensor product. For Rf_*, by contrast, it rests on finite cohomological dimension and on continuity, and for $f^!$ it rests on a second computation of $f^!$, for closed immersions and for $\mathbf{P}^1$. These results are collected in Proposition 7.10.

A change of coefficients is derived extension of scalars along a map of condensed rings, pulled back to X. For a map $\Lambda \to \Lambda'$ write

$$K_{\Lambda'} = \Lambda'_X \otimes^{\mathbf{L}}_{\Lambda_X} K.$$

This functor $D(X_{\text{pro\'et}}, \Lambda) \to D(X_{\text{pro\'et}}, \Lambda')$ is left adjoint to the exact restriction of scalars $D(X_{\text{pro\'et}}, \Lambda') \to D(X_{\text{pro\'et}}, \Lambda)$. We consider the following maps of coefficient rings:

1. any homomorphism $A \to B$ of finite admissible rings;
2. the maps $\mathcal{O}_E \to \mathcal{O}_F$ and $E \to F$ for a finite extension F/E inside $\overline{\mathbf{Q}}_\ell$;
3. localization $\mathcal{O}_E \to E$, inclusions $E \to \overline{\mathbf{Q}}_\ell$, and continuous maps $\mathcal{O}_E \to A$ to finite admissible rings, such as the reductions $\mathcal{O}_E \to \mathcal{O}_E/\pi_E^n$ (continuity is needed for the map to define a map of condensed rings);
4. composites of these maps.

For $E = \mathbf{Q}_\ell$, the list contains $\mathbf{Z}_\ell \to \mathcal{O}_F$, $\mathbf{Q}_\ell \to F$, $\mathbf{Z}_\ell \to \mathbf{Q}_\ell$, $\mathbf{Z}_\ell \to A$ and $\mathbf{Q}_\ell \to \overline{\mathbf{Q}}_\ell$. Thus these maps connect all the admissible coefficient rings. Note that the map $A \to B$ in (1) need not be flat, and for this reason extension of scalars is derived. As a consequence, the comparison isomorphisms below hold for complexes but not degree by degree, as the following example shows.

Example 7.3 (Reducing cohomology modulo ℓ)**.** Let $X = \operatorname{Spec} \mathbf{F}_q$ and consider $\mathbf{Z}/\ell^2 \to \mathbf{F}_\ell$. Let K be the lisse sheaf $\mathbf{Z}/\ell^2$ on which Frobenius, the topological generator of $\operatorname{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q) \simeq \widehat{\mathbf{Z}}$, acts by multiplication by the unit $1 + \ell$. Recall that for a finite module M with a continuous action of $\widehat{\mathbf{Z}}$, cohomology is computed by the complex $M \xrightarrow{F-1} M$ in degrees 0 and 1, where F is the action of the generator. For K this differential is multiplication by ℓ. Hence $H^0(X, K) = \ell\mathbf{Z}/\ell^2$ and $H^1(X, K) = (\mathbf{Z}/\ell^2)/\ell$, both of order ℓ. On the other hand, $K_{\mathbf{F}_\ell} = K/\ell K$ is $\mathbf{F}_\ell$ with trivial action, and $H^0(X, K/\ell K) = \mathbf{F}_\ell$. The map $H^0(X, K) \otimes \mathbf{F}_\ell \to H^0(X, K/\ell K)$ is zero, since the generator ℓ of $H^0(X, K)$ reduces to 0.

The derived comparison is nevertheless an isomorphism. The terms of the complex are free over $\mathbf{Z}/\ell^2$. Therefore $R\Gamma(X, K) \otimes^{\mathbf{L}} \mathbf{F}_\ell$ is $\mathbf{F}_\ell \xrightarrow{0} \mathbf{F}_\ell$, which is $R\Gamma(X, K/\ell K)$. The class $1 \in H^0(X, K/\ell K)$ comes from the element 1 of the degree-zero term, which is not a cocycle before reduction.

Before stating the result we fix the comparison maps. Items (1)–(3) below, for f^*, $j_!$ and $\otimes^{\mathbf{L}}$, are identifications of two composites of left adjoints. The remaining items are derived from these: items (4) and (6), for Rf_* and $R\mathcal{H}om$, are adjoint transposes of the inverse of (1), respectively (3), composed with the extension of scalars of the counit, and item (5) combines (4) for p with (2).

Definition 7.4 (Change-of-coefficients maps)**.** Let $\Lambda \to \Lambda'$ be one of the maps listed above, and let $f : X \to Y$ be a morphism of schemes of finite type over k.

1. For $K \in D(Y_{\text{pro\'et}}, \Lambda)$, the isomorphism $(f^*K)_{\Lambda'} \simeq f^*(K_{\Lambda'})$ is the one given by associativity of derived pullback for ringed topoi [Stacks, 06YV]: both sides are the derived pullback along the morphism of ringed topoi $(X_{\text{pro\'et}}, \Lambda'_X) \to (Y_{\text{pro\'et}}, \Lambda_Y)$, factored in the two possible ways.
2. For an open immersion $j : U \to X$ and $K \in D(U_{\text{pro\'et}}, \Lambda)$, the isomorphism $(j_!K)_{\Lambda'} \simeq j_!(K_{\Lambda'})$ is the isomorphism of left adjoints corresponding to the equality of their right adjoints, restriction of scalars followed by j^* and j^* followed by restriction of scalars.
3. For $K, L \in D(X_{\text{pro\'et}}, \Lambda)$, the isomorphism $(K \otimes^{\mathbf{L}}_\Lambda L)_{\Lambda'} \simeq K_{\Lambda'} \otimes^{\mathbf{L}}_{\Lambda'} L_{\Lambda'}$ is the associativity isomorphism: both sides are $K \otimes^{\mathbf{L}}_\Lambda L \otimes^{\mathbf{L}}_\Lambda \Lambda'$.
4. For $K \in D(X_{\text{pro\'et}}, \Lambda)$, the map $(Rf_*K)_{\Lambda'} \to Rf_*(K_{\Lambda'})$ is the adjoint transpose, under $f^* \dashv Rf_*$ on $D(-_{\text{pro\'et}}, \Lambda')$, of

$$f^*((Rf_*K)_{\Lambda'}) \simeq (f^*Rf_*K)_{\Lambda'} \xrightarrow{(\varepsilon_K)_{\Lambda'}} K_{\Lambda'},$$

where the first arrow is the inverse of (1) and ε is the counit of $f^* \dashv Rf_*$ on $D(-_{\text{pro\'et}}, \Lambda)$.

5. If f is separated, $f = p \circ j$ is a compactification and $K \in D^b_{\text{ctf}}(X, \Lambda)$, the map $(Rf_!K)_{\Lambda'} \to Rf_!(K_{\Lambda'})$ is the composite

$$(Rp_*j_!K)_{\Lambda'} \longrightarrow Rp_*((j_!K)_{\Lambda'}) \simeq Rp_*j_!(K_{\Lambda'})$$

of (4) for p and Rp_* of (2).

6. For $K, L \in D(X_{\text{pro\'et}}, \Lambda)$, the map $R\mathcal{H}om(K, L)_{\Lambda'} \to R\mathcal{H}om(K_{\Lambda'}, L_{\Lambda'})$ is the adjoint transpose, under $- \otimes^{\mathbf{L}} K_{\Lambda'} \dashv R\mathcal{H}om(K_{\Lambda'}, -)$, of

$$R\mathcal{H}om(K, L)_{\Lambda'} \otimes^{\mathbf{L}}_{\Lambda'} K_{\Lambda'} \simeq \left(R\mathcal{H}om(K, L) \otimes^{\mathbf{L}}_{\Lambda} K\right)_{\Lambda'} \xrightarrow{(\text{ev})_{\Lambda'}} L_{\Lambda'},$$

where the first arrow is the inverse of (3) and ev is evaluation.

The map for $f^!$ is defined in the same way, as the transpose of a counit, once we know that (5) is an isomorphism; see (7.1) below. We now show, one operation at a time, that these maps are isomorphisms on D^b_{ctf}.

In the lemmas below it suffices to treat the maps in items (1)–(3) of the list of coefficient maps, that is, all maps except composites. Indeed, extension of scalars along a composite of two maps of the list is the composite of the two extensions. Each map in Definition 7.4 for the composite is the composite of the maps for the two factors. For (1)–(3) this holds by the coherence of the associativity isomorphisms, and for (4)–(6) it holds because the adjoint transpose of a composite of such maps is the composite of the transposes.

Lemma 7.5 (Extension of scalars preserves D^b_{ctf})**.** *For each map $\Lambda \to \Lambda'$ listed above, extension of scalars sends $D^b_{\text{ctf}}(X, \Lambda)$ into $D^b_{\text{ctf}}(X, \Lambda')$.*

Proof. An object that is locally $\Lambda_S \otimes P$ on a stratum S, with P perfect, remains of this form after derived extension of scalars, since $P \otimes^{\mathbf{L}}_{\Lambda} \Lambda'$ is a perfect Λ'-complex. Moreover, extension of scalars commutes with extension by zero and with finite cones. Every object of $D^b_{\text{ctf}}(X, \Lambda)$ is built from such objects by extension by zero from strata and finite cones, and so its extension of scalars lies in $D^b_{\text{ctf}}(X, \Lambda')$. ∎

Lemma 7.6 (Change of coefficients for direct image)**.** *For each map $\Lambda \to \Lambda'$ listed above, every morphism $f : X \to Y$ of schemes of finite type over k and every $K \in D^b_{\text{ctf}}(X, \Lambda)$, the map $(Rf_*K)_{\Lambda'} \to Rf_*(K_{\Lambda'})$ of Definition definition-change-of-coefficients-maps is an isomorphism.*

Proof. The argument depends on the kind of map $\Lambda \to \Lambda'$. For maps of finite rings we use a projection formula, whereas in the other cases extension of scalars is either a finite construction (finite sums and a cone), with which every exact functor commutes, or a filtered colimit, with which Rf_* commutes on uniformly bounded below systems.

Maps $A \to B$ of finite rings. We compute on the étale site using Proposition 5.15, which identifies the pro-étale and étale categories in this case, and we test the comparison on derived sections over quasi-compact étale objects. For a quasi-compact étale $V \to Y$, we apply the projection formula for constant coefficients [Stacks, 0F12] on both V and $X_V = X \times_Y V$ and obtain

$$R\Gamma(V, (Rf_*K)_B) \simeq R\Gamma(X_V, K) \otimes^{\mathbf{L}}_A B \simeq R\Gamma(X_V, K_B).$$

Here the first identification uses the projection formula on V and then $R\Gamma(V, Rf_*K) = R\Gamma(X_V, K)$, and the second is the projection formula on X_V. This projection formula needs only

cohomological bounds for ℓ-primary coefficients, which hold for schemes of finite type over k, and so it applies to every finite A-algebra B. Since these isomorphisms commute with restriction in V, the comparison map is an isomorphism in $D(Y_{\acute{e}t}, B)$, and hence, by Proposition 5.15, in $D(Y_{\text{pro}\acute{e}t}, B)$.

Maps $\mathcal{O}_E \to A$ with A finite. Since $\mathcal{O}_E$ is a discrete valuation ring, A has a finite free $\mathcal{O}_E$-resolution of length at most one. Its condensed version is again exact. Indeed, a continuous map from a profinite set to the finite quotient A is constant on the pieces of a finite clopen partition, and hence lifts to a continuous map to the free module. The pullback of this resolution to X is a finite free resolution of A_X over $\mathcal{O}_{E,X}$. Consequently, tensoring with A_X is computed by finite sums and a cone, and these commute with the exact functor Rf_*.

Finite extensions. For $\mathcal{O}_E \to \mathcal{O}_F$ and $E \to F$, the ring $\mathcal{O}_F$ is free of rank $[F : E]$ over $\mathcal{O}_E$. Hence $K_{\mathcal{O}_F}$ is, as an $\mathcal{O}_E$-complex, a finite direct sum of copies of K. The assertion follows because Rf_* commutes with finite sums.

The maps $\mathcal{O}_E \to E$ and $E \to \overline{\mathbf{Q}}_\ell$. These are not finite, and we use continuity. For $\mathcal{O}_E \to E$, the complex K_E is the colimit of $K \xrightarrow{\ell} K \xrightarrow{\ell} \cdots$, and for $E \to \overline{\mathbf{Q}}_\ell$, the complex $K_{\overline{\mathbf{Q}}_\ell}$ is the filtered colimit of the K_F over finite extensions F/E in $\overline{\mathbf{Q}}_\ell$. Since K is bounded, all terms of these systems are bounded below by a common integer. Hence the colimits commute with Rf_* by Corollary 2.29. ■

Lemma 7.7 (Change of coefficients for compactly supported direct image)**.** *For each map $\Lambda \to \Lambda'$ listed above, every separated morphism $f : X \to Y$ of finite type and every $K \in D^b_{\text{ctf}}(X, \Lambda)$, the map $(Rf_!K)_{\Lambda'} \to Rf_!(K_{\Lambda'})$ of Definition definition-change-of-coefficients-maps is an isomorphism, and it does not depend on the compactification.*

Proof. The map is the composite of the isomorphism of Lemma 7.6 for p, applied to $j_!K \in D^b_{\text{ctf}}$ (Proposition 5.22), and of Rp_* of the isomorphism of Definition definition-change-of-coefficients-maps. For independence, let $h : Z' \to Z$ be a refinement of compactifications. Both comparison maps (4) and (2) commute with the refinement map (5.7). Indeed, they are formed from identifications of pullbacks and adjunction maps. A map out of an extension by zero is determined by its restriction to the open subscheme, and there all maps involved are identities. We also use that (4) is compatible with composition of morphisms. Under $R(ph)_* \simeq Rp_*Rh_*$, the map (4) for ph is Rp_* of (4) for h followed by (4) for p, since transposes of composites are composites. As in the proof of Proposition 5.26, this gives independence for any two compactifications. ■

We can now define the comparison map for $f^!$. Let $f : X \to Y$ be separated of finite type and $K \in D^b_{\text{ctf}}(Y, \Lambda)$. Then $f^!K \in D^b_{\text{ctf}}(X, \Lambda)$ by Theorem 5.2, and we let

$$\beta_{f,K} : (f^!K)_{\Lambda'} \longrightarrow f^!(K_{\Lambda'}) \tag{7.1}$$

be the transpose, under the adjunction $Rf_! \dashv f^!$ on $\text{Loc}(D^b_{\text{ctf}}(-, \Lambda'))$, of the morphism

$$Rf_!((f^!K)_{\Lambda'}) \xrightarrow{\sim} (Rf_!f^!K)_{\Lambda'} \xrightarrow{(\varepsilon_K)_{\Lambda'}} K_{\Lambda'},$$

whose first arrow is the inverse of the isomorphism of Lemma 7.7.

The proof below uses three properties of β. First, the maps β are compatible with composition: under $(gf)^! \simeq f^!g^!$ we have $\beta_{gf,K} = f^!(\beta_{g,K}) \circ \beta_{f,g^!K}$. This holds because the counit of a composite adjunction is the composite of the counits, and because the comparison for $Rf_!$ is compatible with the composition isomorphisms of Proposition 5.28; both are determined by their restrictions

to the open subscheme. Second, let $\Lambda \to \Lambda' \to \Lambda''$ be a composite of two maps of the list. Then β for the composite is β for $\Lambda' \to \Lambda''$ composed with the extension of scalars to Λ'' of β for $\Lambda \to \Lambda'$. Indeed, both are transposes of the composite of the extended counits and the comparisons for $Rf_!$. Third, for an open immersion u, under $u^! \simeq u^*$, the map $\beta_{u,K}$ is the isomorphism of Definition definition-change-of-coefficients-maps. Both are transposes of the same map under $u_! \dashv u^*$.

Lemma 7.8 (Change of coefficients for the exceptional inverse image)**.** *For each map $\Lambda \to \Lambda'$ listed above, every separated morphism $f : X \to Y$ of finite type and every $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$, the map $\beta_{f,K}$ of (7.1) is an isomorphism.*

Proof. Since $f^!$ is only known through its adjunction, there is no formula into which we could substitute $K_{\Lambda'}$ directly. Instead we show that $\beta_{f,K}$ is an isomorphism in two cases where $f^!$ has been computed: closed immersions and the projection $\mathbf{P}^1_Y \to Y$. We then factor a general f locally into such maps, using that β is compatible with composition and with restriction to opens.

Closed immersions. For a closed immersion i with open complement j, the functor $i^!$ is determined by the localization triangle

$$i_* i^! K \longrightarrow K \longrightarrow R j_* j^* K.$$

It expresses $i_* i^! K$ as the fibre of a map between objects whose behaviour under extension of scalars we already know. Extension of scalars commutes with i_*, j^* and, by Lemma 7.6, Rj_*. Hence the extension of scalars of this triangle is the triangle for $K_{\Lambda'}$ over Λ'. Since the first map is the counit, the comparison on fibres is $i_* \beta_{i,K}$. The comparisons on the other two terms are isomorphisms, and therefore $i_* \beta_{i,K}$ is one as well. Since i_* is fully faithful, $\beta_{i,K}$ is an isomorphism.

The projective line. Let $q : \mathbf{P}^1_Y \to Y$. Recall that the adjunction $Rq_* \dashv q^*(1)[2]$ between $D^b_{\mathrm{ctf}}(\mathbf{P}^1_Y, \Lambda)$ and $D^b_{\mathrm{ctf}}(Y, \Lambda)$ is defined by the diagonal class and the trace. Through these two maps, $q^!$ is identified with $q^*(1)[2]$. It therefore suffices to know that the two maps commute with all the maps of coefficient rings listed above. This holds by construction (Propositions 6.7 and Proposition 6.22). Indeed, the classes with finite coefficients are constructed from the Kummer sequence and derived extension of scalars, while the integral classes are their compatible limits, and the classes over fields are obtained from these by extension of scalars. Only one map in the list needs a further remark. For $\mathcal{O}_E \to A$, the proof of Lemma 7.6 shows that A_X is perfect. Hence the derived tensor product with A_X commutes with the inverse limits defining the integral classes, and so the resulting classes are exactly the classes with finite coefficients. Consequently the identification $q^! \simeq q^*(1)[2]$ is compatible with extension of scalars. Under this identification, finally, $\beta_{q,K}$ becomes the isomorphism of Definition definition-change-of-coefficients-maps for q^*, twisted by $(1)[2]$.

General f. The open immersion $\mathbf{A}^1_Y \hookrightarrow \mathbf{P}^1_Y$ is a case of restriction to opens, and so β is an isomorphism for $\mathbf{A}^1_Y \to Y$. Since the projection $\mathbf{A}^d_Y \to Y$ is a composite of d projections of this kind, the compatibility of β with composition gives the case of affine space. In general, whether $\beta_{f,K}$ is an isomorphism may be checked locally on X. Using a finite affine cover, we may therefore assume that f factors as a closed immersion into an affine space over Y followed by the projection. Then the two cases already treated and compatibility with composition give the claim. ∎

Lemma 7.9 (Change of coefficients for internal Hom)**.** *For each map $\Lambda \to \Lambda'$ listed above and all $K, L \in D^b_{\mathrm{ctf}}(X, \Lambda)$, the map $R\mathcal{H}om(K, L)_{\Lambda'} \to R\mathcal{H}om(K_{\Lambda'}, L_{\Lambda'})$ of Definition definition-change-of-coefficients-maps is an isomorphism.*

Proof. Internal Hom is a right adjoint. As for $f^!$, there is no formal reason for it to commute with extension of scalars, and we argue as in the proof of Proposition 7.1. The comparison map for $R\mathcal{H}om(-, L)$ is compatible with triangles in the first argument, so we choose a finite stratification on whose strata K is locally $\Lambda_S \otimes P$ with P perfect. By the five lemma and finite cones we may then reduce to the case $K = a_!P$, with $a : S \hookrightarrow X$ locally closed and P a locally constant perfect complex on S. The identity

$$R\mathcal{H}om_X(a_!P, L) \simeq Ra_*(P^\vee \otimes^{\mathbf{L}} a^!L)$$

commutes with extension of scalars by Lemmas 7.6 and Lemma 7.8, the isomorphism of Definition definition-change-of-coefficients-maps, and the comparison for the dual of a perfect complex. For the last of these, recall that P is locally a finite complex of finite projective modules, and for such modules $R\mathcal{H}om(P, \Lambda)_{\Lambda'} = R\mathcal{H}om(P_{\Lambda'}, \Lambda')$ term by term. Moreover, all of these identifications are defined by the same evaluation pairing. Hence their composite is the map of Definition definition-change-of-coefficients-maps. ∎

The lemmas combine into the following statement, which is the form in which change of coefficients is used later.

Proposition 7.10 (The six operations and change of coefficients)**.** *Let $\Lambda \to \Lambda'$ be one of the maps of coefficient rings listed above. Extension of scalars sends $D^b_{\mathrm{ctf}}(X, \Lambda)$ into $D^b_{\mathrm{ctf}}(X, \Lambda')$. For morphisms $f : X \to Y$ of schemes of finite type over k (separated in the case of $Rf_!$ and $f^!$), the change-of-coefficients maps of Definition 7.4 and (7.1) for f^*, Rf_*, $Rf_!$, $f^!$, $\otimes^{\mathbf{L}}$ and $R\mathcal{H}om$ are isomorphisms on objects of $D^b_{\mathrm{ctf}}(-, \Lambda)$. For a composite $\Lambda \to \Lambda' \to \Lambda''$ of two such maps, each of these isomorphisms is the composite of the isomorphisms for $\Lambda \to \Lambda'$ and $\Lambda' \to \Lambda''$. These isomorphisms are compatible with the composition isomorphisms $(gf)^* \simeq f^*g^*$, $R(gf)_* \simeq Rg_*Rf_*$, $R(gf)_! \simeq Rg_!Rf_!$ and $(gf)^! \simeq f^!g^!$ for composable morphisms f and g. Under these isomorphisms, the extensions of scalars of the units and counits of the adjunctions $f^* \dashv Rf_*$, $Rf_! \dashv f^!$ and $- \otimes^{\mathbf{L}} K \dashv R\mathcal{H}om(K, -)$ (the last counit being evaluation) over Λ are the corresponding units and counits over Λ'.*

Proof. The first assertion is Lemma 7.5. Next, the maps for f^* and $\otimes^{\mathbf{L}}$ are isomorphisms by definition, and those for Rf_*, $Rf_!$, $f^!$ and $R\mathcal{H}om$ by Lemmas 7.6, Lemma 7.7, Lemma 7.8 and Lemma 7.9. As for the compatibilities, the one with composites of coefficient maps was explained before Lemma 7.5, and for $f^!$ after (7.1). The compatibility with composition of morphisms, on the other hand, holds for f^* by the coherence of the associativity isomorphisms for pullbacks. It holds for Rf_* because adjoint transposes of composites are composites. For $Rf_!$ the composition isomorphism is built from $R(qr)_* \simeq Rq_*Rr_*$ and the interchange of an open and a proper direct image. Since the interchange is a map out of an extension by zero, the claim follows from the case of Rf_* and restriction to the open subscheme. Finally, for $f^!$ it was shown after (7.1).

For the units and counits, consider one of the three adjunctions $F \dashv G$ over Λ. We write $F' \dashv G'$ for the same adjunction over Λ', with unit η' and counit ε', and write M' for $M_{\Lambda'}$. Then the comparison for the left adjoint is an isomorphism $c_F : F'(M') \simeq (FM)'$. In each case the comparison $c_G : (GN)' \to G'(N')$ for the right adjoint was defined as the transpose of $(\varepsilon_N)' \circ c_F$. In other words $\varepsilon'_{N'} \circ F'(c_G) = (\varepsilon_N)' \circ c_F$, which is the statement for the counit. For the unit we must show $c_G \circ (\eta_M)' = G'(c_F) \circ \eta'_{M'}$ as maps $M' \to G'((FM)')$. We compare transposes. The transpose of the right side is c_F, whereas the transpose of the left side is $(\varepsilon_{FM})' \circ c_F \circ F'((\eta_M)')$. By naturality of c_F this equals $(\varepsilon_{FM} \circ F\eta_M)' \circ c_F$, which is c_F by a triangle identity. ∎

Without the condition on Tor amplitude, change of coefficients would not preserve the categories. For instance, let X be a geometric point. The sheaf $\mathbf{F}_\ell$ belongs to $D_c^b(X, \mathbf{Z}/\ell^2)$ but not to $D_{\mathrm{ctf}}^b(X, \mathbf{Z}/\ell^2)$. Its extension of scalars $\mathbf{F}_\ell \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell^2} \mathbf{F}_\ell$ is unbounded.

For coefficients in $\overline{\mathbf{Q}}_\ell$, Lemma 3.23 expresses $D_{\mathrm{ctf}}^b(X, \overline{\mathbf{Q}}_\ell)$ in terms of the categories $D_{\mathrm{ctf}}^b(X, E)$ for finite E. As a statement about ordinary categories, it reads as follows.

Proposition 7.11 ($D_{ctf}^b(X, \overline{\mathbf{Q}})$ as a 2-colimit)**.** *Extension of scalars identifies*

$$D_{\mathrm{ctf}}^b(X, \overline{\mathbf{Q}}_\ell) \simeq \underset{E \subset \overline{\mathbf{Q}}_\ell}{2\text{-}colim}\, D_{\mathrm{ctf}}^b(X, E)$$

as ordinary categories. In other words, every object of $D_{\mathrm{ctf}}^b(X, \overline{\mathbf{Q}}_\ell)$ is isomorphic to $K_{\overline{\mathbf{Q}}_\ell}$ for some finite E and some $K \in D_{\mathrm{ctf}}^b(X, E)$, and for $K, L \in D_{\mathrm{ctf}}^b(X, E)$ extension of scalars induces a bijection

$$\varinjlim_{F \supset E} \mathrm{Hom}_{D_{\mathrm{ctf}}^b(X,F)}(K_F, L_F) \xrightarrow{\sim} \mathrm{Hom}_{D_{\mathrm{ctf}}^b(X,\overline{\mathbf{Q}}_\ell)}(K_{\overline{\mathbf{Q}}_\ell}, L_{\overline{\mathbf{Q}}_\ell}),$$

the colimit running over finite extensions F of E in $\overline{\mathbf{Q}}_\ell$.

Proof. An equivalence with a 2-colimit amounts to two statements, essential surjectivity on objects and a bijection on morphisms. Both are parts of Lemma 3.23, and we recall how each is proved before putting them together.

The statement about objects is Lemma lemma-descent-finite-coefficient-field. Its proof first descends the lisse sheaves on the strata, together with their transition matrices. It then descends the connecting morphisms of a finite Postnikov tower, and in this way the complexes themselves descend. The bijection on morphisms is Lemma lemma-descent-finite-coefficient-field with $r = 0$, since D_{ctf}^b is a full subcategory. Its proof uses the continuity of Ext out of a constructible complex (Lemma 3.22), applied to the filtered colimit $\overline{\mathbf{Q}}_{\ell,X} = \varinjlim_F F_X$. Surjectivity says that every morphism descends to a finite stage, and injectivity says that two morphisms with the same image agree at a later finite stage. Now consider the category whose objects are the pairs (E, K) with $K \in D_{\mathrm{ctf}}^b(X, E)$ and whose Hom sets are these colimits. Together with descent of objects, the statements above say exactly that the functor from this category to $D_{\mathrm{ctf}}^b(X, \overline{\mathbf{Q}}_\ell)$ is fully faithful and essentially surjective. ∎

In practice the proposition is used as follows. To prove a statement about finitely many objects and morphisms of $D_{\mathrm{ctf}}^b(X, \overline{\mathbf{Q}}_\ell)$, we may assume that they are extensions of scalars of objects and morphisms of $D_{\mathrm{ctf}}^b(X, E)$ for a finite extension E of $\mathbf{Q}_\ell$. Moreover, two morphisms over E that agree over $\overline{\mathbf{Q}}_\ell$ already agree over a finite extension of E.

7.3 Reduction to finite coefficients

Many later theorems say that a canonical map is an isomorphism. We prove them with finite coefficients and then extend them to all admissible rings in three steps. First, for $\mathcal{O}_E$ we reduce modulo π_E and use derived Nakayama. This step needs derived completeness, since E itself is a nonzero $\mathcal{O}_E$-module with $E/\pi_E E = 0$. Then for E we invert ℓ on integral models, and for $\overline{\mathbf{Q}}_\ell$ we descend to a finite extension. The following proposition makes this argument once and for all. It only asks that the functors F, G and the map θ of the statement commute with change of coefficients, and by Proposition 7.10 this holds for everything built from the six operations.

Proposition 7.12 (Isomorphisms reduce to finite coefficients)**.** *For each admissible Λ let $F_\Lambda, G_\Lambda : D^b_{\mathrm{ctf}}(X,\Lambda) \to D^b_{\mathrm{ctf}}(Y,\Lambda)$ be exact functors and $\theta_\Lambda : F_\Lambda \to G_\Lambda$ a natural transformation of exact functors. Assume that for each coefficient map $\Lambda \to \Lambda'$ of Proposition 7.10 there are isomorphisms $F_\Lambda(K)\otimes\Lambda' \simeq F_{\Lambda'}(K\otimes\Lambda')$ and $G_\Lambda(K)\otimes\Lambda' \simeq G_{\Lambda'}(K\otimes\Lambda')$ carrying $\theta_\Lambda(K)\otimes\Lambda'$ to $\theta_{\Lambda'}(K\otimes\Lambda')$. This holds, by Proposition 7.10, whenever F, G and θ are built from the six operations, their units and counits and base change maps, covariantly in K. If $\theta_A(K)$ is an isomorphism for every finite ring A and every $K \in D^b_{\mathrm{ctf}}(X,A)$, then $\theta_\Lambda(K)$ is an isomorphism for every admissible Λ and every $K \in D^b_{\mathrm{ctf}}(X,\Lambda)$.*

Proof. The three kinds of coefficients are reached from finite ones in three different ways: $\mathcal{O}_E$ by completion, E by inverting ℓ, and $\overline{\mathbf{Q}}_\ell$ as a colimit over finite extensions. We treat them in this order, each case using the previous one.

Let $\Lambda = \mathcal{O}_E$, and put $C = \mathrm{Cone}(\theta(K))$; we must show $C = 0$. Since the complexes $F(K)$ and $G(K)$ lie in $D^b_c(Y,\mathcal{O}_E)$, they are derived π_E-complete by Proposition 3.11. Moreover, derived complete complexes are closed under cones, and so C is derived complete too. Derived Nakayama (Lemma 2.62) therefore reduces $C = 0$ to $C/\pi_E = 0$. Now reduction modulo π_E is a change of coefficients along $\mathcal{O}_E \to \mathcal{O}_E/\pi_E$. By the compatibility assumed in the statement, it identifies $\theta(K)/\pi_E$ with $\theta_{\mathcal{O}_E/\pi_E}(K/\pi_E)$. This is an isomorphism by the hypothesis for finite rings, applied to $A = \mathcal{O}_E/\pi_E$. Thus $C/\pi_E = 0$, and so $C = 0$. In particular, the case $E = \mathbf{Q}_\ell$ gives $\Lambda = \mathbf{Z}_\ell$.

Let $\Lambda = E$. Here completeness is of no help, since a complex over E is killed by reduction modulo π_E. Instead we use integral models. Both F_E and G_E are exact and θ_E is a transformation of exact functors. Hence the objects K for which $\theta_E(K)$ is an isomorphism form a triangulated subcategory. By Lemma lemma-integral-models, every object of $D^b_c(X,E)$ is built by finitely many cones and shifts from objects $K_0[1/\ell]$ with $K_0 \in D^b_c(X,\mathcal{O}_E)$. It therefore suffices to treat these. Inverting ℓ is the change of coefficients along $\mathcal{O}_E \to E$. By hypothesis, $\theta_E(K_0[1/\ell])$ is therefore identified with $\theta_{\mathcal{O}_E}(K_0)[1/\ell]$. This is an isomorphism by the integral case.

Let $\Lambda = \overline{\mathbf{Q}}_\ell$. By Proposition 7.11, every K is isomorphic to $K_E \otimes_E \overline{\mathbf{Q}}_\ell$ for some finite E and some $K_E \in D^b_{\mathrm{ctf}}(X,E)$. By hypothesis, applied to $E \to \overline{\mathbf{Q}}_\ell$, the map $\theta_{\overline{\mathbf{Q}}_\ell}(K)$ is identified with $\theta_E(K_E)\otimes_E \overline{\mathbf{Q}}_\ell$, which is an isomorphism by the case of E. Again $E = \mathbf{Q}_\ell$ is included. ∎

After a rewriting, the proposition also applies to some contravariant constructions. For instance, the double-dual map $K \to \mathbb{D}\mathbb{D}K$ is a natural transformation of covariant exact functors. What does not reduce in this way is the construction of an adjoint or of a map. For this reason the cycle class of the diagonal and the adjunction $Rf_! \dashv f^*(d)[2d]$ for smooth f must be built as single morphisms with $\mathcal{O}_E$-coefficients (Sections 6.7 and Section 9.5). Likewise, a trace with values in Λ is not a map of complexes.

8 Smooth base change

To construct the trace classes in the trace formula we need the Künneth formula for ordinary direct images, $Rf_{1*}M_1 \boxtimes Rf_{2*}M_2 \simeq R(f_1\times f_2)_*(M_1\boxtimes M_2)$ (Proposition 8.18, Section 8.9). For $Rf_!$ this formula follows from proper base change and the projection formula, but for Rf_* this argument does not work, because Rf_* does not commute with base change in general. For example, let $j : U = \mathbf{A}^1 \setminus \{0\} \hookrightarrow \mathbf{A}^1$ over an algebraically closed field, and let n be invertible in it. Then the stalk of $R^1j_*(\mathbf{Z}/n)$ at 0 is H^1 of the punctured strictly henselian disc, which is $\mathbf{Z}/n(-1) \neq 0$, whereas the fibre of j over 0 is empty, so base change along the closed immersion of 0 fails.

The Künneth formula only needs base change along a product projection $Y\times T \to Y$ (Lemma 8.13). The question is whether base change along such a map holds for every f.

In topology it does, at least when T is a manifold. If g is a local homeomorphism, base change along g is restriction to open subsets. If g is the projection $Y \times \mathbf{R}^m \to Y$, then the stalk of $R^q f'_*$ at (y,t) is the colimit of the cohomology groups of $f^{-1}(V) \times B$ over small neighbourhoods $V \times B$ of (y,t), and the contractible ball B does not change them. For schemes the analogue is less clear: a smooth morphism of schemes is, Zariski locally on the source, étale over some $\mathbf{A}^d_Y$, but the affine line has no small contractible neighbourhoods. Nevertheless, the theorem of this chapter, due to Artin (SGA 4, Exposé XVI [SGA4]), says that for coefficients prime to the characteristic the affine line behaves as if it had them. More precisely, let $g : Y' \to Y$ be smooth, let f be quasi-compact and quasi-separated, and let F be a torsion sheaf whose torsion orders are invertible on Y. Then the base change map $g^* R^q f_* F \to R^q f'_* g'^* F$ is an isomorphism (Theorem 8.1). Moreover, on $D^b_{\mathrm{ctf}}(-,\Lambda)$ the same holds for every admissible Λ (Theorem 8.11).

A product projection $Y \times T \to Y$ is not smooth when T is singular. What the Künneth formula uses is a consequence of smooth base change, namely base change under extension of the ground field [Stacks, 0F1C].

The essential case is a change of the algebraically closed ground field, and there the key input is a fact about curves. Let Ω/L be an extension of algebraically closed fields, C a smooth affine curve over L, and n invertible in L. Then pullback $H^1(C, \mathbf{Z}/n) \to H^1(C_\Omega, \mathbf{Z}/n)$ is an isomorphism (Theorem 4.20). In other words, a cyclic cover of C_Ω of degree n does not depend on the extra transcendental parameters of Ω.

For p-torsion coefficients this invariance fails, and so does smooth base change. Let k be algebraically closed of characteristic $p > 0$, and let $f : \mathbf{A}^1_k \to \operatorname{Spec} k$ and $g : \mathbf{A}^1_k \to \operatorname{Spec} k$ be the structure maps, with coordinates x on the source of f and t on the source of g. By the Artin–Schreier sequence, $H^1(\mathbf{A}^1_K, \mathbf{F}_p)$ is the cokernel of $h \mapsto h^p - h$ on $K[x]$, for every field $K \supset k$. At the geometric generic point of the t-line, the stalk of $g^* R^1 f_* \mathbf{F}_p$ is $H^1(\mathbf{A}^1_k, \mathbf{F}_p)$, the stalk of $R^1 f'_* \mathbf{F}_p$ is $H^1(\mathbf{A}^1_K, \mathbf{F}_p)$ with $K = k(t)^{\mathrm{sep}}$, and the base change map is pullback. We claim that the class of tx is not in its image. Suppose $tx = a + h^p - h$ with $a \in k[x]$ and $h \in K[x]$, and let b_e be the coefficient of x^{p^e} in h. Comparing the coefficients of x gives $b_0 \in -t + k$, and comparing those of $x^{p^{e+1}}$ gives $b_{e+1} \in b_e^p + k$. By induction $b_e \in -t^{p^e} + k$, so $b_e \neq 0$ for every $e \geq 0$. This is absurd for a polynomial. In geometric terms, the Artin–Schreier cover $y^p - y = tx$ depends on the parameter t. Hence the hypothesis on the torsion orders cannot be dropped.

8.1 The base change map

Consider a cartesian square, with $X' = Y' \times_Y X$,

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y. \end{array} \tag{8.1}$$

The base change map $g^* R f_* K \to R f'_* g'^* K$, for $K \in D(X_{\mathrm{pro\acute{e}t}}, \Lambda)$, is the map (4.5) of Section 4.1. It comes by adjunction from the counit of $f^* \dashv R f_*$, and it is defined for every such square and every complex, with g smooth or not. If f is quasi-compact and quasi-separated (qcqs), its stalk at a geometric point $\bar{y}'$ of Y' with image $\bar{y}$ in Y is the pullback map from the cohomology of $X \times_Y \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{Y,\bar{y}}$ to that of $X' \times_{Y'} \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{Y',\bar{y}'}$.

In smooth base change for torsion sheaves (Theorem 8.1) we assume that f is qcqs and that g is smooth. In the theorem on D^b_{ctf} all schemes are of finite type over k. Every inverse image below is the exact inverse image for the pullback of the coefficient ring, so that $g^* = Lg^*$ and $g'^* = Lg'^*$.

The proofs of these theorems transport the base change map in three ways: from the pro-étale to the étale site, where the classical arguments are made; along composites of squares, when a smooth morphism is factored into curves; and along a change of the coefficient ring, to reach ℓ-adic coefficients. Since the map is built from units and counits, it is compatible with all three, by Lemma 4.7 for the first two and by Lemma 8.10 in Section 8.8 below for the third.

8.2 Torsion sheaves

Smooth base change for complexes (Section 8.7) and for ℓ-adic coefficients (Section 8.8) follows from the case of a single torsion sheaf, which is the next theorem. Its proof leaves schemes of finite type over k: it passes through base schemes such as spectra of fields and of integral closures. The comparison results of Chapter 2 hold over arbitrary schemes for bounded below complexes, so they apply to these bases as well.

Theorem 8.1 (Smooth base change for classical torsion sheaves)**.** *In (8.1), suppose g is smooth and f is quasi-compact and quasi-separated. Let F be a classical torsion abelian sheaf on $X_{\text{pro}\acute{e}t}$ which is the filtered union of its subsheaves $F[n]$ for the positive integers n invertible on Y. Then for every $q \geq 0$ the degree-q base change map induced by (4.5)*

$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F$$

is an isomorphism.

As for proper base change, we transport the problem to the étale site by the comparison theorems of Chapter 2. Since F is classical, $F = \nu_X^* F_{\acute{e}t}$ for a torsion abelian sheaf $F_{\acute{e}t}$ on $X_{\acute{e}t}$. As f and f' are qcqs, Theorem 2.38 and Lemma 4.7 identify both sides and the base change map with the inverse images of the corresponding étale objects. We may therefore work on the étale sites in the proof below, where g^{-1} denotes inverse image of étale sheaves.

We follow the second proof of [Stacks, 0EYU], an induction on the cohomological degree q. The idea is to concentrate a first failure of base change in a single sheaf $R^q h_* \mathbf{Z}/m$, and then to show that this sheaf vanishes. In degree zero, by contrast, the base change map can be computed directly on sections. Moreover, base change may be checked étale locally and is stable under composition, so that étale coordinates reduce us to the case where g is a smooth affine curve.

Now suppose that base change holds in degrees below q but fails in degree q. The failure persists when we shrink the source of the direct image by limit arguments, and also when we make integral changes of the source and of the base. In this way it is concentrated in a single situation, in which the coefficients are constant, equal to $\mathbf{Z}/m$, and we have a smooth affine curve C over the spectrum S of a normal domain whose fraction field L is algebraically closed, together with the morphism $h : C_\Omega \to C$ for an algebraically closed extension Ω of L. But in this situation $R^q h_* \mathbf{Z}/m = 0$. For $q \geq 2$ the reason is that affine curves over an algebraically closed field have no cohomology in degree at least two (Appendix D). For $q = 1$ it follows from the invariance of H^1 of curves and an extension theorem for covers (Section 8.6).

8.3 Degree zero

In degree zero the base change map can be computed on sections, even for sheaves of sets. Sections of $f_* F$ over V are sections of F over $V \times_Y X$, and the question is whether pulling back to a Y'-scheme U over V creates new sections. It does not when $U \to V$ is flat, quasi-compact and quasi-separated with geometrically connected fibres, because pullback along such a map

does not change sections of the pullback of an étale sheaf. Moreover, a flat g locally of finite presentation with geometrically reduced fibres factors locally into such a map followed by an étale one, and in this way Lemma 8.2 proves the isomorphism for every such g.

Lemma 8.2 (The degree-zero calculation)**.** *Suppose that g in (8.1) is flat and locally of finite presentation with geometrically reduced fibres, and that f is qcqs. Then the base change map $g^{-1}f_*F_{\acute{e}t} \to f'_*g'^{-1}F_{\acute{e}t}$ is an isomorphism for every sheaf of sets $F_{\acute{e}t}$ on $X_{\acute{e}t}$.*

Proof. Both sides are sheaves on $Y'_{\acute{e}t}$, so it is enough to compare their sections on a family of étale objects $U \to Y'$ that cover Y'. We choose these objects so that both sections can be computed from sections of $F_{\acute{e}t}$ over a single scheme $V \times_Y X$.

Such objects exist locally on every étale $U \to Y'$: by [Stacks, 0EY5], locally on U there is a factorization $U \to V \to Y$ in which $V \to Y$ is étale and $U \to V$ is flat and qcqs with geometrically connected fibres. Pullback along such a map does not change the sections of the pullback of an étale sheaf [Stacks, 0A3H, 0EZX], and we use this twice. Applying it to $U \to V$, we compute the left side from $(f_*F_{\acute{e}t})(V)$; applying it to the base change $U \times_Y X \to V \times_Y X$, which again has geometrically connected fibres, we compute the right side. On these objects we therefore have

$$(g^{-1}f_*F_{\acute{e}t})(U) = F_{\acute{e}t}(V \times_Y X) = (f'_*g'^{-1}F_{\acute{e}t})(U),$$

and the base change map is bijective on sections over every such U, because these equalities are induced by it. As these U cover Y', the lemma follows by descent, as in the argument in degree zero of [Stacks, 0EYS]. ∎

8.4 The induction hypothesis

In positive degrees we argue by induction on the degree. This induction hypothesis has to survive the reductions outlined above: composition of base changes, étale localization, and replacement of the source and of the base of the direct image by other schemes. But a statement about the single square (8.1) does not survive them, so instead we fix an integer n invertible on Y and use a property of a morphism $g : Y' \to Y$ which concerns all direct images over all Y-schemes at once, as in [Stacks, 0F02]. For $r \geq 0$, we say that $\mathrm{BC}(g, n, r)$ holds if, for every scheme T over Y, every qcqs morphism $Z \to T$, every sheaf of $\mathbf{Z}/n$-modules M on $Z_{\acute{e}t}$, and every $0 \leq i \leq r$, the base change map in degree i for the square with base change $Y' \times_Y T \to T$ and direct image along $Z \to T$ is an isomorphism. We allow arbitrary schemes T because the reductions below lead to base schemes, such as the spectrum of a normal domain with algebraically closed fraction field, that are in general not Noetherian.

The property $\mathrm{BC}(g, n, r)$ may be checked étale locally on Y and on Y'. Moreover, for composable morphisms g_1 and g_2, $\mathrm{BC}(g_1, n, r)$ and $\mathrm{BC}(g_2, n, r)$ imply $\mathrm{BC}(g_2 \circ g_1, n, r)$. Both statements rest on two properties of the base change map in a fixed degree. First, the map is local for étale coverings of Y' and of Y, since restriction to an étale object is localization of the site. Second, base change maps compose when two base changes are composed [Stacks, 0EZR, 0EZS].

For smooth g, étale coordinates [Stacks, 054L] give, locally on Y' and Y and with all schemes affine, an étale Y-morphism $Y' \to \mathbf{A}^d_Y$, where d is the relative dimension of g. Composing it with the projections $\mathbf{A}^j_Y \to \mathbf{A}^{j-1}_Y$ that forget the last coordinate, we obtain a factorization

$$Y' \longrightarrow \mathbf{A}^{d-1}_Y \longrightarrow \cdots \longrightarrow \mathbf{A}^1_Y \longrightarrow Y$$

of g in which every displayed arrow is smooth of relative dimension one. Indeed, the first is an étale map followed by a projection, and the others are projections. The case $d = 0$ is étale localization. Hence, in the induction on the degree, it suffices to treat affine Y and Y' with g smooth of relative dimension one.

The coefficients reduce to $\mathbf{Z}/n$-modules. Indeed, a sheaf F as in Theorem 8.1 is the filtered union of the subsheaves $F[n]$, ordered by divisibility of integers invertible on Y, and inverse image commutes with this colimit, and so do higher direct images along qcqs morphisms [Stacks, 09Z1]. It therefore suffices to treat $\mathbf{Z}/n$-modules for each such n.

8.5 Reduction to curves

Recall that $\mathrm{BC}(g, n, r)$ asserts base change in degrees $\leq r$, for $\mathbf{Z}/n$-coefficients, along every base change $Y' \times_Y T \to T$ of g. Assume now that $q \geq 1$ and that $\mathrm{BC}(g, n, q-1)$ holds. Under this assumption we show that if $\mathrm{BC}(g, n, q)$ fails, it also fails in a special form: with constant coefficients and with direct image from the spectrum of an algebraically closed field (Lemma 8.6). The next four lemmas, all on the étale sites, carry out this reduction step by step.

We argue by contradiction rather than by dévissage, because a nonzero cokernel of the base change map can be transported along an embedding of coefficient sheaves $M \hookrightarrow N$, whereas an isomorphism cannot.

Lemma 8.3 (Injectivity of the base change map in degree q)**.** *Suppose $q \geq 1$ and $\mathrm{BC}(g, n, q-1)$ holds. For every square obtained by base change as in the definition of* BC *and every sheaf of $\mathbf{Z}/n$-modules M on $Z_{\acute{e}t}$, the base change map $b^q_M : g^{-1}R^q f_* M \to R^q f'_* g'^{-1} M$ in degree q is injective. An inclusion $M \hookrightarrow N$ induces*

$$\operatorname{coker}(b^q_M) \hookrightarrow \operatorname{coker}(b^q_N).$$

For an embedding $M \hookrightarrow I$ with I injective in the category of étale sheaves of $\mathbf{Z}/n$-modules, the cokernel of b^q_M embeds into $R^q f'_* g'^{-1} I$.

Proof. We embed M into an injective sheaf and compare the long exact sequences of the two sides. The induction hypothesis identifies everything in degree $q-1$, and since the injective sheaf has no higher direct images upstairs, a diagram chase shows that b^q_M is injective with cokernel inside $R^q f'_* g'^{-1} I$.

Choose $0 \to M \to I \to Q \to 0$ with I as in the statement. By the long exact sequences, the rows of the following diagram are exact. Here the upper row ends in 0 because $R^q f_* I = 0$ for the injective sheaf I, whereas the lower row does not, because $g'^{-1} I$ need not be injective.

$$\begin{array}{ccccccc}
g^{-1}R^{q-1}f_*I & \longrightarrow & g^{-1}R^{q-1}f_*Q & \longrightarrow & g^{-1}R^q f_* M & \longrightarrow & 0 \\
\sim\downarrow & & \sim\downarrow & & \downarrow & & \downarrow \\
R^{q-1}f'_*g'^{-1}I & \longrightarrow & R^{q-1}f'_*g'^{-1}Q & \longrightarrow & R^q f'_* g'^{-1} M & \longrightarrow & R^q f'_* g'^{-1} I.
\end{array}$$

The first two vertical arrows are isomorphisms by $\mathrm{BC}(g, n, q-1)$, applied to I and to Q. Exactness of the upper row says that $g^{-1}R^q f_* M$ is the cokernel of the map $g^{-1}R^{q-1}f_*I \to g^{-1}R^{q-1}f_*Q$, and exactness of the lower row says that the kernel of $R^q f'_* g'^{-1} M \to R^q f'_* g'^{-1} I$ is the cokernel of the corresponding map in the lower row. Since the first two vertical isomorphisms identify these two maps, b^q_M is an isomorphism onto this kernel. In particular b^q_M is injective, and its cokernel embeds into $R^q f'_* g'^{-1} I$, which is the last assertion. For $M \subset N$, choose an injective I

containing N. What we have just shown, applied to $M \subset I$ and to $N \subset I$, embeds the cokernels of b^q_M and b^q_N compatibly into $R^q f'_* g'^{-1} I$. As in [Stacks, 0F03], the map between the cokernels induced by $M \subset N$ is compatible with these two embeddings and is therefore injective. ∎

We use the lemma in the form of its second assertion: enlarging the coefficient sheaf does not lose a failure. In the next lemma this is how we pass from a scheme to an open subscheme or a closed complement.

Lemma 8.4 (Reduction to a field)**.** *Let $q \geq 1$. If $\mathrm{BC}(g, n, q-1)$ holds but $\mathrm{BC}(g, n, q)$ fails, then some failure occurs for an affine base change $T \to Y$, direct image along a morphism $\operatorname{Spec}\Omega \to T$ with Ω a field, and a sheaf of $\mathbf{Z}/n$-modules on $\operatorname{Spec}\Omega$.*

Proof. We shrink the source of the direct image, keeping a fixed nonzero element of the cokernel, until it becomes the spectrum of a field. A minimal closed subscheme carrying the failure exists by a limit argument, and its generic point still carries it. The one nonformal step is the third property below, which says that a failure on a scheme is seen either on an open subscheme or on its closed complement, and whose proof uses the injectivity of the base change map.

We first fix notation for the cokernels. Let T be a Y-scheme, put $T' = Y' \times_Y T$, and let $g_T : T' \to T$ be the projection. For a qcqs morphism $h : Z' \to T$ we write $h' : Z'_{T'} = Z' \times_T T' \to T'$ and $g_{Z'} : Z'_{T'} \to Z'$ for its base changes, and for a sheaf N of $\mathbf{Z}/n$-modules on $Z'_{\acute{e}t}$ we put

$$b^q_{h,N} : g_T^{-1} R^q h_* N \longrightarrow R^q h'_* g_{Z'}^{-1} N, \qquad C(h, N) = \operatorname{coker}\bigl(b^q_{h,N}\bigr),$$

a sheaf on $T'_{\acute{e}t}$. These cokernels are functorial in two ways. A map of sheaves $N \to N_1$ on $Z'_{\acute{e}t}$ induces $C(h, N) \to C(h, N_1)$, and a morphism $u : Z'' \to Z'$ induces $C(h, N) \to C(hu, u^{-1}N)$, by pulling back along u: upstairs through $N \to u_* u^{-1} N$ and the edge map $R^q h_* u_* \to R^q (hu)_*$, and downstairs in the same way. These pullback maps are compatible with the base change maps, and with composition in u, because base change maps compose [Stacks, 0EZR, 0EZS].

Since $\mathrm{BC}(g, n, q)$ fails, there are a Y-scheme T, a qcqs morphism $f : Z \to T$ and a sheaf M of $\mathbf{Z}/n$-modules on $Z_{\acute{e}t}$ such that $b^q_{f,M}$ is not an isomorphism. By Lemma 8.3 it is injective, so $C(f, M) \neq 0$, and we fix a geometric point $\bar{t}'$ of T' and a nonzero element $\xi \in C(f, M)_{\bar{t}'}$. For a qcqs morphism $u : Z' \to Z$ let $P(Z')$ mean that the image $\xi_{Z'}$ of ξ in $C(fu, u^{-1}M)_{\bar{t}'}$ is nonzero. The verification in [Stacks, 0F07] shows that P has three properties.

1. thickenings preserve P, by topological invariance;
2. limits with affine transition maps preserve P, by continuity;
3. if $P(Z')$ holds and $V \subset Z'$ is a quasi-compact open with closed complement W, then $P(V)$ or $P(W)$ holds.

To prove the third property, write $h = fu : Z' \to T$ and $M' = u^{-1}M$, and let $j : V \to Z'$ and $i : W \to Z'$ be the inclusions. We split M' into an open part and a closed part. Choose an injective sheaf J of $\mathbf{Z}/n$-modules on $V_{\acute{e}t}$ containing $j^{-1}M'$. Then the map

$$M' \longrightarrow j_* J \oplus i_* i^{-1} M',$$

whose components are adjoint to $j^{-1}M' \subset J$ and to the identity of $i^{-1}M'$, is injective. Indeed, at a geometric point of V its first component is the inclusion $M'_{\bar{v}} \subset J_{\bar{v}}$, and at a geometric point of W its second component is the identity of $M'_{\bar{w}}$. By the second assertion of Lemma 8.3, the induced map

$$C(h, M') \longrightarrow C(h, j_*J) \oplus C(h, i_*i^{-1}M')$$

is injective as well. Since $P(Z')$ holds, $\xi_{Z'} \neq 0$, and therefore its image in $C(h, j_*J)_{\bar{t}'}$ or its image in $C(h, i_*i^{-1}M')_{\bar{t}'}$ is nonzero. We show that the second case gives $P(W)$ and the first gives $P(V)$.

The closed summand is formal. The base change $i' : W_{T'} \to Z'_{T'}$ of i is again a closed immersion, and closed direct images are exact (Lemma D.1) and commute with every base change (Lemma 4.27). Hence, for a sheaf N on $W_{\acute{e}t}$,

$$R^q h_*(i_*N) = R^q(hi)_*N, \qquad R^q h'_*(g_{Z'}^{-1}i_*N) = R^q h'_*(i'_*g_W^{-1}N) = R^q(hi)'_*g_W^{-1}N,$$

and these identifications are compatible with the base change maps [Stacks, 0EZR, 0EZS]. Thus $C(h, i_*N) = C(hi, N)$. For $N = i^{-1}M'$ the image of $\xi_{Z'}$ in $C(h, i_*i^{-1}M')_{\bar{t}'}$ becomes ξ_W, and so it is nonzero exactly when $P(W)$ holds.

The open summand needs the induction hypothesis, because j is not finite. Let $j' : V_{T'} \to Z'_{T'}$ and $g_V : V_{T'} \to V$ be the base changes of j, and compare the Leray spectral sequences for $hj = h \circ j$ and $h'j' = h' \circ j'$. Upstairs, $R^b j_*J = 0$ for $b > 0$ because J is injective, so the Leray spectral sequence collapses and gives

$$R^q h_*(j_*J) \xrightarrow{\sim} R^q(hj)_*J.$$

Downstairs we use $\mathrm{BC}(g, n, q-1)$. The scheme Z' is a Y-scheme through T, and $Z'_{T'} = Y' \times_Y Z'$, so the square formed by j, j', $g_{Z'}$ and g_V is one of the squares in the definition of BC. Hence

$$g_{Z'}^{-1}R^b j_*J \xrightarrow{\sim} R^b j'_*g_V^{-1}J \qquad (0 \leq b \leq q-1).$$

For $b = 0$ this identifies $g_{Z'}^{-1}j_*J$ with $j'_*g_V^{-1}J$, and for $0 < b < q$ it shows that $R^b j'_*g_V^{-1}J = 0$. Consequently, in the Leray spectral sequence

$$E_2^{a,b} = R^a h'_*R^b j'_*g_V^{-1}J \implies R^{a+b}(h'j')_*g_V^{-1}J$$

the rows $0 < b < q$ vanish. No differential can reach $E^{q,0}$: the differential d_r into $E_r^{q,0}$ starts at $E_r^{q-r,r-1}$, which is zero for $2 \leq r \leq q$ because then $0 < r-1 < q$, and for $r > q$ because then $q - r < 0$. So $E_2^{q,0} = E_\infty^{q,0}$, which is the bottom step of the filtration of the abutment, and the edge map

$$e : R^q h'_*(j'_*g_V^{-1}J) \longrightarrow R^q(h'j')_*g_V^{-1}J$$

is injective. Since base change maps compose [Stacks, 0EZR, 0EZS] and edge maps are natural, the square

$$\begin{array}{ccc} g_T^{-1}R^q h_*(j_*J) & \xrightarrow{b^q_{h,j_*J}} & R^q h'_*(j'_*g_V^{-1}J) \\ \wr\downarrow & & \downarrow e \\ g_T^{-1}R^q(hj)_*J & \xrightarrow{b^q_{hj,J}} & R^q(h'j')_*g_V^{-1}J \end{array}$$

commutes. Its left arrow is an isomorphism and its right arrow is injective, so it induces an injective map $C(h, j_*J) \to C(hj, J)$. (In the top row we have identified $g_{Z'}^{-1}j_*J$ with $j'_*g_V^{-1}J$ by

the case $b = 0$; this does not change the cokernel.) The vertical arrows are the pullback maps along j, so this map is pullback along j followed by $j^{-1}j_*J = J$.

It remains to pass from J back to $j^{-1}M'$. By naturality, the composite $C(h, M') \to C(h, j_*J) \to C(hj, J)$ equals the composite $C(h, M') \to C(hj, j^{-1}M') \to C(hj, J)$ of pullback along j and the map induced by $j^{-1}M' \subset J$. In the first case the image of $\xi_{Z'}$ in $C(h, j_*J)_{\bar{t}'}$ is nonzero, hence so is its image in $C(hj, J)_{\bar{t}'}$. This image is the image of ξ_V under the second composite, and so $\xi_V \neq 0$, which is $P(V)$.

The field comes out of the three properties as follows. The intersection of a decreasing chain of closed subschemes is the limit of the chain, with affine transition maps. By the second property and Zorn's lemma there is therefore a minimal closed subscheme with P. It is nonempty, since P fails for the empty scheme, and by the first property it is reduced. Let ξ be the generic point of an irreducible component. We claim that every affine open neighbourhood of ξ has P. Indeed, by the third property, either the neighbourhood or its closed complement has P, and the complement is a closed subscheme not containing ξ, hence strictly smaller, so by minimality it does not have P. Since the minimal closed subscheme is reduced, the limit of these neighbourhoods is $\operatorname{Spec}\kappa(\xi)$. It has P by the second property (compare [Stacks, 0F06]), and so $\kappa(\xi)$ is the field asserted in the lemma. ∎

The field Ω produced by this lemma is arbitrary, and the base T is still an arbitrary affine scheme. To bring them into the special situation of Lemma 8.6 we want to replace Ω by its algebraic closure and T by a normal scheme. Both replacements are integral morphisms, and the next lemma shows that a failure survives them.

Lemma 8.5 (Integral changes of source and base)**.** *Assume* $\mathrm{BC}(g, n, q-1)$*. A failure of base change in degree* q *persists after an integral surjective change of the source of the morphism along which we take direct image. It also persists after an integral change of the base through which this morphism factors.*

Proof. Both assertions are [Stacks, 0F04, 0F05]. They use three properties of direct image along an integral morphism π: it is exact and conservative [Stacks, 04C2], even if π is not surjective, and it commutes with arbitrary base change [Stacks, 0EYP]. By exactness, the Leray spectral sequence for a composite with π degenerates, which means that direct images along the composite are computed by applying π_* first.

We use the notation $b^q_{h,N}$ and $C(h, N) = \operatorname{coker} b^q_{h,N}$ of the proof of Lemma 8.4, for a Y-scheme T, its base change $g_T : T' = Y' \times_Y T \to T$, a qcqs morphism $h : Z \to T$ and a sheaf N of $\mathbf{Z}/n$-modules on $Z_{\acute{e}t}$. By Lemma 8.3, $b^q_{h,N}$ is injective, so a failure of base change in degree q means $C(h, N) \neq 0$.

For the first assertion, let $f : Z \to T$ and M give a failure, so $C(f, M) \neq 0$, and let $\pi : \widetilde{Z} \to Z$ be integral and surjective. We show that $C(f\pi, \pi^{-1}M) \neq 0$. Since π is surjective, the unit $M \to \pi_*\pi^{-1}M$ is injective, as one sees on stalks. By the second assertion of Lemma 8.3 it induces an injection $C(f, M) \hookrightarrow C(f, \pi_*\pi^{-1}M)$, and so $C(f, \pi_*\pi^{-1}M) \neq 0$. It remains to identify $C(f, \pi_*\pi^{-1}M)$ with $C(f\pi, \pi^{-1}M)$. Let $\pi' : \widetilde{Z}_{T'} \to Z_{T'}$ be the base change of π, again integral. Upstairs, π_* is exact, so the Leray spectral sequence for $f \circ \pi$ degenerates and gives $R^q f_*(\pi_*\pi^{-1}M) = R^q(f\pi)_*\pi^{-1}M$. Downstairs, π_* commutes with base change, $g_Z^{-1}\pi_*\pi^{-1}M \simeq \pi'_* g_{\widetilde{Z}}^{-1}\pi^{-1}M$, and π'_* is exact, so in the same way

$$R^q f'_*(g_Z^{-1}\pi_*\pi^{-1}M) \simeq R^q f'_*\pi'_*(g_{\widetilde{Z}}^{-1}\pi^{-1}M) = R^q(f\pi)'_*(g_{\widetilde{Z}}^{-1}\pi^{-1}M).$$

These identifications are compatible with the base change maps [Stacks, 0EZR, 0EZS], so they carry $b^q_{f,\pi_*\pi^{-1}M}$ to $b^q_{f\pi,\pi^{-1}M}$. Hence $C(f\pi,\pi^{-1}M) \simeq C(f,\pi_*\pi^{-1}M) \neq 0$, and base change fails for the new source.

For the second assertion, let $f : Z \to T$ factor as $f = \pi \circ f_1$ with $\pi : T_1 \to T$ integral and $f_1 : Z \to T_1$, and suppose base change fails for f and M, that is, $C(f,M) \neq 0$. We show that it fails for f_1 and M over the new base T_1. Put $T_1' = Y' \times_Y T_1 = T' \times_T T_1$, and let $\pi' : T_1' \to T'$ be the base change of π, again integral. The base change map for f_1 is

$$b^q_{f_1,M} : g_{T_1}^{-1} R^q f_{1*} M \longrightarrow R^q f'_{1*} g_Z^{-1} M \qquad \text{on } T'_{1,\acute{e}t}.$$

We push it down to T' by π'_*. Since π'_* is exact and π_* commutes with base change, $\pi'_* g_{T_1}^{-1} R^q f_{1*} M \simeq g_T^{-1} \pi_* R^q f_{1*} M = g_T^{-1} R^q f_* M$, where the last equality is the degenerate Leray spectral sequence for $f = \pi f_1$. In the same way $\pi'_* R^q f'_{1*} g_Z^{-1} M = R^q f'_* g_Z^{-1} M$. Under these identifications $\pi'_*(b^q_{f_1,M}) = b^q_{f,M}$, again because base change maps compose. Since π'_* is exact, it follows that $\pi'_* C(f_1,M) \simeq C(f,M) \neq 0$, so $C(f_1,M) \neq 0$, which is the assertion. ∎

With the two previous lemmas we may take the source of the direct image to be the spectrum of an algebraically closed field. Its cohomology vanishes in positive degrees, so the source of the base change map is zero and the failure sits in the target, a direct image along an extension of the ground field.

Lemma 8.6 (Reduction to a curve over a normal base)**.** *Suppose $g : Y' \to Y$ is affine and smooth of relative dimension one, n is invertible on Y, and* $\mathrm{BC}(g,n,q-1)$ *holds. If* $\mathrm{BC}(g,n,q)$ *fails, there are an affine scheme $S = \operatorname{Spec} R$ over Y with R a normal domain whose fraction field L is algebraically closed, an algebraically closed extension field Ω of L, an integer $m \mid n$, and, for the smooth affine curve $C = Y' \times_Y S$, the cartesian square*

$$\begin{array}{ccc} C_\Omega & \xrightarrow{h} & C \\ \downarrow & & \downarrow \\ \operatorname{Spec}\Omega & \longrightarrow & S \end{array}$$

such that

$$R^q h_* \underline{\mathbf{Z}/m} \neq 0.$$

Here $\underline{\mathbf{Z}/m}$ is the classical constant sheaf.

Proof of Lemma 8.6. We first simplify the source of the direct image and the coefficients, then show that the source of the base change map vanishes, and finally replace the base by a normal one.

By Lemma 8.4 there is a failure with direct image from the spectrum of a field Ω. Since the map from the spectrum of an algebraic closure of Ω to $\operatorname{Spec}\Omega$ is integral and surjective, Lemma 8.5 allows us to replace Ω by an algebraic closure. On this geometric point, a sheaf of $\mathbf{Z}/n$-modules is a $\mathbf{Z}/n$-module, hence a direct sum of cyclic groups $\mathbf{Z}/m$ with $m \mid n$. As the base change map commutes with this direct sum by continuity of qcqs direct images, base change still fails for some summand $\mathbf{Z}/m$.

The source of the base change map is the pullback of $R^q f_* \underline{\mathbf{Z}/m}$ for $f : \operatorname{Spec}\Omega \to T$, and we show that it is zero. The stalk of this sheaf at a geometric point $\bar{t}$ of T is the cohomology of

$\operatorname{Spec}\Omega \times_T \operatorname{Spec}\mathcal{O}^{\mathrm{sh}}_{T,\bar{t}}$. This scheme is a limit of étale Ω-schemes, which, since Ω is algebraically closed, are disjoint unions of copies of $\operatorname{Spec}\Omega$. By continuity of cohomology the stalk therefore vanishes for $q \geq 1$. As the source of the base change map is zero, the failure means that the target $R^q h_* \underline{\mathbf{Z}/m}$ is nonzero.

The extension theorem for covers will require a normal base with algebraically closed fraction field. If the base is $\operatorname{Spec} B$, we replace it by $\operatorname{Spec} R$, where R is the integral closure in Ω of the image of B. This is allowed, because the morphism $\operatorname{Spec}\Omega \to \operatorname{Spec} B$ factors through $\operatorname{Spec} R$, and $\operatorname{Spec} R \to \operatorname{Spec} B$ is integral, so the failure persists by Lemma 8.5. Moreover, the fraction field L of R is the algebraic closure of $\operatorname{Frac}(\operatorname{im}(B \to \Omega))$ in Ω, hence algebraically closed, although in general Ω/L is transcendental. We are now in the situation of [Stacks, 0F08]. ∎

Example 8.7 (Where the reduction ends)**.** The square of the lemma is itself an instance of smooth base change, and the simplest case shows what remains to be proved. Let $Y = S = \operatorname{Spec} L$ with L algebraically closed, let $g : Y' = C = \mathbf{A}^1_L \to Y$ be the structure map, and take direct image along $f : X = \operatorname{Spec}\Omega \to Y$ for an algebraically closed extension Ω of L, so that $X' = \mathbf{A}^1_\Omega$ and $f' = h$. The sheaf $R^q f_* \mathbf{Z}/m$ on $\operatorname{Spec} L$ is $H^q(\operatorname{Spec}\Omega, \mathbf{Z}/m)$, which vanishes for $q \geq 1$. So smooth base change for this square says exactly that $R^q h_* \mathbf{Z}/m = 0$ for $q \geq 1$. For m invertible in L this is the content of Proposition 8.8 below.

Let $q = 1$ and m invertible in L. The sections over C itself are no problem: choosing $\mu_m \simeq \mathbf{Z}/m$, the Kummer sequence gives $H^1(\mathbf{A}^1_\Omega, \mathbf{Z}/m) = \Omega^*/(\Omega^*)^m = 0$, since $\Gamma(\mathbf{A}^1_\Omega, \mathcal{O})^* = \Omega^*$ is divisible and $\operatorname{Pic}(\mathbf{A}^1_\Omega) = 0$. However, the presheaf whose sheafification is $R^1 h_* \mathbf{Z}/m$ has other sections. For example, on the open $U = \mathbf{G}_{m,L}$ of C the group $H^1(\mathbf{G}_{m,\Omega}, \mathbf{Z}/m)$ is cyclic of order m, and such a class has to be killed by an étale covering of U, as we shall see after the statement of the proposition.

For $m = p$ in characteristic p the argument breaks down already over C itself. Indeed, the group $H^1(\mathbf{A}^1_\Omega, \mathbf{F}_p)$ is the cokernel of $u \mapsto u^p - u$ on $\Omega[x]$, and for $\omega \in \Omega$ transcendental over L it contains the class of ωx, which does not come from C (compare the class of tx at the beginning of the chapter).

8.6 Change of base field

Lemma 8.6 turned a failure of $\mathrm{BC}(g, n, q)$ into a single nonvanishing sheaf $R^q h_* \mathbf{Z}/m \neq 0$, where $h : C_\Omega \to C$ is the base change of a smooth affine curve C over a normal base S to a larger algebraically closed field Ω. Proposition 8.8 below shows that this sheaf vanishes, which contradicts the lemma and finishes the induction. As the sheaf $R^q h_* \mathbf{Z}/m$ is the sheafification of $U \mapsto H^q(U_\Omega, \mathbf{Z}/m)$ on $C_{\acute{e}t}$, we must show that classes on U_Ω die after étale localization on U. In degrees $q \geq 2$ there is nothing to kill, since affine curves over an algebraically closed field have cohomological dimension one.

In degree one the problem is about covers. Recall that $S = \operatorname{Spec} R$ with R a normal domain whose fraction field L is algebraically closed, and $L \subset \Omega$ (Lemma 8.6). Over the generic point of S, a class on C_Ω already comes from C_L, by the invariance of H^1 of a curve under extension of algebraically closed fields (Theorem 4.20). In other words, it is given by a $\mathbf{Z}/m$-torsor over C_L, so it suffices to find an étale cover of C on which this torsor becomes trivial. To find one, we take a connected component of the torsor and extend it over C by the extension theorem for Galois covers of order invertible on S. This theorem needs S normal and a section of $C \to S$. The section is available: since L is algebraically closed, the generic fibre C_L has a rational point, and after étale localization on S this point extends to a section.

We use that m is invertible in two places, and both fail for p-torsion. First, the invariance of H^1 fails: the Artin–Schreier class of tx on $\mathbf{A}^1_{k(t)^{\mathrm{sep}}}$ does not come from $\mathbf{A}^1_k$. Second, the extension

theorem can fail for covers of degree divisible by a residue characteristic, because such covers may be wildly ramified along the special fibres.

Proposition 8.8 (Base change to a larger algebraically closed field)**.** *Let $S = \operatorname{Spec} R$ for a normal domain R with algebraically closed fraction field L, and let $a : C \to S$ be affine and smooth of relative dimension one. Let Ω/L be an extension of algebraically closed fields, let m be a positive integer invertible on S, and let $h : C_\Omega \to C$ be the projection. Then*

$$R^q h_* \underline{\mathbf{Z}/m} = 0 \qquad (q \geq 1)$$

on $C_{\text{proét}}$.

The simplest case already shows how the proof goes. Take $C = \mathbf{G}_{m,S}$ with coordinate x. Then $H^1(\mathbf{G}_{m,\Omega}, \mathbf{Z}/m)$ is cyclic of order m, generated by the class of the Kummer cover $y^m = x$. Here we view this cover as a $\mathbf{Z}/m$-torsor through a primitive m-th root of unity, which lies in R because R is normal and L is algebraically closed. The point is that the cover is already defined over $\mathbf{G}_{m,S}$, and it is finite étale there because m is invertible on S, so the class becomes trivial after pullback to it. The proof below produces such a cover for an arbitrary class.

Proof of Proposition 8.8. We describe $R^q h_*$ as a sheafification and show that each local class dies after a suitable étale covering. In degrees at least two this is the cohomological dimension of affine curves. In degree one we first descend the class to C_L, then trivialize it on a finite étale Galois cover of C_L, and finally extend this cover over C.

Step 1: reduction to local classes. We may work on the étale site, because h is qcqs and direct images can be compared (Theorem 2.38). There, [Stacks, 03Q8] identifies $R^q h_* \underline{\mathbf{Z}/m}$ with the sheafification of the presheaf

$$U \longmapsto H^q(U \times_S \operatorname{Spec} \Omega, \mathbf{Z}/m), \qquad U \to C \text{ étale},$$

so we must show that every such class vanishes on an étale covering of U.

Step 2: degrees $q \geq 2$. In these degrees the groups themselves vanish for affine U, which we may assume because the question is local on U. Indeed, U_Ω is then affine and étale over C_Ω. Hence it is empty or a finite disjoint union of smooth affine curves over the algebraically closed field Ω, and (C_2) of Theorem 4.20 kills its cohomology in degrees $q \geq 2$.

Step 3: a section and connected fibres. From now on $q = 1$. The extension theorem we want to use requires a section and geometrically connected fibres. The map $U \to S$ need not have them, but we may arrange them locally. By [Stacks, 0EY4], after an étale covering of U the map $U \to S$ factors as $U \to V \to S$ with U and V affine, $V \to S$ étale, and $U \to V$ smooth with geometrically connected fibres and a section. Moreover, the separated étale map $V \to S$ is a finite disjoint union of open immersions [Stacks, 09Z9], because S is normal and integral with separably closed function field. As we only have to kill classes on U_Ω, we may therefore replace S by one of these opens and C by the part of U over it, which does not change the fraction field L. Afterwards there is a section $s : S \to C$ and the fibres are geometrically connected.

Step 4: descent to C_L and a trivializing cover. Let $\xi \in H^1(C_\Omega, \mathbf{Z}/m)$. It is the pullback of a unique class $\xi_L \in H^1(C_L, \mathbf{Z}/m)$, because the pullback $H^1(C_L, \mathbf{Z}/m) \to H^1(C_\Omega, \mathbf{Z}/m)$ is an isomorphism by (C_6) of Theorem 4.20, applied to the affine curve C_L as in [Stacks, 03SC]; this step uses that m is prime to the characteristic. Next, the classification of H^1 by torsors [Stacks, 03AJ] and Lemma 2.41 show that ξ_L is the class of a finite étale $\mathbf{Z}/m$-torsor $P \to C_L$. Choose a connected component Z of P and let $G \subseteq \mathbf{Z}/m$ be its stabilizer. As the curve C_L is connected,

$Z \to C_L$ is a finite étale Galois cover with group G. The class ξ_L vanishes on Z, because the pullback of P to Z has the diagonal section.

Step 5: extending the cover over C. We use the extension theorem for generic Galois covers [Stacks, 0EZJ]: let S be integral, normal and qcqs with generic point η, and let $C \to S$ be qcqs and smooth with geometrically connected fibres and a section s. If $Z \to C_\eta$ is a finite étale Galois cover with group G of order invertible on S, and Z has a $\kappa(\eta)$-rational point over $s(\eta)$, then Z extends to a finite étale Galois cover $\widetilde{Z} \to C$ with group G. The idea is that a cover of order prime to the residue characteristics ramifies only tamely, and splitting over the section then rules out ramification along the irreducible special fibres, and at points of codimension at least two the cover extends by purity of the branch locus.

The hypotheses hold with $C_\eta = C_L$. Indeed, the order $|G|$ divides m and is therefore invertible on S. Moreover, the fibre of Z over $s(\eta)$ is nonempty, as $Z \to C_L$ is finite étale and surjective, and it has an L-rational point because L is algebraically closed. So $\widetilde{Z}$ exists.

We check that $\widetilde{Z} \to C$ is an étale covering, that is, that it is surjective. The scheme C is connected, since $C \to S$ is open and surjective with connected fibres over a connected base. Hence the image of $\widetilde{Z}$ in C, which is open and closed and contains the generic fibre, is all of C. Finally, ξ dies on this covering. Indeed, $\widetilde{Z} \times_C C_\Omega = Z \times_{C_L} C_\Omega$, so the pullback of ξ to $\widetilde{Z} \times_C C_\Omega$ is the pullback of $\xi_L|_Z = 0$. ∎

Proof of Theorem 8.1. We argue by induction on q. The degree-zero calculation starts the induction. By the reductions above, a hypothetical failure in degree q gives a nonzero sheaf $R^q h_* \underline{\mathbf{Z}/m}$, and Proposition 8.8 says that this sheaf is zero.

A smooth morphism is flat, locally of finite presentation, and has geometrically reduced fibres, and so the case of degree zero is Lemma 8.2. Let $q \geq 1$ and assume the theorem in all degrees less than q. Fix n invertible on Y. Since smoothness is stable under base change, we may apply the induction hypothesis after every base change $T \to Y$, which shows that $\mathrm{BC}(g, n, q-1)$ holds. By étale coordinates and composition of base change maps, it suffices to treat the case where g is affine and smooth of relative dimension one.

If $\mathrm{BC}(g, n, q)$ failed, Lemma 8.6 would give S, C, h and m with $R^q h_* \underline{\mathbf{Z}/m} \neq 0$. These satisfy the hypotheses of Proposition 8.8: by construction S is the spectrum of a normal domain with algebraically closed fraction field, m is invertible on S because it divides n, and C is a smooth affine curve over S. Therefore the conclusion $R^q h_* \underline{\mathbf{Z}/m} = 0$ of the proposition is a contradiction. Hence $\mathrm{BC}(g, n, q)$ holds for affine g of relative dimension one, and by the reduction above it holds for every smooth g.

Since f and f' are qcqs, both sides commute with the filtered union $F = \bigcup_n F[n]$, which gives the theorem for F on the étale sites. Finally, Theorem 2.38 transfers the isomorphism and the base change map from the étale sites to $Y'_{\mathrm{pro\acute{e}t}}$. ∎

8.7 Bounded below complexes

Theorem 8.1 concerns a single torsion sheaf, but its applications concern complexes. Smooth base change on D^b_{ctf} (Theorem 8.11) is reduced, for integral coefficients, to complexes with finite coefficients, and the Künneth formula (Section 8.9) is a statement about complexes. So we need smooth base change for complexes of torsion sheaves, and bounded below complexes suffice, since objects of D^b_{ctf} are bounded.

Theorem 8.9 (Smooth base change for classical torsion complexes)**.** *In (8.1), let g be smooth and f quasi-compact and quasi-separated. Let Λ be a discrete commutative ring annihilated by an integer n invertible on Y. For every*

$$K \in D^{+}_{\mathrm{class}}(X_{\mathrm{pro\acute{e}t}}, \Lambda)$$

the canonical map (4.5) is an isomorphism.

Proof. We first treat a single sheaf, where the only point is Λ-linearity, and then pass to complexes by comparing hypercohomology spectral sequences.

For a classical sheaf the assertion is Theorem 8.1, applied to the underlying abelian sheaf, which is killed by n. Since the base change map is constructed from units and counits, it is Λ-linear, and so it is an isomorphism of Λ-module sheaves.

For a complex, we first note that by Theorem 2.38 and Corollary 2.37 the direct and inverse images involved are classical, and that Lemma 4.7 identifies the comparison with (4.5). So we may argue on the étale sites. Choose an integer m with $K \in D^{\geq m}$. Since both g^* and g'^* are exact, they commute with taking cohomology sheaves, and the two sides of (4.5) are computed by the hypercohomology spectral sequences for Rf_* and Rf'_*, applied to K and to g'^*K. These spectral sequences have E_2 terms

$$g^* R^p f_* H^q(K), \qquad R^p f'_* g'^* H^q(K), \qquad p \geq 0, \quad q \geq m,$$

and the base change map induces on them the comparison for sheaves, which is an isomorphism by the preceding paragraph. Since in each total degree only finitely many pairs (p, q) occur, these shifted first quadrant spectral sequences converge; this is where we use that K is bounded below. A map of convergent spectral sequences that is an isomorphism on E_2 is an isomorphism on abutments, so the base change map induces isomorphisms $g^* R^i f_* K \to R^i f'_* g'^* K$ for all i, and so it is an isomorphism in the derived category. ∎

For the theorem on D^b_{ctf} we only need the case of a finite admissible ring A, where the hypothesis holds with $n = \ell^r$ for $\ell^r A = 0$.

8.8 ℓ-adic coefficients

For every admissible coefficient ring, smooth base change takes the following form. For finite rings it is a case of smooth base change for torsion complexes (Theorem 8.9), and we deduce the other rings by Proposition 7.12.

On the pro-étale site the base change map for $\mathcal{O}_E$-coefficients is a single map of complexes, defined by the same adjunction as for finite rings. Lemma 8.10 identifies its reduction modulo π_E with the base change map for $\mathcal{O}_E/\pi_E$, so derived Nakayama gives the integral case from the torsion theorem. The rational cases then follow by inverting ℓ and by descent to a finite extension.

Lemma 8.10 (Change of coefficients)**.** *Suppose that the schemes in (8.1) are of finite type over k. Let $\Lambda \to \Lambda'$ be a coefficient map of Proposition 7.10 and let $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$. Then the square*

$$\begin{array}{ccc} (g^* R f_* K)_{\Lambda'} & \longrightarrow & (R f'_* g'^* K)_{\Lambda'} \\ \downarrow & & \downarrow \\ g^* R f_* (K_{\Lambda'}) & \longrightarrow & R f'_* g'^* (K_{\Lambda'}) \end{array}$$

commutes. Here the upper arrow is the extension of scalars of the base change map (4.5) for K, the lower arrow is the base change map for $K_{\Lambda'}$, and the vertical arrows are the composites of

the change-of-coefficients isomorphisms of Proposition 7.10 for g^* and Rf_*, respectively for Rf'_* and g'^*.

Proof. The base change map is built from units, counits and composition isomorphisms for inverse images. By Proposition 7.10 the change-of-coefficients maps are compatible with all of them, and so the square commutes. ∎

Theorem 8.11 (Smooth base change for all admissible coefficients)**.** *In (8.1), assume that all schemes are of finite type over k and that g is smooth. For every admissible coefficient ring*

$$\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$$

and every $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$, the base change map (4.5) is an isomorphism

$$g^* Rf_* K \xrightarrow{\sim} Rf'_* g'^* K. \tag{8.2}$$

For integral and rational coefficients we have $D^b_{\mathrm{ctf}} = D^b_c$, so in those cases the theorem concerns all bounded constructible complexes. The isomorphisms (8.2) respect composition of cartesian squares and change of coefficients, because the base change map itself does (Lemmas 4.7 and Lemma 8.10).

Proof of Theorem 8.11. For finite rings the theorem is the torsion theorem once we know that K is classical, and the other rings follow by the reduction principle. By Proposition 7.1, both sides lie in $D^b_{\mathrm{ctf}}(Y', \Lambda)$, so we only have to show that the canonical map (4.5) is invertible.

Finite rings A. The torsion theorem is stated for classical complexes, so we first check that K is one. By Theorem 2.43, constructible pro-étale sheaves with discrete Noetherian coefficients are classical, and since the cohomology sheaves of K are constructible, K is a bounded classical complex. Since A is killed by a power of ℓ, which is invertible on Y, smooth base change for torsion complexes (Theorem 8.9) applies to K.

Other rings. We apply Proposition 7.12 with $F_\Lambda = g^* Rf_*$, $G_\Lambda = Rf'_* g'^*$ and θ_Λ the base change map. Both functors are exact, as the proposition requires, and the base change map commutes with change of coefficients, which is Lemma 8.10. Since θ_A is an isomorphism for every finite A by the first case, the proposition shows that θ_Λ is an isomorphism for every admissible Λ. ∎

Remark 8.12 (Smooth base change, proper base change and base change for $f^!$)**.** In (8.1), smooth base change assumes that g is smooth and allows any f of finite type, whereas proper base change assumes that f is proper and allows any g. Smooth purity (Chapter 9) gives a third base change isomorphism $g'^* f^! \simeq f'^! g^*$, for the exceptional inverse image, when f is smooth and g is arbitrary. By contrast, neither smooth nor proper base change applies to the open immersion $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ of the beginning of this chapter, where base change along the immersion of the origin fails: Theorem 8.11 does not apply because the immersion of the origin is not smooth, and proper base change does not apply because j is not proper.

8.9 The Künneth formula

The trace formula needs the Künneth formula for ordinary direct images (Proposition 8.18), which is a base change statement for Rf_*. Base change for Rf_* fails in general, as the open immersion $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ of the beginning of the chapter shows. Along a product projection $Y \times T \to Y$, however, base change holds (Lemma 8.13). To prove this we compactify, after which only an open immersion remains, and for an open immersion the statement comes down to two results: base change along extensions of the ground field, a consequence of smooth base change, and the Künneth formula with one proper factor, a consequence of proper base change. In this section A is a finite admissible ring, and we write $M \boxtimes N = p_1^* M \otimes^{\mathbf{L}} p_2^* N$ for the external tensor product.

We start with the base change statement, which is the case of the Künneth formula in which the second coefficient is constant.

Lemma 8.13 (Base change along a product projection)**.** *Let $f : X \to Y$ be a morphism of separated k-schemes of finite type, let T be a separated k-scheme of finite type, and let $M \in D_c^b(X, A)$. Let $p_X : X \times T \to X$ and $p_Y : Y \times T \to Y$ be the projections. Then the base change map*

$$p_Y^* Rf_* M \longrightarrow R(f \times \mathrm{id}_T)_* p_X^* M$$

is an isomorphism.

Proof. The idea is to compactify f, so that only an open immersion remains, and for an open immersion base change along a product projection holds even when T is singular. We argue on the étale site, where M is a bounded complex with torsion coefficients, and transport the result to the pro-étale site by Theorem 2.38.

Choose a compactification $f = \bar{f} j$, with $j : X \hookrightarrow \bar{X}$ open and $\bar{f} : \bar{X} \to Y$ proper (Theorem A.2). Since base change maps compose, it suffices to prove the lemma for j and for $\bar{f}$ separately. For the open immersion j, the base change map $p_{\bar{X}}^* Rj_* M \to R(j \times \mathrm{id}_T)_* p_X^* M$ is an isomorphism by [Stacks, 0F1H], applied to the cohomology sheaves of M and combined with the bounded-below spectral sequence. For $\bar{f}$, proper base change, applied to $Rj_* M$, shows that

$$p_Y^* R\bar{f}_* Rj_* M \longrightarrow R(\bar{f} \times \mathrm{id}_T)_* p_{\bar{X}}^* Rj_* M$$

is an isomorphism. By Lemma 4.5 the composite of the two base change maps is the base change map for f.

The step for j does not depend on Proposition 8.18 below, so there is no circularity. Indeed, the proof of [Stacks, 0F1H] has two inputs. The first is base change under extension of the ground field [Stacks, 0F1C], which by [Stacks, 0F09] follows from smooth base change (Theorem 8.9), because after passing to perfect closures the larger field is a filtered union of smooth algebras over the smaller one. The second input, [Stacks, 0F1G], is a Künneth formula with one proper factor, which comes from proper base change and the projection formula. ∎

For $f = j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ and $M = A$, the lemma says that the stalk of $R(j \times \mathrm{id}_T)_* A$ at a point $(0, t)$ has the same cohomology as the stalk of $Rj_* A$ at the origin, namely A in degree 0 and $A(-1)$ in degree 1, whatever T is. The Künneth formula allows an arbitrary constructible second factor in place of the constant one, and we first define the map it concerns.

Definition 8.14 (Künneth map)**.** Let $f_i : X_i \to Y_i$, $i = 1, 2$, be morphisms of separated k-schemes of finite type, put $h = f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2$, and let $M_i \in D^b_{\mathrm{ctf}}(X_i, A)$. The *Künneth map*

$$Rf_{1*}M_1 \boxtimes Rf_{2*}M_2 \longrightarrow Rh_*(M_1 \boxtimes M_2) \tag{8.3}$$

is the adjoint transpose, for $h^* \dashv Rh_*$, of the morphism

$$h^*(Rf_{1*}M_1 \boxtimes Rf_{2*}M_2) \simeq f_1^*Rf_{1*}M_1 \boxtimes f_2^*Rf_{2*}M_2 \xrightarrow{\varepsilon_{f_1} \boxtimes \varepsilon_{f_2}} M_1 \boxtimes M_2,$$

where ε_{f_i} is the counit of $f_i^* \dashv Rf_{i*}$ and the first isomorphism comes from $h^*p_i^* \simeq p_i^*f_i^*$.

The case we prove first is the one in which one factor is an identity. For $f : X \to Y$, T as in Lemma 8.13, $M \in D^b_{\mathrm{ctf}}(X, A)$ and $Q \in D^b_{\mathrm{ctf}}(T, A)$, with $f_1 = f$ and $f_2 = \mathrm{id}_T$, put $h = f \times \mathrm{id}_T$. The Künneth map is then

$$\kappa_{M,Q} : Rf_*M \boxtimes Q \longrightarrow Rh_*(M \boxtimes Q), \tag{8.4}$$

the adjoint transpose of $h^*(Rf_*M \boxtimes Q) = p_X^*f^*Rf_*M \otimes^{\mathbf{L}} p_T^*Q \xrightarrow{\varepsilon_f \otimes^{\mathbf{L}} 1} M \boxtimes Q$. For constant Q it is the base change map of Lemma 8.13, and every constructible Q is built from finite direct images of constant sheaves, which the next lemma treats.

Lemma 8.15 (Finite direct images of constant sheaves)**.** *In the situation of (8.4), let Q be a finite direct sum of sheaves $\pi_*A_{T'}$ with $\pi : T' \to T$ finite. Then the Künneth map $\kappa_{M,Q}$ is an isomorphism.*

Proof. Both sides are additive in Q, so we may assume $Q = \pi_*A_{T'}$. Set

$$v = \mathrm{id}_Y \times \pi, \qquad w = \mathrm{id}_X \times \pi, \qquad h' = f \times \mathrm{id}_{T'}\,.$$

Direct image along a finite morphism is exact and commutes with pullback, and by the projection formula we have

$$Rf_*M \boxtimes \pi_*A_{T'} \simeq v_*p_{Y,T'}^*Rf_*M, \qquad M \boxtimes \pi_*A_{T'} \simeq w_*p_{X,T'}^*M.$$

Since $hw = vh'$, the comparison between these two expressions is v_* of the base change map of Lemma 8.13 for h', hence it is an isomorphism. Moreover, this comparison is $\kappa_{M,Q}$, because under the adjunction $h^* \dashv Rh_*$ and the adjunctions for the finite maps, both correspond to the counit $\varepsilon_f \otimes^{\mathbf{L}} 1$ that defines $\kappa_{M,Q}$. ∎

Over a field every constructible sheaf embeds into a sheaf of this form, and this allows an induction on the cohomological degree.

Lemma 8.16 (Field coefficients)**.** *In the situation of (8.4), suppose that $A = L$ is a finite field of characteristic ℓ. Then the Künneth map $\kappa_{M,Q}$ is an isomorphism.*

Proof. We first treat constructible sheaves M and Q, by the usual argument for comparing a δ-functor with an exact functor, and then pass to complexes by truncation. We may compute with the small étale models of M and Q, since Theorems 2.43 and Theorem 2.38 identify the operations and the adjunction maps defining $\kappa_{M,Q}$ on the two sites.

Sheaves. Let M and Q be constructible sheaves of L-modules. By Stacks, [09Z7](https://stacks.math.columbia.edu applied on T, there is a short exact sequence of constructible L-sheaves

$$0 \longrightarrow Q \longrightarrow P \longrightarrow Q' \longrightarrow 0, \qquad P = \bigoplus_a \pi_{a*} L_{T_a}^{\oplus n_a}, \quad \pi_a \text{ finite}.$$

Indeed, every constructible sheaf on a qcqs scheme embeds into a finite sum of finite direct images of constant sheaves with finite coefficients, and over L these coefficients are finite sums of L. For $r \geq 0$ put

$$F^r(Q) = p_Y^* R^r f_* M \otimes_L p_T^* Q, \qquad G^r(Q) = R^r h_*(p_X^* M \otimes_L p_T^* Q).$$

Since tensoring over L is exact, each F^r is exact in Q, while the G^r form a long exact sequence. On cohomology sheaves the Künneth map induces natural maps $a_Q^r : F^r(Q) \to G^r(Q)$, and a_P^r is an isomorphism for every r by Lemma 8.15. It follows that each a_Q^r is injective. Indeed, in its naturality square for $Q \hookrightarrow P$ the map $F^r(Q) \to F^r(P)$ is injective and a_P^r is an isomorphism. Surjectivity is proved by induction on r, simultaneously for all constructible Q. For $r = 0$, the beginning of the long exact sequence of the G^r identifies $G^0(Q)$ with the kernel of $G^0(P) \to G^0(Q')$, and since F^0 is exact, $F^0(Q)$ is the kernel of $F^0(P) \to F^0(Q')$. As a_P^0 is an isomorphism and $a_{Q'}^0$ is injective, a_Q^0 maps the second kernel onto the first. For $r > 0$, the induction hypothesis for Q' and the surjection $F^{r-1}(P) \to F^{r-1}(Q')$ imply that $G^{r-1}(P) \to G^{r-1}(Q')$ is surjective. By the long exact sequence,

$$G^r(Q) = \ker\bigl(G^r(P) \to G^r(Q')\bigr).$$

Since $F^r(Q) = \ker(F^r(P) \to F^r(Q'))$ by exactness of F^r, the isomorphism a_P^r and the injectivity of $a_{Q'}^r$ identify this kernel with $F^r(Q)$. This completes the induction, so $\kappa_{M,Q}$ is an isomorphism for sheaves.

Complexes. The source and target of $\kappa_{M,Q}$ are exact functors of M and of Q, and $\kappa_{M,Q}$ is natural in both. Using the finitely many truncation triangles of M and of Q and two-out-of-three, we can therefore extend the case of sheaves to $M \in D^b_{\mathrm{ctf}}(X, L)$ and $Q \in D^b_{\mathrm{ctf}}(T, L)$. This is possible because over the field L the truncations stay in D^b_{ctf}, since constructible cohomology sheaves have Tor amplitude zero, and all operations are bounded on the schemes under consideration. ∎

Over a general finite ring the truncations of a complex in D^b_{ctf} need not have finite Tor amplitude, so the argument by truncation does not apply directly. Instead we reduce to the residue fields and use derived Nakayama.

Lemma 8.17 (Künneth formula with one identity factor)**.** *In the situation of (8.4), the Künneth map $\kappa_{M,Q}$ is an isomorphism.*

Proof. We show that the cone N of $\kappa_{M,Q}$ vanishes modulo every maximal ideal of A, and conclude by derived Nakayama on stalks. The source and target of $\kappa_{M,Q}$ are in $D^b_{\mathrm{ctf}}(Y \times T, A)$ by Proposition 7.1, so every geometric stalk of N is a perfect A-complex. Now let $\mathfrak{m} \subset A$ be a maximal ideal and put $k = A/\mathfrak{m}$. The change-of-coefficients maps of Proposition 7.10 for pullback, tensor product and ordinary direct image respect the counit defining the Künneth map.

Hence they identify the reduction $\kappa_{M,Q} \otimes^{\mathbf{L}}_A k$ with $\kappa_{M\otimes^{\mathbf{L}}_A k,\, Q\otimes^{\mathbf{L}}_A k}$, which is an isomorphism by Lemma 8.16. This means that $N \otimes^{\mathbf{L}}_A k = 0$ for every maximal ideal. Since the Artinian ring A is a product of local rings, derived Nakayama for the perfect complex $N_{\bar z}$ at a geometric point $\bar z$ shows that $N_{\bar z} = 0$. Finally, since N lies in $D^b_{\mathrm{ctf}}(Y \times T, A)$, it comes from a complex on the small étale site by Theorem 2.43, where vanishing is detected on geometric stalks, so $N = 0$. ∎

Proposition 8.18 (Ordinary external Künneth)**.** *Let $f_i : X_i \to Y_i$, $i = 1, 2$, be morphisms of separated k-schemes of finite type, and let $M_i \in D^b_{\mathrm{ctf}}(X_i, A)$. Then the Künneth map (8.3)*

$$Rf_{1*}M_1 \boxtimes Rf_{2*}M_2 \longrightarrow R(f_1 \times f_2)_*(M_1 \boxtimes M_2)$$

is an isomorphism.

Proof. We apply Lemma 8.17 once in each factor of the factorization

$$X_1 \times X_2 \xrightarrow{\mathrm{id}_{X_1} \times f_2} X_1 \times Y_2 \xrightarrow{f_1 \times \mathrm{id}_{Y_2}} Y_1 \times Y_2$$

of $f_1 \times f_2$. We apply it to f_1 with second coefficient $Rf_{2*}M_2$, which lies in $D^b_{\mathrm{ctf}}(Y_2, A)$ by Proposition 7.1. After interchanging the factors, we apply it to f_2 with coefficient M_1. Then the composite of $R(f_1 \times \mathrm{id}_{Y_2})_*$ of the second isomorphism with the first is an isomorphism from $Rf_{1*}M_1 \boxtimes Rf_{2*}M_2$ to $R(f_1 \times f_2)_*(M_1 \boxtimes M_2)$. Since its adjoint transpose is $\varepsilon_{f_1} \boxtimes \varepsilon_{f_2}$, it is the Künneth map (8.3). ∎

Part II Duality and the trace formula

At the end of Part I we have, for every separated X_0 of finite type over $\mathbf{F}_q$, cohomology groups H^i_c with an action of Frobenius, and they are finite because $Rf_!$ preserves D^b_{ctf}. Nothing yet connects them with the numbers $\#X_0(\mathbf{F}_{q^m})$ of which the zeta function is made. This part supplies the connection: Weil's axioms (b) and (c), the trace formula and Poincaré duality, and with them rationality and the functional equation (Section 12.7).

For the elliptic curve $y^2 = x^3 - x$ over $\mathbf{F}_3$ these two axioms, together with the single count $N_1 = 4$, determine the zeta function and every N_m (Chapter 12, Example 12.32).

Both axioms are statements about the diagonal. For example, on a compact oriented manifold M, the Künneth components of the class of the diagonal in $H^*(M \times M)$ form the copairing inverse to the cup product pairing, and the Lefschetz number of a map φ is the intersection number of its graph with the diagonal. A scheme, however, comes with no orientation and no class of the diagonal, and smooth purity is what provides them. It says that $f^! \simeq f^*(d)[2d]$ for f smooth of relative dimension d (Theorem 9.3), and this isomorphism gives the trace map.

Verdier duality (Theorems 10.8 and Theorem 10.10) says that $\mathbb{D} = R\mathcal{H}om(-, \omega)$ is an involution of D^b_{ctf} exchanging $Rf_!$ with Rf_*. For smooth proper X it turns the trace map into a perfect pairing between H^i and $H^{2d-i}(d)$ (Corollary 10.17), and this pairing is compatible with Frobenius (Theorem 10.21). Verdier duality also gives the vanishing in (a) for such X (Theorem 10.15).

For the trace formula we use the Künneth formula for Verdier duals (Lemma 11.7), which turns a correspondence $u : c_2^*K \to c_1^!K$ into a class on C; for an endomorphism, C is the graph. Restricting this class to the diagonal gives the Lefschetz–Verdier formula (Theorem 11.13), which

says that on a proper scheme the trace of u on cohomology is a sum of local terms over the fixed locus.

This does not yet count points, for two reasons. A local term is computed from $R\mathcal{H}om(c_2^*K, c_1^!K)$ near a fixed point, and the stalk of K does not determine it (Remark 12.4). Moreover X need not be proper. Both problems are settled by one computation on a curve: the local term of Frobenius at a rational point is the trace on the stalk (Proposition 12.16). The computation works because Frobenius has zero differential, so that its graph meets the diagonal with multiplicity one and is tangent to a coordinate axis. The result is the Grothendieck–Lefschetz trace formula (Theorem 12.24): for every Weil complex K, $\mathrm{tr}(\mathrm{Frob}^m \mid R\Gamma_c(X, K))$ is the sum over $X_0(\mathbf{F}_{q^m})$ of the traces of Frobenius on the stalks.

Smooth purity and Verdier duality hold over any finite or algebraically closed field, and their compatibility with Frobenius follows from their construction (Section 10.5).

9 Smooth purity

Poincaré duality for a smooth variety X of dimension d pairs $H_c^i(X, \Lambda)$ with $H^{2d-i}(X, \Lambda(d))$ through the cup product and a trace map $H_c^{2d}(X, \Lambda(d)) \to \Lambda$. If a is the structure map of X, such a trace map is the counit of $Ra_! \dashv a^!$, once we know that $a^!\Lambda \simeq \Lambda_X(d)[2d]$. But Part I produces $f^!$ (Theorem 5.2) as a right adjoint by Brown representability, with no formula.

A formula exists on charts. Consider $f : X \to Y$ smooth of pure relative dimension d, and an open subscheme $U \subset X$ on which f factors as an étale map $e : U \to \mathbf{A}_Y^d$ followed by the projection p_d. With finite coefficients $f^!K|_U \simeq e^*p_d^*K(d)[2d]$ (Proposition 6.9), and under this isomorphism the counit is integration along d affine lines. This integration comes from the trace map of $\mathbf{P}^1$ (Proposition 6.7), which sends the class of a point to 1. The question is then whether these chart descriptions agree on overlaps.

In topology they need not. For a submersion $f : X \to Y$ of manifolds with fibres of real dimension m, Verdier's theory gives $f^!K \simeq f^*K \otimes \mathrm{or}_{X/Y}[m]$. Here the relative orientation sheaf $\mathrm{or}_{X/Y}$ is locally constant of rank one, and it is trivial exactly when f can be oriented. A linear change of coordinates g acts on $H_c^m(\mathbf{R}^m)$ by the sign of $\det g$, and since $\mathrm{GL}_m(\mathbf{R})$ has two components, the local identifications need not glue. For complex manifolds, by contrast, $\mathrm{GL}_d(\mathbf{C})$ is connected and the orientation is canonical.

The same argument works for us on a chart. Indeed, a linear change of coordinates multiplies the trace map of $\mathbf{A}^d$ by a locally constant unit on GL_d, and this unit is 1 at the identity, hence everywhere, since GL_d has connected geometric fibres. So every vector bundle has a canonical trace map (Proposition 9.6). The trace map has source $H_c^{2d}(-, \Lambda(d))$, so the Tate twist $\Lambda(d)$ plays the role of the orientation sheaf. Over an algebraically closed field a compatible system of roots of unity trivializes $\Lambda(d)$, but not canonically, and over $\mathbf{F}_q$ the twist is visible: there geometric Frobenius acts on $\Lambda(1)$ by q^{-1} (Section 10.5), and this is where the factor q^d in Weil's Poincaré duality comes from.

Connectedness of GL_d does not settle the gluing, because two étale charts $U \to \mathbf{A}_Y^d$ differ by a map that need not be linear. We therefore construct the isomorphism without charts, from the diagonal, in the way a tubular neighbourhood linearizes a submanifold (Section 9.4).

The outcome, Theorem 9.3, is a canonical isomorphism

$$\gamma_f : f^*K(d)[2d] \xrightarrow{\sim} f^!K,$$

natural in $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ for every admissible Λ. It is compatible with base change, composition and change of coefficients, and it is normalized by the trace map of the affine line. This statement is Deligne's [SGA4, XVIII], where it is called Poincaré duality.

9.1 Statement

Smooth purity is stated for six kinds of coefficients at once, because the proof moves between them: the geometry is done with finite rings, the case $\mathcal{O}_E$ is a limit of finite levels, and the fields are obtained by inverting ℓ. Throughout this chapter Λ is an admissible coefficient ring, that is, a finite commutative ring killed by a power of ℓ, $\mathcal{O}_E$, E or $\overline{\mathbf{Q}}_\ell$ (the rings $\mathbf{Z}_\ell$ and $\mathbf{Q}_\ell$ being the cases $E = \mathbf{Q}_\ell$), and Tate twists are those of Definition 2.53.

The theorem asserts one isomorphism on all of X, whereas on a chart an isomorphism of this shape is already given by Proposition 6.9 with finite coefficients. A bare isomorphism for each f separately, however, would not be enough for what follows, and the theorem therefore also lists three properties that the later chapters use. Property (2), compatibility with base change, restricts γ_f to the fibres of f, and it is what makes γ_f commute with Frobenius, because the Frobenius square is a base change (Proposition 10.20). Property (3), compatibility with composition, gives the transitivity of the trace map (Proposition 9.26), the analogue of integrating over the fibres of a composite in two steps. Property (4), compatibility with change of coefficients, lets us construct γ_f with finite coefficients and pass to $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$, and it also makes Poincaré duality compatible with change of coefficients (Theorem 10.12). Properties (1) and (2) involve maps that we define now, for every admissible Λ; the functors are those of Theorem 5.2, restricted to D^b_{ctf}.

The first map compares $f^!$ before and after a change of base. Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms of separated schemes of finite type over k, let $X' = X \times_Y Y'$, and let $f' : X' \to Y'$ and $g' : X' \to X$ be the projections. Write $\varepsilon_f(K) : Rf_! f^! K \to K$ for the counit of $Rf_! \dashv f^!$.

Definition 9.1 (Exceptional base change map)**.** For $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$, the *exceptional base change map* $g'^* f^! K \to f'^! g^* K$ is the adjoint transpose, for $Rf'_! \dashv f'^!$, of the composite

$$Rf'_! g'^* f^! K \xrightarrow{\sim} g^* Rf_! f^! K \xrightarrow{g^* \varepsilon_f(K)} g^* K,$$

whose first arrow is the inverse of the base change isomorphism of Proposition proposition-composition-base-change-rf-shriek.

The second map is integration along the affine line. Let $p : \mathbf{A}^1_Y \to Y$ be the projection, and factor it as $p = q \circ j$, where $j : \mathbf{A}^1_Y \hookrightarrow \mathbf{P}^1_Y$ is the complement of the section at infinity and $q : \mathbf{P}^1_Y \to Y$ is the projection.

Definition 9.2 (Trace map of the affine line)**.** The *trace map of the affine line* is the composite

$$\mathrm{Tr}_p : Rp_! \Lambda_{\mathbf{A}^1_Y}(1)[2] \simeq Rq_* j_! \Lambda_{\mathbf{A}^1_Y}(1)[2] \longrightarrow Rq_* \Lambda_{\mathbf{P}^1_Y}(1)[2] \xrightarrow{\mathrm{Tr}_q} \Lambda_Y, \tag{9.1}$$

where the first arrow is induced by the counit $j_! j^* \to \mathrm{id}$ and Tr_q is the trace map of the projective line: for finite Λ it is the map (6.5) of Lemma 6.4 for the projection q, and otherwise it is the trace of Proposition 6.22.

Over an algebraically closed field the trace map of the affine line sends the class in $H^2_c(\mathbf{A}^1, \Lambda(1))$ of a point to 1, as we check with finite coefficients in Section 9.3.

Theorem 9.3 (Smooth purity for all admissible coefficients)**.** *Let $f : X \to Y$ be smooth of pure relative dimension d. For every admissible coefficient ring $\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, with A finite, there are isomorphisms, natural in $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$,*

$$\gamma_f(K) : f^*K(d)[2d] \xrightarrow{\sim} f^!K, \tag{9.2}$$

with the following properties.

1. *If f is the projection $p : \mathbf{A}^1_Y \to Y$, then $\varepsilon_p(\Lambda_Y) \circ Rp_!(\gamma_p(\Lambda_Y))$ is the trace map (9.1).*
2. *For $g : Y' \to Y$ as in Definition 9.1, with exceptional base change map $\beta : g'^* f^! K \to f'^! g^* K$, we have $\beta \circ g'^* \gamma_f(K) = \gamma_{f'}(g^*K)$, the sources being identified by $g'^* f^* \simeq f'^* g^*$.*
3. *For a smooth morphism $h : Y \to Z$ of pure relative dimension e and $K \in D^b_{\mathrm{ctf}}(Z, \Lambda)$, the map $\gamma_{h\circ f}(K)$ is the composite of $f^*(\gamma_h(K))(d)[2d]$ and $\gamma_f(h^!K)$, the sources being identified by $(h \circ f)^* \simeq f^*h^*$ and the targets by the composition isomorphism $f^!h^! \simeq (h \circ f)^!$ of Proposition proposition-composition-open-restriction-compactifications-f.*
4. *For a coefficient map $\Lambda \to \Lambda'$ of Proposition 7.10, the change-of-coefficients isomorphism $(f^!K)_{\Lambda'} \to f^!(K_{\Lambda'})$ of that proposition carries $(\gamma_{f,\Lambda}(K))_{\Lambda'}$ to $\gamma_{f,\Lambda'}(K_{\Lambda'})$, the sources being identified by the change-of-coefficients isomorphism for f^*.*

We recall what D^b_{ctf} means for each coefficient ring. For integral coefficients and for coefficients in a field, $D^b_{\mathrm{ctf}} = D^b_c$, and we write D^b_c in calculations that only concern these cases. With integral coefficients finite Tor amplitude is automatic, and membership in $D^b_c(T, \mathcal{O}_E)$ can be tested modulo ℓ (Proposition 3.11).

The proof passes through finite rings A with $\ell^r A = 0$, such as $\mathcal{O}_E/\ell^n$. For these, D^b_{ctf} is the category of Definition 3.1, which on the pro-étale site is identified with the pullback of its étale counterpart (Corollary 2.44). Here the Tor condition makes derived change of coefficients preserve boundedness, as the example after Proposition 7.10 shows. When we compare coefficient rings, we write the ring as a subscript of the functor, as in $Rf_{!,A}$ or $f^!_{\mathcal{O}_E}$; otherwise we omit the subscript.

With coefficients A, we have $Rf_! = R\bar{f}_* \circ j_!$ for a compactification $f = \bar{f} \circ j$, and on D^b_{ctf} these functors and their right adjoints are those of Theorem 5.31, restricted in Theorem 6.20. Proper base change, the projection formula and the localization triangle are the identities of Section 6.1, which we may prove on the étale sites by Proposition 5.15. For an étale map, extension by zero is exact on sheaves of A-modules on the pro-étale sites; on the derived categories it is left adjoint to inverse image, and the counit is given on geometric stalks by the sum over the fibre.

Property (2) implies that the exceptional base change map is an isomorphism for smooth f, because γ_f and $\gamma_{f'}$ are isomorphisms.

For étale f we have $d = 0$, and the theorem says that $f^! \simeq f^*$ for every admissible Λ. For a smooth curve C with structure map $a : C \to \operatorname{Spec} k$, it gives $a^!\mathbf{Z}_{\ell,\operatorname{Spec} k} \simeq \mathbf{Z}_{\ell,C}(1)[2]$, a single sheaf in degree -2.

For smooth f the whole functor $f^!$ is determined by its value on the constant sheaf, through the map $f^*K \otimes^{\mathbf{L}} f^!\Lambda_Y \to f^!K$ of Proposition proposition-exceptional-inverse-image-smooth-morphism. Smoothness is needed here, as the following example shows. Let $i : y \to C$ be the inclusion of a rational point of a smooth curve, with open complement $j : U \to C$, and take $K = j_!\Lambda_U$. Then the analogous map $i^*K \otimes^{\mathbf{L}} i^!\Lambda_C \to i^!K$ has source zero, since the stalk of $j_!\Lambda_U$ at y vanishes, but the localization triangle gives $i^!j_!\Lambda_U \simeq (i^*Rj_*\Lambda_U)[-1] \neq 0$.

We give $f^!\Lambda_Y$ a name rather than working with $\Lambda_X(d)[2d]$ directly, because the two are not yet known to be isomorphic. Moreover, the complex $f^!\Lambda_Y$ is attached to f without choices, and it satisfies base change (Proposition proposition-exceptional-inverse-image-smooth-morphism). Proving the theorem thus amounts to constructing an isomorphism between it and $\Lambda_X(d)[2d]$ that

does not depend on charts. In this sense the complex is the analogue of the shifted orientation sheaf $\mathrm{or}_{X/Y}[m]$, and for the structure map of a smooth scheme it is the dualizing complex of Poincaré duality (Corollary 9.28 and Theorem 10.12).

Definition 9.4 (Relative dualizing complex)**.** For smooth $f : X \to Y$ and any admissible Λ, the *relative dualizing complex* is

$$\omega_f = f^!\Lambda_Y \in D^b_{\mathrm{ctf}}(X, \Lambda). \tag{9.3}$$

For finite coefficients, ω_f can be computed on the pro-étale site or, equivalently, on the étale site (Proposition 5.15).

9.2 The tensor product map

With finite coefficients, smooth purity says that for f smooth of pure relative dimension d the exceptional inverse image $f^!$ is the ordinary inverse image, twisted by (d) and shifted by $[2d]$ (Proposition 9.12). The first step is the map $f^*K \otimes^{\mathbf{L}} \omega_f \to f^!K$, an isomorphism by Proposition proposition-exceptional-inverse-image-smooth-morphism, which reduces the problem to the single complex ω_f.

Fix $n \geq 1$. We carry out the geometric calculation with coefficients $\mathbf{Z}/\ell^n\mathbf{Z}$, and until Section 9.5 we omit the subscript, so that f^*, $Rf_!$, $f^!$, ω_f and γ_f refer to this cyclic coefficient ring. Afterwards the calculation extends along $\mathbf{Z}/\ell^n\mathbf{Z} \to A$ when $\ell^n A = 0$, and in this way Proposition 9.12 gives the result for every finite admissible ring A.

Let $f : X \to Y$ be smooth of pure relative dimension d and $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$. From the projection formula and the counit $Rf_!\omega_f \to (\mathbf{Z}/\ell^n\mathbf{Z})_Y$ we get $Rf_!(f^*K \otimes^{\mathbf{L}} \omega_f) \simeq K \otimes^{\mathbf{L}} Rf_!\omega_f \to K$, whose adjoint is the canonical map

$$\mu_{f,K} : f^*K \otimes^{\mathbf{L}}_{(\mathbf{Z}/\ell^n\mathbf{Z})_X} \omega_f \longrightarrow f^!K. \tag{9.4}$$

To construct the orientation, we compare ω on the fibres of a family of sections, and for this we need base change. Consider a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y, \end{array} \tag{9.5}$$

with f smooth of pure relative dimension d. For it we have the exceptional base change map of Definition 9.1. By Proposition proposition-exceptional-inverse-image-smooth-morphism and (3), with $\Lambda = \mathbf{Z}/\ell^n\mathbf{Z}$, the map (9.4) and the exceptional base change map

$$g'^* f^!K \xrightarrow{\sim} (f')^! g^*K \tag{9.6}$$

are isomorphisms for $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$, and they are compatible with each other. Moreover, on a chart $f|_U = p_d \circ e$ we have $\omega_f|_U \simeq (\mathbf{Z}/\ell^n\mathbf{Z})_U(d)[2d]$. Hence $\omega_f[-2d]$ is a locally constant sheaf of free $\mathbf{Z}/\ell^n\mathbf{Z}$-modules of rank one.

9.3 The trace map of affine space

Every identification of $f^!$ in this chapter is ultimately an adjunction whose counit is integration along affine lines, so we first make that integration explicit. The affine line is not proper, so we cannot integrate on it directly; instead we integrate on $\mathbf{P}^1$. There, on a geometric fibre $H^2(\mathbf{P}^1, \mathbf{Z}/\ell^n(1)) \simeq \mathbf{Z}/\ell^n$ is generated by $c_1(\mathcal{O}(1))$, the class of a point, and the trace map of $\mathbf{P}^1$ reads off the coefficient of this class. The trace map of $\mathbf{A}^1$ is then its composite with the map $H^2_c(\mathbf{A}^1, \mathbf{Z}/\ell^n(1)) \to H^2(\mathbf{P}^1, \mathbf{Z}/\ell^n(1))$ given by extension by zero. This construction works over an arbitrary base S and commutes with base change, and iterating it gives the trace map (9.11) of $\mathbf{A}^d$, which is the counit of the adjunction on a chart.

For a scheme T, Chern classes are defined by the Kummer sequence on $T_{\acute{e}t}$,

$$1 \longrightarrow \mu_{\ell^n} \longrightarrow \mathbf{G}_m \xrightarrow{(\cdot)^{\ell^n}} \mathbf{G}_m \longrightarrow 1, \tag{9.7}$$

whose boundary $H^1(T_{\acute{e}t}, \mathbf{G}_m) \to H^2(T_{\acute{e}t}, (\mathbf{Z}/\ell^n\mathbf{Z})_T(1))$ sends a line bundle L to $c_1(L)$. For a scheme S, let $q : \mathbf{P}^1_S \to S$ be the projection. By Lemma 6.4, the unit and $c_1(\mathcal{O}_{\mathbf{P}^1_S}(1))$ give an isomorphism, compatible with base change,

$$Rq_*(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S} \simeq (\mathbf{Z}/\ell^n\mathbf{Z})_S \oplus (\mathbf{Z}/\ell^n\mathbf{Z})_S(-1)[-2], \tag{9.8}$$

and the *trace map* $\mathrm{Tr}_q : Rq_*(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S}(1)[2] \to (\mathbf{Z}/\ell^n\mathbf{Z})_S$ is the projection onto the second summand, after twisting. It sends $c_1(\mathcal{O}_{\mathbf{P}^1_{\bar{s}}}(1))$ on each geometric fibre to 1.

Let p_1, p_2 be the projections of $\mathbf{P}^1_S \times_S \mathbf{P}^1_S$, and let x_0, x_1 and y_0, y_1 be the pullbacks along p_1 and p_2 of the homogeneous coordinates of $\mathbf{P}^1_S$. The diagonal Δ is the zero locus of the section $x_0y_1 - x_1y_0$ of $p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$, so that $\mathcal{O}(\Delta)$ is this line bundle and $x_0y_1 - x_1y_0$ is its canonical section. Lemma 6.3 provides the cycle class of Δ,

$$\delta : \Delta_*(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S} \longrightarrow (\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S \times_S \mathbf{P}^1_S}(1)[2],$$

the Kummer boundary of $\mathcal{O}(\Delta)$ with its canonical section, compatible with every base change on S. Let Tr_{p_i} be the pullback of Tr_q. Then the diagonal integrates to 1 along either factor, that is,

$$\mathrm{Tr}_{p_i} \circ Rp_{i*}\delta = \mathrm{id}_{(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S}} \qquad (i = 1, 2). \tag{9.9}$$

This is (6.6) of Proposition 6.7. To see it directly, note that both sides are endomorphisms of a constant sheaf, so it suffices to check (9.9) on geometric fibres of p_i, where the diagonal is a single rational point with line bundle $\mathcal{O}(1)$, and δ restricts to $c_1(\mathcal{O}(1))$, which Tr_q sends to 1.

For $\Lambda = \mathbf{Z}/\ell^n\mathbf{Z}$, Proposition 6.7 gives the adjunction $Rq_* \dashv q^*(-)(1)[2]$, and its unit and counit, (6.7) and (6.8) there, take the following form. The counit is integration along the fibres,

$$\varepsilon_K : Rq_*(q^*K(1)[2]) \simeq K \otimes^{\mathbf{L}} Rq_*(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{P}^1_S}(1)[2] \xrightarrow{\mathrm{id}_K \otimes \mathrm{Tr}_q} K, \tag{9.10}$$

and the unit is the class of the diagonal,

$$\eta_M : \; M \simeq Rp_{2*}\Delta_*M \xrightarrow{Rp_{2*}(\delta \otimes p_1^*M)} Rp_{2*}p_1^*M(1)[2] \simeq q^*Rq_*M(1)[2],$$

using the projection formulas and proper base change. Both commute with base change on S, and the triangle identities reduce to (9.9).

Let $j : \mathbf{A}^1_S \hookrightarrow \mathbf{P}^1_S$ be the complement of the section at infinity, and let $p = q \circ j$. Composing this adjunction with $j_! \dashv j^*$ we obtain an adjunction $Rp_! \dashv p^*(-)(1)[2]$ between $D^b_{\mathrm{ctf}}(\mathbf{A}^1_{S,\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$ and $D^b_{\mathrm{ctf}}(S_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$, whose counit is the counit for the open immersion followed by ε_K. At the constant sheaf this counit is the trace map of the affine line $\mathrm{Tr}_p : Rp_!(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^1_S}(1)[2] \to (\mathbf{Z}/\ell^n\mathbf{Z})_S$ of (9.1), and it is an isomorphism. Indeed, we apply Rq_* to the localization triangle for the section at infinity. In the resulting triangle, restriction to infinity is the identity on the first summand of (9.8), and it kills the second summand, because the pullback of $\mathcal{O}(1)$ along the section at infinity is the trivial line bundle on S, and so the restriction of $c_1(\mathcal{O}(1))$ is 0. Hence $Rp_!(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^1_S} \simeq (\mathbf{Z}/\ell^n\mathbf{Z})_S(-1)[-2]$ is the second summand. Over a geometric point this is $H^2_c(\mathbf{A}^1, (\mathbf{Z}/\ell^n\mathbf{Z})(1)) \simeq \mathbf{Z}/\ell^n\mathbf{Z}$, generated by the class κ that maps to the class of a point in $H^2(\mathbf{P}^1, (\mathbf{Z}/\ell^n\mathbf{Z})(1))$.

For affine space, let $p_j : \mathbf{A}^j_S \to S$ be the projection and let $r_j : \mathbf{A}^j_S \to \mathbf{A}^{j-1}_S$ be the map forgetting the last coordinate. We define Tr_{p_j} inductively as the composite

$$\begin{aligned} Rp_{j!}(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^j_S}(j)[2j] &\simeq Rp_{j-1!}(Rr_{j!}(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^j_S}(1)[2])(j-1)[2j-2] \\ &\xrightarrow{\mathrm{Tr}_{p_{j-1}} \circ Rp_{j-1!}(\mathrm{Tr}_{r_j})} (\mathbf{Z}/\ell^n\mathbf{Z})_S, \end{aligned}$$

using $Rp_{j!} = Rp_{j-1!}Rr_{j!}$ and the projection formula. For $j = d$ this is the *trace map of affine space*, an isomorphism (the identity when $d = 0$)

$$\mathrm{Tr}_{p_d} : Rp_{d!}(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^d_S}(d)[2d] \xrightarrow{\sim} (\mathbf{Z}/\ell^n\mathbf{Z})_S. \tag{9.11}$$

By Lemma lemma-affine-space-etale-localization, $Rp_{d!} \dashv p_d^*(-)(d)[2d]$, and the counit is the composite of the d counits for the affine lines, which at the constant sheaf is (9.11). Like those of the affine line, the unit and counit commute with base change. On a geometric fibre, the external product $\kappa \boxtimes \cdots \boxtimes \kappa \in H^{2d}_c(\mathbf{A}^d_{\bar{s}}, (\mathbf{Z}/\ell^n\mathbf{Z})(d))$ integrates to 1, since by the projection formula, integrating along the last coordinate replaces the last factor by its integral, which is 1.

Objects of a derived category do not glue in general. For example, suppose $X = U \cup V$ and we know objects on U and on V isomorphic to $f^!K|_U$ and $f^!K|_V$. Then the two isomorphisms restricted to $U \cap V$ may differ by an automorphism, and nothing tells us which identification to glue along. On a chart $U \to \mathbf{A}^d_Y$ we have an explicit candidate for $f^!K|_U$, namely $e^*p_d^*K(d)[2d]$, and we want to compare such candidates with $f^!K$ and later, for towers, to glue $f^!$ from them. For this we keep the counit together with the object. The point is that a pair consisting of an object and a counit that satisfies a universal property is unique up to a unique isomorphism compatible with the counit. Hence local pairs can be compared on overlaps without any choice of transition maps. We use this for the comparison following Definition 9.5 and in the construction of the right adjoint for towers (Proposition 9.18).

Definition 9.5 (Representing objects)**.** Let f be smooth, and let us work in $D(Y, \mathbf{Z}/\ell^n\mathbf{Z})$ and $D(X, \mathbf{Z}/\ell^n\mathbf{Z})$ on either the étale or the pro-étale sites. Let K be an object on Y. We say that a pair (H, ε), where H is an object on X and $\varepsilon : Rf_!H \to K$ is a morphism, *represents* $\mathrm{Hom}(Rf_!(-), K)$ if $b \mapsto \varepsilon \circ Rf_!(b)$ is a bijection $\mathrm{Hom}(M, H) \to \mathrm{Hom}(Rf_!M, K)$ for all $M \in D^b_{\mathrm{ctf}}(X, \mathbf{Z}/\ell^n\mathbf{Z})$.

The geometric computations with finite coefficients in Section 6.1 are stated on the pro-étale site and proved by calculations on the étale site. By Corollary 2.44, Theorem 2.38 with (2.10), and Proposition 5.15, the pro-étale and étale categories $D^b_{\mathrm{ctf}}(-, A)$, and the localizing subcategories they generate, are identified compatibly with pullback, direct image and extension by zero. In particular $Rf_! = R\bar{f}_* \circ j_!$ is computed on the étale sites. Consequently, an adjunction between $Rf_!$ and some functor G on the étale categories $D^b_{\mathrm{ctf}}(-_{\acute{e}t}, A)$ transfers to the pro-étale categories

$D^b_{\mathrm{ctf}}(-_{\mathrm{pro\acute{e}t}}, A)$. Indeed, the Hom groups agree by full faithfulness, and the two functors $Rf_!$ correspond to each other.

By Proposition 5.15, the functor $f^!$ on $\mathrm{Loc}(D^b_{\mathrm{ctf}}(-, \mathbf{Z}/\ell^n\mathbf{Z}))$ of Theorem 5.31 is the étale functor $f^!$ on $D(-_{\acute{e}t}, \mathbf{Z}/\ell^n\mathbf{Z})$ of Section 6.1, and it preserves D^b_{ctf} by Theorem 6.20. Hence, for $f : X \to Y$ smooth of pure relative dimension d, Proposition proposition-exceptional-inverse-image-smooth-morphism gives the local form of $Rf_! \dashv f^!$: on an open $U \subset X$ with $f|_U = p_d \circ e$, where $e : U \to \mathbf{A}^d_Y$ is étale,

$$f^!K|_U \simeq e^* p_d^* K(d)[2d],$$

with counit the sum over the fibres of e followed by (9.11) tensored with K.

Since a right adjoint is unique up to a unique isomorphism preserving the counit, these counits on charts agree on overlaps with the counit of $Rf_! \dashv f^!$. With finite coefficients, then, $f^!$ exists first and the charts describe it. For towers, by contrast, there is no global functor yet, and we glue $f^!$ from models on charts by a Mayer–Vietoris argument, which uses the triangle for an open cover $X = U \cup V$ and $N \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$:

$$j_{U\cap V!}(N|_{U\cap V}) \xrightarrow{(1,-1)} j_{U!}(N|_U) \oplus j_{V!}(N|_V) \xrightarrow{(1,1)} N \longrightarrow . \tag{9.12}$$

Suppose that $H \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$ and $\varepsilon : Rf_!H \to K$ restrict to pairs that represent $\mathrm{Hom}(Rf_!(-), K)$ (Definition 9.5) on U, V and $U \cap V$, where the restriction of ε to U is the composite $R(f|_U)_!(H|_U) = Rf_!j_{U!}(H|_U) \to Rf_!H \to K$. For $M \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$ we apply $\mathrm{Hom}(-, H)$ and $\mathrm{Hom}(Rf_!(-), K)$ to (9.12) with $N = M$. The maps $b \mapsto \varepsilon \circ Rf_!(b)$ form a morphism between the two long exact sequences, which is bijective on the terms coming from U, V and $U \cap V$. By the five lemma it is bijective on $\mathrm{Hom}(M, H)$, which means that (H, ε) represents $\mathrm{Hom}(Rf_!(-), K)$. Since such a pair is unique up to a unique isomorphism compatible with ε, we get $H \simeq f^!K$ compatibly with the counits. In Section 9.5 we use this gluing to realize the compatible counits with finite coefficients as a single map of towers.

On a vector bundle, the choice of a local trivialization transports (9.11) to a trace map. These maps do not depend on the trivialization, so they glue to a global map. This is the model case of the orientation below, because deformation to the normal bundle will reduce the dualizing complex of a smooth morphism to that of a vector bundle.

Proposition 9.6 (Purity for vector bundles)**.** *Let $\pi : V \to S$ be a vector bundle of rank d. For $K \in D^b_{\mathrm{ctf}}(S_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/\ell^n\mathbf{Z})$, the trace map induces an isomorphism*

$$\pi^* K(d)[2d] \xrightarrow{\sim} \pi^! K.$$

Proof. We transport the trace map of affine space along local trivializations, show that the result does not depend on the trivialization, so that the local maps glue to a trace isomorphism Tr_π for V, and take the map $\pi^*K(d)[2d] \to \pi^!K$ to be its adjoint.

Local trace maps. Locally on S the bundle is affine space, where the trace map has already been constructed. Over an open $U \subset S$ on which V is trivial, a trivialization $\tau : V \times_S U \xrightarrow{\sim} \mathbf{A}^d_U$ transports (9.11) to an isomorphism

$$\mathrm{Tr}_\tau : R(\pi|_{V\times_S U})_!(\mathbf{Z}/\ell^n\mathbf{Z})_{V\times_S U}(d)[2d] \xrightarrow{\sim} (\mathbf{Z}/\ell^n\mathbf{Z})_U.$$

These maps commute with base change in U, because the construction uses only $c_1(\mathcal{O}_{\mathbf{P}^1_U}(1))$, the splitting (9.8) and localization at infinity: pullback preserves the Kummer class, and proper base change and localization identify the pullbacks of the constructions with those over the new base.

Independence of the trivialization. A change of trivialization multiplies the trace map by a locally constant unit on GL_d, and this unit is 1. As two trivializations differ by a matrix with values in GL_d, it suffices to study one universal change of coordinates, parametrized by $\mathrm{GL}_{d,S}$ itself, and then pull back. Let

$$\theta : \mathbf{A}^d_{\mathrm{GL}_{d,S}} \longrightarrow \mathbf{A}^d_{\mathrm{GL}_{d,S}}, \qquad (g, v) \longmapsto (g, gv),$$

be the universal linear change of coordinates, an automorphism over $\mathrm{GL}_{d,S}$. It induces an automorphism of the compactly supported direct image of $(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathbf{A}^d_{\mathrm{GL}_{d,S}}}(d)[2d]$ to $\mathrm{GL}_{d,S}$, which the trace map identifies with the constant sheaf $(\mathbf{Z}/\ell^n\mathbf{Z})_{\mathrm{GL}_{d,S}}$ in degree zero, so that this automorphism is multiplication by a section

$$u \in H^0\big((\mathrm{GL}_{d,S})_{\acute{e}t}, (\mathbf{Z}/\ell^n\mathbf{Z})^\times_{\mathrm{GL}_{d,S}}\big),$$

a locally constant function with values in $(\mathbf{Z}/\ell^n\mathbf{Z})^\times$. It is therefore constant on each geometric fibre $\mathrm{GL}_{d,\bar{s}}$, which is connected: for $d \geq 1$ it is a nonempty open subset of the affine space of $d \times d$ matrices, and for $d = 0$ it is a point. As θ is the identity at the identity matrix, which lies on every geometric fibre, $u = 1$.

The local trace maps agree on overlaps of trivializing opens. Indeed, let τ and τ' be two trivializations over U. Then $\tau' \circ \tau^{-1}$ is the pullback of θ along the transition matrix $U \to \mathrm{GL}_{d,S}$, so by base change the equality $u = 1$ implies $\mathrm{Tr}_{\tau'} = \mathrm{Tr}_\tau$. Moreover, morphisms between source and target form a sheaf, as both are locally sheaves in degree zero. Hence the local maps glue over a trivializing cover to an isomorphism

$$\mathrm{Tr}_\pi : R\pi_!(\mathbf{Z}/\ell^n\mathbf{Z})_V(d)[2d] \xrightarrow{\sim} (\mathbf{Z}/\ell^n\mathbf{Z})_S.$$

The map $\pi^*K(d)[2d] \to \pi^!K$. For general K, the projection formula and Tr_π give

$$R\pi_!(\pi^*K(d)[2d]) \simeq K \otimes^{\mathbf{L}} R\pi_!(\mathbf{Z}/\ell^n\mathbf{Z})_V(d)[2d] \xrightarrow{\mathrm{id}_K \otimes \mathrm{Tr}_\pi} K.$$

Its adjoint under $R\pi_! \dashv \pi^!$ (Proposition proposition-exceptional-inverse-image-smooth-morphism) is the asserted map. We show that it is an isomorphism. This may be checked locally on V, because $\pi^!$ commutes with restriction to opens. Over each trivializing open, Tr_π is Tr_τ by construction, so there the map is the isomorphism given by the adjunction for affine space following (9.11). ∎

The trace maps of vector bundles commute with base change on S, because the local trace maps do.

9.4 Finite coefficients

Since $f^!K \simeq f^*K \otimes^{\mathbf{L}} \omega_f$, smooth purity with finite coefficients comes down to a single global isomorphism $\omega_f \simeq (\mathbf{Z}/\ell^n\mathbf{Z})_X(d)[2d]$. We obtain it from the diagonal $\Delta : X \to X \times_Y X$, which is a section of the smooth projection p_1. Deformation to the normal bundle moves Δ to the zero section of the relative tangent bundle $T_{X/Y}$, where Proposition 9.6 gives the isomorphism.

We use the diagonal because it is attached to f without choices, and because ω_f is recovered from it: $\omega_f \simeq \Delta^*\omega_{p_1}$ by base change. Near the diagonal, however, $X \times_Y X$ looks like a neighbourhood of the zero section in $T_{X/Y}$ only to first order, and the deformation makes this precise without choosing a tubular neighbourhood. The comparison it yields, Lemma 9.9, is determined by the requirement that it be the identity at $t = 1$.

The deformation argument works for any section of a smooth morphism. Let $g : V \to S$ be smooth of pure relative dimension d and let $s : S \to V$ be a section of g that is a closed immersion. Then s is a regular immersion of codimension d, and we let N_s be its normal bundle, with projection $\pi : N_s \to S$ and zero section $z : S \to N_s$.

Definition 9.7 (Deformation to the normal bundle)**.** The *deformation to the normal bundle* of s is the open subscheme $\mathcal{D}_{s(S)}V$ of the blowup of $V \times \mathbf{A}^1$ along $s(S) \times \{0\}$ obtained by removing the strict transform of $V \times \{0\}$.

It comes with a smooth morphism $\widetilde{g} : \mathcal{D}_{s(S)}V \to \mathbf{A}^1_S$ and a section $\widetilde{s} : \mathbf{A}^1_S \to \mathcal{D}_{s(S)}V$ that is a closed immersion. Over $t = 1$ the pair $(\widetilde{g}, \widetilde{s})$ restricts to (g, s), and over $t = 0$ to (π, z). To see this in local terms, let t be the coordinate of $\mathbf{A}^1$, and let $x_1, \dots, x_d$ be functions on an open neighbourhood U of $s(S)$ in V which are étale coordinates over S and generate the ideal of $s(S)$. On the part of $\mathcal{D}_{s(S)}V$ lying over $U \times \mathbf{A}^1$, the quotients $u_i = x_i/t$ are regular functions, so that $x_i = tu_i$ there. At $t = 0$, then, the function u_i restricts to the linear function on N_s given by the class of x_i in the conormal sheaf. Since these classes form a basis of the conormal sheaf, the u_i restrict to linear coordinates on N_s.

Example 9.8 (The origin in the affine line)**.** Let $S = \operatorname{Spec} k$, $V = \mathbf{A}^1_k$ with coordinate x, and let s be the origin. The blowup of $V \times \mathbf{A}^1 = \mathbf{A}^2$, with coordinates (x, t), at the origin is covered by two charts. On the chart with coordinates (u, t) and $x = tu$ the exceptional divisor is $t = 0$; on the chart with coordinates (x, v) and $t = xv$ it is $x = 0$, and the strict transform of the line $V \times \{0\} = \{t = 0\}$ is $v = 0$. Since a point of the second chart with $v \neq 0$ lies on the first chart, with $u = 1/v$, removing the strict transform leaves exactly the first chart:

$$\mathcal{D}_{s(S)}V = \operatorname{Spec} k[u, t] \longrightarrow V \times \mathbf{A}^1, \qquad (u, t) \longmapsto (tu, t),$$

with $\widetilde{g}$ the projection to t and $\widetilde{s}$ the line $u = 0$. For $t = c \neq 0$ the fibre is the line V itself, with coordinate $u = x/c$ and the origin as section. For $t = 0$ the fibre is the u-line, mapped to the origin of V; it is the tangent line $N_s = T_0\mathbf{A}^1$, and u is the linear coordinate given by the class of x in $\mathfrak{m}/\mathfrak{m}^2$. So the family rescales the line by $1/t$ and in the limit keeps only the first-order information at the section. Here the family is the product $\mathbf{A}^1_u \times \mathbf{A}^1_t$, so $\widetilde{s}^*\omega_{\widetilde{g}}$ is visibly constant along t. In general this holds only étale locally, after a choice of coordinates, and the point of Lemma 9.9 is that the comparison normalized at $t = 1$ does not depend on that choice.

Lemma 9.9 (A section and its normal bundle)**.** *There is a canonical isomorphism*

$$s^*\omega_g \xrightarrow{\sim} z^*\omega_\pi. \tag{9.13}$$

It is obtained by restricting to $t = 0$ the unique isomorphism $\widetilde{s}^*\omega_{\widetilde{g}} \to a^*s^*\omega_g$ that is the identity at $t = 1$, and inverting. Here $a : \mathbf{A}^1_S \to S$ is the projection.

Proof. The fibres of $\widetilde{s}^*\omega_{\widetilde{g}}$ at $t = 1$ and $t = 0$ are the two sides of (9.13), so it suffices to show that this shifted sheaf of rank one is pulled back from S, étale locally on S. In positive characteristic a locally constant sheaf on $\mathbf{A}^1_S$ need not be pulled back from S, so we use coordinates. Étale

locally on S, near the section the deformation is étale over $\mathbf{A}^d_{\mathbf{A}^1_S} = \mathbf{A}^d_S \times \mathbf{A}^1$, a family pulled back from S, and hence so is $\widetilde{s}^*\omega_{\widetilde{g}}[-2d]$. We then normalize the local comparisons at $t = 1$; since the geometric fibres of $\mathbf{A}^1_S \to S$ are connected, this makes them agree on overlaps. Finally, they glue because all objects involved are shifted locally constant sheaves.

Step 1: the objects. The restrictions of $\widetilde{s}^*\omega_{\widetilde{g}}$ to $t = 1$ and $t = 0$ are $s^*\omega_g$ and $z^*\omega_\pi$, by Proposition proposition-exceptional-inverse-image-smooth-morphism applied to the inclusions of these fibres. Proposition proposition-exceptional-inverse-image-smooth-morphism shows that both $\widetilde{s}^*\omega_{\widetilde{g}}[-2d]$ and $s^*\omega_g[-2d]$ are locally constant sheaves, free of rank one over $\mathbf{Z}/\ell^n\mathbf{Z}$. Hence morphisms between two such complexes are morphisms of sheaves, and they glue.

Step 2: local construction. Étale locally on S we reduce to the zero section of affine space, whose deformation is pulled back from S. After passing to an étale cover $\{S_\alpha \to S\}_\alpha$ we may choose, for the base change of the section $s_\alpha : S_\alpha \to V \times_S S_\alpha$, a neighbourhood U_α of $s_\alpha(S_\alpha)$ and an étale map $e_\alpha : U_\alpha \to \mathbf{A}^d_{S_\alpha}$ sending s_α to the zero section. We may assume that the inverse image of the zero section under e_α is exactly $s_\alpha(S_\alpha)$. Indeed, this inverse image is separated and étale over S_α, so $s_\alpha(S_\alpha)$ is open and closed in it, and we remove the complement, which is closed in U_α. Then e_α is an étale map of pairs whose square of centers is cartesian. Since blowing up and removing the strict transform commute with étale base change, e_α induces an étale map from a neighbourhood of the deformed section in $\mathcal{D}_{s(S)}V$ to the deformation space of the zero section of $\mathbf{A}^d_{S_\alpha}$. The latter is $\mathbf{A}^d_{\mathbf{A}^1_{S_\alpha}}$, with coordinates $u_1, \dots, u_d$ and t, and its map to $\mathbf{A}^d_{S_\alpha} \times \mathbf{A}^1$ sends (u, t) to (tu, t), that is, $x_i = tu_i$, and the deformed zero section is $u = 0$.

The adjunction for affine space identifies the relative dualizing complex of the projection $\mathbf{A}^d_{\mathbf{A}^1_{S_\alpha}} \to \mathbf{A}^1_{S_\alpha}$ with the constant sheaf $(\mathbf{Z}/\ell^n\mathbf{Z})(d)[2d]$. Pulling back along the étale map just constructed, and composing adjunctions for it, gives the same identification for $\omega_{\widetilde{g}}$ near the deformed section. Similarly, the chart e_α gives the analogous identification of $s^*\omega_g$ over S_α. Composing the two identifications we obtain an isomorphism over $\mathbf{A}^1_{S_\alpha}$

$$\widetilde{s}^*\omega_{\widetilde{g}} \xrightarrow{\sim} a^*s^*\omega_g.$$

At $t = 1$ it restricts to an automorphism of $s^*\omega_g$ over S_α. Composing with the pullback along a of the inverse of this automorphism, we may assume that the local isomorphism is the identity at $t = 1$.

Step 3: gluing. The local isomorphisms from different charts agree on overlaps, because of the normalization at $t = 1$. Over $S_\alpha \times_S S_\beta$ the two local isomorphisms that are the identity at $t = 1$ differ by an automorphism of the locally constant sheaf $a^*s^*\omega_g[-2d]$ of rank one, that is, by a section of a locally constant sheaf of units of $\mathbf{Z}/\ell^n\mathbf{Z}$. The locus where this section equals 1 is open and closed, and it contains the point $t = 1$ of each geometric fibre $\mathbf{A}^1_{\bar{b}}$. As these fibres are connected, the section is 1 at every geometric point. So the local isomorphisms glue to a unique isomorphism over $\mathbf{A}^1_S$ that is the identity at $t = 1$. Restricting it to $t = 0$ gives an isomorphism $z^*\omega_\pi \simeq s^*\omega_g$, whose inverse is (9.13). ∎

The orientation of ω_f will be the comparison (9.13) for the diagonal. Two things have to be checked about it: on a coordinate chart it must be the identification given by coordinates, and it must commute with base change, as property (2) of Theorem 9.3 requires. Since a chart is an étale map that preserves the section, both checks need to transport (9.13) along base change on S and along étale maps that preserve the section. In both cases we use the uniqueness in Lemma 9.9: any construction that is the identity at $t = 1$ produces the comparison of that lemma.

Lemma 9.10 (Naturality under base change and étale maps)**.** *Write $\alpha_{g,s} : s^*\omega_g \xrightarrow{\sim} z^*\omega_\pi$ for the comparison of Lemma 9.9.*

1. *Let $h : S' \to S$ be any morphism for which the schemes below satisfy the standing hypotheses. Put $V' = V \times_S S'$, and denote the base changes of the maps by g', s', π', z'. There are canonical identifications*

$$N_{s'} \simeq N_s \times_S S', \qquad \mathcal{D}_{s'(S')}V' \simeq \mathcal{D}_{s(S)}V \times_S S'.$$

Under these identifications and the exceptional base change isomorphisms of Proposition proposition-exceptional-inverse-image-smooth-morphism, the following square commutes:

$$\begin{array}{ccc} h^*s^*\omega_g & \xrightarrow{h^*\alpha_{g,s}} & h^*z^*\omega_\pi \\ \sim\downarrow & & \downarrow\sim \\ (s')^*\omega_{g'} & \xrightarrow{\alpha_{g',s'}} & (z')^*\omega_{\pi'}. \end{array}$$

2. *Let $e : V' \to V$ be an étale S-morphism and let $s' : S \to V'$ be a closed section such that $es' = s$ and $e^{-1}(s(S)) = s'(S)$ scheme-theoretically. Put $g' = ge$. The induced map $N_{s'} \to N_s$ is a linear isomorphism, and the induced map of deformation spaces is étale. The identifications*

$$(s')^*\omega_{g'} \simeq s^*\omega_g, \qquad (z')^*\omega_{\pi'} \simeq z^*\omega_\pi$$

obtained by composing adjunctions carry $\alpha_{g',s'}$ to $\alpha_{g,s}$.

These comparisons respect identities, composition and interchange of base change with an étale map of pairs.

In part (1) the morphism h is arbitrary. Blowups in general commute only with flat base change, but by its description in local coordinates, the deformation space of a section of a smooth morphism commutes with arbitrary base change.

Proof of Lemma 9.10. The geometric identifications follow from the coordinate description of the deformation space. Every comparison of dualizing complexes is then the identity at $t = 1$, and so the uniqueness in Lemma 9.9 identifies it with the comparison of that lemma.

The geometry. Since g is smooth and s is a section, $N_s \simeq s^*T_{V/S}$. Since formation of relative differentials for a smooth map commutes with arbitrary base change, and an étale map induces an isomorphism on relative tangent bundles, this proves both assertions about normal bundles.

For deformation spaces, we use the cartesian étale coordinate chart $U \to \mathbf{A}^d_S$ of the proof of Lemma 9.9. There the deformation is

$$\mathcal{D}_{s(S)}U \simeq U \times_{\mathbf{A}^d_S} \mathbf{A}^d_{\mathbf{A}^1_S}, \qquad x_i = tu_i,$$

because this is the chart of the blowup along $(t, x_1, \ldots, x_d)$ on which the pullback of the ideal is generated by t, and this description commutes with every change of base. Away from the centre, the deformation is $(V \setminus s(S)) \times \mathbf{G}_m$, which also commutes with base change. After passing to an étale cover of S if necessary, these opens cover the deformation space, and their identifications glue to the canonical one. For an étale map of pairs as in (2), we obtain the étale map of deformation spaces by flat base change for the blowup and for the removal of the strict transform, and the coordinate description identifies its special fibre with the map of normal bundles.

The comparison of dualizing complexes. In both parts the strategy is the same: transport the comparison over $\mathbf{A}^1_S$ along the geometric identification, check that the result is still the identity at $t = 1$, and invoke uniqueness. Write

$$\chi_{g,s} : \widetilde{s}^* \omega_{\widetilde{g}} \xrightarrow{\sim} a^* s^* \omega_g$$

for the isomorphism of Lemma 9.9 that is the identity at $t = 1$, so that $\alpha_{g,s}$ is the inverse of its restriction to $t = 0$. For (1), we apply Proposition proposition-exceptional-inverse-image-smooth-morphism to g, π and $\widetilde{g}$. These exceptional base change isomorphisms commute with restriction to $t = 1$ and $t = 0$, because they are adjoint transposes of base change for compactly supported direct image applied to the same counit, and this base change respects composition of cartesian squares. Transported through these isomorphisms, the pullback of $\chi_{g,s}$ is therefore a comparison for (g', s') that is the identity at $t = 1$, and by uniqueness it equals $\chi_{g',s'}$. Restricting to $t = 0$ and inverting proves (1).

For (2), composing $Re_! \dashv e^*$ with $Rg_! \dashv g^!$ gives $e^* \omega_g \simeq \omega_{g'}$, and likewise for the induced étale map of deformation spaces. On the fibres $t = 1$ and $t = 0$ these identifications are the ones for the original pairs and for the normal bundles, since the counit of a composite adjunction is the composite of the counits. They transport $\chi_{g,s}$ to a comparison that is the identity at $t = 1$, which is $\chi_{g',s'}$ by uniqueness.

For the coherence assertions we use the same uniqueness once more. The geometric identifications respect composition, and so do the identifications obtained from adjunctions, as we see by transposing composite counits. Therefore each composite in the coherence assertions is a comparison that is the identity at $t = 1$, and by uniqueness any two of them agree. ∎

We apply Lemma 9.9 to the diagonal. Let $p_1, p_2 : X \times_Y X \to X$ be the projections. Since f is separated, Δ is a closed immersion, and it is a section of the smooth morphism p_1. Its conormal sheaf is $\Omega^1_{X/Y}$, so its normal bundle is the relative tangent bundle $T_{X/Y}$, and we write $\pi : T_{X/Y} \to X$ for its projection and $z : X \to T_{X/Y}$ for its zero section.

Proposition 9.11 (The relative dualizing complex)**.** *Let $f : X \to Y$ be smooth of pure relative dimension d. There is a canonical isomorphism*

$$\omega_f \xrightarrow{\sim} (\mathbf{Z}/\ell^n \mathbf{Z})_X(d)[2d]. \tag{9.14}$$

Proof. We define the isomorphism as a chain of three steps: restrict ω_f to the diagonal of $X \times_Y X$, move the diagonal to the zero section of $T_{X/Y}$, and use the orientation of a vector bundle.

The projection p_1 is the base change of f along f, so Proposition proposition-exceptional-inverse-image-smooth-morphism shows that $\omega_{p_1} \simeq p_2^* \omega_f$. Since $p_2 \circ \Delta$ is the identity of X, we get the first isomorphism in

$$\omega_f \simeq L\Delta^* \omega_{p_1} \xrightarrow{\sim} z^* \omega_\pi \xrightarrow{\sim} z^* (\mathbf{Z}/\ell^n \mathbf{Z})_{T_{X/Y}}(d)[2d] \simeq (\mathbf{Z}/\ell^n \mathbf{Z})_X(d)[2d].$$

The second isomorphism is Lemma 9.9 for the section Δ of p_1, and the third is the inverse of the isomorphism of Proposition 9.6 at the constant sheaf, applied to the vector bundle $T_{X/Y}$ over X. The last one holds because the pullback of a constant sheaf is constant. Since each step is attached to f without choices, the composite is canonical. ∎

The orientation (9.14) is defined without coordinates, but on a coordinate chart it agrees with the identification given by the trace map of affine space (9.11). This is what makes the map $f^* K(d)[2d] \to f^! K$ below the expected one, and the trace map in Section 9.6 relies on it as well.

We prove the agreement for a vector bundle, where both orientations can be computed, and reduce a coordinate chart to the vector bundle $\mathbf{A}^d_Y$ by naturality.

Vector bundles. Let $\pi : V \to S$ be a vector bundle of rank d. Here the orientation obtained from the diagonal is the orientation of Proposition 9.6, because the construction from the diagonal moves the diagonal to the zero section and then deforms the zero section to itself, and neither step changes integration along the fibres. For the first step, regard $V \times_S V$ as a vector bundle over the first factor. The automorphism $(v, w) \mapsto (v, w - v)$ of $V \times_S V$ over the first factor takes the diagonal to the zero section, and it preserves integration in the second factor. To see this, let the translation vector vary over V. Its effect on the top compactly supported direct image, identified with the constant sheaf by integration, is then multiplication by a locally constant unit on V, and this unit is 1 at the zero vector, where the translation is the identity. Since each geometric fibre of $V \to S$ is connected, it is 1 everywhere.

Next, the deformation to the normal bundle of the zero section of V is $V \times \mathbf{A}^1$, with the map $V \times \mathbf{A}^1 \to V \times \mathbf{A}^1$, $(w, t) \mapsto (tw, t)$, to the original family. This map is defined by scalar multiplication, without a trivialization, and in a local trivialization it is $x_i = tu_i$, as in the coordinate description above. Integration on this product family is the pullback of the original integration. Therefore the comparison of Lemma 9.9, which is the identity at $t = 1$, is the identity under these integrations, in particular at $t = 0$. Combined with the translation, this identifies the orientation obtained from the diagonal for π with the orientation of the vector bundle.

Coordinate charts. A chart $e : U \to \mathbf{A}^d_Y$ relates the diagonal of U to the diagonal of $\mathbf{A}^d_Y$, where the case of vector bundles applies, so we have to transport the orientation along e. By Lemma 9.10 it suffices to see that every ingredient of the orientation of a chart obtained from the diagonal is pulled back from affine space. So let $U \subset X$ be open with an étale map $e : U \to \mathbf{A}^d_Y$ over Y. Then the étale map $e \times e : U \times_Y U \to \mathbf{A}^d_Y \times_Y \mathbf{A}^d_Y$ carries the diagonal of U into the diagonal of $\mathbf{A}^d_Y$, whose inverse image is $U \times_{\mathbf{A}^d_Y} U$. In it the diagonal of U is open and closed, because e is separated and étale, and removing the other components, as in the proof of Lemma 9.9, we obtain a cartesian square of centers. The induced map of normal bundles is then the differential $T_{U/Y} \simeq e^* T_{\mathbf{A}^d_Y/Y}$, a pullback of vector bundles. Likewise, the dualizing complex is pulled back from the affine model, because by composing adjunctions the exceptional inverse image along an étale map is its ordinary inverse image. For the comparison obtained from the deformation, we factor $e \times e$ as

$$U \times_Y U \xrightarrow{\mathrm{id}_U \times e} U \times_Y \mathbf{A}^d_Y \xrightarrow{e \times \mathrm{id}} \mathbf{A}^d_Y \times_Y \mathbf{A}^d_Y.$$

The second arrow is the base change of the first projection along e, and it carries the diagonal section to the graph of e over U, so Lemma lemma-naturality-under-base-change-etale applies to it. Part (2), over the common base U, applies to the first arrow after we remove the other components of the inverse image of that graph. In this factorization the map on special fibres is the displayed differential, and the comparison at $t = 1$ is the one obtained from the adjunction in coordinates. Moreover, integration on vector bundles is preserved by the linear isomorphism of normal bundles. By the case of vector bundles, we conclude that the orientation on U obtained from the diagonal is the orientation of affine space given by its adjunction in coordinates.

Combining the orientation with the isomorphism $f^! K \simeq f^* K \otimes^{\mathbf{L}} \omega_f$, we obtain the isomorphism γ_f of Theorem 9.3 with finite coefficients (Proposition 9.12). We state the result for an arbitrary finite admissible ring.

Proposition 9.12 (Smooth purity with finite coefficients)**.** *Let A be any finite admissible coefficient ring, and let $f : X \to Y$ be smooth of pure relative dimension d. For $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, A)$ there is a canonical isomorphism, natural in K,*

$$f_A^* K(d)[2d] \xrightarrow{\gamma_{f,A}(K)} f_A^! K. \tag{9.15}$$

Proof. The calculation so far was made with coefficients $\mathbf{Z}/\ell^n\mathbf{Z}$, and every ingredient in it extends along $\mathbf{Z}/\ell^n\mathbf{Z} \to A$, where we choose n with $\ell^n A = 0$. Once the Kummer and diagonal classes are extended, the calculation works with A throughout. In particular, the splitting for the projective line has two free summands, and its integration and triangle identities keep the same normalization, since (9.9) is an identity of endomorphisms of a constant sheaf and is preserved by extension of scalars. So the local adjunctions compute $f^!_A$, and the arguments for vector bundles and deformations then show that $\omega_{f,A} \simeq A_X(d)[2d]$. We define $\gamma_{f,A}(K)$ as the composite

$$f_A^* K \otimes^{\mathbf{L}} A_X(d)[2d] \xrightarrow{\sim} f_A^* K \otimes^{\mathbf{L}} \omega_{f,A} \xrightarrow{\sim} f_A^! K.$$

These are the isomorphisms given by the orientation and by $\mu_{f,K}$, and both are natural in K. By (9.17) below, $\gamma_{f,A}$ does not depend on the choice of n and is compatible with maps of finite rings. ∎

9.5 ℓ-adic coefficients

Let $f : X \to Y$ be smooth of pure relative dimension d. For $\mathcal{O}_E$-coefficients we want an isomorphism of functors $f^*(-)(d)[2d] \simeq f^!$ on $D^b_c(-, \mathcal{O}_E)$. Here $f^!_{\mathcal{O}_E}$ is defined by Brown representability, not as a limit of the finite $f^!$. The reduction principle of Section 7.3 does not apply, because it checks that a given morphism is an isomorphism, and we have no morphism $f^*(-)(d)[2d] \to f^!$ with $\mathcal{O}_E$-coefficients yet. Instead we deduce Theorem 9.3 for $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ from Proposition 9.12 by building the right adjoint ourselves. The key input is that the finite isomorphisms $\gamma_{f,A}$ commute with change of finite coefficients (9.17), which lets us construct an explicit right adjoint of $Rf_{!,\mathcal{O}_E}$ from towers (Proposition 9.19). Inverting ℓ then gives the cases E and $\overline{\mathbf{Q}}_\ell$ (Proposition 9.20 and Theorem 9.21). Each right adjoint we build is identified with $f^!$ by the uniqueness of adjoints, normalized by the counits.

Proposition 9.13 (Uniqueness of the right adjoint of $Rf_!$)**.** *Let Λ be an admissible coefficient ring and let J be a right adjoint of $Rf_! : D^b_{\mathrm{ctf}}(X, \Lambda) \to D^b_{\mathrm{ctf}}(Y, \Lambda)$ with counit ε^J. There is a unique natural isomorphism $\rho : f^! \xrightarrow{\sim} J$ with $\varepsilon^J(K) \circ Rf_!(\rho(K)) = \varepsilon_f(K)$.*

Proof. Let $\rho(K)$ be the J-transpose of $\varepsilon_f(K)$ and $\rho'(K)$ the $f^!$-transpose of $\varepsilon^J(K)$. Since $\rho'\rho$ and the identity have the same transpose, they are equal, and by symmetry $\rho\rho' = \mathrm{id}$. For uniqueness, note that a map compatible with the counits has J-transpose $\varepsilon_f(K)$, so it is $\rho(K)$. ∎

9.5.1 Change of finite coefficients

The levels $\mathcal{O}_E/\ell^n$ of a tower are related by extension of scalars. The first thing to know is therefore how $Rf_!$ and $f^!$ behave under a map of finite rings $A \to B$. The comparison for $f^!$ is the transpose of a counit, and the counits commute with $- \otimes^{\mathbf{L}} B$, because on a chart they are built from the trace of the projective line and sums over fibres.

Let $A \to B$ be a map of finite commutative rings killed by a power of ℓ.

Definition 9.14 (Extension of scalars)**.** For a scheme T and $K \in D^b_{\mathrm{ctf}}(T_{\mathrm{pro\acute{e}t}}, A)$ put $K_B = K \otimes^{\mathbf{L}}_{A_T} B_T \in D^b_{\mathrm{ctf}}(T_{\mathrm{pro\acute{e}t}}, B)$, and for a morphism u put $u_B = u \otimes^{\mathbf{L}} \mathrm{id}_{B_T}$. For a further map $B \to C$ we have $(K_B)_C \simeq K_C$ by associativity of the tensor product.

With this notation, the comparison for $f^!$ is characterized by its counit: it is the unique map that makes the counit over B agree with the extended counit over A.

Lemma 9.15 (Extension of scalars commutes with $Rf_!$ and $f^!$)**.** *For $M \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, A)$ and $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, A)$ there are natural isomorphisms $b(M) : (Rf_{!,A}M)_B \xrightarrow{\sim} Rf_{!,B}(M_B)$ and $c_{A,B}(K) : (f^!_A K)_B \xrightarrow{\sim} f^!_B(K_B)$, where $c_{A,B}(K)$ is determined by*

$$\varepsilon_B(K_B) \circ Rf_{!,B}(c_{A,B}(K)) = (\varepsilon_A(K))_B \circ b(f^!_A K)^{-1}. \tag{9.16}$$

For $A \to B \to C$, the comparisons for $A \to C$ are the composites of those for $A \to B$ and $B \to C$.

Proof. The comparison b for $Rf_!$ is a direct computation, and $c_{A,B}$ is then determined by the adjunction. To see that $c_{A,B}$ is invertible we use the explicit form of $f^!$ on a chart, where it is an ordinary inverse image.

The map b is the change of coefficients for $Rf_!$ (Proposition 7.10). To see that it is an isomorphism, note that extension by zero commutes with $- \otimes^{\mathbf{L}}_A B$, and for $R\bar{f}_*$ we resolve B by finite free A-modules and use the flat models of Lemma 3.13. The comparison holds on each free term, which is a finite direct sum of copies of A, and since $R\bar{f}_*$ has finite cohomological dimension, only finitely many terms of the resolution contribute in each degree, so the comparison holds for every B. Next, we define $c_{A,B}(K)$ as the transpose of the right-hand side of (9.16). Whether it is an isomorphism is a local question, so we may restrict to a chart $f|_U = p_d \circ e$ of Proposition proposition-exceptional-inverse-image-smooth-morphism. There the counit is built from the trace of the projective line and sums over the fibres of e, which commute with extension of coefficients (Proposition 6.7). On the chart, therefore, $c_{A,B}(K)$ is the transpose of the comparison for pullback $(e^* p_d^* K(d)[2d])_B \simeq e^* p_d^*(K_B)(d)[2d]$, which is an isomorphism. Hence $c_{A,B}(K)$ is an isomorphism on every chart. Transitivity holds because both sides have the same transpose: for $A \to B \to C$, the counit identity (9.16) for $A \to C$ is obtained by applying the one for $B \to C$ after the one for $A \to B$. ∎

It follows that the isomorphisms $\gamma_{f,A}$ are compatible with change of coefficients: for $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, A)$,

$$c_{A,B}(K) \circ (\gamma_{f,A}(K))_B = \gamma_{f,B}(K_B), \tag{9.17}$$

the sources being identified by $(f_A^* K(d)[2d])_B \simeq f_B^*(K_B)(d)[2d]$. Indeed, $\gamma_{f,A}$ is the map $\mu_{f,K}$ composed with the inverse of the orientation, and each factor commutes with change of coefficients.

The map $\mu_{f,K}$ is built from the projection formula and the counit. The projection formula commutes with extension of scalars, which is itself a tensor product, and (9.16) relates the counits over A and B. Hence $\mu_{f,K}$ commutes with change of coefficients.

The orientation is a map between invertible sheaves placed in degree $-2d$, so its compatibility is local. For vector bundles it follows from the construction via the projective line (Proposition 9.6), since the Kummer class and the splitting (9.8) are compatible with extension of scalars. On the deformation family we use the uniqueness in Lemma 9.9 over the base of the family. It applies once we know that the extended comparison is a comparison of the kind considered there, and this holds because c commutes with smooth exceptional base change and with composition with étale maps. Indeed, both sides are transposes of counits, and by the proof of Lemma 9.15 they agree on charts. Since the original comparison is the identity at $t = 1$, so is the extended one, and by uniqueness the extended comparison is the comparison of Lemma 9.9 over B.

9.5.2 Integral coefficients

Fix a finite extension $E/\mathbf{Q}_\ell$. An object M of $D^b_c(T, \mathcal{O}_E)$ is derived complete, $M \simeq R\varprojlim_n M/\ell^n$, and the finite theory applies to each M/ℓ^n. The units, counits and transition maps at the finite levels, however, live in the different categories $D(T_{\text{pro\'et}}, \mathcal{O}_E/\ell^n)$, and a family of them that commutes with the transition maps only up to unspecified isomorphisms does not define a morphism between the derived limits. We therefore work in the derived category $D(\mathrm{Tow}(T))$ of towers $(C_n)_n$, where C_n is a sheaf of $\mathcal{O}_{E,T}/\ell^n$-modules on $T_{\text{pro\'et}}$ and the transition maps are linear over $\mathcal{O}_{E,T}/\ell^{n+1} \to \mathcal{O}_{E,T}/\ell^n$. These are the modules over $(\mathcal{O}_{E,T}/\ell^n)_n$ in the topos of inverse systems [Stacks, 0940]. In this category a compatible family of maps is one morphism. Evaluation $C_\bullet \mapsto C_n$ is exact, and the evaluations jointly detect isomorphisms, but they do not detect equality of morphisms. This is why the maps below are always compared in $D(\mathrm{Tow}(T))$. A tower is *constructible* if each C_n lies in $D^b_{\mathrm{ctf}}(T_{\text{pro\'et}}, \mathcal{O}_E/\ell^n)$ with bounds independent of n, and *normalized* if its reduction maps $C_n \otimes^{\mathbf{L}} \mathcal{O}_{E,T}/\ell^m \to C_m$ are isomorphisms for all $m \le n$.

We need two facts about integral complexes. Part (1) of the next proposition gives the uniform bounds in the definition of a constructible tower, and part (2) shows that the derived limit of such a tower is again an object of D^b_c.

Proposition 9.16 (Integral constructible complexes as towers)**.**

1. *For $K \in D^b_c(T, \mathcal{O}_E)$ there are integers $a \le b$ such that every reduction K/ℓ^n lies in $D^b_{\mathrm{ctf}}(T_{\text{pro\'et}}, \mathcal{O}_E/\ell^n)$, with cohomology and Tor amplitude in $[a, b]$.*
2. *If $K \in D(T_{\text{pro\'et}}, \mathcal{O}_E)$ is derived complete and K/ℓ lies in $D^b_{\mathrm{ctf}}(T_{\text{pro\'et}}, \mathcal{O}_E/\ell)$, then $K \in D^b_c(T, \mathcal{O}_E)$.*

Proof. For (1), by Lemma 5.19, K is a finite extension of shifted sheaves $a_!C$ with C of Tor amplitude $[-1, 0]$ on a stratum. For a single such piece the reductions $(a_!C)/\ell^n = a_!(C/\ell^n)$ are constructible with Tor amplitude in $[-1, 0]$ for every n, since reduction commutes with extension by zero. Moreover, Tor amplitude and constructibility of the reductions are preserved by extensions and shifts, and since the number of steps and the shifts do not depend on n, the bounds do not either. For (2), we reduce further from ℓ to the uniformizer. The reduction $K/\pi_E = (K/\ell) \otimes^{\mathbf{L}}_{\mathcal{O}_E/\ell} \mathcal{O}_E/\pi_E$ is bounded constructible, because K/ℓ has finite Tor amplitude over $\mathcal{O}_E/\ell$. Since K is derived complete, $K \in D^b_c(T, \mathcal{O}_E)$ by the criterion (2)⇒(1) of Proposition 3.11. ∎

Nothing is lost by passing from $\mathcal{O}_E$-complexes to towers: a constructible $\mathcal{O}_E$-complex is recovered from its tower of reductions as the derived limit, and the towers that arise in this way are exactly the normalized constructible ones.

Lemma 9.17 (Reduction to towers is left adjoint to the derived limit)**.** *The functor $\mathrm{red}_T M = (M/\ell^n)_n$ is left adjoint to $R\varprojlim : D(\mathrm{Tow}(T)) \to D(T_{\text{pro\'et}}, \mathcal{O}_E)$, with unit the derived completion map $M \to R\varprojlim_n M/\ell^n$. This adjunction restricts to an equivalence between $D^b_c(T, \mathcal{O}_E)$ and the normalized constructible towers.*

Proof. The adjunction is a composite of two standard adjunctions of ringed topoi. For the equivalence we check that the unit is an isomorphism on D^b_c and that the counit is an isomorphism on normalized constructible towers. The second is the substantial point, since there we must show that the limit of such a tower is again constructible.

The adjunction is $Lh^*p^{-1} \dashv Rp_*h_*$, for the change of rings h from towers to constant inverse systems of $\mathcal{O}_{E,T}$-modules and the morphism p to $T_{\text{pro\'et}}$ with $p_* = \varprojlim$ [Stacks, 06YV, 07A6]. By

Proposition proposition-sp-integral-towers, red_T maps $D^b_c(T, \mathcal{O}_E)$ into normalized constructible towers. On D^b_c the unit is an isomorphism, since objects of D^b_c are derived complete.

Conversely, let $C_\bullet$ be normalized and put $C = R\varprojlim C_n$. The key point is that $C/\ell^m \simeq C_m$. Indeed, the complex C/ℓ^m differs from C_m by the derived limit of the tower $(C_m[1])_n$ with transition maps multiplication by ℓ, and this limit vanishes, because ℓ^m kills C_m and so any m consecutive transition maps compose to zero. Consequently C is derived complete and $C/\ell \simeq C_1$ is constructible. Moreover C is bounded, because the C_n have cohomology in a fixed interval, and on the replete topos $T_{\mathrm{pro\acute{e}t}}$ derived limits of such towers increase the range by at most one [Stacks, 0940]. Therefore $C \in D^b_c(T, \mathcal{O}_E)$ by Proposition proposition-sp-integral-towers. Finally, the counit $\mathrm{red}_T C \to C_\bullet$ is at level m the reduction map $C/\ell^m \to C_m$, which is an isomorphism by what we just showed. Since evaluations detect isomorphisms, the counit is an isomorphism. ∎

All operations on towers are defined levelwise, and the functor Rg_* is computed levelwise as well, since an injective tower is levelwise a summand of finite products of injective sheaves. So proper base change and the projection formula hold for towers.

Moreover $Rf_{!,\bullet}$ preserves constructible towers. Each level lies in D^b_{ctf} by Proposition 7.1, so we only check that the bounds are independent of n. For cohomology, note that for a compactification $f = pj$, $Rf_!$ has amplitude $[0, \mathrm{cd}_\ell(p)]$ for every torsion coefficient ring [Stacks, 0F10]. For Tor amplitude we use the projection formula $(Rf_!C_n) \otimes^{\mathbf{L}} N \simeq Rf_!(C_n \otimes^{\mathbf{L}} N)$ for every $\mathcal{O}_E/\ell^n$-module N, which follows from the comparison b and shows that if all C_n have Tor amplitude in $[a, b]$, then all $Rf_!C_n$ have Tor amplitude in $[a, b + \mathrm{cd}_\ell(p)]$.

We do not use Brown representability for towers. Instead we construct the right adjoint of $Rf_{!,\bullet}$ on constructible towers by repeating the geometric argument with finite coefficients in the category of towers.

Proposition 9.18 (The adjunction $Rf_! \dashv f^!$ on constructible towers)**.** *There is an adjunction*

$$Rf_{!,\bullet} \dashv f^!_\bullet \quad \textit{between the constructible towers on } X \textit{ and on } Y. \tag{9.18}$$

At level n its unit and counit are those of the finite adjunction, so $(f^!_\bullet C_\bullet)_n \simeq f^!_{\mathcal{O}_E/\ell^n} C_n$, and the reduction maps of $f^!_\bullet C_\bullet$ are $f^!$ of those of $C_\bullet$ composed with the comparisons c of Lemma 9.15.

Proof. We repeat the construction of Section 9.3 in the topos of towers. The construction has two parts: on a chart we write down an explicit adjunction, built from the trace of the projective line, and then we glue these local adjunctions over a finite cover. The gluing is done with pairs consisting of an object and a counit, as in Definition 9.5, because such pairs are unique up to unique isomorphism and so need no choice of transition maps.

The Kummer sequence $0 \to (\mu_{\ell^n})_n \to (\mathbf{G}_m, (\cdot)^\ell) \to (\mathbf{G}_m, \mathrm{id}) \to 0$ is exact in towers. Hence Chern classes, the diagonal class and the trace of the projective line are single morphisms of towers. Moreover, the identities (9.9) are equalities of maps between sheaves of towers in degree zero, so they can be checked at each level. The construction of (9.10) then gives the adjunction for the projective line in $D(\mathrm{Tow})$, and extension by zero and the étale adjunction give local adjunctions on charts.

We glue the local adjunctions in $D(\mathrm{Tow})$, by induction on a finite cover. Let $X = U \cup V$, and let (H_U, ε_U) and (H_V, ε_V) be pairs representing $\mathrm{Hom}(Rf_{!,\bullet}(-), K_\bullet)$ on U and on V. Put $W = U \cap V$. Both pairs restrict to pairs representing this functor on W, so by Yoneda there is a unique counit-preserving isomorphism between the restrictions, and we call the common restriction H_W. Let H be the cone of $j_{W!}H_W \to j_{U!}H_U \oplus j_{V!}H_V$, as in the triangle (9.12). Since the counits $\varepsilon_U, \varepsilon_V$ agree on $Rf_{!,\bullet}j_{W!}H_W$, the exact Hom sequence shows that they lift to a counit

$\varepsilon : Rf_{!,\bullet}H \to K_\bullet$. The five lemma, applied as in the finite case, then shows that (H, ε) represents $\mathrm{Hom}(Rf_{!,\bullet}(-), K_\bullet)$ on X. The tower H is also constructible. Indeed, $H|_U \simeq H_U$ and $H|_V \simeq H_V$, so the union of their intervals bounds its cohomology and Tor amplitude uniformly in n.

Repeating this step over a finite cover of X by charts gives a pair representing $\mathrm{Hom}(Rf_{!,\bullet}(-), K_\bullet)$ on all of X, and we let $f^!_\bullet K_\bullet$ be its object. By uniqueness of representing pairs, this is functorial in $K_\bullet$ and gives the adjunction (9.18).

The levels are the finite ones, because evaluation at level n carries the glued pair to a pair representing $\mathrm{Hom}(Rf_!(-), K_n)$ with coefficients $\mathcal{O}_E/\ell^n$. Indeed, on each chart it is the local finite adjunction, and the five lemma argument applies at level n, so by uniqueness this pair is the finite one. As the counit is one morphism of towers, it commutes with the transition maps, and transposing this identity gives the reduction maps. ■

The orientation passes to towers as well. By the finite case each level of $\omega_{f,\bullet}[-2d]$ is an invertible sheaf, so it is an invertible sheaf of towers. Moreover, the comparisons for vector bundles and for the deformation family are maps of sheaves of towers in degree zero, and equality of such maps can be tested level by level, unlike equality of maps in $D(\mathrm{Tow})$ in general. Hence the uniqueness in Lemma 9.9 holds for them, because it holds at each level. Composing the inverse of the orientation with the map $\mu_{f,K}$ on towers, we obtain an isomorphism, natural in the constructible tower $K_\bullet$,

$$\gamma_{f,\bullet}(K_\bullet) : f^*_\bullet K_\bullet(d)[2d] \xrightarrow{\sim} f^!_\bullet K_\bullet, \tag{9.19}$$

whose level n is $\gamma_{f,\mathcal{O}_E/\ell^n}(K_n)$.

Both functors in (9.18) preserve normalized towers. For $Rf_{!,\bullet}$ this follows from the comparison b, and for $f^!_\bullet$ from (9.17) and (9.19).

Since $R\bar{f}_*$ and $j_!$ commute with $R\varprojlim$ [Stacks, 09BP], we have

$$Rf_{!,\mathcal{O}_E} M \simeq R\varprojlim_n Rf_{!,\mathcal{O}_E/\ell^n}(M/\ell^n)$$

for $M \in D^b_c(X, \mathcal{O}_E)$. For $K \in D^b_c(Y, \mathcal{O}_E)$ we put

$$H_{f,\mathcal{O}_E}(K) = R\varprojlim_n f^!_{\mathcal{O}_E/\ell^n}(K/\ell^n) \in D^b_c(X, \mathcal{O}_E), \tag{9.20}$$

the derived limit of the normalized tower $f^!_\bullet \operatorname{red}_Y K$.

The functor $H_{f,\mathcal{O}_E}$ is given by an explicit formula, whereas we only know that $f^!_{\mathcal{O}_E}$ exists. The next proposition shows that the formula defines a right adjoint, and so, by uniqueness, computes $f^!_{\mathcal{O}_E}$.

Proposition 9.19 (The integral right adjoint built from towers)**.** *With the unit and counit obtained from towers, $H_{f,\mathcal{O}_E}$ is right adjoint to $Rf_{!,\mathcal{O}_E}$ on $D^b_c(-, \mathcal{O}_E)$. Hence there is a unique counit-preserving isomorphism $H_{f,\mathcal{O}_E} \simeq f^!_{\mathcal{O}_E}$ with the right adjoint of Theorem 5.31.*

Proof. We move the adjunction on towers down to $\mathcal{O}_E$-complexes along the equivalence of Lemma 9.17. For this we need both functors to stay inside normalized towers, which is where the equivalence holds.

Since $Rf_{!,\bullet}$ and $f^!_\bullet$ preserve normalized towers, the equivalence of Lemma 9.17 gives $\mathrm{red}_Y(Rf_{!,\mathcal{O}_E}M) \simeq Rf_{!,\bullet}\,\mathrm{red}_X M$ and $\mathrm{red}_X H_{f,\mathcal{O}_E}(K) \simeq f^!_\bullet \mathrm{red}_Y K$. Using this lemma and (9.18), we get natural bijections

$$\begin{aligned}\mathrm{Hom}(Rf_{!,\mathcal{O}_E}M, K) &\simeq \mathrm{Hom}(Rf_{!,\bullet}\,\mathrm{red}_X M, \mathrm{red}_Y K)\\ &\simeq \mathrm{Hom}(\mathrm{red}_X M, f^!_\bullet \mathrm{red}_Y K) \simeq \mathrm{Hom}(M, H_{f,\mathcal{O}_E}(K)).\end{aligned}$$

The second statement follows from Proposition 9.13. ∎

Applying $R\varprojlim$ to (9.19) at $\mathrm{red}_Y K$, we obtain $\gamma_{f,\mathcal{O}_E} : f^*_{\mathcal{O}_E}K(d)[2d] \xrightarrow{\sim} f^!_{\mathcal{O}_E}K$, whose reductions are the isomorphisms $\gamma_{f,\mathcal{O}_E/\ell^n}$. This gives the isomorphism (9.2) for $\mathcal{O}_E$, and for $E = \mathbf{Q}_\ell$ the one for $\mathbf{Z}_\ell$.

9.5.3 Field coefficients

With coefficients in E there is no tower to pass to the limit over, but none is needed. Indeed, an object of $D^b_c(T, E)$ comes from an integral complex by inverting ℓ, and its Hom groups are the integral Hom groups tensored with E (9.21). Since an adjunction over $\mathcal{O}_E$ is a natural bijection of Hom groups, tensoring it with E gives an adjunction over E (Proposition 9.20). For $\overline{\mathbf{Q}}_\ell$ the Hom groups are colimits over finite extensions of $\mathbf{Q}_\ell$, and the adjunctions over these fields pass to the colimit (Theorem 9.21). For each coefficient ring we thus construct an explicit right adjoint of $Rf_!$, and we identify it with $f^!$ by the uniqueness of adjoints (Proposition 9.13).

Every object of $D^b_c(T, E)$ is $K_0[1/\ell]$ for some $K_0 \in D^b_c(T, \mathcal{O}_E)$, called an integral model, and for integral models

$$\mathrm{Hom}(K_0, L_0[q]) \otimes_{\mathcal{O}_E} E \xrightarrow{\sim} \mathrm{Hom}_{D^b_c(T,E)}(K_0[1/\ell], L_0[1/\ell][q]). \tag{9.21}$$

Both statements are Proposition 3.27: its essential surjectivity provides the models, and (9.21) is its full faithfulness. To see the latter directly, note that the target is $\mathrm{Hom}(K_0, L_0[1/\ell][q])$ by adjunction, and $L_0[1/\ell]$ is the colimit of $L_0 \xrightarrow{\ell} L_0 \xrightarrow{\ell} \cdots$. The left side is the colimit of the same system of Hom groups, so the claim is that Hom out of K_0 commutes with this colimit. Since both sides are cohomological functors of K_0, by the five lemma and dévissage we may take $K_0 = a_!C$ with C dualizable on a stratum (Lemma lemma-structure-integral-constructible-complexes). The stratum formula of Lemma lemma-sections-support-closed-subscheme, whose proof uses only that C is dualizable, turns Hom out of $a_!C$ into cohomology on a quasi-compact quasi-separated scheme, and this commutes with the colimit by Corollary 2.29. On models f^* and $Rf_!$ are the integral functors with ℓ inverted, again because $R\bar{f}_*$ commutes with the colimit along ℓ.

With (9.21) the adjunction for E becomes a matter of inverting ℓ in the integral adjunction.

Proposition 9.20 (Smooth purity with coefficients in E)**.** *Let $E/\mathbf{Q}_\ell$ be finite. Then $Rf_{!,E} \dashv (K \mapsto f^*_E K(d)[2d])$ as functors between $D^b_c(X, E)$ and $D^b_c(Y, E)$.*

Proof. An adjunction is a natural bijection of Hom groups, and by (9.21) every Hom group over E is an integral Hom group tensored with E. So we compute both sides on integral models. Choose integral models M_0 and K_0, so that $M = M_0[1/\ell]$ and $K = K_0[1/\ell]$. By (9.21) twice and Proposition 9.19 with $\gamma_{f,\mathcal{O}_E}$,

$$\begin{aligned}\mathrm{Hom}(Rf_{!,E}M, K) &\simeq \mathrm{Hom}(Rf_{!,\mathcal{O}_E}M_0, K_0) \otimes E\\ &\simeq \mathrm{Hom}(M_0, f^*_{\mathcal{O}_E}K_0(d)[2d]) \otimes E \simeq \mathrm{Hom}(M, f^*_E K(d)[2d]).\end{aligned}$$

These bijections are given by the integral unit and counit with ℓ inverted, so they are natural and do not depend on the models. ∎

Since $f^*(-)(d)[2d]$ is now a right adjoint of $Rf_{!,E}$, as is $f^!_E$, Proposition 9.13 gives an isomorphism $f^!_E \xrightarrow{\sim} f^*_E(-)(d)[2d]$ compatible with the counits, and we let $\gamma_{f,E} : f^*_E(-)(d)[2d] \xrightarrow{\sim} f^!_E$ be its inverse.

Hom groups over $\overline{\mathbf{Q}}_\ell$ are colimits of Hom groups over finite extensions, so the adjunctions over the finite extensions pass to the colimit provided they correspond under the transition maps. To see that they do, let $E \subset F$ be a finite extension. Then extension of scalars along the finite free map $\mathcal{O}_E \to \mathcal{O}_F$ commutes with $R\varprojlim$, and at level n it is extension along $\mathcal{O}_E/\ell^n \to \mathcal{O}_F/\ell^n$, so we may compare the counits in $D(\mathrm{Tow})$. Extending scalars in the pair of towers $(f^!_\bullet K_\bullet, \varepsilon)$ for $\mathcal{O}_E$ gives a pair over $\mathcal{O}_F$, whose levels represent $\mathrm{Hom}(Rf_!(-), K_n)$ by Lemma 9.15. By the five lemma, applied levelwise, the pair then represents $\mathrm{Hom}(Rf_{!,\bullet}(-), K_\bullet)$, and by uniqueness it is isomorphic to the corresponding pair for $\mathcal{O}_F$ by a unique counit-preserving isomorphism. Applying $R\varprojlim$ and inverting ℓ, we see that the adjunctions over E and over F correspond under extension of scalars, transitively in $E \subset F \subset F'$.

Theorem 9.21 (Smooth purity with coefficients in $\overline{\mathbf{Q}}_\ell$)**.** *We have $Rf_{!,\overline{\mathbf{Q}}_\ell} \dashv (K \mapsto f^*_{\overline{\mathbf{Q}}_\ell} K(d)[2d])$ as functors between $D^b_c(X, \overline{\mathbf{Q}}_\ell)$ and $D^b_c(Y, \overline{\mathbf{Q}}_\ell)$.*

Proof. As for E, we compare Hom groups, now as colimits over finite extensions instead of after inverting ℓ. By Proposition 7.11, $D^b_c(T, \overline{\mathbf{Q}}_\ell)$ is the 2-colimit of the categories $D^b_c(T, F)$ over finite extensions F of $\mathbf{Q}_\ell$, so its Hom groups are the colimits of those over F. The functors $Rf_!$ and f^* commute with extension of scalars (Section 7.2). Choosing models over some E, the adjunctions of Proposition 9.20 over the fields $F \supset E$ are compatible with the transition maps, and their colimit is the asserted bijection. ∎

Applying Proposition 9.13 once more and inverting the resulting isomorphism, we get $\gamma_{f,\overline{\mathbf{Q}}_\ell} : f^*_{\overline{\mathbf{Q}}_\ell}(-)(d)[2d] \xrightarrow{\sim} f^!_{\overline{\mathbf{Q}}_\ell}$, whose counit is obtained by extension of scalars from a finite extension of $\mathbf{Q}_\ell$. What is left of Theorem 9.3 are the compatibilities with change of coefficients, base change and composition, and the trace normalization.

9.6 Compatibilities

The isomorphism γ_f now exists for every admissible coefficient ring, and it remains to prove the properties (2)–(4) of Theorem 9.3. In topology, orientations of a submersion restrict to fibres and multiply under composition, and Propositions 9.22 and Proposition 9.23 are the corresponding statements. These propositions also let us define the trace map Tr_f (Definition 9.24), the analogue of integration over the fibres, on which Poincaré duality rests. With finite coefficients the compatibilities follow from the construction, since every ingredient of the orientation commutes with base change. Passing them to $\mathbf{Z}_\ell$, however, is harder, because there the base change map itself has to be built as a limit.

9.6.1 Base change and composition

Fix a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y, \end{array}$$

with f smooth of pure relative dimension d, and a map $A \to B$ of finite commutative rings killed by a power of ℓ. A subscript B denotes extension of scalars, as in Definition 9.14.

Base change. The statement of Proposition 9.22 needs an exceptional base change isomorphism for $\mathbf{Z}_\ell$. It is obtained as a limit of the finite ones, and the limit exists once the finite isomorphisms commute with the reduction maps, that is, once base change commutes with change of coefficients. To state this we name the four comparison maps involved. For $M \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, A)$ and $K \in D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, A)$, let $b^f(M) : (Rf_{!,A}M)_B \xrightarrow{\sim} Rf_{!,B}M_B$ and $c^f(K) : (f^!_A K)_B \xrightarrow{\sim} f^!_B K_B$ be the change-of-coefficients maps of Lemma 9.15. With $g'^* : D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, A) \to D^b_{\mathrm{ctf}}(X'_{\mathrm{pro\acute{e}t}}, A)$ and $g^* : D^b_{\mathrm{ctf}}(Y_{\mathrm{pro\acute{e}t}}, A) \to D^b_{\mathrm{ctf}}(Y'_{\mathrm{pro\acute{e}t}}, A)$, let $\beta_{f,A}(M) : g^* Rf_{!,A}M \xrightarrow{\sim} Rf'_{!,A}g'^*M$ be the base change map for compact support, and let $e_{f,A}(K) : g'^* f^!_A K \xrightarrow{\sim} f'^!_A g^* K$ be the exceptional base change isomorphism of Proposition proposition-exceptional-inverse-image-smooth-morphism. The claim is

$$\beta_{f,B}(M_B) \circ g^*(b^f(M)) = b^{f'}(g'^*M) \circ (\beta_{f,A}(M))_B, \tag{9.22}$$

$$c^{f'}(g^*K) \circ (e_{f,A}(K))_B = e_{f,B}(K_B) \circ g'^* c^f(K). \tag{9.23}$$

Both identities say that two ways of passing from (A, Y) to (B, Y') give the same map: changing coefficients first and then base, or the other way round. We prove the first by splitting $Rf_!$ into an open and a proper part, and deduce the second from the first by transposition.

To prove (9.22), choose a compactification $f = \bar{f} \circ j$ and pull it back to Y'. Since b and β for f are the composites of those for j and for $\bar{f}$, it suffices to treat these two cases. For the open immersion j, extension by zero commutes with pullback and with extension of scalars, and both sides of (9.22) are this identification. For the proper map $\bar{f}$, both $b^{\bar{f}}$ and proper base change are transposes of the counit of $L\bar{f}^* \dashv R\bar{f}_*$, the first after extension of scalars and the second after pullback; here we work on the derived categories of sheaves of A-modules on the pro-étale sites. Therefore both sides of (9.22) transpose to the pulled-back counit with scalars extended.

For the second identity, recall that c^f and $e_{f,A}$ are defined by transposing the counit composed with $(b^f)^{-1}$, respectively with $\beta^{-1}_{f,A}$ (see (9.16)). Transposing both sides of (9.23) under $Rf'_{!,B} \dashv f'^!_B$ thus gives two maps $Rf'_{!,B}(g'^*((f^!_A K)_B)) \to g^* K_B$. Each is built from the counit $\varepsilon_{f,A}(K)$, the comparisons b and the base change maps β, and the counit commutes with extension of scalars and with pullback. By naturality of b and β, the two maps therefore agree once (9.22) holds.

These two identities let us assemble the finite base change maps into a single map of towers. Its levels are linked by the change-of-coefficients maps along $\mathcal{O}_E/\ell^{n+1} \to \mathcal{O}_E/\ell^n$.

Let $E/\mathbf{Q}_\ell$ be a finite extension and work in the category of towers of (9.18). Extension by zero and proper base change define $\beta_{f,\bullet} : g^* Rf_{!,\bullet} \to Rf'_{!,\bullet} g'^*$, and transposing $g^*(\varepsilon_{f,\bullet}) \circ \beta^{-1}_{f,\bullet}$ under the adjunction on towers for f' defines $e_{f,\bullet} : g'^* f^!_\bullet \to f'^!_\bullet g^*$. At level n these are $\beta_{f,\mathcal{O}_E/\ell^n}$ and the exceptional base change map with coefficients $\mathcal{O}_E/\ell^n$, so they are isomorphisms; for the second, this is Proposition proposition-exceptional-inverse-image-smooth-morphism with $\Lambda = \mathcal{O}_E/\ell^n$. Moreover, the transition maps of $f^!_\bullet$ are induced by the change-of-coefficients maps c (Proposition 9.18), and by (9.23) the levels of $e_{f,\bullet}$ commute with them. So $e_{f,\bullet}$ is an isomorphism of functors on towers. In particular, for a normalized tower $K_\bullet$ both $g'^* f^!_\bullet K_\bullet$ and $f'^!_\bullet g^* K_\bullet$ are normalized.

The functors red of Lemma 9.17 carry the maps over $\mathcal{O}_E$ (base change, composition, counits and transposes) to the maps on towers. We check this in steps. First, the functor red is a derived pullback of ringed topoi, so it commutes with g^* by associativity of derived pullbacks, and it commutes with extension by zero because the right adjoints commute. Next, for proper $\bar{f}$, the comparison $\mathrm{red}_Y R\bar{f}_* M \to R\bar{f}_{*,\bullet}\,\mathrm{red}_X M$ is the transpose of red_X of the counit of $\bar{f}^* \dashv R\bar{f}_*$, and its level n is the change-of-coefficients map of Lemma 7.6 for $\mathcal{O}_E \to \mathcal{O}_E/\ell^n$. As evaluations detect isomorphisms, the comparison is an isomorphism for $M \in D^b_c(X, \mathcal{O}_E)$. Composing with the compatibility with $j_!$ we obtain the comparison $\mathrm{red}_Y Rf_{!,\mathcal{O}_E} M \simeq Rf_{!,\bullet}\,\mathrm{red}_X M$ for a compactification $f = \bar{f} \circ j$. The formal argument in the proof of Proposition 7.10 shows that, under these comparisons, red carries the unit and counit of $\bar{f}^* \dashv R\bar{f}_*$ over $\mathcal{O}_E$ to the unit and counit on towers. Since the base change and composition isomorphisms of Proposition 5.28 are composites of such units and counits and of the identifications of pullbacks and of extensions by zero, red carries them, with coefficients $\mathcal{O}_E$, to $\beta_{f,\bullet}$ and to the corresponding composition isomorphisms on towers. For the counit, the adjunction of Proposition 9.19 is transported from the adjunction on towers, so red_Y carries $\varepsilon_f(K)$ to $\varepsilon_{f,\bullet}(\mathrm{red}_Y K)$, once the sources are identified by this comparison and by the isomorphism $\mathrm{red}_X f^!_{\mathcal{O}_E} K \simeq f^!_\bullet \mathrm{red}_Y K$ of its proof. Since a transpose is determined by the counit, red also carries transposes under $Rf_{!,\mathcal{O}_E} \dashv f^!_{\mathcal{O}_E}$ to transposes on towers. Finally, by Lemma 9.17, red is an equivalence on $D^b_c(-, \mathcal{O}_E)$ whose inverse is $R\varprojlim$. A morphism of $D^b_c(-, \mathcal{O}_E)$ is therefore $R\varprojlim$ of its image under red, up to the unit isomorphisms $M \simeq R\varprojlim \mathrm{red}\, M$.

Let $K \in D^b_c(Y, \mathcal{O}_E)$, and let $\mathrm{red}_Y K = (K/\ell^n)_n$ be its tower of reductions (Lemma 9.17). Since inverse image commutes with reduction modulo ℓ^n, and $f^!_{\mathcal{O}_E} K \simeq R\varprojlim_n f^!_{\mathcal{O}_E/\ell^n}(K/\ell^n)$ by Proposition 9.19 and (9.20), both sides below are $R\varprojlim_n$ of the corresponding terms at level n for K/ℓ^n. Applying $R\varprojlim$ to $e_{f,\bullet}$ at $\mathrm{red}_Y K$ we obtain the integral exceptional base change isomorphism

$$g'^* f^!_{\mathcal{O}_E} K \xrightarrow{\sim} f'^!_{\mathcal{O}_E} g^* K; \tag{9.24}$$

for $E = \mathbf{Q}_\ell$ the coefficients are $\mathbf{Z}_\ell$. With coefficients E and $\overline{\mathbf{Q}}_\ell$ we define exceptional base change by inverting ℓ in (9.24) on integral models and then extending scalars from a finite extension of $\mathbf{Q}_\ell$. This is possible because f^*, $f^!$ and $Rf_!$ with these coefficients are obtained from the integral functors in the same way (Proposition 9.20 and Theorem 9.21).

In each case the map obtained is the exceptional base change map of Definition 9.1. For $\mathcal{O}_E$, the map of the definition is the transpose under $Rf'_{!,\mathcal{O}_E} \dashv f'^!_{\mathcal{O}_E}$ of $g^*(\varepsilon_f(K)) \circ \beta^{-1}$. By the preceding paragraph, red carries this composite to $g^*(\varepsilon_{f,\bullet}) \circ \beta^{-1}_{f,\bullet}$ at $\mathrm{red}_Y K$, and it carries the transpose to the transpose on towers, which is $e_{f,\bullet}(\mathrm{red}_Y K)$. Hence the map of the definition is $R\varprojlim$ of $e_{f,\bullet}(\mathrm{red}_Y K)$, that is, (9.24). For E and $\overline{\mathbf{Q}}_\ell$ we argue in the same way, with extension of scalars along $\mathcal{O}_E \to E$ and $E \to \overline{\mathbf{Q}}_\ell$ in place of red. Indeed, extension of scalars commutes with g^*, extension by zero and $R\bar{f}_*$ through the change-of-coefficients maps of Definition 7.4, and by Proposition 7.10 it carries the units and counits of $\bar{f}^* \dashv R\bar{f}_*$ and of $Rf_! \dashv f^!$ to the corresponding units and counits. So it carries the base change maps for $Rf_!$, the counits, and therefore the exceptional base change maps over $\mathcal{O}_E$ to those over E. In the same way those over a finite extension of $\mathbf{Q}_\ell$ go to those over $\overline{\mathbf{Q}}_\ell$.

Since the exceptional base change map is an isomorphism for every admissible coefficient ring, we can state that γ_f commutes with it. For g the inclusion of a closed point y of Y, this says that γ_f restricts on the fibre X_y to the isomorphism γ of the smooth $\kappa(y)$-scheme X_y.

Proposition 9.22 (Compatibility of purity with base change)**.** *Let Λ be any admissible coefficient ring and let $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$. Identify the sources by $g'^* f^* \simeq (f')^* g^*$ and the targets by the exceptional base change map of Definition 9.1. Then*

$$g'^{*}\gamma_f(K) = \gamma_{f'}(g^*K) \quad \textit{in } D^b_{\mathrm{ctf}}(X', \Lambda).$$

Proof. The map γ_f has two factors, the orientation of ω_f and the map $\mu_{f,K}$, and we show that each commutes with base change. For the orientation this is a local statement about rank one sheaves, and it follows from the naturality of the deformation to the normal bundle. For $\mu_{f,K}$ we compare transposes under the adjunction for f'.

For a finite ring A we compare the orientations and the maps $\mu_{f,K}$ directly with coefficients in A, and for $\mathcal{O}_E$ we compare the maps on towers $\gamma_{f,\bullet}$ of (9.19) and apply $R\varprojlim$. The cases E and $\overline{\mathbf{Q}}_\ell$ follow, since $\gamma_{f,E}$ and $\gamma_{f,\overline{\mathbf{Q}}_\ell}$ are obtained from $\gamma_{f,\mathcal{O}_E}$ by inverting ℓ and by extension of scalars from a finite extension of $\mathbf{Q}_\ell$. We write out the calculation for towers; dropping the index $\bullet$ gives the calculation at finite level. There the map $\gamma_{f,\bullet}$ is the composite of the orientation $(\mathcal{O}_{E,X}/\ell^\bullet)(d)[2d] \xrightarrow{\sim} \omega_{f,\bullet}$ and the map $f^*_\bullet K_\bullet \otimes \omega_{f,\bullet} \to f^!_\bullet K_\bullet$ for $K_\bullet = \mathrm{red}_Y K_\bullet$.

Orientation. After removing the shift $[2d]$, the two orientations to be compared are maps between sheaves of towers of rank one in degree zero. Since maps between sheaves form a sheaf, and a map of sheaves of towers is a compatible family of maps of sheaves, equality may be tested locally on X' and at each level n. At a fixed level, the diagonal, its normal bundle, the deformation to the normal bundle and the trace map of vector bundles all commute with base change (Lemma lemma-naturality-under-base-change-etale), and therefore so does the orientation built from them.

The map $f^*K \otimes \omega_f \to f^!K$. This factor is not a map of sheaves, so we cannot compare the two maps locally. However, a map into $f'^!_\bullet g^* K_\bullet$ is determined by its transpose under the adjunction on towers for f'. By the definition of $e_{f,\bullet}$, the transpose of the pulled-back map $\mu_{f,K}$ for f is the pulled-back counit for f, preceded by the map of the projection formula. Since the projection formula is natural under pullback, the transpose of the map for f' is the same composite, as in the counit calculation in the proof of Proposition proposition-exceptional-inverse-image-smooth-morphism, carried out with towers. ∎

Composition. For composition we again fix the composition isomorphisms for $f^!$ with every coefficient ring before comparing the maps γ. From now on, g no longer denotes the base change in a square: let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be smooth of pure relative dimensions d and e. Composing the finite adjunctions, we identify $(g\circ f)^!_A$ with $f^!_A g^!_A$, compatibly with $R(g\circ f)_{!,A} \simeq Rg_{!,A}Rf_{!,A}$, and the counit of the composite adjunction is the composite of the two counits.

The change-of-coefficients maps are compatible with composition: for $M \in D^b_{\mathrm{ctf}}(X_{\mathrm{pro\acute{e}t}}, A)$ and $K \in D^b_{\mathrm{ctf}}(Z_{\mathrm{pro\acute{e}t}}, A)$,

$$b^{g\circ f}(M) = Rg_{!,B}(b^f(M)) \circ b^g(Rf_{!,A}M), \qquad c^{g\circ f}(K) = f^!_B c^g(K) \circ c^f(g^!_A K).$$

The first identity holds on each free term of a resolution of B over A, since the composition isomorphism is natural, and the totalization in Lemma 9.15 extends it to arbitrary $A \to B$. We get the second by transposing under the composite adjunction.

On towers, the composition isomorphism for $Rf_{!,\bullet}$ is built from the same constructions. The isomorphism $(g\circ f)^!_\bullet \simeq f^!_\bullet g^!_\bullet$ is the isomorphism of right adjoints given by the composite adjunction on towers. Since evaluation at level n carries the counits on towers to the finite counits (Proposition 9.18), the level n of these isomorphisms is the finite one, and by the identities above they commute with the transition maps, which are built from b and c. Applying $R\varprojlim$ gives the integral composition isomorphisms

$$R(g\circ f)_{!,\mathcal{O}_E} \simeq Rg_{!,\mathcal{O}_E}Rf_{!,\mathcal{O}_E}, \qquad (g\circ f)^!_{\mathcal{O}_E} \simeq f^!_{\mathcal{O}_E} g^!_{\mathcal{O}_E},$$

between $D^b_c(X, \mathcal{O}_E)$ and $D^b_c(Z, \mathcal{O}_E)$; for $E = \mathbf{Q}_\ell$ the coefficients are $\mathbf{Z}_\ell$. As for base change, inverting ℓ and extending scalars give the composition isomorphisms with coefficients E and $\overline{\mathbf{Q}}_\ell$. These isomorphisms $(g \circ f)^! \simeq f^! g^!$ are the composition isomorphisms of Proposition proposition-composition-open-restriction-compactifications-f. Indeed, the isomorphism of that proposition is characterized by the following property: it carries the counit of $R(g \circ f)_! \dashv (g \circ f)^!$, composed with the composition isomorphism for $Rf_!$, to the composite of the counits for f and g. By uniqueness of adjoints it therefore suffices to check this property for every coefficient ring. With finite coefficients it holds by definition. On towers the composition isomorphism is the one of the composite adjunction, so in $D(\mathrm{Tow})$ it carries the counit on towers of $g \circ f$ to the composite of the counits on towers. By the comparison following the construction of $e_{f,\bullet}$ in Subsection 9.6.1, red carries the counits and the composition isomorphism for $Rf_!$ with coefficients $\mathcal{O}_E$ to those on towers, and applying $R\varprojlim$ gives the property over $\mathcal{O}_E$. Similarly, the same comparison shows that extension of scalars along $\mathcal{O}_E \to E$ and $E \to \overline{\mathbf{Q}}_\ell$ carries the counits and the composition isomorphisms for $Rf_!$ to the corresponding ones, which gives the property over E and $\overline{\mathbf{Q}}_\ell$.

With these composition isomorphisms in place for every coefficient ring, the statement takes the following form. In topological terms it says that the orientation of a composite of submersions is the product of the orientations.

Proposition 9.23 (Compatibility of purity with composition)**.** *For every admissible Λ and $K \in D^b_{\mathrm{ctf}}(Z, \Lambda)$, the map $\gamma_{g \circ f}(K)$ of (9.2) is the composite*

$$\begin{aligned}(g \circ f)^* K(d+e)[2d+2e] &\simeq f^*\big(g^* K(e)[2e]\big)(d)[2d] \\ &\xrightarrow{f^*(\gamma_g(K))(d)[2d]} f^*(g^! K)(d)[2d] \\ &\xrightarrow{\gamma_f(g^! K)} f^! g^! K \simeq (g \circ f)^! K\end{aligned}$$

in $D^b_{\mathrm{ctf}}(X, \Lambda)$, where the coefficient Λ is omitted from the displayed composite.

Proof. As for base change, we compare orientations and the maps $\mu_{f,K}$, first over any finite admissible ring and then for towers. The orientations are compared on a chart adapted to both f and g, where each is iterated integration along affine lines, with the coordinates ordered so that the two iterated integrals are the same. Applying $R\varprojlim$ then gives the integral identity, and inverting ℓ and descending to a finite extension of $\mathbf{Q}_\ell$ give the identities for coefficient fields.

Orientation, finite coefficients. Both orientations are maps of rank one locally constant sheaves after removing twist and shift, so it suffices to compare them on a cover of X by charts, and we choose charts adapted to both morphisms. That is, choose an open $W \subset Y$ with an étale Z-map $W \to \mathbf{A}^e_Z$, and an open $U \subset f^{-1}(W)$ with an étale W-map $U \to \mathbf{A}^d_W$. Then the composite $U \to \mathbf{A}^d_W \to \mathbf{A}^d_{\mathbf{A}^e_Z} = \mathbf{A}^{d+e}_Z$ is étale, and we order the coordinates of $\mathbf{A}^{d+e}_Z$ with the e coordinates of g first and the d coordinates of f last. Over this chart, both orientations are adjoint to iterated integration along affine lines, in the same order: first the d coordinates of f, then the e coordinates of g. Since the recursive construction preceding (9.11) integrates the last coordinate first, the integration on $\mathbf{A}^{d+e}_Z$ is exactly this composite. Each diagonal orientation agrees on a chart with the integration in its coordinates, so the two orientations agree on the cover.

The maps $\mu_{f,K}$, finite coefficients. The two maps agree because their transposes under the composite adjunction do. Indeed, the transpose of the iterated map applies the projection formula and the counit for f, then the projection formula and the counit for g, whereas the transpose of the map for $g \circ f$ applies the projection formula for $g \circ f$ and its counit. The two isomorphisms given by the projection formula agree by associativity of tensor products, and the counit of the composite adjunction is the composite of the two counits.

Towers. The same two arguments apply in the category of towers, with its maps $\gamma_{f,\bullet}$ and its composite adjunction. Since the orientations are maps of sheaves of towers of rank one, we may test their equality at each level, while the maps (9.4) are compared by the same transposition. ■

9.6.2 The trace map

On a compact oriented manifold of dimension n, Poincaré duality pairs H^i with H^{n-i} by cup product followed by integration $H^n(M) \to \mathbf{R}$, which sends the class of a point to 1. To obtain the Poincaré pairing of Weil's axiom (c) we need the analogue of integration. By adjunction we already have a map, the counit $Rf_!f^!K \to K$, but its source involves $f^!$, which we only know through its universal property. Composing with γ_f replaces $f^!K$ by the twisted and shifted inverse image, so the result is a map out of compactly supported cohomology of an explicit complex. For example, for X smooth of dimension d over an algebraically closed field k and $K = \Lambda$, it is a map $H_c^{2d}(X_{\text{proét}}, \Lambda_X(d)) \to \Lambda$. Here the twist cannot be dropped, for the reason explained before Definition 2.53: geometric Frobenius acts on $H^2(\mathbf{P}^1_k, \mathbf{Z}_\ell)$ by q, so only the twisted trace map commutes with Frobenius (Section 10.5). The trace map is used in the Poincaré pairing (Theorem 10.12) and the trace isomorphism in top degree (Theorem 10.15).

Let $f : X \to Y$ be smooth of pure relative dimension d, and let Λ be any admissible coefficient ring. We omit Λ from the notation and let $\varepsilon_f(K) : Rf_!f^!K \to K$ be the counit of the adjunction $Rf_! \dashv f^!$ between $D^b_{\text{ctf}}(X, \Lambda)$ and $D^b_{\text{ctf}}(Y, \Lambda)$.

Definition 9.24 (Trace map)**.** For $K \in D^b_{\text{ctf}}(Y, \Lambda)$, the *trace map* is the composite

$$\operatorname{Tr}_f(K) : \quad Rf_!(f^*K(d)[2d]) \xrightarrow{Rf_!\gamma_f(K)} Rf_!f^!K \xrightarrow{\varepsilon_f(K)} K. \tag{9.25}$$

Remark 9.25 (The trace map in two special cases)**.**

1. For the projection $\mathbf{A}^d_Y \to Y$ and $K = \Lambda_Y$, the trace map is the trace map of affine space (9.11) with the corresponding coefficients, which on each geometric fibre sends the class of the origin to 1.
2. For a finite étale morphism, where $d = 0$, the trace map is summation over the fibres.

By the first part of the remark, for affine space the trace map is the explicit integration along lines built from $\mathbf{P}^1$. The trace map of a general smooth morphism is glued from such charts, and the next statement says that the gluing respects composition, as integration over the fibres of a composite of submersions is iterated integration.

Proposition 9.26 (Transitivity of the trace map)**.** *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be smooth of pure relative dimensions d and e, and $K \in D^b_{\text{ctf}}(Z, \Lambda)$ for any admissible Λ. Then $\operatorname{Tr}_{g\circ f}(K)$ is the composite*

$$R(g\circ f)_!((g\circ f)^*K(d+e)[2d+2e]) \simeq Rg_!Rf_!(f^*(g^*K(e)[2e])(d)[2d])$$
$$\xrightarrow{Rg_!(\operatorname{Tr}_f(g^*K(e)[2e]))} Rg_!(g^*K(e)[2e]) \xrightarrow{\operatorname{Tr}_g(K)} K.$$

Proof. Each trace map is a map γ followed by a counit. The maps γ compose by Proposition 9.23, and the counits compose because the counit of a composite adjunction is the composite of the counits, so the only thing to arrange is the order of the factors. If we expand both traces using (9.25), naturality of γ_f lets us move the map $\gamma_g(K) : g^*K(e)[2e] \to g^!K$ past the map γ_f of the first trace. The composite then consists of the two maps γ_f and γ_g, which together are $\gamma_{g\circ f}(K)$ by Proposition 9.23, followed by

$$Rg_!Rf_!f^!g^!K \xrightarrow{Rg_!(\varepsilon_f(g^!K))} Rg_!g^!K \xrightarrow{\varepsilon_g(K)} K.$$

This is the counit of the composite adjunction $R(g\circ f)_! \dashv (g\circ f)^!$, so the composite is the map $\gamma_{g\circ f}(K)$ followed by its counit, that is, $\mathrm{Tr}_{g\circ f}(K)$. ∎

9.6.3 Change of coefficients

The last compatibility compares the maps γ_f for different coefficient rings. For a map $A \to B$ of finite rings the comparison $c_{A,B}$ of Lemma 9.15 is the change-of-coefficients map for $f^!$ of Proposition 7.10, since both are defined as the transpose of the extended counit composed with the inverse of the comparison for $Rf_!$.

Proposition 9.27 (Compatibility of purity with change of coefficients)**.** *Let $f : X \to Y$ be smooth of pure relative dimension d, let $\Lambda \to \Lambda'$ be a coefficient map of Proposition 7.10 and let $K \in D^b_{\mathrm{ctf}}(Y,\Lambda)$. Then the change-of-coefficients isomorphism $(f^!K)_{\Lambda'} \to f^!(K_{\Lambda'})$ carries $(\gamma_{f,\Lambda}(K))_{\Lambda'}$ to $\gamma_{f,\Lambda'}(K_{\Lambda'})$, the sources being identified by the change-of-coefficients isomorphism for f^*. The same holds for the trace maps: the change-of-coefficients isomorphism for $Rf_!$ carries $(\mathrm{Tr}_{f,\Lambda}(K))_{\Lambda'}$ to $\mathrm{Tr}_{f,\Lambda'}(K_{\Lambda'})$.*

Proof. Both kinds of comparison maps are compatible with composition of coefficient maps, so it suffices to treat the elementary maps in the list of Proposition 7.10. For a map $A \to B$ of finite rings the assertion for γ_f is (9.17), and the assertion for the traces follows from it and the counit identity (9.16). A continuous map $\mathcal{O}_E \to A$ factors through $\mathcal{O}_E/\ell^n$ for some n, so it suffices to treat $\mathcal{O}_E \to \mathcal{O}_E/\ell^n$. By construction (Subsection 9.5.2), red_X carries $\gamma_{f,\mathcal{O}_E}(K)$ to $\gamma_{f,\bullet}(\mathrm{red}_Y K)$, whose level n is $\gamma_{f,\mathcal{O}_E/\ell^n}(K/\ell^n)$. Here the targets are identified by the isomorphism $\mathrm{red}_X f^!_{\mathcal{O}_E}K \simeq f^!_\bullet \mathrm{red}_Y K$, whose level n is an isomorphism $(f^!_{\mathcal{O}_E}K)/\ell^n \simeq f^!_{\mathcal{O}_E/\ell^n}(K/\ell^n)$, and this identification is the change-of-coefficients map of Proposition 7.10. Indeed, the integral counit reduces to the finite one, namely to level n of the counit on towers (Subsection 9.6.1 and Proposition 9.18), and both maps are transposes of that counit composed with the comparison for $Rf_!$. So this case follows from the finite one. For $\mathcal{O}_E \to E$ and $E \to \overline{\mathbf{Q}}_\ell$ the assertion holds by construction, because $\gamma_{f,E}$ and $\gamma_{f,\overline{\mathbf{Q}}_\ell}$ were defined by inverting ℓ in the integral adjunction and by extension of scalars from finite extensions of $\mathbf{Q}_\ell$ (Subsection 9.5.3). For a finite extension F/E we showed before Theorem 9.21 that extension of scalars carries the pair of towers for $\mathcal{O}_E$ to the pair for $\mathcal{O}_F$ by a counit-preserving isomorphism. By applying $R\varprojlim$ and inverting ℓ we get the cases $\mathcal{O}_E \to \mathcal{O}_F$ and $E \to F$. In every case the statement for the traces follows from the one for γ_f, because the change-of-coefficients maps for $Rf_!$ and $f^!$ are compatible with the counits. ∎

9.7 Proof of smooth purity

The proof of Theorem 9.3 assembles the constructions above; only the trace normalization (1) with integral coefficients needs a separate argument.

Proof of Theorem 9.3. We recall γ_f for each kind of coefficient ring, prove the trace normalization (1) first with finite and then with integral coefficients, and cite (2)–(4).

The isomorphisms γ_f. For finite A it is $\gamma_{f,A}$ of Proposition 9.12. For $\mathcal{O}_E$ (and $\mathbf{Z}_\ell$) it is $R\varprojlim$ of the map on towers (9.19), which is an isomorphism with target $f^!$ by Proposition 9.19. For E (and $\mathbf{Q}_\ell$) and for $\overline{\mathbf{Q}}_\ell$ it is the inverse of the counit-preserving isomorphism of Proposition 9.13

between $f^!$ and the right adjoints of Proposition 9.20 and Theorem 9.21. In every case, therefore, γ_f is natural in K.

Property (1), finite coefficients. Here (1) holds because the orientation obtained from the diagonal agrees on the chart $\mathbf{A}^1_Y$ with the one given by the adjunction in coordinates, whose counit at the constant sheaf is Tr_p (the comparison following Proposition 9.11).

Property (1), integral and rational coefficients. For $\Lambda = \mathcal{O}_E$ we write the left side ϕ of (1) as $u \circ \mathrm{Tr}_p$ and show that $u = 1$ by reducing modulo ℓ^n.

First, the trace map Tr_p of (9.1) is an isomorphism, and this is what makes u well defined. Its reduction modulo ℓ is the trace map with coefficients $\mathcal{O}_E/\ell$, because the trace of Proposition 6.22 lifts the finite ones, and this reduction is an isomorphism. Indeed, it is obtained by extension of scalars along the free map $\mathbf{Z}/\ell \to \mathcal{O}_E/\ell$ from the trace map with coefficients $\mathbf{Z}/\ell$, which is an isomorphism (Section 9.3). So the cone C of Tr_p has $C/\ell = 0$, and hence $C/\pi_E = (C/\ell) \otimes^{\mathbf{L}}_{\mathcal{O}_E/\ell} \mathcal{O}_E/\pi_E = 0$. Since the source and target of Tr_p lie in $D^b_c(Y, \mathcal{O}_E)$, the complex C is derived complete, and derived Nakayama (Lemma 2.62) gives $C = 0$. Thus $\phi = u \circ \mathrm{Tr}_p$ for a unique endomorphism u of $\mathcal{O}_{E,Y}$, that is, for an element $u \in H^0(Y, \mathcal{O}_E) = \varprojlim_n H^0(Y, \mathcal{O}_E/\ell^n)$.

Next, we show that $u \equiv 1$ modulo ℓ^n for every n, so that $u = 1$. The trace part of Proposition 9.27 for $\mathcal{O}_E \to \mathcal{O}_E/\ell^n$ shows that the reduction of ϕ modulo ℓ^n is the left side of (1) with coefficients $\mathcal{O}_E/\ell^n$, which by the finite case equals the trace map with these coefficients. This is also the reduction of Tr_p, and that reduction is an isomorphism. Hence $u \equiv 1$ modulo ℓ^n. For E and $\overline{\mathbf{Q}}_\ell$ both sides are obtained from the integral ones by inverting ℓ and extending scalars.

Properties (2)–(4). These are Propositions 9.22, Proposition 9.23 and Proposition 9.27. ∎

For the structure morphism of a smooth scheme, Theorem 9.3 computes the dualizing complex used in Poincaré duality (Chapter 10).

Corollary 9.28 (The dualizing complex of a smooth scheme)**.** *Let $a : X \to \operatorname{Spec} k$ be smooth of pure dimension d, and let Λ be any admissible coefficient ring. There is a natural isomorphism $a^!\Lambda_{\operatorname{Spec} k} \simeq \Lambda_X(d)[2d]$ in $D^b_{\mathrm{ctf}}(X, \Lambda)$.*

Proof. Apply Theorem 9.3 with coefficients Λ to $K = \Lambda_{\operatorname{Spec} k}$, and use $a^*\Lambda_{\operatorname{Spec} k} = \Lambda_X$. ∎

As a check, take $X = \mathbf{A}^1_k$ with k algebraically closed. The adjunction gives $\mathrm{Hom}(R\Gamma_c(\mathbf{A}^1, \Lambda), \Lambda[-2]) \simeq \mathrm{Hom}(\Lambda_X, a^!\Lambda[-2]) = H^0(\mathbf{A}^1, \Lambda(1)) = \Lambda(1)$, and indeed $R\Gamma_c(\mathbf{A}^1, \Lambda) = \Lambda(-1)[-2]$.

If k is algebraically closed, taking global sections turns the trace $\mathrm{Tr}_a(\Lambda_{\operatorname{Spec} k})$ into a map $R\Gamma_c(X_{\text{pro\'et}}, \Lambda_X(d))[2d] \to \Lambda$, and taking H^0 of it gives the trace map $H^{2d}_c(X_{\text{pro\'et}}, \Lambda_X(d)) \to \Lambda$. For example, for $X = \mathbf{P}^1_k$ it sends $c_1(\mathcal{O}(1))$ to 1.

10 Verdier and Poincaré duality

The main estimate in the proof of the Deligne–Laumon theorem, the goal of Part III, is an upper bound: every Frobenius eigenvalue on H^i of a smooth proper variety has absolute value at most $q^{i/2}$. The lower bound comes from Weil's axiom (c). Suppose that the cup product pairs H^i with H^{2d-i} perfectly and that Frobenius multiplies the pairing by q^d. Then the eigenvalues α in degree i correspond to the eigenvalues q^d/α in degree $2d - i$, so the upper bound for q^d/α is a lower bound for α (Chapter 19). The same relation gives the functional equation of the zeta function (Corollary 12.33). This chapter proves axiom (c).

For the elliptic curve $y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, duality says that the two eigenvalues on H^1 have product 3 (Example 10.23), but it does not bound them separately.

For a compact oriented manifold M of dimension n the model is Verdier's. Let $a : M \to \mathrm{pt}$ be the map to a point. Then $\omega_M = a^!\mathbf{Z}$ is the orientation sheaf placed in degree $-n$, and the functor $\mathbb{D} = R\mathcal{H}om(-, \omega_M)$ is an involution on constructible complexes. Duality is the isomorphism $R\Gamma(M, \mathbb{D}K) \simeq R\mathrm{Hom}(R\Gamma_c(M, K), \mathbf{Z})$.

We follow the same route. For a scheme X with structure map $a_X : X \to s = \operatorname{Spec} k$, put $\omega_X = a_X^!\Lambda_s$ and $\mathbb{D}_X = R\mathcal{H}om(-, \omega_X)$ (Definition 10.2). The adjunction $Ra_{X!} \dashv a_X^!$ gives for free a pairing of $R\Gamma_c(X, K)$ with $R\Gamma(X, \mathbb{D}_X K)$, and it also gives two exchange formulas

$$\mathbb{D}_Y Rf_! \simeq Rf_* \mathbb{D}_X, \qquad f^! \mathbb{D}_Y \simeq \mathbb{D}_X f^*$$

(Lemma 10.3). Over an algebraically closed field the first formula, for a_X, is global duality, $R\Gamma(X, \mathbb{D}_X K) \simeq R\mathrm{Hom}_\Lambda(R\Gamma_c(X, K), \Lambda)$ (Theorem 10.10). For smooth X of pure dimension d, smooth purity gives $\omega_X \simeq \Lambda_X(d)[2d]$, and global duality becomes Poincaré duality: the pairing

$$R\Gamma_c(X, K) \otimes^{\mathbf{L}} R\Gamma(X, K^\vee(d))[2d] \longrightarrow \Lambda, \qquad K^\vee = R\mathcal{H}om(K, \Lambda_X),$$

built from evaluation and the trace map, is perfect (Theorem 10.12).

For X smooth, proper and connected and Λ a field, this will give a perfect cup product pairing $H^i \times H^{2d-i} \to \Lambda(-d)$ compatible with Frobenius (Corollary 10.17 and Theorem 10.21 in Section 10.5).

What the adjunction does not give is that the pairing is perfect. For that we need Verdier biduality: the map $K \to \mathbb{D}_X \mathbb{D}_X K$ is an isomorphism for every $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ and every admissible Λ (Theorem 10.8). Then $\mathbb{D}_X$ is an anti-equivalence, and applying the first two exchange formulas to $\mathbb{D}_X K$ gives the other two, $\mathbb{D}_Y Rf_* \simeq Rf_! \mathbb{D}_X$ and $\mathbb{D}_X f^! \simeq f^* \mathbb{D}_Y$ (Corollary 10.9).

Biduality cannot be proved by dévissage, because dualizing a piece $t_!P$, with P locally constant on a smooth stratum $t : T \hookrightarrow X$, twice needs the exchange formula $\mathbb{D}_X Rt_* \simeq t_! \mathbb{D}_T$, which is itself a consequence of biduality (Section 10.2). We argue instead by induction on dimension, restricting to the geometric generic fibre of a function, where duality commutes with restriction up to the twist $(1)[2]$ (Lemmas 10.6 and Lemma 10.7).

Global duality was announced by Verdier and proved by Deligne [SGA4, XVIII]. Biduality for constructible complexes over a field is due to Deligne [SGA4½, Th. finitude].

A perfect pairing of complexes induces perfect pairings of cohomology modules over a field (Corollary 10.13), but not over $\mathcal{O}_E$ (Remark 10.14). For this reason we formulate the duality statements in the derived category, as perfect pairings of perfect complexes (Definition 10.11).

10.1 Exchange formulas

After defining the dualizing complex ω_X, the Verdier dual $\mathbb{D}_X$ and the double-dual map $\delta_K : K \to \mathbb{D}_X \mathbb{D}_X K$, we prove the two exchange formulas that need only adjunctions (Lemma 10.3). They say that $\mathbb{D}$ turns $Rf_!$ into Rf_* and f^* into $f^!$. For example, for $f = a_X$ and $K = \Lambda_X$ the first says that $R\Gamma(X, \omega_X)$ is the dual of $R\Gamma_c(X, \Lambda)$, as for the homology of a manifold. The other half, with the roles of $!$ and $*$ reversed, needs biduality.

The induction in the proof of Theorem 10.8 ends on the geometric point, the case $\dim X = 0$. There we want duality to be the duality of finite projective modules, and for this constructible

complexes must be perfect. Over $\mathbf{Z}/\ell^n$ this is a real condition; for instance, the module $\mathbf{Z}/\ell$ has infinite projective dimension over $\mathbf{Z}/\ell^2$. This is why we require the coefficients to have finite Tor dimension. With this requirement, Lemma 10.1 shows that D^b_{ctf} of the point is exactly the category of perfect complexes.

Lemma 10.1 (Perfect complexes at the geometric point)**.** *Suppose k is algebraically closed. The category $D^b_{\mathrm{ctf}}(s,\Lambda)$ identifies with $\mathrm{Perf}(\Lambda)$, the category of bounded complexes of finite projective Λ-modules up to quasi-isomorphism. For $K \in D^b_{\mathrm{ctf}}(X,\Lambda)$, both*

$$R\Gamma_c(X,K) = Ra_{X!}K, \qquad R\Gamma(X,K) = Ra_{X*}K$$

and every geometric stalk $K_{\bar{x}}$ are perfect complexes. Their duals $R\mathrm{Hom}_\Lambda(-,\Lambda)$ are perfect, and the evaluation map $P \to R\mathrm{Hom}_\Lambda(R\mathrm{Hom}_\Lambda(P,\Lambda),\Lambda)$ is an isomorphism for each of them.

Proof. Once $D^b_{\mathrm{ctf}}(s,\Lambda)$ is identified with $\mathrm{Perf}(\Lambda)$, the other assertions follow, because the direct images and the stalks of K are objects of $D^b_{\mathrm{ctf}}(s,\Lambda)$.

By the comparisons of Section 2.13, taking global sections identifies $D^b_{\mathrm{ctf}}(s,\Lambda)$ with the full subcategory of $D(\Lambda)$ of complexes with bounded finitely generated cohomology and finite Tor dimension, and we show that these complexes are perfect separately for the three kinds of coefficients. A constructible complex with finite coefficients has bounded finite cohomology and finite Tor amplitude, so it is perfect: take a bounded-above resolution by finite free modules and truncate it below the Tor bound. Then the last syzygy that remains is finitely presented and flat, hence projective, so the truncated resolution is a bounded complex of finite projective modules. With integral coefficients, by contrast, the constructible complexes on the point are those with bounded finitely generated $\mathcal{O}_E$-cohomology, and since a discrete valuation ring has global dimension one, these complexes are perfect. Finally, with field coefficients, bounded complexes with finite-dimensional cohomology are perfect.

By Proposition 7.1, the two direct images of a_X lie in $D^b_{\mathrm{ctf}}(s,\Lambda)$ and are therefore perfect. If we represent such a complex by a bounded complex of finite projective modules, its termwise dual is again of this form, and the double evaluation map is termwise an isomorphism. The stalk $K_{\bar{x}}$ is the complex of global sections of $\bar{x}^*K \in D^b_{\mathrm{ctf}}(s,\Lambda)$ (Proposition 7.1), so it is perfect as well. ∎

On the point, then, duality is the linear duality of perfect complexes, and the double dual is the identity. On a general X, however, we dualize into $a^!_X\Lambda_s$ instead of Λ. By the adjunction $Ra_{X!} \dashv a^!_X$, maps into $a^!_X\Lambda_s$ are the same as linear forms on compactly supported cohomology.

Definition 10.2 (Verdier dual and evaluation)**.** The *dualizing complex* and the *Verdier dual* are

$$\omega_X = a^!_X\Lambda_s, \qquad \mathbb{D}_X K = R\mathcal{H}om_{\Lambda_X}(K,\omega_X).$$

Thus $\mathbb{D}_X : D(X_{\mathrm{pro\acute{e}t}},\Lambda)^{\mathrm{op}} \to D(X_{\mathrm{pro\acute{e}t}},\Lambda)$ is an exact functor. The *double-dual morphism*

$$\delta_K : K \longrightarrow \mathbb{D}_X\mathbb{D}_X K$$

is the tensor–Hom adjoint of the composite $K \otimes^{\mathbf{L}} \mathbb{D}_X K \simeq \mathbb{D}_X K \otimes^{\mathbf{L}} K \to \omega_X$ of the symmetry isomorphism, with its Koszul sign, and evaluation.

On the geometric point we have $\omega_s = \Lambda_s$, so δ is the evaluation map of Lemma 10.1.

The proof of biduality uses only the exchange formulas that follow from adjunctions. We first write down the comparison maps. Let $f : X \to Y$ be a morphism, $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ and $L \in D^b_{\mathrm{ctf}}(Y, \Lambda)$. The map

$$Rf_* R\mathcal{H}om_X(K, f^! L) \longrightarrow R\mathcal{H}om_Y(Rf_! K, L) \tag{10.1}$$

is the tensor–Hom adjoint of the composite

$$\begin{aligned} Rf_* R\mathcal{H}om_X(K, f^! L) \otimes^{\mathbf{L}} Rf_! K &\simeq Rf_!(f^* Rf_* R\mathcal{H}om_X(K, f^! L) \otimes^{\mathbf{L}} K) \\ &\longrightarrow Rf_!(R\mathcal{H}om_X(K, f^! L) \otimes^{\mathbf{L}} K) \longrightarrow Rf_! f^! L \xrightarrow{\varepsilon_f} L \end{aligned}$$

of the projection formula, the counit of $f^* \dashv Rf_*$, evaluation and the counit ε_f of $Rf_! \dashv f^!$. In particular, for $L = \omega_Y$, using $f^! \omega_Y = f^! a_Y^! \Lambda_s = \omega_X$, it is a map $Rf_* \mathbb{D}_X K \to \mathbb{D}_Y Rf_! K$. The map $f^! \mathbb{D}_Y L \to \mathbb{D}_X f^* L$ is the tensor–Hom adjoint of the map $f^! \mathbb{D}_Y L \otimes^{\mathbf{L}} f^* L \to f^! \omega_Y = \omega_X$ that corresponds under $Rf_! \dashv f^!$ to

$$Rf_!(f^! \mathbb{D}_Y L \otimes^{\mathbf{L}} f^* L) \simeq Rf_! f^! \mathbb{D}_Y L \otimes^{\mathbf{L}} L \xrightarrow{\varepsilon_f \otimes 1} \mathbb{D}_Y L \otimes^{\mathbf{L}} L \longrightarrow \omega_Y,$$

the last arrow being evaluation.

Lemma 10.3 (Internal adjunction and two exchange formulas)**.** *Let $f : X \to Y$ be a morphism, $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ and $L \in D^b_{\mathrm{ctf}}(Y, \Lambda)$. The map (10.1) and the two maps derived from it and defined above are isomorphisms*

$$R\mathcal{H}om_Y(Rf_! K, L) \simeq Rf_* R\mathcal{H}om_X(K, f^! L), \tag{10.2}$$

$$e_K : \mathbb{D}_Y Rf_! K \xrightarrow{\sim} Rf_* \mathbb{D}_X K, \tag{10.3}$$

$$f^! \mathbb{D}_Y L \simeq \mathbb{D}_X f^* L. \tag{10.4}$$

If f is proper, then the composite

$$Rf_* K \xrightarrow{Rf_* \delta_K} Rf_* \mathbb{D}_X \mathbb{D}_X K \simeq \mathbb{D}_Y Rf_* \mathbb{D}_X K$$

of $Rf_* \delta_K$ and $e_{\mathbb{D}_X K}$ equals the composite

$$Rf_* K \xrightarrow{\delta_{Rf_* K}} \mathbb{D}_Y \mathbb{D}_Y Rf_* K \simeq \mathbb{D}_Y Rf_* \mathbb{D}_X K$$

of $\delta_{Rf_* K}$ and $\mathbb{D}_Y$ applied to e_K.

Proof. We check each isomorphism by Yoneda. After applying $\mathrm{Hom}(N, -)$, we identify both sides with the same Hom group by the adjunctions and the projection formula. We only have to make sure that the test objects N lie in the localizing category on which $Rf_! \dashv f^!$ is defined.

The adjunction $Rf_! \dashv f^!$ lives on the localizing categories $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ and $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$, and the functors in the argument preserve them. Indeed, pullback is exact, preserves direct sums, and carries the constructible generators to constructible objects by Proposition 7.1, so it maps $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$ into $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$. For the same reasons, tensoring with a fixed constructible complex preserves the localizing category. In particular, for $N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$ the objects $f^* N \otimes^{\mathbf{L}} K$ and $N \otimes^{\mathbf{L}} Rf_! K$ lie in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X, \Lambda))$ and $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y, \Lambda))$, respectively.

The second step of the computation below is the projection formula

$$Rf_!(f^*N \otimes^{\mathbf{L}} K) \simeq N \otimes^{\mathbf{L}} Rf_!K \qquad (N \in \mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))).$$

For finite coefficients it follows from Proposition 5.15 and [Stacks, 0GL5], and for rational coefficients it is Proposition 6.23. For integral coefficients we compactify $f = pj$, with j an open immersion and p proper, and combine Lemma 6.26 with $j_!(M \otimes^{\mathbf{L}} j^*N) \simeq j_!M \otimes^{\mathbf{L}} N$.

By Proposition 7.1, both $R\mathcal{H}om_Y(Rf_!K, L)$ and $Rf_*R\mathcal{H}om_X(K, f^!L)$ belong to $D^b_{\mathrm{ctf}}(Y,\Lambda)$, so we may test the comparison map between them by Yoneda inside the full subcategory $\mathrm{Loc}(D^b_{\mathrm{ctf}}(Y,\Lambda))$. For every object N of this category we obtain, by the adjunctions and the projection formula,

$$\begin{aligned}
\mathrm{Hom}_Y(N, R\mathcal{H}om_Y(Rf_!K, L)) &\simeq \mathrm{Hom}_Y(N \otimes^{\mathbf{L}} Rf_!K, L)\\
&\simeq \mathrm{Hom}_Y(Rf_!(f^*N \otimes^{\mathbf{L}} K), L)\\
&\simeq \mathrm{Hom}_X(f^*N \otimes^{\mathbf{L}} K, f^!L)\\
&\simeq \mathrm{Hom}_Y(N, Rf_*R\mathcal{H}om_X(K, f^!L)).
\end{aligned}$$

The composite bijection is induced by the map (10.1). By Yoneda this map is an isomorphism, which is (10.2). Taking $L = \omega_Y$ and using $f^!\omega_Y = \omega_X$, we get e_K. For (10.4), on the other hand, note that both $f^!\mathbb{D}_Y L$ and $\mathbb{D}_X f^*L$ lie in $D^b_{\mathrm{ctf}}(X,\Lambda)$. Since tensoring with f^*L preserves $\mathrm{Loc}(D^b_{\mathrm{ctf}}(X,\Lambda))$, we may test on objects M of this category:

$$\begin{aligned}
\mathrm{Hom}_X(M, f^!\mathbb{D}_Y L) &\simeq \mathrm{Hom}_Y(Rf_!M, R\mathcal{H}om_Y(L, \omega_Y))\\
&\simeq \mathrm{Hom}_Y(Rf_!M \otimes^{\mathbf{L}} L, \omega_Y)\\
&\simeq \mathrm{Hom}_X(M \otimes^{\mathbf{L}} f^*L, f^!\omega_Y)\\
&\simeq \mathrm{Hom}_X(M, \mathbb{D}_X f^*L).
\end{aligned}$$

Again, the bijection is induced by the map defined before the lemma, and by Yoneda we get (10.4).

The comparison of double-dual maps is used in the proof of biduality for $f = a_X$, where it identifies $R\Gamma(X, \delta_K)$ with the double-dual map of the perfect complex $R\Gamma(X, K)$. Both maps are determined by pairings, and we compare the pairings. For proper f we have $Rf_! = Rf_*$, and the isomorphism e_K is adjoint to the pairing

$$Rf_*K \otimes^{\mathbf{L}} Rf_*\mathbb{D}_X K \longrightarrow Rf_*(K \otimes^{\mathbf{L}} \mathbb{D}_X K) \longrightarrow Rf_*\omega_X \longrightarrow \omega_Y.$$

Using e_K and $e_{\mathbb{D}_X K}$, we view both $Rf_*\delta_K$ and δ_{Rf_*K} as maps $Rf_*K \to \mathbb{D}_Y Rf_*\mathbb{D}_X K$. Up to the symmetry of the tensor product, both are adjoint to this pairing. Indeed, for δ_{Rf_*K} this holds because under e_K evaluation on Y corresponds to the pairing. Similarly, for $Rf_*\delta_K$ it holds because δ_K is adjoint to evaluation on X and $e_{\mathbb{D}_X K}$ is adjoint to the analogous pairing. ∎

10.2 Verdier biduality

On a smooth curve C over an algebraically closed field, with $\omega_C = \Lambda_C(1)[2]$, biduality is easy to see for the two basic kinds of sheaves. Indeed, for a locally constant sheaf L of free Λ-modules the dual is $L^\vee(1)[2]$, and dualizing twice gives back L. For a skyscraper i_*M at a closed point $i : x \to C$, the exchange formula $\mathbb{D}_C i_* \simeq i_*\mathbb{D}_x$ for the proper map i shows that the dual is $i_*\mathbb{D}_x M$, so biduality reduces to biduality on the point. In general, a constructible sheaf is an extension of

i_*M by $j_!L$, where now i is the inclusion of a finite set of closed points and L is locally constant on its complement $j : U \hookrightarrow C$. The open piece is the difficult one: we have $\mathbb{D}_C(j_!L) \simeq Rj_*\mathbb{D}_U L$, and to dualize Rj_* we would need the second exchange formula. That formula, however, depends on biduality. In dimension one this can still be checked by hand, using the cohomology of the henselian punctured disc, whereas in higher dimension the direct images Rj_* along the boundary of a stratum have no such description.

Example 10.4 (The dual of $j_!\Lambda$ on the line)**.** Let k be algebraically closed, $X = \mathbf{A}^1_k$, $i : 0 \to X$ the origin and $j : U = \mathbf{G}_{m,k} \to X$ its complement, and take Λ finite. The sheaf $j_!\Lambda_U$ has stalk zero at 0, but its dual does not. By (10.3) for j and $\omega_U = \Lambda_U(1)[2]$,

$$\mathbb{D}_X(j_!\Lambda_U) \simeq Rj_*\mathbb{D}_U\Lambda_U = Rj_*\Lambda_U(1)[2],$$

and by Example 6.25 its stalk at 0 is $\Lambda(1)$ in degree -2 and Λ in degree -1. The term in degree -1 comes from $R^1j_*\Lambda_U = i_*\Lambda(-1)$. The same answer comes out of the triangle $j_!\Lambda_U \to \Lambda_X \to i_*\Lambda \to$. Dualizing it, with $\mathbb{D}_X\Lambda_X = \Lambda_X(1)[2]$ and $\mathbb{D}_X(i_*\Lambda) = i_*\mathbb{D}_0\Lambda = i_*\Lambda$, gives a triangle

$$i_*\Lambda \longrightarrow \Lambda_X(1)[2] \longrightarrow \mathbb{D}_X(j_!\Lambda_U) \longrightarrow .$$

On stalks at 0 the first map, from degree 0 to degree -2, is zero. Hence the cone has $\Lambda(1)$ in degree -2 and Λ in degree -1.

Dualizing once more, biduality predicts $\mathbb{D}_X\mathbb{D}_X(j_!\Lambda_U) \simeq j_!\Lambda_U$, with zero stalk at 0. In other words, the second dual cancels the stalk that the first created. Here this follows from the two easy cases. The map δ is a morphism of triangles, and it is an isomorphism on Λ_X and on $i_*\Lambda$, so by the five lemma it is an isomorphism on $j_!\Lambda_U$. The argument used, however, that Λ_U extends to the locally constant sheaf Λ_X. A locally constant L on U with nontrivial monodromy around 0 has no such extension, and that is the case which needs the henselian disc.

The proof of Theorem 10.8 needs two preliminary facts. First, Lemma 10.5 shows that $\mathbb{D}_X$ preserves $D^b_{\mathrm{ctf}}(X, \Lambda)$. Second, Lemmas 10.6 and Lemma 10.7 show that $\mathbb{D}_X$ and δ commute with restriction to a geometric generic fibre up to the twist $(1)[2]$, and this twist cancels in the double dual.

Lemma 10.5 (Verdier duality preserves constructibility)**.** *For every admissible Λ, $\mathbb{D}_X$ restricts to a functor $D^b_{\mathrm{ctf}}(X, \Lambda)^{\mathrm{op}} \to D^b_{\mathrm{ctf}}(X, \Lambda)$.*

Proof. The lemma follows from the stability of D^b_{ctf} under the six operations. Proposition 7.1 shows that $a^!_X$ and internal Hom preserve D^b_{ctf}. As Λ_s lies in $D^b_{\mathrm{ctf}}(s, \Lambda)$, both $\omega_X = a^!_X\Lambda_s$ and $\mathbb{D}_X K$ lie in $D^b_{\mathrm{ctf}}(X, \Lambda)$ for $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$. We also give a direct argument with finite coefficients and smooth strata, because the proof of Lemma 10.6 uses its dévissage to generators $t_!P$ again.

Dévissage. The complex K is built by finitely many cones from complexes $t_!P$, for locally closed $t : T \hookrightarrow X$ with T smooth and P a locally constant perfect complex. To see this, choose a finite stratification by smooth strata, disregarding nilpotents, on which the restrictions P of K are locally constant perfect complexes, and use the localization triangles. The bounded flat models of Lemma 3.13 and their stupid filtrations make this precise. Since $\mathbb{D}_X$ is exact and $D^b_{\mathrm{ctf}}(X, \Lambda)$ is closed under cones, it suffices to show that $\mathbb{D}_X(t_!P)$ lies in $D^b_{\mathrm{ctf}}(X, \Lambda)$.

The generators. Internal adjunction (10.3) for t gives

$$\mathbb{D}_X(t_!P) \simeq Rt_*\mathbb{D}_T P.$$

On a smooth stratum of dimension r, Theorem 9.3 identifies the dual $\mathbb{D}_T P$ with $P^\vee(r)[2r]$, which is locally a perfect complex. By Proposition 7.1 its ordinary direct image lies in $D^b_{\mathrm{ctf}}(X,\Lambda)$. ■

We prove biduality by induction on dimension, restricting the double-dual morphism to the geometric generic fibre of a map to $\mathbf{A}^1_k$. This needs comparison maps between the operations on a scheme and on its geometric generic fibre. Let B/k be a smooth integral curve and $\bar\eta = \operatorname{Spec}\overline{k(B)}$, and for a B-scheme T of finite type let $g_T : T_{\bar\eta} = T \times_B \bar\eta \to T$ be the projection, with pullback $g_T^* : D(T_{\mathrm{pro\acute{e}t}}, A) \to D(T_{\bar\eta,\mathrm{pro\acute{e}t}}, A)$, where A is a finite coefficient ring. First, for a morphism $h : T \to X$ of B-schemes the base change map $g_X^* Rh_* L \to Rh_{\bar\eta,*} g_T^* L$ is adjoint to g_T^* of the counit $h^* Rh_* L \to L$. Second, for a closed immersion $i : Z \to X$ of B-schemes, the pullback $i_{\bar\eta,*} g_Z^* i^! M = g_X^* i_* i^! M \to g_X^* M$ of the counit corresponds under $i_{\bar\eta,*} \dashv i^!_{\bar\eta}$ to a map $g_Z^* i^! M \to i^!_{\bar\eta} g_X^* M$. Finally, for internal Hom, $g_X^* R\mathcal{H}om_X(K,M) \to R\mathcal{H}om_{X_{\bar\eta}}(g_X^* K, g_X^* M)$ is the tensor–Hom adjoint of g_X^* of evaluation $R\mathcal{H}om_X(K,M) \otimes^{\mathbf{L}} K \to M$. The next lemma says that these maps are isomorphisms.

Lemma 10.6 (Restriction to a geometric generic fibre)**.** *In the setting above, let X be a B-scheme of finite type, let $h : T \to X$ be a separated morphism of B-schemes of finite type, let $i : Z \to X$ be a closed immersion, and let L on T and K, M on X be bounded constructible complexes, with $K \in D^b_{\mathrm{ctf}}(X, A)$. The three maps*

$$\begin{aligned} g_X^* Rh_* L &\longrightarrow Rh_{\bar\eta,*} g_T^* L, \\ g_Z^* i^! M &\longrightarrow i^!_{\bar\eta} g_X^* M, \\ g_X^* R\mathcal{H}om_X(K,M) &\longrightarrow R\mathcal{H}om_{X_{\bar\eta}}(g_X^* K, g_X^* M) \end{aligned}$$

defined above are isomorphisms.

Proof. We prove the comparisons in the order stated, each using the previous ones: the direct image by continuity along a limit of étale maps, $i^!$ by the localization triangle, and internal Hom by dévissage to generators $t_! P$. Throughout, with finite coefficients Proposition 5.15 lets us compute $D^b_{\mathrm{ctf}}(X, A)$ and the direct images by their classical étale counterparts.

Direct image. The geometric generic point is a limit of étale B-schemes. Indeed, a separable closure of $k(B)$ is the inverse limit of nonempty affine opens in the normalizations of B in its finite separable extensions, with the branch loci removed, and each stage is étale over an open of B. The functor Rh_* commutes with pullback to each stage, because pullback to an étale B-scheme is restriction to an object of the small site. It remains to pass to the limit.

Every affine object étale over the limit descends to a stage. Continuity of cohomology (Corollary 2.29) therefore shows that, degree by degree, the derived sections with finite coefficients commute with this filtered limit along affine transition maps, provided the complexes vanish below one fixed degree. They do, because L is bounded. Moreover, all the direct images lie in one fixed range of degrees. The base changes of h to the stages are of finite type of bounded dimension, and [Stacks, 0F10] bounds their ℓ-cohomological dimensions (Definition 2.45) independently of the stage. This proves the first comparison over the separable closure. It also holds over the algebraic closure, since a purely inseparable extension induces the same classical étale topos.

Exceptional inverse image. Let j be the open complement of i and pull back

$$i_* i^! M \longrightarrow M \longrightarrow Rj_* j^* M \longrightarrow .$$

We compare the pulled-back triangle with the triangle with closed support on the generic fibre, whose first term is $i_{\bar\eta,*} i^!_{\bar\eta} g_X^* M$. Their second terms agree. The third terms agree too, because

the first comparison identifies the pullback of Rj_*j^*M with $Rj_{\bar\eta,*}j_{\bar\eta}^*g_X^*M$, and the map between them is the same. It follows that the first terms are determined functorially. Indeed, there are no nonzero maps, in any degree, from a complex supported on $Z_{\bar\eta}$ to one of the form $Rj_{\bar\eta,*}N$, so the first term of each triangle is the fibre of the second map. Now pullback commutes with direct image along a closed immersion, so the first term of the pulled-back triangle is $i_{\bar\eta,*}g_Z^*i^!M$. By construction, the induced map $i_{\bar\eta,*}g_Z^*i^!M \to i_{\bar\eta,*}i_{\bar\eta}^!g_X^*M$ is adjoint to the pulled-back counit, that is, it is $i_{\bar\eta,*}$ of our comparison map, and it is an isomorphism. Since $i_{\bar\eta,*}$ is fully faithful, the comparison map itself is an isomorphism.

Internal Hom. The complexes K for which the comparison is an isomorphism are closed under cones and shifts, because both sides are exact functors of K. The dévissage of Lemma 10.5 therefore reduces us to $K = t_!P$, where t is the inclusion of a stratum and P is locally perfect on it. For these generators, internal adjunction and the dualizability of the locally perfect complex P express internal Hom through operations we have already compared:

$$R\mathcal{H}om_X(t_!P, M) \simeq Rt_*(P^\vee \otimes^{\mathbf{L}} t^!M).$$

We check that g^* commutes with each operation on the right. For $t^!$ we factor t as a closed immersion followed by an open immersion: for the open immersion $t^!$ is restriction, which commutes with g^*, and for the closed immersion we use the second comparison. Similarly, pullback commutes with Rt_* by the first comparison, and it commutes with tensor products and with $P^\vee$. The first two comparisons apply here, because by Lemma 10.5 all complexes used remain bounded constructible. Hence the comparison holds for $K = t_!P$. ∎

For $B = \mathbf{A}^1_k$ the dualizing complex changes by a twist under restriction to the geometric generic fibre, and this twist cancels in the double dual.

Lemma 10.7 (Double duals on the geometric generic fibre)**.** *In the setting of Lemma 10.6, let $B = \mathbf{A}^1_k$ and $K \in D^b_{\mathrm{ctf}}(X, A)$. For every affine open V of X there is an isomorphism $g_V^*\omega_V \simeq \omega_{V_{\bar\eta}}(1)[2]$, and the isomorphism $g_V^*\mathbb{D}_V\mathbb{D}_V(K|_V) \simeq \mathbb{D}_{V_{\bar\eta}}\mathbb{D}_{V_{\bar\eta}}g_V^*(K|_V)$ obtained from it and from the comparison for internal Hom (applied twice) carries $g_V^*\delta_{K|_V}$ to $\delta_{g_V^*(K|_V)}$.*

Proof. Replacing X by an affine open we may assume X affine. The twist $(1)[2]$ appears because X has one more dimension than its generic fibre, and we see it by embedding both in affine spaces whose dimensions differ by one. To do this, choose a closed immersion i over $\mathbf{A}^1_k$ of X into $\mathbf{A}^r_{\mathbf{A}^1_k}$. Since $\mathbf{A}^r_{\mathbf{A}^1_k}$ is smooth of dimension $r+1$ over k, by composition of exceptional inverse images and Theorem 9.3 we get $\omega_X \simeq i^!A(r+1)[2r+2]$. In the same way, the geometric generic fibre is a closed subscheme of $\mathbf{A}^r_{\bar\eta}$, so $\omega_{X_{\bar\eta}} \simeq i_{\bar\eta}^!A(r)[2r]$. By the second comparison, $g_X^*i^!A(r+1)[2r+2] \simeq i_{\bar\eta}^!A(r+1)[2r+2]$, which is $\omega_{X_{\bar\eta}}(1)[2]$.

For the double-dual maps, note that the comparison for internal Hom is adjoint to the pullback of the evaluation $K \otimes^{\mathbf{L}} R\mathcal{H}om_X(K, M) \to M$, so it carries evaluation to evaluation. Together with the twist it gives $g_X^*\mathbb{D}_X K \simeq \mathbb{D}_{X_{\bar\eta}}(g_X^*K)(1)[2]$. This twist does no harm, because the double-dual map does not change if the dualizing complex is replaced by an isomorphic object ω': an isomorphism $\omega \simeq \omega'$ is used once in each of the two duals, and the two uses cancel in δ. We apply this to $\omega' = \omega_{X_{\bar\eta}}(1)[2]$, using that $R\mathcal{H}om(R\mathcal{H}om(N, \omega(1)[2]), \omega(1)[2]) \simeq R\mathcal{H}om(R\mathcal{H}om(N, \omega), \omega)$ compatibly with evaluation. It follows that the isomorphism $g_X^*\mathbb{D}_X\mathbb{D}_X K \simeq \mathbb{D}_{X_{\bar\eta}}\mathbb{D}_{X_{\bar\eta}}g_X^*K$ obtained by applying the comparison twice carries $g_X^*\delta_K$ to $\delta_{g_X^*K}$. ∎

With these lemmas we can prove biduality.

Theorem 10.8 (Verdier biduality)**.** *For every admissible coefficient ring Λ and every scheme X,*

$$\delta_K : K \xrightarrow{\sim} \mathbb{D}_X \mathbb{D}_X K \qquad (K \in D^b_{\mathrm{ctf}}(X, \Lambda)).$$

Thus $\mathbb{D}_X$ is an exact anti-equivalence of $D^b_{\mathrm{ctf}}(X, \Lambda)$ with itself.

Proof. We first treat finite coefficients A over an algebraically closed field k, and deduce the other coefficient rings and a finite base field at the end. In this main case we reduce to proper X, show by induction that the cone of δ_K has zero stalks at all nonclosed points, and compute global sections to see that its stalks at the finitely many remaining closed points vanish. This induction is on $d = \dim X$. Since the induction step passes to a geometric generic fibre over $\overline{k(t)}$, we run it simultaneously over all algebraically closed fields of characteristic different from ℓ. In dimension zero, X_{red} is a finite disjoint union of copies of $\operatorname{Spec} k$, and by nilpotent invariance we may replace X by it. On each copy, moreover, $\omega = \Lambda$ and δ_K is the evaluation map of a perfect complex, which is an isomorphism by Lemma 10.1.

We may assume that X is proper of dimension d; properness is used only in the last step, to compute global sections. This reduction rests on the fact that the double-dual morphism commutes with restriction to opens. This holds because restriction to an open $j : U \to X$ commutes with internal Hom, satisfies $j^*\omega_X = j^!\omega_X = \omega_U$, and identifies all the evaluation maps. Given this, choose a compactification $j : X \hookrightarrow \overline{X}$, replace the target by the closure of X, which has dimension d, and extend K by zero. Then $j_!K$ is in $D^b_{\mathrm{ctf}}(\overline{X}, A)$, and the restriction of its double-dual morphism to X is δ_K.

Let $Q = \mathrm{Cone}(\delta_K)$. We have to show that $Q = 0$. Since Q is bounded and constructible by Lemma 10.5, it is enough to show that all its geometric stalks vanish.

At nonclosed points the induction hypothesis applies. Let x be a nonclosed point of X. We choose an affine neighbourhood U of x and an algebra generator t whose image in $\kappa(x)$ is transcendental over k. Such a generator exists because a nonclosed point has positive transcendence degree. The induced map to $\mathbf{A}^1_k$ then sends x to the generic point. Moreover, an irreducible component of U that meets the generic fibre dominates $\mathbf{A}^1_k$, and its generic fibre has dimension one less than the component. Hence the geometric generic fibre has dimension at most $d - 1$, and the induction hypothesis applies to it. Let g be the projection from the geometric generic fibre of U. By Lemma 10.7, locally on U the complex g^*Q is isomorphic to the cone of δ_{g^*K}. This cone vanishes by the induction hypothesis over $\overline{k(t)}$. As x lies over the generic point of $\mathbf{A}^1_k$, every geometric point above x factors through the geometric generic fibre. Therefore Q has zero geometric stalks above every nonclosed point of X.

Only finitely many closed points are left. Indeed, the support of each cohomology sheaf of Q is constructible, and by the previous paragraph it consists of closed points. A constructible subset of a Noetherian scheme of finite type which contains only closed points is finite, since each positive-dimensional locally closed piece would contain a nonclosed generic point. Thus the finitely many cohomology sheaves of Q are supported on a single finite set of closed points.

The stalks at these closed points vanish because X is proper and Q has no global cohomology. To see that $R\Gamma(X, Q) = 0$, we apply the exchange for proper maps in Lemma 10.3 to a_X. It identifies $R\Gamma(X, \delta_K)$ with the double-dual morphism of the perfect complex $R\Gamma(X, K)$, which is an isomorphism by Lemma 10.1. On the other hand, a sheaf $\mathcal{F}$ supported on a finite set S of closed points is $\bigoplus_{x \in S} x_* \mathcal{F}_{\bar{x}}$. Since each residue field is algebraically closed, $H^0(X, \mathcal{F}) = \bigoplus_x \mathcal{F}_{\bar{x}}$ and the higher cohomology vanishes. The hypercohomology spectral sequence of the bounded complex Q therefore degenerates, and it shows that

$$0 = H^m R\Gamma(X, Q) = \bigoplus_x \mathcal{H}^m(Q)_{\bar{x}}.$$

All these stalks vanish. Since the cohomology sheaves of Q are supported on this finite set, $Q = 0$.

A finite base field reduces to its algebraic closure. Let $\bar{k}$ be an algebraic closure of k and $g : X_{\bar{k}} = X \times_k \bar{k} \to X$ the projection. The three comparisons of Lemma 10.6 hold for g, with the same proofs. Indeed, about $\bar{\eta} \to B$ these proofs use only that it is a limit of étale B-schemes, and here, with $B = \operatorname{Spec} k$ in place of a curve, $\operatorname{Spec} \bar{k}$ is the limit of the finite étale k-schemes $\operatorname{Spec} k'$, for $k' \subset \bar{k}$ finite over k. Unlike in Lemma 10.7, the dualizing complex does not change. For an affine open V of X, choose a closed immersion $i : V \hookrightarrow \mathbf{A}^r_k$. By composition of exceptional inverse images and Theorem 9.3, $\omega_V \simeq i^! A(r)[2r]$. The base change $i_{\bar{k}} : V_{\bar{k}} \hookrightarrow \mathbf{A}^r_{\bar{k}}$ is a closed immersion into an affine space of the same dimension, so $\omega_{V_{\bar{k}}} \simeq i_{\bar{k}}^! A(r)[2r]$, and the second comparison gives $g_V^* \omega_V \simeq \omega_{V_{\bar{k}}}$, with no twist and no shift. The argument of Lemma 10.7, with this isomorphism in place of $g_V^* \omega_V \simeq \omega_{V_{\bar{\eta}}}(1)[2]$, shows that the isomorphism $g_V^* \mathbb{D}_V \mathbb{D}_V(K|_V) \simeq \mathbb{D}_{V_{\bar{k}}} \mathbb{D}_{V_{\bar{k}}} g_V^*(K|_V)$ it gives carries $g_V^* \delta_{K|_V}$ to $\delta_{g_V^*(K|_V)}$. The latter is an isomorphism by the case already proved, so g^*Q vanishes on every $V_{\bar{k}}$, and $g^*Q = 0$. Finally, every geometric point of X factors through $X_{\bar{k}}$, since its algebraically closed residue field contains an algebraic closure of k. Hence all geometric stalks of Q vanish, and $Q = 0$ over the finite field as well.

For coefficient rings that are not finite we reduce to finite ones. Indeed, the double-dual map $\delta : \mathrm{id} \to \mathbb{D}_X \mathbb{D}_X$ is a natural transformation of exact functors, built from $a^!_X$, internal Hom and evaluation. Its extensions of scalars are identified by Proposition 7.10. By Proposition 7.12 the finite case therefore implies the theorem for every admissible Λ. ∎

Since $\mathbb{D}_X$ is now an anti-equivalence, we can dualize the formulas of Lemma 10.3. This gives the two exchange formulas with the roles of $!$ and $*$ reversed.

Corollary 10.9 (Verdier exchange formulas)**.** *For a morphism $f : X \to Y$ there are natural isomorphisms of functors on $D^b_{\mathrm{ctf}}(X, \Lambda)$, respectively $D^b_{\mathrm{ctf}}(Y, \Lambda)$,*

$$\mathbb{D}_Y R f_! \simeq R f_* \mathbb{D}_X, \quad \mathbb{D}_Y R f_* \simeq R f_! \mathbb{D}_X,$$
$$\mathbb{D}_X f^* \simeq f^! \mathbb{D}_Y, \qquad \mathbb{D}_X f^! \simeq f^* \mathbb{D}_Y.$$

In particular, for an open $j : U \hookrightarrow X$ and a closed $i : Z \hookrightarrow X$,

$$\mathbb{D}_X j_! \simeq R j_* \mathbb{D}_U, \qquad \mathbb{D}_X R j_* \simeq j_! \mathbb{D}_U, \qquad i^* \mathbb{D}_X \simeq \mathbb{D}_Z i^!.$$

The isomorphisms in the left column are (10.3) and (10.4); those in the right column are obtained from them by biduality, as displayed in the proof.

Proof. Two of the formulas are Lemma 10.3, and we obtain the other two by applying these to the Verdier dual of the argument and using biduality on both schemes. Explicitly, for $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ and $L \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ this reads

$$\mathbb{D}_Y R f_* K \simeq \mathbb{D}_Y R f_* \mathbb{D}_X (\mathbb{D}_X K) \simeq \mathbb{D}_Y \mathbb{D}_Y R f_! \mathbb{D}_X K \simeq R f_! \mathbb{D}_X K,$$

$$\mathbb{D}_X f^! L \simeq \mathbb{D}_X f^! \mathbb{D}_Y (\mathbb{D}_Y L) \simeq \mathbb{D}_X \mathbb{D}_X f^* \mathbb{D}_Y L \simeq f^* \mathbb{D}_Y L.$$

The formulas for j and i are the special cases $f = j$ and $f = i$. ∎

10.3 Poincaré duality

Over an algebraically closed field, $D^b_{\mathrm{ctf}}(s,\Lambda)$ is the category of perfect Λ-complexes, and on it $\mathbb{D}_s$ is $R\mathrm{Hom}_\Lambda(-,\Lambda)$. For $f = a_X$ the first two Verdier exchange formulas therefore become dualities between compactly supported and ordinary cohomology. This is global Verdier duality (Theorem 10.10). It holds for every X of finite type, smooth or not.

For smooth X of pure dimension d we have $\omega_X \simeq \Lambda_X(d)[2d]$. For locally perfect K the Verdier dual is therefore the ordinary dual $K^\vee$ twisted by $(d)[2d]$, and the counit of $Ra_{X!} \dashv a^!_X$ becomes the trace map of Definition 9.24. In this way global duality becomes cup product followed by the trace.

Theorem 10.10 (Global Verdier duality)**.** *Suppose k is algebraically closed. For every $K \in D^b_{\mathrm{ctf}}(X,\Lambda)$ there are natural isomorphisms*

$$R\Gamma(X,\mathbb{D}_X K) \simeq R\mathrm{Hom}_\Lambda(R\Gamma_c(X,K),\Lambda), \tag{10.5}$$
$$R\Gamma_c(X,\mathbb{D}_X K) \simeq R\mathrm{Hom}_\Lambda(R\Gamma(X,K),\Lambda). \tag{10.6}$$

The first is the adjoint of the pairing formed by the cup product with mixed supports, evaluation, and the counit ε_{a_X} of $Ra_{X!} \dashv a^!_X$:

$$R\Gamma_c(X,K) \otimes^{\mathbf{L}} R\Gamma(X,\mathbb{D}_X K) \longrightarrow R\Gamma_c(X,\omega_X) \xrightarrow{\varepsilon_{a_X}} \Lambda.$$

Proof. Both isomorphisms are the first two Verdier exchange formulas of Corollary 10.9 for $f = a_X$, so we only have to recognize the Verdier dual on the point as the linear dual. On s the dualizing complex is Λ_s, so $\mathbb{D}_s = R\mathcal{H}om_s(-,\Lambda_s)$. Since k is algebraically closed, Lemma 10.1 identifies $D^b_{\mathrm{ctf}}(s,\Lambda)$ with $\mathrm{Perf}(\Lambda)$. Under this identification, then, $\mathbb{D}_s$ becomes $R\mathrm{Hom}_\Lambda(-,\Lambda)$, and the two exchange formulas read (10.5) and (10.6). As for the pairing, Lemma 10.3 constructs the exchange formula (10.3) from evaluation and the counit of $Rf_! \dashv f^!$. For $f = a_X$ it is therefore the adjoint of the displayed pairing. ∎

Over a finite base field k the same argument gives us isomorphisms $Ra_{X*}\mathbb{D}_X K \simeq \mathbb{D}_s Ra_{X!}K$ and $Ra_{X!}\mathbb{D}_X K \simeq \mathbb{D}_s Ra_{X*}K$ in $D(s_{\mathrm{pro\acute{e}t}},\Lambda)$, with $\mathbb{D}_s = R\mathcal{H}om_s(-,\Lambda_s)$. These are the first two formulas of Corollary 10.9 for $f = a_X$. After base change to an algebraic closure they become the isomorphisms of the theorem.

The first isomorphism of the theorem is the adjoint of a pairing. We give a name to pairings whose adjoint is an isomorphism.

Definition 10.11 (Perfect pairing)**.** A pairing $P \otimes^{\mathbf{L}} Q \to \Lambda$ of perfect complexes is *perfect* if its adjoint $Q \to R\mathrm{Hom}_\Lambda(P,\Lambda)$ is an isomorphism.

For smooth X, Corollary 9.28 identifies ω_X with $\Lambda_X(d)[2d]$, and global duality becomes the Poincaré pairing of Weil's axiom (c).

Theorem 10.12 (Poincaré duality)**.** *Let X/k be smooth, separated of pure dimension d, with k algebraically closed. For every admissible Λ and $K \in D^b_{\mathrm{ctf}}(X,\Lambda)$, Theorem 9.3 gives isomorphisms*

$$\omega_X \simeq \Lambda_X(d)[2d], \qquad \mathbb{D}_X K \simeq R\mathcal{H}om_X(K,\Lambda_X)(d)[2d].$$

Put $K^\vee = R\mathcal{H}om_X(K, \Lambda_X)$. If K is locally perfect over Λ_X, then so is $K^\vee$, and it is the ordinary dual. Let Tr_X be the map induced on $R\Gamma_c$ by the trace map of a_X (Definition 9.24). The pairing

$$\begin{aligned}\beta_K : \quad R\Gamma_c(X,K) \otimes^{\mathbf{L}} R\Gamma(X, K^\vee(d))[2d] &\longrightarrow R\Gamma_c(X, \Lambda_X(d))[2d] \\ &\xrightarrow{\mathrm{Tr}_X} \Lambda\end{aligned} \tag{10.7}$$

is perfect, and its adjoint is the canonical isomorphism

$$R\Gamma(X, K^\vee(d))[2d] \xrightarrow{\sim} R\mathrm{Hom}_\Lambda(R\Gamma_c(X,K), \Lambda). \tag{10.8}$$

If X is proper, $R\Gamma_c$ may be replaced by $R\Gamma$. For a map $\Lambda \to \Lambda'$ of admissible coefficient rings, the change-of-coefficients isomorphisms of Proposition 7.10 for $R\Gamma_c$, $R\Gamma$ and $R\mathcal{H}om$ carry $\beta_K \otimes^{\mathbf{L}}_\Lambda \Lambda'$ to $\beta_{K_{\Lambda'}}$, where $K_{\Lambda'} = K \otimes^{\mathbf{L}}_\Lambda \Lambda'$.

Proof. The theorem is global duality rewritten by means of smooth purity: once ω_X and $\mathbb{D}_X K$ are identified, the pairing of Theorem 10.10 becomes β_K.

Corollary 9.28 identifies ω_X. Since internal Hom commutes with tensoring its second variable by an invertible Tate twist and a shift, we get the displayed description of $\mathbb{D}_X K$. If K is locally perfect, then locally K is represented by a bounded complex of finite projective modules, whose dual is the termwise dual, with the usual evaluation.

Next, β_K is the pairing of Theorem 10.10 for K, transported along the isomorphism $\omega_X \simeq \Lambda_X(d)[2d]$. Indeed, under this isomorphism evaluation $K \otimes^{\mathbf{L}} \mathbb{D}_X K \to \omega_X$ becomes evaluation on coefficients $K \otimes^{\mathbf{L}} K^\vee(d)[2d] \to \Lambda_X(d)[2d]$. Moreover, the trace map of Definition 9.24 is by definition the counit ε_{a_X} precomposed with γ_{a_X}, so the counit becomes Tr_X. Hence the adjoint of β_K is the isomorphism (10.5), which is (10.8). By Lemma 10.1 both factors of β_K are perfect complexes. Since the adjoint of β_K is an isomorphism, β_K is perfect.

For proper X we have $Ra_{X!} = Ra_{X*}$, which gives the assertion on proper X. The last statement holds because the pairing is built from tensor products, the two direct images, evaluation, the isomorphism γ_{a_X} and the trace map, all of which Proposition 7.10 and Theorem 9.3 show to be compatible with every coefficient map. ∎

For $X = \mathbf{A}^1$ and $K = \Lambda$ the theorem pairs $R\Gamma_c(\mathbf{A}^1, \Lambda) = \Lambda(-1)[-2]$ with $R\Gamma(\mathbf{A}^1, \Lambda(1))[2] = \Lambda(1)[2]$, and the pairing is the trace $H^2_c(\mathbf{A}^1, \Lambda(1)) \simeq \Lambda$. Over a field we can pass from this statement about complexes to statements about the individual cohomology groups.

Corollary 10.13 (Degreewise duality over fields)**.** *Let X/k be smooth and separated of pure dimension d, with k algebraically closed, let Λ be a field (a finite coefficient field, $\mathbf{Q}_\ell$, E or $\overline{\mathbf{Q}}_\ell$), let $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$, and put $K^\vee = R\mathcal{H}om_X(K, \Lambda_X)$. For every i the pairings induced by β_K,*

$$H^i_c(X,K) \times H^{2d-i}(X, K^\vee(d)) \longrightarrow \Lambda, \qquad H^i_c(X,K) \times H^{2d-i}(X, K^\vee) \longrightarrow \Lambda(-d),$$

are perfect pairings of finite-dimensional vector spaces. If X is proper, the same holds with H^i_c replaced by H^i.

Proof. We take cohomology of the isomorphism (10.8); over a field this loses nothing, because linear duality is exact. More precisely, a bounded complex with finite-dimensional cohomology is isomorphic in the derived category to the sum of its shifted cohomology spaces, so that $H^{-i}(R\mathrm{Hom}_\Lambda(P, \Lambda)) = \mathrm{Hom}_\Lambda(H^i(P), \Lambda)$. Because of the shift $[2d]$, the source of (10.8) has

$H^{2d-i}(X, K^\vee(d))$ in degree $-i$, and taking H^{-i} we get $H^{2d-i}(X, K^\vee(d)) \simeq \mathrm{Hom}_\Lambda(H^i_c(X, K), \Lambda)$. Since this isomorphism is induced by the pairing in degrees $(i, 2d-i)$, it says exactly that the first pairing is perfect. Here both spaces are finite-dimensional by Lemma 10.1. Finally, the second pairing is the first one with the second factor and the value line twisted by $(-d)$, so it is perfect as well. ∎

The only property of a field used in this proof is that $R\mathrm{Hom}_\Lambda(-, \Lambda)$ commutes with taking cohomology.

Remark 10.14 (Duality on cohomology groups over $\mathcal{O}_E$ and finite rings)**.** Over $\mathcal{O}_E$, derived duality gives only the natural universal coefficient exact sequence

$$0 \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_E}(H^{i+1}_c(X, K), \mathcal{O}_E) \longrightarrow H^{2d-i}(X, K^\vee(d)) \longrightarrow \mathrm{Hom}_{\mathcal{O}_E}(H^i_c(X, K), \mathcal{O}_E) \longrightarrow 0.$$

It comes from applying derived Hom to a bounded finite free model of $R\Gamma_c(X, K)$, since $\mathcal{O}_E$ has global dimension one. The sequence need not split canonically, and because of its Ext^1 term the pairings on individual cohomology modules need not be perfect. For example, on a point the perfect complex $[\mathcal{O}_E \xrightarrow{\pi_E} \mathcal{O}_E]$ in degrees $-1, 0$ has $H^0 = \mathcal{O}_E/\pi_E$, whereas its dual has $H^0 = 0$ and $H^1 = \mathcal{O}_E/\pi_E$. Derived evaluation is an isomorphism, but the degree-zero pairing is not perfect.

Over a finite ring A that is not injective as a module over itself, derived Hom out of a bounded complex of finite projective modules records all the groups $\mathrm{Ext}^i_A(-, A)$ of the cohomology modules, and this information may be lost in their ordinary duals. Moreover, the cohomology modules of a complex in $D^b_{\mathrm{ctf}}(X, A)$ need not have finite Tor dimension, so the derived pairing cannot be reduced to pairings of cohomology sheaves.

10.4 Cohomology in top degree

For constant coefficients on a smooth, proper and connected X, we expect $H^{2d}(X, \Lambda_X(d))$ to be free of rank one, generated by the class of a point, as for a manifold. We use this normalization in the trace formula and in point counting.

For a compact connected oriented manifold this is the statement that integration $H^n(M) \to \mathbf{R}$ is an isomorphism sending a point to 1. Here the statement also has an arithmetic meaning, because geometric Frobenius acts on $\Lambda(-d)$ by q^d. Once we know that the trace map commutes with Frobenius (Theorem 10.21), the theorem below says that Frobenius acts on H^{2d} by q^d and on H^0 trivially. These give the factors $1 - T$ and $1 - q^d T$ of the zeta function.

Combined with Corollary 10.13, the theorem gives the cup product pairing (Corollary 10.17).

Theorem 10.15 (The trace isomorphism in top degree)**.** *Let X/k be smooth, proper and connected of pure dimension d, with k algebraically closed. For every admissible coefficient ring,*

$$H^i(X, \Lambda_X) = 0 \quad (i \notin [0, 2d]), \qquad H^0(X, \Lambda_X) = \Lambda,$$

and the map

$$\mathrm{tr}_X : H^{2d}(X, \Lambda_X) \xrightarrow{\sim} \Lambda(-d),$$

the twist by $(-d)$ of the map $\mathrm{tr}_X(d) : H^{2d}(X, \Lambda_X(d)) \to \Lambda$ induced on H^{2d} by Tr_X of Definition 9.24, is an isomorphism.

Proof. We read off the top degree from Poincaré duality in degree zero over $\mathbf{Z}/\ell^n$, and then change coefficients: to a finite A by extension of scalars, to $\mathcal{O}_E$ by a limit, and to the fields by inverting ℓ.

With finite coefficients, étale cohomology of a d-dimensional scheme of finite type over an algebraically closed field vanishes above degree $2d$, that is, $\mathrm{cd}_\ell(X) \leq 2d$ in the notation of Definition 2.45 [SGA4, X, Cor. 4.3]. This gives the vanishing outside $[0, 2d]$, and $H^0 = \Lambda$ because X is connected.

Now let $\Lambda = \mathbf{Z}/\ell^n$. Since this ring is injective as a module over itself, $R\mathrm{Hom}_{\mathbf{Z}/\ell^n}(-, \mathbf{Z}/\ell^n)$ has cohomology equal to the ordinary dual of the opposite cohomology. Hence Poincaré duality (Theorem 10.12) in degree zero identifies $H^{2d}(X, \mathbf{Z}/\ell^n(d))$ with $\mathrm{Hom}(H^0(X, \mathbf{Z}/\ell^n), \mathbf{Z}/\ell^n) = \mathbf{Z}/\ell^n$. By Theorem 10.12 this identification is given by pairing with H^0 and applying the trace map, so it is $\mathrm{tr}_X(d)$.

A general finite ring A need not be injective over itself, so we reach it by extension of scalars instead. Every finite A is an algebra over some $\mathbf{Z}/\ell^n$, and by Proposition 7.10 the complexes with constant coefficients and their traces over A are obtained by derived extension of scalars from $\mathbf{Z}/\ell^n$. For a complex P with cohomology in degrees $\leq 2d$, right exactness of the tensor product gives $H^{2d}(P \otimes^{\mathbf{L}} A) = H^{2d}(P) \otimes A$. The top cohomology is therefore $\mathbf{Z}/\ell^n \otimes A = A$, and the trace is an isomorphism. Since derived extension of scalars does not raise the top degree, the same upper bound holds.

For $\mathcal{O}_E$, we write $R\Gamma(X, \mathcal{O}_{E,X}(d))$ as the derived inverse limit of the complexes $R\Gamma(X, (\mathcal{O}_E/\pi_E^n)_X(d))$. Its cohomology fits into the Milnor sequence $0 \to \lim_n{}^1 H^{i-1} \to H^i \to \lim_n H^i \to 0$. Since the cohomology groups at each level are finite, they satisfy the Mittag–Leffler condition and there is no $\lim^1$ contribution to the top group. The traces in top degree are compatible, so we get $H^{2d}(X, \mathcal{O}_{E,X}(d)) = \mathcal{O}_E$, with the same vanishing bounds. The lower bound and H^0 also follow directly from ordinary sheaf cohomology and connectedness. Finally, inverting ℓ and then extending scalars along finite extensions of E, we obtain the cases $\mathbf{Q}_\ell$, E and $\overline{\mathbf{Q}}_\ell$. ∎

By the theorem, $H^{2d}(X, \Lambda_X(d))$ is free of rank one. It is generated by the class of a point, which we define by the counit of the exceptional adjunction for the inclusion of the point. Let $x : \operatorname{Spec} k \to X$ be a closed point. Using $x^! \omega_X = (a_X x)^! \Lambda_s = \Lambda$ and $\omega_X \simeq \Lambda_X(d)[2d]$, the counit $x_* x^! \omega_X \to \omega_X$ induces a map $\Lambda \to R\Gamma(X, \omega_X) = R\Gamma(X, \Lambda_X(d))[2d]$, and we let $[x] \in H^{2d}(X, \Lambda_X(d))$ be the image of 1.

Lemma 10.16 (The class of a point)**.** *In the setting of Theorem 10.15, we have* $\mathrm{tr}_X(d)([x]) = 1$ *for every closed point x of X. In particular $[x]$ generates $H^{2d}(X, \Lambda_X(d))$.*

Proof. By definition, $[x]$ is the image of 1 under the map induced by the counit $x_* x^! \omega_X \to \omega_X$, and $\mathrm{tr}_X(d)$ is induced by the counit $R a_{X*} \omega_X \to \Lambda$. Because exceptional adjunctions compose, the composite of these two counits is the counit for $a_X x = \mathrm{id}_s$, which is the identity of Λ. Thus $\mathrm{tr}_X(d)([x]) = 1$. Since $\mathrm{tr}_X(d)$ is an isomorphism onto Λ by Theorem 10.15, $[x]$ is a generator. ∎

For a field Λ and the constant sheaf, the trace isomorphism in top degree turns Poincaré duality into the classical cup product pairing. On $\mathbf{P}^1$, for instance, the only nontrivial instance is $H^0 \times H^2 \to \Lambda(-1)$. After twisting by (1) it pairs the generators 1 and $[x]$ to 1 by Lemma 10.16, so it is perfect.

Corollary 10.17 (The cup product pairing)**.** *Let X/k be smooth, proper and connected of pure dimension d, with k algebraically closed, and let Λ be a field. For every i the pairing*

$$\langle a, b\rangle = \mathrm{tr}_X(a \cup b), \qquad H^i(X, \Lambda) \times H^{2d-i}(X, \Lambda) \longrightarrow \Lambda(-d),$$

is perfect, and $\langle b, a\rangle = (-1)^i \langle a, b\rangle$*. In particular, if* d *is odd and* Λ *has characteristic zero, the pairing on* $H^d(X, \Lambda)$ *is alternating, as for a compact complex manifold of odd dimension.*

Proof. For $K = \Lambda_X$ we have $K^\vee = \Lambda_X$, and since X is proper, Corollary 10.13 says that the pairing $H^i \times H^{2d-i} \to \Lambda(-d)$ induced by β_{Λ_X} is perfect. This pairing is $\langle -, -\rangle$, because it is cup product followed by the trace map in top degree, that is, by the isomorphism tr_X of Theorem 10.15. By graded commutativity of the cup product we have $\langle b, a\rangle = (-1)^{i(2d-i)}\langle a, b\rangle = (-1)^i\langle a, b\rangle$. ∎

10.5 Weil complexes

The Weil conjectures concern varieties over $\mathbf{F}_q$, while the six operations of Part I live on schemes over the algebraic closure. On $X = X_0 \times_{\mathbf{F}_q} k$, what remains of a variety X_0 over $\mathbf{F}_q$ is an action of Frobenius on X and on every complex pulled back from X_0.

Here $k_0 = \mathbf{F}_q$ and k is an algebraic closure of k_0. In this setting the geometric Frobenius $\sigma \in \mathrm{Gal}(k/k_0)$ is the inverse of the arithmetic Frobenius $a \mapsto a^q$. For a scheme T of the form $T_0 \times_{k_0} k$, with T_0 over k_0, let

$$\mathrm{Fr}_T : T \longrightarrow T$$

be the semilinear automorphism induced by σ. It is a morphism of schemes covering the automorphism $\mathrm{Spec}(\sigma)$ of $\mathrm{Spec}\, k$.

We use Fr_T to define Weil structures, but we prove the trace formula for the k-linear Frobenius $\Phi_T = F_{T_0/k_0} \times_{k_0} k$, obtained by base change from the relative q-power Frobenius of T_0, since its graph is a correspondence over k. The two are compared in Section 12.1 (Lemma 12.3).

Since Fr_T is an automorphism of schemes, pullback of sheaves of modules along it is exact and preserves derived completeness. Hence for every admissible coefficient ring Λ it induces an autoequivalence

$$\mathrm{Fr}_T^* : D^b_{\mathrm{ctf}}(T, \Lambda) \longrightarrow D^b_{\mathrm{ctf}}(T, \Lambda),$$

which agrees with the derived inverse image along Fr_T.

Instead of complexes on T_0 pulled back to T, we keep only the part of that datum that the trace formula sees: the complex on T, where all cohomology is computed, together with an identification of it with its Frobenius pullback. Every construction on T (direct images, $f^!$, truncations, duals) is compatible with such identifications by base change, as we check below, so we never have to descend anything to T_0.

There are also more of these pairs than complexes on T_0, and we use the extra ones in Part III. Indeed, for a $\mathbf{Q}_\ell$-sheaf on T_0, the stalk at a geometric point over a closed point is a continuous representation of a Galois group. It follows that Frobenius preserves a lattice in it, and its eigenvalues are ℓ-adic units. By contrast, on $\mathrm{Spec}\, k$ the sheaf $\mathbf{Q}_\ell$ with Frobenius acting by ℓ is a pair of this kind that does not come from $\mathrm{Spec}\, \mathbf{F}_q$. Pulled back to a scheme, such pairs are the sheaves $\mathcal{B}_b$ that normalize weights in Proposition 13.7.

We require an isomorphism so that the Frobenius operator on cohomology is invertible over a field (Theorem 10.21). The trace formula itself only needs a morphism (Definition 12.2), and it is stated for these pairs (Theorem 12.24).

Definition 10.18 (Weil structure)**.** For $K \in D^b_{\mathrm{ctf}}(T, \Lambda)$, a *Weil structure* is an isomorphism

$$\phi_K : \mathrm{Fr}_T^* K \xrightarrow{\sim} K.$$

A *Weil complex* is a pair (K, ϕ_K) with this structure. A morphism $(K, \phi_K) \to (L, \phi_L)$ is a morphism $h : K \to L$ satisfying

$$h \circ \phi_K = \phi_L \circ \mathrm{Fr}_T^*(h).$$

A Weil structure is the datum needed to define a Frobenius operator on cohomology. The basic example is a complex K pulled back from a complex K_0 on T_0. Since the projection $T \to T_0$ composed with Fr_T is again the projection, the pullback of K_0 gives a canonical isomorphism $\mathrm{Fr}_T^* K \simeq K$.

The triangles, truncations and spectral sequences we form in $D^b_{\mathrm{ctf}}(T, \Lambda)$ are compatible with Weil structures. Since Fr_T^* is exact on sheaves, it commutes with standard truncations and with the triangle of a short exact sequence of sheaves. It also commutes with the localization triangle $j_! j^* K \to K \to i_* i^* K$ for an open immersion $j : U \to T$ obtained by base change from an open $U_0 \subset T_0$, with closed complement i. Indeed, for such U we have $\mathrm{Fr}_T^{-1}(U) = U$, and Fr_T restricts to Fr_U and to the Frobenius of the complement. These constructions therefore carry Weil structures to Weil structures, and the induced long exact cohomology sequences are Frobenius-equivariant.

Let (M, ϕ_M) be a Weil complex on X and let $f : X \to Y$ come from f_0 over k_0. The base change map for the cartesian square with horizontal arrows Fr_X, Fr_Y identifies $\mathrm{Fr}_Y^* Rf_! M$ with $Rf_! \mathrm{Fr}_X^* M$, and the composite

$$\mathrm{Fr}_Y^* Rf_! M \simeq Rf_! \mathrm{Fr}_X^* M \xrightarrow{Rf_!(\phi_M)} Rf_! M$$

is a Weil structure on $Rf_! M$. Likewise f^*, Rf_*, $f^!$, $\otimes^{\mathbf{L}}$ and $R\mathcal{H}om$ carry Weil structures, via the corresponding base change maps. These maps are isomorphisms for every f, because Fr_X and Fr_Y are isomorphisms of schemes and base change along an isomorphism is transport of structure.

For $Rf_!$ this needs a word, since Fr_X and Fr_Y are not k-morphisms, while $Rf_!$ is built from compactifications over k. We compute $Rf_! = R\bar{f}_* j_!$ with a compactification $X \overset{j}{\hookrightarrow} \bar{X} \overset{\bar{f}}{\to} Y$ obtained by base change from a compactification of f_0 over k_0 (Theorem A.2). Then $\mathrm{Fr}_{\bar{X}}$ is an automorphism of the scheme $\bar{X}$ with $\mathrm{Fr}_{\bar{X}}^{-1}(X) = X$ and $\bar{f}\,\mathrm{Fr}_{\bar{X}} = \mathrm{Fr}_Y\,\bar{f}$. For an automorphism h of a scheme, $h^* \simeq (h^{-1})_*$, so transport of structure gives $\mathrm{Fr}_Y^* R\bar{f}_* \simeq R\bar{f}_* \mathrm{Fr}_{\bar{X}}^*$ and $\mathrm{Fr}_{\bar{X}}^* j_! \simeq j_! \mathrm{Fr}_X^*$, and their composite is the base change map for $Rf_!$. It does not depend on the compactification. Indeed, two compactifications of f_0 are dominated by a third, the closure of X_0 in their fibre product over Y_0, and the refinement isomorphisms of $Rf_!$ come from proper maps over k_0, which commute with the Frobenii. The map for $f^!$ is the mate of the map for $Rf_!$, and it is an isomorphism because both functors Fr^* are equivalences.

Pullback along Fr_T on compactly supported cohomology is constructed in the same way. For a morphism of schemes $h : S' \to S$ and $L \in D(S_{\mathrm{pro\acute{e}t}}, \Lambda)$, the *pullback map*

$$\mathrm{pb}_{h,L} : R\Gamma(S, L) \longrightarrow R\Gamma(S', h^* L)$$

is $R\Gamma$ of the unit $L \to Rh_* h^* L$, followed by the identification $R\Gamma(S, Rh_* -) = R\Gamma(S', -)$. We write pb_h when L is clear. It is natural in L, and it is transitive: for $g : S' \to S$ and $h : S'' \to S'$

we have $\mathrm{pb}_{gh,L} = \mathrm{pb}_{h,g^*L} \circ \mathrm{pb}_{g,L}$ under $(gh)^* \simeq h^* g^*$. Let $j : T \hookrightarrow \bar{T}$ be the base change of a compactification of T_0 over k_0. Then $\mathrm{Fr}_{\bar{T}}$ is an automorphism of $\bar{T}$ with $\mathrm{Fr}_{\bar{T}}^{-1}(T) = T$, and *pullback along* Fr_T is the composite

$$\mathrm{pb}_{\mathrm{Fr}_T} : R\Gamma_c(T, K) = R\Gamma(\bar{T}, j_! K) \xrightarrow{\mathrm{pb}_{\mathrm{Fr}_{\bar{T}}}} R\Gamma(\bar{T}, \mathrm{Fr}_{\bar{T}}^* j_! K) \simeq R\Gamma(\bar{T}, j_! \mathrm{Fr}_T^* K) = R\Gamma_c(T, \mathrm{Fr}_T^* K).$$

It does not depend on $\bar{T}$. Indeed, a refinement $\pi : \bar{T}' \to \bar{T}$ between two such compactifications comes from k_0, so $\pi \mathrm{Fr}_{\bar{T}'} = \mathrm{Fr}_{\bar{T}} \pi$. Transitivity of pullback maps for this composite, together with the base change $\mathrm{Fr}_{\bar{T}}^* R\pi_* \simeq R\pi_* \mathrm{Fr}_{\bar{T}'}^*$ along the automorphisms Fr and with $R\pi_* j'_! \simeq j_!$, shows that the refinement isomorphism $R\Gamma(\bar{T}', j'_! K) \simeq R\Gamma(\bar{T}, R\pi_* j'_! K) \simeq R\Gamma(\bar{T}, j_! K)$ carries one pullback map to the other. In the same way, the pullback map along Fr_T^m is defined, and by transitivity it is the m-fold composite $\mathrm{pb}_{\mathrm{Fr}_T^m, K} = \mathrm{pb}_{\mathrm{Fr}_T, (\mathrm{Fr}_T^{m-1})^* K} \circ \cdots \circ \mathrm{pb}_{\mathrm{Fr}_T, K}$.

Definition 10.19 (Frobenius operator)**.** For a Weil complex (K, ϕ_K) on T, the *geometric Frobenius operator* on compactly supported cohomology is the endomorphism

$$\begin{aligned} \mathrm{Frob}_{T,K} : R\Gamma_c(T_{\mathrm{pro\acute{e}t}}, K) &\xrightarrow{\mathrm{pb}_{\mathrm{Fr}_T}} R\Gamma_c(T_{\mathrm{pro\acute{e}t}}, \mathrm{Fr}_T^* K) \\ &\xrightarrow{R\Gamma_c(T_{\mathrm{pro\acute{e}t}}, \phi_K)} R\Gamma_c(T_{\mathrm{pro\acute{e}t}}, K). \end{aligned}$$

Here $\mathrm{pb}_{\mathrm{Fr}_T}$ is the pullback along Fr_T constructed above.

When T and K are clear, we write F for $\mathrm{Frob}_{T,K}$ and for the endomorphisms it induces on the groups $H_c^i(T, K)$.

For $r \in \mathbf{Z}$, write $\Lambda(r) = \Gamma((\mathrm{Spec}\, k)_{\mathrm{pro\acute{e}t}}, \Lambda_{\mathrm{Spec}\, k}(r))$ for the Tate coefficient module with its induced Frobenius operator. Arithmetic Frobenius acts on $\mu_{\ell^n}(k)$ by $\zeta \mapsto \zeta^q$. Hence geometric Frobenius acts on $\Lambda(1)$ by q^{-1}, and on $\Lambda(-d)$ by q^d. Let X_0 be smooth of pure dimension d. After removing the twist, the trace map of Definition 9.24 is a Frobenius-equivariant map, by Proposition 10.20 below:

$$H_c^{2d}(X_{\mathrm{pro\acute{e}t}}, \Lambda_X) \longrightarrow \Lambda(-d).$$

For $X = \mathbf{P}_k^1$ this says that geometric Frobenius acts on $H^2(\mathbf{P}_k^1, \Lambda) = \Lambda(-1)$ by q, as checked after Proposition 6.7.

10.6 Frobenius and Poincaré duality

Let X_0 be smooth of pure dimension d and let K be locally perfect, so that $\mathbb{D}_X K \simeq K^\vee(d)[2d]$. In this case cup product pairs $H_c^i(X, K)$ with $H^{2d-i}(X, K^\vee)$ and takes values in $H_c^{2d}(X, \Lambda)$. To use this pairing for the Weil conjectures we need to know how F acts on both sides. The target is not the trivial Weil complex but $\Lambda(-d)$, as for $\mathbf{P}^1$ above. If we forgot the twist, we would predict the relation $\alpha \leftrightarrow 1/\alpha$ between eigenvalues in complementary degrees instead of $\alpha \leftrightarrow q^d/\alpha$.

We built Poincaré duality from the purity isomorphism γ_f and the trace map. It therefore suffices to check that these two maps commute with pullback along Frobenius (Proposition 10.20). This statement about natural transformations is transport of structure, except for the identification of Tate twists. There we use that pullback along Fr_T respects Kummer classes. The compatibility of the duality pairing with Weil structures then follows formally (Theorem 10.21).

Let f be obtained from a smooth morphism $f_0 : X_0 \to Y_0$ over $k_0 = \mathbf{F}_q$ by extension to k, let Fr_X, Fr_Y be the geometric Frobenius maps above, and let γ_f be the isomorphism of (9.2).

Proposition 10.20 (Purity and trace commute with Frobenius)**.** *Fix an admissible Λ, and let $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ and $M \in D^b_{\mathrm{ctf}}(X, \Lambda)$. The square with horizontal arrows $\mathrm{Fr}_X, \mathrm{Fr}_Y$ and vertical arrows f, f is cartesian, and its base change comparisons identify*

$$\mathrm{Fr}_X^*\, f^! K \simeq f^!\, \mathrm{Fr}_Y^*\, K, \qquad \mathrm{Fr}_Y^*\, Rf_! M \simeq Rf_!\, \mathrm{Fr}_X^*\, M. \tag{10.9}$$

With these identifications, and the Kummer identification $\mathrm{Fr}_T^*\, \Lambda_T(1) \simeq \Lambda_T(1)$ for the twists, we have

$$\mathrm{Fr}_X^*\, \gamma_f(K) = \gamma_f(\mathrm{Fr}_Y^*\, K), \qquad \mathrm{Fr}_Y^*\, \mathrm{Tr}_f(K) = \mathrm{Tr}_f(\mathrm{Fr}_Y^*\, K).$$

Proof. Both identities are instances of base change for smooth purity, applied to a square whose horizontal maps are not k-linear. Since γ_f and Tr_f are built from base change maps and from the Kummer class of the twist, we only have to know how Fr_T acts on Kummer classes.

The first identity is Proposition 9.22 for this square. Since Fr_X is not a morphism over k, the only point to check is that the identification of twists used there is the Kummer identification, that is, that pullback along Fr_T sends the Kummer class of a unit to the Kummer class of its pullback. This holds because the Kummer sequence $0 \to \mu_{\ell^n} \to \mathbf{G}_m \xrightarrow{\ell^n} \mathbf{G}_m \to 0$ and its boundary map are natural for every morphism of schemes, k-linear or not.

For the second identity we combine the first with the counit identity defining exceptional base change, which says that after base change for compact support, the pulled-back counit is the counit preceded by the exceptional comparison. As the trace map is the counit composed with γ_f, the two identities together give the statement for Tr_f. ∎

Weil structures are transported along $f^!$, f^* and $Rf_!$. On $f^*K(d)[2d]$ the Weil structure is the pullback of ϕ_K combined with the Kummer identification. By the proposition above and naturality of γ_f and Tr_f applied to ϕ_K, the maps $\gamma_f(K)$ and $\mathrm{Tr}_f(K)$ are morphisms of Weil complexes.

The constant sheaf Λ_Y has the Weil structure given by the identification $\mathrm{Fr}_Y^*\, \Lambda_Y \simeq \Lambda_Y$ of pullbacks of constant sheaves. For $K = \Lambda_Y$ with this structure, the previous paragraph says that the isomorphism $\omega_f \simeq \Lambda_X(d)[2d]$ and the trace $\mathrm{Tr}_f(\Lambda_Y)$ are morphisms of Weil complexes.

Theorem 10.21 (Poincaré duality and Frobenius)**.** *Let $X_0/\mathbf{F}_q$ be smooth and separated of pure dimension d, let $X = X_0 \times \overline{\mathbf{F}}_q$, and let (K, ϕ_K) be a Weil complex on X (Definition 10.18) with K locally perfect. Equip the dual with the contragredient Weil structure*

$$\mathrm{Fr}_X^*\, K^\vee \simeq (\mathrm{Fr}_X^*\, K)^\vee \xrightarrow{(\phi_K^{-1})^\vee} K^\vee.$$

For every admissible Λ the Poincaré pairing β_K of (10.7) is a morphism of Weil complexes. Thus, with F the Frobenius operators defined above, the pairing $H^i_c(X, K) \times H^{2d-i}(X, K^\vee(d)) \to \Lambda$ satisfies $\langle Fx, Fy\rangle = \langle x, y\rangle$, and the pairing $H^i_c(X, K) \times H^{2d-i}(X, K^\vee) \to \Lambda(-d)$ satisfies

$$\langle Fx, Fy\rangle = q^d \langle x, y\rangle.$$

If Λ is a field, then F is invertible on $H^i_c(X, K)$, and with respect to the second pairing F on $H^{2d-i}(X, K^\vee)$ is q^d times the inverse transpose of F on $H^i_c(X, K)$. Hence the two spaces have the

same dimension b, the eigenvalues of F on $H^{2d-i}(X, K^\vee)$ are the q^d/α, with multiplicities, where α ranges over the eigenvalues on $H^i_c(X, K)$, and $\det(F \mid H^i_c(X,K))\det(F \mid H^{2d-i}(X,K^\vee)) = q^{db}$.

Proof. The pairing β_K is a composite of maps each of which commutes with pullback along Frobenius, so it is equivariant. Over a field the remaining statements then follow by linear algebra, because the pairing is perfect.

Cup product with mixed supports and evaluation are natural for pullback by Frobenius, and the isomorphism γ_{a_X} and the trace map are equivariant by Proposition 10.20. Moreover, the contragredient Weil structure on $K^\vee$ is chosen so that evaluation $K \otimes K^\vee \to \Lambda_X$ is a morphism of Weil complexes, with the trivial structure on Λ_X. It follows that the composite β_K is equivariant, which gives $\langle Fx, Fy\rangle = \langle x, y\rangle$ for the first pairing. The second pairing is obtained by moving the twist from the second factor to the values. Now geometric Frobenius acts on $\Lambda(d)$ by q^{-d} and on $\Lambda(-d)$ by q^d, so for y in $H^{2d-i}(X, K^\vee)$ the operator F on $y \otimes \zeta$, with ζ a generator of $\Lambda(d)$, is $q^{-d}(Fy) \otimes \zeta$. Removing the twist, we get the displayed identity $\langle Fx, Fy\rangle = q^d\langle x, y\rangle$.

For field coefficients, Corollary 10.13 identifies $H^{2d-i}(X, K^\vee)$, via the untwisted pairing, with the dual of $H^i_c(X, K)$ tensored with $\Lambda(-d)$. Moreover, Frobenius on $H^i_c(X, K)$ is invertible, because ϕ_K is an isomorphism and pullback along the automorphism Fr_X is invertible on cohomology. In matrix terms, choose dual bases for the perfect pairing, let F_i be the matrix of F on $H^i_c(X, K)$ and F_{2d-i} that on $H^{2d-i}(X, K^\vee)$. Then the identity $\langle Fx, Fy\rangle = q^d\langle x, y\rangle$ reads ${}^tF_iF_{2d-i} = q^d$. In other words, Frobenius on $H^{2d-i}(X, K^\vee)$ is q^d times the inverse transpose of Frobenius on $H^i_c(X, K)$. Since a matrix and its transpose have the same characteristic polynomial, this proves the eigenvalue statement with algebraic multiplicities. Finally, taking determinants of $F_{2d-i} = q^d\,{}^t(F_i)^{-1}$ on spaces of dimension b, we get $\det F_{2d-i} = q^{db}(\det F_i)^{-1}$. ∎

For X_0 proper and geometrically connected and $K = \Lambda_X$, Theorem 10.15 shows that F is the identity on $H^0(X, \Lambda) = \Lambda$ and acts by q^d on $H^{2d}(X, \Lambda) \simeq \Lambda(-d)$.

Example 10.22 (Duality on the punctured line)**.** Let $X_0 = \mathbf{G}_{m,\mathbf{F}_q}$, with coordinate t, let X be its base change to k, and let $\Lambda = \overline{\mathbf{Q}}_\ell$, with the constant Weil sheaf. Since $\mathrm{Pic}(X) = 0$ and the units of $k[t, t^{-1}]$ are the ct^m, the Kummer sequence gives $H^0(X, \Lambda) = \Lambda$ and $H^1(X, \Lambda(1)) = \Lambda$, generated by the Kummer class of t (Proposition D.5, with $g = 0$ and two punctures). Thus $H^1(X, \Lambda) = \Lambda(-1)$, and the other groups vanish. For compact support, on the other hand, we write $X = \mathbf{P}^1 \setminus \{0, \infty\}$. The long exact sequence of $j_!\Lambda_X \to \Lambda_{\mathbf{P}^1} \to \Lambda_{\{0,\infty\}} \to$ reads

$$0 \to H^0_c(X) \to \Lambda \xrightarrow{\text{diag}} \Lambda^2 \to H^1_c(X) \to H^1(\mathbf{P}^1) = 0 \to 0 \to H^2_c(X) \to H^2(\mathbf{P}^1) \to 0,$$

so $H^0_c = 0$, $H^1_c = \Lambda^2/\Lambda \simeq \Lambda$ and $H^2_c = H^2(\mathbf{P}^1, \Lambda) = \Lambda(-1)$. Poincaré duality with $d = 1$ pairs H^1_c with H^1 and H^2_c with H^0, and the dimensions match.

The Frobenius eigenvalues show the twist at work. On H^1_c, which comes from the two rational points 0 and ∞, Frobenius acts by 1, whereas on $H^1 = \Lambda(-1)$ it acts by q. Their product is $1 \cdot q = q^d$, as Theorem 10.21 requires. Without the twist, duality would predict the eigenvalue $1/1 = 1$ on H^1 instead of q. Likewise H^0 and H^2_c have eigenvalues 1 and q.

Example 10.23 (The running example: duality)**.** Let $C_0\colon y^2 = x^3 - x$ over $\mathbf{F}_3$ be the elliptic curve of the introduction, and $C = C_0 \times_{\mathbf{F}_3} k$. Since C has genus one, $H^1(C, \mathbf{Q}_\ell)$ is a plane. Indeed, by the Kummer sequence $H^1(C, \mu_{\ell^n}) = J[\ell^n](k) \simeq (\mathbf{Z}/\ell^n)^{2g}$ with $2g = 2$ (Proposition D.3 and Corollary C.15). Since k is algebraically closed, $\mu_{\ell^n} \simeq \mathbf{Z}/\ell^n$, and passing to the limit and inverting ℓ gives $\dim H^1(C, \mathbf{Q}_\ell) = 2$. For $d = 1$ the middle pairing on H^1 is alternating and perfect, with values in $\mathbf{Q}_\ell(-1)$ (Corollary 10.17). By Theorem 10.21 it satisfies $\langle Fx, Fy\rangle = 3\langle x, y\rangle$. On a plane

an alternating form is unique up to a scalar, so $\langle Fx, Fy\rangle = \det(F)\langle x, y\rangle$. Hence $\det(F \mid H^1) = 3$. If α, β are the eigenvalues, then $\alpha\beta = 3$, and $\det(1 - TF \mid H^1) = 1 - (\alpha + \beta)T + 3T^2$. This is the functional equation of a curve of genus one: replacing T by $1/(3T)$ multiplies the polynomial by $(3T^2)^{-1}$.

Duality alone does not give $|\alpha| = 3^{1/2}$. For example, the pair $\alpha = 3$, $\beta = 1$, with $\det(1 - TF) = 1 - 4T + 3T^2$, consists of algebraic integers and satisfies $\alpha\beta = 3$. Here the count $N_1 = 4$ gives $\alpha + \beta = 0$ and excludes it; in general only purity does.

11 The Lefschetz–Verdier formula

The trace formula has to express the trace of Frobenius on $R\Gamma_c(X, K)$ as a sum over the fixed points of Frobenius, that is, over $X_0(\mathbf{F}_{q^m})$. We do not prove it for Frobenius directly. Instead we first prove a fixed point formula for every correspondence on a proper scheme. In this generality the formula is compatible with proper pushforward, and pushing forward to a point turns it into a tautology.

For a general correspondence, however, the fixed locus need not be discrete (Section 11.4). The local side is then a sum of contributions, one for each connected component of the fixed locus, each computed in a neighbourhood of that component.

We use Grothendieck's formalism of cohomological correspondences, written up by Illusie in [SGA5, III]. In this formalism, a correspondence on X is a map $c = (c_1, c_2) : C \to X \times X$, and a cohomological correspondence on K is a morphism $u : c_2^* K \to c_1^! K$. For example, for the graph of an endomorphism φ this is a map $\varphi^* K \to K$. In particular, for Frobenius it is the Weil structure $\phi_K : \mathrm{Fr}_X^* K \simeq K$ of Section 10.5, up to the comparison of Fr_X with the k-linear Frobenius in Chapter 12. When c_2 is proper, u induces an endomorphism $T_c(u)$ of $R\Gamma_c(X, K)$. In the same way ϕ_K induces the Frobenius operator $\mathrm{Frob}_{X,K}$ on $R\Gamma_c(X, K)$ and on every stalk at a fixed point.

The local contributions come from duality, as $\mathrm{Hom}(V, V) = V \otimes V^\vee$ turns an endomorphism into a tensor whose evaluation is the trace. In this way u gives the trace class $\mathrm{Tr}_c(u) \in H^0(\mathrm{Fix}(c), \omega_{\mathrm{Fix}(c)})$, supported on the fixed locus. Its integral over a proper open and closed part B of $\mathrm{Fix}(c)$ is the local term $\mathrm{LT}_B(u)$. The Lefschetz–Verdier formula (Theorem 11.13) says that if X and C are proper over k, then

$$\mathrm{tr}\big(T_c(u) \mid R\Gamma(X, K)\big) = \sum_B \mathrm{LT}_B(u).$$

The proof involves no computation, because trace classes commute with proper pushforward of correspondences. After pushing forward to the identity correspondence of the point, the trace class becomes the trace of $T_c(u)$ on the perfect complex $R\Gamma(X, K)$ (Lemma 11.10). The price of this generality, however, is that the formula does not compute the individual local terms; for Frobenius that is done in Chapter 12.

With coefficients in a finite ring A both sides are traces of endomorphisms of perfect complexes (Definition 11.1). They are not alternating sums of traces on cohomology modules, since these need not be projective. In fact, the trace is not even determined by the maps induced on cohomology (Example 11.4). Moreover, finite rings such as $\mathcal{O}_E/\pi_E^n$ cannot be avoided, because the trace formula over $\mathcal{O}_E$ is obtained from its reductions.

Throughout, $k_0 = \mathbf{F}_q$, $k = \overline{\mathbf{F}}_q$ and $s = \operatorname{Spec} k$. For a scheme X over k, $a_X : X \to s$ is the structure morphism, and $X = X_0 \times_{k_0} k$ whenever a model X_0 over k_0 is given. From Section 11.1 until Section 12.5 the coefficient ring is an arbitrary finite commutative ℓ-primary ring A, with

$\ell \neq \operatorname{char} k$, and we write $\omega_X = a_X^! A_s$ and $\mathbb{D}_X K = R\mathcal{H}om_A(K, \omega_X)$ for the dualizing complex and the Verdier dual of Definition 10.2 with coefficients A. Some proofs compute on small étale sites and transport the result by means of the comparison theorems for complexes of finite Tor amplitude.

11.1 Traces of perfect complexes

Both sides of the trace formula are traces of endomorphisms of perfect A-complexes, namely of $R\Gamma_c(X, K)$ on the global side and of the stalks of K on the local side.

Over a field the trace of an endomorphism of a complex is the alternating sum of its traces on cohomology. Over $A = \mathbf{Z}/\ell^2$, however, the cohomology modules need not be free, so they have no trace, and Example 11.4 below shows that the maps on cohomology do not determine the trace. We therefore define the trace on the complex. On a bounded complex of finite projective modules it is the alternating sum of the traces of the components, but the definition below expresses this sum through evaluation and coevaluation, so that it depends only on the morphism in $D(A)$. In this form it is compatible with every change of rings (Proposition 11.2), and this is how we pass from the trace formula over $\mathcal{O}_E/\pi_E^n$ to the trace formula over $\mathcal{O}_E$ and E (Section 12.5).

Definition 11.1 (Categorical trace)**.** For $P \in \operatorname{Perf}(A)$ put $P^\vee = R\operatorname{Hom}_A(P, A)$. If $v : P \to P$ is a morphism in $D(A)$, its trace is the scalar

$$\operatorname{tr}_A(v \mid P) = \left[A \xrightarrow{\text{coev}} P \otimes^{\mathbf{L}} P^\vee \xrightarrow{v \otimes^{\mathbf{L}} 1} P \otimes^{\mathbf{L}} P^\vee \xrightarrow{\sim} P^\vee \otimes^{\mathbf{L}} P \xrightarrow{\text{ev}} A \right] \in A.$$

Here coev and ev are the coevaluation and evaluation of the dual pair $(P, P^\vee)$ in $D(A)$, and the unlabelled arrow is the symmetry isomorphism, which includes the Koszul sign.

For $P = A$ in degree zero and v multiplication by a, coevaluation is $1 \mapsto 1 \otimes 1$ and evaluation is multiplication, so the trace is a. The next proposition lists the properties of the trace that we use. Compatibility with change of rings lets us move the trace formula between coefficient rings, and the chain-level formula is how we compute traces in examples.

Proposition 11.2 (Properties of the trace)**.** *The trace is A-linear in v, invariant under conjugacy, and compatible with derived scalar extension: for every ring map $A \to B$,*

$$\operatorname{tr}_A(v \mid P) \longmapsto \operatorname{tr}_B(v \otimes^{\mathbf{L}} B \mid P \otimes^{\mathbf{L}} B).$$

For a bounded complex of finite projective modules and a chain endomorphism $v^\bullet$ representing v one has

$$\operatorname{tr}_A(v \mid P) = \sum_i (-1)^i \operatorname{Tr}_A(v^i \mid P^i).$$

If P is the underlying complex of a two-step filtered object $F^1P \to P \to P/F^1P$ of the filtered derived category, with F^1P and P/F^1P perfect, and v is the underlying map of an endomorphism $\tilde{v}$ of this filtered object, with graded pieces v_1 on F^1P and v_0 on P/F^1P, then

$$\operatorname{tr}_A(v \mid P) = \operatorname{tr}_A(v_1 \mid F^1P) + \operatorname{tr}_A(v_0 \mid P/F^1P).$$

Over a field L it is also

$$\operatorname{tr}_L(v \mid P) = \sum_i (-1)^i \operatorname{Tr}_L(H^i(v) \mid H^i(P)).$$

Proof. Everything follows from the chain formula, which we prove first by writing evaluation and coevaluation in a finite dual basis. If $e_1, \dots, e_n$ and $e^1, \dots, e^n$ are dual bases of A^n and its dual, then $\operatorname{coev}(1) = \sum_j e_j \otimes e^j$, and the composite of the definition sends 1 to $\sum_j e^j(v(e_j))$, the ordinary trace of a matrix. The same computation applies to a finite projective module, which is a direct summand of such an A^n. On a complex, the symmetry interchanging a term of degree i and its dual of degree $-i$ contributes $(-1)^i$, which proves the chain formula. Its right-hand side is invariant under chain homotopy, since the supertrace of $dh + hd$ is zero by cyclicity and a shift of index.

Tensoring a dual basis with B preserves evaluation and coevaluation. Hence the trace is compatible with extension of scalars for every ring map $A \to B$. Linearity and invariance under conjugacy follow from the same description. For additivity, note that a two-step filtered object of the filtered derived category may be represented by a filtered complex of projective modules whose filtration is split in each degree. The endomorphism is then block triangular, so its supertrace is the sum of the supertraces on the two graded pieces. Finally, over a field the boundaries B^i and cycles Z^i are v-invariant subspaces $B^i \subset Z^i \subset P^i$, with $Z^i/B^i = H^i(P)$, and the differential induces a v-equivariant isomorphism $P^i/Z^i \simeq B^{i+1}$. Hence

$$\operatorname{Tr}(v^i \mid P^i) = \operatorname{Tr}(v \mid B^i) + \operatorname{Tr}(H^i(v) \mid H^i(P)) + \operatorname{Tr}(v \mid B^{i+1}),$$

and in the alternating sum the terms for the B^i cancel in pairs, leaving the cohomological formula. ∎

In the additivity statement of the proposition we ask for an endomorphism of the filtered object, and not merely of the three vertices of a triangle. The reason is that the middle component is not determined by the outer ones, and even for a morphism of triangles the trace of the middle component need not be the sum of the outer traces.

Remark 11.3 (Additivity needs a filtered endomorphism)**.** The coherence condition is needed over a general ring: three endomorphisms of a distinguished triangle whose squares commute only in the triangulated category need not satisfy trace additivity [Stacks, 087J]. Every localization sequence we use below is obtained functorially from the Frobenius correspondence and the filtration by an open subscheme and its closed complement, and hence it lifts to the filtered derived category.

The following example shows why the trace has to be taken on the complex.

Example 11.4 (A trace not determined by cohomology)**.** Let $A = \mathbf{Z}/\ell^2$ and let $P = [A \xrightarrow{\ell} A]$, with the terms in degrees 0 and 1. This is the perfect complex from Section 3.2 whose cohomology modules $H^0(P) = \ell A$ and $H^1(P) = A/\ell A$ are not perfect. The chain maps $1_P = (1, 1)$ and $v = (1 + \ell, 1)$ commute with the differential, since $\ell(1 + \ell) = \ell$ in A, and both induce the identity on $H^0(P)$ and on $H^1(P)$. By the chain formula of Proposition 11.2,

$$\operatorname{tr}_A(1_P \mid P) = 1 - 1 = 0, \qquad \operatorname{tr}_A(v \mid P) = (1 + \ell) - 1 = \ell \neq 0.$$

Thus, over a finite ring, the trace of an endomorphism of a perfect complex is not determined by the maps it induces on cohomology. The trace formula has to be stated for the operator on the complex.

11.2 Correspondences

As in topology, a pair of maps $c_1, c_2 : C \to X$ should act on $R\Gamma_c(X, K)$ by pulling back along c_2 and pushing forward along c_1. The pullback $R\Gamma_c(X, K) \to R\Gamma_c(C, c_2^* K)$ exists when c_2 is proper. To push forward we then need a map $Rc_{1!}L \to K$, where L is the complex on C we have arrived at. By the adjunction $Rc_{1!} \dashv c_1^!$ there is a canonical such map exactly when $L = c_1^! K$, namely the counit. What connects the two steps is therefore a map $u : c_2^* K \to c_1^! K$. The resulting endomorphism $T_c(u)$ is defined in Definition 11.11.

With c_1^* in place of $c_1^!$ the second step fails, since there may be no nonzero map $Rc_{1!}c_1^* K \to K$. For example, for the inclusion i of a rational point y into a smooth curve X,

$$\mathrm{Hom}(i_* i^* \Lambda_X, \Lambda_X) = \mathrm{Hom}(\Lambda_y, i^! \Lambda_X) = \mathrm{Hom}(\Lambda_y, \Lambda_y(-1)[-2]) = 0,$$

so the only map $Ri_! i^* \Lambda_X \to \Lambda_X$ is zero. When c_1 is finite étale, smooth purity gives $c_1^! = c_1^*$, and the counit is summation over the fibres. Then $T_c(u)$ is pullback along c_2 followed by a sum over the fibres of c_1, as for a Hecke operator.

The Lefschetz–Verdier formula (Theorem 11.13) is stated in this generality. For the trace formula we apply it to the graph of Frobenius (Definition 12.2).

Definition 11.5 (Correspondences)**.** Let X be a separated k-scheme of finite type. A *correspondence* on X is a morphism of k-schemes $c = (c_1, c_2) : C \to X \times_s X$ from a separated k-scheme of finite type (similarly over a base S). For $K \in D^b_{\mathrm{ctf}}(X, A)$, a *cohomological correspondence* on (X, K) is a morphism

$$u : c_2^* K \longrightarrow c_1^! K.$$

The *fixed locus* of c is the closed subscheme

$$\mathrm{Fix}(c) = C \times_{X \times X, \Delta_X} X, \qquad i_c : \mathrm{Fix}(c) \longrightarrow C, \quad d_c : \mathrm{Fix}(c) \longrightarrow X,$$

where i_c and d_c are the two projections.

Here c_2 is the leg along which K is pulled back and c_1 is the leg along which we take $c_1^!$; in [SGA5, III, §3] the two names are interchanged. Since X is separated, Δ_X and i_c are closed immersions.

The example to keep in mind is the graph of an endomorphism $\varphi : X \to X$, that is, $C = X$ with $c_1 = \mathrm{id}_X$ and $c_2 = \varphi$. Then $c_1^! K = K$, and a cohomological correspondence is a morphism $u : \varphi^* K \to K$. In this case the fixed locus $\mathrm{Fix}(c)$ is the fixed point scheme of φ, which we also denote $\mathrm{Fix}(\varphi)$. For the Frobenius Φ_X and a Weil complex this is the correspondence whose trace formula we want.

11.3 The exceptional Künneth formula

A cohomological correspondence $u : c_2^* K \to c_1^! K$ is a morphism on C, so to obtain from it a class on the fixed locus we must first view it as a section of $c^!$ of a single complex on $X \times X$, which can then be restricted to the diagonal. In linear algebra this is the identification $\mathrm{Hom}(V, W) = W \otimes V^\vee$, under which the trace becomes evaluation. Its sheaf version is the isomorphism $R\mathcal{H}om(p_2^* K, p_1^! K) \simeq K \boxtimes \mathbb{D}_X K$ of Lemma 11.7. Here $p_1, p_2 : X \times X \to X$ are the projections and $M \boxtimes N = p_1^* M \otimes^{\mathbf{L}} p_2^* N$ is the external tensor product in $D((X \times X)_{\mathrm{pro\acute{e}t}}, A)$. Applying $c^!$ to it, we obtain $R\mathcal{H}om(c_2^* K, c_1^! K)$. On the point the isomorphism is exactly $\mathrm{Hom}(V, V) = V \otimes V^\vee$.

The isomorphism comes from Künneth formulas for $Rf_!$, Rf_*, $f^!$, $R\mathcal{H}om$ and Verdier duals. For $Rf_!$ the formula follows directly from base change for compact supports and the projection formula, and for ordinary direct images from smooth and proper base change (Proposition 8.18, Section 8.9). The formula for $f^!$ (Proposition 11.6), by contrast, needs a dévissage on the coefficients, starting from constant coefficients, where for smooth f the functor $f^!$ is a twist and a shift.

Let $f_i : X_i \to Y_i$, $i = 1, 2$, be morphisms of separated k-schemes of finite type. For $M_i \in D^b_{\mathrm{ctf}}(X_i, A)$, using base change for compact supports and the projection formula, we obtain the Künneth isomorphism

$$R(f_1 \times f_2)_!(M_1 \boxtimes M_2) \simeq Rf_{1!}M_1 \boxtimes Rf_{2!}M_2.$$

For the structure maps it reads $R\Gamma_c(X_1 \times X_2, M_1 \boxtimes M_2) \simeq R\Gamma_c(X_1, M_1) \otimes^{\mathbf{L}} R\Gamma_c(X_2, M_2)$. For $L_i \in D^b_{\mathrm{ctf}}(Y_i, A)$ let

$$\alpha : f_1^! L_1 \boxtimes f_2^! L_2 \longrightarrow (f_1 \times f_2)^!(L_1 \boxtimes L_2)$$

be the morphism adjoint, under $R(f_1 \times f_2)_! \dashv (f_1 \times f_2)^!$, to the composite

$$\begin{aligned} R(f_1 \times f_2)_!(f_1^! L_1 \boxtimes f_2^! L_2) &\simeq Rf_{1!}f_1^! L_1 \boxtimes Rf_{2!}f_2^! L_2 \\ &\xrightarrow{\varepsilon_{f_1} \boxtimes \varepsilon_{f_2}} L_1 \boxtimes L_2, \end{aligned}$$

where ε_{f_i} is the counit of $Rf_{i!} \dashv f_i^!$.

Proposition 11.6 (Exceptional external Künneth)**.** *For all $L_i \in D^b_{\mathrm{ctf}}(Y_i, A)$, the morphism α is an isomorphism.*

Proof. Both sides of α are exact in each variable, so by dévissage it is enough to check α on generators. First, for constant coefficients we compute $f_i^!$ by Theorem 9.3 for smooth maps and by localization, and check α directly. Next, general coefficients are built from extensions by zero $j_!F$ of lisse sheaves. Since $f^!$ does not commute with $j_!$, we replace $j_!$ by Rj_*, which does. The difference has smaller support, and we treat it by induction. For the classical statements we refer to [SGA5, III, (1.6.4), (1.7.3), Proposition 1.7.4 and (3.1.1)].

Constant and lisse coefficients. First let the coefficients be constant. For smooth f_i of relative dimensions d_i, Theorem 9.3 identifies $f_i^! L_i$ with $f_i^* L_i(d_i)[2d_i]$, and $(f_1 \times f_2)^!$ with $(f_1 \times f_2)^*(d_1 + d_2)[2d_1 + 2d_2]$, so both sides of α become the same twisted pullback. That α is this identification follows from the compatibility of the trace map (Definition 9.24) with products. Indeed, α is defined through the counits, and for a smooth map the trace map is the counit composed with γ_f.

For closed immersions we use the localization triangle and the case of ordinary external Künneth where one factor is the identity, applied to the complementary open immersions. In general, every map factors locally as a closed immersion followed by a smooth projection of affine spaces. Since α is compatible with composition of morphisms and with restriction to open subschemes, Zariski localization gives the constant case for general maps. Next, lisse finite flat coefficients become constant with finite projective values on an étale cover, so descent gives the lisse case, and then the case of bounded perfect lisse models. In each of these steps the isomorphism we obtain is α itself: for smooth maps because γ_f identifies the counit with the trace map, and for localization and descent because α is natural and compatible with restriction.

General coefficients. A constructible complex is lisse on a dense open, and the rest has smaller support. We therefore argue by induction on the sum of the dimensions of the supports of L_1, L_2. Pushforward along closed immersions is compatible with α by proper base change and the exceptional exchange of Corollary 10.9. Hence we may first assume that the support is the entire base. As both sides of α are exact in each variable, a finite constructible stratification, bounded flat models and two-out-of-three reduce us to generators $j_{i!}F_i$ for dense open $j_i : U_i \hookrightarrow Y_i$, with F_i a lisse sheaf with finite projective stalks (Lemma 3.13).

Since $f_i^!$ does not commute with $j_{i!}$ in general, we pass through Rj_{i*}. Let j_i' and $f_{i,U}$ be the base changes of j_i and f_i. The exceptional exchange $f_i^! Rj_{i*} \simeq Rj_{i*}' f_{i,U}^!$ is the mate, under the right adjoints, of base change for $Rf_{i!}$ along the open immersion. Hence it is compatible with the counits used to define α. By this exchange and the ordinary external Künneth formula, the comparison on $(Rj_{1*}F_1, Rj_{2*}F_2)$ is the pushforward of the lisse comparison. Let $z_i : Z_i \hookrightarrow Y_i$ be the closed complement of U_i. In the triangles

$$j_{i!}F_i \longrightarrow Rj_{i*}F_i \longrightarrow z_{i*}z_i^* Rj_{i*}F_i \xrightarrow{+1},$$

the third term is still constructible of finite Tor amplitude by Proposition 7.1, and its support has smaller dimension. Since α is natural, it maps the corresponding triangles to each other in each variable. By induction and two-out-of-three, first in one variable and then in the other, we obtain α for $(j_{1!}F_1, j_{2!}F_2)$. ∎

Using the Künneth formulas for both kinds of direct image and for $f^!$, we now identify cohomological correspondences with sections of $c^!(K \boxtimes \mathbb{D}_X K)$.

Lemma 11.7 (External duality)**.** *Let $K \in D^b_{\mathrm{ctf}}(X, A)$, let $p_1, p_2 : X \times X \to X$ be the projections, and put $P_X = K \boxtimes \mathbb{D}_X K \in D^b_{\mathrm{ctf}}(X \times X, A)$. For a correspondence $c : C \to X \times X$ there are canonical isomorphisms*

$$P_X \xrightarrow{\sim} R\mathcal{H}om_A(p_2^* K, p_1^! K), \qquad \mathcal{H}_c(K) := R\mathcal{H}om_A(c_2^* K, c_1^! K) \xrightarrow{\sim} c^! P_X.$$

Proof. The exceptional Künneth formula (Proposition 11.6) and the ordinary one give Künneth formulas for internal Hom and for Verdier duals, and these give the two displayed isomorphisms.

Internal Hom and Verdier duals. On the point this is the identity $\mathrm{Hom}(V_1, W_1) \otimes \mathrm{Hom}(V_2, W_2) = \mathrm{Hom}(V_1 \otimes V_2, W_1 \otimes W_2)$ for finite free modules. For $E_i, L_i \in D^b_{\mathrm{ctf}}(X_i, A)$ the external comparison $R\mathcal{H}om_A(E_1, L_1) \boxtimes R\mathcal{H}om_A(E_2, L_2) \to R\mathcal{H}om_A(E_1 \boxtimes E_2, L_1 \boxtimes L_2)$ is an isomorphism. By dévissage it suffices to check this on generators $E_i = Rh_{i!}F_i$, with h_i the inclusion of a stratum and F_i a lisse sheaf with finite projective stalks (Lemma 3.13), for which

$$R\mathcal{H}om_A(Rh_{i!}F_i, L_i) \simeq Rh_{i*}(F_i^\vee \otimes^{\mathbf{L}} h_i^! L_i).$$

By the ordinary Künneth formula and α, the exterior product of these expressions is identified with internal Hom on the product. Taking $L_i = \omega_{X_i}$ and using α for the structure maps, we get, for $M_i \in D^b_{\mathrm{ctf}}(X_i, A)$,

$$\mathbb{D}_{X_1 \times X_2}(M_1 \boxtimes M_2) \simeq \mathbb{D}_{X_1} M_1 \boxtimes \mathbb{D}_{X_2} M_2.$$

The two isomorphisms. The first isomorphism is the sheaf version of $\mathrm{Hom}(V, W) \simeq W \otimes V^\vee$, and we obtain it by writing internal Hom as a dual of a tensor product. Indeed, for $M \in D^b_{\mathrm{ctf}}(U, A)$, $N \in D^b_{\mathrm{ctf}}(V, A)$ and the projections p_U, p_V of $V \times U$, Verdier exchange and biduality let us

rewrite $R\mathcal{H}om_A(p_U^*M, p_V^!N)$ as $\mathbb{D}_{V\times U}(\mathbb{D}_V N \boxtimes M)$. By the formula for duals of exterior products this is $N \boxtimes \mathbb{D}_U M$. Taking $M = N = K$, with the factors of the product ordered as above, we obtain the first isomorphism. The second isomorphism follows from the canonical exchange of exceptional inverse image with Hom, for $P, Q \in D^b_{\mathrm{ctf}}(X \times X, A)$,

$$c^! R\mathcal{H}om_A(P, Q) \simeq R\mathcal{H}om_A(c^*P, c^!Q),$$

applied to $P = p_2^*K$ and $Q = p_1^!K$. To prove the exchange, we use Yoneda inside $\mathrm{Loc}(D^b_{\mathrm{ctf}}(C, A))$, the category on which $c_! \dashv c^!$ is defined (Theorem 5.2). Test objects in all of $D(C_{\mathrm{pro\acute{e}t}}, A)$ would not do, since $c^!$ is not defined there. Both sides of the exchange lie in $D^b_{\mathrm{ctf}}(C, A)$ by Proposition 7.1, so these test objects suffice. By Proposition 5.15 the localizing categories are $D(C_{\acute{e}t}, A)$ and $D((X \times X)_{\acute{e}t}, A)$, where the projection formula for $c_!$ holds for all objects [Stacks, 0GL5], and tensoring with c^*P preserves $\mathrm{Loc}(D^b_{\mathrm{ctf}}(C, A))$. For M in $\mathrm{Loc}(D^b_{\mathrm{ctf}}(C, A))$ we obtain

$$\begin{aligned}\mathrm{Hom}(M, c^! R\mathcal{H}om_A(P, Q)) &\simeq \mathrm{Hom}(c_!M \otimes^{\mathbf{L}} P, Q) \simeq \mathrm{Hom}(c_!(M \otimes^{\mathbf{L}} c^*P), Q)\\ &\simeq \mathrm{Hom}(M \otimes^{\mathbf{L}} c^*P, c^!Q) \simeq \mathrm{Hom}(M, R\mathcal{H}om_A(c^*P, c^!Q)),\end{aligned}$$

by $c_! \dashv c^!$ and the tensor–Hom adjunction, the projection formula, $c_! \dashv c^!$ again, and the tensor–Hom adjunction. ∎

In linear algebra the trace is evaluation $W \otimes W^\vee \to A$ on the diagonal. Here evaluation takes values in ω_X on the diagonal, and by applying $c^!$ we move it to the fixed locus.

Definition 11.8 (Trace morphism)**.** In the setting of Lemma 11.7, let $e_K : P_X \to R\Delta_{X*}\omega_X$ be adjoint to the evaluation $\Delta_X^*P_X = K \otimes^{\mathbf{L}} \mathbb{D}_X K \to \omega_X$. The *trace morphism* is the composite

$$\mathcal{H}_c(K) \simeq c^!P_X \xrightarrow{c^!e_K} c^!R\Delta_{X*}\omega_X \xrightarrow{\sim} Ri_{c*}d_c^!\omega_X \simeq Ri_{c*}\omega_{\mathrm{Fix}(c)}$$

of the second isomorphism of the lemma, $c^!e_K$, exceptional base change for the proper map Δ_X, and the composition of exceptional inverse images $d_c^!a_X^! = a_{\mathrm{Fix}(c)}^!$.

11.4 Trace classes and local terms

We want to write $\mathrm{tr}(T_c(u))$ as a sum of contributions from the fixed locus. The first guess, a sum over fixed points of the traces of u on stalks, does not make sense in general, because the fixed locus need not be finite. For example, for the identity correspondence of $\mathbf{P}^1$ and $K = A$, every point is fixed. The trace on $R\Gamma(\mathbf{P}^1, A) = A \oplus A(-1)[-2]$ is $1 + 1 = 2$, the self-intersection of the diagonal.

What does make sense is a class on $\mathrm{Fix}(c)$. By Lemma 11.7, u is a section of $c^!(K \boxtimes \mathbb{D}_X K)$, and composing with the trace morphism we obtain a section of $\omega_{\mathrm{Fix}(c)}$. Such a section can be integrated over a proper piece B of the fixed locus. Indeed, $H^0(B, \omega_B) = \mathrm{Hom}(Ra_{B!}A_B, A)$, and $Ra_{B*}A_B = Ra_{B!}A_B$ for proper B. Composing with the unit $A \to Ra_{B*}A_B$ therefore gives an element of A. In the example above, $H^0(\mathbf{P}^1, \omega_{\mathbf{P}^1}) = H^2(\mathbf{P}^1, A(1))$ and the degree map $H^2(\mathbf{P}^1, A(1)) \to A$ is an isomorphism, so Theorem 11.13 shows that the trace class is 2 times the class of a point.

We will see that the trace class is additive in K (Proposition 11.19) and compatible with proper pushforward (Theorem 11.13). However, at an isolated fixed point the local term can still differ from the trace on the stalk (Remark 12.4). For Frobenius the two are shown to agree in Proposition 12.16.

Definition 11.9 (Trace class and local terms)**.** Let $u : c_2^* K \to c_1^! K$ be a cohomological correspondence on (X, K). By adjunction, u is a section $A_C \to \mathcal{H}_c(K)$. Its composite with the trace morphism of Definition 11.8, transposed by $i_c^* \dashv Ri_{c*}$, is the *trace class*

$$\mathrm{Tr}_c(u) : A_{\mathrm{Fix}(c)} \longrightarrow \omega_{\mathrm{Fix}(c)}.$$

For a proper k-scheme B, integration is the counit map

$$\int_B : H^0(B, \omega_B) = H^0(Ra_{B!}a_B^! A_s) \longrightarrow A.$$

If $B \subset \mathrm{Fix}(c)$ is proper and open and closed, the *local term* of u on B is

$$\mathrm{LT}_B(u) = \int_B \mathrm{Tr}_c(u)|_B.$$

On the point, the local term is the categorical trace of Definition 11.1, and we use this later to identify the local terms of Frobenius with traces on stalks.

Lemma 11.10 (Local terms on the point)**.** *Let $P \in \mathrm{Perf}(A)$, viewed as an object of $D^b_{\mathrm{ctf}}(s, A)$, and let $u : P \to P$ be a cohomological correspondence on (s, P) for the identity correspondence of s, whose fixed locus is s. Then*

$$\mathrm{LT}_s(u) = \mathrm{tr}_A(u \mid P).$$

Proof. On the point each step of the construction of the trace class is a step of the categorical trace. Here $\omega_s = A_s$, so $\mathbb{D}_s P = P^\vee$, and $P_s = P \boxtimes P^\vee = P \otimes^{\mathbf{L}} P^\vee$. First, under the first isomorphism of Lemma 11.7, the map u becomes the element of $P \otimes^{\mathbf{L}} P^\vee$ obtained by applying $u \otimes 1$ to the coevaluation. Next, the diagonal of s is the identity, so the trace morphism is the evaluation $P \otimes^{\mathbf{L}} P^\vee \to A$, taken after the symmetry that puts the factors in the order $P^\vee \otimes P$. That symmetry is precisely the symmetry in Definition 11.1, and integration over s is the identity of A. The composite is therefore the categorical trace. ∎

The global side of the trace formula is the trace of the following endomorphism.

Definition 11.11 (Action of a correspondence on cohomology)**.** Let u be a cohomological correspondence on (X, K) for a correspondence c with c_2 proper. The endomorphism $T_c(u)$ of $R\Gamma_c(X, K)$ is the composite

$$\begin{aligned} T_c(u) : R\Gamma_c(X, K) &\xrightarrow{\eta_{c_2}} R\Gamma_c(C, c_2^* K) \xrightarrow{u} R\Gamma_c(C, c_1^! K) \\ &\simeq R\Gamma_c(X, Rc_{1!}c_1^! K) \xrightarrow{\varepsilon_{c_1}} R\Gamma_c(X, K). \end{aligned}$$

In the first arrow we use the ordinary unit together with $Rc_{2*} = Rc_{2!}$, which holds since c_2 is proper, and the displayed isomorphism comes from the composition of direct images with compact support.

The Lefschetz–Verdier formula computes the trace of $T_c(u)$ in terms of the trace class. We obtain it as the special case, for the map to a point, of the compatibility of trace classes with proper pushforward. This works because $T_c(u)$ is itself a cohomological correspondence: for proper X and C it is a correspondence on the point with coefficients $R\Gamma(X, K)$, and on the point the trace class is the trace by Lemma 11.10. The global trace is therefore the trace class

of a pushed-forward correspondence, and it remains to show that trace classes commute with pushforward along proper maps.

To push a cohomological correspondence forward along proper morphisms, we must produce from u a map $c_2'^{*}Rf_*K \to c_1'^{!}Rf_*K$. The only maps available for this are the units and counits of the adjunctions on the two legs, used as in the construction of $T_c(u)$. Here properness of f and g gives $Rf_* = Rf_!$ and $Rg_* = Rg_!$, and this allows us to push forward the counit for $Rc_{1!} \dashv c_1^{!}$. We also push forward along closed immersions (Section 11.5).

Definition 11.12 (Pushforward of a correspondence)**.** Let $f : X \to Y$ and $g : C \to C'$ be proper morphisms, and let $c' = (c_1', c_2') : C' \to Y \times Y$ be a correspondence with $c_i' g = f c_i$ for $i = 1, 2$. Put $K' = Rf_*K \in D^b_{\mathrm{ctf}}(Y, A)$. For a cohomological correspondence u on (X, K), the *pushforward* $u' : c_2'^{*}K' \to c_1'^{!}K'$ of u is the composite $u' = \beta\, Rg_*u\, \alpha$, where α is built from the unit and counit of $(\,)^* \dashv R(\,)_*$,

$$\alpha : c_2'^{*}Rf_*K \xrightarrow{\eta_g} Rg_*c_2^{*}f^{*}Rf_*K \xrightarrow{\varepsilon_f} Rg_*c_2^{*}K,$$

and $\beta : Rg_*c_1^{!}K \to c_1'^{!}Rf_*K$ is built from the counit of $R(\,)_! \dashv (\,)^!$: it is adjoint to

$$Rc_{1!}'Rg_*c_1^{!}K \simeq Rf_*Rc_{1!}c_1^{!}K \xrightarrow{Rf_*\varepsilon_{c_1}} Rf_*K.$$

The map g induces a proper morphism $h : \mathrm{Fix}(c) \to \mathrm{Fix}(c')$ of fixed loci, and $h_* : H^0(\mathrm{Fix}(c), \omega_{\mathrm{Fix}(c)}) \to H^0(\mathrm{Fix}(c'), \omega_{\mathrm{Fix}(c')})$ denotes the map given by the ordinary unit, proper pushforward and the exceptional counit.

For $Y = s$, $C' = s$ and $f = a_X$, $g = a_C$ with X and C proper, the pushforward u' is an endomorphism of $R\Gamma(X, K)$. Unwinding α and β shows that it is $T_c(u)$, and the following theorem compares trace classes before and after pushforward. In this special case it is the Lefschetz–Verdier formula.

Theorem 11.13 (Proper pushforward and Lefschetz–Verdier)**.** *Let $f : X \to Y$ and $g : C \to C'$ be proper, and let $c' = (c_1', c_2') : C' \to Y \times Y$ satisfy $c_i' g = f c_i$. Set $K' = Rf_*K$, and let u' be the pushforward of u to (Y, K') of Definition 11.12. With the induced proper map $h : \mathrm{Fix}(c) \to \mathrm{Fix}(c')$ one has*

$$\mathrm{Tr}_{c'}(u') = h_*\, \mathrm{Tr}_c(u).$$

If X and C are proper over k, this gives

$$\mathrm{tr}_A(T_c(u) \mid R\Gamma(X, K)) = \int_{\mathrm{Fix}(c)} \mathrm{Tr}_c(u) = \sum_B \mathrm{LT}_B(u),$$

for any finite open-and-closed decomposition $\mathrm{Fix}(c) = \coprod B$.

Proof. The trace class of u is built in three steps: u becomes a map into $P_X = K \boxtimes \mathbb{D}_X K$, evaluation maps P_X to the dualizing complex on the diagonal, and $c^!$ moves the result to the fixed locus. We check that each step commutes with pushforward. For the first two steps this means identifying the map adjoint to u' with the pushforward of the map adjoint to u, and evaluation for K' with the pushforward of evaluation for K. The third step needs more care. The squares $c_i' g = f c_i$ are only required to commute, so in general $\mathrm{Fix}(c) \neq C \times_{C'} \mathrm{Fix}(c')$, and we compare the two fixed loci through an intermediate scheme.

Maps into P. Put $P_Y = K' \boxtimes \mathbb{D}_Y K'$, and let $\tau_f : Rf_*\omega_X \to \omega_Y$ be the counit. By the Künneth formula for the proper map $f \times f$ and the exchange $\mathbb{D}_Y Rf_* \simeq Rf_*\mathbb{D}_X$ of Verdier duality, we obtain an isomorphism

$$\rho_f : R(f \times f)_* P_X \xrightarrow{\sim} P_Y.$$

A cohomological correspondence u defines a map $\theta_u : Rc_! A_C \to P_X$, obtained by exceptional adjunction from the section $A_C \to c^! P_X$. Under $Rc'_! Rg_* \simeq R(f \times f)_* Rc_!$, the map defined by the pushforward u' is

$$\theta_{u'} = \rho_f \, R(f \times f)_* \theta_u \, (Rc'_! Rg_* \simeq R(f \times f)_* Rc_!) \, Rc'_! \eta_g.$$

To verify this identity, we apply tensor–Hom adjunction and then the exceptional adjunction for c'_1. After this both sides become $Rc'_{1!}\alpha$, followed by $Rf_* Rc_{1!} u$ and $Rf_* \varepsilon_{c_1}$. The two sides use the projection formulas for c_1 and for c'_1 in different orders, but they agree because $c'_1 g = f c_1$ and projection formulas are compatible with composition.

Evaluation. Evaluation is compatible with ρ_f, in the sense that

$$e_{K'} \rho_f = R\Delta_{Y*} \tau_f \, R(f \times f)_* e_K, \qquad (f \times f)\Delta_X = \Delta_Y f.$$

After restricting to the diagonal and taking tensor–Hom adjoints, this says that Verdier duality for the proper map f identifies evaluation against Rf_*K with the pushforward of evaluation against K, followed by τ_f. This holds because that duality is defined by means of the counit τ_f.

Fixed loci. The two maps into P_Y and the two evaluations therefore match. The trace classes, however, live on two different fixed loci, and we must see that passing from $\mathrm{Fix}(c)$ to $\mathrm{Fix}(c')$ is the map h_* of Definition 11.12. Recall that the fixed loci are the pullbacks of the diagonals, $\mathrm{Fix}(c) = C \times_{X \times X} X$ and $\mathrm{Fix}(c') = C' \times_{Y \times Y} Y$. To compare them, we factor h through the fibre product $C \times_{C'} \mathrm{Fix}(c')$:

$$\mathrm{Fix}(c) \xrightarrow{i'} C \times_{C'} \mathrm{Fix}(c') \xrightarrow{\pi} \mathrm{Fix}(c'),$$

where the closed immersion i' is the pullback of $X \to X \times_Y X$ and π is the base change of g. We use proper base change for the square defining $C \times_{C'} \mathrm{Fix}(c')$, and the exceptional counit for i'. Since counits are compatible with composition, the integration map for h is $\tau_h = \tau_\pi R\pi_* \tau_{i'}$. Applying the identity for $\theta_{u'}$ and the compatibility of evaluation, together with these base change maps, we identify the map $A_{C'} \to Ri_{c'*}\omega_{\mathrm{Fix}(c')}$ defining the left-hand trace with

$$A_{C'} \xrightarrow{\eta_g} Rg_* A_C \longrightarrow Rg_* Ri_{c*}\omega_{\mathrm{Fix}(c)} = Ri_{c'*} Rh_* \omega_{\mathrm{Fix}(c)} \xrightarrow{Ri_{c'*}\tau_h} Ri_{c'*}\omega_{\mathrm{Fix}(c')}.$$

Taking the $i_{c'}^*$-adjoint by means of the identity of composite units $Rg_*\eta_{i_c}\,\eta_g = Ri_{c'*}\eta_h\,\eta_{i_{c'}}$, we obtain the first formula (compare [SGA5, III, Theorem 4.4 and Corollary 4.7]).

The trace formula. For proper X and C, we push forward to the identity correspondence on s, using $f = a_X$ and $g = a_C$. Then the pushed cohomological correspondence is $T_c(u)$. Indeed, for $g = a_C$ the map α is the pullback $R\Gamma(X, K) \to R\Gamma(C, c_2^* K)$ and β is the counit of c_1 after $R\Gamma$, and these are the outer arrows in the definition of $T_c(u)$. Moreover, the fixed locus of the identity correspondence of s is s itself. Hence the left side of the first formula is the local term of $T_c(u)$ on s, which by Lemma 11.10 is the trace of $T_c(u)$. On the right, h_* is integration over $\mathrm{Fix}(c)$, and since integration over a disjoint union is the sum of the integrations, this gives the second assertion. ∎

Example 11.14 (A correspondence on the point)**.** Let $X = s$, let $C = \coprod_{i=1}^{n} s$ be n points, with $c_1 = c_2$ the structure map, and let $u : c_2^* A = A_C \to c_1^! A = A_C$ be the identity, using $c_1^! = c_1^*$ for the finite étale map c_1. Then pullback along c_2 is the diagonal $A \to A^n$, and the counit $Rc_{1!}c_1^! A = A^n \to A$ is the sum. Hence $T_c(u)$ is multiplication by n on $R\Gamma(s, A) = A$, with trace n. Since $X \times X = s$, the fixed locus is all of C. On each of these n points u restricts to the identity correspondence of s on A, whose local term is $\mathrm{tr}_A(1 \mid A) = 1$ by Lemma 11.10. The right side of the theorem is therefore n as well. Note that the local terms are indexed by the points of C, not of X: here X has one point and the fixed locus has n.

11.5 Additivity of trace classes

To compute local terms, we cut K into pieces along a decomposition of X into an open and a closed part, both invariant under the correspondence. For this the trace class must be additive. For the trace of an endomorphism of a perfect complex, additivity says that a block triangular matrix has trace equal to the sum of the traces of its diagonal blocks. By Remark 11.3 it can only be expected for filtrations lifted to the filtered derived category. The trace class needs the same care, and we lift each step of the construction of the trace morphism to filtered complexes.

In a two-step filtration, for example, a filtered correspondence is a block triangular matrix $\left(\begin{smallmatrix} u_0 & * \\ 0 & u_1 \end{smallmatrix}\right)$ whose off-diagonal entry has weight 1. Since ω_X sits in filtration degree 0, the trace class does not see this entry (Lemma 11.16 below).

The operations that enter the trace morphism have filtered versions, with explicit formulas for their graded pieces. We use finite decreasing filtrations, the convolution filtration on tensor products, and the filtration on internal Hom given by

$$F^r \mathcal{H}om(\mathcal{M}, \mathcal{N}) = \{v : v(F^a \mathcal{M}) \subset F^{a+r} \mathcal{N} \text{ for every } a\}.$$

Write $R\mathcal{H}om_F$, $\otimes_F$ and $\mathbb{D}_F = R\mathcal{H}om_F(-, \omega_X)$ for the derived functors on the filtered derived category, where ω_X carries the filtration concentrated in weight zero, and M_a, N_a, K_a for the graded pieces of $\mathcal{M}$, $\mathcal{N}$, $\mathcal{K}$.

Lemma 11.15 (Graded pieces of filtered operations)**.** *For finitely filtered complexes $\mathcal{M}, \mathcal{N}, \mathcal{K}$ on X with graded pieces in $D^b_{\mathrm{ctf}}(X, A)$ there are isomorphisms*

$$\begin{aligned} \mathrm{gr}^r R\mathcal{H}om_F(\mathcal{M}, \mathcal{N}) &\simeq \bigoplus_a R\mathcal{H}om(M_a, N_{a+r}), \\ \mathrm{gr}^r \mathbb{D}_F \mathcal{K} &\simeq \mathbb{D}_X K_{-r}, \\ \mathrm{gr}^r(\mathcal{M} \otimes_F \mathcal{N}) &\simeq \bigoplus_{a+b=r} M_a \otimes^{\mathbf{L}} N_b. \end{aligned}$$

Under these isomorphisms, the restriction of the filtered evaluation $\mathcal{K} \otimes_F \mathbb{D}_F \mathcal{K} \to \omega_X$ to the summand $K_a \otimes^{\mathbf{L}} \mathbb{D}_X K_a$ of gr^0 is the evaluation of K_a.

Proof. The operations are derived using filtered flat and filtered injective resolutions. For finite filtrations these resolutions can be built successively on the graded pieces by mapping cylinders, and on them the three formulas hold termwise. For the evaluation, recall that the filtered evaluation is the tensor–Hom adjoint of $\mathrm{id}_{\mathcal{K}}$, whose graded map in weight zero is id_{K_a} on each summand. Hence on $K_a \otimes^{\mathbf{L}} \mathbb{D}_X K_a$ it is the ordinary evaluation. Moreover, the Koszul signs are those of the underlying complexes, and the filtration weights contribute no signs. ∎

With these formulas we lift the trace morphism of Definition 11.8 to filtered complexes.

Lemma 11.16 (Filtered trace morphism)**.** *Let $c : C \to X \times X$ be a correspondence and let $\mathcal{K}$ be a finitely filtered complex on X with underlying complex K and graded pieces K_a in $D^b_{\mathrm{ctf}}(X, A)$. There is a morphism of filtered complexes*

$$\mathrm{tr}_F : R\mathcal{H}om_F(c_2^*\mathcal{K}, c_1^!\mathcal{K}) \longrightarrow Ri_{c*}\omega_{\mathrm{Fix}(c)},$$

with the target in weight zero, whose underlying morphism is the trace morphism for K, and whose restriction to F^0 factors through

$$\mathrm{gr}^0\, R\mathcal{H}om_F(c_2^*\mathcal{K}, c_1^!\mathcal{K}) \simeq \bigoplus_a R\mathcal{H}om(c_2^*K_a, c_1^!K_a)$$

by a map whose restriction to the summand a is the trace morphism for K_a.

Proof. The trace morphism is the composite of four steps: the exceptional tensor–Hom adjunction, external duality, the adjoint of evaluation along the diagonal, and exceptional proper base change for the closed diagonal. Since each is defined by units, counits and tensor evaluation, the composite has a filtered version

$$R\mathcal{H}om_F(c_2^*\mathcal{K}, c_1^!\mathcal{K}) \simeq c^!(\mathcal{K} \boxtimes_F \mathbb{D}_F\mathcal{K}) \longrightarrow Ri_{c*}\omega_{\mathrm{Fix}(c)}$$

whose graded pieces, computed by Lemma 11.15, are the unfiltered versions. In particular the filtered comparison for external duality is an isomorphism, because each of its graded maps is a finite sum of the isomorphisms in Lemma 11.7, and graded pieces detect isomorphisms of finitely filtered complexes. Forgetting the filtration we recover the trace morphism for K.

The target is pure of weight zero, so tr_F vanishes on F^1, and its restriction to F^0 factors through gr^0. This factorization holds on filtered dg models, hence also in the filtered derived category. By Lemma 11.15, gr^0 of the source is $\bigoplus_a R\mathcal{H}om(c_2^*K_a, c_1^!K_a)$. Now the steps for internal Hom and external duality preserve the two indices, and on a pair of equal indices evaluation along the diagonal is the evaluation of K_a. Hence the map on the summand a is the trace morphism for K_a. ∎

To apply this to an open subscheme U and its closed complement Z, we must lift u to the localization triangle. For this the correspondence must respect the decomposition: we need $c_1^{-1}(Z) \subset c_2^{-1}(Z)$. For the graph of an endomorphism φ, where $c_1 = \mathrm{id}_X$ and $c_2 = \varphi$, this says that $\varphi(Z) \subset Z$.

It is convenient to work with exceptional adjoints. Put $\Theta = Rc_{1!}c_2^* : D^b_{\mathrm{ctf}}(X, A) \to D^b_{\mathrm{ctf}}(X, A)$. In these terms a cohomological correspondence u on (X, K) is the same as a map $\widehat{u} : \Theta K \to K$, its adjoint under $Rc_{1!} \dashv c_1^!$. Under the invariance condition there are no nonzero maps from ΘK_U to K_Z, so u restricts to K_U in an essentially unique way. Let

$$C_U = c_2^{-1}(U), \qquad C_Z = (c_1^{-1}(Z))_{\mathrm{red}}, \qquad i' : C_Z \longrightarrow C,$$

with i' the closed immersion. By the invariance condition both legs of C_U land in U and both legs of C_Z land in Z; we write $c_U = (c_{1U}, c_{2U}) : C_U \to U \times U$ and $c_Z = (c_{1Z}, c_{2Z}) : C_Z \to Z \times Z$ for the restricted correspondences.

Lemma 11.17 (Lifting a correspondence to an invariant decomposition)**.** *Let $c : C \to X \times X$ be a correspondence, $i : Z \to X$ a closed immersion with open complement $j : U \to X$, and suppose*

$$c_1^{-1}(Z) \subset c_2^{-1}(Z), \qquad \text{equivalently } c_2^{-1}(U) \subset c_1^{-1}(U).$$

Let u be a cohomological correspondence on (X, K), and write $K_U = j_! j^* K$ and $K_Z = Ri_* i^* K$ for the two pieces of the localization triangle $K_U \to K \to K_Z$. Then u has a lift, unique up to contractible choice, to a cohomological correspondence $\tilde{u}$ in the filtered derived category on the two-step filtered complex given by this triangle. Its graded pieces are cohomological correspondences u_U on (X, K_U) and u_Z on (X, K_Z), and $\widehat{u}_Z$ restricted to Z is

$$Rc_{1Z!} c_{2Z}^* i^* K \simeq i^* \Theta K \xrightarrow{i^* \widehat{u}} i^* K.$$

Proof. By the hypothesis $c_1^{-1}(Z) \subset c_2^{-1}(Z)$, base change of $Rc_{1!}$ along i, and the adjunction $i^* \dashv Ri_*$, we have

$$i^* \Theta K_U = 0, \qquad R\mathrm{Hom}_X(\Theta K_U, K_Z) \simeq R\mathrm{Hom}_Z(i^* \Theta K_U, i^* K) = 0.$$

The second vanishing is that of the whole mapping complex, including all shifts. Hence the composite $\Theta K_U \to \Theta K \xrightarrow{\widehat{u}} K$ factors through $K_U \to K$, with a contractible space of choices. The coherent cofibre of this factorization is a morphism of two-step filtered objects

$$\begin{array}{ccccc} \Theta K_U & \longrightarrow & \Theta K & \longrightarrow & \Theta K_Z \\ \downarrow & & \downarrow \widehat{u} & & \downarrow \\ K_U & \longrightarrow & K & \longrightarrow & K_Z. \end{array}$$

Since this cofibre includes the homotopy, it is a morphism in the filtered derived category, and its adjoint under the filtered adjunction $Rc_{1!} \dashv c_1^!$ is the lift $\tilde{u}$. By base change, composition and nilpotent invariance (which lets us use the reduced scheme C_Z),

$$\Theta K_U \simeq j_! Rc_{1U!} c_{2U}^* j^* K, \qquad i^* \Theta K \simeq Rc_{1Z!} c_{2Z}^* i^* K.$$

Applying i^* to the right-hand square and using $i^* \Theta K_U = 0$, so that $i^* \Theta K \simeq i^* \Theta K_Z$, we obtain the formula for $\widehat{u}_Z$ restricted to Z. ∎

The graded piece on the closed part is a correspondence on Z pushed forward to X. This is the form in which Proposition 12.16 uses it, combined with Theorem 11.13.

Lemma 11.18 (The closed piece as a pushforward)**.** *In the situation of Lemma 11.17, put $M = i^* K \in D^b_{\mathrm{ctf}}(Z, A)$, and let $u_{|Z}$ be the cohomological correspondence on (Z, M) for c_Z whose exceptional adjoint is $Rc_{1Z!} c_{2Z}^* M \simeq i^* \Theta K \xrightarrow{i^* \widehat{u}} M$. Then u_Z is the pushforward of $u_{|Z}$ along the proper maps i and i' (Definition 11.12).*

Proof. The maps α and β of Definition 11.12 are here

$$c_2^* Ri_* M \xrightarrow{\eta_{i'}} Ri'_* i'^* c_2^* Ri_* M = Ri'_* c_{2Z}^* i^* Ri_* M \xrightarrow{Ri'_* c_{2Z}^* \varepsilon_i} Ri'_* c_{2Z}^* M$$

and the map $Ri'_* c_{1Z}^! M \to c_1^! Ri_* M$ adjoint to

$$Rc_{1!}Ri'_*c_{1Z}^!M = Ri_*Rc_{1Z!}c_{1Z}^!M \xrightarrow{Ri_*\varepsilon_{c_{1Z}}} Ri_*M.$$

The exceptional adjoint of the pushforward is thus a map $\Theta Ri_*M \to Ri_*M$. After applying i^* and the base change identity $i^*\Theta Ri_*M \simeq Rc_{1Z!}c_{2Z}^*M$, it becomes the exceptional adjoint of $u_{|Z}$. By Lemma 11.17 the same holds for $\widehat{u}_Z$. Since $K_Z = Ri_*M$ and a map into Ri_*M is determined by its adjoint under $i^* \dashv Ri_*$, the two correspondences agree. ∎

Additivity takes the following form; in Proposition 12.16 we apply it with Z a single fixed point of Frobenius.

Proposition 11.19 (Additivity of trace classes)**.** *Let $c : C \to X \times X$ be a correspondence.*

1. *Let $\mathcal{K}$ be a finitely filtered complex on X with underlying complex K and graded pieces K_a in $D^b_{\mathrm{ctf}}(X, A)$, and let $\tilde{u} : c_2^*\mathcal{K} \to c_1^!\mathcal{K}$ be a cohomological correspondence in the filtered derived category, with underlying correspondence u on K and graded pieces u_a on K_a. Then*

$$\mathrm{Tr}_c(u) = \sum_a \mathrm{Tr}_c(u_a) \quad \text{in } H^0(\mathrm{Fix}(c), \omega_{\mathrm{Fix}(c)}).$$

2. *Let $i : Z \to X$ be a closed immersion with open complement U such that $c_1^{-1}(Z) \subset c_2^{-1}(Z)$, let $K \in D^b_{\mathrm{ctf}}(X, A)$, and let u be a cohomological correspondence on (X, K). Then*

$$\mathrm{Tr}_c(u) = \mathrm{Tr}_c(u_U) + \mathrm{Tr}_c(u_Z),$$

where u_U and u_Z are the graded pieces of the lift of Lemma 11.17, and u_Z is the pushforward along i and $C_Z \to C$ of the correspondence $u_{|Z}$ on (Z, i^*K) for $c_Z : C_Z = (c_1^{-1}(Z))_{\mathrm{red}} \to Z \times Z$ (Lemma 11.18).

Proof. For (1), the section $A_C \to R\mathcal{H}om_F(c_2^*\mathcal{K}, c_1^!\mathcal{K})$ defined by $\tilde{u}$ comes from the unit in weight zero, so it lands in F^0, and its image in gr^0 is the tuple $(u_a)_a$. Composing with the filtered trace morphism of Lemma 11.16 and forgetting the filtration gives on the one hand $\mathrm{Tr}_c(u)$, and on the other hand, by the factorization through gr^0, the sum of the trace classes $\mathrm{Tr}_c(u_a)$. For (2) we apply (1) to the lift $\tilde{u}$ of Lemma 11.17. ∎

Restricting the formula to a proper open and closed part B of the fixed locus and integrating, we get $\mathrm{LT}_B(u) = \mathrm{LT}_B(u_U) + \mathrm{LT}_B(u_Z)$.

12 The trace formula

Let X_0 be a separated scheme of finite type over $\mathbf{F}_q$, let $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let $\ell \nmid q$. By Lemma 12.5, the points of X_0 over $\mathbf{F}_{q^m}$ are the fixed points of the m-th power of the Frobenius Φ_X on X. The Lefschetz–Verdier formula (Theorem 11.13) writes a trace on cohomology as a sum of local terms, but it says nothing about what a local term is. To count points we need the local term of Frobenius at a fixed point to be the trace on the stalk.

For a self-map φ of a compact manifold with transverse fixed points, the Lefschetz formula counts each fixed point x with the sign of $\det(1 - d\varphi_x)$. The differential of Frobenius is zero, so we expect every sign to be $+1$. On a curve this can be seen directly. If y is a fixed point of Φ_X^m, u is a uniformizer at y defined over $\mathbf{F}_{q^m}$, and v is the same function on the second factor, then near (y, y) the graph of Φ_X^m is $v = u^{q^m}$, and it meets the diagonal $v = u$ with multiplicity one, since $u^{q^m} - u$ has a simple zero at $u = 0$.

Question 12.1 (The Lefschetz formula for Frobenius)**.** Does the Lefschetz formula hold for Frobenius?

For constant coefficients the local term at a transverse fixed point is 1, so on a proper X transversality would be enough. Two things stand in the way in general. First, X need not be proper. If we compactify, the new fixed points at the boundary have zero stalk for $j_!K$, and we need their local terms to vanish. Second, a local term is computed from $R\mathcal{H}om(c_2^*K, c_1^!K)$ near the fixed point, where c_1, c_2 are the two projections of the graph of Frobenius to X (Definition 11.5). The stalk $K_{\bar{y}}$ does not determine this complex (Remark 12.4), so the vanishing of the local term at a point where the stalk vanishes is a real statement, which transversality alone does not give.

One computation on a curve settles both problems: Proposition 12.16 says that for any morphism $\psi : (\Phi_X^m)^*K \to K$ on a smooth curve, the local term at a rational point y is $\operatorname{tr}(\psi_{\bar{y}} \mid K_{\bar{y}})$. The proof, which is Grothendieck's [SGA5, XV, §3], uses $d\Phi_X = 0$ a second time: the graph $v = u^{q^m}$ is tangent to the axis $v = 0$ and the diagonal is not, so blowing up (y, y) separates them, and near the exceptional divisor the local term becomes a restriction map that vanishes when $K_{\bar{y}} = 0$ (Remark 12.6). Since both sides of the formula are additive along an open subscheme and its complement (Lemma 12.19) and compatible with $Rf_!$ (Lemma 12.18), we never need to compactify anything but $\mathbf{A}^1 \subset \mathbf{P}^1$.

The answer is Theorem 12.24: for every Weil complex $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ (Section 10.5), where $\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$ and A is a finite commutative ℓ-primary ring, and for every $m \geq 1$, we have in Λ

$$\operatorname{tr}_\Lambda\big(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)\big) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \operatorname{tr}_\Lambda\big(\mathrm{Frob}_{y,m} \mid K_{\bar{y}}\big).$$

Here $\operatorname{tr}_\Lambda$ is the trace of an endomorphism of a perfect complex (Definition 11.1), and $\mathrm{Frob}_{y,m}$ is the m-th iterate of the Weil structure on the stalk at the fixed point $\bar{y}$. For $K = \Lambda_X$ and Λ a field the formula is Weil's axiom (b), and it gives the rationality of the zeta function (Corollary 12.31).

Purity, however, needs the formula for general K. For $\Lambda \in \{\mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, in the form of an Euler product (Theorem 12.30), it reads

$$\prod_{x \in |X_0|} \det\big(1 - T^{\deg x}\,\mathrm{Frob}_x \mid K_{\bar{x}}\big)^{-1} = \prod_i \det\big(1 - T\,\mathrm{Frob}_{X,K} \mid H_c^i(X, K)\big)^{(-1)^{i+1}},$$

which relates the local eigenvalues of K to the eigenvalues on cohomology.

The conventions of Chapter 11 remain in force.

12.1 Frobenius correspondences

Two Frobenius maps act on $X = X_0 \times_{k_0} k$: the k-linear scheme Frobenius Φ_X and the semilinear geometric Frobenius Fr_X that defines Weil structures. We prove the trace formula for Φ_X, whose graph is a correspondence over k, and Lemma 12.3 translates it to Fr_X.

The operator T_{c_m} on cohomology with compact support defined below makes sense even when X is not proper, because the second leg Φ_X^m of the graph c_m is proper. Indeed, Φ_X and all its iterates are finite, since a finitely generated algebra over a perfect field is finite over its subring of q-th powers.

We cannot work with Fr_X directly. Indeed, the Weil structure ϕ_K is a map $\mathrm{Fr}_X^* K \to K$, but Fr_X lies over the automorphism $\mathrm{Spec}(\sigma)$ of s. Hence its graph is not a morphism $X \to X \times_s X$ of k-schemes. Since the formalism of Chapter 11 uses $c_1^!$ only for k-morphisms, it does not apply to this graph. The k-linear $\Phi_X = F_X \circ \mathrm{Fr}_X$, on the other hand, has a graph of the required kind. Pullback along the absolute Frobenius F_X is the identity on étale sheaves, and this is how Lemma 12.3 turns ϕ_K into a map $\Phi_X^* K \to K$. For example, on $X = \mathbf{A}^1_k$ the map Φ_X is $t \mapsto t^q$, and the fixed points of Φ_X^m are the q^m elements of $\mathbf{F}_{q^m}$. Lemma 12.5 describes them in general.

We allow maps $\Phi_X^* K \to K$ that are not isomorphisms, because the proof over E clears denominators: it multiplies a Weil structure by a power of π_E (Lemma 12.22). The operators below are the two sides of the trace formula in Theorem 12.20.

Definition 12.2 (Frobenius correspondences and their operators)**.** Let X_0/k_0 be separated and of finite type, and put $X = X_0 \times_{k_0} k$. Denote by $\Phi_X : X \to X$ the base change of the q-power scheme Frobenius of X_0. It is a finite k-morphism and a universal homeomorphism. A *Frobenius correspondence* on K is a morphism, not necessarily invertible,

$$\psi : \Phi_X^* K \longrightarrow K, \qquad K \in D^b_{\mathrm{ctf}}(X, A).$$

Its iterates are the morphisms

$$\psi_m = \psi \circ \Phi_X^* \psi \circ \cdots \circ (\Phi_X^{m-1})^* \psi : (\Phi_X^m)^* K \longrightarrow K.$$

Let $c_m = (\mathrm{id}_X, \Phi_X^m) : X \to X \times_s X$ be the graph of Φ_X^m. Since its first leg is the identity, ψ_m is a cohomological correspondence on (X, K) for c_m. Its operator on cohomology with compact support (Definition 11.11) is the endomorphism

$$T_{c_m}(\psi_m) = R\Gamma_c(X, \psi_m) \circ (\Phi_X^m)^*,$$

where $(\Phi_X^m)^* : R\Gamma_c(X, K) \to R\Gamma_c(X, (\Phi_X^m)^* K)$ is the pullback map induced by the unit of $(\Phi_X^m)^* \dashv R(\Phi_X^m)_*$. For $x \in X_0(\mathbf{F}_{q^m})$ let $\bar{x} : s \to X$ be the associated fixed point. The stalk of ψ_m at $\bar{x}$ is an endomorphism $\psi_{m,\bar{x}} : K_{\bar{x}} \to K_{\bar{x}}$.

The operator $T_{c_m}(\psi_m)$ is defined with the k-linear Φ_X, while the Frobenius operator of Section 10.5, whose trace the theorem is about, is defined with Fr_X. The next lemma says that the two agree, so that the trace formula proved for Φ_X is a statement about $\mathrm{Frob}_{X,K}$.

Lemma 12.3 (Scheme Frobenius and geometric Frobenius)**.** *Let Fr_X be the semilinear geometric Frobenius of Definition 10.18, and for a morphism $\phi : \mathrm{Fr}_X^* K \to K$ put $\phi_m = \phi \circ \mathrm{Fr}_X^* \phi \circ \cdots \circ (\mathrm{Fr}_X^{m-1})^* \phi$. There are natural isomorphisms*

$$\kappa_m : (\Phi_X^m)^* K \xrightarrow{\sim} (\mathrm{Fr}_X^m)^* K$$

compatible with iteration. For a Weil structure $\phi : \mathrm{Fr}_X^* K \xrightarrow{\sim} K$, put $\psi = \phi\kappa_1$. Then

$$\psi_m = \phi_m \kappa_m, \qquad T_{c_m}(\psi_m) = \mathrm{Frob}_{X,K}^m,$$

where $\mathrm{Frob}_{X,K}$ is the geometric Frobenius operator on compact cohomology of Definition 10.19. The same identity of operators holds for an arbitrary morphism $(\mathrm{Fr}_X^m)^* K \to K$ in place of ϕ_m.

Proof. The maps Φ_X and Fr_X differ by the absolute Frobenius, and pullback along the absolute Frobenius is naturally the identity on étale sheaves. This gives κ_m, and we then compare the two pullback maps on one compactification. On coordinate rings the two maps are

$$\Phi_X^{\#}(a \otimes b) = a^q \otimes b, \qquad \mathrm{Fr}_X^{\#}(a \otimes b) = a \otimes b^{1/q}$$

so that $\Phi_X = F_X \mathrm{Fr}_X$, where F_X is the absolute q-power Frobenius of X. The absolute Frobenius commutes with every morphism of schemes over $\mathbf{F}_p$, so also $\Phi_X^m = \mathrm{Fr}_X^m F_X^m$. On the classical constructible category, pullback along absolute Frobenius is naturally isomorphic to the identity, because for an étale object its relative Frobenius is an isomorphism, functorial in that object. Transporting this natural isomorphism to $D^b_{\mathrm{ctf}}(X, A)$ by the comparison with the pro-étale site, we obtain $\theta : F_X^* \simeq \mathrm{id}$. It is compatible with pullback along every morphism h, that is, $h^*\theta = \theta h^*$ under $h^*F^* = F^*h^*$, since relative Frobenius of étale objects is compatible with base change. Let $\theta^m : (F_X^m)^* \simeq \mathrm{id}$ be its m-fold composite and put

$$\kappa_m = \theta^m_{(\mathrm{Fr}_X^m)^* K} : (\Phi_X^m)^* K = (F_X^m)^* (\mathrm{Fr}_X^m)^* K \xrightarrow{\sim} (\mathrm{Fr}_X^m)^* K.$$

Compatibility with iteration means that $\kappa_{m+1} = \kappa_{m,\mathrm{Fr}_X^* K} \circ (\Phi_X^m)^* \kappa_1$, and it follows from the compatibility of θ with pullback along Φ_X^m. Now $\psi_m = \phi_m \kappa_m$ follows by induction on m. If it holds for m, then

$$\begin{aligned} \psi_{m+1} &= \psi_m \circ (\Phi_X^m)^* \psi = \phi_m \circ \kappa_m \circ (\Phi_X^m)^* \phi \circ (\Phi_X^m)^* \kappa_1 \\ &= \phi_m \circ (\mathrm{Fr}_X^m)^* \phi \circ \kappa_{m,\mathrm{Fr}_X^* K} \circ (\Phi_X^m)^* \kappa_1 = \phi_{m+1} \kappa_{m+1}, \end{aligned}$$

where the third equality is the naturality of κ_m in K.

For the pullback maps on compact cohomology, let $j : X \hookrightarrow \bar{X}$ be the base change of a compactification of X_0 over k_0, and use the pullback maps pb_h of Section 10.5. The endomorphisms $\Phi_{\bar{X}}$, $\mathrm{Fr}_{\bar{X}}$ and $F_{\bar{X}}$ of the scheme $\bar{X}$ restrict to Φ_X, Fr_X and F_X. Each of them has inverse image X of X, so each commutes with $j_!$, and $\Phi_{\bar{X}}^m = \mathrm{Fr}_{\bar{X}}^m F_{\bar{X}}^m$. Put $L = j_! K$, so that $R\Gamma_c(X, K) = R\Gamma(\bar{X}, L)$.

First, the map $(\Phi_X^m)^*$ of Definition 12.2 is $\mathrm{pb}_{\Phi_{\bar{X}}^m, L}$. Indeed, Φ_X^m is finite, so $R(\Phi_X^m)_! = R(\Phi_X^m)_*$, and the square $j\Phi_X^m = \Phi_{\bar{X}}^m j$ is cartesian, so $j_! R(\Phi_X^m)_* \simeq R(\Phi_{\bar{X}}^m)_* j_!$. Under this isomorphism the identification $R\Gamma_c(X, R(\Phi_X^m)_* -) = R\Gamma_c(X, -)$ used there is the composition of direct images $R\Gamma(\bar{X}, R(\Phi_{\bar{X}}^m)_* -) = R\Gamma(\bar{X}, -)$, and the unit for Φ_X^m corresponds to the unit for $\Phi_{\bar{X}}^m$.

Second, by transitivity, $\mathrm{pb}_{\Phi_{\bar{X}}^m, L} = \mathrm{pb}_{F_{\bar{X}}^m, (\mathrm{Fr}_{\bar{X}}^m)^* L} \circ \mathrm{pb}_{\mathrm{Fr}_{\bar{X}}^m, L}$. The absolute Frobenius $F_{\bar{X}}^m$ satisfies the hypothesis of [Stacks, 03SP], since its relative Frobenius over each étale object is an isomorphism, functorial in that object. Since A is finite, we may compute on the étale site (Proposition 5.15). There the formal argument of [Stacks, 03SP] identifies $(F_{\bar{X}}^m)_*$ and $(F_{\bar{X}}^m)^{-1}$ with the identity functors of the topos, compatibly with the unit, and this identification is θ^m, since both come from the relative Frobenius isomorphisms of étale objects. Hence for every $N \in D^b_{\mathrm{ctf}}(\bar{X}, A)$ the pullback map $\mathrm{pb}_{F_{\bar{X}}^m, N}$ followed by $R\Gamma(\bar{X}, \theta_N^m)$ is the identity of $R\Gamma(\bar{X}, N)$. With $N = (\mathrm{Fr}_{\bar{X}}^m)^* L$ this gives

$$R\Gamma_c(X, \kappa_m) \circ (\Phi_X^m)^* = \mathrm{pb}_{\mathrm{Fr}_X^m}.$$

Finally, the naturality of $\mathrm{pb}_{\mathrm{Fr}_X}$ moves the maps $R\Gamma_c(X, (\mathrm{Fr}_X^i)^* \phi)$ past the pullback maps, and transitivity gives $\mathrm{Frob}_{X,K}^m = R\Gamma_c(X, \phi_m) \circ \mathrm{pb}_{\mathrm{Fr}_X^m}$. Hence

$$T_{c_m}(\psi_m) = R\Gamma_c(X, \phi_m) \circ R\Gamma_c(X, \kappa_m) \circ (\Phi_X^m)^* = R\Gamma_c(X, \phi_m) \circ \mathrm{pb}_{\mathrm{Fr}_X^m} = \mathrm{Frob}_{X,K}^m.$$

The same computation with an arbitrary morphism $\chi : (\mathrm{Fr}_X^m)^* K \to K$ in place of ϕ_m gives $T_{c_m}(\chi \kappa_m) = R\Gamma_c(X, \chi) \circ \mathrm{pb}_{\mathrm{Fr}_X^m}$. ∎

For example, if X_0 is smooth and geometrically connected of dimension d and $K = A_X$ with its constant Weil structure, then $T_{c_1}(\psi)$ acts on $H_c^{2d}(X, A) = A(-d)$ by q^d, which agrees with the action of geometric Frobenius on $A(-d)$ recorded after Definition 10.19.

12.2 Local terms on a curve

The local terms of the trace formula live on the fixed point scheme $\mathrm{Fix}(\Phi_X^m) = \mathrm{Fix}(c_m)$ of the graph c_m. More precisely, a local term is defined on each open and closed part of this scheme that is proper over k. The next lemma shows that the fixed scheme is a finite reduced set of points, one for each $\mathbf{F}_{q^m}$-rational point of X_0, so every such point has its own local term. For Frobenius on a smooth curve we then show that this local term is the trace on the stalk.

On $\mathbf{A}^1$ the fixed scheme of Φ^m is $\operatorname{Spec} k[t]/(t^{q^m} - t)$. Its q^m points are reduced because $t^{q^m} - t$ has derivative -1. This is the algebraic form of the transversality seen at the start of Chapter 12. Reducedness alone does not determine the local term, because the stalk does not determine $R\mathcal{H}om(c_2^* K, c_1^! K)$ near the fixed point (Remark 12.4 below).

Remark 12.4 (Local terms are not computed stalk by stalk)**.** The vanishing of the local term at a rational point y with $K_{\bar{y}} = 0$, proved below, is a geometric fact about the restriction from cohomology with supports in the graph of Φ_X^m to cohomology with supports in y, and it does not follow from a locality property of internal Hom. For example, if $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ and $K = j_! A$, then the stalk of K at the origin is zero, but $R\mathcal{H}om_A(j_! A, j_! A) \simeq R j_* A$ has stalk A in degree zero there. Hence a local term cannot be computed by commuting a geometric stalk functor past this internal Hom.

Lemma 12.5 (The Frobenius fixed scheme)**.** *For $m \geq 1$, the fixed point scheme of Φ_X^m satisfies*

$$\mathrm{Fix}(\Phi_X^m) \simeq \coprod_{x \in X_0(\mathbf{F}_{q^m})} s.$$

In particular, the fixed scheme is finite, reduced and proper over k.

Proof. We reduce to a closed subscheme of affine space, where the equations $t^{q^m} = t$ have simple roots.

For every open $U_0 \subset X_0$ the restriction to $U = U_0 \times_{k_0} k$ of the fixed scheme of Φ_X^m is the fixed scheme of Φ_U^m, because Φ_X is the base change of a morphism of X_0 and so maps U into itself. As X_0 is quasi-compact, we may therefore assume that X_0 is a closed subscheme of $\mathbf{A}^r_{\mathbf{F}_q}$, with coordinates $t_1, \ldots, t_r$.

In this case the fixed scheme is the pullback of the diagonal along the graph. Since the equations of X_0 have coefficients in $\mathbf{F}_q$, we have $(\Phi_X^m)^{\#} t_i = t_i^{q^m}$, and the fixed scheme is cut out by the defining equations of X_0 together with $t_i^{q^m} - t_i = 0$. Each polynomial $t^{q^m} - t$ is separable, so the algebra $k[t_1, \ldots, t_r]/(t_i^{q^m} - t_i)_i$ is a product of copies of k indexed by $\mathbf{F}^r_{q^m}$. Its quotient by the defining equations of X_0 retains exactly the factors at the rational solutions of those equations, as asserted. In particular the fixed scheme is a closed subscheme of a finite étale k-scheme, hence étale. The argument uses only the equations of X_0, so X_0 need not be reduced. ∎

Thus every fixed point y is an open and closed point of the fixed scheme, and its local term LT_y is defined.

Remark 12.6 (Outline of the local computation)**.** Let X_0 be a smooth curve and $y \in X_0(\mathbf{F}_{q^m})$. The trace formula on $\mathbf{P}^1$ needs Proposition 12.16 below, which says that the local term at y of any morphism $\psi : (\Phi_X^m)^* K \to K$ is the naive trace $\mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}})$, also when K is not lisse at y. To prove it, we filter K by the open complement of y and the point y. By additivity of traces this reduces the proposition to two cases: the point itself, where Lemma 11.10 applies, and the case $K_{\bar{y}} = 0$, where we must show that the local term vanishes.

The second case needs a geometric argument. Indeed, the local term is defined on a neighbourhood of (y, y) in $X \times X$, where the Frobenius graph meets the diagonal, and the vanishing of the stalk at y does not by itself control it. For this we follow [SGA5, III B, §§2–4]. In this approach the local term is a restriction map from the graph to the diagonal, and blowing up (y, y) separates the two curves. After the blowup the local term becomes a restriction to punctured curves near the point where two transverse divisors meet. There it vanishes when the monodromy is an ℓ-group. In general we reduce to that case by passing to finite covers of degree prime to ℓ (Appendix D); for such a cover the trace map splits the pullback.

What vanishes is a morphism in $D(A)$, before any trace is taken. For this reason the vanishing holds for every ψ, for every finite ℓ-primary ring A, and for every K with $K_{\bar{y}} = 0$. In particular it holds for $j_! K$ on $\mathbf{P}^1$, which is not lisse at infinity. We forget the A-action only in Lemma 12.9, to detect a zero object.

12.2.1 The local term as a restriction map

The vanishing of a local term when $K_{\bar{y}} = 0$ is hard to see from the definition through the trace morphism (Definition 11.8), so we first replace that definition by a description familiar from topology. There the intersection number of two submanifolds Γ and Δ is computed by restricting the Thom class of Γ to Δ. Since the Thom class is supported on Γ, the result is a class supported on $\Gamma \cap \Delta$. Similarly, in our setting ψ defines a class supported on the graph. A restriction map ρ takes it to cohomology of the diagonal supported at y, and the local term is the evaluation of the result. Note that the map ρ does not involve ψ. It therefore suffices to show that ρ itself is zero when $K_{\bar{y}} = 0$ (Lemma 12.15), and then the vanishing holds for every ψ, invertible or not.

Lemma 12.7 (The local term as a restriction map)**.** *Let X_0 be a smooth curve over k_0, let $X = X_0 \times_{k_0} k$, let $K \in D^b_{\mathrm{ctf}}(X, A)$, and let y be a fixed point of Φ_X^m. Let $\Gamma \subset X \times X$ be the transposed graph of Φ_X^m, the image of the closed immersion $\gamma = (\Phi_X^m, \mathrm{id}_X)$, and let $\Delta \subset X \times X$ be the diagonal, the image of $\delta = (\mathrm{id}_X, \mathrm{id}_X)$. On the strict localization T of $X \times X$ at (y, y) put*

$$P = (\mathbb{D}_X K \boxtimes K)|_T.$$

A morphism $\psi : (\Phi_X^m)^* K \to K$ determines a class $[\psi] \in H^0_\Gamma(T, P)$, and the local term $\mathrm{LT}_y(\psi)$ of Definition 11.9 is the image of this class under the composite

$$R\Gamma_\Gamma(T, P) \xrightarrow{\rho} R\Gamma_y(\Delta, \delta^* P) \xrightarrow{\mathrm{ev}_K} R\Gamma_y(X, \omega_X) \simeq R\Gamma(y, \omega_y) \xrightarrow{\int_y} A,$$

where ρ is restriction from Γ-supported to y-supported cohomology.

In this lemma the two factors of the product are interchanged compared with the convention $c_m = (\mathrm{id}_X, \Phi_X^m)$ of Definition 12.2.

Proof of Lemma 12.7. We show that the local term and the displayed composite are both adjoint to one map $R\gamma_*\gamma^!P \to R\delta_*\omega_X$. Once the class $[\psi]$ and the map ρ are built, the comparison is formal, and computing on étale models changes nothing.

By external duality, $\gamma^!P$ is identified with $R\mathcal{H}om((\Phi_X^m)^*K, K)$, so exceptional adjunction identifies ψ with $[\psi]$. The map ρ is constructed from the two canonical arrows

$$R\gamma_*\gamma^!P \longrightarrow P \longrightarrow R\delta_*\delta^*P.$$

After restriction to Δ, the source is supported on $\Gamma \cap \Delta$. By Lemma 12.5 the scheme $\Gamma \cap \Delta$ is reduced, and on T it is the single point (y, y). Hence the resulting map factors through $R\Gamma_y(\Delta, \delta^*P)$. Equivalently, it is the ordinary pullback of the object with supports, followed by the base change map to local cohomology with support in the intersection.

The adjoint of $\mathrm{ev}_K : \delta^*P \to \omega_X$ is the map $P \to R\delta_*\omega_X$ used to define the trace morphism in Definition 11.8. Hence the displayed composite, before $\int_y$, is adjoint to the map

$$R\gamma_*\gamma^!P \longrightarrow P \longrightarrow R\delta_*\omega_X.$$

The local term is built from the same map. Indeed, by exceptional proper base change for δ, the adjoint of this map, which is supported on y, is the trace map $\gamma^!P \to Ri_*\omega_y$ of Definition 11.9. The triangle identities of the ordinary and exceptional adjunctions then show that the two constructions give the same element of A.

The constructions above also make sense on the constructible étale models, where restriction to the strict localization is the filtered limit over étale neighbourhoods. The comparison theorems identify inverse images, exceptional adjunctions, duality and the support maps on the two sites, and hence they identify the class obtained on the étale models with the pro-étale local term of Definition 11.9. ∎

12.2.2 Vanishing on a strictly local surface

The vanishing of the local term is explained by the position of the curves at (y, y). Since $d\Phi_X = 0$, the transposed graph Γ is tangent to the vertical axis $V = \{y\} \times X$. The diagonal Δ and the horizontal axis H have other tangent directions. This persists on suitable covers of degree prime to ℓ (Lemma 12.13 below). Blowing up the closed point separates the curves: the strict transforms of $\Gamma \cup V$, of Δ and of H meet the exceptional divisor $E \simeq \mathbf{P}^1$ in different points.

Let M be the direct image of the coefficients from the complement of E. We consider M on E minus the points of $\Gamma \cup V$, and let z be the point where H meets E. There M is locally constant away from z, and its stalk at z is zero. Both facts use that the monodromy is an ℓ-group (Lemma 12.9). Now a branch of Δ meets E in one point w, and the restriction to this branch factors through the stalk M_w. Composing the restriction from E to M_w with the specialization isomorphism from M_w to the generic stalk of E, we get the restriction to the zero stalk M_z followed by specialization. Hence the restriction to M_w is zero.

The transverse case is Lemma 12.9, and Lemma 12.11 reduces to it by blowing up the closed point. We compute on classical étale models of strictly local schemes, which are filtered limits of schemes of finite type, and in the proof of Lemma 12.15 we transport the result to the pro-étale site.

Lemma 12.8 (A punctured surface and a punctured curve)**.** *Let S be the strict localization of a smooth surface over an algebraically closed field k of characteristic p at a closed point z, let $\ell \neq p$, and let $D_1, D_2 \subset S$ be regular divisors meeting transversely at z. Then the restriction map*

$$R\Gamma(S - D_2, \mathbf{F}_\ell) \longrightarrow R\Gamma(D_1 - z, \mathbf{F}_\ell)$$

is an isomorphism.

Proof. Both complexes have cohomology $\mathbf{F}_\ell$ in degree zero, $\mathbf{F}_\ell(-1)$ in degree one, and zero in all other degrees. For the surface this follows from localization, the acyclicity of the strictly local schemes S, D_2, and the purity identity $i^!_{D_2}\mathbf{F}_\ell = \mathbf{F}_\ell(-1)[-2]$. This identity, in turn, comes by transitivity from the dualizing complexes of the smooth schemes S and D_2.

For the punctured trait $D_1 - z$, vanishing in degrees ≥ 2 follows from Proposition D.5, by passing to the filtered limit of punctured neighbourhoods that are curves. In degree one the Kummer sequence gives $H^1(-, \mu_\ell) = k'^*/k'^{*\ell}$, where k' is the fraction field. Moreover, the units of the strictly henselian trait are ℓ-divisible. Hence the valuation induces $k'^*/k'^{*\ell} \simeq \mathbf{Z}/\ell$, and the class of a uniformizer is a generator.

To see that the restriction map itself, and not only the two sides, is an isomorphism, we find a generator of H^1 on each side. The restriction is clearly the identity in degree zero. In degree one, let g be a local equation of D_2. Since the regular local ring of S is factorial, $\mathrm{Pic}(S - D_2) = 0$ and the units on $S - D_2$ are the units of S times the powers of g. As the units of S are ℓ-divisible, the Kummer sequence shows that $H^1(S - D_2, \mu_\ell) \simeq \mathbf{Z}/\ell$, generated by the class of g. Since D_1 and D_2 meet transversely, g restricts to a uniformizer of D_1, whose class generates $H^1(D_1 - z, \mu_\ell)$. So the restriction is an isomorphism in degree one as well, and since both complexes vanish in the other degrees, it is an isomorphism. ∎

Lemma 12.9 (Cohomology near two transverse divisors)**.** *Let X be a smooth surface over an algebraically closed field of characteristic p, or an étale limit with affine transition maps of such surfaces. Let D_1, D_2 be regular divisors meeting transversely at exactly one point z, and let*

$$j : X - D_2 \hookrightarrow X, \qquad j' : X - (D_1 \cup D_2) \hookrightarrow X - D_2.$$

Let $\ell \neq p$, let A be a finite commutative ℓ-primary ring, and let N be a bounded complex of A-modules on $X - (D_1 \cup D_2)$ with finite locally constant cohomology whose monodromy factors through a finite ℓ-group. Put $M = Rj_*j'_!N$. Then $M_z = 0$, and $M|_{D_2 - z}$ has finite locally constant cohomology. If D_2 is connected, then for every closed geometric point $w \in D_2 - z$ the restriction map

$$R\Gamma(D_2, M|_{D_2}) \longrightarrow M_w$$

is zero in $D(A)$.

Proof. The idea is that, since the monodromy is an ℓ-group, we may filter the coefficients until they become the constant sheaf $\mathbf{F}_\ell$. For constant coefficients, $M_z = 0$ is a statement on the strict localization of X at z: restriction from the complement of D_2 to the punctured curve $D_1 - z$ is an isomorphism on cohomology. This holds because a local equation of D_2 gives the generator of H^1 on both sides. Once we know that M is locally constant along $D_2 - z$, the vanishing of the restriction map follows from $M_z = 0$ by specialization along D_2.

For the assertion at z, pass to the strict localization S of X at z and write $D_{1,S}, D_{2,S}$ for the two transverse divisors. By the formula for stalks of a direct image along an open immersion [Stacks, 03Q7] and localization we have

$$M_z = R\Gamma(S - D_{2,S}, j'_!N).$$

To prove that this object is zero we may forget the A-action, since the exact forgetful functor is conservative and the cohomology of sheaves of modules agrees with the cohomology of the underlying abelian sheaves [Stacks, 03FD]. A finite abelian ℓ-group with an action of a finite ℓ-group has a finite filtration whose quotients are $\mathbf{F}_\ell$ with trivial action. To see this, filter by powers of ℓ and use the nilpotence of the augmentation ideal of the group algebra over $\mathbf{F}_\ell$. Since $j'_!$ and $R\Gamma(S - D_{2,S}, -)$ are exact, the class of N for which $R\Gamma(S - D_{2,S}, j'_!N) = 0$ is closed under extensions. Using truncations and these composition series we reduce the assertion to $N = \mathbf{F}_\ell$ on $S - (D_{1,S} \cup D_{2,S})$.

By Lemma 12.8, applied to S with the divisors $D_{1,S}, D_{2,S}$, the restriction map $R\Gamma(S - D_{2,S}, \mathbf{F}_\ell) \to R\Gamma(D_{1,S} - z, \mathbf{F}_\ell)$ is an isomorphism. The localization sequence for $D_{1,S} - z$ and its complement in $S - D_{2,S}$ identifies $R\Gamma(S - D_{2,S}, j'_!\mathbf{F}_\ell)$ with the cone of this map, shifted by -1. Hence this complex is zero, and $M_z = 0$ in $D(A)$ by conservativity.

We turn to the local constancy of M along $D_2 - z$, where only the divisor D_2 remains. Since D_1 meets D_2 only at z, we have $j'_!N = N$ near $D_2 - z$, so $M = Rj_*N$ there. To compute Rj_*N we use Abhyankar's lemma along the regular divisor D_2 [Stacks, 0EYG]: étale locally on D_2, a finite tame local system is trivialized by a Kummer cover of a local parameter of D_2 and an unramified cover. The lemma applies here because our finite ℓ-group monodromy is tame, wild inertia being pro-p. Consequently, the stalks of Rj_*N are the continuous cohomology of the same representation of tame inertia, and their specialization maps along $D_2 - z$ are isomorphisms. Explicitly, the pro-ℓ part of tame inertia is $\mathbf{Z}_\ell(1)$. Its cohomology with coefficients in a finite ℓ-primary module is computed by the two-term complex given by a topological generator minus one. On the other hand, the finite quotients of the remaining factors of tame inertia have order prime to ℓ, so taking invariants under them is exact on these coefficients. Finally, we may choose the parameter and the unramified cover on an étale neighbourhood. Hence this description, including the specialization maps, is locally constant. Applying the argument to the cohomology sheaves of N and using the bounded spectral sequence proves the assertion for N.

The last assertion combines the two facts just proved. A global section of $M|_{D_2}$ specializes to every stalk. Its image at z is zero, and along $D_2 - z$ specialization is an isomorphism on stalks. To make this precise, choose a geometric generic point $\bar{\eta}$ of the connected regular curve D_2, together with specialization maps from $\bar{\eta}$ to z and to w. By naturality of restriction we get a commutative square

$$\begin{array}{ccc} R\Gamma(D_2, M|_{D_2}) & \longrightarrow & M_w \\ \downarrow & & \downarrow\sim \\ M_z = 0 & \longrightarrow & M_{\bar{\eta}}. \end{array}$$

The right arrow is an isomorphism by the local constancy just proved, so the upper arrow is zero. ∎

Since the restriction to a curve meeting D_2 in one point factors through the stalk of M at that point, the lemma implies that restrictions to punctured curves vanish.

Lemma 12.10 (Restriction to a punctured curve)**.** *In the situation of Lemma 12.9 with D_2 connected, let $C \subset X$ be a closed curve with $C \cap D_2 = \{w\}$, and let $C_{(w)}$ be its strict localization at w. Then the restriction map*

$$R\Gamma(X - D_2, j'_!N) \longrightarrow R\Gamma(C_{(w)} - w, (j'_!N)|_{C_{(w)}-w})$$

is zero in $D(A)$.

Proof. The source is $R\Gamma(X - D_2, j'_!N) = R\Gamma(X, M)$, by the definition $M = Rj_*j'_!N$. Since C meets D_2 only at w, the punctured curve $C_{(w)} - w$ lies over the strict localization of X at w and avoids D_2. The map therefore factors as the restriction to $R\Gamma(D_2, M|_{D_2})$, followed by the restriction to the stalk $M_w = R\Gamma(X_{(w)} - D_2, j'_!N)$ (by the stalk formula), followed by the restriction to the displayed target. Since all these maps are restrictions of the same complex, the composite is the given map. If $w \neq z$, the middle map is zero by Lemma 12.9. If $w = z$, the intermediate object M_z is zero. ∎

Near (y, y) the curves we care about do not meet transversely, since the graph, the diagonal and the axis H all pass through the same point. The next lemma reduces this configuration to the transverse one by a single blowup.

Lemma 12.11 (Vanishing of restrictions after blowing up)**.** *Let T be the strict localization of a smooth surface at a closed point t. Let H be a regular curve through t, and let D and C be reduced closed curves through t such that the tangent cone of D at t meets the tangent cones of C and of H only in 0. Let $j' : T - (D \cup H) \hookrightarrow T - D$ be the open immersion and let N be a bounded complex on $T - (D \cup H)$ with finite locally constant cohomology and finite ℓ-group monodromy. Then the restriction map*

$$R\Gamma(T - D, j'_!N) \longrightarrow R\Gamma(C - t, (j'_!N)|_{C-t})$$

is zero in $D(A)$.

Proof. After blowing up t, the three curves are separated along the exceptional divisor, which then plays the role of D_2 in the previous lemma.

Let $\widetilde{T} \to T$ be the blowup in t, with exceptional divisor $E \simeq \mathbf{P}^1_k$, and let D', H', C' be the strict transforms. In these terms, the hypotheses on tangent cones say that $D' \cap E$ is disjoint from both $C' \cap E$ and $H' \cap E$. We therefore apply Lemmas 12.9 and Lemma 12.10 to the surface $X = \widetilde{T} - D'$ with the divisors $D_1 = H' \cap X$ and $D_2 = E \cap X = E - D'$. Here the divisor $E - D'$ is connected, since it is a projective line with finitely many points removed, and $H' \cap X$ meets it transversely in the single point z corresponding to the tangent direction of H. Since the blowup is an isomorphism off E, the source of the map is $R\Gamma(X - E, j'_!N')$ for the pulled-back complex N'.

The strict transform C' is proper and quasi-finite over the strictly local curve C, hence finite. Its closed fibre is the finite set of tangent directions of its branches, so C' is a finite disjoint union of strictly local curves C'_w, each meeting E at a single point w of $E - D'$. Each restriction to $C'_w - w$ is zero by Lemma 12.10. Since $C' - E \simeq C - t$, the given restriction map is the direct sum of these zero maps. ∎

12.2.3 The local term at a zero stalk

In the setting of Lemma 12.7 we compute on the classical étale models. This is legitimate because the classical comparison identifies tensor products, duality, exceptional adjunctions and the pairing between graph and diagonal with their versions there. Let X_1 and X_2 be two copies of

the strict localization of X at y. The strict localization of $X_1 \times X_2$ at (y, y) is the strictly local surface T of Lemma 12.7. Write

$$H = X_1 \times \{y\}, \qquad V = \{y\} \times X_2,$$

and keep Γ and Δ for the transposed Frobenius graph and the diagonal in T. Let u and v be the pullbacks to the two factors X_1 and X_2 of the same uniformizer of X at y, so that Δ, V and H have equations $u - v = 0$, $u = 0$ and $v = 0$. Write $\varphi(v) = (\Phi_X^m)^{\#} u$, a power series in v; the graph Γ has equation $u - \varphi(v) = 0$. Since $d\Phi_X = 0$, the graph Γ is tangent to V, while Δ and H have other tangent directions.

The next three lemmas prepare the proof of Lemma 12.15.

Lemma 12.12 (Covers of degree prime to ℓ)**.** *Let S be the strict localization of a smooth curve over the algebraically closed field k at a closed point s, and let $\mathcal{F}_1, \dots, \mathcal{F}_r$ be finite locally constant sheaves on $(S - s)_{\text{ét}}$. There is an integer d_1 prime to ℓ with the following property. For every multiple d of d_1 that is prime to ℓ there is a finite flat morphism $S' \to S$ of degree d such that*

1. *S' is the strict localization of a smooth curve over k at a closed point s', and s' is the only point of S' over s; more precisely, $S' \to S$ is the base change of a finite flat morphism $C' \to C$ of smooth curves over k, where C is an étale neighbourhood of s, which is étale away from s and has a single point over s,*
2. *$S' - s' \to S - s$ is finite étale, and*
3. *the monodromy of the pullback of each $\mathcal{F}_i$ to $S' - s'$ factors through a finite ℓ-group.*

Proof. We take the fixed field of an ℓ-Sylow subgroup of the monodromy, adjust the degree by an Eisenstein extension, and descend to curves.

Let F be the fraction field of the discrete valuation ring of S. Then the punctured trait $S - s$ is $\operatorname{Spec}(F)$, so the $\mathcal{F}_i$ are finite sets with a continuous action of $\operatorname{Gal}(F^{\text{sep}}/F)$. We choose a finite Galois extension L/F over which all of them become constant, let P be an ℓ-Sylow subgroup of $G = \operatorname{Gal}(L/F)$, and put $F_1 = L^P$ and $d_1 = [F_1 : F] = [G : P]$, which is prime to ℓ.

Let $d = d_1 e$ be a multiple of d_1 prime to ℓ. The normalization S_1 of S in the separable extension F_1 is finite over S, and it is local because it is a domain finite over a henselian local ring. Hence S_1 is a strictly henselian discrete valuation ring with residue field k, and it has degree d_1 over S. If $e > 1$, let t be a uniformizer of S_1. The Eisenstein polynomial $z^e + tz - t$ is irreducible, and its derivative $ez^{e-1} + t$ is nonzero, also when $p \mid e$, so it is separable. Hence $S' = S_1[z]/(z^e + tz - t)$ is a discrete valuation ring, totally ramified of degree e over S_1, with separable fraction field F' over F. If $e = 1$ we put $S' = S_1$. Then $S' \to S$ is finite flat of degree d, with a single point s' over s, and $S' - s' = \operatorname{Spec}(F') \to \operatorname{Spec}(F)$ is finite étale. The group $\operatorname{Gal}(F^{\text{sep}}/F')$ acts on the stalks of the $\mathcal{F}_i$ through its image in G, which lies in $\operatorname{Gal}(L/F_1) = P$ since $F_1 \subset F'$.

Finally, by a limit argument $S' \to S$ descends to a finite flat morphism $C' \to C$ onto an étale neighbourhood C of s, which we may take with normal source. Since k is perfect, the normal curve C' is smooth. Moreover, the morphism $C' \to C$ is étale over a dense open of C, and after shrinking C around s it is étale away from s. Since k is algebraically closed, $C' \times_C S$ is the product of the strict localizations of C' at the points over s. As it equals the local scheme S', there is a single point over s, and S' is the strict localization of C' there. ∎

Lemma 12.13 (Tangent cones on the cover)**.** *For $i = 1, 2$ let $X_i' \to X_i$ be a finite flat morphism of degree d as in Lemma 12.12, with y_i' the point over y. Let T' be the strict localization of $X_1' \times X_2'$ at $y' = (y_1', y_2')$, and write Γ', V', Δ', H' for the inverse images of Γ, V, Δ, H in T'. Then*

T' is the strict localization of a smooth surface over k at a closed point, the morphism $T' \to T$ is finite flat of degree d^2 and étale over $T - (H \cup V)$, the reduction H'_{red} is regular, and the tangent cone of $(\Gamma' \cup V')_{\mathrm{red}}$ at y' meets the tangent cones of Δ'_{red} and of H'_{red} only in 0.

Proof. The two covers descend to finite flat morphisms $C'_i \to C_i$ of degree d between smooth curves, where C_i is an étale neighbourhood of y in X, étale away from y and with a single point y'_i over y (Lemma lemma-covers-degree-prime-to-ell). Their product $C'_1 \times C'_2 \to C_1 \times C_2$ is a finite flat morphism of degree d^2 of smooth surfaces, étale away from the two fibres through y, with y' the only point over (y, y). Base change to the strict localization T of $C_1 \times C_2$ at (y, y) gives a scheme finite over T with one point over the closed point, hence local and strictly henselian, and this is T'. This proves the assertions about T' and $T' \to T$.

Choose uniformizers ξ of X'_1 and η of X'_2, so that the completed local ring of T' at y' is $k[[\xi, \eta]]$. Since the residue field k is algebraically closed, both covers are totally ramified, so u is a unit of $k[[\xi]]$ times ξ^d, and similarly for v. Multiplying ξ and η by suitable dth roots of nonzero constants, we may assume that

$$u \equiv \xi^d, \qquad v \equiv \eta^d \pmod{(\xi, \eta)^{d+1}}.$$

Since Φ_X^m is purely inseparable of degree q^m and fixes y, we have $\operatorname{ord}_v \varphi = q^m > 1$, and hence $\varphi(v) \in (\eta)^{dq^m} \subseteq (\xi, \eta)^{d+1}$. It follows that the pulled-back equations of Γ', V', Δ' and H' have initial forms

$$\xi^d, \qquad \xi^d, \qquad \xi^d - \eta^d, \qquad \eta^d,$$

respectively. The support of the tangent cone of a curve is the zero set of the initial form of its equation, and it does not change when we pass to the reduction, since if J is the radical of I and $J^n \subset I$ then the ideals of initial forms satisfy $\mathrm{in}(J)^n \subset \mathrm{in}(I) \subset \mathrm{in}(J)$. Thus the tangent cone of $(\Gamma' \cup V')_{\mathrm{red}}$ is supported on the line $\xi = 0$, that of H'_{red} on $\eta = 0$, and that of Δ'_{red} on the lines $\xi = \zeta\eta$ with $\zeta^d = 1$. This holds also when $p \mid d$ and $\xi^d - \eta^d$ is a pth power. Away from the origin the first support meets neither of the others. Finally H'_{red} is the curve $\eta = 0$, which is regular. ∎

Lemma 12.14 (Descent of vanishing along a finite étale cover)**.** *Let $f : W' \to W$ be a finite étale morphism of schemes of constant degree n, with n invertible in A, and let $L \in D(W_{\acute{e}t}, A)$. Then the pullback map $R\Gamma(W, L) \to R\Gamma(W', f^{-1}L)$ has a left inverse in $D(A)$. In particular, a morphism $\alpha : M \to R\Gamma(W, L)$ in $D(A)$ is zero if its composite with this pullback map is zero.*

Proof. Since f is finite étale, $f_* = f_!$ is exact, and the trace map $f_* f^{-1}\mathcal{F} \to \mathcal{F}$ composed with the adjunction unit $\mathcal{F} \to f_* f^{-1}\mathcal{F}$ is multiplication by n for every sheaf of A-modules $\mathcal{F}$ [Stacks, 03SH]. Since both maps are natural and the functors are exact, applying them termwise to a complex representing L gives maps $L \to f_* f^{-1} L \to L$ in $D(W_{\acute{e}t}, A)$ whose composite is n. Applying $R\Gamma(W, -)$ and using $R\Gamma(W, f_* f^{-1} L) = R\Gamma(W', f^{-1}L)$, we see that n^{-1} times the map induced by the trace is a left inverse of the pullback. ∎

The map ρ of Lemma 12.7 depends on K and on the geometry of Γ and Δ near (y, y), but not on ψ. We therefore state the vanishing of local terms at a zero stalk as a statement about ρ alone.

Lemma 12.15 (Vanishing of the local term at a zero stalk)**.** *In the setting of Lemma 12.7, let y be a fixed point of Φ_X^m, and suppose $K_{\bar{y}} = 0$. Then*

$$\rho : R\Gamma_\Gamma(T, P) \longrightarrow R\Gamma_y(\Delta, \delta^* P)$$

is zero in $D(A)$*. Consequently* $\mathrm{LT}_y(\psi) = 0$ *for every morphism* $\psi : (\Phi_X^m)^* K \to K$*.*

Proof. We use the notation introduced before Lemma 12.12 and compute on the étale model: we identify ρ with a restriction map r away from y, kill r on a cover with ℓ-group monodromy, and descend.

Step 1: reduction to r*.* Let $N = (\mathbb{D}_X K)|_{X_1 - y}$, the dual of $K|_{X_1 - y}$ with respect to the restriction of ω_X, and let $Q = K|_{X_2 - y}$. Write $j_1 : X_1 - y \hookrightarrow X_1$ and $j_2 : X_2 - y \hookrightarrow X_2$ for the inclusions. As $K_{\bar{y}} = 0$, localization gives $K|_{X_2} = j_{2!} Q$, and $(\mathbb{D}_X K)|_{X_1} = R j_{1*} N$ because the dual of an extension by zero is the direct image of the dual. The ordinary external Künneth comparison therefore identifies $P = (\mathbb{D}_X K \boxtimes K)|_T$ with $R j_* j'_!(N \boxtimes Q)$, where

$$j : T - V \hookrightarrow T, \qquad j' : T - (H \cup V) \hookrightarrow T - V.$$

Both $R\Gamma(T, P)$ and $R\Gamma(\Delta, \delta^* P)$ are zero, because the schemes are strictly local and the stalk at the closed point is $(\mathbb{D}_X K)_{\bar{y}} \otimes^{\mathbf{L}} K_{\bar{y}} = 0$. So the localization triangles identify ρ, shifted by one, with the ordinary restriction $R\Gamma(T - \Gamma, P) \to R\Gamma(\Delta - y, \delta^* P)$, which under the Künneth identification is

$$r : R\Gamma(T - (V \cup \Gamma), j'_!(N \boxtimes Q)) \longrightarrow R\Gamma(\Delta - y, (N \boxtimes Q)|_{\Delta - y}).$$

Step 2: the restriction vanishes on a cover. Choose bounded finite projective lisse models of N and Q by Lemma 3.13; their terms are finite locally constant, as a punctured strictly local trait is its generic point. Then Lemma 12.12, applied to the terms of the model of N on $X_1 - y$ and to those of Q on $X_2 - y$, gives integers d_1, d_2 prime to ℓ. With $d = \mathrm{lcm}(d_1, d_2)$ we then obtain covers $X'_1 \to X_1$ and $X'_2 \to X_2$ of the same degree d. We want equal degrees so that u and v become dth powers for the same d in Lemma 12.13. Let T', Γ', V', Δ', H' be as in that lemma, and let N' and Q' be the pullbacks of N and Q to $X'_1 - y'_1$ and $X'_2 - y'_2$. By construction, the complex $N' \boxtimes Q'$ has finite ℓ-group monodromy on $T' - (H' \cup V')$, hence also on the smaller open set obtained by removing Γ'. Then the restriction r' on T', defined like r, is zero by Lemma 12.11, applied to T' with $D = (\Gamma' \cup V')_{\mathrm{red}}$, the regular curve H'_{red} and $C = \Delta'_{\mathrm{red}}$; its hypotheses hold by Lemma 12.13, and passing to reductions changes neither the open complements nor the étale cohomology.

Step 3: descent. The map r followed by pullback to $\Delta' - y'$ equals pullback to T' followed by $r' = 0$, because pullback along $T' \to T$ commutes with restriction and with extension by zero. As Δ meets $H \cup V$ only at y, Lemma 12.13 shows that $\Delta' - y' \to \Delta - y$ is finite étale of degree d^2. This degree is invertible in A, so Lemma 12.14 gives $r = 0$, and hence $\rho = 0$. By the comparison theorems, this identity on the classical étale models of these strictly local schemes also holds in the pro-étale formulation. Finally, by Lemma 12.7, $\mathrm{LT}_y(\psi)$ is the image of $[\psi]$ under a composite that begins with ρ, so it is zero. ∎

12.2.4 The local term at a fixed point

The point y is stable under Φ_X^m, so additivity of trace classes applies to the localization triangle for $X \setminus \{y\}$ and $\{y\}$. Here the open piece has zero stalk at y, so its local term vanishes by Lemma 12.15. The closed piece, in turn, is pushed forward from the point, where the local term is a trace by Lemma 11.10. Note that the argument never uses that K is lisse at y. This is why, in the proof of Theorem 12.20, we can apply the result to $j_! K$ at the point at infinity of $\mathbf{P}^1$.

Proposition 12.16 (The Frobenius local term on a curve)**.** *Let X_0 be a smooth curve over k_0, let $X = X_0 \times_{k_0} k$, let $K \in D^b_{\mathrm{ctf}}(X, A)$ and let $\psi : (\Phi_X^m)^* K \to K$ be any morphism. For $y \in X_0(\mathbf{F}_{q^m})$ one has*

$$\mathrm{LT}_y(\psi) = \mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}}).$$

Proof. Let $i : \{y\} \hookrightarrow X$ and $j : U = X \setminus \{y\} \hookrightarrow X$. We split ψ along the localization triangle and compute the two local terms separately.

Splitting ψ. Both pieces are stable under Φ_X^m, and in fact $((\Phi_X^m)^{-1}\{y\})_{\mathrm{red}} = \{y\}$. In the enhanced derived category, the functorial localization triangle is a two-step filtered object,

$$K_U = j_! j^* K \longrightarrow K \longrightarrow i_* i^* K \xrightarrow{+1},$$

and ψ induces a coherent map from the $(\Phi_X^m)^*$-pullback of this filtration to the filtration itself, with induced correspondences ψ_U and $i_*\psi_{\bar{y}}$ on the pieces. This rests on two base change identities. For the open part, $(\Phi_X^m)^* j_! \simeq j_!(\Phi_X^m|_U)^*$ by base change along inverse images in the cartesian square. The square for the point, by contrast, is cartesian only after passing to reductions, so there proper base change followed by nilpotent invariance gives $(\Phi_X^m)^* i_* \simeq i_*(\Phi_X^m|_y)^*$. Both comparisons are mates of the adjunction units for j and i used in the triangle, which is why the induced maps are compatible with the filtration.

Additivity. The second half of Proposition 11.19 applies to the graph c_m, because its hypothesis $\Phi_X^m(\{y\}) \subset \{y\}$ holds for the fixed point y. It gives

$$\mathrm{LT}_y(\psi) = \mathrm{LT}_y(\psi_U) + \mathrm{LT}_y(i_*\psi_{\bar{y}}).$$

The two local terms. The first term is zero by Lemma 12.15, since $(K_U)_{\bar{y}} = 0$. For the second, the description of the closed piece u_Z in Proposition 11.19 shows that $i_*\psi_{\bar{y}}$ is the pushforward along i of the cohomological correspondence $\psi_{\bar{y}}$ on $(y, K_{\bar{y}})$ for the identity correspondence of y. Its local term is therefore $\mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}})$, the trace on the perfect complex $K_{\bar{y}}$, by the compatibility with proper pushforward in Theorem 11.13 and Lemma 11.10. ∎

For $K = A_X$ on $X = \mathbf{A}^1_k$ the proposition gives $\mathrm{LT}_y(\psi_m) = 1$ at each of the q^m fixed points of Φ_X^m. This matches the intersection count at the start of the chapter. For general K the proposition is not a formal consequence of the definitions, as Remark 12.4 explains.

Example 12.17 (Frobenius on the projective line)**.** Let $X = \mathbf{P}^1_k$, $K = A_X$, and let ψ be the constant Frobenius correspondence, so that ψ_m is the identity on every stalk. Both sides of the Lefschetz–Verdier formula for the graph c_m can be computed explicitly. The fixed points of Φ_X^m are the $q^m + 1$ points of $\mathbf{P}^1(\mathbf{F}_{q^m})$. By the proposition each has local term $\mathrm{tr}_A(1 \mid A) = 1$, so the local side is $q^m + 1$. On the global side, $R\Gamma(\mathbf{P}^1, A) = A \oplus A(-1)[-2]$. By Lemma 12.3 the operator $T_{c_m}(\psi_m)$ is the m-th power of geometric Frobenius. It acts by 1 on H^0 and by q^m on $H^2 = A(-1)$, and both degrees are even, so its trace is $1 + q^m$ with no signs. Compare the identity correspondence at the start of Section 11.4. There the fixed locus is all of $\mathbf{P}^1$ and the answer 2 is a self-intersection. Here the fixed locus is finite and each point counts once.

12.3 Frobenius and $Rf_!$

The trace formula with finite coefficients is proved by induction on the number of affine coordinates. We project a closed subscheme of $\mathbf{A}^d$ to $\mathbf{A}^{d-1}$, apply the induction hypothesis to $Rf_!K$, and

identify each stalk of $Rf_!K$ at a rational point with the cohomology of the fibre, a closed subscheme of $\mathbf{A}^1$ over a larger finite field. For this, $Rf_!K$ must carry a Frobenius correspondence with two properties: its operator on $R\Gamma_c$ is that of ψ, and its stalk maps are the Frobenius operators on the cohomology of the fibres.

The square formed by f and the two Frobenii commutes, but it is usually not cartesian. Indeed, already for $f : \mathbf{A}^1 \to s$ the Frobenius of s is the identity, so the fibre product is $\mathbf{A}^1$ itself, while $\Phi_{\mathbf{A}^1}$ has degree q. The difference is measured by the relative Frobenius. It is a universal homeomorphism, so pullback along it is an equivalence on étale sheaves, and the comparison map built from it is an isomorphism.

Lemma 12.18 (Comparison via relative Frobenius)**.** *Let $f_0 : X_0 \to Y_0$ be a morphism of separated schemes of finite type over k_0, write $f : X \to Y$ for its geometric base change, and let $m \geq 1$. Put*

$$X'_m = Y \times_{\Phi_Y^m, Y, f} X, \quad p_m : X'_m \to X, \quad f'_m : X'_m \to Y, \quad r_m = (f, \Phi_X^m) : X \to X'_m,$$

where p_m and f'_m are the two projections.

1. *The relative Frobenius r_m is a finite universal homeomorphism, $p_m r_m = \Phi_X^m$ and $f'_m r_m = f$, and there are natural isomorphisms*

$$\begin{aligned}\theta_{f,m} : (\Phi_Y^m)^* Rf_!K &\simeq Rf'_{m!} p_m^* K \xrightarrow{Rf'_{m!}\eta_{r_m}} Rf'_{m!} Rr_{m*} r_m^* p_m^* K \\ &\simeq Rf_!(\Phi_X^m)^* K,\end{aligned}$$

   compatible with iteration in m.
2. *For a Frobenius correspondence $\psi : \Phi_X^* K \to K$ on $K \in D^b_{\mathrm{ctf}}(X, A)$, the object $L = Rf_!K$ of $D^b_{\mathrm{ctf}}(Y, A)$ carries the Frobenius correspondence $\psi' = Rf_!\psi \circ \theta_{f,1} : \Phi_Y^* L \to L$, and its iterates are*

$$\psi'_m = Rf_!\psi_m \circ \theta_{f,m}.$$

3. *The composition isomorphism $R\Gamma_c(Y, L) \simeq R\Gamma_c(X, K)$ intertwines the endomorphisms $T_{c_m}(\psi'_m)$ on Y and $T_{c_m}(\psi_m)$ on X of Definition 12.2.*
4. *At $y \in Y_0(\mathbf{F}_{q^m})$, base change for compact supports identifies $(L_{\bar{y}}, \psi'_{m,\bar{y}})$ with*

$$\left(R\Gamma_c(X_{\bar{y}}, K|_{X_{\bar{y}}}), T_{c_1}(\psi_m|_{X_{\bar{y}}})\right),$$

   where the fibre is regarded as a scheme over $\mathbf{F}_{q^m}$, with the q^m-power Frobenius as its scheme Frobenius.

Proof. The only geometric fact we use is that, since r_m is a universal homeomorphism, pullback along it is an equivalence and its unit is an isomorphism. The rest is base change and composition of units, and each part uses the previous ones.

Proof of (1). The identities $p_m r_m = \Phi_X^m$ and $f'_m r_m = f$ hold by the definition of $r_m = (f, \Phi_X^m)$, since $f\Phi_X^m = \Phi_Y^m f$. The map r_m is the geometric base change of the relative q^m-power Frobenius of X_0/Y_0. It is integral, radicial and surjective [Stacks, 0CCB], and finite because it is integral and of finite type. For affine $Y_0 = \operatorname{Spec} R$ and $X_0 = \operatorname{Spec} S$, with F the q-power Frobenius of R, it is given on coordinate rings by

$$R \otimes_{F^m, R} S \longrightarrow S, \qquad b \otimes a \longmapsto f^{\#}(b) a^{q^m}.$$

By topological invariance for classical objects of finite Tor amplitude, r_m^* is an equivalence, so the unit η_{r_m} in the construction is an isomorphism. In the construction, the first arrow is base

change for compact supports, and the last comes from the composition of direct images with compact support, since r_m is proper. Moreover, pasting the cartesian squares defining X'_m and X'_n, we get the comparison for $m+n$, because the relative Frobenii compose and the unit of their composite is the composite of the units. Thus

$$\theta_{f,m+n,K} = \theta_{f,m,(\Phi_X^n)^*K} \circ (\Phi_Y^m)^* \theta_{f,n,K},$$

which is the compatibility with iteration.

Proof of (2). The formula for ψ'_m follows by induction on m from the displayed iteration identity and the naturality of θ.

Proof of (3). Both endomorphisms are a pullback map followed by $R\Gamma_c$ of the correspondence, so we must see that the pullback along Φ_Y^m on $R\Gamma_c(Y, L)$ becomes the pullback along Φ_X^m on $R\Gamma_c(X, K)$. For a finite map h we also write h^* for the pullback map on cohomology with compact support, defined by the unit of $h^* \dashv Rh_*$. Since base change is compatible with these units, the pullback along Φ_Y^m, followed by the base change isomorphism, is the pullback along p_m, and the next arrow in $\theta_{f,m}$ becomes r_m^*. Their composite is $r_m^* p_m^* = (p_m r_m)^* = (\Phi_X^m)^*$, and composing with $R\Gamma_c(\psi_m)$ we see that the operators are intertwined.

Proof of (4). We pull the construction back to the fixed point y. Since Φ_Y^m fixes $\bar{y}$, the fibre of $X'_m \to Y$ over $\bar{y}$ is again the fibre $X_{\bar{y}}$ of f, and r_m restricts to a self-map of $X_{\bar{y}}$. For $f : \mathbf{A}^1 \to s$ and $m = 1$, for instance, $X'_1 = \mathbf{A}^1$ and r_1 is $t \mapsto t^q$. In general, the base change of r_m to this fibre is its q^m-power scheme Frobenius, and by proper base change its unit restricts to the unit on the fibre. Hence the same calculation identifies $\psi'_{m,\bar{y}}$ with the Frobenius operator on the cohomology with compact support of the fibre. Moreover, at a rational point x of the fibre, the restricted correspondence $\psi_m|_{X_{\bar{y}}}$ has stalk map $\psi_{m,\bar{x}}$, by composition of pullbacks. ■

To reduce the trace formula to pieces of X_0, we cut X_0 into an open subscheme and its closed complement.

Lemma 12.19 (Additivity for an open subscheme and its complement)**.** *Let $j_0 : U_0 \hookrightarrow X_0$ be open with reduced closed complement $i_0 : Z_0 \to X_0$, and let $K \in D^b_{\mathrm{ctf}}(X, A)$. A Frobenius correspondence ψ on K induces Frobenius correspondences on j^*K, i^*K, $j_!j^*K$ and Ri_*i^*K, whose stalk maps are the restrictions of ψ_m. Writing T_{c_m} also for the endomorphisms of $R\Gamma_c$ induced by the Frobenius correspondences on U and Z, for every $m \geq 1$*

$$\mathrm{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X, K)) = \mathrm{tr}_A(T_{c_m} \mid R\Gamma_c(U, j^*K)) + \mathrm{tr}_A(T_{c_m} \mid R\Gamma_c(Z, i^*K)).$$

Proof. The four induced correspondences come from base change, and $R\Gamma_c$ of the localization triangle is a two-step filtered object with an endomorphism, on which traces add.

Since Frobenius preserves opens defined over the finite field, its square with j is cartesian, and by base change we get the correspondences on j^*K and $j_!j^*K$. The Frobenius square of the closed complement, however, is cartesian only after passing to reductions, and we obtain the correspondences on i^*K and Ri_*i^*K from Lemma lemma-comparison-via-relative-frobenius–(2) and nilpotent invariance. We then apply the functorial localization triangle $j_!j^*K \to K \to Ri_*i^*K$ to the Frobenius correspondence and take direct images with compact support. Since the triangle is functorial, the Frobenius correspondence becomes an endomorphism of a two-step filtered object, and hence the traces add by Proposition 11.2. Finally, Lemma lemma-comparison-via-relative-frobenius for j and i identifies the endomorphisms T_{c_m} of the cohomology of $j_!j^*K$ and Ri_*i^*K with those on U and Z. ■

12.4 Finite coefficients

The proof uses a single compactification, $\mathbf{A}^1 \subset \mathbf{P}^1$, so the Lefschetz–Verdier formula and the curve computation of Proposition 12.16 are applied only on $\mathbf{P}^1$. This suffices because every affine scheme of finite type over $\mathbf{F}_q$ is a closed subscheme of some $\mathbf{A}^d$, and the fibres of a coordinate projection lie in $\mathbf{A}^1$. On $\mathbf{P}^1$ the only new fixed point is the rational point ∞, where $j_!K$ has zero stalk. The fibres over points of $Y_0(\mathbf{F}_{q^m})$ are schemes over $\mathbf{F}_{q^m}$, so we prove the theorem for all finite fields at once. The other coefficient rings are treated in Section 12.5.

The base of the induction is the case of finite schemes. There we can see directly what the formula counts. For example, take $X_0 = \operatorname{Spec} \mathbf{F}_{q^2}$ and $K = A$. Then the scheme X consists of two points interchanged by Φ_X. Hence T_{c_1} is the swap of $A \oplus A$, with trace $0 = \#X_0(\mathbf{F}_q)$, and T_{c_2} is the identity, with trace $2 = \#X_0(\mathbf{F}_{q^2})$. In general T_{c_m} is a block permutation matrix, and only the points fixed by Φ_X^m contribute to its trace.

The iterates ψ_m, the endomorphisms $T_{c_m}(\psi_m)$ and the stalk maps $\psi_{m,\bar{x}}$ are those of Definition 12.2.

Theorem 12.20 (Trace formula for a Frobenius correspondence)**.** *Let $X_0/\mathbf{F}_q$ be separated and of finite type, put $X = X_0 \times_{\mathbf{F}_q} k$, let A be any finite commutative ℓ-primary ring and $K \in D^b_{\mathrm{ctf}}(X, A)$, and let $\psi : \Phi_X^* K \to K$ be any Frobenius correspondence on K. For every $m \geq 1$, in A,*

$$\operatorname{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X_{\text{pro\'et}}, K)) = \sum_{x \in X_0(\mathbf{F}_{q^m})} \operatorname{tr}_A(\psi_{m,\bar{x}} \mid K_{\bar{x}}).$$

Proof. Both sides are defined, because the complexes are perfect by Lemma 10.1. We prove the assertion simultaneously for all finite fields, all Frobenius correspondences and all m, so that $\mathbf{F}_q$ may be replaced by $\mathbf{F}_{q^m}$ when we pass to a rational fibre. We go in four steps: finite schemes; the affine line, where we use the Lefschetz–Verdier formula on $\mathbf{P}^1$; closed subschemes of $\mathbf{A}^d$, by projecting away one coordinate; and arbitrary X_0, by cutting it into affine pieces.

Finite schemes. If X_0 is finite, both sides of the formula can be read off from a block matrix. Indeed, up to nilpotents, X is the finite disjoint union of its geometric points. By nilpotent invariance, $R\Gamma_c(X, K) = \bigoplus_{z \in X(k)} K_z$. In this decomposition, the pullback along Φ_X^m given by the unit permutes these summands, and the block of $T_{c_m}(\psi_m)$ from $K_{\Phi_X^m(z)}$ to K_z is the stalk of ψ_m at z, a map $K_{\Phi_X^m(z)} = ((\Phi_X^m)^* K)_z \to K_z$. On a finite projective model only the diagonal blocks contribute to the trace. These occur exactly at the points $z \in X_0(\mathbf{F}_{q^m})$, with the required stalk maps.

The affine line. This is the only step where we compute a local term. We pass to $\mathbf{P}^1$, where the Lefschetz–Verdier formula applies, and extend K by zero, so that the new fixed point ∞ contributes nothing. Let $j : \mathbf{A}^1_k \hookrightarrow \mathbf{P}^1_k$. The correspondence ψ_m extends to a correspondence $\bar{\psi}_m$ on $j_!K$, compatible with iteration. Indeed, j is defined over $\mathbf{F}_q$, so its square with the Frobenius maps is cartesian, and base change for this square gives $(\Phi^m_{\mathbf{P}^1})^* j_! K \simeq j_!(\Phi^m_{\mathbf{A}^1})^* K$. Lemma lemma-comparison-via-relative-frobenius, applied to j, shows that the composition isomorphism

$$R\Gamma_c(\mathbf{A}^1, K) \simeq R\Gamma(\mathbf{P}^1, j_! K)$$

intertwines the endomorphisms T_{c_m} induced by ψ and $\bar{\psi}$. The graph is proper on the proper curve $\mathbf{P}^1$, so by Theorem 11.13 the trace on the right is the sum of the local terms at the fixed points. By Proposition 12.16, each local term is a trace on a stalk. In particular, the local term at infinity vanishes, because the stalk of $j_!K$ there is zero. The other fixed points and stalk maps are those of $\mathbf{A}^1_{\mathbf{F}_q}$. This proves the formula for the affine line.

Closed subschemes of the affine line. Let $X_0 = \operatorname{Spec} \mathbf{F}_q[t]/I$. If $I = 0$, the previous paragraph applies. Otherwise I contains a nonzero polynomial, so X_0 is finite over $\mathbf{F}_q$ and the case of finite schemes applies. This also covers nonreduced subschemes.

Induction on affine coordinates. We push K forward along a coordinate projection. This does not change the global side. The local side over a point of the base becomes the global side of a fibre lying in $\mathbf{A}^1$. Suppose that the assertion holds for closed subschemes of $\mathbf{A}^{d-1}$ over every finite field, and let $X_0 \hookrightarrow \mathbf{A}^d_{\mathbf{F}_q}$ be closed. Let $f_0 : X_0 \to Y_0 = \mathbf{A}^{d-1}_{\mathbf{F}_q}$ forget the last coordinate, and let $f : X \to Y$ be its base change to k. The complex $L = Rf_!K$ lies in $D^b_{\mathrm{ctf}}(Y, A)$ by Proposition 7.1, and we equip it with the Frobenius correspondence ψ' of Lemma lemma-comparison-via-relative-frobenius. As Lemma lemma-comparison-via-relative-frobenius identifies the endomorphisms T_{c_m}, the induction hypothesis shows that

$$\operatorname{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X, K)) = \sum_{y \in Y_0(\mathbf{F}_{q^m})} \operatorname{tr}_A(\psi'_{m,\bar{y}} \mid L_{\bar{y}}).$$

By Lemma lemma-comparison-via-relative-frobenius, each summand is the trace on the cohomology with compact support of the fibre $X_{0,y}$, a closed subscheme of $\mathbf{A}^1_{\mathbf{F}_{q^m}}$. Here the endomorphism is given by the restriction of ψ_m to the fibre and the first power of the Frobenius over $\mathbf{F}_{q^m}$. The case of closed subschemes of the affine line then turns this summand into the sum of the traces on stalks over $X_{0,y}(\mathbf{F}_{q^m})$. As y varies, these sets partition $X_0(\mathbf{F}_{q^m})$, which proves the affine case.

General schemes. Both sides are additive for a decomposition into an open subscheme and its closed complement: the global side by Lemma 12.19, and the local side because the rational points split. It is therefore enough to cut X_0 into affine pieces. Choose a finite affine open cover $U_1, \ldots, U_N$ of X_0, let $W_i = \bigcup_{j \le i} U_j$ and $V_i = W_i \setminus W_{i-1}$, with the reduced closed structure. Each V_i is closed in the affine U_i, hence affine, and it is the closed complement of W_{i-1} in W_i. Applying Lemma 12.19 successively, we see that the traces add over this stratification. The affine case applied to each V_i now gives the equality for all separated X_0 of finite type. ∎

12.5 ℓ-adic coefficients

To pass from finite coefficient rings to $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ we follow the steps of Proposition 7.12. Since the statement is now an equality of traces in Λ rather than an isomorphism, we carry the steps out directly. First, for $\mathcal{O}_E$ we reduce modulo π_E^n. This works because both sides are traces of perfect complexes, traces commute with reduction, and $\mathcal{O}_E$ embeds in $\varprojlim_n \mathcal{O}_E/\pi_E^n$. Next, for E we clear denominators: a Frobenius correspondence ψ_E with coefficients E need not come from an integral one; only some multiple $\pi_E^N \psi_E$ does, and this integral model need not be invertible even when ψ_E is (Lemma 12.22). This is why we prove the formula for correspondences that are not invertible. Finally, for $\overline{\mathbf{Q}}_\ell$ we descend to a finite extension of $\mathbf{Q}_\ell$.

The Frobenius comparison κ_m of Lemma 12.3 extends to the coefficient rings $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$. Let $K \in D^b_c(X, \mathcal{O}_E)$ have tower of reductions (K_n) modulo π_E^n (Definition 2.61). Both $(\Phi_X^m)^*$ and Fr_X^{m*} preserve derived completeness and commute with reduction, so each is the derived limit of its values on the K_n. The finite comparisons κ_m come from a single functorial identification on the classical site. Hence they are compatible with reduction of coefficients, and their derived limit is the integral κ_m. The finite comparisons are compatible with iteration and with the units of the pullbacks, and therefore so is the integral one. It follows that the integral κ_m identifies the endomorphisms T_{c_m} with the geometric Frobenius operators as in Lemma 12.3. For E we localize on integral models. By the localized Hom comparison (9.21) the result does not depend on the choice of model. For $\overline{\mathbf{Q}}_\ell$ we descend from finite extensions.

Proposition 12.21 (Integral trace formula)**.** *Let $K \in D_c^b(X, \mathcal{O}_E)$ and let $\theta : \Phi_X^* K \to K$ be any morphism, that is, a Frobenius correspondence with integral coefficients. Define its iterates θ_m, the endomorphisms $T_{c_m}(\theta_m)$ and the stalk maps $\theta_{m,\bar{y}}$ by the formulas of Definition 12.2. Then for every $m \geq 1$ we have*

$$\mathrm{tr}_{\mathcal{O}_E}(T_{c_m}(\theta_m) \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \mathrm{tr}_{\mathcal{O}_E}(\theta_{m,\bar{y}} \mid K_{\bar{y}}), \tag{12.1}$$

where both sides are categorical traces of perfect complexes over $\mathcal{O}_E$. Their reductions modulo π_E^n are the corresponding traces of Theorem 12.20 for K_n and the reduction $\bar{\theta}_n : \Phi_X^* K_n \to K_n$ of θ.

Proof. We show that both sides reduce modulo π_E^n to the two sides of the finite formula, by identifying the reductions of the complexes, of the operators and of the traces. This suffices, since an element of $\mathcal{O}_E$ that vanishes modulo every π_E^n is zero.

Complexes and operators. By Lemma 10.1, $R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)$ and the geometric stalks are perfect, and Proposition 7.10 provides canonical isomorphisms

$$\begin{aligned} R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K) \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n &\simeq R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K_n), \\ K_{\bar{y}} \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n &\simeq (K_n)_{\bar{y}}. \end{aligned}$$

Under these isomorphisms the reductions of the integral operators are the operators of $\bar{\theta}_n$, because extension of scalars commutes with the pullbacks, with the units defining the endomorphisms T_{c_m}, and with composition, and hence with the iterates θ_m. Moreover Theorem 12.20 applies for every n, because $K_n \in D^b_{\mathrm{ctf}}(X, \mathcal{O}_E/\pi_E^n)$.

Traces. Represent each of the two integral complexes by a bounded complex $P^\bullet$ of finite projective $\mathcal{O}_E$-modules and its endomorphism by a chain map $v^\bullet$. By Proposition 11.2 its trace is

$$\mathrm{tr}_{\mathcal{O}_E}(v) = \sum_i (-1)^i \,\mathrm{Tr}(v^i \mid P^i).$$

The reduction of this expression modulo π_E^n is the trace on $P^\bullet \otimes_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n$, as each P^i is finite projective.

Conclusion. The reduction of each side of (12.1) is therefore the corresponding side of the finite formula for $(K_n, \bar{\theta}_n)$. Hence the difference of the two sides lies in $\pi_E^n \mathcal{O}_E$ for every n. It is zero because $\mathcal{O}_E \hookrightarrow \varprojlim_n \mathcal{O}_E/\pi_E^n$ is injective. ∎

To pass from $\mathcal{O}_E$ to E we need an integral Frobenius correspondence whose rationalization is the given one, up to a power of π_E. The next lemma provides it and records how the operators of both sides of the trace formula change under this rescaling.

Lemma 12.22 (Integral models for rational Frobenius correspondences)**.** *Let $K_E \in D_c^b(X, E)$ and let $\psi_E : \Phi_X^* K_E \to K_E$ be any morphism. There exist $K_0 \in D_c^b(X, \mathcal{O}_E)$, an isomorphism $K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_E} E \simeq K_E$, an integer $N \geq 0$, and a Frobenius correspondence $\theta : \Phi_X^* K_0 \to K_0$ such that*

$$\psi_E = \pi_E^{-N}(\theta \otimes^{\mathbf{L}}_{\mathcal{O}_E} E).$$

For every $m \geq 1$ the operators of Definition 12.2 satisfy

$$\begin{aligned} T_{c_m}((\psi_E)_m) &= \pi_E^{-Nm}(T_{c_m}(\theta_m) \otimes^{\mathbf{L}}_{\mathcal{O}_E} E), \\ (\psi_E)_{m,\bar{y}} &= \pi_E^{-Nm}(\theta_{m,\bar{y}} \otimes^{\mathbf{L}}_{\mathcal{O}_E} E). \end{aligned} \tag{12.2}$$

Proof. Morphisms in the rational category are integral morphisms divided by a power of π_E. This gives K_0 and θ. The formulas for the operators hold because the operators of Definition 12.2 commute with extension of scalars.

By Proposition 3.27, extension of scalars induces an equivalence

$$D_c^b(X, \mathcal{O}_E)[1/\ell] \simeq D_c^b(X, E),$$

where the source has the objects of $D_c^b(X, \mathcal{O}_E)$ and Hom groups tensored with E. By essential surjectivity, K_E has an integral model K_0. The model constructed there is built by finite cones and shifts from the integral generators of Lemma lemma-integral-models, and hence in general it is not a lattice in a lisse sheaf.

Since inverse image preserves integral constructibility, $\Phi_X^* K_0$ is an integral model of $\Phi_X^* K_E$. By the description of the Hom groups above, applied to the pair $(\Phi_X^* K_0, K_0)$, we may write $\psi_E = \pi_E^{-N}(\theta \otimes^{\mathbf{L}}_{\mathcal{O}_E} E)$. Since pullback is $\mathcal{O}_E$-linear, each of the m factors of the iterate contributes π_E^{-N}, and so $(\psi_E)_m = \pi_E^{-Nm}(\theta_m \otimes^{\mathbf{L}} E)$. By Proposition 7.10, the change-of-coefficients maps for cohomology with compact support and for stalks respect the units of the pullback adjunctions. Hence the pullback map $(\Phi_X^m)^*$ on $R\Gamma_c$ for E is the extension of scalars of the integral one. The same holds for the identification $((\Phi_X^m)^* K_E)_{\bar{y}} \simeq (K_E)_{\bar{y}}$ at a fixed point $\bar{y}$ of Φ_X^m, because the comparison for inverse images is compatible with composition and with extension of scalars. Applying T_{c_m} and taking stalks at $\bar{y}$ in the formula for $(\psi_E)_m$ gives (12.2). ∎

Remark 12.23 (Integral Frobenius correspondences need not be invertible)**.** In Lemma 12.22, K_0 need not be a lattice in a lisse sheaf, and θ need not be invertible even when ψ_E is. The factor π_E^{-N} is needed because ψ_E itself need not be integral: on a point, the automorphism ℓ^{-1} of E preserves no lattice. Even with this factor, θ cannot always be chosen invertible, already on a point. On E^2, the two eigenvalues of $\mathrm{diag}(1, \ell)$ cannot both become units after multiplying the operator by a single scalar. So Lemma 12.22 only makes $\pi_E^N \psi_E$ integral, and the reductions of θ may fail to be invertible. For this reason Theorem 12.20 is stated for arbitrary Frobenius correspondences.

With the integral formula and integral models of rational Frobenius correspondences, we obtain the trace formula stated at the beginning of this chapter for all coefficient rings.

Theorem 12.24 (Grothendieck–Lefschetz trace formula)**.** *Let X_0 be separated and of finite type over $\mathbf{F}_q$, put $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and suppose that $\ell \nmid q$. Let*

$$\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}, \qquad K \in D^b_{\mathrm{ctf}}(X, \Lambda),$$

with a Weil structure as in Definition 10.18. Here A is any finite commutative ℓ-primary ring, and for integral and rational coefficients, $D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda)$. Let $\mathrm{Frob}_{X,K}$ be the geometric Frobenius on $R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)$, and let $\mathrm{Frob}_{y,m}$ be the Frobenius acting on the stalk $K_{\bar{y}}$ at $y \in X_0(\mathbf{F}_{q^m})$. Then for every $m \geq 1$ we have, in Λ,

$$\mathrm{tr}_\Lambda\big(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)\big) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \mathrm{tr}_\Lambda\big(\mathrm{Frob}_{y,m} \mid K_{\bar{y}}\big). \tag{12.3}$$

The complexes inside the traces are perfect over Λ, and the formula is compatible with the change-of-coefficients maps of Proposition 7.10. For $\Lambda = \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, it reads

$$\sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_{X,K}^m \mid H_c^i(X_{\mathrm{pro\acute{e}t}}, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \sum_j (-1)^j \operatorname{Tr}(\mathrm{Frob}_{y,m} \mid H^j(K_{\bar{y}})), \tag{12.4}$$

where all sums over degrees are finite.

For finite and integral rings, (12.3) is an identity of categorical traces of perfect complexes, which makes sense even when the cohomology modules are not projective.

Proof of Theorem 12.24. In each case we turn the Weil structure into a Frobenius correspondence by composing with κ_1, and we use Lemma 12.3 to see that this does not change the operators. We then treat the coefficient rings in the order finite, integral, rational, algebraic, each case using the previous one.

For finite A, we apply Theorem 12.20 to $\psi = \phi\kappa_1$. Then Lemma 12.3 identifies $T_{c_m}(\psi_m)$ with $\mathrm{Frob}_{X,K}^m$ and, at each fixed point, $\psi_{m,\bar{y}}$ with $\mathrm{Frob}_{y,m}$. Similarly, for $\mathcal{O}_E$, composing the Weil structure with κ_1 we get a Frobenius correspondence $\Phi_X^* K \to K$. By the integral form of Lemma 12.3, the endomorphisms it induces on $R\Gamma_c$ and on the stalks are those induced by the Weil structure. The identity therefore follows from Proposition 12.21. In particular, taking $E = \mathbf{Q}_\ell$ gives the case of $\mathbf{Z}_\ell$.

For coefficients E, the Weil structure composed with κ_1 is a Frobenius correspondence ψ_E, and by the rational form of Lemma 12.3 the endomorphisms it induces on $R\Gamma_c$ and on the stalks are those induced by the Weil structure. We apply Lemma 12.22 to ψ_E, apply (12.1) to (K_0, θ), extend scalars to E, and multiply by π_E^{-Nm}. By linearity of categorical traces and their compatibility with extension of scalars (Proposition 11.2), together with (12.2), this gives the identity for every m. The case $E = \mathbf{Q}_\ell$ gives the case of $\mathbf{Q}_\ell$.

For $\overline{\mathbf{Q}}_\ell$, complexes and morphisms are colimits of their versions over finite extensions, so it is enough to find the Weil complex over some finite E and to check that the answer does not depend on the choice. By Proposition 7.11 the complex K, its Weil structure ϕ and ϕ^{-1} descend to a common finite extension $E/\mathbf{Q}_\ell$. After a further finite extension the two identities saying that these maps are inverse to each other hold as well. Thus we get a Weil complex at a finite stage, and by extension of scalars its formula over E gives (12.3). If we choose another representative, any isomorphism compatible with the Weil structures descends to a common finite extension, so the resulting operators are conjugate. Since the trace is compatible with extension of scalars and invariant under conjugation, the result does not depend on the stage or on the representative.

The complexes are perfect by Lemma 10.1. Compatibility with the change-of-coefficients maps holds because the pullbacks, the endomorphisms T_{c_m} and the iterated Frobenius correspondences commute with the comparisons of Proposition 7.10. Finally, over a field the categorical trace is the alternating trace on cohomology by Proposition 11.2, and this proves (12.4). ∎

12.6 The function–sheaf dictionary

Under the trace formula, $Rf_!$ becomes summation over fibres and $\otimes$ becomes multiplication of functions on rational points. This is Grothendieck's function–sheaf dictionary [SGA4½, Sommes trig.] (Section 1.3), and in Part III we compute with sheaves through their functions in this way.

Definition 12.25 (Trace function)**.** Let (K, ϕ) be a Weil complex on X as in Theorem 12.24, and let $m \geq 1$. The *trace function* of K is

$$t_{K,m} : X_0(\mathbf{F}_{q^m}) \longrightarrow \Lambda, \qquad t_{K,m}(x) = \operatorname{tr}_\Lambda(\mathrm{Frob}_{x,m} \mid K_{\bar{x}}).$$

For Λ a field, $t_{K,m}(x)$ is the alternating sum of the traces of Frobenius on the stalks $\mathcal{H}^i(K)_{\bar{x}}$ (Proposition 11.2). For example, the constant sheaf Λ_X with its constant Weil structure has trace function 1. In terms of trace functions, the theorem says that $\mathrm{tr}_\Lambda(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\text{proét}}, K))$ is the sum of $t_{K,m}$ over $X_0(\mathbf{F}_{q^m})$. Applied to the fibres of a morphism, it shows that direct image with compact support sums trace functions over the fibres.

Corollary 12.26 (Trace functions and direct image with compact support)**.** *Let $f_0 : X_0 \to Y_0$ be a morphism of separated $\mathbf{F}_q$-schemes of finite type, with geometric base change $f : X \to Y$, and let (K, ϕ) be a Weil complex on X with $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ for an admissible Λ. Give $Rf_!K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ the Weil structure induced by ϕ (after Definition 10.18). Then for every $m \geq 1$ and $y \in Y_0(\mathbf{F}_{q^m})$,*

$$t_{Rf_!K,m}(y) = \sum_{\substack{x \in X_0(\mathbf{F}_{q^m}) \\ f_0(x) = y}} t_{K,m}(x).$$

Proof. The left side is a trace on a stalk of $Rf_!K$. By base change that stalk is the compactly supported cohomology of a fibre. Hence the corollary is the trace formula for the fibre, once we check that the Frobenius operators correspond.

Fix y and let $X_{0,y}$ be the fibre of f_0 over y. It is a separated scheme of finite type over $\mathbf{F}_{q^m}$, and its rational points are the $x \in X_0(\mathbf{F}_{q^m})$ with $f_0(x) = y$. Restrict K to its geometric fibre $X_{\bar{y}}$ and give the restriction the Weil structure over $\mathbf{F}_{q^m}$ obtained from the m-th iterate of ϕ. By base change for compact supports, $(Rf_!K)_{\bar{y}}$ is $R\Gamma_c(X_{\bar{y}}, K|_{X_{\bar{y}}})$, and $\mathrm{Frob}_{y,m}$ is the geometric Frobenius operator of this Weil complex. For finite A this is part (4) of Lemma 12.18, applied to the Frobenius correspondence $\phi\kappa_1$ and translated by Lemma 12.3. By the comparisons of Proposition 7.10 it extends to the other coefficient rings. Since the Frobenius acting on the stalk of the Weil complex on the fibre at a rational point x of $X_{0,y}$ is $\mathrm{Frob}_{x,m}$, the stated identity is Theorem 12.24 for $X_{0,y}$ over $\mathbf{F}_{q^m}$, with exponent 1. ∎

Example 12.27 (The squaring map)**.** Let q be odd, let $f_0 : \mathbf{A}^1_{\mathbf{F}_q} \to \mathbf{A}^1_{\mathbf{F}_q}$ be $x \mapsto x^2$, and let $K = \Lambda_X$ with its constant Weil structure, so that $t_{K,m} = 1$. The fibre of f_0 over $y \in \mathbf{F}_{q^m}$ has $1 + \chi_m(y)$ rational points, where χ_m is the quadratic character of $\mathbf{F}^*_{q^m}$, extended by $\chi_m(0) = 0$. Hence

$$t_{Rf_!\Lambda,m}(y) = 1 + \chi_m(y).$$

The trace function of $Rf_!\Lambda$ is not constant, although the sheaf we pushed forward is: over $\mathbf{G}_m$ the map is a finite étale double cover, and $\chi_m(y)$ records whether the fibre over y splits into two rational points or is a single point of degree 2. Summing over y recovers the trace formula on the source. Since χ_m takes the values ± 1 equally often on $\mathbf{F}^*_{q^m}$, the sum is $q^m = \#\mathbf{A}^1(\mathbf{F}_{q^m})$.

The other operations on functions also have counterparts on sheaves. These counterparts, together with the corollary, are the entries of the dictionary that we use. In this dictionary, morphisms come from morphisms of separated $\mathbf{F}_q$-schemes of finite type, and the operations carry the Weil structures described after Definition 10.18.

Proposition 12.28 (The function–sheaf dictionary)**.** *Let Λ be as in Theorem 12.24, let $m \geq 1$, and let K, L be Weil complexes in $D^b_{\mathrm{ctf}}(X, \Lambda)$.*

1. *For $g_0 : W_0 \to X_0$ and $w \in W_0(\mathbf{F}_{q^m})$, $t_{g^*K,m}(w) = t_{K,m}(g_0(w))$.*
2. *$t_{K \otimes^{\mathbf{L}}_\Lambda L,m} = t_{K,m}\, t_{L,m}$.*

3. *For $f_0 : X_0 \to Y_0$ and $y \in Y_0(\mathbf{F}_{q^m})$, $t_{Rf_!K,m}(y) = \sum_{x \in X_0(\mathbf{F}_{q^m}),\, f_0(x)=y} t_{K,m}(x)$.*
4. *For $U_0 \subset X_0$ open with closed complement Z_0, and $j : U \to X$, $i : Z \to X$ the inclusions, $t_{K,m} = t_{j_!j^*K,m} + t_{i_*i^*K,m}$, where $t_{j_!j^*K,m}$ is $t_{K,m}$ on $U_0(\mathbf{F}_{q^m})$ and 0 on $Z_0(\mathbf{F}_{q^m})$.*
5. *$\sum_{x \in X_0(\mathbf{F}_{q^m})} t_{K,m}(x) = \operatorname{tr}_\Lambda(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K))$.*

Proof. Part (5) is Theorem 12.24 and part (3) is Corollary 12.26.

For (1), the stalk of g^*K at $\bar{w}$ is $K_{\bar{x}}$ with $x = g_0(w)$, and the Weil structure of g^*K is $g^*\phi$ composed with the base change isomorphism. Under this identification of stalks the m-th iterate of the Weil structure of g^*K at $\bar{w}$ is the m-th iterate of ϕ at $\bar{x}$, so the two Frobenius endomorphisms agree.

For (2), the stalk of $K \otimes^{\mathbf{L}}_\Lambda L$ at $\bar{x}$ is $K_{\bar{x}} \otimes^{\mathbf{L}}_\Lambda L_{\bar{x}}$, since pullback to $\bar{x}$ commutes with derived tensor products, and its Frobenius is the tensor product of the two Frobenius endomorphisms. These stalks are perfect (Lemma 10.1), so we may represent them by bounded complexes P, Q of finite projective modules and the endomorphisms by chain maps v, w. Now the term of degree n of $P \otimes Q$ is $\bigoplus_{i+j=n} P^i \otimes Q^j$, the trace of $v^i \otimes w^j$ on $P^i \otimes Q^j$ is $\operatorname{Tr}(v^i)\operatorname{Tr}(w^j)$, and $(-1)^n = (-1)^i(-1)^j$. By the chain formula of Proposition 11.2 the trace on $P \otimes Q$ is therefore the product of the traces on P and on Q.

For (4), note that $j_! = Rj_!$ and $i_* = Ri_!$, with the same Weil structures. By (3), $t_{j_!j^*K,m}(x)$ is the sum of $t_{j^*K,m}$ over the fibre of j_0 at x. This fibre is the point x if $x \in U_0$, and it is empty otherwise. By (1), $t_{j^*K,m}(x) = t_{K,m}(x)$ for $x \in U_0(\mathbf{F}_{q^m})$. The same argument applies to i_*i^*K, and the two functions add up to $t_{K,m}$. ∎

The Fourier transform of Chapter 16 is one more entry, and the proposition shows why it corresponds to the discrete transform. Let p_1, p_2 be the projections of $\mathbf{A}^1 \times \mathbf{A}^1$, let $\mu(x,y) = xy$, let $\psi : \mathbf{F}_q \to E^\times$ be a nontrivial additive character, and let $\mathcal{K}_\psi = \mu^*\mathcal{L}_\psi$, where $\mathcal{L}_\psi$ is the Artin–Schreier sheaf of Lemma 16.1 (proved in Chapter 16 below). Here ψ is a character, not a Frobenius correspondence. With this kernel the transform is $\mathrm{FT}_\psi K = Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[1]$. For $y \in \mathbf{F}_{q^m}$ we compute

$$\begin{aligned}
t_{\mathrm{FT}_\psi K,m}(y) &= -\,t_{Rp_{2!}(p_1^*K\otimes\mathcal{K}_\psi),m}(y) && \text{(shift)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{p_1^*K\otimes\mathcal{K}_\psi,m}(x,y) && \text{by (3)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{p_1^*K,m}(x,y)\, t_{\mu^*\mathcal{L}_\psi,m}(x,y) && \text{by (2)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{K,m}(x)\,\psi(\mathrm{Tr}_{\mathbf{F}_{q^m}/\mathbf{F}_q}(xy)) && \text{by (1) and Lemma 16.1.1.}
\end{aligned}$$

The first equality holds because a shift changes the sign of a trace, by the chain formula of Proposition 11.2. Each of the other three steps is one operation in the formula $\hat{t}(y) = \sum_x t(x)\psi(xy)$: pull t back to the plane, multiply by the kernel, and sum over the fibres of p_2.

12.7 Rationality and the functional equation

Over a field of characteristic zero, the traces of all powers of an endomorphism determine its characteristic polynomial. Taking the trace formula for all $m \geq 1$, we therefore obtain an identity of power series: a product over closed points equals an alternating product of characteristic polynomials of Frobenius on cohomology. This step divides by m, so in the determinant formulas below Λ is one of the fields $\mathbf{Q}_\ell$, E, $\overline{\mathbf{Q}}_\ell$. Characteristic zero is needed: on $\mathbf{F}_p^{\oplus p}$ every power of the

identity has trace $p = 0$, as for the zero endomorphism, but $\det(1 - T \cdot \mathrm{id}) = (1-T)^p = 1 - T^p$ is not 1.

For $K = \Lambda_X$ the product over cohomology is finite, so $Z(X_0, T)$ is rational, and for smooth proper X_0 duality pairs the factors in degrees i and $2d - i$ (Corollary 12.33).

Lemma 12.29 (Logarithm of a determinant)**.** *Let L be a field of characteristic zero. For an endomorphism v of a finite-dimensional L-vector space V we have*

$$\det(1 - Tv \mid V)^{-1} = \exp\left(\sum_{m \geq 1} \mathrm{Tr}(v^m \mid V) \frac{T^m}{m}\right) \quad \text{in } 1 + TL[[T]]. \tag{12.5}$$

For an endomorphism v of a perfect complex P in $D(L)$, put

$$\det(1 - Tv \mid P) := \prod_i \det(1 - Tv \mid H^i(P))^{(-1)^i}.$$

Then (12.5) holds for P, with $\mathrm{Tr}(v^m \mid V)$ replaced by the categorical trace $\mathrm{tr}_L(v^m \mid P)$.

Proof. For a single eigenvalue α the identity is $-\log(1 - \alpha T) = \sum_m \alpha^m T^m / m$, and the general case could be reduced to this by triangularizing v over an algebraic closure. Instead we use logarithmic derivatives, which allows us to stay over L. In $L[[T]]$, the matrix $1 - Tv$ is invertible, with inverse $\sum_{r \geq 0} T^r v^r$. Moreover, the logarithmic derivative of the determinant of an invertible matrix M over $L[[T]]$ is $\mathrm{Tr}(M^{-1} M')$, and hence

$$\frac{d}{dT} \log \det(1 - Tv) = -\mathrm{Tr}\big(v(1 - Tv)^{-1}\big) = -\sum_{m \geq 1} \mathrm{Tr}(v^m) T^{m-1}.$$

Since both logarithms have constant coefficient zero and L has characteristic zero, we may integrate and exponentiate, and we obtain (12.5). For a perfect complex, the categorical trace over a field is the alternating sum of the traces on the finitely many nonzero cohomology spaces (Proposition 11.2), and the assertion follows by multiplying the identities for the $H^i(P)$. ∎

The local factors of the Euler product are indexed by closed points. A closed point $x \in |X_0|$ of degree d over $\mathbf{F}_q$ corresponds to a Frobenius orbit of d geometric points, and its Frobenius on the stalk is the Frobenius going once around this orbit.

Theorem 12.30 (The trace formula as an Euler product)**.** *Let X_0 be separated and of finite type over $\mathbf{F}_q$, let $\Lambda \in \{\mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, and let $K \in D_c^b(X, \Lambda)$ carry a Weil structure, with stalk Frobenius maps $\mathrm{Frob}_{y,m}$ as in Theorem 12.24. For each closed point $x \in |X_0|$, let $\deg(x) = [\kappa(x) : \mathbf{F}_q]$ be its degree, choose one of the $\deg(x)$ points $y \in X_0(\mathbf{F}_{q^{\deg(x)}})$ lying over x, write $\bar{x} = \bar{y}$ for the corresponding geometric point, and let $\mathrm{Frob}_x = \mathrm{Frob}_{y,\deg(x)}$ be its action on $K_{\bar{x}}$. Another choice of y replaces Frob_x by a conjugate. Define*

$$L(X_0, K, T) := \prod_{x \in |X_0|} \det(1 - T^{\deg(x)} \mathrm{Frob}_x \mid K_{\bar{x}})^{-1}. \tag{12.6}$$

For every n, only finitely many factors differ from 1 modulo T^n, so the product converges T-adically in $1 + T\Lambda[[T]]$, and

$$L(X_0, K, T) = \prod_i \det\bigl(1 - T \operatorname{Frob}_{X,K} \mid H^i_c(X_{\text{pro\'et}}, K)\bigr)^{(-1)^{i+1}}. \tag{12.7}$$

In particular, $L(X_0, K, T)$ is the expansion at zero of a rational function in $\Lambda(T)$ with value 1 at zero. If X_0 is proper, we may replace H^i_c by H^i.

Proof. Both sides are of the form exp of a power series, and we compare the logarithms coefficient by coefficient. The coefficient of T^m on the cohomological side is the global trace of Frob^m divided by m, and on the side of closed points it is the sum of the local traces over $X_0(\mathbf{F}_{q^m})$ divided by m. Hence the theorem is the trace formula for all m at once.

For convergence, note that X_0 has only finitely many closed points of degree at most any given bound n, because each of them lies under a point of the finite set $X_0(\mathbf{F}_{q^{n!}})$. The factor at x in (12.6) is 1 modulo $T^{\deg(x)}$, so the product converges.

Before comparing logarithms we rewrite the sum over points in Theorem 12.24 in terms of closed points. A closed point of degree d has exactly d points over $\mathbf{F}_{q^m}$ when $d \mid m$, and none otherwise. For instance, $\operatorname{Spec} \mathbf{F}_{q^2}$ has one closed point, of degree 2, with no $\mathbf{F}_q$-points and two $\mathbf{F}_{q^2}$-points. These d points form one Frobenius orbit, and the Weil structure gives isomorphisms between the stalks at consecutive points of the orbit. The Frobenius on the stalk at a point of the orbit is then the composite of these isomorphisms taken once around the orbit, starting at that point. Moving the starting point cyclically therefore conjugates this composite by one of these isomorphisms. Hence the operators at the $\deg(x)$ points are conjugate. In the same way, iterating the Weil structure shows that for $\deg(x) \mid m$ the operator $\operatorname{Frob}_{y,m}$ at each of the $\deg(x)$ points $y \in X_0(\mathbf{F}_{q^m})$ over x is conjugate to $\operatorname{Frob}_x^{m/\deg(x)}$. Since traces are invariant under conjugation, (12.3) reads

$$\operatorname{tr}_\Lambda(\operatorname{Frob}^m_{X,K} \mid R\Gamma_c(X_{\text{pro\'et}}, K)) = \sum_{\substack{x \in |X_0| \\ \deg(x) \mid m}} \deg(x)\, \operatorname{tr}_\Lambda(\operatorname{Frob}_x^{m/\deg(x)} \mid K_{\bar{x}}). \tag{12.8}$$

We multiply (12.8) by T^m/m and sum over $m \geq 1$. By Lemma 12.29, the left side becomes the logarithm of the right side of (12.7). In the sum over closed points on the right, we substitute $m = r \deg(x)$. Then the multiplicity $\deg(x)$ cancels the factor $\deg(x)$ in m, and Lemma 12.29 applied to Frob_x on $K_{\bar{x}}$, with $T^{\deg(x)}$ in place of T, shows that

$$\sum_{x \in |X_0|} \sum_{r \geq 1} \operatorname{tr}_\Lambda(\operatorname{Frob}_x^r \mid K_{\bar{x}}) \frac{T^{r \deg(x)}}{r} = \log L(X_0, K, T).$$

Each coefficient of T receives only finitely many contributions, so we may regroup the sums. Exponentiating proves (12.7). Rationality holds because the cohomological product has finitely many factors, each a polynomial or the reciprocal of one, with value 1 at zero. If X_0 is proper, compact and ordinary cohomology agree. ∎

For the constant sheaf every local factor is $(1 - T^{\deg(x)})^{-1}$, and the Euler product becomes the zeta function of X_0. This proves rationality of $Z(X_0, T)$ over $\mathbf{Q}_\ell$. A short argument with linear recurrences descends it to $\mathbf{Q}$.

Corollary 12.31 (Point counting and the zeta function)**.** *Give Λ_X the Weil structure of constant sheaves defined after Definition 10.18. For every coefficient ring Λ of Theorem 12.24 and every $m \geq 1$ we have*

$$\operatorname{tr}_\Lambda(\operatorname{Frob}_{X,\Lambda_X}^m \mid R\Gamma_c(X_{\text{pro\'et}}, \Lambda_X)) = \#X_0(\mathbf{F}_{q^m}) \cdot 1_\Lambda. \tag{12.9}$$

For $\Lambda = \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, this gives

$$\#X_0(\mathbf{F}_{q^m}) = \sum_i (-1)^i \operatorname{Tr}(\operatorname{Frob}_{X,\Lambda_X}^m \mid H_c^i(X_{\text{pro\'et}}, \Lambda_X)),$$

and

$$\begin{aligned} Z(X_0, T) &:= \exp\left(\sum_{m \geq 1} \#X_0(\mathbf{F}_{q^m}) \frac{T^m}{m}\right) \\ &= \prod_{x \in |X_0|} (1 - T^{\deg(x)})^{-1} = \prod_i \det(1 - T \operatorname{Frob}_{X,\Lambda_X} \mid H_c^i(X_{\text{pro\'et}}, \Lambda_X))^{(-1)^{i+1}}. \end{aligned}$$

The first two series lie in $1 + T\mathbf{Z}[[T]]$, and the last equality holds in $\Lambda[[T]]$. Moreover, $Z(X_0, T)$ belongs to $\mathbf{Q}(T)$.

Proof. The identities over Λ are the trace formula and the Euler product for the constant sheaf. What needs an argument is that $Z(X_0, T)$ has rational coefficients, which we deduce at the end from the rationality over $\mathbf{Q}_\ell$ by linear algebra.

For $K = \Lambda_X$, every stalk is Λ, and every stalk Frobenius $\operatorname{Frob}_{y,m}$ is its identity, with categorical trace 1_Λ. Thus (12.9) follows from Theorem 12.24, and the formula over the fields follows from its cohomological form (12.4). By the orbit count in the proof of Theorem 12.30, we have $\#X_0(\mathbf{F}_{q^m}) = \sum_{\deg(x) \mid m} \deg(x)$. Expanding each $(1 - T^{\deg(x)})^{-1}$ as a geometric series shows that the product over closed points has integer coefficients. By the orbit count, its logarithm is the series defining $Z(X_0, T)$. The last equality is Theorem 12.30 for $K = \Lambda_X$.

Rationality descends from $\mathbf{Q}_\ell$ to $\mathbf{Q}$ for the following reason. A power series is rational exactly when its coefficients eventually satisfy a linear recurrence with constant coefficients. The existence of such a recurrence is a system of linear equations, and a system with rational coefficients has a rational solution as soon as it has any solution. More precisely, write $Z(X_0, T) = \sum_{n \geq 0} z_n T^n$, with $z_n \in \mathbf{Z}$. Rationality over $\mathbf{Q}_\ell$ means that there are integers d, N and $b_1, \ldots, b_d \in \overline{\mathbf{Q}}_\ell$ such that $z_n + b_1 z_{n-1} + \cdots + b_d z_{n-d} = 0$ for all $n > N$. These are linear equations with rational coefficients in the d unknowns b_i. Their coefficient rows span a finite-dimensional subspace of $\mathbf{Q}^{d+1}$, and hence finitely many of the equations imply all the others. This finite rational linear system has a solution over $\mathbf{Q}_\ell$, and hence one over $\mathbf{Q}$ by Gaussian elimination. With these rational b_i, the product of $1 + b_1 T + \cdots + b_d T^d \in \mathbf{Q}[T]$ and $Z(X_0, T)$ is a polynomial. Therefore $Z(X_0, T) \in \mathbf{Q}(T)$. ∎

For example, let $X_0 = \mathbf{A}^1_{\mathbf{F}_q}$. Then $\#X_0(\mathbf{F}_{q^m}) = q^m$, so that $Z(X_0, T) = (1 - qT)^{-1}$. On the cohomological side, $H_c^i(X, \mathbf{Q}_\ell)$ vanishes except for $i = 2$, and $H_c^2(X, \mathbf{Q}_\ell) \simeq \mathbf{Q}_\ell(-1)$ is a line on which geometric Frobenius acts by q. The only factor is therefore $(1 - qT)^{-1}$, as it should be.

For $X_0 = \mathbf{P}^1_{\mathbf{F}_q}$ the cohomology is $\mathbf{Q}_\ell$ in degree 0 and $\mathbf{Q}_\ell(-1)$ in degree 2, so $Z(\mathbf{P}^1, T) = 1/((1 - T)(1 - qT))$. This matches $\#\mathbf{P}^1(\mathbf{F}_{q^m}) = 1 + q^m$. For a curve of positive genus the middle cohomology contributes as well, and for the running example a single point count determines it.

Example 12.32 (The running example: the trace formula)**.** For the elliptic curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$, geometric Frobenius acts on $H^0(C, \mathbf{Q}_\ell) = \mathbf{Q}_\ell$ by 1 and on $H^2(C, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$ by 3, and $H^1(C, \mathbf{Q}_\ell)$ is a plane (Example 10.23). The point-counting formula reads

$$N_m = 1 - \operatorname{Tr}(F^m \mid H^1(C, \mathbf{Q}_\ell)) + 3^m.$$

From $N_1 = 4$ we get $\operatorname{Tr}(F \mid H^1) = 0$. Duality gives $\det(F \mid H^1) = 3$, so $\det(1 - TF \mid H^1) = 1 + 3T^2$. Hence

$$Z(C_0, T) = \frac{1 + 3T^2}{(1 - T)(1 - 3T)}.$$

The formula predicts $N_2 = 1 + 9 + 6 = 16$ and $N_3 = 1 + 27 - 0 = 28$, and direct counts over $\mathbf{F}_9$ and $\mathbf{F}_{27}$ confirm this. In general, for a genus-one curve over $\mathbf{F}_q$ the single number N_1 determines $P_1 = 1 - (1 + q - N_1)T + qT^2$, and therefore every N_m.

For smooth proper X_0, Poincaré duality relates the factors of the determinant formula in complementary degrees, and this gives a functional equation.

Corollary 12.33 (Functional equation)**.** *Let $X_0/\mathbf{F}_q$ be smooth and proper of pure dimension d, and let $\ell \nmid q$. Put $H^i = H^i(X_{\text{pro\'et}}, \mathbf{Q}_\ell)$, let F_i be the Frobenius operator F on H^i (Section 10.5), and put*

$$b_i = \dim_{\mathbf{Q}_\ell} H^i, \qquad \chi = \sum_i (-1)^i b_i, \qquad D = \prod_i \det(F_i)^{(-1)^i}.$$

Then $b_{2d-i} = b_i$, $d\chi$ is even, $D^2 = q^{d\chi}$, and in $\mathbf{Q}_\ell(T)$

$$Z\left(X_0, \frac{1}{q^d T}\right) = (-1)^\chi D\, T^\chi\, Z(X_0, T) = \varepsilon q^{d\chi/2} T^\chi Z(X_0, T), \qquad \varepsilon = (-1)^\chi D q^{-d\chi/2} \in \{1, -1\}.$$

Proof. Duality gives the relations $b_{2d-i} = b_i$ and $D^2 = q^{d\chi}$. If we substitute $1/(q^d T)$ into each factor of the cohomological expression for $Z(X_0, T)$, duality turns the factor in degree i into the factor in degree $2d - i$ times an explicit scalar.

We apply Theorem 10.21 with $K = \mathbf{Q}_\ell$ and X proper. It gives $b_{2d-i} = b_i$, and it shows that the eigenvalues of F_{2d-i} are the q^d/α for the eigenvalues α of F_i, with multiplicities. Hence $\det F_i \cdot \det F_{2d-i} = q^{db_i}$. Since $(-1)^i = (-1)^{2d-i}$, we get

$$D^2 = \prod_i (\det F_i \cdot \det F_{2d-i})^{(-1)^i} = q^{d\chi}.$$

Since $b_{2d-i} = b_i$, the terms of χ for i and $2d - i$ are equal and cancel modulo 2, so $\chi \equiv b_d \pmod 2$. If d is even, $d\chi$ is even anyway. If d is odd, the pairing on H^d of each connected component of X is perfect and alternating (Corollary 10.17), so b_d is even. In both cases $d\chi$ is even. Then $\varepsilon^2 = 1$ because $D^2 = q^{d\chi}$.

Now consider a single factor. Since F_i is invertible, its eigenvalues α are nonzero. Pulling $-\alpha/(q^d T)$ out of each linear factor gives

$$\det(1 - q^{-d} T^{-1} F_i) = \prod_\alpha \left(-\frac{\alpha}{q^d T}\right)\left(1 - \frac{q^d}{\alpha} T\right) = (-1)^{b_i} \frac{\det F_i}{(q^d T)^{b_i}} \det(1 - T F_{2d-i}).$$

By Corollary 12.31, and since $H^i_c = H^i$ as X_0 is proper, $Z(X_0, T)$ is the product of the $\det(1 - TF_i)^{(-1)^{i+1}}$. We substitute $1/(q^d T)$ for T and apply the last display to each factor. Since

$(-1)^i = (-1)^{2d-i}$, the factors $\det(1 - TF_{2d-i})^{(-1)^{i+1}}$ multiply to $Z(X_0, T)$, and the scalar factors multiply to

$$(-1)^{\chi} D^{-1} (q^d T)^{\chi} = (-1)^{\chi} D\, T^{\chi},$$

by $D^2 = q^{d\chi}$. The second form of the equation is the definition of ε. ∎

For the elliptic curve of Example 12.32 we have $d = 1$ and $b_0 = b_2 = 1$, $b_1 = 2$. So $\chi = 0$ and $D = \det(F_1)^{-1} \det(F_2) = 3/3 = 1$. The corollary says $Z(C_0, 1/(3T)) = Z(C_0, T)$, and indeed

$$\frac{1 + 3(3T)^{-2}}{(1 - (3T)^{-1})(1 - T^{-1})} = \frac{(3T^2 + 1)/(3T^2)}{(3T - 1)(T - 1)/(3T^2)} = \frac{1 + 3T^2}{(1 - 3T)(1 - T)}.$$

Example 12.34 (The functional equation of the projective line)**.** For $X_0 = \mathbf{P}^1_{\mathbf{F}_q}$ we have $d = 1$, $b_0 = b_2 = 1$, $b_1 = 0$, so $\chi = 2$, and $D = \det(F_0) \det(F_2) = 1 \cdot q = q$, so that $\varepsilon = 1$. The corollary predicts $Z(\mathbf{P}^1, 1/(qT)) = qT^2 Z(\mathbf{P}^1, T)$. Directly,

$$\frac{1}{(1 - (qT)^{-1})(1 - T^{-1})} = \frac{qT^2}{(qT - 1)(T - 1)} = \frac{qT^2}{(1 - T)(1 - qT)}.$$

Here the factor T^{χ} is not trivial, unlike for the elliptic curve, and the constant $q^{d\chi/2} = q$ comes from the twist (-1) on H^2.

Remark 12.35 (Duality and weights)**.** The functional equation is an identity of rational functions, not of power series at $T = 0$, since $1/(q^d T)$ is not a power series in T.

What duality does not give is the absolute value of the eigenvalues α. In Chapters 13–Chapter 19 we prove that the eigenvalues of F_i have absolute value $q^{i/2}$ under every embedding of $\overline{\mathbf{Q}}_\ell$ into $\mathbf{C}$. It follows that the factors $\det(1 - TF_i)$ for different i have no common zeros, and hence we can read them off from $Z(X_0, T)$ alone. In Chapter 19 we use this to show that they are polynomials with integer coefficients that do not depend on ℓ.

Part III Weights and the Riemann hypothesis

After Parts I and II we can write the number of points of a smooth proper variety as an alternating sum of powers of Frobenius eigenvalues. But we know nothing about the size of these eigenvalues, so the formula alone estimates nothing. For a curve of genus g the formula reads $N_m = 1 + q^m - \sum_j \alpha_j^m$. It gives Weil's bound $|N_m - 1 - q^m| \le 2gq^{m/2}$ only once we know that $|\alpha_j| = q^{1/2}$. This Part proves the analogous statement in every dimension, Weil's axiom (d) (Corollary 19.2): for smooth proper X_0 of pure dimension d, every eigenvalue α of Frobenius on $H^i(X, \mathbf{Q}_\ell)$ satisfies $|\iota(\alpha)| = q^{i/2}$ for every complex embedding ι. Theorem 19.7 deduces from it the Riemann hypothesis, the integrality of the factors P_i and their independence of ℓ.

For curves Weil proved this with the Jacobian. There the eigenvalues on H^1 are those of Frobenius on a Tate module, and the positivity of the Rosati involution forces $|\alpha| = q^{1/2}$. In higher dimension, however, no such structure is available. Instead, Deligne's first proof [Del74] fibred X_0 by a Lefschetz pencil and used the monodromy of the pencil. In his second paper [Del80], by contrast, he proved a theorem about all sheaves:

if K has weights at most w, then so does $Rf_!K$

(Theorem 13.11; weights are defined in Definition 13.1). In particular, for the constant sheaf on a smooth proper X_0, the theorem gives the upper bound $|\iota(\alpha)| \le q^{i/2}$ on H^i. Since Poincaré duality pairs α with q^d/α on H^{2d-i}, the upper bound on H^{2d-i} gives the lower bound on H^i. Here we follow Laumon's proof [Lau87] of the Deligne–Laumon theorem, which replaces the Lefschetz pencil by the ℓ-adic Fourier transform.

The theorem has to be about sheaves because the proof fibres the variety, locally, in curves over a variety of smaller dimension, and along such a fibration the direct image of the constant sheaf is no longer constant. Everything therefore comes down to a pure lisse sheaf F of weight w on a curve C. There we must show that $H^r_c(C, F)$ has weights at most $w + r$ (Theorem 17.15), and only H^1_c is hard.

On a curve the one tool is the trace formula. It writes the L-function, a product of local factors over the closed points, as the ratio of the characteristic polynomials on H^1_c and H^2_c. The difficulty is cancellation: a large local eigenvalue gives a local factor with a pole near the origin, but the other factors might cancel it. Rankin's trick from the theory of modular forms rules this out when the traces are real, because then the even tensor powers have Euler factors with nonnegative coefficients. By Lemma 14.24, no factor of such a product can have a pole closer to the origin than the product itself. On an affine curve, moreover, only H^2_c contributes poles, and their position is controlled by the determinants. These determinants are pure because a lisse sheaf of rank one has finite geometric monodromy (Section 14.1). The outcome is the bound $|\iota(\alpha)|^2 \le q_x^\beta$, where β is the largest determinant weight (Proposition 14.25). Applied to exterior powers, this bound shows that every irreducible constituent of a real sheaf is pure (Theorem 14.26, Chapter 14).

This bounds eigenvalues at points of a curve. What we must bound, however, is a cohomology group $H^1_c(\mathbf{A}^1, j_*M)$, for a pure lisse sheaf M on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$. The Fourier transform (Chapter 16) turns this group into the stalk of a sheaf on the dual line, and a twist of that sheaf is a quotient of a real sheaf. The Plancherel formula, together with two facts about pure sheaves on curves (Chapter 15), shows that the sheaf has weight $w + 1$.

In higher dimension the same positivity shows that real sheaves on any base are mixed (Theorem 18.8). This lets us apply the curve case along a fibration (Chapter 18). The Weil conjectures are then a short deduction (Chapter 19).

13 Weights

For the elliptic curve $y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, Frobenius acts on H^1 with eigenvalues $\pm i\sqrt{3}$. The Riemann hypothesis for this curve is then the statement $|\pm i\sqrt{3}| = 3^{1/2}$. But the eigenvalues live in $\overline{\mathbf{Q}}_\ell$, and an ℓ-adic number has no complex absolute value until we choose a field embedding $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$. Since no choice is preferred, we fix one ι throughout, and the final theorems hold for every choice.

An eigenvalue α of Frobenius at a closed point of degree n has weight w if $|\iota(\alpha)| = q^{nw/2}$. The factor n makes the weight independent of the field over which we look at the point: over $\mathbf{F}_{q^{mn}}$ the eigenvalue becomes α^m and q^n becomes q^{mn}. For example, on $\mathbf{P}^1$ Frobenius acts on H^0 by 1, of weight 0, and on $H^2 = \overline{\mathbf{Q}}_\ell(-1)$ by q, of weight 2. Similarly, the eigenvalues on H^1 of our elliptic curve have weight 1. Weil's purity axiom says that this pattern persists: H^i of a smooth proper variety is pure of weight i. The name comes from Hodge theory, where H^i of a compact Kähler manifold carries a Hodge structure of weight i.

The proof cannot stay with smooth proper varieties. It fibres them in curves, and the cohomology of the fibres forms sheaves on the base. So we need weights for sheaves, and for cohomology that is not pure. Without smoothness and properness only an upper bound survives. For example, let $\mathbf{G}_m = \mathbf{P}^1 \setminus \{0, \infty\}$. The localization sequence gives $H^1_c(\mathbf{G}_m, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell$ with trivial Frobenius and $H^2_c(\mathbf{G}_m, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell(-1)$. This agrees with the trace formula $q - 1 = \#\mathbf{G}_m(\mathbf{F}_q)$. Thus H^1_c has weight 0, strictly less than 1.

A sheaf is pure of weight w if all its eigenvalues at all closed points have weight w. It is mixed if it has a finite filtration whose quotients are pure. A bounded complex K has weights at most w if each $\mathcal{H}^i(K)$ is mixed with all weights at most $w + i$ (Definition 13.1). Here the shift by i puts the constant sheaf and its cohomology into one statement: $R\Gamma_c(X, \overline{\mathbf{Q}}_\ell)$ has weights at most 0 exactly when H^i_c has weights at most i, as on $\mathbf{G}_m$. In these terms the Deligne–Laumon theorem (Theorem 13.11) says that $Rf_!$ does not increase weights.

Two warnings will matter later. The first is that weights depend on ι. For example, consider the rank-one Weil sheaf on $\operatorname{Spec}\mathbf{F}_q$ on which Frobenius acts by $1 + \sqrt{2} \in \overline{\mathbf{Q}}_\ell$. It has weight $2\log_q(1+\sqrt{2}) > 0$ for an embedding with $\iota(\sqrt{2}) = \sqrt{2}$, and weight $2\log_q(\sqrt{2}-1) = -2\log_q(1+\sqrt{2})$ for the other one. For this reason Theorem 19.7 needs the Deligne–Laumon theorem for every ι: only then do the reciprocal roots of P_i have absolute value $q^{i/2}$ under every complex embedding.

The second warning is that being mixed is not a pointwise condition: it asks for a filtration with pure quotients, and such a filtration has to be constructed. Most of Chapter 18 is spent constructing one for direct images.

We define weights on Weil sheaves, sheaves on X with a Frobenius structure, rather than on sheaves on X_0. The reason is that normalizing weights requires twisting by the geometrically constant rank-one Weil sheaves $\mathcal{B}_b$, on which a degree-one Frobenius acts by a scalar $b \in \overline{\mathbf{Q}}_\ell^\times$. These do not come from X_0 when b is not an ℓ-adic unit.

13.1 Weil sheaves and weights

Weights are defined by the absolute values of the Frobenius eigenvalues on stalks at closed points, so we first fix how Frobenius acts on sheaves.

On a smooth curve U_0 we want to treat lisse Weil sheaves as representations, just as a lisse sheaf on U_0 is a continuous representation of $\pi_1(U_0)$. However, the profinite group $\pi_1(U_0)$ is an extension of $\widehat{\mathbf{Z}}$ by $\pi_1(U)$, and it has too few representations for this. Indeed, a continuous representation of a profinite group preserves a lattice, so a lift s of Frobenius acts with eigenvalues that are ℓ-adic units. Hence the rank-one Weil sheaf $\mathcal{B}_\ell$, on which s acts by ℓ and $\pi_1(U)$ acts trivially, is not a representation of $\pi_1(U_0)$. Concretely, $n \mapsto \ell^n$ does not extend continuously to $\widehat{\mathbf{Z}}$: the integers $m!$ tend to 0 in $\widehat{\mathbf{Z}}$, whereas $\ell^{m!}$ tends to 0 in $\mathbf{Z}_\ell$ and not to 1.

The Weil group keeps $\pi_1(U)$ with its topology and replaces $\widehat{\mathbf{Z}}$ by the discrete group $\mathbf{Z}$ of integer powers of Frobenius. Continuity then constrains only the geometric part. These representations of the Weil group are the language of Theorem 14.5 and of the monodromy groups of Section 14.2.

Convention. Put $k_0 = \mathbf{F}_q$, $k = \overline{\mathbf{F}}_q$, and fix a prime $\ell \nmid q$ and a field embedding $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$. For X_0/k_0 separated of finite type, we write $X = X_0 \times_{k_0} k$, and for a closed point x of X_0 write $\deg(x) = [\kappa(x) : k_0]$ for its degree and $q_x = q^{\deg(x)} = \#\kappa(x)$. Moreover, geometric Frobenius, Weil complexes and their Frobenius operators are those of Section 10.5. As there, a Weil sheaf is a Weil complex concentrated in degree zero, and the six operations carry Weil structures to Weil structures. In this Part Weil sheaves and complexes have coefficients in $\overline{\mathbf{Q}}_\ell$; a Weil complex K with coefficients in a finite extension $E \subset \overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$ is called pure, mixed, or of weights at most w if $K \otimes_E \overline{\mathbf{Q}}_\ell$ is (see Lemma 13.2 and the remark after it for why scalars must be extended).

On a connected normal scheme of finite type over a field, we identify a lisse E-sheaf with the corresponding continuous representation of the profinite fundamental group, by Proposition 13.10, proved in Section 13.2 below. This covers all arguments with representations below, since they take place on smooth schemes. Moreover, a lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaf has a lisse Weil model over some finite extension $E/\mathbf{Q}_\ell$, and two models become identified after a finite extension of scalars. This model includes the Weil isomorphism, which need not preserve a geometric $\mathcal{O}_E$-lattice.

For a geometrically connected smooth curve U_0, put $U = U_0 \times_{k_0} k$ and choose a geometric point $\bar\eta$ of U. We then write $\pi_1(U)$ and $\pi_1(U_0)$ for $\pi_1(U, \bar\eta)$ and $\pi_1(U_0, \bar\eta)$. The inverse image of the integer powers of geometric Frobenius in $\pi_1(U_0)$ is its Weil group $W(U_0)$. It contains the geometric fundamental group, with an exact sequence

$$1 \longrightarrow \pi_1(U) \longrightarrow W(U_0) \xrightarrow{\deg} \mathbf{Z} \longrightarrow 1.$$

A degree-one lift s splits this sequence as a group extension. In these terms, lisse Weil sheaves are the same as representations ρ of $W(U_0)$, defined over a finite extension of $\mathbf{Q}_\ell$, whose restriction to $\pi_1(U)$ is continuous. For $x \in |U_0|$, the local Frobenius Frob_x at x (Theorem 12.30) is a conjugacy class of degree $\deg(x)$.

Every sheaf defined over k_0 is a Weil sheaf. By means of the Weil isomorphism, Frobenius acts on the stalk at a geometric point over each closed point x, so we can speak of the eigenvalues of Frob_x. In the description by representations the Weil isomorphism is transport by s, and conversely it determines the action of s.

The definition below contains two kinds of bounds. The first, being punctually of weights $\le w$, bounds the eigenvalues at each closed point; it is what the curve case yields on each stalk of a direct image. But it cannot serve as input for the curve case, which needs pure lisse sheaves. The second, mixedness, has the opposite virtue: a mixed sheaf can be treated one pure quotient at a time, and for a mixed sheaf the bound on weights is again a pointwise condition (Lemma 13.3). In Chapter 18 we use this to turn the stalkwise bounds into the bound of Theorem 13.11.

Definition 13.1 (Weights, purity and mixedness)**.** Let K be a constructible Weil sheaf on X, let x be a closed point of X_0, and let α be an eigenvalue of Frob_x on $K_{\bar x}$. We say α has *ι-weight* w if $|\iota(\alpha)| = q_x^{w/2}$.

1. K is *punctually ι-pure of weight w*, or simply *pure of weight w*, if every such eigenvalue, at every closed point, has ι-weight w.
2. K is *punctually of weights $\le w$* if $|\iota(\alpha)| \le q_x^{w/2}$ for every such eigenvalue at every closed point.
3. K is *ι-mixed* if it has a finite filtration by Weil subsheaves whose nonzero quotients are pure, of possibly different real weights. It is *mixed of weights at most w*, or has *weights at most w*, if it has such a filtration whose nonzero quotients have weights at most w.

A bounded constructible Weil complex K has *weights at most w* if $\mathcal{H}^i(K)$ has weights at most $w + i$ for all i, and it is *pure of weight w* if each nonzero $\mathcal{H}^i(K)$ is pure of weight $w + i$.

Deligne calls a complex K pure of weight w if K has weights at most w and its Verdier dual $\mathbb{D}K$ has weights at most $-w$. Over a point the two notions agree. In general neither implies the other, and we use ours only for sheaves and over a point.

The cohomology groups of Theorem 13.11 are Weil sheaves on $\operatorname{Spec} k_0$. For them, having weights at most w is just a condition on the eigenvalues of Frobenius. Indeed, by Lemma 13.2 every Weil $\overline{\mathbf{Q}}_\ell$-sheaf over a point is mixed, and Lemma 13.3 reads the bound of a mixed sheaf off its eigenvalues.

Lemma 13.2 (Weil sheaves on a point are mixed)**.** *Every constructible Weil $\overline{\mathbf{Q}}_\ell$-sheaf on* $\operatorname{Spec} k_0$ *is mixed. A constructible Weil E-sheaf on* $\operatorname{Spec} k_0$ *becomes mixed after extending scalars to a finite extension of E containing the eigenvalues of Frobenius.*

Proof. A constructible Weil sheaf on $\operatorname{Spec} k_0$ is a finite-dimensional vector space V with an automorphism F, the geometric Frobenius. In both cases we may assume that V is defined over a finite extension E' of $\mathbf{Q}_\ell$ containing the eigenvalues of F. For $w \in \mathbf{R}$ let $V_w \subset V$ be the sum of the generalized eigenspaces of F for the eigenvalues α with $|\iota(\alpha)| = q^{w/2}$. These subspaces are F-stable, only finitely many are nonzero, and $V = \bigoplus_w V_w$. Let $w_1 < \cdots < w_m$ be the weights that occur. Then the subspaces $V_{w_1} \oplus \cdots \oplus V_{w_j}$ form a finite filtration by Weil subsheaves with quotients V_{w_j}, and V_{w_j} is pure of weight w_j. ∎

Extending scalars is needed for E-sheaves. Suppose 2 is not a square in $\mathbf{Q}_\ell$. Then the rank-two $\mathbf{Q}_\ell$-sheaf on which Frobenius has eigenvalues $1 \pm \sqrt{2}$ is irreducible. Its two eigenvalues have different weights, so it has no filtration with pure quotients over $\mathbf{Q}_\ell$.

Lemma 13.3 (Bounding the weights of a mixed sheaf)**.** *A mixed sheaf has weights at most w if and only if it is punctually of weights $\leq w$.*

Proof. At every closed point x the eigenvalues of Frob_x on the stalk of a mixed sheaf are those on the stalks of the quotients of its filtration. If the quotients have weights at most w, the sheaf is therefore punctually of weights $\leq w$. Conversely, each nonzero quotient has a nonzero stalk at some closed point x, and the eigenvalues of Frob_x there are eigenvalues on the stalk of the sheaf. So if the sheaf is punctually of weights $\leq w$, the weight of the quotient is at most w. ∎

The lemma does not say that a sheaf punctually of weights $\leq w$ is mixed; the filtration has to be found. Nor need the filtration split: weights see only the characteristic polynomials of Frobenius, so a Jordan block and its semisimplification have the same weights. The following example is a mixed sheaf that is not pure and whose filtration does not split.

Example 13.4 (The Kummer extension)**.** On $\mathbf{G}_{m,k_0}$ with coordinate t, the μ_{ℓ^n}-torsors $u^{\ell^n} = t$ give the Kummer cocycle κ, a continuous map from $\pi_1(\mathbf{G}_{m,k_0})$ to $\mathbf{Z}_\ell(1)$. Let χ be the cyclotomic character, by which π_1 acts on $\mathbf{Z}_\ell(1)$. The matrices

$$\sigma \longmapsto \begin{pmatrix} \chi(\sigma) & \kappa(\sigma) \\ 0 & 1 \end{pmatrix}$$

define a lisse sheaf $\mathcal{E}$ on $\mathbf{G}_{m,k_0}$, an extension $0 \to \overline{\mathbf{Q}}_\ell(1) \to \mathcal{E} \to \overline{\mathbf{Q}}_\ell \to 0$. Geometric Frobenius acts on $\overline{\mathbf{Q}}_\ell(1)$ by q_x^{-1}. Hence at a closed point x the eigenvalues of Frob_x are q_x^{-1} and 1, of weights -2 and 0, and $\mathcal{E}$ is mixed of weights at most 0 and not pure. The filtration does not split. Indeed, on $\pi_1(\mathbf{G}_{m,k})$ the character χ is trivial, and κ is a homomorphism onto $\mathbf{Z}_\ell(1)$ because the covers $u^{\ell^n} = t$ of $\mathbf{G}_{m,k}$ are connected. So the line $\overline{\mathbf{Q}}_\ell(1)$ is the only stable line. At each closed point, by contrast, the two eigenvalues are distinct and the stalk is the direct sum of its eigenlines.

The behaviour of weights under the usual operations comes down to the eigenvalues of an operator on a filtered space, on duals and on tensor and exterior powers.

Lemma 13.5 (Filtrations and spectra)**.** *Let V have a finite filtration preserved by an automorphism u. Then*

$$\det(1 - Tu \mid V) = \prod_a \det(1 - Tu \mid \operatorname{gr}_a V).$$

In particular, the eigenvalues, with algebraic multiplicity, are the union of the eigenvalues on the graded pieces. If the eigenvalues of u on V are $\alpha_1, \dots, \alpha_d$, then those on $V^\vee$ are α_i^{-1}, and those on $\bigwedge^r V$ are $\alpha_{i_1} \cdots \alpha_{i_r}$ for $i_1 < \cdots < i_r$. If v on V' has eigenvalues β_j, those of $u \otimes v$ are $\alpha_i \beta_j$.

Proof. In a basis adapted to the filtration, the matrix of u is block upper triangular, so its characteristic polynomial is the product of those of the diagonal blocks. For the other constructions we may choose bases over $\overline{\mathbf{Q}}_\ell$ in which u and v are upper triangular. Then, in the induced tensor basis and in the ordered wedge basis, the resulting matrices are upper triangular with the stated diagonal entries. Finally, the matrix on the dual is the inverse transpose. ∎

At a closed point, each operation in the next lemma and proposition acts on stalks by one of the constructions of the lemma: a subobject, a quotient, a tensor product or a dual, or, for an inverse image, a power of Frobenius. So the behaviour of weights is a matter of eigenvalues. Mixedness needs in addition some bookkeeping to produce the filtrations.

Lemma 13.6 (Subobjects and quotients of Weil sheaves)**.** *Subobjects and quotients of a pure Weil sheaf of weight w are pure of weight w. Subquotients of a mixed sheaf are mixed, and a subquotient of a sheaf that is mixed of weights at most w is mixed of weights at most w.*

Proof. By Lemma 13.5, the eigenvalues on a subobject or a quotient occur among those of the whole. Hence subobjects and quotients of pure sheaves are pure. For mixed sheaves, we intersect a mixed filtration with a subobject, or push it into a quotient. The graded pieces are then subquotients of the original pure pieces, hence pure of the same weights, and so the upper bound persists. ∎

Proposition 13.7 (Weights under the usual operations)**.** *Extensions of mixed sheaves are mixed, and an extension of two sheaves with a common upper bound has the same bound. Tensor products of pure sheaves of weights a and b are pure of weight $a + b$, and the dual of a pure lisse sheaf of weight a is pure of weight $-a$. A tensor product of mixed sheaves of weights at most a and at most b is mixed of weights at most $a + b$. The dual of a mixed lisse sheaf is mixed, of the negated weights; when its filtration has lisse graded pieces, their duals are the graded pieces of the annihilator filtration of the dual. Inverse image preserves purity, mixedness and their upper bounds. If a complex K has weights at most w, then $K[r]$ has weights at most $w + r$.*

For $b \in \overline{\mathbf{Q}}_\ell^\times$, let $\mathcal{B}_b$ be the geometrically constant rank-one Weil sheaf on which a degree-one Frobenius acts by b. Tensoring by $\mathcal{B}_b$ adds $\lambda_\iota(b) = 2 \log_q |\iota(b)|$ to every weight. The Tate sheaf $\overline{\mathbf{Q}}_\ell(1)$ has geometric Frobenius q^{-1}, so twisting by (r) subtracts $2r$ from every weight.

Proof. Each assertion has two parts: a statement about eigenvalues at closed points, which follows from Lemma 13.5, and, for mixedness, the construction of a filtration with pure quotients.

An extension is filtered by a filtration of its subobject followed by the inverse images of a filtration of its quotient. The graded pieces of the result are those of the two given filtrations, so a common upper bound persists.

For tensor products we use that the coefficients form a field, so that the associated graded of the tensor product of two filtrations is the tensor product of the two associated gradeds. By Lemma 13.5, the eigenvalues of tensor products and duals are the products $\alpha_i \beta_j$ and the inverses α_i^{-1}, whose weights are $a + b$ and $-a$.

For the dual of a mixed lisse sheaf G with mixed filtration (G_i), suppose first that the graded pieces G_i/G_{i-1} are lisse. Dualizing is exact on lisse sheaves, so the annihilator filtration $G_i^{\perp} = \ker(G^{\vee} \to G_i^{\vee})$ has graded pieces $(G_i/G_{i-1})^{\vee}$, which are pure of the negated weights by the computation above. In general the graded pieces need not be lisse: on a curve, a pure lisse G has the filtration $j_!j^*G \subseteq G$, whose quotient i_*i^*G is a skyscraper with dual zero. We then argue by Noetherian induction on X. Let $U \subseteq X$ be a dense open, stable under Frobenius, on which all graded pieces are lisse, and let $i : Z \to X$ be its closed complement. On U the first case applies to j^*G. Since G is lisse, $i^*(G^{\vee}) \simeq (i^*G)^{\vee}$, and i^*G is a mixed lisse sheaf on Z, so $i^*(G^{\vee})$ is mixed by induction, of the negated weights. Extension by zero and direct image along a closed immersion are exact and do not change stalks, so they carry mixed filtrations to mixed filtrations of the same weights, and the exact sequence

$$0 \to j_!j^*(G^{\vee}) \to G^{\vee} \to i_*i^*(G^{\vee}) \to 0$$

shows that $G^{\vee}$ is mixed, of the negated weights.

Inverse images do not change weights, because the weight is normalized by the degree. Indeed, at a closed point of an inverse image the Frobenius is a positive power of the Frobenius at the image point, and the residue field degree is multiplied by the same integer. For shifts, $\mathcal{H}^i(K[r]) = \mathcal{H}^{i+r}(K)$ is mixed of weights at most $w + i + r$, which is the bound $w + r$ for $K[r]$.

The local eigenvalue of $\mathcal{B}_b$ at x is $b^{\deg(x)}$, of weight $2\log_q|\iota(b)|$, and tensoring by $\mathcal{B}_b$ multiplies every eigenvalue by it. For the Tate sheaf, recall that arithmetic Frobenius acts on roots of unity by raising to the qth power. Geometric Frobenius is its inverse, so it acts on $\overline{\mathbf{Q}}_\ell(1)$ by q^{-1}, of weight -2. ∎

13.2 Lisse sheaves as representations

Let X be a connected normal scheme of finite type over a field and L a lisse E_X-module. Lemma 3.24 gives lattices in L only on the strata of a finite stratification, and the nodal curve of Section 3.8 shows that without normality a global lattice need not exist. On a normal X, however, we find a lattice on all of X, and as a consequence lisse sheaves are the same as continuous representations of the fundamental group. Here normality is used to show that the étale X-scheme representing lattices in L has a component that is finite étale and surjective. Once we have such a component, we argue as in Lemma 3.24.

Lemma 13.8 (Geometric coverings of a normal scheme)**.** *Let X be a nonempty connected normal scheme of finite type over a field. Let $p : Y \to X$ be separated and étale. Suppose that for every valuation ring V, with fraction field K, every commutative diagram*

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & Y \\ \downarrow & & \downarrow p \\ \operatorname{Spec} V & \longrightarrow & X \end{array}$$

has a lift $\operatorname{Spec} V \to Y$*. Then every connected component of Y is finite étale and surjective over X.*

Proof. We compare a component Y_1 with the normalization N of X in the function field of Y_1. By Zariski's main theorem, Y_1 is an open subscheme of N, and by the valuative hypothesis this open subscheme contains every point of N.

Since Y is étale over X, it is locally noetherian and normal [Stacks, 033C]. Hence its connected components are integral, open and closed [Stacks, 033N]. Fix a nonempty component Y_1. Since a valuative lift whose generic point lies in Y_1 stays in Y_1, the hypothesis also holds for $Y_1 \to X$.

The image of Y_1 is nonempty and open, hence contains the generic point η of X. The generic fibre of Y_1 is integral and étale over $k(X)$, so it is $\operatorname{Spec} k(Y_1)$. Thus the function field $k(Y_1)$ is a finite separable extension of $k(X)$, and we let N be the normalization of X in $k(Y_1)$. Since X is Nagata, N is finite over X [Stacks, 035B, 0AVK]. We claim that $Y_1 \to X$ factors through a map $f : Y_1 \to N$. Indeed, since Y_1 is normal, an element of $k(Y_1)$ that is integral over the coordinate ring of an affine open of X lies in every local ring of Y_1 over that open. These regular functions agree in $k(Y_1)$, and hence they glue. The map f is separated, locally of finite type and locally quasi-finite, because its fibres lie in the zero-dimensional fibres of $Y_1 \to X$.

We show that f is an open immersion. Choose affine opens $W \subseteq Y_1$ and $N_0 \subseteq N$ with $f(W) \subseteq N_0$. By Zariski's main theorem [Stacks, 05K0], W is an open subscheme of a finite N_0-scheme, and the reduced closure of W in it is integral, finite and birational over N_0, with function field $k(Y_1)$. Since N_0 is normal, this closure is N_0 [Stacks, 0AB1], and therefore $W \to N_0$ is an open immersion. These local inverses glue: on the intersection of two such open images, the inverse sections agree generically, hence everywhere by separatedness, and so they glue to an inverse on the union of the images.

Finally, let $n \in N$ and choose a valuation ring $V \subseteq k(Y_1)$ dominating $\mathcal{O}_{N,n}$ [Stacks, 00IA]. By hypothesis, the generic map $\operatorname{Spec} k(Y_1) \to Y_1$ extends to a map $\operatorname{Spec} V \to Y_1$, whose composite with f agrees with the canonical map $\operatorname{Spec} V \to N$ generically and over X. By separatedness of $N \to X$ the two maps agree, so n lies in the open image of Y_1. We conclude that $Y_1 = N$ is finite étale over X. In particular its image is open and closed, and being nonempty, it is all of X. ∎

We apply the lemma to the scheme of lattices in a lisse sheaf. There the valuative hypothesis says that a lattice defined at the generic point of a valuation ring extends over the whole ring. This holds because, locally, lattices form a constant sheaf on a discrete set.

Lemma 13.9 (Global lattices on a normal scheme)**.** *Let X be a nonempty connected normal scheme of finite type over a field, and let $E/\mathbf{Q}_\ell$ be finite. Every lisse E_X-module L, in the pro-étale coefficient category of Section 2.12, contains a lisse $\mathcal{O}_{E,X}$-submodule M with $M[1/\ell] = L$. The reductions $M_n = M/\pi_E^n M$ are finite locally constant étale sheaves, and $M \simeq \varprojlim_n M_n$.*

Proof. The scheme of lattices satisfies the valuative condition of Lemma 13.8, so it has a finite étale surjective component, and we push the universal lattice of this component down to X by the trace.

We may assume that L has positive rank r. Let Y be the X-scheme representing lattices in L (Lemma 3.15), a separated surjective étale X-scheme with a universal lattice. Pro-étale locally on X, the scheme Y is constant: the proof of that lemma shows that, for a quasi-compact pro-étale covering $U \to X$, with U a pro-étale affine on which L is free,

$$Y \times_X U \simeq \coprod_{\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)} U,$$

where $\mathrm{GL}_r(E)/\mathrm{GL}_r(\mathcal{O}_E)$ is the discrete set of lattices in E^r.

We check the valuative property of Lemma 13.8. Let V be a valuation ring with fraction field K, and suppose given $\operatorname{Spec} V \to X$ and a lift $\xi : \operatorname{Spec} K \to Y$. Since $U \to X$ is flat and surjective, there are a local extension of valuation rings $V \to V'$ and a compatible map $\operatorname{Spec} V' \to U$ [Stacks, 089F]. This extension is faithfully flat, because it is local and V' is torsion-free, hence flat, over

V [Stacks, 0539]. By the displayed description of $Y \times_X U$, the pullback of ξ to the spectrum of the fraction field of V' picks out a lattice in E^r, and the constant section with this value extends it to a map $\xi' : \operatorname{Spec} V' \to Y$.

We descend ξ' by fpqc descent, which needs its two pullbacks to $R = V' \otimes_V V'$ to agree. Over the domain $V' \otimes_V K$, the maps ξ' and ξ agree at the generic point, hence everywhere by separatedness, so the two pullbacks agree after base change to K. This suffices: since R is flat over V, the map $R \to R \otimes_V K$ is injective, and therefore $\operatorname{Spec}(R \otimes_V K)$ is schematically dense in $\operatorname{Spec} R$. As the equalizer of the two pullbacks is closed by separatedness, they agree on $\operatorname{Spec} R$. Descent then gives a lift $\operatorname{Spec} V \to Y$ restricting to ξ, which is unique by separatedness and the density of $\operatorname{Spec} K$.

By Lemma 13.8, every nonempty component $p_1 : Y_1 \to X$ of Y is therefore finite étale and surjective. We fix one, and let $N \subseteq p_1^* L$ be its universal lattice. As in Lemma 3.24, set

$$M = \operatorname{im}\bigl(p_{1*}N \longrightarrow p_{1*}p_1^*L \xrightarrow{\mathrm{Tr}} L\bigr).$$

The sheaf M is lisse with $M[1/\ell] = L$. Indeed, after the cover is split into d sheets, this image is the sum of d lattices. On a common pro-étale refinement these become constant lattices in E^r, and their sum is a finite free $\mathcal{O}_E$-lattice spanning E^r.

By Theorem 2.43 the finite reductions $M_n = M/\pi_E^n M$ are finite locally constant étale sheaves, and $M = \varprojlim_n M_n$ by Lemma 2.51, which we may check where M is free. ∎

Galois theory of finite étale covers turns statements about the finite reductions of such an M into statements about representations.

Proposition 13.10 (Lattices and representations on a normal scheme)**.** *Let X be a nonempty connected normal scheme of finite type over a field, let $E/\mathbf{Q}_\ell$ be finite, and fix a geometric point $\bar{x}$. By Lemma 13.9, every lisse E_X-module L contains a lisse $\mathcal{O}_{E,X}$-submodule M with $M[1/\ell] = L$ and $M \simeq \varprojlim_n M/\pi_E^n M$, whose reductions are finite locally constant étale sheaves. The fibre functor at $\bar{x}$, $L \mapsto (\varprojlim_n (M/\pi_E^n M)_{\bar{x}}) \otimes_{\mathcal{O}_E} E$, does not depend on the choice of M up to canonical isomorphism, and it gives an equivalence between lisse E_X-modules and finite-dimensional continuous E-representations of the ordinary profinite group $\pi_1(X, \bar{x})$.*

Proof. Let $M \subset L$ be as in the statement, with reductions $M_n = M/\pi_E^n M$. The fibres of these reductions define a continuous action of $\pi_1(X, \bar{x})$ on the finite free completed fibre

$$M_{\bar{x}} := \varprojlim_n (M_n)_{\bar{x}}.$$

The fibre functor of the proposition takes L to the rational fibre $M_{\bar{x}} \otimes_{\mathcal{O}_E} E$; in other words, we apply the fibre functor of finite covers to the finite reductions and complete the result.

Full faithfulness comes from the Galois theory of finite étale covers of the connected scheme X. It says that taking the fibre at $\bar{x}$ is an equivalence between finite locally constant sheaves on $X_{\acute{e}t}$ and finite sets with continuous $\pi_1(X, \bar{x})$-action, compatible with module structures. To pass from the finite reductions to L we need to clear denominators. Let $M \subseteq L$ and $M' \subseteq L'$ be lattices in two lisse E_X-modules. Then a morphism $\phi : L \to L'$ has a uniform denominator: $\pi_E^c \phi$ maps M into M' for some $c \geq 0$. To see this, bound the finitely many matrix entries on a quasi-compact pro-étale covering trivializing both lattices.

For faithfulness, suppose the fibre of ϕ vanishes. Then the fibres of all the reductions of $\pi_E^c \phi$ vanish, so these reductions vanish by the Galois correspondence. Since $M' = \varprojlim_n M'_n$, it

follows that $\pi_E^c \phi = 0$. For fullness, take an equivariant map between the rational fibres. After multiplication by some π_E^c, it maps $M_{\bar{x}}$ into $M'_{\bar{x}}$. Its reductions are then equivariant maps of finite fibres, so by the Galois correspondence they come from compatible maps $M_n \to M'_n$. The inverse limit of these is a map $M \to M'$ whose rationalization, divided by π_E^c, has the given fibre.

The same uniform denominator, applied to the identity of L, shows that the rational fibre does not depend on the lattice. Indeed, if M_1 and M_2 are two lattices in L, then $\pi_E^c \mathrm{id}_L$ sends M_1 into M_2 for some $c \geq 0$, and symmetrically. On fibres these maps, divided by π_E^c, therefore give mutually inverse equivariant isomorphisms between the two rational fibres.

For essential surjectivity we build a stable lattice and glue its reductions. A continuous representation of the profinite group $\pi_1(X, \bar{x})$ on E^r has compact image. Hence the $\mathcal{O}_E$-span of the translates of a lattice is bounded and contains that lattice. As $\mathcal{O}_E$ is noetherian, this span is a stable finite free lattice, and its reductions define compatible finite locally constant étale sheaves. These sheaves become constant simultaneously on the surjective pro-étale cover obtained as the inverse limit of the finite étale covers corresponding to their kernels. Their inverse limit is therefore lisse over $\mathcal{O}_E$, and its rationalization is lisse over E. Finally, this construction is inverse to the fibre functor. ∎

13.3 The Deligne–Laumon theorem

For a separated k_0-scheme of finite type X_0, we write $D_{\leq w}(X_0)$ for the category of bounded constructible Weil complexes K on X_0 of weights at most w in the sense of Definition 13.1, that is, such that $\mathcal{H}^i(K)$ is mixed of weights at most $w + i$ for every i. The Deligne–Laumon theorem says that for a separated morphism of finite type $f : X_0 \to Y_0$, the functor $Rf_!$ maps $D_{\leq w}(X_0)$ into $D_{\leq w}(Y_0)$. The shift by i makes this one statement for all degrees: for the structure morphism of X_0, one condition on $R\Gamma_c(X, K)$ bounds all the groups $H_c^n(X, K)$ at once. Moreover, by Proposition 13.7, replacing K by $K[r]$ only moves the bound by r. The proof occupies Chapter 18 and, for a single curve, the sections before it, and Corollary 19.2 deduces from the theorem the purity of the cohomology of a smooth proper variety.

Theorem 13.11 (Weights of $Rf_!$)**.** *For every separated morphism of finite type $f : X_0 \to Y_0$,*

$$Rf_! D_{\leq w}(X_0) \subset D_{\leq w}(Y_0).$$

In particular, if $K \in D_{\leq w}(X_0)$, then $H_c^n(X, K)$ has weights at most $w + n$.

14 Purity of real sheaves

In Chapter 17 the Fourier transform of a pure sheaf produces a lisse sheaf M' whose Frobenius eigenvalues we cannot compute. What we do know is that a twist of M' is a quotient of a sheaf whose traces are real, and from this alone we must conclude that M' is pure. In Chapter 18 the same need returns for the direct images of a family of curves, which we can make real but do not know to be mixed. Both arguments rest on one theorem (Theorem 14.26):

> on a smooth geometrically connected curve, every irreducible constituent of a real lisse Weil sheaf is pure, of weight equal to its determinant weight.

Before this theorem we know the weights of one kind of sheaf only: a lisse sheaf of rank one on a curve has finite geometric monodromy, and therefore it is pure (Section 14.1). In particular the determinant of every lisse sheaf is pure, and its weight divided by the rank, the determinant

weight, is the average weight of the eigenvalues at each point. The theorem says that on an irreducible constituent of a real sheaf no eigenvalue deviates from this average.

The tool is the trace formula. On an affine curve U_0 it expresses the L-function of a lisse sheaf $\mathcal{G}$ as

$$L(U_0, \mathcal{G}, t) = \prod_{x \in |U_0|} \det(1 - \mathrm{Frob}_x \, t^{\deg(x)} \mid \mathcal{G}_{\bar{x}})^{-1} = \frac{\det(1 - Ft \mid H^1_c(U, \mathcal{G}))}{\det(1 - Ft \mid H^2_c(U, \mathcal{G}))}.$$

The poles of the right-hand side are controlled by H^2_c, which is easy to compute (Lemma 14.20). On the left, a large eigenvalue at a point x gives a pole of the local factor at x close to the origin. But this pole could be cancelled by a zero of the product of the other factors, as in $(1-2t)^{-1} \cdot (1-2t) = 1$. Positive factors, however, cannot cancel in this way. Indeed, suppose all factors are power series with nonnegative coefficients and constant term 1. Then the coefficients of each factor are bounded by those of the product, and Lemma 14.24 shows that no factor can converge on a smaller disc than the product.

This idea is Rankin's. In 1939 he improved the bound for Ramanujan's function $\tau(n)$ to $O(n^{29/5})$ by studying the Dirichlet series $\sum_n \tau(n)^2 n^{-s}$, whose coefficients are nonnegative. Deligne used the same argument in his first proof [Del74].

For the Euler factors to be positive, the traces must be real. We therefore call a lisse Weil sheaf real if its local characteristic polynomials have real coefficients under the fixed embedding ι of Convention `situation-weil` (Definition 14.21). For example, every sheaf with rational characteristic polynomials is real, such as a Tate twist of the constant sheaf. On an even tensor power $\mathcal{G}^{\otimes 2m}$ of a real sheaf the traces are $2m$th powers of real numbers, and this makes the Euler factors positive. From this, Proposition 14.25 bounds each local eigenvalue α of $\mathcal{G}$ by $|\iota(\alpha)|^2 \le q_x^{\beta}$, where β is the largest determinant weight of the constituents.

This is not yet the theorem, because a real sheaf need not be pure. For example, the sheaf $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ is real, with constituents of weights 0 and 2, but the bound only gives 2 for both. To separate the constituents we pass to exterior powers. In $\bigwedge^2(\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)) = \det \overline{\mathbf{Q}}_\ell(-1) \otimes \overline{\mathbf{Q}}_\ell$ the factor $\overline{\mathbf{Q}}_\ell$ appears multiplied by the known eigenvalue q_x of $\det \overline{\mathbf{Q}}_\ell(-1)$, so the bound 2 for this summand leaves the bound 0 for $\overline{\mathbf{Q}}_\ell$ itself. In general, Proposition proposition-determinant-weights-tensor-operations finds the exterior power in which a given group of constituents carries the largest determinant weight, and the same argument applies there. Positivity, however, only ever gives upper bounds. The lower bound comes from applying the same argument to the dual, and in Chapter 19 this role is played by Poincaré duality.

14.1 Sheaves of rank one

Rankin's argument bounds the local eigenvalues of a sheaf by its determinant weights, so it needs the determinant of every lisse sheaf to be pure (Corollary 14.8). Since a determinant has rank one, we prove more generally that every rank-one lisse Weil sheaf on a smooth geometrically connected curve is pure (Corollary 14.6), and the later sections Sections 14.2–Section 14.4 are built on this fact.

In rank one the question is about characters. Let U_0 be a smooth geometrically connected curve and $s \in W(U_0)$ a lift of Frobenius of degree one (Convention `situation-weil`). Then a rank-one lisse Weil sheaf is a character $\chi : W(U_0) \to E^\times$, continuous on $\pi_1(U)$. If χ is trivial on $\pi_1(U)$, the sheaf is one of the sheaves $\mathcal{B}_b$ with $b = \chi(s)$; at a closed point x its eigenvalue is $b^{\deg(x)}$, so it is pure of weight $2 \log_q |\iota(b)|$. A Kummer sheaf is the next simplest case. If $n \mid q-1$, the cover $t \mapsto t^n$ of $\mathbf{G}_m$ is a μ_n-torsor, and so a character of $\mu_n(k_0)$ gives a rank-one sheaf on $\mathbf{G}_{m,k_0}$ whose

geometric monodromy is cyclic of order dividing n. Its eigenvalues are values of the character on $\mu_n(k_0)$, hence roots of unity, and it is pure of weight 0.

Up to a twist, every rank-one sheaf behaves like these two. Indeed, by Theorem 14.5 the image $\chi(\pi_1(U))$ is finite. It follows that $\chi(\mathrm{Frob}_x)$ is $\chi(s)^{\deg(x)}$ times a root of unity, so the sheaf is pure of weight $2\log_q|\iota(\chi(s))|$.

Finiteness is a statement about Weil sheaves, because on the geometric curve there are rank-one local systems with infinite monodromy. For example, the tame fundamental group of $\mathbf{G}_{m,k}$ has the quotient $\mathbf{Z}_\ell(1) \simeq \mathbf{Z}_\ell$, and $a \mapsto (1+\ell)^a$ is a character of infinite order. But it has no Weil structure. Indeed, a Weil structure forces $\chi(s^{-1}gs) = \chi(g)$ for $g \in \pi_1(U)$, while conjugation by s acts on $\mathbf{Z}_\ell(1)$ by multiplication by a power $q^{\pm 1}$. A character of $\mathbf{Z}_\ell(1)$ that extends to the Weil group therefore satisfies $\chi(g)^{q-1} = 1$, and it has finite order.

The general proof follows the same pattern. Since there is no explicit quotient to compute with, it uses a logarithm instead: a power χ^N takes values in $1 + \mathfrak{m}_E^r$, where the ℓ-adic logarithm converges, and then $\log \chi^N$ is an additive character of $\pi_1(U)$ invariant under conjugation by s, that is, a class in $H^1(U, E)$ fixed by Frobenius (Lemma 14.4). Such a class is zero (Proposition 14.3), so χ^N is trivial on $\pi_1(U)$.

For a proper curve, Frobenius fixes no class in H^1 because the Jacobian has only finitely many rational points. If C_0 is a proper curve with Jacobian J_0, Kummer theory and Poincaré duality identify $\varprojlim_n H^1(C, \mathbf{Z}/\ell^n)$ with the dual of the Tate module $T_\ell J$ (Proposition 14.2). A fixed vector in the dual therefore means that 1 is an eigenvalue of Frobenius on $V_\ell J = T_\ell J \otimes \mathbf{Q}_\ell$, and after scaling, Frobenius would then fix a nonzero element of $T_\ell J$. Such an element is a compatible system of ℓ-power torsion points of J defined over k_0, of unbounded order. These points cannot all lie in the finite group $J_0(k_0)$, and so there is no such element (Proposition B.15).

For an open curve, the residue sequence contains in addition the boundary term $H^0(D, E(-1))$. A power of Frobenius acts on it by a power of $q \neq 1$, so it has no fixed vector either. For example, for $\mathbf{G}_m$ this is the computation above, since there $H^1(\mathbf{G}_{m,k}, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$, with Frobenius acting by q.

14.1.1 Frobenius invariants

Over $\mathbf{C}$ the link between H^1 of a curve and its Jacobian is classical. There the Jacobian is a torus V/Λ with $\Lambda = H_1(C, \mathbf{Z})$, and $H^1(C, \mathbf{Z}) = \mathrm{Hom}(\Lambda, \mathbf{Z})$. Proposition 14.2 is the ℓ-adic version, with $T_\ell J$ in place of $\Lambda \otimes \mathbf{Z}_\ell$. Here the link with the Jacobian is Kummer theory, which is a statement about μ_{ℓ^n} and line bundles. For this reason we work with the coefficients $\mathbf{Z}/\ell^n$ and pass to E only at the end. Apart from this, the only arithmetic input is the finiteness of $J_0(k_0)$, which holds because J_0 is of finite type over a finite field.

Lemma 14.1 (A proper smooth model)**.** *A smooth geometrically connected curve U_0/k_0 has a smooth projective geometrically connected compactification $j_0 : U_0 \hookrightarrow C_0$. The reduced boundary D_0 is finite étale over k_0.*

Proof. We compactify by Nagata's theorem and then normalize. On a curve, normality is the same as regularity, and over a perfect field regularity is smoothness, so the normalization is the model we want.

By Theorem A.2, U_0 is an open subscheme of a proper k_0-scheme, and we take the reduced closure of U_0 in it and normalize this closure in $k_0(U_0)$. Since a scheme of finite type over a field is Nagata, the normalization is finite [Stacks, 035B, 035S], and it is an isomorphism over the normal open U_0. The resulting proper normal curve is regular, because a one-dimensional

noetherian normal local domain is a discrete valuation ring. Hence it is smooth over the perfect field k_0 [Stacks, 0B8Y]. Its boundary is zero-dimensional and proper over k_0, hence finite. Since k_0 is perfect, the reduced boundary is étale. After base change to k, the curve U is dense and connected, and therefore its closure C is connected. Finally, a proper scheme of dimension one over a field is projective [Stacks, 0A26]. ∎

Once we have a smooth projective model C_0, the Jacobian J_0 of C_0 exists (Appendix C), and we can compare the cohomology of C with its Tate module. We first make the comparison with finite coefficients, where Kummer theory applies directly. The proposition ends with a matrix description of Frobenius, which is the form used in Proposition 14.3.

Proposition 14.2 (Kummer, duality and the integral Tate module)**.** *Let C_0/k_0 be a smooth proper geometrically connected curve with Jacobian J_0, and put $A_n = \mathbf{Z}/\ell^n$. There are canonical compatible Galois-equivariant identifications*

$$H^1(C, A_n(1)) \simeq J[\ell^n](k), \qquad \varprojlim_n H^1(C, A_n) \simeq \mathrm{Hom}_{\mathbf{Z}_\ell}(T_\ell J, \mathbf{Z}_\ell).$$

Here the transition maps are multiplication by ℓ on $J[\ell^n](k)$ and reduction of coefficients on $H^1(C, A_n)$. If arithmetic Frobenius on $T_\ell J$ has matrix B, then geometric Frobenius on $\mathrm{Hom}_{\mathbf{Z}_\ell}(T_\ell J, \mathbf{Z}_\ell)$ has matrix B^t in the dual basis.

Proof. By the Kummer sequence, $H^1(C, \mu_{\ell^n})$ is the ℓ^n-torsion of the Picard group, and Poincaré duality turns this into a statement about the untwisted H^1.

The first identification is Proposition C.17, which also gives its Galois equivariance, and it shows that the coefficient map $z \mapsto z^\ell$ from $\mu_{\ell^{n+1}}$ to μ_{ℓ^n} corresponds to $[\ell]$ on the Jacobian.

For the dual Tate module, Poincaré duality at finite level pairs the untwisted H^1 with the twisted one that we have just identified with $J[\ell^n](k)$. Derived duality gives a statement about the single group H^1 because all the groups involved are free A_n-modules. Indeed, we have $H^0(C, A_n) = A_n$ and $H^2(C, A_n(1)) = A_n$ by Proposition D.3, while the group $H^1(C, A_n(1))$ is free of rank $2g$ by the identification above and Corollary C.15. Since $A_n(1) \simeq A_n$ over k, the untwisted H^1 is free of rank $2g$ as well. Finally, a bounded complex with projective cohomology modules is the sum of its shifted cohomology modules in the derived category, so A_n-duality acts degree by degree. In this situation derived Poincaré duality, Theorem 10.12, applied to the constant sheaf, induces in degree one an isomorphism

$$H^1(C, A_n) \xrightarrow{\sim} \mathrm{Hom}_{A_n}(H^1(C, A_n(1)), A_n).$$

This map is the adjoint of cup product followed by the trace map of the duality theorem, and both cup product and trace commute with reduction of coefficients and with the action of $\mathrm{Gal}(k/k_0)$. Because of the twist on the second factor, Frobenius acts trivially on the target A_n.

In the limit, the finite-level dualities combine into a duality with the Tate module. Since the transition maps $[\ell]$ are surjective, we have $J[\ell^n](k) = T_\ell J/\ell^n T_\ell J$. Moreover $T_\ell J$ is finite free by Proposition B.14, and a homomorphism of finite free $\mathbf{Z}_\ell$-modules is determined by its reductions modulo all ℓ^n, and every compatible system of reductions lifts. Hence the inverse limit of the adjoints at finite level is the dual of $T_\ell J$. We obtain a perfect pairing

$$\varprojlim_n H^1(C, A_n) \times T_\ell J \longrightarrow \mathbf{Z}_\ell,$$

invariant under Frobenius. Since geometric Frobenius acts on $T_\ell J$ by B^{-1}, its matrix on $\varprojlim_n H^1(C, A_n)$ in the dual basis is $(B^{-1})^{-t} = B^t$. ∎

For the elliptic curve $y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, arithmetic Frobenius B has eigenvalues $\pm i\sqrt{3}$ on $T_\ell J$. By the proposition, geometric Frobenius acts on $\varprojlim_n H^1(C, A_n)$ with the same eigenvalues. Since $\det(B^t - 1) = 4 \neq 0$, it has no fixed vector. The next proposition proves the same for every smooth curve, including open ones.

Proposition 14.3 (No fixed class in degree one)**.** *For a smooth geometrically connected curve U_0/k_0, geometric Frobenius has no fixed vector on $H^1(U, E)$ for every finite $E/\mathbf{Q}_\ell$. More precisely,*

$$\left(\varprojlim_n H^1(U, \mathbf{Z}/\ell^n)\right)^{F=1} = 0.$$

Proof. On the compactification the vanishing holds because $J_0(k_0)$ is finite, and on the boundary it holds because Frobenius acts by a power of q, which is not 1. The residue sequence reduces the open case to the proper one, and at the end we pass from integral to rational coefficients.

The proper case. Let $U_0 = C_0$ be proper, with Jacobian J_0. Here the cohomology is the dual of the Tate module, and the question is whether 1 is an eigenvalue of Frobenius on it. By Proposition B.15, arithmetic Frobenius has no fixed vector on $T_\ell J$; indeed, a fixed compatible tuple would consist of ℓ-power torsion points of the finite group $J_0(k_0)$. This means that $B - 1$ is invertible on $V_\ell J$. On the other side, geometric Frobenius acts on the untwisted H^1 by B^t (Proposition 14.2). Since $\det(B^t - 1) = \det(B - 1) \neq 0$, it fixes neither a nonzero rational vector nor a nonzero integral compatible tuple. The argument applies to every power of Frobenius, so the same holds over every finite extension of k_0, which we need in the open case.

The open case. A fixed class has zero residues at the boundary points, so it extends to the compactification, where the proper case shows that it is zero. To make this precise, choose $j : U \hookrightarrow C$ as in Lemma 14.1, with boundary $i : D \hookrightarrow C$. At each finite level there is then a canonical residue sequence

$$0 \longrightarrow H^1(C, A_n) \longrightarrow H^1(U, A_n) \longrightarrow H^0(D, A_n(-1)).$$

To obtain it with the correct twist and Frobenius action, we use the dualizing complexes. By transitivity, $i^!\omega_C = \omega_D = A_n$. Since $\omega_C = A_n(1)[2]$, it follows that $i^!A_{n,C} = A_{n,D}(-1)[-2]$. The localization triangle $i_*i^!A_{n,C} \to A_{n,C} \to Rj_*A_{n,U}$ now yields the displayed sequence, and all its maps are natural in the coefficients and under automorphisms of the model.

Choose $m > 0$ such that F^m fixes all geometric boundary points, and put $Q = q^m$. On $\varprojlim_n H^0(D, A_n(-1))$, which is finite free over $\mathbf{Z}_\ell$, the operator F^m acts as Q. Since $Q - 1 \neq 0$, it has no nonzero fixed element. A fixed compatible class on U therefore has zero residues at every level, so by exactness it lifts uniquely to $H^1(C, A_n)$. By uniqueness the lifts are compatible and fixed by F^m, and the proper case over $\mathbf{F}_Q$ shows that they vanish.

Rational coefficients. To pass from compatible systems of classes to $H^1(U, E)$, we identify $H^1(U, \mathbf{Q}_\ell)$ with the rationalization of the inverse limit and check that rationalizing does not create fixed vectors. By Proposition 2.55, applied to the constant sheaves A_n, we have $R\Gamma(U, \mathbf{Z}_\ell) = R\varprojlim_n R\Gamma(U, A_n)$, and hence an exact sequence

$$0 \to \varprojlim{}^1_n H^0(U, A_n) \to H^1(U, \mathbf{Z}_\ell) \to \varprojlim_n H^1(U, A_n) \to 0.$$

The $\varprojlim^1$-term vanishes because the groups $H^0(U, A_n) = A_n$ have surjective transition maps. On $U_{\text{pro\'et}}$ the sheaf $\mathbf{Q}_\ell$ is $\mathbf{Z}_\ell[1/\ell]$ (Lemma 2.51), a filtered colimit of copies of $\mathbf{Z}_\ell$, and cohomology on the quasi-compact and quasi-separated scheme U commutes with filtered colimits of sheaves (Corollary 2.29, applied to sheaves placed in degree zero). So $H^1(U, \mathbf{Q}_\ell) = H^1(U, \mathbf{Z}_\ell) \otimes \mathbf{Q}_\ell$ is the rationalization of the inverse limit.

The inverse limit of the residue sequences is exact on the left, and rationalization is exact, so we obtain an exact sequence of finite-dimensional $\mathbf{Q}_\ell$-spaces $0 \to H^1(C, \mathbf{Q}_\ell) \to H^1(U, \mathbf{Q}_\ell) \to I \to 0$, where I is the image of the residue map inside the rationalized boundary term. By the proper case, $F^m - 1$ is invertible on $H^1(C, \mathbf{Q}_\ell)$. On the boundary term, $F^m - 1$ acts by $Q - 1$, so it is injective on the F^m-stable subspace I. Now let v be a vector of $H^1(U, \mathbf{Q}_\ell)$ killed by $F^m - 1$. Its image in I is killed by $F^m - 1$, hence zero. Thus v comes from $H^1(C, \mathbf{Q}_\ell)$, where it is zero by the proper case. Since a vector fixed by F is fixed by F^m, it follows that Frobenius has no fixed rational vector, and extending scalars to E gives the assertion over E. ∎

14.1.2 Finite monodromy

With Proposition 14.3 in hand, the logarithm argument sketched at the start of Section 14.1 proves Theorem 14.5. Corollaries Corollary 14.6 and Corollary 14.8 then translate finiteness of monodromy into weights. For a lisse sheaf of higher rank, however, the determinant weight (Definition 14.7) is the only weight we can attach until Rankin's argument. Throughout, U_0 is a smooth geometrically connected curve over k_0 and $s \in W(U_0)$ is a lift of degree one, as in Convention `situation-weil`. In this setting, the first step identifies the additive characters to which the logarithm leads.

Lemma 14.4 (Invariant additive characters)**.** *Every continuous additive homomorphism $a : \pi_1(U) \to E$ satisfying $a(s^{-1}gs) = a(g)$ is zero.*

Proof. After scaling, the coordinates of a reduce modulo ℓ^n to a compatible system of Frobenius-fixed classes in $H^1(U, \mathbf{Z}/\ell^n)$, which vanish by Proposition 14.3. This needs H^1 to be identified with homomorphisms.

Recall that for a finite abelian group G the group $H^1(U, G)$ classifies G-torsors. By [Stacks, 0DV5], finite locally constant sheaves on the connected scheme U correspond to finite continuous $\pi_1(U)$-sets, and under this correspondence a G-torsor becomes a finite $\pi_1(U)$-set with a commuting simply transitive action of G, that is, a continuous homomorphism $\pi_1(U) \to G$. Hence $H^1(U, G) = \mathrm{Hom}_{\mathrm{cont}}(\pi_1(U), G)$. In this description Frobenius acts on $H^1(U, G)$ through the action of s on $\pi_1(U)$ by conjugation. The Frobenius-fixed classes are therefore the homomorphisms b with $b(s^{-1}gs) = b(g)$.

Choose a $\mathbf{Q}_\ell$-basis of E, with coordinate maps t_i. Since $\pi_1(U)$ is compact, each $t_i a(\pi_1(U))$ is bounded, and after multiplying all coordinates by one power of ℓ we may assume that they take values in $\mathbf{Z}_\ell$. Each coordinate is then a continuous homomorphism $\pi_1(U) \to \mathbf{Z}_\ell$ invariant under conjugation by s. Its reductions modulo ℓ^n are compatible classes in $H^1(U, \mathbf{Z}/\ell^n)$ fixed by Frobenius. By the proposition they vanish, and hence every coordinate of a is zero. ∎

The invariance hypothesis of the lemma is automatic for characters of the Weil group. If $\chi : W(U_0) \to E^\times$ is a character, then $\chi(s^{-1}gs) = \chi(s)^{-1}\chi(g)\chi(s) = \chi(g)$, because $E^\times$ is commutative. So the lemma applies as soon as χ has been made additive on $\pi_1(U)$.

Theorem 14.5 (Rank-one sheaves have finite geometric monodromy)**.** *Let U_0/k_0 be a smooth geometrically connected curve, and let $\chi : W(U_0) \to E^\times$ be the character of a rank-one lisse Weil sheaf. Then $\chi(\pi_1(U))$ is finite.*

Proof. We show that a fixed power of χ is trivial on $\pi_1(U)$, by applying Lemma 14.4 to its logarithm. For this we need a subgroup $1+\mathfrak{m}_E^r$ of finite index in $\mathcal{O}_E^\times$ on which the logarithm is an injective homomorphism, and a power of χ that lands in it.

Step 1: the logarithm. The compact group $\chi(\pi_1(U))$ lies in $\mathcal{O}_E^\times$, because its image under the normalized valuation is a finite subgroup of $\mathbf{Z}$ and therefore zero. Now write $e = v_E(\ell)$ and choose an integer $r > e/(\ell-1)$. We claim that log is injective on $1+\mathfrak{m}_E^r$. For $x \in \mathfrak{m}_E^r$ and $n \geq 1$ we have $v_E(n) = e\, v_\ell(n)$ and, by Legendre's formula, $v_\ell(n!) \leq (n-1)/(\ell-1)$, so that

$$v_E(x^n/n) \geq nr - e\, v_\ell(n), \qquad v_E(x^n/n!) \geq r + (n-1)(r - e/(\ell-1)).$$

Both bounds tend to infinity with n and are at least r: for the second this is clear, and for the first it follows from $n-1 \geq \ell^{v_\ell(n)} - 1 \geq (\ell-1)v_\ell(n)$ and $(\ell-1)r > e$, which give $(n-1)r \geq e\, v_\ell(n)$. The logarithm and exponential series therefore converge on $\mathfrak{m}_E^r$; the logarithm maps $1+\mathfrak{m}_E^r$ into $\mathfrak{m}_E^r$, and the exponential maps $\mathfrak{m}_E^r$ into $1+\mathfrak{m}_E^r$. By the usual identities of formal power series, $\log : 1+\mathfrak{m}_E^r \to \mathfrak{m}_E^r$ and exp are mutually inverse continuous group homomorphisms.

Step 2: a power of χ. The quotient $\mathcal{O}_E^\times/(1+\mathfrak{m}_E^r)$ is finite, being isomorphic to $(\mathcal{O}_E/\mathfrak{m}_E^r)^\times$; let $N > 0$ be an integer that kills it. As $\chi(\pi_1(U)) \subset \mathcal{O}_E^\times$, we have $\chi(g)^N \in 1+\mathfrak{m}_E^r$ for $g \in \pi_1(U)$. Hence $a(g) = \log(\chi(g)^N)$ defines a continuous homomorphism $\pi_1(U) \to E$, which is invariant under conjugation by s because χ takes values in an abelian group.

Step 3: conclusion. Lemma 14.4 gives $a = 0$, so $\chi(g)^N = 1$ for every $g \in \pi_1(U)$, the logarithm being injective. The theorem follows, since $T^N - 1$ has only finitely many roots in E. ∎

Finiteness gives the sheaf a weight, and since the lift s is a choice, part of the proof checks that the weight does not depend on it.

Corollary 14.6 (Rank-one weights)**.** *Every rank-one lisse Weil sheaf on a smooth geometrically connected curve is punctually ι-pure. For $c = \chi(s)$ its weight is $\lambda_\iota(c) = 2\log_q|\iota(c)|$.*

Proof. We separate from χ the part that depends only on the degree. Put $\varepsilon(w) = \chi(w)c^{-\deg w}$, a character of $W(U_0)$ that is trivial on s. Every element of $W(U_0)$ can be written as $w = gs^d$ with $g \in \pi_1(U)$ and $d = \deg w$, and then $\varepsilon(w) = \chi(g)\chi(s)^d c^{-d} = \chi(g)$. So ε has the same image as $\chi|_{\pi_1(U)}$, which is finite by Theorem 14.5. At a closed point x the eigenvalue is $\varepsilon(\mathrm{Frob}_x)c^{\deg(x)}$, where the first factor has finite order, hence complex absolute value one. Therefore the eigenvalue has absolute value $q_x^{\lambda_\iota(c)/2}$. This number does not depend on s, because changing s multiplies c by a value of χ on $\pi_1(U)$, which has finite order. ∎

For a Kummer sheaf on $\mathbf{G}_{m,k_0}$ the corollary gives weight 0, since $\chi(s)$ is then a root of unity. For $\mathcal{B}_b$ it gives $\lambda_\iota(b)$, the weight found in Proposition 13.7.

Before purity is known, the eigenvalues of a lisse sheaf $\mathcal{V}$ of rank r at a closed point are not controlled individually. Their product, however, is controlled, since it is the eigenvalue of the rank-one sheaf $\det\mathcal{V}$, whose weight is given by Corollary 14.6. Dividing by r we get the average weight of the eigenvalues, which is the only candidate for a weight of $\mathcal{V}$. Indeed, if $\mathcal{V}$ is pure of weight w, then $\det\mathcal{V}$ is pure of weight rw, and the number defined below is w. For a sheaf that is not pure, on the other hand, the number is only an average. For example, on $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ the determinant is $\overline{\mathbf{Q}}_\ell(-1)$, of weight 2, and the number is 1, the weight of no eigenvalue. For this reason we apply it to irreducible constituents: Proposition 14.25 bounds the eigenvalues of a real sheaf by the largest determinant weight of its constituents, and Theorem 14.26 then shows that each irreducible constituent is pure of weight equal to its determinant weight.

Definition 14.7 (Determinant weight)**.** Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve U_0, let ρ be the corresponding representation of $W(U_0)$, and let $s \in W(U_0)$ be a lift of degree one. The *determinant weight* of $\mathcal{V}$ is

$$\mathrm{dw}_\iota(\mathcal{V}) = \frac{\lambda_\iota(\det \rho(s))}{r}$$

(this does not depend on s, Corollary 14.8). For a lisse Weil sheaf $\mathcal{V}$ we write $\mathrm{DW}(\mathcal{V})$ for the set of determinant weights of its irreducible constituents.

Corollary 14.8 (Determinant weights)**.** *Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve. Its determinant has finite geometric monodromy and is punctually pure of weight $r\,\mathrm{dw}_\iota(\mathcal{V})$. The number $\mathrm{dw}_\iota(\mathcal{V})$ does not depend on the choice of s or of a finite coefficient field over which $\mathcal{V}$ is defined.*

Proof. The sheaf $\det \mathcal{V}$ is a rank-one lisse Weil sheaf defined over a finite extension of E, so Theorem 14.5 and Corollary 14.6 apply to it. By the formula of Corollary 14.6, with $c = \det \rho(s)$, the weight of $\det \mathcal{V}$ is $r\,\mathrm{dw}_\iota(\mathcal{V})$, and it does not depend on s. Finally, a finite extension of scalars changes neither the character nor its absolute value under the fixed embedding. ∎

The corollary applies to every irreducible constituent of $\mathcal{V}$ and of every tensor or exterior power of $\mathcal{V}$. Indeed, a finite-dimensional representation has a composition series, and each factor is again lisse and defined over a finite extension of $\mathbf{Q}_\ell$, because after enlarging E the invariant subspaces and quotients have bases defined over E.

Remark 14.9 (Determinant weight of a sheaf with pure constituents)**.** Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve. If the irreducible constituents $\mathcal{V}_1, \dots, \mathcal{V}_n$ of $\mathcal{V}$ are pure of weights $w_1, \dots, w_n$, then $\mathrm{dw}_\iota(\mathcal{V}) = \sum_a \mathrm{rank}(\mathcal{V}_a) w_a / r$. In particular a sheaf pure of weight w has determinant weight w.

14.2 Determinant weights

Before purity is known, the only weight we can attach to an irreducible lisse Weil sheaf Q on a curve is its determinant weight $\mathrm{dw}_\iota(Q)$, the weight of $\det Q$ divided by the rank (Definition 14.7 and Corollary 14.8). This section proves the two facts about determinant weights that Rankin's argument in Section 14.3 needs for the tensor and exterior powers of a real sheaf $\mathcal{G}$. First, every simple factor of $\mathcal{G}^{\otimes 2m}$ has determinant weight at most $2m$ times the largest determinant weight of the constituents of $\mathcal{G}$, and this bounds the weights of H^2_c. Second, the determinant weights of the simple factors of $\bigwedge^r \mathcal{G}$ are known exactly, and this is what lets exterior powers separate the constituents.

The difficulty is that a determinant only sees an average. For irreducible A and B of ranks a and b, we have $\det(A \otimes B) = (\det A)^{\otimes b} \otimes (\det B)^{\otimes a}$. Hence the determinant weight of $A \otimes B$ as a whole is $\mathrm{dw}_\iota(A) + \mathrm{dw}_\iota(B)$. But $A \otimes B$ is usually reducible, and which subquotients occur depends on the monodromy. The rank-weighted average of the determinant weights of its simple factors is the number above, but nothing so far prevents one factor from lying above it and another below. If the factors were pure, the determinant weight of each would be its weight, but purity is what we are trying to prove.

The model to keep in mind is a central character. If a group acts on V and some central element z acts on V by the scalar c, then z acts on every subquotient of $V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ by c^{a-b}, without

any knowledge of how these representations decompose. We therefore look for a central element of positive degree in the monodromy group; its scalars will record weights.

Let $H \subseteq G$ be the Zariski closures in $\mathrm{GL}(V)$ of the images of $\pi_1(U)$ and of $W(U_0)$, the geometric and arithmetic monodromy groups. If H were trivial, the image $\rho(s)$ of a degree-one lift would itself be central. In general we use the structure of the identity component,

$$G^\circ = Z(G^\circ)^\circ H^\circ$$

for V semisimple (Lemma 14.15). The reason is that the Lie algebra of G° is reductive, hence the sum of its center and its derived algebra, and since G/H is abelian, the derived algebra lies in the Lie algebra of H. Accordingly, in Proposition 14.17 we write a power of $\rho(s)$ as a central torus element times an element of H, and then we take the product of its conjugates over G/G°. The result is an element $z = h\rho(s)^N$, central in G, with $h \in H$ and $N > 0$.

On a simple factor Q of any tensor construction, z acts by a scalar c_Q, and taking determinants gives $c_Q^{\dim Q} = \det(h \mid Q)\det(s \mid Q)^N$. Since determinants have finite geometric monodromy, $\det(h \mid Q)$ is a root of unity, and from this Lemma 14.18 concludes that $\mathrm{dw}_\iota(Q) = \lambda_\iota(c_Q)/N$, where $\lambda_\iota(a) = 2\log_q|\iota(a)|$ is the additive function of Proposition 13.7. Since scalars multiply under tensor products, we obtain Proposition 14.19.

The facts on algebraic groups that we use, namely the relation between closed subgroups, their identity components and their Lie algebras, and the structure of a Lie algebra with a faithful semisimple representation, are proved in Appendix F.

14.2.1 Representations and filtrations

We treat lisse sheaves as representations of the Weil group. From the geometry of the curve we use only one fact, proved in Section 14.1: the geometric fundamental group acts through a finite group on every determinant. Everything else is linear algebra and the structure of algebraic groups.

Throughout, U_0 is a smooth geometrically connected curve over k_0 and $s \in W(U_0)$ is a lift of degree one, as in Convention `situation-weil`, so that $W(U_0) = \pi_1(U) \rtimes \langle s \rangle$.

Definition 14.10 (Weil representation)**.** A *Weil representation* is a finite-dimensional $\overline{\mathbf{Q}}_\ell$-representation of $W(U_0)$ whose restriction to $\pi_1(U)$ is continuous.

By Convention `situation-weil` the Weil representations are the lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaves on U_0, and so, by Corollary 14.8, $\pi_1(U)$ acts through a finite group on the determinant of every Weil representation. A Weil representation V is semisimple if it is a direct sum of simple representations, and the semisimplification V^{ss} of V is the direct sum of its composition factors.

Two elementary facts are needed. First, semisimplicity is preserved by restriction to a normal subgroup, which lets us apply Lie theory to the identity component of the monodromy group below. Second, the composition factors of tensor and exterior powers are computed from those of the given representations, so it suffices to prove the formulas for semisimple representations.

Lemma 14.11 (Restriction to a normal subgroup)**.** *Let N be a normal subgroup of a group Γ. The restriction to N of a finite-dimensional semisimple $\overline{\mathbf{Q}}_\ell$-representation of Γ is semisimple.*

Proof. It suffices to consider a simple Γ-module V. A nonzero N-submodule A of least dimension is simple, and since N is normal, each translate gA is again a simple N-submodule. The sum of

all these translates is a nonzero Γ-submodule, hence equals V. It remains to see that a finite sum of simple submodules is semisimple. Indeed, the intersection of a simple submodule with the sum of those already chosen is either zero or the whole submodule, so we add it in the first case and omit it in the second, and after finitely many steps we obtain a direct-sum decomposition. ∎

The second fact concerns representations that are not semisimple. A nonsplit extension of $\overline{\mathbf{Q}}_\ell(-1)$ by $\overline{\mathbf{Q}}_\ell$, such as the twist $\mathcal{E}(-1)$ of the Kummer extension of Example 13.4, does not decompose. Yet the composition factors of its tensor and exterior powers are determined by those of the sheaf, as the next lemma shows by filtering. This is what lets us pass to semisimplifications in Proposition 14.19.

Lemma 14.12 (Tensor and exterior powers of filtered representations)**.** *Suppose V and V' have composition factors $A_1, \dots, A_t$ and $B_1, \dots, B_u$. There is a $W(U_0)$-stable filtration of $V \otimes V'$ with graded pieces $A_a \otimes B_b$, one for each pair (a, b). There is a $W(U_0)$-stable filtration of $\bigwedge^r V$ with graded pieces*

$$\bigotimes_{a=1}^{t} \bigwedge^{n_a} A_a, \qquad 0 \le n_a \le \dim A_a, \qquad \sum_a n_a = r.$$

The simple factors of $V \otimes V'$ and $\bigwedge^r V$, counted with multiplicity, are those of the respective graded pieces. Every graded piece of $\bigwedge^r V$ displayed above is nonzero.

Proof. We construct the two filtrations and then compare simple factors by the Jordan–Hölder argument. The tensor product is filtered by the filtration on V, and each quotient $A_a \otimes V'$ by the filtration on V'; since tensoring over a field is exact, the quotients are as asserted.

For the exterior power we cannot simply filter factor by factor, because a wedge product may use several vectors from the same composition factor. Instead we sort wedges of basis vectors by how many vectors they take from each factor, after lifting bases of the A_a to a basis of V adapted to its filtration. Thus a wedge of basis vectors determines a tuple (n_a), by counting how many vectors it uses from each lifted basis. Since the basis is adapted, the group $W(U_0)$ carries a lifted basis vector of A_a into the span of the lifted bases of $A_1, \dots, A_a$. Hence the action is block upper triangular: each term of the image of a basis wedge either has the same tuple or has a tuple obtained from it by moving some of its vectors to smaller indices a. Choose a total order on the tuples compatible with these moves. Then the spans of the initial segments of basis wedges form a stable filtration, and on the quotient indexed by (n_a) the diagonal blocks act as on $\bigotimes_a \bigwedge^{n_a} A_a$, a space of dimension $\prod_a \binom{\dim A_a}{n_a} > 0$. For example, for V an extension of A_2 by A_1, both of dimension one, and $r = 1$, the tuples are $(1, 0)$ and $(0, 1)$, and the filtration is the given one. For $r = 2$ there is the single tuple $(1, 1)$, and $\bigwedge^2 V = A_1 \otimes A_2$.

The graded pieces of the filtrations just built need not be simple, so to compare simple factors we refine. By the Schreier refinement theorem, any two finite filtrations have refinements with isomorphic graded pieces, matched by the Zassenhaus isomorphism $(V_{i-1} + V_i \cap W_j)/(V_{i-1} + V_i \cap W_{j-1}) \simeq (W_{j-1} + W_j \cap V_i)/(W_{j-1} + W_j \cap V_{i-1})$; for two composition series this is the Jordan–Hölder theorem. Applying the same refinement to a composition series and to the tensor or exterior filtration, we obtain the statement about simple factors. In particular, these multisets do not change when V and V' are replaced by their semisimplifications. ∎

14.2.2 Central elements

In the arithmetic monodromy group we construct a central element of positive degree (Proposition 14.17). It acts by a scalar on each simple factor, and this scalar determines the determinant

weight of the factor. As explained above, we construct the element for semisimple representations, and then use stable filtrations to obtain the formulas for arbitrary representations.

The degree must be positive. Indeed, an element of H that is central in G also acts by a scalar on each simple factor Q. But, as for every element of H, its determinant on Q is a root of unity, so under ι the scalar has absolute value one, and it records no weight. By contrast, an element $z = h\rho(s)^N$ with $N > 0$ contains N factors of Frobenius, and its scalar c_Q on Q satisfies $\lambda_\iota(c_Q) = N \deg_\iota(Q)$ (Lemma 14.18).

We produce the central element by Lie theory, which applies to algebraic groups and not to the abstract images of $\pi_1(U)$ and $W(U_0)$. For $\mathcal{B}_b$ with b not a root of unity, the image of $W(U_0)$ in GL_1 is the infinite cyclic group generated by b, which is not an algebraic subgroup; its Zariski closure is all of GL_1, since the proper closed subgroups of $\mathbf{G}_m$ are finite. Passing to the closure loses nothing that we need. Indeed, stabilizers are closed, so a subspace of V is stable under a group of matrices if and only if it is stable under its Zariski closure. For this reason Lemma 14.15 and Proposition 14.17 are statements about these closures.

Definition 14.13 (Monodromy groups)**.** Let $\rho : W(U_0) \to \mathrm{GL}(V)$ be a Weil representation. Its *geometric* and *arithmetic monodromy groups* are the Zariski closures H of $\rho(\pi_1(U))$ and G of $\rho(W(U_0))$ in $\mathrm{GL}(V)$.

We keep the letters H and G for these two groups in the rest of this section.

Lemma 14.14 (Normality of geometric monodromy)**.** *The groups H and G are reduced closed algebraic subgroups of $\mathrm{GL}(V)$, and $H \triangleleft G$. Moreover every commutator of two elements of G lies in H.*

Proof. Each assertion holds for the images of $\pi_1(U)$ and $W(U_0)$, and we pass to Zariski closures by continuity. First, multiplication and inversion are continuous for the Zariski topology, so the closure of a subgroup is a subgroup. Next, conjugation by $\rho(W(U_0))$ preserves H because $\pi_1(U)$ is normal in $W(U_0)$, and since the normalizer of a closed subgroup is closed, conjugation by G preserves H as well. Finally, since $W(U_0)/\pi_1(U) \simeq \mathbf{Z}$ is abelian, commutators of elements of $\rho(W(U_0))$ lie in $\rho(\pi_1(U))$. As the commutator map $G \times G \to \mathrm{GL}(V)$ is continuous, the same holds on the closures. ∎

For example, for $V = \mathcal{B}_b$ of Proposition 13.7 the group H is trivial. Hence $\rho(s) = b$ is central in G, and the determinant weight of $\mathcal{B}_b$ is $\lambda_\iota(b)$.

Lemma 14.15 (The identity component)**.** *If V is a semisimple $W(U_0)$-representation, then*

$$G^\circ = Z(G^\circ)^\circ H^\circ.$$

Proof. A connected algebraic group is determined by its Lie algebra among its closed connected subgroups, so it suffices to show that the Lie algebras of $Z(G^\circ)^\circ$ and H° together span that of G°. We show that this Lie algebra is reductive, hence the sum of its center and its derived algebra, and that the derived algebra lies in the Lie algebra of H because commutators lie in H.

Reductivity. Write $\mathfrak{g} = \mathrm{Lie}(G^\circ)$. Then

$$\mathfrak{g} = Z(\mathfrak{g}) + [\mathfrak{g}, \mathfrak{g}],$$

by Proposition F.8, once we know that V is a faithful semisimple $\mathfrak{g}$-module. The $\rho(W(U_0))$-stable subspaces are precisely the G-stable subspaces, because the stabilizer of a subspace is closed.

Hence V is semisimple as a $G(\overline{\mathbf{Q}}_\ell)$-module, and by Lemma 14.11 it is also semisimple as a $G^\circ(\overline{\mathbf{Q}}_\ell)$-module. Then Lemma F.3 makes it a faithful semisimple $\mathfrak{g}$-module.

The center and the derived algebra. Put $T = Z(G^\circ)^\circ$ and $\mathfrak{h} = \mathrm{Lie}(H^\circ)$. The comparison between groups and Lie algebras in Lemma F.3 gives $\mathrm{Lie}(T) = Z(\mathfrak{g})$. Moreover $[\mathfrak{g},\mathfrak{g}] \subseteq \mathfrak{h}$, because the commutator morphism $G \times G \to G$ factors through the closed subgroup H: it does so on $\overline{\mathbf{Q}}_\ell$-points by Lemma 14.14, and $G \times G$ is reduced. To deduce the inclusion, let $X, Y \in \mathfrak{g}$ and put $R = \overline{\mathbf{Q}}_\ell[\varepsilon,\eta]/(\varepsilon^2,\eta^2)$. The commutator of $1+\varepsilon X$ and $1+\eta Y$ in $G(R)$ is $1+\varepsilon\eta[X,Y]$, and it lies in $H(R)$. Here the subring $\overline{\mathbf{Q}}_\ell[\varepsilon\eta] \subset R$ is a ring of dual numbers, and the equations of H vanish on $1+\varepsilon\eta[X,Y]$ already over this subring. Hence $[X,Y] \in \mathrm{Lie}(H) = \mathfrak{h}$, and

$$\mathfrak{g} = \mathrm{Lie}(T) + \mathfrak{h}.$$

Back to groups. The multiplication $T \times H^\circ \to G^\circ$ is a homomorphism, because T is central. By Lemma F.2 its image is a closed connected subgroup, whose Lie algebra contains both summands on the right, hence equals that of G°. By the same lemma the image is therefore G°. ∎

Rankin's argument needs the determinant weights of the simple factors of $\mathcal{G}^{\otimes 2m}$, of the exterior powers of $\mathcal{G}$ and of the dual, so we give a name to the representations built in this way; the monodromy group acts on all of them.

Definition 14.16 (Tensor construction)**.** A *tensor construction* on a Weil representation V is a representation obtained from V and $V^\vee$ by finitely many tensor products, direct sums, exterior powers and subquotients by $W(U_0)$-stable subspaces.

Since the stabilizer of a subspace is closed, a $W(U_0)$-stable subspace is G-stable, and so G acts on every tensor construction through its action on V.

Proposition 14.17 (A central element of the arithmetic monodromy group)**.** *Let $V_1,\dots,V_n$ be semisimple Weil representations and form the monodromy groups $H \subseteq G$ of $V = \bigoplus_i V_i$, and let $s \in W(U_0)$ have degree one. There are an integer $N > 0$ and $h \in H(\overline{\mathbf{Q}}_\ell)$ such that*

$$z = h\,\rho(s)^N \in Z(G)(\overline{\mathbf{Q}}_\ell).$$

The action of z on each tensor construction on V preserves every $W(U_0)$-stable subquotient and is scalar on each simple subquotient.

Proof. We write a power of $\rho(s)$ as a central torus element times an element of H, by Lemma 14.15, and average the torus part over the components of G so that it becomes central in all of G. The factors from H that accumulate in this process are collected in h; their determinants are roots of unity, so they do not affect determinant weights (see Lemma 14.18 below). Note that the direct sum V of the V_i is again semisimple, so Lemma 14.15 applies to it.

Since the component group of G is finite (Lemma F.2), we may choose $m > 0$ with $\rho(s)^m \in G^\circ$. By Lemma 14.15, we can write $\rho(s)^m = ct$ with $c \in Z(G^\circ)^\circ$ and $t \in H^\circ$. Choose representatives $g_1,\dots,g_b$ of the components of G and set

$$z = \prod_{j=1}^{b} g_j c g_j^{-1}, \qquad N = mb.$$

The element c is central in G° but perhaps not in G, since the other components may conjugate it nontrivially. The product z of its conjugates, one for each component, is however central

in G. Indeed, the factors lie in the abelian group $Z(G^\circ)^\circ$, and since replacing g_j by $g_j u$ with $u \in G^\circ$ does not change the corresponding factor, conjugation by an element of G permutes the factors. Next, every commutator lies in H, so each factor has the same image as c in G/H. Since $c \in H\rho(s)^m$, we get $z \in H\rho(s)^{mb}$, which gives h and $N = mb > 0$.

Being central in G, the element z commutes with the action of $W(U_0)$ on tensor constructions. Moreover, as noted after the definition of tensor constructions, every $W(U_0)$-stable subspace of a tensor construction is G-stable. Hence z acts on the $W(U_0)$-stable subquotients and commutes with the action of $W(U_0)$ on them. On a simple subquotient, therefore, an eigenspace of z is nonzero and $W(U_0)$-stable, hence is the whole space; in other words, z acts by a scalar. ∎

14.2.3 Tensor operations

The central element acts on each simple factor of a tensor construction by a scalar, and from these scalars we read off determinant weights. Since Weil representations are the lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaves, Definition 14.7 applies to them.

Lemma 14.18 (Determinant weight from a central scalar)**.** *In the setting of Proposition 14.17, let Q be a nonzero subquotient of a tensor construction on V on which the element z acts by a scalar c_Q. Then*

$$\mathrm{dw}_\iota(Q) = \frac{\lambda_\iota(c_Q)}{N}.$$

Proof. The determinant of Q is an algebraic character of H, and its image on the dense subgroup $\rho(\pi_1(U))$ is finite, and hence its image on all of H lies in the same finite group of roots of unity. Taking determinants in $z = h\rho(s)^N$, we find

$$c_Q^{\dim Q} = \det(h \mid Q)\det(s \mid Q)^N.$$

Now apply λ_ι, which vanishes on the first factor, and divide by $N \dim Q$. ∎

The lemma reduces determinant weights of tensor constructions to a computation with the scalars on the composition factors of V. By Proposition 14.17, the element z acts by a scalar on every simple subquotient of a tensor construction on V, so the formula of the lemma applies to each of them. Since the scalars by which a single element acts multiply under tensor products and exterior powers, this gives the following formulas.

Proposition 14.19 (Determinant weights of tensor operations)**.** *Let V and V' be nonzero Weil representations, let $A_1, \dots, A_t$ be the composition factors of V, and put $\gamma_a = \mathrm{dw}_\iota(A_a)$.*

1. $\mathrm{DW}(V \otimes V') = \mathrm{DW}(V) + \mathrm{DW}(V')$.
2. $\mathrm{DW}(\bigwedge^r V) = \{\sum_a n_a\gamma_a : 0 \le n_a \le \dim A_a,\ \sum_a n_a = r\}$ *for* $0 \le r \le \dim V$.
3. $\mathrm{DW}(V^\vee) = -\mathrm{DW}(V)$.

Proof. We use one central element for V and V' simultaneously, so that its scalars on the factors of $V \otimes V'$ are products of its scalars on the factors of V and of V'. Since Proposition 14.17 requires semisimple representations, we apply it to $V^{\mathrm{ss}} \oplus V'^{\mathrm{ss}}$. This changes nothing, because by Lemma 14.12 the simple factors of $V \otimes V'$ and $\bigwedge^r V$ depend only on the composition factors of V and V'. Now let c_a and c'_b be the scalars by which its central element of degree N acts on the composition factors A_a of V and A'_b of V'. By Lemma 14.18, $\gamma_a = \lambda_\iota(c_a)/N$ and $\mathrm{dw}_\iota(A'_b) = \lambda_\iota(c'_b)/N$.

(1) The central element acts by $c_a c'_b$ on $A_a \otimes A'_b$, and hence on every simple factor of it. By the same lemma, each such factor has determinant weight $\gamma_a + \mathrm{dw}_\iota(A'_b)$. Moreover, each $A_a \otimes A'_b$ is nonzero, and Lemma 14.12 identifies the simple factors of these tensor products with those of $V \otimes V'$.

(2) The central element acts on $\bigotimes_a \bigwedge^{n_a} A_a$ by $\prod_a c_a^{n_a}$. So each simple factor of this nonzero representation has determinant weight $\sum_a n_a \gamma_a$, and Lemma 14.12 identifies these factors with those of $\bigwedge^r V$.

(3) We have $\det(s \mid A^\vee) = \det(s \mid A)^{-1}$, and duality carries a composition series of V to one of $V^\vee$ whose factors are the duals $A_a^\vee$. ∎

In particular every simple factor of $V^{\otimes m}$ has determinant weight at most $m \max \mathrm{DW}(V)$. In Section 14.4 we use this bound, and part (2), for the constituents of a real sheaf before we know that they are pure.

14.3 Rankin's bound

We bound the local eigenvalues of a real sheaf by the eigenvalues of Frobenius on H_c^2 of its even tensor powers (Proposition 14.25). Only H_c^2 is needed, and Lemma 14.20 computes it by Poincaré duality as the coinvariants of the geometric monodromy.

Rankin's argument uses one fact from each side of the trace formula. On the cohomological side, an affine curve has $H_c^0 = 0$, so only H_c^2 can produce poles of the L-function, and we can compute this group without knowing anything about H_c^1: its eigenvalues are q times eigenvalues of Frobenius on a quotient of the generic fibre. On the side of the Euler product, if the local factors have nonnegative coefficients, then each of them is holomorphic wherever the product is.

For example, for the constant sheaf on $\mathbf{G}_m$ we have

$$L(\mathbf{G}_m, \overline{\mathbf{Q}}_\ell, t) = \prod_{x \in |\mathbf{G}_m|} (1 - t^{\deg(x)})^{-1} = \frac{1-t}{1-qt},$$

where $1 - t$ comes from H_c^1 and the pole at $t = q^{-1}$ from $H_c^2 = \overline{\mathbf{Q}}_\ell(-1)$. Each local factor has its poles on the circle $|t| = 1$, outside the disc $|t| < q^{-1}$, as Lemma 14.24 predicts. In general we apply the two facts to the even tensor powers of a real sheaf, whose H_c^2 is bounded by the determinant weights of Section 14.2, and this gives Proposition 14.25.

Lemma 14.20 (Degree zero and top cohomology)**.** *Let $U_0/\mathbf{F}_q$ be a smooth geometrically connected curve, let $\mathcal{F}$ be a lisse Weil sheaf on U_0, and let V be its geometric generic fibre, a representation of the geometric fundamental group $\pi_1(U)$. There are canonical Frobenius-equivariant identifications*

$$H^0(U, \mathcal{F}) = V^{\pi_1(U)}, \qquad H_c^2(U, \mathcal{F}) = V_{\pi_1(U)}(-1).$$

There is no compactly supported cohomology outside degrees $0, 1, 2$. If U_0 is affine, $H_c^0(U, \mathcal{F}) = 0$. If $\mathcal{F}$ is punctually of weights $\leq a$, the weights of H^0 are at most a and those of H_c^2 at most $a + 2$. If $\mathcal{F}$ is pure of weight a, the two groups, when nonzero, are pure of weights a and $a + 2$, respectively.

Proof. We deduce both formulas from the description of a lisse sheaf by its representation, using Poincaré duality to reduce H_c^2 to H^0, and then we read off the weights from the stalks.

A global section of a local system is determined by its value at $\bar{\eta}$, and a vector of the fibre extends to a global section exactly when it is fixed by the monodromy. This is the first formula. For the second, by degreewise Poincaré duality (Corollary 10.13) we have

$$H^2_c(U,\mathcal{F})^\vee = H^0(U,\mathcal{F}^\vee(1)) = (V^\vee)^{\pi_1(U)}(1).$$

Since the dual of $(V^\vee)^{\pi_1(U)}$ is $V_{\pi_1(U)}$, this gives the second formula, with its twist. Moreover, the identification is induced by the canonical duality pairing and is therefore Frobenius equivariant.

Duality also gives the vanishing outside degrees $0, 1, 2$. Indeed, for $i < 0$ there is nothing to prove, and for $i > 2$ the space $H^i_c(U,\mathcal{F})$ is dual to $H^{2-i}(U,\mathcal{F}^\vee(1))$, which is zero because $2 - i < 0$. For torsion coefficients the vanishing comes instead from the bound on cohomological dimension of Appendix D.

Suppose that U_0 is affine, and choose a smooth proper completion of U. Its boundary is nonempty, because a connected proper curve has only constant global functions and hence is not affine. A compactly supported section extends by zero to the completion, so it vanishes on a neighbourhood of the boundary. Since a section of a lisse sheaf on a connected curve is determined by its generic value, the section is zero, and $H^0_c(U,\mathcal{F}) = 0$.

For the weights we compare the global Frobenius on these two spaces with the local Frobenius at a closed point. Since $\pi_1(U)$ acts trivially on $V^{\pi_1(U)}$ and on $V_{\pi_1(U)}$, the action of $W(U_0)$ on them factors through the degree quotient. If β is an eigenvalue of the operator of degree one, then $\beta^{\deg(x)}$ is an eigenvalue of Frob_x on $\mathcal{F}_{\bar{x}}$ for every closed point x. Indeed, $V^{\pi_1(U)}$ is a subspace and $V_{\pi_1(U)}$ a quotient of $V \simeq \mathcal{F}_{\bar{x}}$, both stable under $\mathrm{Frob}_x \in \pi_1(U) s^{\deg(x)}$, which acts on them as $s^{\deg(x)}$. If $\mathcal{F}$ is punctually of weights $\leq a$, it follows that $|\iota(\beta)| \leq q^{a/2}$, and if $\mathcal{F}$ is pure of weight a, equality holds. Finally, the twist (-1) multiplies each global eigenvalue by q, which gives the stated weights for H^2_c. ∎

For the constant sheaf on $\mathbf{G}_m$ the lemma gives $H^2_c = \overline{\mathbf{Q}}_\ell(-1)$, of weight 2, as in the computation at the start of the section. Now take a Kummer sheaf of a nontrivial character on $\mathbf{G}_m$. Then $H^0_c = 0$ because $\mathbf{G}_m$ is affine, and $H^2_c = 0$ because the coinvariants vanish.

The positivity we want is $a^{2m} \geq 0$ for the traces a of all powers Frob_x^n on $\mathcal{G}$, and for this all these traces must be real. It is not enough to ask that the trace of Frob_x itself be real. For example, if the eigenvalues at x are i and $1 - i$, the trace is 1, but the trace of the square is $-1 + (1 - i)^2 = -1 - 2i$. By the Newton identities, on the other hand, all the power sums are real exactly when the characteristic polynomial is real, and this is the condition below. Its basic examples are the constant sheaf, its Tate twists and Kummer sheaves of quadratic characters. More generally, in Chapter 15, Proposition 15.9 makes every pure sheaf a summand of a real one after a twist. The condition enters the proof of Theorem 14.26 through Lemma 14.22.

Definition 14.21 (Real sheaves)**.** A Weil sheaf $\mathcal{G}$ is ι-real if $\iota \det(1 - T \,\mathrm{Frob}_x \mid \mathcal{G}_{\bar{x}})$ has real coefficients for every closed point; equivalently, the multiset of its complex eigenvalues at each point is stable under complex conjugation. In particular, the traces of all positive powers of local Frobenius are real.

The next lemma is the ℓ-adic form of Rankin's trick. Since the even powers of a real trace are nonnegative, the Euler factors of $\mathcal{G}^{\otimes 2m}$ have nonnegative coefficients.

Lemma 14.22 (Positivity of Euler factors)**.** *Let $\mathcal{G}$ be an ι-real Weil sheaf, let $m \geq 1$, and let x be a closed point. Then the power series $\iota \det(1 - \mathrm{Frob}_x\, t^{\deg(x)} \mid \mathcal{G}_{\bar{x}}^{\otimes 2m})^{-1}$ has nonnegative real coefficients and constant coefficient one.*

Proof. We write an inverse characteristic polynomial as the exponential of a series in the traces of powers, whose coefficients are then visibly nonnegative. For a square matrix A, triangularizing A and expanding the logarithm of each factor we get the formal identity

$$\det(1-tA)^{-1} = \exp\left(\sum_{n\geq 1} \operatorname{Tr}(A^n)\frac{t^n}{n}\right).$$

On an even tensor power, the trace of the nth power of Frobenius is $\operatorname{Tr}(\mathrm{Frob}_x^n \mid \mathcal{G})^{2m}$. Since $\mathcal{G}$ is real, this is nonnegative after applying ι. Every coefficient of the exponential is then a finite sum of products of nonnegative numbers. This remains true after replacing t by $t^{\deg(x)}$. ∎

Example 14.23 (Why even tensor powers)**.** The sheaf $\mathcal{B}_{-1}$ is real, since Frob_x acts by $(-1)^{\deg(x)}$, but its Euler factor at a point of degree one is $(1+t)^{-1} = 1-t+t^2-\cdots$, with negative coefficients. Its square $\mathcal{B}_{-1}\otimes\mathcal{B}_{-1}$ is the constant sheaf, whose Euler factors $(1-t^{\deg(x)})^{-1}$ have nonnegative coefficients, as the lemma predicts for $m=1$.

Without positivity a factor can converge on a smaller disc than the product, as in the cancellation at the start of this chapter. With positivity, however, the coefficients of each factor are bounded by those of the product.

Lemma 14.24 (Positivity and convergence)**.** *Let*

$$L(t) = \prod_j f_j(t), \qquad f_j \in 1 + t\mathbf{R}_{\geq 0}[[t]],$$

be a coefficientwise locally finite product. If L is the Taylor series of a function holomorphic on $|t|<R$, then every f_j has radius of convergence at least R, and if $f_j = P_j^{-1}$ with $P_j(0)=1$, then P_j has no zero in that disc.

Proof. We compare coefficients and then apply Cauchy's estimate. To compare, write $f_j = \sum_n a_{j,n}t^n$ and $L = \sum_n A_n t^n$. Since the product is coefficientwise locally finite, A_n is already the coefficient of t^n in a finite subproduct containing f_j, which is f_j times a series g with constant coefficient one and nonnegative coefficients. The coefficient of t^n in $f_j g$ is $a_{j,n}$ plus a sum of products of nonnegative numbers, so that $0 \leq a_{j,n} \leq A_n$; positivity enters only here. Next, for every $0<r<R$, Cauchy's estimate gives $|A_n| \leq C_r r^{-n}$. Hence the radius of convergence of f_j is at least r for every such r, and so at least R. A zero of P_j in the disc would be a pole of $f_j = 1/P_j$, which is impossible. ∎

In the proposition below, the twist (-1) on H_c^2 costs a factor q, and this loss disappears when we take $2m$th roots and let m grow.

Proposition 14.25 (A bound by the largest determinant weight)**.** *Let $U_0/\mathbf{F}_q$ be a smooth geometrically connected affine curve and let $\mathcal{G}\neq 0$ be a lisse ι-real Weil sheaf. If β is the largest determinant weight of its irreducible Weil constituents, then*

$$|\iota(\alpha)|^2 \leq q_x^{\beta} \quad (x\in|U_0|,\ \alpha \text{ an eigenvalue on } \mathcal{G}_{\bar{x}}).$$

Proof. We bound the poles of $L(U_0, \mathcal{G}^{\otimes 2m}, t)$ by determinant weights through H_c^2, and positivity then passes the bound to each local factor.

We may assume that $\mathcal{G}$ is semisimple. Indeed, by Lemma 13.5, replacing $\mathcal{G}$ by its semisimplification changes neither its local characteristic polynomials nor the determinant weights of its constituents.

The top cohomology. Fix $m \geq 1$ and put $\mathcal{T} = \mathcal{G}^{\otimes 2m}$. We show that every eigenvalue μ of F on $H^2_c(U, \mathcal{T})$ satisfies $|\iota(\mu)| \leq q^{m\beta+1}$; these eigenvalues determine the possible poles of the L-function. First, every constituent of $\mathcal{T}$ has determinant weight at most $2m\beta$, by Proposition proposition-determinant-weights-tensor-operations. Next, Lemma 14.20 identifies $H^2_c(U, \mathcal{T})$ with $T_{\pi_1(U)}(-1)$, where T is the generic fibre of $\mathcal{T}$. The geometric fundamental group acts trivially on $T_{\pi_1(U)}$, so its simple Weil factors are one-dimensional, as one sees by triangularizing the operator induced by a degree-one lift. Let b be the eigenvalue on such a factor. Since $T_{\pi_1(U)}$ is a Weil quotient of T, the factor is a constituent of $\mathcal{T}$, and its determinant weight is $\lambda_\iota(b)$. From $\lambda_\iota(b) = 2\log_q |\iota(b)| \leq 2m\beta$ we get $|\iota(b)| \leq q^{m\beta}$. The twist (-1) multiplies eigenvalues by q, and the claim follows.

The local factors. The bound on H^2_c makes the L-function holomorphic on a disc, and positivity passes this to each local factor. As $H^0_c = 0$ by Lemma 14.20, Theorem 12.30 gives

$$L(U_0, \mathcal{T}, t) = \prod_{x \in |U_0|} \det(1 - \mathrm{Frob}_x \, t^{\deg(x)} \mid \mathcal{T}_{\bar{x}})^{-1} = \frac{\det(1 - Ft \mid H^1_c(U, \mathcal{T}))}{\det(1 - Ft \mid H^2_c(U, \mathcal{T}))}.$$

This is an identity of formal power series, and the product is coefficientwise locally finite because there are only finitely many closed points of bounded degree. Now put $R_m = q^{-m\beta-1}$. By the bound on H^2_c, the denominator has no zero on the disc $|t| < R_m$ after applying ι, so the rational function, and with it its Taylor series, is holomorphic on this disc.

Each local factor $\det(1 - \mathrm{Frob}_x \, t^{\deg(x)} \mid \mathcal{T}_{\bar{x}})$ has no zero on the same disc. Indeed, after applying ι the local factors of $L(U_0, \mathcal{T}, t)$ are power series with nonnegative coefficients and constant coefficient one by Lemma 14.22, so that Lemma 14.24 applies.

Now let α be a local eigenvalue of $\mathcal{G}$. Then α^{2m} is an eigenvalue on $\mathcal{T}$ by Lemma 13.5, and $1 - \iota(\alpha)^{2m} t^{\deg(x)}$ divides the local factor. Its roots have absolute value $|\iota(\alpha)|^{-2m/\deg(x)}$, so by the previous paragraph this number must be at least R_m, that is,

$$|\iota(\alpha)|^2 \leq q_x^{\beta + 1/m}.$$

Letting m tend to infinity proves the bound. ∎

14.4 Exterior powers and purity

Proposition 14.25 only bounds the local eigenvalues by the largest determinant weight. To bound the eigenvalues of each constituent by its own determinant weight, we apply the proposition to exterior powers, which separate the determinant weights of the simple factors.

Tensor powers cannot do this. For example, suppose $\mathcal{G} = \mathcal{G}_1 \oplus \mathcal{G}_2$ with determinant weights $\gamma_1 > \gamma_2$. In $\mathcal{G}^{\otimes m}$ the largest determinant weight $m\gamma_1$ belongs to the factors of $\mathcal{G}_1^{\otimes m}$, while every factor involving $\mathcal{G}_2$ lies strictly below it. So the bound of Proposition 14.25 says nothing sharp about $\mathcal{G}_2$.

An exterior power, by contrast, uses each basis vector of $\mathcal{G}_1$ at most once. Thus if r_1 is the rank of $\mathcal{G}_1$, every graded piece of $\bigwedge^{r_1+1} \mathcal{G}$ involves $\mathcal{G}_2$, and the summand $\det \mathcal{G}_1 \otimes \mathcal{G}_2$ has the largest determinant weight $r_1\gamma_1 + \gamma_2$. Since Corollary 14.8 gives the eigenvalues of $\det \mathcal{G}_1$, a bound for this summand is a bound for $\mathcal{G}_2$. We argue on an affine curve, where the proposition applies, and then pass to an arbitrary curve by restricting to affine opens, which preserves irreducibility.

Theorem 14.26 (Purity of real sheaves on curves)**.** *On a smooth geometrically connected curve over $\mathbf{F}_q$, every irreducible constituent of an ι-real lisse Weil sheaf is punctually ι-pure of weight equal to its determinant weight. In particular, every ι-real lisse Weil sheaf on the curve is mixed.*

Proof. We group the constituents by determinant weight and place each group, tensored with the determinants of the groups of larger weight, in an exterior power where it carries the largest determinant weight. There Proposition 14.25 bounds the eigenvalues of the group by its own determinant weight, and the same argument applied to the dual gives the reverse bound.

Let $\mathcal{G}$ be an ι-real lisse Weil sheaf on the curve. We first prove the assertion when the curve is affine. There, replacing $\mathcal{G}$ by its semisimplification changes neither its local characteristic polynomials nor its irreducible constituents, and it keeps the sheaf real, so we may assume that $\mathcal{G}$ is semisimple and write

$$\mathcal{G} = \bigoplus_{i=1}^{t} \mathcal{G}_i, \qquad \gamma_1 > \cdots > \gamma_t, \qquad r_i = \operatorname{rank} \mathcal{G}_i,$$

where $\mathcal{G}_i$ is the sum of the irreducible constituents with determinant weight γ_i.

Fix i, and let us find the exterior power in which $\mathcal{G}_i$, together with the determinants of $\mathcal{G}_1, \dots, \mathcal{G}_{i-1}$, carries the largest determinant weight. To this end set $m = \sum_{j<i} r_j$, and put $B = \sum_{j<i} r_j \gamma_j$. By Proposition proposition-determinant-weights-tensor-operations, the largest determinant weight of a constituent of $\bigwedge^{m+1} \mathcal{G}$ is $B + \gamma_i$. Indeed, it is obtained by taking $n_j = r_j$ for every $j < i$ and $n_i = 1$, and every other choice gives a smaller sum, because the γ_j are strictly decreasing. Moreover, exterior powers of a real sheaf are real. Indeed, their eigenvalues are the products of eigenvalues with distinct indices, and complex conjugation permutes this multiset. Hence Proposition 14.25 applies to $\bigwedge^{m+1} \mathcal{G}$.

The eigenvalues of $\mathcal{G}_i$ appear in $\bigwedge^{m+1} \mathcal{G}$ not by themselves but multiplied by the eigenvalues of the determinants of the groups of larger weight, and those eigenvalues are known. To make this precise, let x be a closed point of the curve and α an eigenvalue of Frob_x on $(\mathcal{G}_i)_{\bar{x}}$. The decomposition of the exterior power of a direct sum shows that $\bigwedge^{m+1} \mathcal{G}$ contains the summand

$$(\det \mathcal{G}_1 \otimes \cdots \otimes \det \mathcal{G}_{i-1}) \otimes \mathcal{G}_i.$$

If β_j is the eigenvalue of Frob_x on $(\det \mathcal{G}_j)_{\bar{x}}$, then the product $\alpha\beta_1 \cdots \beta_{i-1}$ is an eigenvalue of Frob_x on the stalk of this summand at $\bar{x}$. Since $\det \mathcal{G}_j$ is pure of weight $r_j\gamma_j$, we have $|\iota(\beta_1 \cdots \beta_{i-1})|^2 = q_x^B$. Now Proposition 14.25 gives

$$|\iota(\alpha)|^2 q_x^B \le q_x^{B+\gamma_i}, \qquad |\iota(\alpha)|^2 \le q_x^{\gamma_i}.$$

For the reverse bound, note that the dual $\mathcal{G}^\vee$ is real as well, with determinant weights $-\gamma_t > \cdots > -\gamma_1$. Applying the same argument to the summand $\mathcal{G}_i^\vee$, we get $|\iota(\alpha^{-1})|^2 \le q_x^{-\gamma_i}$, and hence $|\iota(\alpha)|^2 = q_x^{\gamma_i}$. Since every irreducible constituent of $\mathcal{G}$ is a summand of exactly one $\mathcal{G}_i$, this proves the theorem on an affine curve.

On an arbitrary curve we restrict to an affine neighbourhood of a given closed point x, which suffices because purity is a condition at each closed point. It remains to check that the restriction changes neither the constituents nor their determinant weights. The neighbourhood is a nonempty open subscheme of a smooth geometrically integral curve, so it is geometrically connected, and restriction to it preserves irreducibility. Indeed, a connected finite étale cover of a normal connected curve is integral, and hence remains connected after restriction to a dense open. So the maps of geometric and of arithmetic fundamental groups are surjective, and since restriction

preserves the degree map, the map of Weil groups is surjective too. Hence a subspace of the fibre is stable on the open if and only if it is stable on the curve.

The restriction of a real sheaf is real, and by Corollary 14.6 the determinant keeps its weight. The affine case therefore applies at x, and since x was arbitrary, we obtain purity on the whole curve. Finally, the sheaf is mixed because a composition series is a finite filtration with pure quotients. ■

15 Pure sheaves on curves

How much does the trace function of a sheaf know about its Frobenius eigenvalues? For a sheaf of rank one, pure of weight w, on a dense open U_0 of the line, it knows the weight. Indeed, every trace over $\mathbf{F}_{q^n}$ has absolute value $q^{nw/2}$, so the sum S_n of the $|t(x)|^2$ over $U_0(\mathbf{F}_{q^n})$ is $q^{nw}\#U_0(\mathbf{F}_{q^n})$, about $q^{n(w+1)}$, and the growth rate q^{w+1} of these sums therefore determines w. In higher rank, however, a trace is a sum of eigenvalues, and the eigenvalues can cancel. For example, on $\mathcal{B}_b \oplus \mathcal{B}_{-b}$ the trace at a point of degree n is $b^n + (-b)^n$, which is zero for every odd n. We show that the cancellation cannot persist: for a nonzero pure lisse sheaf of weight w on an affine curve, $\limsup_n \frac{1}{n} \log_q S_n = w + 1$ (Theorem 15.14).

The reason is positivity once more. Recall that a sheaf $\mathcal{G}$ is real if the characteristic polynomials of Frobenius on its stalks have real coefficients under ι (Definition 14.21). For such $\mathcal{G}$, pure of weight w, $|t_{\mathcal{G}}|^2$ is the trace function of $\mathcal{G} \otimes \mathcal{G}$, and hence the generating series of the S_n is the logarithmic derivative of $L(\mathcal{G} \otimes \mathcal{G}, t)$. Positivity of the Euler factors (Lemma 14.22) shows that $L(\mathcal{G} \otimes \mathcal{G}, t)$ is not a polynomial, and Lemma 14.20 puts its poles on $|t| = q^{-w-1}$. Together these force the sums S_n to reach the size $q^{n(w+1)}$ infinitely often. The restriction to real sheaves costs nothing, since an arbitrary pure sheaf becomes real after a twist to weight zero and the addition of its dual (Proposition 15.9).

This is what the Fourier argument of Chapter 17 needs. That argument starts from a pure lisse sheaf M of weight w on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$ and produces a pure lisse sheaf M' on a dense open j' of the dual line. It knows the trace function of M' exactly, as the discrete Fourier transform of that of M, but it cannot see the weight of M' directly. Since Plancherel's formula multiplies the sums S_n by q^n, the theorem shows that M' has weight $w + 1$.

The group we finally want to bound, $H^1_c(\mathbf{A}^1, j_*M)$, is the stalk of j'_*M' at the origin. But the origin may be one of the points where M' is not lisse, and there we need to know that j'_* does not create larger weights (Proposition 15.5).

15.1 Weights of middle extensions

Let j be the inclusion of a dense open in a smooth curve, and let $\mathcal{F}$ be a lisse sheaf on the open. The Fourier argument needs an extension of $\mathcal{F}$ to the whole curve that loses nothing at the boundary points. The ordinary direct image $j_*\mathcal{F}$ is such an extension, whereas extension by zero loses something. For example, for the constant sheaf on $\mathbf{G}_m \subset \mathbf{A}^1$, $j_!\overline{\mathbf{Q}}_\ell$ has stalk 0 at the origin, while the constant sheaf on $\mathbf{A}^1$ is the natural extension. In general, by Lemma 15.2, $j_*\mathcal{F}$ recovers a lisse extension whenever one exists, and its stalk at a boundary point is the space of inertia invariants.

If $\mathcal{F}$ is punctually of weights $\leq a$, then so is $j_*\mathcal{F}$, at the boundary points as well (Proposition 15.5). In Chapter 17 this bounds the weights of the boundary terms in the Fourier transform of a middle extension.

Definition 15.1 (Middle extension)**.** Let $j : U \hookrightarrow X$ be a dense open in a smooth curve and let $\mathcal{F}$ be a lisse sheaf on U. The *middle extension* of $\mathcal{F}$ is the ordinary direct image $j_*\mathcal{F}$ of sheaves.

Lemma 15.2 (The boundary stalk)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open in a smooth curve, let $\mathcal{F}$ be a lisse Weil sheaf on U_0, and let j_* be the ordinary direct image of sheaves. Let $\bar{z}$ be a geometric point over a point z of $X_0 - U_0$, and let I_z be the inertia group at z, acting on the local generic fibre V of $\mathcal{F}$. There is a canonical Frobenius-equivariant identification*

$$(j_*\mathcal{F})_{\bar{z}} = V^{I_z}.$$

If $\mathcal{F}$ extends to a lisse sheaf on X_0, the adjunction map from that extension to $j_*\mathcal{F}$ is an isomorphism.

Proof. We prove the formula for finite coefficients, where it is Galois theory of the punctured henselian neighbourhood, and pass to the limit.

The stalk of $j_*\mathcal{F}$ at $\bar{z}$ is the colimit of the sections of $\mathcal{F}$ over the inverse images of U in étale neighbourhoods of $\bar{z}$, that is, the group of sections over the punctured spectrum of the strict henselization at $\bar{z}$. This punctured spectrum is the spectrum of a field, whose absolute Galois group is the inertia group I_z, so finite locally constant sheaves there correspond to finite continuous representations of the inertia group, and their sections are the invariant vectors. Applying this to the finite reductions of a stable lattice and passing to the inverse limit, we obtain the formula with integral coefficients.

Two compatibilities take us to rational coefficients and to $\overline{\mathbf{Q}}_\ell$. Rationalization commutes with taking invariants, since a power of the uniformizer carries an invariant rational vector into the lattice. A finite extension of scalars also commutes with invariants. Indeed, it is exact, hence commutes with finite intersections of kernels, and in a finite-dimensional space the intersection of the kernels of all $g - 1$ is the intersection of finitely many of them. This gives the formula with rational coefficients, after finite extension of scalars, and hence also over $\overline{\mathbf{Q}}_\ell$. Since all these maps commute with the action of the decomposition group, the identification is Frobenius equivariant. Finally, if $\mathcal{F}$ extends to a lisse sheaf on X_0, the local inertia acts trivially, and so the adjunction map is an isomorphism on every geometric stalk. ∎

Example 15.3 (Boundary stalks on $\mathbf{G}_m \subset \mathbf{A}^1$)**.** Let $j : \mathbf{G}_{m,k_0} \hookrightarrow \mathbf{A}^1_{k_0}$ and $z = 0$. The covers $u^n = t$ are totally ramified at 0, so the inertia I_0 maps onto the monodromy of a Kummer sheaf. For the Kummer sheaf of a nontrivial character, I_0 acts on the line V through that character. Hence $V^{I_0} = 0$, and the stalk of j_* of this sheaf at $\bar{0}$ is zero, so here j_* and $j_!$ agree. For the Kummer extension $\mathcal{E}$ of Example 13.4, on the other hand, I_0 acts by the matrices $\left(\begin{smallmatrix} 1 & \kappa \\ 0 & 1 \end{smallmatrix}\right)$ with κ running over $\mathbf{Z}_\ell(1)$, whose common fixed vectors form the line $\overline{\mathbf{Q}}_\ell(1)$. So the stalk of $j_*\mathcal{E}$ at 0 is $\overline{\mathbf{Q}}_\ell(1)$, of weight -2, while $\mathcal{E}$ has weights -2 and 0; the stalk at the boundary keeps only part of the weights.

To bound $j_*\mathcal{F}$ we compare the L-functions of $\mathcal{F}$ and of $j_*\mathcal{F}$ on a disc. To see that the Euler product converges there, we need a crude bound for the number of points of a curve.

Lemma 15.4 (A point-count bound)**.** *For a smooth affine curve U_0/k_0 there is a constant D such that $\#U_0(k_n) \leq Dq^n$ for all $n \geq 1$. For a finite k_0-scheme, the number of k_n-points is bounded independently of n.*

Proof. We map U_0 finitely and flatly onto the affine line, whose number of k_n-points is q^n, and bound the size of the fibres.

By Noether normalization [Stacks, 00OY] there is a finite map $U_0 \to \mathbf{A}^1_{k_0}$, and the coordinate ring of U_0 is a finite torsion-free module over $k_0[t]$. To see this, let $f \neq 0$ in $k_0[t]$ and $fa = 0$. On each irreducible component of U_0 the image of f is nonzero, since otherwise the component would be finite over $k_0[t]/(f)$ and hence of dimension zero. Since the coordinate ring of each component is a domain, $fa = 0$ forces a to vanish on every component, and then $a = 0$ because U_0 is reduced. As $k_0[t]$ is a principal ideal domain, it follows that the coordinate ring of U_0 is free of some rank D. Hence, after any extension of the base field, every fibre has a coordinate ring of dimension D, and by the Chinese remainder theorem it has at most D rational points. Summing over $\mathbf{A}^1(k_n)$, we obtain the first bound. The second holds because the number of rational points of a finite scheme is at most its number of geometric points. ∎

In the next proof we use the lemma to show that the logarithm of the Euler product of $\mathcal{F}$ converges on a disc of radius $q^{-a/2-1}$, and on this disc we compare it with $j_*\mathcal{F}$.

Proposition 15.5 (Weights of j_* at the boundary points)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open of a smooth geometrically connected curve. If a lisse Weil sheaf $\mathcal{F}$ on U_0 is punctually of weights $\leq a$, so is $j_*\mathcal{F}$ on X_0.*

Proof. Comparing the L-functions of $\mathcal{F}$ and of $j_*\mathcal{F}$ gives the weaker bound $a+2$ at the boundary, and tensor powers remove the extra 2.

The assertion is local at a point of $X_0 - U_0$, so we may replace X_0 by an affine neighbourhood of that point. The complement of U_0 is an effective Cartier divisor, so after shrinking this neighbourhood it is cut out by one equation f, and then $U_0 = D(f)$ is affine as well. Write $i : Z_0 = X_0 - U_0 \hookrightarrow X_0$. The hypotheses are preserved by this replacement. Indeed, forming j_* commutes with restriction to open subschemes, and the restriction of $\mathcal{F}$ is still punctually of weights $\leq a$. Moreover, a nonempty open of X_0 is still geometrically connected.

The boundary Euler factors of $j_*\mathcal{F}$ are the ratio L_X/L_U of the two L-functions defined below. We show that L_U has no zero and L_X no pole on the disc $|t| < q^{-a/2-1}$, so that the boundary factors have no pole there, and this bounds their eigenvalues.

The L-function on U_0. Put $L_U(t) = L(U_0, \mathcal{F}, t)$. We show that ιL_U is holomorphic and nonvanishing on the disc, by bounding the coefficients of its logarithm. The logarithm of the Euler product has coefficients

$$\log \iota L_U(t) = \sum_{n \geq 1} \frac{c_n}{n} t^n, \qquad c_n = \sum_{d \mid n} d \sum_{\deg(x) = d} \iota \operatorname{Tr}(\operatorname{Frob}_x^{n/d} \mid \mathcal{F}_{\bar{x}}).$$

Let $r = \operatorname{rank} \mathcal{F}$. For a closed point x of degree $d \mid n$, the bound on the weights gives $|\iota \operatorname{Tr}(\operatorname{Frob}_x^{n/d} \mid \mathcal{F}_{\bar{x}})| \leq r q_x^{(n/d)a/2} = r q^{na/2}$. Since $\sum_{d \mid n} d \cdot \#\{x \in |U_0| : \deg(x) = d\} = \#U_0(k_n)$, Lemma 15.4 gives $|c_n| \leq rDq^{n(a/2+1)}$. Hence the logarithm converges on the disc $|t| < R = q^{-a/2-1}$, and its exponential is a nonvanishing holomorphic function there. By the trace formula, its Taylor series is that of the rational function $\iota L_U(t)$, so the two agree on the whole disc.

The L-function on X_0. Let $L_X(t) = L(X_0, j_*\mathcal{F}, t)$. We show that ιL_X is holomorphic on the same disc. By the trace formula its poles come from H^0_c and H^2_c of $j_*\mathcal{F}$, and we compute both groups.

The group H^0_c vanishes. Indeed, the support of a compactly supported section of $j_*\mathcal{F}$ on X is closed and proper over k, and since X is an affine curve, the support is finite. So the section

vanishes on a dense open subset of U, and as $\mathcal{F}$ is lisse and U is connected, it vanishes on all of U. Since a section of $j_*\mathcal{F}$ is determined by its restriction to U, the section is zero. For H^2_c, note that the quotient sheaf in

$$0 \longrightarrow j_!\mathcal{F} \longrightarrow j_*\mathcal{F} \longrightarrow i_*i^*j_*\mathcal{F} \longrightarrow 0$$

is supported on the finite boundary, whose cohomology is concentrated in degree zero. From the long exact sequence we obtain $H^2_c(X, j_*\mathcal{F}) = H^2_c(U, \mathcal{F})$, and by Lemma 14.20 its eigenvalues have absolute value at most $q^{a/2+1}$. By the trace formula, every pole of ιL_X is the inverse of a Frobenius eigenvalue on H^0_c or H^2_c. Since $H^0_c = 0$, the poles have absolute value at least $q^{-a/2-1} = R$. Hence ιL_X is holomorphic on $|t| < R$.

The boundary factors. On U_0 the sheaves $\mathcal{F}$ and $j_*\mathcal{F}$ have the same stalks, so the two Euler products differ only by the factors at the points of Z_0, and we get

$$L_X(t) = L_U(t) \prod_{z \in Z_0} \det(1 - \mathrm{Frob}_z\, t^{\deg(z)} \mid (j_*\mathcal{F})_{\bar{z}})^{-1}.$$

Since L_U has no zero on the disc, the finite product on the right is holomorphic there. As it is the reciprocal of a product of polynomials with constant term one, none of these polynomials has a zero on the disc. Now, for an eigenvalue α at z, the roots of $1 - \iota(\alpha)t^{\deg(z)}$ have absolute value $|\iota(\alpha)|^{-1/\deg(z)}$. This number is therefore at least R, which is the preliminary bound

$$|\iota(\alpha)|^2 \leq q_z^{a+2}.$$

Tensor powers. The error 2 in the exponent comes from the twist (-1) on H^2_c. Since it does not grow with the weight, it becomes negligible for high tensor powers. We therefore apply the preliminary bound to $\mathcal{F}^{\otimes m}$ for every $m \geq 1$, a sheaf punctually of weights $\leq ma$. Let V be the local generic fibre of $\mathcal{F}$ at z. By Lemma 15.2, the eigenvalue α of Frob_z on $(j_*\mathcal{F})_{\bar{z}}$ has an eigenvector $v \in V^{I_z}$, and then $v^{\otimes m}$ is a nonzero invariant eigenvector in $V^{\otimes m}$ with eigenvalue α^m. By the same lemma, α^m is therefore an eigenvalue on the stalk of $j_*(\mathcal{F}^{\otimes m})$ at $\bar{z}$, and the preliminary bound for $\mathcal{F}^{\otimes m}$ gives $|\iota(\alpha)|^{2m} \leq q_z^{ma+2}$. Taking mth roots and letting m tend to infinity, we obtain $|\iota(\alpha)|^2 \leq q_z^a$ at every boundary point. On U_0, finally, the bound is the hypothesis. ∎

The proposition gives only an upper bound at the boundary, and in general nothing better is true. For example, for a family of elliptic curves with multiplicative reduction at z, the first cohomology of the fibres is pure of weight 1, while its inertia invariants at z form a line of weight 0. When the inertia acts trivially, however, we get the lower bound by duality.

Corollary 15.6 (An unramified boundary point)**.** *Let $\mathcal{F}$ be a lisse sheaf, pure of weight a, on a smooth geometrically connected curve with a point removed. If the inertia at the removed point acts trivially, then the stalk at that point of the direct image of $\mathcal{F}$ to the curve is pure of weight a. In particular, a lisse Weil sheaf on a smooth geometrically connected curve that is pure on a dense open is pure on the whole curve.*

Proof. Purity follows from the upper bounds for the sheaf and for its dual, provided the stalk of the dual is the dual of the stalk, and trivial inertia guarantees this. So we apply Proposition 15.5 to the sheaf and to its dual, which is pure of weight $-a$. Since the inertia acts trivially, the stalk of the extended dual is the dual of the stalk of the extension, and so every eigenvalue α satisfies $|\iota\alpha|^2 \leq q_z^a$ and $|\iota\alpha^{-1}|^2 \leq q_z^{-a}$, and equality holds.

For a sheaf that is already lisse on the curve, the inertia is trivial, and Lemma 15.2 identifies the stalk of j_* with the original stalk. The first assertion, applied at each of the finitely many points outside the dense open, then gives the second. ∎

The proof of Proposition 15.5, together with Lemma 14.20, also gives the extreme degrees of the cohomology of a middle extension on an affine curve.

Lemma 15.7 (The extreme degrees on an affine curve)**.** *Let $j : U_0 \hookrightarrow C_0$ be a dense open in a smooth geometrically connected affine curve, let F be lisse on U_0, and put $V = F_{\bar{\eta}}$. Then $H^i_c(C, j_*F) = 0$ for $i \neq 1, 2$, and there is a canonical identification*

$$H^2_c(C, j_*F) = V_{\pi_1(U)}(-1).$$

If F is pure of weight w, the group in degree two, when nonzero, is pure of weight $w + 2$.

Proof. The vanishing of H^0_c was shown in the proof of Proposition 15.5. For the higher degrees, let i be the inclusion of $C_0 - U_0$. Since sheaves with finite support have compact-support cohomology only in degree zero, the localization sequence for $0 \to j_!F \to j_*F \to i_*i^*j_*F \to 0$ identifies $H^r_c(C, j_*F)$ with $H^r_c(U, F)$ for $r \geq 2$. Lemma 14.20 gives the value and the weight of this group for $r = 2$, and its vanishing for $r > 2$. ∎

For the perverse t-structure in Chapter 16 we need the same range of degrees for arbitrary constructible sheaves.

Lemma 15.8 (Cohomological range on a curve)**.** *Let C be a smooth curve over an algebraically closed field and let F be a constructible sheaf on C with coefficients in E or $\overline{\mathbf{Q}}_\ell$. Then $H^i_c(C, F) = 0$ for $i \notin \{0, 1, 2\}$.*

Proof. Choose a dense open $j : U \hookrightarrow C$ on which F is lisse, and let i be the inclusion of the finite complement. The sequence $0 \to j_!j^*F \to F \to i_*i^*F \to 0$ gives a long exact sequence in which $H^r_c(C, i_*i^*F)$ vanishes for $r \neq 0$, because the support is finite. It therefore suffices to show $H^r_c(U, j^*F) = 0$ for $r \notin \{0, 1, 2\}$. For $r < 0$ there is nothing to prove, and for $r > 2$, degreewise Poincaré duality (Corollary 10.13, with $d = 1$) shows that $H^r_c(U, j^*F)$ is dual to $H^{2-r}(U, (j^*F)^\vee(1))$, which is zero because $(j^*F)^\vee$ is a sheaf and $2 - r < 0$. ∎

A constructible Weil sheaf on a curve C_0 over k_0 gives a constructible sheaf on $C = C_0 \times_{k_0} k$, so the lemma applies to it as well.

15.2 Growth of squared traces

In Chapter 17 we know the trace function of the Fourier transform M' exactly, but not its weight. For a pure sheaf the trace function determines the weight: it is the growth rate of the sums of squared traces (Theorem 15.14).

Proposition 15.9 (Twisting to weight zero and adding the dual)**.** *Let $\mathcal{F} \neq 0$ be lisse and pure of weight w on a nonempty finite-type scheme over k_0. There is $b \in \overline{\mathbf{Q}}_\ell^\times$ with $|\iota(b)| = q^{-w/2}$. Put*

$$\mathcal{F}^0 = \mathcal{F} \otimes \mathcal{B}_b, \qquad \mathcal{E} = \mathcal{F}^0 \oplus (\mathcal{F}^0)^\vee.$$

Then $\mathcal{E}$ is real and pure of weight zero, contains $\mathcal{F}^0$ as a direct summand, and at points over k_n its trace is twice the real part of the trace of $\mathcal{F}^0$.

Proof. A twist to weight zero puts all local eigenvalues on the unit circle, where inversion is complex conjugation. Adding the dual therefore makes each multiset of eigenvalues stable under conjugation.

To find the twist, choose a closed point x, of degree d, at which the stalk is nonzero, and a Frobenius eigenvalue α at x. Let $b \in \overline{\mathbf{Q}}_\ell$ be a dth root of α^{-1}. By purity, $|\iota(b)| = |\iota(\alpha)|^{-1/d} = q^{-w/2}$, and Proposition 13.7 shows that $\mathcal{F}^0$ is pure of weight zero.

Its local eigenvalues β_j therefore have absolute value one after applying ι. The local eigenvalues of its dual are the β_j^{-1}, and $\iota(\beta_j^{-1}) = \overline{\iota(\beta_j)}$. So the union of the two multisets is stable under complex conjugation, and $\mathcal{E}$ is real and pure. In the same way, at a point x over k_n the eigenvalues of $\mathrm{Frob}_{x,n}$ on $(\mathcal{F}^0)^\vee$ become, after ι, the complex conjugates of those on $\mathcal{F}^0$. Hence ι of the trace of $(\mathcal{F}^0)^\vee$ is the complex conjugate of ι of the trace of $\mathcal{F}^0$, and the trace of $\mathcal{E}$ is twice its real part. ∎

The number ν below is the growth exponent of the sums of squared traces, shifted by 1 so that it equals the weight. We take a lim sup because the sums can vanish for infinitely many n, as the example of $\mathcal{B}_b \oplus \mathcal{B}_{-b}$ at the start of this chapter shows.

Definition 15.10 (Growth of squared traces)**.** Let K be a constructible Weil complex on an affine curve X_0. For $n \geq 1$ and $x \in X_0(k_n)$, let $\mathrm{Frob}_{x,n}$ be the Frobenius acting on the stalk $K_{\bar{x}}$ (Theorem 12.24), and set

$$t_{K,n}(x) = \sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_{x,n} \mid \mathcal{H}^i(K)_{\bar{x}}), \qquad S_n(K) = \sum_{x \in X_0(k_n)} |\iota t_{K,n}(x)|^2.$$

Put

$$\nu(K) = \limsup_{n \to \infty} \frac{\log_q S_n(K)}{n} - 1, \qquad \log_q 0 = -\infty.$$

A cohomological shift changes the weight of a complex in the sense of Definition 13.1. But it only multiplies the trace function by a sign, and so it leaves ν unchanged.

In the theorem below we find ν from the radius of convergence of a generating series, which is a logarithmic derivative of an L-function.

Lemma 15.11 (Radius of convergence)**.** *Let K be a constructible Weil complex on an affine curve X_0. The series $\varphi_K(t) = \sum_{n \geq 1} S_n(K) t^{n-1}$ has radius of convergence $q^{-\nu(K)-1}$, where this number is read as $+\infty$ if $\nu(K) = -\infty$ and as 0 if $\nu(K) = +\infty$.*

Proof. By the Cauchy–Hadamard formula the radius is the inverse of $\limsup_n S_n(K)^{1/n} = q^{\nu(K)+1}$; the shift of the index by one does not change it. ∎

To compute ν we also need upper bounds for lisse sheaves and for complexes with finite support, as well as the effect of twisting by $\mathcal{B}_b$.

Lemma 15.12 (Bounds for squared traces)**.** *Let X_0 be a smooth affine curve over k_0, and let a be a real number.*

1. *If F is a lisse Weil sheaf of rank r on X_0, punctually of weights $\leq a$, then $S_n(F) \leq Dr^2 q^{n(a+1)}$ for all n, with D as in Lemma 15.4, and hence $\nu(F) \leq a$.*

2. *If K is a constructible Weil complex on X_0 supported at finitely many closed points, and every eigenvalue α of Frob_x on every $\mathcal{H}^i(K)_{\bar{x}}$ satisfies $|\iota(\alpha)| \leq q_x^{a/2}$, then there is a constant C with $S_n(K) \leq Cq^{na}$ for all n, and hence $\nu(K) \leq a-1$.*
3. *For every constructible Weil complex K on X_0 and every $b \in \overline{\mathbf{Q}}_\ell^\times$, we have $\nu(K \otimes \mathcal{B}_b) = \nu(K) + \lambda_\iota(b)$, with $\mathcal{B}_b$ as in Proposition 13.7.*

Proof. For (1), the trace at each point of $X_0(k_n)$ has absolute value at most $rq^{na/2}$, and Lemma 15.4 gives $S_n \leq Dr^2q^{n(a+1)}$. For (2), Lemma 15.4 bounds the number of points of the support with values in k_n independently of n, so it suffices to bound the trace at each of them. Since the support consists of finitely many points with finite-dimensional stalks, the sum c of the ranks of the stalks of the $\mathcal{H}^i(K)$ is finite, and at each such point the trace has absolute value at most $cq^{na/2}$. For (3), the trace of $\mathcal{B}_b$ at a k_n-point is b^n, so tensoring by $\mathcal{B}_b$ multiplies S_n by $|\iota(b)|^{2n} = q^{n\lambda_\iota(b)}$. ∎

A correction at finitely many points of small weight does not change ν. We use this to discard the boundary terms in Corollary 15.15.

Lemma 15.13 (Finite corrections)**.** *Let K and L be constructible Weil complexes on an affine curve X_0, and let a be a real number. Suppose that there is a constant C with $\sum_{x \in X_0(k_n)} |\iota(t_{K,n} - t_{L,n})(x)|^2 \leq Cq^{na}$ for all n. If $\max(\nu(K), \nu(L)) > a-1$, then $\nu(K) = \nu(L)$.*

Proof. The inequality $|z+u|^2 \leq 2|z|^2 + 2|u|^2$, applied to $t_{L,n}$ and the difference $t_{K,n} - t_{L,n}$, gives $S_n(K) \leq 2S_n(L) + 2Cq^{na}$, so that $\nu(K) \leq \max(\nu(L), a-1)$, and by symmetry $\nu(L) \leq \max(\nu(K), a-1)$. If one of $\nu(K)$, $\nu(L)$ exceeds $a-1$, these two inequalities force equality. ∎

By counting points, Lemma lemma-bounds-squared-traces gives $\nu \leq w$ for a sheaf pure of weight w. A priori, however, the squared traces could be much smaller than this trivial bound, through cancellation among the eigenvalues at each point. The next theorem shows that on average they are not.

Theorem 15.14 (Weights and sums of squared traces)**.** *Let $X_0/\mathbf{F}_q$ be a smooth geometrically connected affine curve. For every nonzero pure lisse Weil sheaf $\mathcal{G}$ of weight w,*

$$\nu(\mathcal{G}) = w.$$

Proof. Lemma lemma-bounds-squared-traces gives $\nu(\mathcal{G}) \leq w$, so, by Lemma 15.11, we must show that the radius of $\varphi_\mathcal{G}$ is not larger than q^{-w-1}. For a real sheaf we find a pole of the logarithmic derivative of an L-function on that circle, and Proposition 15.9 reduces the general case to this one by twisting to weight zero and adding the dual.

The real case. Suppose first that $\mathcal{G}$ is real, and put $\mathcal{T} = \mathcal{G} \otimes \mathcal{G}$. The squared traces of $\mathcal{G}$ are then the traces of $\mathcal{T}$, and we express their generating series through the L-function of $\mathcal{T}$. Since $\iota t_{\mathcal{G},n}(x)$ is real, we have $\iota t_{\mathcal{T},n}(x) = |\iota t_{\mathcal{G},n}(x)|^2$ for $x \in X_0(k_n)$. The Euler product and the trace formula then give

$$\varphi_\mathcal{G}(t) = \frac{L'(t)}{L(t)}, \qquad L(t) = \frac{\iota \det(1 - Ft \mid H_c^1(X, \mathcal{T}))}{\iota \det(1 - Ft \mid H_c^2(X, \mathcal{T}))}.$$

For the first equality, group the local logarithms by degree. Then the coefficient of t^{n-1} in L'/L is the sum of the traces over $X_0(k_n)$, which is $S_n(\mathcal{G})$. As $\mathcal{T}$ is pure of weight $2w$, Lemma 14.20 shows that all roots of the denominator have absolute value $R = q^{-w-1}$.

We show that the function L has a pole on the circle $|t| = R$. This is not automatic, since a priori all the eigenvalues on H_c^2 could cancel against eigenvalues on H_c^1, but positivity rules this out. Indeed, by Lemma 14.22, each local Euler factor of $\mathcal{T}$ has nonnegative Taylor coefficients, so the coefficients of the full product dominate those of any one factor. Now choose a closed point of the curve. Since $\mathcal{G}$ has positive rank and Frobenius is invertible, the Euler factor of $\mathcal{T}$ at this point is the reciprocal of a nonconstant polynomial. It has infinitely many nonzero Taylor coefficients, because a polynomial cannot be inverse to a nonconstant polynomial. Hence the full Euler product is not a polynomial, and the rational function L has a pole, which by the bound on the denominator lies on $|t| = R$.

A pole of L is also a pole of its logarithmic derivative: if $L(t) = (t - t_0)^{-m}u(t)$ near t_0, with $m > 0$ and $u(t_0) \neq 0$, then $L'/L = -m/(t - t_0) + u'/u$. So the radius of convergence of $\varphi_{\mathcal{G}}$ is exactly R, and Lemma 15.11 gives $\nu(\mathcal{G}) = w$.

The general case. Let $\mathcal{G}$ be an arbitrary pure sheaf, and let $\mathcal{G}^0 = \mathcal{G} \otimes \mathcal{B}_b$ and $\mathcal{E}$ be as in Proposition 15.9. The sheaf $\mathcal{G}^0$ is pure of weight zero, and by Lemma lemma-bounds-squared-traces we have $\nu(\mathcal{G}^0) = \nu(\mathcal{G}) + \lambda_\iota(b) = \nu(\mathcal{G}) - w$, so it suffices to show that $\nu(\mathcal{G}^0) = 0$. The sheaf $\mathcal{E} = \mathcal{G}^0 \oplus (\mathcal{G}^0)^\vee$ is nonzero, real and pure of weight zero, so by the case already treated $\nu(\mathcal{E}) = 0$.

This lower bound passes to $\mathcal{G}^0$, because the trace of $\mathcal{E}$ is twice the real part of that of $\mathcal{G}^0$. In other words, for every n and $x \in X_0(k_n)$ we have $\iota t_{\mathcal{E},n}(x) = 2\,\mathrm{Re}(\iota t_{\mathcal{G}^0,n}(x))$. Hence $S_n(\mathcal{E}) \leq 4S_n(\mathcal{G}^0)$, and so $\nu(\mathcal{G}^0) \geq 0$. Since the opposite inequality comes from Lemma lemma-bounds-squared-traces, we obtain $\nu(\mathcal{G}^0) = 0$, that is, $\nu(\mathcal{G}) = w$. ∎

Under the hypothesis of Proposition 16.17, the Fourier transform of $(j_*M)[1]$ is again a middle extension. But Plancherel applies only to complexes on the whole line, boundary points included, so we need the theorem for j_* and $j_!$ on the larger curve. This costs nothing: the boundary points do not change ν, because their weights are small.

Corollary 15.15 (Sums of squared traces of $j_!$ and j_*)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open of a smooth geometrically connected affine curve, and let $\mathcal{G}$ be a nonzero pure lisse Weil sheaf of weight w on U_0. Then*

$$\nu(j_!\mathcal{G}) = \nu(j_*\mathcal{G}) = w.$$

Proof. For $j_!$ the trace function is that of $\mathcal{G}$ extended by zero, and for j_* it differs from this at finitely many points of small weight, which Lemma 15.13 allows us to ignore.

The open curve U_0 is again smooth, geometrically connected and affine, and Theorem 15.14 on U_0 gives $\nu(\mathcal{G}) = w$. Extension by zero only adds zero traces, so $\nu(j_!\mathcal{G}) = w$. The trace functions of j_* and of the extension by zero differ only at the points of $X_0 - U_0$, by the trace function of the boundary stalks. These stalks have weights at most w by Proposition 15.5, so if i denotes the inclusion of $X_0 - U_0$, then Lemma lemma-bounds-squared-traces, applied to the sheaf $i_*i^*j_*\mathcal{G}$, bounds the squared sum of the difference by Cq^{nw}. Since $\nu(j_!\mathcal{G}) = w > w - 1$, Lemma 15.13 shows that $\nu(j_*\mathcal{G})$ is still w. ∎

16 The Fourier transform

Theorem 14.26 controls the Frobenius eigenvalues of a real sheaf at the points of a curve. The bound we need next, however, is of a different kind. Let M be an irreducible lisse sheaf, pure of weight w, on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$. The curve case of the Deligne–Laumon theorem comes down to showing that $H_c^1(\mathbf{A}^1, j_*M)$ has weights at most $w + 1$. This is a cohomology group, not

the stalk of a sheaf at a point, and Rankin's argument does not reach it. Is there a sheaf on another curve whose stalk at one point is this group, and whose weight we can compute from that of M? Up to a shift, the Fourier transform of $(j_*M)[1]$ is such a sheaf.

For functions on $\mathbf{F}_q$ the discrete Fourier transform already does what we want. Fix a nontrivial additive character ψ and put $\hat{t}(y) = \sum_x t(x)\psi(xy)$. Two properties matter. First, the value $\hat{t}(0)$ is the sum of all values of t, so a sum over the line becomes a single value. Second, Plancherel's formula $\sum_y |\hat{t}(y)|^2 = q \sum_x |t(x)|^2$ holds, by orthogonality of characters.

To see Plancherel compute an absolute value, let χ be a nontrivial character of $\mathbf{F}_q^\times$, extended by $\chi(0) = 0$, and let $g = \sum_x \chi(x)\psi(x)$ be its Gauss sum. Substituting $x \mapsto x/y$ shows that $\hat{\chi}(y) = \overline{\chi}(y)g$ for $y \neq 0$, so the transform has the same absolute value $|g|$ at all $q-1$ nonzero points, while $\hat{\chi}(0) = 0$. Plancherel therefore reads $(q-1)|g|^2 = q(q-1)$, and we find $|g| = \sqrt{q}$. In the language of weights, this means that the transform carries a function of weight 0 to one of weight 1. For the quadratic character of $\mathbf{F}_3$, moreover, Example 16.2 finds this Gauss sum, up to sign, as a Frobenius eigenvalue on a cohomology group.

The proof in Chapter 17 follows the same two steps for M: first we show that the transform is pure, and then Plancherel finds its weight.

Grothendieck's function–sheaf dictionary (Proposition 12.28) turns the transform into a functor. Let $\mathcal{K}_\psi = \mu^*\mathcal{L}_\psi$ be the pullback of the Artin–Schreier sheaf (Lemma 16.1) along the multiplication $\mu(x, y) = xy$. Its trace function on $\mathbf{F}_q$-points is $\psi(xy)$, and we put

$$\mathrm{FT}_\psi K = Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[1].$$

On trace functions it is the discrete transform up to sign: for $y \in \mathbf{F}_q$,

$$t_{\mathrm{FT}_\psi K}(y) = -\sum_{x \in \mathbf{F}_q} t_K(x)\,\psi(xy),$$

because under the dictionary (Section 12.6) pullback, tensor product and $Rp_{2!}$ become composition, multiplication and summation over the fibres, the three operations from which $\hat{t}$ is built, while the shift [1] contributes the sign. Both properties of the discrete transform carry over. The first becomes the statement that the stalk at the origin of the dual line is $R\Gamma_c(\mathbf{A}^1, K)[1]$ (Proposition 16.5), so for $K = j_*M$ this complex contains the group we want. Plancherel becomes an identity of trace functions (Corollary 16.8), and it is what will compute the weight of the transform. In addition, Fourier inversion, $\mathrm{FT}_{\psi^{-1}} \mathrm{FT}_\psi K \simeq K(-1)$, holds because $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi) = 0$, the sheaf version of $\sum_x \psi(x) = 0$. The ℓ-adic transform is due to Deligne, and Laumon's proof [Lau87] of the Deligne–Laumon theorem is built on it.

To speak of the weight of the transform we need it to be a lisse sheaf on a dense open. But the transform of a sheaf need not be a sheaf. For example, the transform of the constant sheaf is a skyscraper at the origin in degree one, $\mathrm{FT}_\psi(E) = \delta_0(-1)[-1]$. This is the sheaf version of $\hat{1} = q\,\delta_0$, up to the sign coming from the shift. What the transform does preserve, however, is the category of perverse sheaves on the line (Proposition 16.16, proved in Section 16.5 below). Since the transform is an exact equivalence of perverse sheaves, it sends the simple object $(j_*M)[1]$ to a simple object, which is either punctual or a middle extension. Moreover, only character sheaves $\mathcal{L}_{\psi(ax)}$ are sent to skyscrapers. Hence, if no geometric constituent of the irreducible sheaf M is such a character, we have

$$\mathrm{FT}_\psi((j_*M)[1]) \simeq (j'_*M')[1]$$

for an irreducible lisse M' on a dense open j' of the dual line (Proposition 16.17 below). Comparing stalks at the origin then identifies $H^1_c(\mathbf{A}^1, j_*M)$ with a stalk of j'_*M', and this is the sheaf we asked for.

We use the six operations on the pro-étale site. At first the coefficients are a finite extension $E/\mathbf{Q}_\ell$ that contains the pth roots of unity, where p is the characteristic of k_0, so that E contains the values of the additive characters of k_0. Later the coefficients are $\overline{\mathbf{Q}}_\ell$.

16.1 The Artin–Schreier sheaf

The kernel of the transform is a lisse sheaf whose trace function is the additive character ψ, just as $\psi(xy)$ is the kernel of the discrete transform. It is cut out of the Artin–Schreier cover of the line as an eigenspace of the translations. This cover is Galois with group the additive group $\mathbf{F}_p$, acting by $u \mapsto u + a$. Let $u^p - u = x$ with $x \in \mathbf{F}_{q^n}$. Raising to the powers $p, p^2, \dots$ and adding gives $u^{q^n} = u + \operatorname{Tr}(x)$. In other words, Frobenius acts on the fibre over x by the translation $\operatorname{Tr}(x) \in \mathbf{F}_p$. A character ψ_0 of $\mathbf{F}_p$ then turns this translation into the number $\psi_0(\operatorname{Tr} x)$. Similarly, the Kummer cover $u^{q-1} = x$ of $\mathbf{G}_m$ produces the multiplicative characters in the same way.

Besides the sheaf itself (Lemma 16.1), we need the sheaf version of $\sum_x \psi(x) = 0$, namely the vanishing of $R\Gamma_c(\mathbf{A}^r, \lambda^*\mathcal{L}_\psi)$ for a nonzero linear form λ (Lemma 16.4). This is stronger than the statement on trace functions, since the trace formula only says that the alternating sum of the traces vanishes over every $\mathbf{F}_{q^n}$. Fourier inversion and the stalk computations below, however, need each cohomology group to vanish. This vanishing comes from the eigenspace decomposition of the cover: the total space of the cover is again an affine line, and the translations act trivially on the compactly supported cohomology of a line. Hence the eigenspace of a nontrivial character is zero.

Lemma 16.1 (The Artin–Schreier character sheaf)**.** *Let $\psi : k_0 \to E^\times$ be a nontrivial additive character. There is a lisse rank-one E-sheaf $\mathcal{L}_\psi$ on $\mathbf{A}^1_{k_0}$, with a specified trivialization at zero, such that*

$$m^*\mathcal{L}_\psi \simeq \operatorname{pr}_1^*\mathcal{L}_\psi \otimes \operatorname{pr}_2^*\mathcal{L}_\psi, \qquad \mathcal{L}_\psi^\vee \simeq \mathcal{L}_{\psi^{-1}}.$$

Here m is the addition map, and the isomorphisms respect the trivializations at zero. For $x \in \mathbf{F}_{q^n}$, the Frobenius trace is

$$t_{\mathcal{L}_\psi, n}(x) = \psi(\operatorname{Tr}_{\mathbf{F}_{q^n}/k_0}(x)).$$

In particular $\mathcal{L}_\psi$ is pure of weight zero.

Proof. We define $\mathcal{L}_\psi$ as an eigensheaf in the direct image of the constant sheaf under an Artin–Schreier cover, and read off its properties from the corresponding properties of the cover: addition of roots for the first isomorphism, a fibrewise pairing for the second, and the action of Frobenius on roots for the trace.

Fix a nontrivial character $\psi_0 : \mathbf{F}_p \to E^\times$. Since the trace pairing on k_0 is nondegenerate, there is $a \in k_0^\times$ such that $\psi(z) = \psi_0(\operatorname{Tr}_{k_0/\mathbf{F}_p}(az))$. The cover $u^p - u = ax$ of the affine line is finite étale, with group of translations $\mathbf{F}_p$. Let $\mathcal{L}_\psi$ be the image of

$$e_{\psi_0} = \frac{1}{p} \sum_{c \in \mathbf{F}_p} \psi_0(c)^{-1} \tau_c^*$$

on the direct image of the constant sheaf E under this cover. It is lisse of rank one, since its stalk is the ψ_0-eigenspace in the regular representation, and the root $u = 0$ over $x = 0$ specifies the trivialization at zero. For the first isomorphism, adding roots of two Artin–Schreier equations identifies the contracted product of the two torsors with the torsor for the sum, and passing to the character eigenspaces we obtain the isomorphism. For the second, the pairing on the direct image given by summing products over the fibres is perfect and invariant under the translations. Since, for an invariant pairing, the χ- and χ'-eigenspaces of two characters are orthogonal unless $\chi\chi' = 1$, our pairing pairs the ψ_0-eigenspace perfectly with the ψ_0^{-1}-eigenspace. Since $\psi^{-1}(z) = \psi_0^{-1}(\mathrm{Tr}_{k_0/\mathbf{F}_p}(az))$, the ψ_0^{-1}-eigenspace of the same cover is $\mathcal{L}_{\psi^{-1}}$, and we obtain $\mathcal{L}_\psi^\vee \simeq \mathcal{L}_{\psi^{-1}}$.

For a root u above $x \in \mathbf{F}_{q^n}$ we have

$$u^{q^n} = u + \mathrm{Tr}_{\mathbf{F}_{q^n}/\mathbf{F}_p}(ax).$$

Geometric Frobenius acts on functions on the torsor by pullback along this map, and hence on the chosen eigenspace by $\psi_0(\mathrm{Tr}(ax))$. By transitivity of field traces, this is the asserted trace. Since these scalars are roots of unity, their images under any field embedding have absolute value one. ■

We check the lemma by hand at two points. Over $\mathbf{F}_p$ with $a = 1$, the fibre of the cover over $x = 1$ is $u^p - u = 1$, which has no root in $\mathbf{F}_p$. Indeed, Frobenius acts on the p roots by $u \mapsto u^p = u + 1$, a translation of order p, and the trace of $\mathcal{L}_\psi$ at $x = 1$ is therefore $\psi(1) = \psi_0(1) \neq 1$. Over $x = 0$, by contrast, the roots are the elements of $\mathbf{F}_p$, so they are all fixed and the trace is 1.

Example 16.2 (A Gauss sum over $\mathbf{F}_3$)**.** Take $q = 3$, a primitive cube root of unity $\omega \in \overline{\mathbf{Q}}_\ell$ and $\psi(x) = \omega^x$ on $\mathbf{F}_3$. Pull $\mathcal{L}_\psi$ back along $h : \mathbf{A}^1 \to \mathbf{A}^1$, $x \mapsto x^2$. By the lemma, the trace of Frobenius on the stalk of $h^*\mathcal{L}_\psi$ at $x \in \mathbf{F}_3$ is $\psi(x^2)$, so the trace formula gives

$$\sum_r (-1)^r \,\mathrm{Tr}(F \mid H_c^r(\mathbf{A}^1_k, h^*\mathcal{L}_\psi)) = \sum_{x \in \mathbf{F}_3} \omega^{x^2} = 1 + 2\omega,$$

and $(1 + 2\omega)^2 = -3$. The sheaf $h^*\mathcal{L}_\psi$ is lisse of rank one. It is geometrically nontrivial, since the cover $u^3 - u = x^2$ is geometrically irreducible. Hence its geometric coinvariants vanish, and $H_c^0 = H_c^2 = 0$ (Lemma 15.7). The sum is therefore $-\mathrm{Tr}(F \mid H_c^1)$.

To see that H_c^1 is a line, we use the running example. The cover $u^3 - u = x^2$ is the affine part Y_0 of the elliptic curve $C_0 \colon y^2 = x^3 - x$ of the introduction, with the coordinates renamed $(x^2 = u^3 - u)$. It lies over $\mathbf{A}^1$ by $\pi(u, x) = x$. Since π is the pullback along h of the Artin–Schreier cover, $\pi_* E$ is the direct sum of $h^*\mathcal{L}_{\psi'}$ over the three characters ψ' of $\mathbf{F}_3$. Over $k = \overline{\mathbf{F}}_3$ we obtain

$$H_c^1(Y, E) = H_c^1(\mathbf{A}^1, E) \oplus H_c^1(\mathbf{A}^1, h^*\mathcal{L}_\psi) \oplus H_c^1(\mathbf{A}^1, h^*\mathcal{L}_{\psi^{-1}}).$$

The left side is $H^1(C, E)$, because C minus Y is the single point at infinity, and it has dimension 2. Since the first summand on the right is zero, it remains to see that the last two summands have the same dimension. For this, note that the automorphism $(u, x) \mapsto (-u, ix)$ of Y, with $i^2 = -1$ in $\mathbf{F}_9$, conjugates the translation by c into the translation by $-c$. It therefore exchanges the last two summands, and each of them is a line. This automorphism is defined only over $\mathbf{F}_9$, but we use it only to count dimensions. The decomposition itself is stable under Frobenius, because the translations are defined over $\mathbf{F}_3$.

Thus Frobenius acts on the line $H_c^1(\mathbf{A}^1, h^*\mathcal{L}_\psi)$ by $-(1 + 2\omega)$. This is a square root of -3, and it is one of the two reciprocal roots $\pm i\sqrt{3}$ of $P_1(T) = 1 + 3T^2$ found in the introduction. Moreover,

the number $1+2\omega=\omega-\omega^2$ is the Gauss sum g of the quadratic character of $\mathbf{F}_3$. The classical fact $|g|=\sqrt{q}$ is thus the statement that this H^1_c is pure of weight one. Chapter 17 proves the analogue for every such sheaf.

Fourier inversion integrates the kernel over a whole fibre $\mathbf{A}^r$ of a vector bundle, so we need the vanishing on $\mathbf{A}^r$ and not only on the line. In coordinates in which the linear form is the first coordinate, the integral over $\mathbf{A}^r$ is a product of an integral over the line and one over $\mathbf{A}^{r-1}$, and the Künneth formula makes this precise.

Let $f : X \to S$ and $g : Y \to S$ be separated morphisms of finite type, let $p_1 : X \times_S Y \to X$ and $p_2 : X \times_S Y \to Y$ be the projections, and let $h = f \circ p_1 : X \times_S Y \to S$ be the structure map. For complexes K on X and L on Y we put $K \boxtimes_S L = p_1^*K \otimes p_2^*L$.

Lemma 16.3 (Künneth formula for compact supports)**.** *In this situation, let K, L be bounded constructible complexes with coefficients in a field. Then the maps in*

$$Rh_!(p_1^*K \otimes p_2^*L) \simeq Rf_!(K \otimes Rp_{1!}p_2^*L) \simeq Rf_!(K \otimes f^*Rg_!L) \simeq Rf_!K \otimes Rg_!L$$

of the projection formula for p_1, the proper base change isomorphism $Rp_{1!}p_2^*L \to f^*Rg_!L$ and the projection formula for f are isomorphisms. We call their composite $Rh_!(K \boxtimes_S L) \simeq Rf_!K \otimes Rg_!L$ the Künneth isomorphism*.*

Proof. We integrate first along the fibres of p_1, which are copies of the fibres of g. This is possible because $h = f \circ p_1$, so that $Rh_! = Rf_!Rp_{1!}$. Since the square formed by p_1, p_2, f and g is cartesian, proper base change gives $Rp_{1!}p_2^*L \simeq f^*Rg_!L$. The two projection formulas are isomorphisms as well, and hence so is the composite. ∎

Over a point, this is the Künneth isomorphism for cohomology with compact support. All three isomorphisms in the composite are natural, so when X, Y, S, f, g, K and L are defined over k_0, the Künneth isomorphism commutes with the Weil structures.

The case $\lambda = 0$ of the next lemma is needed as well, since Fourier inversion integrates the kernel along the diagonal, where the character is trivial.

Lemma 16.4 (The integral of a linear character)**.** *Let λ be a linear form on $\mathbf{A}^r$ over an algebraically closed field of characteristic p. Then*

$$R\Gamma_c(\mathbf{A}^r, \lambda^*\mathcal{L}_\psi) \simeq \begin{cases} E(-r)[-2r], & \lambda = 0, \\ 0, & \lambda \neq 0. \end{cases}$$

For $\lambda = 0$, the isomorphism is the trace map for cohomology with compact support.

Proof. On the line, from the cohomology of $\mathbf{P}^1$ and localization at the point at infinity, we get $R\Gamma_c(\mathbf{A}^1, E) = E(-1)[-2]$, identified by the trace map (Appendix D). Every translation acts trivially on it, since it preserves the trace and the cohomology is concentrated in degree two. Now let $\pi : \mathbf{A}^1 \to \mathbf{A}^1$, $u \mapsto u^p - u$, be the Artin–Schreier cover, which is finite étale and whose total space is again an affine line. Let $\mathcal{L}_{\psi_0}$ be the ψ_0-eigenspace of π_*E, that is, the sheaf of the proof of Lemma 16.1 for $a = 1$. The projector e_{ψ_0} of that proof cuts out $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_{\psi_0})$ from $R\Gamma_c(\mathbf{A}^1, \pi_*E) = R\Gamma_c(\mathbf{A}^1, E)$ as the ψ_0-eigenspace of the deck translations. As we have just seen, these translations act trivially on the cohomology of the upper line, so, since ψ_0 is nontrivial, the eigenspace is zero, that is, $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_{\psi_0}) = 0$. Finally, by its construction, $\mathcal{L}_\psi$ is the pullback of $\mathcal{L}_{\psi_0}$ by the automorphism $x \mapsto ax$ of the line, and hence $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi) = 0$ as well.

On $\mathbf{A}^r$ with $\lambda \neq 0$, we choose linear coordinates in which λ is the first coordinate. Then $\lambda^*\mathcal{L}_\psi$ is the external product of $\mathcal{L}_\psi$ on the first factor with the constant sheaf on $\mathbf{A}^{r-1}$, and the Künneth isomorphism of Lemma 16.3 writes its cohomology as a tensor product with the first factor $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi) = 0$. For $\lambda = 0$, on the other hand, the sheaf is constant, so the trace map of Definition 9.24 for the projection $\mathbf{A}^r \to \operatorname{Spec} k$ is an isomorphism $R\Gamma_c(\mathbf{A}^r, E)(r)[2r] \simeq E$. On constant coefficients this map is the trace map of affine space (9.11), obtained by integrating along one coordinate line at a time. Finally, all these calculations take place with coefficients in E and commute with extension of scalars. ∎

16.2 Fourier inversion

We integrate with compact supports, $Rp_{2!}$ rather than Rp_{2*}, for two reasons. First, proper base change computes the stalks of $Rp_{2!}$ as compactly supported cohomology of the fibres, and this puts $H^1_c(\mathbf{A}^1, j_*M)$ at the origin. Second, the trace formula holds for $Rp_{2!}$, so that on trace functions the transform becomes the discrete Fourier transform (16.3) (proved in Section 16.3).

The shift $[r]$ is the one for which the transform preserves the perverse heart. For example, on the line we will find $\mathrm{FT}_\psi(E[1]) = \delta_0(-1)$, whereas without the shift the perverse sheaf $E[1]$ would go to $\delta_0(-1)[-1]$, which is not perverse.

We work with a vector bundle $V \to S$ rather than the line alone, because the proofs move along the base. For instance, the stalk formula is base change to a point of the dual bundle, and Fourier inversion is a computation on $V \times_S V^\vee \times_S V$, which reduces, fibre by fibre, to Lemma 16.4 on $\mathbf{A}^r$. We prove these two properties in Propositions 16.5 and Proposition 16.6, while the preservation of the heart is Proposition 16.16.

Let S be a scheme of finite type over k_0 or over k, let $V \to S$ be a vector bundle of rank r, and let $V^\vee$ be its dual bundle. Write p_1, p_2 for the projections of $V \times_S V^\vee$, and set

$$\mathcal{K}_\psi = \langle\ ,\ \rangle^*\mathcal{L}_\psi, \qquad \mathrm{FT}_\psi(K) = Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[r]. \tag{16.1}$$

The kernel $\mathcal{K}_\psi$ is lisse of rank one, and since the six operations preserve constructibility, this defines a triangulated functor $\mathrm{FT}_\psi : D^b_c(V, E) \to D^b_c(V^\vee, E)$, normalized by the shift $[r]$.

Proposition 16.5 (Base change and stalks)**.** *Let $g : S' \to S$ be a morphism of schemes with S' of finite type over k_0 or over k, let $V' = V \times_S S'$, and let $g_V : V' \to V$ and $g_{V^\vee} : V'^\vee \to V^\vee$ be the projections.*

1. *The proper base change isomorphism for p_2 induces an isomorphism $g^*_{V^\vee}\,\mathrm{FT}_\psi(K) \simeq \mathrm{FT}_\psi(g_V^*K)$.*
2. *For $N \in D^b_c(S, E)$, the projection formula for p_2 induces an isomorphism $\mathrm{FT}_\psi(K \otimes \pi^*N) \simeq \mathrm{FT}_\psi(K) \otimes \pi^{\vee *}N$, where $\pi : V \to S$ and $\pi^\vee : V^\vee \to S$ are the projections.*
3. *For a geometric point $\bar{y}$ of $V^\vee$, the proper base change map gives*

$$(\mathrm{FT}_\psi K)_{\bar{y}} \simeq R\Gamma_c(V_{\bar{y}}, K|_{V_{\bar{y}}} \otimes \mathcal{L}_{\psi\langle x,y\rangle})[r]. \tag{16.2}$$

Proof. For (1), the formation of $V \times_S V^\vee$, of the pairing and of $\mathcal{K}_\psi$ commutes with base change along g, and proper base change for $Rp_{2!}$ gives the isomorphism. For (2), we have $p_1^*\pi^*N = p_2^*\pi^{\vee *}N$, and the projection formula for p_2 gives the isomorphism. For (3), the fibre of p_2 over $\bar{y}$ is $V_{\bar{y}}$, on which $\mathcal{K}_\psi$ restricts to $\mathcal{L}_{\psi\langle x,y\rangle}$, and proper base change for $Rp_{2!}$ along $\bar{y} \to V^\vee$ gives the formula. ∎

Proposition 16.6 (Fourier inversion)**.** *Under $V^{\vee\vee} = V$ there are natural isomorphisms of functors $D_c^b(V,E) \to D_c^b(V,E)$*

$$\mathrm{FT}_{\psi^{-1}}\,\mathrm{FT}_\psi(K) \simeq K(-r), \qquad \mathrm{FT}_\psi\,\mathrm{FT}_\psi(K) \simeq [-1]_V^* K(-r).$$

Proof. The composite of two transforms is again an integral transform, and we compute its kernel. It will turn out to be the constant sheaf on the diagonal, twisted by $(-r)$, which up to this twist is the kernel of the identity functor. Let r_{ij} be the projections of $V \times_S V^\vee \times_S V$ onto pairs of factors. By base change and the projection formula, the composite of two transforms is the transform with the convolution kernel

$$Rr_{13!}(r_{12}^*\mathcal{K}_\psi \otimes r_{23}^*\mathcal{K}_{\psi^{-1}})[2r].$$

At a point (x,z), the unshifted integral over the middle factor is the cohomology of the character sheaf attached to $y \mapsto \psi\langle x-z, y\rangle$, and hence it vanishes for $x \neq z$ by Lemma 16.4. Thus the unshifted convolution is supported on the diagonal, so the localization triangle identifies it canonically with the direct image of its restriction to the diagonal. There the unit trivialization of the character sheaf and the trace map for the vector bundle identify the restriction with $E(-r)[-2r]$. Since the shift $[2r]$ cancels $[-2r]$, we obtain the first inversion formula. If instead the two kernels use the same character, then $x-z$ is replaced by $x+z$, the support is the antidiagonal, and we obtain the second formula. Since the traces and unit trivializations used here are defined over k_0, these isomorphisms commute with the Weil structures. ∎

On the line, the transform exchanges the two simplest perverse sheaves, the shifted constant sheaf and the skyscraper at the origin, up to a twist. This is the sheaf version of $\hat{1} = q\,\delta_0$ and $\hat{\delta}_0 = 1$.

Lemma 16.7 (Transform of the constant sheaf)**.** *Let $V = \mathbf{A}^1_{k_0}$ over $S = \operatorname{Spec} k_0$, and let δ_0 be the constant skyscraper sheaf E at the origin. There are isomorphisms of Weil complexes*

$$\mathrm{FT}_\psi(E) \simeq \delta_0(-1)[-1], \qquad \mathrm{FT}_\psi(\delta_0) \simeq E[1].$$

Proof. For $K = E$, the stalk formula (16.2) at y is the integral of the character sheaf $\mathcal{L}_{\psi(xy)}$ over the line, shifted by one. By Lemma 16.4 it vanishes for $y \neq 0$ and equals $E(-1)[-2]$ at $y = 0$. Hence $\mathrm{FT}_\psi(E)$ is supported at the origin, and with the shift $[1]$ it is $\delta_0(-1)[-1]$. For $K = \delta_0$, the pullback by the first projection is supported on $\{0\} \times \mathbf{A}^1$. Since the kernel is trivial there and the second projection is an isomorphism, this gives $E[1]$. Moreover, the trace map and the unit trivialization are defined over k_0, so both identifications are Frobenius-equivariant. ∎

16.3 The Plancherel formula

On trace functions, the Fourier transform is, up to the sign $(-1)^r$, the discrete Fourier transform over $\mathbf{F}_{q^n}$ for the character $\psi \circ \mathrm{Tr}_{\mathbf{F}_{q^n}/k_0}$ (equation (16.3) below). The Plancherel formula then says how it changes the growth of squared trace sums, and for a pure sheaf this growth is the weight (Theorem 15.14). This is how we find the weight of the transform in Chapter 17: on the line, Plancherel multiplies the sums of squared traces by q^n, and so it raises the weight read off from their growth by one.

In this section $S = \operatorname{Spec} k_0$ and $V = \mathbf{A}^r_{k_0}$. For a Weil complex K on a k_0-scheme X_0, we write $t_{K,n}(x)$ for the alternating sum of the Frobenius traces at $x \in X_0(\mathbf{F}_{q^n})$, as in Definition

15.10. With this notation, the dictionary (Proposition 12.28) computes the trace function of the transform step by step. For $y \in \mathbf{F}_{q^n}^r$, the shift contributes $(-1)^r$ (chain formula of Proposition 11.2), part (3) sums over the fibre of p_2, and part (2) splits the trace of the tensor product:

$$t_{\mathrm{FT}_\psi K,n}(y) = (-1)^r\, t_{Rp_{2!}(p_1^* K \otimes \mathcal{K}_\psi),n}(y) = (-1)^r \sum_{x \in \mathbf{F}_{q^n}^r} t_{p_1^* K,n}(x,y)\, t_{\mathcal{K}_\psi,n}(x,y).$$

By part (1), $t_{p_1^* K,n}(x,y) = t_{K,n}(x)$, and by part (1) and Lemma 16.1, $t_{\mathcal{K}_\psi,n}(x,y) = \psi(\mathrm{Tr}_{\mathbf{F}_{q^n}/k_0}\langle x,y\rangle)$. Hence

$$t_{\mathrm{FT}_\psi K,n}(y) = (-1)^r \sum_{x \in \mathbf{F}_{q^n}^r} t_{K,n}(x)\, \psi(\mathrm{Tr}_{\mathbf{F}_{q^n}/k_0}\langle x,y\rangle). \tag{16.3}$$

Corollary 16.8 (Plancherel)**.** *For every $n \geq 1$,*

$$\sum_y |\iota t_{\mathrm{FT}_\psi K,n}(y)|^2 = q^{nr} \sum_x |\iota t_{K,n}(x)|^2.$$

Consequently, for $r = 1$, the number ν of Definition 15.10 satisfies $\nu(\mathrm{FT}_\psi K) = \nu(K) + 1$ whenever these numbers are finite.

Proof. As for the classical transform, the proof is by orthogonality of characters. We expand the square of the sum in (16.3), using that the complex conjugate of a character value is its inverse. This writes the left side as a double sum over $x, z \in \mathbf{F}_{q^n}^r$ of $\iota t_{K,n}(x)\overline{\iota t_{K,n}(z)}$ times a character sum over y, which we now compute. For $x, z \in \mathbf{F}_{q^n}^r$, the orthogonality of characters gives

$$\sum_y \iota\psi(\mathrm{Tr}\langle x - z, y\rangle) = \begin{cases} q^{nr}, & x = z, \\ 0, & x \neq z. \end{cases}$$

Indeed, for $x \neq z$ the linear form $y \mapsto \mathrm{Tr}_{\mathbf{F}_{q^n}/k_0}\langle x - z, y\rangle$ is nonzero, since the trace pairing is nondegenerate, and hence surjective onto k_0. As ψ is nontrivial, there is y_0 with $c = \psi(\mathrm{Tr}\langle x - z, y_0\rangle) \neq 1$, and substituting $y + y_0$ for y multiplies the sum by $\iota(c)$, so the sum is zero. Therefore only the diagonal terms remain, and they give the asserted identity. On a line, finally, the factor q^n raises ν by one. ∎

For the constant sheaf E on the line and $n = 1$, formula (16.3) gives the trace $-\sum_x \psi(xy)$, which is $-q$ at $y = 0$ and 0 elsewhere, in agreement with $\mathrm{FT}_\psi(E) = \delta_0(-1)[-1]$. Both sides of Plancherel are then q^2, and over $\mathbf{F}_{q^n}$ they are q^{2n}. This gives $\nu(E) = 0$ and $\nu(\mathrm{FT}_\psi E) = 1$, as the corollary predicts.

16.4 Perverse sheaves on a curve

To apply Theorem 15.14 to the transform of a pure lisse sheaf M, we need to know that the transform of $(j_* M)[1]$ is again a lisse sheaf on a dense open, placed in degree -1. Although the transform sends the constant sheaf to a skyscraper (Lemma 16.7), it preserves the perverse heart, which on a smooth curve has an elementary description in terms of ordinary cohomology sheaves and punctual support.

We can read off the shape of this description from the stalk formula. The stalk of the transform of $G[1]$ at y is $R\Gamma_c(\mathbf{A}^1, G \otimes \mathcal{L}_{\psi(yx)})[2]$. For a sheaf G without punctual sections H_c^0 vanishes,

and the proof of Proposition 16.16 below shows that H^2_c is nonzero only at finitely many y. So the transform lives in degrees -1 and 0, with a punctual sheaf in degree 0, and this suggests allowing skyscrapers in degree 0 in ${}^pD^{\leq 0}$. Orthogonality then forces the condition on the other side: a sheaf in degree 0 lies in ${}^pD^{\geq 1}$ only if it receives no nonzero map from a skyscraper, that is, only if it has no punctual subsheaf. The perverse t-structure on a curve is the t-structure built from these two conditions.

What we finally use about it is the list of its simple objects (Lemma 16.14). By this list, an exact equivalence must send an irreducible middle extension to an irreducible middle extension or to an irreducible punctual sheaf.

Recall that a *t-structure* on a triangulated category D is a pair of full subcategories $D^{\leq 0} \supset D^{\leq -1}$ and $D^{\geq 0} \supset D^{\geq 1}$, where $D^{\leq n} = D^{\leq 0}[-n]$ and $D^{\geq n} = D^{\geq 0}[-n]$, such that $\mathrm{Hom}(A, B) = 0$ for $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$, and every object sits in a triangle $A \to K \to B \to$ with $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$. Its *heart* is $D^{\leq 0} \cap D^{\geq 0}$; for the ordinary t-structure it is the category of sheaves. We write $D^{[a,b]} = D^{\geq a} \cap D^{\leq b}$. The triangle is unique up to unique isomorphism, so, as for the ordinary t-structure, it gives truncation functors and cohomology objects of K in the heart. Finally, a functor is *t-exact* if it preserves both subcategories.

Let C be a smooth curve over a perfect field. On C the perverse t-structure is built from punctual sheaves.

Definition 16.9 (Punctual sheaf)**.** A constructible sheaf on C is *punctual* if its support is a finite set of closed points.

The truncation triangle of the perverse t-structure splits a sheaf into a punctual part and a part without punctual subsheaves, so we first construct the largest punctual subsheaf. In the following lemma "sheaf" means a constructible sheaf with coefficients E or $\overline{\mathbf{Q}}_\ell$.

Lemma 16.10 (The largest punctual subsheaf)**.** *A lisse sheaf on C has no nonzero punctual subsheaf. Let F be a constructible sheaf on C, let $j : U \hookrightarrow C$ be a dense open on which F is lisse, and put*

$$F_{\mathrm{p}} = \ker(F \longrightarrow j_* j^* F).$$

Then F_{p} is the largest punctual subsheaf of F, and F/F_{p} has no nonzero punctual subsheaf. In particular, F_{p} does not depend on U.

Proof. Let L be a lisse sheaf. For a geometric point $\bar{x}$ and a geometric generic point $\bar{\eta}$ specializing to it, the generization map $L_{\bar{x}} \to L_{\bar{\eta}}$ is injective. Indeed, for a sheaf defined over a finite extension E of $\mathbf{Q}_\ell$, this holds for the finite locally constant reductions of the lattice in Proposition 13.10, and we pass to the inverse limit, invert the uniformizer and extend scalars. Now a punctual subsheaf $G \subset L$ has $G_{\bar{\eta}} = 0$, and since the generization maps of G and L are compatible, $G_{\bar{x}} = 0$.

Now let F and U be as in the statement. The sheaf F_{p} is supported on the finite set $C \setminus U$, hence punctual. Conversely, a punctual subsheaf G of F restricts to a punctual subsheaf of the lisse sheaf j^*F, hence to zero. Since its map to j_*j^*F is adjoint to $j^*G \to j^*F$, this map vanishes, and $G \subset F_{\mathrm{p}}$. Finally, j_*j^*F has no nonzero punctual subsheaf. Indeed, if $H \subset j_*j^*F$ is punctual, then $j^*H = 0$, and the inclusion is zero since $\mathrm{Hom}(H, j_*j^*F) = \mathrm{Hom}(j^*H, j^*F) = 0$. As F/F_{p} embeds in j_*j^*F, it has no nonzero punctual subsheaf either. ∎

Definition 16.11 (Perverse sheaves on a curve)**.** Let Λ be E or $\overline{\mathbf{Q}}_\ell$. Define two classes of objects of $D^b_c(C_{\mathrm{pro\acute{e}t}}, \Lambda)$ by

$$
\begin{aligned}
{}^{p}D^{\leq 0} &= \{K : \mathcal{H}^a K = 0 \ (a > 0),\ \mathcal{H}^0 K \text{ is punctual}\},\\
{}^{p}D^{\geq 0} &= \{K : \mathcal{H}^a K = 0 \ (a < -1),\ \mathcal{H}^{-1} K \text{ has no nonzero punctual subsheaf}\},
\end{aligned}
$$

and ${}^{p}D^{\leq n} = {}^{p}D^{\leq 0}[-n]$, ${}^{p}D^{\geq n} = {}^{p}D^{\geq 0}[-n]$. A *perverse sheaf* on C is an object of ${}^{p}D^{\leq 0} \cap {}^{p}D^{\geq 0}$. For $C = C_0 \times_{k_0} k$, a *perverse Weil sheaf* on C_0 is a perverse sheaf on C with a Weil structure, with morphisms as in Definition 10.18.

By Lemma 16.12 below these classes form a t-structure, the perverse t-structure. We write ${}^{p}D^{[a,b]}$ and ${}^{p}\mathcal{H}^n$ for its subcategories and cohomology objects.

On the line, the two basic examples are $E[1]$ and the skyscraper δ_0, and by Lemma 16.7 the Fourier transform exchanges them up to a Tate twist. This is the simplest instance of the exactness proved in Proposition 16.16.

Lemma 16.12 (Perverse sheaves on a curve)**.** *The classes ${}^{p}D^{\leq 0}$ and ${}^{p}D^{\geq 0}$ of Definition 16.11 form a bounded t-structure on $D^b_c(C_{\text{pro\'et}}, \overline{\mathbf{Q}}_\ell)$, and its heart, the category of perverse sheaves on C, is abelian. A punctual sheaf in degree zero is perverse, and so is $(j_*M)[1]$ for a lisse sheaf M on a dense open $j : U \hookrightarrow C$.*

Proof. The truncation triangle of K splits $\mathcal{H}^0 K$ into its largest punctual subsheaf, which goes into ${}^{p}D^{\leq 0}$, and the quotient, which goes into ${}^{p}D^{\geq 1}$, using Lemma 16.10.

The shift axioms are immediate from the definitions. For orthogonality, if $A \in {}^{p}D^{\leq 0}$ and $B \in {}^{p}D^{\geq 1}$, then ordinary truncation identifies $\mathrm{Hom}(A, B)$ with $\mathrm{Hom}(\mathcal{H}^0 A, \mathcal{H}^0 B)$, which is zero because the source is punctual and the target has no punctual subsheaf. To construct the truncation triangle for K, put $K' = \tau^{\leq 0} K$ and $F = \mathcal{H}^0 K$, and let A be the fibre of $K' \to F/F_{\mathrm{p}}$. Then the cohomology of A is that of K below zero, and $\mathcal{H}^0 A = F_{\mathrm{p}}$, so that $A \in {}^{p}D^{\leq 0}$. By the octahedral axiom, applied to $A \to K' \to K$, we obtain a triangle $A \to K \to B \to$, in which the cone B is an extension of $\tau^{\geq 1} K$ by F/F_{p} in degree zero, so it has no negative ordinary cohomology and $\mathcal{H}^0 B = F/F_{\mathrm{p}}$. Since F/F_{p} has no punctual subsheaf by Lemma 16.10, $B \in {}^{p}D^{\geq 1}$. Finally, the t-structure is bounded because the ordinary one is.

The heart of any t-structure is abelian, with kernels and cokernels computed from the perverse cohomology of a cone [BBD82, Théorème 1.3.6]; the proof uses only the octahedral axiom and orthogonality. Punctual sheaves in degree zero clearly lie in the heart. So does $(j_*M)[1]$, since j_*M has no punctual subsheaf, as follows from Lemma 16.10 applied to $F = j_*M$, for which $F_{\mathrm{p}} = 0$. ∎

Unwinding the definition, a perverse sheaf on C is a complex in ordinary degrees $-1, 0$ whose $\mathcal{H}^0$ is punctual and whose $\mathcal{H}^{-1}$ has no nonzero punctual subsheaf.

Lemma 16.13 (Perverse Weil sheaves)**.** *The perverse Weil sheaves on C_0 form an abelian category, and the functor forgetting the Weil structure, to perverse sheaves on C, is exact and reflects exactness.*

Proof. Frobenius pullback is an exact equivalence for the ordinary t-structure, and it preserves punctual support and the absence of punctual subsheaves. It is therefore t-exact for the perverse t-structure, which means that it carries the kernel and cokernel of a compatible morphism of perverse sheaves to the kernel and cokernel of the pulled-back morphism. The Weil isomorphisms and their inverses therefore induce inverse isomorphisms on the kernel and cokernel, and by naturality the coimage–image map is compatible with them. So perverse Weil sheaves form an

abelian category, with kernels and cokernels computed in perverse sheaves on C, and this is the assertion about the forgetful functor. ■

An exact equivalence of abelian categories sends simple objects to simple objects. Once we know that the transform is exact on the heart, the following list of simple objects will tell us what the transform of a middle extension can be.

Lemma 16.14 (The simple objects)**.** *In the category of perverse sheaves on C, and in that of perverse Weil sheaves on C_0, the simple objects are the irreducible punctual objects and the objects $(j_*M)[1]$ with M nonzero irreducible lisse on a dense open, where irreducibility is taken in the geometric or in the Weil category, respectively. These middle extensions have no nonzero punctual subobject or quotient in the perverse heart.*

Proof. Since a middle extension has no punctual subobject or quotient and M is irreducible on the open, the middle extension is simple. Conversely, ordinary truncation splits an arbitrary perverse sheaf into a punctual part and a part of the form $F[1]$, and simplicity forces one of them to vanish.

We first show that $P = (j_*M)[1]$ has no punctual subobject or quotient. The adjunction calculations below are geometric, so they apply equally to compatible Weil morphisms. Let i be the inclusion of a finite set of closed points and A a sheaf on it. A map $P \to i_*A$ is zero, because by adjunction it corresponds to a map from $i^*P = (i^*j_*M)[1]$ to the sheaf A, that is, to a negative Ext. A map in the other direction is also zero, and to see this we first enlarge the boundary. Since M is lisse, j_*M does not change when we replace U by a smaller dense open and M by its restriction, so we may assume that the support of A lies in $Z = C \setminus U$, and we let i now denote the inclusion of Z. Then the adjunction $j^* \dashv Rj_*$ shows that $i^!Rj_*M = 0$. In the triangle

$$j_*M \longrightarrow Rj_*M \longrightarrow \tau^{\geq 1}Rj_*M \longrightarrow$$

the last term is supported on Z, so that $i^!$ agrees with i^* on it. Applying $i^!$ and using $i^!Rj_*M = 0$, we find $i^!j_*M \simeq (i^*\tau^{\geq 1}Rj_*M)[-1]$, which has ordinary cohomology only in degrees at least two. Hence $i^!P$ has ordinary cohomology only in degrees at least one. Therefore $\mathrm{Hom}(i_*A, P) = \mathrm{Hom}(A, i^!P) = 0$.

To see that P is simple, note that on a dense open a perverse subobject of P is a shifted lisse subsheaf of M. By irreducibility it is 0 or everything there, and the difference is then punctual. In detail, consider a perverse exact sequence $0 \to Q \to P \to R \to 0$, restrict it to a smaller dense open on which all ordinary cohomology sheaves are lisse. There $P = M[1]$, and the sequence is a shifted exact sequence of lisse sheaves. Moreover, restriction to a dense open preserves irreducibility, because the fundamental group of a dense open of a normal connected curve surjects onto that of the curve, as in the proof of Theorem 14.26, and the same holds for the Weil groups. Hence Q or R has zero generic restriction. Such a perverse object is punctual. Indeed, its $\mathcal{H}^{-1}$ is then supported on a finite set, and a sheaf without punctual subsheaves with this property is zero, while its $\mathcal{H}^0$ is punctual by definition. Thus Q or R is a punctual subobject or quotient of P, which by the vanishing just proved is zero, and so P is simple.

Conversely, let P be simple. We show that P is punctual or a middle extension of an irreducible lisse sheaf. Ordinary truncation gives a perverse exact sequence

$$0 \longrightarrow (\mathcal{H}^{-1}P)[1] \longrightarrow P \longrightarrow \mathcal{H}^0P \longrightarrow 0.$$

Hence a simple P is either punctual or of the form $F[1]$, where F has no punctual subsheaf. In the second case, choose a dense open U on which F is lisse and put $M = F|_U$. Then the map $F \to j_*M$ is injective, since its kernel is $F_{\mathrm{p}} = 0$ (Lemma 16.10), and its cokernel B is punctual. Rotating the corresponding triangle of sheaves we obtain the perverse exact sequence $0 \to B \to F[1] \to (j_*M)[1] \to 0$, whose last term is nonzero, so $B = 0$ by simplicity. It remains to see that M is irreducible. Suppose that M had a proper nonzero lisse subsheaf M_1. Then the object $j_!M_1[1]$ is perverse, because $j_!M_1 \subset j_*M_1$ has no punctual subsheaf, and the image of $j_!M_1[1] \to P$ in the abelian heart is proper and nonzero, as one sees on U. This contradicts simplicity. Finally, if P is punctual, then by the long exact cohomology sequence its subobjects in the heart are punctual sheaves in degree zero, so P is simple exactly when it is irreducible as a punctual sheaf. ∎

Example 16.15 (Extension by zero is not simple)**.** Let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ and $i : 0 \hookrightarrow \mathbf{A}^1$ be the inclusions over k, and put $\delta_0 = i_*E$. The sheaf $j_!E$ has no punctual subsheaf, as its stalk at the origin is zero, so $j_!E[1]$ is perverse. It is not simple. Rotating the triangle of the exact sequence $0 \to j_!E \to E \to \delta_0 \to 0$ gives the triangle $\delta_0 \to j_!E[1] \to E[1] \to$. Its three terms are perverse, and so

$$0 \longrightarrow \delta_0 \longrightarrow j_!E[1] \longrightarrow E[1] \longrightarrow 0$$

is exact in the perverse heart. Thus, while among sheaves the skyscraper is a quotient of E, in the perverse heart it becomes a subobject of $j_!E[1]$. By contrast, the middle extension of $E|_{\mathbf{G}_m}$ is E itself (Lemma 15.2), and $E[1]$ is simple. On the other hand $(E \oplus \delta_0)[1]$ is not perverse, because $\mathcal{H}^{-1}$ has the punctual subsheaf δ_0, which is the subsheaf F_{p} of Lemma 16.10, the kernel of $E \oplus \delta_0 \to j_*j^*(E \oplus \delta_0) = E$.

16.5 Perverse t-exactness

The transform FT_ψ is t-exact for the perverse t-structure on the line (Proposition 16.16). By the stalk formula (16.2), this comes down to the compactly supported cohomology of sheaves on an affine curve in the extreme degrees zero and two.

Proposition 16.16 (Fourier transform and the perverse heart)**.** *On the geometric affine line,* FT_ψ *is a t-exact equivalence of bounded constructible derived categories for the perverse t-structure of Lemma 16.12, with inverse* $\mathrm{FT}_{\psi^{-1}}(-)(1)$*. Hence it is an exact equivalence of the perverse hearts, and its canonical Frobenius compatibilities induce an exact equivalence of the categories of perverse Weil sheaves.*

Proof. We show that the geometric transform and its inverse preserve ${}^pD^{\leq 0}$. Being mutually inverse, they then preserve ${}^pD^{\geq 0}$ by orthogonality, and the Weil statement follows by Lemma 16.13, since the forgetful functor on perverse Weil sheaves is exact and reflects exactness. For the first step, the key point is that the stalks of the transform are compactly supported cohomology groups on the affine line, in degrees $0, 1, 2$, and that the group in degree two is nonzero only at the finitely many points y for which the twisted sheaf has a geometrically constant quotient.

Right exactness. Since the transform is triangulated and ${}^pD^{\leq 0}$ is stable under extensions and positive shifts, ordinary truncation reduces this to punctual sheaves in degree zero and to objects $G[1]$ with G constructible. Indeed, an object of ${}^pD^{\leq 0}$ is an iterated extension of its shifted ordinary cohomology sheaves $\mathcal{H}^a(K)[-a]$ with $a \leq 0$; the one with $a = 0$ is punctual, and the others are of the form $G[1]$ shifted by a nonnegative amount. Punctual sheaves map into ${}^pD^{\leq 0}$, since by the stalk formula the transform of a geometric skyscraper is a lisse sheaf shifted by one.

For $G[1]$ we show that the transform lies in ordinary degrees $-2, -1, 0$ and that its degree-zero sheaf has finite support. The first assertion follows from the stalk formula and Lemma 15.8. For the second, note that the degree-zero stalk at a closed geometric point y is

$$H^2_c(\mathbf{A}^1, G \otimes \mathcal{L}_{\psi(yx)}).$$

On a dense open where G is lisse with generic representation V, write χ_y for the character of π_1 given by the restriction of $\mathcal{L}_{\psi(yx)}$. By localization we may discard the finitely many remaining points, which contribute only in degree zero, and Lemma 15.7 (whose proof of the formula for H^2_c uses only the geometric curve) shows that this stalk is $(V \otimes \chi_y)_{\pi_1}(-1)$. In particular, it is nonzero only if V has the rank-one quotient χ_y^{-1}, and we show that this happens for only finitely many y. First, the characters χ_y are pairwise distinct. Indeed, if $\chi_y = \chi_{y'}$ with $y \neq y'$, then $\mathcal{L}_{\psi((y-y')x)}$ is trivial on the dense open, hence on the line by the surjectivity of fundamental groups. This is impossible, because its compactly supported cohomology vanishes by Lemma 16.4, whereas the constant sheaf has nonzero top cohomology with compact support. Second, if V has pairwise nonisomorphic rank-one quotients $\chi_1, \dots, \chi_n$, then the induced map $V \to \chi_1 \oplus \cdots \oplus \chi_n$ is surjective. To see this, note that its image is a subrepresentation of a sum of pairwise nonisomorphic simple representations, hence the sum of some of the χ_i, and it maps onto each χ_i. So $n \leq \dim V$, and χ_y^{-1} is a quotient of V for at most $\dim V$ points y. The degree-zero sheaf therefore has only finitely many nonzero closed geometric stalks. Since it is constructible, it has finite support, and the transform of $G[1]$ belongs to ${}^pD^{\leq 0}$.

Exactness. The argument for FT_ψ applies equally to ψ^{-1}, and a Tate twist does not affect supports or degrees. Both FT_ψ and its quasi-inverse $\mathrm{FT}_{\psi^{-1}}(-)(1)$ of Proposition 16.6 therefore preserve ${}^pD^{\leq 0}$. Since ${}^pD^{\geq 0}$ is the right orthogonal of ${}^pD^{\leq -1}$, the two mutually inverse equivalences also preserve ${}^pD^{\geq 0}$. Both functors are therefore t-exact, and they induce exact equivalences of the geometric hearts. With their canonical Frobenius compatibilities and the equivariant inversion isomorphisms of Proposition 16.6, they then give quasi-inverse functors on perverse Weil sheaves, which are exact because the forgetful functors are exact and reflect exactness. ∎

16.6 Middle extensions

Let $j : U_0 \hookrightarrow \mathbf{A}^1_{k_0}$ be a dense open, and let M be an irreducible lisse Weil sheaf on U_0. The argument of Chapter 17 needs the transform of the middle extension of M to be again a middle extension, that is, to come from a lisse sheaf M' on a dense open of the dual line. This holds unless M has a geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, and the hypothesis of the next proposition excludes these. They behave differently: for $a \neq 0$ the sheaf $\mathcal{L}_{\psi(ax)}$ has no cohomology with compact support on the line by Lemma 16.4, and its transform is a skyscraper rather than a lisse sheaf of weight one. Excluding them, however, costs nothing in the application, because a sheaf unramified at infinity has no such constituent other than the constant one, and constant constituents contribute nothing, since $H^1_c(\mathbf{A}^1, \overline{\mathbf{Q}}_\ell) = 0$.

Proposition 16.17 (The Fourier transform of a middle extension)**.** *Suppose that M is nonzero and irreducible, and that none of its geometric irreducible constituents is the restriction of $\mathcal{L}_{\psi(ax)}$ for some $a \in \bar{k}_0$ (the case $a = 0$ being the constant sheaf). Then there are a dense open $j' : U_0' \hookrightarrow \mathbf{A}^{1\vee}_{k_0}$ and a nonzero irreducible lisse Weil sheaf M' on it such that*

$$\mathrm{FT}_\psi((j_* M)[1]) \simeq (j'_* M')[1].$$

Proof. By Fourier inversion, a punctual transform would force every geometric constituent of M to be a linear character sheaf, and the hypothesis excludes this.

By Proposition 16.16, the transform of the simple object $(j_*M)[1]$ is simple, and so by Lemma 16.14 it is either punctual or of the displayed form. Restricting the kernel to $\{a\} \times \mathbf{A}^1$ shows that the inverse transform of a geometric skyscraper at a is $\mathcal{L}_{\psi(-ax)}(1)[1]$. Now, if the transform were punctual, it would geometrically be an iterated extension of skyscrapers at finitely many points a, so by Fourier inversion and exactness $(j_*M)[1]$ would be geometrically an iterated extension of the objects $\mathcal{L}_{\psi(-ax)}(1)[1]$. Restricting to U, we see that every geometric constituent of M would be one of these linear characters, which the hypothesis excludes. Finally, the open U_0' and the sheaf M' are defined over k_0, since we may choose a Weil-stable open on which the constructible transform is lisse. ■

The sheaf M' is determined by M up to isomorphism and shrinking of U_0', since it is the restriction to U_0' of $\mathcal{H}^{-1}\,\mathrm{FT}_\psi((j_*M)[1])$. In the application we check the hypothesis of the proposition with the following criterion, whose proof is a computation of ramification at infinity.

Lemma 16.18 (Sheaves unramified at infinity)**.** *Let M be an irreducible lisse Weil sheaf on a dense open $U_0 \subset \mathbf{A}^1_{k_0}$. If M is unramified at infinity and is not geometrically constant, then no geometric constituent of M is of the form $\mathcal{L}_{\psi(ax)}$ with $a \in \bar{k}_0$.*

Proof. For $a \neq 0$ the sheaf $\mathcal{L}_{\psi(ax)}$ is ramified at infinity, and the trivial character is excluded because M is irreducible and not geometrically constant, its geometric restriction being semisimple. For the first point, write $t = 1/x$ and express the character by the Artin–Schreier equation $u^p - u = c/t$ with $c \neq 0$. Let v be the normalized valuation of the field generated by a root u over the completion $k((t))$ at infinity, and let e be the ramification index, so that $v(t) = e$ and $e \leq p$. Since $v(c/t) = -e < 0$, we must have $v(u) < 0$, so that $v(u^p) = pv(u) < v(u)$ and $v(u^p - u) = pv(u)$. Now $pv(u) = -e$ forces $p \mid e$, and so $e = p$: the cover is totally ramified. Inertia at infinity therefore maps onto the Galois group $\mathbf{F}_p$ of this cover, on which ψ is nontrivial. Since M is unramified at infinity, this proves the first point, because a character that is nontrivial on inertia cannot be a constituent of a representation on which inertia acts trivially.

For the trivial character, we apply Lemma 14.11 to the geometric fundamental group, which is normal in the Weil group: the restriction of M to it is semisimple, and so a trivial geometric constituent gives nonzero geometric invariants. Since this space of invariants is Weil-stable, the irreducibility of M shows that M is geometrically constant. ■

17 The Deligne–Laumon theorem for curves

Let F be a pure lisse Weil sheaf of weight w on a smooth curve C. Why should $H^1_c(C, F)$ have weights at most $w + 1$? For the constant sheaf on a smooth projective curve this is Weil's theorem, proved with the Jacobian, and for an arbitrary F there is no such object to work with. Yet the Deligne–Laumon theorem comes down to exactly this bound. The theorem says that $H^r_c(C, F)$ has weights at most $w + r$ (Theorem 17.15), and only $r = 1$ requires work. Indeed, on an affine curve H^0_c vanishes, while by Lemma 15.7 H^2_c is a twist of the coinvariants of geometric monodromy, pure of weight $w + 2$. For $r = 1$ the bound can be strict: on $\mathbf{G}_m$ the group H^1_c comes from the boundary and has weight 0 (see the example of Chapter 13).

For a Gauss sum the answer takes two steps. The transform $\hat{\chi}(y) = \overline{\chi}(y)g$ has the same absolute value for all $y \neq 0$, and Plancherel finds the common value $\sqrt{q}$. On an open of the affine line, Chapter 16 makes the same two steps available for sheaves. More precisely, let M be irreducible on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$, with no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, and let M' be the lisse sheaf with $\mathrm{FT}_\psi((j_*M)[1]) \simeq (j'_*M')[1]$ (Proposition 16.17). Then

$$H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'}$$

(Corollary 17.6). Since a stalk of a middle extension has weights at most the weight of the lisse sheaf (Proposition 15.5), it suffices to show that M' is pure of weight $w+1$. Here the second step is the easy one, because once M' is pure, the sums of squared traces of Chapter 15 give its weight $w+1$ (Theorem 17.5).

The first step, purity of M', is the deep one. The only purity theorem we have is Theorem 14.26, for real sheaves. But the transform of a real sheaf is not real, since ψ takes nonreal values. For example, over $\mathbf{F}_3$ let M be the Kummer sheaf of the quadratic character χ on $\mathbf{G}_m$, which is real. By the computation of Chapter 16 and the signs in (16.3), the trace of M' at $y \in \mathbf{F}_3^\times$ is $-\chi(y)g$. Here $g = \sum_x \chi(x)\psi(x) = \omega - \omega^2 = 1 + 2\omega$ is a square root of -3, which is not real. In this case, admittedly, a twist by $\mathcal{B}_{1/g}$ makes M' real, but in general no twist is known to do this.

The plan is to realize M', up to a twist, as a quotient of a real sheaf on the dual line, to which that theorem applies. To build it we add a conjugate. First we twist M to weight zero, $M^0 = M \otimes \mathcal{B}_b$. Then we transform $j_!M^0$ with the kernel $\mathcal{K}_\psi$ and $j_!(M^0)^\vee$ with the kernel $\mathcal{K}_{\psi^{-1}}$. At each point the traces of the two transforms are complex conjugate, so their sum is a real sheaf H on the dual line. Since the twist $M' \otimes \mathcal{B}_b$ is a quotient of H (Proposition 17.4), purity of real sheaves then makes M' pure. We use $j_!$ rather than j_* because its transform is a sheaf, and by Lemma 17.1 this transform differs from the transform of j_*M by a lisse sheaf of weights at most w.

Passing from one irreducible M to every pure sheaf needs care, because j_* is not exact in general. For example, the extension $0 \to \overline{\mathbf{Q}}_\ell(1) \to F \to \overline{\mathbf{Q}}_\ell \to 0$ on $\mathbf{G}_m$ given by the Kummer class of the coordinate is lisse, but $j_*F \to j_*\overline{\mathbf{Q}}_\ell$ is not surjective at 0, since the monodromy of F around 0 is a nontrivial unipotent matrix. For sheaves of one weight w, however, exactness holds by a weight gap (Proposition 17.9): by local duality at the boundary points (Lemma 17.7), the connecting map $j_*F_2 \to R^1j_*F_1$ goes from weights at most w to weights at least $w+2$, and so it vanishes.

17.1 Purity of the lisse part

Throughout this section and the next, $j : U_0 \hookrightarrow \mathbf{A}^1_{k_0}$ is a dense open and M is a nonzero irreducible lisse Weil sheaf on U_0, pure of weight w. We assume that no geometric irreducible constituent of M is the restriction of $\mathcal{L}_{\psi(ax)}$ for some $a \in \bar{k}_0$, and we let M' be the lisse Weil sheaf on a dense open $j' : U'_0 \hookrightarrow \mathbf{A}^{1\vee}_{k_0}$ with $\mathrm{FT}_\psi((j_*M)[1]) \simeq (j'_*M')[1]$ (Proposition 16.17).

We build the real sheaf from extensions by zero, because for them we can check reality on the trace function. Indeed, at a point of U_0 the eigenvalues of M^0 and of its dual have absolute value one, and hence they are complex conjugate. At a boundary point, however, the stalk of j_*M^0 is the space of inertia invariants, and the stalk of $j_*(M^0)^\vee$ is the space of invariants of the dual representation, but nothing we know at this stage makes their Frobenius eigenvalues conjugate. With $j_!$, by contrast, both stalks vanish there, and the trace function of the sum of the two transforms is clearly real.

The price is that $j_!M$ is not a simple perverse sheaf, so its transform G is not j'_*M' but an extension of j'_*M' by the transform B' of the boundary. This B' is a lisse sheaf of weights at most w (Lemma 17.1), and in the proof of Proposition 17.4 we use only that G is a sheaf and that it surjects onto j'_*M'.

Lemma 17.1 (Fourier transform of the boundary sequence)**.** *Put $B = j_*M/j_!M$. There are sheaves G and B' on the dual line such that $\mathrm{FT}_\psi(j_!M) = G$, $\mathrm{FT}_\psi(B) = B'[1]$, and the sequence*

$$0 \longrightarrow B' \longrightarrow G \longrightarrow j'_*M' \longrightarrow 0$$

is exact. The sheaf B' is lisse on the entire dual line and has weights at most w.

Proof. The three assertions are of different kinds. That the transform of $j_!M$ is a single sheaf is a vanishing statement for the cohomology of the twisted sheaves $M \otimes \mathcal{L}_{\psi(yx)}$ on U, whereas the exact sequence comes from transforming the boundary sequence $0 \to j_!M \to j_*M \to B \to 0$ term by term. Finally, the bound on B' comes from the stalks of B, which are known from Proposition 15.5.

For G, the stalk formula of Proposition 16.5 says that the stalk of $\mathrm{FT}_\psi(j_!M)$ at y is $R\Gamma_c(U, M \otimes \mathcal{L}_{\psi(yx)})[1]$, so we must show that only H^1_c is nonzero. We apply Lemma 15.7 to the lisse sheaf $M \otimes \mathcal{L}_{\psi(yx)}$ on the affine curve U, with j the identity. It shows that the group in degree zero vanishes and that the group in degree two is a twist of the coinvariants of geometric monodromy on $M \otimes \mathcal{L}_{\psi(yx)}$. But a nonzero coinvariant would be a geometric quotient of M isomorphic to $\mathcal{L}_{\psi(-yx)}$, and our assumption on M excludes such a constituent, so the group in degree two vanishes as well. The shift then places the remaining group H^1_c in degree zero, and the transform is a sheaf G.

The transform of B is computed point by point. After geometric base change, the punctual sheaf B is a finite sum of skyscrapers with coefficient modules, and the transform of each skyscraper is a linear character sheaf tensored with that module and shifted by one. Since a linear character sheaf is lisse on the whole dual line, $\mathrm{FT}_\psi(B) = B'[1]$ with B' lisse.

Applying the Fourier transform to the triangle obtained from $0 \to j_!M \to j_*M \to B \to 0$, and using Proposition 16.17 for the middle term, we obtain a triangle $G \to j'_*M' \to B'[1] \to$. Here G and j'_*M' are sheaves in degree zero, and $B'[1]$ is a sheaf in degree -1. Hence the long exact sequence of cohomology sheaves has only three nonzero terms, and it is the stated exact sequence.

The weights of B' come from those of B, because a character sheaf has weight zero, and the weights of B are at most w by Proposition 15.5. Only at boundary points that are not rational does this need an argument. So let s be a boundary point of degree d and y a rational point of the dual line. The geometric points over s contribute d summands to $B'_{\bar{y}}$, which Frob_y permutes cyclically, and its dth power acts on one summand as Frob_s on $B_{\bar{s}}$ times a root of unity coming from the character sheaf. Every eigenvalue β of Frob_y on these summands therefore satisfies $\beta^d = \zeta\alpha$, with ζ a root of unity and α an eigenvalue of Frob_s. Since $|\iota\alpha| \le q^{dw/2}$, we get $|\iota\beta| \le q^{w/2}$. At a point y of higher degree we replace q by the cardinality of $\kappa(y)$.

For a filtration of B' with pure quotients, choose for each boundary point, over its residue field, a Frobenius-stable full flag of the coefficient module, which exists because $\overline{\mathbf{Q}}_\ell$ is algebraically closed. Each successive quotient is then a line, on which Frobenius acts by one of its eigenvalues. Tensoring with the linear character sheaves and summing over the Frobenius orbit, we obtain a finite Weil filtration of B' whose quotients are pure of the weights of the coefficient eigenvalues, and these weights are at most w. ∎

Example 17.2 (A Kummer sheaf has no boundary term)**.** Let M be the Kummer sheaf of a nontrivial character χ on $U_0 = \mathbf{G}_m$. It is irreducible and pure of weight 0. Since it is tame at infinity, it has no geometric constituent $\mathcal{L}_{\psi(ax)}$, and so it satisfies the hypotheses of this section. Inertia at the origin acts on M through χ, nontrivially, and hence $(j_*M)_{\bar{0}} = 0$ by Lemma 15.2. This means that $j_!M = j_*M$ and $B = 0$, and the lemma gives $G \simeq j'_*M'$. On $\mathbf{F}_q$-points this is visible: by (16.3) the trace of G at $y \neq 0$ is $-\sum_x \chi(x)\psi(xy) = -\overline{\chi}(y)g$, with $g = \sum_x \chi(x)\psi(x)$, and for the quadratic character this is the trace of M' computed at the start of Chapter 17. At $y = 0$, on the other hand, the trace of G is $-\sum_x \chi(x) = 0$, in agreement with $H^1_c(\mathbf{G}_m, M) = 0$.

The argument of Proposition 15.9 also applies to twists by the Fourier kernel, and we need this in the proof of Proposition 17.4.

Lemma 17.3 (Twisting and adding the dual)**.** *Let $\mathcal{A}$ be lisse and pure of weight zero, and let $\mathcal{L}$ be lisse of rank one with local eigenvalues of finite order. Then*

$$(\mathcal{A}\otimes\mathcal{L})\oplus(\mathcal{A}^\vee\otimes\mathcal{L}^{-1})$$

is real and pure of weight zero, and its extension by zero along an open immersion is real as well.

Proof. As in the proof of Proposition 15.9, the local eigenvalues of $\mathcal{A}\otimes\mathcal{L}$ have absolute value one after applying ι, and those of the second summand are their inverses, hence their complex conjugates. The extension by zero is real as well, because its stalks outside the open are zero, and so its local characteristic polynomials there are equal to 1. ∎

In the next proof we use Lemma 17.1 twice. Applying its vanishing argument to M^0 and to its dual, we see that the transform of the real sheaf is a single sheaf. Then, using the surjection $G\to j'_*M'$ of the lemma, we exhibit a twist of M' as a quotient of that real sheaf.

Proposition 17.4 (Purity of the lisse part of the Fourier transform)**.** *The sheaf M' is pure of some real weight v on the whole open where it is lisse.*

Proof. We twist to weight zero and add the dual, as in Proposition 15.9, before taking the transform, so that a twist of M' becomes a quotient of a real sheaf, to which Theorem 14.26 applies.

Choose b as in Proposition 15.9, so that $M^0=M\otimes\mathcal{B}_b$ is pure of weight zero.

On the product of the line and the dual line, consider the sheaf

$$F=(p_1^*j_!M^0\otimes\mathcal{K}_\psi)\ \oplus\ (p_1^*j_!(M^0)^\vee\otimes\mathcal{K}_{\psi^{-1}}).$$

At each closed point, the Frobenius eigenvalues of the second summand are the complex conjugates of those of the first, because the eigenvalues of M^0 have absolute value one and those of the character sheaves are roots of unity, while at the boundary both zero extensions have zero stalks. Hence F is real, as in Lemma 17.3. Moreover, since duality replaces a by $-a$, the dual $(M^0)^\vee$ also has no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, so the argument of Lemma 17.1 shows that the direct image with compact support of F along p_2 is concentrated in degree one. We set $H=R^1p_{2!}F$.

Let y be a point with values in a finite field. Since $Rp_{2!}F=H[-1]$, the trace formula for $Rp_{2!}$ (Corollary 12.26) gives $-t_H(y)=\sum_x t_F(x,y)$. For each x the traces of the two summands of F at (x,y) are complex conjugate, so $t_H(y)$ is real. Over the finite extensions of $\kappa(y)$ the same argument shows that the trace of every power of the local Frobenius is real. These traces are the power sums of the local eigenvalues, and the Newton identities express the coefficients of the characteristic polynomial through the power sums with rational coefficients. Hence the local characteristic polynomials of H have real coefficients, that is, H is real.

The first summand of H is $R^1p_{2!}(p_1^*j_!M\otimes\mathcal{K}_\psi)\otimes\mathcal{B}_b$ by the projection formula for the constant twist, that is, $G\otimes\mathcal{B}_b$ with $G=\mathrm{FT}_\psi(j_!M)$ as in Lemma 17.1. Tensoring the surjection $G\to j'_*M'$ of that lemma with $\mathcal{B}_b$, we see that $M'\otimes\mathcal{B}_b$ is a quotient of a direct summand of H. To apply the purity theorem we restrict to a dense open on which both H and M' are lisse. There $M'\otimes\mathcal{B}_b$ is a lisse quotient of the real lisse sheaf H, and it is irreducible by Lemma 16.14, so by Theorem 14.26 it is pure of some real weight. Removing the twist by $\mathcal{B}_b$, we find that M' is pure of some weight v on that open. Since the open where M' is lisse is a smooth geometrically connected curve, the last assertion of Corollary 15.6 extends the purity to all of it. ∎

17.2 The weight of the Fourier transform

By Proposition 17.4, M' is pure of some weight v, and Plancherel now finds v. Purity had to come first, since the growth exponent ν of Definition 15.10 is the weight only for pure sheaves.

Theorem 17.5 (The weight of the Fourier transform)**.** *The sheaf M' is pure of weight exactly $w+1$.*

Proof. Let v be the weight of M' given by Proposition 17.4. Shifts do not change the squared absolute values of traces, and, if a nonzero lisse sheaf on a dense open of the line is pure of weight u, then its direct image j_* has $\nu = u$ (Corollary 15.15). Applying this to M' and to M, and using the Plancherel formula (Corollary 16.8), we get

$$v = \nu(j'_*M') = \nu(\mathrm{FT}_\psi((j_*M)[1])) = \nu((j_*M)[1]) + 1 = w+1.$$

The first equality holds because M' is pure of weight v, the second is the defining isomorphism of M', the third is Plancherel, and the last holds because M is pure of weight w. ∎

In the example of the chapter opening, M is the Kummer sheaf of the quadratic character over $\mathbf{F}_3$, of weight 0, and the trace of M' at $y \in \mathbf{F}_3^\times$ is $-\chi(y)g$ with $g^2 = -3$. Its absolute value is $\sqrt{3} = 3^{1/2}$, which is weight $1 = w+1$, as the theorem predicts.

The weight of M' bounds H^1_c, because the stalk of the transform at the origin of the dual line is the compactly supported cohomology of j_*M.

Corollary 17.6 (H^1_c as a stalk of the Fourier transform)**.** *We have $H^2_c(\mathbf{A}^1, j_*M) = 0$, and there is a Frobenius-equivariant isomorphism $H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'}$ with the stalk at the origin $0'$ of the dual line. In particular $H^1_c(\mathbf{A}^1, j_*M)$ has weights at most $w+1$.*

Proof. We compute the stalk at $0'$ of $\mathrm{FT}_\psi((j_*M)[1])$ in two ways: by the stalk formula, which gives the cohomology of j_*M, and through the isomorphism with $(j'_*M')[1]$, which gives a stalk of a middle extension.

At the rational origin $0'$ of the dual line, the kernel $\mathcal{K}_\psi$ restricts to the pullback of $\mathcal{L}_\psi$ along the constant map to 0, which is canonically trivial (on traces this is $\psi(0) = 1$). The stalk formula of Proposition 16.5, which is compatible with Frobenius, therefore gives

$$\left(\mathrm{FT}_\psi((j_*M)[1])\right)_{0'} = R\Gamma_c(\mathbf{A}^1, j_*M)[2].$$

We compare this with the stalk of $(j'_*M')[1]$, which is concentrated in degree -1. In degree 0 we therefore see that $H^2_c(\mathbf{A}^1, j_*M) = 0$, and in degree -1 we obtain

$$H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'} = (M'_{\bar\eta})^{I_{0'}},$$

where $I_{0'}$ is the inertia group at the origin $0'$ of the dual line acting on the generic fibre of M', and the last equality is the formula of Lemma 15.2 for the stalk of a middle extension. By Proposition 15.5 the weights of this stalk are at most the weight of M', which is $w+1$ by Theorem 17.5. ∎

17.3 The affine line

To pass from irreducible sheaves to arbitrary ones we show that j_* is exact on sequences of pure sheaves of one weight. The reason is a gap in weights. Let $0 \to F_1 \to F \to F_2 \to 0$ be an exact sequence of lisse sheaves on a dense open U of a curve C. Applying j_* gives an exact sequence $0 \to j_*F_1 \to j_*F \to j_*F_2 \to R^1j_*F_1$, and $R^1j_*F_1$ is supported on $C \setminus U$. Now suppose that F_1 and F_2 are pure of weight w. Then, by Proposition 15.5, the source of the last map has weights at most w at a point of $C \setminus U$. On the other hand, by local duality, which we prove first, the stalk of $R^1j_*F_1$ at such a point is the dual of the invariants of $F_1^\vee$, twisted by (-1), and this space has weights at least $w+2$. Since a Frobenius-equivariant map between spaces with disjoint weights is zero, j_* is exact on such sequences (Proposition 17.9).

The hypothesis of equal weights is needed. In the Kummer extension $0 \to \overline{\mathbf{Q}}_\ell(1) \to F \to \overline{\mathbf{Q}}_\ell \to 0$ of the chapter opening, $F_1 = \overline{\mathbf{Q}}_\ell(1)$ has weight -2 and $F_2 = \overline{\mathbf{Q}}_\ell$ has weight 0, so at the origin the stalks of $j_*\overline{\mathbf{Q}}_\ell$ and of $R^1j_*\overline{\mathbf{Q}}_\ell(1)$ both have weight 0, and the connecting map is indeed nonzero.

Exactness lets us extend Corollary 17.6 from one irreducible sheaf with no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$ to a pure sheaf unramified at infinity (Theorem 17.10) by induction along a composition series. Each factor of the series is either geometrically constant, and then $H^1_c = 0$, or it satisfies the hypotheses of the corollary.

Lemma 17.7 (Local duality at a point of $C \setminus U$)**.** *Let $j_0 : U_0 \hookrightarrow C_0$ be a dense open in a smooth curve over k_0, and let F be a lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaf on U_0. For a geometric point $\bar{s}$ over a closed point $s \in C_0 \setminus U_0$, there is a canonical perfect pairing, equivariant for the local geometric Frobenius at s,*

$$(j_*F^\vee)_{\bar{s}} \otimes (R^1j_*F)_{\bar{s}} \longrightarrow \overline{\mathbf{Q}}_\ell(-1).$$

Proof. The two stalks are cohomology sheaves of the complexes $i^*Rj_*F^\vee$ (in degree zero) and i^*Rj_*F (in degree one) on the boundary. We compute the Verdier dual of the second complex by duality on the curve, and we find the first one, shifted and twisted.

Let $i : Z \hookrightarrow C$ be the inclusion of the whole finite complement, and put $K_F = i^*Rj_*F$. For a lisse sheaf G on U, the adjunction $j^* \dashv Rj_*$ shows that $i^!Rj_*G = 0$, and applying $i^!$ to the localization triangle for Rj_*G we obtain $i^!j_!G \simeq i^*Rj_*G[-1]$. Since duality exchanges Rj_* with $j_!$ and i^* with $i^!$, this formula lets us write the result again in the form i^*Rj_*. Indeed, by Verdier duality on the smooth curve, $\mathbb{D}_U F = F^\vee(1)[2]$ for the lisse sheaf F, and we find

$$\begin{aligned}\mathbb{D}_Z K_F &\simeq i^!\mathbb{D}_C Rj_*F \simeq i^!j_!\mathbb{D}_U F \\ &\simeq i^*Rj_*(F^\vee(1)[2])[-1] = K_{F^\vee}(1)[1].\end{aligned}$$

On the finite geometric scheme Z, Verdier duality is the exact duality of coefficient vector spaces, so taking degree -1 at $\bar{s}$ we get $(R^1j_*F)^\vee_{\bar{s}} = (j_*F^\vee)_{\bar{s}}(1)$, which is the pairing. It is Frobenius-equivariant because the adjunction, localization and duality maps are natural in F and respect its Weil structure. ∎

For F pure lisse of weight w, the pairing shows that $(R^1j_*F)_{\bar{s}}$ has weights at least $w+2$. Indeed, it puts this stalk in duality with a stalk of $j_*F^\vee$, which has weights at most $-w$, and the pairing takes values in $\overline{\mathbf{Q}}_\ell(-1)$, which has weight 2. The next proposition uses this gap between the source and the target of the connecting map.

Example 17.8 (Local duality at the origin of the line)**.** Let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ over k_0, let $F = \overline{\mathbf{Q}}_\ell$, which is pure of weight 0, and let $s = 0$. Since inertia at the origin acts trivially on the constant sheaf, $(j_*\overline{\mathbf{Q}}_\ell)_{\bar{0}} = \overline{\mathbf{Q}}_\ell$ by Lemma 15.2, with Frobenius acting by 1, of weight 0. On the other side, the Kummer covers $t \mapsto t^{\ell^n}$ of a punctured neighbourhood of the origin give $(R^1 j_*\overline{\mathbf{Q}}_\ell)_{\bar{0}} = \overline{\mathbf{Q}}_\ell(-1)$ (Chapter 6), on which geometric Frobenius acts by q, of weight 2. The pairing of Lemma 17.7 is the multiplication $\overline{\mathbf{Q}}_\ell \otimes \overline{\mathbf{Q}}_\ell(-1) \to \overline{\mathbf{Q}}_\ell(-1)$. Thus the two stalks have weights 0 and 2, and the gap of two comes entirely from the Tate twist. Indeed, with coefficients $\overline{\mathbf{Q}}_\ell(1)$, H^1 of a punctured disc is spanned by the Kummer class of the coordinate, which Frobenius fixes, so with coefficients $\overline{\mathbf{Q}}_\ell = \overline{\mathbf{Q}}_\ell(1) \otimes \overline{\mathbf{Q}}_\ell(-1)$ Frobenius acts by q.

Proposition 17.9 (Exactness at a fixed pure weight)**.** *Let $j_0 : U_0 \hookrightarrow C_0$ be as in Lemma 17.7. If $0 \to F_1 \to F \to F_2 \to 0$ is a short exact sequence of lisse Weil sheaves on U_0, with F_1 and F_2 pure of the same weight w, then the sequence*

$$0 \longrightarrow j_*F_1 \longrightarrow j_*F \longrightarrow j_*F_2 \longrightarrow 0$$

is exact.

Proof. Applying j_* to the short exact sequence gives the long exact sequence

$$0 \to j_*F_1 \to j_*F \to j_*F_2 \xrightarrow{\delta} R^1 j_*F_1,$$

so it suffices to show that the connecting map δ vanishes. It is Frobenius-equivariant, and we show that its source and target have disjoint weights. Since $R^1 j_*F_1$ is zero on U, we look at a point of $C \setminus U$. There the stalk of the source has weights at most w by Proposition 15.5. For the target, note that the dual $F_1^\vee$ is pure of weight $-w$, so the same result shows that the stalk of $j_*F_1^\vee$ has weights at most $-w$. By Lemma 17.7 the stalk of the target is the dual of this stalk twisted by (-1), and hence the target has weights at least $w + 2$. Now the image of δ is a quotient of the source and a submodule of the target, so its eigenvalues would have weights both at most w and at least $w + 2$. Hence δ vanishes at each point of $C \setminus U$. As $R^1 j_*F_1$ is zero on U, it follows that $\delta = 0$. ∎

All factors of a composition series of a pure lisse sheaf are pure of the same weight, so the proposition applies to every step of such a series. In this way we pass from the irreducible sheaves of Corollary 17.6 to arbitrary ones.

Theorem 17.10 (The weight bound on the affine line)**.** *Let $j : U_0 \hookrightarrow \mathbf{A}^1_{k_0}$ be a dense open, and let F be a lisse sheaf on U_0, pure of weight w and unramified at infinity. Then $H^1_c(\mathbf{A}^1, j_*F)$ has weights at most $w + 1$.*

Proof. We prove the bound for the irreducible factors of F and pass to F by induction. The irreducible factors are of two kinds: geometrically constant ones, for which H^1_c vanishes, and the others, to which Corollary 17.6 applies.

Take a finite composition series of the finite-dimensional Weil representation of F. Each irreducible lisse factor is pure of weight w, since its local eigenvalues are eigenvalues of a subquotient. It is again unramified at infinity, since inertia at infinity acts trivially on every subquotient of a representation on which it acts trivially. An irreducible factor that is geometrically constant is a sheaf $\mathcal{B}_b$ of Proposition 13.7, because the geometric action is trivial and an invertible linear operator over $\overline{\mathbf{Q}}_\ell$ has an eigenline. Its middle extension is then constant on the affine line, and so its H^1_c vanishes by the trivial case of Lemma 16.4. Every other factor F_i is irreducible, unramified

at infinity and not geometrically constant, so by Lemma 16.18 it has no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, and we may apply Corollary 17.6 with $M = F_i$.

The induction needs j_* to turn the composition series into a filtration of j_*F whose quotients are the j_*F_i. Since all factors are pure of the same weight w, Proposition 17.9 shows that the short exact sequences of the series remain exact after applying j_*. So consider a step $0 \to F' \to F'' \to F_i \to 0$ of the series, where we may assume that the bound holds for F'. In the resulting long exact sequence of compactly supported cohomology, the image of $H^1_c(\mathbf{A}^1, j_*F')$ in $H^1_c(\mathbf{A}^1, j_*F'')$ is a quotient of the former, and the cokernel embeds in $H^1_c(\mathbf{A}^1, j_*F_i)$. Since the bound $w+1$ is preserved under subquotients and extensions, we conclude by induction that it holds for F. ∎

The condition at infinity enters only through Lemma 16.18, to verify the hypothesis of Proposition 16.17: it excludes the constituents $\mathcal{L}_{\psi(ax)}$ with $a \neq 0$, whose transforms are skyscrapers rather than middle extensions. In the last section we remove this hypothesis, together with the restriction to opens of the line.

17.4 Arbitrary curves

Theorem 17.10 assumes that the curve is an open of the affine line and that the sheaf is unramified at infinity. We remove the second hypothesis by moving a point where the sheaf is lisse to infinity, and the first by taking the direct image along a finite map to the line.

For an upper bound it is enough to replace H^1_c by a group that surjects onto it, and each move does this. For the first move, let s be a point where the sheaf is lisse. Since a point has no H^1_c, deleting s gives a surjection $H^1_c(W \setminus \{s\}, F) \to H^1_c(W, F)$, and after a finite extension of the constant field and an automorphism of $\mathbf{P}^1$ sending s to infinity, the sheaf is unramified there. For the second, a finite map $\pi : C \to \mathbf{A}^1$ does not change compactly supported cohomology at all, since $R\Gamma_c(\mathbf{A}^1, \pi_* j_! F) = R\Gamma_c(C, F)$. However, $\pi_* j_! F$ is lisse and pure only off the branch locus, so we remove the bad points by one more localization. Such a map π exists by Riemann–Roch, because a function with a single pole of order prime to p defines a finite generically étale map (Lemma 17.13). Curves that are not affine, and the degrees $r \neq 1$, are treated afterwards in Theorem 17.15.

Lemma 17.11 (Weight bounds and finite constant extensions)**.** *Let X_0 be a separated scheme of finite type over k_0, let F be a constructible Weil sheaf on X_0, let $k_m \supset k_0$ be the extension of degree m, and let $u \in \mathbf{R}$. The group $H^r_c(X, F)$ is the same for X_0 and for $X_0 \otimes_{k_0} k_m$, and the geometric Frobenius of k_m acts on it as the mth power of that of k_0. Every eigenvalue α of the Frobenius of k_0 on $H^r_c(X, F)$ satisfies $|\iota\alpha| \leq q^{u/2}$ if and only if every eigenvalue β of the Frobenius of k_m satisfies $|\iota\beta| \leq q^{mu/2}$.*

Proof. An extension of degree m of the constant field does not change the geometric scheme X, the sheaf on it, or its cohomology, and it replaces the Frobenius of k_0 by its mth power. So the eigenvalues β are the α^m, and $|\iota\alpha^m| \leq q^{mu/2}$ holds if and only if $|\iota\alpha| \leq q^{u/2}$. ∎

To move a point to infinity we need a rational point, and hence a finite extension of the constant field. By the lemma this does not affect a weight bound.

Lemma 17.12 (Moving a lisse point to infinity)**.** *Let W_0 be a dense open of $\mathbf{P}^1_{k_0}$, and let F be a lisse Weil sheaf on W_0, pure of weight w. There are a finite extension k_m of k_0, a dense open V_0 of $\mathbf{A}^1_{k_m}$, a lisse Weil sheaf G on V_0, pure of weight w and unramified at infinity, and a surjection*

$$H^1_c(V, G) \twoheadrightarrow H^1_c(W, F)$$

that commutes with the geometric Frobenius of k_m.

Proof. We delete a rational point where F is lisse, which replaces H^1_c by a group surjecting onto it, and move that point to infinity by an automorphism of $\mathbf{P}^1$. First choose a closed point of W_0 and a finite extension k_m of the constant field over which it has a rational point s. The restriction of F to $W_0 \otimes_{k_0} k_m$ is again pure of weight w, because a residue extension raises an eigenvalue and the residue cardinality to the same power. Next, some element of $\mathrm{PGL}_2(k_m)$ sends s to infinity, and we transport $F|_{W\setminus\{s\}}$ by this automorphism. As F is lisse at s, its local inertia there is trivial, so the transported sheaf G extends to a lisse sheaf across the new infinity. It is defined on the image V_0 of $W_0 \setminus \{s\}$, a dense open of the affine line, and $H^1_c(V, G) = H^1_c(W \setminus \{s\}, F)$ because the automorphism identifies the two pairs. The surjection follows from the localization sequence

$$H^1_c(W \setminus \{s\}, F) \to H^1_c(W, F) \to H^1_c(\{s\}, F_{\bar{s}}) = 0,$$

in which the last group vanishes because a point has no cohomology in positive degrees. Since every construction is defined over k_m, the map commutes with the Frobenius of k_m. ∎

By the lemma, for any open of the projective line and any pure lisse sheaf on it, we may bound H^1_c with Theorem 17.10. To reach an arbitrary affine curve we map it to the line, and we have to choose the map so that the direct image is again pure and lisse on a dense open.

Lemma 17.13 (A finite generically étale map to the line)**.** *Let C_0 be a smooth geometrically connected affine curve. There are a smooth curve C_0' containing C_0 with finite complement and a finite generically étale map $\pi : C_0' \to \mathbf{A}^1_{k_0}$. If F is a pure lisse sheaf of weight w on C_0, then the sheaf $K = \pi_* j_! F$, where $j : C_0 \hookrightarrow C_0'$, is lisse and pure of weight w outside the branch locus and the image of $C_0' \setminus C_0$. Moreover, $R\Gamma_c(\mathbf{A}^1, K) = R\Gamma_c(C, F)$ canonically.*

Proof. The map is given by a function with a single pole, at a point at infinity, and we choose the order of this pole prime to p so that the map is generically étale. We find the function by Riemann–Roch, we show by a local computation at the pole that its differential is nonzero, and we check the assertions about K stalk by stalk.

Let $\overline{C}_0$ be the smooth projective completion, choose a closed point s of $\overline{C}_0$ outside C_0, and choose m prime to p with $(m-1)\deg(s) > 2g(\overline{C}_0) - 2$. Both divisors ms and $(m-1)s$ then have degree $> 2g-2$, so Riemann–Roch (Theorem C.3) gives $h^0(\mathcal{O}(ns)) = n\deg(s) + 1 - g$ for $n = m, m-1$, and hence

$$h^0(\mathcal{O}(ms)) - h^0(\mathcal{O}((m-1)s)) = \deg(s) > 0.$$

Therefore there is a function f whose only pole is at s, of exact order m. The sections $1, f$ define a nonconstant map $\overline{C}_0 \to \mathbf{P}^1$ whose inverse image of infinity is $\{s\}$, and this map is proper with finite fibres, hence finite. To see that it is generically étale, choose a uniformizer t at s and write $f = t^{-m}u$ with u a unit. Since $\kappa(s)/k_0$ is separable and the curve is smooth, dt generates the module of differentials near s. Writing $du = u'\,dt$, we get

$$df = t^{-m-1}(-mu + tu')\,dt \neq 0,$$

since $-mu + tu'$ is congruent to the unit $-mu$ modulo t, and mu is a unit because $p \nmid m$. The differential of this finite map between smooth curves is therefore invertible at the generic point, and the map is generically étale. We set $C_0' = \overline{C}_0 \setminus \{s\}$ and let π be the restriction of the map to C_0'.

Over the open in the statement, π is finite étale and its inverse image lies in C_0, where $j_!F = F$. Moreover, the direct image of a lisse sheaf under a finite étale map is lisse, because étale locally the map is a disjoint union of copies of the base. Hence K is lisse there. For purity, note that at a closed point y of this open, the stalk of K is a sum over the geometric points of the fibre, which Frobenius permutes. Consider such a cycle of length e, corresponding to a point over y with residue extension of degree e. On it the eth power of the operator is the local Frobenius upstairs, so an eigenvalue β satisfies $|\iota\beta|^e = q_y^{ew/2}$, and the weight is still w. Since $R\pi_! = \pi_*$ for the finite map π, the identification of cohomology follows by composing direct images with compact support. It commutes with Frobenius because the construction is defined over k_0. ∎

For example, take for C_0 the affine curve $y^2 = x^3 - x$ over $\mathbf{F}_3$, the complement of the point at infinity of the projective curve. There $g = 1$, the point $s = \infty$ is rational, and $m = 2$ satisfies $(m-1)\deg(s) > 0$. Accordingly, the function x has a single pole, of order two, which is prime to 3, and the map is the double cover $(x, y) \mapsto x$ of the line, whose extension to the projective curve is branched over 0, ± 1 and ∞.

With both lemmas the bound on the line extends to every affine curve.

Theorem 17.14 (The weight bound on an affine curve)**.** *For a pure lisse Weil sheaf F of weight w on a smooth geometrically connected affine curve C_0, the group $H^1_c(C, F)$ has weights at most $w + 1$.*

Proof. We push the sheaf forward to the line and realize H^1_c as a quotient of a group bounded by Theorem 17.10. This uses two surjections: one removes the points where the direct image fails to be lisse and pure, and the other moves a lisse point to infinity.

Choose π as in Lemma 17.13 and put $K = \pi_* j_! F$. Let W_0 be the dense open on which K is lisse and pure. Since its complement is finite, localization gives

$$H^1_c(W, K|_W) \twoheadrightarrow H^1_c(\mathbf{A}^1, K) = H^1_c(C, F).$$

By Lemma 17.12, after a finite extension of the constant field there is a further surjection from $H^1_c(V, G)$ onto $H^1_c(W, K|_W)$, with G pure of weight w and unramified at infinity. Let $v : V \hookrightarrow \mathbf{A}^1$ be the inclusion and $i : Z \hookrightarrow \mathbf{A}^1$ its finite complement. The localization sequence for v_*G reads

$$H^0_c(Z, i^* v_* G) \longrightarrow H^1_c(V, G) \longrightarrow H^1_c(\mathbf{A}^1, v_* G) \longrightarrow 0.$$

The first group has weights at most w. Indeed, the stalks of $i^* v_* G$ at the closed points of Z_0 have weights at most w by Proposition 15.5, and $H^0_c(Z, i^* v_* G)$ is the direct sum, over the closed points z of Z_0, of the stalks at the $e = [\kappa(z) : k_m]$ geometric points over z, where k_m is the constant field of V_0. The geometric Frobenius of k_m permutes these e stalks cyclically, and its eth power on each of them is conjugate to the local Frobenius F_z. An eigenvalue β on such a cycle therefore satisfies $\beta^e = \alpha$ for an eigenvalue α of F_z, so that $|\iota\beta|^e = |\iota\alpha| \le q_z^{w/2} = (q^m)^{ew/2}$, and $|\iota\beta| \le (q^m)^{w/2}$, which is weight at most w over k_m. This is the case of the finite map $Z_0 \to \operatorname{Spec} k_m$ in Lemma 18.1, whose proof is the same computation. The last group has weights at most $w + 1$ by Theorem 17.10. As an extension of a subgroup of the last by a quotient of the first, the middle group therefore has weights at most $w + 1$. This bound passes to quotients under the two surjections, and by Lemma 17.11 it holds over the original constant field. ∎

To reach a curve that is not affine, we pass to a dense affine open and view the sheaf on the whole curve as a middle extension.

Theorem 17.15 (The weight bound on a curve)**.** *Let $j : U_0 \hookrightarrow C_0$ be a dense open in a smooth curve over k_0, and let F be a pure lisse Weil sheaf of weight w on U_0. Then for every $r \in \mathbf{Z}$ the group $H^r_c(C, j_*F)$ has weights at most $w + r$. In particular, for $U_0 = C_0$, the group $H^r_c(C, F)$ has weights at most $w + r$.*

Proof. On a geometrically connected curve we compare j_*F with its restriction to a dense affine open, where the affine case applies, and we deduce the bounds from the localization sequence. The general case reduces to this one, because a curve that is not geometrically connected becomes a disjoint union of geometrically connected ones after a finite constant extension.

Suppose first that C_0 is geometrically connected, and choose a dense affine open $v : V_0 \hookrightarrow U_0$. Since F is lisse on U_0, the unit $F \to v_*F|_V$ is an isomorphism by the last assertion of Lemma 15.2. Hence j_*F is also the middle extension of $F|_V$ from V_0 to C_0. On V, by Theorem 17.14 and Lemma 15.7, $H^r_c(V, F)$ has weights at most $w + r$ for every r, and $H^0_c(V, F) = 0$. On the complement, write $Z_0 = C_0 \setminus V_0$ and i for its inclusion; then the restriction of j_*F to the finite set Z_0 has weights at most w by Proposition 15.5, and its cohomology is concentrated in degree zero. These two pieces are joined by the localization sequence for $V \subset C$, which reads

$$0 = H^0_c(V, F) \to H^0_c(C, j_*F) \to H^0(Z, i^*j_*F) \to H^1_c(V, F) \to H^1_c(C, j_*F) \to 0,$$

followed by an isomorphism $H^2_c(V, F) \simeq H^2_c(C, j_*F)$, and all higher groups vanish. Over $\operatorname{Spec} k_0$ the bounds are conditions on eigenvalues (Lemmas 13.2 and Lemma 13.3), and we read them off degree by degree. In degree zero, the group $H^0_c(C, j_*F)$ injects into $H^0(Z, i^*j_*F)$, which has weights at most w: Frobenius permutes the stalks at the geometric points over each closed point of Z_0 cyclically, with the local Frobenius as the power that returns to a stalk, as in the proof of Theorem 17.14. In degree one, $H^1_c(C, j_*F)$ is a quotient of $H^1_c(V, F)$, which has weights at most $w + 1$. In degree two the groups are isomorphic, and the bound $w + 2$ holds for V.

In general, the connected components of C are defined over a finite extension k_m of k_0, and over k_m each of them is a smooth geometrically connected curve. For example, $\mathbf{P}^1_{\mathbf{F}_{q^2}}$, viewed as a curve over $\mathbf{F}_q$, is connected, but C has two components, which Frobenius exchanges and which are defined over $\mathbf{F}_{q^2}$. As Frobenius permutes the components, only its mth power acts on the cohomology of each of them. On such a component, j_*F is the middle extension of the restriction of F to the dense open where the component meets U, and this restriction is pure of weight w. Moreover, the group $H^r_c(C, j_*F)$ is the direct sum of the groups of these components, compatibly with the mth power of Frobenius. Hence the bound holds over k_m by the first case, and by Lemma 17.11 it holds over k_0. ∎

18 The Deligne–Laumon theorem in general

At this point we know the Deligne–Laumon theorem on curves: a pure lisse sheaf of weight w on a curve has H^r_c of weights at most $w + r$. For the Weil conjectures we need the same bound for the cohomology of a variety of any dimension. In this chapter we prove Theorem 13.11, which says that for a separated morphism of finite type $f : X_0 \to Y_0$, the functor $Rf_!$ maps $D_{\leq w}(X_0)$ into $D_{\leq w}(Y_0)$.

Take a surface fibred in curves over a curve, $f : X_0 \to Y_0$. Here the Leray spectral sequence $H^a_c(Y, R^b f_! \overline{\mathbf{Q}}_\ell) \Rightarrow H^{a+b}_c(X, \overline{\mathbf{Q}}_\ell)$ suggests applying the curve case twice, once on the fibres and

once on the base. On the fibres there is no difficulty: at a closed point y the stalk of $R^b f_! \overline{\mathbf{Q}}_\ell$ is $H^b_c(X_{\bar{y}}, \overline{\mathbf{Q}}_\ell)$, and the curve case says that it has weights at most b. On the base, however, the curve case applies only to pure sheaves. So we need a finite filtration of $R^b f_! \overline{\mathbf{Q}}_\ell$ with pure quotients, that is, we need $R^b f_! \overline{\mathbf{Q}}_\ell$ to be mixed in the sense of Definition 13.1, and bounds at each point separately do not produce such a filtration. In his first proof [Del74] Deligne avoided the problem with a Lefschetz pencil, where the Picard–Lefschetz formula describes the direct image in the middle degree, but for an arbitrary fibration the sheaves on the base are not explicit. Once a mixed filtration exists, on the other hand, the pointwise bound also bounds the weights of its quotients (Lemma 13.3). The question is therefore whether the direct images are mixed.

The answer comes once more from positivity (Theorem 18.8):

> every constructible real Weil sheaf on a separated scheme of finite type over k_0 is mixed.

To see why, let S be an irreducible constituent of a real lisse sheaf on a smooth affine X_0. In order to compare the eigenvalues of S at a fixed point x_* with those at an arbitrary point x, we restrict to a curve through both. On the curve each irreducible constituent is pure by Theorem 14.26, so if the restriction of S stays irreducible, its eigenvalues at x have the weight read off at x_*. In order to find curves on which S stays irreducible, we first show that irreducibility can be detected in one finite quotient of the fundamental group (Lemma 18.2), and then Bertini's theorem gives a curve through the given points that stays connected in the corresponding finite étale cover (Lemma 18.3 and Corollary 18.4). For constructible sheaves, finally, we stratify.

By Noether normalization over strata (Lemma 18.14), an arbitrary f is built from quasi-finite maps and projections of relative affine lines. The essential case is a smooth family of affine curves $h : U_0 \to Y_0$ with a pure lisse sheaf F on U_0. To apply the theorem on real sheaves we need direct images that are real, and here the curve case helps a second time. After a twist to weight zero and the addition of the dual, F becomes a real sheaf G, and the curve case on the fibres makes the direct images of G real, hence mixed. A twist of F is a direct summand of G, so the direct images of F are mixed as well (Theorem 18.9).

18.1 Quasi-finite morphisms and curves

To prove Theorem 18.8 we restrict a sheaf to curves, reassemble it over strata and descend it along constant extensions. For the reassembly and the descent we need that quasi-finite direct images preserve weights. For the restriction we need curves on which given lisse sheaves stay geometrically irreducible.

The first fact is elementary, because a finite direct image only permutes stalks. Let y be a point downstairs and x a point over y with residue extension of degree e. Frobenius at y cycles the e copies of the stalk at x, and its eigenvalues are eth roots of the eigenvalues upstairs. Measured against the residue field, absolute values are therefore unchanged (Lemma 18.1).

The second fact is geometric, since restricting to a curve can lose monodromy. On $\mathbf{G}_m \times \mathbf{G}_m$ with coordinates x, y, in odd characteristic, the double cover $u^2 = x$ is connected, but over the curve $x = 1$ it splits into two copies. Accordingly, the Kummer sheaf of the quadratic character in x is geometrically nontrivial on the surface but constant on that curve. In the same way an irreducible sheaf of higher rank can become reducible on a badly chosen curve. On the other hand, a representation stays absolutely irreducible on a curve if the image of the fundamental group of the curve is large enough, and we show that this can be checked in one finite quotient (Lemma 18.2). It then suffices that the curve stays connected in one finite étale cover, and we find such curves with Bertini's irreducibility theorem (Lemma 18.3).

Lemma 18.1 (Finite and quasi-finite direct images)**.** *Let $g : Z_0 \to Y_0$ be separated and quasi-finite. Then $Rg_!$ is exact on sheaves and carries mixed sheaves of weights at most w to mixed sheaves of weights at most w. It also carries $D_{\leq w}(Z_0)$ into $D_{\leq w}(Y_0)$. For a finite map, direct image carries a pure sheaf of weight w to a pure sheaf of weight w.*

Proof. We first treat a finite map and then reduce to it by Zariski's main theorem, which writes a quasi-finite map as an open immersion followed by a finite map. All the computation is in the finite case, since extension by zero does not change any nonzero stalk.

For a finite map, exactness follows from Lemma D.1 and the comparison of coefficients. The same lemma shows that the stalk at a geometric point $\bar{y}$ over a closed point y is the direct sum of the stalks at the geometric points of Z_0 over $\bar{y}$, and the local Frobenius at y permutes these summands. Let z be a point of residue degree $e = [\kappa(z) : \kappa(y)]$. Frobenius cycles through the e summands over z, and its eth power on each of them is conjugate to the local Frobenius at z. Each eigenvalue β on this cycle therefore satisfies $\beta^e = \alpha$ for an eigenvalue α at z. If the input has weight w, then

$$|\iota(\beta)|^e = |\iota(\alpha)| = q_z^{w/2} = q_y^{ew/2},$$

and we find $|\iota(\beta)| = q_y^{w/2}$. This proves that finite direct image preserves purity. For example, for the constant sheaf on $\operatorname{Spec} \mathbf{F}_{q^2}$ over $\operatorname{Spec} \mathbf{F}_q$, the direct image has rank two and Frobenius swaps the two summands. Its eigenvalues ± 1 have weight 0, as the constant sheaf does.

In general, Y_0 is Noetherian, hence quasi-compact and quasi-separated, and Zariski's main theorem in the form [Stacks, 05K0] applies. It factors g as an open immersion $j : Z_0 \hookrightarrow \overline{Z}_0$ followed by a finite map $\bar{g} : \overline{Z}_0 \to Y_0$. By Proposition 5.28 we have $Rg_! = \bar{g}_* j_!$, and both functors are exact. Extension by zero preserves purity, since its other stalks are zero. Now take a mixed sheaf of weights at most w on Z_0, with a finite filtration by Weil subsheaves whose quotients are pure of weights at most w. Since j and $\bar{g}$ are defined over k_0, the functors $j_!$ and $\bar{g}_*$ take Weil subsheaves to Weil subsheaves. Being exact, they take the filtration to a finite filtration of the direct image whose quotients are the direct images of the pure quotients. By the first part these are pure of the same weights, which proves the assertion on sheaves. Since $Rg_!$ is exact, it commutes with taking cohomology sheaves, and the assertion on $D_{\leq w}$ follows. ∎

A lisse sheaf stays geometrically irreducible on a curve C in X as soon as the image of $\pi_1(C)$ is large enough. The next lemma shows that it suffices for this image to surject onto one finite quotient of $\pi_1(X)$, and in Lemma 18.3 we construct curves with this property.

Lemma 18.2 (Detecting irreducibility in a finite quotient)**.** *Let Π be profinite and let $\rho : \Pi \to \mathrm{GL}_E(V)$ be continuous and absolutely irreducible, with $\dim_E V = n$. There are a Π-stable lattice $L \subset V$ and an integer N such that every subgroup $\Pi' \subseteq \Pi$ with the same image as Π in $\mathrm{GL}(L/\pi_E^N L)$ acts absolutely irreducibly on V.*

Proof. Recall Burnside's theorem: an algebra of endomorphisms of a finite-dimensional vector space over an algebraically closed field that acts irreducibly is the full endomorphism algebra. Absolute irreducibility thus means that the group spans the full matrix algebra. We detect spanning by a nonzero determinant of n^2 group elements. Such a determinant stays nonzero when we change the elements by a congruence modulo a high power of π_E.

The compact group $\rho(\Pi)$ stabilizes a lattice L. Indeed, the stabilizer of any lattice L_0 is open, hence of finite index in $\rho(\Pi)$. So L_0 has only finitely many translates, and their sum is a $\rho(\Pi)$-stable lattice. By absolute irreducibility and Burnside's theorem, there are elements $g_1, \dots, g_{n^2}$

whose matrices span $\mathrm{End}_E(V)$. In a basis of L, the determinant Δ of their matrices, written as column vectors, is nonzero and integral. Choose $N > v_{\pi_E}(\Delta)$. Suppose that $g'_1, \dots, g'_{n^2}$ have matrices congruent to those of the g_i modulo π_E^N. Since the determinant is a polynomial with integer coefficients in the entries, their determinant Δ' satisfies $\Delta' \equiv \Delta \pmod{\pi_E^N}$. Hence $v_{\pi_E}(\Delta') = v_{\pi_E}(\Delta)$, and in particular $\Delta' \neq 0$. A subgroup as in the statement contains such elements. Its matrices therefore span the whole matrix algebra, also after any field extension, and by Burnside's theorem it acts absolutely irreducibly. Finally, the image of Π in $\mathrm{GL}(L/\pi_E^N L)$ is a finite quotient of Π, since by continuity the kernel is open. ∎

For a lisse sheaf on X the finite quotient corresponds to a connected finite Galois cover $T \to X$. If $C \subset X$ is a curve with $T \times_X C$ connected, then the fundamental group of C surjects onto the Galois group of T. What we need, then, are curves through given points whose inverse image in a given connected cover stays connected.

Lemma 18.3 (A curve preserving a finite cover)**.** *Let X be a smooth connected affine scheme over $\overline{\mathbf{F}}_q$, of dimension at least two. Let $T \to X$ be a connected finite étale cover and let $S \subset X$ be a nonempty finite set of closed geometric points. There is a smooth connected closed affine curve $C \subset X$ containing S, with $T \times_X C$ connected. The curve and its inclusion are defined over a finite field.*

Proof. We cut X by a general hypersurface through S, show that it is smooth and that both it and its inverse image in T are connected, and then repeat.

The linear system. Embed X in a projective closure $\overline{X}$. For large n, the sections of $\mathcal{O}_{\overline{X}}(n)$ vanishing on S have base locus exactly S on X and define an immersion off S ([Stacks, 0FD5]). They also surject onto the first-order jets $\bigoplus_{s \in S} \mathfrak{m}_s/\mathfrak{m}_s^2 \otimes \mathcal{O}_{\overline{X}}(n)$ at S. Indeed, if $\mathcal{I}$ is the ideal sheaf of S, then $H^1(\overline{X}, \mathcal{I}^2(n)) = 0$ for large n by Serre vanishing ([Stacks, 0B5U]), and the cohomology sequence of $0 \to \mathcal{I}^2(n) \to \mathcal{I}(n) \to (\mathcal{I}/\mathcal{I}^2)(n) \to 0$ shows that $H^0(\overline{X}, \mathcal{I}(n))$ surjects onto the jets.

Smoothness. A general divisor of these sections is smooth on X. The immersion off S says nothing about the points of S themselves, and there smoothness is controlled by the jets. A divisor through s is smooth at s exactly when its equation has nonzero image in $\mathfrak{m}_s/\mathfrak{m}_s^2$. Since the sections surject onto the jets, the condition that a section have nonzero derivative at each s is nonempty and open. On $X \setminus S$ the divisors are the pullbacks of hyperplanes under the immersion of $X \setminus S$ into a projective space that these sections define. By Bertini's theorem ([Stacks, 0FD6]) a general one is smooth there.

Connectedness. A general divisor and its inverse image in T are irreducible. We use Bertini's irreducibility theorem [Stacks, 0G4F]: if Z is an irreducible scheme of finite type over an algebraically closed field and $Z \to \mathbf{P}^N$ is a morphism whose image has closure of dimension at least two, then the inverse image of a general hyperplane is irreducible. We apply it to the immersion of $X \setminus S$ above and to its composition with the restriction of $T \to X$ over $X \setminus S$. In both cases the hyperplanes correspond to the sections vanishing on S and their pullbacks to T. Both sources are smooth and connected, hence irreducible. Both images have closure of dimension $\dim X \geq 2$, because $T \to X$ is finite and surjective. So for a general divisor D, both $D \cap (X \setminus S)$ and its inverse image in T are irreducible. The finitely many points over S cannot be components, since every component of a divisor has dimension $\dim X - 1 \geq 1$, and the claim follows.

Induction. Intersecting these nonempty open conditions in the space of sections, we obtain a smooth connected affine hypersurface H containing S whose inverse image in T is connected. It is smooth, connected and affine of dimension $\dim X - 1$, it contains S, and $T \times_X H \to H$ is a connected finite étale cover. Thus H again satisfies the hypotheses of the lemma, and we repeat

the construction until the dimension is one. The curve is defined over a finite field, because it has finitely many equations, whose coefficients lie in a finite extension of $\mathbf{F}_q$. ■

Combining the two lemmas, we obtain the curves we need for real sheaves. We can treat several sheaves at once, because a single cover dominates the finite quotients attached to all of them.

Corollary 18.4 (A curve preserving geometric irreducibility)**.** *Let X_0 be a smooth geometrically connected affine scheme of positive dimension. Given finitely many geometrically irreducible lisse Weil sheaves on X_0 and finitely many closed geometric points, there is, after a finite extension of the constant field, a smooth geometrically connected affine curve through those points on which all the sheaves remain geometrically irreducible.*

Proof. We translate irreducibility into the surjectivity of $\pi_1(C)$ onto one finite group, and then find C with Bertini's theorem applied to the corresponding cover.

Choose a finite extension E of $\mathbf{Q}_\ell$ over which all the sheaves are defined, and apply Lemma 18.2 to their geometric representations. Let $T \to X$ be the connected Galois cover corresponding to the intersection of the kernels of the resulting finite quotients of $\pi_1(X)$, and let Q be its group. In dimension one, we take $C = X$. Otherwise $\dim X \geq 2$, and we apply Lemma 18.3 to T and the given points. If no points are given we add one, since the lemma needs S nonempty. As $T \times_X C$ is connected, $\pi_1(C)$ acts transitively on the fibre of the Q-torsor $T \times_X C \to C$, so its image in Q is all of Q. Each of the finite quotients above is a quotient of Q. Hence $\pi_1(C)$ has the same image as $\pi_1(X)$ in each of them, and Lemma 18.2, applied with $\Pi = \pi_1(X)$ and Π' the image of $\pi_1(C)$, shows that the restrictions remain geometrically irreducible. By Lemma 18.3 the curve and the points are defined over a finite extension of the constant field. The curve C is geometrically connected because it is connected over $\overline{\mathbf{F}}_q$. ■

18.2 Mixedness of real sheaves

With these curves we pass from the purity theorem for real sheaves on curves (Theorem 14.26) to the mixedness of real sheaves on any base. The curves preserve geometric irreducibility, so we first split a lisse sheaf into geometrically irreducible pieces. These pieces need not be Weil sheaves, because an irreducible Weil sheaf need not be geometrically irreducible. For example, let χ be a character of $\mathbf{F}_{q^2}^\times$ with $\chi^q \neq \chi$, and let g be the map from $\mathbf{G}_m$ over $\mathbf{F}_{q^2}$ to $\mathbf{G}_m$ over $\mathbf{F}_q$. Geometrically, the direct image under g of the Kummer sheaf of χ splits into the Kummer sheaves of χ and χ^q, which are distinct. Frobenius exchanges them, and so the direct image is an irreducible Weil sheaf of rank two. After the quadratic constant extension the two pieces are Weil sheaves themselves. In general, the geometric constituents become Weil subquotients after a finite constant extension (Lemma 18.5). In the proof of Theorem 18.8 we descend mixedness back along this extension by a trace retraction. We also reduce constructible sheaves to lisse sheaves on smooth affine strata, and glue the strata using Lemma 18.1.

Lemma 18.5 (Geometric constituents after a constant extension)**.** *A lisse Weil sheaf on a smooth geometrically connected scheme admits, after one finite constant extension, a finite Weil filtration with geometrically irreducible quotients.*

Proof. An irreducible Weil representation is geometrically semisimple. A power of Frobenius fixes each isotypic component, and on it we can triangularize this power. The flags obtained in this way are stable under the Weil group of the constant extension, and by them we refine a composition series to the required filtration.

In detail, take a composition series for the Weil representation. By Lemma 14.11, applied to the geometric fundamental group, which is normal in the Weil group, the geometric restriction of each irreducible factor is semisimple. Frobenius permutes the finitely many geometric isotypic components of the factor, and we choose a lift s of geometric Frobenius to the Weil group and a power s^m fixing all of them. On a component $S \otimes M$, geometric monodromy acts irreducibly on S and trivially on M. Since s^m fixes the isomorphism type of S, there is an isomorphism u from S to its conjugate by s^m. We claim that the action of s^m on the component has the form $u \otimes v$ with v invertible on M. Indeed, the composite of the action of s^m with $u^{-1} \otimes 1$ commutes with geometric monodromy, and by Schur's lemma the endomorphisms of $S \otimes M$ commuting with it are $1 \otimes \mathrm{End}(M)$. Triangularizing v, we obtain a stable flag whose geometric factors are S.

We take a common m for the finitely many factors. Then the subgroup generated by geometric monodromy and s^m is the Weil group after the constant extension of degree m. Choose a finite extension E' of the coefficient field over which these flags are defined. Since a subspace defined over E' is stable under this group if and only if its extension to $\overline{\mathbf{Q}}_\ell$ is, the flags define lisse Weil subsheaves, and refining the composition series by them gives the required filtration. ■

In the example above, after the quadratic extension the sheaf $g_*\mathcal{K}_\chi$ becomes the direct sum of the Kummer sheaves of χ and χ^q, and either summand gives a filtration as in the lemma. Beyond this example, we use the lemma, together with the curves of Corollary 18.4, to prove that real sheaves are mixed (Theorem 18.8). In that proof, however, the filtrations with pure quotients exist only after a constant extension and only on the pieces of a stratification, so the next two lemmas recover mixedness of the original sheaf.

Lemma 18.6 (Mixedness and finite constant extensions)**.** *Let X_0 be a separated scheme of finite type over k_0, let $g : X_0' = X_0 \otimes_{k_0} k_m \to X_0$ be a finite constant extension, and let G be a constructible Weil sheaf on X_0. If g^*G is mixed, then G is mixed.*

Proof. Since g^*G is mixed, so is g_*g^*G by Lemma 18.1. The unit $G \to g_*g^*G$ has the Weil-equivariant retraction $(\deg g)^{-1}\,\mathrm{Tr}$. Indeed, étale locally the unit is the diagonal and the trace is the sum over the copies. These local maps glue because the sum is invariant under permutations of the copies. So G is a Weil subsheaf of g_*g^*G. A Weil subobject of a mixed sheaf is mixed (Proposition 13.7), and therefore G is mixed. ■

Lemma 18.7 (Mixedness on an open and its complement)**.** *Let $j : U_0 \to X_0$ be an open immersion with closed complement $i : Z_0 \to X_0$, and let G be a constructible Weil sheaf on X_0. If j^*G and i^*G are mixed, then G is mixed.*

Proof. The sheaf G is an extension

$$0 \longrightarrow j_!j^*G \longrightarrow G \longrightarrow i_*i^*G \longrightarrow 0,$$

whose outer terms are mixed by Lemma 18.1, since j and i are quasi-finite. An extension of mixed sheaves is mixed (Proposition 13.7). ■

Theorem 18.8 (Real constructible sheaves are mixed)**.** *Every constructible ι-real Weil sheaf on a separated finite-type k_0-scheme is ι-mixed.*

Proof. We join a fixed point x_* to an arbitrary point x by a curve on which the geometrically irreducible constituents stay irreducible. On that curve each constituent is pure by Theorem 14.26. Its weight can be read off at x_*, so it does not depend on the curve. This handles a lisse sheaf on a smooth affine scheme, and stratifying handles an arbitrary constructible sheaf. Descent along constant extensions and gluing of strata preserve mixedness (Lemmas 18.6 and Lemma 18.7), so nothing about curves is used beyond Theorem 14.26.

The lisse case. Suppose that X_0 is smooth, geometrically connected and affine, and that the sheaf G is lisse. Since the curves of Corollary 18.4 preserve geometric irreducibility, we first split G into geometrically irreducible pieces. To do so, we make the constant extension of Lemma 18.5 and write $S_1, \dots, S_r$ for the quotients of its Weil filtration. Note that the sheaf G remains real after this extension, because the powers of a conjugation-stable multiset of eigenvalues again form a conjugation-stable multiset.

Fix a closed point x_* of X_0, and let x be any other closed point. By Corollary 18.4, there is a curve through chosen geometric points above x_* and x on which every S_j remains irreducible. The restriction of G to this curve is real, and Theorem 14.26 says that the irreducible constituents of a real sheaf on a curve are pure. Since the restriction of S_j is irreducible, it is one of them, and so it is pure. Let α be an eigenvalue of S_j at x_*, and let e be the residue degree over $\kappa(x_*)$ of the point of the curve above x_*. At that point, then, the corresponding eigenvalue is α^e and the residue cardinality is $q_{x_*}^e$. If the restriction is pure of weight u, then $|\iota(\alpha)|^e = q_{x_*}^{eu/2}$, so the weight of the restriction of S_j must be

$$w_j = 2 \log_{q_{x_*}} |\iota(\alpha)|,$$

which does not depend on the curve. In particular, all eigenvalues of S_j at x_* have this modulus. At x, the same calculation with powers shows that S_j is pure of weight w_j. As x was arbitrary, each S_j is pure of weight w_j on X_0. The chosen Weil filtration then shows that G is mixed after the constant extension, and by Lemma 18.6, G is mixed.

The general case. Now let G be an arbitrary constructible real sheaf. Since nilpotents do not change the étale site, we may assume that X_0 is reduced. In each irreducible component, choose a dense smooth affine open on which G is lisse and which avoids the other components. Since the complement has smaller dimension, iterating we obtain a finite chain of closed subschemes whose differences are disjoint unions of smooth connected affine pieces on which G is lisse. After one finite constant extension these pieces are geometrically connected, and such an extension is allowed by Lemma 18.6, and it keeps G real. The lisse case then applies to the positive-dimensional pieces, while at a zero-dimensional piece a Frobenius-stable full flag of each stalk has quotients of rank one, which are pure. The restriction of G to every piece is therefore mixed, and Lemma 18.7, applied successively along the chain of closed subschemes, shows that G is mixed. ∎

18.3 Relative affine curves

Let $h : U_0 \to Y_0$ be a smooth family of affine curves and F a pure lisse sheaf on U_0. On each stalk of $R^i h_! F$, Theorem 17.15 bounds the eigenvalues by $w + i$. Suppose that $R^i h_! F$ is mixed. Each pure quotient of its filtration has a nonzero stalk somewhere, and its eigenvalues there are among those of $R^i h_! F$. By Lemma 13.3 the quotients therefore have weights at most $w + i$. Most of the proof of Theorem 18.9 is devoted to showing that $R^i h_! F$ is mixed.

Theorem 18.9 (Pure lisse sheaves over relative affine curves)**.** *Let $h : U_0 \to Y_0$ be smooth separated of relative dimension one, with affine geometric fibres, and let F be a lisse Weil sheaf on U_0, pure of weight w. Then $R^i h_! F$ is mixed of weights at most $w + i$.*

Proof. We bound the stalks of $R^i h_! F$ by the curve case, we show that these sheaves are mixed by twisting to weight zero and adding the dual, and we combine the two facts using Lemma 13.3.

The stalks. Fix a closed point y of Y_0. Base change for direct images with compact support (Proposition 5.28) gives an identification

$$(R^i h_! F)_{\bar{y}} = H^i_c(U_{\bar{y}}, F)$$

compatible with the local Frobenius operators. The restriction of F to the fibre is again pure of weight w, because a residue extension raises both an eigenvalue and the residue cardinality to the same power. The fibre is a smooth affine curve over $\kappa(y)$, so Theorem 17.15 shows that $H^i_c(U_{\bar{y}}, F)$ has weights at most $w + i$. After a finite extension of $\kappa(y)$ the components of the fibre are geometrically connected. Lemma 15.7, applied to each of them with j the identity, shows that these groups vanish outside degrees 1 and 2.

Mixedness. We prove that $R^i h_! F$ is mixed. This is trivial for $F = 0$, so we may assume $F \neq 0$. Let $G = F^0 \oplus (F^0)^\vee$, with $F^0 = F \otimes \mathcal{B}_b$, be the sheaf of Proposition 15.9. It is lisse, real and pure of weight zero. At a closed point y write

$$P_{i,y}(T) = \det(1 - T \operatorname{Frob}_y \mid H^i_c(U_{\bar{y}}, G)).$$

Applying Lemma 15.7 to the components of the fibre after a finite extension of $\kappa(y)$, as above, we see that $P_{0,y} = 1$ and that every reciprocal root of $P_{2,y}$ has modulus q_y. By the bound just proved, every reciprocal root of $P_{1,y}$ has modulus at most $q_y^{1/2}$.

We claim that $\iota P_{1,y}$ and $\iota P_{2,y}$ have real coefficients. By the Euler product and the determinant identity of Theorem 12.30, after applying ι we have

$$L_\iota(U_y, G, T) = \frac{\iota P_{1,y}(T)}{\iota P_{2,y}(T)}.$$

Since the Euler factors are real, this rational function has real Taylor coefficients. Both it and its complex conjugate are regular at zero, so they are equal as rational functions. Moreover, the numerator and the denominator are coprime: a common root would give an eigenvalue having both permitted moduli, whereas $q_y^{1/2} < q_y$. Since a reduced fraction whose numerator and denominator have constant coefficient one is unique, comparing it with its conjugate we find $\iota P_{1,y}, \iota P_{2,y} \in \mathbf{R}[T]$, which proves the claim. Without the curve case this step would fail, because a common factor of numerator and denominator could cancel, and we would only know that the quotient is real, not each polynomial.

The sheaves $R^i h_! G$ are constructible by the finiteness theorems, and we have shown that they are real at every closed point. By Theorem 18.8 each of them has a finite mixed filtration.

To return from G to F, note that by additivity $R^i h_! F^0$ is a direct summand of $R^i h_! G$. A direct summand is a Weil subsheaf, and so it is mixed by Proposition 13.7. By the projection formula for the constant twist, it is isomorphic to $R^i h_! F \otimes \mathcal{B}_b$. Removing the twist preserves the finite Weil filtration and shifts the weight of every pure quotient by w. Hence $R^i h_! F$ is mixed.

Conclusion. The sheaf $R^i h_! F$ is mixed, and by the bound on the stalks all its local eigenvalues have modulus at most $q_y^{(w+i)/2}$. Lemma 13.3, with $w + i$ in place of w, shows that every nonzero quotient in the filtration has weight at most $w + i$. ∎

18.4 Proof of the Deligne–Laumon theorem

An arbitrary f reduces to relative affine curves and quasi-finite maps. Over a field, Noether normalization makes an affine variety of dimension d finite over $\mathbf{A}^d$, and $\mathbf{A}^d \to \operatorname{Spec} k$ is a composite of d projections of affine lines. For a separated morphism f of finite type the same works over a dense open of the target, and we treat the complement by induction on its dimension (Lemma 18.14). So we must check the weight bound for two kinds of maps and show that it survives the gluing.

Quasi-finite maps were treated in Lemma 18.1. For the projection $\mathbf{A}^1_Y \to Y$ (Proposition 18.12), consider a sheaf on the relative line. Over a dense open of the base it is lisse off a closed subscheme finite over the base. It is therefore glued from a relative affine curve and a quasi-finite piece. The bound is stable under composition by the Leray spectral sequence (Theorem 18.15). It is stable under partitions of the source and of the target by the localization sequences below. These gluing steps use only that mixed sheaves of weights at most u are closed under subquotients and extensions.

Lemma 18.10 (Weights in an exact sequence)**.** *Let $F' \to F \to F''$ be an exact sequence of Weil sheaves. If F' and F'' are mixed of weights at most u, then so is F.*

Proof. Let $a : F' \to F$ and $b : F \to F''$ be the maps. Exactness at F gives a short exact sequence

$$0 \to \operatorname{im}(a) \to F \to \operatorname{im}(b) \to 0,$$

where $\operatorname{im}(a)$ is a quotient of F' and $\operatorname{im}(b)$ a subobject of F''. Since mixed sheaves of weights at most u are closed under subquotients by Lemma 13.6 and under extensions by Proposition 13.7, F is mixed of weights at most u. ∎

By the next lemma, to prove a weight bound for $R^i f_!$ of a morphism f, it suffices to prove it over the pieces of finite partitions of the source and of the target.

Lemma 18.11 (Weight bounds and partitions)**.** *Let $f : X_0 \to Y_0$ be separated of finite type, and let K be a constructible Weil sheaf on X_0, mixed of weights at most w.*

1. *Let $j : U_0 \to X_0$ be an open immersion with closed complement $i : Z_0 \to X_0$. If $R^q(fj)_! j^*K$ and $R^q(fi)_! i^*K$ are mixed of weights at most $w+q$ for every q, then so is $R^q f_! K$.*
2. *Let $V_0 \subset Y_0$ be open with closed complement W_0, and let f_V and f_W be the base changes of f to V_0 and W_0. If $R^q f_{V!}(K|_{f^{-1}V_0})$ and $R^q f_{W!}(K|_{f^{-1}W_0})$ are mixed of weights at most $w+q$ for every q, then so is $R^q f_! K$.*

Proof. In both cases the restrictions of K are again mixed of weights at most w (Proposition 13.7), and we place $R^q f_! K$ in the middle of an exact sequence of Weil sheaves whose outer terms are bounded.

For (1), applying $Rf_!$ and the composition isomorphism to the triangle $j_! j^*K \to K \to i_* i^*K \to$ gives an exact sequence

$$R^q(fj)_! j^*K \longrightarrow R^q f_! K \longrightarrow R^q(fi)_! i^*K,$$

which is Frobenius-equivariant by naturality, and Lemma 18.10 gives the bound for $R^q f_! K$.

For (2), let $j' : V_0 \to Y_0$ and $i' : W_0 \to Y_0$ be the inclusions. Base change for direct images with compact support (Proposition 5.28) identifies $j'^* R^q f_! K$ with $R^q f_{V!}(K|_{f^{-1}V_0})$ and $i'^* R^q f_! K$ with $R^q f_{W!}(K|_{f^{-1}W_0})$. By Lemma 18.1, the outer terms of

$$0 \longrightarrow j'_! j'^* R^q f_! K \longrightarrow R^q f_! K \longrightarrow i'_* i'^* R^q f_! K \longrightarrow 0$$

are mixed of weights at most $w + q$, and Lemma 18.10 gives the bound for the middle term. ∎

Proposition 18.12 (A relative affine line)**.** *Let $p : \mathbf{A}^1_{Y_0} \to Y_0$ be the projection, and let G be a constructible Weil sheaf on $\mathbf{A}^1_{Y_0}$, mixed of weights at most w. Then $R^i p_! G$ is mixed of weights at most $w + i$ for every i.*

Proof. We stratify the target so that over each stratum, G is lisse off a subscheme that is finite over the base. On the lisse part the projection is a relative affine curve, and we apply Theorem 18.9. On the finite part it is quasi-finite, and we apply Lemma 18.1.

Take a finite mixed filtration of G. In the long exact sequences of $R^i p_!$ for its steps, the bound $w + i$ passes from the pure quotients to G by Lemma 18.10. We may therefore assume that G is pure and constructible of weight $v \leq w$.

Suppose first that the target $Y_0 = \operatorname{Spec} R$ is integral and affine, so that $\mathbf{A}^1_{Y_0}$ is integral. We construct the strata as follows. Choose a dense open U on which G is lisse. Since the locus where G is lisse is stable under Frobenius, we may replace U by the intersection of its finitely many Frobenius conjugates. Then U is defined over k_0, and so are the ideal $I \subset R[T]$ of its reduced complement and the choices below. If the complement is nonempty, choose a nonzero $g \in I$ and invert its nonzero leading coefficient b. Then over R_b the polynomial $b^{-1}g$ is monic and vanishes in $R_b[T]/I$. Hence the image of T is integral over R_b, and $R_b[T]/I$ is generated as an R_b-module by the powers T^k with $k < \deg g$. In particular $R_b[T]/I$ is finite over R_b; if g has degree zero, the complement is empty over $D(b)$.

Over $D(b)$ the complement is therefore finite, hence quasi-finite. The open U is smooth of relative dimension one over $D(b)$, and each of its fibres is an affine line minus finitely many points. So $U \to D(b)$ satisfies the hypotheses of Theorem 18.9. Applying that theorem on U and Lemma 18.1 on the complement, we obtain the bound over $D(b)$ from Lemma lemma-weight-bounds-partitions.

For a general target we argue by induction on $\dim Y_0$. Since nilpotents do not change the étale site, we may replace Y_0 by its reduction. Take disjoint integral affine opens containing the generic points of its components, and perform this construction on each of them. Their union is open, and its complement has smaller dimension. The case of dimension zero is contained in the construction, because a reduced zero-dimensional scheme is a finite disjoint union of integral affine pieces. In general, the restriction of G over the closed complement is again mixed of weights at most w, and the inductive hypothesis gives the bound there. Lemma lemma-weight-bounds-partitions, applied to the open union and its closed complement in the target, gives the bound over all of Y_0. ∎

Example 18.13 (Why the leading coefficient is inverted)**.** Let $Y_0 = \mathbf{A}^1_{k_0} = \operatorname{Spec} k_0[y]$, and suppose that G is lisse on the relative line $\mathbf{A}^1_{Y_0} = \operatorname{Spec} k_0[y, T]$ off the curve $Z = V(yT - 1)$. This curve is not finite over the base. Indeed, over a point $y \neq 0$ the fibre of Z is the single point $T = 1/y$, and over $y = 0$ it is empty, since the equation becomes $-1 = 0$. So $Z \to Y_0$ is quasi-finite, but its image is $D(y)$, which is not closed, whereas a finite map is closed; geometrically, as y tends to 0 the point $1/y$ runs off to infinity. To repair this, the construction in the proof takes $g = yT - 1$ with leading coefficient $b = y$. Over $D(y)$ the polynomial $T - y^{-1}$ is monic, and

indeed $k_0[y, y^{-1}, T]/(yT - 1) = k_0[y, y^{-1}]$ is finite over the base. The closed stratum $y = 0$, which has smaller dimension, is treated by induction, and on it the complement of the lisse locus happens to be empty.

To apply the proposition to an arbitrary morphism, we factor it, over strata, as a quasi-finite map followed by projections of relative affine lines.

Lemma 18.14 (Relative Noether normalization over a stratification)**.** *Let $f : X_0 \to Y_0$ be a morphism of finite type. There are finite partitions of X_0 and of Y_0 into locally closed pieces such that each piece of X_0 lies over a piece V of Y_0, and its map to V factors through a finite morphism to $\mathbf{A}^d_V$ for some d.*

Proof. We spread out Noether normalization from the generic points of the target, and then argue by induction on the dimension of the complement. Spreading out is possible because a normalization over the fraction field involves finitely many equations with finitely many denominators. Recall Noether normalization over a field k' [Stacks, 00OY]: a nonzero finitely generated k'-algebra is finite over a polynomial subalgebra.

Let R be an integral domain with fraction field k', let $\operatorname{Spec} R$ be an affine open of the target, and let B be a finitely generated R-algebra. If $B \otimes_R k' = 0$, then $B_s = 0$ for some nonzero $s \in R$. Otherwise, Noether normalization over k' provides elements $t_1, \dots, t_d$ and monic equations for each of a fixed set of algebra generators of B over $k'[t_1, \dots, t_d]$. Choose a nonzero $s \in R$ divisible by all denominators of the t_i and of the coefficients of these equations. Each equation, read in B_s, has a left-hand side that vanishes in $B \otimes_R k'$ and is therefore killed by a nonzero element of R. Multiplying s by these finitely many elements, we may assume that the equations hold in B_s. Then bounded powers of the generators span B_s over $R_s[t_1, \dots, t_d]$, and the resulting map to $\mathbf{A}^d_{R_s}$ is finite.

To obtain the partitions, choose disjoint integral affine opens containing the generic points of the reduced target. Partition their inverse images into finitely many reduced affine locally closed pieces and apply the preceding construction to each piece. On each target open, a single product of the finitely many denominators then works for all pieces. Since the complement of the resulting open has smaller dimension, by induction on this complement we obtain finite partitions of the whole source and target. Moreover, all choices are made over k_0. ∎

On each piece the map to V is thus a finite map followed by d projections of relative affine lines. The bound holds for the finite map by Lemma 18.1 and for each projection by Proposition 18.12, and the next theorem puts the pieces together.

Theorem 18.15 (Direct image of a mixed sheaf)**.** *Let $f : X_0 \to Y_0$ be separated of finite type and let F be mixed of weights at most w. Then $R^i f_! F$ is mixed of weights at most $w + i$ for every i.*

Proof. Say that a morphism has the property if the conclusion holds for it, for every w and every mixed F of weights at most w. Quasi-finite maps and projections of relative affine lines have the property, and by Lemma 18.14 every f is built from these two kinds of maps. So it suffices to show that the property is stable under composition and under finite partitions.

For composition, let $X_0 \xrightarrow{h} Z_0 \xrightarrow{g} Y_0$ and consider the Leray spectral sequence for compact supports,

$$E_2^{a,b} = R^a g_!(R^b h_! F) \implies R^{a+b}(gh)_! F.$$

It is Frobenius-equivariant by the naturality of the truncation tower and of the composition isomorphism for direct image with compact support. If both maps have the property, then $E_2^{a,b}$ is mixed of weights at most $w+a+b$. Since each later page is a subquotient of the previous one, the same holds for $E_\infty^{a,b}$. The abutment filtration of $R^n(gh)_!F$ is finite, because the operations preserve bounded constructible complexes. Its quotients are the $E_\infty^{a,n-a}$, and by Lemma 18.10 $R^n(gh)_!F$ is mixed of weights at most $w+n$.

For partitions, we apply Lemma 18.14 and refine the partitions to filtrations by open and closed pieces. After this refinement we can order the pieces so that each is open in the union of itself and the later ones. This lets us apply the localization sequences one piece at a time. On the pieces, the restriction of F to each source piece is again mixed of weights at most w, and on every refined source piece the finite map to $\mathbf{A}_V^d$ restricts to a separated quasi-finite map, which has the property by Lemma 18.1. Moreover, the projection $\mathbf{A}_V^d \to V$ is a composition of d projections of relative affine lines. Each of them has the property by Proposition 18.12, and by the composition argument the map from the piece to V has the property. Finally, we assemble the pieces one at a time in the chosen order. By Lemma lemma-weight-bounds-partitions on the source and Lemma lemma-weight-bounds-partitions on the target, the bound for the pieces implies the bound for $R^i f_! F$ on Y_0. ∎

For Theorem 13.11, stated in Section 13.3, the case of sheaves is Theorem 18.15, and the spectral sequence of the truncation tower passes to complexes.

Proof of Theorem 13.11. Let $K \in D_{\le w}(X_0)$, so that each $\mathcal{H}^b(K)$ is mixed of weights at most $w+b$. Applying $Rf_!$ to the ordinary truncation tower of K, which is bounded and functorial, we obtain the spectral sequence

$$E_2^{a,b} = R^a f_! \mathcal{H}^b(K) \implies \mathcal{H}^{a+b}(Rf_!K),$$

By naturality and the Frobenius compatibility of $Rf_!$, its differentials and its finite abutment filtration are compatible with the Weil structures. By Theorem 18.15, $E_2^{a,b}$ is mixed of weights at most $w+a+b$, and so is its subquotient $E_\infty^{a,b}$. The quotients of the filtration of $\mathcal{H}^n(Rf_!K)$ are the $E_\infty^{a,n-a}$. By Lemma 18.10 and induction on the length of the filtration, $\mathcal{H}^n(Rf_!K)$ is mixed of weights at most $w+n$. In other words, $Rf_!K \in D_{\le w}(Y_0)$. The last assertion is the case of the structure map of X_0, for which the geometric stalk of $R^n f_!K$ is $H_c^n(X, K)$ with its Frobenius operator. ∎

19 Proof of the Weil conjectures

The Deligne–Laumon theorem is an upper bound for weights, stated with ℓ-adic coefficients and one embedding $\iota : \overline{\mathbf{Q}}_\ell \to \mathbf{C}$. Weil's conjectures ask for more: an equality $|\alpha| = q^{i/2}$, for polynomials P_i with integer coefficients, under every complex embedding. This chapter closes both gaps.

The lower bound comes from Poincaré duality, and a curve shows how. Let X_0 be a smooth projective curve and α an eigenvalue of Frobenius on $H^1(X, \overline{\mathbf{Q}}_\ell)$. Then Theorem 13.11, applied to the constant sheaf, gives $|\iota(\alpha)| \le q^{1/2}$. On the other hand, cup product is a perfect pairing $H^1 \times H^1 \to H^2 = \overline{\mathbf{Q}}_\ell(-1)$, which Frobenius multiplies by q. Hence q/α is an eigenvalue as well, and the same bound for q/α says that $|\iota(\alpha)| \ge q^{1/2}$. In dimension d the argument is the same: an eigenvalue α on $H^i(X, \overline{\mathbf{Q}}_\ell)$ satisfies $|\iota(\alpha)| \le q^{i/2}$, while duality makes q^d/α an eigenvalue on H^{2d-i}, of absolute value at most $q^{(2d-i)/2}$, and we find $|\iota(\alpha)| = q^{i/2}$. This equality is Weil's axiom (d), and we prove it in Section 19.1, more generally for the higher direct images of a pure

lisse sheaf under a smooth proper morphism (Theorem 19.1); the case of the constant sheaf over a point is Corollary 19.2. Since the Deligne–Laumon theorem holds for every ι, purity holds for all complex embeddings at once. Indeed, every embedding of a number field into $\mathbf{C}$ extends to an isomorphism $\overline{\mathbf{Q}}_\ell \simeq \mathbf{C}$.

The polynomials $P_{i,\ell} = \det(1 - TF \mid H^i(X, \mathbf{Q}_\ell))$ are defined with ℓ-adic coefficients, and a priori they could depend on ℓ. The zeta function does not mention ℓ. It has a unique expression $Z(X_0, T) = A/B$ with $A, B \in \mathbf{Z}[T]$ coprime and $A(0) = B(0) = 1$ (Lemma 19.4; Sections 19.2–Section 19.3). Can each $P_{i,\ell}$ be read off from $Z(X_0, T)$? The trace formula writes Z as the alternating product of the $P_{i,\ell}$, but an alternating product can cancel. For example, $X_0 = \mathbf{G}_m \sqcup \operatorname{Spec} \mathbf{F}_q$ is not proper, and it has q^m points over $\mathbf{F}_{q^m}$, as many as $\mathbf{A}^1$. Hence $Z(X_0, T) = 1/(1 - qT)$. Yet H^0_c and H^1_c are both one-dimensional with eigenvalue 1, and the factor $1 - T$ cancels between them. For a smooth proper X_0 purity rules this out. The reciprocal roots of $P_{i,\ell}$ have absolute value $q^{i/2}$, so no root of $P_{i,\ell}$ is a root of $P_{j,\ell}$ for $i \neq j$. It follows that the product of the $P_{i,\ell}$ with i odd is A and the product with i even is B. Grouping the reciprocal roots of A and B by absolute value then recovers each $P_{i,\ell}$ from $Z(X_0, T)$ alone (Theorem 19.7). In particular $P_{i,\ell}$ has integer coefficients and does not depend on ℓ.

In terms of point counts, purity gives the classical estimates. For a smooth proper geometrically connected X_0 we obtain $|\#X_0(\mathbf{F}_{q^m}) - (1 + q^{dm})| \leq \sum_{i=1}^{2d-1} b_i q^{mi/2}$ (Corollary 19.10). In particular, for a curve of genus g this is Weil's bound $2gq^{m/2}$, and for elliptic curves it is Hasse's theorem. As an illustration, Example 19.9 works this out for genus-one curves over $\mathbf{F}_3$.

19.1 The Riemann hypothesis

For a proper map we have $Rf_! = Rf_*$, so the upper bound of Theorem 18.15 applies to ordinary cohomology. If the map is also smooth, Poincaré duality on each fibre pairs $H^i(X_{\bar{y}}, F)$ perfectly with $H^{2d-i}(X_{\bar{y}}, F^\vee(d))$. The eigenvalues of the second group are the inverses of those of the first. The second group has weights at most $(-w - 2d) + (2d - i) = -w - i$, and this upper bound becomes the lower bound $w + i$ for the first.

Both hypotheses are needed. Without properness, H^1_c of $\mathbf{G}_m$ has weight 0 instead of 1. Without smoothness duality fails. For example, the nodal cubic is proper, and its H^1 is a line on which Frobenius acts by ± 1, of weight 0. The relative statement is no harder than the absolute one, because we check the bound at one closed point y at a time, on the smooth proper fibre over $\kappa(y)$. In the proof of Theorem 19.7 we use only the case of a point as base and the constant sheaf (Corollary 19.2).

Theorem 19.1 (Purity of higher direct images under a smooth proper morphism)**.** *Let $f_0 : X_0 \to Y_0$ be smooth proper of pure relative dimension d, and let F be a lisse Weil sheaf, punctually ι-pure of weight w. For every i, the canonical identification*

$$R^i f_{0!} F = R^i f_{0*} F$$

is Weil-equivariant, and this constructible sheaf is punctually ι-pure of weight $w + i$. In particular, it is mixed of the single weight $w + i$.

Proof. Since f_0 is proper, the definition of $Rf_{0!}$ by compactification identifies it with Rf_{0*}. This identification commutes with the base change maps that define the Weil structures, which proves the first assertion.

Fix a closed point y. By proper base change, the stalk at $\bar{y}$ with its local Frobenius is $H^i(X_{\bar{y}}, F)$. By Theorem 18.15, $R^i f_{0*} F$ is mixed of weights at most $w + i$, so every eigenvalue α there satisfies

$$|\iota(\alpha)| \le q_y^{(w+i)/2}.$$

The opposite inequality comes from duality. The dual $F^\vee$ is lisse of weight $-w$, so $F^\vee(d)$ has weight $-w-2d$. On the smooth proper geometric fibre, Corollary 10.13 and Theorem 10.21 give a Frobenius-equivariant perfect pairing

$$H^i(X_{\bar y}, F) \times H^{2d-i}(X_{\bar y}, F^\vee(d)) \longrightarrow \overline{\mathbf{Q}}_\ell.$$

Since the pairing is perfect and takes values in the trivial Weil line, it identifies the second factor with the dual of the first as a Frobenius module, so the eigenvalues on the second factor are the α^{-1}, with multiplicities. Applying Theorem 18.15 once more, now to the structure map of the fibre over $\kappa(y)$ and to $F^\vee(d)$, of weight $-w-2d$, and to its cohomology in degree $2d-i$, we get

$$|\iota(\alpha)|^{-1} \le q_y^{(-w-2d+2d-i)/2} = q_y^{(-w-i)/2}.$$

The two inequalities give $|\iota(\alpha)| = q_y^{(w+i)/2}$ at every closed point y. Every nonzero pure quotient in a mixed filtration of $R^i f_{0*}F$ has a nonzero stalk at some closed point, so its weight is $w+i$. This proves the last assertion. ∎

For the Weil conjectures we need only the case where the base is a point and the sheaf is constant. Then the fibre is X itself, and the stalk of $R^i f_* \overline{\mathbf{Q}}_\ell$ is $H^i(X, \overline{\mathbf{Q}}_\ell)$.

Corollary 19.2 (Purity of the cohomology of a smooth proper variety)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, put $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, let $\ell \nmid q$ be a prime, and let $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$ be a field embedding. Then every geometric Frobenius eigenvalue α on $H^i(X_{\text{proét}}, \overline{\mathbf{Q}}_\ell)$ satisfies*

$$|\iota(\alpha)| = q^{i/2}.$$

Proof. The constant sheaf has weight zero, since its local Frobenius is the identity. So Theorem 19.1 applies to the structure map $f_0 : X_0 \to \operatorname{Spec} \mathbf{F}_q$ with $w = 0$. The base has a single closed point, with $q_y = q$. The stalk of $R^i f_{0*} \overline{\mathbf{Q}}_\ell$ there is $H^i(X, \overline{\mathbf{Q}}_\ell)$ with its geometric Frobenius, and its eigenvalues therefore satisfy $|\iota(\alpha)| = q^{i/2}$. ∎

19.2 Rational power series

The zeta function has integer coefficients by its definition, but the trace formula proves its rationality only over $\mathbf{Q}_\ell$, as an alternating product of ℓ-adic characteristic polynomials. To compare these polynomials for different primes ℓ, we need a form of the zeta function that does not mention ℓ. We show that a rational power series with integer coefficients is a quotient of integral polynomials in a unique reduced form.

Lemma 19.3 shows that rationality is a condition of linear algebra on the coefficients, namely that the Hankel matrix (c_{r+s}) has finite rank. Such a condition descends from $\mathbf{Q}_\ell$ to $\mathbf{Q}$. For integrality of the reduced fraction, Lemma 19.4 uses a lattice in the row span of the Hankel matrix that is stable under the shift. The characteristic polynomial of the shift on this lattice is then integral. For example, the coefficients $c_n = 2^{n+1} - 1$ of $1/((1-2T)(1-T))$ satisfy $c_{n+2} = 3c_{n+1} - 2c_n$. The shift matrix S on the row span has $\det(1 - TS) = (1-2T)(1-T)$, which is the reduced denominator. In the next section we compare the ℓ-adic factorizations with the reduced fraction A/B.

Lemma 19.3 (Hankel rank and rationality)**.** *For a field K and $C(T) = \sum_{n\geq 0} c_n T^n \in K[[T]]$, the following conditions are equivalent:*

1. *$C(T)$ is rational and regular at zero;*
2. *the finite submatrices of $(c_{r+s})_{r,s\geq 0}$ have bounded rank;*
3. *the span of the rows $R_j = (c_{j+s})_{s\geq 0}$ is finite-dimensional.*

If $K \subset K'$ and C has coefficients in K, rationality over K' implies rationality over K.

Proof. The equivalence of (2) and (3) is linear algebra. For the equivalence of (1) and (3), note that a denominator Q gives a linear recurrence among the rows. Conversely, the characteristic polynomial of the shift on a finite-dimensional row span is a denominator. The descent statement holds because (2) does not change when we extend the field.

For example, $C = 1/(1-T)$ has all $c_n = 1$, every row is $(1, 1, \ldots)$, and the row span is a line. For $C = 1/(1-T)^2$ we have $c_n = n+1$, and $R_j = R_0 + j(R_1 - R_0)$, so the row span is a plane.

For (3)⇒(2), a nonzero $k \times k$ minor of a finite submatrix gives k linearly independent rows. So every finite submatrix has rank at most the dimension of the row span. For (2)⇒(3), let $R_{j_1}, \ldots, R_{j_m}$ be linearly independent. On their span the coordinate functionals have common kernel zero. They therefore span its dual, and m of them already do. The corresponding $m \times m$ minor is nonzero, and so m is at most the bound in (2).

For (1)⇒(3), suppose that $QC = P$ with $Q(0) = 1$. Comparing coefficients gives a fixed linear recurrence for all sufficiently large indices, and the same recurrence expresses every sufficiently late row in terms of a fixed finite list of earlier rows.

For (3)⇒(1), the left shift $S(v)_n = v_{n+1}$ preserves the finite-dimensional row span V, since $S(R_j) = R_{j+1}$. Let e be evaluation at coordinate zero and $v = R_0$. In any basis of V, with M the matrix of the shift, we have

$$C(T) = e(I - TM)^{-1}v = \frac{e\,\mathrm{adj}(I - TM)v}{\det(I - TM)}.$$

The first equality holds coefficient by coefficient. Indeed, $(I - TM)^{-1} = \sum_n T^n M^n$, and $eM^n v$ is the coordinate zero of $S^n(R_0) = R_n$, that is, c_n. As the denominator has constant coefficient one, this proves (1).

For the descent, note that condition (2) involves only the coefficients $c_n \in K$, and the rank of a matrix with entries in K does not change when we pass to K'. So (2) holds over K if and only if it holds over K'. Applying (1)⇒(2) over K' and (2)⇒(1) over K gives the descent of rationality. ∎

Applied with $K = \mathbf{Q}$ and $K' = \mathbf{Q}_\ell$, the lemma shows that the zeta function, which the trace formula writes as a quotient of polynomials over $\mathbf{Q}_\ell$, is a quotient of polynomials over $\mathbf{Q}$. The next lemma shows that the integrality of its coefficients passes to the reduced fraction.

Lemma 19.4 (Reduced fractions with integer coefficients)**.** *If $C(T) \in 1 + T\mathbf{Z}[[T]]$ is rational over $\mathbf{Q}$, then there are unique coprime $A, B \in \mathbf{Z}[T]$ such that*

$$C(T) = A(T)/B(T), \qquad A(0) = B(0) = 1.$$

Proof. We repeat the construction in the proof of Lemma 19.3 with an integral basis of the row span. This gives an integral fraction, which we then reduce with Gauss' lemma.

Let V be the rational row span of Lemma 19.3, and put $L = V \cap \mathbf{Z}^{\mathbf{Z}_{\geq 0}}$. Then L contains all rows, spans V, and is preserved by the left shift. Since V is finite-dimensional, some r of the coordinate functionals span its dual, as in the proof of (2)$\Rightarrow$(3) in Lemma 19.3, and so they are jointly injective on V. These coordinates embed L into $\mathbf{Z}^r$. Since a subgroup of $\mathbf{Z}^r$ is free of finite rank, L is a shift-stable lattice in V, free of rank $\dim V$.

In a basis of L, the shift matrix M, the row e and the column $v = R_0$ are integral. The matrix identity in the proof of Lemma 19.3 yields integral polynomials

$$A_0 = e\,\mathrm{adj}(I - TM)v, \qquad B_0 = \det(I - TM), \qquad C = A_0/B_0.$$

Both constant coefficients are one. Let G be a primitive integral polynomial that is a greatest common divisor of A_0 and B_0 in $\mathbf{Q}[T]$. By Gauss' lemma the quotients A_0/G and B_0/G are integral. Since $G(0)$ divides $B_0(0) = 1$, we may choose the sign of G so that $G(0) = 1$. Dividing by G, we obtain polynomials A and B that are coprime and have constant coefficients one.

For uniqueness, let $A/B = A'/B'$ be two such fractions, so that $AB' = A'B$ in $\mathbf{Q}[T]$. Since B is coprime to A, it divides B', and in the same way B' divides B. So $B' = cB$ for a scalar c. Comparing constant coefficients gives $c = 1$, and then $AB = A'B$ forces $A = A'$. ∎

By Corollary 12.31, $Z(X_0, T) \in \mathbf{Q}(T) \cap (1 + T\mathbf{Z}[[T]])$ for every separated X_0 of finite type over $\mathbf{F}_q$. So Lemma 19.4 applies. We write $Z(X_0, T) = A/B$ for the resulting fraction, with A and B coprime in $\mathbf{Z}[T]$ and $A(0) = B(0) = 1$.

19.3 Integrality and independence of ℓ

Let X_0 be smooth and proper over $\mathbf{F}_q$, of pure dimension d, and let A/B be the reduced fraction of $Z(X_0, T)$ (Lemma 19.4). We now carry out the argument sketched at the start of the chapter. The point is that purity makes the characteristic polynomials of Frobenius in different degrees pairwise coprime, so no cancellation can occur, and the polynomials can be read off from A and B. For a prime $\ell \nmid q$ put

$$H^i_\ell = H^i(X_{\mathrm{pro\acute{e}t}}, \mathbf{Q}_\ell), \qquad P_{i,\ell}(T) = \det(1 - TF \mid H^i_\ell), \qquad b_{i,\ell} = \dim_{\mathbf{Q}_\ell} H^i_\ell.$$

Since X_0 is proper, $H^i_c = H^i$, and the determinant identity of Corollary 12.31 for $\Lambda = \mathbf{Q}_\ell$ reads

$$Z(X_0, T) = \prod_{i=0}^{2d} P_{i,\ell}(T)^{(-1)^{i+1}}. \tag{19.1}$$

Since Frobenius is invertible, $\deg P_{i,\ell} = b_{i,\ell}$.

Definition 19.5 (Reciprocal roots)**.** A *reciprocal root* of a polynomial P over a field is an element α of an algebraic closure such that $1 - \alpha T$ divides P.

Thus the reciprocal roots of $P_{i,\ell}$ are the Frobenius eigenvalues on H^i_ℓ, counted with multiplicity. The next lemma shows that A and B sort these eigenvalues by degree, independently of ℓ and of the complex embedding.

Lemma 19.6 (Reciprocal roots of the zeta function)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, let $Z(X_0,T) = A/B$ be the reduced fraction fixed above, and let $\ell \nmid q$ be a prime. Then*

$$A = \prod_{i \text{ odd}} P_{i,\ell}, \qquad B = \prod_{i \text{ even}} P_{i,\ell} \quad \text{in } \mathbf{Q}_\ell[T]. \tag{19.2}$$

Every reciprocal root $a \in \overline{\mathbf{Q}}$ of A or of B is a nonzero algebraic integer, and there is an integer i, odd for A and even for B, such that $|\tau(a)| = q^{i/2}$ for every field embedding $\tau : \mathbf{Q}(a) \hookrightarrow \mathbf{C}$ and such that $\theta(a)$ is a Frobenius eigenvalue on H^i_ℓ for every field embedding $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$.

Proof. The identity (19.2) holds because both sides are reduced fractions for $Z(X_0,T)$ over $\mathbf{Q}_\ell$. By Corollary 19.2, the reciprocal roots of $P_{i,\ell}$ have absolute value $q^{i/2}$ under every embedding $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$. Since $q > 1$, distinct degrees give distinct absolute values, and so the polynomials $P_{i,\ell}$ are pairwise coprime. Their products over odd and over even degrees are therefore coprime, and they have constant coefficients one. By (19.1) their quotient is $Z(X_0,T)$. On the other hand, A and B remain coprime in $\mathbf{Q}_\ell[T]$, since a Bézout relation over $\mathbf{Q}$ remains valid over $\mathbf{Q}_\ell$. The two reduced fractions agree by the uniqueness argument in the proof of Lemma 19.4, which works over any field.

Write $A(T) = \prod_a (1 - aT)$ and $B(T) = \prod_b (1 - bT)$ over $\overline{\mathbf{Q}}$. The reciprocal roots a and b are roots of the monic integral polynomials $U^{\deg A} A(U^{-1})$ and $U^{\deg B} B(U^{-1})$. Hence they are nonzero algebraic integers. Let $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$ be an embedding, and let a be a reciprocal root of A or B. By (19.2) and the pairwise coprimality of the $P_{i,\ell}$, there is exactly one i such that $\theta(a)$ is a Frobenius eigenvalue on H^i_ℓ. It is odd for A and even for B.

Corollary 19.2 bounds $|\iota(\theta(a))|$ for embeddings ι of $\overline{\mathbf{Q}}_\ell$. The lemma, however, concerns the embeddings of the number field $\mathbf{Q}(a)$. We pass from one to the other by extending field embeddings. Let $\tau : \mathbf{Q}(a) \hookrightarrow \mathbf{C}$ be any embedding. The map $\tau \circ \theta^{-1}$, defined on $\theta(\mathbf{Q}(a))$, extends to a field isomorphism $\iota' : \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} \mathbf{C}$, because both are algebraically closed fields of characteristic zero with transcendence degree $2^{\aleph_0}$ over $\mathbf{Q}$. We have $\iota'(\theta(a)) = \tau(a)$. Applying Corollary 19.2 with this ι' to the eigenvalue $\theta(a)$ on H^i_ℓ, we find

$$|\tau(a)| = q^{i/2}.$$

Since $q > 1$, this absolute value determines i, and so i does not depend on θ. ∎

Theorem 19.7 (The Weil conjectures for smooth proper schemes)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, and let $P_{i,\ell}$ be as above.*

1. *There are unique polynomials $P_i(T) \in \mathbf{Z}[T]$ with $P_i(0) = 1$ whose images in $\mathbf{Q}_\ell[T]$ are $P_{i,\ell}(T)$ for every prime $\ell \nmid q$, and $Z(X_0,T) = \prod_{i=0}^{2d} P_i(T)^{(-1)^{i+1}}$.*
2. *The reciprocal roots of P_i are algebraic integers, and each of them has absolute value $q^{i/2}$ under every complex embedding.*
3. *The degrees $b_i = \deg P_i = \dim_{\mathbf{Q}_\ell} H^i_\ell$ do not depend on ℓ, and $b_i = b_{2d-i}$. If X_0 is geometrically connected, then $P_0 = 1 - T$, $P_{2d} = 1 - q^d T$, and $b_0 = b_{2d} = 1$.*
4. *Put $\chi = \sum_i (-1)^i b_i$. Then $d\chi$ is even and there is $\varepsilon \in \{1, -1\}$ such that*

$$Z\left(X_0, \frac{1}{q^d T}\right) = \varepsilon q^{d\chi/2} T^\chi Z(X_0,T) \quad \text{in } \mathbf{Q}(T).$$

Proof. For (1) and (2), we define P_i using only the reduced fraction A/B of Lemma 19.4, by sorting the reciprocal roots of A and B by absolute value, and then we compare P_i with $P_{i,\ell}$ by Lemma 19.6.

For odd i, let R_i be the submultiset of the reciprocal roots of A whose absolute value under every complex embedding is $q^{i/2}$, and for even i define R_i in the same way using the reciprocal roots of B. By Lemma 19.6, every reciprocal root of A or B lies in exactly one R_i. The multisets of reciprocal roots of A and B are Galois-stable, with multiplicities. The condition on all complex embeddings is Galois-invariant, and so each R_i is Galois-stable. Put

$$P_i(T) = \prod_{a \in R_i} (1 - aT).$$

The coefficients of P_i are integers, being rational by Galois invariance and algebraic integers because the roots satisfy monic integral equations.

Now let $\ell \nmid q$ be any prime and $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$ an embedding. By (19.2) and Lemma 19.6, the embedding θ maps R_i onto the multiset of Frobenius eigenvalues on H^i_ℓ. Since both polynomials have constant coefficient one, $P_i = P_{i,\ell}$ in $\overline{\mathbf{Q}}_\ell[T]$, hence in $\mathbf{Q}_\ell[T]$. As ℓ was arbitrary, this proves the existence in (1), and uniqueness follows by comparing with a single $P_{i,\ell}$. The product formula for $Z(X_0, T)$ in (1) is then (19.1), and assertion (2) is Lemma 19.6.

The remaining assertions follow from Poincaré duality and the functional equation of Chapter 12. For (3), since $P_i = P_{i,\ell}$, we have $b_i = \deg P_i = \dim_{\mathbf{Q}_\ell} H^i_\ell$, which does not depend on ℓ, and $b_i = b_{2d-i}$ by Corollary 12.33.

For the second part of (3), assume that X_0 is geometrically connected, so that X is connected. By Theorem 10.15, $H^0(X, \mathbf{Q}_\ell) = \mathbf{Q}_\ell$, with Frobenius acting as the identity, and the trace map tr_X identifies $H^{2d}(X, \mathbf{Q}_\ell)$ with $\mathbf{Q}_\ell(-d)$, on which Frobenius acts by q^d. So $b_0 = b_{2d} = 1$, $P_0 = 1 - T$ and $P_{2d} = 1 - q^d T$.

For (4), Poincaré duality pairs the eigenvalues on H^i_ℓ with those on H^{2d-i}_ℓ by $\alpha \mapsto q^d/\alpha$. Substituting $T \mapsto 1/(q^d T)$ in the product formula therefore exchanges the corresponding factors up to a monomial, and Corollary 12.33 carries out this computation and gives (4) in $\mathbf{Q}_\ell(T)$. Since the zeta function is rational over $\mathbf{Q}$, and the scalar $\varepsilon q^{d\chi/2}$ is a rational number, the identity holds in $\mathbf{Q}(T)$. ∎

In the following examples we read the theorem backwards. For projective space the point count determines Z, and the grouping of the reciprocal roots by absolute value recovers the P_i. For an elliptic curve the theorem turns one point count into an inequality.

Example 19.8 (Projective space and elliptic curves)**.** We have $\#\mathbf{P}^d(\mathbf{F}_{q^m}) = 1 + q^m + \cdots + q^{dm}$, so that $Z(\mathbf{P}^d_{\mathbf{F}_q}, T) = \prod_{i=0}^{d} (1 - q^i T)^{-1}$. Grouping the reciprocal roots by absolute value as in (2), we find $P_{2i} = 1 - q^i T$ for $0 \le i \le d$ and $P_i = 1$ for odd i. For an elliptic curve C_0, we have $P_0 = 1 - T$ and $P_2 = 1 - qT$ by (3). Since $b_1 = 2$ (Appendix D), the polynomial P_1 has two reciprocal roots α, β. Poincaré duality matches them as $\beta = q/\alpha$, so $P_1 = 1 - aT + qT^2$ with $a = \alpha + \beta$. The point-counting formula for $m = 1$ gives $\#C_0(\mathbf{F}_q) = 1 - a + q$, that is, $a = 1 + q - \#C_0(\mathbf{F}_q) \in \mathbf{Z}$. By (2) both α and β have absolute value $q^{1/2}$, and we get Hasse's theorem $|a| \le 2q^{1/2}$.

Example 19.9 (The running example: purity)**.** For the curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, the reciprocal roots $\pm i\sqrt{3}$ of $P_1 = 1 + 3T^2$ (Example 12.32) have absolute value $\sqrt{3}$, as (2) predicts. What purity adds to the earlier examples is a bound: for every genus-one curve over $\mathbf{F}_3$ we get $|4 - N_1| \le 2\sqrt{3}$, that is, $1 \le N_1 \le 7$. In the proof, the upper bound $|\alpha| \le \sqrt{3}$

comes from the weight bound for $Rf_!$ (Theorem 13.11) applied to the constant sheaf, and $\alpha\beta = 3$ (Example 10.23) turns it into the lower bound.

The argument of Hasse's theorem works in every dimension. The point-counting formula expresses $\#X_0(\mathbf{F}_{q^m})$ through the traces of F^m, and purity bounds each trace by the dimension of the group times the common absolute value of its eigenvalues.

Corollary 19.10 (Point-count estimates)**.** *Let $X_0/\mathbf{F}_q$ be smooth, proper and geometrically connected of dimension $d \geq 1$, and let b_i be as in Theorem 19.7. Then for every $m \geq 1$,*

$$\left|\#X_0(\mathbf{F}_{q^m}) - (1 + q^{dm})\right| \leq \sum_{i=1}^{2d-1} b_i q^{mi/2}.$$

For a smooth proper geometrically connected curve of genus g, this is

$$|\#X_0(\mathbf{F}_{q^m}) - (1 + q^m)| \leq 2gq^{m/2}.$$

Proof. Since X_0 is proper, the point-counting formula of Corollary 12.31 reads

$$\#X_0(\mathbf{F}_{q^m}) = \sum_{i=0}^{2d} (-1)^i \operatorname{Tr}(F^m \mid H^i_\ell).$$

The terms in degrees 0 and $2d$ are 1 and q^{dm} by Theorem theorem-weil-conjectures-smooth-proper-schemes. The left side of the estimate is therefore the absolute value of the sum of the remaining terms. In degree i, the trace is a sum of b_i eigenvalues of F^m, each of complex modulus $q^{mi/2}$ by Theorem theorem-weil-conjectures-smooth-proper-schemes. Its absolute value is thus at most $b_i q^{mi/2}$, and the triangle inequality proves the first bound. For a curve, $b_1 = 2g$ by the cohomology calculation of Appendix D. ∎

Appendices

To define $Rf_!$ in Part I we must compactify every separated morphism of finite type, and to prove proper base change we must dominate every proper morphism by a projective one. These are Nagata's theorem and Chow's lemma (Appendix A). Nagata's theorem, followed by normalization, also gives the smooth compactification of a curve used in Part III (Lemma 14.1).

Proper base change ends with a change of separably closed ground field. There we need the cohomology of curves for all torsion coefficients, including torsion divisible by the characteristic (Theorem 4.20). Purity starts from sheaves of rank one. Their geometric monodromy is finite because Frobenius fixes no class in H^1 of a curve (Proposition 14.3). For coefficients prime to the characteristic, both facts come from the Jacobian. Indeed, $H^1(C, \mu_n)$ is the n-torsion of the Jacobian, and a fixed class would give ℓ-power torsion points of unbounded order among its finitely many rational points. Appendices Appendix B, Appendix C and Appendix D prove both facts.

Rankin's argument in Chapter 14 rests on the determinant weights of tensor and exterior powers (Proposition 14.19). These come from the structure of the monodromy group (Lemma 14.15), and Appendix F proves the facts about algebraic groups over $\overline{\mathbf{Q}}_\ell$ behind it.

Appendix E is of a different kind. Following Varshavsky [Var05], it computes the local terms of contracting correspondences in a second way; Chapter 12 does not depend on it.

A Nagata compactification and Chow's lemma

The main text defines cohomology with compact support as in topology: to push forward along $f : X \to Y$, embed X as an open subset of something proper over Y, extend by zero and push forward, so that $Rf_! = Rp_* j_!$ (Proposition 5.26). This definition needs every morphism to have a compactification, and it needs any two compactifications to be dominated by a third, so that proper base change can show that $Rf_!$ does not depend on the choice. The second point is elementary once we use compactifications in which X is scheme-theoretically dense: the closure of X in the fibre product of two of them dominates both (Proposition A.7). The first point is Nagata's theorem, Theorem A.2: every separated morphism of finite type $X \to S$ with S quasi-compact and quasi-separated factors as a quasi-compact open immersion $X \to \overline{X}$ followed by a proper morphism $\overline{X} \to S$.

Why is this hard? Locally a compactification is easy to find: if S and X are affine, we take the closure of X in projective space over S. The difficulty is to glue. For example, cover $X = \mathbf{A}^1_k$ by $U_1 = \mathbf{A}^1 \setminus \{0\}$ and $U_2 = \mathbf{A}^1 \setminus \{1\}$, compactify each by $\mathbf{P}^1$, and glue the two copies of $\mathbf{P}^1$ along $U_1 \cap U_2$. The result is then $\mathbf{P}^1$ with the three points 0, 1 and ∞ doubled, which is not separated. For curves blow-ups are not needed, and they would not help either, since blowing up a point on a regular curve does nothing. In dimension at least two, however, the closures of $U_1 \setminus U_2$ and $U_2 \setminus U_1$ can meet in the boundary, and we pull them apart by blow-ups with centers in the boundary, which leave X unchanged (Lemma A.20 and Proposition A.23). Induction over an affine cover then gives the theorem over a Noetherian base (Section A.4), and Noetherian approximation gives the general case, which we need because proper base change is proved over arbitrary qcqs bases.

Proper base change itself is computed on projective space, and Chow's lemma carries it to every proper morphism (Section 4.5). We use it in the form of Theorem A.25: for a proper morphism $X \to S$ there is a proper surjection $\widetilde{X} \to X$ with $\widetilde{X}$ projective over S.

A.1 Compactifications

To show that $Rp_* j_!$ does not depend on the compactification, we must compare two of them, and in general neither maps to the other. The plane $\mathbf{A}^2$ is open in $\mathbf{P}^2$ and in $\mathbf{P}^1 \times \mathbf{P}^1$, but the identity of $\mathbf{A}^2$ extends to a morphism in neither direction: the map $(x : y : z) \mapsto ((x : z), (y : z))$ is undefined at $(1 : 0 : 0)$ and $(0 : 1 : 0)$, and its inverse is undefined at (∞, ∞). What exists is a third compactification mapping to both, here the closure of $\mathbf{A}^2$ in $\mathbf{P}^2 \times \mathbf{P}^1 \times \mathbf{P}^1$. So we compare two compactifications through a common refinement. A refinement h is required to satisfy $h^{-1}(X) = X$. This is what lets us compute $Rh_* j'_!$ by restriction to X in the proof of Proposition 5.26.

Without density, refinements are not unique. For example, the point $X = \operatorname{Spec} k$ is open in the compactification $\operatorname{Spec} k \sqcup \mathbf{P}^1_k$, which is not dense. Every automorphism of $\mathbf{P}^1_k$ gives a refinement of it to itself. Dense compactifications do not have this problem: morphisms between them are unique, and with their refinements they form a cofiltered category (Proposition A.7).

Definition A.1 (Compactification and refinement)**.** Let $f : X \to S$ be a morphism. A compactification is a factorization

$$X \xrightarrow{j} \overline{X} \xrightarrow{p} S$$

with j a quasi-compact open immersion and p proper. We say that it is *dense* if j is scheme-theoretically dense, i.e., if the map $\mathcal{O}_{\overline{X}} \to j_* \mathcal{O}_X$ is injective. A refinement of compactifications is

an S-morphism $h : \overline{X}' \to \overline{X}$ with $hj' = j$ and $h^{-1}(X) = X$, under the specified identifications with X.

For example, $\mathbf{A}^1_k \subset \mathbf{P}^1_k$ is a dense compactification of $\mathbf{A}^1_k \to \operatorname{Spec} k$, since a regular function on an open of $\mathbf{P}^1_k$ that vanishes on $\mathbf{A}^1_k$ is zero. The theorem says that such a factorization always exists.

Theorem A.2 (Nagata's compactification theorem)**.** *Let S be quasi-compact and quasi-separated. Every separated morphism $f : X \to S$ of finite type has a dense compactification.*

This is [Stacks, 0F41], and we give the proof in Section A.4. We need it in this generality because proper base change is proved over arbitrary qcqs bases; for schemes of finite type over k the Noetherian case of the proof suffices.

Density is a condition on structure sheaves. For instance, a nilpotent section supported on the boundary is invisible on X, but it prevents X from being schematically dense. The condition does two things for us. First, morphisms between dense compactifications are unique and common refinements exist (Proposition A.7). Second, passing to a dense compactification does not change the direct image with proper support (Lemma A.5).

Lemma A.3 (Schematic closure)**.** *Let $i : U \to T$ be a quasi-compact immersion, with T quasi-separated. Its scheme-theoretic image $\overline{U} \subseteq T$ is a closed subscheme, $U \to \overline{U}$ is an open immersion, and U is scheme-theoretically dense in $\overline{U}$. The formation of $\overline{U}$ commutes with flat base change $T' \to T$, in particular with restriction to opens of T.*

Proof. The scheme-theoretic image is the smallest closed subscheme through which i factors. Its ideal is $\ker(\mathcal{O}_T \to i_*\mathcal{O}_U)$, which is quasi-coherent because i is quasi-compact and quasi-separated. Its formation commutes with flat base change $T' \to T$: pushforward along the quasi-compact quasi-separated i commutes with flat base change, and pullback along a flat morphism is exact, so the pullback of the ideal is the ideal of the base change [Stacks, 081I]. Restriction to an open of T is a special case. On an open where the immersion is closed the image is therefore U itself, and this shows that U is open in its image. By construction, a section of $\mathcal{O}_{\overline{U}}$ vanishing on U is zero, which is the definition of schematic density. For these statements see [Stacks, 01R8, 01RG]. ∎

Density is what makes extensions from U unique.

Lemma A.4 (Uniqueness of extensions from a dense open)**.** *Let T be an S-scheme, let $U \subseteq T$ be a scheme-theoretically dense open, and let $Y \to S$ be separated. Two S-maps $T \to Y$ that agree on U are equal.*

Proof. The equalizer of the two maps is a closed subscheme of T, since it is the inverse image of the diagonal of Y/S. Its ideal restricts to zero on U, because the two maps agree there. As a section of $\mathcal{O}_T$ that vanishes on a scheme-theoretically dense open is zero, the ideal is zero and the equalizer is all of T. ∎

By Lemma A.3, every compactification contains a dense one, the schematic closure of X. For this replacement to be harmless we need that it does not change $Rp_*j_!$.

Lemma A.5 (Passing to the schematic closure)**.** *Let $X \xrightarrow{j} \overline{X} \xrightarrow{p} S$ be any compactification with $\overline{X}$ Noetherian, and let $i : Z \hookrightarrow \overline{X}$ be the schematic closure of X, with $j_Z : X \hookrightarrow Z$ the induced*

open immersion. For a condensed coefficient ring Λ, we use the pullbacks of its coefficient sheaf to these schemes as coefficients. For every $K \in D(X_{\mathrm{pro\acute{e}t}}, \Lambda_X)$ there are canonical identifications

$$j_! K \simeq i_* j_{Z,!} K, \qquad Rp_* j_! K \simeq R(p \circ i)_* j_{Z,!} K.$$

Proof. The first identification is a statement about supports, and the second follows from it by applying Rp_*. The only point that needs care on the pro-étale site is the exactness of i_*.

Since $j_! K$ vanishes outside $X \subseteq Z$, it should be pushed forward from Z. To say this in the derived category we need i_* to be exact, so that $Ri_* = i_*$. By Proposition 2.58, the functors attached to open and closed immersions on the pro-étale site are exact when the complementary open is quasi-compact. Since $\overline{X}$ is Noetherian, the complement of Z is quasi-compact, so all functors below are exact and may be applied degreewise to unbounded complexes. The inverse image of $j_! K$ on Z is $j_{Z,!} K$, so we have the natural map $j_! K \to i_* i^* j_! K = i_* j_{Z,!} K$. It is the identity on the open X, and both sides restrict to zero on the closed complement of X. By the localization sequence for an open and its closed complement, this map is therefore an isomorphism. The second identification then follows by applying Rp_* and using $Rp_* Ri_* \simeq R(p \circ i)_*$. ∎

Remark A.6 (Non-Noetherian compactifications)**.** The same proof works for qcqs $\overline{X}$ whenever $\overline{X} \setminus Z$ is quasi-compact. For a non-Noetherian compactification this can fail, and then Proposition 2.58 does not apply to pushforward along the closed immersion on the pro-étale site. The schematic closure and dense refinements exist in the generality of Theorem A.2.

For dense compactifications the comparison is formal. We obtain common refinements by Lemma A.3, and morphisms are unique by Lemma A.4.

Proposition A.7 (Dense compactifications form a cofiltered category)**.** *Let S be quasi-compact and quasi-separated, and let $f : X \to S$ be separated and of finite type. Any finite family of dense compactifications of f has a common dense refinement. Morphisms between dense compactifications that restrict to the identity on X are unique, proper, and automatically satisfy $h^{-1}(X) = X$. The category of dense compactifications of f is therefore cofiltered.*

Proof. A common refinement of two compactifications is the closure of X in their fibre product, as in the example of $\mathbf{A}^2$ above. The properties of morphisms then follow from density.

Given two compactifications, let Z be the schematic closure of the image of X in $\overline{X}_1 \times_S \overline{X}_2$ under the diagonal map. The map from X is a quasi-compact immersion, because over the open $X \subseteq \overline{X}_1$ it is the graph of a morphism to $\overline{X}_2$, and $\overline{X}_2$ is separated over S. By Lemma A.3, X is an open subscheme of Z and is scheme-theoretically dense in Z. Since the product is proper over S, the scheme Z and its projections are proper.

The projections are refinements, that is, X is its own inverse image in Z. By Lemma A.3 the formation of the schematic closure commutes with restriction to opens. Over $X \subseteq \overline{X}_1$ we may therefore compute it on the open $X \times_S \overline{X}_2$. There the immersion is the graph of $X \to \overline{X}_2$, which is closed, so the closure is this graph. This means that the inverse image of X under the first projection is X, on which the first projection is the identity. The same holds for the second projection. The assertion for finite families follows by induction.

Let $h : \overline{X}' \to \overline{X}$ be a morphism of dense compactifications restricting to the identity on X. To see that h is proper, note that it factors through its graph. This graph is closed in $\overline{X}' \times_S \overline{X}$ because $\overline{X}$ is separated over S, and the projection of this product to $\overline{X}$ is proper because $\overline{X}'$ is proper over S.

Next, the inclusion $X \to h^{-1}(X)$ is a section of the separated map $h^{-1}(X) \to X$. It is therefore closed, as well as open. Since X is schematically dense in $\overline{X}'$, it is schematically dense in the open $h^{-1}(X)$. A closed subscheme containing it must be all of $h^{-1}(X)$, and we conclude that $h^{-1}(X) = X$. Two parallel morphisms are equal by Lemma A.4. Finally, the category is nonempty by Nagata's theorem. Since any two objects have a common refinement and parallel morphisms are equal, it is cofiltered. ∎

Proper base change is applied after a change of base $S' \to S$, so we also need to know what happens to a compactification under base change, where density need not survive.

Remark A.8 (Base change of a compactification)**.** For every $S' \to S$, the base change of a compactification is an open immersion followed by a proper morphism, which suffices to define the direct image with compact support of $X \times_S S' \to S'$. When the base-changed ambient scheme is Noetherian, as for schemes of finite type over k, Lemma A.5 shows that this direct image agrees with the one defined by the schematic closure.

Schematic closure commutes with flat base change (Lemma A.3), and so does schematic density, but density can be lost under an arbitrary base change. In that case the schematic closure of $X \times_S S'$ in $\overline{X} \times_S S'$ is a dense compactification, and a base-changed common refinement can be replaced by its closure in the same way. The comparisons in the main text are made on a further common refinement, where they are canonical by uniqueness.

A.2 Elimination of indeterminacy

Every modification in the proof of Nagata's theorem is a blow-up whose center avoids a fixed open, so that this open is not changed. Here the open U is the scheme being compactified, or a piece of it, and only the boundary may be modified. For instance, the coordinate axes minus the origin are disjoint closed subsets of $U = \mathbf{A}^2 \setminus \{0\}$, but their closures in $\mathbf{A}^2$ meet at the origin. The blow-up of the origin is an isomorphism over U, and in it the strict transforms of the axes meet the exceptional line in the two distinct points of slope 0 and ∞, so they become disjoint. Lemma A.20 proves the same in general.

We use blow-ups rather than arbitrary proper modifications that are isomorphisms over U, because a blow-up is given by an ideal. This makes them easy to combine. Two of them are dominated by the blow-up of the product ideal, and a composite is again a blow-up (Lemma A.13). The ideal is required to be of finite type, so that the blow-up is proper.

Definition A.9 (Admissible blow-up)**.** Let T be quasi-compact and quasi-separated, and let $U \subseteq T$ be a quasi-compact open. For a quasi-coherent ideal of finite type $\mathcal{I} \subseteq \mathcal{O}_T$ with $\mathcal{I}|_U = \mathcal{O}_U$, the blow-up

$$b : \mathrm{Bl}_{\mathcal{I}}(T) = \mathrm{Proj}_T\Big(\bigoplus_{m \geq 0} \mathcal{I}^m\Big) \longrightarrow T$$

is called U-admissible. Its center is disjoint from U.

On an affine open $\mathrm{Spec}(R)$, write $I = (a_1, \ldots, a_r)$. The charts of the blow-up have rings

$$R[I/a_i] = \big(\big(\bigoplus_{m \geq 0} I^m t^m\big)_{a_i t}\big)_0.$$

This is the degree-zero part of the localization at $a_i t$ of the Rees algebra, where t is a variable of degree one. It is the quotient of $R[x_1, \ldots, x_r]/(a_i x_j - a_j)_j$ by its a_i-power torsion, that is, by the

elements killed by some power of a_i, and this torsion can be nonzero. In the chart $R[I/a_i]$, the element a_i is a nonzerodivisor and generates the extended ideal $IR[I/a_i]$. In other words, the center becomes an effective Cartier divisor. The blow-up is an isomorphism away from the center. Since I is finitely generated, the blow-up of an affine base is a closed subscheme of a projective space over it, and therefore b is proper. These constructions glue under restriction to opens.

We first use admissible blow-ups to restore density. A compactification may contain components or embedded components supported in the boundary. Blowing up the boundary removes them, because after the blow-up the boundary is cut out locally by a nonzerodivisor.

Lemma A.10 (Making an open schematically dense)**.** *Let T be quasi-compact and quasi-separated, and let $U \subseteq T$ be a quasi-compact open. There is a U-admissible blow-up $T' \to T$ such that the inverse image of U is schematically dense in T' and its complement is the support of an effective Cartier divisor.*

Proof. We blow up an ideal that cuts out the complement of U. Its pullback is then locally generated by a nonzerodivisor a, and an open whose complement is cut out by a nonzerodivisor is schematically dense, because a section killed by a power of a is zero.

Choose a quasi-coherent ideal $\mathcal{I}$ of finite type with $V(\mathcal{I}) = T \setminus U$. Such an ideal exists because U is a quasi-compact open in a qcqs scheme [Stacks, 01PH]. We blow up this ideal, and the extended ideal defines the exceptional Cartier divisor. On the affine charts its local equation a is a nonzerodivisor. Hence each chart ring injects into its localization at a. The complement of the divisor is the inverse image of U, and it is therefore schematically dense. ∎

Example A.11 (Blowing up removes an embedded point)**.** Let $T = \operatorname{Spec} R$ with $R = k[x, y]/(xy, y^2)$, the line $y = 0$ with an embedded point at the origin, and let $U = D(x)$. On U the element y is zero, since $xy = 0$ and x is invertible, so U is not schematically dense in T. On the other hand, the ideal $I = (x)$ is the unit ideal on U, so its blow-up is U-admissible. Since the ideal is principal, the chart formula above gives $R[I/x] = R/\{b : x^n b = 0 \text{ for some } n\}$. Now every element of R is uniquely $f(x) + cy$ with $c \in k$, and $x^n(f + cy) = x^n f$, so the x-power torsion is ky and the blow-up is $\operatorname{Spec} k[x]$. Thus the embedded point is gone, U is dense, and the boundary is the Cartier divisor $x = 0$, as in Lemma A.10.

After a blow-up, we replace a scheme over T by the part of its base change that is not concentrated on the exceptional divisor.

Definition A.12 (Strict transform)**.** Let $b : T' \to T$ be the blow-up of T in a quasi-coherent ideal of finite type, with exceptional divisor $E \subseteq T'$, and let $W \to T$ be a morphism. The *strict transform* of W is the closed subscheme of $W \times_T T'$ whose ideal consists of the local sections of $\mathcal{O}_{W \times_T T'}$ supported on the inverse image of E.

Over a chart of T' where E has equation a and $W \times_T T'$ is affine with ring B, the ring of the strict transform is

$$B/\{b \in B : \ a^n b = 0 \text{ for some } n \geq 0\}.$$

If U is the complement of the center, this is exactly the schematic closure of W_U in $W \times_T T'$, the object that occurs in the elimination of indeterminacy. By [Stacks, 080E], the strict transform is the blow-up of W in the pulled-back ideal, so it is proper over W.

In the proofs below we construct blow-ups locally and one after another. By the next lemma, finitely many such constructions combine into a single U-admissible blow-up of T.

Lemma A.13 (Products and composites of U-admissible blow-ups)**.** *Finitely many U-admissible blow-ups of T admit a common U-admissible blow-up. A composite of two U-admissible blow-ups is again such a blow-up. A U-admissible blow-up on a quasi-compact open $V \subseteq T$ containing U extends to a U-admissible blow-up of T.*

Proof. All three assertions are statements about ideals, because products of ideals dominate blow-ups, composites are computed by products of ideals, and extension from an open amounts to extending a quasi-coherent ideal of finite type. Let $\mathcal{I}, \mathcal{J}$ be ideals, both equal to $\mathcal{O}$ on U, so that their product is too. Then the blow-up of their product factors through both blow-ups. Indeed, on it the product $\mathcal{I}\mathcal{J}$ is invertible, hence so is each factor, and the universal property of each blow-up then gives the factorization. The first assertion follows by induction. For the second, blowing up first $\mathcal{I}$ and then the pullback of $\mathcal{J}$ yields the blow-up of $\mathcal{I}\mathcal{J}$ [Stacks, 080A], which proves the composition assertion when the second center is pulled back from T. In general, however, the second center lives only on the first blow-up, and the composite is a blow-up of T by the calculation with ideals of finite type in [Stacks, 080B]. For the extension assertion we extend a quasi-coherent ideal of finite type from a quasi-compact open, keeping the center outside U; such an extension exists because on a qcqs scheme every quasi-coherent sheaf is the union of its quasi-coherent subsheaves of finite type, and the precise result on admissible extensions is [Stacks, 080M]. ∎

Blow-ups commute with flat base change [Stacks, 0805], because flat pullback preserves the inclusions $\mathcal{I}^m \subseteq \mathcal{O}_T$. Under a base change that is not flat, the pullbacks of these inclusions can have kernels. In that case the blow-up of the pulled-back ideal is the strict transform of the base change, which can be strictly smaller than the base change itself.

The gluing argument has to extend a morphism defined on a dense open $U \subseteq X$ as far as possible. When the target Y is not proper, the extension can stop before reaching all of X. We need a way to say that a partial map is already extended as far as it can be.

Take $X = Y = \mathbf{A}^1_k$ and $U = \mathbf{A}^1 \setminus \{0\}$. The inclusion $t \mapsto t$ extends over 0. Its graph, the diagonal minus the origin, is not closed in $X \times Y$. The map $t \mapsto 1/t$ does not extend, even after a blow-up, since blowing up a point on a regular curve does nothing. Its graph $V(ty - 1)$ is closed. In general, a closed graph means that no limit point lies over $X \setminus U$. A map with closed graph is therefore not the restriction of a map on a larger open. Both the hypothesis and the conclusion of Theorem A.16 are stated in this form. For proper Y the condition forces the domain to be all of X, which gives Corollary A.17.

Definition A.14 (Quasi-domination)**.** Let X and Y be S-schemes, with $Y \to S$ separated. A *quasi-domination of X over Y* consists of a dense open $U \subseteq X$ and an S-map $f : U \to Y$ whose graph is closed in $X \times_S Y$.

Since $Y \to S$ is separated, the graph is always closed in $U \times_S Y$; the condition is about its behaviour over $X \setminus U$. Every map $X \to Y$ is a quasi-domination with domain X. Conversely, if $Y \to S$ is proper, every quasi-domination has domain X, because the projection of the closed graph to X is closed with image the dense open U.

The extension theorem below is proved first for targets that embed in affine space over the base. We reduce a general target to this form by Zariski's Main Theorem.

Proposition A.15 (The relative form of Zariski's Main Theorem)**.** *A separated, quasi-compact, locally quasi-finite morphism $Z \to T$ locally of finite type factors as an open immersion $Z \hookrightarrow T_0$ followed by an affine morphism $T_0 \to T$. Consequently $Z \to T$ is quasi-affine.*

Proof. The scheme T_0 is the normalization of T in Z, that is, the relative spectrum of the integral closure of $\mathcal{O}_T$ in the direct image of $\mathcal{O}_Z$; it is affine over T by construction. By Zariski's Main Theorem in the form [Stacks, 02LR], locally on T the canonical map from Z to the normalization T_0 of T in Z is an open immersion, and $T_0 \to T$ is affine. These factorizations agree on overlaps and glue to the stated factorization. Over an affine open of T the scheme T_0 is affine, so Z is a quasi-compact open of an affine scheme there, which means that $Z \to T$ is quasi-affine. ∎

Quasi-affine targets are the ones where a partial section can be extended by hand, by clearing denominators of finitely many coordinates. Using the proposition, we reduce the general extension theorem to this case, once we know that the relevant closure is quasi-finite over the base.

Theorem A.16 (Extension of a partial section)**.** *Let T be qcqs, let $U \subseteq V \subseteq T$ be quasi-compact opens with U dense in T, and let $p : Y \to T$ be separated and of finite type. Suppose that $f : U \to Y$ is a section of p over U and that its image is closed in $p^{-1}(V)$. There are a V-admissible blow-up $\pi : T' \to T$, an open $W \subseteq T'$ containing $\pi^{-1}(U)$, and a map $g : W \to Y$ such that*

$$p \circ g = \pi|_W, \qquad g|_{\pi^{-1}(U)} = f \circ \pi|_{\pi^{-1}(U)},$$

and the graph of g is closed in $Y \times_T T'$.

We quote this theorem from [Con07, Theorem 2.4], with the auxiliary base equal to T, and explain below the structure of its proof. Note that Y is qcqs, since p is separated and of finite type and T is qcqs. The hypothesis on f says that f is a quasi-domination of V over Y. We apply the theorem only when $V = U$, the case of [Con07, Remark 2.5]. Then the hypothesis holds automatically, because a section of a separated morphism is a closed immersion.

The proof first treats a quasi-affine $Y \to T$. If $Y \to T$ is quasi-affine and T is affine, then Y is a quasi-affine scheme of finite type over T, and it embeds in some $\mathbf{A}^n_T$ [Stacks, 04II]. The coordinates of f are then finitely many fractions, and a blow-up clearing their denominators extends f. One first makes U schematically dense, so that the extended coordinates still satisfy the equations of Y. Using a finite affine cover of T, we then obtain the quasi-affine case over a qcqs base. In fact the proof needs this case for opens $U \subseteq V$ with U not necessarily dense, because the reduction below applies it to the opens $f^{-1}(Y_i) \subseteq V$, which need not be dense. For this, a further construction in Conrad's proof keeps the centers outside V.

The general case reduces to the quasi-affine one, and this reduction is where Zariski's Main Theorem enters. The plan is to extend the section on each affine open of Y separately. The closure of the section is then quasi-finite over the base, hence quasi-affine, and the quasi-affine case applies once more.

In detail, choose a finite affine open cover $Y = \bigcup_i Y_i$. Since T is quasi-separated, each $Y_i \to T$ is quasi-compact, and as Y_i is affine it is quasi-affine. Hence the quasi-affine case applies to each $f|_{f^{-1}(Y_i)}$, and we take a common V-admissible blow-up $T_1 \to T$. Let Z be the schematic closure of the image of f in $Y \times_T T_1$. It meets each $Y_i \times_T T_1$ inside the closed graph of an extension, which has at most one point over each point of T_1. Since there are finitely many Y_i, the fibres of $Z \to T_1$ are finite. As $Z \to T_1$ is separated and of finite type, it is quasi-finite, and it is quasi-affine by Proposition A.15. The quasi-affine case then extends the section into Z, and the composite of the two blow-ups is V-admissible by Lemma A.13.

Corollary A.17 (Extension to a proper target)**.** *Let X be qcqs over S, let $U \subseteq X$ be a dense quasi-compact open and let $Y \to S$ be proper. Every S-map $f : U \to Y$ extends to an S-map $X' \to Y$ for some U-admissible blow-up $X' \to X$.*

Proof. A map $U \to Y$ is the same as a section over U of the projection $Y \times_S X \to X$. The theorem extends this section to an open of a blow-up, and properness of Y forces this open to be everything.

Apply Theorem A.16, with $T = X$ and $V = U$, to the section $(f, \mathrm{id}) : U \to Y \times_S X$ of the proper projection to X. This gives an extension on an open $W \subseteq X'$ whose graph is closed in $Y \times_S X'$. The open W is dense, because it contains the inverse image of U, which is dense in X'. Since $Y \to S$ is proper, the observation after Definition A.14 shows that $W = X'$. ∎

When the target is not proper, the extension is defined only on an open of the blow-up, and we can still say what it looks like. Suppose the target is a separated map $W \to T$ of finite type that is an isomorphism over U, and the section is its inverse over U. Then the extension identifies a blow-up of W with an open of a blow-up of T.

Theorem A.18 (Elimination of indeterminacy)**.** *Let T be qcqs, let $U \subseteq T$ be a quasi-compact open, and let $W \to T$ be separated and of finite type, with $W_U \to U$ an isomorphism. There is a U-admissible blow-up $T' \to T$ such that the inverse image of U is schematically dense in T' and the strict transform $W' \to T'$ of $W \to T$ is an open immersion.*

Proof. This is [Stacks, 081S]. The idea is to view the inverse of $W_U \to U$ as a partial section of $W \to T$ and to extend it by Theorem A.16; the strict transform is then the graph of the extension.

Choose a finite type ideal $\mathcal{I}_0$ cutting out $T \setminus U$. Its blow-up $T_1 \to T$ makes U schematically dense by Lemma A.10. Over T_1, the base change of $W \to T$ to T_1 is separated and of finite type, and the inverse of $W_U \to U$ is a section of it over U. We may therefore apply Theorem A.16 over T_1, with $V = U$, which gives a further blow-up and an extension of this section, with closed graph, over an open containing the inverse image of U. By Lemma A.13 the composite is a U-admissible blow-up $\mathrm{Bl}_{\mathcal{I}}(T)$. Let $b : T' \to T$ be the blow-up of $\mathcal{I}\mathcal{I}_0$, which dominates both. Pulling back to T' gives an open $W_0 \subseteq T'$ containing $b^{-1}(U)$ and a map $W_0 \to W$ over T whose graph Γ is closed in $W \times_T T'$. As the zero set of $\mathcal{I}\mathcal{I}_0$ is exactly $T \setminus U$, $b^{-1}(U)$ is schematically dense in T', hence in W_0.

We included the factor $\mathcal{I}_0$ so that the center of b is exactly $T \setminus U$. Then the strict transform W' is the schematic closure of W_U in $W \times_T T'$. Over U, the subscheme W_U is the graph of the original section, so it lies in Γ and $W' \subseteq \Gamma$. Conversely, $\Gamma \simeq W_0$ contains the inverse image of U as a schematically dense open. Hence the schematic closure of W_U in Γ is Γ. We conclude that $W' = \Gamma \simeq W_0$, and $W' \to T'$ is an open immersion. ∎

The blow-up is not uniquely determined. However, on any such modification with U schematically dense, the graph of the extension is forced to be the closure of the original section. By Lemma A.4, the identifications obtained on common admissible refinements therefore agree.

In the gluing argument below, a compactification of an open V of M is related to M itself by a proper map defined on a larger open of the compactification.

Lemma A.19 (Extending a graph from a compactification)**.** *Let S be qcqs, let $M \to S$ be separated of finite type, and let $V \subseteq M$ be a quasi-compact open with a compactification $V \subseteq Y$ over S. There are a V-admissible blow-up $Y' \to Y$, an open $V \subseteq V' \subseteq Y'$ and a proper map $\psi : V' \to M$ with $\psi^{-1}(V) = V$ and $\psi|_V = \mathrm{id}_V$.*

Proof. The map ψ is the projection to M of the closure of the graph of $V \to M$. By elimination of indeterminacy this closure becomes an open of a blow-up of Y, and ψ is proper because the graph is closed.

Let G be the schematic closure of the graph of $V \to M$ in $Y \times_S M$. Its projection to Y is separated of finite type, and it is an isomorphism over V. By Theorem A.18, after a V-admissible blow-up $Y' \to Y$ its strict transform G' is an open $V' \subseteq Y'$. This strict transform is the closure of the graph of $V \to M$ in $Y' \times_S M$, since both are the closure of the same graph outside the exceptional divisor [Stacks, 0F3Y]. So G' is closed in $Y' \times_S M$, and since the projection of $Y' \times_S M$ to M is proper, the map $V' \simeq G' \to M$ is proper. It remains to identify the inverse image of V. For this, note that over $V \subseteq M$ the graph of the inclusion $V \to Y'$ is already closed in $Y' \times_S V$, because Y' is separated over S. The inverse image of V under ψ is therefore this graph, that is, V itself. See also [Stacks, 0F3Z]. ∎

A.3 Separation by blow-ups

The inductive step of Nagata's theorem is Proposition A.23. Suppose that $M = M_1 \cup M_2$ with $M_1 \cap M_2$ dense, and that each M_i has a compactification X_i; we want a compactification of M. The natural candidate glues M to each X_i along M_i, after we remove from X_i the closure of the part lying over $M \setminus M_i$. For the result to be separated, however, and for every valuation to extend into one of the two pieces, the closures in X_i of the two disjoint closed subsets over $M \setminus M_1$ and $M \setminus M_2$ must be disjoint. They need not be, and this is where Lemma A.20 enters: it makes them disjoint by an admissible blow-up, as with the coordinate axes in Section A.2.

We need two other lemmas. Lemma A.21 keeps an isomorphism near such a closure, and Lemma A.22 glues two modifications into one separated scheme. Both use elimination of indeterminacy from the previous section. Recall that it turns a map that is an isomorphism over a quasi-compact open U into an open immersion after a U-admissible blow-up.

Lemma A.20 (Separation of closures)**.** *Let T be qcqs and let $V \subseteq T$ be a quasi-compact open. If $F_1, F_2 \subseteq V$ are disjoint constructible closed subsets, there is a V-admissible blow-up of T on which their closures are disjoint.*

We quote this from [Stacks, 0F3V] and sketch the argument, which works with closed subschemes of T rather than closed subsets of V. Let $\mathcal{I}_1, \mathcal{I}_2$ be ideals of finite type on T cutting out closed subschemes Z_1, Z_2 that are disjoint over V and restrict to F_1, F_2 there. Producing such ideals on all of T is where constructibility and finite presentation are needed. Since $\mathcal{I}_1 + \mathcal{I}_2$ is the unit ideal over V, its blow-up is V-admissible. Moreover, on each affine chart the extended ideal is generated by a single element of $\mathcal{I}_1$ or of $\mathcal{I}_2$. Now, on a chart where it is generated by $f \in \mathcal{I}_1$, the strict transform of Z_1 is obtained from its inverse image by dividing out the f-power torsion. Since f vanishes on Z_1, every element of the ring of the inverse image is killed by f. Hence the strict transform of Z_1 does not meet that chart. The same holds for Z_2 on the other charts, and therefore the strict transforms of Z_1 and Z_2 are disjoint [Stacks, 080P]. The constructibility needed to produce such ideals causes no trouble for us, because in the Noetherian applications below every closed subset is constructible and its reduced ideal is of finite type.

For the gluing we also need to know that a proper map that is an isomorphism near a closed subset of V remains one near its closure in T, after a blow-up.

Lemma A.21 (Extending an isomorphism over the closure of a closed subset)**.** *Let $q : P \to T$ be a proper morphism of qcqs schemes, let $V \subseteq T$ be a quasi-compact open, and let $F \subseteq V$ be closed. Suppose that q is an isomorphism over a neighbourhood of F in V. After a V-admissible blow-up, the strict transform of q is an isomorphism over a neighbourhood of the closure of F.*

Proof. We separate the closure of F from the closure of the locus where q fails to be an isomorphism, and then apply elimination of indeterminacy on the complement of the second closure.

Inside V, the set F is disjoint from the closed complement of the largest open over which q is an isomorphism. Since V is spectral, both closed subsets can be enlarged to disjoint constructible closed subsets. To these we apply Lemma A.20 so that their closures in T become disjoint. Then, on the complement T_0 of the second closure, the restriction of q is an isomorphism over an open V_0 containing F. After making this open schematically dense, we apply Theorem A.18 on T_0. The resulting strict transform is both open and proper over the modification, and hence it is an isomorphism there. By the extension part of Lemma A.13, this blow-up of T_0 extends to a V-admissible blow-up of T, which is an isomorphism over an open containing the closure of F. We have only sketched the argument; the precise version, including the choice of constructible enlargements, is [Stacks, 0F3W]. ∎

The last ingredient replaces two separated schemes containing a common open by modifications that glue to a separated scheme.

Lemma A.22 (Gluing after admissible blow-ups)**.** *Let S be qcqs, and let $U \subseteq T_i$ $(i = 1, 2)$ be open immersions of separated S-schemes of finite type, with U quasi-compact. There are U-admissible blow-ups $T_i' \to T_i$ and open immersions $T_i' \to T$ agreeing on U, with T separated and of finite type over S. If U is dense in both T_i, then it is dense in T.*

Proof. We glue the T_i along the closure of U in $T_1 \times_S T_2$, after blow-ups that make this closure open in both. The glued scheme is separated because the closure is closed in the product.

Let G be the schematic closure of U in $T_1 \times_S T_2$. Then Theorem A.18, applied to the two projections, makes the strict transforms of G opens in suitable blow-ups of the two sides. To obtain a single modification, we blow up G in the product of the two pulled-back centers, which gives a common refinement. By Lemma A.13, these further centers extend from the opens to each side, and the resulting scheme G' is an open in both T_i'.

The map $G' \to T_1' \times_S T_2'$ is a closed immersion. It is an immersion, and it is proper because $G' \to G$ is proper, $G \subseteq T_1 \times_S T_2$ is closed, and the modified product is separated over the original product. We glue the two T_i' along G', and the glued scheme T is covered by T_1' and T_2'. To see that T is separated, note that the diagonal of T restricted to $T_i' \times_S T_j'$ is the diagonal of T_i' for $i = j$ and the graph G' for $i \neq j$. Both are closed, so T is separated. It is of finite type because it has a finite cover by opens of finite type.

Suppose that U is dense in both T_i. The complement of the exceptional divisor of a blow-up is schematically dense, as in the proof of Lemma A.10, and it contains the inverse image of U as a dense open. Hence U is dense in each T_i', and so in T. This construction is [Stacks, 0F3X]. ∎

The gluing lemma produces a separated scheme but says nothing about properness. In the inductive step we prove properness by the valuative criterion, using the separation of closures.

Proposition A.23 (Compactifying a union of two opens)**.** *Let S be Noetherian and let $M \to S$ be separated of finite type. Suppose that $M = M_1 \cup M_2$, that each M_i has a compactification over S, and that $D = M_1 \cap M_2$ is dense in M. Then M has a compactification over S.*

Proof. The idea is to glue M to each compactification X_i of M_i after removing from X_i the closure of the part lying over $M \setminus M_i$. This gives separated schemes W_i containing M, and a common modification combines them into one separated W. It remains to check that W is

proper, and for this we use the valuative criterion. Separating the closures of the boundaries, we obtain (A.1), and from it we deduce that every valuation of D extends into one of the W_i.

Separating the boundaries. Since the subsets to be removed from X_i are defined by points of M outside M_i, we must relate each X_i to M itself. Choose dense compactifications $M_i \subseteq X_i$. By Lemma A.19, after admissible blow-ups there are opens $M_i \subseteq V_i \subseteq X_i$ and proper maps $\psi_i : V_i \to M$ that are isomorphisms above M_i. Put, for $\{i, j\} = \{1, 2\}$,

$$Z_i = M \setminus M_j, \qquad Z_{i,r} = \psi_i^{-1}(Z_r), \qquad B_{i,r} = \overline{Z_{i,r}} \subseteq X_i.$$

The subsets $Z_{i,1}, Z_{i,2}$ are disjoint and closed in V_i. By Lemma A.20, after a further V_i-admissible blow-up we may assume that $B_{i,1} \cap B_{i,2} = \varnothing$, and subsequent V_i-admissible blow-ups preserve this disjointness.

Let X_{12} be the schematic closure of $V_1 \times_M V_2$ in $X_1 \times_S X_2$, with proper projections p_i. Above V_1, the first projection is the base change of ψ_2, so it is an isomorphism near $Z_{1,2}$. By Lemma A.21 it becomes an isomorphism near $B_{1,2}$ after a blow-up. We replace X_{12} by its strict transform, or equivalently by the closure of the new graph. Since p_1 is now an isomorphism over a neighbourhood of $B_{1,2}$ and $Z_{1,2}$ is dense in $B_{1,2}$, the inverse image of $Z_{1,2}$ is dense in $p_1^{-1}(B_{1,2})$. The second projection maps that inverse image into $Z_{2,2}$. By density it maps all of $p_1^{-1}(B_{1,2})$ into $B_{2,2}$, which is disjoint from $B_{2,1}$. This gives the disjointness used in the valuative argument below:

$$X_{12} \cap (B_{1,2} \times_S B_{2,1}) = \varnothing. \tag{A.1}$$

The separated schemes W_i. We glue

$$W_i = M \coprod_{M_i} (X_i \setminus B_{i,j}).$$

We claim that these schemes are separated over S. The scheme W_i is covered by M and $X_i \setminus B_{i,j}$, both separated, and they are glued along M_i. It therefore suffices to show that the graph of the inclusion $M_i \to X_i \setminus B_{i,j}$, which is the mixed part of the diagonal, is closed in $M \times_S (X_i \setminus B_{i,j})$.

Suppose that a point of the graph of M_i specializes in $M \times_S (X_i \setminus B_{i,j})$. Since ψ_i is proper, its graph is closed in $M \times_S X_i$. Hence the specialized point lies in the graph of ψ_i, over a point of V_i. If its image in M lay in $M \setminus M_i = Z_j$, the lifted point would lie in $Z_{i,j} \subseteq B_{i,j}$, contradicting the second coordinate. So the specialization remains in the graph. The graph is an immersion with constructible image, and in the Noetherian situation a constructible subset that is closed under specialization is closed. Therefore the graph is a closed immersion, which proves separatedness.

Valuations of D. For properness we use the valuative criterion, and we show first that every valuation of D extends into one of the W_i. Let R be a valuation ring with fraction field K, and let $\operatorname{Spec}(K) \to D$ be an S-map over a given $\operatorname{Spec}(R) \to S$. It extends to both X_i, since they are proper, and the pair of extensions factors through X_{12}, because the closure of the graph contains its generic point and $\operatorname{Spec}(K)$ is schematically dense in $\operatorname{Spec}(R)$. By (A.1), the closed point cannot map into both of the removed closed subsets $B_{1,2}$ and $B_{2,1}$. We may therefore choose i such that the closed point does not map into $B_{i,j}$. Since an open containing the closed point of a local scheme contains the whole scheme, the extension factors through $X_i \setminus B_{i,j} \subseteq W_i$.

Gluing and properness. Lemma A.22, applied to $M \subseteq W_i$, gives a separated W of finite type, covered by admissible modifications W_i'. Since the modification is proper, the lift of the valuation into W_i lifts further into W_i', hence into W. Such lifts are unique because W is separated. The valuative criterion [Stacks, 0894], in the form that only requires lifts of maps from $\operatorname{Spec}(K)$ into the dense open D, then shows that $W \to S$ is universally closed. As W is also of finite type and

separated, it is proper over S. Its open M yields the desired compactification, which becomes dense after we take the schematic closure of M. This is [Stacks, 0F40]. ∎

A.4 Proof of Nagata's theorem

Over a Noetherian base, the theorem follows from Proposition A.23 by induction over an affine cover whose members overlap densely. Each affine piece is compactified by its closure in a projective space. The proposition then compactifies the union of the first r pieces with the next one. It requires the overlap to be dense, because its valuative criterion only tests maps from spectra of valuation rings whose generic point maps into the overlap. Lemma A.24 provides a cover whose total intersection contains all generic points of X.

Over a general qcqs base we use Noetherian approximation. This works because a morphism of finite presentation is the base change of one over a Noetherian scheme, and compactifications are stable under base change. A morphism that is only of finite type, however, need not descend in this way. We therefore first embed X as a closed subscheme of a scheme of finite presentation over S, and take the closure of X in a compactification of the latter.

Lemma A.24 (Affine covers with dense intersection)**.** *For a scheme X of finite type over a Noetherian scheme S, there is a finite affine cover $X = \bigcup_{r=1}^{m} U_r$ whose total intersection is dense in X. If $X \to S$ is separated, then each $U_r \to S$ has a projective compactification.*

Proof. The scheme X has finitely many irreducible components. By [Stacks, 01ZX], every point of X has an affine open neighbourhood that also contains all the generic points of those components, and since X is quasi-compact, finitely many such opens cover X. Their total intersection still contains all generic points, so it is dense. For the second assertion, note that each U_r is affine of finite type over S, so by [Stacks, 04II] we may embed it as a locally closed subscheme of some $\mathbf{A}_S^{n_r}$. Its schematic closure in $\mathbf{P}_S^{n_r}$ is proper, and by Lemma A.3 the scheme U_r is open in that closure. ∎

We prove the theorem in three steps of increasing generality: a Noetherian base, a finitely presented morphism over any qcqs base, and a morphism of finite type.

Proof of Theorem A.2. If S is Noetherian, we choose a cover as in Lemma A.24 and argue by induction on the unions $X_r = U_1 \cup \cdots \cup U_r$. For $r = 1$, the lemma gives a compactification of U_1. If X_r has a compactification, we apply Proposition A.23 to $M = X_{r+1}$, with $M_1 = X_r$ and $M_2 = U_{r+1}$. Each piece has a compactification, by induction and by the lemma. The overlap $X_r \cap U_{r+1}$ is dense in X_{r+1}, since it contains the total intersection, which contains all generic points of X. Hence X_{r+1} has a compactification. The last union is X itself, and taking the schematic closure of X makes the compactification dense.

Now let S be a general qcqs scheme, and suppose first that $X \to S$ is separated and of finite presentation. By absolute Noetherian approximation we can write

$$S = \varprojlim_i S_i$$

with S_i Noetherian and affine transition maps [Stacks, 01ZA]. Since $X \to S$ is of finite presentation, it descends to a model $X_i \to S_i$ for some i, and after increasing i the model is separated [Stacks, 01ZM, 01ZQ]. We then compactify this model by the Noetherian case and base change the compactification to S. The result is a compactification, because open immersions and proper

morphisms are stable under base change, and taking the schematic closure of X makes the result dense.

Finally, let $X \to S$ be separated of finite type. By [Stacks, 01ZJ], there is a closed immersion $X \hookrightarrow X_0$ with $X_0 \to S$ separated of finite presentation. We compactify X_0 by the case already proved, and take the schematic closure of X in that compactification. This closure is proper over S. Its restriction to the open X_0 is the closed subscheme X itself, so X is open in the closure, and it is schematically dense there by Lemma A.3. ∎

A.5 Chow's lemma

Let $f : X \to S$ be proper. In Section 4.5 proper base change is first proved for projective morphisms. By Lemma 4.29 it then descends along a proper surjection $\pi : \widetilde{X} \to X$ as soon as it holds for π and for the composite $f\pi$. Chow's lemma says precisely that such a π exists with both π and $f\pi$ projective. Some modification of X is unavoidable here, since there are proper varieties that are not projective.

Theorem A.25 (Chow's lemma, projective form)**.** *Let $f : X \to S$ be proper with S qcqs. There are a proper surjection $\pi : \widetilde{X} \to X$ and a closed immersion $i_S : \widetilde{X} \hookrightarrow \mathbf{P}^n_S$ such that $f\pi = q_S i_S$, where $q_S : \mathbf{P}^n_S \to S$ is the projection. Moreover,*

$$i_X = (\pi, i_S) : \widetilde{X} \hookrightarrow X \times_S \mathbf{P}^n_S = \mathbf{P}^n_X$$

is a closed immersion. Both π and $f\pi$ are projective.

Proof. We take the immersion into projective space given by the Stacks Project and show that properness of X makes it closed. The second assertion then follows by writing i_X as a graph followed by a base change of i_S.

The version of Chow's lemma in [Stacks, 0202] gives a proper surjection $\pi : \widetilde{X} \to X$ and an immersion $i_S : \widetilde{X} \to \mathbf{P}^n_S$ for every separated $X \to S$ of finite type with S qcqs. It is deduced from the Noetherian case by the approximation argument in the proof of Theorem A.2, using the embedding of X into a scheme of finite presentation [Stacks, 01ZJ]. In our situation, moreover, f and π are proper, so $\widetilde{X}$ is proper over S. Now a map from a proper S-scheme to a separated S-scheme is proper, because its graph is closed and the projection of the product to the target is proper. Hence i_S is a proper immersion. Its image is therefore closed, and i_S is a closed immersion.

For the second assertion, we factor i_X as the graph of π followed by the base change of i_S:

$$\widetilde{X} \longrightarrow \widetilde{X} \times_S X \longrightarrow \mathbf{P}^n_S \times_S X.$$

The first arrow is closed because $X \to S$ is separated, and the second is closed because i_S is. Their composite is i_X. Since π is i_X followed by the projection $\mathbf{P}^n_X \to X$, it is projective, and so is $f\pi = q_S i_S$. ∎

B Abelian varieties

Two steps of the main proof come down to the torsion points of a Jacobian. By the Kummer sequence, $H^1(C, \mu_n)$ of a curve is the n-torsion of its Jacobian. The first step, the finiteness of H^1 needed for proper base change, is therefore a count of torsion points. The second step is the

finiteness of geometric monodromy in rank one, where purity begins. It needs Frobenius to fix no class in H^1 of a curve over $\mathbf{F}_q$ (Proposition 14.3).

For a complex torus $\mathbf{C}^g/\Lambda$ the count is trivial: the n-torsion is $\frac{1}{n}\Lambda/\Lambda \simeq (\mathbf{Z}/n)^{2g}$. For an abelian variety A of dimension g over an arbitrary field k, however, there is no lattice, and the torsion points must be counted algebraically. This is what we do: we prove that for n invertible in k the group scheme $A[n]$ is finite étale of rank n^{2g} (Proposition B.12). It follows that the Tate module $T_\ell A = \varprojlim_m A[\ell^m](\bar{k})$ is free of rank $2g$ over $\mathbf{Z}_\ell$ (Proposition B.14). Moreover, over $\mathbf{F}_q$, no power of Frobenius has a nonzero fixed vector on $T_\ell A$ (Proposition B.15). The reason is short: a fixed vector would be a compatible system of ℓ-power torsion points of unbounded order in the finite group $A_0(\mathbf{F}_{q^m})$.

The count rests on one formula. If $\mathcal{N}$ is a symmetric ample invertible sheaf on A, then

$$[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$$

(Proposition B.11). On an elliptic curve with $\mathcal{N} = \mathcal{O}(e)$ this says that the divisor $[n]^*(e)$ is linearly equivalent to $n^2(e)$. When n is prime to the characteristic, $[n]$ is étale and $[n]^*(e)$ is the reduced divisor $E[n]$. Comparing degrees then gives $\#E[n] = n^2$.

In general the formula is used twice. First, it forces $[n]$ to have finite fibres. Indeed, $\mathcal{N}^{\otimes n^2}$ has positive degree on every complete curve in A, while $[n]^*\mathcal{N}$ is trivial on a curve contracted by $[n]$. Second, we compute the Euler characteristics $\chi(A, [n]^*\mathcal{N}^{\otimes m})$ in two ways, and comparing their leading coefficients gives $\deg[n] = n^{2g}$.

The formula itself comes from the theorem of the cube (Theorem B.6). Following [Stacks, 0BEZ], we prove the theorem of the cube together with the rigidity lemma. As a by-product, from the rigidity lemma we deduce that A is commutative; see also [Mil86a].

In this appendix all group schemes are of finite type over a field k of arbitrary characteristic.

B.1 The theorem of the cube

The theorem of the cube says that an invertible sheaf $\mathcal{L}$ on $X \times Y \times Z$, with X, Y complete varieties and Z connected, that is trivial on the three slices through a point (x, y, z) is trivial. To prove it, we view $\mathcal{L}$ as a family of invertible sheaves on $X \times Y$ parametrized by Z. Following [Stacks, 0BEZ], we then consider the set of points of Z over which the sheaf is trivial. This set is closed, by semicontinuity. It is also open, by a computation of H^0 and H^1 near such a point. Since it contains z and Z is connected, it is all of Z. The openness uses the triviality on the two slices $x \times Y \times Z$ and $X \times y \times Z$, and with only two factors it fails.

Example B.1 (Two factors do not suffice)**.** Let E be an elliptic curve over an algebraically closed field, with origin O, and let $\Delta \subset E \times E$ be the diagonal. On $E \times E$ put

$$\mathcal{L} = \mathcal{O}_{E\times E}(\Delta - E \times \{O\} - \{O\} \times E).$$

Consider the slice $E \times \{t\}$. It meets the diagonal in the point t, and the divisor $\{O\} \times E$ meets it in the point O. The divisor $E \times \{O\}$, on the other hand, is pulled back from the second factor, so it contributes the trivial sheaf for every t. Hence the restriction is $\mathcal{O}_E(t - O)$. By symmetry the restriction to $\{t\} \times E$ is also $\mathcal{O}_E(t - O)$. In particular $\mathcal{L}$ is trivial on both slices through (O, O).

However, $\mathcal{L}$ is not trivial, because for $t \neq O$ the sheaf $\mathcal{O}_E(t - O)$ is nontrivial. Indeed, a rational function with divisor $t - O$ would be a map $E \to \mathbf{P}^1$ of degree one, which is impossible in genus

one. The set of t over which the fibre is trivial is the point O. It is closed, as Lemma B.2 below predicts, but not open.

Lemma B.2 (Triviality on fibres)**.** *Let P be a proper geometrically integral k-scheme, let T be a k-scheme of finite type, let $f : P \times T \to T$ be the projection, and let $\mathcal{L}$ be an invertible sheaf on $P \times T$. For $t \in T$ write $P_t = P \times_k \kappa(t)$ and $\mathcal{L}_t = \mathcal{L}|_{P_t}$.*

1. *$\mathcal{L}_t$ is trivial if and only if $H^0(P_t, \mathcal{L}_t) \neq 0$ and $H^0(P_t, \mathcal{L}_t^{-1}) \neq 0$.*
2. *The set of $t \in T$ such that $\mathcal{L}_t$ is trivial is closed.*

Proof. The proof uses only that P_t is proper and integral with $H^0(P_t, \mathcal{O}_{P_t}) = \kappa(t)$, which holds by [Stacks, 0BUG] because P is proper and geometrically integral.

For (1), a trivial sheaf and its inverse both have the nonzero section 1, so only the converse needs an argument. On $\mathbf{P}^1$, for instance, $\mathcal{O}(d)$ and $\mathcal{O}(-d)$ both have nonzero sections only when $d = 0$, and the general argument is the same. Let s and s' be nonzero sections of $\mathcal{L}_t$ and $\mathcal{L}_t^{-1}$. Since P_t is integral, the product ss' is a nonzero element of $H^0(P_t, \mathcal{O}) = \kappa(t)$, hence a unit. Since ss' would vanish wherever s does, the section s vanishes nowhere, and it trivializes $\mathcal{L}_t$.

For (2), note that by (1) the set in question is the intersection of the loci where $\dim H^0(P_t, \mathcal{L}_t)$ and $\dim H^0(P_t, \mathcal{L}_t^{-1})$ are positive. These loci are closed because the functions $t \mapsto \dim H^0(P_t, \mathcal{L}_t^{\pm 1})$ are upper semicontinuous [Stacks, 0BDN], as $\mathcal{L}$ is flat over T and f is proper. ∎

Part (2) is the closed half of the theorem of the cube. If the sheaf is trivial over the members of an open cover of T, then the evaluation map identifies it with a pullback from T, since sections along the proper factor P are constant.

Corollary B.3 (Triviality over an open cover of the base)**.** *With the notation of Lemma B.2, suppose that $\mathcal{L}$ is trivial on $P \times U_\alpha$ for an open cover (U_α) of T. Then $f_*\mathcal{L}$ is invertible and the evaluation map $f^* f_* \mathcal{L} \to \mathcal{L}$ is an isomorphism.*

Proof. Both assertions are local on T, so we may replace T by one of the U_α and assume that $\mathcal{L} \simeq \mathcal{O}_{P \times T}$. Since $H^0(P, \mathcal{O}_P) = k$ [Stacks, 0BUG], and T is flat over k, flat base change gives $f_* \mathcal{O}_{P \times T} = \mathcal{O}_T$. Hence the chosen trivialization identifies $f_*\mathcal{L}$ with the invertible sheaf $\mathcal{O}_T$. Under the same identification the evaluation map $f^* f_* \mathcal{L} \to \mathcal{L}$ becomes the identity of $\mathcal{O}_{P \times T}$. ∎

The open half of the theorem is the substantial step. It uses the fact that H^1 of the structure sheaf of a product is the sum of the H^1 of the factors.

Lemma B.4 (H^1 of the structure sheaf of a product)**.** *Let K be a field, let P and Q be proper K-schemes with $H^0(P, \mathcal{O}_P) = H^0(Q, \mathcal{O}_Q) = K$, and let $p_0 \in P(K)$ and $q_0 \in Q(K)$. Restriction to the slices $P \times q_0$ and $p_0 \times Q$ is an isomorphism*

$$H^1(P \times Q, \mathcal{O}) \longrightarrow H^1(P, \mathcal{O}) \oplus H^1(Q, \mathcal{O}).$$

Proof. By the Künneth formula [Stacks, 0BED], and since H^0 of $\mathcal{O}$ is K on both factors, the source is $\mathrm{pr}_1^* H^1(P, \mathcal{O}) \oplus \mathrm{pr}_2^* H^1(Q, \mathcal{O})$. Restriction to $P \times q_0$ is the identity on the first summand. It kills the second, because on the second summand it factors through H^1 of the point q_0. Restriction to $p_0 \times Q$ does the opposite. ∎

Lemma B.5 (Triviality near a fibre)**.** *Let P and Q be proper geometrically integral k-schemes with points $p_0 \in P(k)$ and $q_0 \in Q(k)$, let T be a k-scheme of finite type, and let $\mathcal{L}$ be an invertible sheaf on $P \times Q \times T$ whose restrictions to $p_0 \times Q \times T$ and to $P \times q_0 \times T$ are trivial. If $\mathcal{L}$ is trivial on the fibre over a point $t \in T$, then $\mathcal{L}$ is trivial on $P \times Q \times U$ for an open neighbourhood U of t.*

Proof. We want a section of $\mathcal{L}$ over $X \times U$, where $X = P \times Q$, that does not vanish on the fibre over t. Its zero locus then misses a neighbourhood of that fibre. Such a section exists as soon as $f_*\mathcal{L}$ is free of rank one near t, with formation compatible with passage to the fibre.

To get this, we represent $Rf_*\mathcal{L}$, and the pushforwards of the restrictions of $\mathcal{L}$ to the two slices, by complexes of free modules whose differentials vanish at t. By the Künneth formula, restriction from H^1 of the fibre to H^1 of the two slices is an isomorphism. This forces the first differential of $Rf_*\mathcal{L}$ to vanish after shrinking T. Then H^0 of $Rf_*\mathcal{L}$ is free of rank one, as we need.

Setup. Since $X_{\bar{k}} = P_{\bar{k}} \times Q_{\bar{k}}$ is a product of integral varieties, X is geometrically integral [Stacks, 05P3]. We may assume that $T = \operatorname{Spec} R$ is affine, and we let $\mathfrak{p}$ be the prime of t. Below, "shrinking T" means replacing R by R_g for some $g \notin \mathfrak{p}$; everything already constructed survives this. Consider the projections

$$f : X \times T \to T, \qquad f_1 : P \times T \to T, \qquad f_2 : Q \times T \to T,$$

and the restrictions $\mathcal{L}_1$ and $\mathcal{L}_2$ of $\mathcal{L}$ to $P \times q_0 \times T$ and $p_0 \times Q \times T$, which are trivial. The complexes $E = Rf_*\mathcal{L}$ and $E_i = Rf_{i*}\mathcal{L}_i$ are perfect, and their formation commutes with base change [Stacks, 0B91]. Restriction of sections gives a map $E \to E_1 \oplus E_2$, also compatible with base change. Since T is affine, $H^0(X \times T, \mathcal{L}) = H^0(E)$ and likewise for E_i.

Minimal complexes. A perfect complex over a local ring can be represented by a complex of free modules whose differentials vanish modulo the maximal ideal; the ranks are then the dimensions of the cohomology of the fibre. We use this to read off the ranks from the fibre over t, where everything is trivial. To do so, put $h^j = \dim H^j(X_t, \mathcal{O})$ and $h_i^j = \dim H^j$ of $\mathcal{O}$ on the fibres P_t and Q_t. Since $\mathcal{L}$ is trivial on X_t and the $\mathcal{L}_i$ are trivial, $E \otimes^{\mathbf{L}}_R \kappa(t)$ and $E_i \otimes^{\mathbf{L}}_R \kappa(t)$ have cohomology of dimensions h^j and h_i^j, with $h^0 = h_i^0 = 1$. After shrinking T we may therefore represent E and E_i by complexes of finite free modules

$$R \xrightarrow{d} R^{h^1} \to R^{h^2} \to \cdots, \qquad R \xrightarrow{d_i} R^{h_i^1} \to R^{h_i^2} \to \cdots$$

whose differentials vanish modulo $\mathfrak{p}$ [Stacks, 0BCD]. We represent the map $E \to E_i$ by a map of complexes [Stacks, 064B], with components $a_i : R \to R$ in degree 0 and $b_i : R^{h^1} \to R^{h_i^1}$ in degree 1. Since the differentials vanish modulo $\mathfrak{p}$, the map $b_i \otimes \kappa(t)$ is the restriction map on H^1 of the fibre over t.

We want to show that $d = 0$ after shrinking T, since then $H^0(E) = \ker d = R$ is free of rank one. The maps b_i will let us transfer this question to the d_i, which are easier because $\mathcal{L}_i$ is trivial.

The differentials d_i vanish. Fix a trivialization $\mathcal{L}_1 \simeq \mathcal{O}$, and let $u \in H^0(E_1) = \ker d_1 \subset R$ be the section corresponding to 1. Its image in $H^0(E_1 \otimes^{\mathbf{L}} \kappa(t)) = H^0(P_t, \mathcal{O})$ is $1 \neq 0$, so $u \notin \mathfrak{p}$. Since $d_1(u) = u\, d_1(1) = 0$, after inverting u we get $d_1 = 0$. The same argument gives $d_2 = 0$.

The differential d vanishes. This step uses the two slices together. If the map (b_1, b_2) is an isomorphism, then d is determined by d_1 and d_2, and this condition can be checked on the fibre. The map $(b_1, b_2) \otimes \kappa(t)$ is restriction from $H^1(P_t \times Q_t, \mathcal{O})$ to the slices $P_t \times q_0$ and $p_0 \times Q_t$. It is an isomorphism by Lemma B.4, and in particular $h^1 = h_1^1 + h_2^1$. Hence the square matrix (b_1, b_2) has determinant outside $\mathfrak{p}$. After inverting this determinant, (b_1, b_2) is an isomorphism. Since the maps form a map of complexes, we have $b_i \circ d = d_i \circ a_i = 0$, and therefore $d = 0$.

Conclusion. A generator of H^0 gives the trivialization. We have $H^0(X \times T, \mathcal{L}) = H^0(E) = \ker d = R$, and its generator σ maps to a generator of $H^0(E \otimes^{\mathbf{L}} \kappa(t)) = H^0(X_t, \mathcal{L}_t)$. Since $\mathcal{L}_t \simeq \mathcal{O}_{X_t}$ and $H^0(X_t, \mathcal{O}) = \kappa(t)$, the section σ vanishes nowhere on X_t. Its zero locus is closed in $X \times T$ and disjoint from X_t. Since f is proper, its image in T is a closed set not containing t. On the complement U of this image, σ trivializes $\mathcal{L}$. ∎

With the closed and the open half in hand, it remains to glue the local trivializations.

Theorem B.6 (Theorem of the cube)**.** *Let X and Y be proper geometrically integral k-schemes with points $x \in X(k)$ and $y \in Y(k)$, let Z be a connected k-scheme of finite type with a point $z \in Z(k)$, and let $\mathcal{L}$ be an invertible sheaf on $X \times Y \times Z$. If the restrictions of $\mathcal{L}$ to $x \times Y \times Z$, $X \times y \times Z$ and $X \times Y \times z$ are trivial, then $\mathcal{L}$ is trivial.*

Proof. The local trivializations are glued by Corollary B.3, which shows that $\mathcal{L}$ is pulled back from $Y \times Z$; the slice $x \times Y \times Z$ then identifies the sheaf on $Y \times Z$.

Let $Z_0 \subset Z$ be the set of points over which $\mathcal{L}$ is trivial on the fibre of $p : X \times Y \times Z \to Z$. It contains z. It is closed by Lemma lemma-triviality-fibres applied to $X \times Y$, which is proper and geometrically integral [Stacks, 05P3]. It is open by Lemma B.5. Since Z is connected, $Z_0 = Z$. Now Lemma B.5 gives an open cover of Z over whose members $\mathcal{L}$ is trivial.

Let $\pi : X \times Y \times Z \to Y \times Z$ be the projection and σ its section through x. We apply Corollary B.3 with $P = X$, $T = Y \times Z$ and the open cover by the products of Y with these opens. It shows that $\mathcal{M} = \pi_* \mathcal{L}$ is invertible and that $\pi^* \mathcal{M} \to \mathcal{L}$ is an isomorphism. Hence $\mathcal{M} \simeq \sigma^* \pi^* \mathcal{M} \simeq \sigma^* \mathcal{L}$. This is the restriction of $\mathcal{L}$ to $x \times Y \times Z$, which is trivial by hypothesis. We conclude that $\mathcal{L} \simeq \pi^* \mathcal{M}$ is trivial. ∎

The second tool, which we use to show that an abelian variety is commutative (Lemma B.9), is a rigidity statement. It says that a family of maps from a proper variety that is constant on one member is constant on all nearby members.

Lemma B.7 (Rigidity)**.** *Let X be a proper geometrically integral k-scheme, let Y be a k-scheme of finite type, let Z be a k-scheme, and let $h : X \times Y \to Z$ be a morphism. If h maps $X \times y_0$ to a single point for some $y_0 \in Y$, then there are an open neighbourhood V of y_0 and a morphism $g : V \to Z$ with $h = g \circ \mathrm{pr}_2$ on $X \times V$.*

Proof. Near $X \times y_0$ the map h lands in an affine open of Z, and a map to an affine scheme is given by functions. Since X is proper and geometrically integral, functions on $X \times V$ are functions on V. Hence h factors through V.

Let $W \subset Z$ be an affine open containing $h(X \times y_0)$. Since X is proper, the image in Y of the closed set $h^{-1}(Z \setminus W)$ is closed, and its complement contains y_0. Let V be an affine open neighbourhood of y_0 in this complement. Then h maps $X \times V$ into the affine scheme W. It is therefore determined by the ring map $\mathcal{O}(W) \to H^0(X \times V, \mathcal{O})$ [Stacks, 01I1]. By flat base change [Stacks, 02KH] and $H^0(X, \mathcal{O}_X) = k$ [Stacks, 0BUG], we have $H^0(X \times V, \mathcal{O}) = \mathcal{O}(V)$. The ring map is thus pr_2^* composed with a ring map $\mathcal{O}(W) \to \mathcal{O}(V)$, and this ring map defines g. ∎

B.2 Multiplication by n

Let $\mathcal{L}$ be an invertible sheaf on an abelian variety A. We pull $\mathcal{L}$ back to $A \times A \times A$ along the partial sums of the coordinates. There the theorem of the cube gives a second order recursion for $n \mapsto [n]^* \mathcal{L}$, whose solution is

$$[n]^*\mathcal{L} \simeq \mathcal{L}^{\otimes n(n+1)/2} \otimes ([-1]^*\mathcal{L})^{\otimes n(n-1)/2}$$

(Proposition B.11). For symmetric $\mathcal{L}$ the formula gives $\mathcal{L}^{\otimes n^2}$. If $[-1]^*\mathcal{L} \simeq \mathcal{L}^{-1}$, as for an invertible sheaf of degree zero on an elliptic curve, it gives $\mathcal{L}^{\otimes n}$.

We apply the formula to a symmetric ample sheaf (Lemma B.10). It shows that $[n]$ is finite and flat of degree n^{2g}, and that $[n]$ is étale when n is invertible in k (Proposition B.12). The hypothesis on n cannot be dropped. For example, in characteristic p the differential of $[p]$ is zero, and as a consequence an elliptic curve has at most p points of order dividing p over $\bar{k}$, although $E[p]$ has rank p^2.

Definition B.8 (Abelian variety)**.** An *abelian variety* over k is a smooth proper geometrically connected k-group scheme. If A is an abelian variety, $[n]$ denotes multiplication by the integer n, and $A[n] = \ker[n]$ for $n > 0$. An invertible sheaf $\mathcal{L}$ on A is *symmetric* if $[-1]^*\mathcal{L} \simeq \mathcal{L}$.

We use three facts: commutativity, which follows from rigidity; a symmetric ample sheaf, which follows from projectivity; and the formula for $[n]^*\mathcal{L}$, which follows from the cube. Rigidity and the cube require A to be geometrically integral, which we check together with commutativity.

Lemma B.9 (Commutativity)**.** *An abelian variety A over k is geometrically integral and commutative.*

Proof. An abelian variety is geometrically reduced because it is smooth, and geometrically irreducible because the identity component of a group scheme over a field is geometrically irreducible [Stacks, 0B7R] and A is connected.

For commutativity, consider $h : A \times A \to A$, $h(x, y) = xyx^{-1}y^{-1}$. It maps $A \times \{e\}$ to e. By Lemma B.7, applied with the first factor as the proper one, there are an open $V \ni e$ and $g : V \to A$ with $h|_{A\times V} = g \circ \mathrm{pr}_2$. Composing with $(e, \mathrm{id}) : V \to A \times V$ gives $g = h(e, -) = e$, so $h = e$ on $A \times V$. To pass from $A \times V$ to $A \times A$, we compare h with the constant map e. Since $A \times A$ is smooth, hence reduced, and irreducible, the nonempty open $A \times V$ is scheme-theoretically dense. Two maps to a separated scheme that agree on a scheme-theoretically dense open agree everywhere (Lemma A.4). Hence $h = e$ [Stacks, 056D, 01RH]. ∎

From now on we write the group law additively. Projectivity is the one structural fact that we quote.

Lemma B.10 (A symmetric ample sheaf)**.** *An abelian variety A over k is projective, and it has a symmetric ample invertible sheaf.*

Proof. Every group scheme of finite type over a field is quasi-projective [Stacks, 0BF7]. A quasi-projective proper scheme is projective [Stacks, 0B45]. If $\mathcal{L}$ is ample, then so is $[-1]^*\mathcal{L}$, since $[-1]$ is an automorphism, and the tensor product $\mathcal{N} = \mathcal{L} \otimes [-1]^*\mathcal{L}$ of two ample sheaves is ample. It is symmetric because $[-1]$ is an involution, which interchanges the two factors. ∎

Proposition B.11 (Pullback of invertible sheaves by $\{[n]\}$)**.** *Let A be an abelian variety over k and let $\mathcal{L}$ be an invertible sheaf on A. For every integer $n \geq 0$,*

$$[n]^*\mathcal{L} \simeq \mathcal{L}^{\otimes n(n+1)/2} \otimes ([-1]^*\mathcal{L})^{\otimes n(n-1)/2}.$$

In particular $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ if $\mathcal{N}$ is symmetric.

Proof. The idea is to produce one trivial sheaf $\mathcal{M}$ on $A \times A \times A$ by the theorem of the cube, and to pull it back along maps $A \to A \times A \times A$ chosen so that the result is a relation between $[n+1]^*\mathcal{L}$, $[n]^*\mathcal{L}$ and $[n-1]^*\mathcal{L}$. The theorem of the cube applies because A is geometrically integral (Lemma B.9).

Consider on $A \times A \times A$ the invertible sheaf

$$\mathcal{M} = m_{123}^*\mathcal{L} \otimes m_{12}^*\mathcal{L}^{-1} \otimes m_{13}^*\mathcal{L}^{-1} \otimes m_{23}^*\mathcal{L}^{-1} \otimes p_1^*\mathcal{L} \otimes p_2^*\mathcal{L} \otimes p_3^*\mathcal{L},$$

where the p_i are the projections and m_{12}, m_{13}, m_{23}, m_{123} are the sums of the indicated coordinates. To apply the theorem of the cube, we restrict $\mathcal{M}$ to the three slices. On $e \times A \times A$, with coordinates (x_2, x_3), the seven factors restrict to $\mathcal{L}$ pulled back along $x_2 + x_3$, x_2, x_3, $x_2 + x_3$, the constant map e, x_2 and x_3, with exponents $1, -1, -1, -1, 1, 1, 1$. Of these, the pullback along the constant map is trivial, and the others cancel. The same holds on $A \times e \times A$ and $A \times A \times e$. Hence $\mathcal{M}$ is trivial by Theorem B.6. Finally, pulling the trivial sheaf $\mathcal{M}$ back along $x \mapsto (x, x, [n-1]x)$ for $n \geq 0$ yields

$$[n+1]^*\mathcal{L} \otimes \mathcal{L}^{\otimes 2} \otimes [n-1]^*\mathcal{L} \simeq [2]^*\mathcal{L} \otimes ([n]^*\mathcal{L})^{\otimes 2}.$$

To see this, note that the composites of $x \mapsto (x, x, [n-1]x)$ with m_{123}, m_{12}, m_{13}, m_{23}, p_1, p_2, p_3 are $[n+1]$, $[2]$, $[n]$, $[n]$, id, id, $[n-1]$. The cases $n = 0$ and $n = 1$ of the formula are clear. For $n = 0$, since $[0]^*\mathcal{L}$ is trivial, the relation reads $\mathcal{L}^{\otimes 3} \otimes [-1]^*\mathcal{L} \simeq [2]^*\mathcal{L}$. This is the case $n = 2$ of the formula. In general, the relation expresses $[n+1]^*\mathcal{L}$ in terms of $[n]^*\mathcal{L}$ and $[n-1]^*\mathcal{L}$, and the exponents in the formula satisfy the same second order recursion. Induction on n therefore gives the formula [Stacks, 0BFF]. In particular, for symmetric $\mathcal{N}$ the two exponents add up to n^2. ∎

For a symmetric ample $\mathcal{N}$, the degree of $[n]$ can be read off from the growth of $\chi(A, \mathcal{N}^{\otimes m})$, since pulling back by $[n]$ multiplies m by n^2 and χ grows like m^g.

Proposition B.12 (Multiplication by n)**.** *Let A be an abelian variety of dimension g over k. For every $n > 0$, the morphism $[n]$ is surjective and finite locally free of degree n^{2g}, and its kernel $A[n]$ has rank n^{2g} over k. If n is invertible in k, then $[n]$ and $A[n] \to \operatorname{Spec} k$ are finite étale.*

Proof. We use the isomorphism $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ twice: on curves in the fibres of $[n]$ it shows that $[n]$ is finite, and in Euler characteristics it computes the degree. In between, translation invariance shows that all fibres have the same length, which makes $[n]$ finite locally free. Finally, whether $[n]$ is étale is decided by its differential.

Since finiteness, flatness and the degree may be checked after base change to an algebraic closure, we may assume that k is algebraically closed. Choose a symmetric ample $\mathcal{N}$ (Lemma B.10); then $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ by Proposition B.11. If a fibre of $[n]$ had positive dimension, it would contain a proper integral curve. Since $[n]$ maps that curve to a point, the restriction of $[n]^*\mathcal{N}$ to it is trivial, whereas the restriction of $\mathcal{N}^{\otimes n^2}$ has positive degree. Hence every fibre has dimension zero, and since $[n]$ is proper, it is finite. Its image is a closed subset of dimension g, hence equal to the integral variety A (Lemma B.9). After base change to an algebraic closure of its residue field, every fibre of $[n]$ becomes a translate of $A[n]$. Indeed, if $y = nx_0$, then translation by x_0 carries $A[n] = [n]^{-1}(0)$ isomorphically onto $[n]^{-1}(y)$, because $[n]$ is a homomorphism. Hence all fibres have the same degree $r = \dim_k \Gamma(A[n], \mathcal{O}_{A[n]})$.

To get flatness from the constancy of the fibre degree, we use the following general fact. A finite morphism f to a reduced Noetherian scheme whose fibres all have degree r is finite locally free of rank r. Indeed, near any point Nakayama's lemma gives a surjection $\mathcal{O}^{\oplus r} \to f_*\mathcal{O}$, since the

fibre at that point has dimension r. This surjection is an isomorphism at each generic point, where the local ring is a field and both sides are vector spaces of dimension r. Hence on an affine open every coordinate of a section of its kernel lies in all minimal primes, and is therefore zero because the ring is reduced. Since A is smooth, hence reduced, $[n]$ is finite locally free.

Recall that for a locally free sheaf $\mathcal{E}$ of rank r on A, the function $m \mapsto \chi(A, \mathcal{E} \otimes \mathcal{N}^{\otimes m})$ is a polynomial of degree g with leading coefficient $r \deg_{\mathcal{N}}(A)/g!$. Here $\deg_{\mathcal{N}}(A) = (\mathcal{N}^g)$ is the degree of A with respect to $\mathcal{N}$, a positive integer because $\mathcal{N}$ is ample. We compare the coefficients of m^g in two expressions for $\chi(A, [n]^*\mathcal{N}^{\otimes m})$. Since $[n]$ is finite, this Euler characteristic equals $\chi(A, [n]_*[n]^*\mathcal{N}^{\otimes m})$, which by the projection formula is $\chi(A, \mathcal{N}^{\otimes m} \otimes [n]_*\mathcal{O}_A)$. As $[n]_*\mathcal{O}_A$ is locally free of rank $\deg[n]$, the coefficient is $\deg[n] \deg_{\mathcal{N}}(A)/g!$. Since $[n]^*\mathcal{N}^{\otimes m} \simeq \mathcal{N}^{\otimes n^2 m}$, it is also $n^{2g} \deg_{\mathcal{N}}(A)/g!$. Hence

$$\deg[n] \deg_{\mathcal{N}}(A) = n^{2g} \deg_{\mathcal{N}}(A).$$

Since $\deg_{\mathcal{N}}(A) > 0$, we conclude that $\deg[n] = n^{2g}$; compare [Stacks, 0BFG]. It remains to treat étaleness. The differential of the group law at $(0, 0)$ is addition of tangent vectors. By induction, the differential of $[n]$ at the identity is $n \,\mathrm{id}$. Since $[n]$ commutes with translations, the same holds at every point. If n is invertible, $[n]$ is therefore étale. Base change along the identity section gives the assertion for $A[n]$. ∎

It follows that the group $A(\bar{k})$ is divisible. Indeed, since $[n]$ is finite and surjective, its fibre over a point of $A(\bar{k})$ is a nonempty finite $\bar{k}$-scheme, and so it has a $\bar{k}$-point.

B.3 Tate modules

Fix a prime ℓ invertible in k. With this prime, compatible systems of ℓ-power torsion points of A form the Tate module $T_\ell A$, a $\mathbf{Z}_\ell$-module with an action of $\mathrm{Gal}(\bar{k}/k)$. For example, for a complex torus $\mathbf{C}^g/\Lambda$ it is $\Lambda \otimes \mathbf{Z}_\ell$, the first homology with ℓ-adic coefficients. In general, Proposition B.14 shows that $T_\ell A$ is free of rank $2g$ over $\mathbf{Z}_\ell$. The input is the orders of the groups $A[\ell^n](\bar{k})$, which are known from Proposition B.12, and these orders force each of these groups to be free over $\mathbf{Z}/\ell^n$ of rank $2g$. Moreover, the transition maps are surjective because $A(\bar{k})$ is divisible.

Although Frobenius can have many fixed points on the finite levels, no power of it has a nonzero fixed vector in the Tate module (Proposition B.15). For example, for the elliptic curve $E_0 : y^2 = x^3 - x$ over $\mathbf{F}_3$, the three roots of $x^3 - x$ lie in $\mathbf{F}_3$. So all of $E[2]$ is rational, and Frobenius acts trivially on it. On $T_2 E$, however, a fixed vector would be an unbounded compatible system of 2-power torsion points in the finite group $E_0(\mathbf{F}_{3^m})$. For the Jacobian of a curve, Proposition 14.3 turns this into the statement that Frobenius has no fixed class in H^1 of the curve.

Definition B.13 (Tate module)**.** Let A be an abelian variety over k and let ℓ be a prime invertible in k. The *Tate module* of A is

$$T_\ell A = \varprojlim_n A[\ell^n](\bar{k}),$$

the limit taken along the maps $[\ell] : A[\ell^{n+1}](\bar{k}) \to A[\ell^n](\bar{k})$, and we put $V_\ell A = T_\ell A \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell$.

Proposition B.14 (The integral Tate module)**.** *Let A be an abelian variety of dimension g over k, and let ℓ be a prime invertible in k. There are noncanonical isomorphisms*

$$A[\ell^n](\bar{k}) \simeq (\mathbf{Z}/\ell^n)^{2g}, \qquad T_\ell A \simeq \mathbf{Z}_\ell^{2g}.$$

The canonical projection induces an isomorphism $T_\ell A/\ell^n T_\ell A \simeq A[\ell^n](\bar{k})$. The groups $A[\ell^n](\bar{k})$, the transition maps and the projections from $T_\ell A$ are $\mathrm{Gal}(\bar{k}/k)$-equivariant, and they do not change when $\bar{k}$ is replaced by a larger algebraically closed field.

Proof. A finite abelian group of order ℓ^{2gn} killed by ℓ^n need not be free over $\mathbf{Z}/\ell^n$; for $g \geq 1$ the group $(\mathbf{Z}/\ell^2)^{2g-2} \oplus (\mathbf{Z}/\ell)^4$ has order ℓ^{4g} and is killed by ℓ^2. For $A[\ell^n]$, however, this cannot happen because we know the orders at every level together with the exact sequences given by $[\ell]$. Indeed, we use them to lift a basis of the ℓ-torsion step by step and check by counting that the lifts form a basis at each level.

The ℓ-torsion and the exact sequences. The group $A[\ell](\bar{k})$ is a vector space of dimension $2g$ over $\mathbf{F}_\ell$, since it is killed by ℓ and has order ℓ^{2g} by Proposition B.12. Moreover there are exact sequences

$$0 \longrightarrow A[\ell](\bar{k}) \longrightarrow A[\ell^{n+1}](\bar{k}) \xrightarrow{[\ell]} A[\ell^n](\bar{k}) \longrightarrow 0,$$

whose surjectivity holds because $A(\bar{k})$ is divisible: every point of $A[\ell^n](\bar{k})$ is ℓ times a point, which then lies in $A[\ell^{n+1}](\bar{k})$.

Lifting bases. Choose a basis of $A[\ell](\bar{k})$ and lift it successively along $[\ell]$; the lifts form a basis at each level. Suppose that $x_1, \dots, x_{2g} \in A[\ell^{n+1}](\bar{k})$ lift a basis of $A[\ell^n](\bar{k})$ obtained in this way, so that $\ell^n x_1, \dots, \ell^n x_{2g}$ is the chosen basis of $A[\ell](\bar{k})$. To see that the x_i are independent, note first that all coefficients of a relation between them are divisible by ℓ^n, as we see by applying $[\ell]$ to it. The relation then becomes a relation between the elements $\ell^n x_i$ of the chosen basis of $A[\ell](\bar{k})$, so the coefficients are even divisible by ℓ^{n+1}. As $A[\ell^{n+1}](\bar{k})$ has order $\ell^{2g(n+1)}$, the x_i therefore form a basis of it over $\mathbf{Z}/\ell^{n+1}$. Passing to the inverse limit of these compatible bases, we obtain both isomorphisms and the description of $T_\ell A/\ell^n T_\ell A$.

Equivariance and independence of $\bar{k}$. The groups and maps in the statement commute with the action of $\mathrm{Gal}(\bar{k}/k)$, because the group schemes $A[\ell^n]$ and the maps $[\ell]$ are defined over k. In particular the projection $T_\ell A/\ell^n T_\ell A \to A[\ell^n](\bar{k})$ is equivariant. The chosen bases need not be equivariant, which is why the isomorphisms with $(\mathbf{Z}/\ell^n)^{2g}$ and $\mathbf{Z}_\ell^{2g}$ are noncanonical. Finally, the points of $A[\ell^n]$ and the multiplication maps do not change under a further algebraically closed extension, because $A[\ell^n]$ is finite étale and hence a finite disjoint union of points over $\bar{k}$. ∎

Over a finite field, Frobenius acts on $T_\ell A$, and by the last proposition this action is compatible with its action on each $A[\ell^n](\bar{k})$. The next proposition shows that no power of Frobenius has a nonzero fixed vector. For the Jacobian of a curve, this is the input to Proposition 14.3.

Proposition B.15 (No Frobenius-fixed vector)**.** *Let $A_0/\mathbf{F}_q$ be an abelian variety, let $A = A_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let F be geometric Frobenius on $T_\ell A$. For every $m > 0$,*

$$(T_\ell A)^{F^m} = 0, \qquad (V_\ell A)^{F^m} = 0.$$

Consequently $F^m - 1$ is an automorphism of the finite-dimensional $\mathbf{Q}_\ell$-vector space $V_\ell A$, and hence of $V_\ell A \otimes_{\mathbf{Q}_\ell} E$ for every field extension E of $\mathbf{Q}_\ell$.

Proof. A fixed vector is a compatible system of torsion points, each rational over $\mathbf{F}_{q^m}$. But the rational points form a finite group, so the orders of the ℓ-power torsion points in it are bounded,

and compatibility under $[\ell]$ then forces the system to vanish. The statement for $V_\ell A$ then follows by clearing denominators.

Since an automorphism and its inverse have the same fixed vectors, we may use arithmetic Frobenius instead. On $A(\overline{\mathbf{F}}_q)$ its mth power fixes exactly $A_0(\mathbf{F}_{q^m})$. This group is finite, because on each affine open of a finite affine cover a rational point is determined by finitely many coordinates in the finite field. Hence the ℓ-primary part of $A_0(\mathbf{F}_{q^m})$ is killed by some ℓ^e. Let (t_n) be a fixed compatible sequence. Each t_{n+e} lies in this ℓ-primary part, and since the transition maps are $[\ell]$, we find

$$t_n = \ell^e t_{n+e} = 0$$

for every n. This proves that $(T_\ell A)^{F^m} = 0$.

Since $T_\ell A$ is free by Proposition B.14, it injects into $V_\ell A$. A suitable power of ℓ times a fixed vector of $V_\ell A$ is a fixed vector of $T_\ell A$, and so it vanishes. Hence $(V_\ell A)^{F^m} = 0$. In other words, $F^m - 1$ is injective on the finite-dimensional space $V_\ell A$, and therefore bijective. ∎

C The Picard scheme of a curve

For a curve C proper and connected over a separably closed field and n invertible in that field, the Kummer sequence shows that $H^1(C, \mu_n)$ is the n-torsion of the Picard group of C. But the Picard group is an abstract group. In particular, nothing about it says that its n-torsion is finite, let alone isomorphic to $(\mathbf{Z}/n)^{2g}$. Nor does it say why Frobenius should have no fixed vector on the ℓ-adic limit, as Proposition 14.3 requires. Both questions become statements of Appendix B once we know that $\mathrm{Pic}^0(C)$, the group of invertible sheaves of degree zero, is the group of points of an abelian variety.

Over $\mathbf{C}$ this is the Abel–Jacobi theorem, $\mathrm{Pic}^0(C) \simeq H^0(C, \Omega^1)^\vee / H_1(C, \mathbf{Z})$. Over an arbitrary field we construct the Jacobian as a Picard scheme, which represents invertible sheaves in families.

Let C be a smooth proper connected curve of genus g over a separably closed field k. The idea is that most invertible sheaves of degree g are given by divisors. Suppose $\mathcal{L}$ has degree g and $h^0(\mathcal{L}) = 1$. Then the zeros of its unique section up to scalars form an effective divisor of degree g, and this divisor determines $\mathcal{L}$. Effective divisors of degree g are the points of the Hilbert scheme $\mathrm{Hilb}^g_{C/k}$, a smooth proper variety of dimension g (Theorem C.6).

By Riemann–Roch, every invertible sheaf of degree g has $h^0 \geq 1$, with equality exactly when $h^1 = 0$. This condition is open in families, by semicontinuity, and it defines an open chart $W \subset \mathrm{Hilb}^g_{C/k}$ of the Picard functor (Proposition C.7). Riemann–Roch also moves every invertible sheaf into this chart by a twist defined over k (Lemma C.8). Hence the translates of W cover the Picard functor, and they glue to the Picard scheme. Its components are proper because the Abel map $\mathrm{Hilb}^g_{C/k} \to \mathrm{Pic}^g$ is surjective and its source is proper.

The Picard functor itself is a sheafified quotient of $T \mapsto \mathrm{Pic}(C \times T)$. We have to divide out the invertible sheaves pulled back from T, since these are trivial locally on T. When C has a rational point, we describe the Picard functor by invertible sheaves trivialized along that point (Lemma C.2).

The outcome is Theorem C.11: the component of degree zero, the Jacobian $J(C)$, is an abelian variety of dimension g with $J(C)(k) = \mathrm{Pic}^0(C)$ and tangent space $H^1(C, \mathcal{O}_C)$ at the origin. Over a finite field we descend the Jacobian to an abelian variety J_0 over $\mathbf{F}_q$. Then Propositions C.16 and Proposition C.17 give an isomorphism $\varprojlim_n H^1(C, \mathbf{Z}/\ell^n(1)) \simeq T_\ell J_0$ compatible with Frobenius, which is the form in which Proposition 14.3 uses the Jacobian; see also [Mil86b].

C.1 The Picard functor

We want a scheme whose points over T are families of invertible sheaves on C parametrized by T. The obvious candidate $T \mapsto \operatorname{Pic}(C_T)$, however, contains too much. For example, for $C = \mathbf{P}^1_k$ and connected T, every invertible sheaf on $\mathbf{P}^1_T$ is $\mathcal{O}(d) \otimes f_T^*\mathcal{M}$ for a unique $d \in \mathbf{Z}$ and a unique $\mathcal{M} \in \operatorname{Pic}(T)$. In other words $\operatorname{Pic}(\mathbf{P}^1_T) = \mathbf{Z} \oplus \operatorname{Pic}(T)$, whereas the Picard scheme should be the constant scheme $\mathbf{Z}$. In fact the functor $T \mapsto \operatorname{Pic}(C_T)$ is not even a Zariski sheaf, because the pullback $f_T^*\mathcal{M}$ of a nontrivial invertible sheaf $\mathcal{M}$ on T is trivial locally on T but not globally, so no scheme represents this functor.

The Picard functor is therefore defined as the sheafification of $T \mapsto \operatorname{Pic}(C_T)/f_T^*\operatorname{Pic}(T)$ (Definition C.1). When C has a section, Lemma C.2 removes the pullbacks from T in another way, by fixing a trivialization along the section. We use this rigidified description to construct the chart of Proposition C.7 and the Picard scheme of Theorem C.11.

Definition C.1 (Relative Picard functor)**.** For a morphism of schemes $f : C \to S$, let

$$\operatorname{Pic}_{C/S} = \left(T \longmapsto \operatorname{Pic}(C_T)/f_T^*\operatorname{Pic}(T)\right)^{\text{fppf}}.$$

Here $C_T = C \times_S T$, and the superscript denotes sheafification on the big fppf site of S. When this functor is represented by a scheme, we call the representing scheme the *Picard scheme*.

Equivalently, one may sheafify $T \mapsto \operatorname{Pic}(C_T)$, since an invertible sheaf on T becomes trivial on a Zariski cover. The sheafification is needed: invertible sheaves satisfy effective descent, but their automorphisms can prevent their isomorphism classes from forming a sheaf. For this reason neither the displayed quotient nor $T \mapsto \operatorname{Pic}(C_T)$ need be an fppf sheaf [Stacks, 0B9L]. If f has a section, the next lemma describes the sheafification directly. We also write $\operatorname{Pic}_{C/S}$ for the Picard scheme when it exists.

Lemma C.2 (Rigidification along a section)**.** *Suppose that f has a section $\sigma : S \to C$ and that*

$$\mathcal{O}_T \longrightarrow f_{T,*}\mathcal{O}_{C_T} \quad \text{is an isomorphism for every } T \to S. \tag{$*$}$$

Then $\operatorname{Pic}_{C/S}(T)$ is identified, functorially in T, with the set of isomorphism classes of pairs $(\mathcal{L}, \alpha)$, with $\mathcal{L}$ invertible on C_T and $\alpha : \sigma_T^*\mathcal{L} \simeq \mathcal{O}_T$. Every such pair has only the identity automorphism. There is a split exact sequence

$$0 \longrightarrow \operatorname{Pic}(T) \xrightarrow{f_T^*} \operatorname{Pic}(C_T) \longrightarrow \operatorname{Pic}_{C/S}(T) \longrightarrow 0,$$

whose retraction is σ_T^*. The normalization map is

$$\mathcal{L} \longmapsto \mathcal{L} \otimes f_T^*(\sigma_T^*\mathcal{L})^{-1}.$$

Proof. The point is that rigidified pairs have no automorphisms other than the identity. Because of this they form a sheaf, and the quotient presheaf maps isomorphically onto this sheaf.

An automorphism of an invertible sheaf is multiplication by a global unit. By condition $(*)$ this unit comes from a unit on T, and compatibility with a rigidification forces it to be 1. To see that the rigidified functor is a sheaf, suppose we are given rigidified classes on an fppf cover that agree on overlaps. We then choose local isomorphisms and rescale them so that they preserve the rigidifications. Such isomorphisms are unique, since rigidified pairs have no

nontrivial automorphisms. In particular they satisfy the cocycle condition, and the pairs glue by effective descent for invertible sheaves.

The normalization map identifies the quotient presheaf $T \mapsto \operatorname{Pic}(C_T)/f_T^*\operatorname{Pic}(T)$ with this rigidified functor. Since the latter is already a sheaf, the quotient presheaf needs no sheafification and equals $\operatorname{Pic}_{C/S}$. The sequence splits since $\sigma_T^* f_T^* = \mathrm{id}$; for this argument see [Stacks, 0B9N]. ■

For a smooth proper geometrically connected curve C over a field k, we have $H^0(C, \mathcal{O}_C) = k$. For every field extension k'/k the curve $C_{k'}$ is again smooth, proper and integral, and flat base change for coherent cohomology gives $H^0(C_{k'}, \mathcal{O}_{C_{k'}}) = k'$ and $\dim_{k'} H^1(C_{k'}, \mathcal{O}_{C_{k'}}) = \dim_k H^1(C, \mathcal{O}_C)$. Every k-scheme is flat over k, so the same flat base change shows that $(*)$ holds for $C \to \operatorname{Spec} k$. If C has a k-point P, the preceding lemma applies with $\sigma = P$ (see [Stacks, 0B9U]).

C.2 Riemann–Roch and Serre duality

The chart of the Picard scheme in degree g is cut out by the condition $h^1 = 0$. Let $\mathcal{L}$ be an invertible sheaf of degree g on a curve of genus g. Then Riemann–Roch gives $h^0(\mathcal{L}) - h^1(\mathcal{L}) = 1$, so $h^1(\mathcal{L}) = 0$ if and only if $\mathcal{L}$ has exactly one section up to scalars. In a family over T, coherent base change shows that the condition "$Rf_{T,*}\mathcal{L}$ is an invertible sheaf in degree zero" is open on T and compatible with base change. It therefore defines an open subfunctor of the Picard functor.

The condition is not closed, because h^0 can jump up under specialization. For example, on a curve of genus two a divisor D of degree two has $h^0(\mathcal{O}(D)) = 1$ unless $\mathcal{O}(D) \simeq \omega_{C/k}$, in which case $h^0 = 2$. The chart therefore misses the line of canonical divisors in $\operatorname{Hilb}^2_{C/k}$, and we need its translates to cover the Picard scheme in degree two.

Fix a smooth proper geometrically connected curve C over k, with structure morphism $f : C \to \operatorname{Spec} k$, and set $g = \dim_k H^1(C, \mathcal{O}_C)$. Such a curve is projective [Stacks, 0A26].

Theorem C.3 (Riemann–Roch and coherent Serre duality)**.** *For an invertible sheaf $\mathcal{L}$ on C we have*

$$\chi(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g, \qquad H^1(C, \mathcal{L})^\vee \simeq H^0(C, \omega_{C/k} \otimes \mathcal{L}^{-1}), \qquad \deg \omega_{C/k} = 2g - 2, \qquad \dim_k H^0(C, \omega_{C/k}) = g.$$

In particular, if $\deg \mathcal{L} > 2g - 2$, then

$$H^1(C, \mathcal{L}) = 0, \qquad h^0(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g.$$

For $C = \mathbf{P}^1_k$ and $\mathcal{L} = \mathcal{O}(d)$ with $d \geq 0$, the sections are the binary forms of degree d, a space of dimension $d + 1 = \deg \mathcal{L} + 1 - 0$, and $\omega = \mathcal{O}(-2)$ has degree $-2 = 2g - 2$.

We quote the theorem from [Stacks, 0B5B], where it is proved for smooth curves and the duality is given by the residue trace on differentials. We prove its compatibility with field extension in Corollary C.4 below. For the displayed vanishing, recall that a nonzero section of an invertible sheaf on an integral curve has an effective divisor of zeros. Hence an invertible sheaf of negative degree has no nonzero section. If $\deg \mathcal{L} > 2g - 2$, then $\omega_{C/k} \otimes \mathcal{L}^{-1}$ has negative degree, and $H^1(C, \mathcal{L}) = 0$ by duality. The dimension formula then follows from Riemann–Roch. For $\mathcal{L} = \mathcal{O}_C$ the duality gives $\dim_k H^0(C, \omega_{C/k}) = \dim_k H^1(C, \mathcal{O}_C) = g$.

The Picard scheme is constructed over a separably closed field, but its points are tested over arbitrary extensions. Hence Riemann–Roch and coherent base change must be compatible with

extension of the ground field, and for duality we need this for the pairing itself, not only for the dimensions.

Corollary C.4 (Serre duality and field extension)**.** *Let k'/k be a field extension. Under flat base change for coherent cohomology and the identification $\omega_{C_{k'}/k'} \simeq \omega_{C/k}|_{C_{k'}}$, the Serre duality pairing for $\mathcal{L}|_{C_{k'}}$ is obtained from that for $\mathcal{L}$ by extension of scalars. In particular, the genus is unchanged by field extension.*

Proof. We factor the duality on C through a finite map to $\mathbf{P}^1_k$, where the trace is an explicit residue, and check that each step is compatible with extension of scalars.

A nonconstant rational function on C extends to a finite map $\phi : C \to \mathbf{P}^1_k$, which is flat because $\phi_*\mathcal{O}_C$ is torsion free over the discrete valuation rings of $\mathbf{P}^1_k$. We may construct the duality on C as duality for ϕ, given by the dual of the finite locally free module $\phi_*\mathcal{L}$, followed by duality on $\mathbf{P}^1_k$, whose trace $H^1(\mathbf{P}^1_k, \omega) \to k$ takes the coefficient of $t^{-1}dt$ in a Čech cocycle for the standard cover. Both steps commute with extension of scalars, hence so does the resulting pairing. Flat base change shows that $g = \dim_k H^1(C, \mathcal{O}_C)$ is unchanged by field extension. ∎

Recall that the chart of the Picard scheme consists of the invertible sheaves $\mathcal{L}$ for which $Rf_{T,*}\mathcal{L}$ is an invertible sheaf in degree 0. We use coherent base change in the following form.

Lemma C.5 (Coherent base change for invertible sheaves)**.** *Let T be a k-scheme and let $\mathcal{L}$ be invertible on C_T. Then $Rf_{T,*}\mathcal{L}$ is locally represented by a bounded complex of finite locally free $\mathcal{O}_T$-modules. For any $a : T' \to T$, the base change map $La^*Rf_{T,*}\mathcal{L} \to Rf_{T',*}\mathcal{L}_{T'}$ is an isomorphism in the derived category of $\mathcal{O}_{T'}$-modules. The locus where this complex is an invertible module in degree 0 is open. On this locus, $f_{T,*}\mathcal{L}$ is invertible, its formation commutes with arbitrary base change, and $R^1 f_{T,*}\mathcal{L} = 0$.*

Proof. This is base change for coherent cohomology along a proper morphism, applied to a flat finitely presented module, together with the openness criterion for a perfect complex (see [Stacks, 0B91, 0B9S]). To construct the complex, one may use a projective embedding, sufficiently large twists and a finite affine Čech complex; by Serre vanishing we then obtain models that are finite locally free. Given such a model, openness follows from a calculation with matrices. Indeed, near a point t where the derived fibre of the complex has cohomology only in degree 0, of dimension 1, we may cancel the differentials that are invertible at t, and what remains is an invertible module in degree 0.

This proves the lemma for locally Noetherian T. If T is an arbitrary k-scheme, then locally on T the sheaf $\mathcal{L}$ is pulled back from C_{T_0} for some k-scheme T_0 of finite type, and the base change isomorphism for $T \to T_0$ reduces every assertion to T_0. The open locus is the preimage of the locus for T_0, and by base change it does not depend on T_0. ∎

C.3 Hilbert schemes of points

On a smooth curve a finite subscheme of length d is the same as an effective divisor of degree d. These divisors are parametrized by the Hilbert scheme of points $\mathrm{Hilb}^d_{C/k}$, which is the symmetric power $C^{(d)}$. Iterated addition of points gives a finite surjection $C^d \to \mathrm{Hilb}^d_{C/k}$, from which $\mathrm{Hilb}^d_{C/k}$ inherits properness, irreducibility and dimension d (Theorem C.6). For example, for $C = \mathbf{P}^1$, an effective divisor of degree d is the zero set of a binary form of degree d up to scalars, and $\mathrm{Hilb}^d_{C/k} = \mathbf{P}^d$.

A divisor D gives the invertible sheaf $\mathcal{O}(D)$ with its canonical section. Conversely, an invertible sheaf with a section up to scalars gives a divisor. When $h^0 = 1$ the section is unique up to scalars, and then the divisor is determined by the sheaf. In this way the open part of $\mathrm{Hilb}^g_{C/k}$ where $h^0(\mathcal{O}(D)) = 1$ and $h^1 = 0$ becomes an open subfunctor of the Picard functor (Proposition C.7).

For $d \geq 0$, let $\mathrm{Hilb}^d_{C/k}$ be the functor on k-schemes sending T to the set of closed subschemes $D \subset C_T$ that are finite locally free of rank d over T.

Theorem C.6 (Hilbert schemes of points on a smooth curve)**.** *The functor $\mathrm{Hilb}^d_{C/k}$ is represented by a smooth proper geometrically integral k-scheme of dimension d, with $\mathrm{Hilb}^0_{C/k} = \operatorname{Spec} k$. Its T-points are precisely the relative effective Cartier divisors that are finite locally free of degree d. It has a universal divisor $D_d \subset C \times_k \mathrm{Hilb}^d_{C/k}$ and an associated invertible sheaf $\mathcal{O}(D_d)$.*

Proof. Representability is a general theorem, once we know that finite sets of points lie in affine opens. We then obtain the geometric properties from the addition map $\mathrm{Hilb}^d_{C/k} \times C \to \mathrm{Hilb}^{d+1}_{C/k}$, which we show to be finite locally free. Starting from $\mathrm{Hilb}^1_{C/k} = C$, we get smoothness and the dimension by induction on d, and properness and irreducibility from the finite surjection $C^d \to \mathrm{Hilb}^d_{C/k}$.

Every finite set of points of C lies in an affine open. Indeed, if Q is a closed point outside the set, then $\mathcal{O}_C(Q)$ has positive degree, hence is ample by Riemann–Roch, and so $C \setminus \{Q\}$ is affine. Hence $\mathrm{Hilb}^d_{C/k}$ is represented by a scheme, separated and locally of finite type over k [Stacks, 0B9A]. The universal closed subscheme is the divisor D_d.

Let T be a k-scheme and $t \in T$ a point. Since the local rings of $C_{\kappa(t)}$ at closed points are discrete valuation rings, every finite closed subscheme of a fibre is an effective Cartier divisor. By the fibrewise criterion for relative effective Cartier divisors, every $D \in \mathrm{Hilb}^d_{C/k}(T)$ is a relative effective Cartier divisor on C_T/T. This gives the description of T-points and the invertible sheaf $\mathcal{O}(D_d)$. Multiplying ideals adds such divisors, and we obtain the addition map

$$\mathrm{Hilb}^d_{C/k} \times C \longrightarrow \mathrm{Hilb}^{d+1}_{C/k}, \qquad (D, x) \longmapsto D + \Gamma_x.$$

It identifies $\mathrm{Hilb}^d_{C/k} \times C$ with the universal divisor over $\mathrm{Hilb}^{d+1}_{C/k}$: if a divisor E contains the graph Γ_x, then dividing a local equation of E by one of Γ_x gives a relative effective Cartier divisor D with $E = D + \Gamma_x$, of degree d by the exact sequence for a sum of divisors. In other words, the addition map is the projection to $\mathrm{Hilb}^{d+1}_{C/k}$ of the universal divisor $D_{d+1} \subset C \times \mathrm{Hilb}^{d+1}_{C/k}$. By definition of the Hilbert functor, D_{d+1} is finite locally free of degree $d+1$ over $\mathrm{Hilb}^{d+1}_{C/k}$. Hence the addition map is finite locally free of degree $d+1$ [Stacks, 0B9G], in particular faithfully flat.

We now argue by induction on d, starting from $\mathrm{Hilb}^0_{C/k} = \operatorname{Spec} k$ and $\mathrm{Hilb}^1_{C/k} = C$. The scheme $\mathrm{Hilb}^{d+1}_{C/k}$ is smooth because regularity descends along faithfully flat maps, also after extending k. It has dimension $d+1$ because the addition map is finite and surjective [Stacks, 0B9H].

Properness comes from the finite surjective map $C^d \to \mathrm{Hilb}^d_{C/k}$ given by iterated addition. This map makes $\mathrm{Hilb}^d_{C/k}$ quasi-compact. It also makes $\mathrm{Hilb}^d_{C/k}$ universally closed, because C^d is proper and surjectivity is stable under base change. Since C^d is geometrically integral, $\mathrm{Hilb}^d_{C/k}$ is geometrically irreducible. Being smooth, it is then geometrically integral [Stacks, 0B9I].

For $C = \mathbf{P}^1$, the map $C^d \to \mathrm{Hilb}^d_{C/k} = \mathbf{P}^d$ sends d points to the binary form with these zeros, that is, the product of d linear forms; it is the quotient by the symmetric group, of degree $d!$. ∎

Assume now that $C(k)$ contains a point P, and let us describe $\mathrm{Pic}_{C/k}$ by rigidified invertible sheaves as in Lemma C.2. We define a subfunctor $F \subset \mathrm{Pic}_{C/k}$ as follows. A rigidified class $\mathcal{L}$ on C_T lies in $F(T)$ if, in the derived category of $\mathcal{O}_T$-modules,

$$Rf_{T,*}\mathcal{L} \simeq \mathcal{N}[0] \quad \text{for some invertible } \mathcal{N} \text{ on } T.$$

By Lemma C.5, this condition defines an open locus of T compatible with base change, so that the inclusion $F \subset \mathrm{Pic}_{C/k}$ is representable by open immersions.

Proposition C.7 (Divisors of degree g as an open chart)**.** *The functor F is represented by an open subscheme $W \subset \mathrm{Hilb}^g_{C/k}$. The morphism $W \to \mathrm{Pic}_{C/k}$ associates to a divisor its rigidified invertible sheaf. Consequently, F is smooth of dimension g.*

Proof. We construct mutually inverse maps: a divisor gives its invertible sheaf, normalized along P, and a family in F has, up to units on T, a unique section in each fibre, whose zero scheme is a family of divisors of degree g.

From divisors to sheaves. On $C \times \mathrm{Hilb}^g_{C/k}$, normalize the universal invertible sheaf as

$$\mathcal{L}_g = \mathcal{O}(D_g) \otimes f^*_{\mathrm{Hilb}^g}(P^*_{\mathrm{Hilb}^g}\mathcal{O}(D_g))^{-1}.$$

Let W be the open locus where $Rf_{\mathrm{Hilb}^g,*}\mathcal{L}_g$ is a line bundle in degree 0; then $\mathcal{L}_g$ defines a morphism $W \to F$.

From sheaves to divisors. Let $\mathcal{L}$ be a rigidified class in $F(T)$ and set $\mathcal{N} = f_{T,*}\mathcal{L}$. By assumption $Rf_{T,*}\mathcal{L} \simeq \mathcal{N}'[0]$ for an invertible $\mathcal{N}'$, so $\mathcal{N} \simeq \mathcal{N}'$ is invertible. The evaluation map $f_T^*\mathcal{N} \to \mathcal{L}$ is a section s of $\mathcal{L} \otimes f_T^*\mathcal{N}^{-1}$. It is determined by $\mathcal{L}$ up to a unit on T, and such a unit does not change its zero scheme D.

This D is a relative effective Cartier divisor. Indeed, the formation of $\mathcal{N}$ commutes with base change (Lemma C.5). Hence on each fibre $C_{\kappa(t)}$ the section s is a nonzero element of the one-dimensional space of sections. The fibre is integral because C is geometrically integral, so the zero scheme of s on the fibre is an effective Cartier divisor. Moreover, the fibres of D have degree g: for a point t of T we have $h^0(\mathcal{L}_t) = 1$ and $H^1(C_t, \mathcal{L}_t) = 0$, so Riemann–Roch gives

$$\deg \mathcal{L}_t = \chi(\mathcal{L}_t) - (1-g) = 1 - (1-g) = g.$$

The morphism $D \to T$ is proper, since D is closed in the proper scheme C_T. It has finite fibres, and it is flat, as is every relative effective Cartier divisor. A proper, flat, finitely presented morphism with finite fibres is finite locally free. Its rank at t is the degree g of the fibre D_t, the divisor of zeros of a section of $\mathcal{L}_t$. Hence D defines a morphism $T \to \mathrm{Hilb}^g_{C/k}$.

The constructions are inverse. Normalizing $\mathcal{O}(D)$ along P_T recovers $\mathcal{L}$, because $\mathcal{O}(D) \simeq \mathcal{L} \otimes f_T^*\mathcal{N}^{-1}$. The two constructions are therefore inverse to each other, and they commute with base change, which proves that $W \simeq F$ (see [Stacks, 0B9X]). ∎

To cover the Picard functor by translates of F, we have to move every invertible sheaf into F by a twist defined over k.

Lemma C.8 (Twisting to $h^0 = 1$ and $h^1 = 0$)**.** *Suppose that k is separably closed. For every field extension k'/k and every invertible sheaf $\mathcal{L}$ on $C_{k'}$, there is an invertible sheaf $\mathcal{M}_0$ on C such that*

$$h^0(C_{k'}, \mathcal{L} \otimes \mathcal{M}_{0,k'}) = 1, \qquad H^1(C_{k'}, \mathcal{L} \otimes \mathcal{M}_{0,k'}) = 0.$$

Proof. We first twist by a large power of an ample sheaf on C, which kills H^1 and leaves many sections. Then we remove k-rational points one at a time, each time lowering h^0 by one and keeping $H^1 = 0$, until $h^0 = 1$. The twist is defined over k because the points are.

Choose an ample invertible sheaf $\mathcal{M}$ on C. For $r \gg 0$, Serre vanishing and Riemann–Roch show that $\mathcal{L} \otimes \mathcal{M}_{k'}^{\otimes r}$ has vanishing H^1 and a positive-dimensional space of sections. As long as h^0 is larger than 1, we choose a nonzero section and a point $x \in C(k)$ where that section is nonzero. Such a point exists: the zero divisor of the section is finite, so its image in C is a finite set, and the complement of this set is a nonempty open of C. This open contains a point of $C(k)$, because the k-rational points of a smooth k-scheme are dense when k is separably closed [Stacks, 056U]. For the current sheaf $\mathcal{F}$, consider the sequence

$$0 \longrightarrow \mathcal{F}(-x_{k'}) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}|_{x_{k'}} \longrightarrow 0$$

The chosen section does not vanish at $x_{k'}$, so the map $H^0(C_{k'}, \mathcal{F}) \to \mathcal{F}|_{x_{k'}}$ is onto the one-dimensional fibre. By the long exact sequence, subtracting $x_{k'}$ lowers h^0 by exactly 1 and preserves $H^1 = 0$. We continue in this way until $h^0 = 1$. Since all subtracted points come from k, the resulting twist has the form $\mathcal{M}_0 = \mathcal{M}^{\otimes r}(-x_1 - \cdots - x_s)$ on C. Finally, normalizing along P does not change the dimensions of fibre cohomology, so the rigidified twist satisfies the same conditions (see [Stacks, 0B9Y]). ∎

C.4 The Jacobian

We glue the translates of the chart of Proposition C.7 to the Picard scheme, whose component of degree zero is the Jacobian. Gluing uses twice that k is separably closed: to find a rational point for the rigidification, and to obtain the twists of Lemma C.8 over k. For an elliptic curve, the Abel map $C = \mathrm{Hilb}^1_{C/k} \to \mathrm{Pic}^1_{C/k}$ is an isomorphism, and translation by $\mathcal{O}_C(-P)$ identifies C with its own Jacobian. In general, the tangent space of the Jacobian at the origin is computed by deforming the trivial sheaf over the dual numbers.

Lemma C.9 (Representability of the Picard functor)**.** *Let k be separably closed and let C/k be a smooth proper connected curve. The fppf Picard functor $\mathrm{Pic}_{C/k}$ is represented by a scheme locally of finite type over k, which is covered by the open subschemes representing the translates $M^{-1}F$, for $M \in \mathrm{Pic}_{C/k}(k)$, of the chart F of Proposition C.7, defined using a point $P \in C(k)$.*

Proof. The preceding lemmas assume C geometrically integral. To see that this holds, note that $\bar{k}/k$ is purely inseparable. Hence $C_{\bar{k}} \to C$ is a homeomorphism, and $C_{\bar{k}}$ is connected and smooth, hence integral. We rigidify using a point $P \in C(k)$. Such a point exists because k is separably closed and the k-points of the smooth curve C are dense [Stacks, 056U].

By Proposition C.7 we have a representable open subfunctor $F \subset \mathrm{Pic}_{C/k}$. For a fixed class $M \in \mathrm{Pic}_{C/k}(k)$, the translate $M^{-1}F$, consisting of the classes $\mathcal{L}$ with $\mathcal{L} \otimes M$ in F, is another representable open subfunctor.

We show that these translates cover $\mathrm{Pic}_{C/k}$. Let a class on a k-scheme T and a point $t \in T$ be given. We apply Lemma C.8 to its fibre over the residue field $k' = \kappa(t)$. This gives an invertible sheaf $\mathcal{M}_0$ on C, defined over k, such that the twisted fibre has $h^0 = 1$ and $H^1 = 0$. By Lemma C.5, the twisted class then lies in F over an open neighbourhood of t. Letting t vary, we see that the open loci where the corresponding translated class lies in F cover T. Note that the translates

are indexed by classes over k, while the points of T have arbitrary residue fields. This is why Lemma C.8 asks for a twist defined over k for every extension k'.

The intersections of the charts are open in either chart, and their transition isomorphisms satisfy the cocycle condition by functoriality. An fppf sheaf covered by representable open subfunctors is representable [Stacks, 0B9Q, 0B9Z]. Hence gluing the charts gives a scheme representing $\mathrm{Pic}_{C/k}$. It is locally of finite type because each chart is an open of $\mathrm{Hilb}^g_{C/k}$. ∎

We compute the tangent space with the dual numbers $k[\varepsilon]$, $\varepsilon^2 = 0$; the computation works for any scheme over k.

Lemma C.10 (First-order deformations of invertible sheaves)**.** *Let X be a scheme over a field k and $X_\varepsilon = X \times_k \operatorname{Spec} k[\varepsilon]$. The kernel of the restriction map $\mathrm{Pic}(X_\varepsilon) \to \mathrm{Pic}(X)$ is $H^1(X, \mathcal{O}_X)$.*

Proof. Because $\varepsilon^2 = 0$, the map $a \mapsto 1 + \varepsilon a$ is a homomorphism, and it gives an exact sequence of sheaves of groups on X

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{X_\varepsilon}^{\times} \longrightarrow \mathcal{O}_X^{\times} \longrightarrow 1,$$

split by the inclusion $\mathcal{O}_X^{\times} \subset \mathcal{O}_{X_\varepsilon}^{\times}$. The schemes X and X_ε have the same underlying space, so $\mathrm{Pic}(X_\varepsilon) = H^1(X, \mathcal{O}_{X_\varepsilon}^{\times})$. Since the sequence is split, taking H^1 gives $H^1(X, \mathcal{O}_{X_\varepsilon}^{\times}) = H^1(X, \mathcal{O}_X) \oplus \mathrm{Pic}(X)$, in which the restriction map is the projection to the second summand. Hence the kernel is $H^1(X, \mathcal{O}_X)$. ∎

Theorem C.11 (The Picard scheme and the Jacobian over a separably closed field)**.** *Let k be separably closed and let C/k be a smooth proper connected curve of genus g. The fppf Picard functor is represented by a smooth separated group scheme, locally of finite type over k, with a decomposition*

$$\mathrm{Pic}_{C/k} = \coprod_{d \in \mathbf{Z}} \mathrm{Pic}^d_{C/k}\,.$$

Each component is a smooth proper geometrically connected variety of dimension g, parametrizing invertible sheaves of degree d. The identity component $J(C) = \mathrm{Pic}^0_{C/k}$ is an abelian variety, and

$$J(C)(k) = \mathrm{Pic}^0(C), \qquad T_0 J(C) = H^1(C, \mathcal{O}_C).$$

The formation of this representing scheme, its degree components and $J(C)$ commutes with extension of the ground field.

Proof. Representability is Lemma C.9. The tangent space is given by Lemma C.10. Fix the point $P \in C(k)$ used to define F.

For an invertible sheaf $\mathcal{L}$ on C_T, the Euler characteristic $\chi(C_t, \mathcal{L}_t)$ is a locally constant function of $t \in T$. This holds because $C_T \to T$ is proper and flat and $Rf_{T,*}\mathcal{L}$ is perfect (Lemma C.5). By Riemann–Roch the degree of $\mathcal{L}_t$ is locally constant as well. This gives the displayed decomposition into open and closed pieces. The charts in $\mathrm{Hilb}^g_{C/k}$ and their translates show that the scheme is smooth of dimension g.

It remains to see that the scheme is separated. A k-rational point of a scheme locally of finite type over a field is closed. Hence the unit section of the group scheme is a closed immersion. The diagonal is the preimage of the unit section under $(x, y) \mapsto xy^{-1}$, so it is closed as well.

The universal divisor defines the Abel map

$$\mathrm{Hilb}^g_{C/k} \longrightarrow \mathrm{Pic}^g_{C/k}, \qquad D \longmapsto \mathcal{O}_C(D).$$

For a line bundle $\mathcal{L}$ of degree g on $C_{k'}$, with k' algebraically closed, Riemann–Roch gives $h^0(\mathcal{L}) \geq \chi(\mathcal{L}) = g + 1 - g = 1$. A nonzero section has an effective divisor D of degree g with $\mathcal{O}(D) \simeq \mathcal{L}$. Hence the Abel map is surjective on points over algebraically closed fields, and this suffices for surjectivity. It follows that $\mathrm{Pic}^g_{C/k}$ is quasi-compact. It is of finite type since the charts are locally of finite type. A separated scheme of finite type that is the surjective image of the proper scheme $\mathrm{Hilb}^g_{C/k}$ is proper. It is geometrically irreducible because $\mathrm{Hilb}^g_{C/k}$ is. Translation by $\mathcal{O}_C((d-g)P)$ identifies $\mathrm{Pic}^g_{C/k}$ with the component of each degree d. In particular, the component of degree zero is smooth, proper, connected and commutative. Such a group scheme is an abelian variety (see [Stacks, 0BA0]).

The tangent space $T_0 J(C)$ is the kernel of $\mathrm{Pic}_{C/k}(k[\varepsilon]) \to \mathrm{Pic}_{C/k}(k)$, where $\varepsilon^2 = 0$. Since $\mathrm{Pic}(k[\varepsilon]) = 0$, this is the kernel of $\mathrm{Pic}(C_{k[\varepsilon]}) \to \mathrm{Pic}(C)$, which is $H^1(C, \mathcal{O}_C)$ by Lemma C.10.

For a field extension k'/k, the Picard functor of $C_{k'}$ is the restriction of $\mathrm{Pic}_{C/k}$ to k'-schemes. Hence its representing scheme is the base change. The degree of $\mathcal{L}_t$ is computed on the fibre C_t, so the decomposition by degree, and with it $J(C)$, is compatible with this base change. ∎

We use the theorem in the following form: over an algebraically closed field, the Picard group of C is an extension of $\mathbf{Z}$ by the group of points of an abelian variety.

Corollary C.12 (Degree and the Jacobian)**.** *Let C be a smooth proper connected curve over an algebraically closed field k. There is an exact sequence*

$$0 \longrightarrow J(C)(k) \longrightarrow \mathrm{Pic}(C) \xrightarrow{\deg} \mathbf{Z} \longrightarrow 0,$$

and a point $P \in C(k)$ splits it by $d \mapsto \mathcal{O}_C(dP)$.

Proof. The degree is additive under tensor product, so it is a homomorphism. By Theorem C.11, its kernel is $\mathrm{Pic}^0(C) = J(C)(k)$. Since $\deg \mathcal{O}_C(dP) = d$, the map $d \mapsto \mathcal{O}_C(dP)$ is a section of the degree, which is therefore surjective. ∎

Example C.13 (The degree sequence in genus zero and one)**.** On $C = \mathbf{P}^1_k$ a divisor $\sum_i n_i a_i + n_\infty \infty$ of degree zero, with $a_i \in k$, is the divisor of the rational function $\prod_i (x - a_i)^{n_i}$, whose order at ∞ is $-\sum_i n_i = n_\infty$. So $\mathrm{Pic}^0(\mathbf{P}^1) = 0$, and the degree is an isomorphism $\mathrm{Pic}(\mathbf{P}^1) \simeq \mathbf{Z}$. This agrees with $J(\mathbf{P}^1) = 0$ for $g = 0$.

On an elliptic curve E with origin O the kernel is as large as the curve. Let $\mathcal{L}$ have degree zero. Then $\mathcal{L}(O)$ has degree one, and by Riemann–Roch with $g = 1$ it has a one-dimensional space of sections. Hence $\mathcal{L}(O) \simeq \mathcal{O}_E(P)$ for a unique point P, and $P \mapsto \mathcal{O}_E(P - O)$ is a bijection $E(k) \to \mathrm{Pic}^0(E)$. In particular $\mathcal{O}_E(P - Q)$ is nontrivial for $P \neq Q$. One can also see this directly: a rational function with divisor $P - Q$ would be a map $E \to \mathbf{P}^1$ of degree one, that is, an isomorphism.

C.5 Torsion points

The Kummer sequence on a curve C over an algebraically closed field identifies $H^1(C,\mu_n)$ with $\operatorname{Pic}(C)[n]$ and $H^2(C,\mu_n)$ with $\operatorname{Pic}(C)/n$ (Proposition D.3). To compute these groups, we combine two facts. First, the exact sequence of Corollary C.12 separates the discrete part $\mathbf{Z}$ of the Picard group from the Jacobian. Second, Appendix B describes multiplication by n on an abelian variety over an algebraically closed field: it is surjective on points, and for n invertible in k its kernel is $(\mathbf{Z}/n)^{2g}$. Now apply the snake lemma to multiplication by n on the degree sequence. Since $\mathbf{Z}$ is torsion free, the n-torsion of $\operatorname{Pic}(C)$ is $J(C)[n](k) \simeq (\mathbf{Z}/n)^{2g}$. Since $J(C)(k)$ is divisible, the cokernel of n on $\operatorname{Pic}(C)$ is $\mathbf{Z}/n$, given by the degree.

For proper base change we also need these groups to be unchanged under extension of the algebraically closed field. This holds because $J(C)[n]$ is finite étale (Corollary C.15).

Example C.14 (Two-torsion of an elliptic curve)**.** Let k be algebraically closed of characteristic different from two and let $E\colon y^2 = (x-e_1)(x-e_2)(x-e_3)$ with distinct $e_i \in k$, with origin O at infinity and $P_i = (e_i, 0)$. The function $x - e_i$ has a double zero at P_i, because y is a local parameter there. It has a double pole at O. Hence $\operatorname{div}(x-e_i) = 2P_i - 2O$, and the classes $\mathcal{O}_E(P_i - O)$ are killed by 2. They are nonzero and pairwise distinct, since $\mathcal{O}_E(P_i - P_j)$ is trivial only for $i = j$ (Example C.13). The divisor $\operatorname{div}(y) = P_1 + P_2 + P_3 - 3O$ shows that the three classes add up to zero. So 0 and the classes of $P_i - O$ form a subgroup $\{0, P_1 - O, P_2 - O, P_3 - O\} \simeq (\mathbf{Z}/2)^2$ of $\operatorname{Pic}(E)[2]$. By Corollary C.15 below, $\operatorname{Pic}(E)[2]$ has order $2^{2g} = 4$, so this subgroup is all of it.

Under the Kummer sequence, the class $P_i - O$, with the trivialization of its square given by $x - e_i$, is the μ_2-torsor obtained by adjoining a square root of $x - e_i$. This torsor is unramified everywhere because the divisor of $x - e_i$ is even. For the running example $y^2 = x^3 - x$ over $\overline{\mathbf{F}}_3$ the e_i are $0, 1, -1 \in \mathbf{F}_3$. Hence the three torsors are defined over $\mathbf{F}_3$, and Frobenius fixes all of $E[2]$, as noted in Section B.3.

Corollary C.15 (Torsion of the Jacobian and field extension)**.** *Let k be algebraically closed, let C/k be a smooth proper connected curve of genus g, and let $n > 0$ be invertible in k. Then there is a noncanonical isomorphism*

$$J(C)[n](k) \simeq (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

For every algebraically closed extension k'/k, the natural map

$$J(C)[n](k) \longrightarrow J(C_{k'})[n](k')$$

is an isomorphism. Multiplication by n on $J(C)(k)$ is surjective, and consequently

$$\operatorname{Pic}(C)[n] = J(C)[n](k), \qquad \operatorname{Pic}(C)/n\operatorname{Pic}(C) \xrightarrow[\deg]{\sim} \mathbf{Z}/n\mathbf{Z}.$$

Proof. The assertions about $J(C)$ are those of Appendix B for this abelian variety, and the degree sequence carries them over to $\operatorname{Pic}(C)$.

By Proposition B.12, the kernel $J(C)[n]$ is finite étale of rank n^{2g}. A finite étale scheme of rank r over the algebraically closed field k is a disjoint union of r copies of $\operatorname{Spec} k$, a description stable under base extension. By Theorem C.11 the formation of the Jacobian and of $[n]$ commutes with base change, so the natural map $J(C)[n](k) \to J(C_{k'})[n](k')$ is a bijection.

If ℓ^m is the exact power of a prime ℓ dividing n, then $J(C)[\ell^m](k) \simeq (\mathbf{Z}/\ell^m)^{2g}$ by Proposition B.14 applied to $J(C)$. Combining the parts for the primes dividing n, we get the stated $\mathbf{Z}/n\mathbf{Z}$-module structure.

By Proposition B.12, the group $J(C)(k)$ is divisible. In the exact sequence of Corollary C.12, the group $\mathbf{Z}$ has no n-torsion and $J(C)(k)/n = 0$. This proves the two assertions for $\mathrm{Pic}(C)$. ■

C.6 Descent to the ground field

To compare the Galois action on the cohomology of a curve over $\bar{k}$ with a Tate module, we need the Jacobian over the ground field k. For example, in Proposition 14.3 we work over a finite field, and Frobenius acts on $\varprojlim_n H^1(C, \mathbf{Z}/\ell^n(1))$. It has no fixed vector there because this module is $T_\ell J_0$ for an abelian variety J_0 over $\mathbf{F}_q$ (Proposition B.15). The construction so far works over a separably closed field and uses a rational point for the rigidification. However, a curve over k need not have a k-point; for instance, the conic $x^2 + y^2 + z^2 = 0$ over $\mathbf{R}$ has none. We therefore define J_0 as the scheme representing the degree-zero part of the fppf Picard sheaf. Concretely, we obtain it from the Jacobian over k_s by Galois descent through a finite Galois extension k'/k (Proposition C.16). The Kummer sequence then identifies $H^1(C, \mathbf{Z}/\ell^n(1))$ with $J_0[\ell^n](\bar{k})$ compatibly with the Galois action and the transition maps (Proposition C.17).

Proposition C.16 (The Jacobian over the ground field)**.** *Let C_0/k be a smooth proper geometrically connected curve of genus g, and put $C = C_0 \times_k \bar{k}$. There is an abelian variety J_0/k of dimension g representing the degree-zero relative Picard fppf sheaf of C_0. For every field extension k'/k, the base change $J_0 \times_k k'$ represents the degree-zero Picard sheaf of $C_0 \times_k k'$; in particular $J_0 \times_k \bar{k}$ represents $\mathrm{Pic}^0_{C/\bar{k}}$, and the canonical identification*

$$J_0(\bar{k}) = \mathrm{Pic}^0(C)$$

is Galois-equivariant.

Proof. We descend the Jacobian of C_{0,k_s}, where $k_s \subseteq \bar{k}$ is a separable closure of k. In general k_s/k is infinite, hence not an fppf cover. For this reason we move the descent datum on the Jacobian over k_s by limit arguments to a finite Galois extension, and descend along that extension. A second limit argument shows that the descended scheme represents the degree-zero Picard sheaf.

Let $F = \mathrm{Pic}^0_{C_0/k}$ be the degree-zero part of the fppf Picard sheaf. Here we use that the fibrewise degree is locally constant, by coherent base change and Riemann–Roch. Since the Picard sheaf and the degree commute with base change, Theorem C.11 gives an abelian variety J_s over k_s representing the restriction of F to k_s-schemes; J_s is projective. Over $k_s \otimes_k k_s$ the two pullbacks of J_s represent the same functor, so by the Yoneda lemma we get an isomorphism between them. All identifications are induced by F. Hence this isomorphism satisfies the cocycle condition over $k_s \otimes_k k_s \otimes_k k_s$ and respects the group structures.

To move this descent datum to a finite extension, note that k_s is the filtered union of its finite separable subextensions, and $k_s \otimes_k k_s$ and the triple tensor product are the corresponding unions. By the limit theorems for schemes of finite presentation [Stacks, 01ZM], the defining equations of J_s in a projective space, the descent isomorphism, its inverse and the group operations are defined over some finite stage. The finitely many identities among them (inverse, cocycle, group axioms) hold after enlarging the stage. Finally, we enlarge it further to a Galois extension. In this way we obtain a finite Galois extension K/k in k_s and a projective K-scheme J_K with a group structure and a compatible descent datum for K/k, whose base change to k_s is the datum on J_s. Since $\mathrm{Spec}\, K \to \mathrm{Spec}\, k$ is finite locally free and surjective, the descent datum on the

projective scheme J_K is effective [Stacks, 0CCJ]. This gives a commutative group scheme J_0 over k with $J_0 \times_k k_s \simeq J_s$. Properness and smoothness descend along K/k, and connectedness and dimension may be checked over $\bar{k}$. Hence J_0 is an abelian variety of dimension g.

The functors J_0 and F are fppf sheaves that agree after base change to k_s. Since k_s/k is not finite, we need a limit argument to show that they agree. Both functors are locally of finite presentation. For J_0 this is [Stacks, 01ZC]. For F it follows from Lemma C.2 once the finite stages contain a field of definition of a point of C_0, because invertible sheaves, rigidifications and their isomorphisms are of finite presentation. Hence, for affine T over k, a section of either functor over T_{k_s}, or over $T_{k_s \otimes_k k_s}$, comes from a finite stage. Likewise, two sections that agree over T_{k_s} agree over a finite stage.

The identification $J_0(T_{k_s}) \simeq F(T_{k_s})$ is compatible with the descent data. Let a section of one functor over T be given. We transport its image over T_{k_s} and descend the result to some T_L with L/k finite. After enlarging L, its two pullbacks to $T_{L \otimes_k L}$ agree. Since both functors are fppf sheaves, it then descends to T. The same argument applied to equalities shows that the result is unique. We obtain inverse bijections $J_0(T) \simeq F(T)$, natural in affine T, and hence an isomorphism of sheaves.

The Picard sheaf and the degree commute with field extension. Hence $J_0 \times_k k'$ represents the degree-zero Picard sheaf of $C_0 \times_k k'$ for every k'/k. Over $\bar{k}$ the value of the Picard sheaf is the Picard group, by rigidification and $\mathrm{Pic}(\bar{k}) = 0$. This gives the displayed identification. For $\gamma \in \mathrm{Gal}(\bar{k}/k)$, naturality with respect to $\mathrm{Spec}(\gamma)$ says that this group isomorphism intertwines the Galois action on points with pullback of invertible sheaves. Here $\mathrm{Gal}(\bar{k}/k) = \mathrm{Gal}(k_s/k)$ because $\bar{k}/k_s$ is purely inseparable. ∎

Since J_0 is defined over k, the Galois group acts on its torsion points. Using the Kummer sequence on C, we compare this action with the Galois action on cohomology.

Proposition C.17 (H^1 of a curve and the Tate module of its Jacobian)**.** *In the preceding setting, suppose ℓ is invertible in k. For every $n \geq 1$ there is a canonical Galois-equivariant isomorphism, with cohomology computed on the pro-étale site,*

$$H^1(C, \mathbf{Z}/\ell^n(1)) \simeq J_0[\ell^n](\bar{k}).$$

The coefficient map $\mu_{\ell^{n+1}} \to \mu_{\ell^n}$, $z \mapsto z^\ell$, corresponds to multiplication by ℓ on the torsion points. Consequently

$$\varprojlim_n H^1(C, \mathbf{Z}/\ell^n(1)) \simeq T_\ell J_0$$

as $\mathbf{Z}_\ell$-modules with Galois action.

Proof. The isomorphism comes from the Kummer sequence and the identification of the preceding proposition. The point is that both are compatible with the Galois action and with the transition maps. Then they pass to the inverse limit.

We compute on the étale site and transport the result to the pro-étale site by Theorem 2.35. Since that comparison is natural, it preserves all the maps and Galois actions below. The Kummer sequence gives

$$0 \longrightarrow H^0(C, \mathbf{G}_m)/\ell^n \longrightarrow H^1(C, \mu_{\ell^n}) \longrightarrow \mathrm{Pic}(C)[\ell^n] \longrightarrow 0.$$

The first term vanishes. Indeed, C is proper and connected, so $H^0(C, \mathbf{G}_m) = \bar{k}^\times$, and this group is divisible. For the last term, note that the degree map takes values in the torsion-free group $\mathbf{Z}$, so $\mathrm{Pic}(C)[\ell^n] = \mathrm{Pic}^0(C)[\ell^n]$, and Proposition C.16 identifies this group with $J_0[\ell^n](\bar{k})$. It remains to check Galois equivariance. The map induced by $\mu_{\ell^n} \to \mathbf{G}_m$ and the identification of Proposition C.16 are natural in the curve and in the field, and hence Galois-equivariant.

To check the transition maps, we map the Kummer sequence for ℓ^{n+1} to that for ℓ^n by $[\ell]$ on the first $\mathbf{G}_m$ and by the identity on the second. On the kernels this is $z \mapsto z^\ell$. On the Picard group it sends an invertible sheaf to its ℓ-th tensor power, which is $[\ell]$ on J_0. Therefore the displayed isomorphisms identify the two inverse systems, including their transition maps. ∎

D Cohomology of torsion sheaves on curves

The last step of proper base change for $\mathbf{P}^1$ compares the cohomology of a curve over two separably closed fields, one inside the other. Smooth base change needs the same comparison for affine curves, with coefficients prime to the characteristic. Both rest on Theorem 4.20, whose assertions we recall. Let k be algebraically closed, let C be a separated scheme of finite type over k with $\dim C \leq 1$, and let F be a classical torsion abelian sheaf on $C_{\text{proét}}$. Then:

- (C_1) $H^q(C, F) = 0$ for $q > 2$;
- (C_2) if C is affine, $H^q(C, F) = 0$ for $q > 1$;
- (C_3) if $\mathrm{char}\, k = p > 0$ and F is p-primary, $H^q(C, F) = 0$ for $q > 1$;
- (C_4) if F is constructible with finite stalks and prime to $\mathrm{char}\, k$, all $H^q(C, F)$ are finite;
- (C_5) if C is proper and F is constructible with finite stalks, all $H^q(C, F)$ are finite;
- (C_6) if k'/k is an extension of algebraically closed fields and F is prime to $\mathrm{char}\, k$, the pullback maps $H^q(C, F) \to H^q(C_{k'}, F|)$ are isomorphisms;
- (C_7) if C is proper, the maps in (C_6) are isomorphisms for every torsion F;
- (C_8) for any open $U \subseteq C$, restriction $H^2(C, F) \to H^2(U, F|_U)$ is surjective.

Over $\mathbf{C}$ the vanishing and finiteness assertions are statements about a Riemann surface, which has the homotopy type of a CW complex of dimension two, or of one if it is not compact.

In algebraic geometry we compute the groups. Let C be a smooth proper connected curve of genus g and let n be invertible in k. Three ingredients compute the cohomology of μ_n: the Kummer sequence $0 \to \mu_n \to \mathbf{G}_m \xrightarrow{n} \mathbf{G}_m \to 0$, the vanishing of $H^q(C, \mathbf{G}_m)$ for $q \geq 2$, which is a consequence of Tsen's theorem, and the Jacobian of Appendix C. They give

$$H^0(C, \mu_n) = \mu_n(k), \qquad H^1(C, \mu_n) = J(C)[n](k) \simeq (\mathbf{Z}/n)^{2g}, \qquad H^2(C, \mu_n) = \mathbf{Z}/n$$

(Proposition D.3). These groups do not change under extension of algebraically closed fields, because torsion points of an abelian variety do not. For $\mathbf{P}^1$, which has $g = 0$, the computation fixes the normalization of the class of $\mathcal{O}(1)$ that Remark D.4 provides for the proof of proper base change. Removing a nonempty finite set of r points kills H^2 and raises the rank of H^1 to $2g + r - 1$ (Proposition D.5).

Proper base change, however, is needed for all torsion coefficients, including those divisible by the characteristic p, and there the answer changes. On the affine line, $H^1(\mathbf{A}^1_k, \mathbf{F}_p) = k[t]/\{h^p - h\}$ is infinite, and it grows when k is enlarged (Example D.8). Indeed, the Artin–Schreier torsors $y^p - y = at$, for $a \in k$, have pairwise distinct classes, and over a larger field k' there are new ones with $a \in k' \setminus k$. So for p-torsion coefficients, finiteness and invariance under field extension hold only for proper curves. This is why (C_5) and (C_7) of Theorem 4.20 assume properness, and why Lemma 4.21 is stated for proper curves.

The Artin–Schreier sequence $0 \to \mathbf{F}_p \to \mathbf{G}_a \xrightarrow{\varphi-1} \mathbf{G}_a \to 0$ reduces the cohomology of $\mathbf{F}_p$ to the coherent cohomology of $\mathcal{O}_C$. On the affine line it presents H^1 as the cokernel of $h \mapsto h^p - h$ on $k[t]$, as above. On every curve it gives $H^q(C, \mathbf{F}_p) = 0$ for $q \geq 2$ (Theorem 4.20, (C_3)).

A general torsion sheaf is reduced to a constant one by two observations. First, a sheaf supported on finitely many points has no cohomology in positive degrees. Hence passing to the normalization of a curve, or extending a sheaf by zero from a dense open subset, changes cohomology only in degrees zero and one, and only by finite groups (Lemmas D.11 and Lemma D.12). Second, a finite $\mathbf{F}_\ell$-local system becomes a successive extension of constant sheaves on a finite étale cover of degree d prime to ℓ. This cover is given by an ℓ-Sylow subgroup of the monodromy. Dividing the trace of the cover by d shows that the local system is a direct summand of the pushforward of its pullback (Lemma D.13). We follow the proof of [Stacks, 03SC].

Throughout this appendix, cohomology means cohomology on $C_{\mathrm{pro\acute{e}t}}$, and the sheaves are classical in the sense of Definition 2.30: by Theorem 2.32 such a sheaf is the inverse image $\nu_C^* G$ of a sheaf G on $C_{\acute{e}t}$ along the comparison morphism. By Theorem 2.35, its cohomology, together with the restriction and pullback maps, agrees with étale cohomology, so we carry out all calculations on $C_{\acute{e}t}$ and transport them back by ν_C^*.

Let $p \geq 0$ be the characteristic of k. In this appendix ℓ is an arbitrary prime, possibly equal to p. In characteristic zero the hypotheses in (C_4) and (C_6) of Theorem 4.20 that F be prime to the characteristic hold automatically. We begin with direct images along finite morphisms, which all the reductions below use.

Lemma D.1 (Finite direct images)**.** *Let $u : V \to W$ be finite. For an abelian sheaf F on $V_{\acute{e}t}$ and a geometric point $\bar{w} : \mathrm{Spec}(\Omega) \to W$, with Ω algebraically closed, one has*

$$(u_* F)_{\bar{w}} \simeq \prod_{\substack{\bar{v}:\mathrm{Spec}(\Omega)\to V \\ u\bar{v}=\bar{w}}} F_{\bar{v}}.$$

The product is finite. The functor u_* from abelian sheaves on $V_{\acute{e}t}$ to abelian sheaves on $W_{\acute{e}t}$ is exact. The same holds for sheaves of modules over a constant coefficient ring.

Proof. Over a strict localization the source splits into strictly henselian local pieces, and on such a piece taking sections is the same as taking a stalk.

After base change to the strict henselization of W at $\bar{w}$, the finite scheme V becomes a finite disjoint union of strictly henselian local schemes. On each such local scheme, the sections of an étale sheaf are its stalk at the closed geometric point. Indeed, every étale neighbourhood of that point admits a section, so a germ extends to a global section, and two sections with the same germ agree after pulling back along such a section. In particular, taking sections is exact on abelian sheaves. Combining this with the description of stalks of direct images via strict localizations [Stacks, 03QP], we obtain the displayed formula. Since geometric stalks are conservative and finite products of abelian groups are exact, u_* is exact. The same argument applies to sheaves of modules. ∎

D.1 Kummer theory

Let C be a smooth curve over an algebraically closed field k, and let $\nu_C^* \mathbf{G}_m$ be the pullback of the multiplicative group $\mathbf{G}_m$ of $C_{\acute{e}t}$ to $C_{\mathrm{pro\acute{e}t}}$. For n invertible in k, the Kummer sequence $0 \to \mu_n \to \mathbf{G}_m \xrightarrow{n} \mathbf{G}_m \to 0$ reduces the cohomology of μ_n to that of $\nu_C^* \mathbf{G}_m$, so we compute the latter first. In degree one, $H^1(C, \mathbf{G}_m)$ is the Picard group, as in the Zariski topology, while the

groups $H^q(C, \mathbf{G}_m)$ vanish for $q \geq 2$ (Lemma D.2). In particular the Brauer group of C is zero. This matches, over $\mathbf{C}$, the topology of a Riemann surface, whose integral cohomology vanishes above degree two, and whose H^2 is accounted for by the degree of line bundles. In algebraic geometry the vanishing comes from Tsen's theorem, which says that forms of low degree over the function field $k(C)$ have nontrivial zeros. Over a finite field Tsen's theorem fails: by class field theory the function field of a curve has nonzero Brauer group. This is one reason why we work over an algebraically closed field. With the lemma and the Jacobian, the Kummer sequence then computes the cohomology of μ_n on a proper curve (Proposition D.3).

Lemma D.2 (Cohomology of the multiplicative group)**.** *For a smooth separated curve C of finite type over an algebraically closed field, one has*

$$H^1(C, \nu_C^* \mathbf{G}_m) = \operatorname{Pic}(C), \qquad H^q(C, \nu_C^* \mathbf{G}_m) = 0 \quad (q \geq 2).$$

The identifications are natural under restriction and extension of the algebraically closed ground field. In particular $\operatorname{Br}(C) = H^2(C_{ét}, \mathbf{G}_m) = 0$.

Proof. We compute on $C_{ét}$. In degree one, descent for invertible modules identifies $\mathbf{G}_m$-torsors with line bundles. In degrees at least two, by contrast, the vanishing comes from Tsen's theorem, which we prove first (see also [Stacks, 03RM]). There are three steps: Tsen's theorem, proved by counting dimensions with Riemann–Roch; the vanishing of the Brauer group and of the Galois cohomology of $k(C)$; and the passage from the generic point to C by the divisor sequence.

Tsen's theorem says that $k(C)$ is C_1, that is, a homogeneous form of degree d in $m > d$ variables with coefficients in $k(C)$ has a nonzero zero. To see this, let $\overline{C}$ be the smooth proper model of $k(C)$, of genus g, choose an effective divisor D on $\overline{C}$ bounding the poles of all coefficients, and let the unknowns range over $H^0(\overline{C}, \mathcal{O}(NH))$ for an effective divisor H of positive degree. The value of the form then lies in $H^0(\overline{C}, \mathcal{O}(dNH + D))$. Once $N \deg H > 2g - 2$, we know both dimensions by Theorem C.3. In terms of bases, the m unknowns amount to $m(N \deg H + 1 - g)$ unknowns in k. Each coordinate of the value, in a basis of the target space, is a homogeneous polynomial of degree d in these unknowns, so the vanishing of the value amounts to $dN \deg H + \deg D + 1 - g$ homogeneous polynomial equations in them. The number of unknowns minus the number of equations is $(m - d)N \deg H + (m - 1)(1 - g) - \deg D$, which is positive for large N because $m > d$. Since k is algebraically closed, homogeneous equations in more unknowns than equations have a nonzero solution [Stacks, 03RD].

The reduced norm form of a division algebra of degree $d > 1$ over $k(C)$ has degree d in $d^2 > d$ variables and does not vanish on a nonzero element. Therefore Tsen's theorem implies that $\operatorname{Br}(k(C)) = 0$ [Stacks, 03RF]. The same holds for every finite extension of $k(C)$, since it is again the function field of a curve over k. This is enough for Galois cohomology, because a field all of whose finite extensions have trivial Brauer group has $H^q(K, (K^{\mathrm{sep}})^\times) = 0$ for all $q \geq 1$ [Stacks, 03R8]; hence this holds for $K = k(C)$. To pass from $k(C)$ to C we use the divisor sequence, which relates $\mathbf{G}_m$ on $C_{ét}$ to the pushforward of the multiplicative group at the generic point and to skyscraper sheaves $\mathbf{Z}$ at the closed points. Of these, the former has no higher cohomology by this Galois vanishing, and the latter have none because the residue fields are algebraically closed. Therefore, from the long exact cohomology sequence, we conclude that $H^q(C_{ét}, \mathbf{G}_m) = 0$ for $q \geq 2$.

By Theorem 2.35 these identifications carry over to $C_{\mathrm{proét}}$, and they are natural because the comparison maps, the identification of torsors and the restriction maps are functorial. ∎

On a proper curve, the lemma and the Kummer sequence reduce the cohomology of μ_n to the groups $\operatorname{Pic}(C)[n]$ and $\operatorname{Pic}(C)/n$, both computed in Appendix C.

Proposition D.3 (Smooth proper curves, torsion prime to the characteristic)**.** *Let C be smooth, proper and connected over an algebraically closed field k, with genus g, and let $n \geq 1$ be invertible in k. There are canonical identifications*

$$H^0(C, \mu_n) = \mu_n(k), \quad H^1(C, \mu_n) = J(C)[n](k),$$
$$H^2(C, \mu_n) = \mathbf{Z}/n, \quad H^q(C, \mu_n) = 0 \quad (q > 2).$$

The isomorphism in degree two sends the Kummer boundary of a line bundle L to $\deg L \bmod n$*. These identifications commute with extension of algebraically closed fields.*

Proof. Pulling back the Kummer sequence of $C_{\acute{e}t}$ by ν_C^*, we get the exact sequence of sheaves on $C_{\mathrm{pro\acute{e}t}}$

$$0 \longrightarrow \mu_n \longrightarrow \nu_C^* \mathbf{G}_m \xrightarrow{(\cdot)^n} \nu_C^* \mathbf{G}_m \longrightarrow 0.$$

Here surjectivity of the nth power map is a local question. For a unit a the equation $T^n = a$ defines an étale cover, since its derivative nT^{n-1} is a unit. Moreover, the global units on the proper connected curve form the group $k^\times$, which is n-divisible because k is algebraically closed. Hence the long exact sequence starts with $H^0(C, \mu_n) = \mu_n(k)$. Since the map $H^0(C, \mathbf{G}_m) \to H^0(C, \mathbf{G}_m)$ is onto, $H^1(C, \mu_n)$ injects into $H^1(C, \mathbf{G}_m)$. Using Lemma D.2, we read off from the long exact sequence that

$$H^1(C, \mu_n) = \mathrm{Pic}(C)[n], \qquad H^2(C, \mu_n) = \mathrm{Pic}(C)/n,$$

with all higher groups zero.

Corollary C.12 gives the exact sequence

$$0 \longrightarrow J(C)(k) \longrightarrow \mathrm{Pic}(C) \xrightarrow{\deg} \mathbf{Z} \longrightarrow 0.$$

Multiplication by n on $J(C)(k)$ is surjective, with kernel $J(C)[n](k)$, and $J(C)[n]$ is finite étale of rank n^{2g} (Corollary C.15). Applying the snake lemma to multiplication by n on this sequence, and using that $\mathbf{Z}$ is torsion free, we obtain $\mathrm{Pic}(C)[n] = J(C)[n](k)$ and $\mathrm{Pic}(C)/n = \mathbf{Z}/n$ via the degree. In particular, $H^1(C, \mu_n)$ is noncanonically isomorphic to $(\mathbf{Z}/n)^{2g}$ [Stacks, 03RQ].

Under an algebraically closed extension k'/k, the schemes $J(C)$ and $J(C)[n]$ base change to the corresponding objects for $C_{k'}$. The map $J(C)[n](k) \to J(C)[n](k')$ is bijective by Corollary C.15. The same argument applies to the finite étale scheme μ_n. Finally, pullback of line bundles preserves the degree. Hence the pullback maps on cohomology are isomorphisms. ∎

Remark D.4 (Twists and the projective line)**.** The degree identifies $H^2(C, \mu_n)$ with $\mathbf{Z}/n$ without any choice. Since $\mu_n = (\mathbf{Z}/n)(1)$, twisting by -1 gives

$$H^2(C, \mathbf{Z}/n) = (\mathbf{Z}/n)(-1),$$

and identifying this group with $\mathbf{Z}/n$ requires the choice of a primitive nth root of unity. For $C = \mathbf{P}^1_k$ the Jacobian is zero, so the cohomology with constant coefficients is concentrated in degrees zero and two, and the Kummer boundary of $\mathcal{O}(1)$ corresponds to $1 \in \mathbf{Z}/n$. Section 4.4 uses this normalization on the geometric fibre.

D.2 Affine curves

Every smooth connected curve over an algebraically closed field that is not proper is affine and is the complement of a finite nonempty set of closed points in a smooth proper connected curve. To see this, take a projective model of its function field and normalize it, and recall that regular curves over a perfect field are smooth.

On the affine curve $U = \overline{C} \setminus D$ the Picard group is a quotient of $\mathrm{Pic}^0(\overline{C})$. It is therefore divisible, and $H^2(U, \mu_n) = 0$; this matches the fact that, topologically, a punctured Riemann surface is homotopy equivalent to a graph. In degree one, on the other hand, new torsors appear, since a Kummer torsor on U need not extend over $\overline{C}$. The obstruction to extending it at each puncture is the residue of the torsor, that is, the order modulo n of the corresponding rational section. For example, on $\mathbf{G}_m = \mathbf{P}^1 \setminus \{0, \infty\}$ the torsor $y^n = x$ has residues 1 and -1 at the two punctures. It generates $H^1(\mathbf{G}_m, \mu_n) \simeq \mathbf{Z}/n$. In general the residues sum to zero. The residue sequence of Proposition D.5 shows that $H^1(U, \mu_n)$ is free of rank $2g + \#D - 1$ over $\mathbf{Z}/n$. This is the first Betti number of a surface of genus g with $\#D$ punctures.

Proposition D.5 (Cohomology of a smooth affine curve)**.** *Let $\overline{C}$ be a smooth proper connected curve of genus g over an algebraically closed field k. Let $U = \overline{C} \setminus D$, where D is a finite nonempty set of closed points, and let n be invertible in k. There is a natural exact sequence*

$$0 \longrightarrow H^1(\overline{C}, \mu_n) \longrightarrow H^1(U, \mu_n) \xrightarrow{\partial} (\mathbf{Z}/n)^D \xrightarrow{\sum} \mathbf{Z}/n \longrightarrow 0. \tag{D.1}$$

Moreover, $H^0(U, \mu_n) = \mu_n(k)$ and $H^q(U, \mu_n) = 0$ for $q > 1$. The group $H^1(U, \mu_n)$ is non-canonically $(\mathbf{Z}/n)^{2g+\#D-1}$, and the pullback maps on all these groups are isomorphisms under algebraically closed extension.

Proof. By the Kummer sequence, the vanishing above degree one follows from the divisibility of $\mathrm{Pic}(U)$ and the vanishing of $H^q(U, \mathbf{G}_m)$ for $q \geq 2$ (Lemma D.2). For H^1 we describe μ_n-torsors as line bundles L with a trivialization of $L^{\otimes n}$. After extending L to $\overline{C}$, the orders at the punctures measure the failure of the torsor to extend. These orders give the residue map ∂. We prove exactness at each place, and then deduce the rank and the invariance under field extension.

Since $\overline{C}$ is regular, we may use divisors to see that

$$\mathrm{Pic}(U) = \mathrm{Pic}(\overline{C}) / \langle \mathcal{O}_{\overline{C}}(x) : x \in D \rangle.$$

One of these line bundles has degree one. Every class in $\mathrm{Pic}(\overline{C})$ is therefore congruent, modulo the subgroup generated by it, to a class of degree zero. Hence $J(\overline{C})(k)$ surjects onto $\mathrm{Pic}(U)$. As $J(\overline{C})(k)$ is n-divisible (Corollary C.15), so is $\mathrm{Pic}(U)$. By the Kummer sequence and Lemma D.2, we have $H^2(U, \mu_n) = \mathrm{Pic}(U)/n = 0$ and $H^q(U, \mu_n) = 0$ for $q \geq 3$.

We construct the sequence (D.1) using line bundles. Recall that a μ_n-torsor on U is given by a pair $(L, \alpha : L^{\otimes n} \simeq \mathcal{O}_U)$ of a line bundle and a trivialization of its nth power. To define its residues, we extend L to a line bundle $\overline{L}$ on $\overline{C}$ and let $s = \alpha^{-1}(1)$ be the corresponding rational section of $\overline{L}^{\otimes n}$. The divisor of s is supported on D, since s generates on U. Its orders modulo n do not depend on the extension. Indeed, two extensions differ by $\mathcal{O}(E')$ with E' supported on D, and this changes the divisor of s by nE'. These orders define $\partial(L, \alpha)$. Their sum is zero modulo n, since the divisor of s has degree $\deg \overline{L}^{\otimes n} = n \deg \overline{L}$.

Suppose all orders are divisible by n, and write $\mathrm{div}(s) = nE$ with E supported on D. Then s is a nowhere vanishing section of $(\overline{L}(-E))^{\otimes n} = \overline{L}^{\otimes n}(-nE)$. So the trivialization extends to a torsor

with line bundle $\overline{L}(-E)$ on $\overline{C}$, and this torsor restricts to (L, α) on U. This proves exactness at $H^1(U, \mu_n)$.

To prove injectivity, note that a trivialization on U of a torsor on $\overline{C}$ gives a rational section of its line bundle whose nth power has no zero or pole on $\overline{C}$. Therefore the divisor of the section is zero, and the section trivializes the torsor on $\overline{C}$.

To see that ∂ maps onto the kernel of $\sum$, we lift a residue vector with sum zero modulo n to an integral divisor B supported on D. Its degree is divisible by n because the residues sum to zero. We claim that, as a consequence, there is a line bundle $\overline{L}$ with $\overline{L}^{\otimes n} \simeq \mathcal{O}_{\overline{C}}(B)$. To find it, choose $P \in \overline{C}(k)$ and put $L_1 = \mathcal{O}_{\overline{C}}((\deg B/n)P)$, so that the line bundle $\mathcal{O}_{\overline{C}}(B) \otimes L_1^{\otimes -n}$ has degree zero. By divisibility of $J(\overline{C})(k)$ it is $N^{\otimes n}$ for some N of degree zero, and $\overline{L} = L_1 \otimes N$ works. It remains to produce the torsor. The canonical section of $\mathcal{O}_{\overline{C}}(B)$ has divisor B and is nowhere zero on U. It therefore defines a trivialization of $\overline{L}^{\otimes n}|_U$, that is, a Kummer torsor on U whose residues are the orders of B. Finally, the summation map is surjective since D is nonempty. This proves that the sequence is exact (see also [Stacks, 03RR]).

The left term and the kernel of summation are free $\mathbf{Z}/n$-modules of ranks $2g$ and $\#D - 1$. Hence the resulting short exact sequence splits noncanonically. For naturality, note that the formation of the compactification, the boundary points, the line bundles and the orders of vanishing commutes with base change. The leftmost pullback map is an isomorphism by Proposition D.3, and the residue groups and their summation maps are unchanged. By the five lemma applied to the two exact sequences, the middle pullback map is an isomorphism. ∎

For the constant sheaf $\mathbf{F}_\ell$, with ℓ a prime different from $\operatorname{char} k$, the two preceding propositions prove (C_1)–(C_8) on smooth curves. Indeed, a primitive ℓth root of unity in k identifies $\mathbf{F}_\ell$ with μ_ℓ, compatibly with the pullback maps to $C_{k'}$, since the root of unity stays primitive in k'. For (C_8), note that an open in a smooth connected proper curve is either the whole curve or an affine curve, which has $H^2 = 0$, and that an open in a smooth affine curve is affine.

D.3 The Artin–Schreier sequence

Since the Kummer cover $T^n = a$ is étale only when n is invertible, p-torsion in characteristic p needs a different sequence. In the Artin–Schreier sequence the role of $\mathbf{G}_m$ is played by the additive group, whose étale cohomology is the coherent cohomology of the structure sheaf. Since coherent cohomology of a curve vanishes above degree one, this gives $H^q(C, \mathbf{F}_p) = 0$ for $q \geq 3$. For $q = 2$ we also need that $\varphi - 1$ is surjective on $H^1(C, \mathcal{O}_C)$, which holds over an algebraically closed field. So $H^2(C, \mathbf{F}_p) = 0$ even on a proper curve, and there is no analogue of the class of a point in H^2.

On a proper curve, $H^1(C, \mathbf{F}_p)$ is the fixed space of the Frobenius-linear map φ on the g-dimensional space $H^1(C, \mathcal{O}_C)$. Hence it has dimension at most g (Proposition D.6). The bound can be strict. For example, for an elliptic curve, φ on the line $H^1(E, \mathcal{O}_E)$ is bijective when E is ordinary, and zero when E is supersingular, as is $y^2 = x^3 - x$ over $\overline{\mathbf{F}}_3$. Accordingly $H^1(E, \mathbf{F}_p)$ has dimension one or zero. On an affine curve, by contrast, the cokernel of $\varphi - 1$ on the ring of functions contributes to H^1, and this group is infinite (Example D.8).

Proposition D.6 (p-torsion on a smooth proper curve in characteristic p)**.** *Let C be a smooth proper connected curve of genus g over an algebraically closed field of characteristic $p > 0$, and let φ be the Frobenius-linear endomorphism of $H^1(C, \mathcal{O}_C)$ induced by $a \mapsto a^p$ on $\mathcal{O}_C$. Then*

$$H^0(C, \mathbf{F}_p) = \mathbf{F}_p, \qquad H^1(C, \mathbf{F}_p) = \ker(\varphi - 1 : H^1(C, \mathcal{O}_C) \to H^1(C, \mathcal{O}_C)),$$

and $H^q(C, \mathbf{F}_p) = 0$ for $q > 1$. The group $H^1(C, \mathbf{F}_p)$ is finite, of $\mathbf{F}_p$-dimension at most g, and its pullback map is an isomorphism under algebraically closed extension.

Proof. The Artin–Schreier sequence turns the question into one about $\varphi - 1$ on $H^0(C, \mathcal{O}_C) = k$ and on $H^1(C, \mathcal{O}_C)$. The key fact is that on a finite-dimensional space over an algebraically closed field, a Frobenius-linear map minus the identity is surjective with finite kernel.

Step 1: the Artin–Schreier sequence. Write also φ for the endomorphism $a \mapsto a^p$ of the sheaf $\mathbf{G}_a$ on $C_{\acute{e}t}$. The Artin–Schreier sequence

$$0 \longrightarrow \mathbf{F}_p \longrightarrow \mathbf{G}_a \xrightarrow{\varphi - 1} \mathbf{G}_a \longrightarrow 0$$

is exact, because surjectivity is a local question and $T^p - T - a$ has derivative -1. The étale cohomology of $\mathbf{G}_a$ is the coherent cohomology of $\mathcal{O}_C$. Indeed, quasi-coherent sheaves have the same cohomology on the étale and on the Zariski site, by faithfully flat descent on affine opens and the Čech resolution. This coherent cohomology vanishes in degrees above one, since C is a curve. Since C is proper, $H^1(C, \mathcal{O}_C)$ is a finite-dimensional k-vector space.

Step 2: linear algebra. Let φ be a Frobenius-linear endomorphism of a finite-dimensional k-vector space V; we show that $\varphi - 1$ is surjective and that its kernel is the set of k-points of a finite étale scheme of order p^f with $f \leq \dim_k V$. To prove this, we reduce along a finite filtration to the part where φ is nilpotent and the part where φ is bijective. On the nilpotent part, $\varphi - 1$ is an automorphism given by an additive polynomial, with inverse $-(1 + \varphi + \varphi^2 + \cdots)$, a finite sum; in particular its kernel there is zero. On a bijective part of dimension f, by contrast, we choose a basis, invert the matrix of φ, and write the equations of the fibres as

$$x_i^p - \sum_j b_{ij} x_j = c_i \qquad (1 \leq i \leq f),$$

where (b_{ij}) is invertible. The coordinate algebra is free of dimension p^f, with basis the monomials having each exponent less than p. Moreover, the Jacobian matrix of these equations is $-(b_{ij})$, because $p = 0$ in k, and it is invertible; so each fibre is finite étale of degree p^f and has p^f points over the algebraically closed field k [Stacks, 0A3L]. In particular every fibre is nonempty, so $\varphi - 1$ is surjective, and the kernel has p^f elements.

Step 3: conclusion. We apply Step 2 to $H^0(C, \mathcal{O}_C) = k$ and to $H^1(C, \mathcal{O}_C)$ in the long exact sequence associated to the Artin–Schreier sequence. Surjectivity of $\varphi - 1$ on H^0 gives $H^1(C, \mathbf{F}_p) = \ker(\varphi - 1)$ on $H^1(C, \mathcal{O}_C)$, with the bound on its dimension, and surjectivity on H^1 together with $H^2(C, \mathcal{O}_C) = 0$ gives $H^q(C, \mathbf{F}_p) = 0$ for $q > 1$. Finally, the pullback map under an extension k'/k is an isomorphism [Stacks, 0HAN, 0A3P]: flat base change for coherent cohomology identifies the vector space and its Frobenius-linear map with their scalar extensions, and the kernel is a finite étale scheme, whose k-points and k'-points agree. As always, Theorem 2.35 carries the statement over to $C_{\text{pro\'et}}$. ∎

On an affine curve the same sequence is even simpler to use, because coherent cohomology vanishes in positive degrees. On the other hand, H^1 is now the cokernel of $\varphi - 1$ on an infinite-dimensional space.

Lemma D.7 (p-torsion on an affine curve in characteristic p)**.** *If U is affine over an algebraically closed field k of characteristic p, then, for the classical constant sheaf $\mathbf{F}_p$ and $\varphi(a) = a^p$ on $\Gamma(U, \mathcal{O}_U)$, we have*

$$H^1(U, \mathbf{F}_p) = \Gamma(U, \mathcal{O}_U)/(\varphi - 1)\Gamma(U, \mathcal{O}_U), \qquad H^q(U, \mathbf{F}_p) = 0 \quad (q > 1).$$

Proof. Quasi-coherent cohomology on an affine scheme vanishes in positive degrees, and, as in the proof of the proposition, étale and Zariski cohomology of $\mathbf{G}_a$ agree. In the long exact sequence associated to the Artin–Schreier sequence on $U_{ét}$, the terms $H^q(U, \mathbf{G}_a)$ for $q \geq 1$ are therefore zero. What remains is the exact sequence $0 \to H^0(U, \mathbf{F}_p) \to \Gamma(U, \mathcal{O}_U) \xrightarrow{\varphi - 1} \Gamma(U, \mathcal{O}_U) \to H^1(U, \mathbf{F}_p) \to 0$, together with $H^q(U, \mathbf{F}_p) = 0$ for $q > 1$. By Theorem 2.35 these statements carry over to $U_{\text{proét}}$. ∎

For the affine line we can compute the cokernel explicitly, and we see that it is infinite and grows with the field, unlike in the proper case.

Example D.8 (The affine line in characteristic p)**.** For $U = \mathbf{A}^1_k$ in characteristic p we get

$$H^1(U, \mathbf{F}_p) = k[t]/\{h^p - h : h \in k[t]\}.$$

Every class has a unique representative with zero constant term and all positive exponents prime to p. Indeed, we can remove exponents divisible by p using pth roots in the perfect field, and we can remove the constant term since $a \mapsto a^p - a$ is surjective on k. For uniqueness, note that a nonconstant polynomial $h^p - h$ has degree divisible by p. In particular the quotient is infinite, because the classes of t^m for positive m prime to p are $\mathbf{F}_p$-linearly independent. For an algebraically closed extension $k \subsetneq k'$ and $a \in k' \setminus k$, the class of at is not in the image of the quotient over k. Indeed, the reduction of a polynomial in $k[t]$ uses only pth roots and roots of $x^p - x = c$ in k, so its reduced representative lies in $k[t]$, whereas at is already reduced. Thus finiteness and invariance under field extension in degree one fail for p-torsion coefficients on an affine curve.

For the constant sheaf $\mathbf{F}_p$ on smooth curves, we obtain (C_1), (C_3), (C_5) and (C_7) from the proposition on proper curves, and (C_1), (C_2) and (C_3) from the lemma on affine curves, while (C_8) holds because H^2 vanishes in both cases.

D.4 Normalization

So far we have computed with constant coefficients on smooth curves, whereas Theorem 4.20 concerns arbitrary curves of dimension at most one and arbitrary torsion sheaves. Two reductions close the gap: normalization and removing finitely many points. Both rest on Lemma D.10, which says that a sheaf supported on finitely many points has no higher cohomology. Hence passing between a curve, its normalization and a dense open subset changes cohomology only in degrees zero and one, and only by finite groups that do not change under field extension.

For example, let $C : y^2 = x^3 + x^2$ be the nodal cubic. Its normalization $\mathbf{A}^1 \to C$ has two points over the node. For $M = \mathbf{Z}/n$ the normalization sequence of Lemma D.11 has cokernel $M^2/\Delta M \simeq M$ at the node. If n is invertible in k, then $H^1(\mathbf{A}^1, \mathbf{Z}/n) = 0$, and the long exact sequence gives $H^1(C, \mathbf{Z}/n) \simeq \mathbf{Z}/n$. This is the analogue of the loop created by the node.

Lemma D.9 (Finite morphisms and cohomology)**.** *Let $h : Y \to C$ be a finite morphism of schemes. The direct image h_* of classical sheaves is exact, and $H^q(C, h_*G) = H^q(Y, G)$ for every classical abelian sheaf G on $Y_{\text{proét}}$ and every q.*

Proof. On the étale sites, Lemma D.1 shows that h_* is exact and that its stalks are finite products of stalks of G. Since h_* is exact, the Leray spectral sequence gives $H^q(C, h_*G) = H^q(Y, G)$. The comparison of classical direct images and Theorem 2.35 transport these statements to the pro-étale site; see [Stacks, 03QP, 0959]. ∎

Since a sheaf supported on finitely many points is a direct image from those points, it has no higher cohomology. All the reductions below use this vanishing.

Lemma D.10 (Sheaves with finite support)**.** *Let F be a classical abelian sheaf supported on finitely many closed points of a scheme C of finite type over an algebraically closed field k. Then $H^q(C, F) = 0$ for $q > 0$, and $H^0(C, F)$ is the direct sum of the stalks of F at these points; the pullback map on H^0 is an isomorphism under algebraically closed extension of k.*

Proof. The sheaf F is h_* of a sheaf on a finite disjoint union of copies of $\operatorname{Spec} k$, where h is the inclusion of the points. On these points taking sections is exact because k is algebraically closed, and the sections are the product, here a finite direct sum, of the stalks, which do not change under extension of k. Finally, Lemma D.9 carries these assertions over to C. ∎

The normalization of a curve is finite and is an isomorphism away from finitely many points, so the constant sheaf on C differs from the direct image of the constant sheaf on C^{nor} only by a sheaf with finite support.

Lemma D.11 (Normalization of a constant sheaf)**.** *Let C be a reduced curve of finite type over an algebraically closed field, and let $\pi : C^{\mathrm{nor}} \to C$ be its normalization. For a finite abelian group M, the natural map*

$$\underline{M}_C \longrightarrow \pi_* \underline{M}_{C^{\mathrm{nor}}}$$

is injective with cokernel supported on finitely many points. At a point with b geometric branches, the cokernel is $M^b/\Delta M$, where Δ is the diagonal. These cokernels and the resulting exact sequences commute with algebraically closed field extension.

Proof. The map is computed on stalks by the formula for stalks of finite direct images, and the cokernel has finite support because π is an isomorphism away from finitely many points. Compatibility with field extension holds because normalization commutes with algebraically closed extension.

Normalization is finite for reduced schemes of finite type over a field. On each one-dimensional component, its normal model is regular, hence smooth over the perfect field, while the isolated components of dimension zero are reduced points, that is, copies of the spectrum of the ground field. Since the normalization is an isomorphism over a dense open of each component, the locus where it is not an isomorphism is finite.

By the calculation of stalks in Lemma D.1, the displayed map at a geometric point is the diagonal $M \to M^b$, with $b \geq 1$, so its kernel is zero and its cokernel has the asserted form. Where π is an isomorphism we have $b = 1$ and the cokernel vanishes; hence the cokernel is supported on the finite locus found above.

After algebraically closed extension, the normalization remains smooth and is a finite birational normal model of the reduced base change, so it is the normalization of the base change. Base change for finite morphisms identifies the fibres and their sets of branches, and therefore the diagonal maps and their cokernels are obtained by extension of scalars. ∎

The normalization sequence reduces constant coefficients on a reduced curve to the smooth case. By Lemma D.10, its cokernel term has finite sections, unchanged under field extension, and no positive cohomology. Moreover, by Lemma D.9, $H^q(C, \pi_* \underline{M}) = H^q(C^{\mathrm{nor}}, \underline{M})$. The long exact sequence therefore yields isomorphisms $H^q(C, \underline{M}) \simeq H^q(C^{\mathrm{nor}}, \underline{M})$ for $q \geq 2$, while in degrees

zero and one it gives a natural exact sequence relating C and C^{nor}, whose remaining term is a finite group. From these sequences, vanishing and finiteness pass from C^{nor} to C, and by the five lemma so does invariance under field extension.

The normalization of an affine or proper curve is again affine or proper. Hence all assertions of Theorem 4.20 for constant $\mathbf{F}_\ell$ follow from the calculations on smooth curves. For (C_8), we apply this comparison to C and to the inverse image of an open subset. The commutative square of restriction maps then reduces surjectivity to the smooth components.

D.5 Extension by zero

A constructible sheaf on a curve is locally constant on a dense open subset. Removing the finitely many remaining points replaces the sheaf by its extension by zero from this open, and changes cohomology only in degrees zero and one. Let $j : U \hookrightarrow C$ be a dense open in a reduced curve, and let $i : Z \hookrightarrow C$ be its finite complementary closed subscheme. For a classical sheaf F we have the exact sequence

$$0 \longrightarrow j_!j^*F \longrightarrow F \longrightarrow i_*i^*F \longrightarrow 0$$

of an open and its closed complement. We check its exactness on geometric stalks on $C_{\acute{e}t}$, and then pull it back.

Lemma D.12 (Removing finitely many points)**.** *Let C be a reduced separated curve of finite type over an algebraically closed field k, let $j : U \hookrightarrow C$ be a dense open with finite complement $i : Z \hookrightarrow C$, let ℓ be a prime, and let F be a classical constructible sheaf with $\ell F = 0$. For each of the assertions (C_1)–(C_8) listed at the beginning of this appendix, under its hypotheses on C and on the prime ℓ, the assertion holds for F if and only if it holds for $j_!j^*F$.*

Proof. The third term of the exact sequence displayed above is supported on the finite set Z, so by Lemma D.10 it has finite sections and no higher cohomology. Comparing the long exact sequences of F and $j_!j^*F$ then shows that each assertion passes in both directions.

The sheaf $j_!j^*F$ is again killed by ℓ, so the hypotheses on the coefficient prime are the same for both sheaves. Since the last term of the exact sequence has no positive cohomology, the long exact sequence yields

$$\begin{aligned} 0 \to H^0(C, j_!j^*F) \to H^0(C, F) \to H^0(Z, i^*F) \to \\ H^1(C, j_!j^*F) \to H^1(C, F) \to 0, \end{aligned}$$

and isomorphisms in degrees at least two. Since the sequence is natural under field extension and its term supported on Z is unchanged, the two long exact sequences show that the pullback maps are isomorphisms for F in all degrees if and only if they are isomorphisms for $j_!j^*F$ in all degrees. For (C_8), we apply the degree-two identifications after restriction to every open subset of C. See also [Stacks, 0GJA]. ∎

When j is dense, Lemma D.12, together with the calculation for constant $\mathbf{F}_\ell$ on C via Lemma D.11, shows that $j_!\underline{\mathbf{F}_\ell}$ satisfies the assertions of Theorem 4.20. An arbitrary open is dense in the reduced union of the components it meets, a closed subscheme of C, so we can apply the dense case there and pass back to C: direct image along a finite morphism is exact [Stacks, 0A3Q] and, by Lemma D.9, does not change cohomology.

D.6 Reduction to constant coefficients

We reduce a local system to constant sheaves by pulling it back along a cover of degree prime to ℓ, on which it becomes unipotent. To find such a cover, note that a local system of finite $\mathbf{F}_\ell$-vector spaces has finite monodromy: the fundamental group acts on a fibre through a finite quotient. The local system then becomes a successive extension of constant sheaves on the cover defined by an ℓ-Sylow subgroup of the monodromy group.

On the full cover of its monodromy the local system even becomes constant. But that cover can have degree divisible by ℓ, and then the local system need not be a direct summand of the pushforward of its pullback. For example, let $v : y^2 = x$ be the double cover of $\mathbf{G}_m$, in characteristic different from two, and let $\ell = 2$. The composite of the unit $\mathbf{F}_2 \to v_*\mathbf{F}_2$ and the trace is multiplication by 2, which is zero. In fact $\mathbf{F}_2$ is not a direct summand of $v_*\mathbf{F}_2$ at all. The monodromy representation of $v_*\mathbf{F}_2$ is the regular representation $\mathbf{F}_2[\mathbf{Z}/2]$, which is indecomposable.

For a cover of degree prime to ℓ, on the other hand, Lemma D.13 shows that the trace divided by the degree splits the unit. It therefore suffices to prove the vanishing, finiteness and invariance statements for the pullback. The pullback is unipotent, that is, built from constant sheaves by extensions.

Lemma D.13 (Reduction to a unipotent local system)**.** *Let ℓ be a prime, let U be a connected scheme, and let G be a finite classical local system of $\mathbf{F}_\ell$-vector spaces. There is a finite étale cover $v : V \to U$ of degree d prime to ℓ such that v^*G has a finite filtration with successive quotients the constant sheaf $\mathbf{F}_\ell$. Moreover, G is a direct summand of v_*v^*G.*

Proof. The filtration comes from the representation theory of ℓ-groups, whose only simple module over $\mathbf{F}_\ell$ is the trivial one, and the splitting from the trace, whose composite with the unit is multiplication by the degree.

By Proposition 2.42, G is the pullback of a finite étale local system, which corresponds to a finite continuous representation of the fundamental group. Let H be the finite image of this representation, choose an ℓ-Sylow subgroup P, and take the cover associated to its inverse image in the fundamental group. Its degree is $[H : P]$, which is prime to ℓ, and the restricted action factors through P. Since the augmentation ideal of $\mathbf{F}_\ell[P]$ is nilpotent, the simple modules of the finite ℓ-group P are trivial of dimension one, and a composition series yields the required filtration [Stacks, 0A3R].

On a fibre, the unit $G \to v_*v^*G$ is the diagonal and the trace map $\mathrm{Tr} : v_*v^*G \to G$ adds the d entries. Their composite is therefore multiplication by d, which is invertible on an $\mathbf{F}_\ell$-sheaf, so $d^{-1}\,\mathrm{Tr}$ is a retraction. ∎

To apply this to $j_!G$ on a reduced curve C, we take for U a dense smooth open on which G is locally constant. Normalizing C in the finite extensions of function fields defined by V, we get a finite morphism $\bar{v} : Y \to C$ with V the inverse image of U, and an open immersion $j' : V \hookrightarrow Y$. From the cartesian square formed by these maps and the formula for stalks of direct images along finite morphisms, we find that

$$j_!v_*v^*G = \bar{v}_*j'_!v^*G,$$

which is also the exchange formula of [Stacks, 0GJ1]. If C is proper, then Y is proper, and if C is affine, then Y is affine.

The assertions travel from constant sheaves to $j'_!v^*G$ on Y, then to $j_!v_*v^*G$ on C, and from there to the direct summand $j_!G$. Since $j'_!$ is exact, the filtration of v^*G gives a filtration of $j'_!v^*G$ with graded pieces $j'_!\mathbf{F}_\ell$. The assertions of Theorem 4.20 hold for these pieces by the calculation for constant sheaves extended by zero. As we show at the start of the next section, they pass along extensions, so they hold for $j'_!v^*G$. Since $\bar{v}_*$ is exact, they hold for $j_!v_*v^*G$. By the splitting given by the trace, they hold for its direct summand $j_!G$. For invariance under field extension we use that the unit and the trace commute with extension of the ground field. Then the pullback map for $j_!G$ is a retract of the pullback map for $j_!v_*v^*G$, and a retract of an isomorphism is an isomorphism [Stacks, 03SD].

If C is not equidimensional, we choose a dense open that is smooth and is the disjoint union of the regular loci of the one-dimensional components and the isolated points. On the isolated points taking sections is exact, each connected one-dimensional piece is treated as above, and finite direct sums preserve all the assertions.

D.7 Dévissage

The assertions of Theorem 4.20 pass along short exact sequences. To see this, let $0 \to F_1 \to F \to F_2 \to 0$ be exact, and suppose that the assertions are known for F_1 and F_2. Vanishing and finiteness follow from the long exact sequence, and invariance under field extension follows from the map between the two long exact sequences and the five lemma. For (C_8) an extra argument is needed. Let $b \in H^2(U, F)$, and lift its image in $H^2(U, F_2)$ to $c_2 \in H^2(C, F_2)$. The obstruction to lifting c_2 to $H^2(C, F)$ lies in $H^3(C, F_1)$, which vanishes by (C_1), so such a lift c exists. The difference $b - c|_U$ comes from $H^2(U, F_1)$, and by (C_8) for F_1 it lifts to $H^2(C, F_1)$; adding that lift to c gives the required global class [Stacks, 0A5C].

Proof of Theorem 4.20. For constructible sheaves we reduce to the explicit calculations, and arbitrary torsion sheaves are filtered colimits of these. The reduction has four steps: dévissage to constructible $\mathbf{F}_\ell$-sheaves, extension by zero from an open on which the sheaf is locally constant, a cover of degree prime to ℓ that makes it unipotent, and normalization, which leaves constant coefficients on smooth curves.

By Theorem 2.35, we may work on $C_{\acute{e}t}$, and since a universal homeomorphism does not change the étale topos, we may replace C by C_{red}. A constructible torsion sheaf has a finite exponent, because it has finitely many isomorphism types of finite stalks. It is therefore the direct sum of its finitely many primary parts, and a finite filtration of an ℓ-primary part has constructible $\mathbf{F}_\ell$-sheaves as graded pieces. By the stability under extensions just proved, it suffices to treat such sheaves.

Choose a dense smooth open $j : U \hookrightarrow C$ on which the sheaf is finite locally constant. By Lemma D.12 we may replace the sheaf by its extension by zero. On the connected components of U, Lemma D.13 and finite direct images reduce us to constant $\mathbf{F}_\ell$ extended by zero on curves finite over C; this is allowed because the assertions pass to direct summands and because, by Lemma D.9, finite direct images are exact and do not change cohomology. Finally, normalization and Lemma D.10 reduce these sheaves to the calculations with constant coefficients on smooth curves, namely the Kummer sequence for $\ell \neq p$ and the Artin–Schreier sequence for $\ell = p$. Since all reductions preserve the hypothesis that C is proper or affine, the theorem holds for constructible torsion sheaves.

An arbitrary torsion sheaf on the qcqs scheme C is a filtered colimit of constructible torsion sheaves, and cohomology commutes with filtered colimits [Stacks, 03SA, 03Q5]. For torsion prime to the characteristic or p-primary torsion, we may take the constructible sheaves with the same kind of torsion, by splitting off primary parts before passing to the colimit. Pullback and

restriction also commute with filtered colimits. Hence vanishing and the pullback isomorphisms pass to the colimit. So does surjectivity in (C_8), because a representative of a class already occurs at some stage. Finiteness is asserted only for constructible sheaves. Finally, Theorem 2.35 carries these conclusions and their maps over to $C_{\text{proét}}$. ∎

E Specialization and contracting correspondences

The trace formula needs the local term at an isolated fixed point to be the trace on the stalk. This is not a formal fact. A local term is computed from $R\mathcal{H}om(c_2^*K, c_1^!K)$ near the fixed point, and the stalk does not determine this complex (Remark 12.4). Instead, Chapter 12 proves the equality by a computation for Frobenius on a smooth curve (Proposition 12.16) and reduces everything else to that case. In particular, its trace formula (Theorem 12.24) does not depend on this appendix.

The computation leaves open why Frobenius behaves so well. Topology suggests that the reason is contraction. If a map φ of a manifold contracts a neighbourhood of a fixed point y into itself, then the local index at y is 1. With coefficients in a sheaf, the local index is the trace of φ on the stalk at y. Frobenius contracts in the strongest possible sense: its differential is zero. In this appendix we make this explanation into a theorem, in any dimension and for a whole class of correspondences at once.

In algebraic geometry we measure contraction by powers of ideals. Let $c = (c_1, c_2) : C \to X \times X$ be a correspondence, $Z \subset X$ a closed subscheme with ideal I, and put $I_i = c_i^{-1}(I)\mathcal{O}_C$. We say, as in [Var05], that c contracts towards Z if $I_2 \subset I_1$ and $I_2^n \subset I_1^{n+1}$ for some $n \geq 1$. Frobenius is the model case. Indeed, take $c = (\mathrm{id}, \Phi_X^m)$ and Z a point $y \in X_0(\mathbf{F}_{q^m})$, and choose coordinates x_i of an affine neighbourhood of y defined over $\mathbf{F}_q$. Then y has coordinates $a_i \in \mathbf{F}_{q^m}$, and $(\Phi_X^m)^*(x_i - a_i) = x_i^{q^m} - a_i = (x_i - a_i)^{q^m}$. Hence $I_2 \subset I_1^{q^m}$, and $n = 1$ works.

The main theorem, Theorem E.30, says that near a component of its fixed locus that meets Z, the trace class of a contracting correspondence can be computed on Z alone, from the correspondence restricted to Z. For Z a point, Corollary E.31 says that the local term at an isolated fixed point is the trace on the stalk, even when X is singular. The theorem and its proof are due to Varshavsky [Var05]. They generalize Fujiwara's theorem, conjectured by Deligne, that the local terms of a correspondence composed with a sufficiently high power of Frobenius are the naive ones.

The proof makes the contraction literal by deforming X to the normal cone of Z. Over the trait $\operatorname{Spec} k[t]_{(t)}$, the deformation to the normal cone $\mathcal{D}_Z X$ has generic fibre X and special fibre the normal cone $N_Z X$. On the special fibre, the contraction hypothesis says that the second leg maps everything into the zero section (Lemma E.26). There the part of K away from Z contributes nothing. What remains is the specialization of K restricted to the zero section, and by Verdier's theorem (Theorem E.8) this is $K|_Z$.

It remains to transport the answer back to X. We identify the trace class on the special fibre with the specialization of the trace class of c (Proposition E.22). The specialization map is an isomorphism, because the fixed locus does not move in the family up to nilpotents (Lemma E.28). Before any of this, we must be able to specialize a trace class at all. A trace class is built from $K \boxtimes \mathbb{D}K$, so we need nearby cycles to commute with external products and with Verdier duality. Propositions E.9 and Proposition E.18 prove both, by a dimension argument of Illusie [Ill94].

We use the six operations and the notation for correspondences and trace classes of Chapter 11, and classical results on nearby cycles over a trait, which we recall where they are first needed. Throughout this appendix, k is algebraically closed, ℓ is invertible in k, and A is a finite commutative ring annihilated by a power of ℓ. The only exception to these conventions

is the base of nearby cycles in Sections E.2–Section E.5, which may be any excellent strictly henselian trait, with residue field only separably closed. As in the rest of the notes, statements about constructible complexes on k-schemes are made on the pro-étale site, but we construct nearby cycles on the classical étale site and transport them by the comparison for constructible complexes (Theorem 2.43 and Corollary 2.44).

E.1 Specialization to the normal cone

Let S be the strict henselization of $\operatorname{Spec} k[t]_{(t)}$, with generic point η. Nearby cycles move a complex from the generic fibre of a family over S to its special fibre. Since the special fibre does not lie in the generic fibre, we cannot simply restrict. The obvious attempt is to extend by Rj_* along the open generic fibre and restrict by i^* to the special fibre. This fails already for S itself, because the fraction field of S is not separably closed. Its absolute Galois group is the inertia group, whose tame quotient is $\prod_{\ell' \neq p} \mathbf{Z}_{\ell'}(1)$. For n invertible in k, the stalk of $R^1 j_*(\mathbf{Z}/n)$ at the closed point is therefore $H^1(\eta, \mathbf{Z}/n) = \mathbf{Z}/n(-1) \neq 0$. This class comes from the arithmetic of η, not from the geometry of the family. Passing first to the geometric generic fibre removes such Galois classes. After this change, the constant sheaf specializes to itself, in degree zero, by the first calculation below.

Both specialization to the normal cone and the comparison of trace classes in Proposition E.22 are built from nearby cycles on the deformation to the normal cone. The key input is Verdier's theorem, Theorem E.8 (see [Var05, §§1.3–1.4 and 3]).

The basic computation is that nearby cycles of a constant family $Y \times S$ give back the complex we started with (Lemma E.2). We use it twice. It makes the map $\tau_{X,Z}$ below land in i^*L, and it makes the specialization of trace classes the identity when the fixed scheme does not move (Corollary E.25). Its proof uses only the definition of nearby cycles, their base change maps and two calculations, which we now recall.

Choose a geometric generic point $\bar{\eta}$ of S, and let $\mathcal{Y} \to S$ be separated of finite type. To define nearby cycles, we pull $\mathcal{Y}$ back to the integral closure of S in the chosen separable closure of its fraction field, and let $\bar{i}$ and $\bar{j}$ be the inclusions of the special fibre and of the geometric generic fibre of this pullback. Then, for L on $\mathcal{Y}_\eta$ with pullback $L_{\bar{\eta}}$ to the geometric generic fibre, the geometric nearby cycles are

$$R\Psi_{\mathcal{Y}}(L) = \bar{i}^* R\bar{j}_* L_{\bar{\eta}}.$$

This defines a functor $R\Psi_{\mathcal{Y}} : D_c^b(\mathcal{Y}_{\eta,\acute{e}t}, A) \to D(\mathcal{Y}_{s,\acute{e}t}, A)$. We suppress the inertia action, except where invariants under the wild inertia group of a finite extension of traits are taken below.

Example E.1 (A node)**.** Nearby cycles see more than the special fibre when the family degenerates. For example, for $\mathcal{Y} = V(xy - t) \subset \mathbf{A}^2_S$ the special fibre is the pair of axes $xy = 0$, while the generic fibre is smooth. To see what to expect, note that over $\mathbf{C}$ the part of the fibre $xy = \varepsilon$ near the origin is an annulus $\{(x, \varepsilon/x)\}$, which retracts onto a circle that vanishes as $\varepsilon \to 0$. Correspondingly, in our setting the stalk of $R\Psi_{\mathcal{Y}}(A)$ at the node is A in degree zero and $A(-1)$ in degree one; the twist is that of $H^1(\mathbf{G}_m)$ (see [SGA7II, Exposé XV]). At every other point of the special fibre, by contrast, the family is smooth over S, and there the nearby cycles are A in degree zero, by smooth base change and the first calculation below.

For a morphism $g : \mathcal{Y}' \to \mathcal{Y}$ of separated S-schemes of finite type, there is a base change map for pullback

$$g_s^* R\Psi_{\mathcal{Y}} \longrightarrow R\Psi_{\mathcal{Y}'} g_\eta^*,$$

which is invertible for smooth g. For proper g there is a base change map for proper direct image

$$Rg_{s*}R\Psi_{\mathcal{Y}'} \xrightarrow{\sim} R\Psi_{\mathcal{Y}}Rg_{\eta*}.$$

These base change maps commute with composition and with the specialization map defined below. Their invertibility follows from two base change theorems for torsion coefficients. First, proper base change [Stacks, 095T] says that restriction to a fibre commutes with proper direct image. Second, smooth base change [Stacks, 0F09] (for sheaves [Stacks, 0EYU]) says that pullback along a smooth map commutes with direct image along a qcqs morphism, when the coefficients are torsion of order invertible on the base. Suppose first that g is proper. The two direct images on the generic fibres agree because direct image is compatible with composition, and $\bar{\imath}^*$ commutes with the proper direct image by proper base change. Suppose next that g is smooth. Since $R\bar{\jmath}_*$ is direct image along a qcqs morphism, we may apply smooth base change to it and then restrict to the special fibre. Smooth base change applies because the coefficients are annihilated by a power of ℓ, which is invertible on S. The same base change maps appear in [SGA7II], Exposé XIII, (2.1.7.1)–(2.1.7.3).

We also use two calculations for a finite A-module M. The first is $R\Psi_S(M_\eta) = M$. It holds because the integral closure $\bar{S}$ of S in the separable closure is strictly henselian, so the stalk of $R\bar{\jmath}_*M$ at its closed point is $R\Gamma(\bar{\eta}, M) = M$. The second is the usual calculation for the affine line with coefficients prime to the characteristic: $H^r(\mathbf{A}^1_k, M) = 0$ for $r > 0$ and $H^0(\mathbf{A}^1_k, M) = M$. It follows from [Stacks, 03RR] applied to the compactification $\mathbf{A}^1 \subset \mathbf{P}^1$, which has genus zero and one point at infinity. To apply that result, note that every finite M is a finite direct sum of cyclic modules, and over the algebraically closed field k the cyclic constant coefficients are isomorphic to the corresponding roots of unity. By truncation the two calculations extend to a bounded complex $M^\bullet$ of finite A-modules, viewed as a constant complex:

$$R\Psi_S(M^\bullet_\eta) \simeq M^\bullet, \qquad R\Gamma(\mathbf{A}^1_k, M^\bullet_{\mathbf{A}^1_k}) \simeq M^\bullet.$$

Finally, all complexes stay bounded and constructible. By the finiteness theorem for nearby cycles with torsion coefficients ([SGA4½], *Théorèmes de finitude*, §3; [illusie94, §4.1, pp. 44–45]), $R\Psi_{\mathcal{Y}}$ takes bounded constructible complexes of finite Tor dimension to complexes of the same kind. Finite Tor dimension is what we need when we take traces. The same holds for an arbitrary bounded constructible complex, without the Tor condition. To see this, we may forget the A-action, since the underlying torsion sheaves of A-linear nearby cycles are the usual nearby cycles. Bounded constructible complexes are stable under extensions and truncation, so the filtration by powers of ℓ reduces us to constructible $\mathbf{F}_\ell$-sheaves. These have finite Tor dimension over $\mathbf{F}_\ell$, and the finiteness theorem applies.

For a separated k-scheme Y of finite type, the constant family $Y \times S$ has special fibre Y. For L on Y, let L_η be its pullback to the generic fibre. Restricting the adjunction unit of $\bar{\jmath}$ on the pullback of L to the special fibre, we get the specialization map $\sigma_Y(L)$ below.

Lemma E.2 (Nearby cycles of a constant family)**.** *For a separated k-scheme Y of finite type and $L \in D^b_c(Y, A)$, the specialization map*

$$\sigma_Y(L) : L \longrightarrow R\Psi_{Y\times S}(L_\eta)$$

is an isomorphism. It is natural in L, and it is compatible with the base change maps above in the following sense.

1. *If $f : Y' \to Y$ is proper and $L' \in D^b_c(Y', A)$, then the composite of $Rf_*\sigma_{Y'}(L')$ with the base change map for proper direct image is $\sigma_Y(Rf_*L')$, where $(Rf_*L')_\eta$ is identified with $Rf_{\eta *}(L'_\eta)$ by proper base change.*
2. *If $f : Y' \to Y$ is smooth, then the composite of $f^*\sigma_Y(L)$ with the base change map for pullback is $\sigma_{Y'}(f^*L)$.*

Proof. The compatibilities (1) and (2) are formal. For the isomorphism we show that the cone $C_Y(L)$ of $\sigma_Y(L)$ vanishes. To do this we use that, like nearby cycles, the cone commutes with smooth pullback and proper direct image. Smooth pullback lets us check vanishing locally, and it also shows that the cone vanishes generically, by reduction to constant coefficients and the calculation $R\Psi_S(M_\eta) = M$. Proper direct image along a compactified linear projection then shows that the remaining support is empty, by induction on the dimension of an affine space containing Y.

By the comparison for constructible complexes, we may work on the small étale site. There σ is $\bar{i}^*$ of the unit of $\bar{j}^* \dashv R\bar{j}_*$ and the base change maps are $\bar{i}^*$ of mates, so (1), (2) and naturality follow from the triangle identities.

We take cofibres in the stable ∞-category underlying the derived category, where they are functorial, unlike cones in a triangulated category, and set $C_Y(L) = \mathrm{cofib}(\sigma_Y(L))$. This is an exact functor with bounded constructible values. By (1) and (2), the base change maps yield

$$C_Y(Rf_*L') \simeq Rf_*C_{Y'}(L') \quad (f \text{ proper}), \qquad C_{Y'}(f^*L) \simeq f^*C_Y(L) \quad (f \text{ smooth}).$$

By the base change isomorphism for étale maps, the question is local on Y. For a closed immersion $Y \hookrightarrow \mathbf{A}^n$, the base change isomorphism for proper maps identifies the direct image of $C_Y(L)$ with $C_{\mathbf{A}^n}$ of the direct image of L, and since direct image along a closed immersion is conservative, it suffices to treat affine spaces.

For $Y = \mathbf{A}^n$, the complex $C_Y(L)$ vanishes on a dense open subset. Indeed, by the truncation triangles we may assume that L is a sheaf. After shrinking, it is locally constant, and after pulling back to a finite étale cover, it is constant. The pullback does no harm: C commutes with étale pullback, so vanishing can be checked after a surjective étale pullback. For a constant sheaf M on a smooth k-scheme, the base change isomorphism for its smooth structure map identifies C with the pullback of $C_{\mathrm{Spec}\, k}(M)$, which is zero because $R\Psi_S(M_\eta) = M$.

We argue by induction on n, the case of a point being the calculation above. By the preceding paragraph, the closure T of the support of $C_{\mathbf{A}^n}(L)$ is a proper closed subset. By the linear form of Noether normalization there is a linear projection $q : \mathbf{A}^n \to \mathbf{A}^{n-1}$ that is finite on T: we choose its kernel line so that it avoids the intersection of the projective closure of T with the hyperplane at infinity. Compactifying that line, we get

$$\bar{q} : \bar{Y} = \mathbf{P}^1 \times \mathbf{A}^{n-1} \longrightarrow \mathbf{A}^{n-1}, \qquad v : \mathbf{A}^n \hookrightarrow \bar{Y}, \qquad \bar{L} = v_!L.$$

We compactified the line so that the support becomes finite over $\mathbf{A}^{n-1}$ and can be detected by direct image there. As C commutes with restriction to the open subscheme $\mathbf{A}^n$ of $\bar{Y}$, the complex $C_{\bar{Y}}(\bar{L})$ is supported on $T \cup (\bar{Y} \setminus \mathbf{A}^n)$. This union is closed in $\bar{Y}$ and finite over $\mathbf{A}^{n-1}$. Indeed, T is finite, hence proper, over $\mathbf{A}^{n-1}$, and therefore closed in $\bar{Y}$; the boundary $\bar{Y} \setminus \mathbf{A}^n$ is a section. By the base change isomorphism for proper maps and the induction hypothesis,

$$R\bar{q}_*C_{\bar{Y}}(\bar{L}) \simeq C_{\mathbf{A}^{n-1}}(R\bar{q}_*\bar{L}) = 0.$$

Direct image is conservative on complexes supported on a closed subscheme finite over the base, since on a geometric stalk the direct image of each cohomology sheaf is the finite product of its stalks on the finite fibre. Therefore $C_{\bar{Y}}(\bar{L}) = 0$, and restricting to $\mathbf{A}^n$ gives $C_{\mathbf{A}^n}(L) = 0$. ∎

The contracting argument needs a complex on the normal cone $N_Z X$ that still remembers K near Z. Pulling $K|_Z$ back along the projection $N_Z X \to Z$ is not enough, since it forgets the directions leaving Z. For example, take $X = \mathbf{A}^1$, Z the origin, and $K = j_! A$ for $j : \mathbf{A}^1 \setminus \{0\} \hookrightarrow \mathbf{A}^1$. Then $K|_Z = 0$, so the pullback is zero. By contrast, the deformation to the normal cone is $\operatorname{Spec} k[x, t, x/t]$ over the trait, that is, the constant family $\mathbf{A}^1 \times S$ in the coordinate $x' = x/t$, and on its generic fibre K is $j_! A$ for $x' \neq 0$. Hence, by Lemma E.2, the nearby cycles are $j_! A$ on $N_Z X = \mathbf{A}^1$, which is not zero. This construction is the specialization functor $\operatorname{Sp}_{X,Z}$ defined below, and by Verdier's theorem (Theorem E.8) it still recovers $K|_Z$ on the zero section. Theorem E.30 uses both properties.

Let X be a separated k-scheme of finite type, and let $i : Z \hookrightarrow X$ be a closed subscheme with ideal I. The deformation to its normal cone, over the chosen trait, is

$$\mathcal{D}_Z X = \operatorname{Spec}_{X \times S}(\mathcal{O}_{X \times S}[I/t]).$$

Here $\mathcal{O}_{X \times S}[I/t]$ is the base change to S of the subalgebra of $\mathcal{O}_X[t, t^{-1}]$ generated by $\mathcal{O}_X[t]$ and the fractions a/t with $a \in I$. Its generic fibre is X_η and its special fibre is

$$N_Z X = \operatorname{Spec}_Z \left(\bigoplus_{r \geq 0} I^r / I^{r+1} \right).$$

The quotient by I and all a/t defines a closed immersion $\tilde{z} : Z \times S \hookrightarrow \mathcal{D}_Z X$, whose special fibre $z : Z \hookrightarrow N_Z X$ is the zero section. For L on X, let L_η be its pullback to the generic fibre X_η of $\mathcal{D}_Z X$, and define the specialization functor $\operatorname{Sp}_{X,Z} : D_c^b(X_{\acute{e}t}, A) \to D_c^b((N_Z X)_{\acute{e}t}, A)$ by

$$\operatorname{Sp}_{X,Z}(L) = R\Psi_{\mathcal{D}_Z X}(L_\eta).$$

The base change map for pullback along $\tilde{z}$, followed by the inverse of the isomorphism of Lemma E.2, defines a natural transformation

$$\tau_{X,Z}(L) : z^* \operatorname{Sp}_{X,Z}(L) \longrightarrow R\Psi_{Z \times S}((i^* L)_\eta) \xrightarrow{\sim} i^* L. \tag{E.1}$$

We use this map, in this direction, to specialize trace classes.

Example E.3 (Normal cones of a point)**.** If X is smooth at a point $Z = \{y\}$, then $\bigoplus_r I^r/I^{r+1}$ is the symmetric algebra of I/I^2 and $N_Z X$ is the tangent space at y. At a singular point, however, the normal cone is the tangent cone, and it need not be reduced. For example, for the cusp $X = V(y^2 - x^3) \subset \mathbf{A}^2$ and Z the origin, with $I = (x, y)$, the graded algebra is $k[\xi, \eta]/(\eta^2)$, where ξ, η are the classes of x, y in I/I^2 and η^2 is the lowest-degree part of $y^2 - x^3$; so $N_Z X$ is a double line. Similarly, for a nonreduced point $Z = V(x^m) \subset \mathbf{A}^1$, each $I^r/I^{r+1} = (x^{mr})/(x^{m(r+1)})$ is free of rank one over $\mathcal{O}_Z = k[x]/(x^m)$, and $N_Z X = Z \times \mathbf{A}^1$ is a line thickened along Z; this is the case treated in Lemma E.6.

We will apply the base change maps for nearby cycles to the deformations of different pairs, and for this we need to know how the deformation depends on the pair.

Lemma E.4 (Functoriality of the deformation)**.** *Let $f : X' \to X$ be a morphism of separated k-schemes of finite type, let $Z' \subset X'$ be a closed subscheme with ideal I', and suppose that $f^{-1}(I)\mathcal{O}_{X'} \subset I'$. Then f induces $\widetilde{f} : \mathcal{D}_{Z'}X' \to \mathcal{D}_Z X$. If $Z' = f^{-1}Z$ schematically, then*

$$\mathcal{D}_{Z'}X' \longrightarrow X' \times_X \mathcal{D}_Z X$$

is a closed immersion, and it is an isomorphism if f is flat. Moreover, $\widetilde{f}_s^{-1}(Z) = Z'$ schematically. In this case $\widetilde{f}$ is proper when f is proper and smooth when f is smooth.

Proof. All assertions are statements about the algebras $\mathcal{O}[I/t]$, and we check them on generators. The first morphism is defined by $a/t \mapsto f^{\#}(a)/t$, which makes sense because $f^{\#}(a) \in I'$ for $a \in I$. If $I' = I\mathcal{O}_{X'}$, the natural map from the base change of the Rees algebra onto $\mathcal{O}_{X' \times S}[I'/t]$ is surjective, and it is injective if f is flat. On the associated graded algebras, the generators of gr_I of positive degree generate the ideal of elements of positive degree of $\mathrm{gr}_{I'}$. The quotient by them is $\mathcal{O}_{Z'}$, and this proves the assertion about the zero section. If f is proper, then $\widetilde{f}$ is the closed immersion followed by the base change of f, which is proper. If f is smooth, it is flat, so $\widetilde{f}$ is the base change $X' \times_X \mathcal{D}_Z X \to \mathcal{D}_Z X$ of f, which is smooth. ∎

Theorem E.8 asserts that $\tau_{X,Z}$ is an isomorphism, that is, that the cofibre

$$C_{X,Z}(L) = \mathrm{cofib}(\tau_{X,Z}(L))$$

vanishes. As for C_Y, we compute the cofibre in the stable ∞-category, so that $C_{X,Z}$ is an exact functor. The proof has three steps. First, the base change maps of the next lemma reduce us to the case $i^*L = 0$ and, after a blow-up, to the case of an effective Cartier divisor. Then, for a principal divisor in a normal scheme, an explicit contraction of the affine line shows that $C_{X,Z}(L)$ vanishes at the generic points of Z. Finally, after an embedding into affine space, projection along a line and induction on the dimension show that the remaining support is empty.

Lemma E.5 (Base change for the cone of $\tau_{X,Z}$)**.** *Let $f : X' \to X$ be a morphism of separated k-schemes of finite type, let $Z' = f^{-1}Z$ schematically, and let $f_Z : Z' \to Z$ be the restriction of f. There are isomorphisms*

$$\begin{aligned} C_{X,Z}(Rf_*L') &\simeq Rf_{Z*}C_{X',Z'}(L') \quad (f \text{ proper}), \\ C_{X',Z'}(f^*L) &\simeq f_Z^* C_{X,Z}(L) \qquad (f \text{ smooth}). \end{aligned}$$

Moreover, $C_{X,Z}(i_*M) = 0$ for every $M \in D_c^b(Z_{\acute{e}t}, A)$. Consequently, writing $j : X \setminus Z \hookrightarrow X$, we have

$$C_{X,Z}(j_!j^*L) \xrightarrow{\sim} C_{X,Z}(L).$$

Proof. Both the source and the target of τ are compatible with proper direct image and smooth pullback, so the cone is as well. To prove the vanishing on i_*M, we reduce to the pair (Z, Z), where the deformation is a constant family.

By Lemma E.4, f induces a morphism $\widetilde{f}$ of deformations, which is proper if f is proper and smooth if f is smooth. To this morphism we apply the base change maps of nearby cycles, together with proper or smooth base change for the restriction to the zero section. The resulting squares for τ then commute because the base change map for pullback is compatible with composition, and their cofibres give the two isomorphisms. In the special case $f = i$, the source

pair is (Z, Z) and $\mathcal{D}_Z Z = Z \times S$, so $\tau_{Z,Z}$ is the inverse of the isomorphism of Lemma E.2. Hence $C_{Z,Z} = 0$, and since i is proper, the first isomorphism gives $C_{X,Z}(i_* M) = 0$. The last assertion then follows by applying the exact functor $C_{X,Z}$ to $j_! j^* L \to L \to i_* i^* L$. ∎

To prove the vanishing at the generic points of Z, we first treat a nonreduced point on the affine line, where an explicit contraction is available.

Lemma E.6 (The case $Z = V(x^m) \subset \mathbf{A}^1$)**.** *For $X = \mathbf{A}^1_k$, $Z = V(x^m)$ with $m \geq 1$, and every finite A-module M, one has $C_{X,Z}(M_X) = 0$.*

Proof. We give a homotopy argument. The contraction $x \mapsto ax$ joins the identity ($a = 1$) to a map that factors through Z ($a = 0$), and the endomorphisms of $z^* \operatorname{Sp}_{X,Z}(M_X)$ that these maps induce form a family over $\mathbf{A}^1$. Since $\mathbf{A}^1$ has the cohomology of a point, the family is constant, and comparing its two ends shows that $\tau_{X,Z}(M_X)$ has an inverse.

Consider the contraction $h : X \times \mathbf{A}^1 \to X$, $h(x, a) = ax$, and write $h_a = h(-, a) : X \to X$ for $a \in k$. It maps $Z \times \mathbf{A}^1$ schematically into Z, since $h^*(x^m) = a^m x^m \in (x^m)$, and by Lemma E.4, whose hypothesis is this inclusion of ideals, all h_a lift to the deformations. Moreover, $h_1 = \mathrm{id}$, and h_0 factors through Z and restricts to the identity on Z_{red}. Set $N = z^* \operatorname{Sp}_{X,Z}(M_X)$. The base change maps for pullback along these lifts induce endomorphisms $\phi_a : N \to N$, with $\phi_1 = \mathrm{id}_N$.

There is one endomorphism ϕ of the pullback of N to $Z \times \mathbf{A}^1$ whose fibre at a is ϕ_a. To construct it, we apply the base change map for pullback along $h : X \times \mathbf{A}^1 \to X$ and then invert the base change map for the projection to X, which is smooth; by flat base change, the deformation of $X \times \mathbf{A}^1$ is $\mathcal{D}_Z X \times \mathbf{A}^1$. The restrictions of h and of the projection to $Z \times \mathbf{A}^1$ induce the same map to Z on reduced schemes, so invariance under nilpotent thickenings identifies $h_Z^* N$ with $\mathrm{pr}_Z^* N$, naturally in N, and we may view the family as an endomorphism of a single pullback. By compatibility with composition, its fibres are the endomorphisms ϕ_a constructed above.

The family ϕ is constant; this is the algebraic form of homotopy invariance. Since Z_{red} is a point, invariance under nilpotent thickenings identifies $D_c^b(Z_{\acute{e}t}, A)$ with the bounded derived category of finite A-modules. Pullback along the projection $p : Z \times \mathbf{A}^1 \to Z$ is fully faithful on this category. Indeed, the calculation for the affine line and the hypercohomology spectral sequence show that $R\Gamma(\mathbf{A}^1, p^* B) = B$ for every such bounded complex B, and full faithfulness follows by adjunction. Thus ϕ is the pullback of a single endomorphism of N. Its two restrictions ϕ_0 and ϕ_1 therefore coincide, and $\phi_0 = \mathrm{id}_N$.

Because h_0 factors through Z, we can also express ϕ_0 in terms of $\tau_{X,Z}(M_X)$. Let $\alpha : M_Z \to N$ be the restriction of the unit of nearby cycles for the constant sheaf on $\mathcal{D}_Z X$. By naturality, and since h_0 factors as $X \to Z \hookrightarrow X$, we get

$$\phi_0 = \alpha \tau_{X,Z}(M_X), \qquad \tau_{X,Z}(M_X)\alpha = \mathrm{id}_{M_Z}\,.$$

The second equality holds because the composite is the specialization map for the constant family $Z \times S$, followed by its fixed inverse. Since $\phi_0 = \phi_1 = \mathrm{id}_N$, the maps α and $\tau_{X,Z}(M_X)$ are inverse isomorphisms. ∎

Near a generic point of a principal divisor in a normal scheme, the pair (X, Z) is a smooth pullback of $(\mathbf{A}^1, V(x^m))$, so by Lemma E.6 $C_{X,Z}$ vanishes there on constant coefficients. The next lemma extends this to all sheaves that vanish on Z.

Lemma E.7 (Vanishing at the generic points of a principal divisor)**.** *Suppose that X is normal, that $Z \subset X$ is an effective principal Cartier divisor, and that $i^* L = 0$. Then $C_{X,Z}(L)$ vanishes at every generic point of Z.*

Proof. For a constant sheaf extended by zero, a smooth map $X \to \mathbf{A}^1$ near the generic point reduces the claim to Lemma E.6. Near the generic point, a general sheaf is $j_!G$ with G locally constant. After a finite étale cover, G becomes constant, while the cokernel of the unit of the cover is again of the form $j_!G'$. Passing to this cokernel shifts the cohomological degree by one, so an induction on the degree proves the claim for all G at once.

Since the question is local at a generic point ζ of Z, we may discard the other irreducible components of X and Z. Consider first $L = j_!M_U$, where $U = X \setminus Z$ and M is a finite A-module. Since X is normal, it is regular in codimension one. After shrinking around ζ, both X and Z_{red} are smooth over k, and a local equation h of Z_{red} defines a smooth map $h : X \to \mathbf{A}^1$. The local ring $\mathcal{O}_{X,\zeta}$ is a discrete valuation ring with uniformizer h, so a local equation g of Z can be written $g = wh^m$ with $m \geq 1$ and w a unit of $\mathcal{O}_{X,\zeta}$. The function w is regular and invertible on a neighbourhood of ζ, and there the ideal of Z is (h^m), that is, $Z = h^{-1}(V(x^m))$ schematically. By the base change isomorphism for smooth maps and Lemma E.6, we see that $C_{X,Z}(M_X) = 0$ there, and the last assertion of Lemma E.5 yields the same vanishing for $j_!M_U$.

For general L, truncation triangles reduce us to a constructible sheaf F with $F|_Z = 0$. Shrinking around ζ, we may assume that $F = j_!G$ for a finite locally constant sheaf G on U, since the closure in X of the locus in U where F is not locally constant cannot contain a generic point of the divisor Z. We prove by induction on r, simultaneously for all such normal pairs and all G, that $H^r(C_{X,Z}(j_!G))$ vanishes generically. As nearby cycles are left t-exact and $i^*j_!G = 0$, we have

$$C_{X,Z}(j_!G) = z^* \operatorname{Sp}_{X,Z}(j_!G)[1] \in D^{\geq -1}.$$

Therefore the statement holds for every $r < -1$, which starts the induction.

For the induction step, we compare $F = j_!G$ with a sheaf to which the first part of the proof applies. To construct it, choose a finite étale surjection $f_U : U' \to U$ trivializing G, and let $f : X' \to X$ be the normalization of X in U'; it is finite since X is excellent, and X' is normal. With $Z' = f^{-1}Z$ and $j' : U' \hookrightarrow X'$, the sheaf $f^*F = j'_!f_U^*G$ is $j'_!$ of a sheaf that is constant on each component of U', and Z' is a principal divisor in the normal scheme X'. Since f is finite, the points of Z' over a generic point of Z are generic points of Z', so the first part of the proof, applied to (X', Z'), and the base change isomorphism for proper maps show that $C_{X,Z}(f_*f^*F)$ vanishes generically on Z.

On a geometric stalk, the unit of the finite étale cover is the diagonal inclusion into a nonempty finite product, so it is injective and there is a short exact sequence

$$0 \longrightarrow F \longrightarrow f_*f^*F \longrightarrow j_!G' \longrightarrow 0,$$

where G' is again finite locally constant on U. Applying $C_{X,Z}$ and taking the long exact sequence at a generic point where the middle term vanishes, we find

$$H^r(C_{X,Z}(F)) \simeq H^{r-1}(C_{X,Z}(j_!G')).$$

By the induction hypothesis, the right side vanishes. ∎

On $D^b_{\mathrm{ctf}}(X, A)$ we define the functor $\operatorname{Sp}_{X,Z}$ and the map (E.1) by transport of the étale constructions above along the comparison for constructible complexes with finite coefficients (Theorem 2.43).

Theorem E.8 (Verdier's theorem on specialization to the zero section)**.** *For every separated k-scheme X of finite type, every closed subscheme $Z \subset X$, and $K \in D^b_{\mathrm{ctf}}(X, A)$, the map (E.1)*

$$\tau_{X,Z}(K) : z^* \operatorname{Sp}_{X,Z}(K) \longrightarrow i^* K$$

is an isomorphism. In particular, $i^*K = 0$ implies $z^* \operatorname{Sp}_{X,Z}(K) = 0$.

Proof. The main step is that $C_{X,Z}(L) = 0$ for every $L \in D^b_c(X_{\acute{e}t}, A)$, which the last paragraph transports to D^b_{ctf}. We need the vanishing for all bounded constructible complexes, because the cokernels $j_!G'$ in the dimension shifting of Lemma E.7 may have infinite Tor dimension.

By Lemma E.5, we may replace L by $j_!j^*L$ and so assume that $i^*L = 0$. We may then replace X by its blow-up $\pi : X' = \operatorname{Bl}_Z X \to X$ along Z. The reason is that a complex that vanishes on the centre is recovered from its pullback to the blow-up, that is, the unit $L \to R\pi_*\pi^*L$ is an isomorphism. Indeed, away from Z this holds because π is an isomorphism there, and on Z it holds because proper base change identifies the target with the cohomology of a fibre with zero coefficients. Using this, the base change isomorphism for proper maps gives $C_{X,Z}(L) \simeq C_{X,Z}(R\pi_*\pi^*L) \simeq R\pi_{Z*}C_{X',\pi^{-1}Z}(\pi^*L)$, and it suffices to treat the pair $(X', \pi^{-1}Z)$. Since the pullback of the ideal is invertible on the blow-up, we are reduced to the case of a Cartier divisor.

To have linear projections available, we move the pair into affine space. By the base change isomorphism for open immersions the assertion is local, so we may assume that X is affine and $Z = V(h)$ for a function h. Choosing algebra generators for $\mathcal{O}(X)$ and appending h as the last coordinate gives a closed immersion

$$\iota : X \hookrightarrow \mathbf{A}^n, \qquad Z = \iota^{-1}(\mathbf{A}^{n-1} \times \{0\})$$

schematically. The base change isomorphism for proper maps identifies $R\iota_{Z*}C_{X,Z}(L)$ with $C_{\mathbf{A}^n,\mathbf{A}^{n-1}\times\{0\}}(R\iota_*L)$, and since direct image along a closed immersion is conservative, it suffices to treat the pair $(\mathbf{A}^n, \mathbf{A}^{n-1} \times \{0\})$. This is a principal divisor in a normal scheme, so Lemma E.7 applies to it.

We argue by induction on $n \geq 1$, again assuming that $L|_Z = 0$. By Lemma E.7, the closure Y of the support of $C_{X,Z}(L)$ is a proper closed subset of the hyperplane Z; for $n = 1$ it is empty, and the assertion follows. For $n > 1$, choose a line λ in the vector space Z such that the linear projection

$$q : X \longrightarrow X' = X/\lambda \simeq \mathbf{A}^{n-1}$$

is finite on Y. Such a line exists: if the projective point $\mathbf{P}(\lambda)$ lies outside the part at infinity of the projective closure of Y in $\mathbf{P}(Z \oplus k)$, the usual proof of linear Noether normalization shows that the projection is finite on Y. Set $Z' = Z/\lambda$, a coordinate hyperplane in X', and compactify the λ-direction:

$$\bar{q} : \bar{X} = \mathbf{P}^1 \times X' \longrightarrow X', \qquad \bar{Z} = \bar{q}^{-1}(Z'), \qquad v : X \hookrightarrow \bar{X}, \qquad \bar{L} = v_!L.$$

Let $\bar{q}_Z : \bar{Z} \to Z'$ be the restriction of $\bar{q}$. By the base change isomorphism for proper maps and the induction hypothesis,

$$R\bar{q}_{Z*}C_{\bar{X},\bar{Z}}(\bar{L}) \simeq C_{X',Z'}(R\bar{q}_*\bar{L}) = 0.$$

Since the cone commutes with restriction to the open subscheme X of $\bar{X}$, the complex $C_{\bar{X},\bar{Z}}(\bar{L})$ is supported on $Y \cup (\bar{Z} \setminus Z)$. This is a closed subset finite over Z'. Indeed, Y is finite over Z' and therefore remains closed in $\bar{Z}$, and $\bar{Z} \setminus Z$ is the section at infinity of $\bar{Z} \to Z'$. As in Lemma E.2, direct image along a finite map is conservative on complexes with this support. Hence $C_{\bar{X},\bar{Z}}(\bar{L}) = 0$, and restriction to X gives $C_{X,Z}(L) = 0$ on the étale site.

By the comparison for finite coefficients, K and its maps descend to the classical étale constructible category, where $C_{X,Z}$ vanishes. The comparison is exact and fully faithful, and it identifies pullback, proper direct image and their adjunction units, so the étale map (E.1) corresponds to the map in the statement, which is therefore an isomorphism. ∎

E.2 Künneth formula for nearby cycles

To specialize a trace class we have to specialize the external product $K \boxtimes \mathbb{D}K$, so we need the comparison map from the external product of nearby cycles to the nearby cycles of the external product to be an isomorphism. We prove this on the classical étale site by the dimension argument of [Ill94, Theorem 4.7, pp. 51–53].

In this section and the next three, S is an excellent strictly henselian trait, as announced at the start of this appendix, with separably closed residue field k, generic point η, and a chosen geometric generic point $\bar{\eta}$. Let p be the characteristic exponent of k. The coefficients lie in a finite commutative ℓ-primary ring A, with ℓ invertible on S. On the geometric fibres we use the ordinary external Künneth formula, Proposition 8.18. It applies over the separably closed field k, since passing to an algebraic closure is a purely inseparable extension and does not change the étale topos.

We use two further facts about traits, both from [SGA4½], *Théorèmes de finitude*, §§3.4 and 3.7. First, geometric nearby cycles are invariant under extension of traits, with the corresponding restriction of the inertia action (loc. cit., 3.7). Second, the calculation in loc. cit., 3.4, describes them by restriction to the generic point of a coordinate line, as follows. Let s' be a geometric generic point of the special fibre $\mathbf{A}^1_k$ of $\mathbf{A}^1_S$, and let $S' = (\mathbf{A}^1_S)_{(s')}$ be the strict localization of $\mathbf{A}^1_S$ at s'. This is a strictly henselian trait with the same uniformizer as S. With compatible geometric generic points, its inertia group fits into an exact sequence

$$1 \longrightarrow P \longrightarrow I' \longrightarrow I \longrightarrow 1,$$

where P is pro-p (and is trivial when $p = 1$). For $a : W \to \mathbf{A}^1_S$, put $W' = W \times_{\mathbf{A}^1_S} S'$, and let N' be the pullback of $N \in D^b_{\mathrm{ctf}}(W_\eta, A)$. There is a canonical identification

$$(R\Psi_{W/S} N)|_{W_s \times_{\mathbf{A}^1_k} s'} \simeq (R\Psi_{W'/S'} N')^P. \tag{E.2}$$

Here $(-)^P = R\Gamma(P, -)$ is exact on the discrete torsion coefficients prime to p under consideration. The formula comes from factoring the map from the geometric generic fibre through the geometric generic point of S', which lies over $\bar{\eta}$. The Galois group of the second step is P. It is pro-p because the common uniformizer identifies the tame inertia quotients. The formula is compatible with the base change maps for pullback and with the product maps obtained from $\bar{i}^* R\bar{j}_*$. To see this, we factor the maps of generic fibres in the same way and use that the adjunctions for direct image are compatible with composition.

The functor of P-invariants commutes with tensoring by an object on which P acts trivially: for a complex Q with continuous P-action and a complex B with trivial P-action, the projection map is an isomorphism

$$(Q \otimes^{\mathbf{L}}_{A} B)^{P} \simeq Q^{P} \otimes^{\mathbf{L}}_{A} B. \tag{E.3}$$

Indeed, P-invariants are exact on discrete torsion A-modules. For a finite quotient G of P, the invariants are the image of the averaging idempotent $|G|^{-1}\sum_{g\in G} g$, since the p-power $|G|$ is invertible in A, and this idempotent commutes with tensoring by B. By continuity we pass from the finite quotients to P, and resolving B by flat A-modules gives the derived identity.

For X, Y separated of finite type over S, L on X_η and M on Y_η, the comparison map $\Pi_{L,M}$ below is the restriction to the special fibre of the map

$$R\bar{j}_{X*}L_{\bar\eta} \boxtimes R\bar{j}_{Y*}M_{\bar\eta} \longrightarrow R\bar{j}_{X\times Y,*}(L_{\bar\eta} \boxtimes M_{\bar\eta})$$

defined by the base change maps for pullback and the lax monoidal structure of direct image.

Proposition E.9 (External Künneth formula for nearby cycles)**.** *Let X, Y be separated of finite type over S, and let $L \in D^b_{\mathrm{ctf}}(X_\eta, A)$ and $M \in D^b_{\mathrm{ctf}}(Y_\eta, A)$. The comparison map*

$$\Pi_{L,M} : R\Psi_{X/S}L \boxtimes R\Psi_{Y/S}M \longrightarrow R\Psi_{(X\times_S Y)/S}(L \boxtimes M) \tag{E.4}$$

is an isomorphism.

The proof has the same shape as that of Lemma E.2: the cone of $\Pi_{L,M}$ has no global sections and finite support, so it vanishes. When X and Y are projective, the ordinary Künneth formula on the geometric generic fibres gives zero global sections, once we know how Π behaves under proper direct image. Restricting to the generic point of a coordinate line by (E.2) lowers the dimension of the generic fibre, and induction on the dimension shows that the support is finite. The next four lemmas supply these ingredients.

Lemma E.10 (The Künneth map and proper direct image)**.** *Let $f : X \to X_1$ and $g : Y \to Y_1$ be proper morphisms of separated S-schemes of finite type, and let $L \in D^b_{\mathrm{ctf}}(X_\eta, A)$ and $M \in D^b_{\mathrm{ctf}}(Y_\eta, A)$. Identify direct images of external products by the ordinary Künneth formula, and direct images of nearby cycles by the base change isomorphism for proper maps. Then $R(f_s \times g_s)_*\Pi_{L,M}$ becomes $\Pi_{Rf_{\eta*}L, Rg_{\eta*}M}$.*

Proof. Write the two product maps in terms of the tensor maps of $R\bar{j}_*$, commute proper direct image past these maps on the generic fibres, and take the adjoints of the ordinary Künneth arrows. Both adjoints are $f^*_\eta Rf_{\eta*}L \boxtimes g^*_\eta Rg_{\eta*}M \to L \boxtimes M$, the tensor product of the two counits. By the triangle identities of the adjunctions and their compatibility with composition, the maps themselves agree. ∎

A complex supported on finitely many closed points is determined by its stalks at these points, and we can read off these stalks from global sections.

Lemma E.11 (Complexes with finite support)**.** *Let Y be a scheme of finite type over a separably closed field, and let $Q \in D^b_c(Y_{\acute{e}t}, A)$ be supported on a finite set of closed points. If $R\Gamma(Y, Q) = 0$, then $Q = 0$.*

Proof. Let $i_T : T \hookrightarrow Y$ be the finite closed reduced subscheme with underlying set the support of Q, so that $Q \simeq i_{T*}i_T^*Q$. Since the residue fields of the points of T are finite and purely inseparable over the separably closed base field, T is étale locally a finite disjoint union of geometric points, and $R\Gamma(Y, Q) = \bigoplus_{t\in T} Q_{\bar t}$. Every stalk of Q therefore vanishes, and $Q = 0$. ∎

To restrict to coordinate lines we need coordinates in which the special fibre is quasi-finite over an affine space of the right dimension.

Lemma E.12 (Quasi-finite coordinates)**.** *Let X be flat, separated and of finite type over S, and let x be a closed point of X_s. Put $q = \dim_x X_s$. Then $q \leq \dim X_{\bar\eta}$, and there are an open neighbourhood V of x in X and a quasi-finite morphism $b : V \to \mathbf{A}^q_S$. In particular $\dim V_{\bar\eta} \leq q$.*

Proof. The inequality $q \leq \dim X_{\bar\eta}$ holds by flatness. Choose a system of parameters $f_1, \dots, f_q$ of the local ring $\mathcal{O}_{X_s,x}$ of the special fibre, defined on an affine neighbourhood, and lift them to functions on the corresponding affine open of X. The resulting map $b : X \to \mathbf{A}^q_S$ is quasi-finite at x, because its fibre through x is the fibre of $b_s = (f_1, \dots, f_q)$, in which x is isolated. Let V be the open locus where b is quasi-finite; as b_η is quasi-finite on V_η, the generic fibre of V has dimension at most q. ∎

In the slicing step we compare Π with the Künneth map over the trait S' of (E.2). Let X, Y be separated of finite type over S, let $b : X \to \mathbf{A}^1_S$ be a morphism, and put

$$X' = X \times_{\mathbf{A}^1_S} S', \qquad Y' = Y \times_S S', \qquad L' = L|_{X'_{\eta'}}, \quad M' = M|_{Y'_{\eta'}}.$$

Lemma E.13 (The Künneth map on a slice)**.** *In the situation just described, let $L \in D^b_{\mathrm{ctf}}(X_\eta, A)$ and $M \in D^b_{\mathrm{ctf}}(Y_\eta, A)$. Identify the restrictions over s' of the nearby cycles of L, M and $L \boxtimes M$ by*

$$\begin{aligned} R\Psi_{X/S} L|_{s'} &\simeq (R\Psi_{X'/S'} L')^P, \\ R\Psi_{Y/S} M|_{s'} &\simeq R\Psi_{Y'/S'} M', \\ R\Psi_{(X\times Y)/S}(L \boxtimes M)|_{s'} &\simeq (R\Psi_{(X'\times Y')/S'}(L' \boxtimes M'))^P, \end{aligned}$$

given by (E.2) for b, by invariance of nearby cycles under change of trait, and by (E.2) for $b \circ \mathrm{pr}_1$. Then the restriction of $\Pi_{L,M}$ over s' is $(\Pi_{L',M'})^P$. In particular, if $\Pi_{L',M'}$ is an isomorphism, then so is the restriction of $\Pi_{L,M}$ over s'.

Proof. In the middle line P acts trivially, because the action on this complex of nearby cycles is the restriction of an action of I and P is the kernel of $I' \to I$. By (E.3), therefore, $(-)^P$ of the source of $\Pi_{L',M'}$ is $(R\Psi_{X'/S'} L')^P \boxtimes R\Psi_{Y'/S'} M'$, the restriction of the source of $\Pi_{L,M}$. By the compatibility of (E.2) with the product maps obtained from $\bar i^* R\bar j_*$, the map $(\Pi_{L',M'})^P$ is the restriction of $\Pi_{L,M}$. ∎

Proof of Proposition E.9. We argue by induction on $n = \dim(X_{\bar\eta} \times_{\bar\eta} Y_{\bar\eta})$, simultaneously for all such traits. Before slicing we need two reductions: global sections require proper families, and bounding fibre dimensions after slicing requires flatness, so we arrange both. For properness, note first that the assertion is local on $X_s \times_k Y_s$. We may therefore pass to affine neighbourhoods of the two factors, take projective closures over S and extend the coefficients by zero on their generic fibres. By the base change isomorphism for nearby cycles along open immersions, the restriction of the new map is the original map, so we may assume that X and Y are projective. For flatness, the base change isomorphism for the closed immersions of the closures of their generic fibres, together with invariance under nilpotent thickenings, lets us also assume that they are reduced with dense generic fibres, hence flat over the trait. Moreover, none of these reductions increases n.

Let Q be the cone of (E.4). For proper families, global sections of nearby cycles on the special fibre are cohomology of the geometric generic fibre, so by the base change isomorphism for the

proper structure maps and Lemma E.10, the map of global complexes induced by $\Pi_{L,M}$ is the ordinary Künneth isomorphism

$$R\Gamma(X_{\bar\eta}, L) \otimes_A^{\mathbf{L}} R\Gamma(Y_{\bar\eta}, M) \xrightarrow{\sim} R\Gamma(X_{\bar\eta} \times Y_{\bar\eta}, L \boxtimes M),$$

and therefore

$$R\Gamma(X_s \times_k Y_s, Q) = 0. \tag{E.5}$$

By Lemma E.11, $Q = 0$ as soon as Q has finite support. When $n = 0$, the special fibres are zero-dimensional by flatness, so the support is finite. Let $n > 0$. By Lemma E.12, $X_s \times_k Y_s$ is covered by finitely many products $V_s \times W_s$ of open subschemes with quasi-finite maps $b : V \to \mathbf{A}^q_S$ and $c : W \to \mathbf{A}^r_S$, where $q + r \leq n$ and $\dim V_{\bar\eta} \leq q$, $\dim W_{\bar\eta} \leq r$. The restriction of Q to $V_s \times W_s$ is the cone of $\Pi_{L|_V, M|_W}$. If $q + r < n$, it vanishes by the induction hypothesis. By symmetry we may therefore assume $q + r = n$ and $q > 0$, and compose b with one of the coordinate projections $\mathbf{A}^q_S \to \mathbf{A}^1_S$. In the notation of Lemma E.13 for V, W and this map, the map $V' \to \mathbf{A}^q_S \times_{\mathbf{A}^1_S} S' = \mathbf{A}^{q-1}_{S'}$ is a base change of b, hence quasi-finite, and therefore

$$\dim V'_{\bar\eta'} \leq q - 1, \qquad \dim W'_{\bar\eta'} \leq r.$$

By the induction hypothesis $\Pi_{L',M'}$ is an isomorphism. By Lemma E.13, the restriction of Q over s' is therefore zero. In other words, the support of Q has no point over the generic point of this coordinate line in the special fibre. Its image in that line is constructible and does not contain the generic point, so it is a finite set of closed points.

We repeat this argument for each of the q coordinates of b. After interchanging the factors, we do the same for each of the r coordinates of c; the analogue of Lemma E.13 with the roles of X and Y exchanged holds by symmetry of $\boxtimes$. The support of Q on $V_s \times W_s$ is then contained in the inverse image under $b_s \times c_s$ of a finite set. This inverse image is finite, because $b_s \times c_s$ is quasi-finite. With finite support, (E.5) and Lemma E.11 show that $Q = 0$. ∎

E.3 Base change for nearby cycles

Let c be a correspondence over the trait S and u a cohomological correspondence on its generic fibre. Nearby cycles turn u into a map between Ψ of $c_2^* K$ and Ψ of $c_1^! K$. To obtain a cohomological correspondence u_s on the special fibre, we must move Ψ past c_2^* and $c_1^!$. This needs base change maps between nearby cycles and the operations f^*, Rf_*, $Rf_!$ and $f^!$. We make these maps explicit here. We also prove the identities between them that Proposition E.22 uses to show that the trace of u_s is the specialization of the trace of u.

We write $\Psi_X = R\Psi_{X/S}$ for a separated S-scheme X of finite type. For a morphism $f : V \to W$ of separated S-schemes of finite type, put

$$\begin{aligned} \alpha_f : f_s^* \Psi_W &\longrightarrow \Psi_V f_\eta^*, \\ \chi_f : \Psi_W R f_{\eta *} &\longrightarrow R f_{s*} \Psi_V, \\ \nu_f : R f_{s!} \Psi_V &\longrightarrow \Psi_W R f_{\eta !}, \\ \beta_f : \Psi_V f_\eta^! &\longrightarrow f_s^! \Psi_W. \end{aligned}$$

The map α_f is the base change map for pullback of $\bar i^* R\bar j_*$, and χ_f is its adjoint mate for $f_s^* \dashv Rf_{s*}$ and $f_\eta^* \dashv Rf_{\eta *}$.

To define ν_f, we choose a relative Nagata compactification $f = ph$, with $h : V \hookrightarrow \overline{V}$ open and p proper. We define $\nu_h : h_{s!}\Psi_V M \to \Psi_{\overline{V}} h_{\eta!} M$ using the adjunction for the open immersion, by requiring that its adjoint be

$$\Psi_V M = \Psi_V h_\eta^* h_{\eta!} M \xrightarrow{\alpha_h^{-1}} h_s^* \Psi_{\overline{V}} h_{\eta!} M.$$

Since p is proper, χ_p is invertible, and we put $\nu_f = \chi_p^{-1} Rp_{s*}(\nu_h)$, using $Rf_{s!} = Rp_{s*}h_{s!}$ and $Rf_{\eta!} = Rp_{\eta*}h_{\eta!}$. The map ν_f does not depend on the compactification. Indeed, if q is a common proper refinement of two compactifications, proper base change identifies $Rq_* h'_!$ with $h_!$ on both fibres. After taking adjoints for the open immersions, compatibility with this identification reduces to the composition rule for α and the triangle identity of the ordinary adjunction. Common refinements also show that ν is compatible with composition.

We can also describe ν_f on the total families. Write i_V and j_V for the inclusions $\bar{i}$ and $\bar{j}$ of the special and geometric generic fibres of V, so that $\Psi_V = i_V^* R j_{V*}$. By proper base change, $i_W^* Rf_! \simeq Rf_{s!} i_V^*$ and $j_W^* Rf_! \simeq Rf_{\eta!} j_V^*$. Consider the map $Rf_! Rj_{V*} \to Rj_{W*} Rf_{\eta!}$ whose j_W^*-adjoint is

$$j_W^* Rf_! Rj_{V*} \simeq Rf_{\eta!} j_V^* Rj_{V*} \xrightarrow{Rf_{\eta!}\varepsilon_{j_V}} Rf_{\eta!}$$

Its restriction by i_W^* is a map $Rf_{s!}\Psi_V \to \Psi_W Rf_{\eta!}$, and we claim that this map is ν_f. As both constructions are compatible with composition, it suffices to check this for p and for h. For proper p the restricted map is χ_p^{-1}, by proper base change for i_W^*. For the open immersion h we have $h_\eta^* h_{\eta!} = 1$, and the h_s^*-adjoint of the restricted map is α_h^{-1}, as in the definition of ν_h.

The map $\beta_f(M)$ is the adjoint, for $Rf_{s!} \dashv f_s^!$, of

$$Rf_{s!}\Psi_V f_\eta^! M \xrightarrow{\nu_f} \Psi_W Rf_{\eta!} f_\eta^! M \xrightarrow{\Psi_W \varepsilon_{f_\eta}} \Psi_W M. \tag{E.6}$$

For proper f, the maps χ_f and ν_f are inverse isomorphisms after the identification $Rf_* = Rf_!$.

The lax monoidal map $\mu_Y : \Psi_Y M \otimes^{\mathbf{L}} \Psi_Y N \to \Psi_Y(M \otimes^{\mathbf{L}} N)$ of $R\bar{j}_*$ is adjoint to the tensor product of the two counits $j^* R j_* \to 1$. It is associative, symmetric and unital, with unit $\sigma_Y : A_{Y_s} \to \Psi_Y A_{Y_\eta}$, and α_f is monoidal. For $f : V \to W$ and $g : W \to Y$, we have

$$\alpha_{gf} = \alpha_f(g_\eta^*) f_s^* \alpha_g, \qquad \sigma_V = \alpha_f(A) f_s^* \sigma_W, \qquad \mu(\sigma \otimes 1) = 1. \tag{E.7}$$

For the structure morphism $a_Y : Y \to S$, set $\theta_Y = \beta_{a_Y}(A) : \Psi_Y \omega_{Y_\eta} \to \omega_{Y_s}$, using $\Psi_S A = A$.

The proof of Proposition E.22 moves the base change maps past the projection formula, composition, external duality and the cartesian square defining the fixed point scheme. The following four lemmas give the identities it needs. Each is proved by taking adjoints until both sides become the same composite of base change maps and counits, and then using the composition laws for the base change maps and for the counits. For a morphism f on either fibre, write $t_f : f^! M \otimes^{\mathbf{L}} f^* N \to f^!(M \otimes^{\mathbf{L}} N)$ for the projection-formula mate of the exceptional counit.

Lemma E.14 (Base change maps and the projection formula)**.** *Let $f : V \to W$ be a morphism of separated S-schemes of finite type, and let $M, N \in D_c^b(W_\eta, A)$. Then the two maps $\Psi_V f_\eta^! M \otimes^{\mathbf{L}} f_s^* \Psi_W N \to f_s^! \Psi_W (M \otimes^{\mathbf{L}} N)$*

$$\beta_f \Psi_V(t_{f_\eta}) \mu_V (1 \otimes \alpha_f) = f_s^! \mu_W \; t_{f_s}(\beta_f \otimes 1) \tag{E.8}$$

are equal.

Proof. The identity follows, after taking $Rf_{s!}$-adjoints, from the compatibility of ν_f with the projection formula: for L on V_η and N on W_η, the composite

$$Rf_{s!}(\Psi_V L \otimes^{\mathbf{L}} f_s^* \Psi_W N) \xrightarrow{\mu_V(1\otimes\alpha_f)} Rf_{s!}\Psi_V(L\otimes^{\mathbf{L}} f_\eta^* N) \xrightarrow{\nu_f} \Psi_W Rf_{\eta!}(L\otimes^{\mathbf{L}} f_\eta^* N) \simeq \Psi_W(Rf_{\eta!}L \otimes^{\mathbf{L}} N)$$

equals the projection formula on the special fibre followed by $\mu_W(\nu_f \otimes 1)$. To check this, we use the description of ν_f on the total families and compare j_W^*-adjoints. On both sides the adjoint is the counit $j_V^* R j_{V*} \to 1$ tensored with the identity, followed by the projection formula on the generic fibre.

Taking $Rf_{s!}$-adjoints in (E.8) and applying the projection formula, the left side becomes the following composite by this compatibility, and the right side becomes it by naturality of μ_W:

$$\begin{aligned} Rf_{s!}\Psi_V f_\eta^! M \otimes^{\mathbf{L}} \Psi_W N &\xrightarrow{\nu_f\otimes 1} \Psi_W(Rf_{\eta!} f_\eta^! M) \otimes^{\mathbf{L}} \Psi_W N \\ &\xrightarrow{\mu_W} \Psi_W(Rf_{\eta!} f_\eta^! M \otimes^{\mathbf{L}} N) \xrightarrow{\Psi_W(\varepsilon_{f_\eta}\otimes 1)} \Psi_W(M \otimes^{\mathbf{L}} N). \end{aligned}$$

Since the adjoints agree, so do the maps. ∎

Lemma E.15 (Composition of the exceptional base change maps)**.** *Let $f : V \to W$ and $g : W \to Y$ be morphisms of separated S-schemes of finite type. Then*

$$\beta_{gf} = f_s^! \beta_g \; \beta_f(g_\eta^!), \qquad \theta_V = f_s^! \theta_W \; \beta_f(\omega_{W_\eta}). \tag{E.9}$$

Proof. The first identity follows by taking adjoints for the adjunctions $Rh_{s!} \dashv h_s^!$ $(h = f, g, gf)$ and using the compatibility of ν with composition together with the composition of counits $\varepsilon_{gf} = \varepsilon_g \, Rg_! \varepsilon_f$. The second is the case $g = a_W$, applied to A, of the first, since $a_V = a_W f$. ∎

Let p_1, p_2 be the projections of $X \times_S X$. On either fibre, let $\xi : p_2^* \omega_X \to p_1^! A_X$ be the isomorphism of Lemma 11.7 for $K = A_X$, and for a complex M put $\kappa_M = t_{p_1}(\xi \otimes 1) : p_2^*\omega_X \otimes^{\mathbf{L}} p_1^* M \to p_1^! M$, where $t_{p_1} : p_1^! A_X \otimes^{\mathbf{L}} p_1^* M \to p_1^! M$ is the projection-formula mate of the exceptional counit.

The next identity says that, under the base change maps, the map κ_M of external duality on the generic fibre corresponds to $\kappa_{\Psi_X M}$ on the special fibre, up to θ_X.

Lemma E.16 (Base change maps and external duality)**.** *Let X be a separated S-scheme of finite type with projections p_1, p_2 of $X \times_S X$, and let $M \in D_c^b(X_\eta, A)$. Then the two maps $p_{2s}^* \Psi_X \omega_{X_\eta} \otimes^{\mathbf{L}} p_{1s}^* \Psi_X M \to p_{1s}^! \Psi_X M$*

$$\beta_{p_1} \Psi(\kappa_M) \mu(\alpha_{p_2} \otimes \alpha_{p_1}) = \kappa_{\Psi_X M}(p_{2s}^* \theta_X \otimes 1) \tag{E.10}$$

are equal.

Proof. Since $\kappa_M = t_{p_1}(\xi \otimes 1)$, the main point is the case of the constant complex,

$$\beta_{p_1} \Psi(\xi) \alpha_{p_2} = p_{1s}^! \sigma_X \, \xi_s \, p_{2s}^* \theta_X, \tag{C}$$

and the general case follows from (C) by Lemma E.14. Since the proof uses proper base change, we prove (C) for a proper family and then extend it by compactification.

Suppose first that $a = a_X : X \to S$ is proper. On the fibre over η, let

$$Q_\eta(L) : a_\eta^* R a_{\eta *} L \xrightarrow{\sim} R p_{1\eta *} p_{2\eta}^* L$$

be the proper base change isomorphism, and define Q_s in the same way. On each fibre, the $Rp_{1!}$-adjoint of ξ is $Q(\omega_X)^{-1}$ followed by $a^*\varepsilon_a$. We compare these adjoints using the identity

$$\nu_{p_1} R p_{1s*}(\alpha_{p_2}) Q_s(\Psi_X L) = \Psi_X(Q_\eta(L)) \alpha_a(R a_{\eta *} L)\, a_s^* \nu_a(L).$$

To prove it, we take adjoints for $p_{1s}^* \dashv R p_{1s*}$. By the composition rule for α and $ap_1 = ap_2$, both adjoints are the composite of $a_s^*\nu_a$, the base change map $\alpha_{ap_1} = \alpha_{ap_2}$, and the pullback by $p_{2\eta}$ of the counit $a_\eta^* R a_{\eta *} L \to L$. Substituting this identity into the $Rp_{1s!}$-adjoint of the left side of (C), and using naturality of α_a with respect to ε_{a_η} and $\alpha_a(A) = \sigma_X$, we find that this adjoint is the composite

$$a_s^* R a_{s!} \Psi_X \omega_{X_\eta} \xrightarrow{a_s^* \nu_a} a_s^* R a_{\eta !} \omega_{X_\eta} \xrightarrow{a_s^* \varepsilon_{a_\eta}} A_{X_s} \xrightarrow{\sigma_X} \Psi_X A_{X_\eta}.$$

By (E.6) for a, this composite equals $\sigma_X a_s^*(\varepsilon_{a_s} R a_{s!} \theta_X)$, which is the $Rp_{1s!}$-adjoint of the right side. This proves (C) for proper X.

If X is not proper over S, we choose an open immersion $j : X \hookrightarrow \overline{X}$ into a proper S-scheme and restrict both factors of the product from $\overline{X}$ to X. Both sides of (C) are compatible with this restriction. For the maps ξ this follows from the composition rule for exceptional inverse images. For the base change maps it holds because $j^! = j^*$, the map α_j is invertible by smooth base change, and $\beta_j = \alpha_j^{-1}$. So (C) on $\overline{X}$ implies it on X. Applying Lemma E.14 to p_1 and using $\mu(\sigma \otimes 1) = 1$, we deduce (E.10). ∎

For the last identity, let $c : C \to X \times_S X$ be a correspondence of separated S-schemes of finite type, with fixed scheme F, closed immersion $i : F \hookrightarrow C$, map $d : F \to X$, and closed diagonal $\Delta : X \hookrightarrow X \times_S X$, so that $F = C \times_{X \times X} X$. Let $\zeta : c^! R\Delta_* \xrightarrow{\sim} R i_* d^!$ be the base change isomorphism for exceptional inverse image and proper direct image in this cartesian square, on either fibre.

Lemma E.17 (Base change maps and the fixed point scheme)**.** *In the situation just described,*

$$R i_{s*} \beta_d \; \chi_i \; \Psi_C(\zeta_\eta) = \zeta_s \; c_s^!(\chi_\Delta) \; \beta_c(R\Delta_{\eta *}). \tag{E.11}$$

Proof. The inverse of the base change map, $R i_* d^! \to c^! R\Delta_*$, is adjoint to the explicit map

$$R c_! R i_* d^! = R\Delta_* R d_! d^! \xrightarrow{R\Delta_* \varepsilon_d} R\Delta_*.$$

This identification uses that i and Δ are proper. Substituting it in both sides, taking $Rc_{s!}$-adjoints, and replacing the comparison maps χ_i, χ_Δ for proper maps by the inverses of ν_i, ν_Δ, we find by the composition rules for the base change maps and for the counits that both sides are the base change map for the composite $ci = \Delta d$, followed by $\Psi_X \varepsilon_d$. ∎

E.4 Duality for nearby cycles

Since the second factor of $K \boxtimes \mathbb{D}K$ is a Verdier dual, specializing a trace class also requires nearby cycles to commute with Verdier duality, by an explicit map $\delta_{X,M}$ defined below [Ill94, Theorem 4.2].

The trait S and the coefficient ring A are those of Proposition E.9, and $\mathbb{D}_{X_{\bar\eta}}$ and $\mathbb{D}_{X_s}$ are the Verdier duals relative to the geometric generic point $\bar\eta$ and the closed point s of S. We work on the classical étale site and transport the result to the pro-étale site at the end.

We use the base change maps χ_f, ν_f, β_f, the lax monoidal map μ_X and the map $\theta_X : \Psi_X\omega_{X_\eta} \to \omega_{X_s}$ of the preceding section. Recall that β_f is the adjoint, for $Rf_{s!} \dashv f_s^!$, of (E.6) and that $\nu_f = \chi_f^{-1}$ for proper f. For $M \in D^b_{\mathrm{ctf}}(X_\eta, A)$, define the morphism $\delta_{X,M} : \Psi_X\mathbb{D}_{X_\eta}M \to \mathbb{D}_{X_s}(\Psi_X M)$ as the tensor–Hom adjoint of the pairing

$$\Psi_X M \otimes \Psi_X\mathbb{D}_{X_\eta}M \xrightarrow{\mu_X} \Psi_X(M \otimes \mathbb{D}_{X_\eta}M) \xrightarrow{\Psi_X\,\mathrm{ev}} \Psi_X\omega_{X_\eta} \xrightarrow{\theta_X} \omega_{X_s}. \tag{E.12}$$

Proposition E.18 (Nearby cycles commute with Verdier duality)**.** *For every separated S-scheme X of finite type and $M \in D^b_{\mathrm{ctf}}(X_\eta, A)$, the map*

$$\delta_{X,M} : \Psi_X\mathbb{D}_{X_\eta}M \longrightarrow \mathbb{D}_{X_s}(\Psi_X M)$$

defined by (E.12) is an isomorphism.

The argument is the one used for Proposition E.9. As there, on a projective X global duality shows that the cone of δ has zero cohomology on X_s. Restriction to the generic point of a coordinate line (E.2) and induction on dimension show that the cone has finite support. For this we need two compatibilities of δ: with proper direct image, and with restriction to the generic point of a coordinate line. In the second we keep track of the twist $A(1)[2]$ on both fibres.

Lemma E.19 (Duality for nearby cycles and proper direct image)**.** *Let $f : X \to Y$ be a proper morphism of separated S-schemes of finite type, and let $M \in D^b_{\mathrm{ctf}}(X_\eta, A)$. Under the isomorphism χ_f and the Verdier duality isomorphisms $Rf_*\mathbb{D} \simeq \mathbb{D}Rf_*$ of Chapter 10 on both fibres,*

$$Rf_{s*}\delta_{X,M} \quad \text{is identified with} \quad \delta_{Y,Rf_{\eta*}M}. \tag{E.13}$$

For an open immersion $j : U \to X$, the restriction $j_s^*\delta_{X,M}$ is δ_{U,j_η^*M} under the base change isomorphism α_j. If X is proper over S, then under global Verdier duality on both fibres

$$R\Gamma(X_s, \delta_{X,M}) \quad \text{is identified with the identity of} \quad R\operatorname{Hom}_A(R\Gamma(X_{\bar\eta}, M), A). \tag{E.14}$$

Proof. Under Verdier duality for the proper map f_η, evaluation on $Rf_{\eta*}M$ is the composite of the lax monoidal map of $Rf_{\eta*}$, the map $Rf_{\eta*}\,\mathrm{ev}_M$, and the trace $Rf_{\eta*}\omega_{X_\eta} \to \omega_{Y_\eta}$. Hence, after taking tensor–Hom adjoints, both sides of (E.13) are the pairing

$$\begin{aligned} Rf_{s*}\Psi_X M \otimes Rf_{s*}\Psi_X\mathbb{D}M &\xrightarrow{\chi_f^{-1}\otimes\chi_f^{-1}} \Psi_Y Rf_{\eta*}M \otimes \Psi_Y Rf_{\eta*}\mathbb{D}M \xrightarrow{\mu_Y} \Psi_Y(Rf_{\eta*}M \otimes Rf_{\eta*}\mathbb{D}M) \\ &\longrightarrow \Psi_Y Rf_{\eta*}(M \otimes \mathbb{D}M) \longrightarrow \Psi_Y Rf_{\eta*}\omega_{X_\eta} \longrightarrow \Psi_Y\omega_{Y_\eta} \xrightarrow{\theta_Y} \omega_{Y_s}. \end{aligned}$$

To identify the pairing obtained from $Rf_{s*}\delta_{X,M}$ with this one, we move the lax monoidal maps past the base change maps for proper morphisms, and we use the identity $\theta_X = f_s^!\theta_Y \circ \beta_f(\omega_{Y_\eta})$ of (E.9). The lax monoidal maps are compatible with the base change maps because, after taking ordinary adjoints, both become the tensor product of the two counits for the open immersions of the generic fibres. This proves (E.13).

For an open immersion j, the map α_j is an isomorphism, $j^! = j^*$ and $\beta_j = \alpha_j^{-1}$. The argument above therefore applies with α_j in place of χ_f^{-1}.

For (E.14) we apply (E.13) to the structure morphism a of X. On the trait, Ψ_S takes the stalk at the geometric generic point, and the pairing (E.12) is ordinary evaluation. So $\delta_{S,Ra_{\eta*}M}$ is the identity of $R\operatorname{Hom}_A(R\Gamma(X_{\bar\eta}, M), A)$. Global Verdier duality on both fibres now gives (E.14). ■

For the slice we need that taking invariants under the pro-p group P of (E.2) commutes with duality.

Lemma E.20 (Duality and invariants under P)**.** *Let Y be a scheme of finite type over a separably closed field, and let $L \in D^b_{\mathrm{ctf}}(Y_{\acute{e}t}, A)$ carry a continuous action of P. There is a natural isomorphism*

$$(\mathbb{D}L)^P \xrightarrow{\sim} \mathbb{D}(L^P), \tag{E.15}$$

under which the evaluation pairing of L^P is the restriction of the evaluation pairing of L to the direct summands L^P and $(\mathbb{D}L)^P$.

Proof. Let G be a finite quotient of P. Since G is a p-group, the order $|G|$ is invertible in A, and the invariants are the image of the idempotent $e = |G|^{-1}\sum_{g\in G} g$. Verdier duality sends this idempotent to the same averaging idempotent for the contragredient action, since inversion permutes the summands. Hence the image of $\mathbb{D}(e)$ is the dual of $\operatorname{im} e$, and the evaluation pairing restricts as stated. As invariants are exact, continuity gives the assertion for the pro-p group P. ■

We fix the notation for the slice. Let $b : X \to \mathbf{A}^1_S$ be a coordinate map and let $S' = (\mathbf{A}^1_S)_{(s')}$ be the trait at the generic point of the special fibre used in (E.2). Set $X' = X \times_{\mathbf{A}^1_S} S'$, $M' = M|_{X'_{\bar\eta'}}$, and write g_s, g_η for the two pullbacks to the geometric generic point of the coordinate line. Restriction to the geometric generic point of the coordinate line is compatible with internal Hom, and Lemma 10.6 and Lemma 10.7, applied on each geometric fibre, give, locally on X, isomorphisms

$$g_\eta^*\mathbb{D}_{X_{\bar\eta}}M \simeq \mathbb{D}_{X'_{\bar\eta'}}M' \otimes A(1)[2], \qquad g_s^*\mathbb{D}_{X_s}L \simeq \mathbb{D}_{X'_{s'}}(g_s^*L) \otimes A(1)[2] \tag{E.16}$$

for $L \in D^b_{\mathrm{ctf}}(X_s, A)$. The twist comes from the dualizing complexes. For a closed embedding over $\mathbf{A}^1$ into $\mathbf{A}^r_{\mathbf{A}^1}$, smooth purity gives $A(r+1)[2r+2]$ before generic restriction and $A(r)[2r]$ after it. Moreover, the comparison of the complexes with closed support for the embedding preserves evaluation. Thus on $\mathbf{A}^r_{\mathbf{A}^1_S}$ the identifications of the dualizing complexes before and after slicing differ by $A(1)[2]$ on both fibres. Finally, the group P acts trivially on $A(1)[2]$, because roots of unity of order prime to the residue characteristic lift over a strictly henselian trait. So the twist commutes with taking P-invariants.

Lemma E.21 (Duality for nearby cycles on a slice)**.** *In the situation just described, suppose that X is affine, and let $i : X \hookrightarrow \mathbf{A}^r_{\mathbf{A}^1_S}$ be a closed immersion over $\mathbf{A}^1_S$; we use the identifications (E.16) defined by i. Under the identifications (E.2), (E.16) and (E.15), we have*

$$g_s^*\delta_{X,M} \;=\; (\delta_{X',M'})^P \otimes A(1)[2]. \tag{E.17}$$

Proof. The map δ is the adjoint of the pairing (E.12). By Lemma E.20 it therefore suffices to compare the three arrows of this pairing with the corresponding arrows on the slice, restricted to P-invariants. Since the slicing formula comes from factoring the inclusion of the geometric generic point and composing the adjunctions for the direct images, the lax monoidal map and the evaluation correspond under it to the maps on the slice, restricted to P-invariants.

For the last arrow θ_X, we claim that under (E.2) and (E.16), the map $g_s^*\theta_X$ corresponds to $(\theta_{X'})^P \otimes A(1)[2]$. We check this on a smooth ambient space, where θ is the identity by smooth purity, and descend along the closed immersion $i : X \hookrightarrow \mathbf{A}^r_{\mathbf{A}^1_S}$ of the statement, which we abbreviate to $i : X \hookrightarrow \mathbf{A}^r$. First, on $\mathbf{A}^r$ and on its slice, smooth purity and the base change isomorphism of nearby cycles for smooth pullback identify θ with the identity of the dualizing complex of a smooth scheme, and these identifications differ by $A(1)[2]$. To descend to X, we use the identity (E.9) for i, which gives $\theta_X = i_s^!\theta_{\mathbf{A}^r} \circ \beta_i(\omega_{\mathbf{A}^r_\eta})$. By (E.6), θ_X is then the adjoint, for $Ri_{s*} \dashv i_s^!$, of ν_i, followed by Ψ of the exceptional counit $Ri_{\eta*}i_\eta^! \to 1$ and by $\theta_{\mathbf{A}^r}$. Now the base change map ν_i for the proper map i and the factorization used for slicing commute with this counit, because their ordinary adjoints are the same composite of the counit for the open immersion of the generic fibre and the counit for i. Since generic restriction is compatible with $i^!$ and with evaluation, the claim for $\mathbf{A}^r$ gives it for X before taking invariants.

It remains to pass to P-invariants. For a finite quotient G of P, the additive functor $i_{s'}^!$ commutes with the action of G, and so it preserves the image of the averaging idempotent. Thus $i_{s'}^!$ commutes with $(-)^P$, and the same descent works after taking P-invariants. Since the shift [2] is even, no additional Koszul sign appears. ∎

Proof of Proposition E.18. We argue by induction on $n = \dim X_{\bar\eta}$, simultaneously for all traits S as above, and show that the cone of $\delta_{X,M}$ has zero global sections and finite support.

Reductions. We may assume that X is projective, reduced, and flat over S, by reductions none of which increases n. The assertion is local and, by Lemma E.19, compatible with restriction to opens. So we may pass to a projective compactification and extend M by zero on its generic fibre. Next we use the compatibility with proper direct image (E.13) for the closure of that generic fibre, together with invariance under nilpotent thickenings. This lets us assume that X is reduced and flat over S.

Put $Q = \mathrm{Cone}(\delta_{X,M})$. It is bounded and constructible by the finiteness theorem for nearby cycles and ordinary Verdier duality. By (E.14), $R\Gamma(X_s, Q) = 0$.

Finite support. By Lemma E.11, it remains to show that the support of Q is finite. By Lemma E.12, X_s is covered by finitely many opens V_s with quasi-finite coordinate maps $V \to \mathbf{A}^q_S$, $q \le n$, and $\dim V_{\bar\eta} \le q$. As the vanishing of Q is local, we may replace each V by finitely many affine opens covering it, with the restricted coordinate maps. Then for each coordinate map $V \to \mathbf{A}^1_S$ there is a closed immersion of V into some $\mathbf{A}^r_{\mathbf{A}^1_S}$ over $\mathbf{A}^1_S$, as required in Lemma E.21.

We claim that Q vanishes over the generic point of each coordinate line in the special fibre. The slice V' has generic fibre of dimension at most $q - 1 < n$. Hence the claim follows from the induction hypothesis and Lemma E.21. The support of Q therefore lies over finitely many closed points of each coordinate line. Since the coordinate map is quasi-finite, the support is finite. This argument also covers $n = 0$, where $q = 0$ and V_s is finite.

Pro-étale formulation. The comparison for constructible complexes with finite coefficients transports both the pairing and this isomorphism to the small pro-étale site. ∎

E.5 Specialization of the trace class

In a continuous family of maps of a compact manifold, the Lefschetz number does not change. The algebraic counterpart is the following proposition: the trace class of the specialized correspondence u_s is the specialization sp_F of the trace class of u. Here sp_F is a map of sections of dualizing complexes on the fixed scheme. A trace class is built from three pieces: the cohomological correspondence u, the external product $K \boxtimes \mathbb{D}K$ in which u becomes a section, and the evaluation map along the diagonal to the dualizing complex. Nearby cycles commute with each piece. They commute with external products by Proposition E.9, with Verdier duality by Proposition E.18, and with the four operations through the base change maps of Section E.3. We prove the proposition by putting these compatibilities together. Theorem E.30 uses it to transfer the computation to the normal cone. For this it also needs Corollary E.25: sp_F is the identity when the fixed scheme is, up to nilpotents, a constant family $Y \times S$.

Proposition E.22 (Specialization of the trace class)**.** *Let $c = (c_1, c_2) : C \to X \times_S X$ be a correspondence of separated schemes of finite type over the trait S, and let $F = \mathrm{Fix}(c)$ be its fixed scheme, with $i = i_c : F \hookrightarrow C$ and $d : F \to X$. For $K \in D^b_{\mathrm{ctf}}(X_\eta, A)$ and a morphism $u : c_{2\eta}^* K \to c_{1\eta}^! K$, put $K' = \Psi_X K$ and let u_s be the composite*

$$u_s : c_{2s}^* K' \xrightarrow{\alpha_{c_2}} \Psi_C c_{2\eta}^* K \xrightarrow{\Psi_C u} \Psi_C c_{1\eta}^! K \xrightarrow{\beta_{c_1}} c_{1s}^! K'.$$

For a section $z : A_{F_\eta} \to \omega_{F_\eta}$ of the dualizing complex, its specialization is

$$\mathrm{sp}_F(z) = \theta_F \Psi_F(z) \sigma_F : A_{F_s} \longrightarrow \omega_{F_s}.$$

Then

$$\mathrm{Tr}_{c_s}(u_s) = \mathrm{sp}_F(\mathrm{Tr}_{c_\eta}(u)).$$

Neither properness of the legs c_1, c_2 nor smoothness of F is assumed. This matters, because we apply the proposition to the deformation of a correspondence to the normal cone. Its generic fibre is the original correspondence, but its fixed scheme, which we describe in Lemma E.28 of Section E.6, may be singular and nonreduced.

A trace class has two ingredients besides u: the external duality isomorphism, which turns u into a section of $c^!(K \boxtimes \mathbb{D}K)$, and the evaluation along the diagonal. In the proof we compare both once and for all on $X \times_S X$, for the universal kernel $K \boxtimes \mathbb{D}K$, and then pull the comparison back along c. After taking adjoints, each comparison reduces to the pairing (E.12) that defines δ_K.

We fix the maps. In the setting of Proposition E.22, let $P_X = K \boxtimes \mathbb{D}K$ be the external product of Lemma 11.7. The comparison map for external products is

$$\Pi_{M,N} = \mu(\alpha_{p_1} \otimes \alpha_{p_2}) : \Psi_X M \boxtimes \Psi_X N \longrightarrow \Psi_{X \times_S X}(M \boxtimes N),$$

and it is invertible by Proposition E.9. Using it and the isomorphism $\delta_K = \delta_{X,K}$ of Proposition E.18, we define the isomorphism

$$\rho : \Psi_{X \times_S X} P_X \xrightarrow{\Pi^{-1}_{K,\mathbb{D}K}} K' \boxtimes \Psi_X \mathbb{D}K \xrightarrow{1 \boxtimes \delta_K} K' \boxtimes \mathbb{D}K'.$$

Write $\lambda_K : P_X \xrightarrow{\sim} R\mathcal{H}om_A(p_2^* K, p_1^! K)$ for the external duality isomorphism of Lemma 11.7, and $e_K : P_X \to R\Delta_* \omega_X$ for the evaluation along the diagonal (Definition 11.8). Let

$$\mathfrak{h} : \Psi_{X\times_S X} R\mathcal{H}om_A(p_2^*K, p_1^!K) \longrightarrow R\mathcal{H}om_A(p_{2s}^*K', p_{1s}^!K')$$

be the comparison obtained from the tensor–Hom mate of $\Psi(\mathrm{ev})\mu$ by precomposing with α_{p_2} and postcomposing with β_{p_1}. For the correspondence c, let $\lambda_{c,K} : c^!P_X \xrightarrow{\sim} R\mathcal{H}om_A(c_2^*K, c_1^!K)$ be the external duality isomorphism on C, and define $\mathfrak{h}_c$ as $\mathfrak{h}$, with α_{c_2} and β_{c_1} in place of α_{p_2} and β_{p_1}.

Lemma E.23 (Nearby cycles of the universal kernel)**.** *In the setting above,*

$$\mathfrak{h}\,\Psi(\lambda_K) = \lambda_{K'}\rho, \qquad e_{K'}\rho = R\Delta_{s*}\theta_X\,\chi_\Delta\,\Psi(e_K). \tag{E.18}$$

Proof. For the first equality, we precompose with $\Pi_{K,\mathbb{D}K}$ and take tensor–Hom adjoints. Both maps then have source

$$(p_{1s}^*K' \otimes^{\mathbf{L}} p_{2s}^*\Psi_X\mathbb{D}K) \otimes^{\mathbf{L}} p_{2s}^*K'$$

and target $p_{1s}^!K'$. The adjoint defining λ_K is

$$(p_1^*K \otimes^{\mathbf{L}} p_2^*\mathbb{D}K) \otimes^{\mathbf{L}} p_2^*K \xrightarrow{1\otimes\widetilde{\mathrm{ev}}_K} p_1^*K \otimes^{\mathbf{L}} p_2^*\omega_X \xrightarrow{\sim} p_2^*\omega_X \otimes^{\mathbf{L}} p_1^*K \xrightarrow{\kappa_K} p_1^!K,$$

where $\widetilde{\mathrm{ev}}_K$ is $\mathbb{D}K \otimes^{\mathbf{L}} K \xrightarrow{\sim} K \otimes^{\mathbf{L}} \mathbb{D}K \xrightarrow{\mathrm{ev}_K} \omega_X$. In both displays the arrows marked $\sim$ are the symmetry isomorphisms of the tensor product, and all Koszul signs come from these symmetries. Since α is monoidal, we can move the ordinary comparison maps past μ. By (E.10) we know what β_{p_1} does to the last arrow κ_K. What remains on the factors K' and $\Psi_X\mathbb{D}K$ is the composite $\theta_X\Psi_X(\mathrm{ev}_K)\mu_X$. This is the pairing (E.12), that is, evaluation against δ_K. The result is the adjoint of $\lambda_{K'}(1 \boxtimes \delta_K)$. Since Π is invertible, the first equality follows.

For the second equality, we again precompose with $\Pi_{K,\mathbb{D}K}$ and take adjoints for $\Delta_s^* \dashv R\Delta_{s*}$. Since α is monoidal and compatible with composition, the map $\alpha_\Delta\Delta_s^*\Pi_{K,\mathbb{D}K}$ is $\mu_X : K' \otimes^{\mathbf{L}} \Psi_X\mathbb{D}K \to \Psi_X(K \otimes^{\mathbf{L}} \mathbb{D}K)$. Hence both adjoints are the pairing (E.12) defining δ_K. Since Δ is proper, the comparison map for direct image is χ_Δ. ∎

By pulling the first identity back along c, we compute the section of $c_s^!P_{X_s}$ that corresponds to u_s.

Lemma E.24 (The section defined by the specialized correspondence)**.** *In the setting above,*

$$\mathfrak{h}_c\Psi_C(\lambda_{c_\eta,K}) = \lambda_{c_s,K'}\,c_s^!\rho\,\beta_c(P_X). \tag{E.19}$$

Consequently, if $k_u : A_{C_\eta} \to c_\eta^!P_X$ is the section corresponding to u, then the section corresponding to u_s is

$$k_{u_s} = c_s^!\rho\,\beta_c\,\Psi_C(k_u)\,\sigma_C.$$

Proof. After taking tensor–Hom adjoints, we factor the ordinary comparison map through $c_2 = p_2c$ and the exceptional comparison map through $c_1 = p_1c$, and we apply (E.8) for $f = c$. What remains is the first equality of (E.18) pulled back by $c_s^!$. The formula for k_{u_s} follows from (E.19), the definition of u_s and $\mu(\sigma \otimes 1) = 1$. ∎

Proof of Proposition E.22. We compare the trace morphisms on the two fibres and then evaluate them on the sections of Lemma E.24. By Definition 11.8, the trace morphism is $t_{c,K} = \zeta\, c^! e_K$. The second equality of (E.18), together with (E.11) and with (E.9) for d, gives

$$t_{c_s,K'} c_s^! \rho\, \beta_c = Ri_{s*}\theta_F\, \chi_i\, \Psi_C(t_{c_\eta,K}).$$

If we apply this to the section k_{u_s} and take adjoints for $i_s^* \dashv Ri_{s*}$, the equality of trace classes reduces to the identity for the units

$$\chi_i \Psi_C(\eta_{i_\eta})\sigma_C = Ri_{s*}(\sigma_F)\eta_{i_s}.$$

Its ordinary adjoint is the identity $\sigma_F = \alpha_i i_s^* \sigma_C$ of (E.7), followed by the ordinary triangle identity. ∎

The proposition is useful only if sp_F loses no information. In the situation of the contracting argument the fixed scheme does not move in the family, and then specialization is the identity.

Corollary E.25 (Specialization when the fixed point scheme is a constant family)**.** *If $F_{\mathrm{red}} = Y \times S$, with Y separated and of finite type over k, then the specialization sp_F of Proposition E.22 is the identity under the identifications for constant families and invariance under nilpotent thickenings. In particular, it is injective.*

Proof. By naturality it suffices to show that $\theta_Y \sigma_Y(\omega_Y)$ is the identity of ω_Y. For proper Y this follows from the adjunction defining θ_Y. We deduce the case of general Y by a compactification, and the case of nonreduced F by invariance under nilpotent thickenings.

Proper Y and $F = Y \times S$. Here $\theta_Y \sigma_Y(\omega_Y) = \mathrm{id}_{\omega_Y}$. The unit $\sigma_Y(\omega_Y) : \omega_Y \to \Psi_Y \omega_{Y_\eta}$ is an isomorphism by Lemma E.2. For the proper structure map a of Y, the base change isomorphism for proper maps and its compatibility with σ give

$$\nu_a\, Ra_!\sigma_Y(\omega_Y) = \sigma_{\mathrm{Spec}\, k}(Ra_!\omega_Y) = \mathrm{id}_{Ra_!\omega_Y}.$$

By the defining adjunction (E.6), $\theta_Y \sigma_Y(\omega_Y)$ is therefore the adjoint of the counit $Ra_!\omega_Y \to A$, which is id_{ω_Y}. Naturality of σ then gives $\theta_Y \Psi_Y(z)\sigma_Y(A) = z$ for every section z of the dualizing complex.

General Y. Choose a proper compactification $j : Y \hookrightarrow \overline{Y}$, so that $\omega_Y = j^*\omega_{\overline{Y}}$. We compare the two sides of the identity under restriction along j. The units for the constant families correspond by base change along the open immersion, and the map θ_Y corresponds to $j^*\theta_{\overline{Y}}$ by the composition rule (E.9) together with $\beta_j = \alpha_j^{-1}$ for the open immersion. Restricting the identity $\theta_{\overline{Y}}\sigma_{\overline{Y}}(\omega_{\overline{Y}}) = 1$ therefore gives $\theta_Y \sigma_Y(\omega_Y) = 1$, and by naturality the assertion follows again for all sections of the dualizing complex.

Nonreduced F. If only $F_{\mathrm{red}} = Y \times S$, we replace F by its reduction. This does not change sp_F, because invariance under nilpotent thickenings of the total family and of its fibres shows that the six operations, their units and counits, and nearby cycles agree with those of the reduction. ∎

E.6 Contracting correspondences

Let $c = (c_1, c_2) : C \to X \times X$ be a correspondence, and let $Z \hookrightarrow X$ be a closed subscheme. Suppose that the leg c_2, along which we take ordinary pullback, contracts towards Z faster than c_1, as measured by powers of ideals. Then the trace class near a connected component of the fixed point scheme meeting Z can be computed by restricting the correspondence to Z. With our convention $u : c_2^* K \to c_1^! K$ for the legs, the leg that contracts is c_2; in [Var05] it is the first leg.

The contraction becomes geometric on the deformation to the normal cone (Lemma E.26 below). Let I be the ideal of Z and $I_i = c_i^{-1}(I)\mathcal{O}_C$. Under the hypothesis $I_2^n \subset I_1^{n+1}$, the special fibre of c_2 maps into the zero section, while the inverse image of the zero section under c_1 is the zero section. On the special fibre the trace class therefore only sees Z.

Lemma E.26 (Normal cones and contracting ideals)**.** *Let $c = (c_1, c_2) : C \to X \times X$ be a correspondence, let $Z \hookrightarrow X$ have ideal I, and put*

$$I_1 = c_1^{-1}(I)\mathcal{O}_C, \qquad I_2 = c_2^{-1}(I)\mathcal{O}_C, \qquad Z' = V(I_1).$$

Assume that $I_2 \subset I_1$ and $I_2^n \subset I_1^{n+1}$ for some $n \geq 1$. Over the trait $T = \operatorname{Spec} k[t]_{(t)}$, whose strict henselization is the trait S used for nearby cycles, the two legs lift to the deformations to the normal cones,

$$\widetilde{c}_i : \mathcal{D}_{Z'}C \longrightarrow \mathcal{D}_Z X, \qquad \mathcal{D}_Z X = \operatorname{Spec}_{X_T} \mathcal{O}_{X_T}[I/t],$$

where $\mathcal{D}_Z X$ is written over T and $\mathcal{D}_{Z'}C$ is defined in the same way from I_1. Their special fibres $\gamma_i : N_{Z'}C \to N_Z X$ satisfy:

1. *$\gamma_1^{-1}(Z) = Z'$ scheme-theoretically, where Z and Z' are the zero sections of the two normal cones;*
2. *the image of γ_2 is set-theoretically contained in the zero section $Z \subset N_Z X$.*

Proof. We read everything off from the graded algebras. The first leg sends I onto generators of I_1, so it identifies the ideals of the zero sections. The second leg, on the other hand, sends I into I_2, and by the hypothesis $I_2^n \subset I_1^{n+1}$ the images of degree-one classes are nilpotent in $\operatorname{gr}_{I_1} \mathcal{O}_C$.

The algebra of the normal cone is $\operatorname{gr}_I \mathcal{O}_X = \bigoplus_{r \geq 0} I^r / I^{r+1}$, and the algebra of the deformation is the subalgebra of $\mathcal{O}_{X_T}[t^{-1}]$ generated by $\mathcal{O}_{X_T}$ and the fractions a/t, $a \in I$. Since $c_i^{-1}(I)\mathcal{O}_C \subset I_1$ for both legs (for $i = 2$ this is the hypothesis $I_2 \subset I_1$), the maps on generic fibres preserve these subalgebras. This gives the lifts.

The zero section of $N_Z X$ is defined by the ideal of elements of positive degree of $\operatorname{gr}_I \mathcal{O}_X$. Since I_1 is the ideal generated by $c_1^{-1}(I)$, the images of elements of degree one under the first leg generate I_1/I_1^2 as an $\mathcal{O}_{Z'}$-module, and hence they generate the ideal of elements of positive degree of $\operatorname{gr}_{I_1} \mathcal{O}_C$. Therefore the quotient by the pullback of the ideal of the zero section is $\mathcal{O}_C / I_1 = \mathcal{O}_{Z'}$, which proves (1).

Under the second leg, a product of n images of elements of degree one is the class of an element of I_2^n in I_1^n / I_1^{n+1}, and this class is zero since $I_2^n \subset I_1^{n+1}$. The pullback of the ideal of the zero section is therefore nilpotent. Its vanishing set is all of $N_{Z'}C$, which means that $(N_{Z'}C)_{\mathrm{red}} \to N_Z X$ factors through the zero section. This proves (2). ∎

Example E.27 (Squaring on the line)**.** Let $X = C = \mathbf{A}^1 = \operatorname{Spec} k[x]$, let Z be the origin, with $I = (x)$, and let $c_1 = \mathrm{id}$ and $c_2(x) = x^2$, so that $c_2^* x = x^2$. Then $I_1 = (x)$ and $I_2 = (x^2) = I_1^2$, so

c contracts towards Z with $n = 1$. The fixed scheme is $V(x^2 - x)$. It consists of the two reduced points 0 and 1, and only the first meets Z.

The deformation to the normal cone is $\operatorname{Spec} k[t]_{(t)}[x, x/t]$. This is the constant family $\mathbf{A}^1_T$ in the coordinate $x' = x/t$. In this coordinate the first leg is the identity and the second is $x' \mapsto x^2/t = tx'^2$. Hence, on the special fibre $t = 0$, the second leg is the constant map to 0, as (2) of the lemma predicts, and the deformed fixed scheme there is the point $x' = 0$. On the generic fibre, by contrast, there are two fixed points: one is $x' = 0$, the constant family of the origin, and the other is $x' = 1/t$, the point $x = 1$. This second point leaves every bounded region of the x'-line as $t \to 0$, and so it does not reach the special fibre.

Corollary E.31 below therefore gives $\mathrm{LT}_0(u) = \mathrm{tr}_A(u_{\bar{0}} \mid K_{\bar{0}})$ for every K and u. For instance, the local term is zero if $K = j_! A$ with $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$, as $K_{\bar{0}} = 0$. The condition fails for the identity correspondence. There $I_2 = I_1$, the inclusion $I_1^n \subset I_1^{n+1}$ is false for every n, and the fixed scheme is all of X.

As in Proposition E.22, write $F = \mathrm{Fix}(c)$ for the fixed scheme, and let $d = c_1|_F = c_2|_F : F \to X$ be the common map. We will show that a connected component of F that meets Z lies in Z set-theoretically, and that the deformed fixed scheme is constant up to nilpotents. Both follow from the hypothesis $I_2^n \subset I_1^{n+1}$, because the two legs agree on F.

The deformation of F to the normal cone of $d^{-1}(Z)$ embeds in the fixed scheme of the deformed correspondence. Put $I_F = d^{-1}(I)\mathcal{O}_F$, so that $d^{-1}(Z) = V(I_F)$. Since $d = c_1|_F$, the ideal I_F^r is the image of I_1^r in $\mathcal{O}_F$ for every r. Hence the closed immersion $F \hookrightarrow C$ induces a surjection of deformation algebras

$$\mathcal{O}_{C_T}[I_1/t] \twoheadrightarrow \mathcal{O}_{F_T}[I_F/t],$$

that is, a closed immersion $\mathcal{D}_{d^{-1}(Z)}F \hookrightarrow \mathcal{D}_{Z'}C$. Its two composites with $\widetilde{c}_1$ and $\widetilde{c}_2$ are the maps induced by $c_1|_F$ and $c_2|_F$, which are equal. So it factors through the fixed scheme $\widetilde{F} = \mathrm{Fix}(\widetilde{c})$ of $\widetilde{c} = (\widetilde{c}_1, \widetilde{c}_2)$. We write

$$\iota_F : \mathcal{D}_{d^{-1}(Z)}F \longrightarrow \widetilde{F} \tag{E.20}$$

for the resulting closed immersion.

Lemma E.28 (The fixed point scheme of the deformation)**.** *In the situation of Lemma E.26, suppose that F is connected and that $d(F)$ meets Z. Then $d^{-1}(Z)_{\mathrm{red}} = F_{\mathrm{red}}$, the reduction of $\mathcal{D}_{d^{-1}(Z)}F$ is $(F_T)_{\mathrm{red}}$, and the map $(\iota_F)_{\mathrm{red}} : (F_T)_{\mathrm{red}} \to \widetilde{F}_{\mathrm{red}}$ induced by (E.20) is an isomorphism.*

Proof. Restricting $I_2^n \subset I_1^{n+1}$ to F, where the two legs agree, gives $I_F^n \subset I_F^{n+1}$. The reverse inclusion always holds, so the coherent ideal $M = I_F^n$ satisfies $M = I_F M$. Hence, at a point of $V(I_F)$, Nakayama's lemma shows that $M = 0$ in the local ring, and hence on a neighbourhood of the point, whereas at a point outside $V(I_F)$ we have $M = \mathcal{O}_F$. The closed set $V(I_F)$ is therefore also open. It is nonempty by assumption, and F is connected, so $V(I_F) = F$ and $I_F^n = 0$. This proves the first assertion. As I_F is nilpotent, each generator h/t of $\mathcal{O}_{F_T}[I_F/t]$ over $\mathcal{O}_{F_T}$ is nilpotent. Hence the reduction of $\mathcal{D}_{d^{-1}(Z)}F$ is $(F_T)_{\mathrm{red}}$.

It remains to see that $\widetilde{F}$ has no further points. On the generic fibre the deformed correspondence is c itself, so the generic fibre of $\widetilde{F}$ is that of F_T. Let x be a point of the special fibre of $\widetilde{F}$. Then $\gamma_2(x)$ lies in the zero section by Lemma lemma-normal-cones-contracting-ideals, and so does $\gamma_1(x) = \gamma_2(x)$. By Lemma lemma-normal-cones-contracting-ideals, the point x lies on the zero section $Z' \subset N_{Z'}C$. On Z' the two maps are the restrictions of c_1 and c_2. Their equalizer is

$F \cap Z'$, whose underlying space is F by the first assertion. Thus every fixed point in the special fibre lies in the closed subscheme above, and the closed immersion of reduced schemes is surjective. Its ideal has zero radical and is therefore zero, so the immersion is an isomorphism. ∎

Up to nilpotents, then, the fixed scheme of the deformed correspondence is the constant family F_T. This is why the specialization of trace classes can be inverted in the proof of Theorem E.30. The hypothesis that F is connected and meets Z is needed. For instance, in Example E.27, the component $F = \{0\}$ gives the constant family $x' = 0$, but the fixed point $x = 1$ does not meet Z, and there the deformation gives only the generic-fibre point $x' = 1/t$, not the constant family $\{1\}_T$.

Remark E.29 (Reduction to a connected fixed point scheme)**.** The hypothesis in the preceding lemma that F is connected is no loss of generality. Indeed, let $F_0 \subset \mathrm{Fix}(c)$ be a connected component that is open and closed. If we replace C by an open neighbourhood of F_0 on which the ideal inclusions hold, and then remove $\mathrm{Fix}(c) \setminus F_0$, we obtain an open subscheme of C, with the same X and K, whose fixed scheme is F_0. Restriction of trace classes along this open immersion identifies the trace class of u on F_0 with the full trace class of the restricted correspondence.

The proof of the main theorem combines the two lemmas on the deformation with Verdier's theorem and with the specialization of trace classes. The statement of the main theorem involves the restriction of a correspondence to Z, which we construct first.

Let $c = (c_1, c_2) : C \to X \times X$ be a correspondence, let $Z \hookrightarrow X$ have ideal I, and suppose that $c_2^{-1}(I)\mathcal{O}_C \subset c_1^{-1}(I)\mathcal{O}_C$. Then $c_1^{-1}(Z) \subset c_2^{-1}(Z)$. As before Lemma 11.17, we put

$$C_Z = (c_1^{-1}Z)_{\mathrm{red}}, \qquad c_Z = (c_{Z,1}, c_{Z,2}) : C_Z \longrightarrow Z_{\mathrm{red}} \times Z_{\mathrm{red}},$$

with $i : Z_{\mathrm{red}} \hookrightarrow X$ and $i' : C_Z \hookrightarrow C$. For $K \in D^b_{\mathrm{ctf}}(X, A)$ put $K_Z = i^*K$. The square formed by i', i, c_1 and $c_{Z,1}$ is cartesian up to reduction. Hence invariance under nilpotent thickenings and base change for direct image with compact support give $i^*Rc_{1!} \simeq Rc_{Z,1!}i'^*$. Since the second leg maps C_Z into Z_{red}, we also have $i'^*c_2^*K \simeq c_{Z,2}^*K_Z$. For a cohomological correspondence $u : c_2^*K \to c_1^!K$ with adjoint $\widehat{u} : Rc_{1!}c_2^*K \to K$ under $Rc_{1!} \dashv c_1^!$, the *restriction* $u_Z : c_{Z,2}^*K_Z \to c_{Z,1}^!K_Z$ of u to Z is the adjoint under $Rc_{Z,1!} \dashv c_{Z,1}^!$ of

$$Rc_{Z,1!}c_{Z,2}^*K_Z \simeq i^*Rc_{1!}c_2^*K \xrightarrow{i^*\widehat{u}} K_Z. \tag{E.21}$$

Equivalently, u_Z is i'^*u followed by the base change map $i'^*c_1^!K \to c_{Z,1}^!K_Z$, which in general is not an isomorphism. This is the restriction to Z that occurs in Lemma 11.18. If F_0 is an open and closed subscheme of $\mathrm{Fix}(c)$, put $F_{0,Z} = \mathrm{Fix}(c_Z) \times_{\mathrm{Fix}(c)} F_0$, and let $h_{F_0} : F_{0,Z} \to F_0$ be the projection.

Theorem E.30 (Contracting correspondences and local terms)**.** *Let $c = (c_1, c_2) : C \to X \times X$ be a correspondence, and let $Z \hookrightarrow X$ have ideal I. Let F_0 be a connected component of* $\mathrm{Fix}(c)$ *that is open and closed and whose image meets Z. Choose an open neighbourhood $W \subset C$ of F_0 on which*

$$I_2 = c_2^{-1}(I)\mathcal{O}_W \subset I_1 = c_1^{-1}(I)\mathcal{O}_W, \qquad I_2^n \subset I_1^{n+1} \quad \text{for some } n \geq 1.$$

Let $K \in D^b_{\mathrm{ctf}}(X, A)$, let $u : c_2^*K \to c_1^!K$ be a cohomological correspondence, and let u_Z be the restriction (E.21) to Z of $u|_W$, for the correspondence $c|_W$, with $c_Z : C_Z = (W \cap c_1^{-1}Z)_{\mathrm{red}} \to Z_{\mathrm{red}} \times Z_{\mathrm{red}}$. Then h_{F_0} is an isomorphism after reduction, and

$$\mathrm{Tr}_c(u)|_{F_0} = (h_{F_0})_*(\mathrm{Tr}_{cZ}(u_Z)|_{F_{0,Z}}),$$

where $(h_{F_0})_* : H^0(F_{0,Z}, \omega_{F_{0,Z}}) \to H^0(F_0, \omega_{F_0})$ *is the induced isomorphism. In particular, if* F_0 *is proper, then* $\mathrm{LT}_{F_0}(u) = \mathrm{LT}_{F_{0,Z}}(u_Z)$.

For Z a point, the theorem specializes to Corollary E.31. Under its hypotheses the local term at F_0 is the categorical trace of u_Z. For $c = (\mathrm{id}_X, h)$ this is $\mathrm{tr}(u_{\bar{y}} \mid K_{\bar{y}})$.

Proof of Theorem E.30. We split K by the triangle $j_!j^*K \to K \to Ri_*i^*K$ and use additivity of trace classes. The piece supported on Z gives the trace of u_Z. The piece $j_!j^*K$, on the other hand, has zero trace, which we show by deforming to the normal cone.

Reductions. We may assume that $W = C$, that $\mathrm{Fix}(c) = F_0$ and that the ideal inclusions hold on C. Indeed, by Remark E.29, restricting C to W and removing the other connected components of $\mathrm{Fix}(c)$ changes neither the trace class on F_0 nor u_Z near $F_{0,Z}$. By Lemma E.28, the common map $\mathrm{Fix}(c) \to X$ maps set-theoretically into Z. The same lemma shows that after base change from T to S the reduction of the fixed scheme of the deformed correspondence is $(\mathrm{Fix}(c) \times S)_{\mathrm{red}}$, as Corollary E.25 requires.

The case $i^*K = 0$. We show that $\mathrm{Tr}_c(u) = 0$. First, the specialized complex vanishes on the zero section,

$$\mathrm{Sp}_{X,Z}(K)|_Z = 0,$$

because Verdier's theorem on the zero section (Theorem E.8) identifies the restriction of $\mathrm{Sp}_{X,Z}(K)$ to the zero section with the restriction of K to Z, and invariance under the nilpotent thickening $Z_{\mathrm{red}} \to Z$ passes from Z to Z_{red}.

Next, consider the leg γ_2 of the correspondence on the special fibre, along which we take ordinary pullback. It maps set-theoretically into the zero section by Lemma lemma-normal-cones-contracting-ideals. Invariance under nilpotent thickenings therefore gives $\gamma_2^* \mathrm{Sp}_{X,Z}(K) = 0$. So the specialized morphism has zero source, and its trace is zero.

Finally, we relate this trace to $\mathrm{Tr}_c(u)$. The generic fibre of the deformed correspondence is c over η. So Proposition E.22 identifies the zero trace with the specialization of the pullback of $\mathrm{Tr}_c(u)$ to the generic fibre. By Corollary E.25, specialization is the identity under the identifications for constant families, and these carry that pullback to $\mathrm{Tr}_c(u)$. Hence $\mathrm{Tr}_c(u) = 0$.

General K. We apply Proposition proposition-additivity-trace-classes to the triangle

$$j_!j^*K \longrightarrow K \longrightarrow Ri_*i^*K, \qquad j : X \setminus Z_{\mathrm{red}} \hookrightarrow X,$$

whose hypothesis $c_1^{-1}(Z_{\mathrm{red}}) \subset c_2^{-1}(Z_{\mathrm{red}})$ follows from $I_2 \subset I_1$. The open piece has zero trace by the case just treated, while the closed graded piece is the proper pushforward of u_Z along i and i', by Lemma 11.18, and Theorem 11.13 identifies its trace with the pushforward of the trace class of u_Z. To compare with F_0, note that the map h_{F_0} is an isomorphism after reduction, because $\mathrm{Fix}(c)$ lies in $c_1^{-1}(Z)$ set-theoretically by Lemma E.28. This proves the displayed formula. If F_0 is proper, then so is $F_{0,Z}$, and pushing both sides forward to $\mathrm{Spec}\, k$ and composing with the trace map of Definition 11.9 gives the equality of local terms. ∎

If Z is a point, the restricted correspondence is a correspondence of the point, and the theorem computes the local term as a trace on a stalk.

Corollary E.31 (Contracting local terms at a point)**.** *Let $y \in X(k)$, and suppose that $c_1^{-1}(y)$ is a single reduced point z, with $c_2(z) = y$. Suppose that $F_0 = \{z\}$ is an isolated reduced component of* $\mathrm{Fix}(c)$*, open and closed, and that on a neighbourhood of F_0 the ideal I_y satisfies*

$$c_2^{-1}(I_y)\mathcal{O}_C \subset c_1^{-1}(I_y)\mathcal{O}_C, \qquad \left(c_2^{-1}(I_y)\mathcal{O}_C\right)^n \subset \left(c_1^{-1}(I_y)\mathcal{O}_C\right)^{n+1} \quad \text{for some } n \geq 1.$$

Then for every $K \in D^b_{\mathrm{ctf}}(X, A)$ and every morphism $u : c_2^* K \to c_1^! K$, the local term at F_0 is the categorical trace of the restriction u_Z to $Z = \{y\}$ of u restricted to that neighbourhood, defined in (E.21). In particular, for $c = (\mathrm{id}_X, h)$,

$$\mathrm{LT}_y(u) = \mathrm{tr}_A(u_{\bar{y}} \mid K_{\bar{y}}).$$

Here X may be singular.

Proof of Corollary E.31. We apply Theorem E.30 with $Z = \{y\}$ and W the neighbourhood of the statement; F_0 meets Z because $d(z) = c_1(z) = y$. Here C_Z is the reduced point z and $F_{0,Z} = F_0$, so that c_Z is the identity correspondence of the point, with coefficient $K_{\bar{y}}$, and by Definition 11.9 its local term is the categorical trace. If the first leg is the identity, the base change map in the construction of u_Z is the identity, and hence the restricted morphism is the stalk map $u_{\bar{y}}$. ∎

F Algebraic groups in characteristic zero

Rankin's argument (Chapter 14) bounds the eigenvalues of a real sheaf by the largest determinant weight of its constituents, that is, the weight of the determinant divided by the rank. For this it needs the determinant weights of the factors of tensor and exterior powers of a lisse sheaf (Proposition 14.19), before anyone knows how these powers decompose. The tool is an element of the arithmetic monodromy group, of positive degree, that acts by a scalar on each simple factor of every tensor construction. This scalar then gives the determinant weight.

In a simple example Frobenius itself does this. Take $V = \overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ on a curve over $\mathbf{F}_q$, on which a Frobenius element s of degree one acts by $\mathrm{diag}(1, q)$. Here the geometric monodromy is trivial, so the arithmetic monodromy group G is the torus $\{\mathrm{diag}(1, t)\}$, and $\rho(s)$ is central in it. Hence on every tensor construction $\rho(s)$ acts by a scalar on each simple factor. For instance, it acts by 1 and q on the factors of V, of weights 0 and 2, and by q on $\bigwedge^2 V$.

In general $\rho(s)$ is not central, and we must correct a power of it by an element of the geometric monodromy group H. An element of H alone would not do, since its determinants on these factors are roots of unity. What we want instead is a central element of the form $h\rho(s)^N$ with $h \in H$ and $N > 0$.

Such an element exists by Lemma 14.15. For a semisimple Weil representation V, the identity component G° is generated by $Z(G^\circ)^\circ$ and the identity component H°. So a power of Frobenius lying in G° is an element of $Z(G^\circ)^\circ$ times an element of H°. Averaging over the components of G then makes it central in G. This is the algebraic form of a familiar fact about compact Lie groups: a connected compact group is the product of the identity component of its center and its derived group.

For algebraic groups over $\overline{\mathbf{Q}}_\ell$ we argue with the Lie algebra $\mathfrak{g}$ of G°. Since $\mathfrak{g}$ acts faithfully and semisimply on V, it is reductive, that is,

$$\mathfrak{g} = Z(\mathfrak{g}) + [\mathfrak{g}, \mathfrak{g}]$$

(Proposition F.8). The derived algebra lies in the Lie algebra of H, because commutators of G lie in H. It remains to use that a closed connected subgroup of G° with the full Lie algebra is all of G°.

The tools behind this argument need characteristic zero. Cartier's theorem that algebraic groups are smooth (Proposition F.1) lets us recognize a closed subgroup by its Lie algebra; this fails for $\mu_p \subset \mathbf{G}_m$ in characteristic p. Similarly, the decomposition of $\mathfrak{g}$ rests on Cartan's criterion, which fails there too. Since the monodromy groups are defined over $\overline{\mathbf{Q}}_\ell$, this does not restrict us. For the theorems of Lie, Engel and Cartan we refer to Humphreys [Hum72].

Throughout, Ω is an algebraically closed field of characteristic zero, and all Lie algebras and representations are finite-dimensional. By an algebraic group we mean a group scheme of finite type over Ω. In particular, stabilizers and centralizers are taken as subgroup schemes, and by Cartier's theorem below they are smooth. The Lie algebra $\mathrm{Lie}(G)$ of an algebraic group G is its tangent space at the identity; for a closed subgroup of $\mathrm{GL}(V)$ we describe it by dual numbers in Section F.2.

F.1 Smoothness and components

To pass from Lie algebras back to groups we use the following principle: a closed subgroup of a connected group G with the same Lie algebra is all of G (Lemma F.2). For reduced groups this is a dimension count: the dimension of a smooth group equals that of its tangent space at 1, and a closed subset of the irreducible variety G of the same dimension is all of G. The difficulty is that stabilizers and centralizers are defined by equations and need not be reduced. For example, in characteristic p, the subgroup $\mu_p \subset \mathbf{G}_m$ defined by $x^p = 1$ is a single nonreduced point with tangent space $\mathrm{Lie}(\mathbf{G}_m)$, and the principle fails. However, Cartier's theorem (Proposition F.1) shows that this cannot happen in characteristic zero: every algebraic group is smooth. In the proof we apply the comultiplication to a nilpotent function at the identity and divide by an integer, and this division is where we use characteristic zero. The facts about connected components and images of homomorphisms in Lemma F.2 are standard consequences of smoothness and of Chevalley's theorem.

Proposition F.1 (Cartier's theorem)**.** *Every algebraic group over Ω is smooth and reduced. In particular*

$$\dim \mathrm{Lie}(G) = \dim G.$$

Proof. We first show, using translations, that the reduced subscheme G_{red} is a smooth group scheme. It then suffices to show that G has the same tangent space at 1 as G_{red}, that is, that nilpotent functions at 1 lie in $\mathfrak{m}^2$, and this is where we divide by an integer.

Step 1: G_{red} is a smooth subgroup scheme of G. Multiplication and inversion restrict to G_{red}. This is because products of reduced schemes of finite type over the perfect field Ω are reduced [Stacks, 035Z], and a morphism from a reduced scheme to G factors through G_{red}. The reduced scheme G_{red} contains a dense smooth open subscheme [Stacks, 056V]. It is smooth everywhere, since a translation moves a closed point of this open to any other closed point of G_{red}.

Step 2: nilpotent functions at 1 lie in $\mathfrak{m}^2$. Let $R = \mathcal{O}_{G,1}$, with maximal ideal $\mathfrak{m}$ and nilradical I, and let $\widehat{R}$ be its completion. At the point $(1,1)$ the multiplication induces a continuous comultiplication

$$\Delta : \widehat{R} \longrightarrow \widehat{R}\,\widehat{\otimes}_\Omega \widehat{R}.$$

The counit is evaluation at 1, and composing Δ with evaluation at 1 in either factor gives the identity (the counit identities), because 1 is a two-sided unit for the group law. We claim that $I \subseteq \mathfrak{m}^2$. As completion is faithfully flat, it suffices to prove that every nilpotent $b \in \widehat{R}$ lies in $\widehat{\mathfrak{m}}^{\,2}$. Suppose that $b \notin \widehat{\mathfrak{m}}^{\,2}$, and let $n \geq 2$ be the least integer with $b^n = 0$. By the counit identities,

$$\Delta(b) = b \otimes 1 + 1 \otimes b + w, \qquad w \in \widehat{\mathfrak{m}} \,\widehat{\otimes}_{\Omega} \widehat{\mathfrak{m}}.$$

Now reduce the second factor modulo $\widehat{\mathfrak{m}}^{\,2}$. Every term in the binomial expansion of $\Delta(b)^n$ with at least two factors from $1 \otimes b + w$ vanishes, because $1 \otimes b + w$ has its second factor in $\widehat{\mathfrak{m}}$, and as $b^n = 0$ we are left with

$$n b^{n-1} \otimes b + n(b^{n-1} \otimes 1) w = 0.$$

If we now also reduce the first factor modulo $b^{n-1}\widehat{\mathfrak{m}}$, the second term vanishes, and we get

$$n\, \overline{b^{n-1}} \otimes \overline{b} = 0 \quad \text{in} \quad (\widehat{R}/b^{n-1}\widehat{\mathfrak{m}}) \otimes_{\Omega} (\widehat{R}/\widehat{\mathfrak{m}}^{\,2}).$$

This identity is impossible, which proves the claim. Indeed, the class $\overline{b^{n-1}}$ is nonzero by Nakayama's lemma, because the module (b^{n-1}) is nonzero by the choice of n. The class $\overline{b}$ is nonzero by assumption. A tensor product of nonzero vectors over a field is nonzero, and $n \neq 0$ in characteristic zero.

Step 3: G is smooth. The schemes G and G_{red} have the same cotangent space at 1. Indeed, by Step 2 the kernel of $\mathfrak{m}/\mathfrak{m}^2 \to (\mathfrak{m}/I)/(\mathfrak{m}/I)^2$ is $(I + \mathfrak{m}^2)/\mathfrak{m}^2 = 0$. They also have the same local dimension at 1, because a nilpotent ideal does not change the underlying space. As the local ring R/I of G_{red} at 1 is regular by Step 1, we get

$$\dim_{\Omega} \mathfrak{m}/\mathfrak{m}^2 = \dim R,$$

so R is regular, and in particular a domain [Stacks, 00NP]. By translation the local rings of G at all closed points are regular. Since Ω is perfect, G is smooth at every closed point. The nonsmooth locus is closed and contains no closed point, and hence it is empty. Being smooth, G is reduced, and its tangent space at 1 has dimension $\dim G$. Finally, the open subgroup G° has the same tangent space at 1. ∎

Since algebraic groups over Ω are smooth, their topology is as simple as that of a Lie group: irreducible and connected components coincide, and the dimension of a closed subgroup can be computed from its Lie algebra.

Lemma F.2 (Connected components and images of homomorphisms)**.** *Let G be an algebraic group.*

1. *G has finitely many connected components. Its identity component G° is an open, closed, irreducible normal subgroup, the other components are its cosets, and* $\mathrm{Lie}(G^{\circ}) = \mathrm{Lie}(G)$.
2. *The image of a homomorphism $G \to G'$ of algebraic groups is a closed subgroup of G'.*
3. *If G is connected and $K \subseteq G$ is a closed subgroup scheme with* $\mathrm{Lie}(K) = \mathrm{Lie}(G)$*, then $K = G$.*

Proof. For (1) we use smoothness to see that irreducible and connected components coincide; for (2), Chevalley's theorem gives an open subset of the closure inside the image, and translations spread it over the closure; (3) is a dimension count.

Components. The irreducible components of G are finite in number, as G is Noetherian, and they are exactly the connected components. Indeed, G is smooth by Proposition F.1, and a regular local ring is a domain. So distinct irreducible components of G do not meet, and they are open and closed. Let G° be the component containing 1. It is a subgroup. To see this, note that $G^\circ \times G^\circ$ is irreducible, Ω being algebraically closed. Hence the image of the morphism $G^\circ \times G^\circ \to G$, $(x, y) \mapsto xy^{-1}$, is irreducible. It contains 1, so it lies in G°. For the same reason the image of G° under conjugation by a point of G is irreducible and contains 1, so G° is normal. Translating G° by points of G, we see that the other components are its cosets. Since G° is open in G and contains 1, both have the same local ring at 1, so $\mathrm{Lie}(G^\circ) = \mathrm{Lie}(G)$.

For example, the homomorphism $\mathbf{G}_m \to \mathbf{G}_m$, $x \mapsto x^2$, has image all of $\mathbf{G}_m$ and kernel μ_2, a closed subgroup with two components.

Images. Let $f : G \to G'$ be a homomorphism. We show that its image equals the closure J of the image, which is a subgroup by continuity of multiplication and inversion. First, the subgroup $f(G(\Omega))$ of $J(\Omega)$ contains the Ω-points of a nonempty open $U \subseteq J$. Indeed, $f(G)$ is constructible by Chevalley's theorem [Stacks, 054K], a dense constructible subset contains a nonempty open subset of its closure, and Ω is algebraically closed. Translating U by the inverse of one of its points in the image, we obtain an open neighbourhood of 1 contained in the image. Translating further by points of the image, we see that the image is open in J. It is then also closed, since its complement is a union of cosets of this open subgroup. As it is dense in J, it is equal to J.

Lie algebras. Let K be as in (3). Cartier's theorem gives $\dim K = \dim \mathrm{Lie}(K) = \dim G$, so K and G have the same underlying closed subset, as a proper closed subset of the irreducible variety G has smaller dimension. They are equal as schemes, since both are reduced by Proposition F.1. ∎

F.2 Lie algebras

Let G be a connected algebraic group with Lie algebra $\mathfrak{g}$. We need two facts (Lemma F.3). First, in a representation, G and $\mathfrak{g}$ have the same stable subspaces. Second, the Lie algebra of the identity component of the center of G is the center of $\mathfrak{g}$. In the proof of Lemma 14.15 we compare the monodromy groups through their Lie algebras, and both facts are used there. By the first, a representation that is semisimple for G° is semisimple for its Lie algebra. This is what lets Proposition F.8 apply to the Weil representation. With the second we translate the decomposition of the Lie algebra into the identity $G^\circ = Z(G^\circ)^\circ H^\circ$.

We prove both facts by the principle of the previous section, applied to a stabilizer and to the kernel of the adjoint representation. Their Lie algebras are computed with dual numbers. The first fact needs G to be connected. For example, the finite group $\{\pm 1\} \subset \mathrm{GL}_1$ has zero Lie algebra, so every subspace of a representation is stable under its Lie algebra. But if -1 acts on $M = \Omega^2$ by $\mathrm{diag}(1, -1)$, then it does not stabilize the line spanned by $(1, 1)$.

For a closed subgroup $G \subseteq \mathrm{GL}(V)$, the tangent space $\mathrm{Lie}(G) \subseteq \mathrm{End}(V)$ is described by dual-number points at 1:

$$X \in \mathrm{Lie}(G) \quad \Longleftrightarrow \quad 1 + \varepsilon X \in G(\Omega[\varepsilon]/\varepsilon^2).$$

The bracket is $[X, Y] = XY - YX$. To see that $\mathrm{Lie}(G)$ is closed under the bracket, note that over $\Omega[\varepsilon, \eta]/(\varepsilon^2, \eta^2)$ the commutator of $1 + \varepsilon X$ and $1 + \eta Y$ is $1 + \varepsilon\eta[X, Y]$. Its entries lie in the subring $\Omega[\varepsilon\eta]$, which is a ring of dual numbers. The equations of G hold for this matrix in the larger ring, hence in this subring. Therefore $[X, Y] \in \mathrm{Lie}(G)$. The differential of a representation $r : G \to \mathrm{GL}(M)$ is characterized by

$$r(1+\varepsilon X) = 1 + \varepsilon\, dr(X).$$

Applying r to the commutator of $1+\varepsilon X$ and $1+\eta Y$, we see that $dr[X,Y] = [dr(X), dr(Y)]$, so dr is a Lie algebra homomorphism.

Lemma F.3 (Stable subspaces and the center)**.** *Let $G \subseteq \mathrm{GL}(V)$ be a connected closed subgroup and $r : G \to \mathrm{GL}(M)$ an algebraic representation. A subspace of M is G-stable if and only if it is $\mathrm{Lie}(G)$-stable. Moreover*

$$\mathrm{Lie}(Z(G)^\circ) = Z(\mathrm{Lie}(G)).$$

Proof. In both assertions we compare a closed subgroup with G, namely the stabilizer of a subspace in the first and the kernel of the adjoint representation in the second. We compute the Lie algebra of this subgroup with dual numbers and conclude with Lemma lemma-connected-components-images-homomorphisms.

Let $U \subseteq M$ be a subspace. In a basis of M extending a basis of U, the scheme-theoretic stabilizer $S \subseteq G$ of U is the closed subgroup defined by the vanishing of the lower-left block of $r(g)$. Since the lower-left block of $r(1+\varepsilon X) = 1 + \varepsilon\, dr(X)$ is ε times that of $dr(X)$, we find

$$\mathrm{Lie}(S) = \{X \in \mathrm{Lie}(G) : dr(X)U \subseteq U\}.$$

If U is $\mathrm{Lie}(G)$-stable, then $\mathrm{Lie}(S) = \mathrm{Lie}(G)$, and Lemma lemma-connected-components-images-homomorphisms shows that $S = G$. Conversely, if U is G-stable, then $S = G$, and the displayed formula shows that U is $\mathrm{Lie}(G)$-stable.

Put $\mathfrak{g} = \mathrm{Lie}(G)$. We claim that the kernel of the adjoint representation $G \to \mathrm{GL}(\mathfrak{g})$ is the center. Indeed, if g acts trivially on $\mathfrak{g}$, consider the centralizer $C_G(g)$. It is a closed subgroup scheme whose Lie algebra is the fixed space of $\mathrm{Ad}(g)$, that is, all of $\mathfrak{g}$. So $C_G(g) = G$ by Lemma lemma-connected-components-images-homomorphisms. Conversely, a central element acts trivially. Since the kernel and the center are reduced by Cartier's theorem, their equality on Ω-points is an equality of schemes.

The Lie algebra of the center is now the kernel of a differential. The formula

$$\mathrm{Ad}(1+\varepsilon X)(Y) = Y + \varepsilon[X,Y]$$

shows that the differential of Ad is ad. The center is a closed subgroup of $\mathrm{GL}(V)$, so the dual-number description applies to it. By this description, the Lie algebra of the kernel of a representation is the kernel of its differential. Hence $\mathrm{Lie}(Z(G)) = \ker(\mathrm{ad}) = Z(\mathfrak{g})$. By Proposition F.1, $Z(G)^\circ$ has the same Lie algebra as $Z(G)$. ∎

F.3 Cartan's criterion

The decomposition $\mathfrak{l} = Z(\mathfrak{l}) + [\mathfrak{l},\mathfrak{l}]$ fails for Lie algebras in general. Consider the algebra $\mathfrak{b}$ of upper triangular 2×2 matrices. Its center consists of the scalars, and $[\mathfrak{b},\mathfrak{b}]$ consists of the strictly upper triangular matrices. So their sum has dimension two and misses $\mathrm{diag}(1,0)$. Here $\mathfrak{b}$ is solvable and is its own radical, much larger than its center. Its standard representation on Ω^2 is faithful but not semisimple, since the line Ωe_1 has no stable complement. In general the obstruction is the radical, that is, the largest solvable ideal. In Proposition F.8 we show that a faithful semisimple representation forces the radical to be central. This gives the decomposition.

The derived series of a Lie algebra is $D^0\mathfrak{l} = \mathfrak{l}$, $D^{n+1}\mathfrak{l} = [D^n\mathfrak{l}, D^n\mathfrak{l}]$, and $\mathfrak{l}$ is solvable if this series reaches zero. The radical $\operatorname{rad}\mathfrak{l}$ is the sum of all solvable ideals of $\mathfrak{l}$. To see that it is itself solvable, note that an extension of solvable Lie algebras is solvable. Indeed, if the quotient of $\mathfrak{l}$ by a solvable ideal is solvable, the derived series of $\mathfrak{l}$ lies in that ideal after finitely many steps, and then reaches zero. Hence the sum of two solvable ideals is solvable, and $\operatorname{rad}\mathfrak{l}$ is the largest solvable ideal of $\mathfrak{l}$ [Hum72, §3.1]. From the same book we use three classical theorems, and we only sketch their proofs.

Lemma F.4 (Lie's theorem)**.** *A solvable Lie algebra acting on a nonzero vector space M has a common eigenvector. Equivalently, there are $v \neq 0$ and a linear functional λ such that $av = \lambda(a)v$ for every a.*

See [Hum72, §4.1]. By induction one finds a common eigenvector for an ideal of codimension one, and then shows by a trace computation, which we redo in the proof of Proposition F.8, that the corresponding eigenspace is stable under the remaining direction.

Lemma F.5 (Engel's theorem)**.** *Let $\mathfrak{n} \subseteq \operatorname{End}(M)$ be a Lie algebra consisting of nilpotent endomorphisms. If $M \neq 0$, then the common kernel of $\mathfrak{n}$ is nonzero. There is a basis of M in which all elements of $\mathfrak{n}$ are strictly upper triangular. In particular, $\mathfrak{n}$ is nilpotent and solvable.*

This is Engel's theorem in its matrix form [Hum72, §3.3]. The first assertion is proved by induction on $\dim\mathfrak{n}$, using that a maximal proper subalgebra is an ideal of codimension one. The triangular basis then follows by induction on $\dim M$.

Lemma F.6 (Cartan's criterion)**.** *Let M be a finite-dimensional Ω-vector space, and let $\mathfrak{a} \subseteq \operatorname{End}(M)$ be a Lie subalgebra satisfying*

$$\operatorname{tr}(XY) = 0 \qquad (X \in [\mathfrak{a}, \mathfrak{a}],\ Y \in \mathfrak{a}).$$

Then $\mathfrak{a}$ is solvable.

See [Hum72, §4.3]. It suffices to show that every element X of $[\mathfrak{a}, \mathfrak{a}]$ is nilpotent. Then $[\mathfrak{a}, \mathfrak{a}]$ is solvable by Lemma F.5, and hence so is $\mathfrak{a}$. To see that X is nilpotent, one pairs X with operators that are polynomials in its semisimple part. Let λ_i be the eigenvalues of the semisimple part, and let f be a $\mathbf{Q}$-linear functional on their $\mathbf{Q}$-span with values in $\mathbf{Q}$. The hypothesis gives $\sum_i \lambda_i f(\lambda_i) = 0$. Applying f to this identity gives $\sum_i f(\lambda_i)^2 = 0$. This is a vanishing sum of squares of rational numbers, so f vanishes on all λ_i, and $f = 0$. As f was arbitrary, all λ_i are zero. This step is where we use characteristic zero.

F.4 Reductive Lie algebras

For $\mathfrak{gl}_n$ the decomposition into scalars and trace-zero matrices, $\mathfrak{gl}_n = \Omega \cdot 1 \oplus \mathfrak{sl}_n$, has the form $\mathfrak{l} = Z(\mathfrak{l}) + [\mathfrak{l}, \mathfrak{l}]$. We prove this decomposition for every Lie algebra with a faithful semisimple representation (Proposition F.8), as Lemma 14.15 requires. The proof has two steps. First, the radical acts by scalars on each simple summand. To see this, we take a common eigenvector given by Lie's theorem. A trace computation, in which we divide by the dimension of a subspace, shows that its eigenspace is stable under all of $\mathfrak{l}$. Since the representation is faithful, the radical is then central. Second, the quotient by the center has zero radical. For such a Lie algebra the Killing form is nondegenerate by Cartan's criterion. This forces the algebra to equal its derived algebra (Lemma F.7). The upper triangular algebra $\mathfrak{b}$ of the previous section shows that the hypothesis of semisimplicity cannot be dropped.

Lemma F.7 (Killing form)**.** *If* $\operatorname{rad}\mathfrak{l} = 0$*, the Killing form* $B(x,y) = \operatorname{tr}(\operatorname{ad} x \operatorname{ad} y)$ *is nondegenerate and* $[\mathfrak{l},\mathfrak{l}] = \mathfrak{l}$*.*

This is the semisimplicity criterion of [Hum72, §§5.1–5.2]; we include the short deduction from Cartan's criterion.

Proof of Lemma F.7. We first use Cartan's criterion to show that the kernel of B is a solvable ideal, hence zero; nondegeneracy then identifies the orthogonal of the derived algebra with the center. By invariance of B, that is, $B([x,y],z) = B(x,[y,z])$, the kernel $\mathfrak{k}$ of B is an ideal. For $x \in [\mathfrak{k},\mathfrak{k}]$ and $y \in \mathfrak{k}$, the operators $\operatorname{ad} x$ and $\operatorname{ad} y$ map $\mathfrak{l}$ into $\mathfrak{k}$, so the trace of $\operatorname{ad} x \operatorname{ad} y$ may be computed on $\mathfrak{k}$. Since $x \in \mathfrak{k}$, we obtain

$$\operatorname{tr}_{\mathfrak{k}}(\operatorname{ad}(x)|_{\mathfrak{k}} \operatorname{ad}(y)|_{\mathfrak{k}}) = B(x,y) = 0.$$

By Cartan's criterion (Lemma F.6), the image of $\mathfrak{k}$ in $\operatorname{End}(\mathfrak{k})$ is solvable. The kernel of $\mathfrak{k} \to \operatorname{End}(\mathfrak{k})$ is the center of $\mathfrak{k}$, which is abelian. So $\mathfrak{k}$ is a solvable ideal of $\mathfrak{l}$. Since $\operatorname{rad}\mathfrak{l} = 0$, we get $\mathfrak{k} = 0$, that is, B is nondegenerate. Invariance and nondegeneracy then give

$$[\mathfrak{l},\mathfrak{l}]^{\perp} = Z(\mathfrak{l}).$$

The center is a solvable ideal, hence zero, and a subspace with zero orthogonal complement is all of $\mathfrak{l}$. ∎

For $\mathfrak{sl}_2$, with basis e, h, f, the brackets $[h,e] = 2e$, $[h,f] = -2f$ and $[e,f] = h$ already show that $[\mathfrak{sl}_2,\mathfrak{sl}_2] = \mathfrak{sl}_2$. The lemma gives the second half of the next proposition, once we know that the quotient by the center has zero radical.

Proposition F.8 (Lie algebras with a faithful semisimple representation)**.** *If* $\mathfrak{l}$ *has a faithful semisimple representation, then*

$$\operatorname{rad}\mathfrak{l} = Z(\mathfrak{l}), \qquad \mathfrak{l} = Z(\mathfrak{l}) + [\mathfrak{l},\mathfrak{l}].$$

Proof. We follow the two halves described at the start of this section: first the radical is central, and then the quotient by the center equals its derived algebra.

Put $\mathfrak{r} = \operatorname{rad}\mathfrak{l}$, and write the faithful module as a direct sum of simple modules. Let M be one of them. By Lemma F.4, there is a nonzero simultaneous λ-eigenspace M_λ for $\mathfrak{r}$. We claim that M_λ is $\mathfrak{l}$-stable. To prove this, fix $x \in \mathfrak{l}$ and $0 \neq v \in M_\lambda$, and consider $U = \operatorname{span}\{v, xv, x^2v, \ldots\}$. For every $a \in \mathfrak{r}$ we have $[a,x] \in \mathfrak{r}$, and the identity

$$ax^j = x^j a + \sum_{i=0}^{j-1} x^i [a,x] x^{j-1-i}$$

shows, by induction on j, that a preserves U and acts on it by a triangular matrix with constant diagonal $\lambda(a)$ in the basis $v, xv, x^2v, \ldots$ of U. Indeed, the identity gives $ax^jv = \lambda(a)x^jv$ plus terms $x^i[a,x]x^{j-1-i}v$, and by induction applied to $[a,x] \in \mathfrak{r}$ these lie in the span of $v, \ldots, x^{j-1}v$. With this, the trace of $[x,a]$ on U can be computed in two ways. Since x and a preserve U, the restriction of $[x,a]$ to U is a commutator, whose trace is zero. On the other hand, $[x,a] \in \mathfrak{r}$ acts on U with constant diagonal $\lambda([x,a])$. Comparing,

$$0 = \operatorname{tr}_U[x,a] = \dim(U)\lambda([x,a]),$$

and $\lambda([x,a]) = 0$ because Ω has characteristic zero. It follows that $a(xv) = \lambda(a)xv + \lambda([a,x])v = \lambda(a)xv$, which proves the claim.

Since M is simple, the claim gives $M_\lambda = M$. Thus $\mathfrak{r}$ acts on M by scalars, and in particular $[\mathfrak{l},\mathfrak{r}]$ acts as zero on M. This holds for every simple summand, and the module is faithful, so $[\mathfrak{l},\mathfrak{r}] = 0$, that is, $\mathfrak{r} \subseteq Z(\mathfrak{l})$. The reverse inclusion holds because the center is an abelian ideal.

For the second equality, we show that the quotient $\bar{\mathfrak{l}} = \mathfrak{l}/Z(\mathfrak{l})$ has zero radical and apply Lemma F.7. The inverse image of a solvable ideal of the quotient is a central extension of a solvable Lie algebra, hence solvable. So it is contained in $\mathfrak{r} = Z(\mathfrak{l})$. The lemma now gives $[\bar{\mathfrak{l}},\bar{\mathfrak{l}}] = \bar{\mathfrak{l}}$. In other words, every element of $\mathfrak{l}$ is congruent modulo $Z(\mathfrak{l})$ to a sum of brackets, and $\mathfrak{l} = Z(\mathfrak{l}) + [\mathfrak{l},\mathfrak{l}]$. ■

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